SWAFL Desktop 1.1.0 — User Guide and Technical Reference

Volume 3 of 3: screenshot atlas by soil and by crop — every layer for each of the ten soils, the three designs and their virtual farms for each soil, and the crop water requirement of every crop

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Version 1.1.0

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About this volume

Each chapter of the first part sets one of the ten soils in Properties and shows every layer, drawn from top to bottom, with the other settings at their defaults; the three irrigation designs are shown after their run with their inputs at the defaults, and the virtual farm of each design on an irrigation day. The equations of each layer are printed once, in Volume 2. The second part shows the crop water requirement for every crop of the library on the Coarse Loam.

Chapter 1

Sand

1.1 Surface irrigation design (Sand)

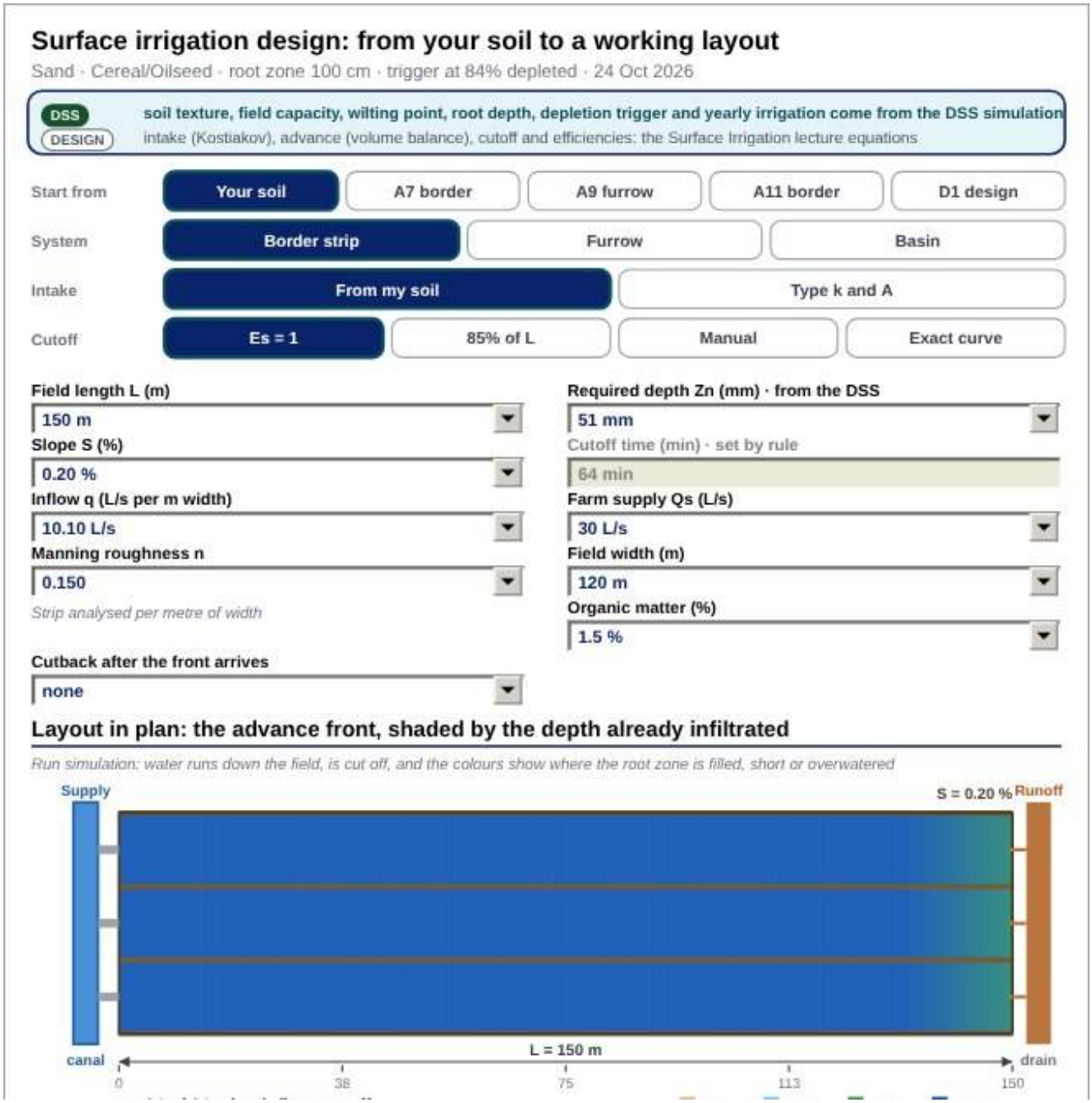


Figure 1.1: Surface irrigation design, Sand, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

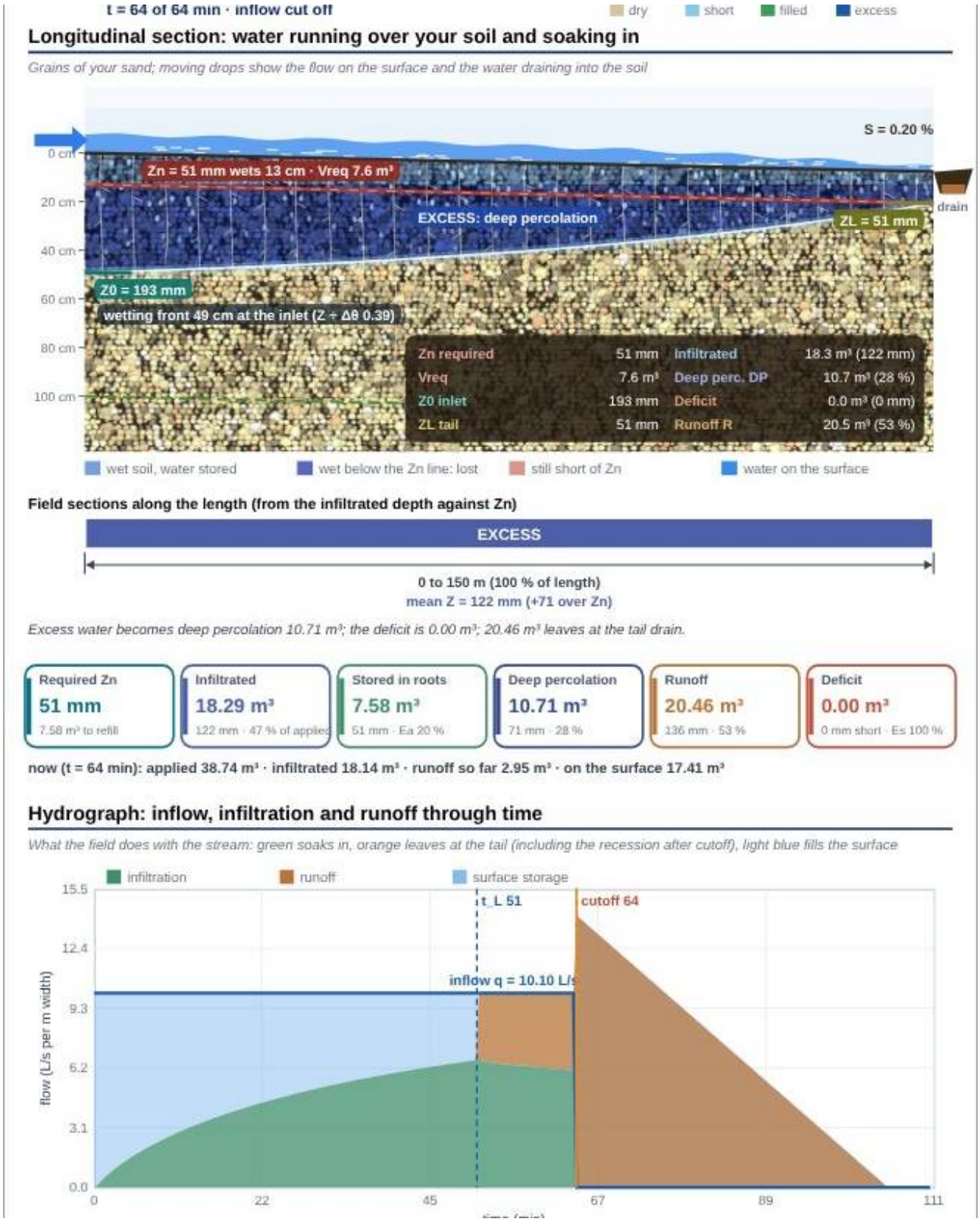


Figure 1.2: Surface irrigation design, Sand, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

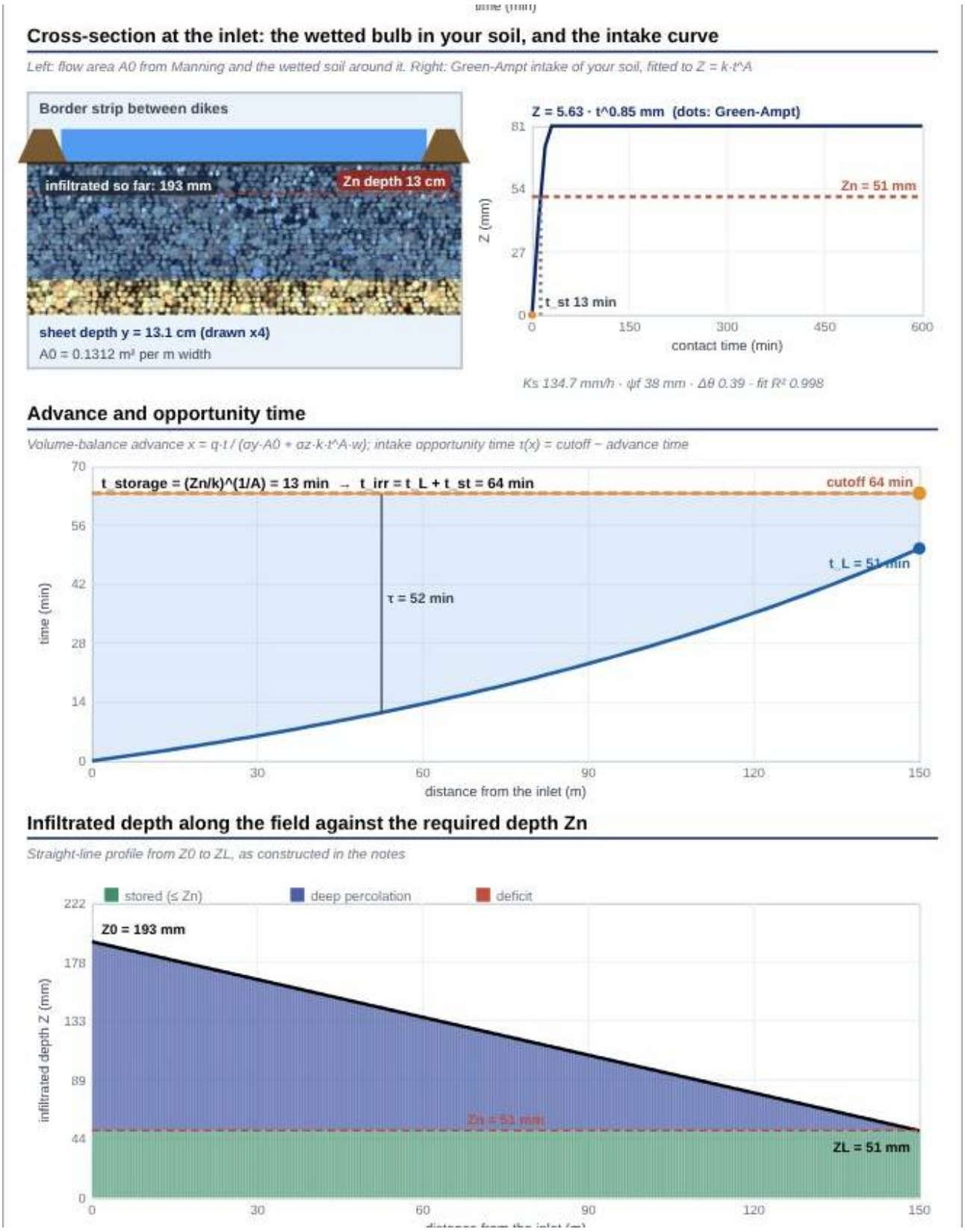


Figure 1.3: Surface irrigation design, Sand, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

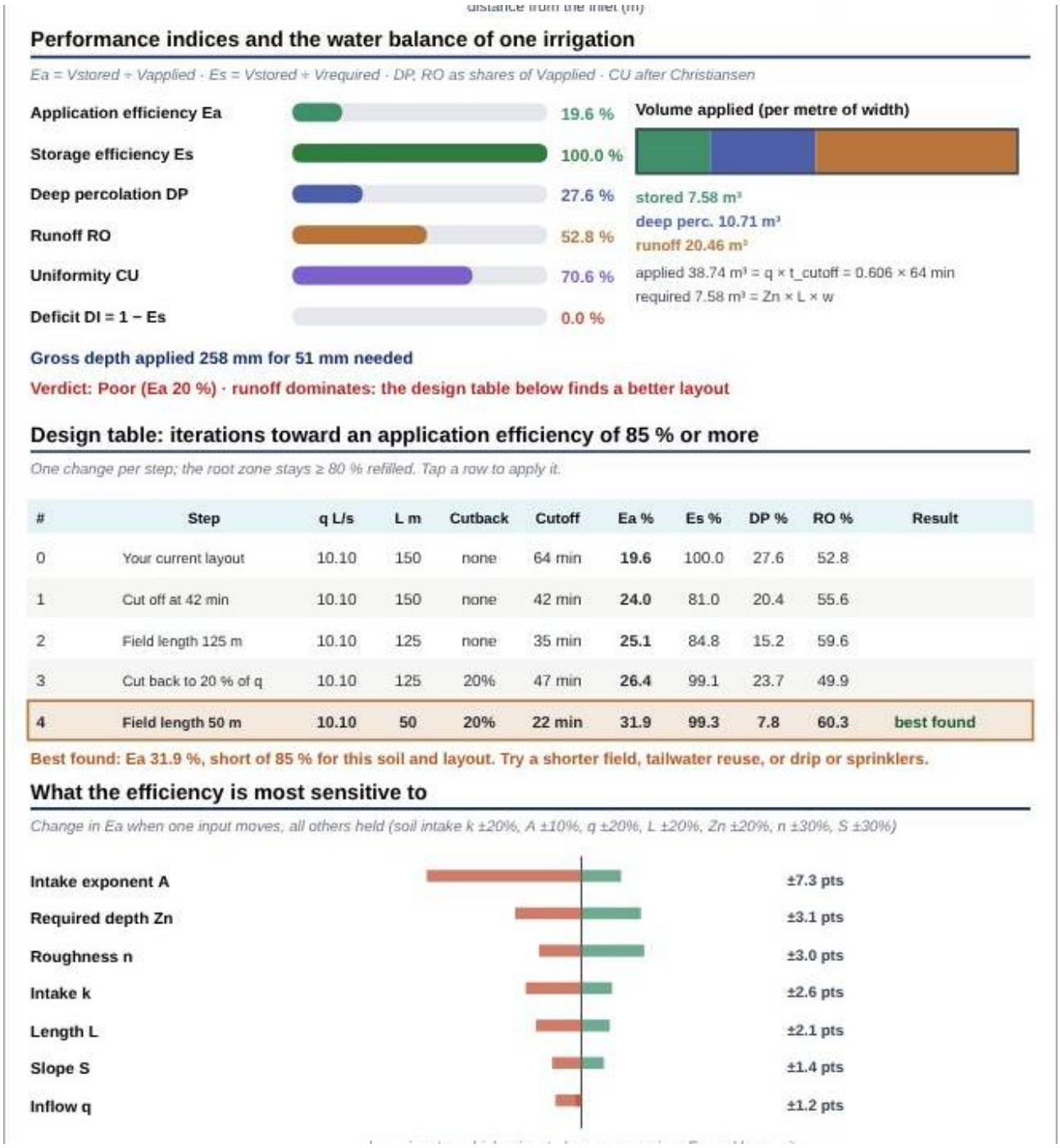


Figure 1.4: Surface irrigation design, Sand, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

Figure 1.5: Surface irrigation design, Sand, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

1.2 Drip irrigation design (Sand)

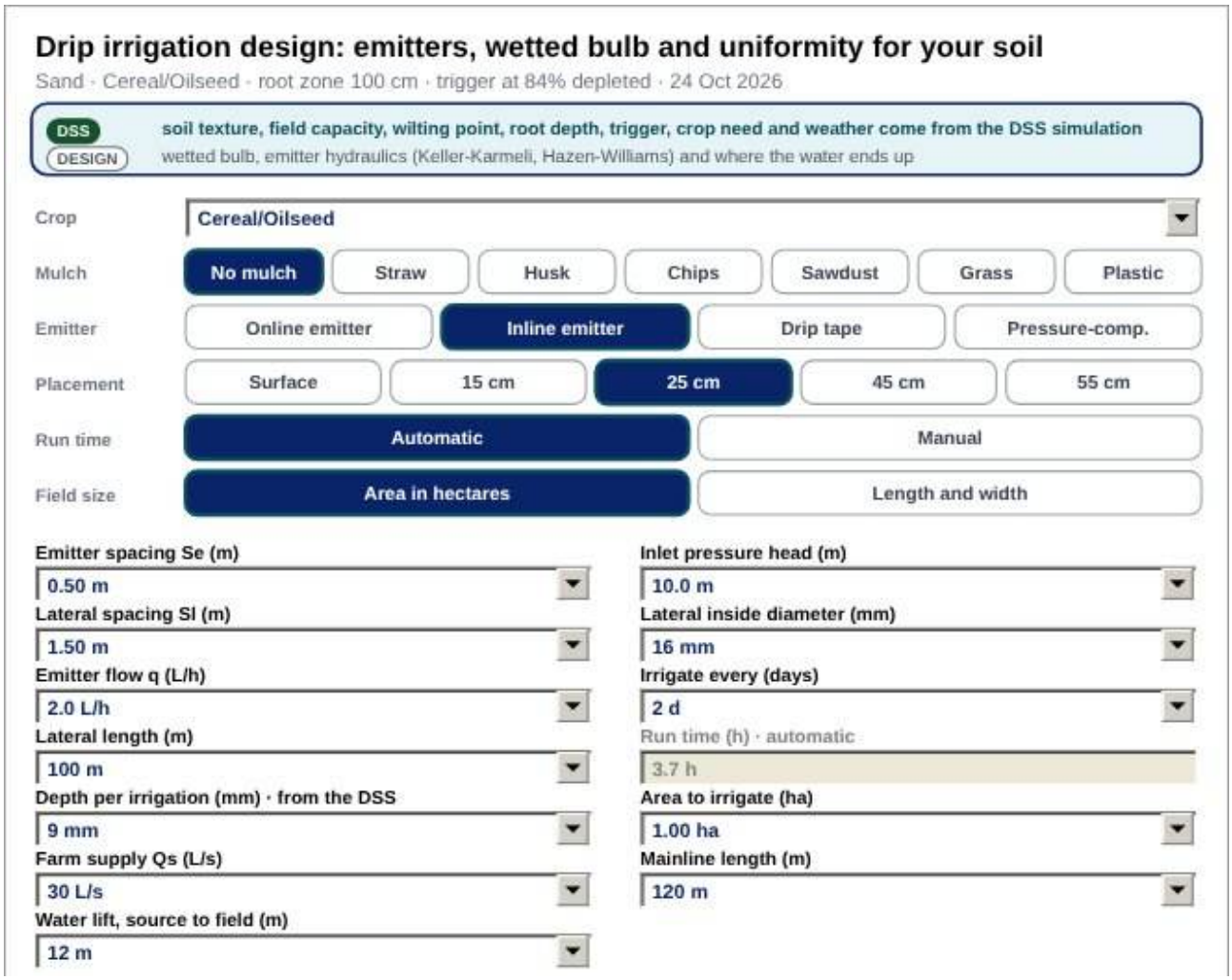


Figure 1.6: Drip irrigation design, Sand, part 1 of 6.

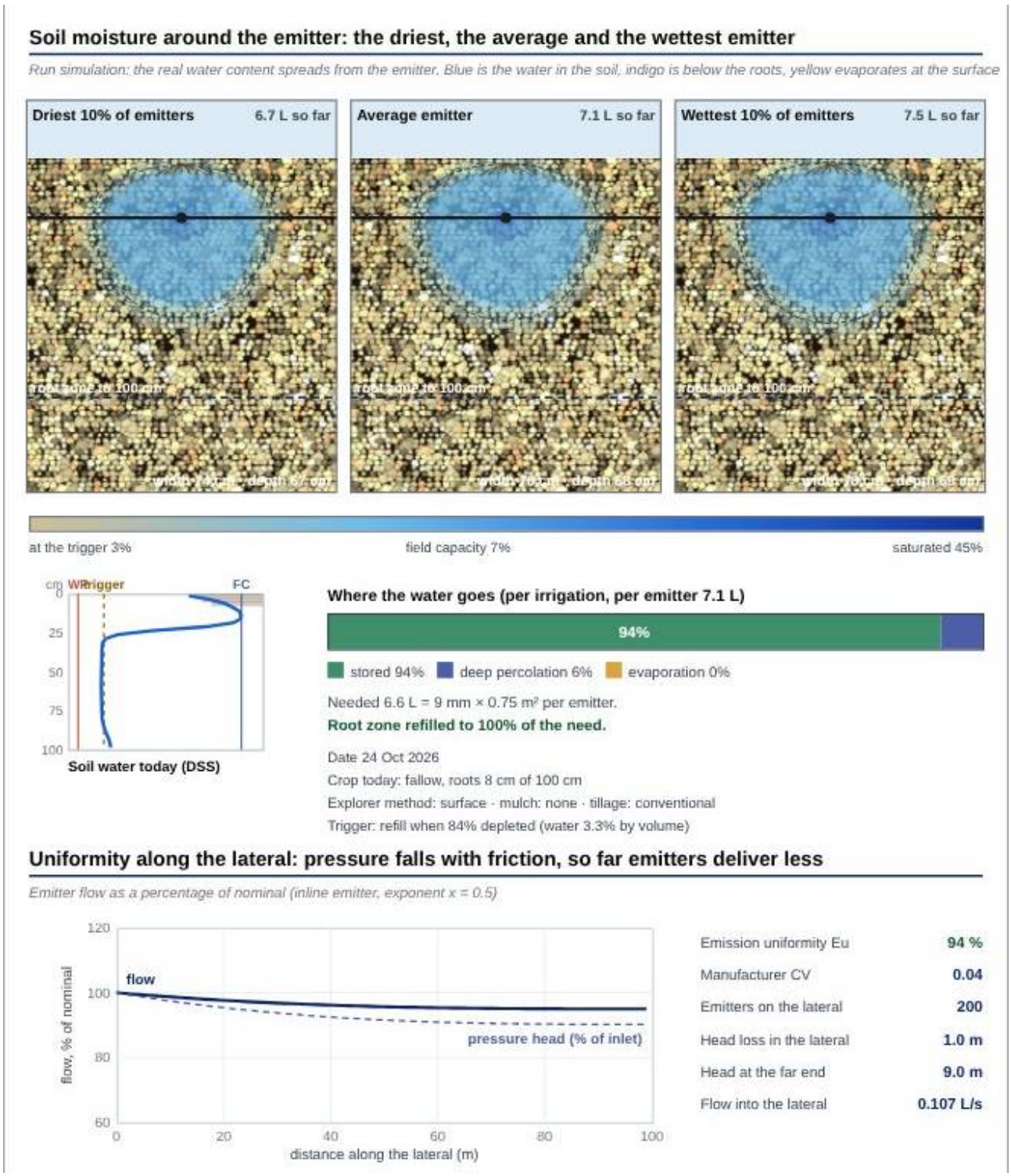


Figure 1.7: Drip irrigation design, Sand, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

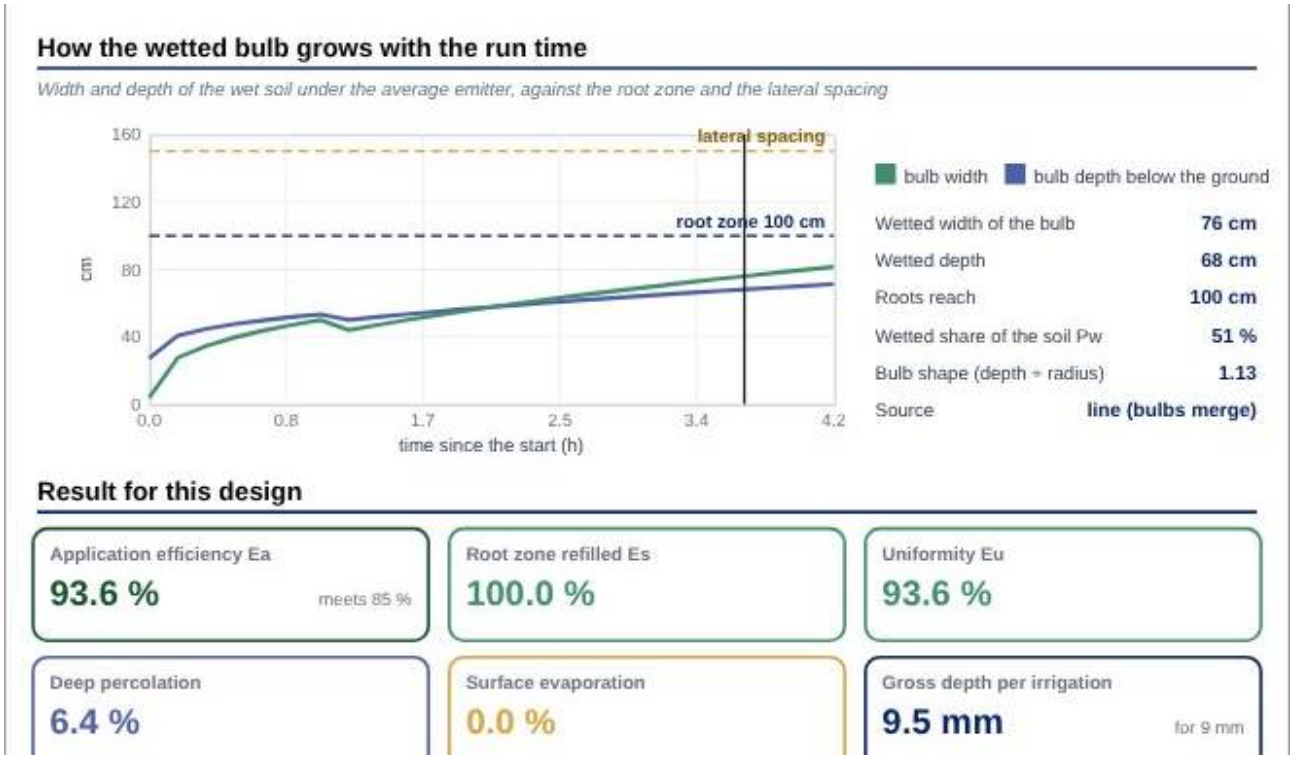
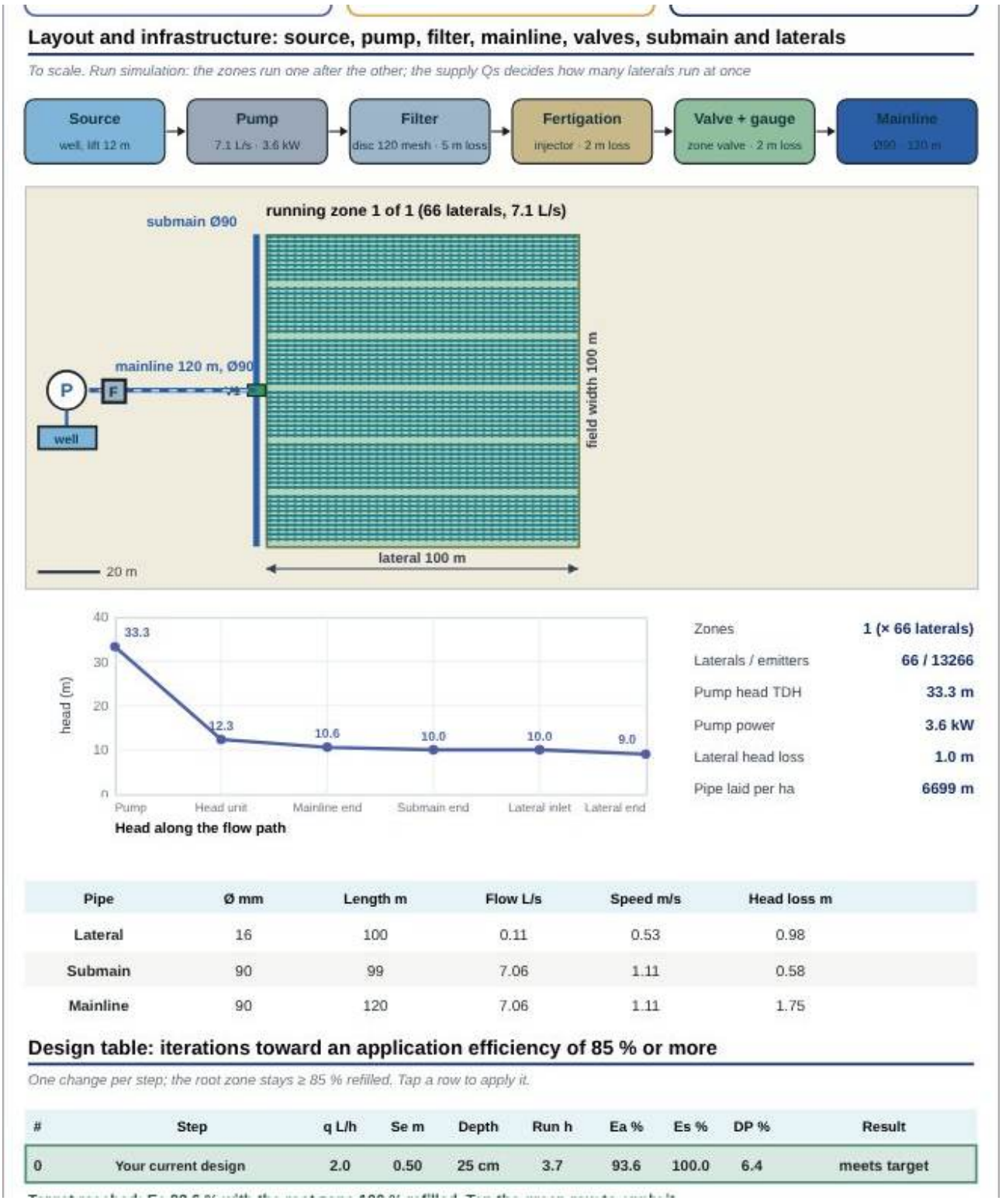


Figure 1.8: Drip irrigation design, Sand, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.7	93.6	100.0	6.4	meets target

Figure 1.9: Drip irrigation design, Sand, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

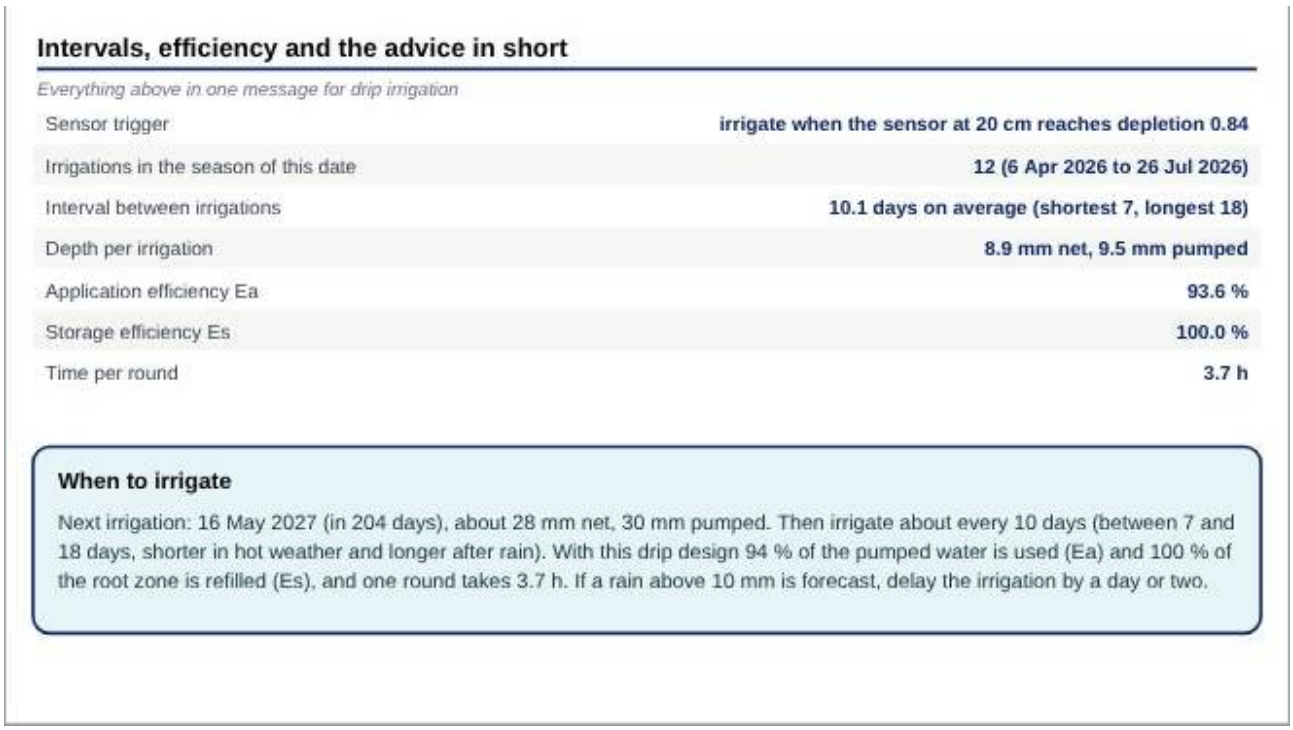
Net irrigation need in a season (DSS)	**334 mm → 38 irrigations of 9 mm**
Gross water per irrigation	**9.5 mm (Ea 94 %)**
Gross water per season	**357 mm = 3565 m³ per hectare**
Peak crop need / interval	**4.4 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.7 h = 3.7 h per round**
Pumping per season	**139 hours · 493 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	28	30	3.7 h	17 days
2	**2 Jun 2027**	221 days	28	30	3.7 h	12 days
3	**14 Jun 2027**	233 days	28	30	3.7 h	9 days
4	**23 Jun 2027**	242 days	28	30	3.7 h	12 days
5	**5 Jul 2027**	254 days	28	30	3.7 h	9 days
6	**14 Jul 2027**	263 days	28	30	3.7 h	13 days
7	**27 Jul 2027**	276 days	28	30	3.7 h	253 days
8	**5 Apr 2028**	529 days	28	30	3.7 h	-

Figure 1.10: Drip irrigation design, Sand, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.84**
Irrigations in the season of this date	**12 (6 Apr 2026 to 26 Jul 2026)**
Interval between irrigations	**10.1 days on average (shortest 7, longest 18)**
Depth per irrigation	**8.9 mm net, 9.5 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**3.7 h**

When to irrigate

Next irrigation: 16 May 2027 (in 204 days), about 28 mm net, 30 mm pumped. Then irrigate about every 10 days (between 7 and 18 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 3.7 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 1.11: Drip irrigation design, Sand, part 6 of 6: Intervals, efficiency and the advice in short.

1.3 Sprinkler design and pipes (Sand)

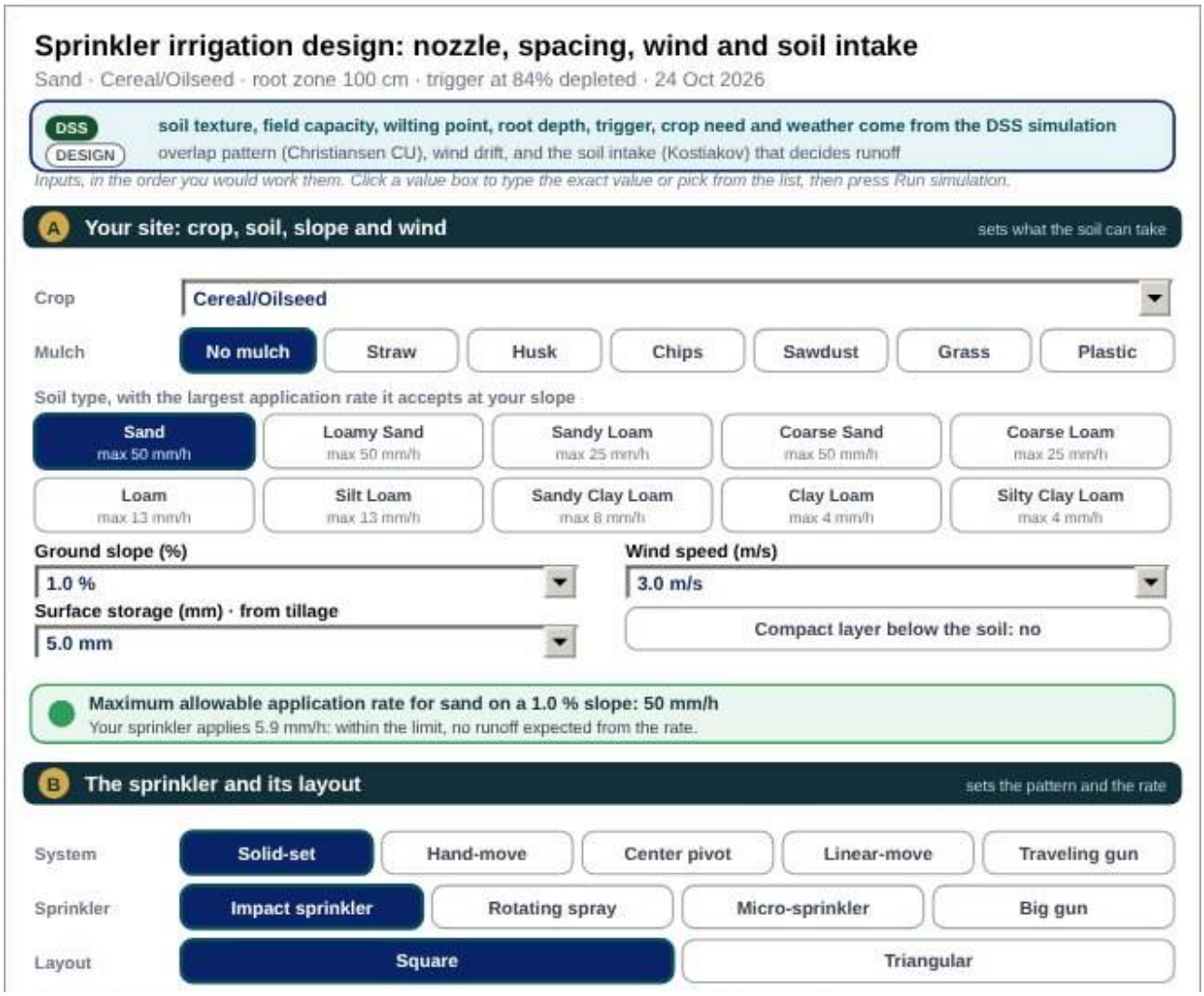


Figure 1.12: Sprinkler design and pipes, Sand, part 1 of 12.

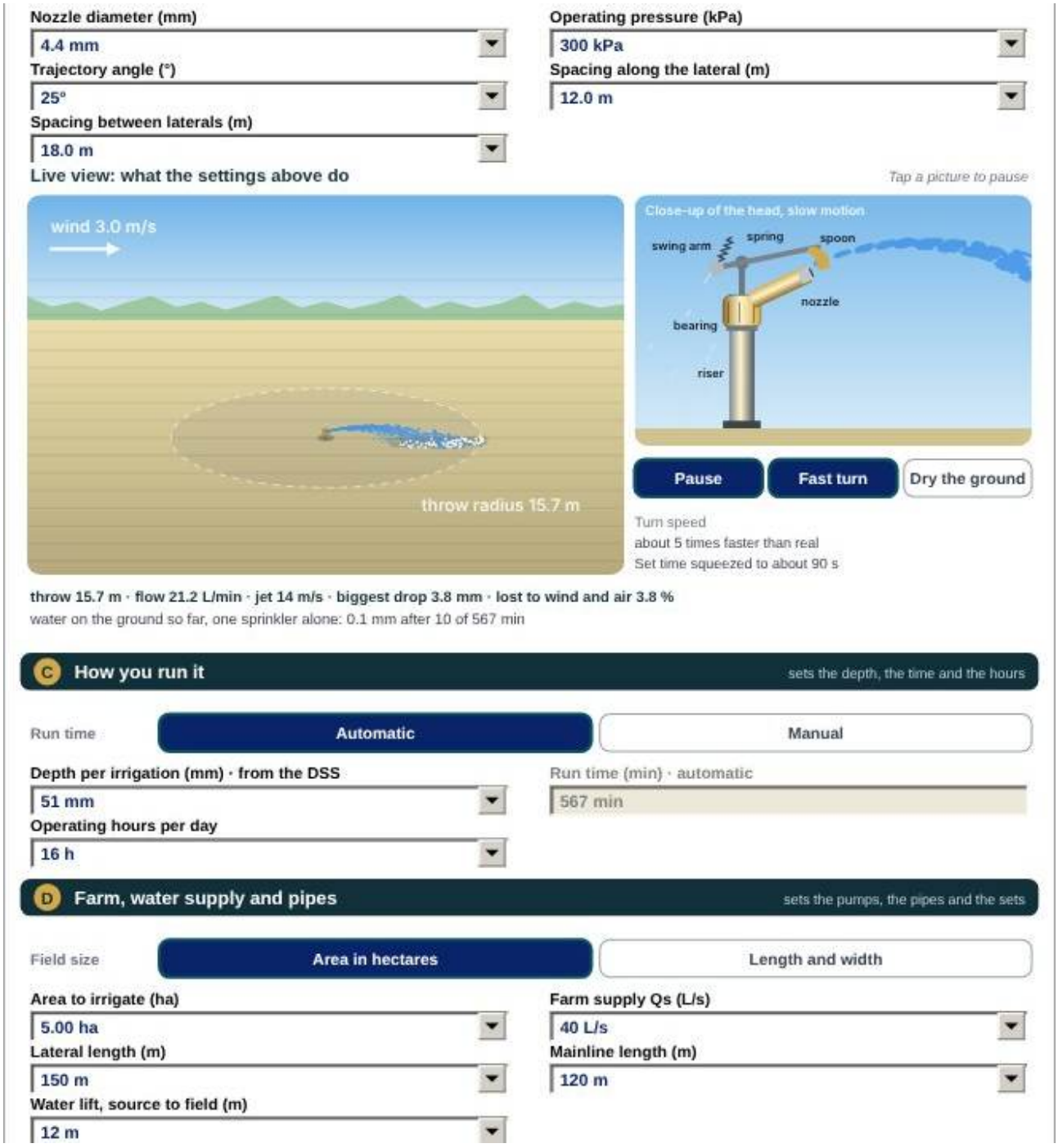


Figure 1.13: Sprinkler design and pipes, Sand, part 2 of 12.

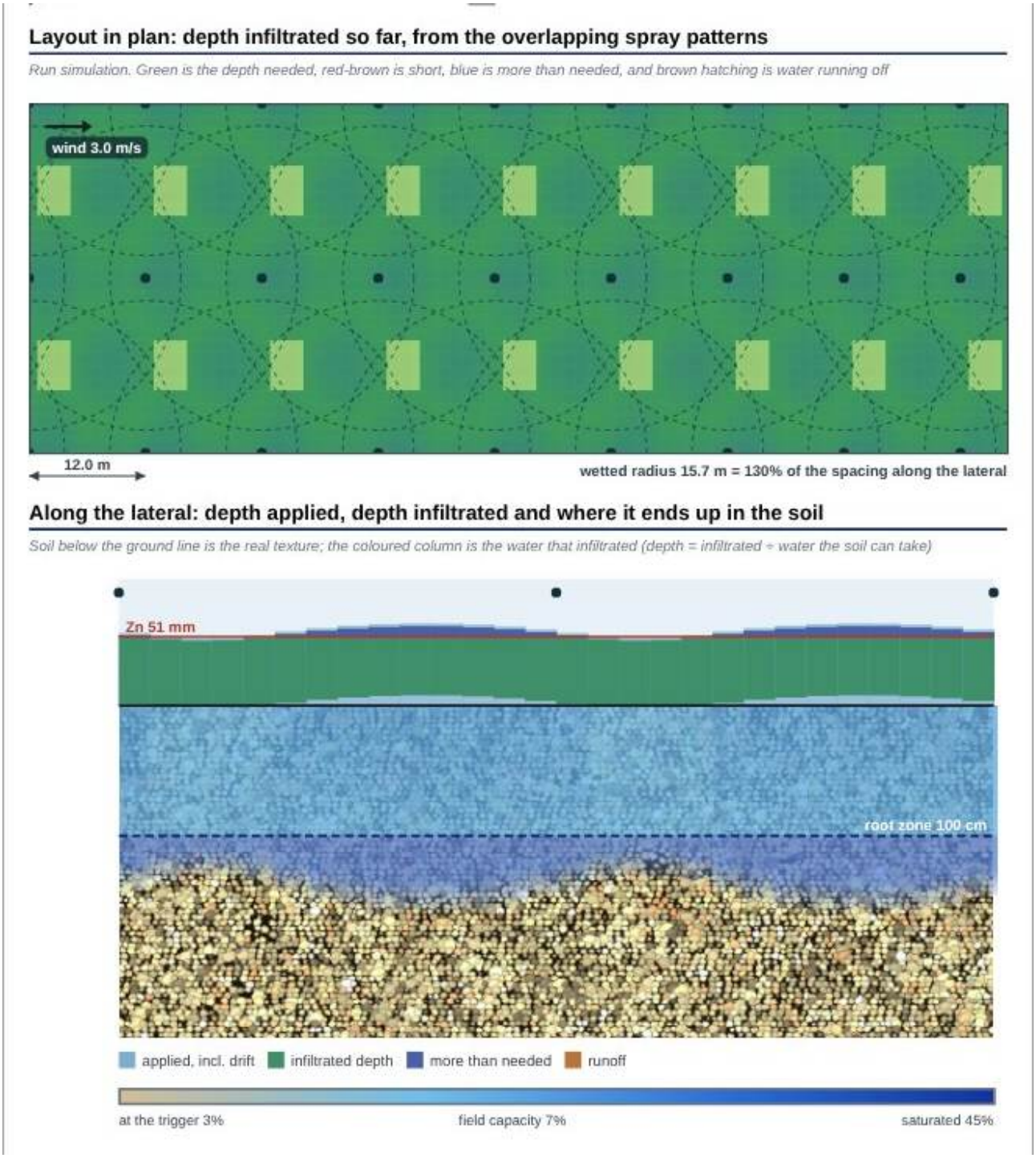


Figure 1.14: Sprinkler design and pipes, Sand, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

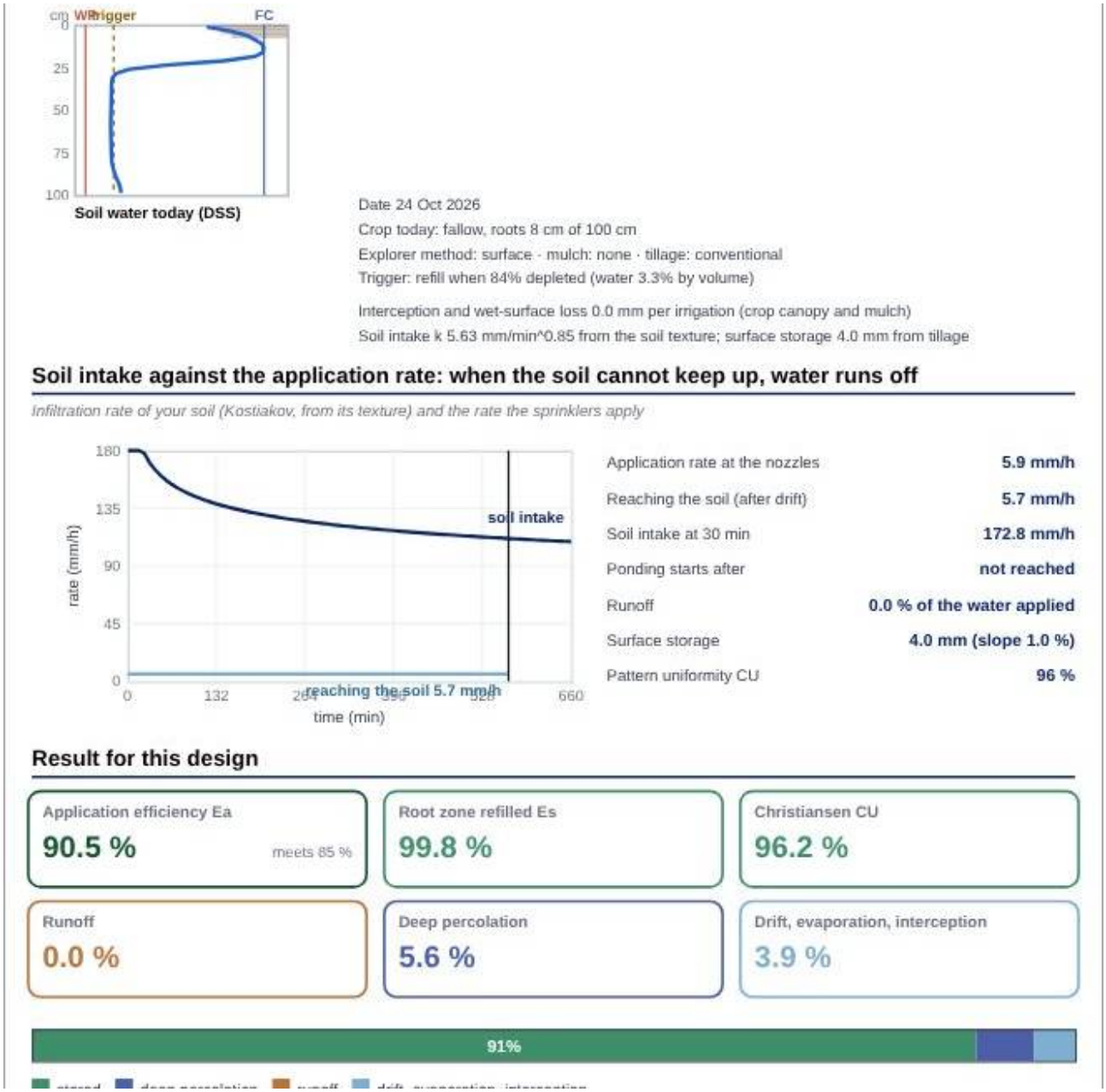


Figure 1.15: Sprinkler design and pipes, Sand, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

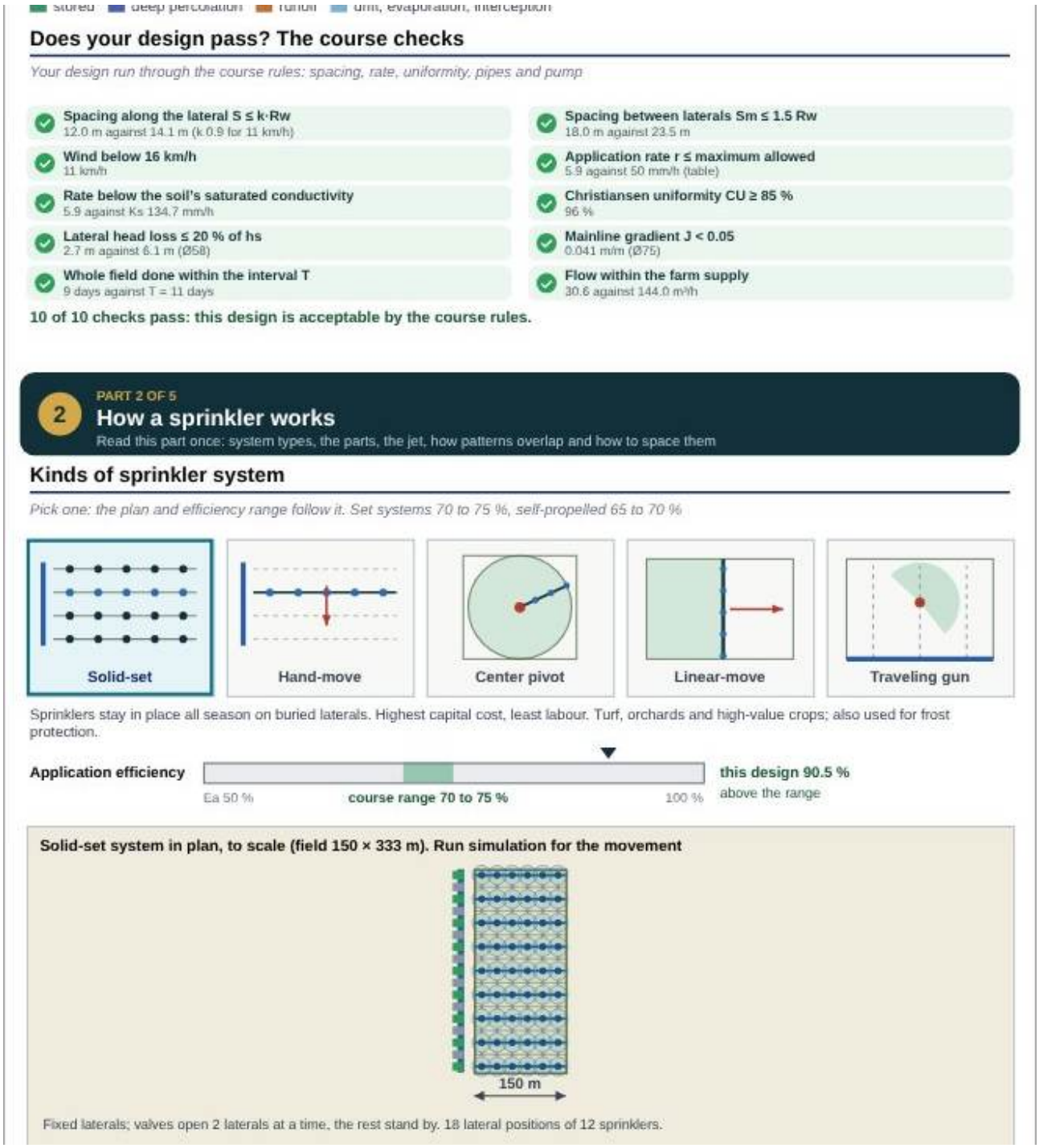


Figure 1.16: Sprinkler design and pipes, Sand, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

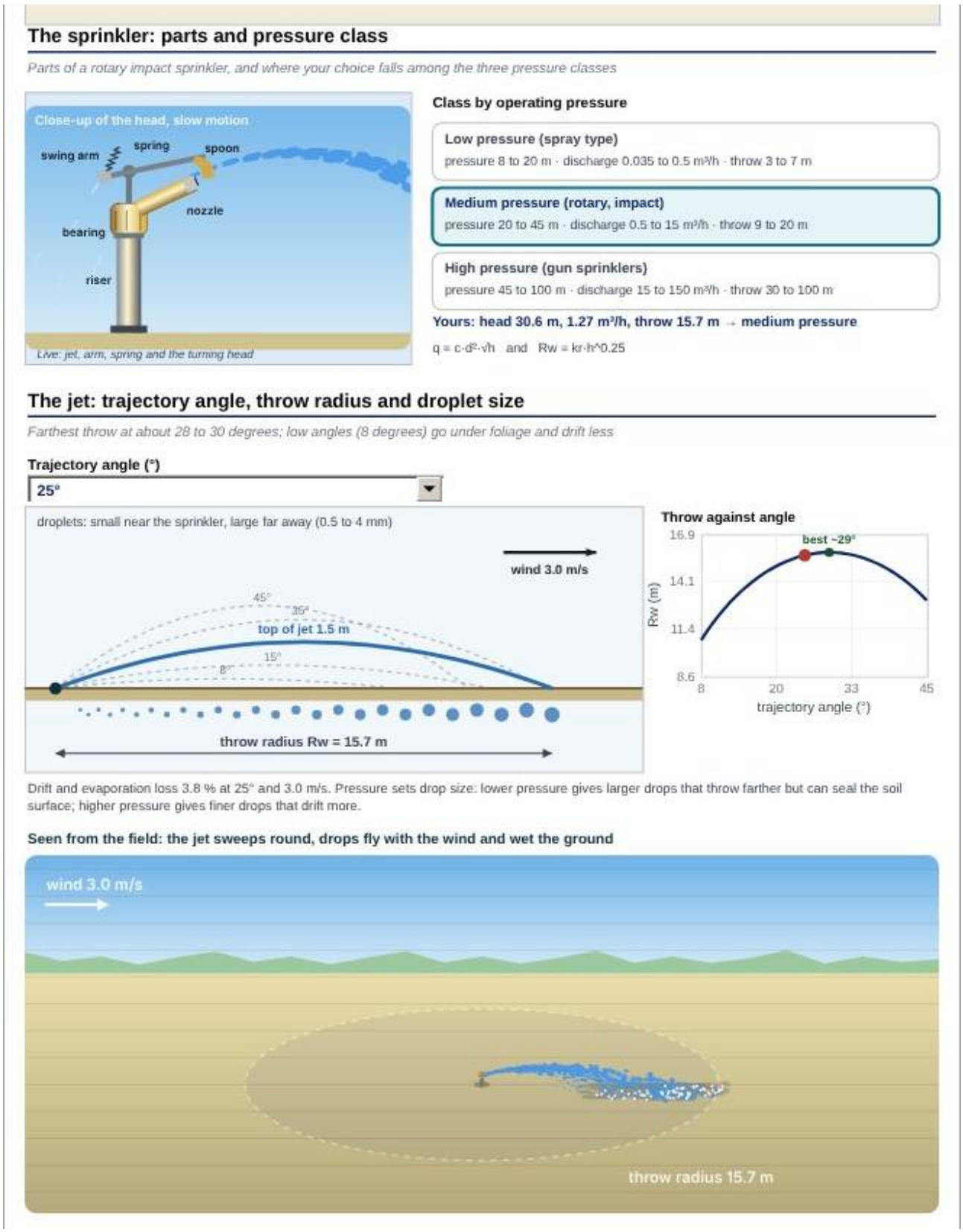


Figure 1.17: Sprinkler design and pipes, Sand, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

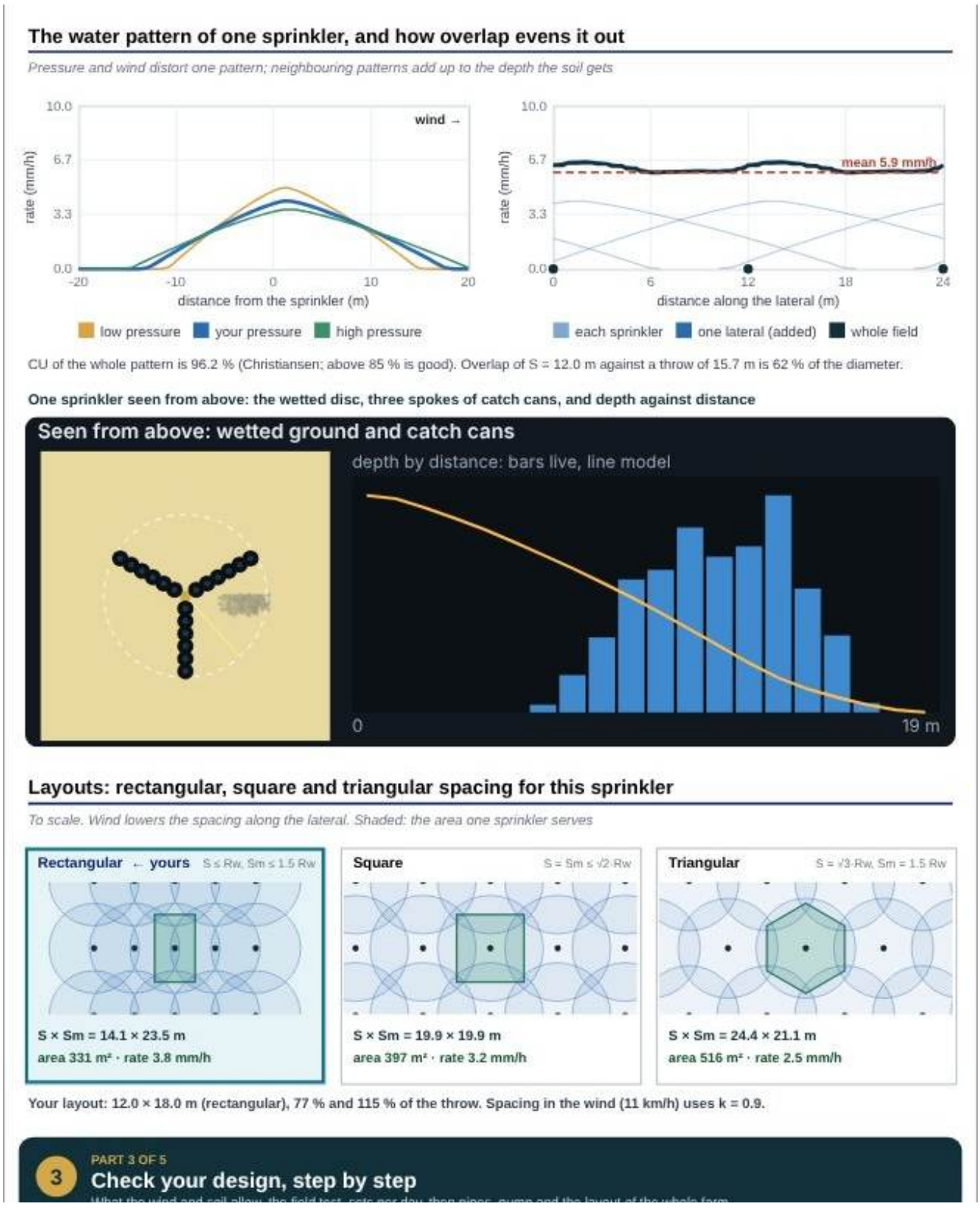


Figure 1.18: Sprinkler design and pipes, Sand, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

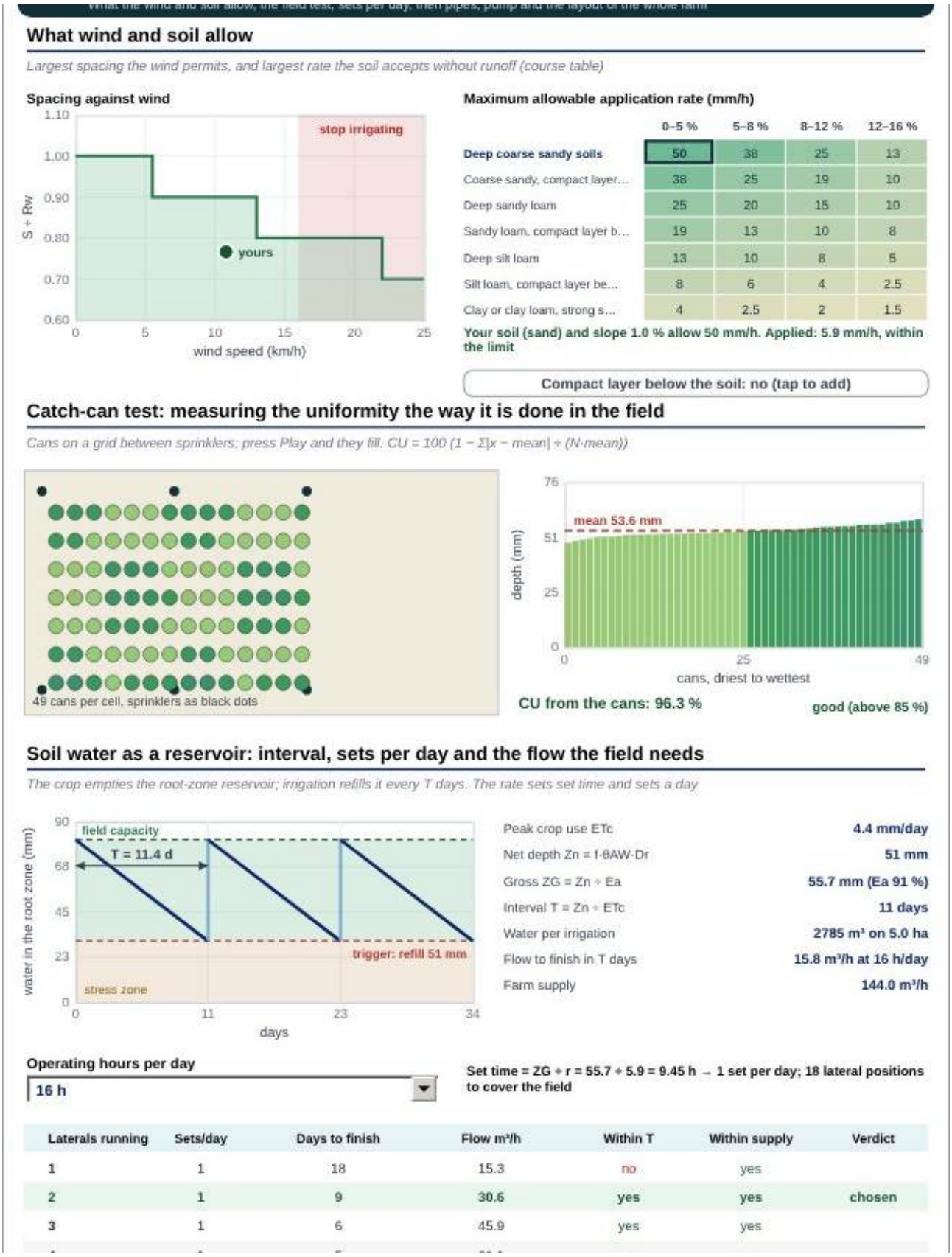


Figure 1.19: Sprinkler design and pipes, Sand, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

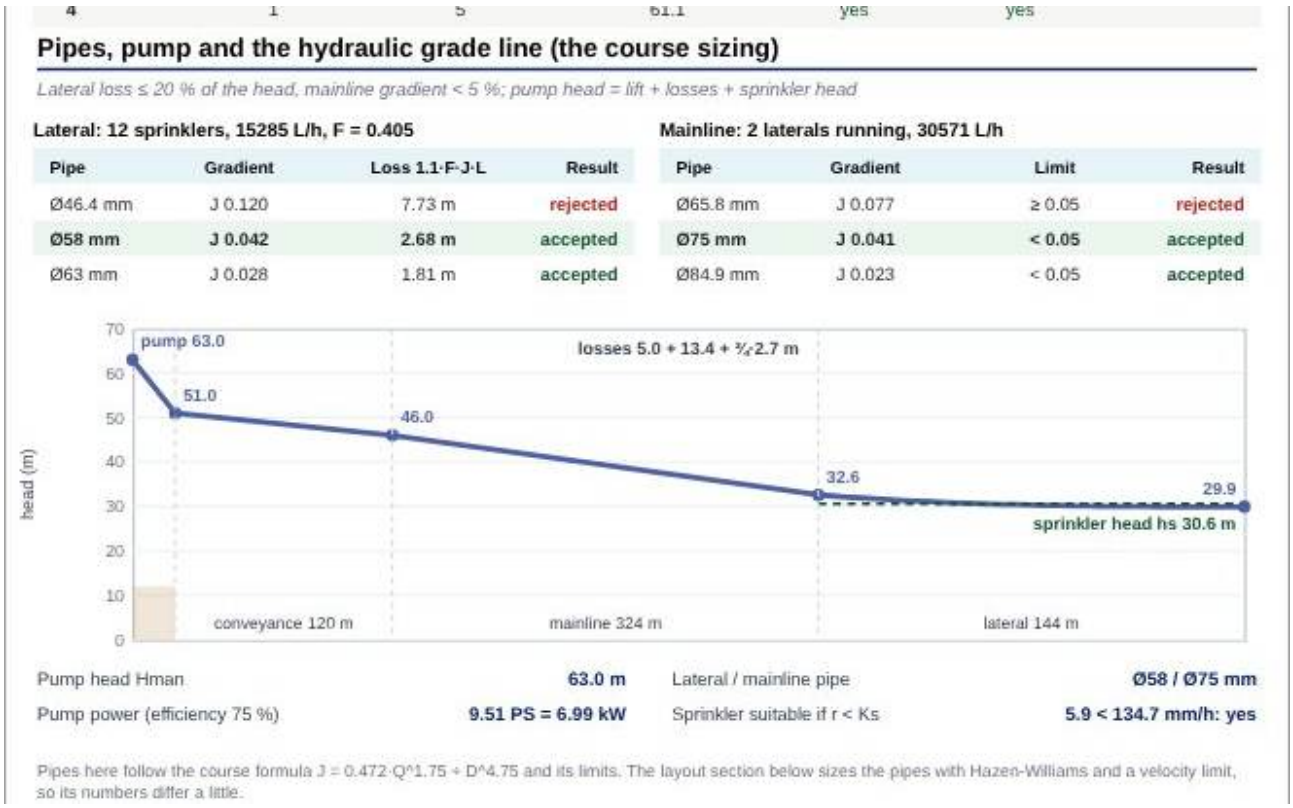
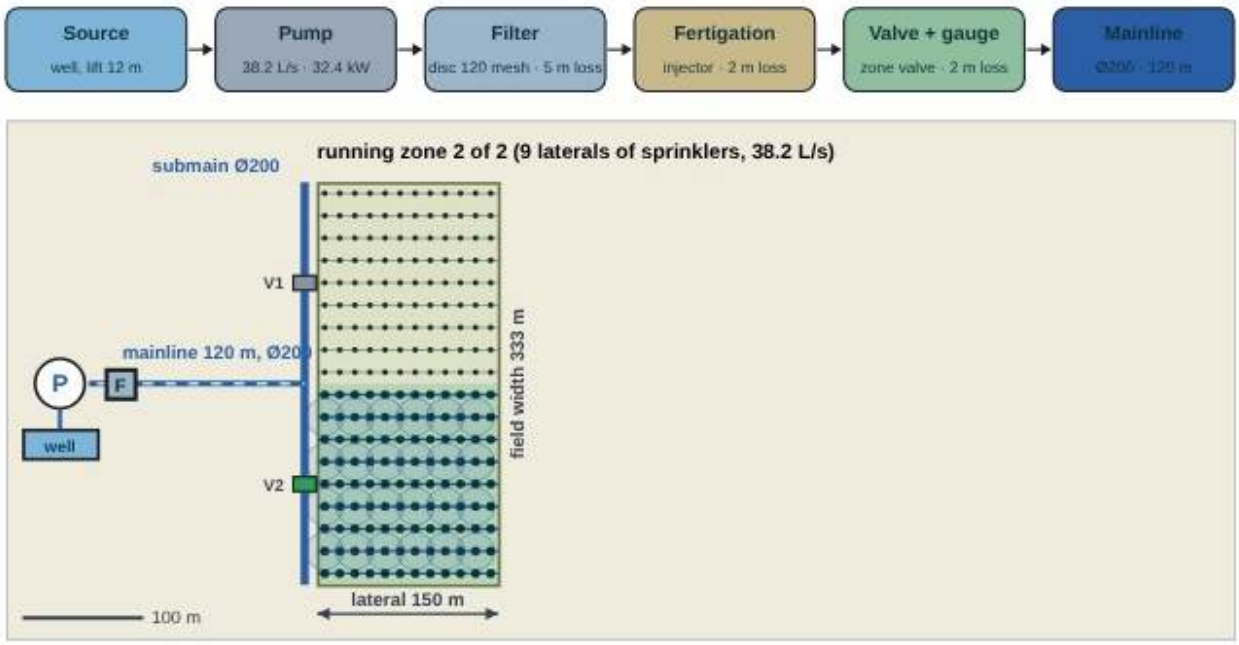
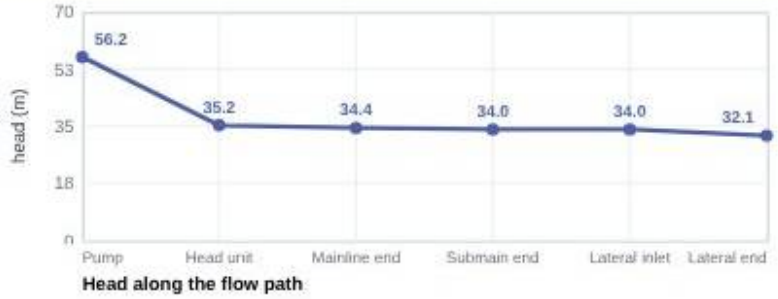


Figure 1.20: Sprinkler design and pipes, Sand, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 **Improve it**

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

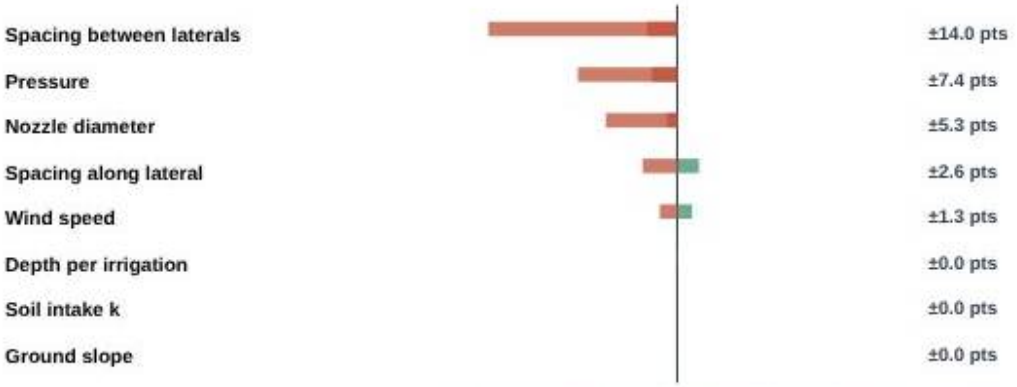
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	567	90.5	99.8	0.0	meets target

Figure 1.21: Sprinkler design and pipes, Sand, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

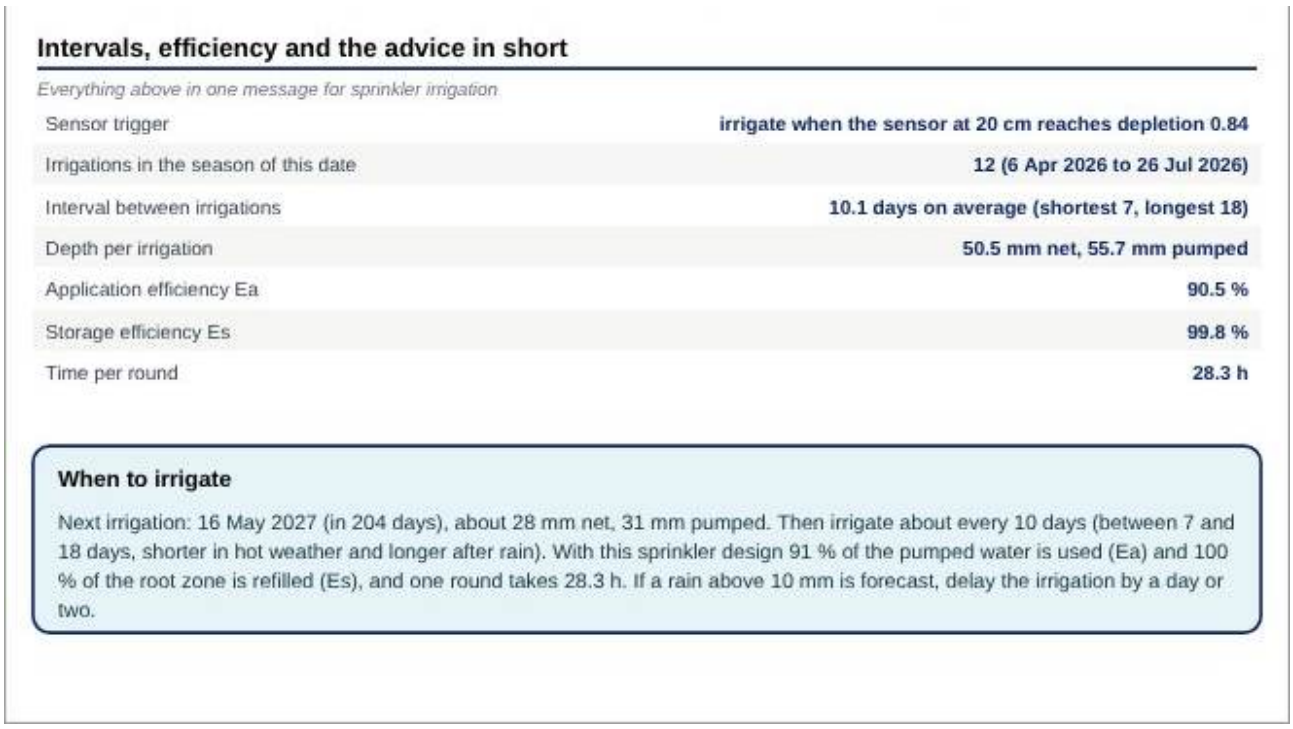
Net irrigation need in a season (DSS)	**334 mm → 7 irrigations of 51 mm**
Gross water per irrigation	**55.7 mm (Ea 91 %)**
Gross water per season	**368 mm = 3679 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 567 min = 28.3 h per round**
Pumping per season	**187 hours · 6072 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	28	31	28.3 h	17 days
2	**2 Jun 2027**	221 days	28	31	28.3 h	12 days
3	**14 Jun 2027**	233 days	28	31	28.3 h	9 days
4	**23 Jun 2027**	242 days	28	31	28.3 h	12 days
5	**5 Jul 2027**	254 days	28	31	28.3 h	9 days
6	**14 Jul 2027**	263 days	28	31	28.3 h	13 days
7	**27 Jul 2027**	276 days	28	31	28.3 h	253 days
8	**5 Apr 2028**	529 days	28	31	28.3 h	-

Figure 1.22: Sprinkler design and pipes, Sand, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.84**
Irrigations in the season of this date	**12 (6 Apr 2026 to 26 Jul 2026)**
Interval between irrigations	**10.1 days on average (shortest 7, longest 18)**
Depth per irrigation	**50.5 mm net, 55.7 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**28.3 h**

When to irrigate

Next irrigation: 16 May 2027 (in 204 days), about 28 mm net, 31 mm pumped. Then irrigate about every 10 days (between 7 and 18 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 28.3 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 1.23: Sprinkler design and pipes, Sand, part 12 of 12: Intervals, efficiency and the advice in short.

1.4 Virtual farm of the surface design (Sand)

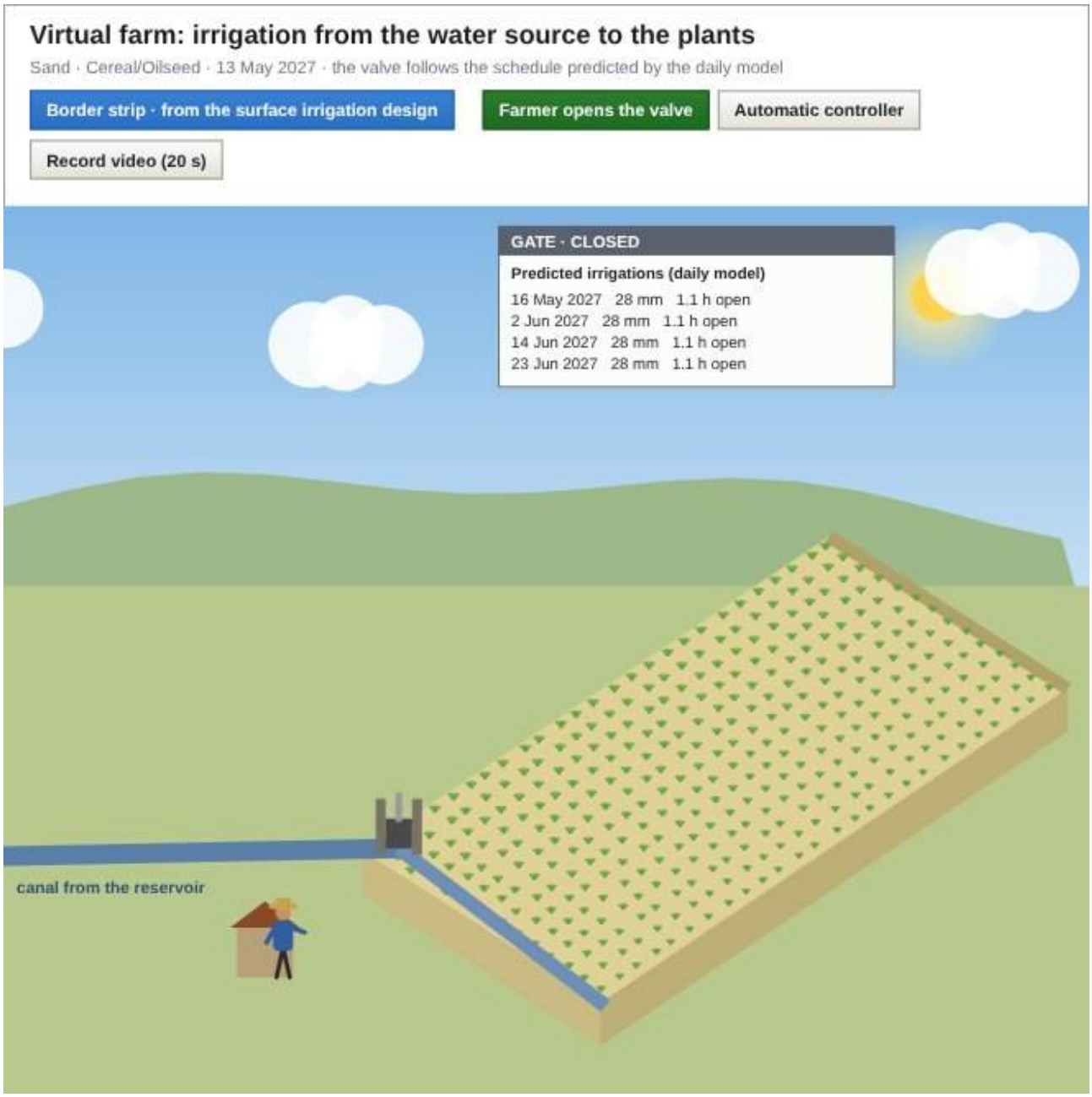


Figure 1.24: Virtual farm, surface design, Sand, part 1 of 3.

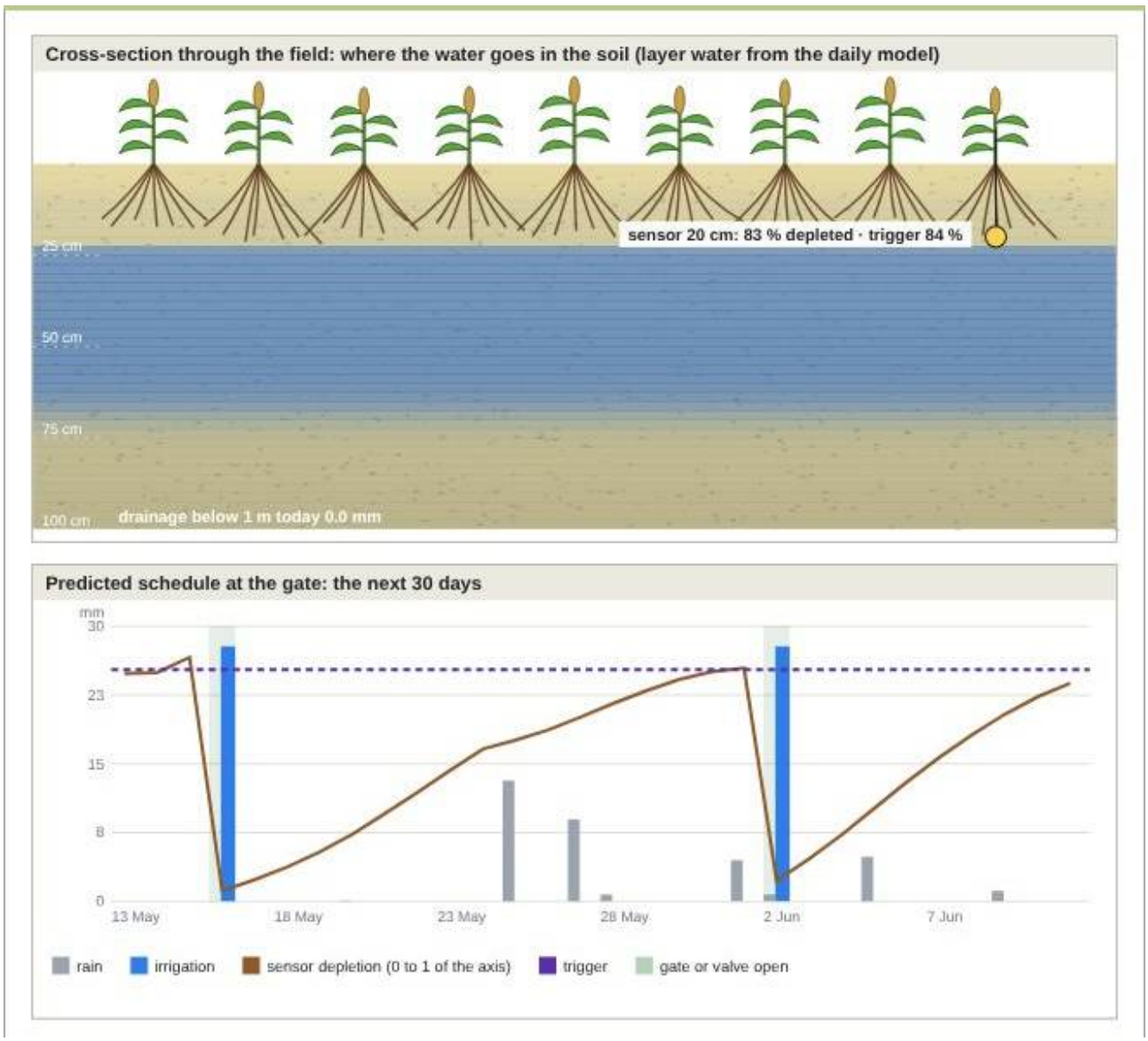


Figure 1.25: Virtual farm, surface design, Sand, part 2 of 3.

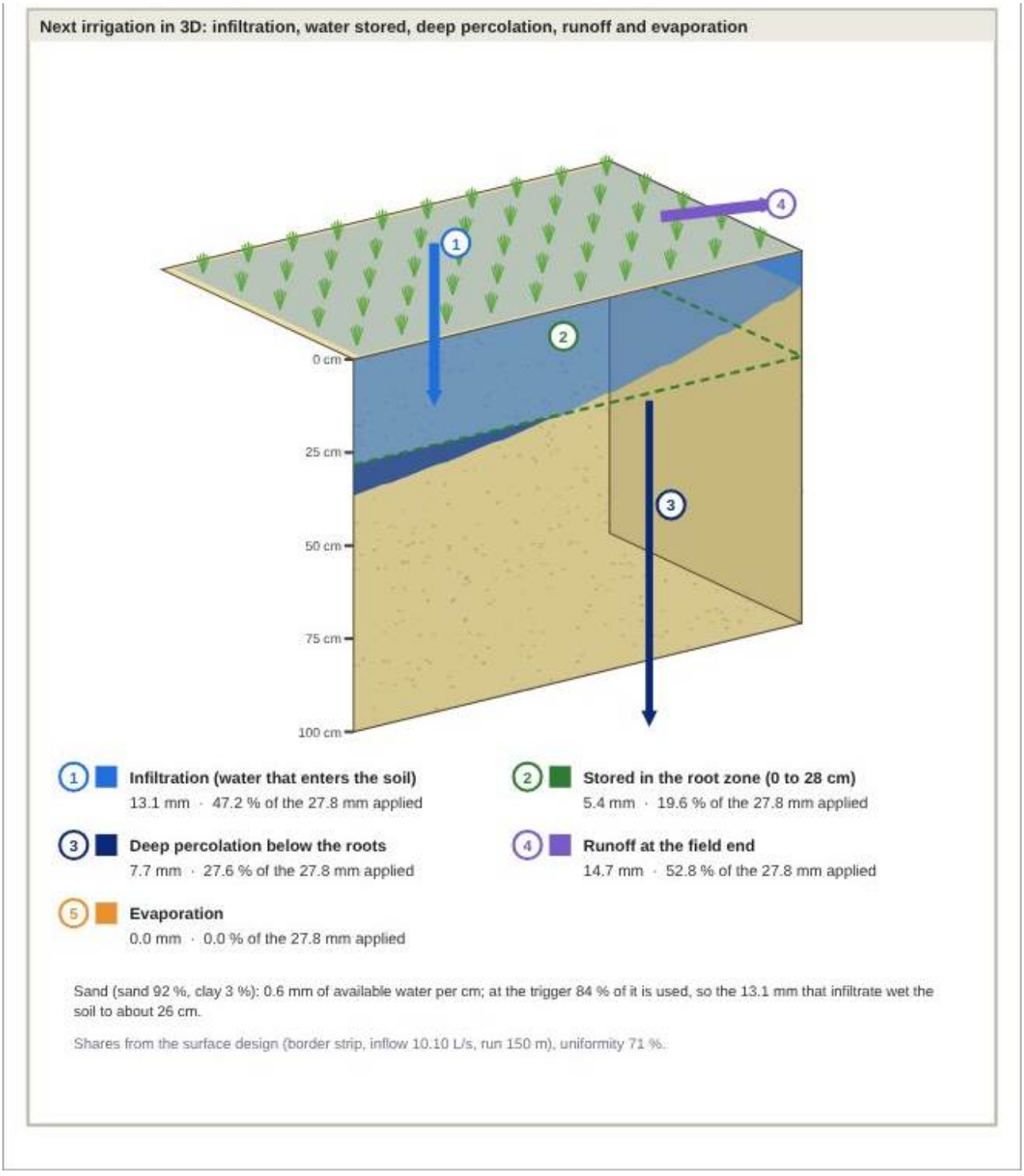


Figure 1.26: Virtual farm, surface design, Sand, part 3 of 3.

1.5 Virtual farm of the drip design (Sand)

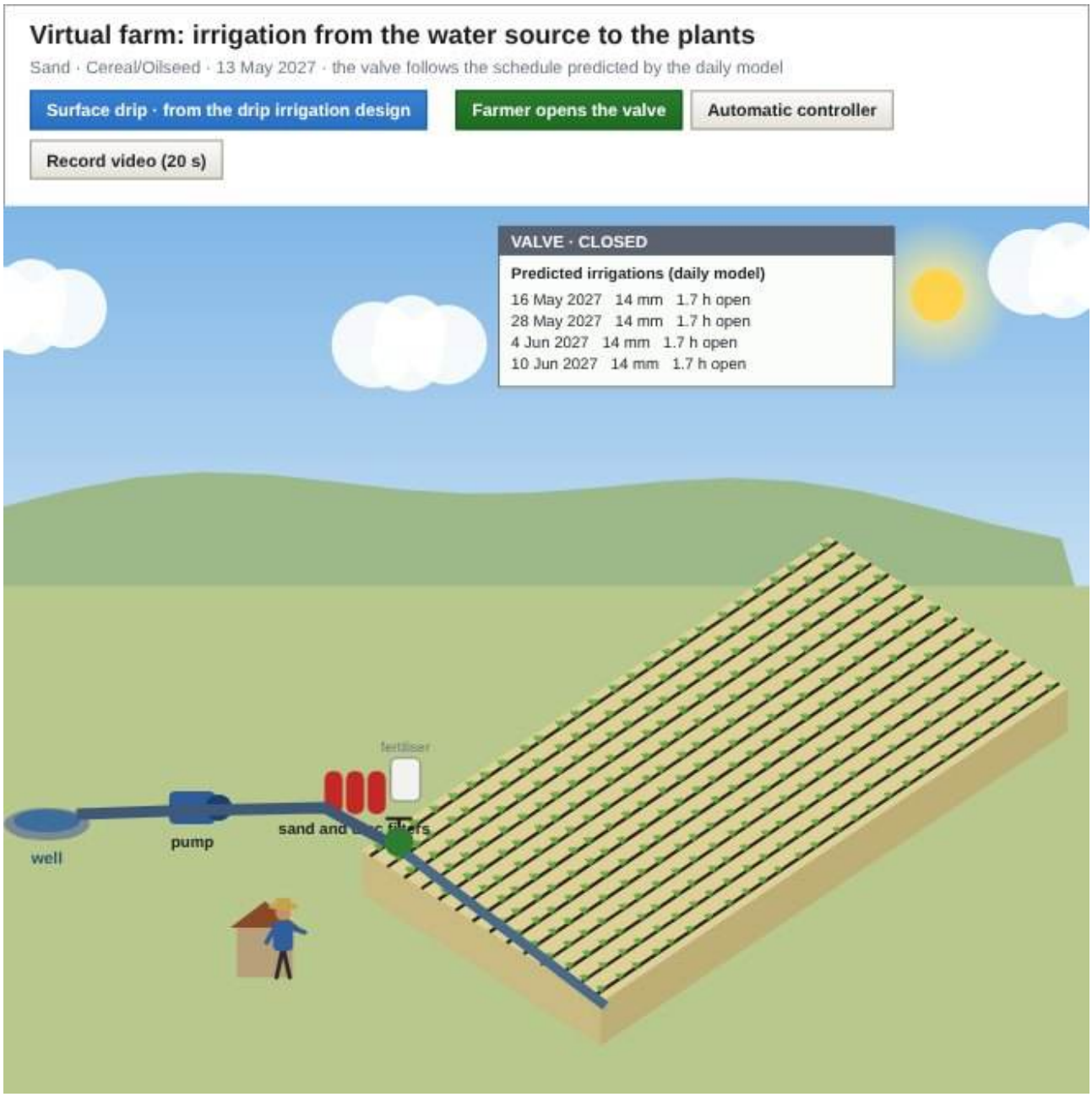


Figure 1.27: Virtual farm, drip design, Sand, part 1 of 3.

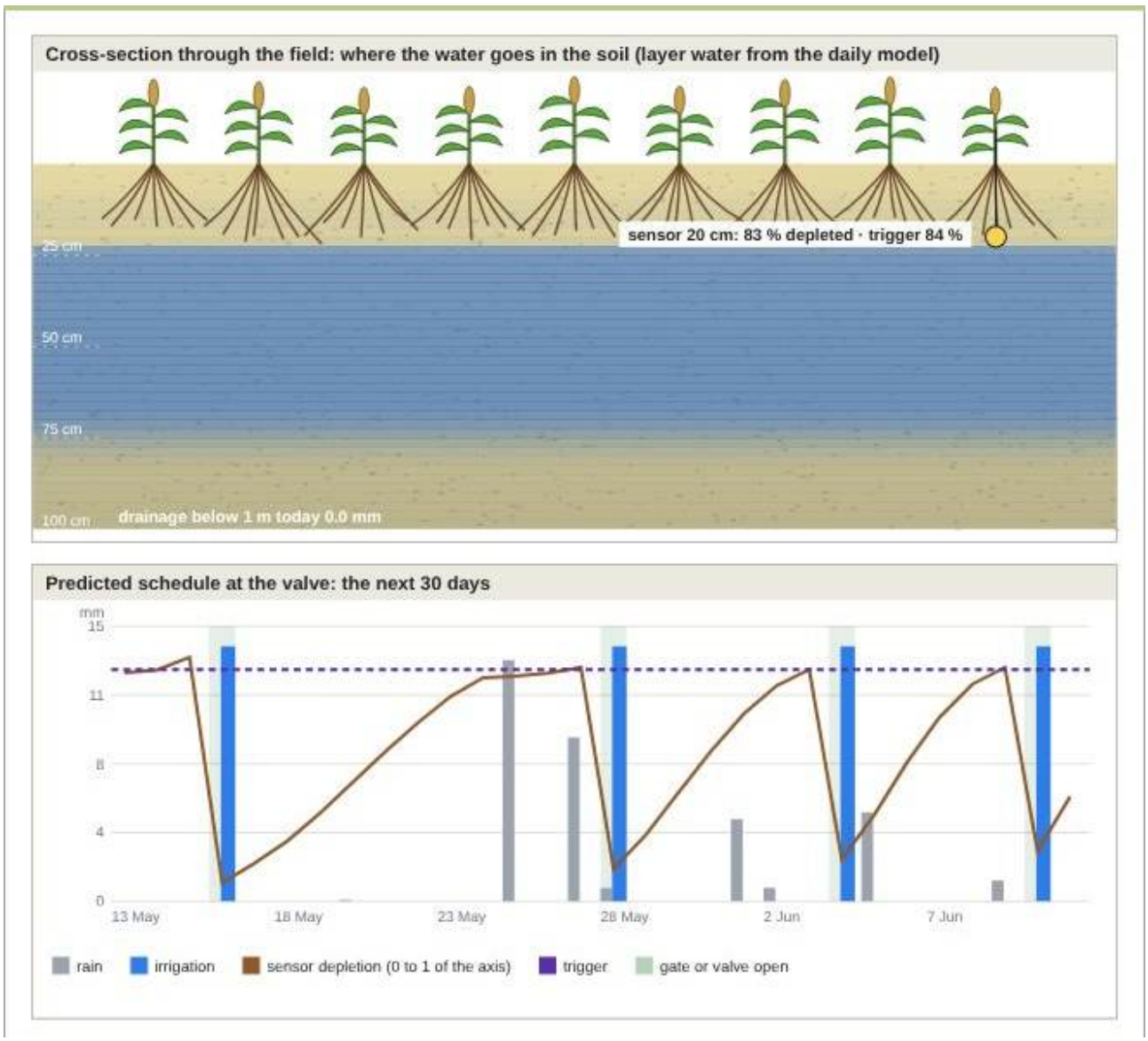


Figure 1.28: Virtual farm, drip design, Sand, part 2 of 3.

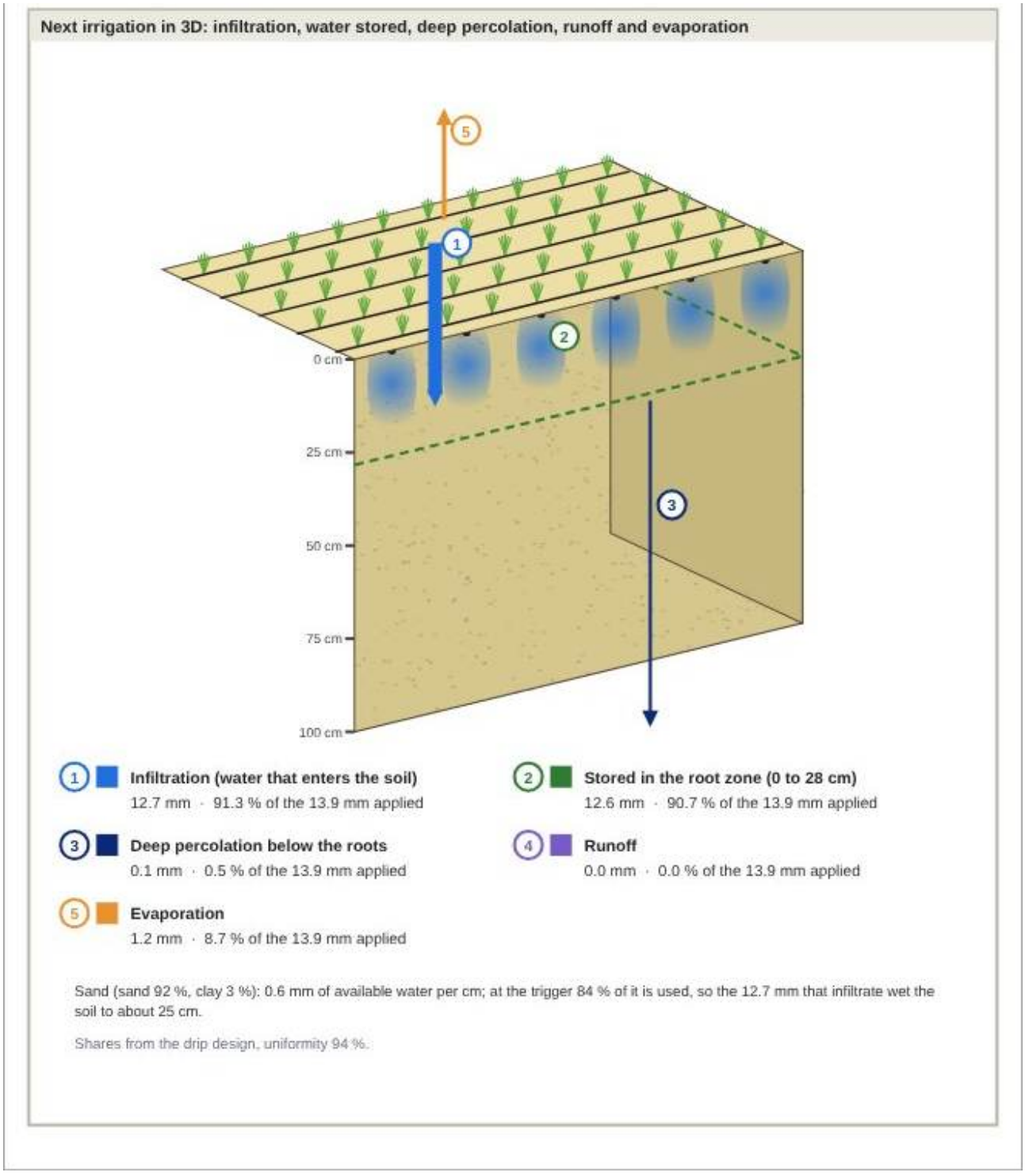


Figure 1.29: Virtual farm, drip design, Sand, part 3 of 3.

1.6 Virtual farm of the sprinkler design (Sand)

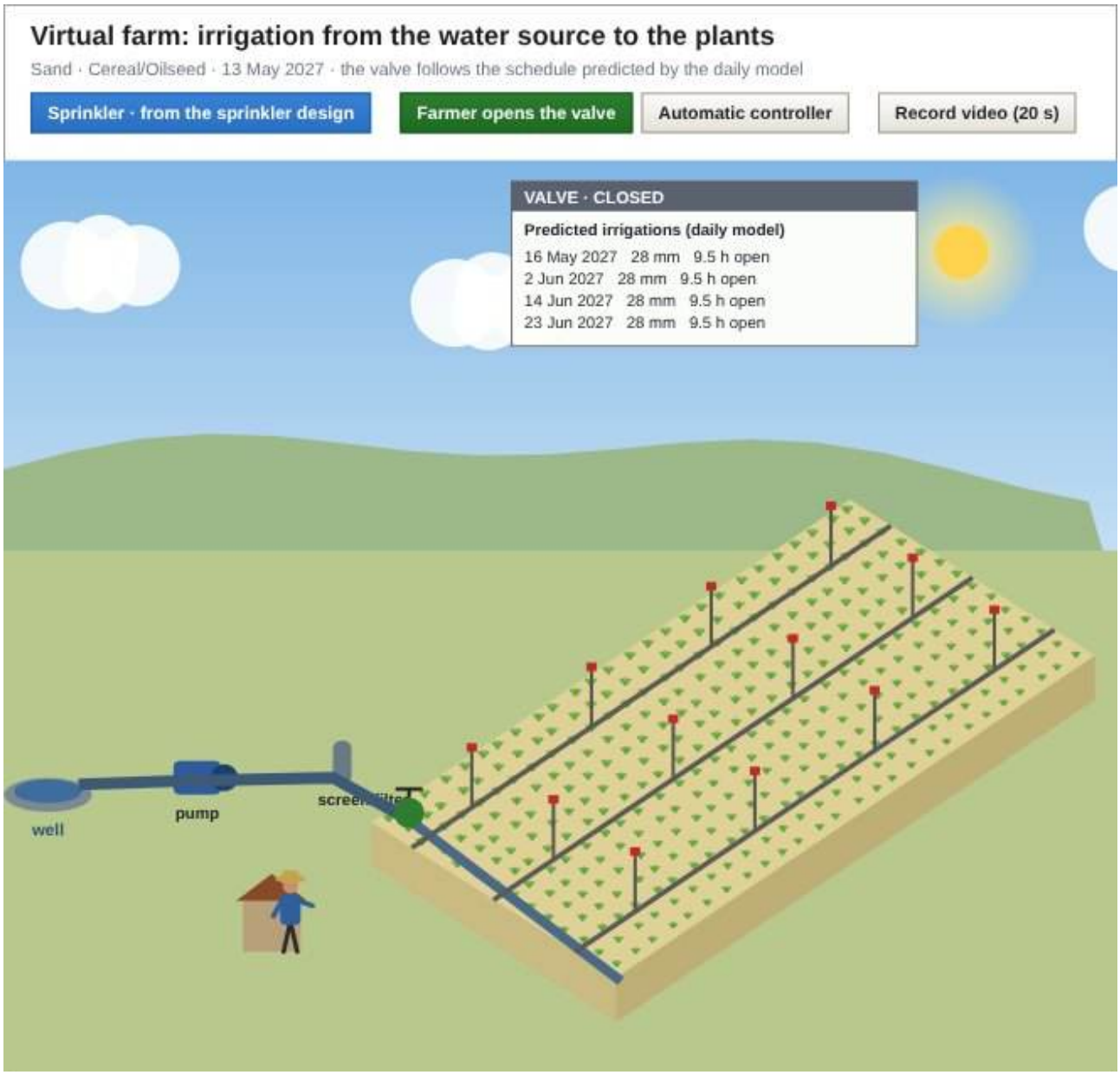


Figure 1.30: Virtual farm, sprinkler design, Sand, part 1 of 3.

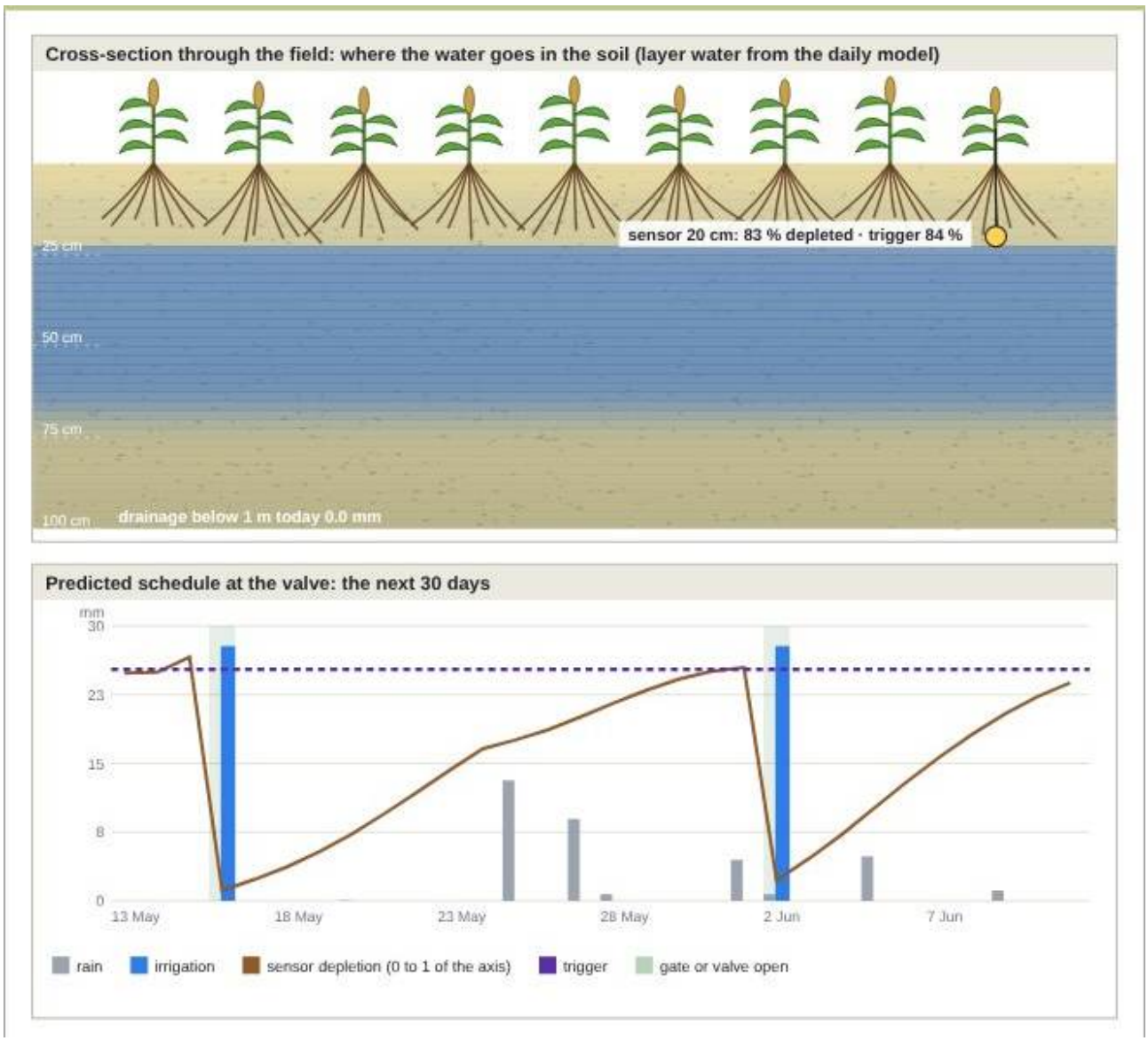


Figure 1.31: Virtual farm, sprinkler design, Sand, part 2 of 3.

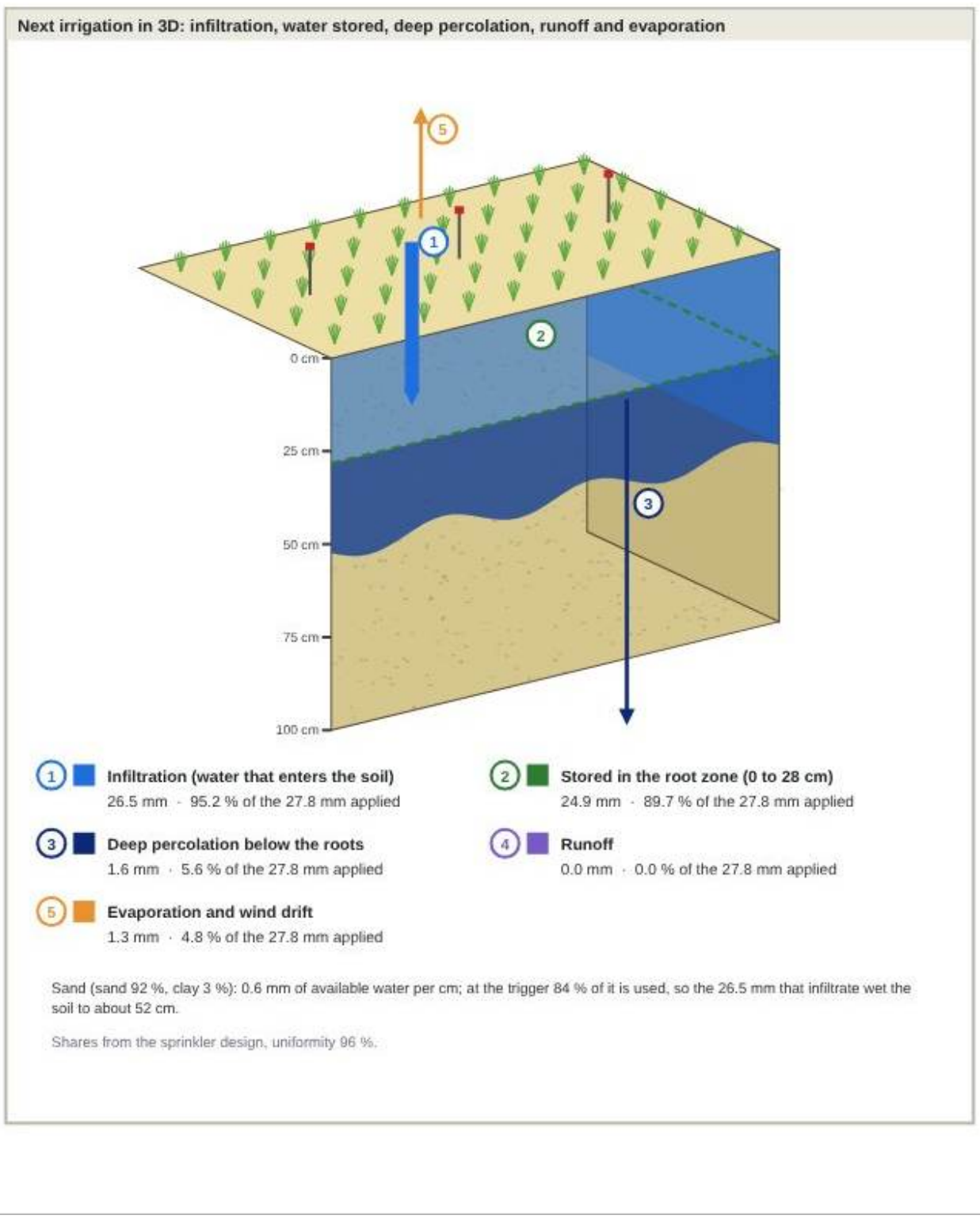


Figure 1.32: Virtual farm, sprinkler design, Sand, part 3 of 3.

1.7 Sensor and trigger (Sand)

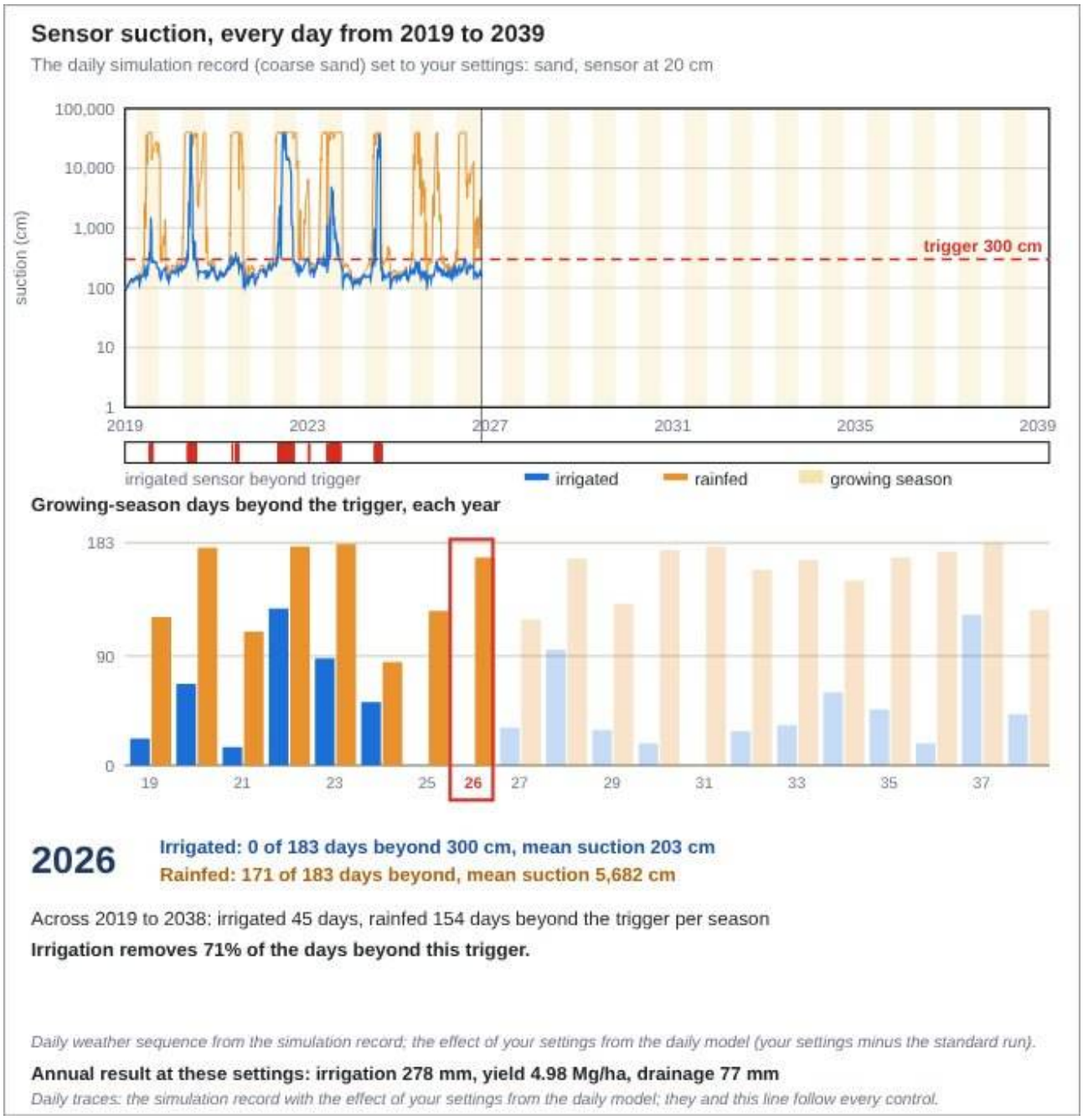


Figure 1.33: Sensor and trigger, Sand, part 1 of 1.

1.8 Trigger law (Sand)

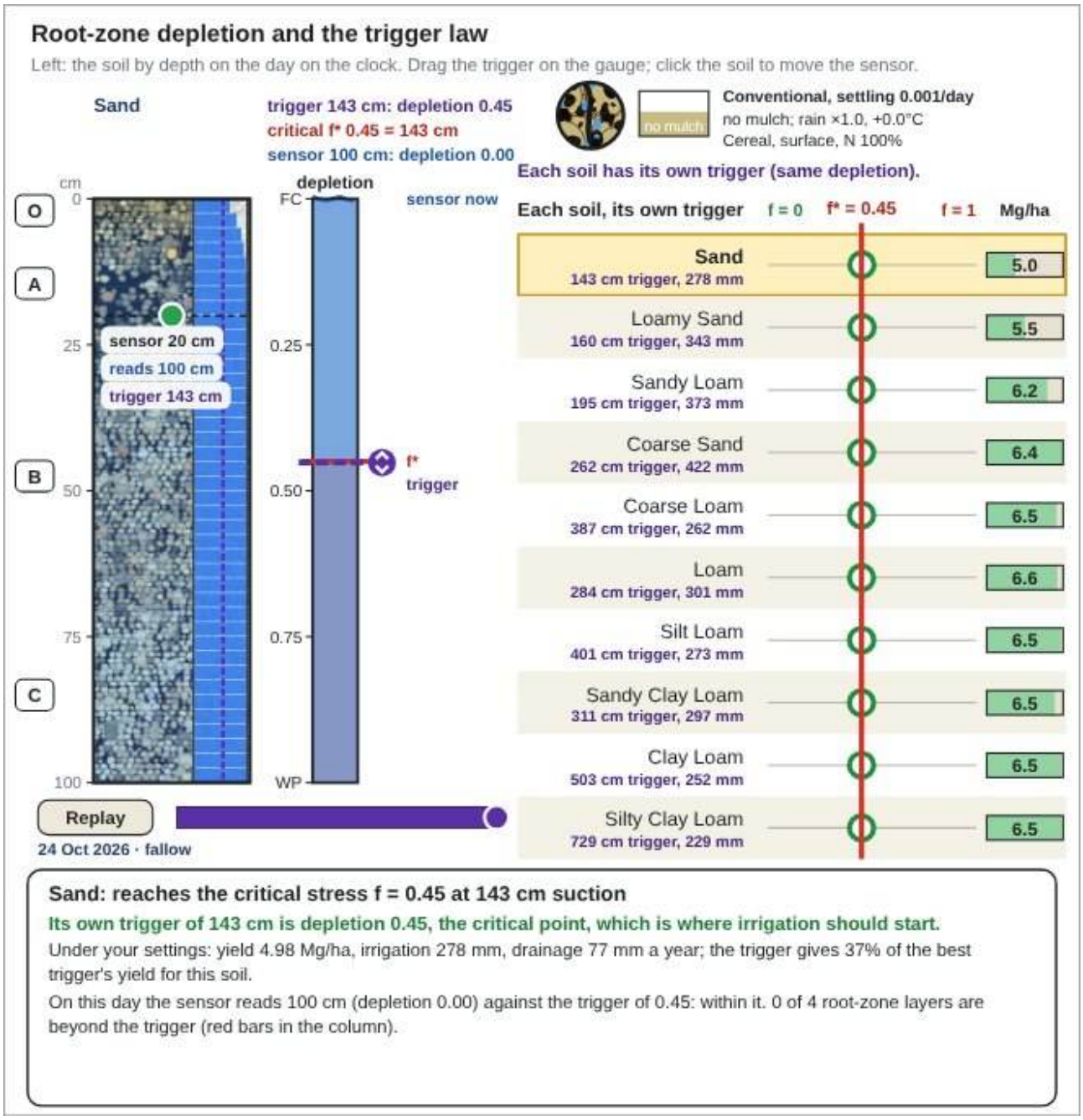


Figure 1.34: Trigger law, Sand, part 1 of 1.

1.9 Mulch against bare soil (Sand)

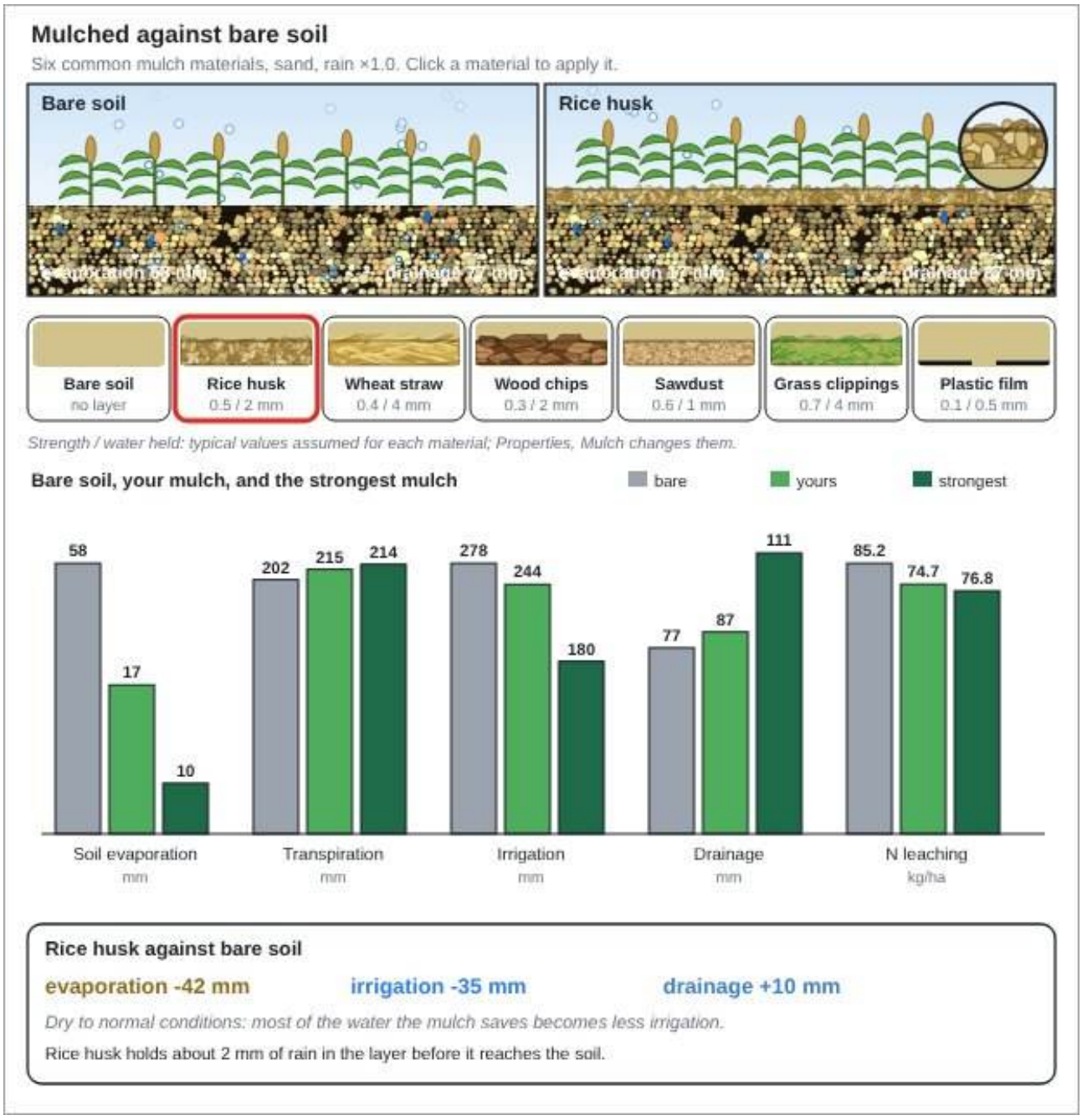


Figure 1.35: Mulch against bare soil, Sand, part 1 of 1.

1.10 Mulch control (Sand)

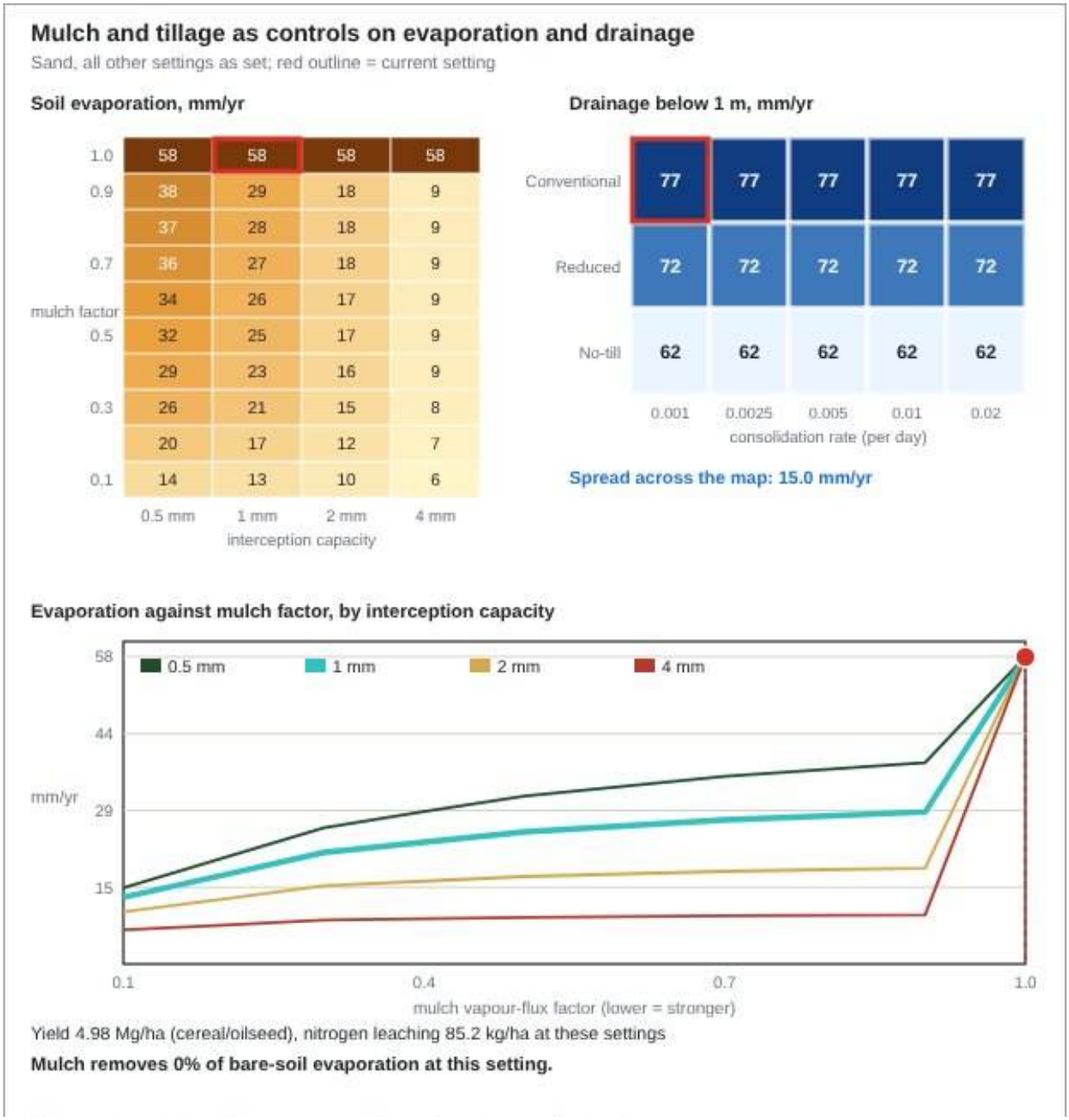


Figure 1.36: Mulch control, Sand, part 1 of 2.

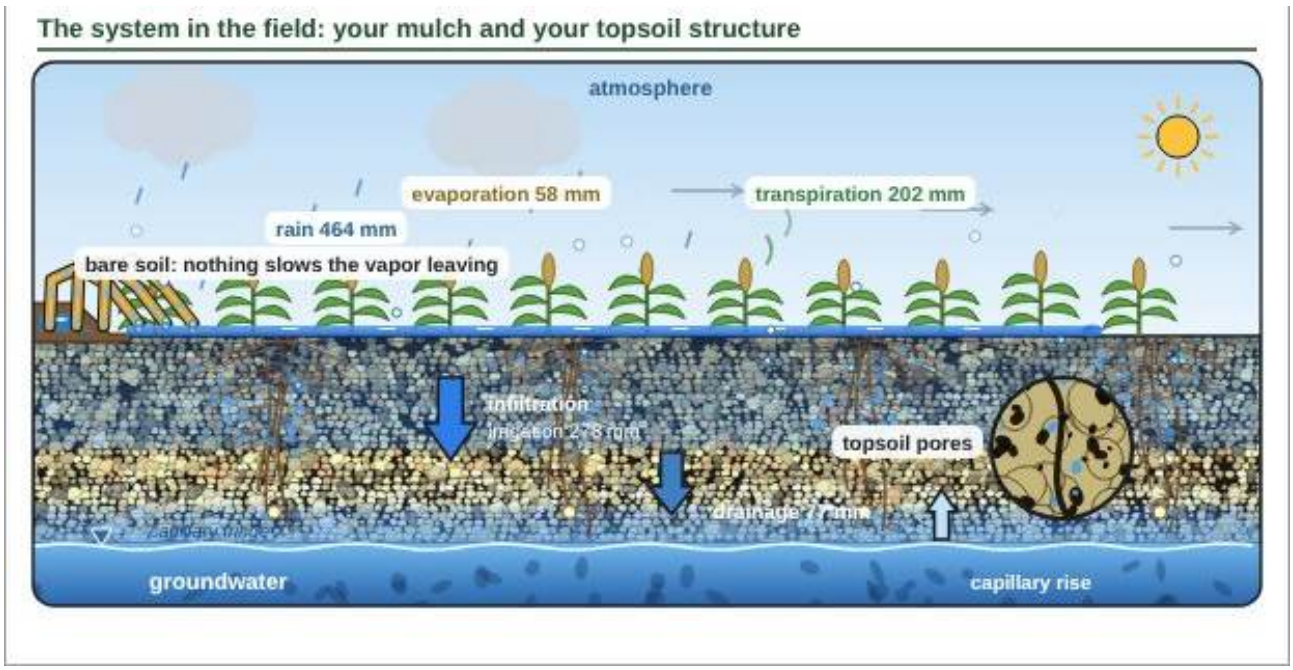


Figure 1.37: Mulch control, Sand, part 2 of 2.

1.11 Evaporation physics (Sand)

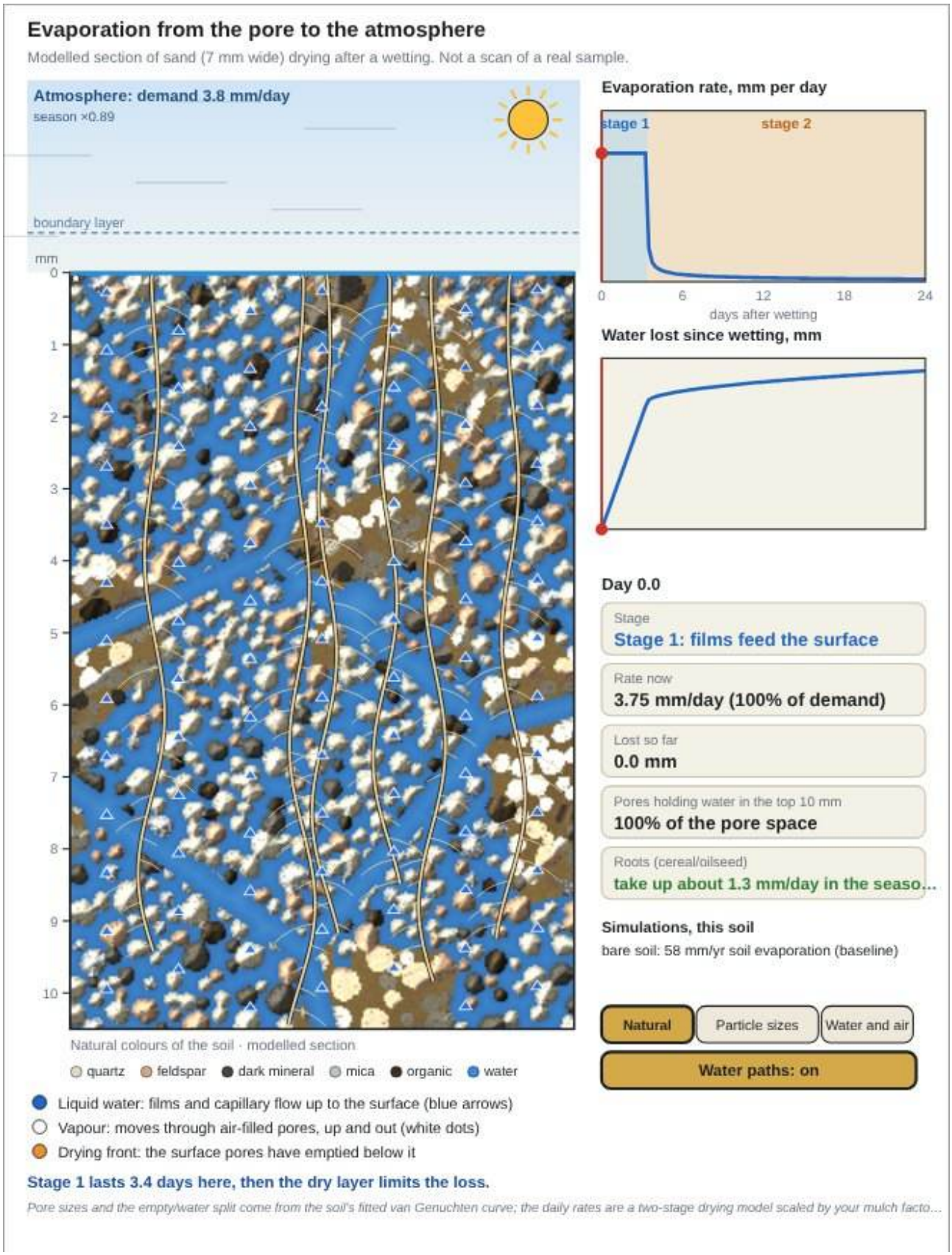


Figure 1.38: Evaporation physics, Sand, part 1 of 3.

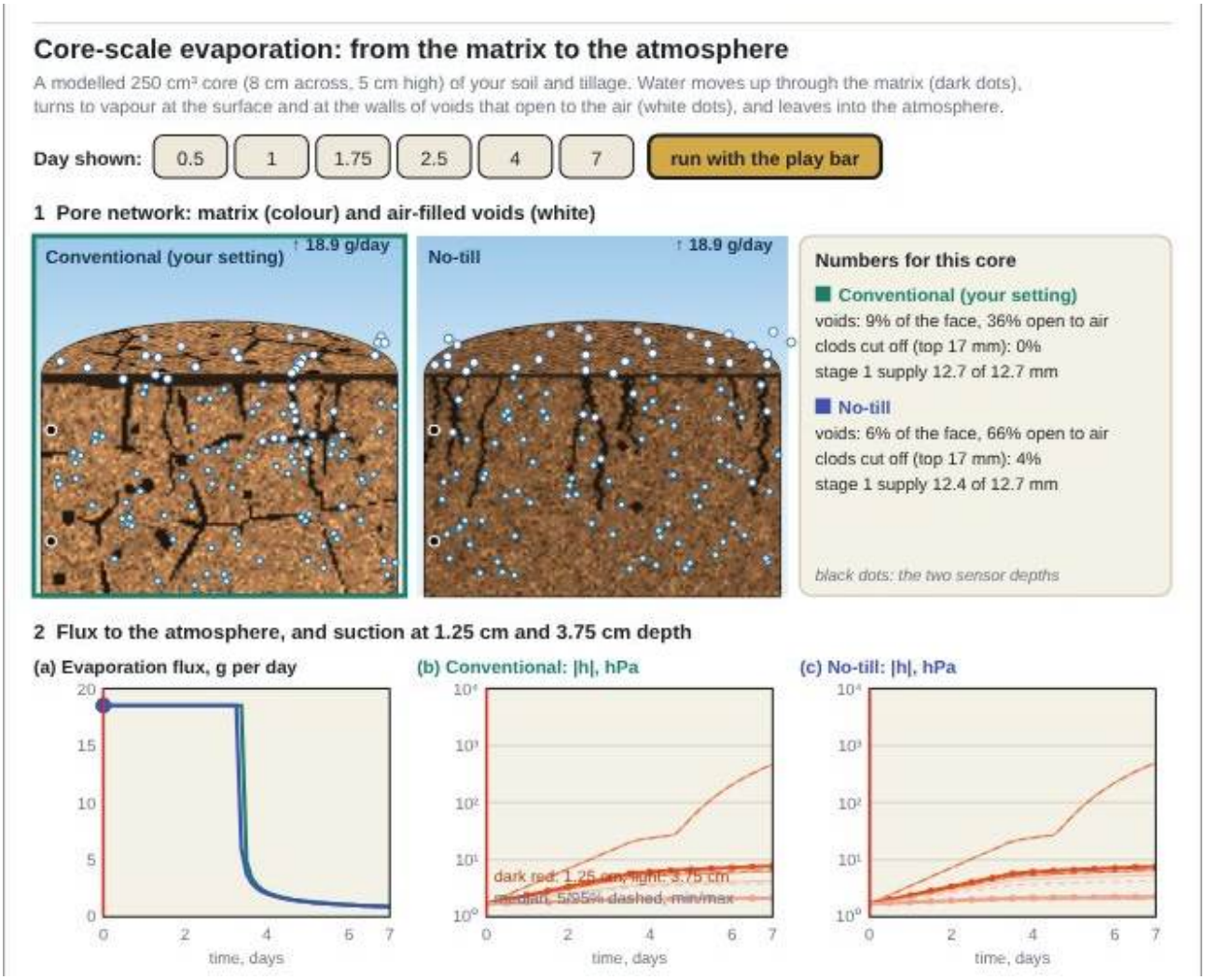


Figure 1.39: Evaporation physics, Sand, part 2 of 3.

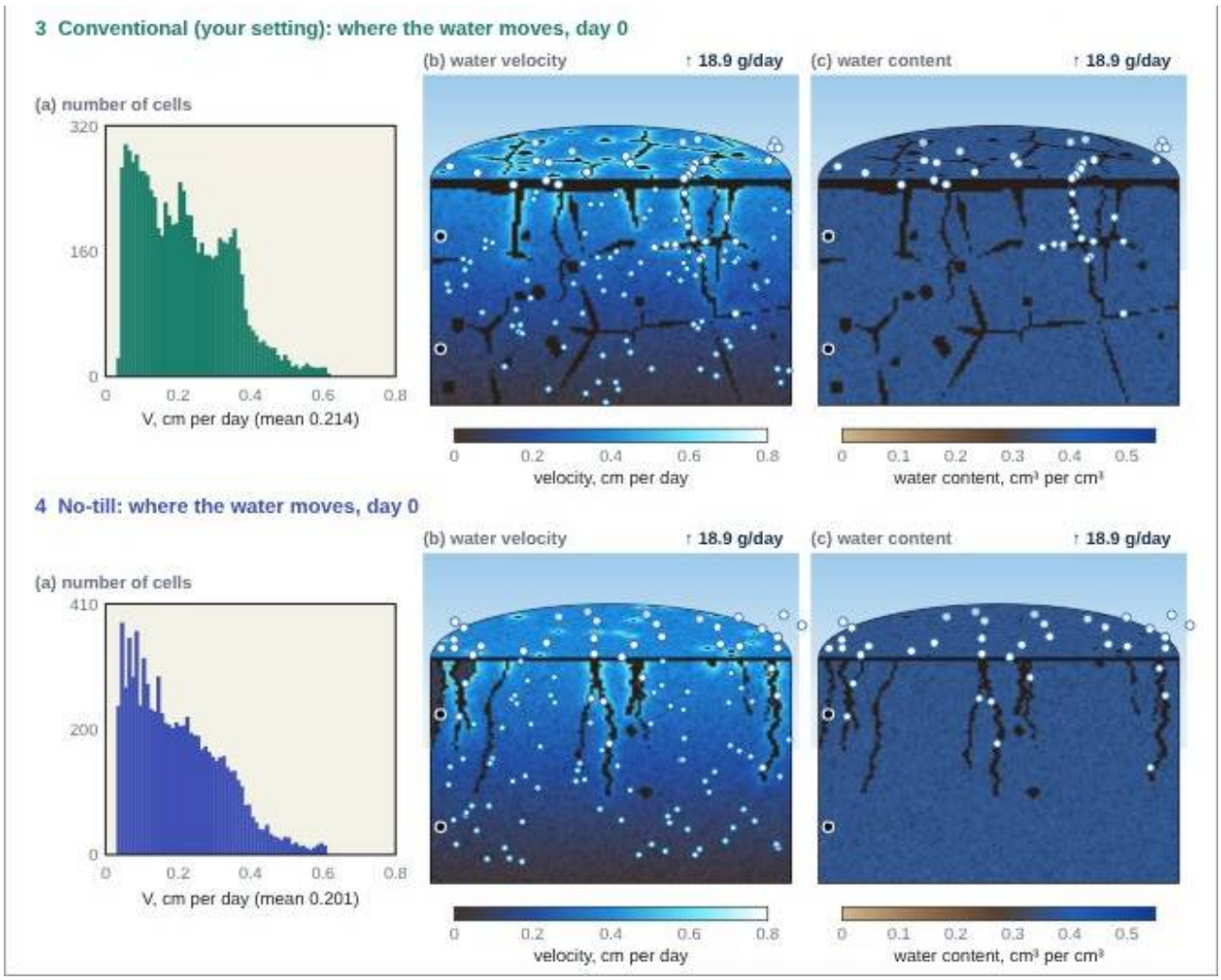


Figure 1.40: Evaporation physics, Sand, part 3 of 3.

1.12 Fate of saved water (Sand)

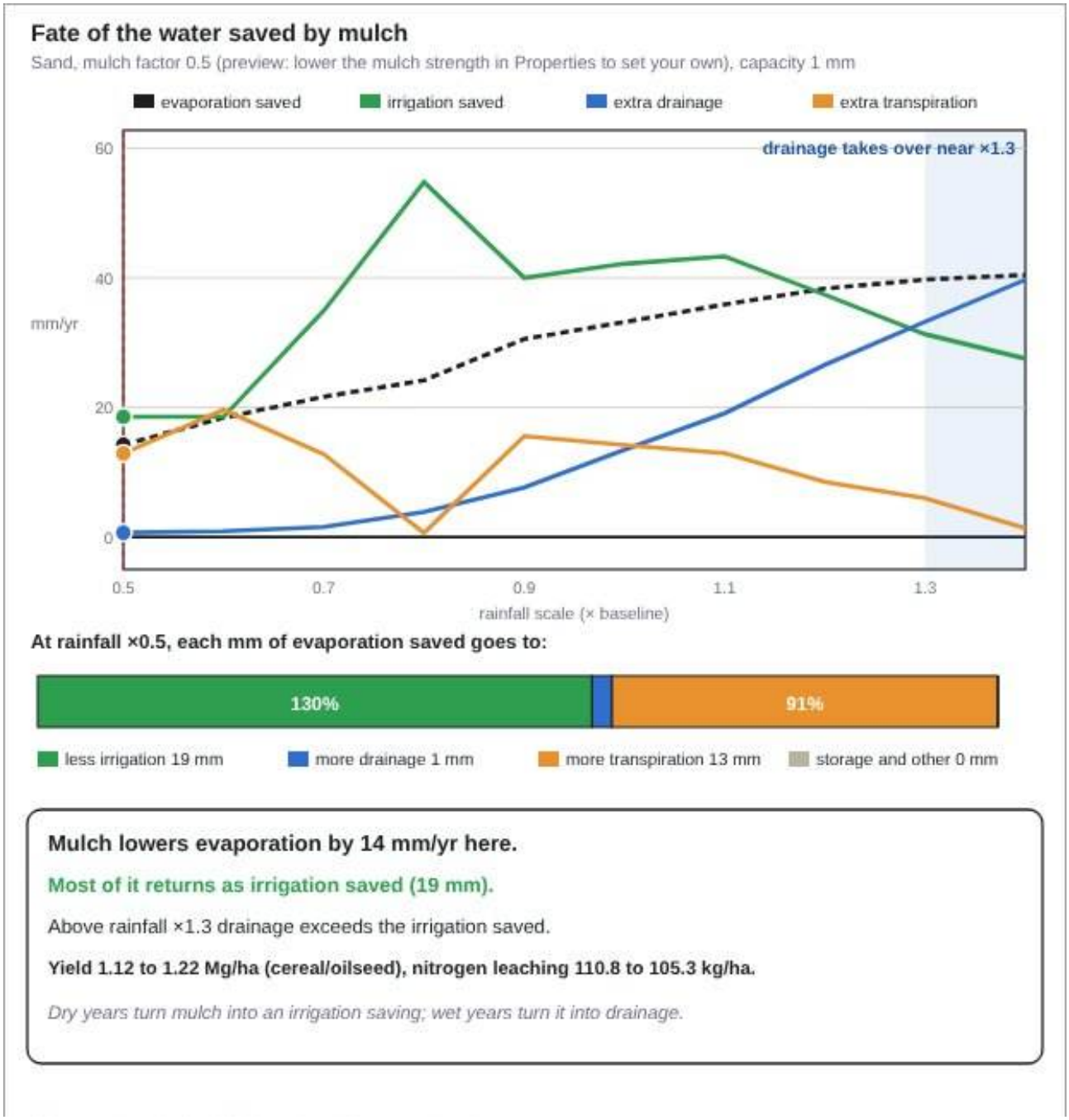


Figure 1.41: Fate of saved water, Sand, part 1 of 2.

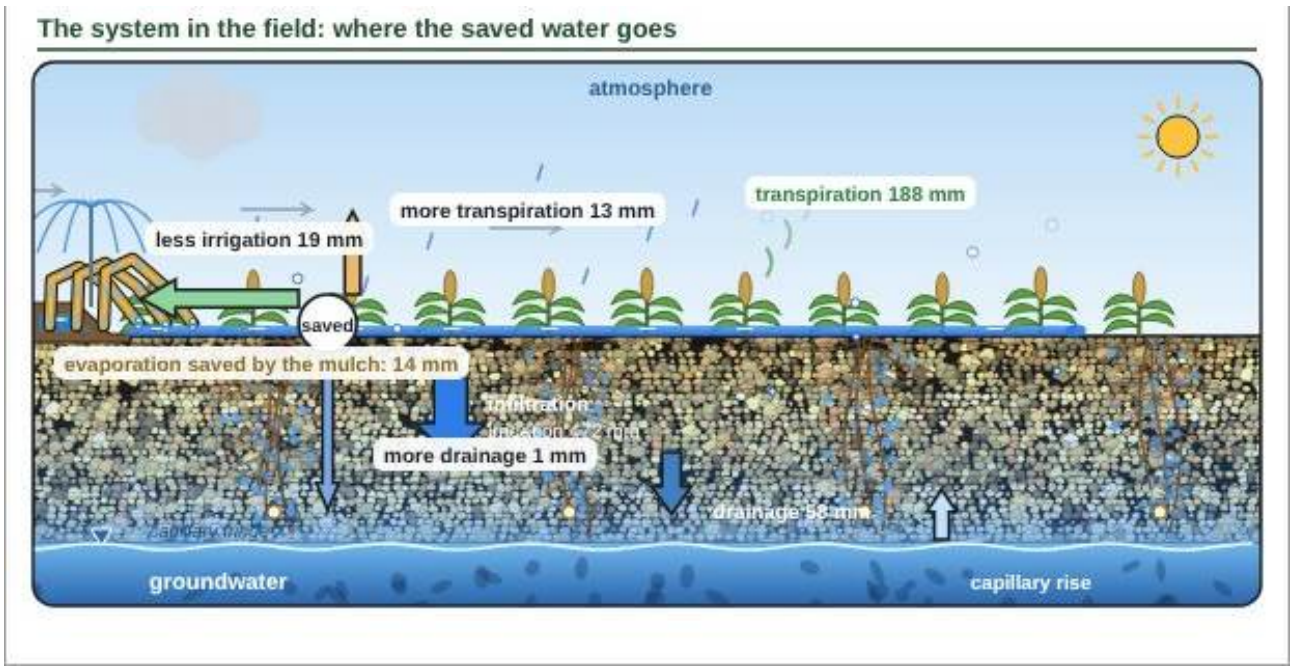


Figure 1.42: Fate of saved water, Sand, part 2 of 2.

1.13 Regime map (Sand)

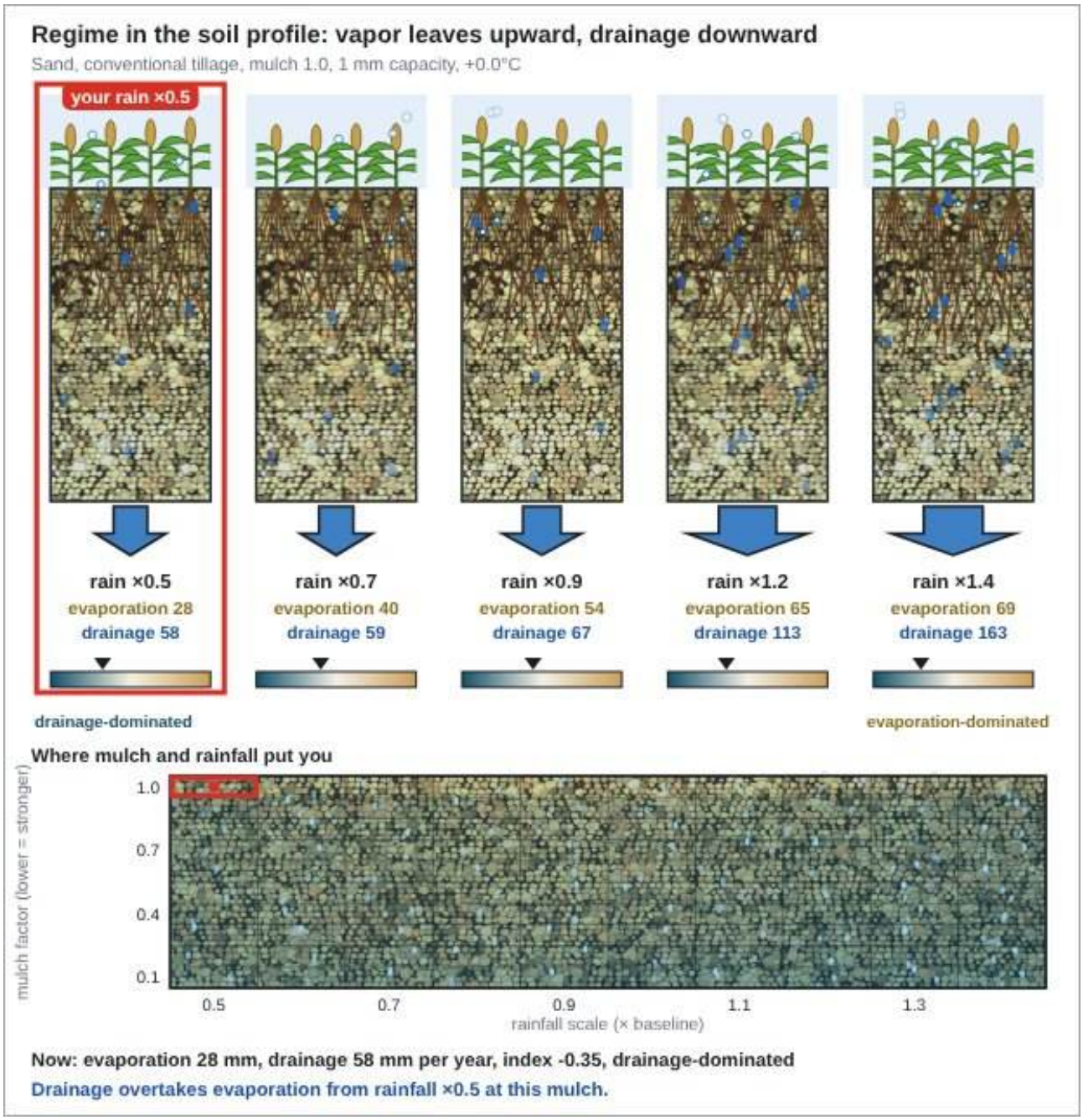


Figure 1.43: Regime map, Sand, part 1 of 1.

1.14 Tillage and structure (Sand)

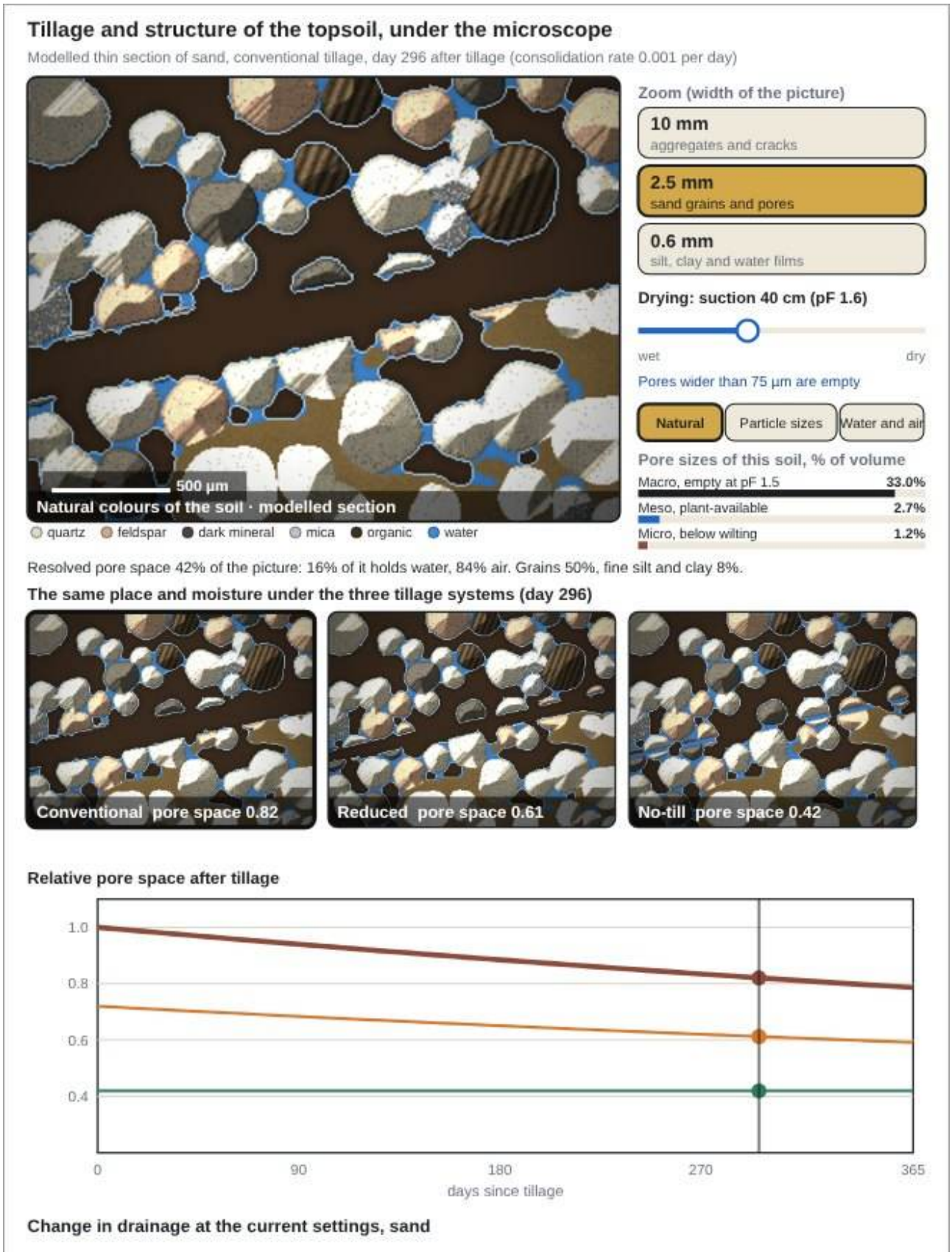


Figure 1.44: Tillage and structure, Sand, part 1 of 2.

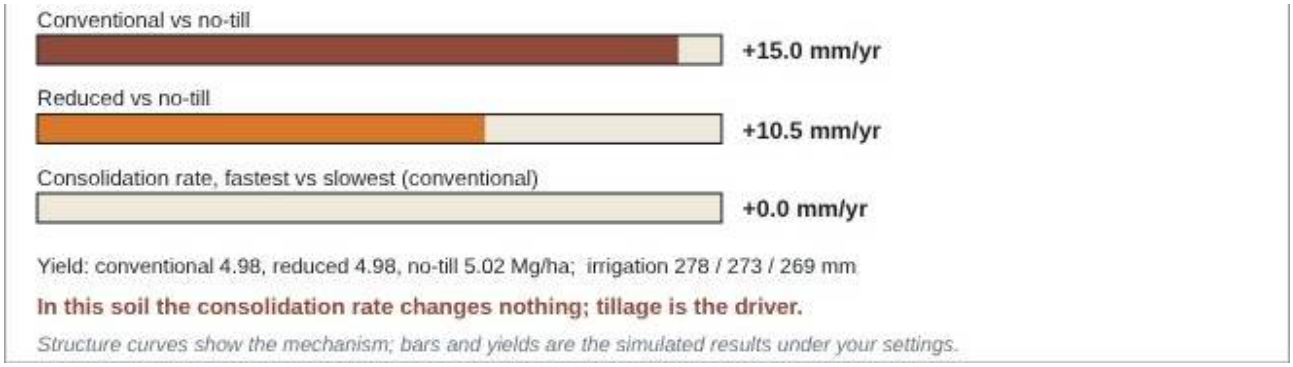


Figure 1.45: Tillage and structure, Sand, part 2 of 2.

1.15 Soil hydraulic drivers (Sand)

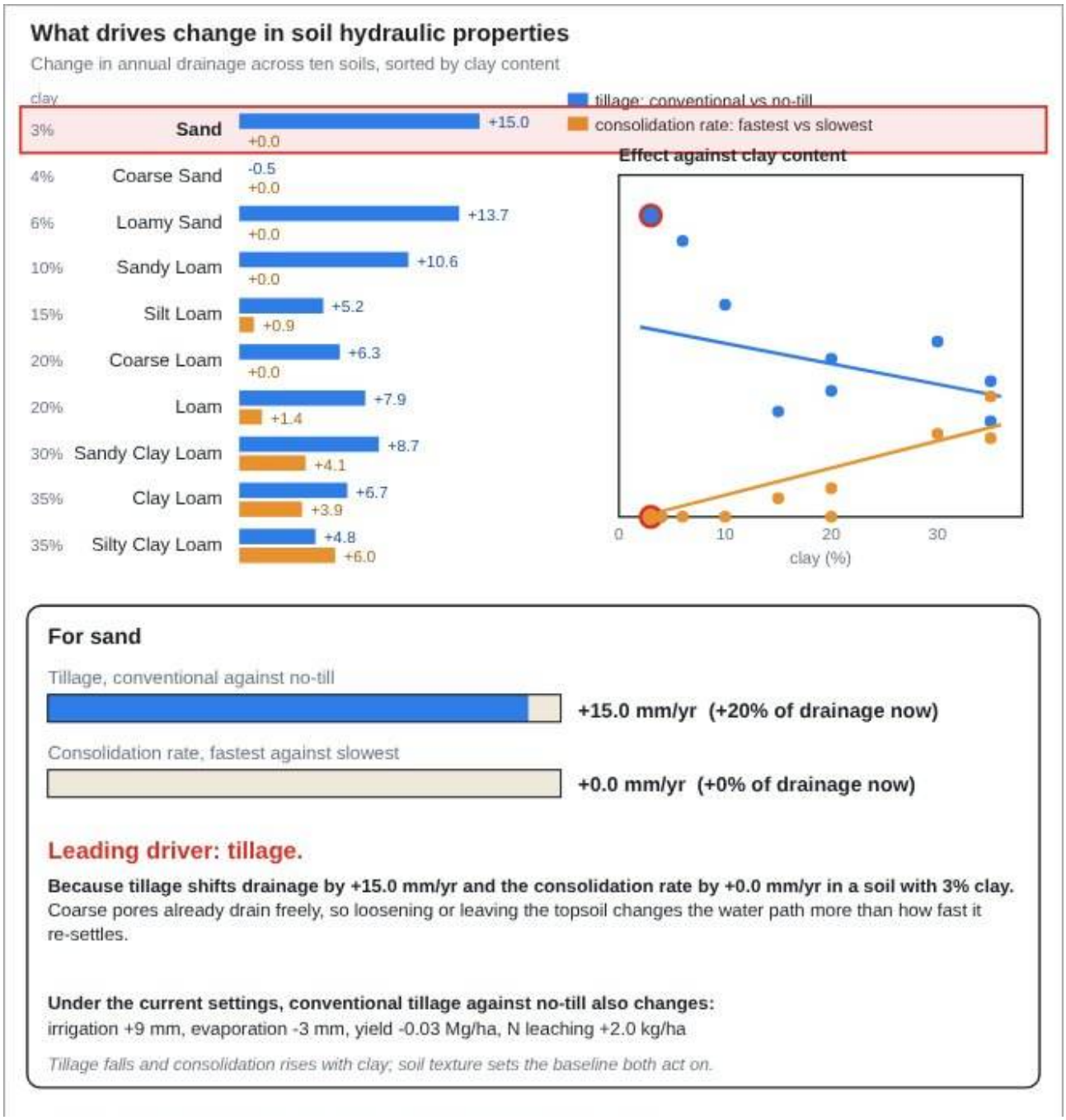


Figure 1.46: Soil hydraulic drivers, Sand, part 1 of 3.

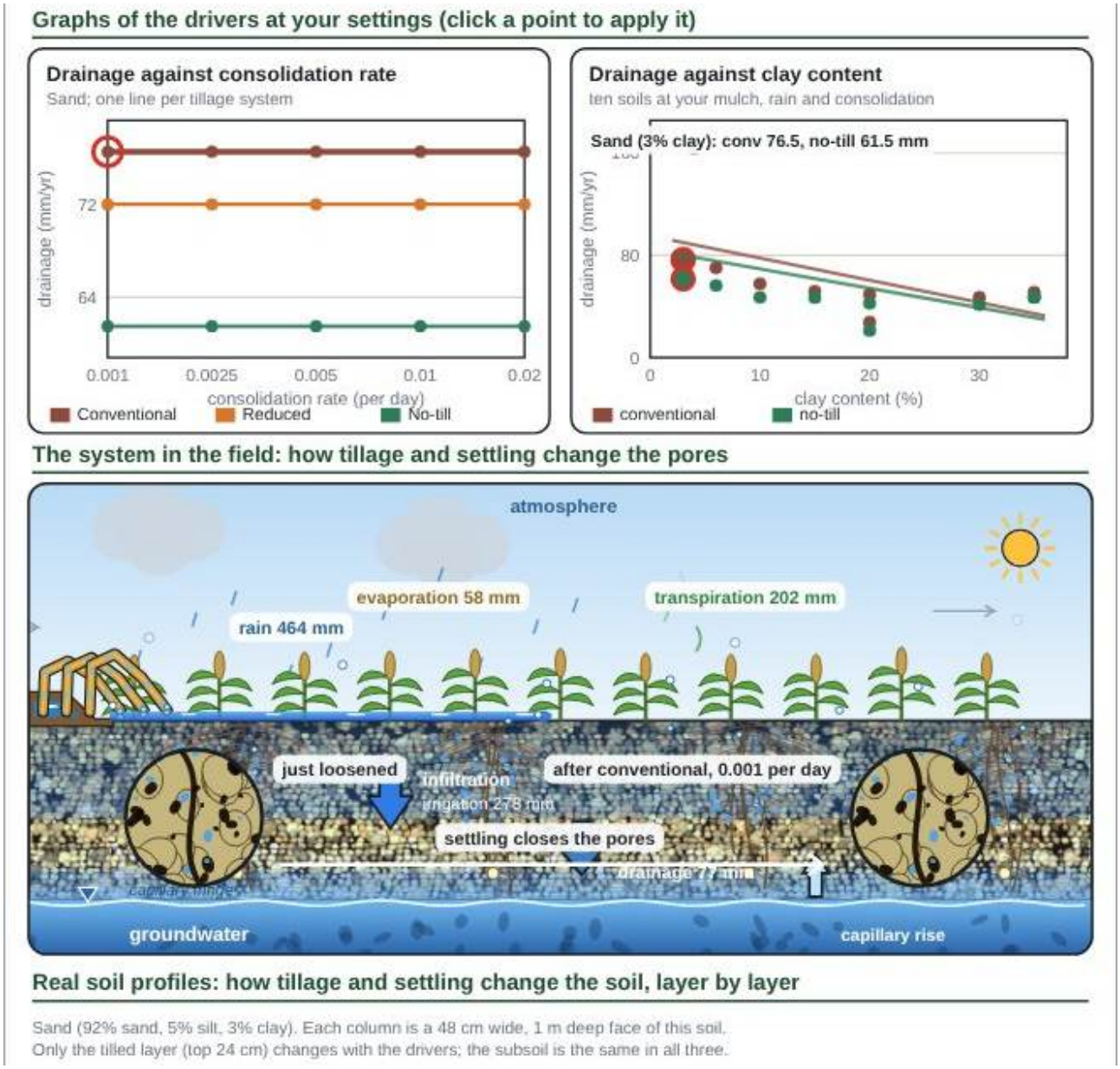


Figure 1.47: Soil hydraulic drivers, Sand, part 2 of 3.

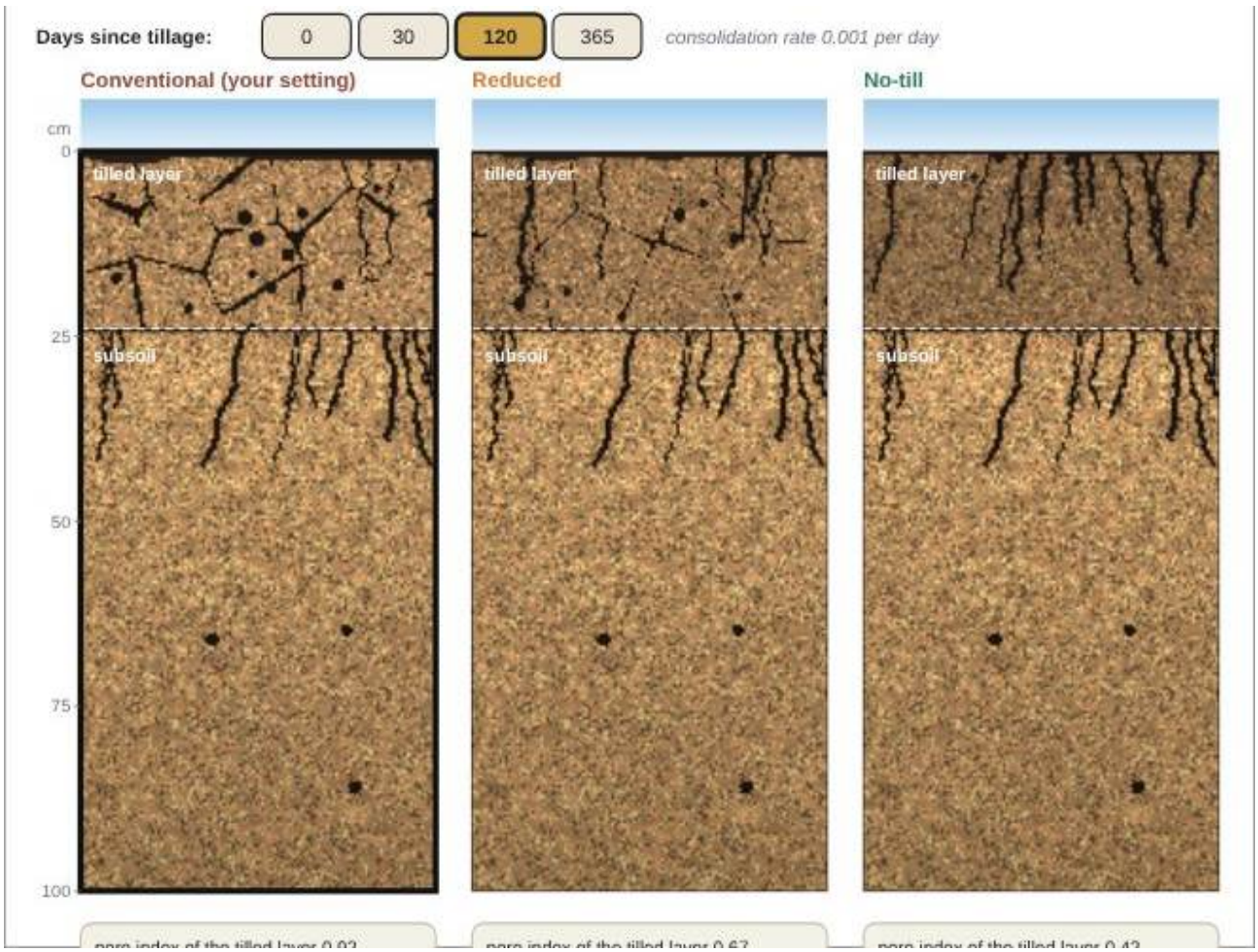


Figure 1.48: Soil hydraulic drivers, Sand, part 3 of 3.

1.16 Root structure (Sand)

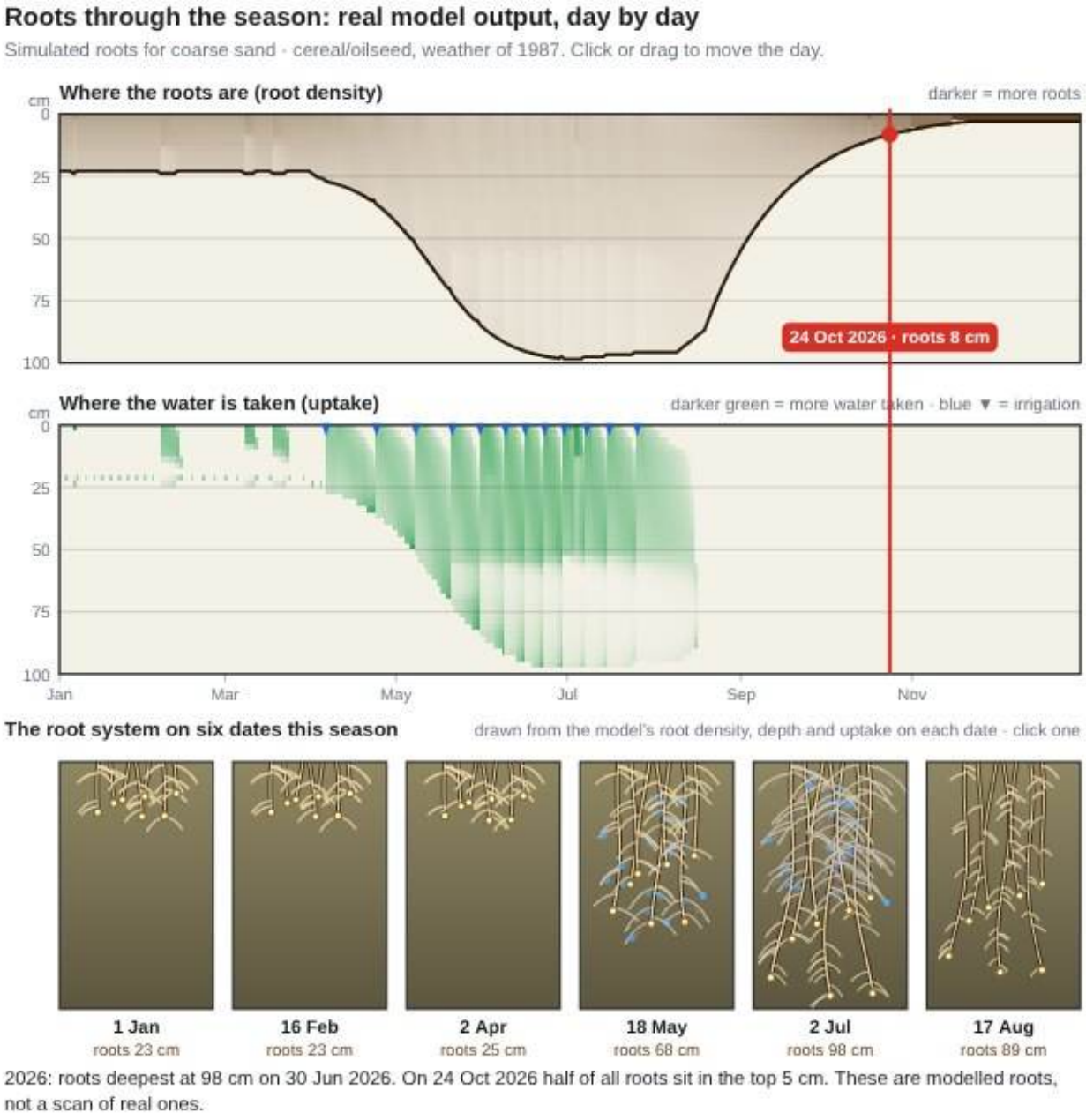


Figure 1.49: Root structure, Sand, part 1 of 4.

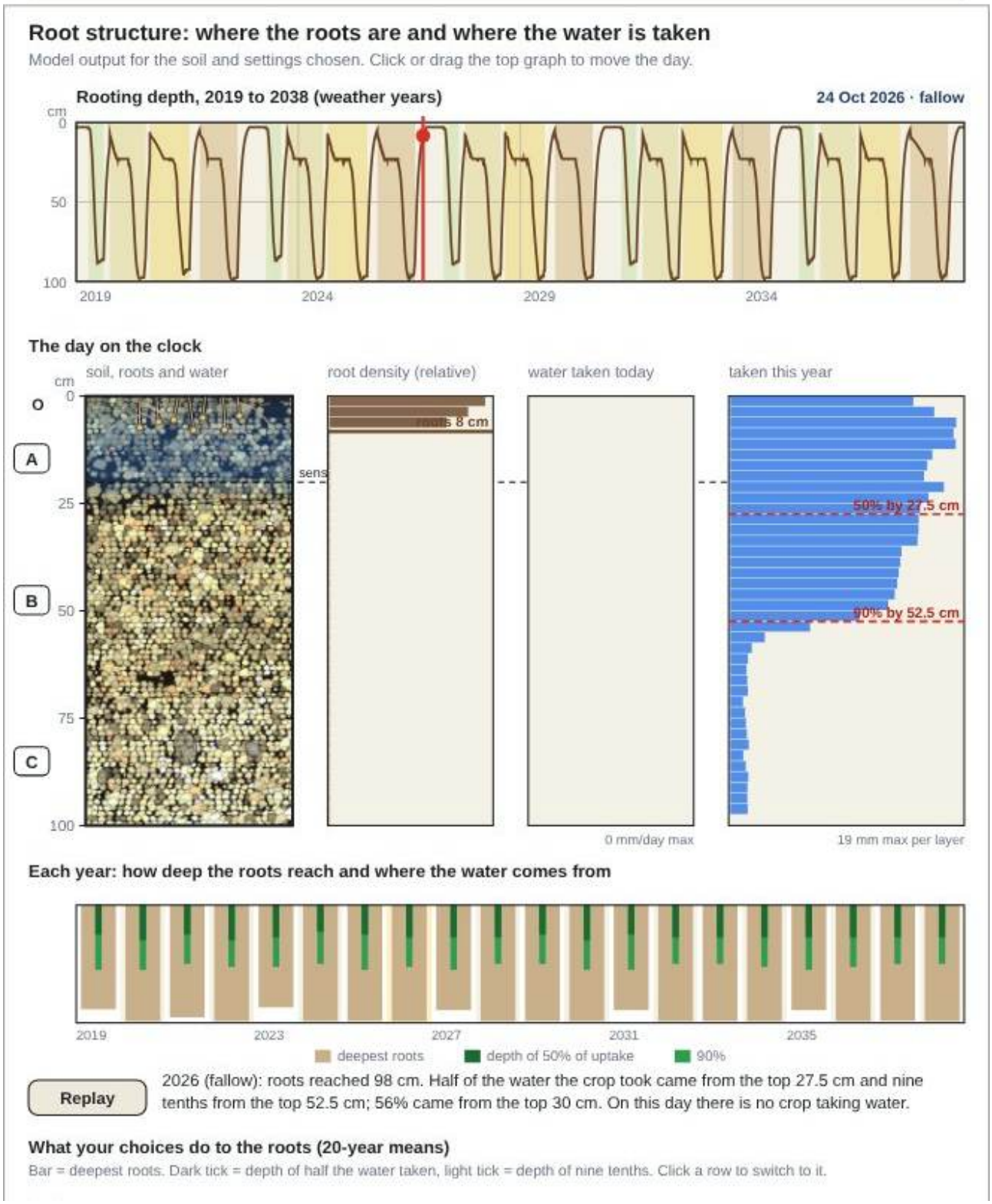


Figure 1.50: Root structure, Sand, part 2 of 4.

Soil

Sand	working it out
Loamy Sand	working it out
Sandy Loam	working it out
Coarse Sand	working it out
Coarse Loam	working it out
Loam	working it out
Silt Loam	working it out
Sandy Clay Loam	working it out
Clay Loam	working it out
Silty Clay Loam	working it out

Tillage

Conventional	working it out
Reduced	working it out
No-till	working it out

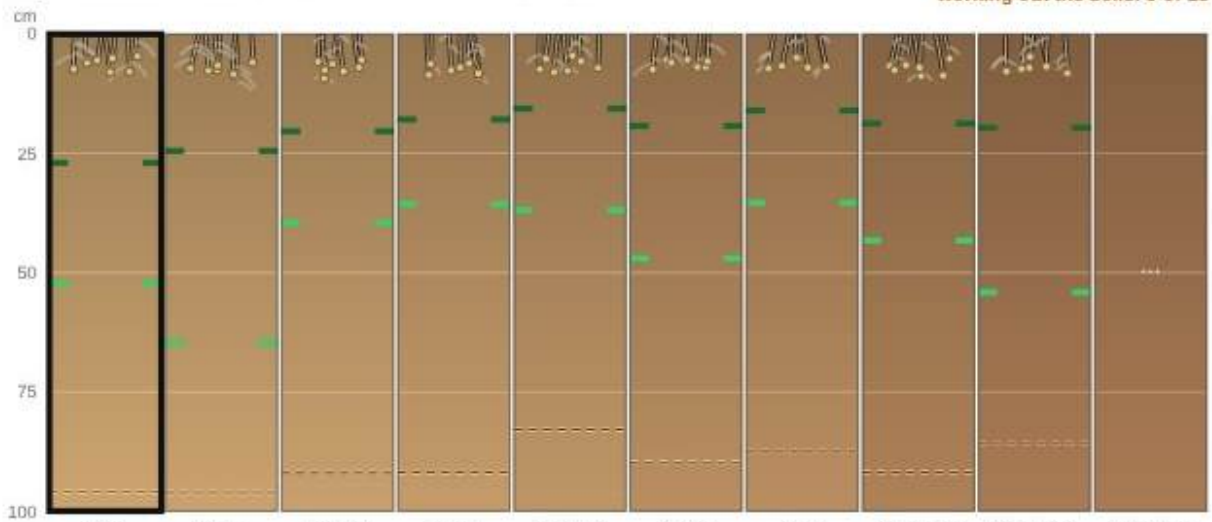
Crop

Cereal/Oilseed	working it out

Rooting depth in all ten soils

Same day and weather for every soil. Crop, tillage, consolidation, mulch, rain, temperature, trigger, sensor depth, irrigation method and nitrogen all act on every column. Brown dashed line: deepest roots (20-year mean). Green ticks: depth holding 50% and 90% of the water taken. Click a column to use that soil.

working out the soils: 9 of 10



	Sand	Coarse Sand	Loamy Sand	Sandy Loam	Silt Loam	Coarse Loam	Loam	Sandy Clay Loam	Clay Loam	Silty Clay Loam
	3% clay	4% clay	6% clay	10% clay	15% clay	20% clay	20% clay	30% clay	35% clay	35% clay
deepest roots (cm)	**96**	**96**	**92**	**92**	**83**	**90**	**88**	**92**	**86**	
50% of water by (cm)	27	25	20	18	16	19	16	19	20	
90% of water by (cm)	52	65	40	36	37	47	35	43	54	
deepest in any year (cm)	98	98	98	98	85	92	91	95	88	

Across the soils the deepest roots differ by 13 cm (83 to 96 cm) at these settings.

The model stops roots at 1 m; the simulations reach 120 cm.

Figure 1.51: Root structure, Sand, part 3 of 4.

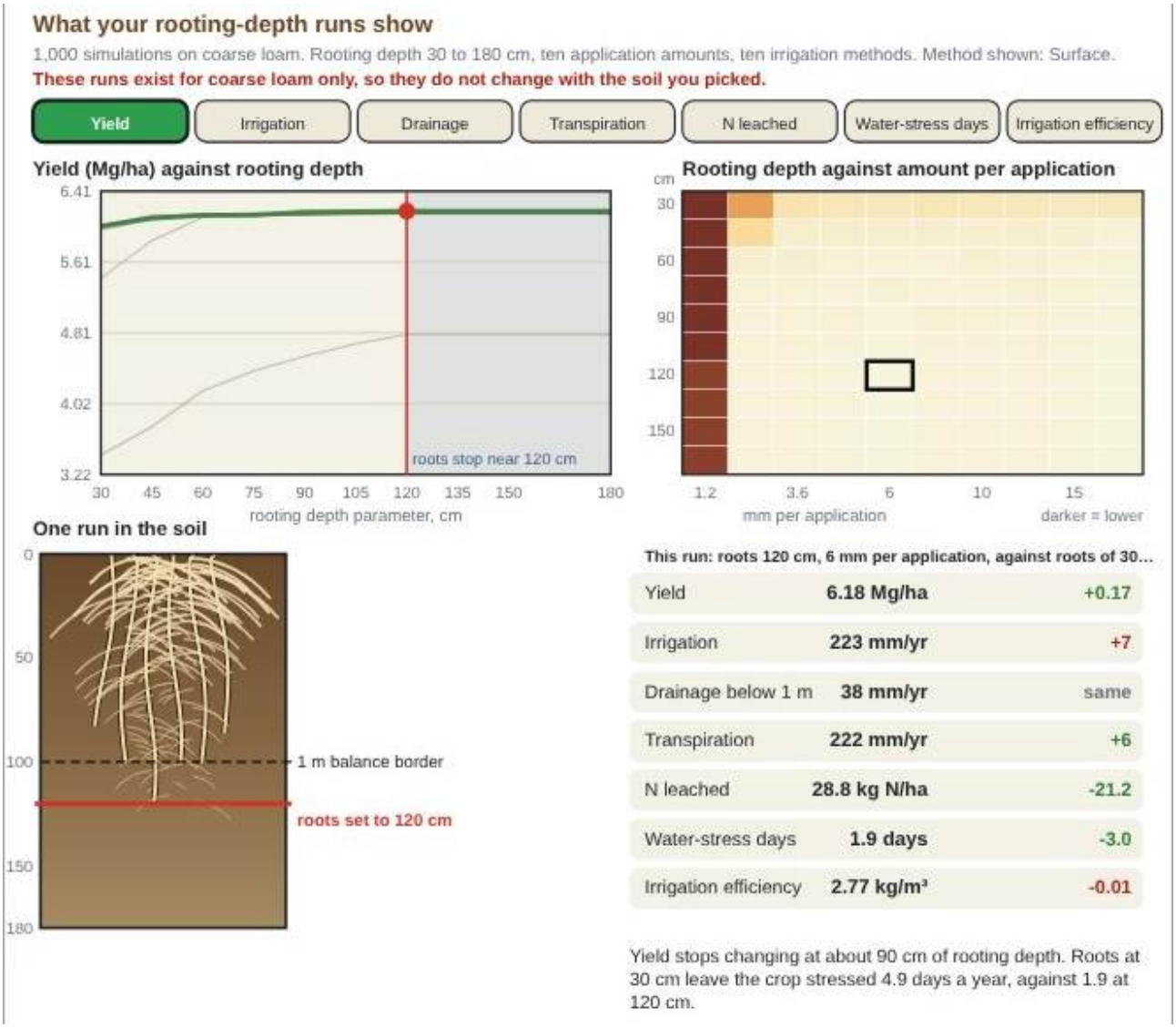


Figure 1.52: Root structure, Sand, part 4 of 4.

1.17 Field balance (Sand)

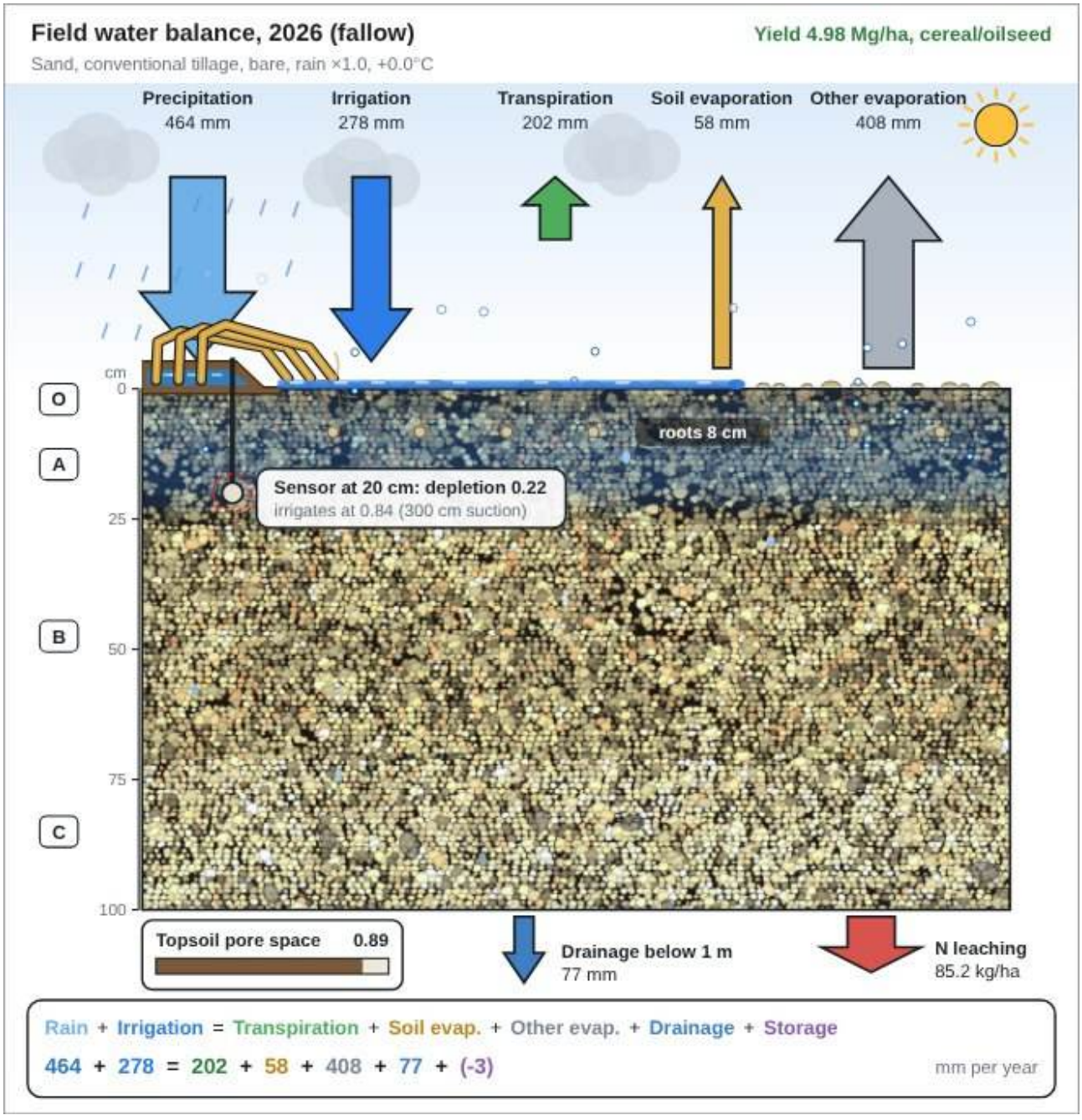


Figure 1.53: Field balance, Sand, part 1 of 1.

1.18 Soil, plant and air (Sand)

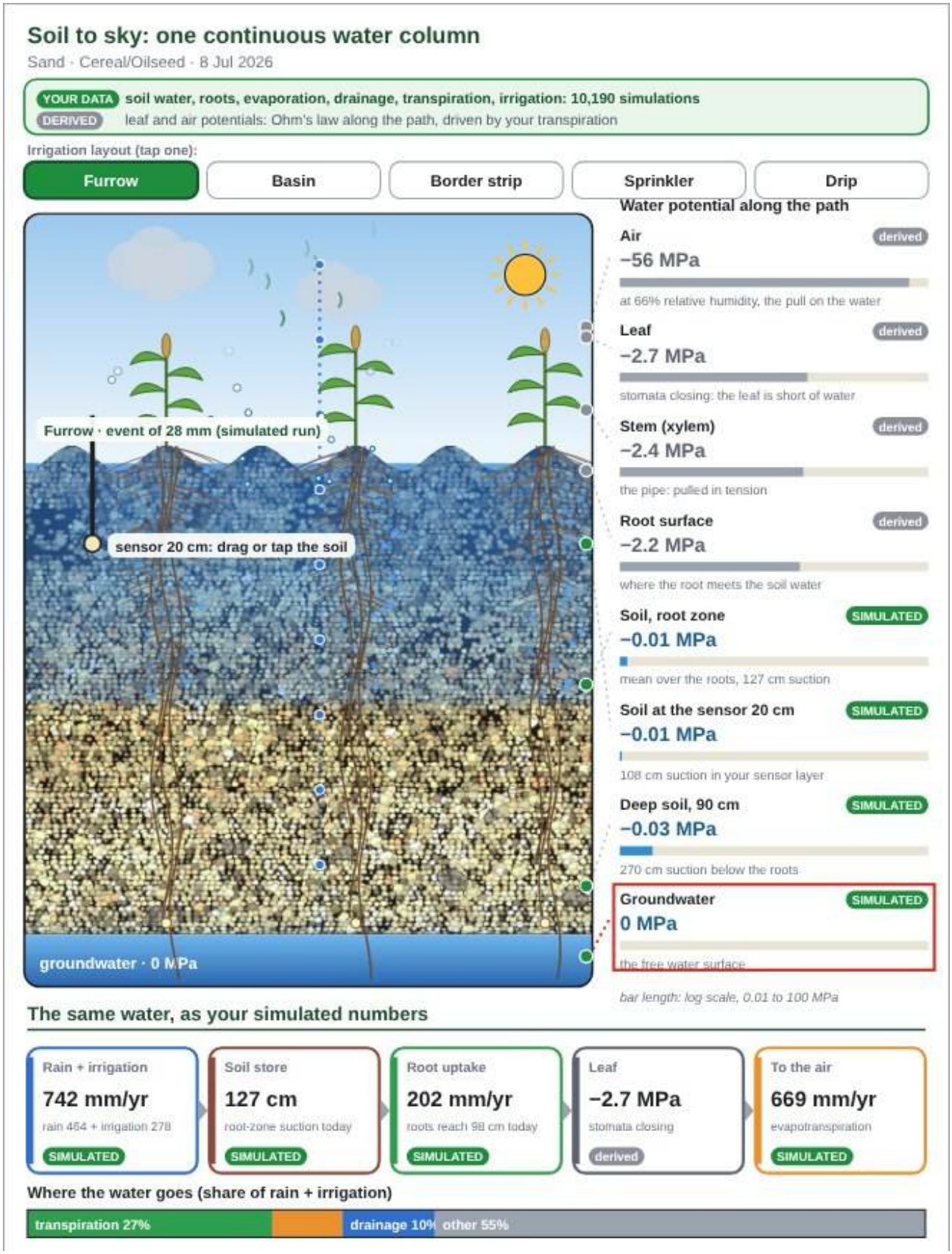


Figure 1.54: Soil, plant and air, Sand, part 1 of 3.

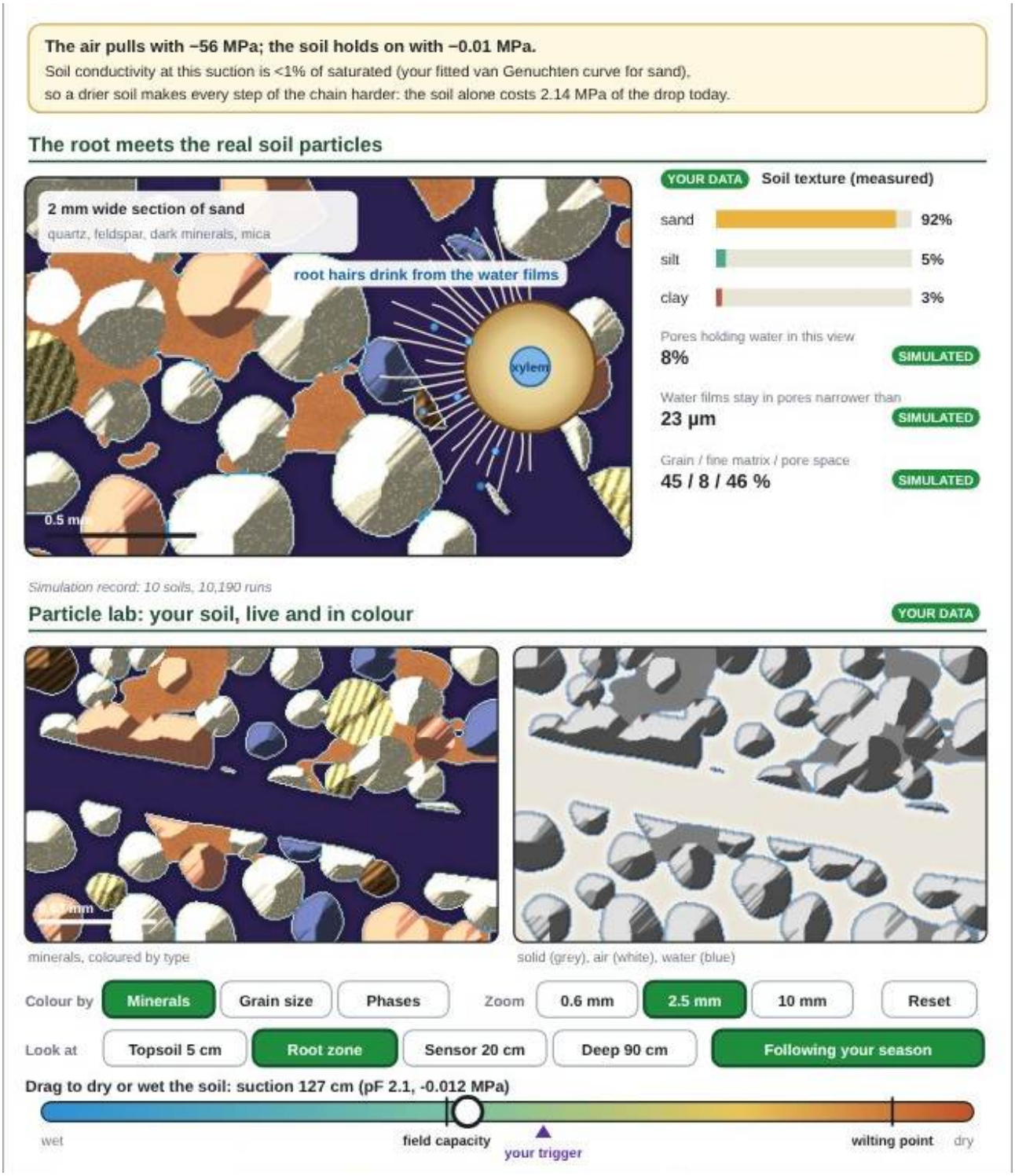


Figure 1.55: Soil, plant and air, Sand, part 2 of 3.

Figure 1.56: Soil, plant and air, Sand, part 3 of 3.

1.19 Profile evolution (Sand)

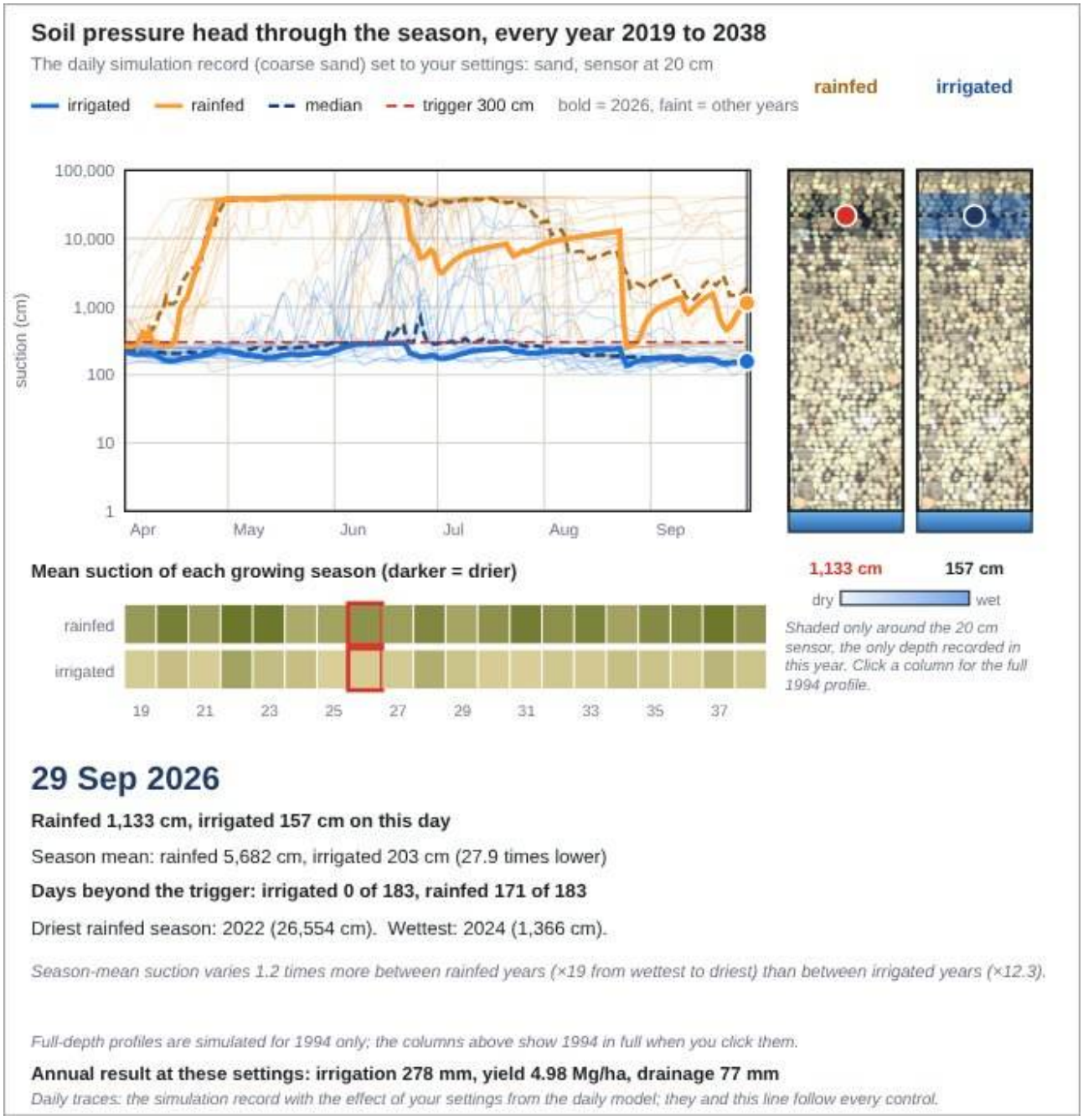


Figure 1.57: Profile evolution, Sand, part 1 of 1.

1.20 Reliability over 20 seasons (Sand)

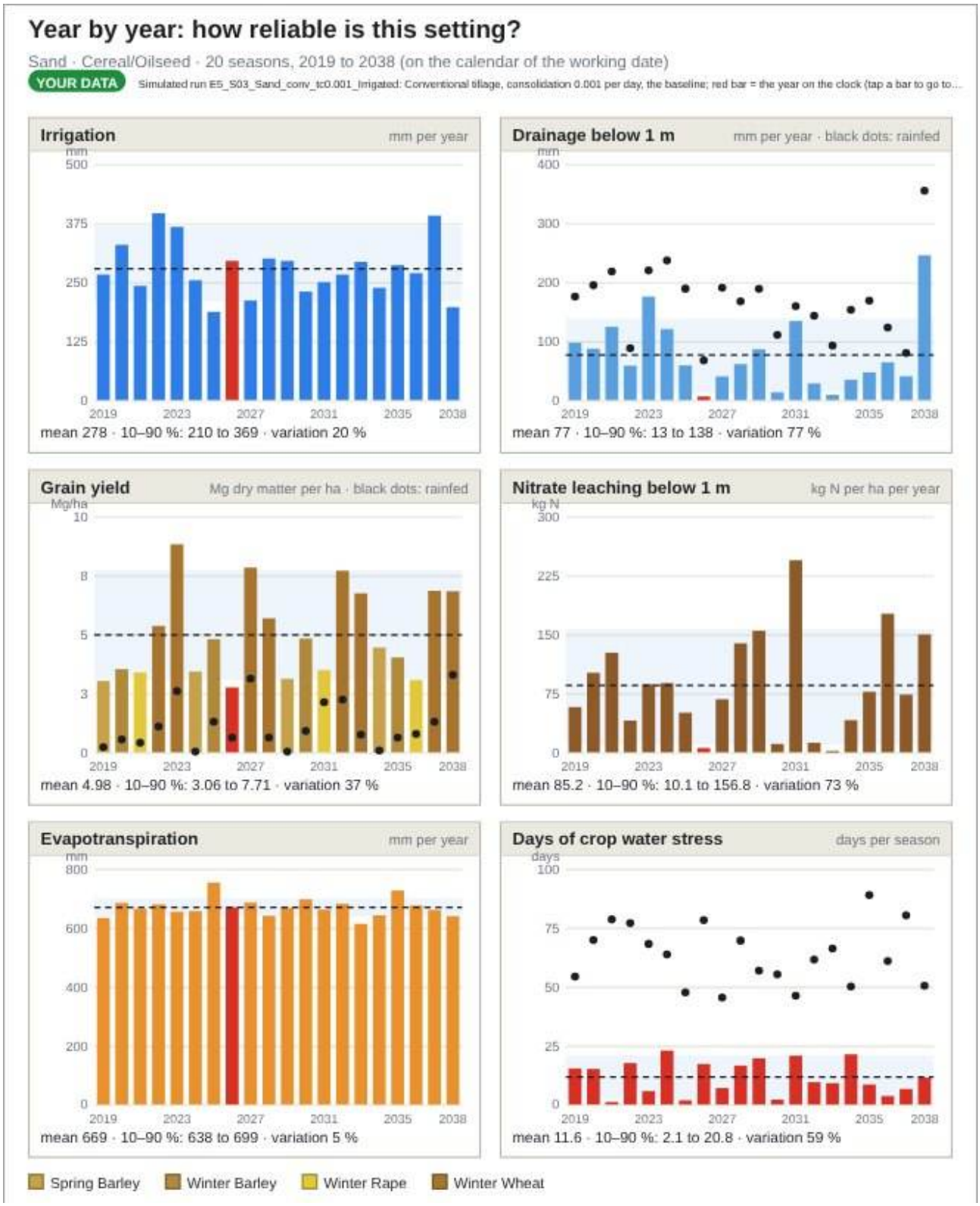


Figure 1.58: Reliability over 20 seasons, Sand, part 1 of 2.

Reliability of this setting over the 20 seasons — from the yearly values above

Irrigation needed in a dry year (9 years in 10 need less)	**369 mm**	against 278 mm in an average year
Years that need 20 % more irrigation than average	**3 of 20**	plan pump, labour and water rights for these
Years with more than 50 mm drainage below the roots	**12 of 20**	water and nitrate lost to groundwater
Years with a yield below 80 % of the average	**8 of 20**	yield risk at this setting
Nitrate leaching in the worst year	**244.3 kg N/ha**	mean 85.2 kg N/ha

How often a given irrigation is exceeded — the 20 seasons sorted from wettest to driest

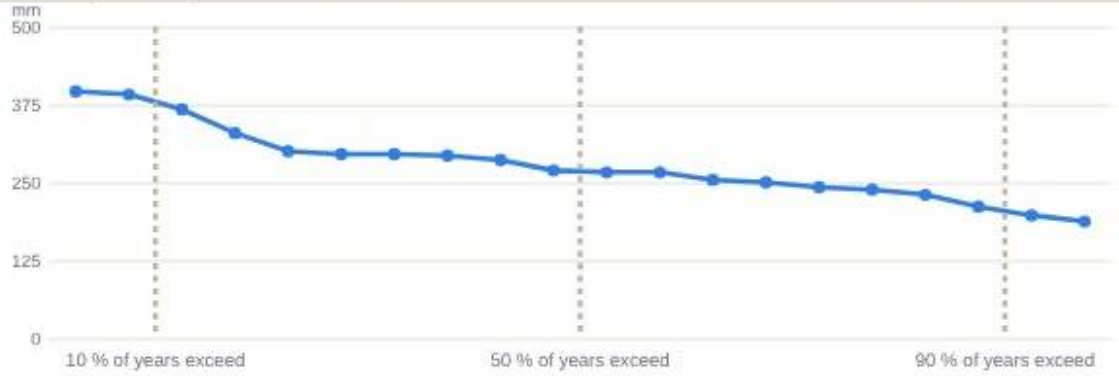


Figure 1.59: Reliability over 20 seasons, Sand, part 2 of 2.

1.21 Full water balance (Sand)

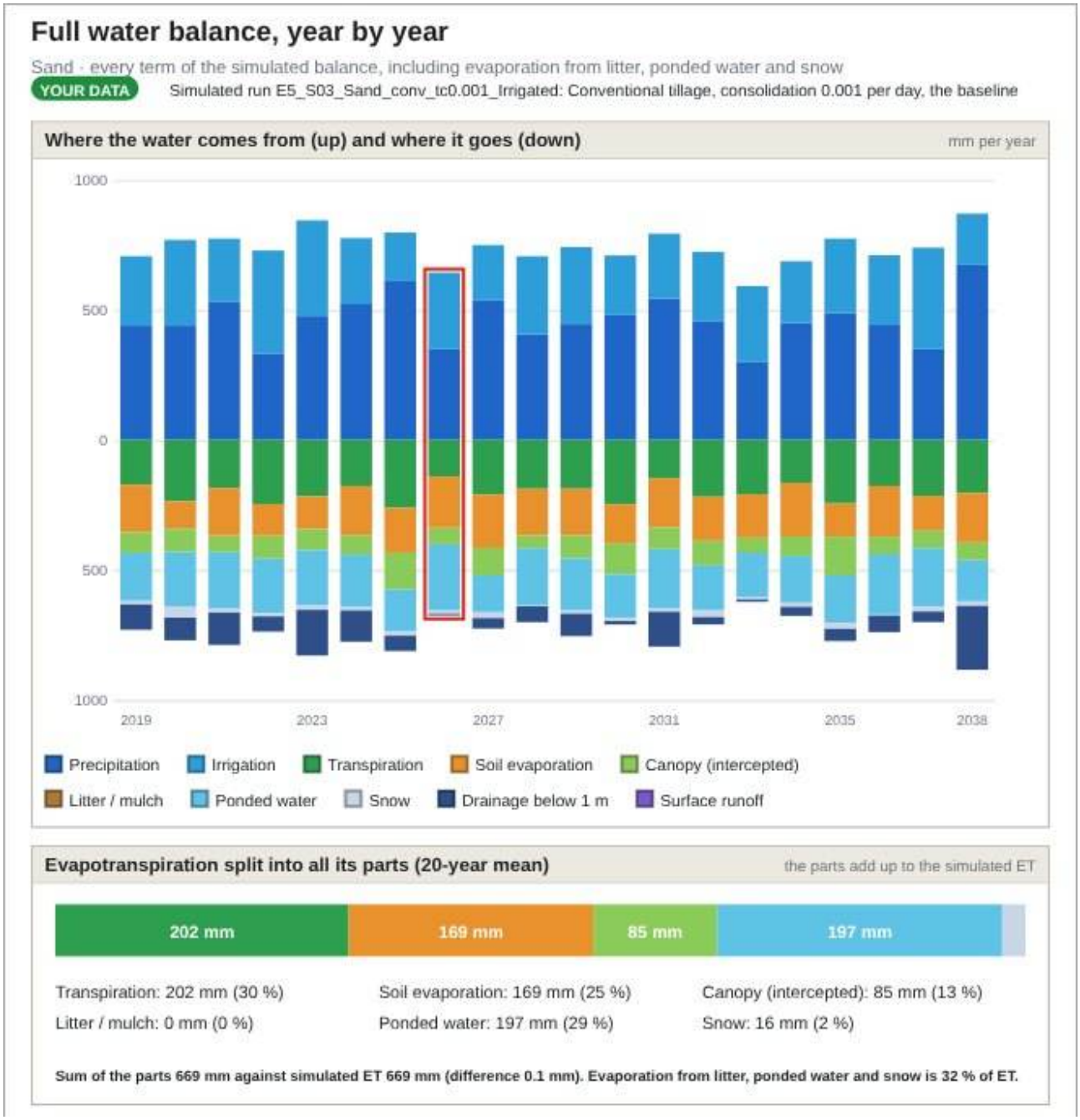


Figure 1.60: Full water balance, Sand, part 1 of 2.

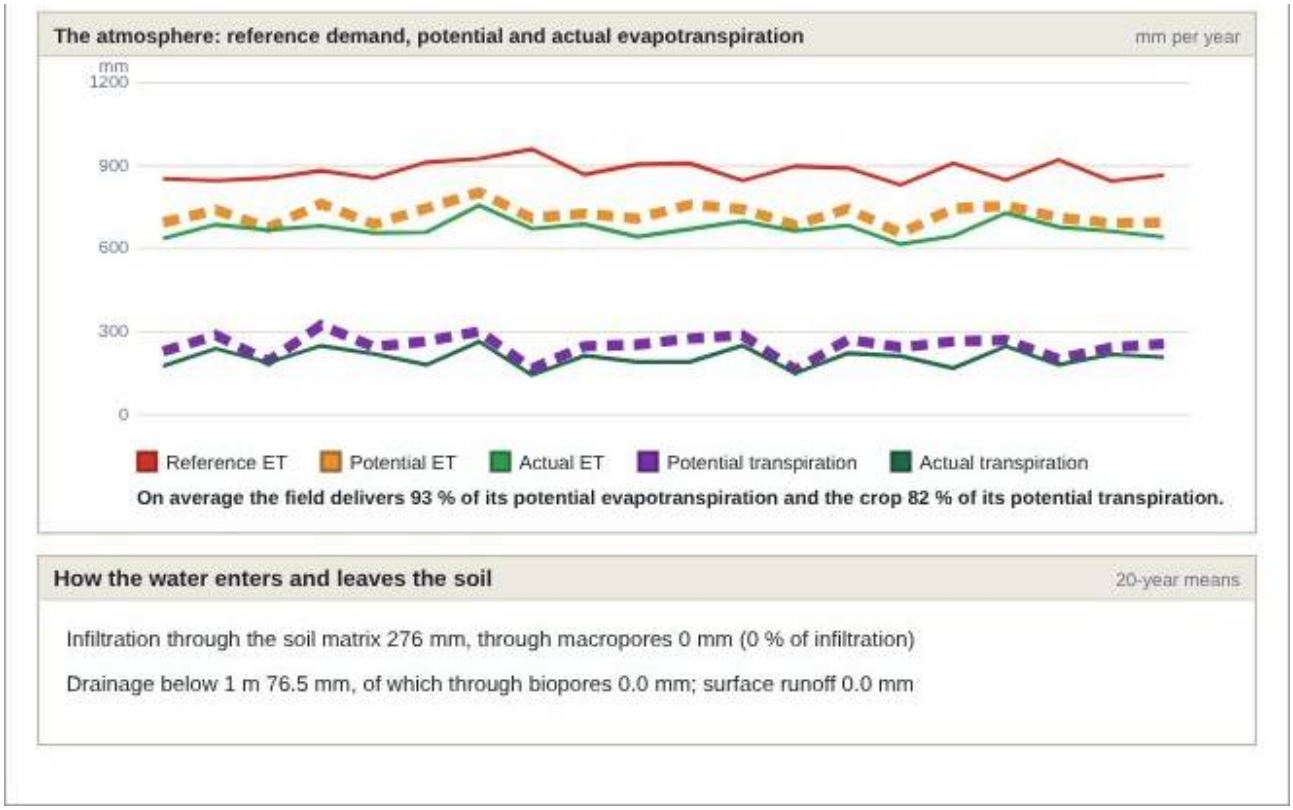


Figure 1.61: Full water balance, Sand, part 2 of 2.

1.22 Nitrogen budget (Sand)

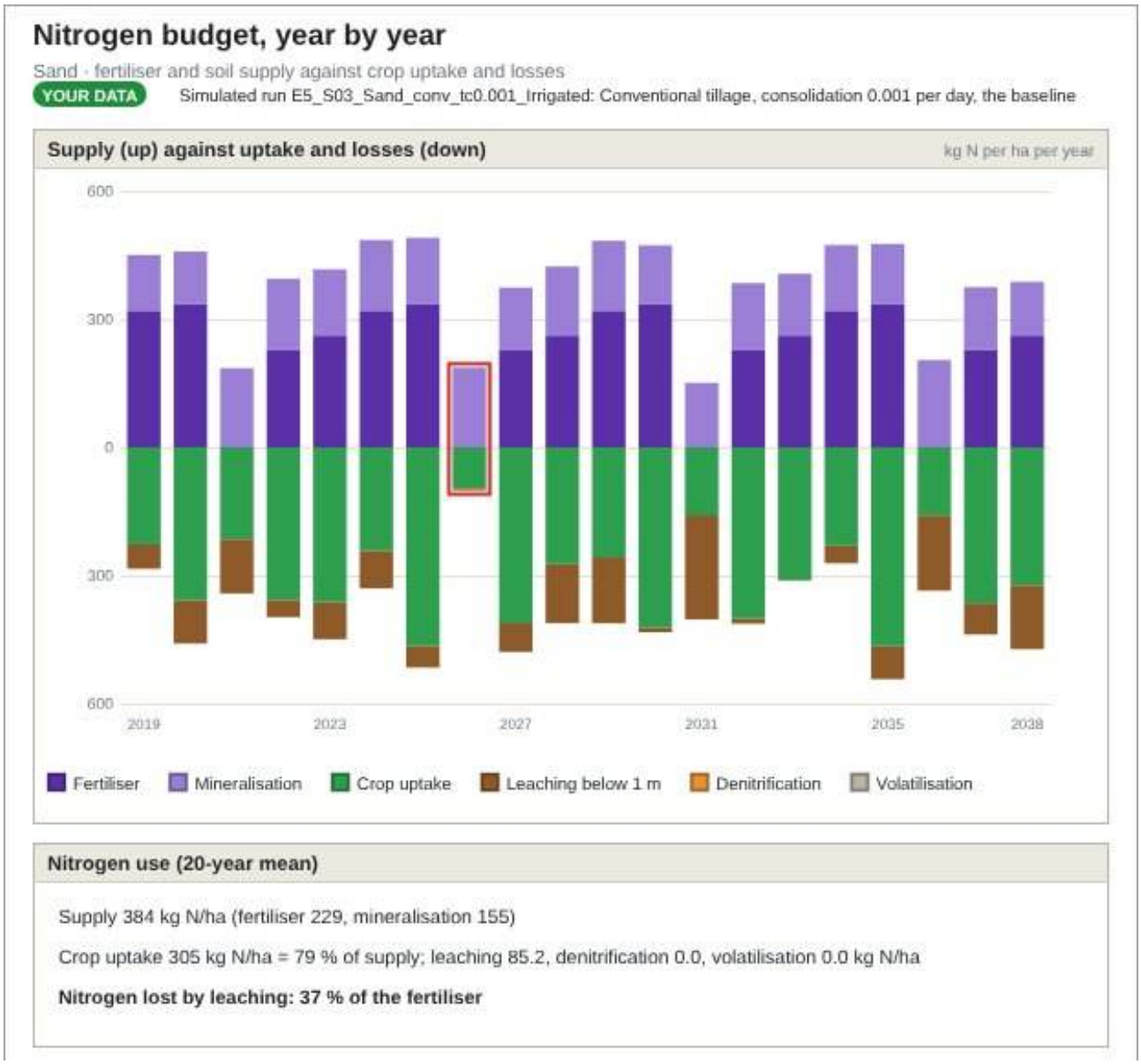


Figure 1.62: Nitrogen budget, Sand, part 1 of 2.

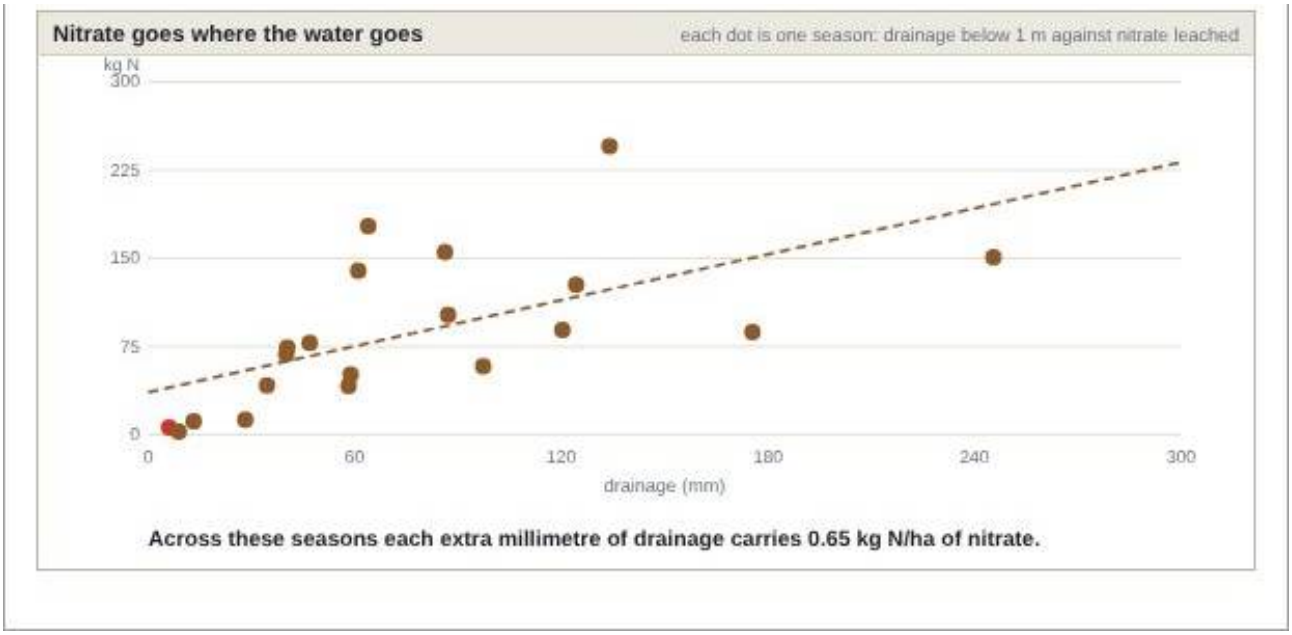


Figure 1.63: Nitrogen budget, Sand, part 2 of 2.

1.23 Crop water requirement (Sand)

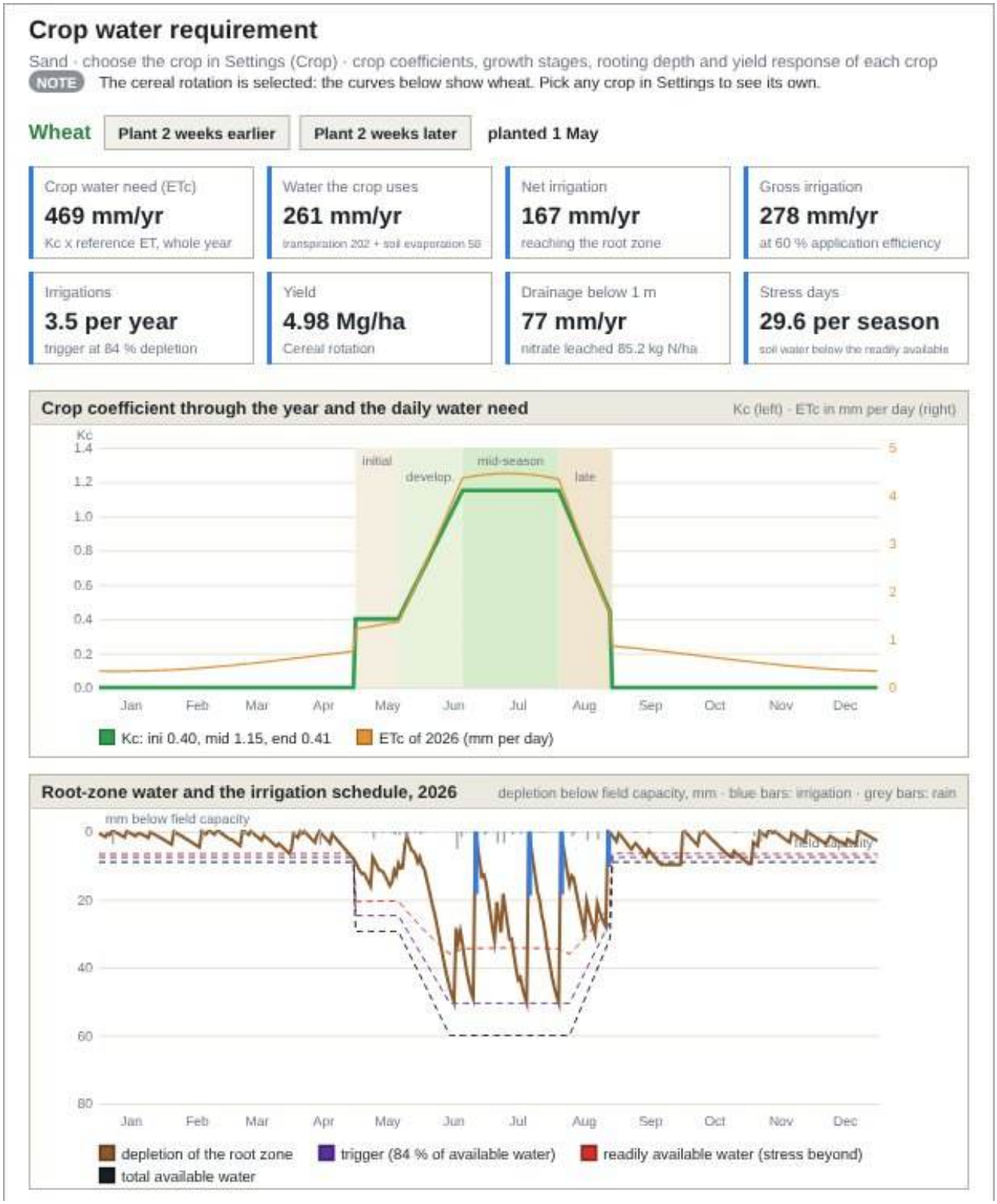


Figure 1.64: Crop water requirement, Sand, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Oil palm (young)	790	**437**	9.5	98 %
Coconut (high)	834	**378**	7.4	89 %
Papaya (high)	814	**369**	8.1	88 %
Avocado (medium)	677	**339**	8.7	100 %
Cotton	571	**316**	6.4	97 %
Cocoa (under high shade canopy)	913	**266**	6.1	72 %
Yam	649	**256**	8.7	82 %
Guava (high)	651	**233**	4.6	91 %
Cabbage	559	**230**	7.2	84 %
Mustard	468	**218**	4.6	93 %
Wheat	469	**202**	4.2	90 %
Walnut (medium)	523	**202**	4.0	90 %
Mango (medium)	620	**200**	3.9	92 %
Pear (medium)	528	**195**	3.9	90 %
Hop (v trellis)	520	**193**	3.8	87 %
Onion (dry)	565	**191**	8.4	70 %
Groundnut	471	**186**	5.1	91 %
Date palm (medium)	602	**184**	3.6	92 %
Maize (grain)	451	**182**	3.7	86 %
Soybean	473	**180**	3.9	88 %
Cherry (medium)	503	**180**	3.6	91 %
Potato	508	**169**	6.8	73 %
Tomato	521	**164**	3.3	78 %
Chickpea	436	**163**	4.2	89 %
Kiwi (pergola)	626	**159**	3.1	76 %
Sorghum (grain)	429	**158**	3.2	93 %
Hazelnut (low)	481	**152**	3.0	91 %
Pecan (low)	481	**152**	3.0	91 %
Cherimoya (medium)	545	**147**	2.9	94 %
Citrus (medium density, short)	552	**144**	2.8	94 %
Millet	395	**139**	3.0	93 %
Table grapes (horizontal trellis)	597	**138**	2.7	77 %
Rice	573	**137**	3.6	62 %
Pistachio (medium)	503	**128**	2.5	83 %
Cassava	466	**126**	3.9	81 %

Figure 1.65: Crop water requirement, Sand, part 2 of 3.

Apple (medium)	459	**122**	2.4	91 %
Cowpea (beans)	398	**116**	3.0	85 %
Peach (low)	435	**116**	2.3	93 %
Sweet pepper	475	**107**	2.8	68 %
Wine grapes (pergola)	457	**104**	2.0	91 %
Green gram (mung bean)	372	**91**	2.4	86 %
Almond (medium)	435	**79**	1.6	87 %
Olive (traditional, medium density)	446	**74**	1.4	100 %
Apricot (young)	380	**73**	1.4	95 %
Highbush blueberry (low)	338	**54**	1.2	96 %
Persimmon (young, low)	358	**43**	0.8	96 %
Pomegranate (low)	358	**43**	0.8	96 %
Fig tree (young, low)	327	**28**	0.5	98 %

Figure 1.66: Crop water requirement, Sand, part 3 of 3.

1.24 Feedback loops (Sand)

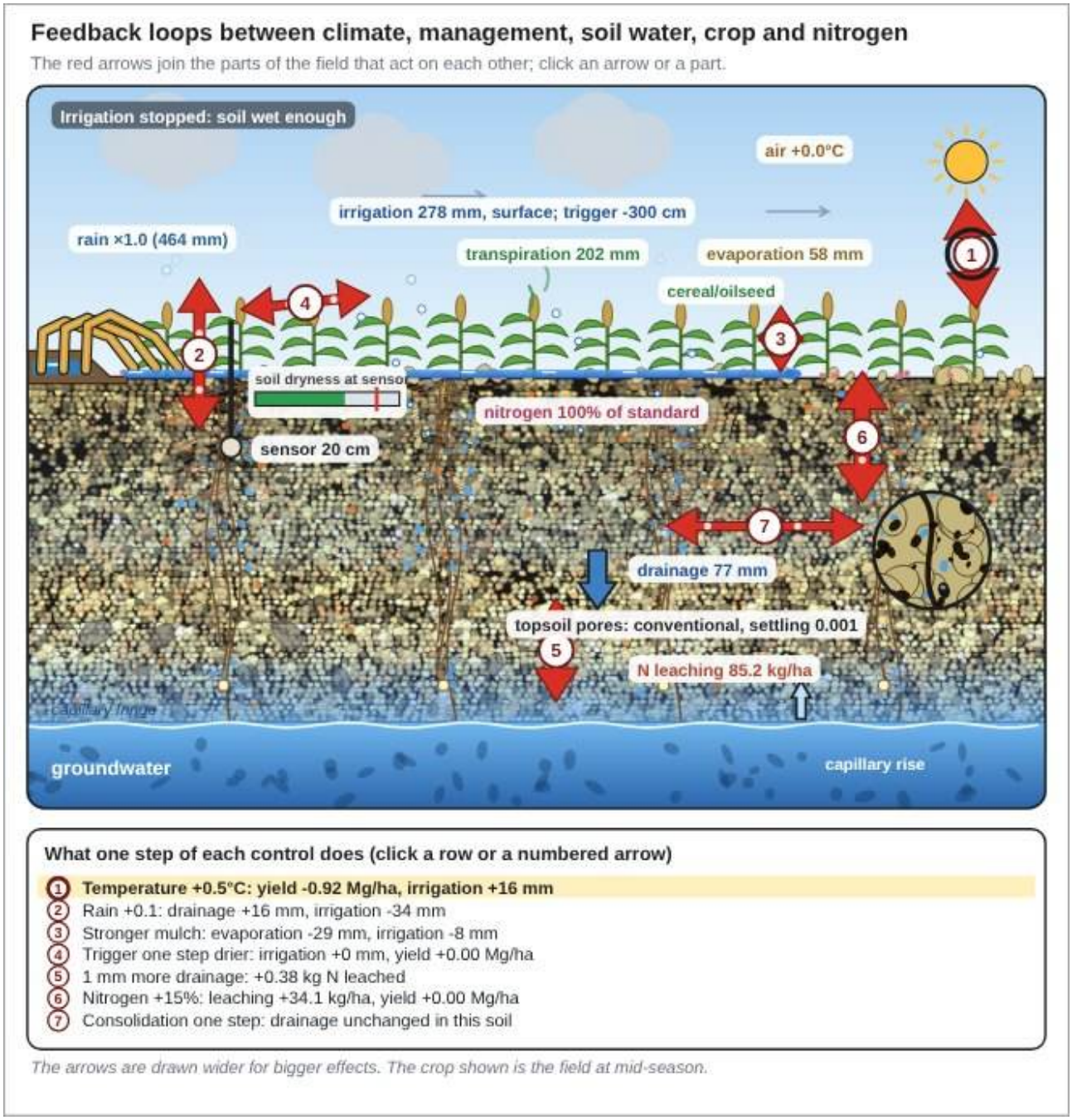


Figure 1.67: Feedback loops, Sand, part 1 of 1.

Chapter 2

Loamy Sand

2.1 Surface irrigation design (Loamy Sand)

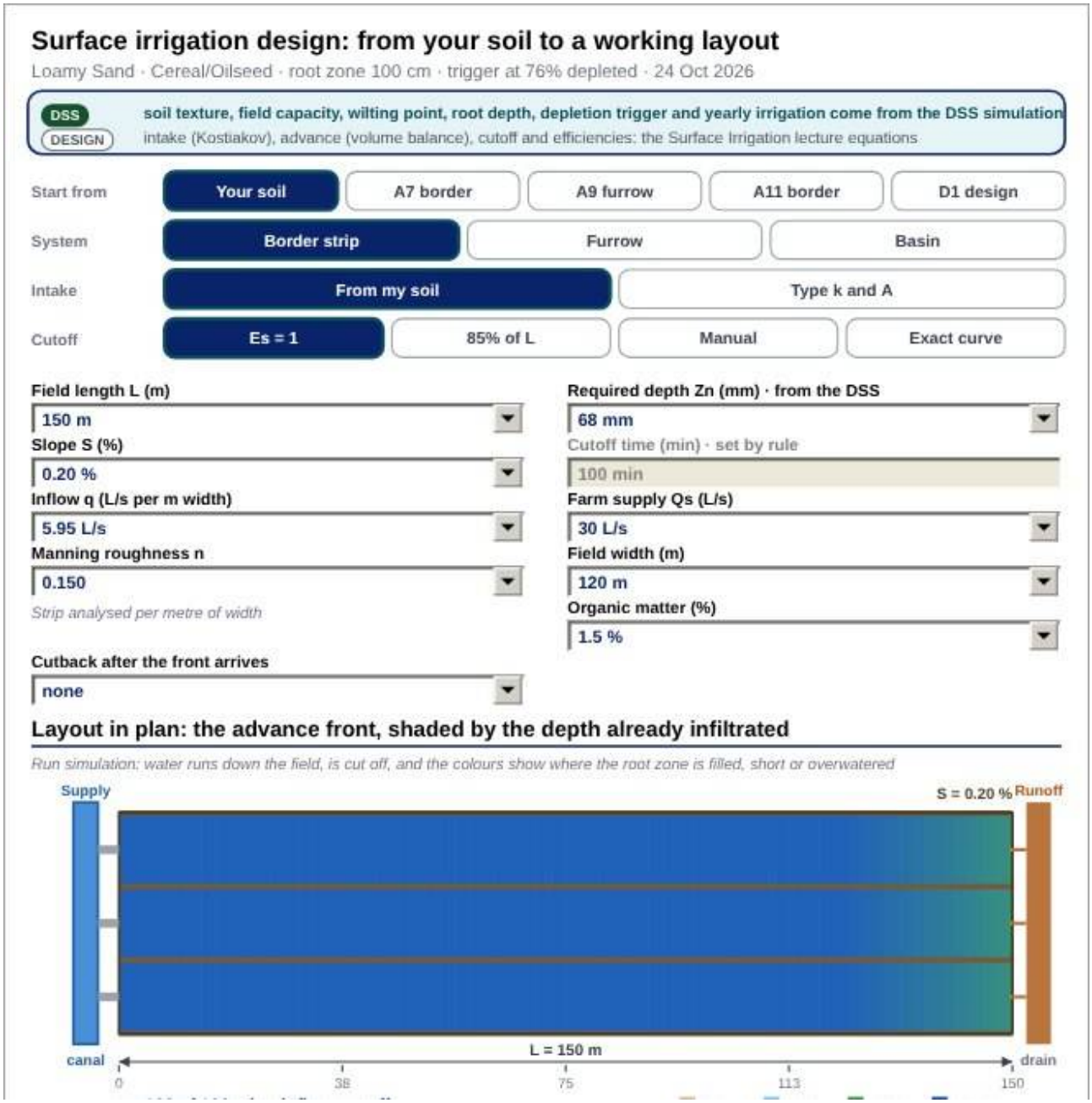


Figure 2.1: Surface irrigation design, Loamy Sand, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

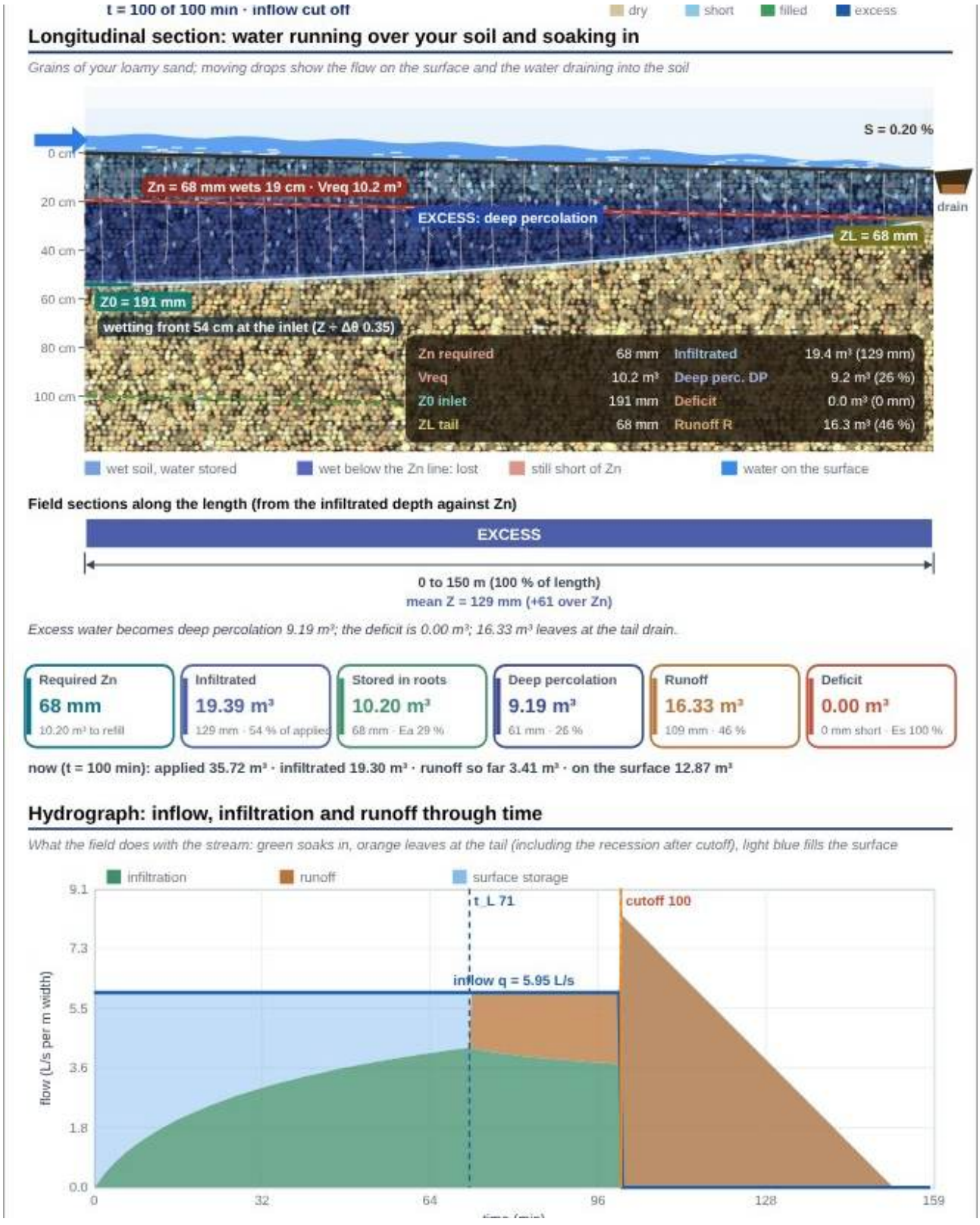


Figure 2.2: Surface irrigation design, Loamy Sand, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

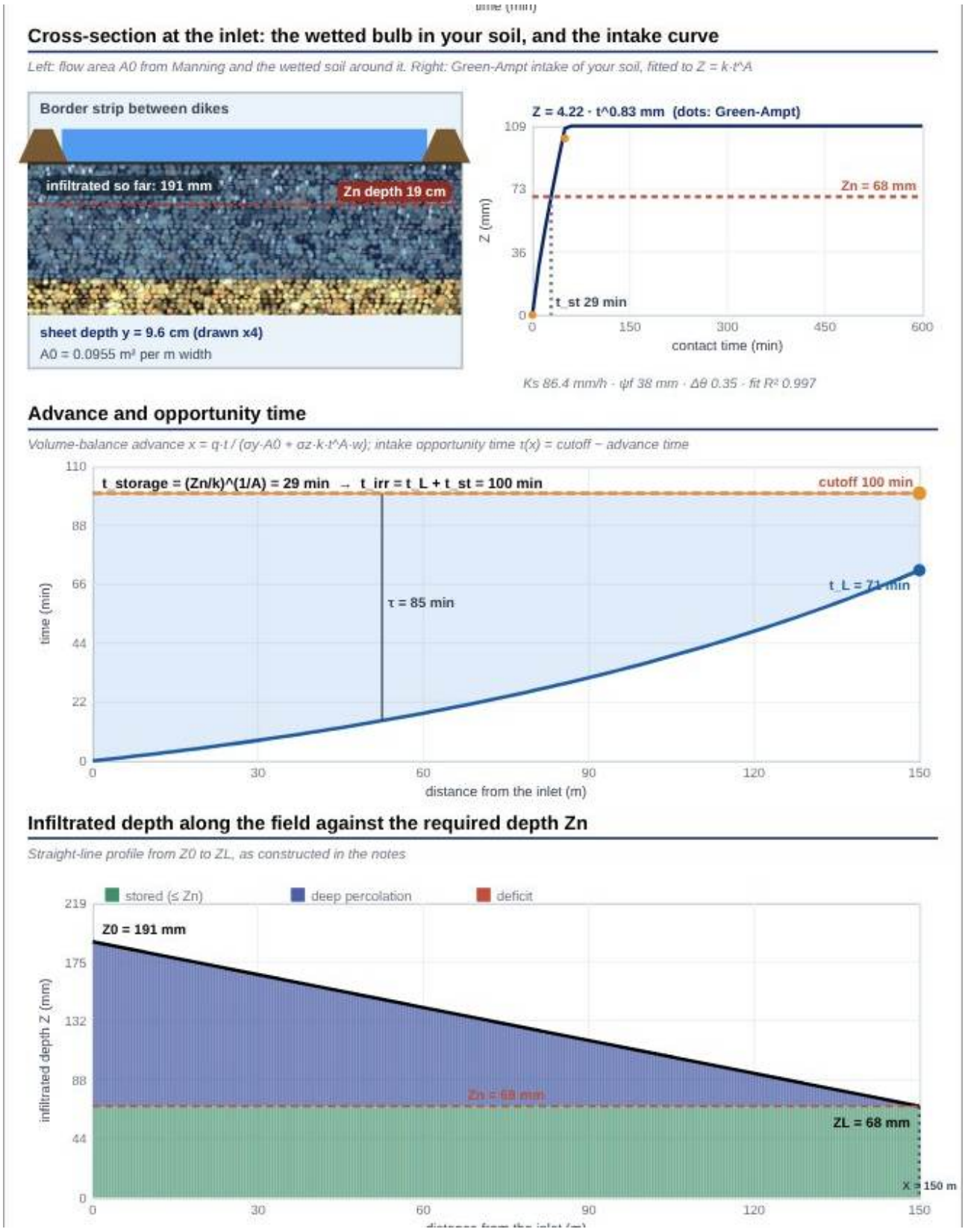


Figure 2.3: Surface irrigation design, Loamy Sand, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

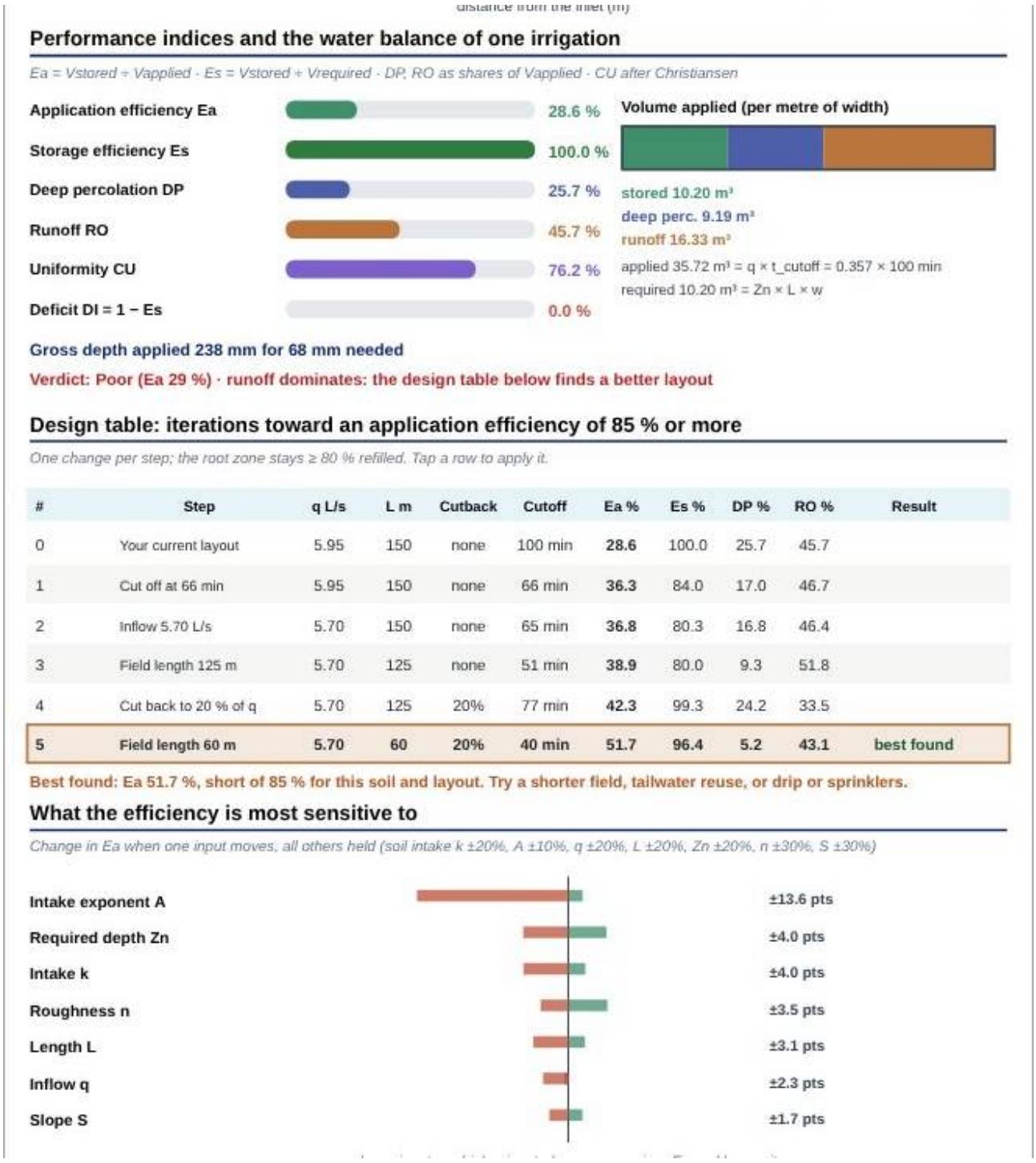


Figure 2.4: Surface irrigation design, Loamy Sand, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

lower input ◄ ►higher input - bars: green raises Ea, red lowers it

What this means for the season and the farm

Change the date or the year in Properties. The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**322 mm → 4.7 irrigations of 68 mm**
Gross water per irrigation	**238 mm (Ea 29 %)**
Gross water per season	**1129 mm = 11287 m³ per hectare**
Units run at once N0M = Qs ÷ q	**5.0 (30 L/s ÷ 5.95 L/s)**
Sets to cover 120 m of width	**23.8 sets × 100 min = 39.7 h per irrigation**
Pumping per season	**188 hours**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**23 May 2027**	211 days	29	103	39.7 h	13 days
2	**5 Jun 2027**	224 days	29	103	39.7 h	10 days
3	**15 Jun 2027**	234 days	29	103	39.7 h	8 days
4	**23 Jun 2027**	242 days	29	103	39.7 h	7 days
5	**30 Jun 2027**	249 days	29	103	39.7 h	8 days
6	**8 Jul 2027**	257 days	29	103	39.7 h	14 days
7	**22 Jul 2027**	271 days	29	103	39.7 h	258 days
8	**5 Apr 2028**	529 days	29	103	39.7 h	-

Intervals, efficiency and the advice in short

Everything above in one message for surface irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.76**
Irrigations in the season of this date	**11 (6 Apr 2026 to 27 Jul 2026)**
Interval between irrigations	**11.2 days on average (shortest 7, longest 22)**
Depth per irrigation	**68.0 mm net, 238.2 mm pumped**
Application efficiency Ea	**28.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**39.7 h**

When to irrigate

Next irrigation: 23 May 2027 (in 211 days), about 29 mm net, 103 mm pumped. Then irrigate about every 11 days (between 7 and 22 days, shorter in hot weather and longer after rain). With this surface design 29 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 39.7 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 2.5: Surface irrigation design, Loamy Sand, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

2.2 Drip irrigation design (Loamy Sand)

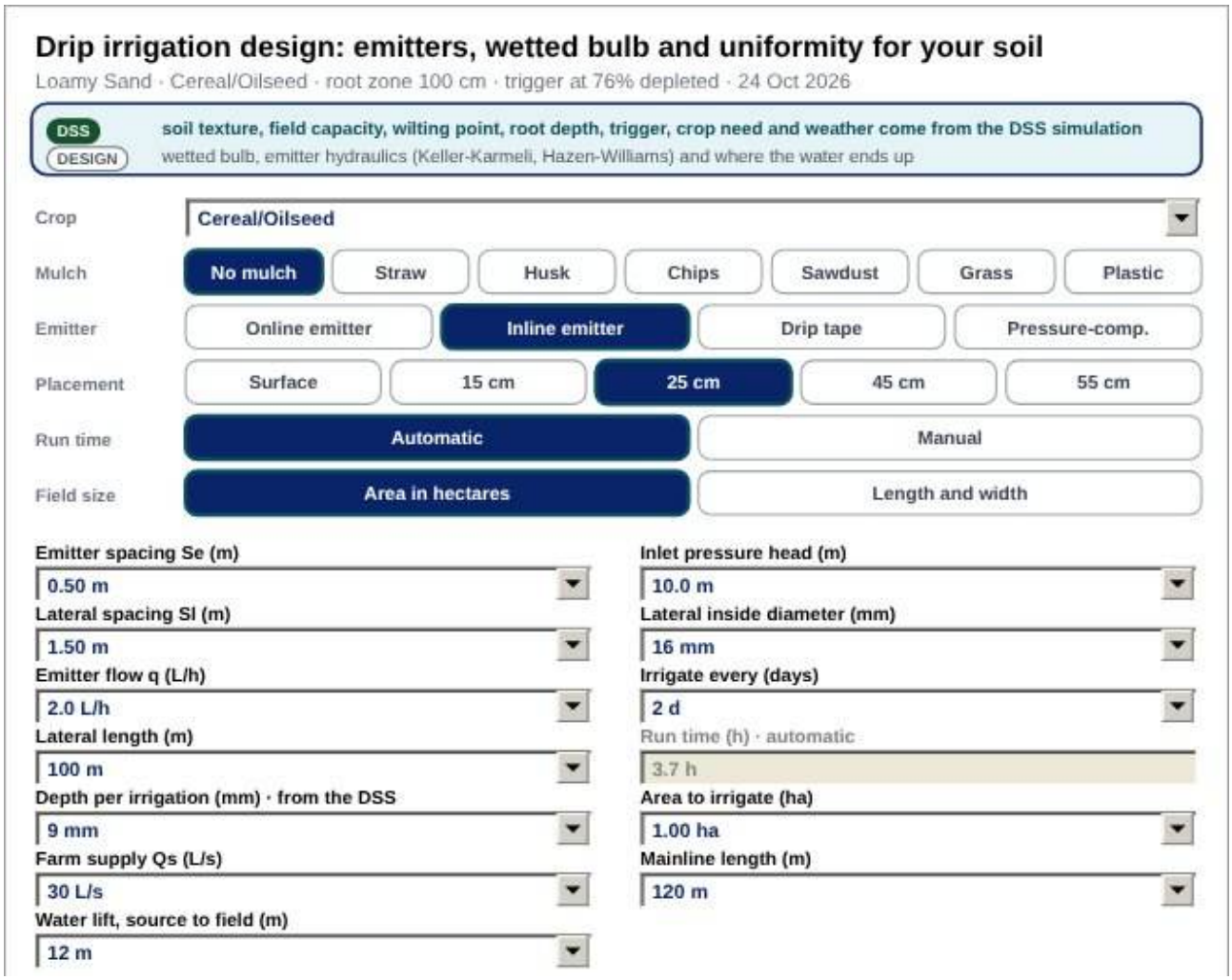


Figure 2.6: Drip irrigation design, Loamy Sand, part 1 of 6.

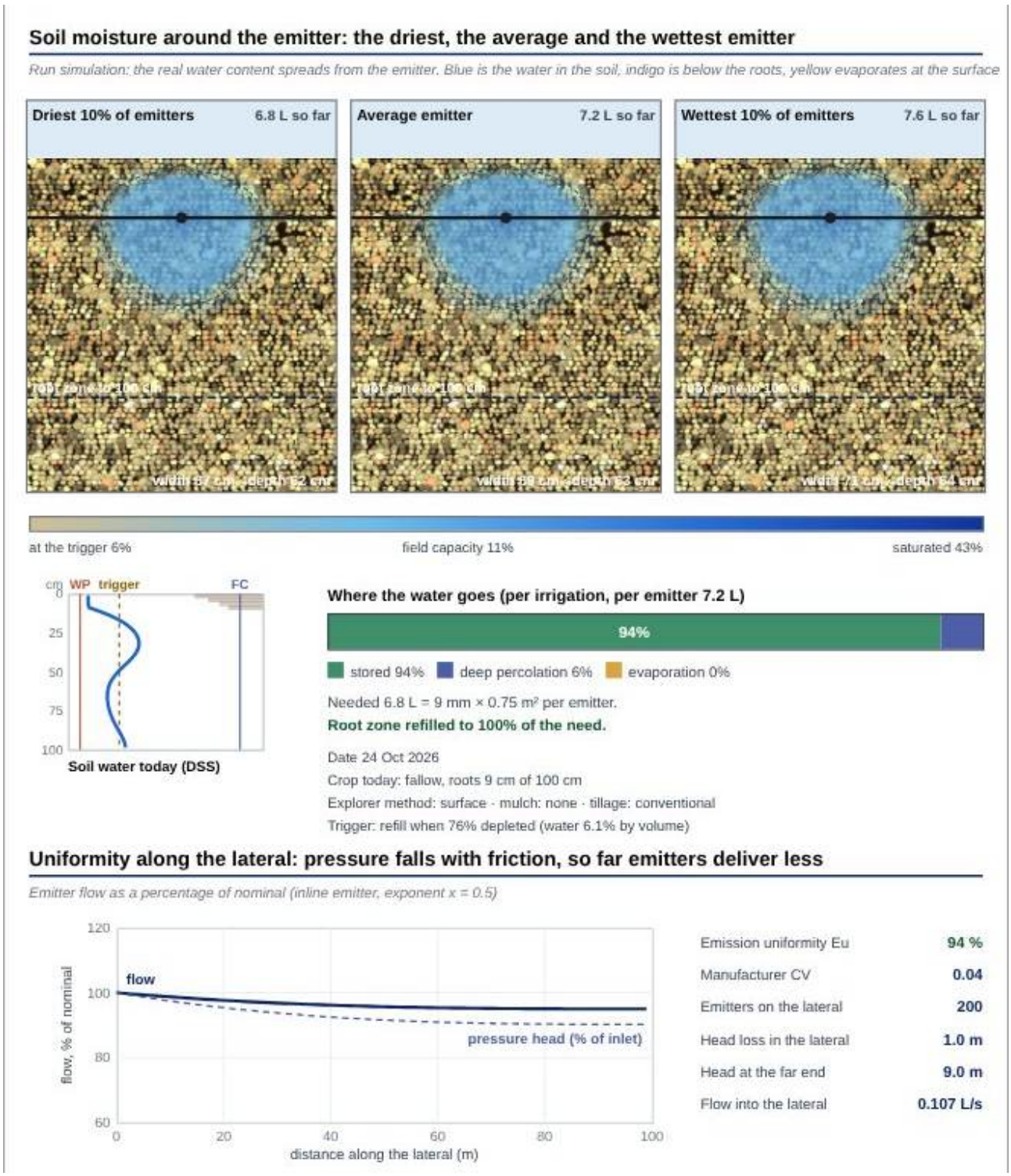


Figure 2.7: Drip irrigation design, Loamy Sand, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

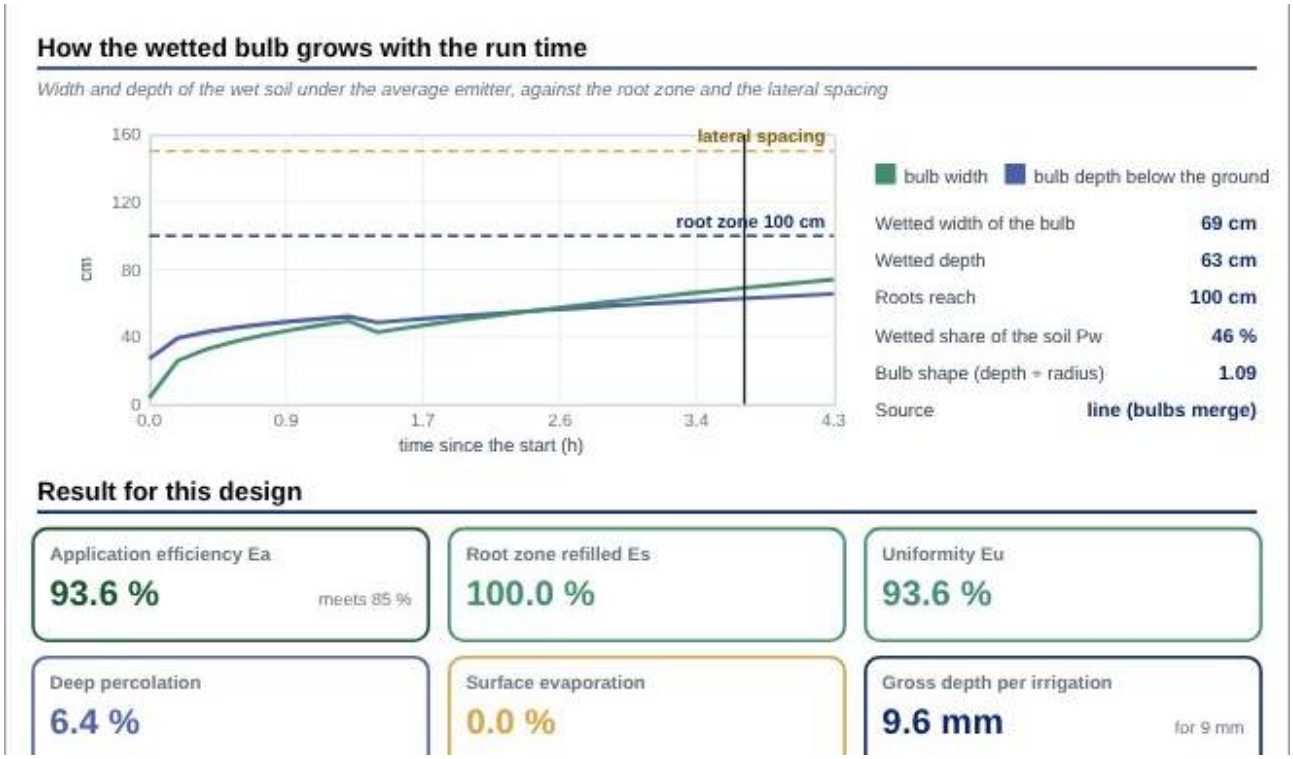
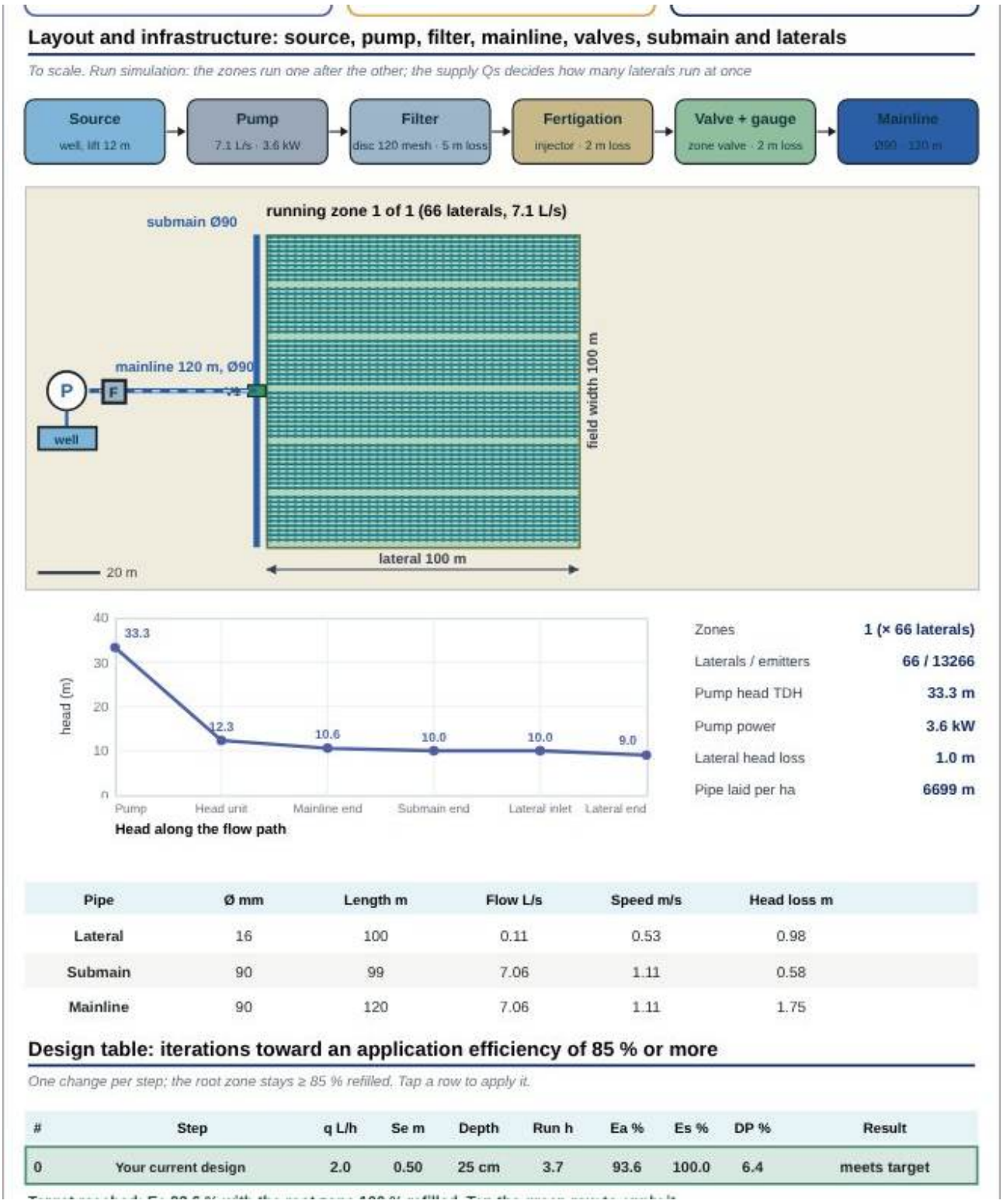


Figure 2.8: Drip irrigation design, Loamy Sand, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.7	93.6	100.0	6.4	meets target

Figure 2.9: Drip irrigation design, Loamy Sand, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

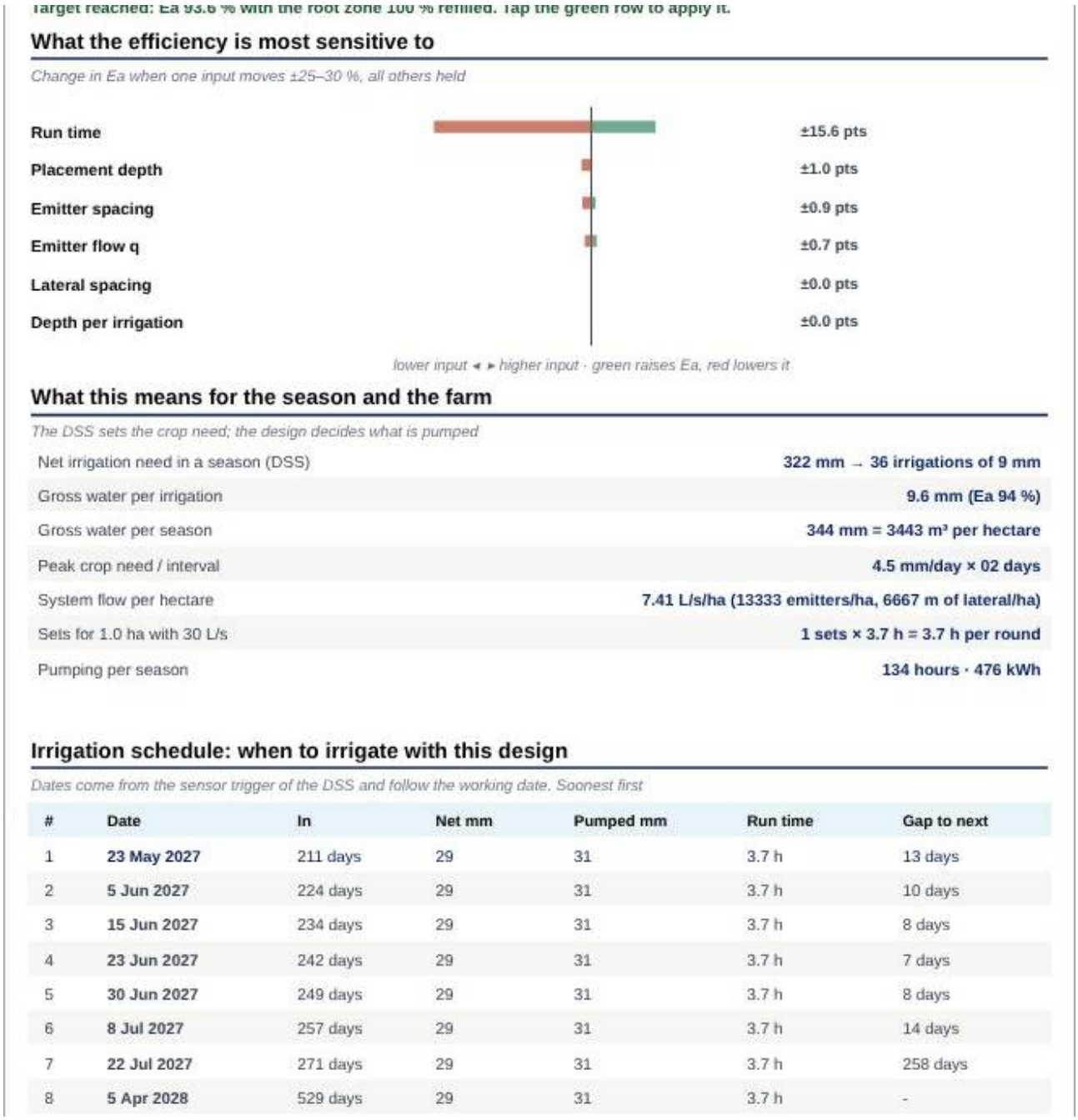
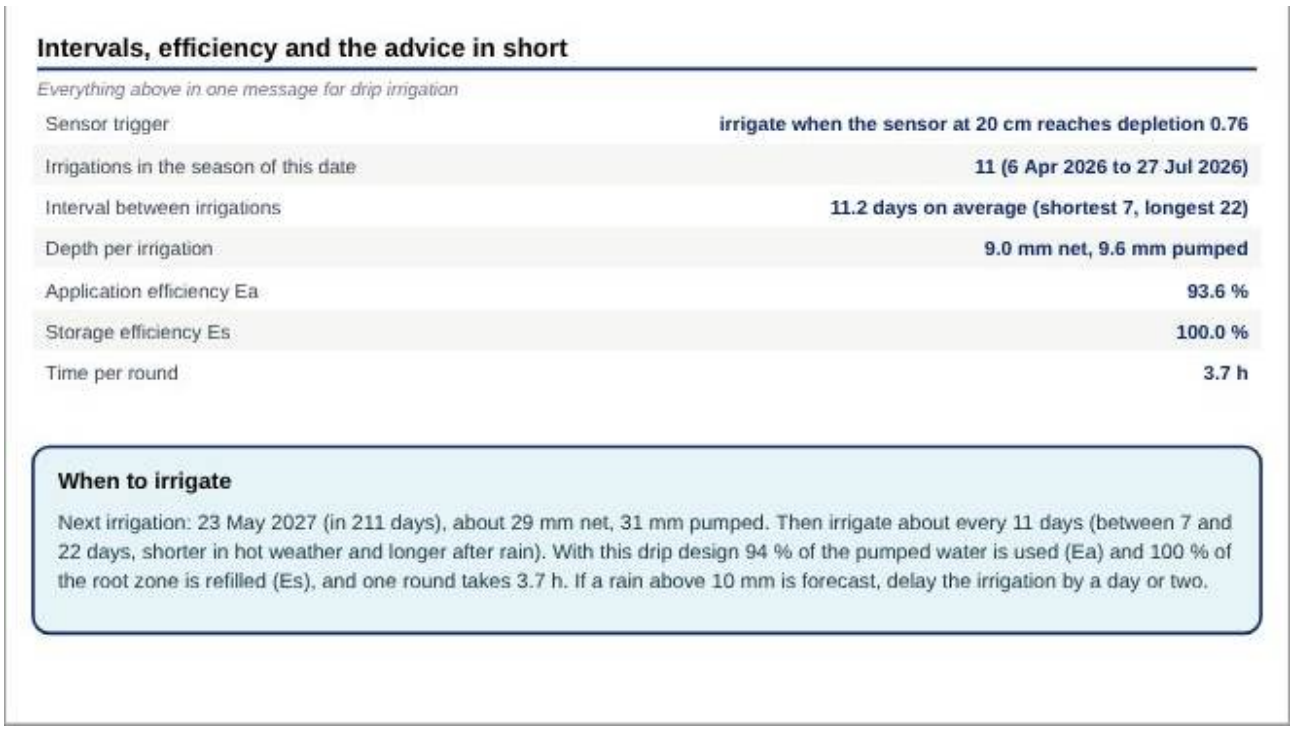


Figure 2.10: Drip irrigation design, Loamy Sand, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.76**
Irrigations in the season of this date	**11 (6 Apr 2026 to 27 Jul 2026)**
Interval between irrigations	**11.2 days on average (shortest 7, longest 22)**
Depth per irrigation	**9.0 mm net, 9.6 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**3.7 h**

When to irrigate

Next irrigation: 23 May 2027 (in 211 days), about 29 mm net, 31 mm pumped. Then irrigate about every 11 days (between 7 and 22 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 3.7 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 2.11: Drip irrigation design, Loamy Sand, part 6 of 6: Intervals, efficiency and the advice in short.

2.3 Sprinkler design and pipes (Loamy Sand)

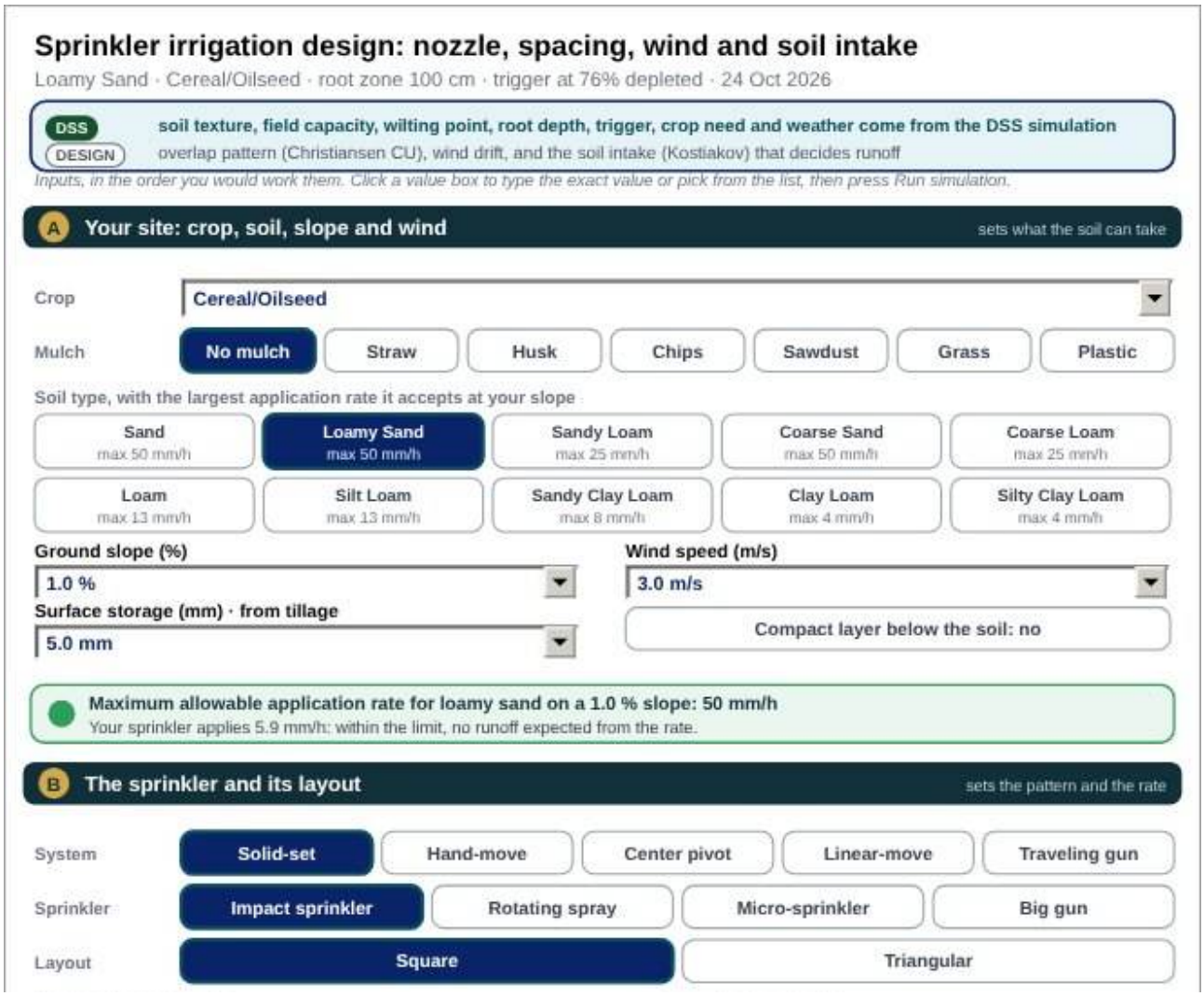


Figure 2.12: Sprinkler design and pipes, Loamy Sand, part 1 of 12.

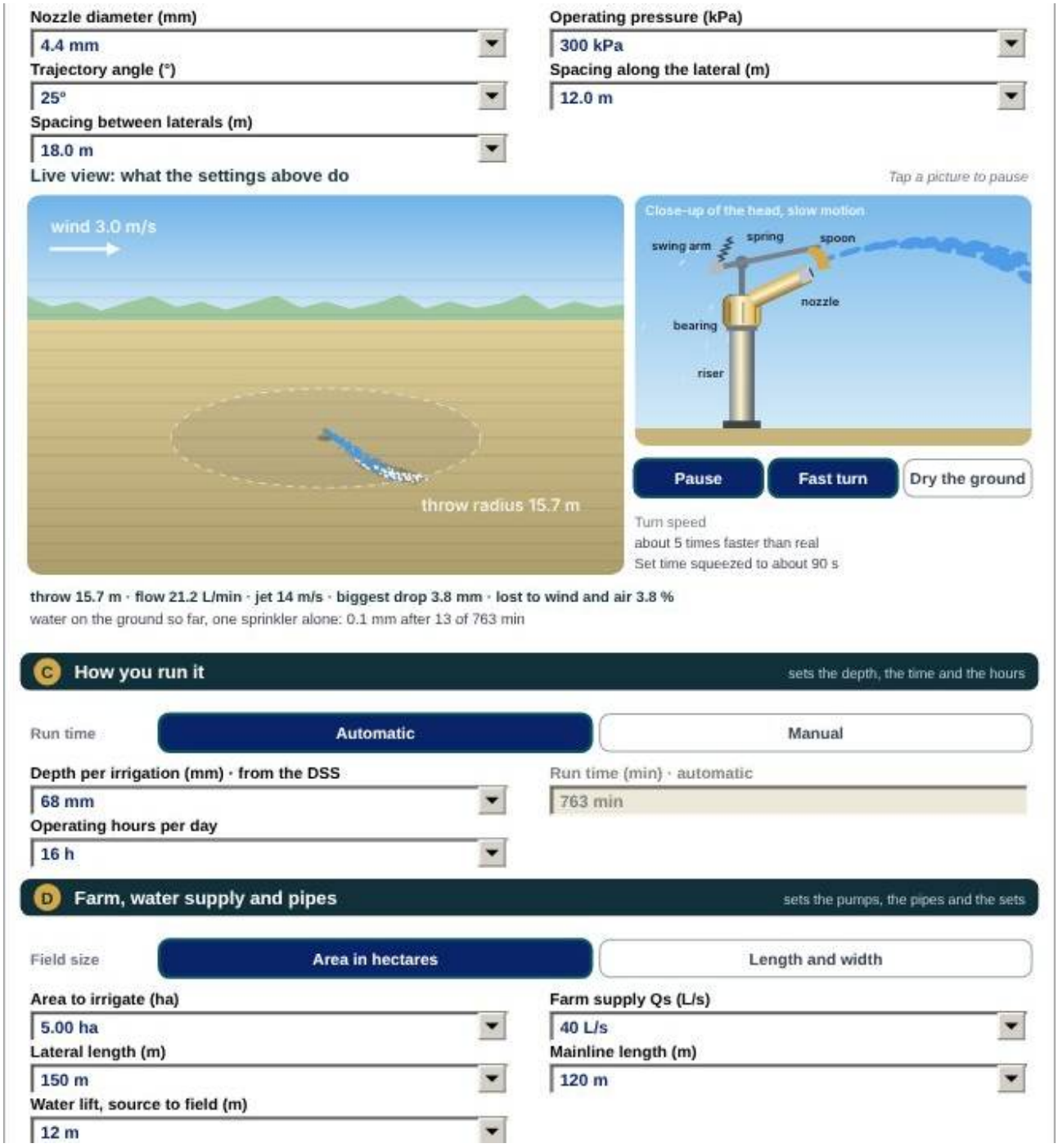


Figure 2.13: Sprinkler design and pipes, Loamy Sand, part 2 of 12.

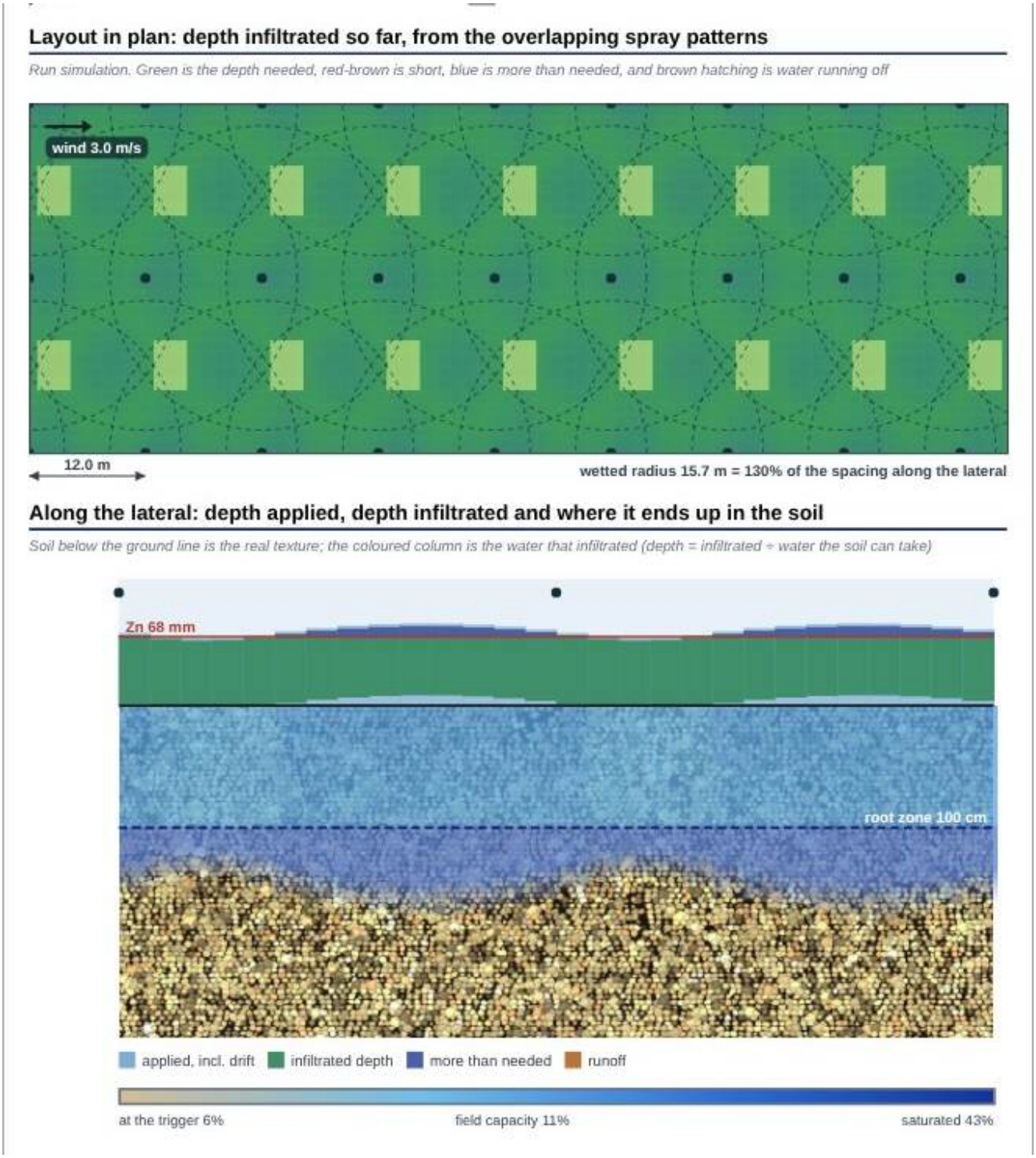


Figure 2.14: Sprinkler design and pipes, Loamy Sand, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

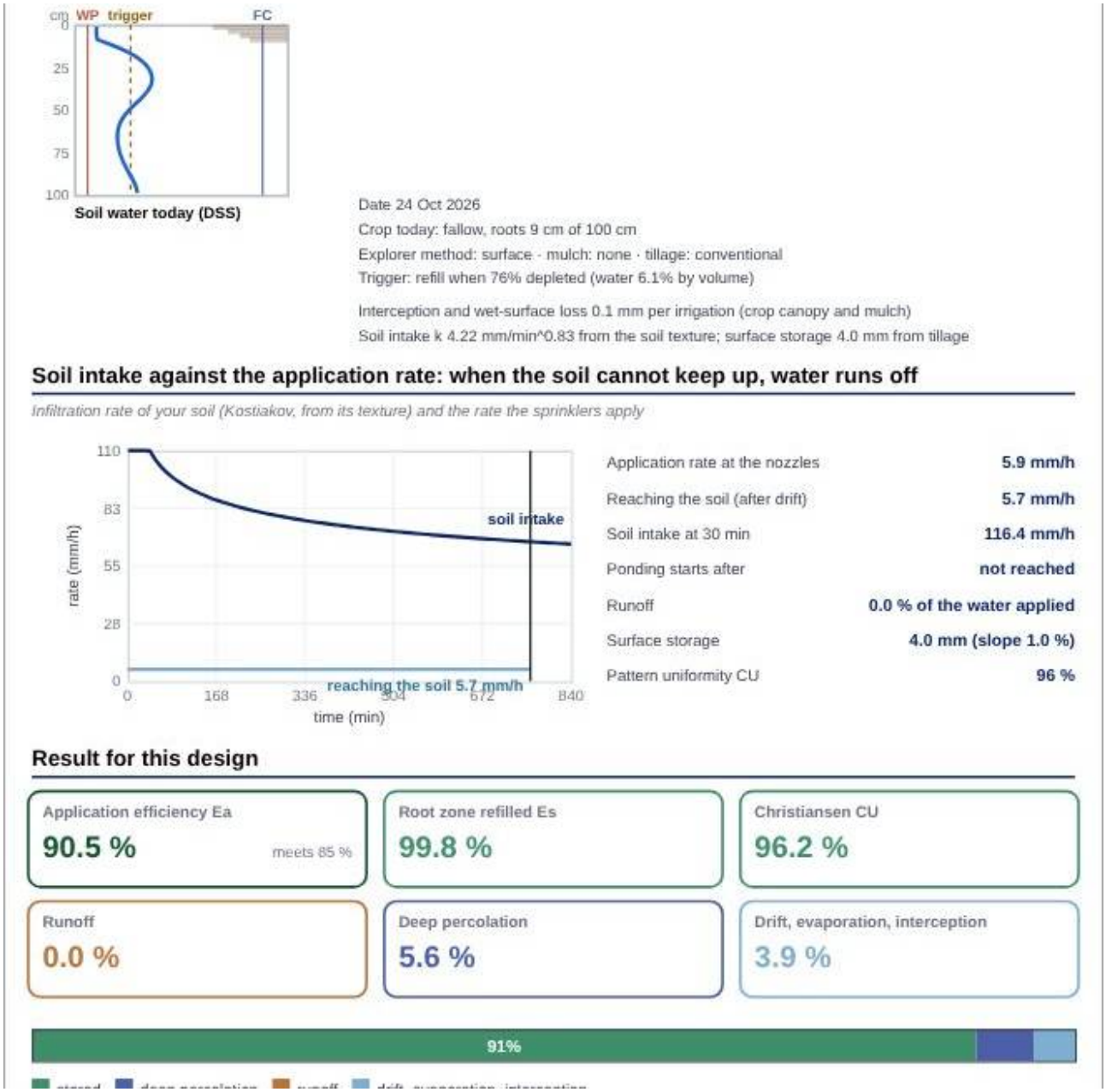


Figure 2.15: Sprinkler design and pipes, Loamy Sand, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

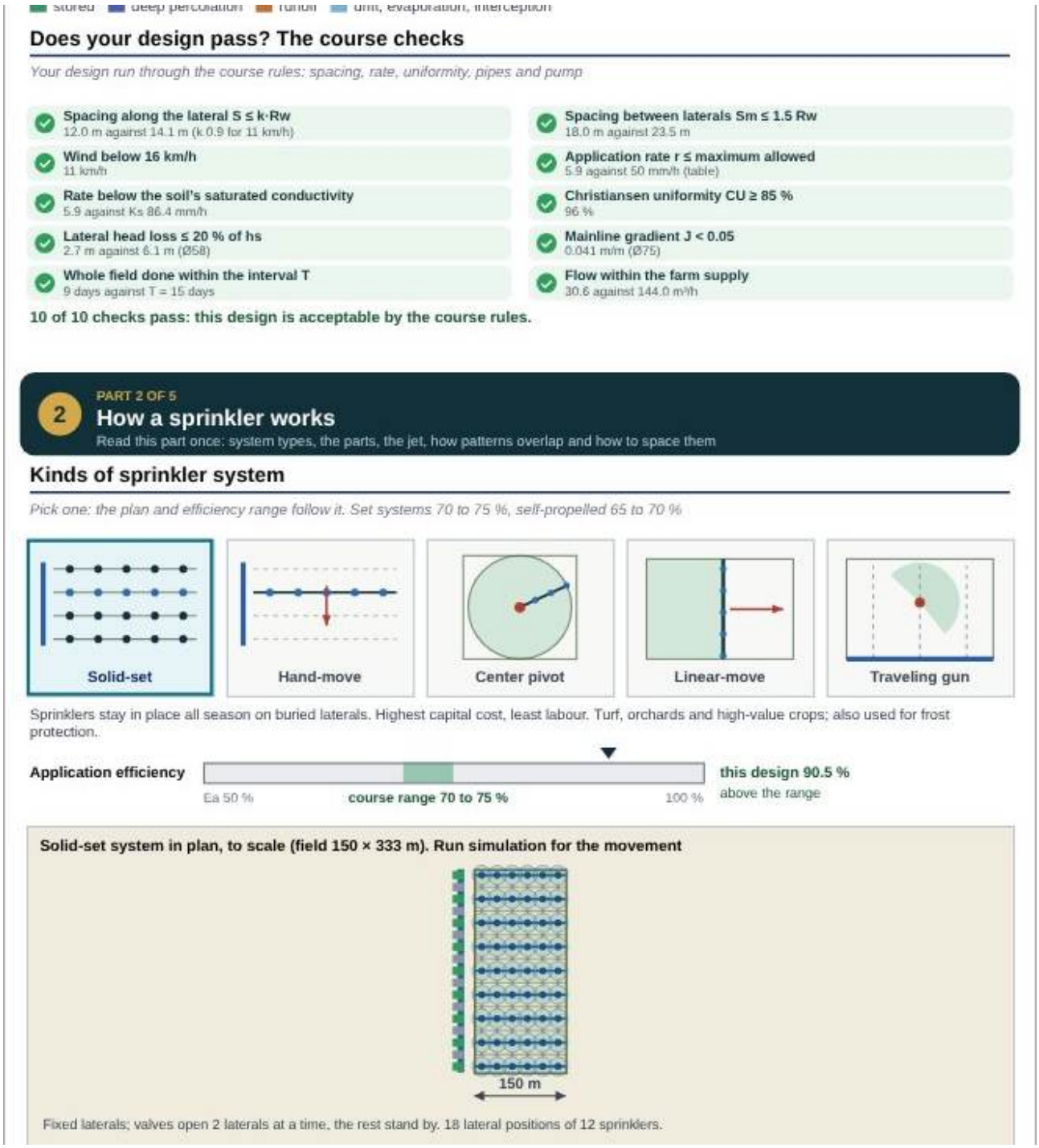


Figure 2.16: Sprinkler design and pipes, Loamy Sand, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

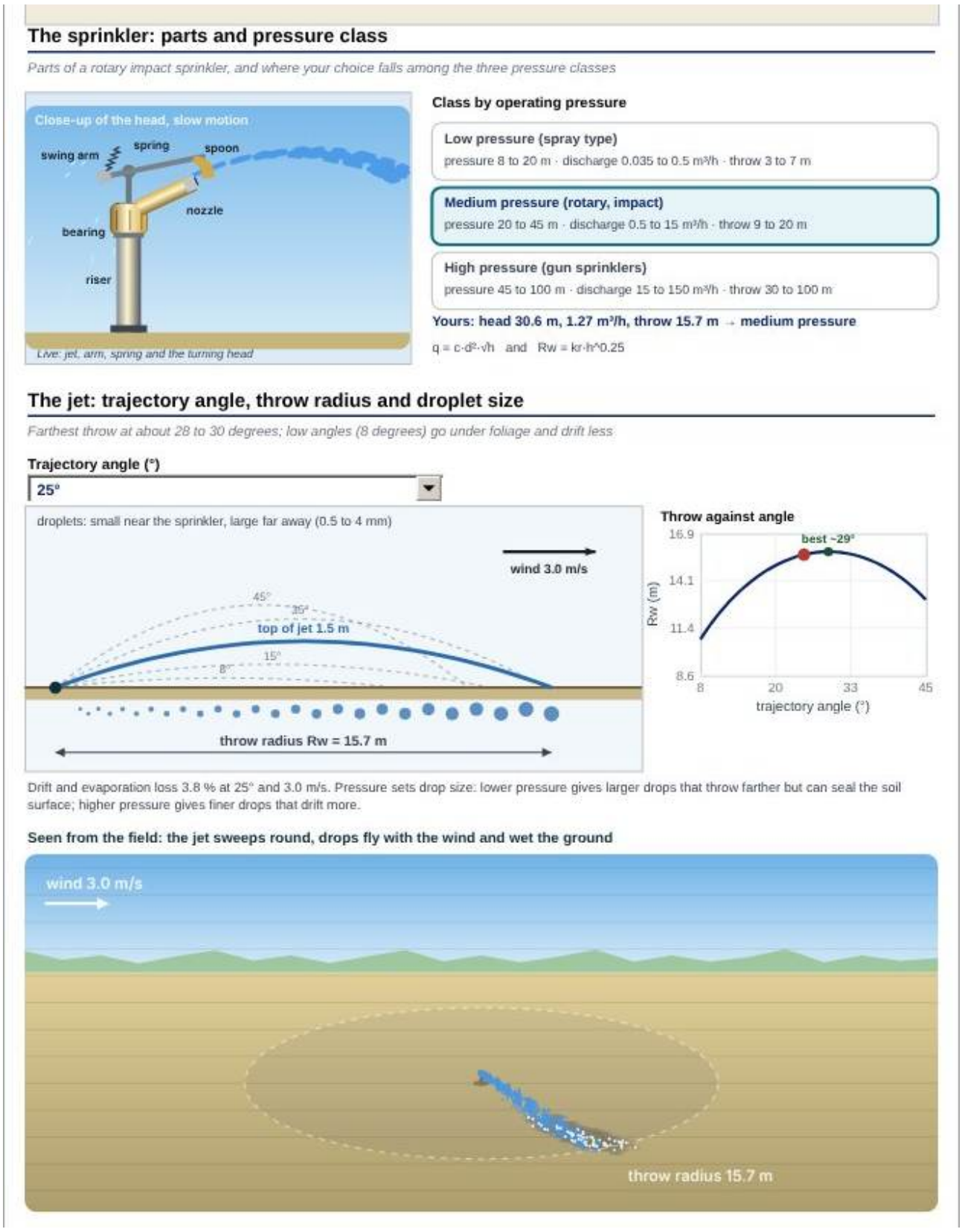


Figure 2.17: Sprinkler design and pipes, Loamy Sand, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

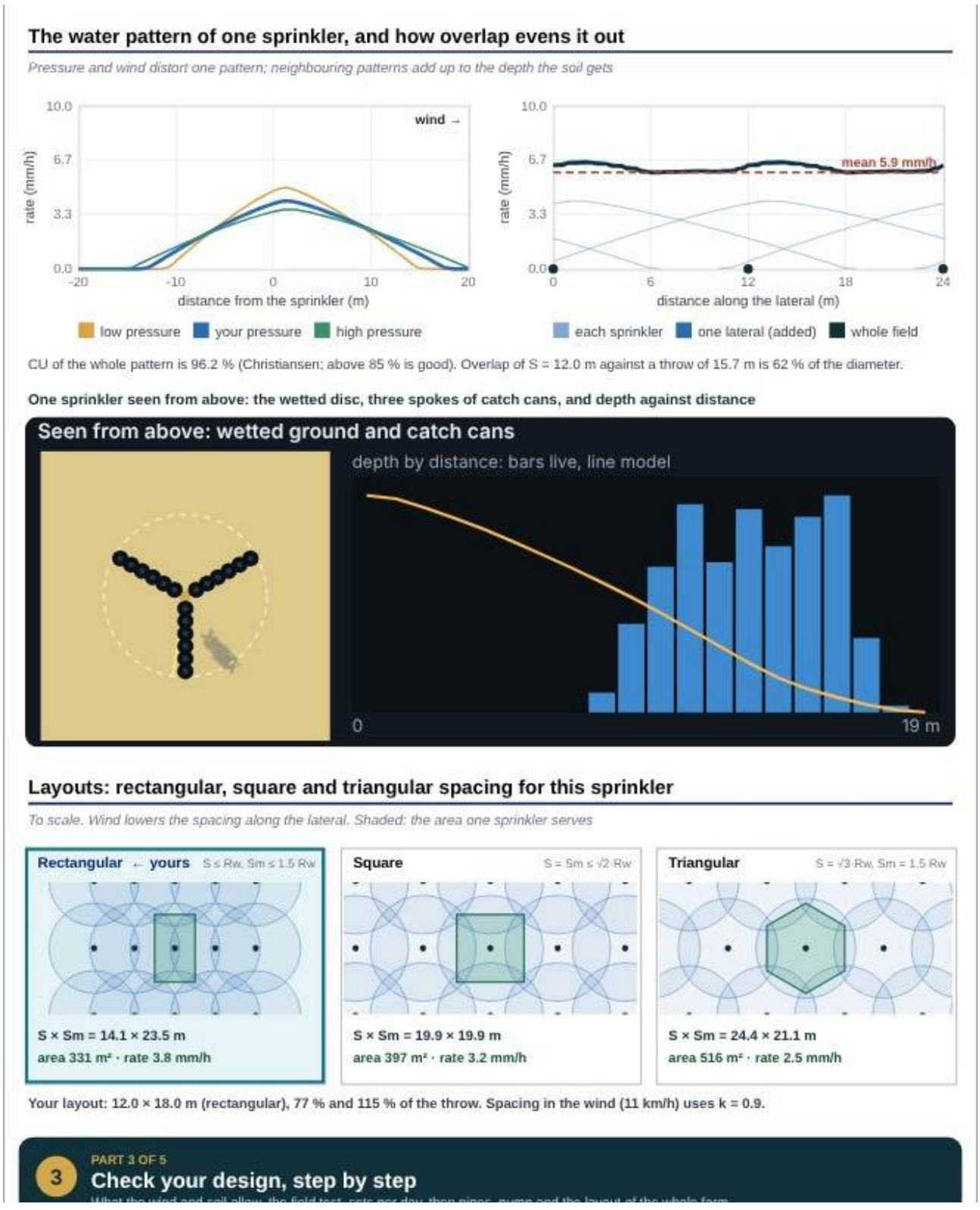


Figure 2.18: Sprinkler design and pipes, Loamy Sand, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

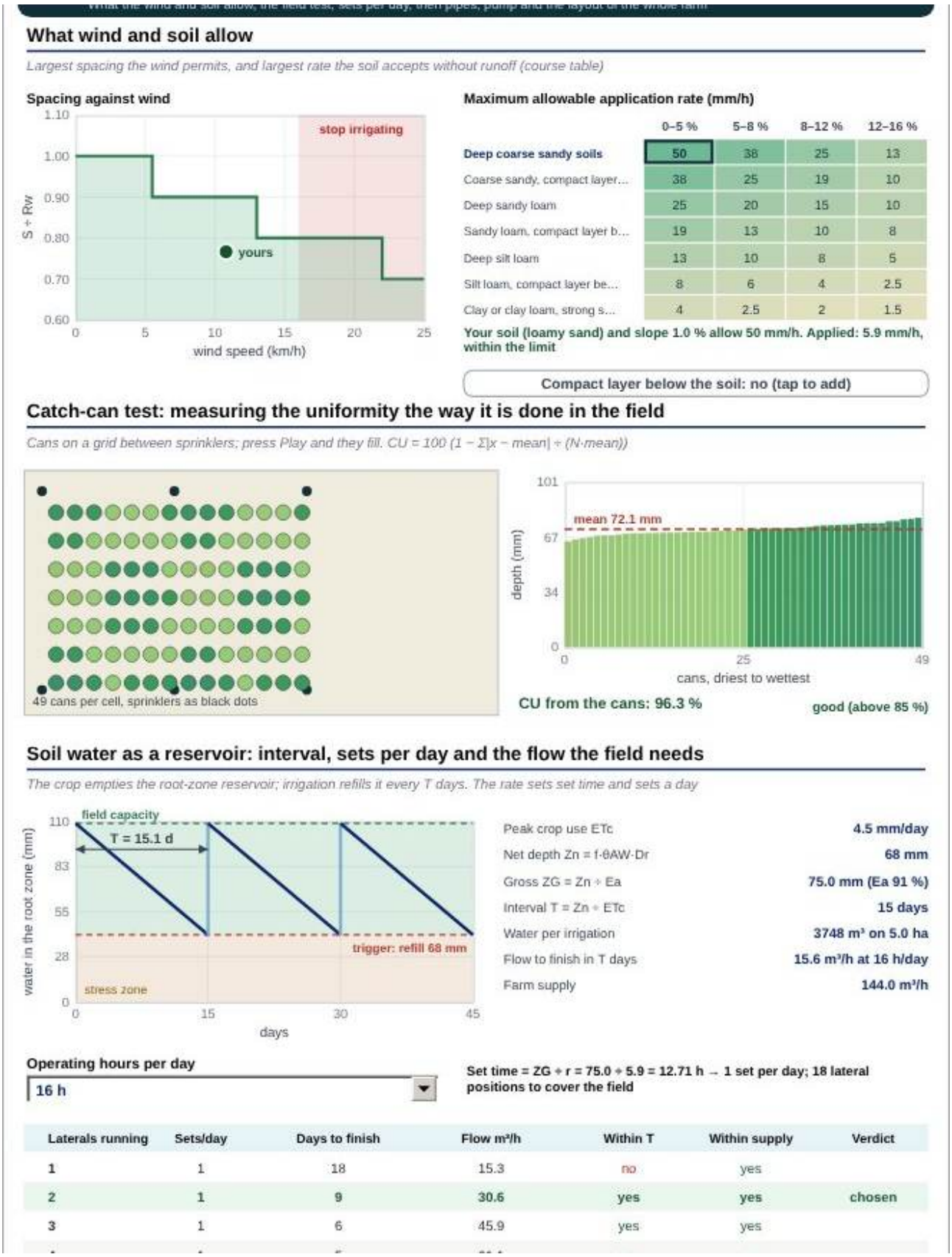


Figure 2.19: Sprinkler design and pipes, Loamy Sand, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

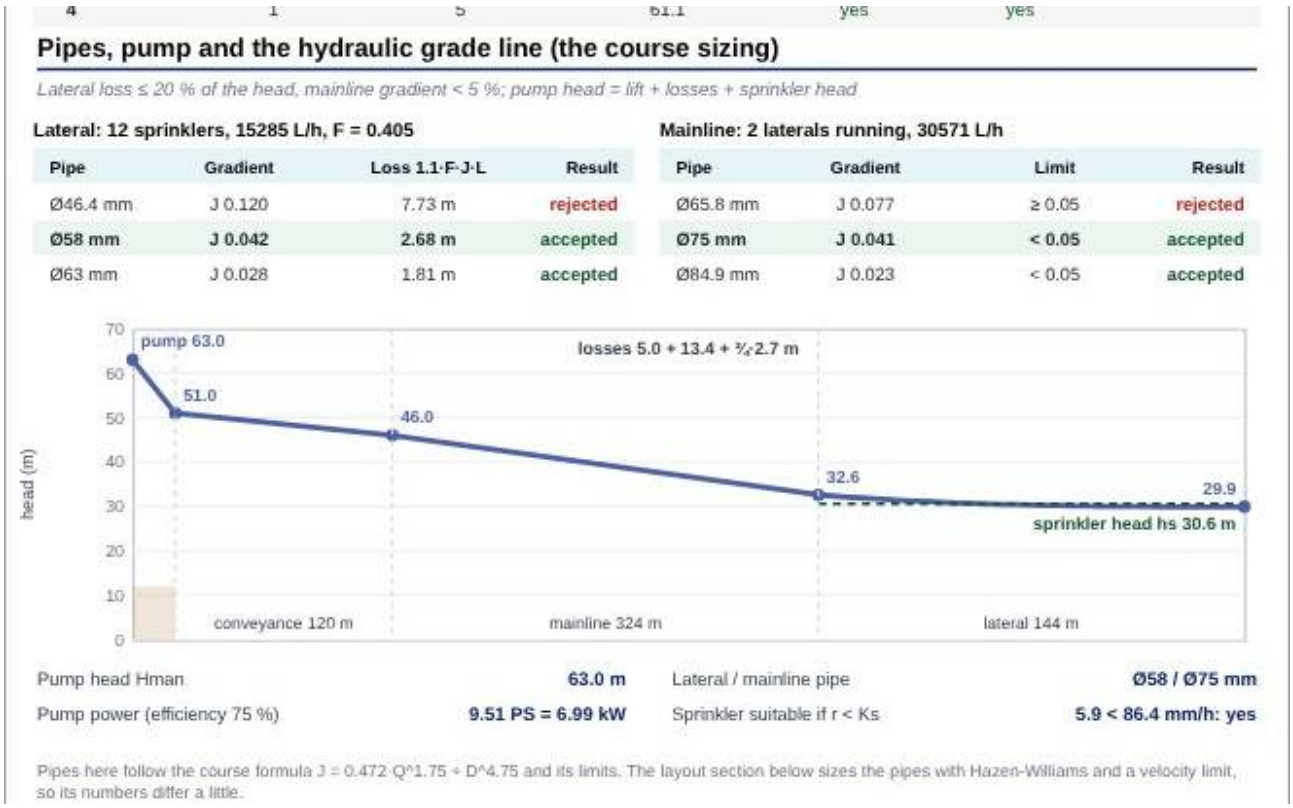
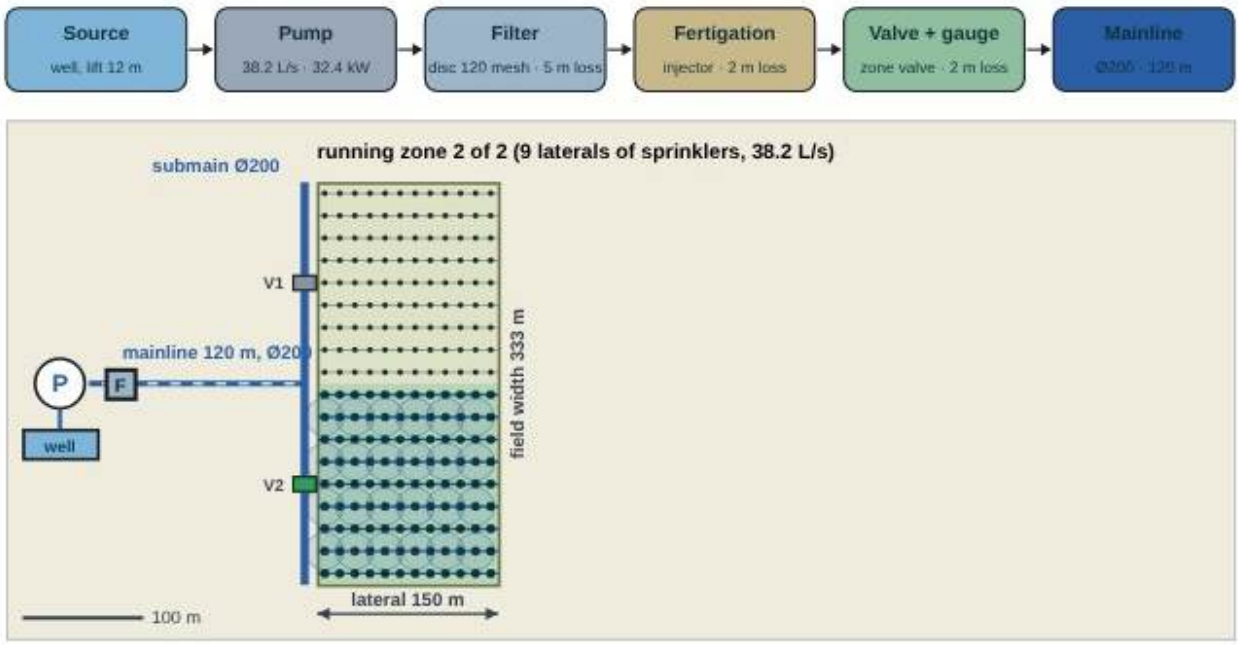
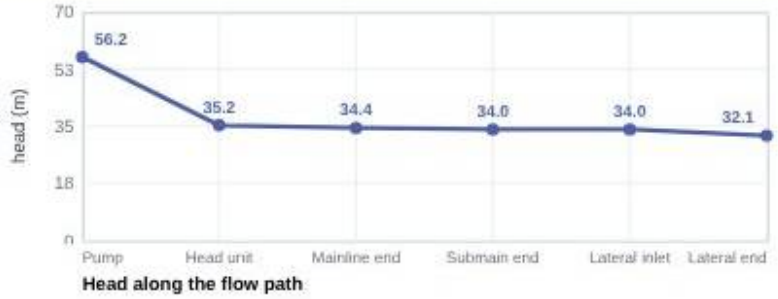


Figure 2.20: Sprinkler design and pipes, Loamy Sand, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

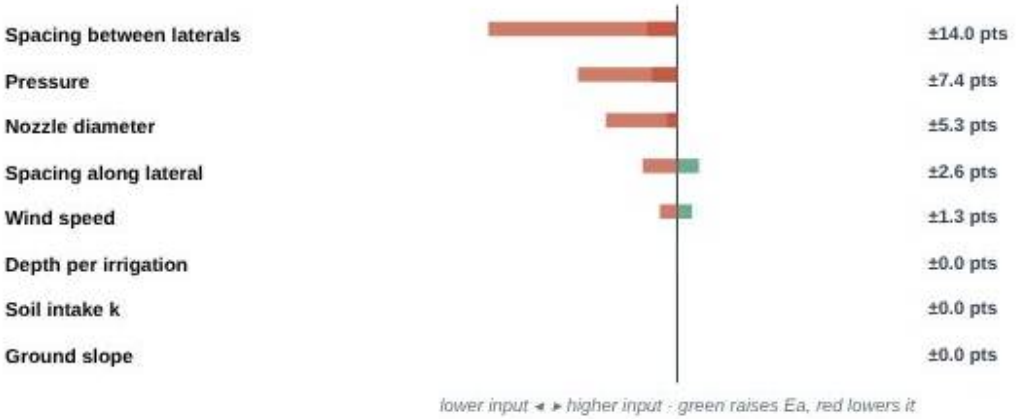
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	763	90.5	99.8	0.0	meets target

Figure 2.21: Sprinkler design and pipes, Loamy Sand, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

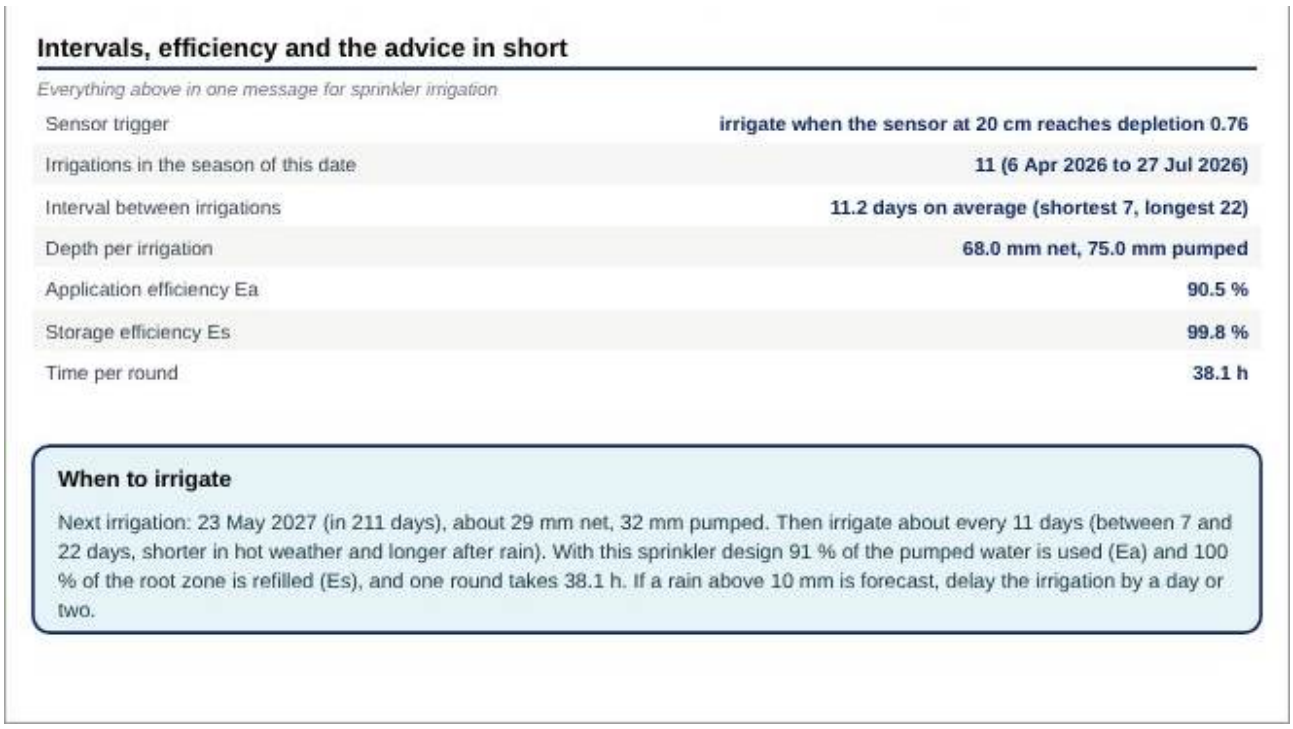
Net irrigation need in a season (DSS)	**322 mm → 5 irrigations of 68 mm**
Gross water per irrigation	**75.0 mm (Ea 91 %)**
Gross water per season	**355 mm = 3553 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 763 min = 38.1 h per round**
Pumping per season	**181 hours · 5863 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**23 May 2027**	211 days	29	32	38.1 h	13 days
2	**5 Jun 2027**	224 days	29	32	38.1 h	10 days
3	**15 Jun 2027**	234 days	29	32	38.1 h	8 days
4	**23 Jun 2027**	242 days	29	32	38.1 h	7 days
5	**30 Jun 2027**	249 days	29	32	38.1 h	8 days
6	**8 Jul 2027**	257 days	29	32	38.1 h	14 days
7	**22 Jul 2027**	271 days	29	32	38.1 h	258 days
8	**5 Apr 2028**	529 days	29	32	38.1 h	-

Figure 2.22: Sprinkler design and pipes, Loamy Sand, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.76**
Irrigations in the season of this date	**11 (6 Apr 2026 to 27 Jul 2026)**
Interval between irrigations	**11.2 days on average (shortest 7, longest 22)**
Depth per irrigation	**68.0 mm net, 75.0 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**38.1 h**

When to irrigate

Next irrigation: 23 May 2027 (in 211 days), about 29 mm net, 32 mm pumped. Then irrigate about every 11 days (between 7 and 22 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 38.1 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 2.23: Sprinkler design and pipes, Loamy Sand, part 12 of 12: Intervals, efficiency and the advice in short.

2.4 Virtual farm of the surface design (Loamy Sand)

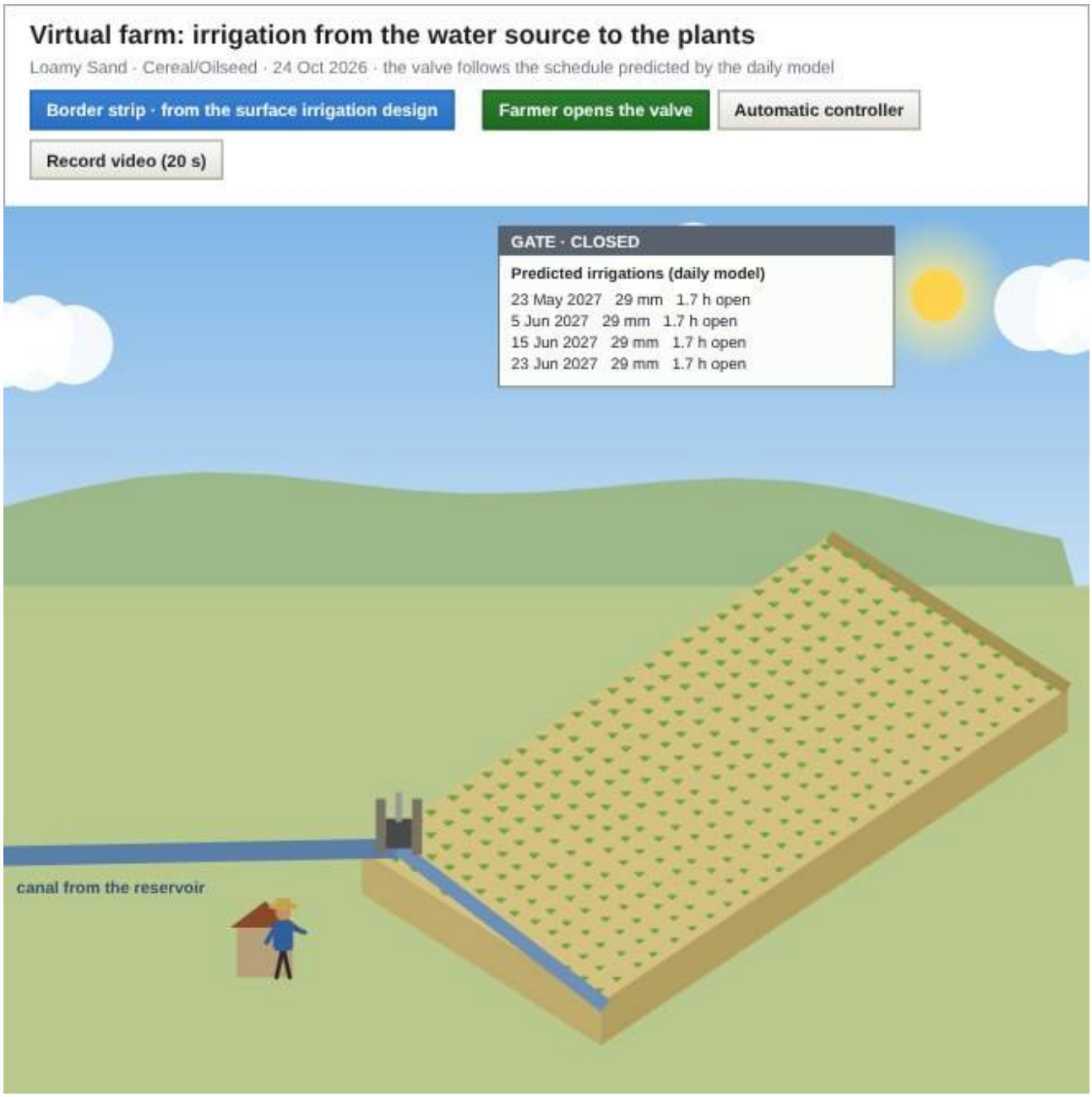


Figure 2.24: Virtual farm, surface design, Loamy Sand, part 1 of 3.

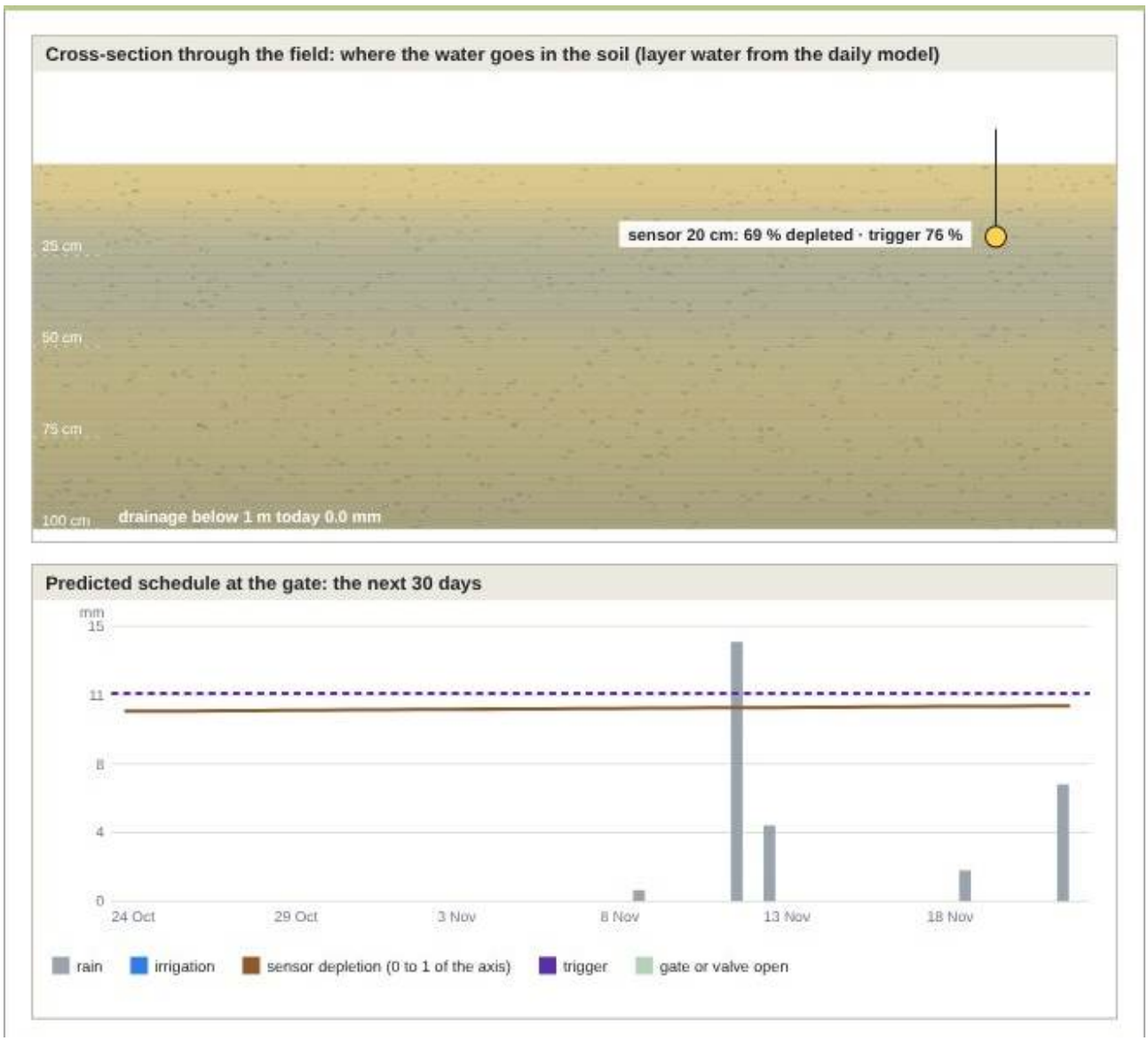


Figure 2.25: Virtual farm, surface design, Loamy Sand, part 2 of 3.

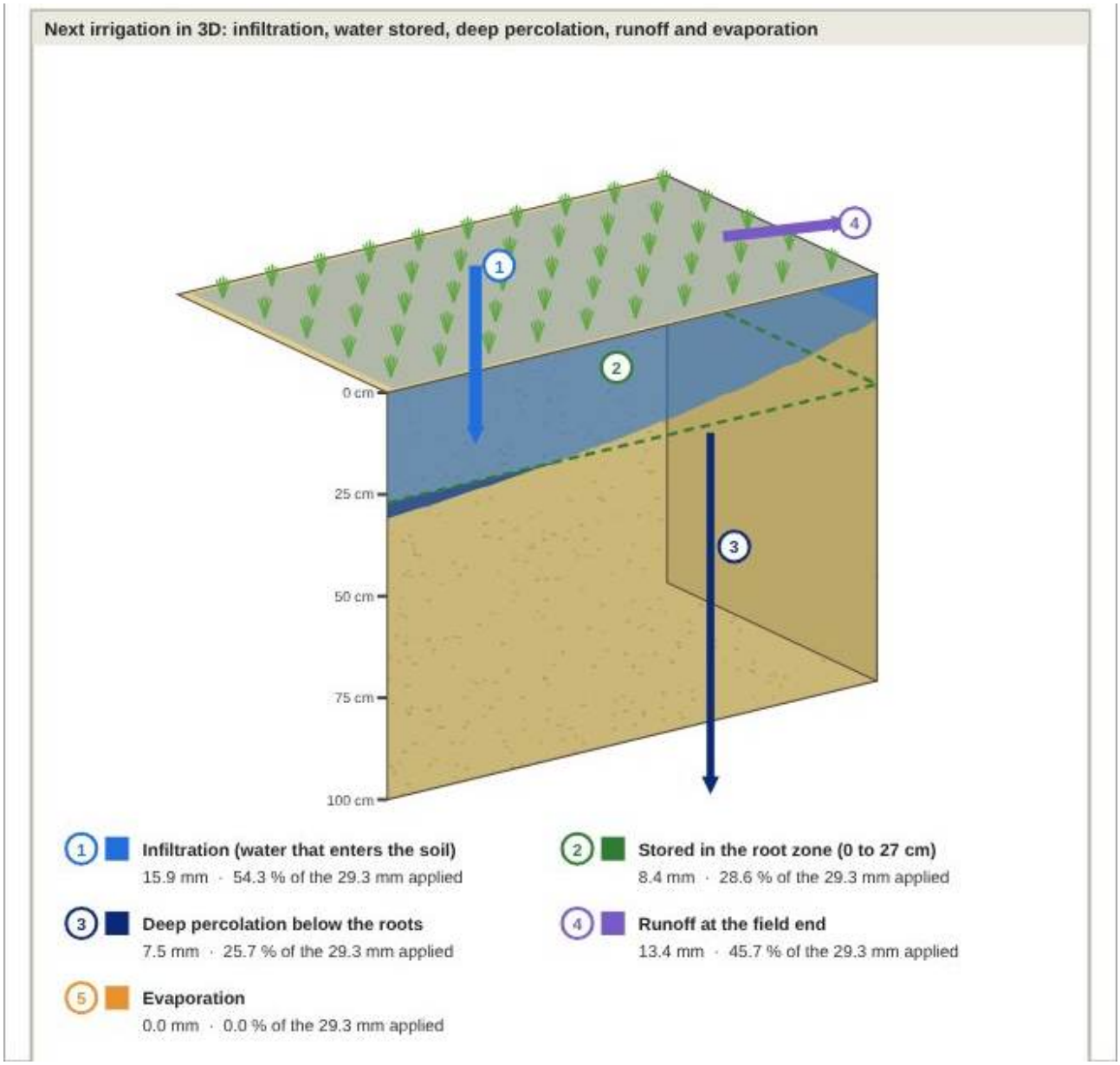


Figure 2.26: Virtual farm, surface design, Loamy Sand, part 3 of 3.

2.5 Virtual farm of the drip design (Loamy Sand)

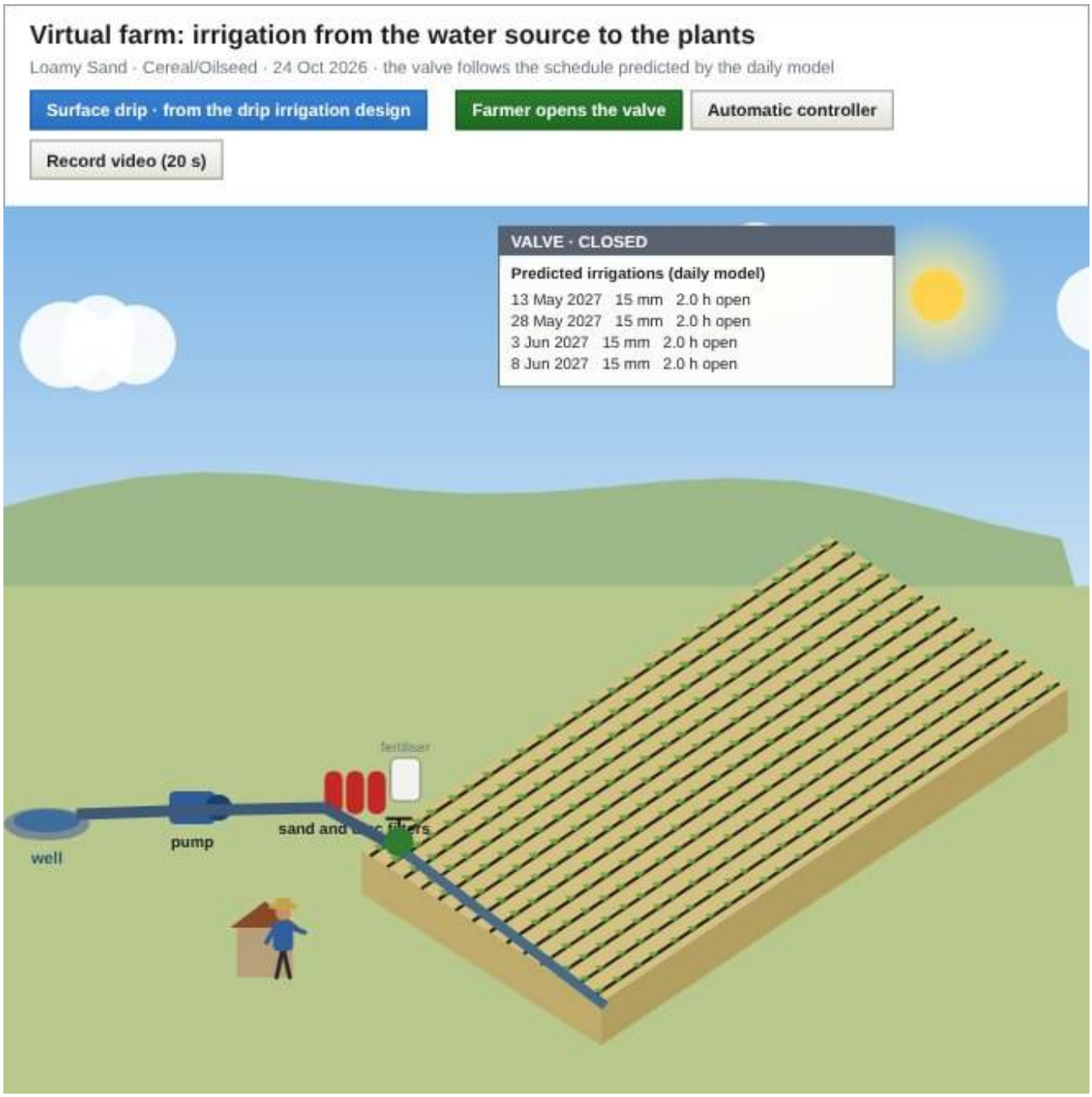


Figure 2.27: Virtual farm, drip design, Loamy Sand, part 1 of 3.

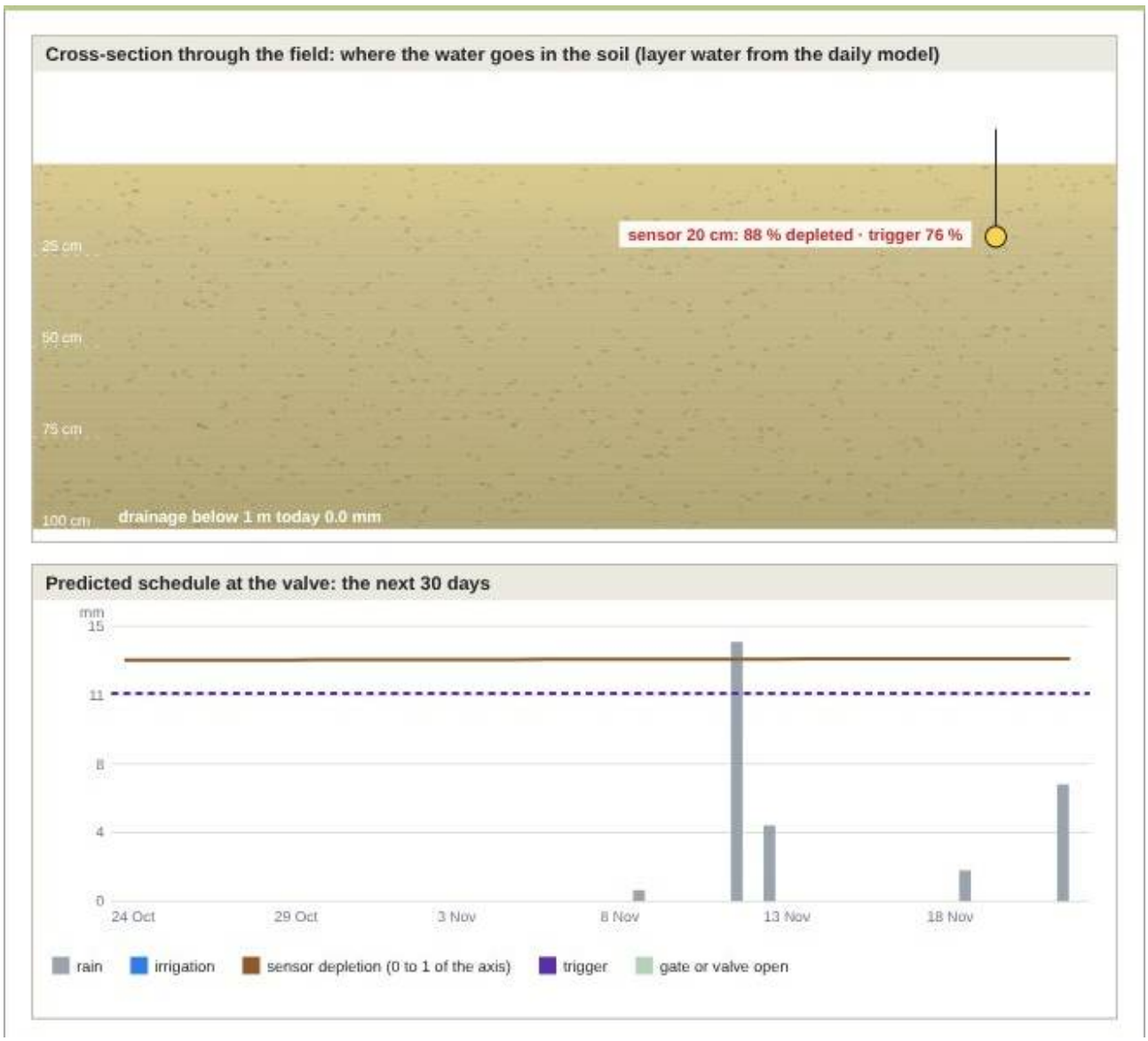


Figure 2.28: Virtual farm, drip design, Loamy Sand, part 2 of 3.

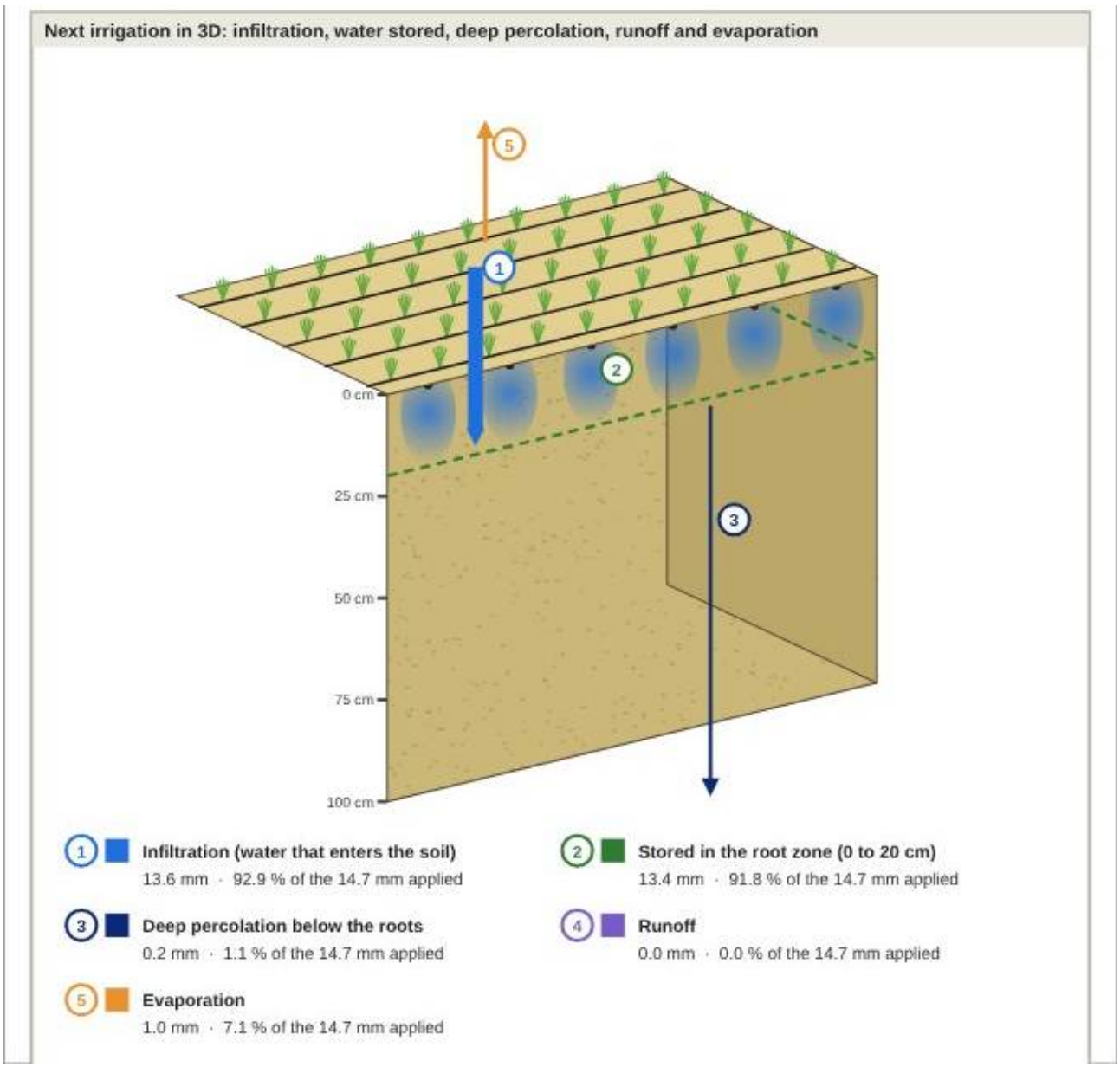


Figure 2.29: Virtual farm, drip design, Loamy Sand, part 3 of 3.

2.6 Virtual farm of the sprinkler design (Loamy Sand)

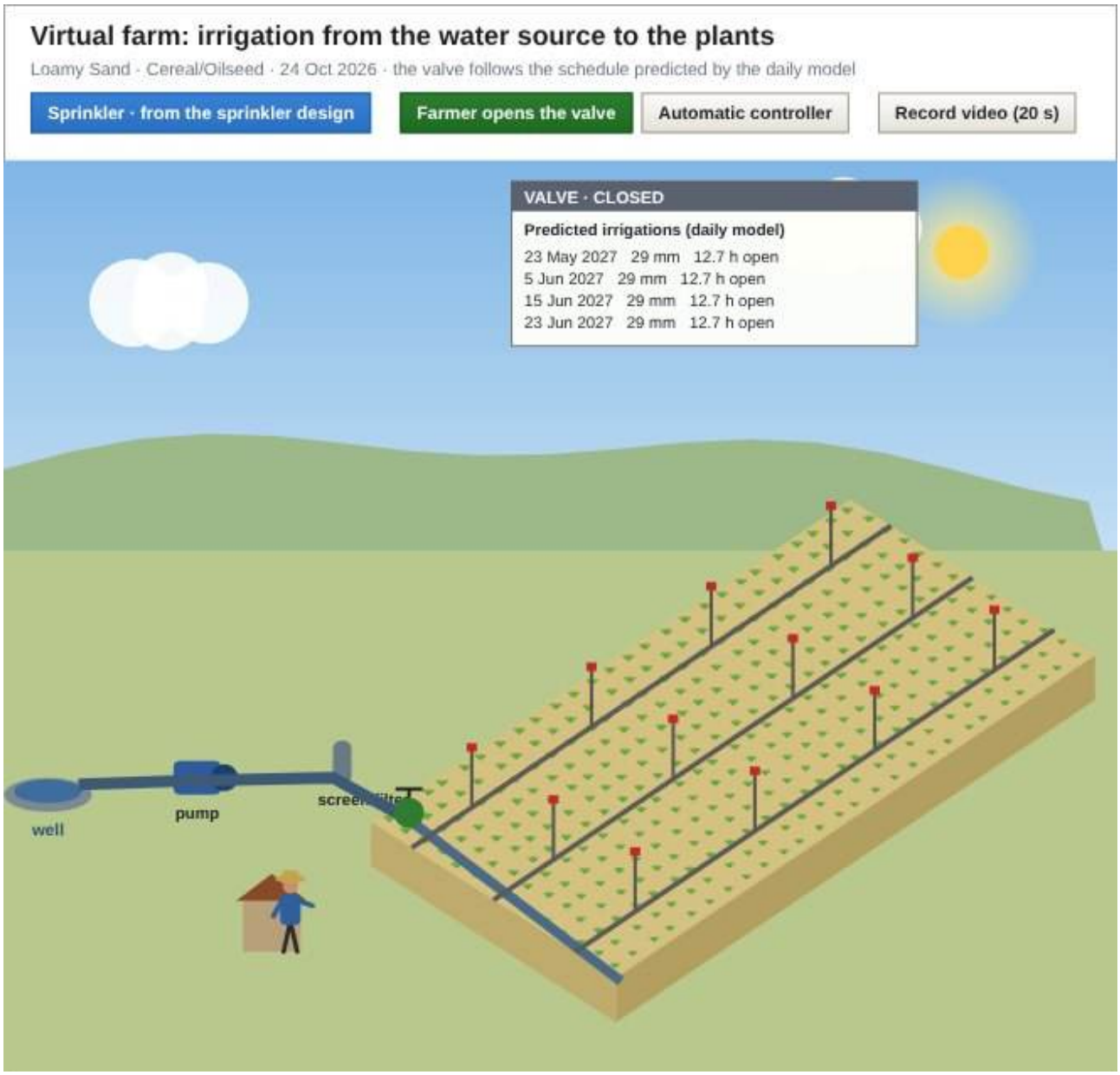


Figure 2.30: Virtual farm, sprinkler design, Loamy Sand, part 1 of 3.

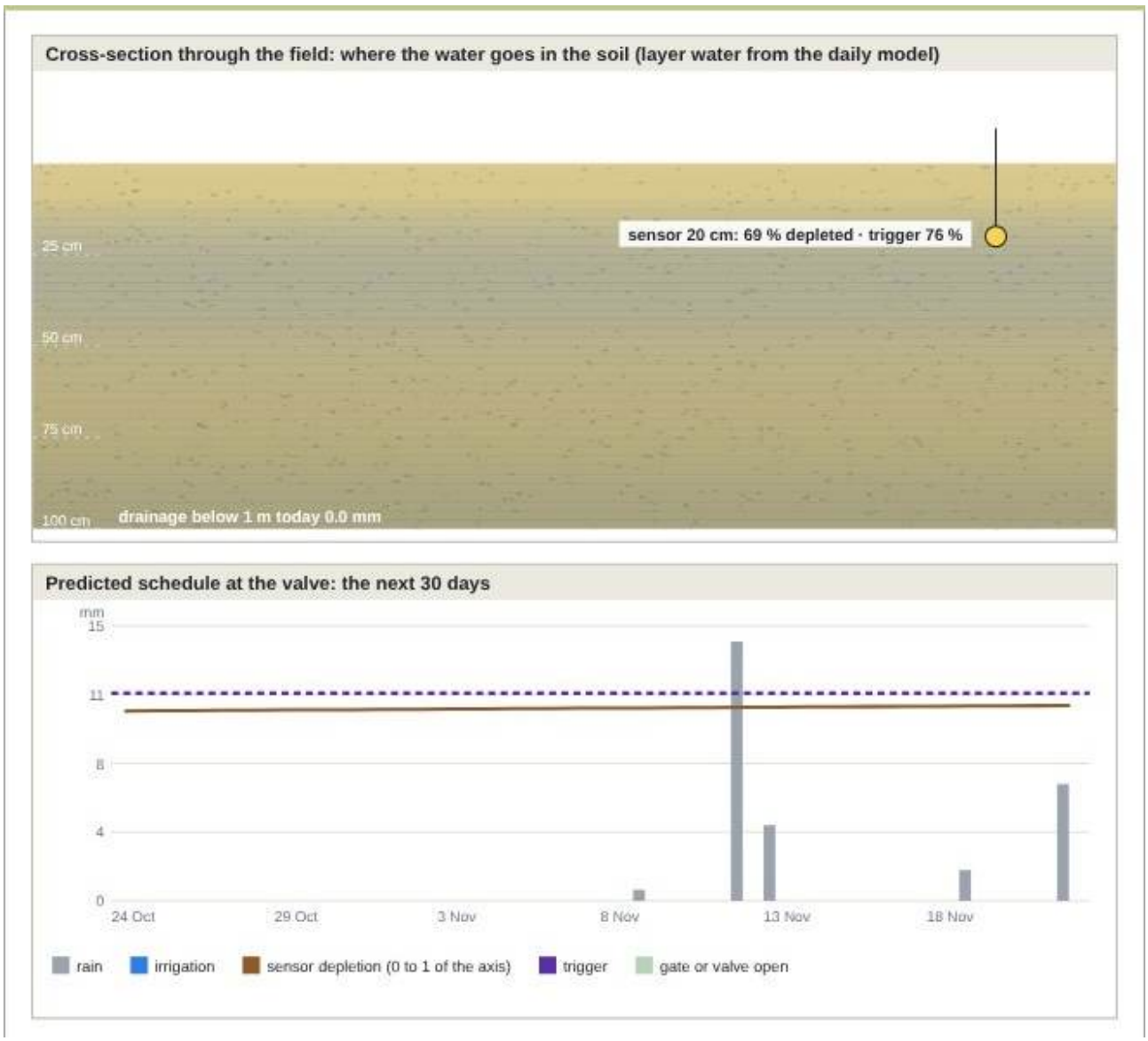


Figure 2.31: Virtual farm, sprinkler design, Loamy Sand, part 2 of 3.

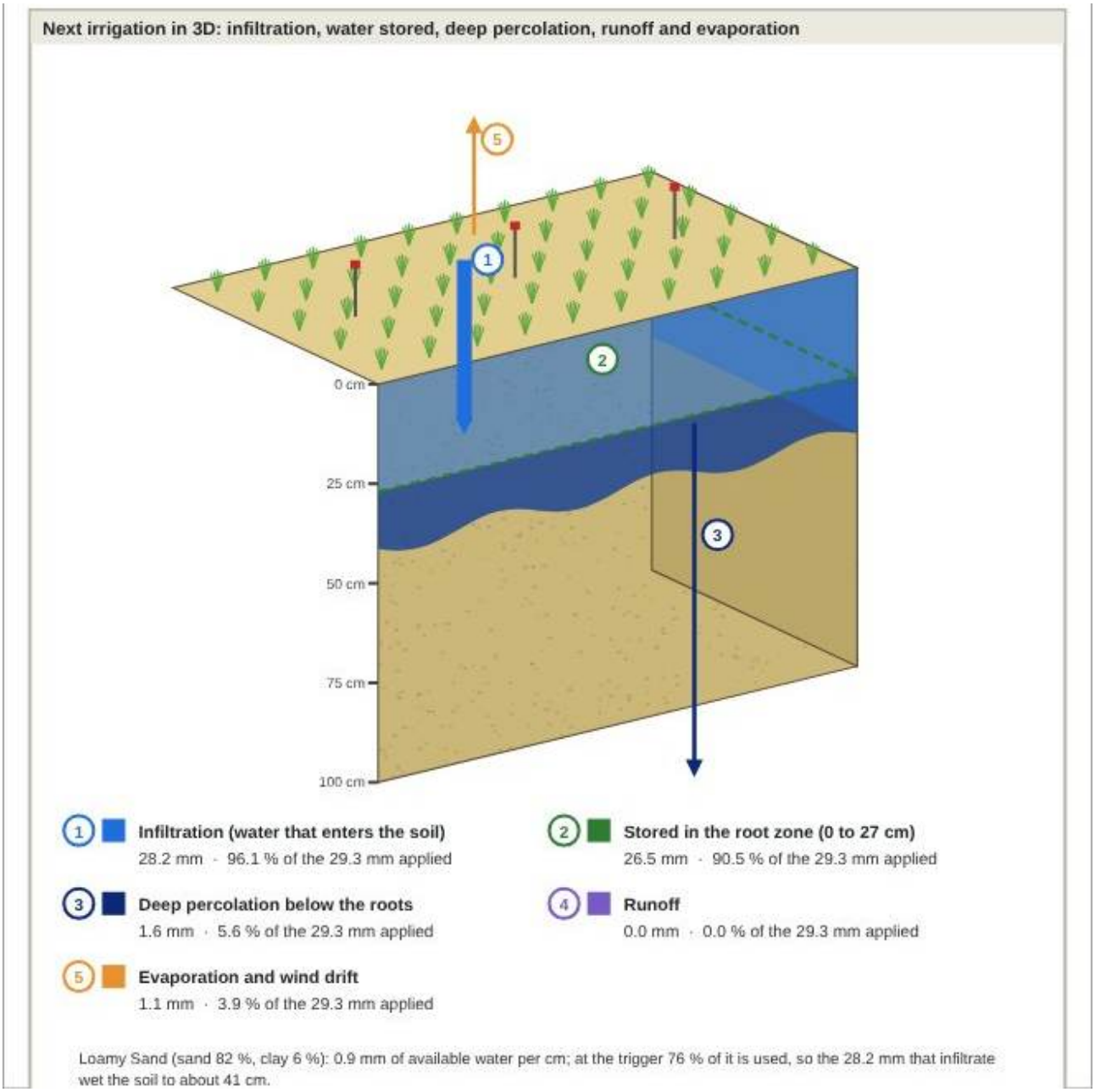


Figure 2.32: Virtual farm, sprinkler design, Loamy Sand, part 3 of 3.

2.7 Sensor and trigger (Loamy Sand)

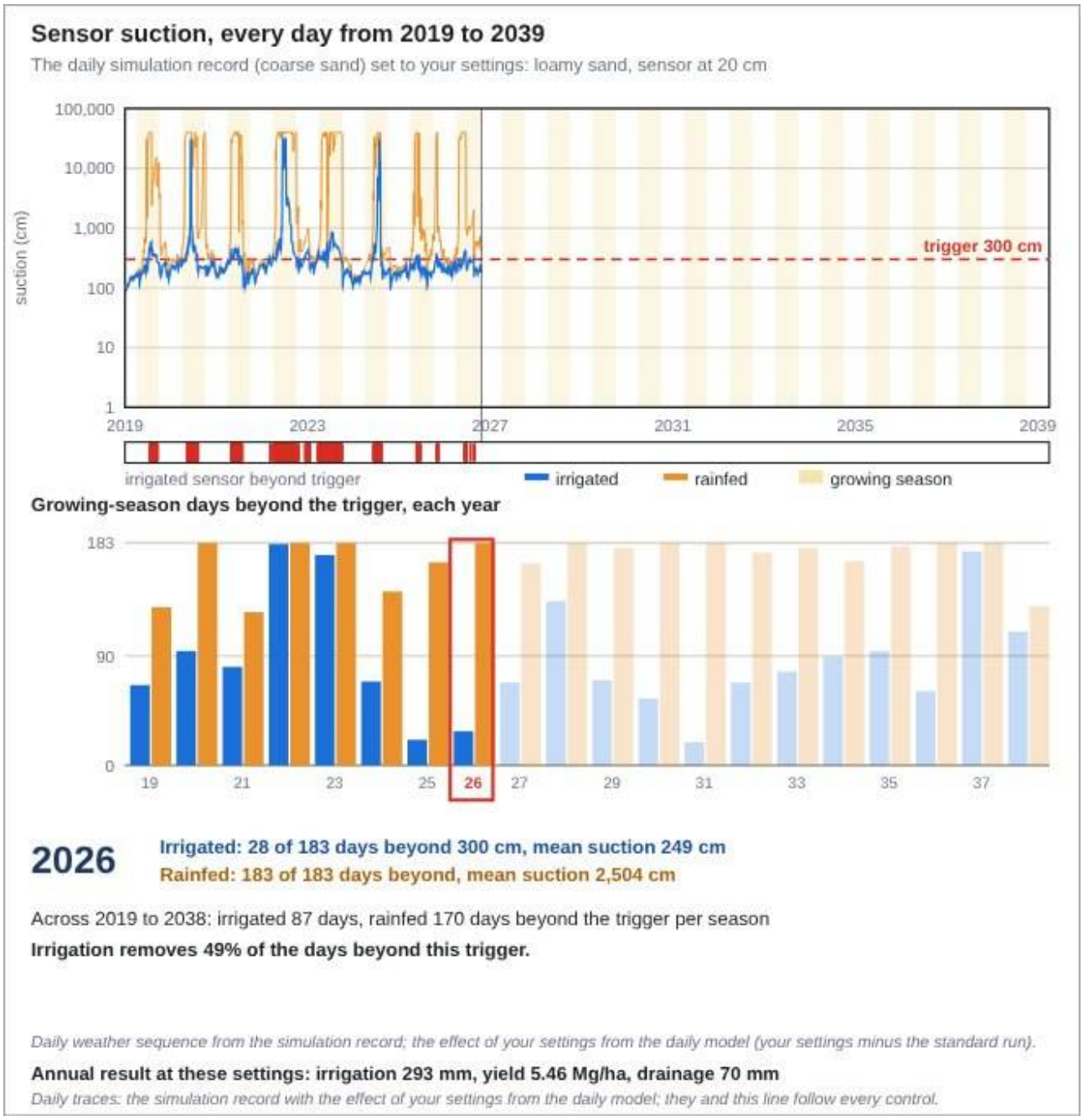


Figure 2.33: Sensor and trigger, Loamy Sand, part 1 of 1.

2.8 Trigger law (Loamy Sand)

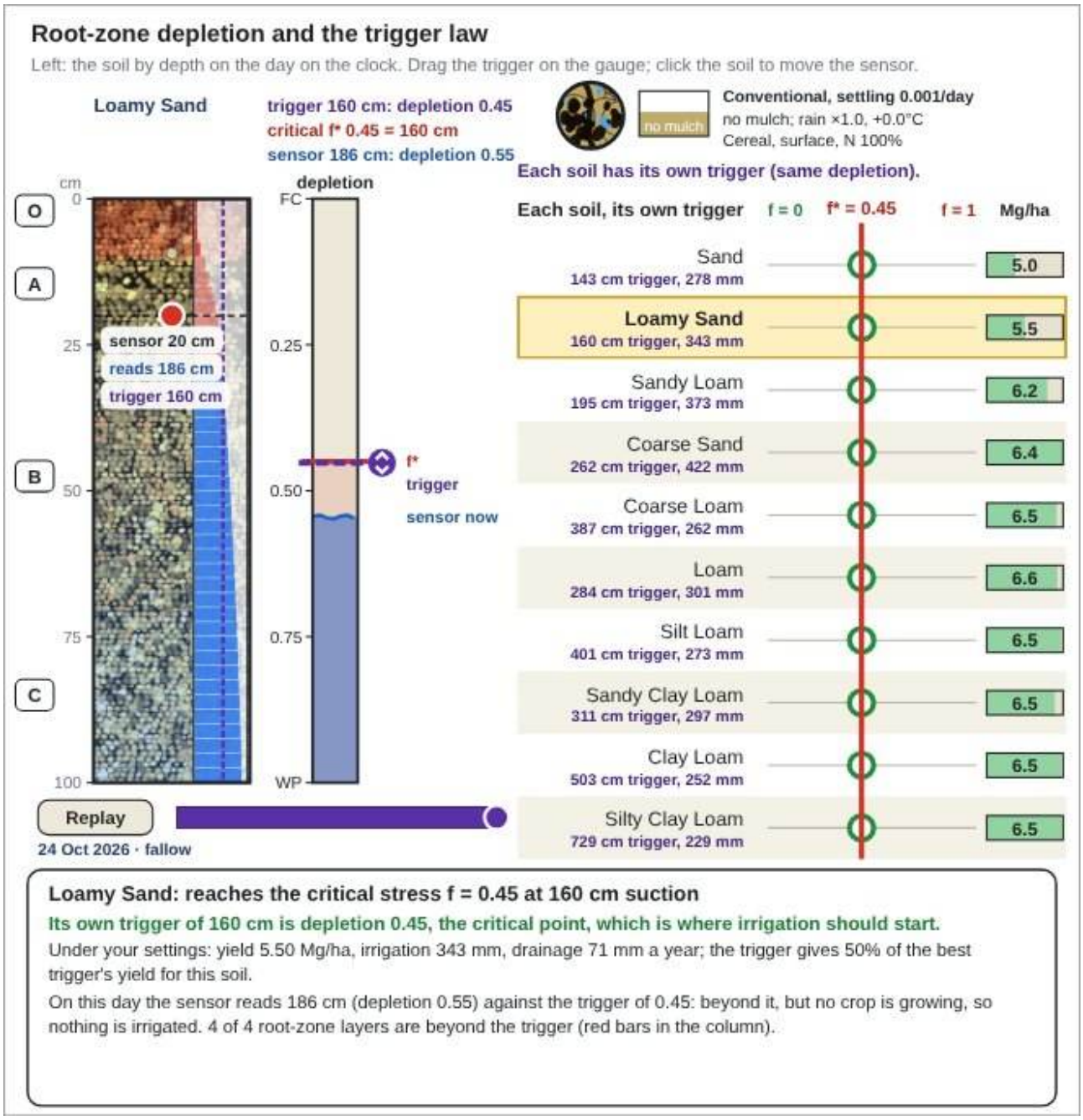


Figure 2.34: Trigger law, Loamy Sand, part 1 of 1.

2.9 Mulch against bare soil (Loamy Sand)

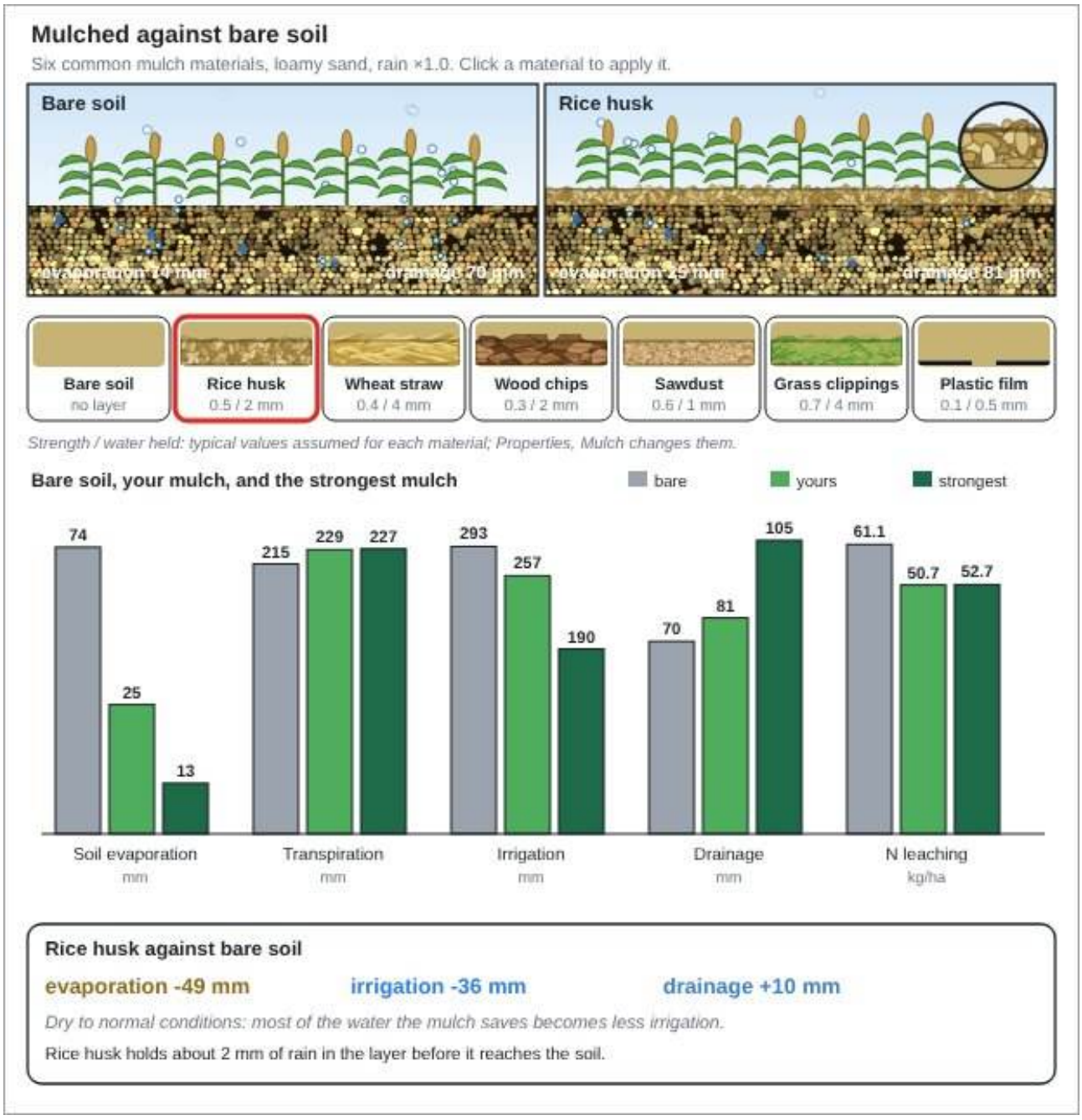


Figure 2.35: Mulch against bare soil, Loamy Sand, part 1 of 1.

2.10 Mulch control (Loamy Sand)

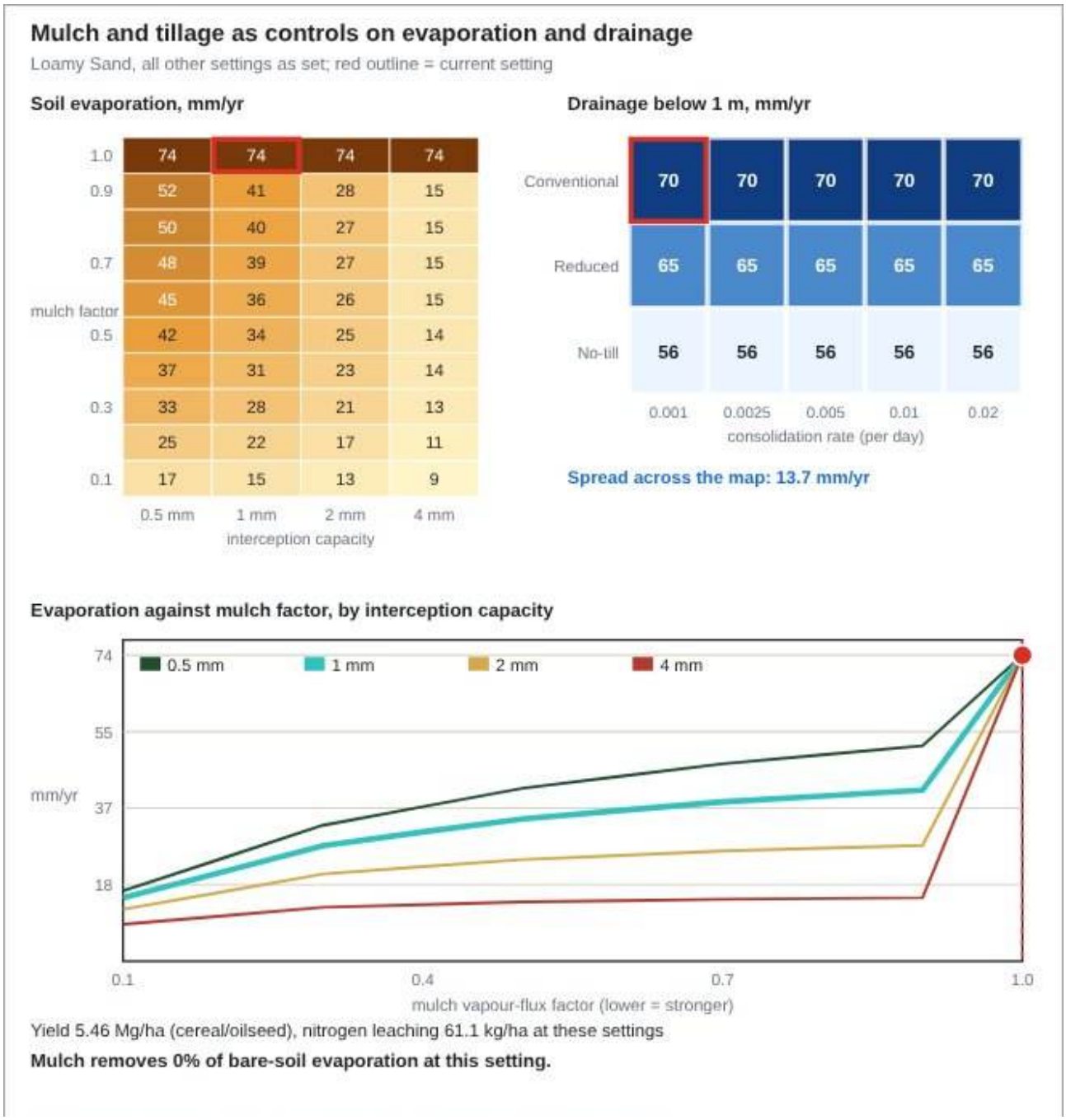


Figure 2.36: Mulch control, Loamy Sand, part 1 of 2.

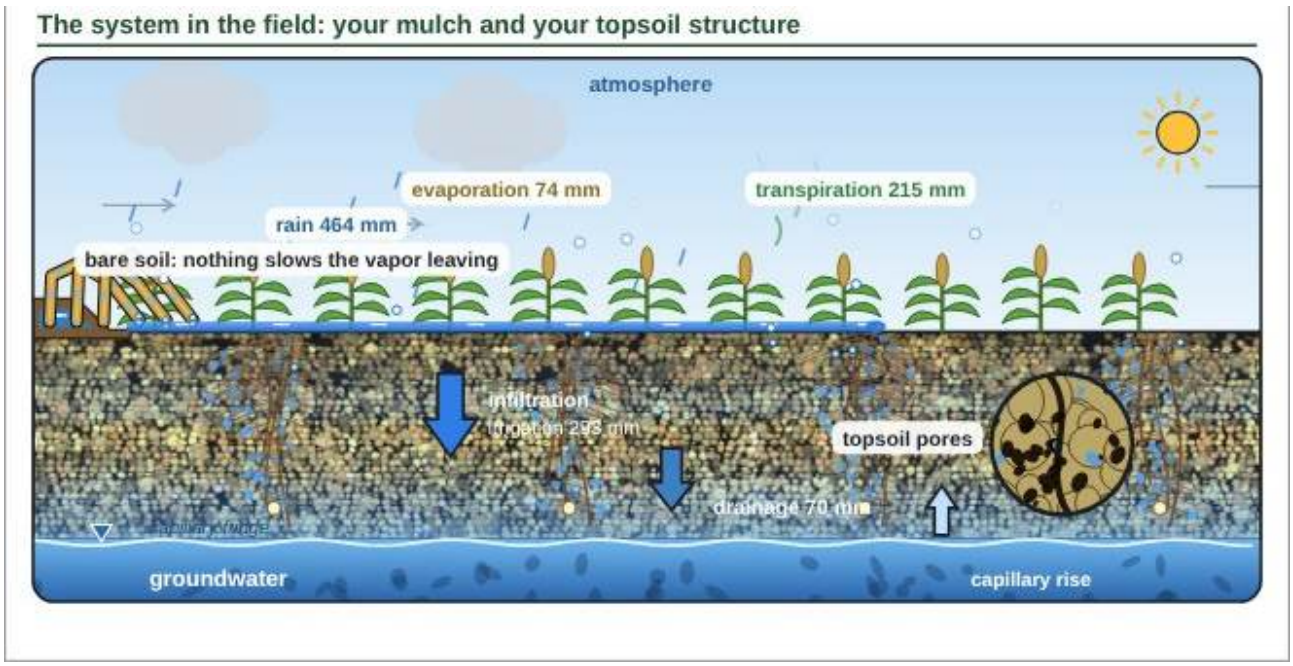


Figure 2.37: Mulch control, Loamy Sand, part 2 of 2.

2.11 Evaporation physics (Loamy Sand)

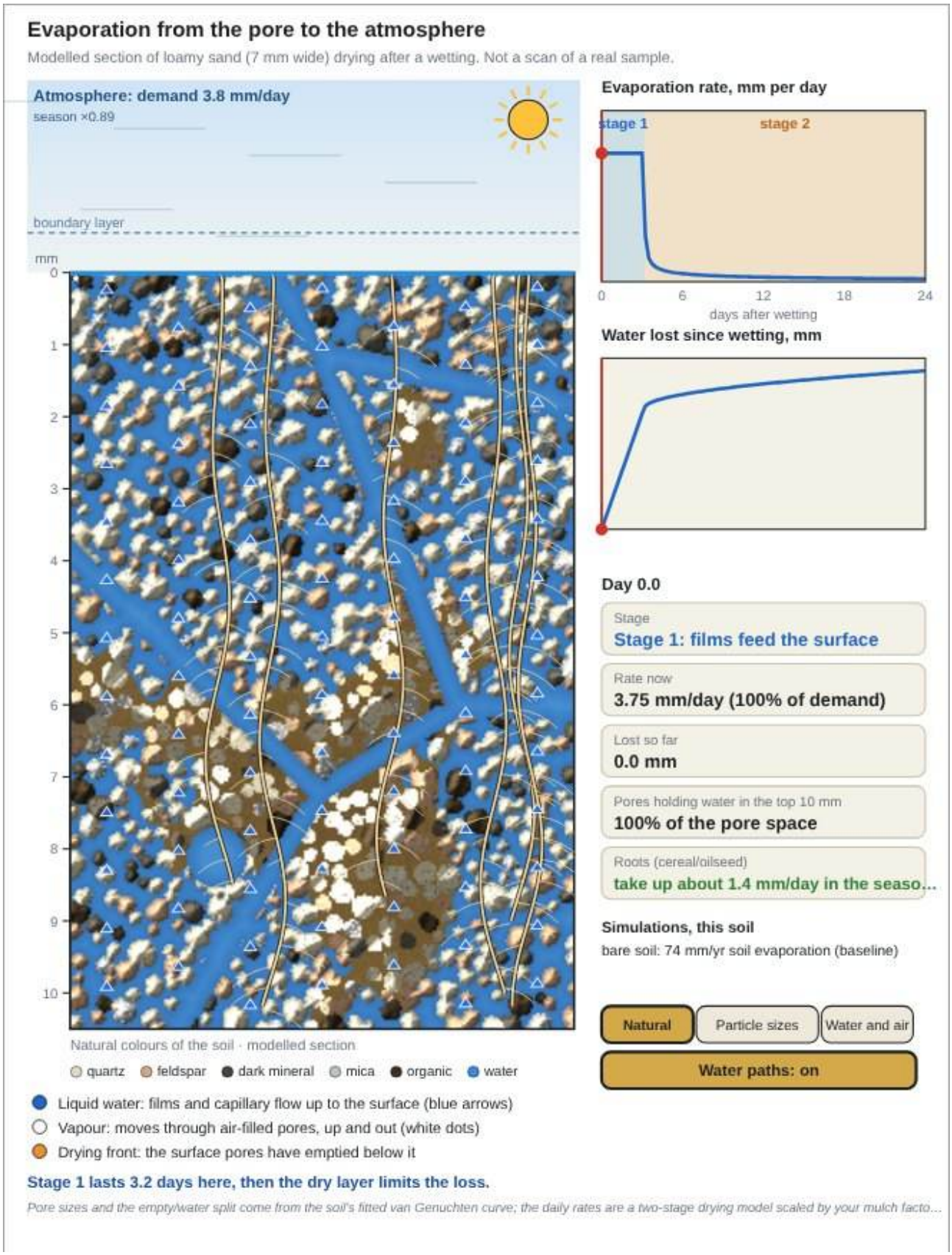


Figure 2.38: Evaporation physics, Loamy Sand, part 1 of 3.

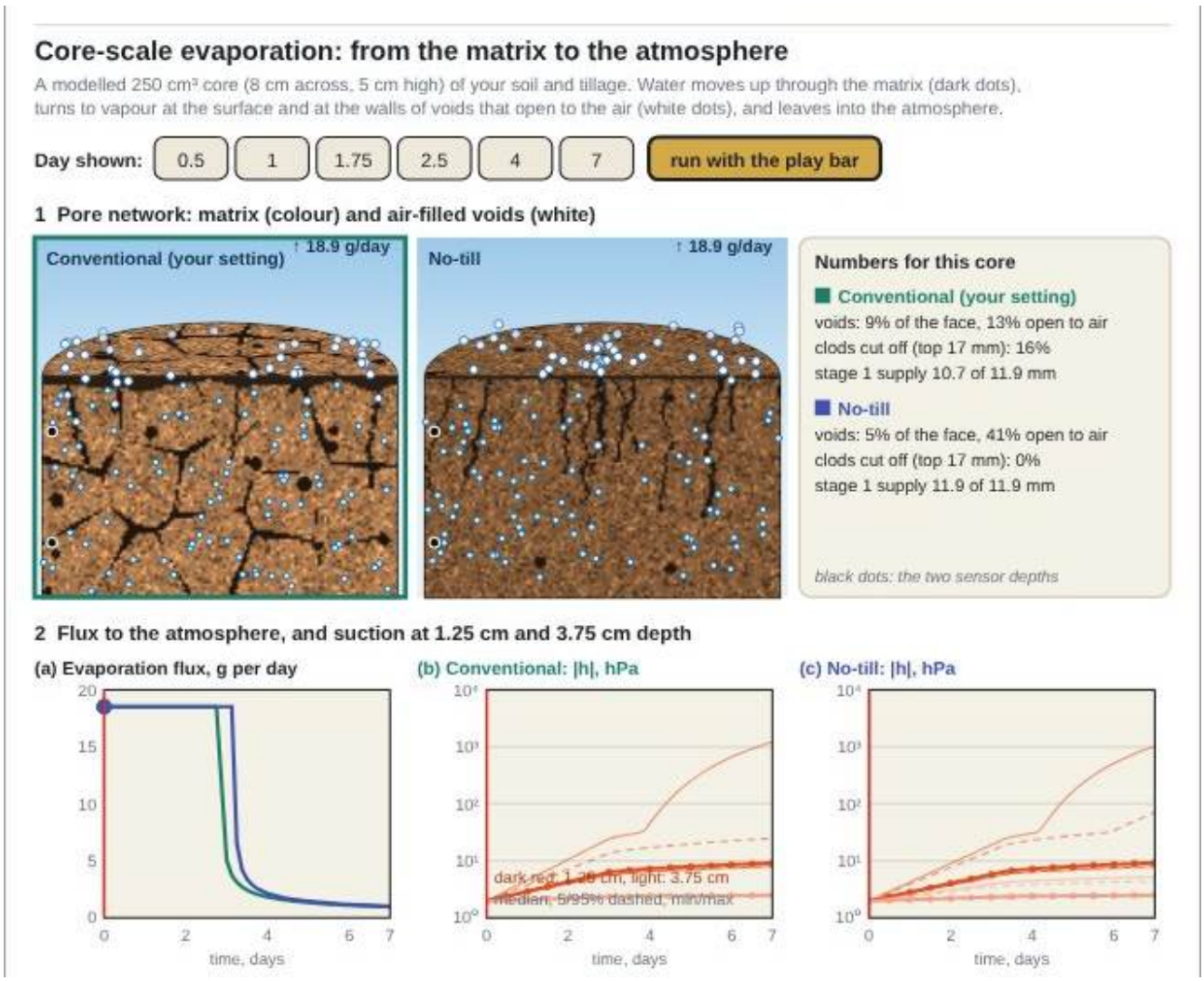


Figure 2.39: Evaporation physics, Loamy Sand, part 2 of 3.

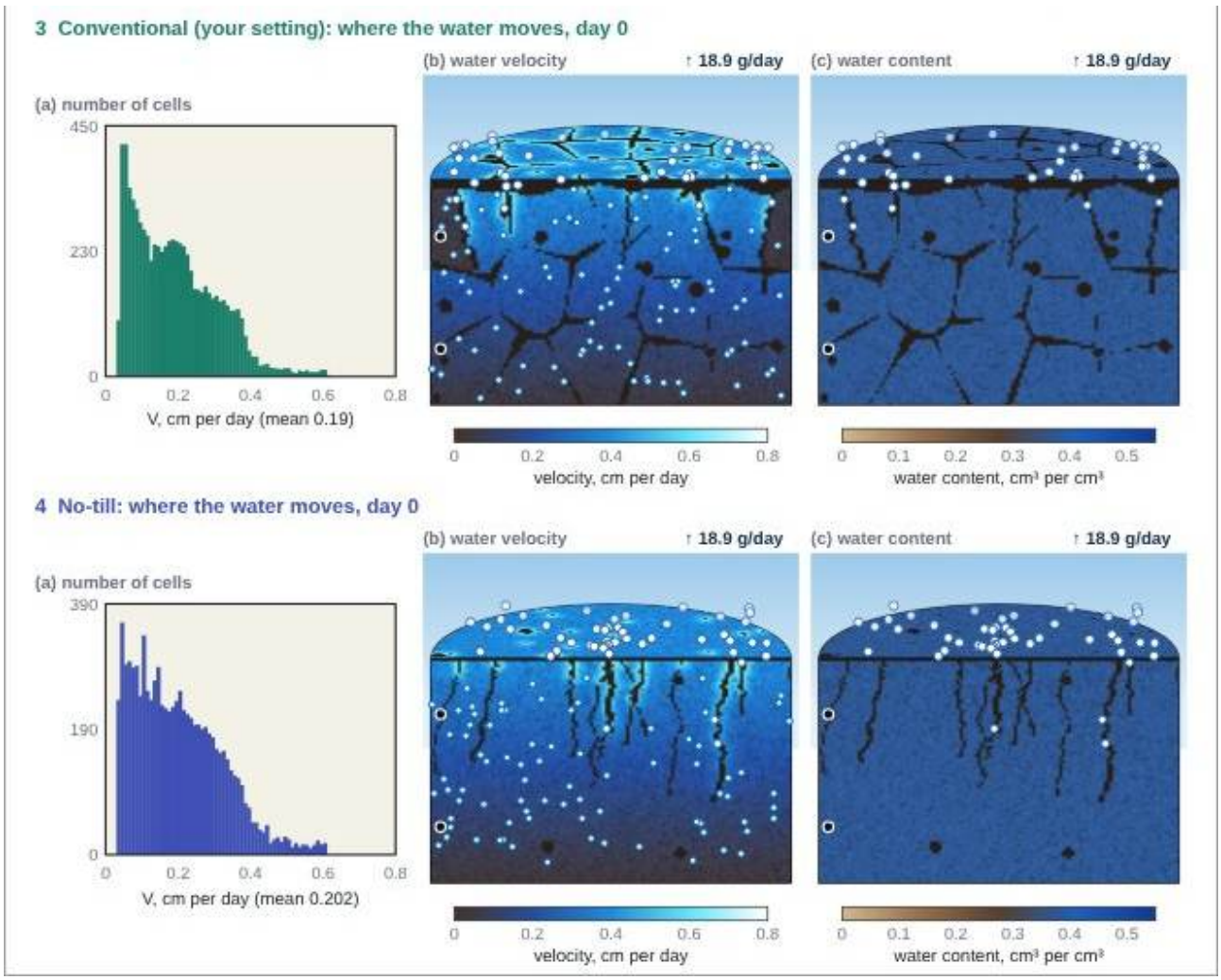


Figure 2.40: Evaporation physics, Loamy Sand, part 3 of 3.

2.12 Fate of saved water (Loamy Sand)

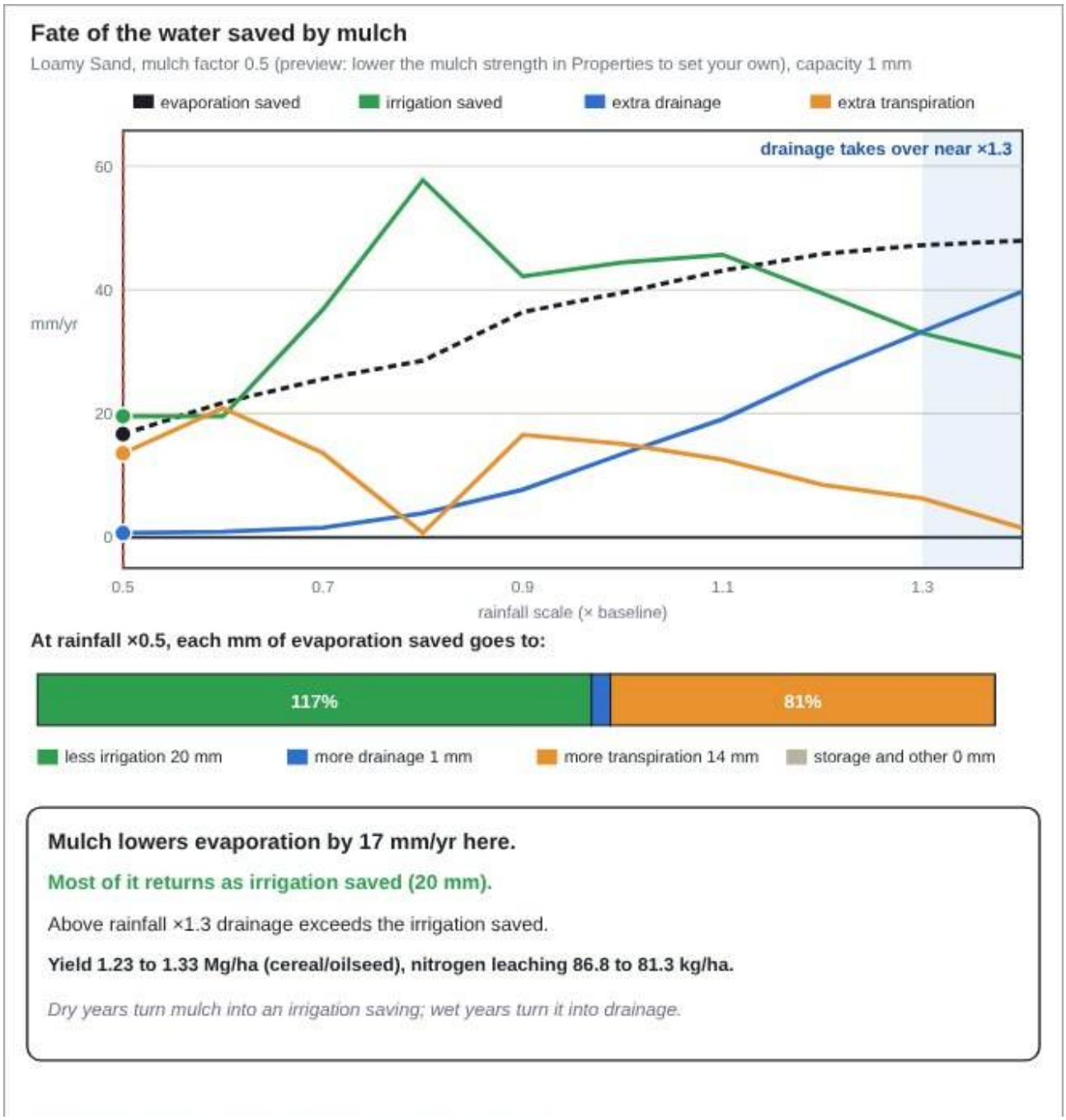


Figure 2.41: Fate of saved water, Loamy Sand, part 1 of 2.

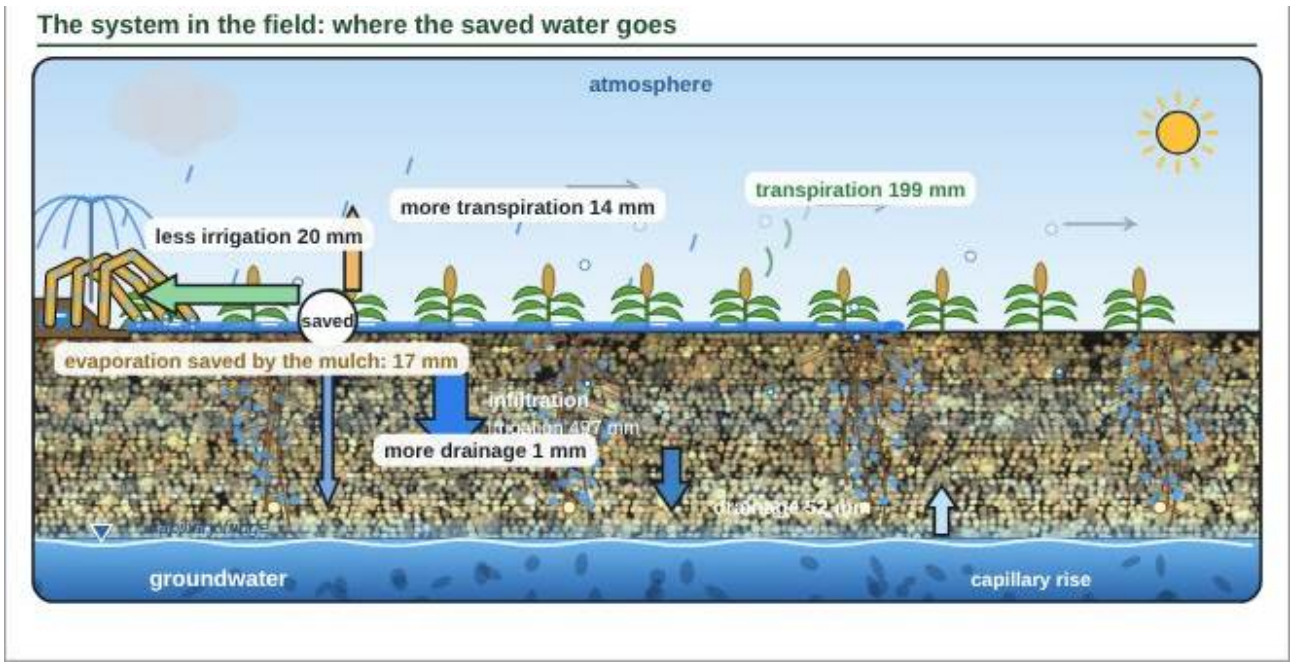


Figure 2.42: Fate of saved water, Loamy Sand, part 2 of 2.

2.13 Regime map (Loamy Sand)

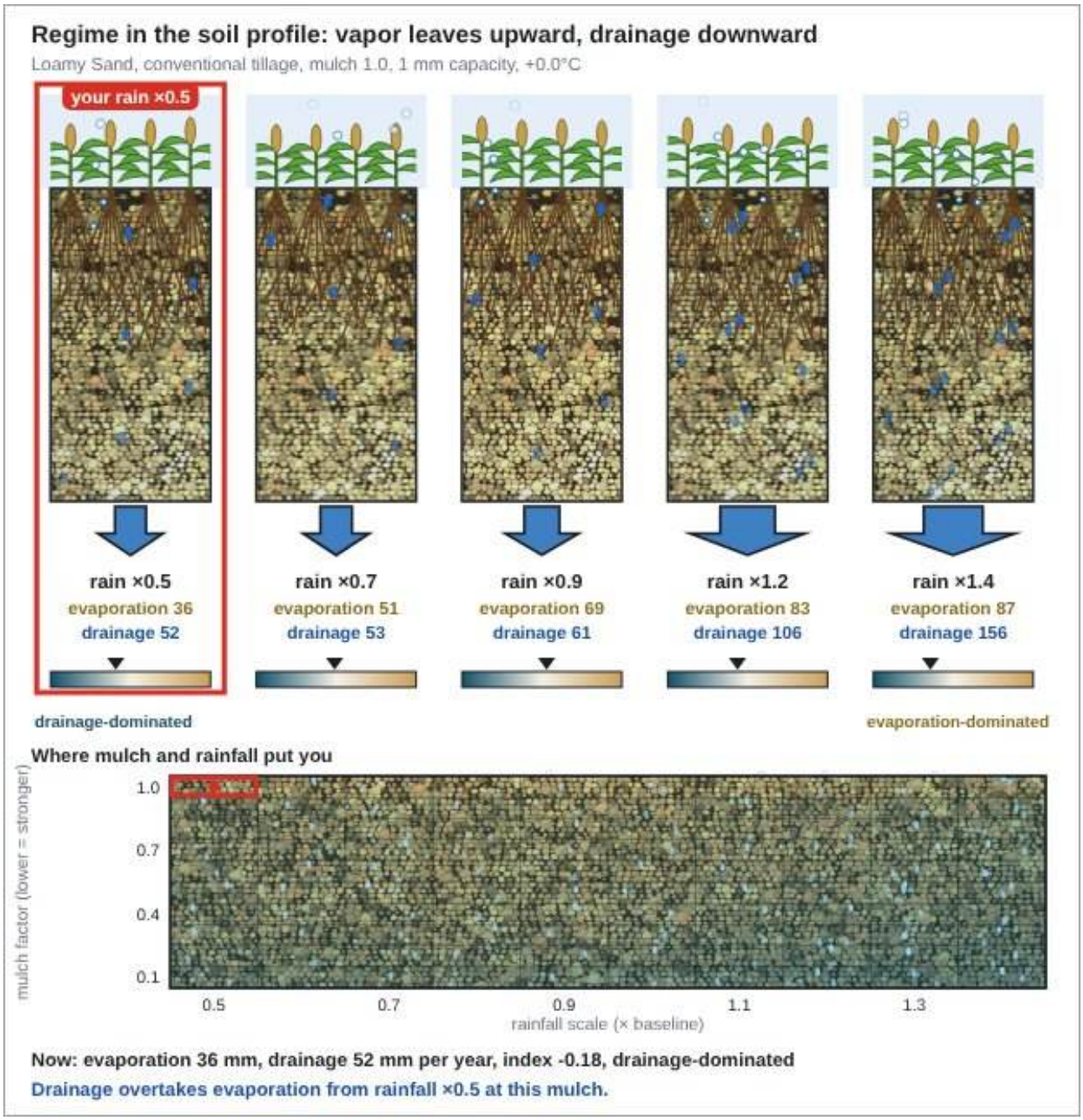


Figure 2.43: Regime map, Loamy Sand, part 1 of 1.

2.14 Tillage and structure (Loamy Sand)

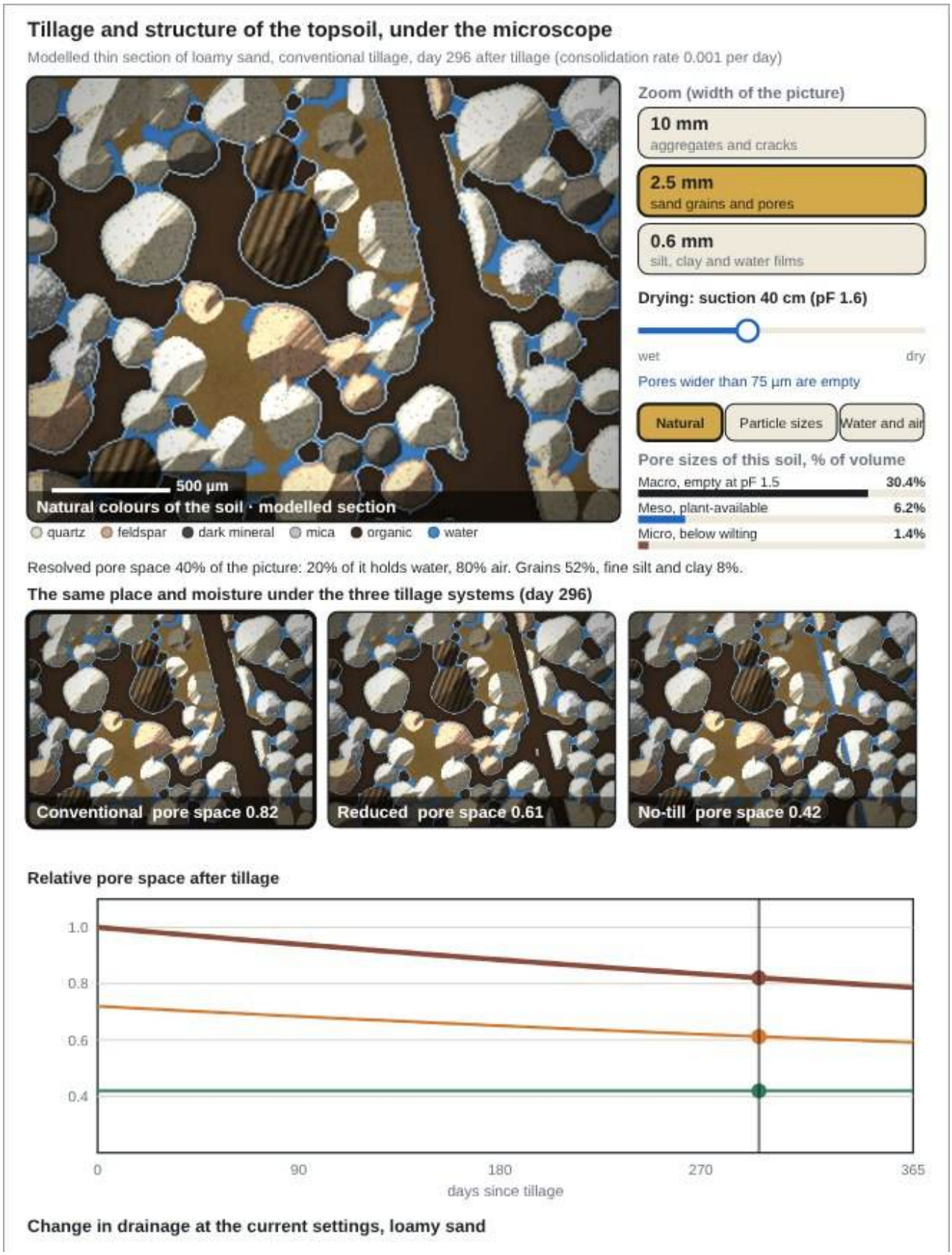


Figure 2.44: Tillage and structure, Loamy Sand, part 1 of 2.

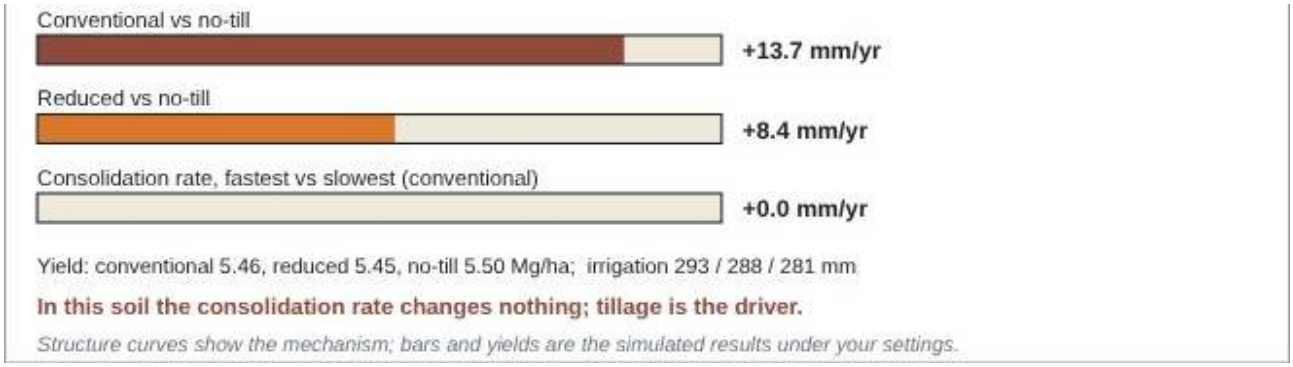


Figure 2.45: Tillage and structure, Loamy Sand, part 2 of 2.

2.15 Soil hydraulic drivers (Loamy Sand)

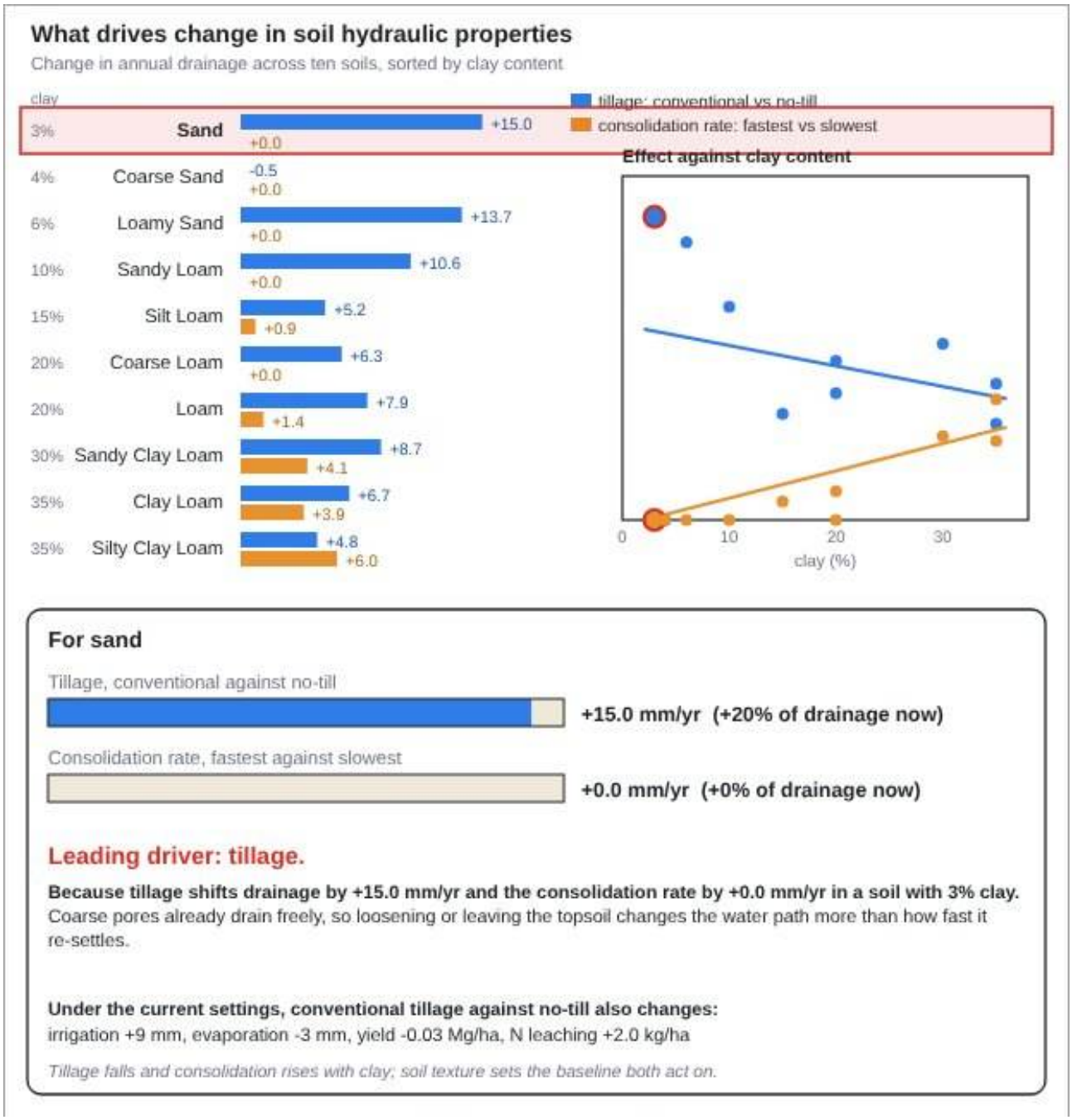


Figure 2.46: Soil hydraulic drivers, Loamy Sand, part 1 of 3.

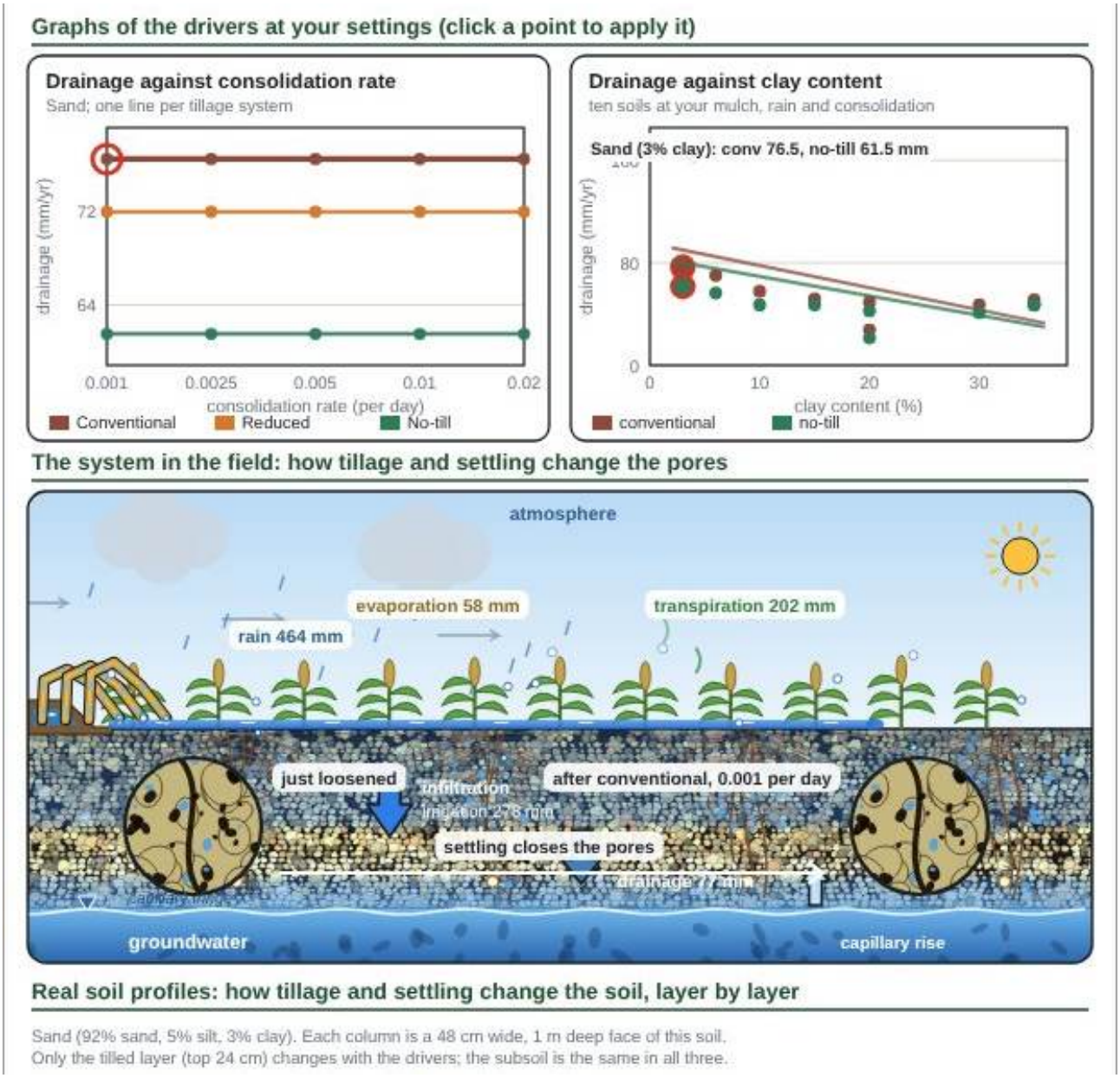


Figure 2.47: Soil hydraulic drivers, Loamy Sand, part 2 of 3.

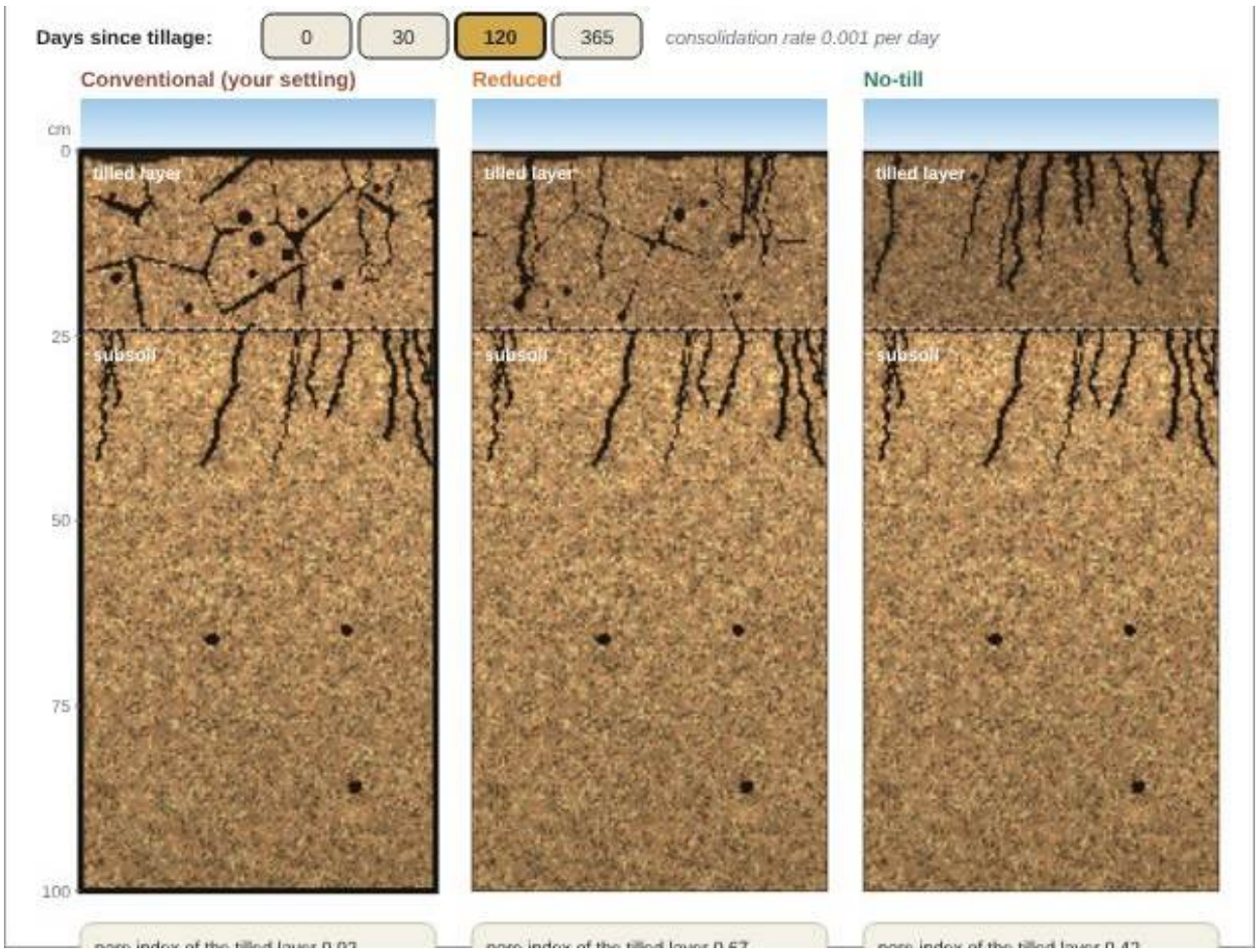


Figure 2.48: Soil hydraulic drivers, Loamy Sand, part 3 of 3.

2.16 Root structure (Loamy Sand)

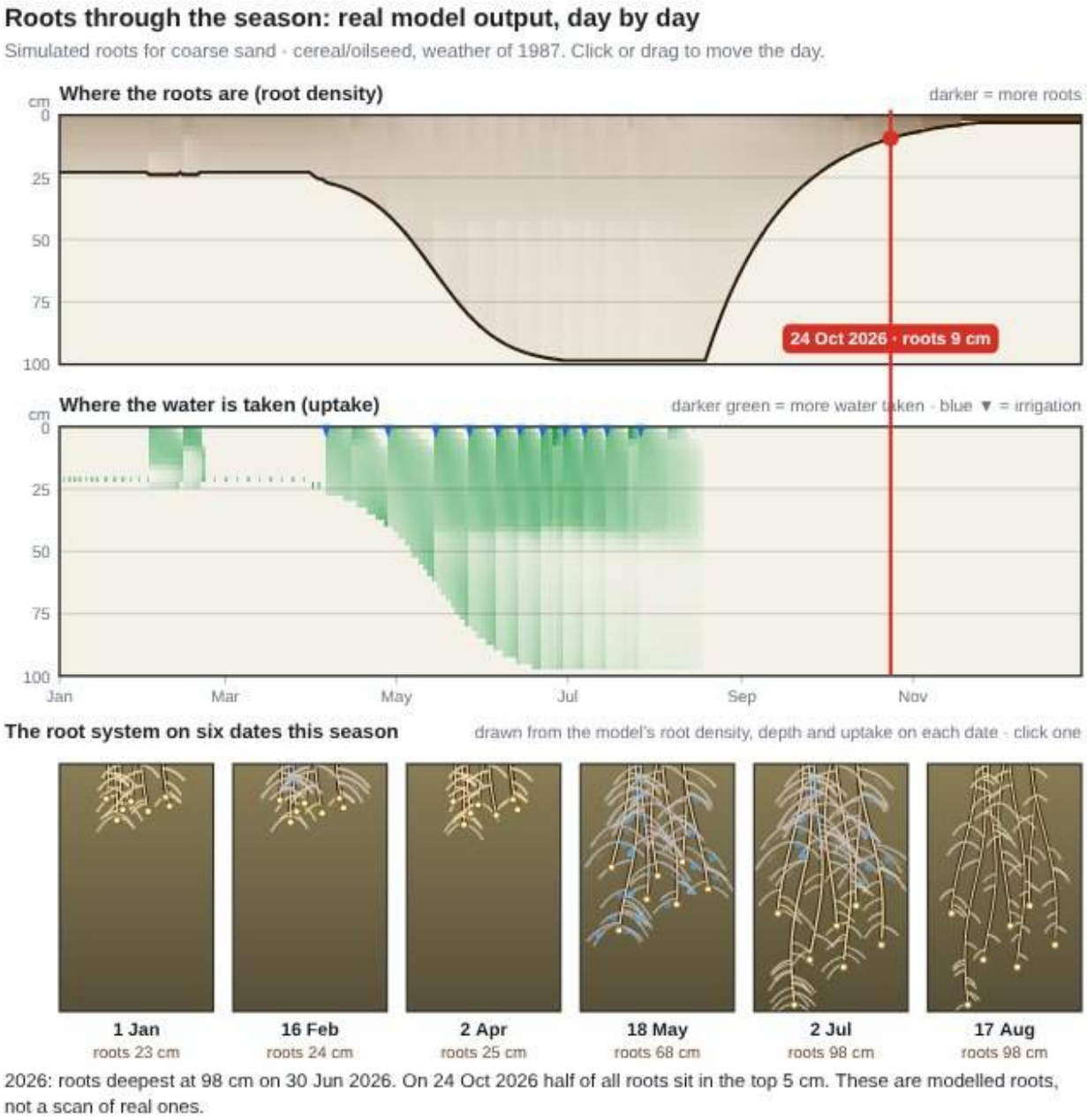


Figure 2.49: Root structure, Loamy Sand, part 1 of 4.

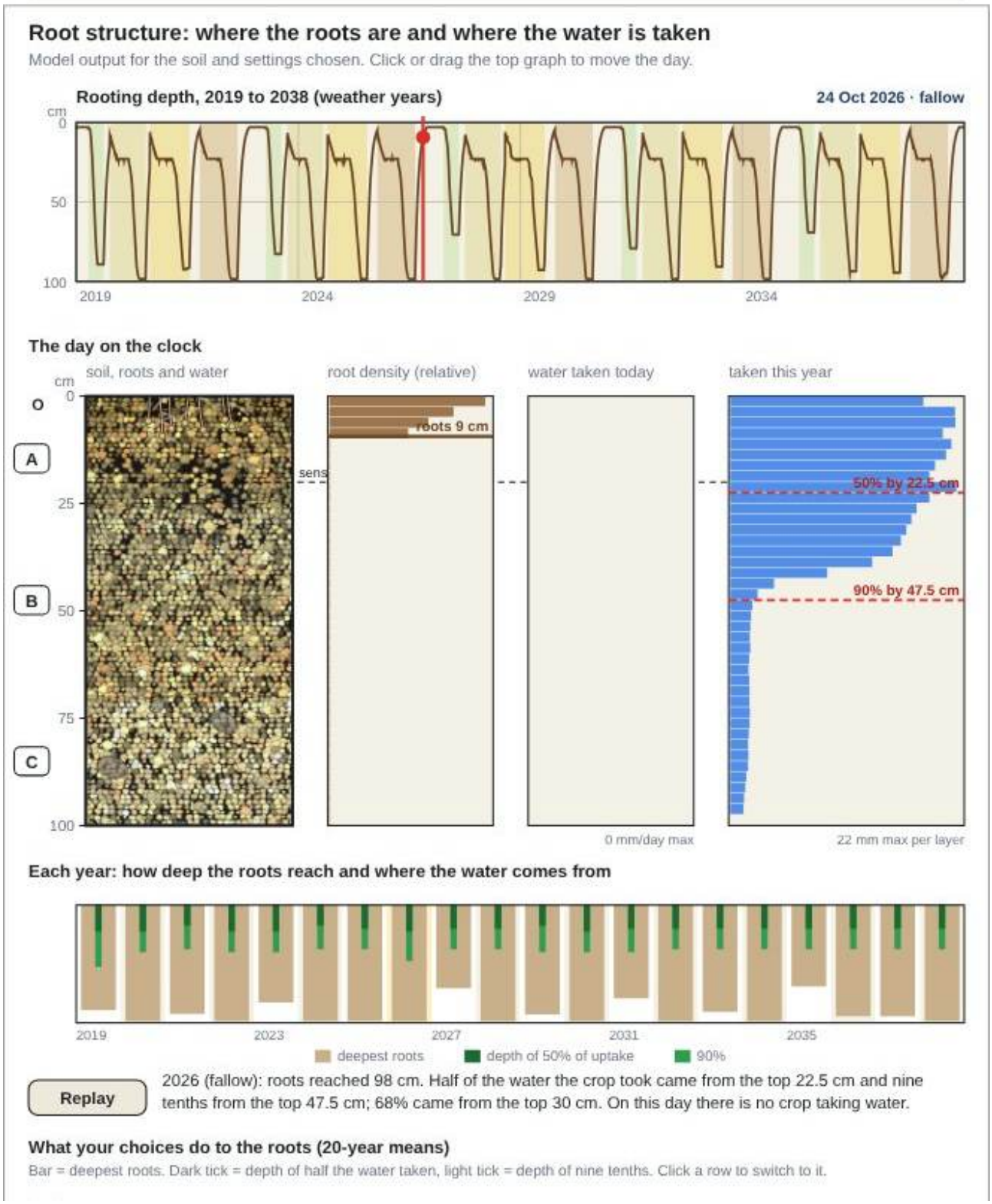


Figure 2.50: Root structure, Loamy Sand, part 2 of 4.

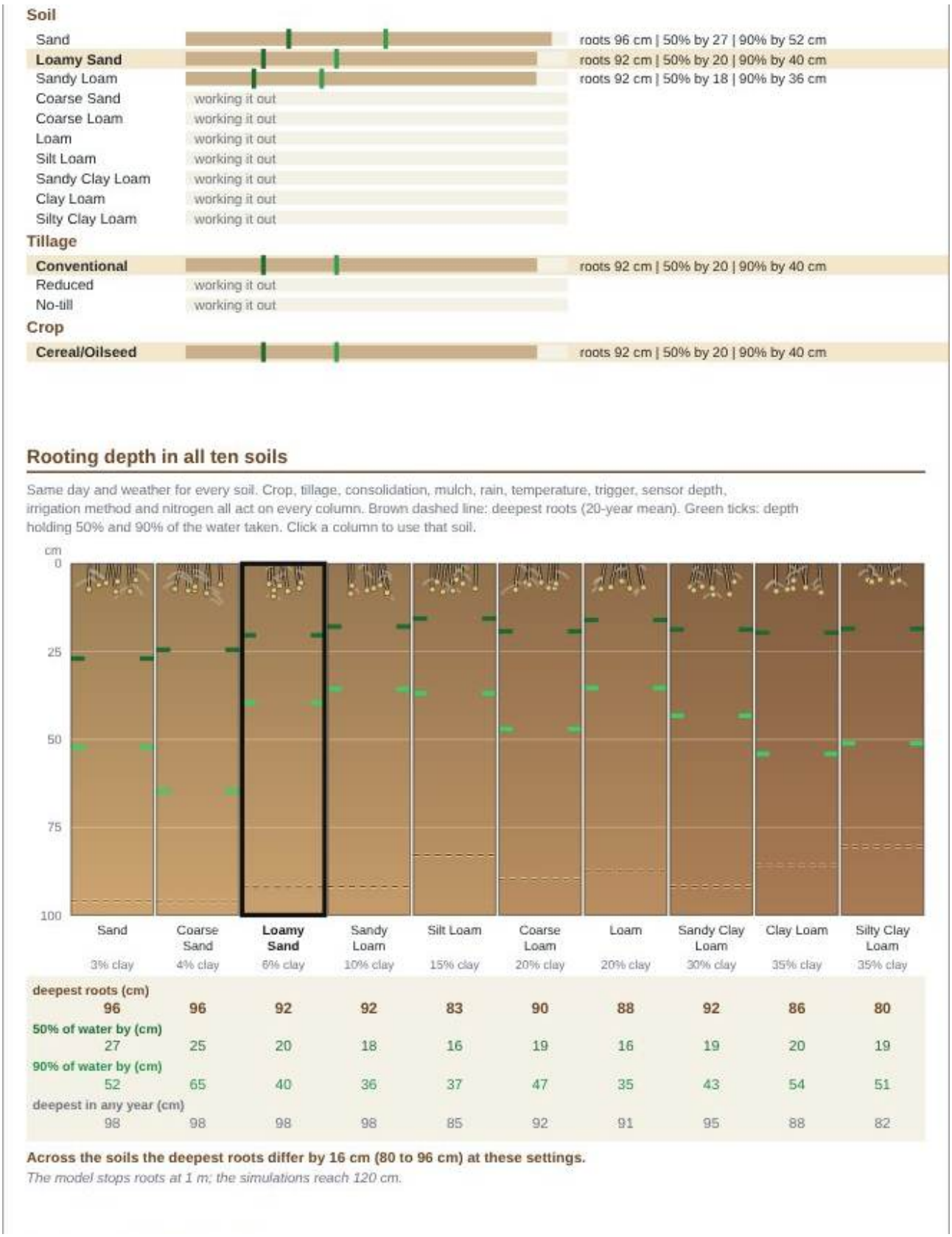


Figure 2.51: Root structure, Loamy Sand, part 3 of 4.

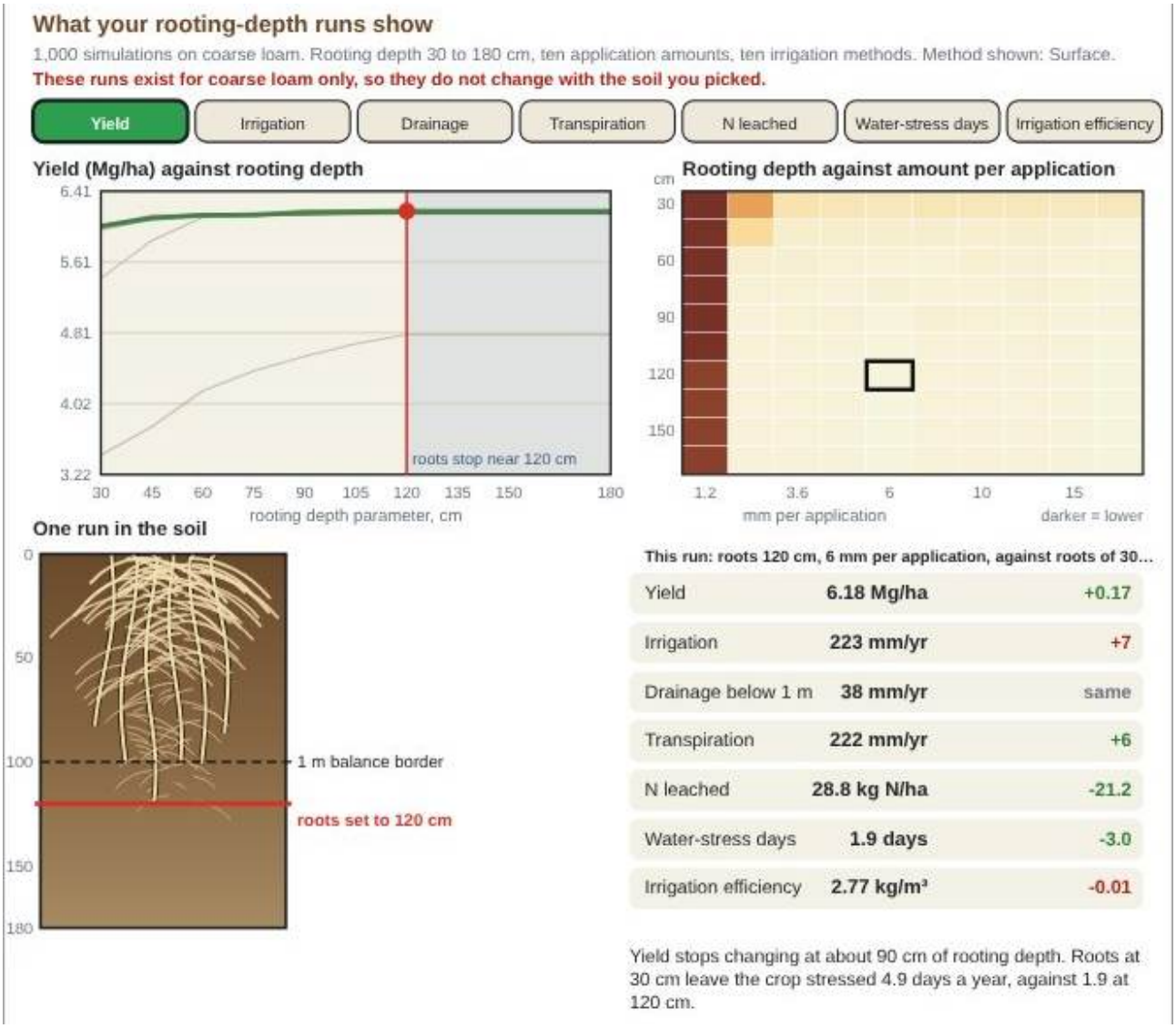


Figure 2.52: Root structure, Loamy Sand, part 4 of 4.

2.17 Field balance (Loamy Sand)

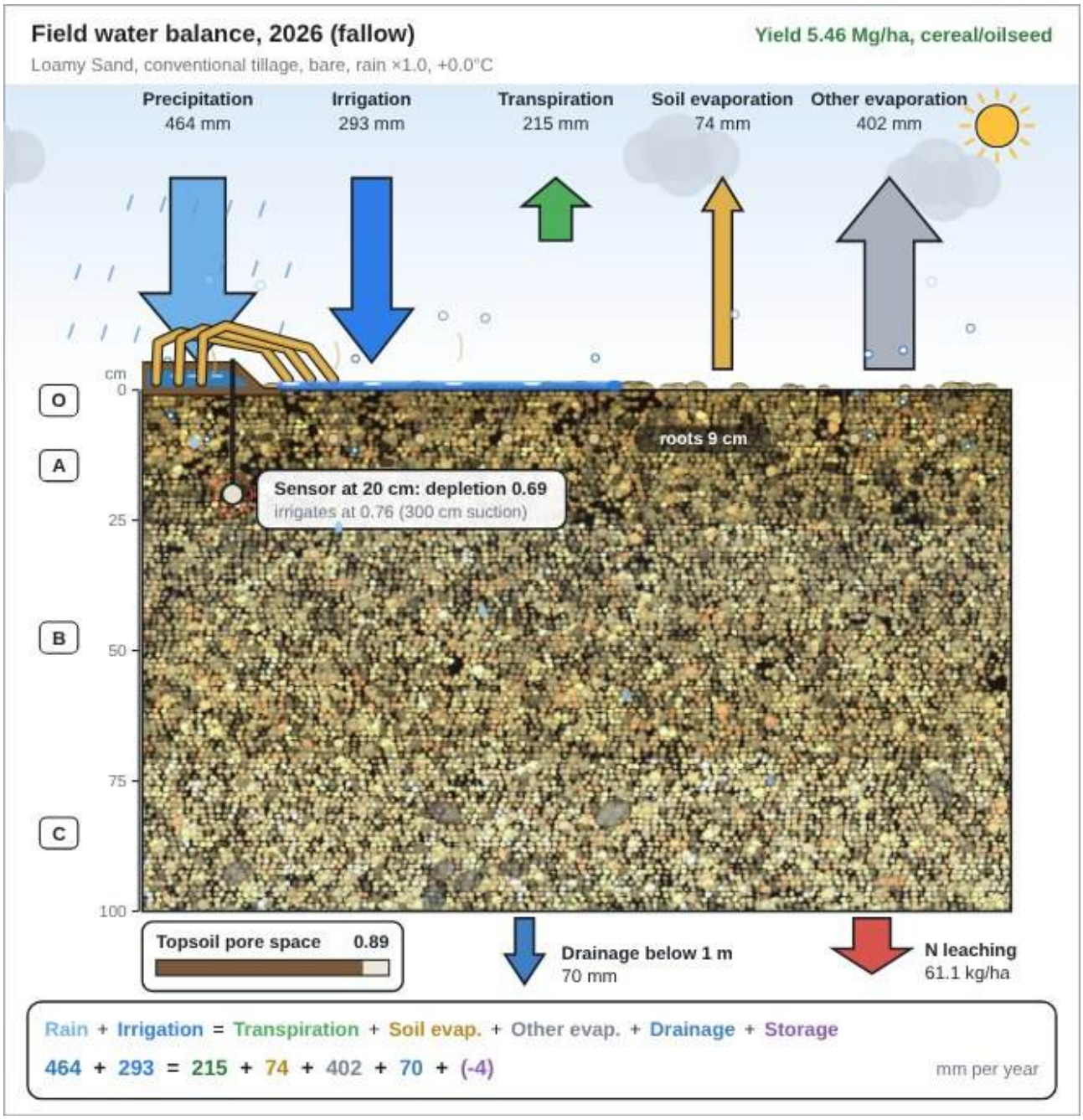


Figure 2.53: Field balance, Loamy Sand, part 1 of 1.

2.18 Soil, plant and air (Loamy Sand)

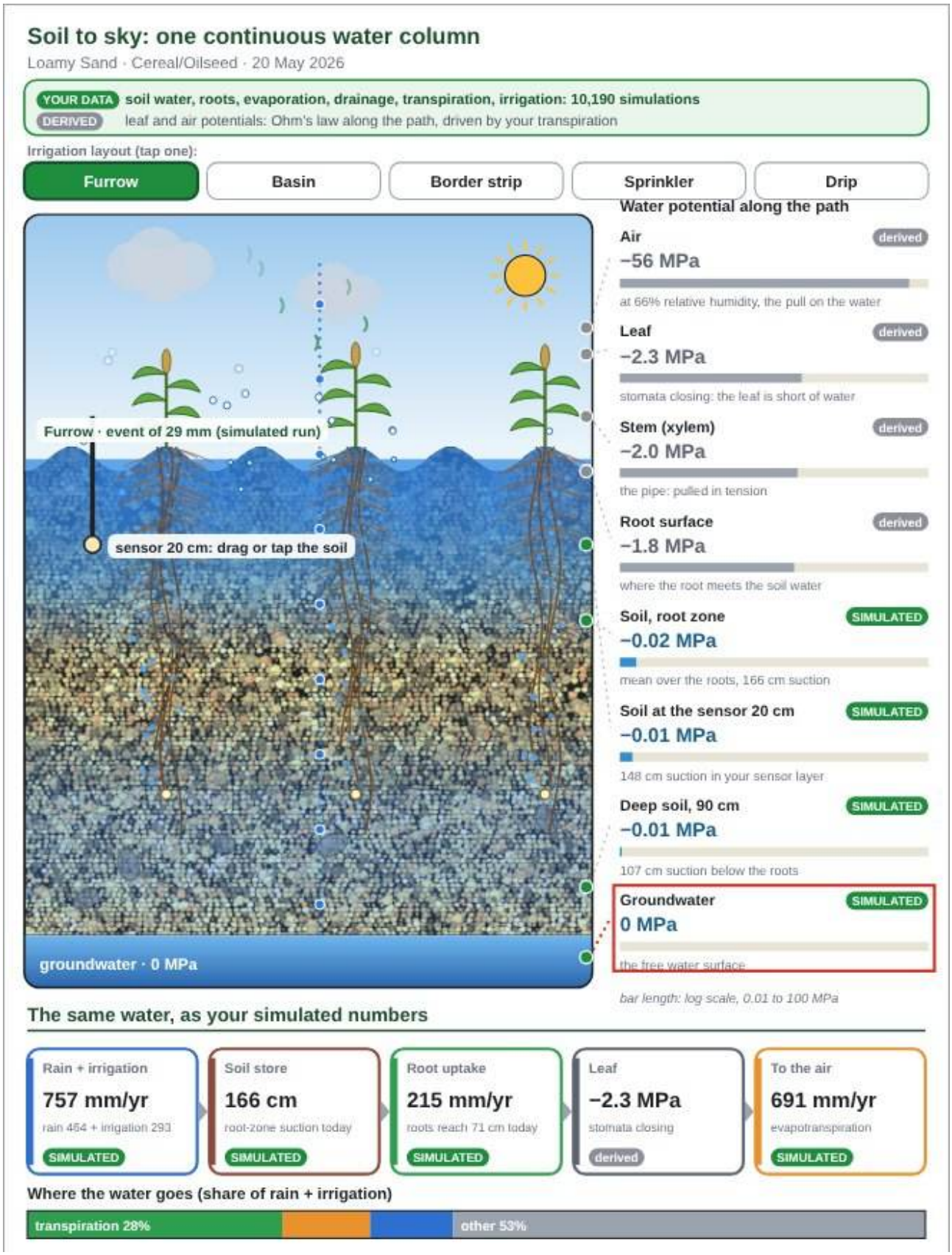


Figure 2.54: Soil, plant and air, Loamy Sand, part 1 of 3.

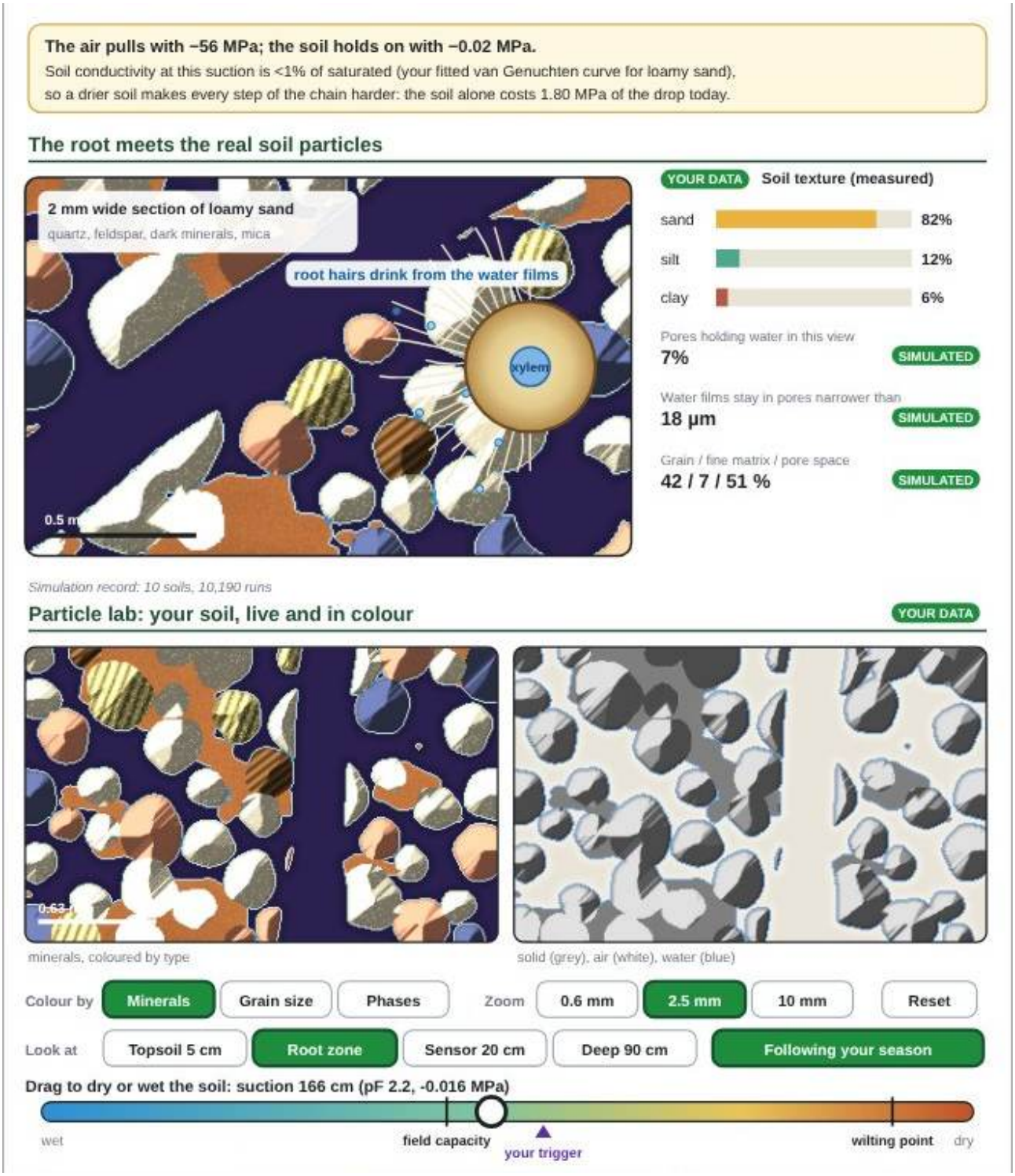


Figure 2.55: Soil, plant and air, Loamy Sand, part 2 of 3.

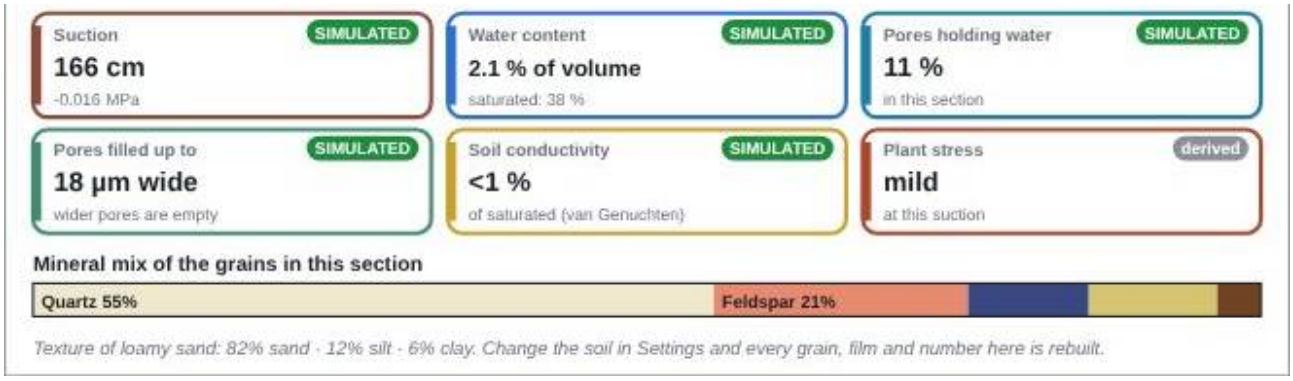


Figure 2.56: Soil, plant and air, Loamy Sand, part 3 of 3.

2.19 Profile evolution (Loamy Sand)

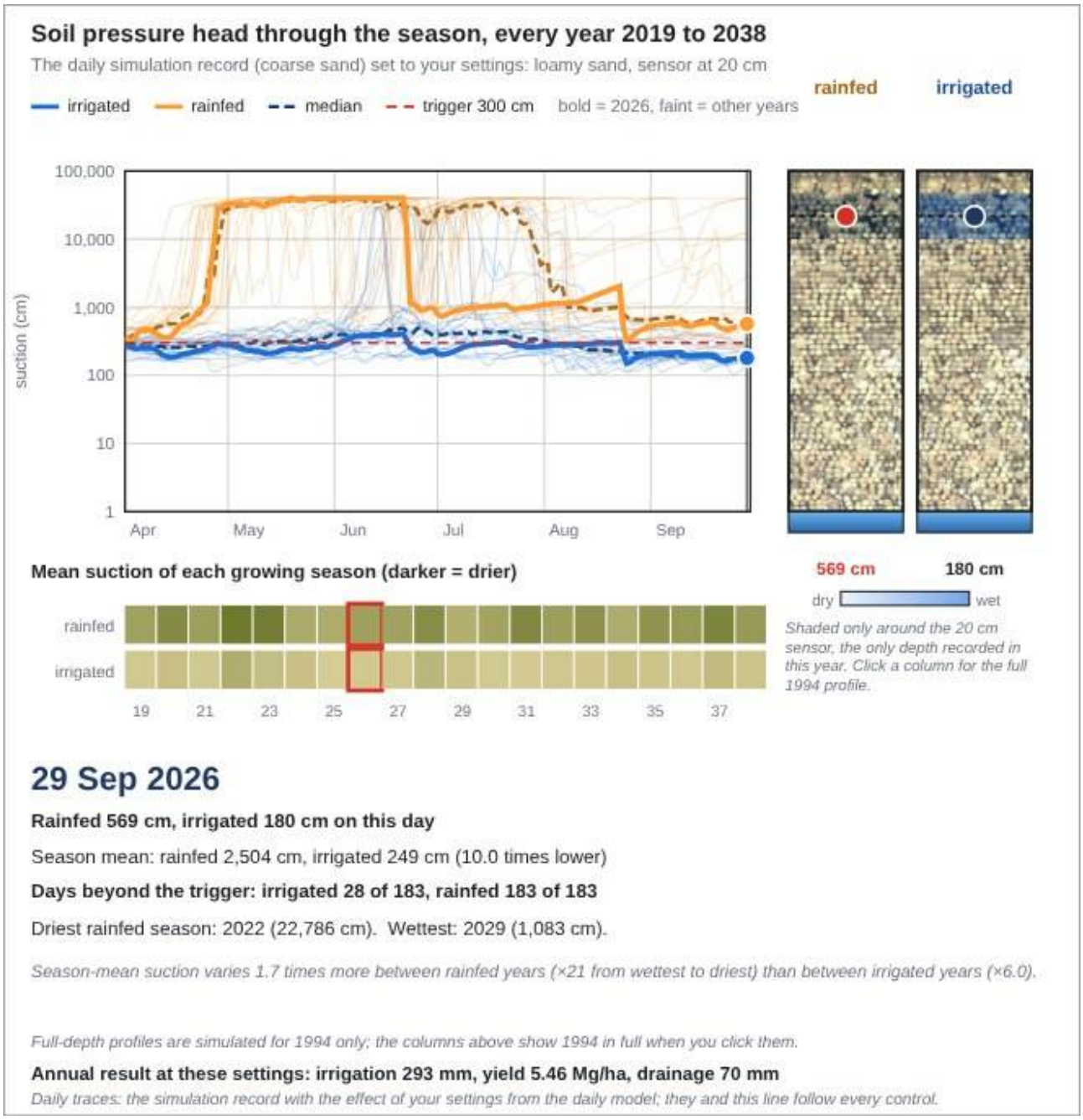


Figure 2.57: Profile evolution, Loamy Sand, part 1 of 1.

2.20 Reliability over 20 seasons (Loamy Sand)

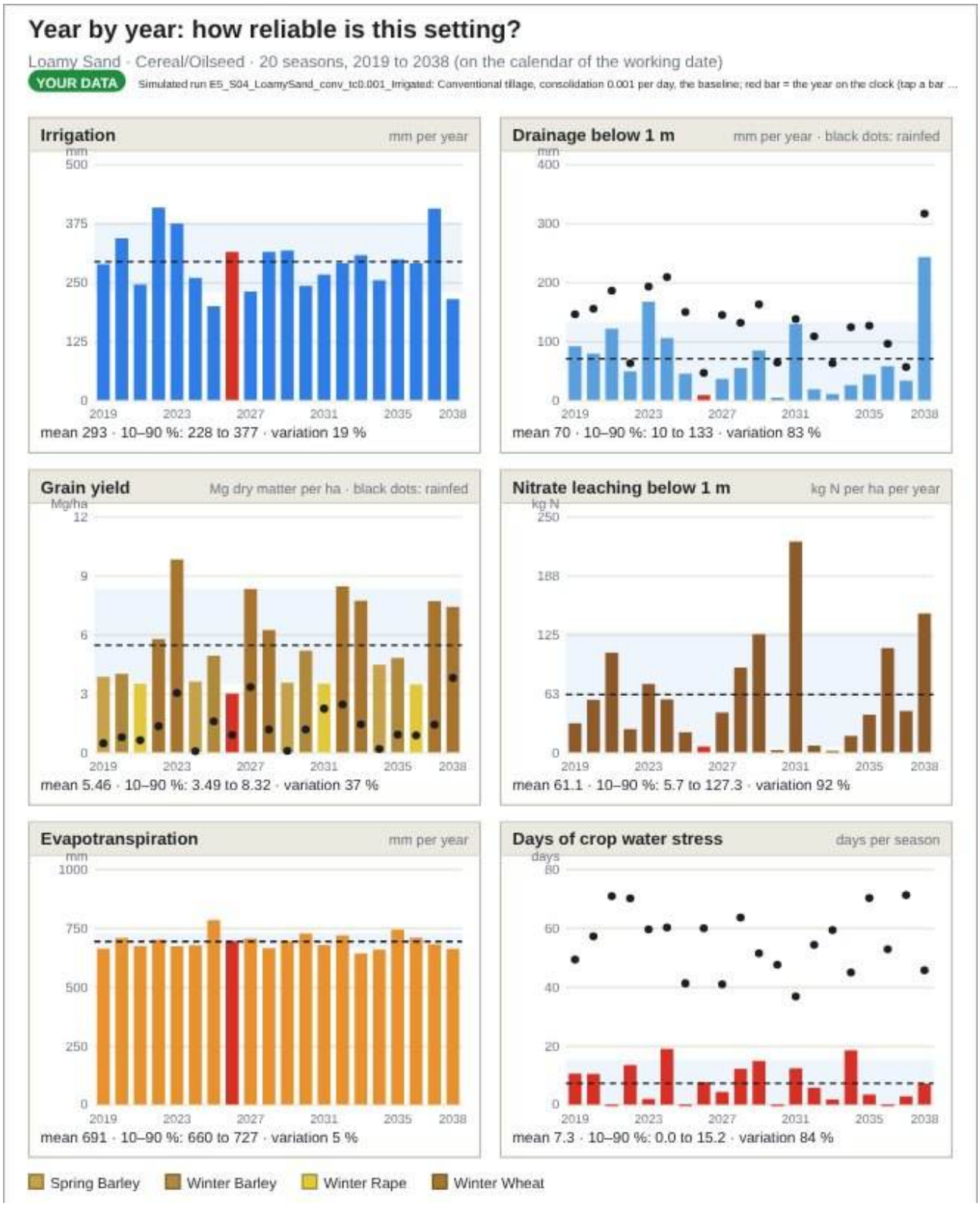


Figure 2.58: Reliability over 20 seasons, Loamy Sand, part 1 of 2.

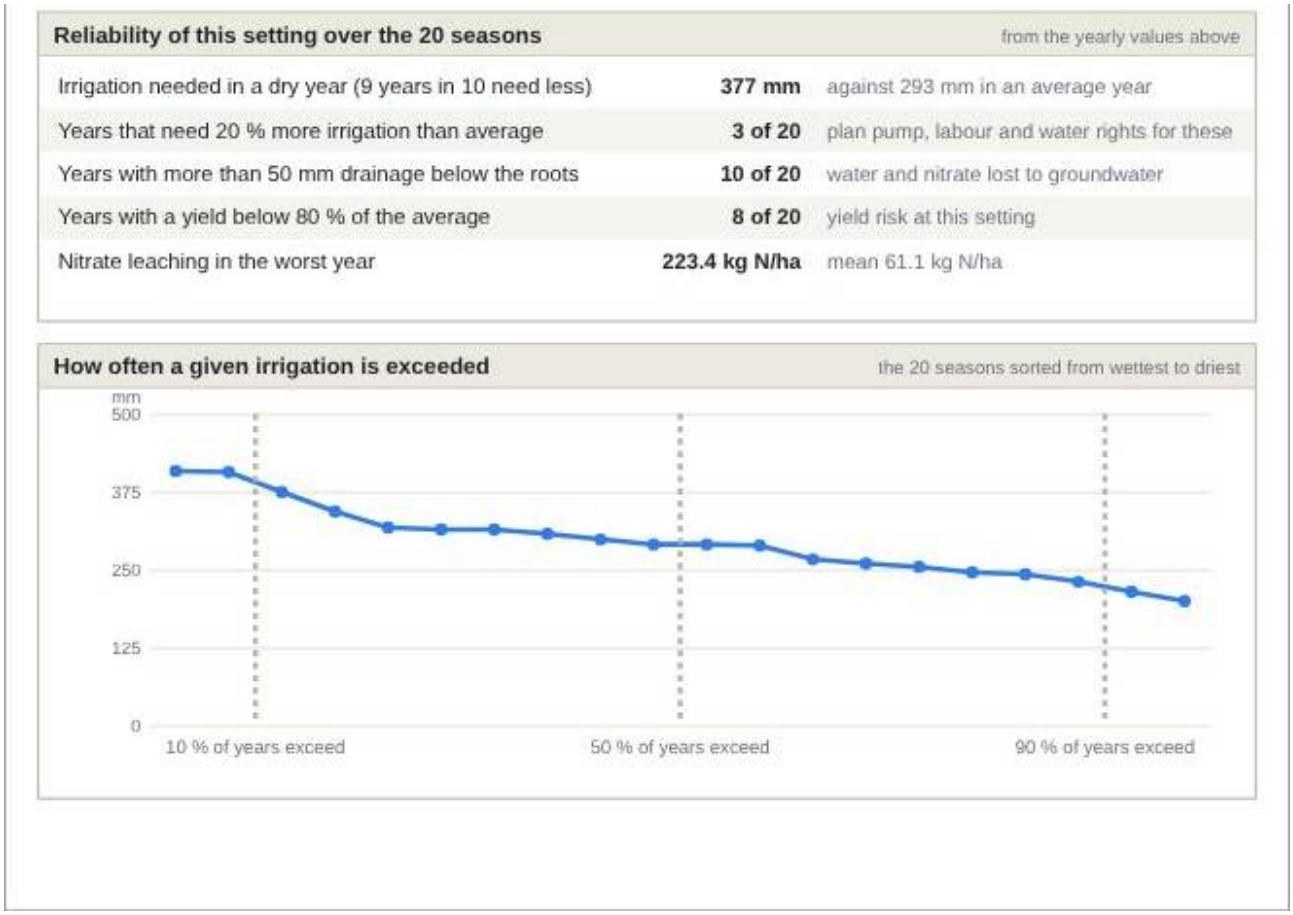


Figure 2.59: Reliability over 20 seasons, Loamy Sand, part 2 of 2.

2.21 Full water balance (Loamy Sand)

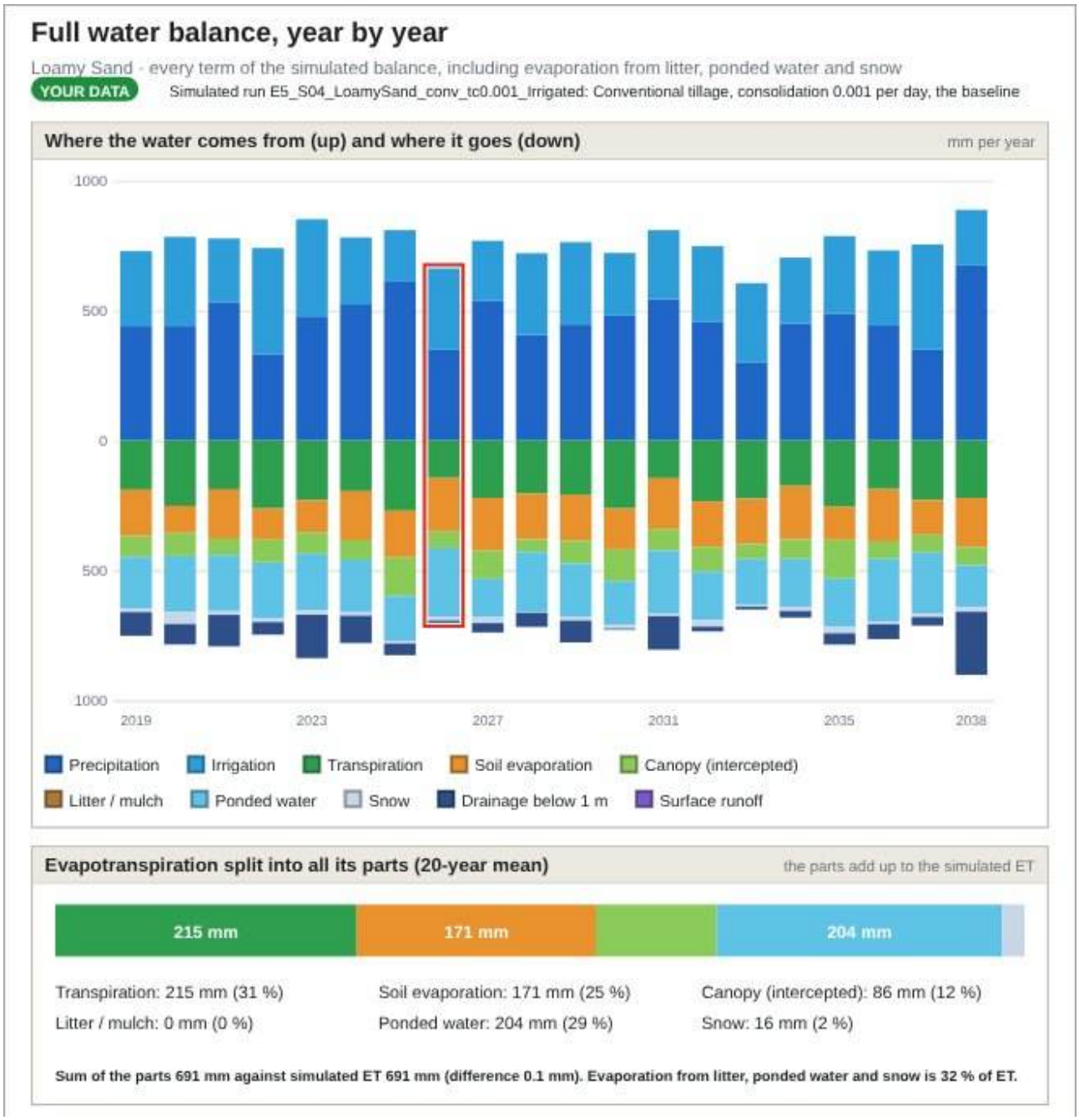


Figure 2.60: Full water balance, Loamy Sand, part 1 of 2.

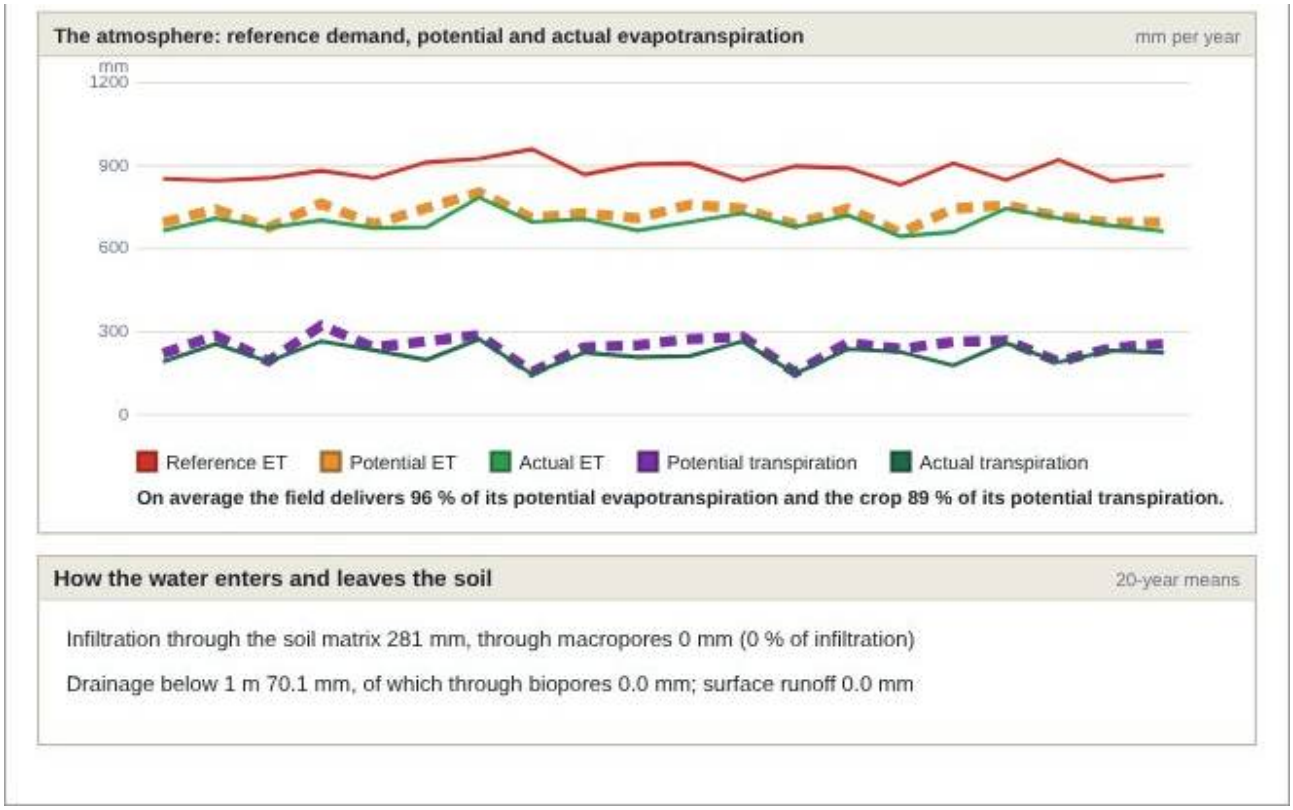


Figure 2.61: Full water balance, Loamy Sand, part 2 of 2.

2.22 Nitrogen budget (Loamy Sand)

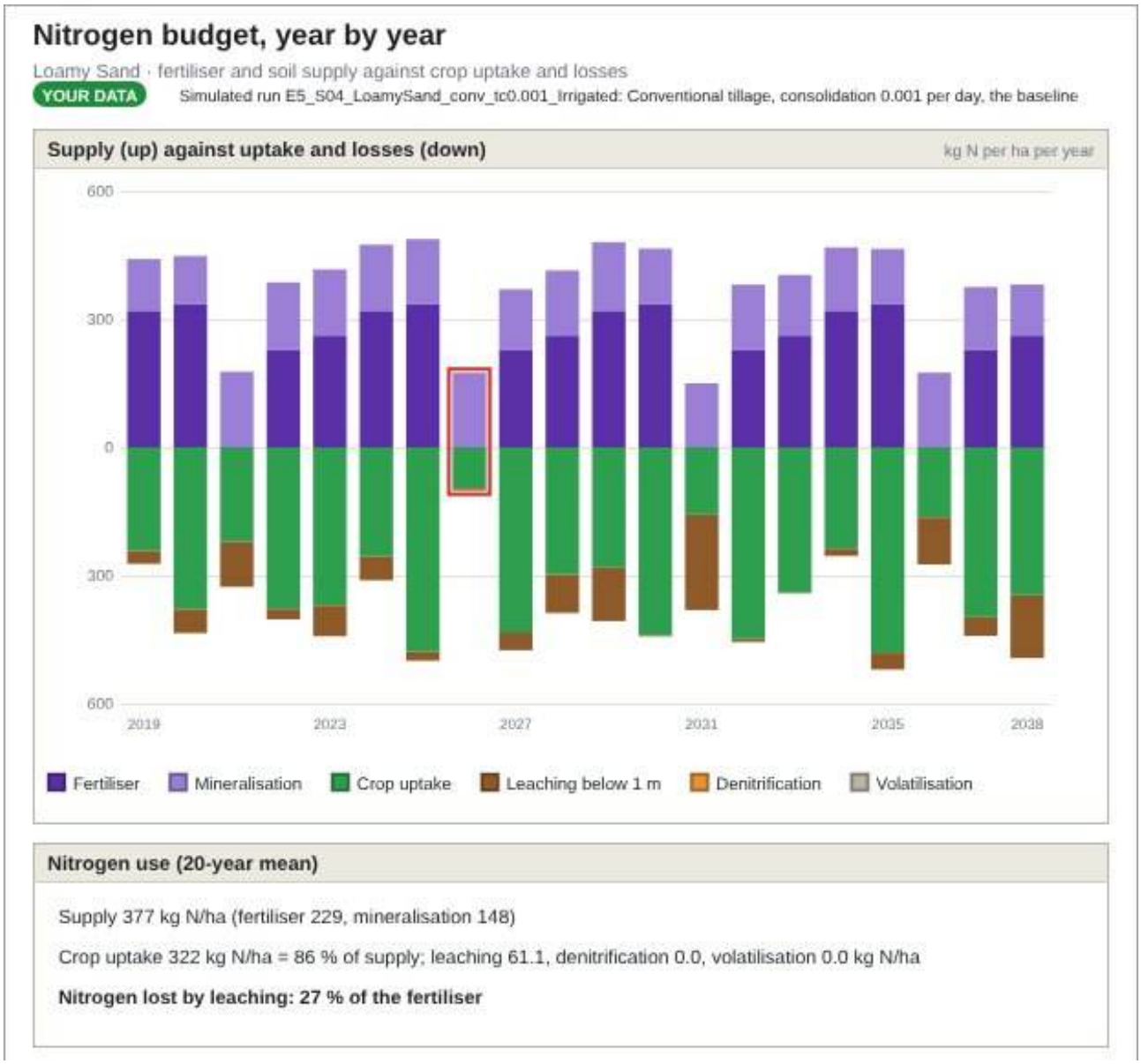


Figure 2.62: Nitrogen budget, Loamy Sand, part 1 of 2.

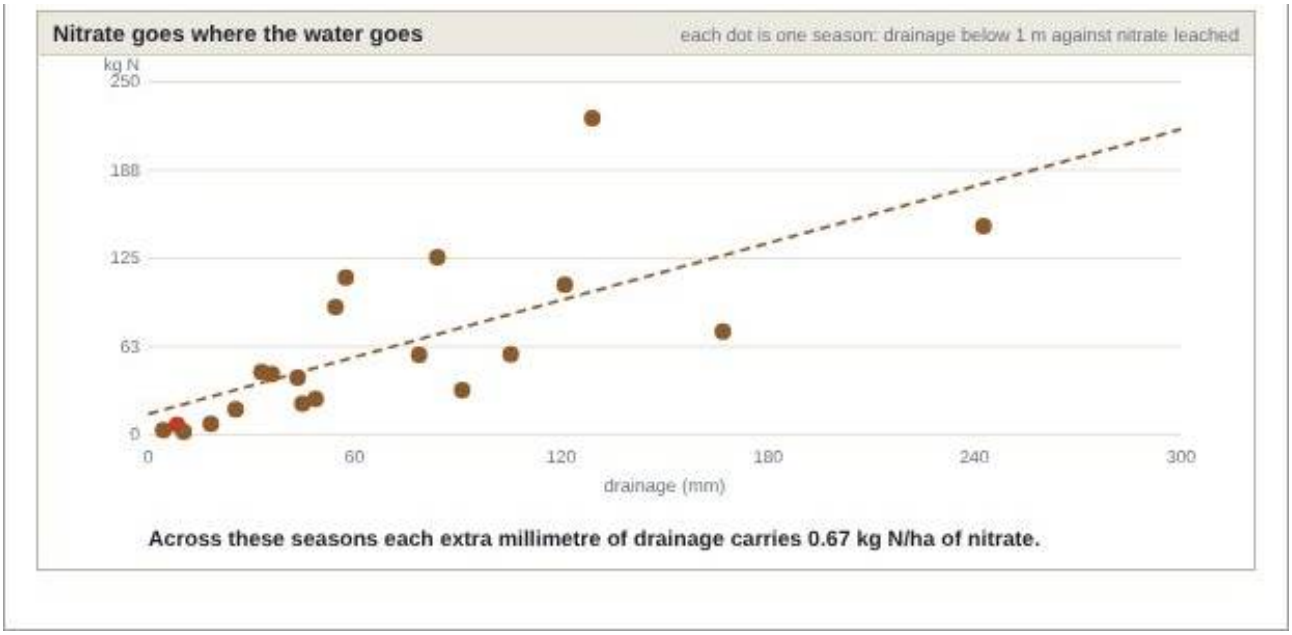


Figure 2.63: Nitrogen budget, Loamy Sand, part 2 of 2.

2.23 Crop water requirement (Loamy Sand)

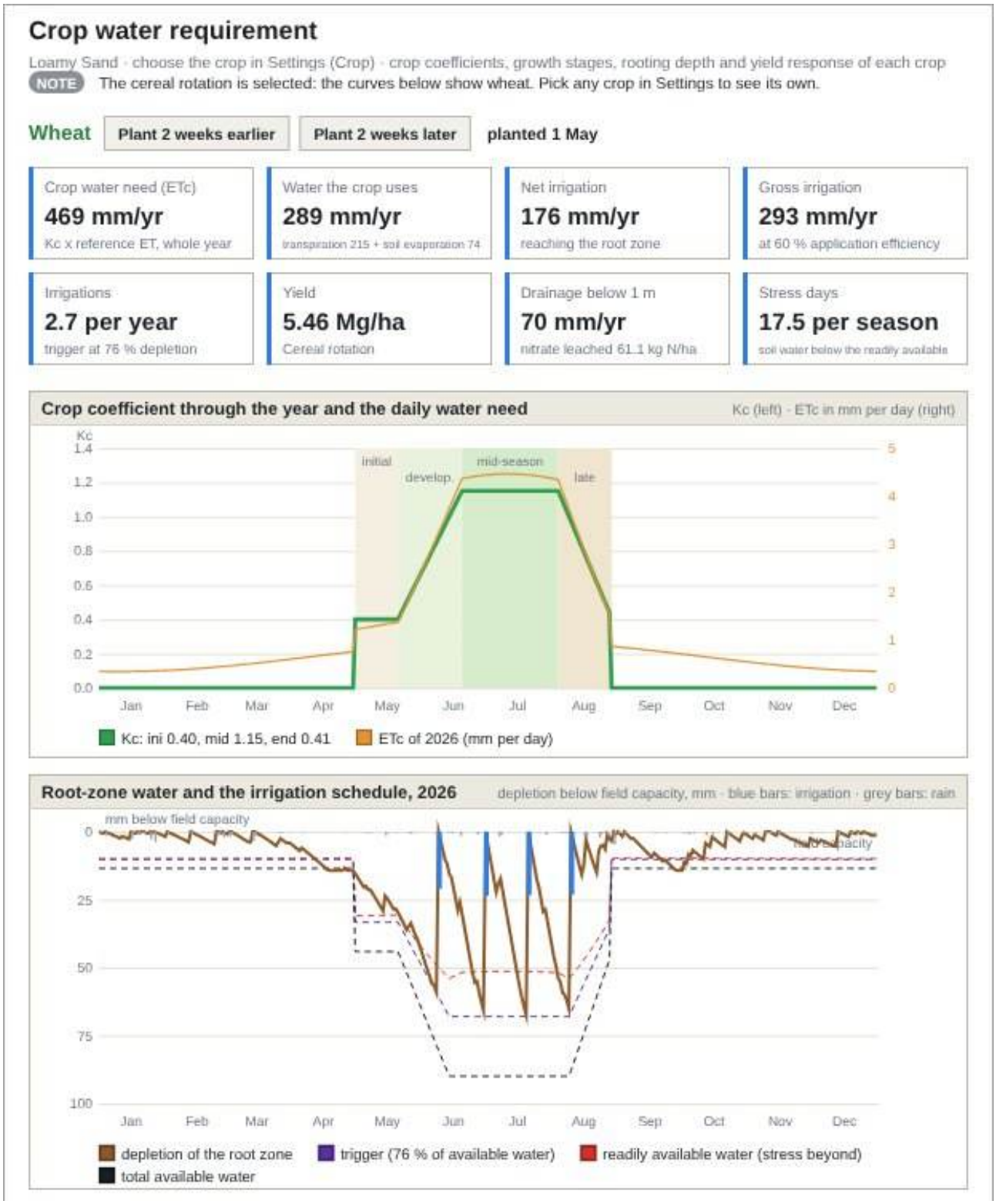


Figure 2.64: Crop water requirement, Loamy Sand, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Coconut (high)	834	**384**	5.6	96 %
Papaya (high)	814	**375**	6.1	96 %
Oil palm (young)	790	**375**	6.0	100 %
Cocoa (under high shade canopy)	913	**301**	5.2	80 %
Avocado (medium)	677	**288**	5.5	100 %
Cotton	571	**274**	4.1	100 %
Yam	649	**256**	6.4	89 %
Cabbage	559	**237**	5.5	93 %
Guava (high)	651	**232**	3.4	97 %
Mustard	468	**213**	3.3	98 %
Onion (dry)	565	**203**	6.6	80 %
Hop (v trellis)	520	**202**	3.0	94 %
Wheat	469	**200**	3.1	96 %
Mango (medium)	620	**199**	2.9	97 %
Groundnut	471	**198**	4.0	95 %
Maize (grain)	451	**194**	3.0	94 %
Soybean	473	**193**	3.1	95 %
Kiwi (pergola)	626	**191**	2.8	86 %
Walnut (medium)	523	**189**	2.8	97 %
Potato	508	**185**	5.6	83 %
Date palm (medium)	602	**184**	2.7	98 %
Tomato	521	**180**	2.7	86 %
Pear (medium)	528	**177**	2.6	97 %
Cherry (medium)	503	**165**	2.4	97 %
Chickpea	436	**163**	3.1	95 %
Sorghum (grain)	429	**158**	2.4	97 %
Table grapes (horizontal trellis)	597	**155**	2.3	85 %

Figure 2.65: Crop water requirement, Loamy Sand, part 2 of 3.

Rice	573	**154**	3.0	69 %
Hazelnut (low)	481	**147**	2.1	97 %
Pecan (low)	481	**147**	2.1	97 %
Millet	395	**142**	2.1	97 %
Cherimoya (medium)	545	**136**	2.0	98 %
Pistachio (medium)	503	**135**	2.0	91 %
Cowpea (beans)	398	**129**	2.5	92 %
Citrus (medium density, short)	552	**129**	1.9	98 %
Cassava	466	**127**	3.0	89 %
Apple (medium)	459	**110**	1.6	97 %
Sweet pepper	475	**107**	2.1	76 %
Green gram (mung bean)	372	**104**	2.0	92 %
Peach (low)	435	**99**	1.4	98 %
Wine grapes (pergola)	457	**96**	1.4	96 %
Almond (medium)	435	**77**	1.1	93 %
Apricot (young)	380	**59**	0.9	98 %
Olive (traditional, medium density)	446	**51**	0.8	100 %
Persimmon (young, low)	358	**40**	0.6	99 %
Pomegranate (low)	358	**40**	0.6	99 %
Highbush blueberry (low)	338	**39**	0.6	99 %
Fig tree (young, low)	327	**22**	0.3	100 %

Figure 2.66: Crop water requirement, Loamy Sand, part 3 of 3.

2.24 Feedback loops (Loamy Sand)

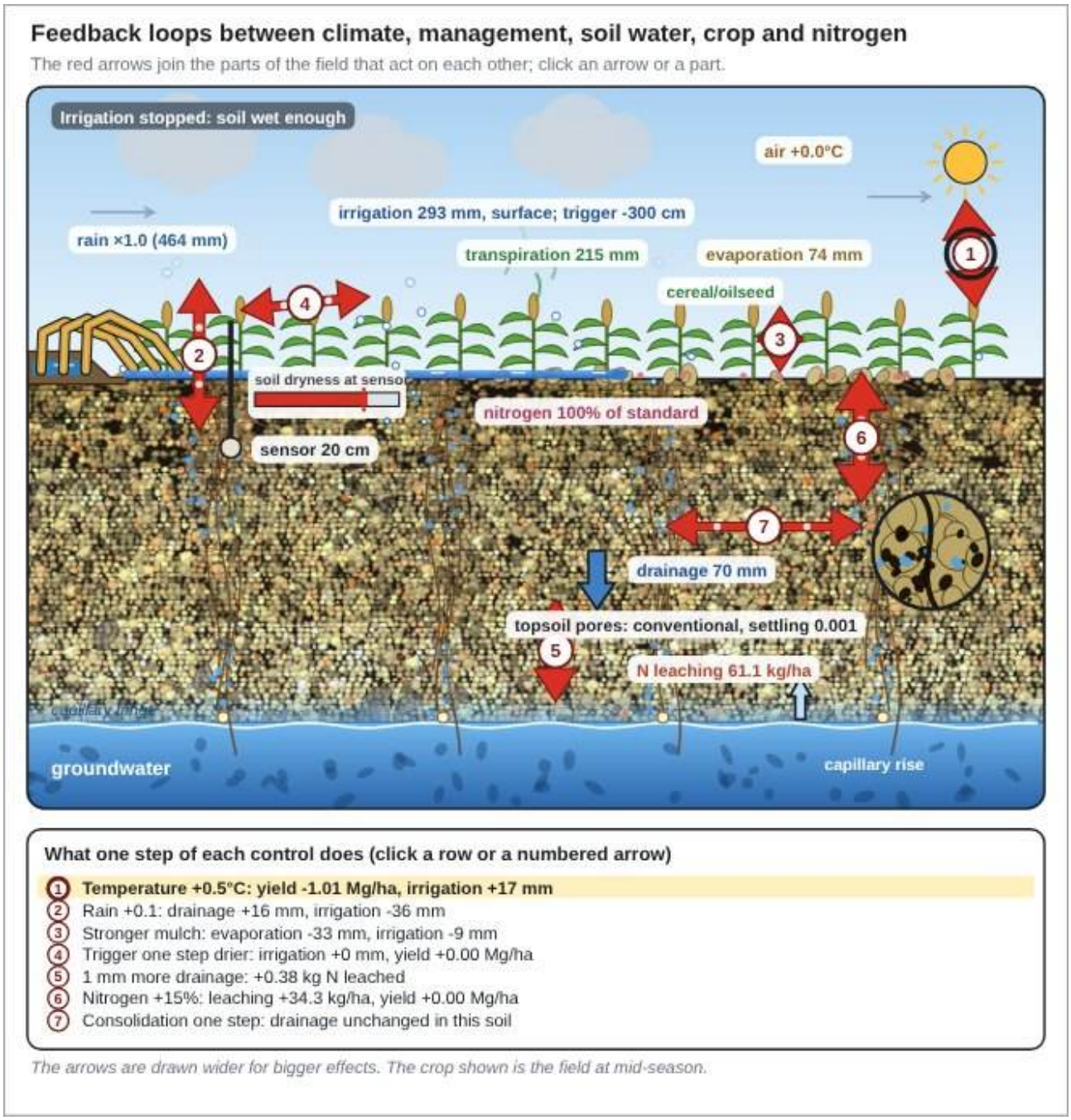


Figure 2.67: Feedback loops, Loamy Sand, part 1 of 1.

Chapter 3

Sandy Loam

3.1 Surface irrigation design (Sandy Loam)

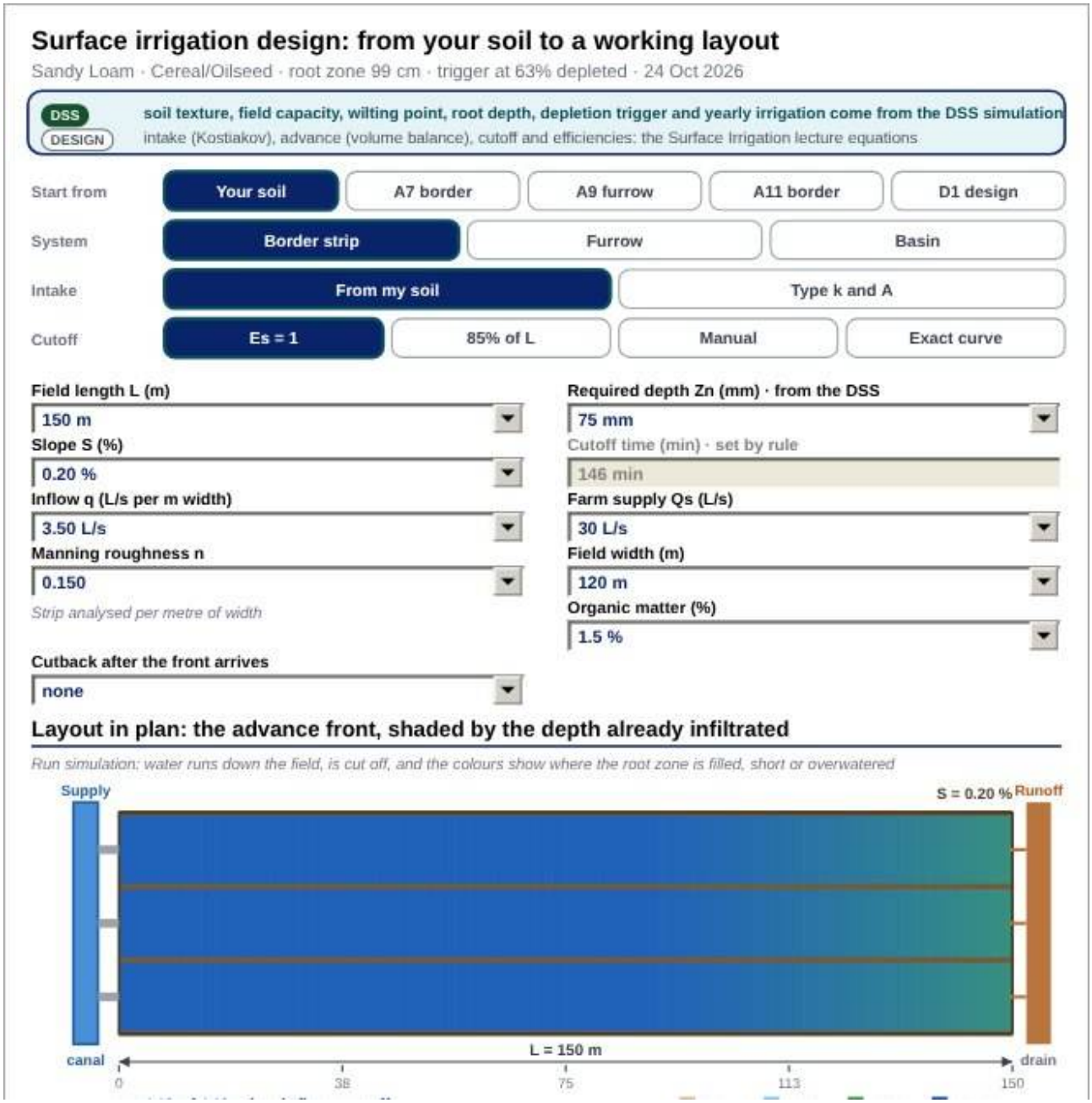


Figure 3.1: Surface irrigation design, Sandy Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

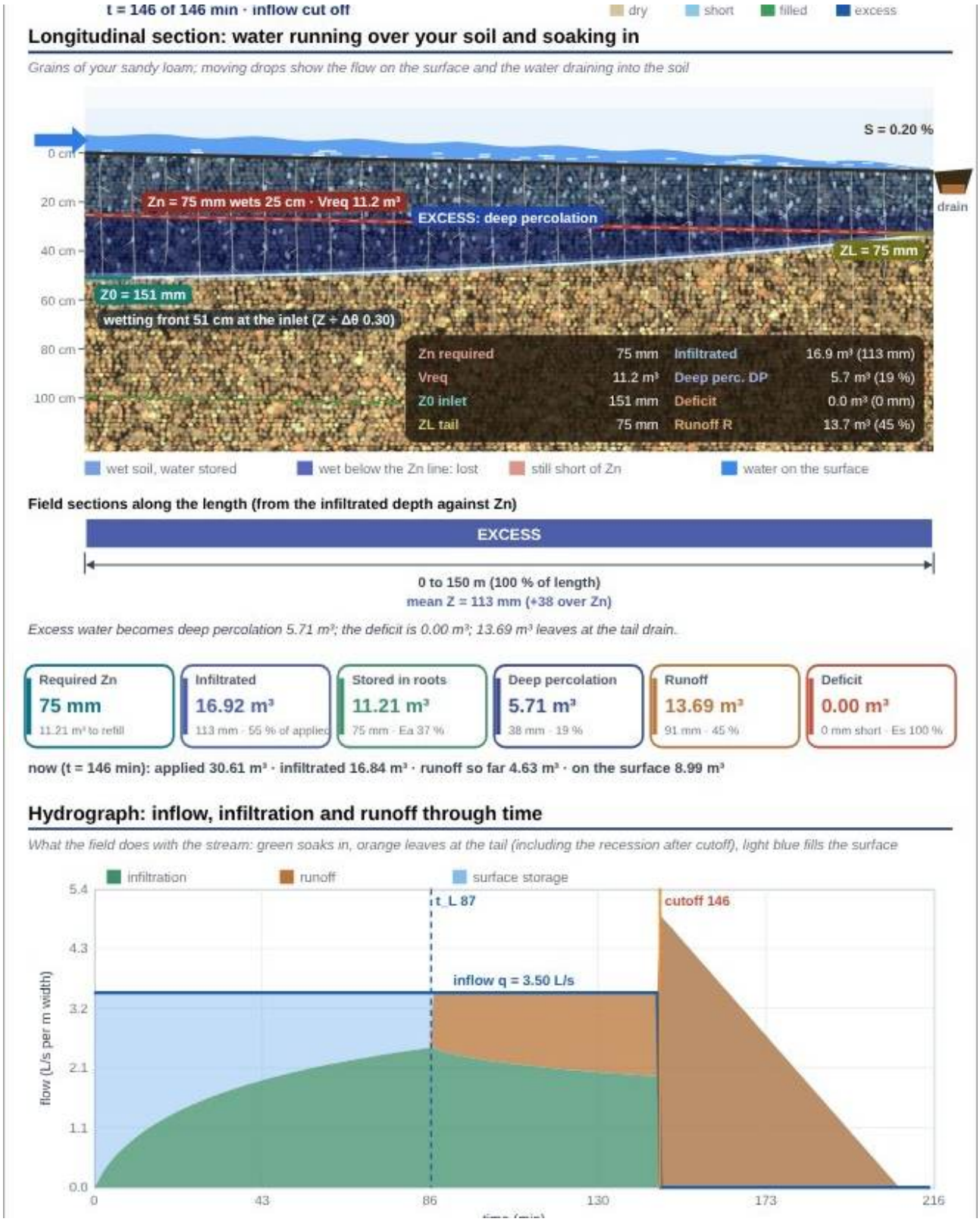


Figure 3.2: Surface irrigation design, Sandy Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

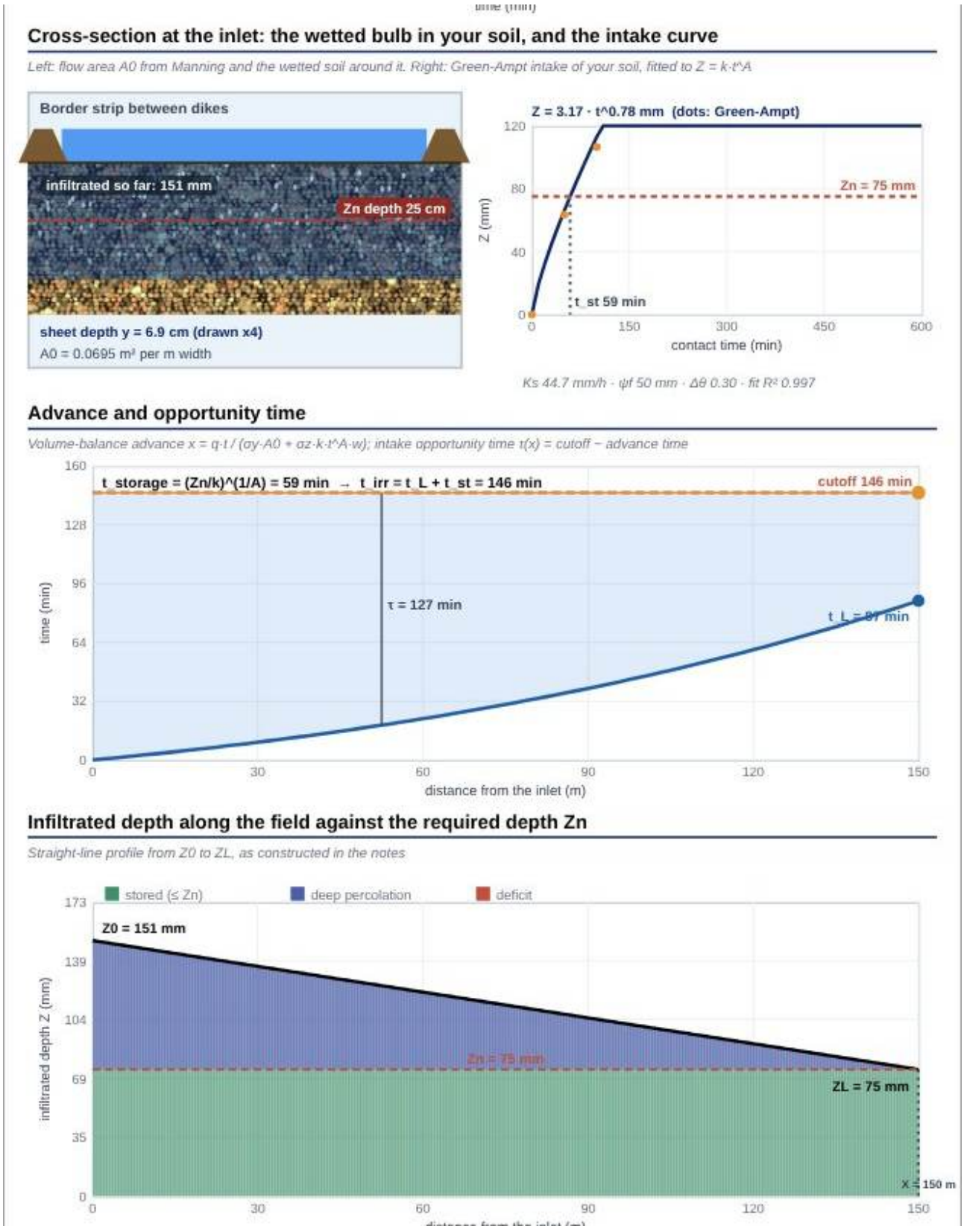


Figure 3.3: Surface irrigation design, Sandy Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

Figure 3.4: Surface irrigation design, Sandy Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

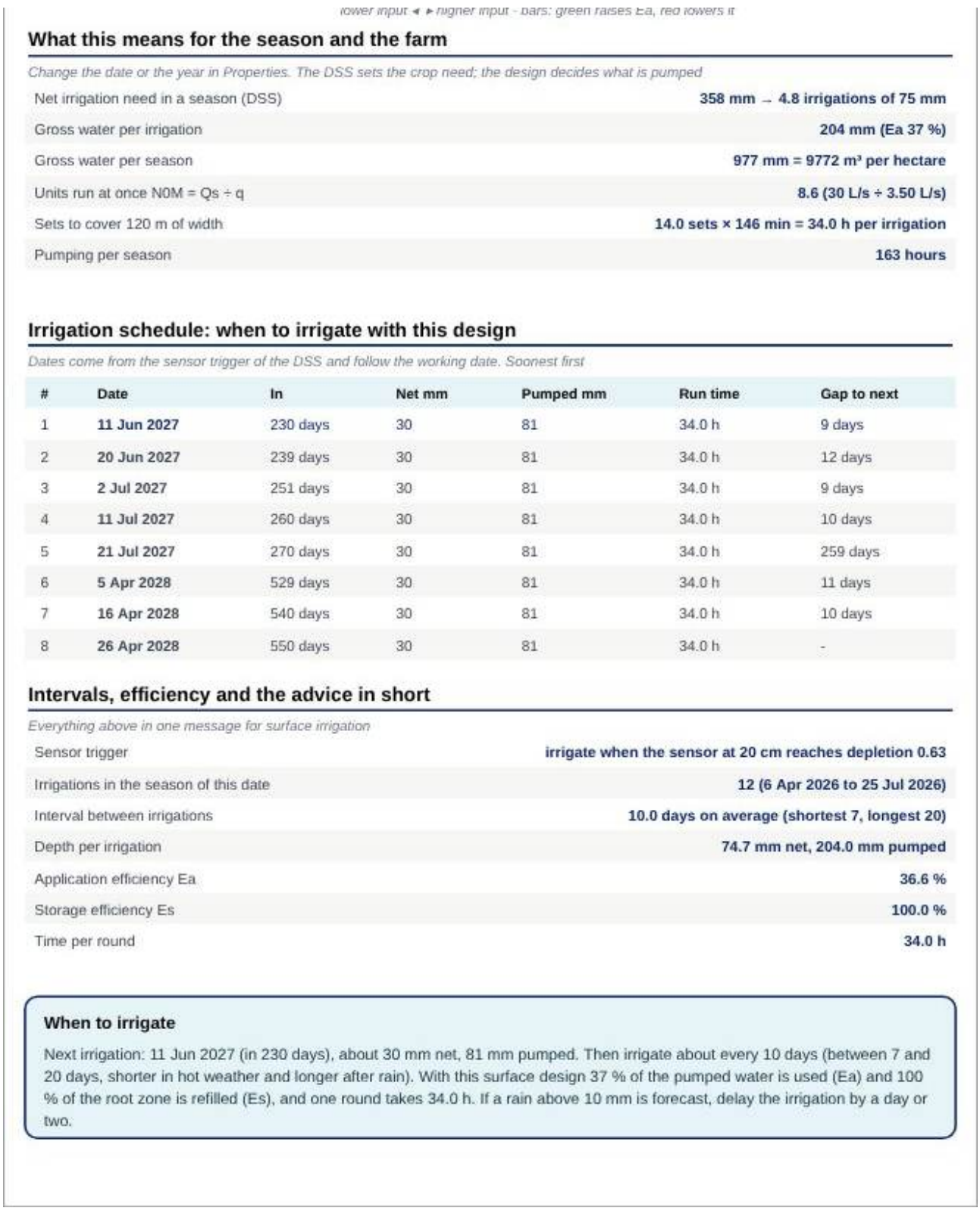


Figure 3.5: Surface irrigation design, Sandy Loam, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

3.2 Drip irrigation design (Sandy Loam)

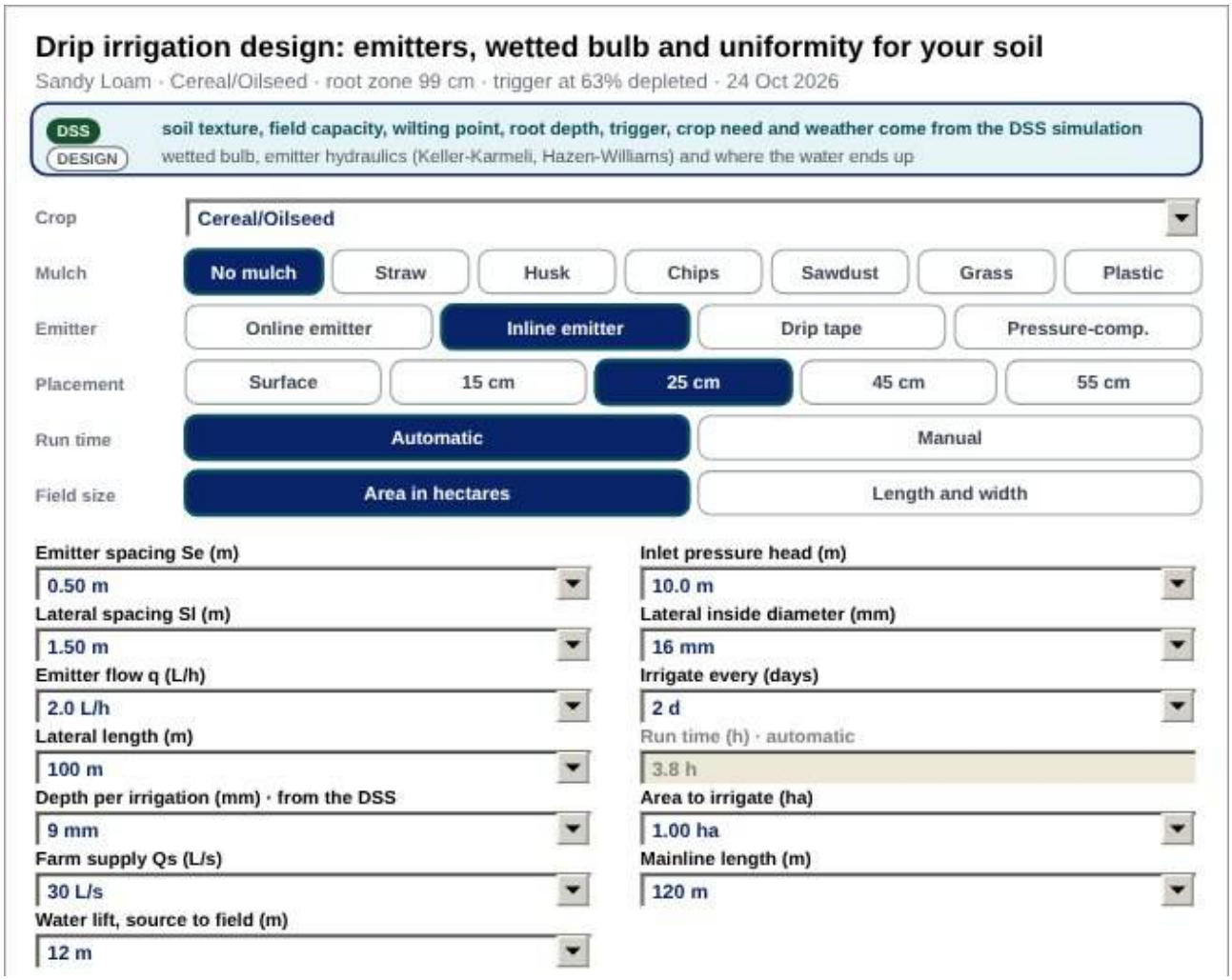


Figure 3.6: Drip irrigation design, Sandy Loam, part 1 of 6.

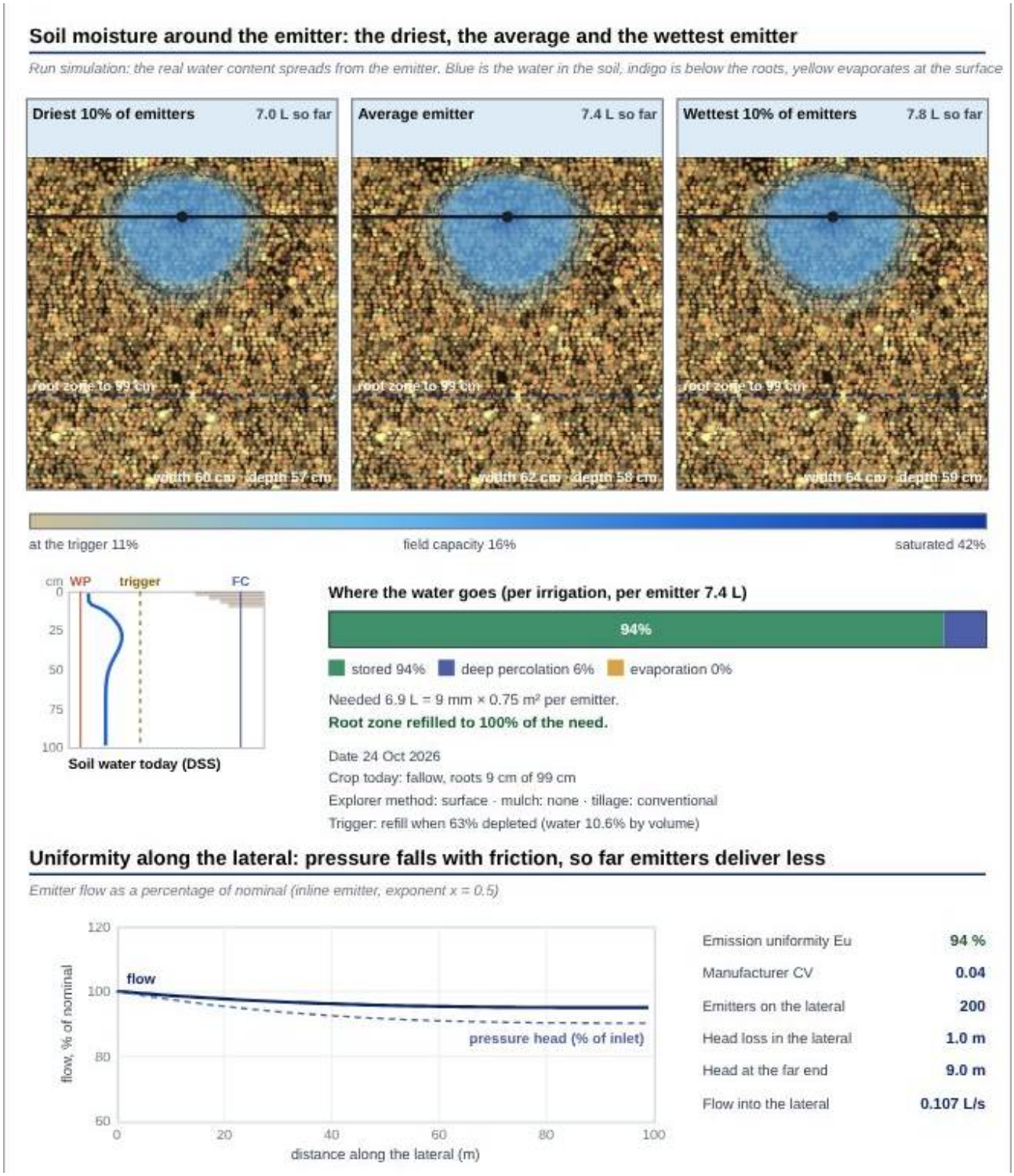


Figure 3.7: Drip irrigation design, Sandy Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

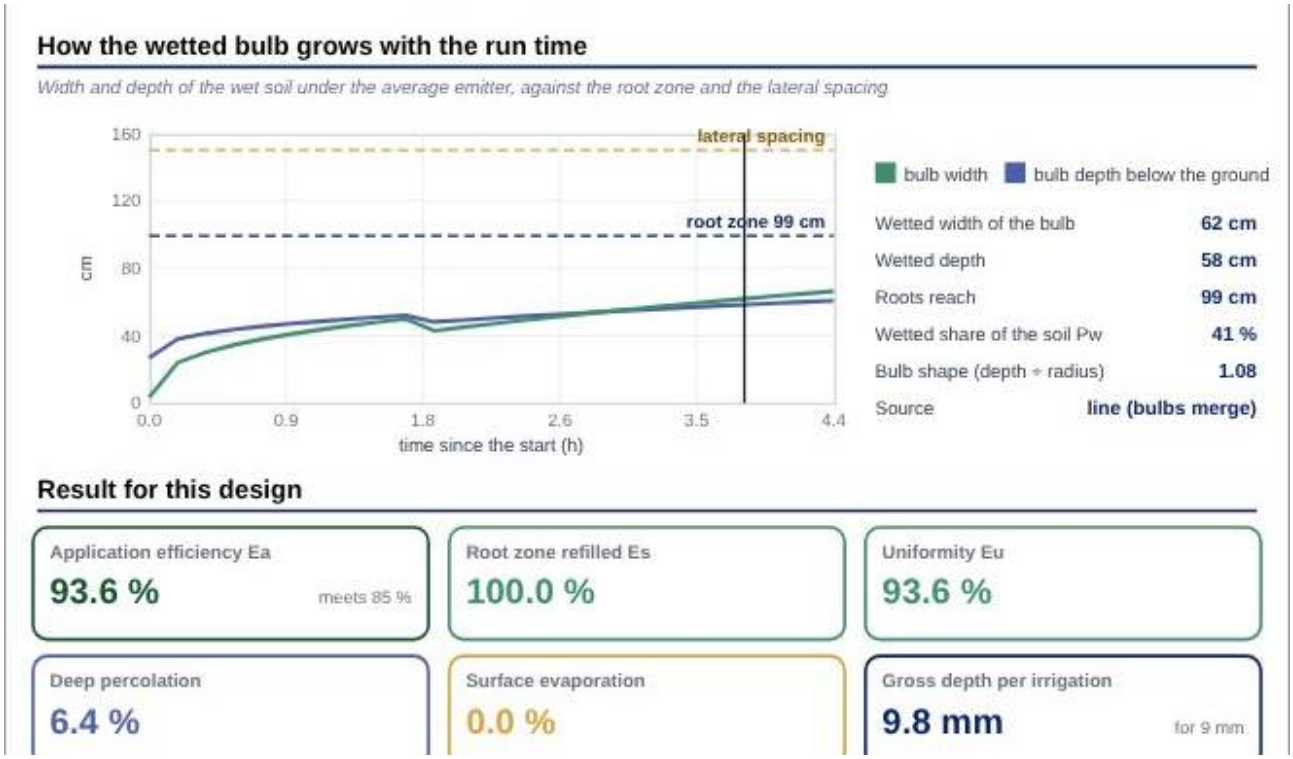
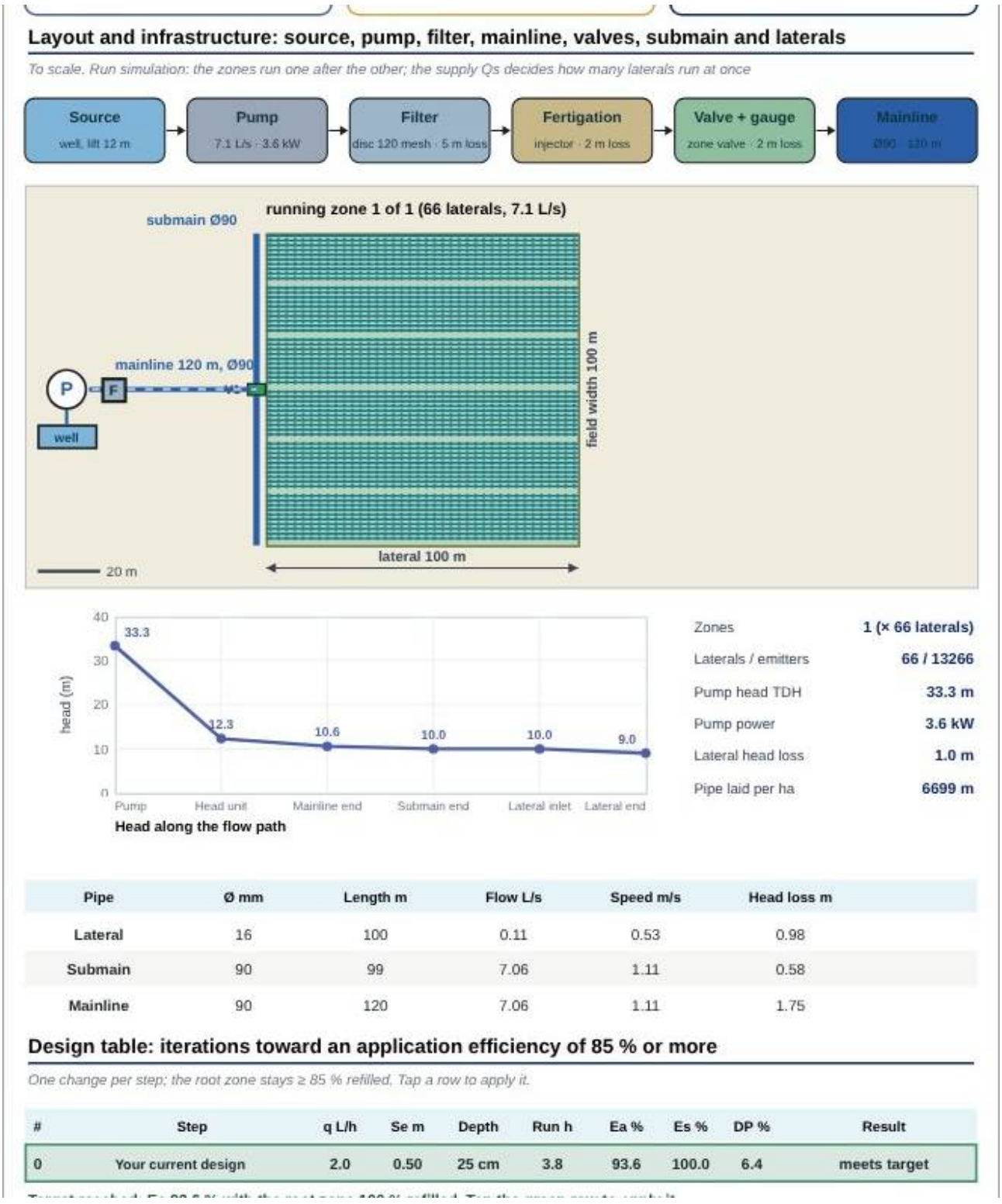


Figure 3.8: Drip irrigation design, Sandy Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.8	93.6	100.0	6.4	meets target

Figure 3.9: Drip irrigation design, Sandy Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**358 mm → 39 irrigations of 9 mm**
Gross water per irrigation	**9.8 mm (Ea 94 %)**
Gross water per season	**382 mm = 3824 m³ per hectare**
Peak crop need / interval	**4.6 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.8 h = 3.8 h per round**
Pumping per season	**149 hours · 529 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**11 Jun 2027**	230 days	30	32	3.8 h	9 days
2	**20 Jun 2027**	239 days	30	32	3.8 h	12 days
3	**2 Jul 2027**	251 days	30	32	3.8 h	9 days
4	**11 Jul 2027**	260 days	30	32	3.8 h	10 days
5	**21 Jul 2027**	270 days	30	32	3.8 h	259 days
6	**5 Apr 2028**	529 days	30	32	3.8 h	11 days
7	**16 Apr 2028**	540 days	30	32	3.8 h	10 days
8	**26 Apr 2028**	550 days	30	32	3.8 h	-

Figure 3.10: Drip irrigation design, Sandy Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.

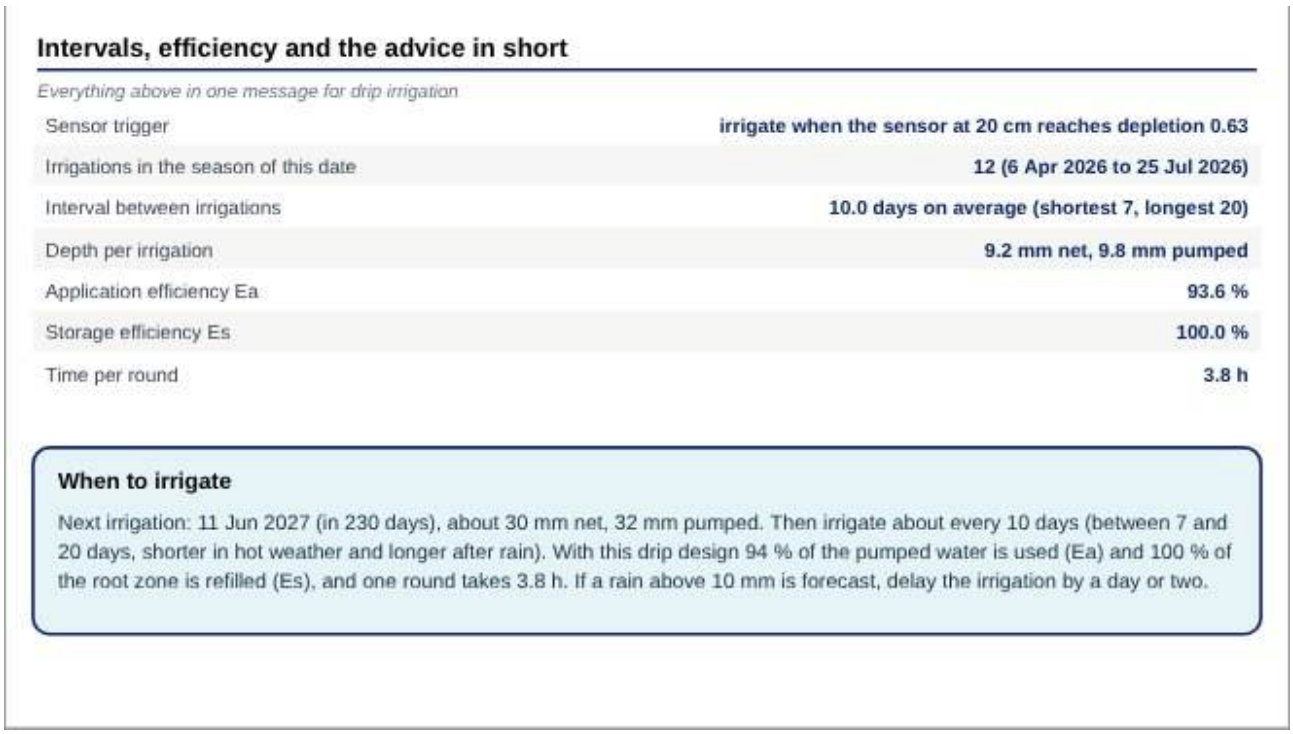


Figure 3.11: Drip irrigation design, Sandy Loam, part 6 of 6: Intervals, efficiency and the advice in short.

3.3 Sprinkler design and pipes (Sandy Loam)

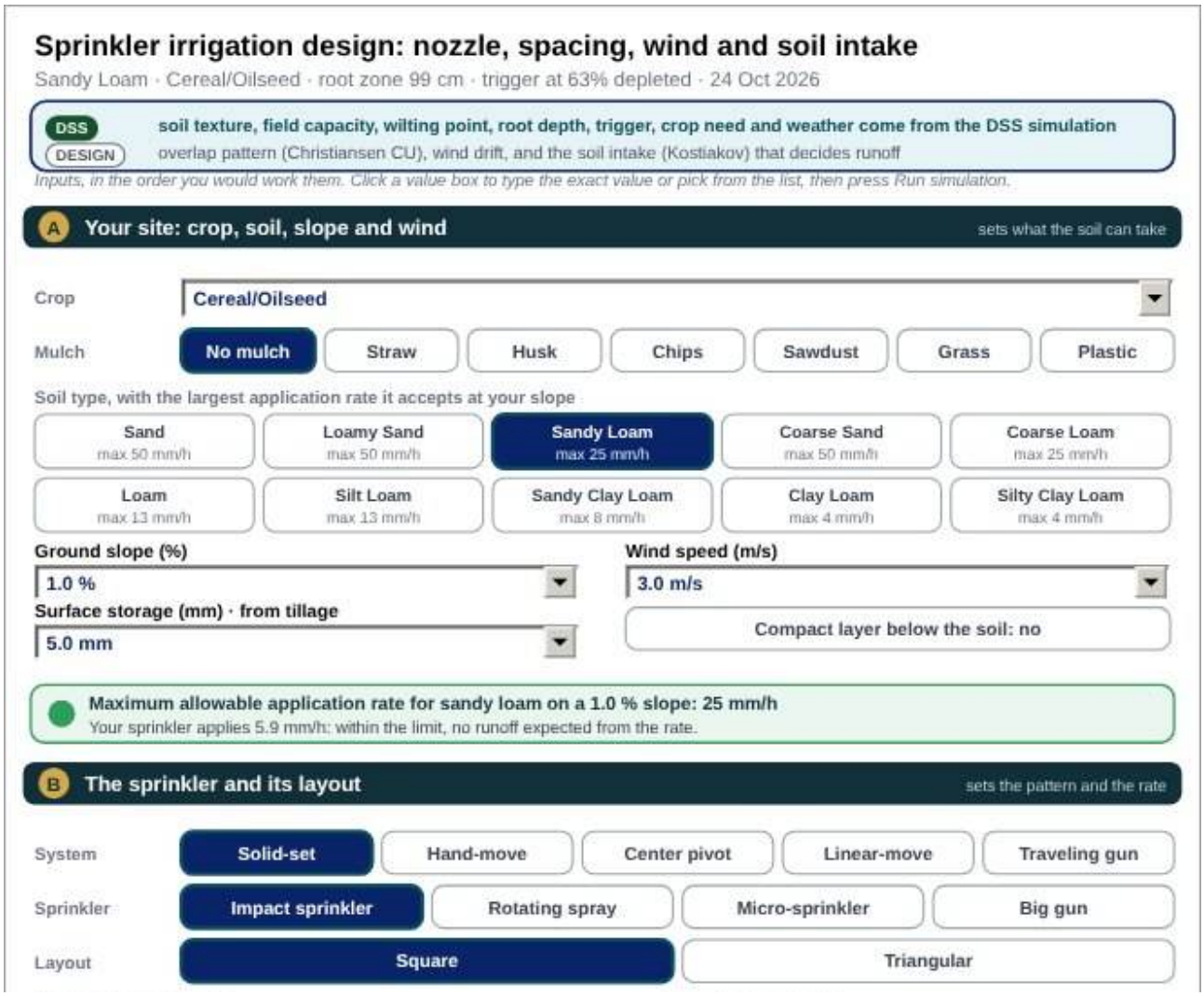


Figure 3.12: Sprinkler design and pipes, Sandy Loam, part 1 of 12.

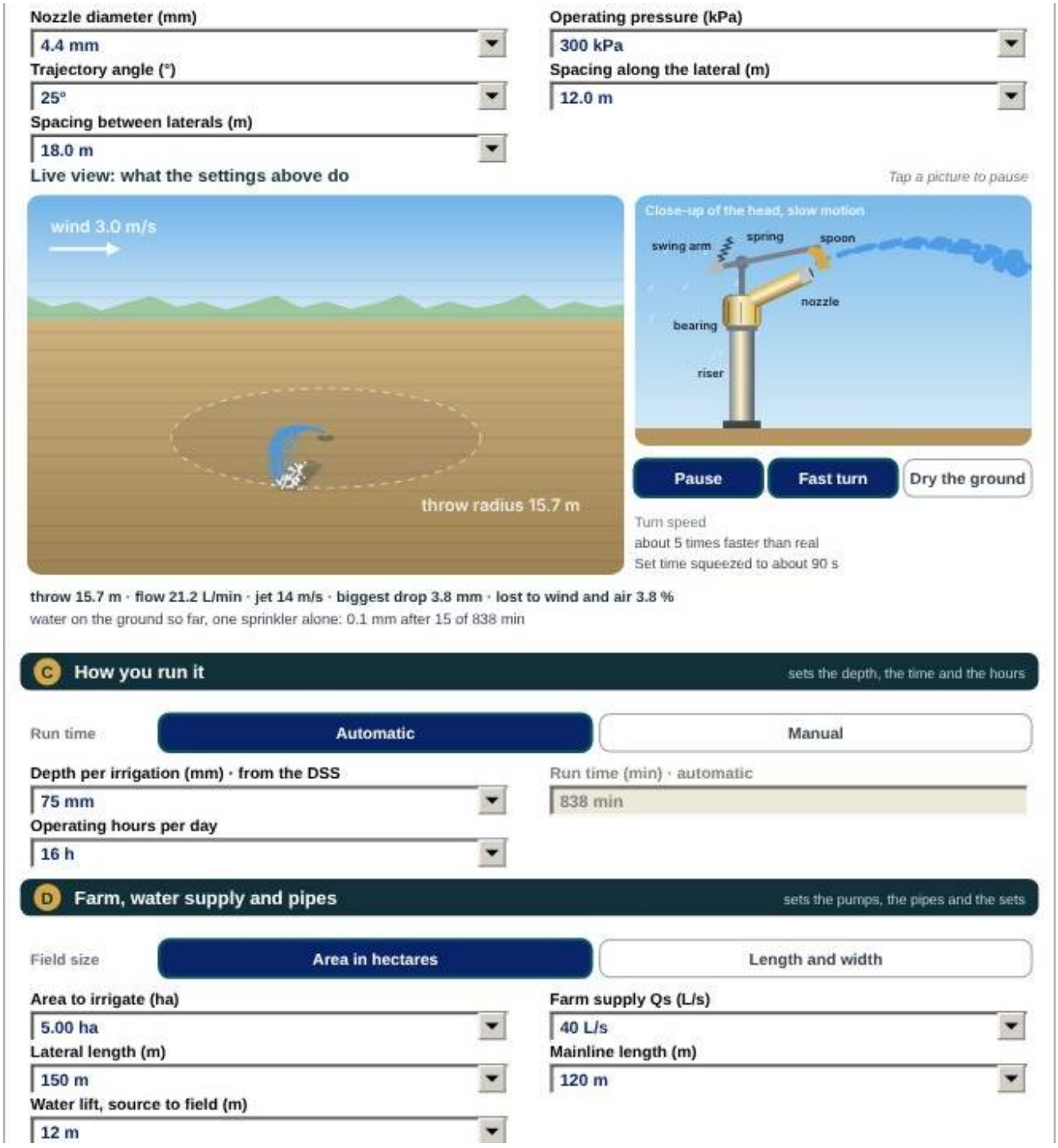


Figure 3.13: Sprinkler design and pipes, Sandy Loam, part 2 of 12.

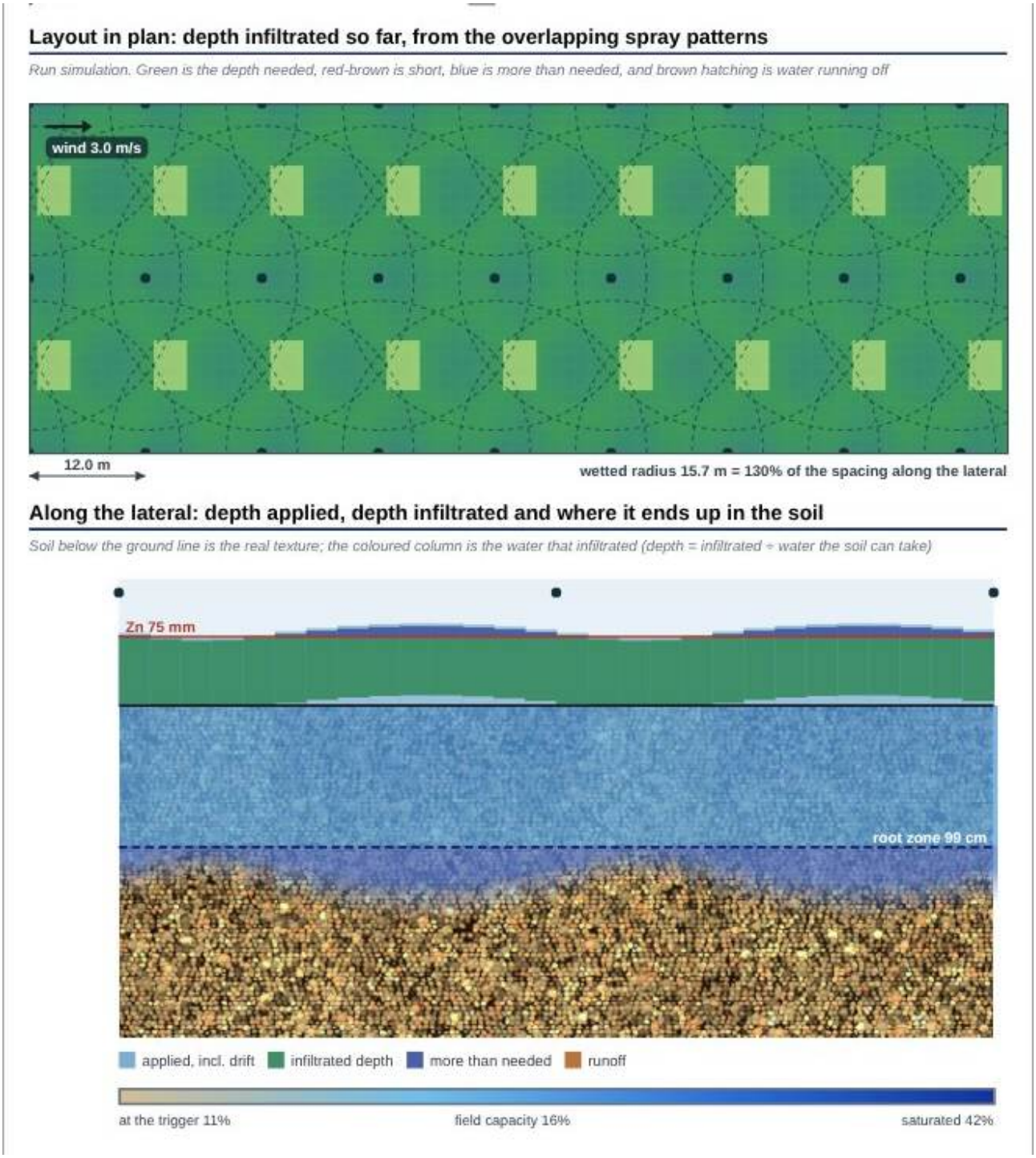


Figure 3.14: Sprinkler design and pipes, Sandy Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

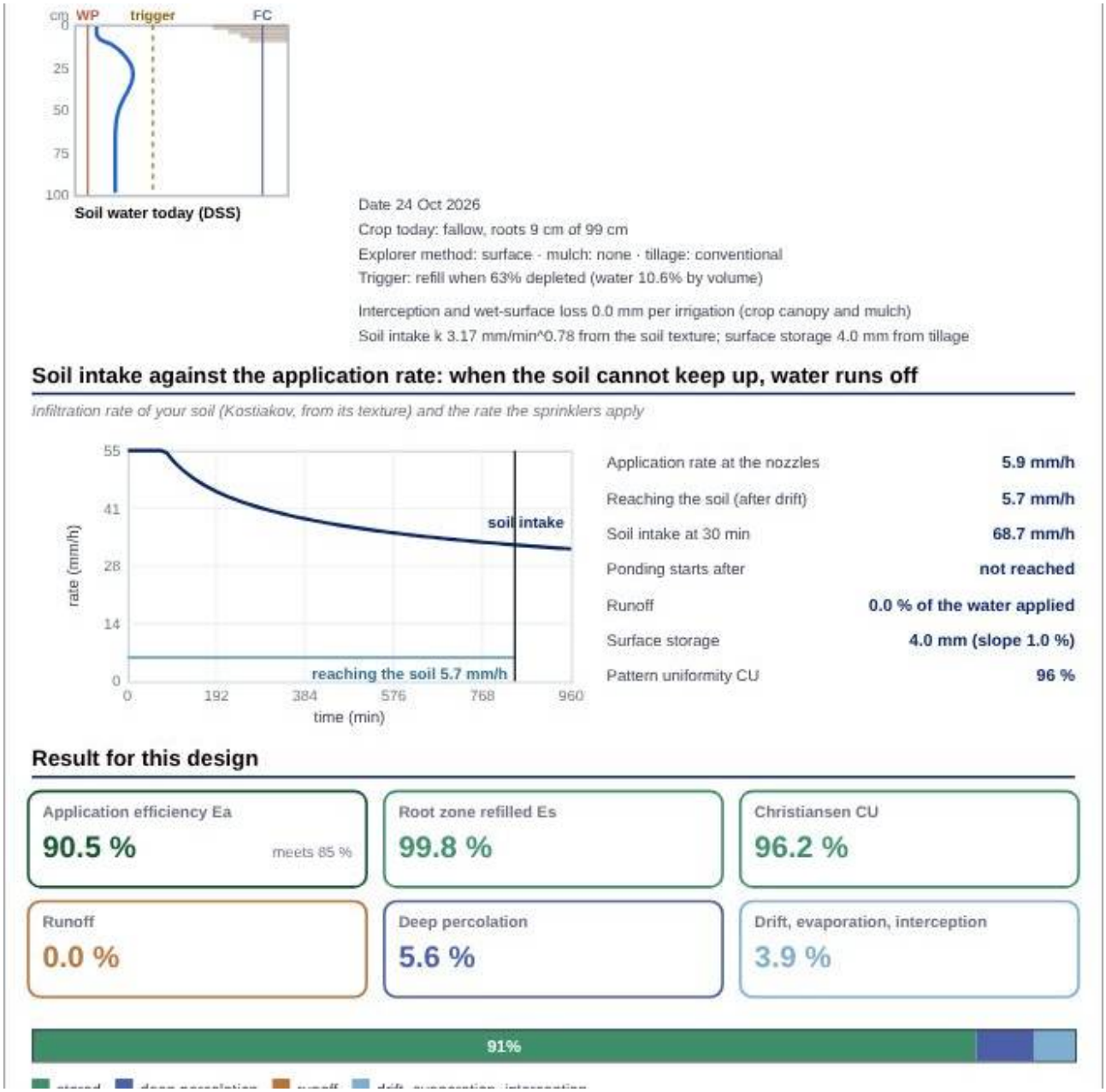


Figure 3.15: Sprinkler design and pipes, Sandy Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

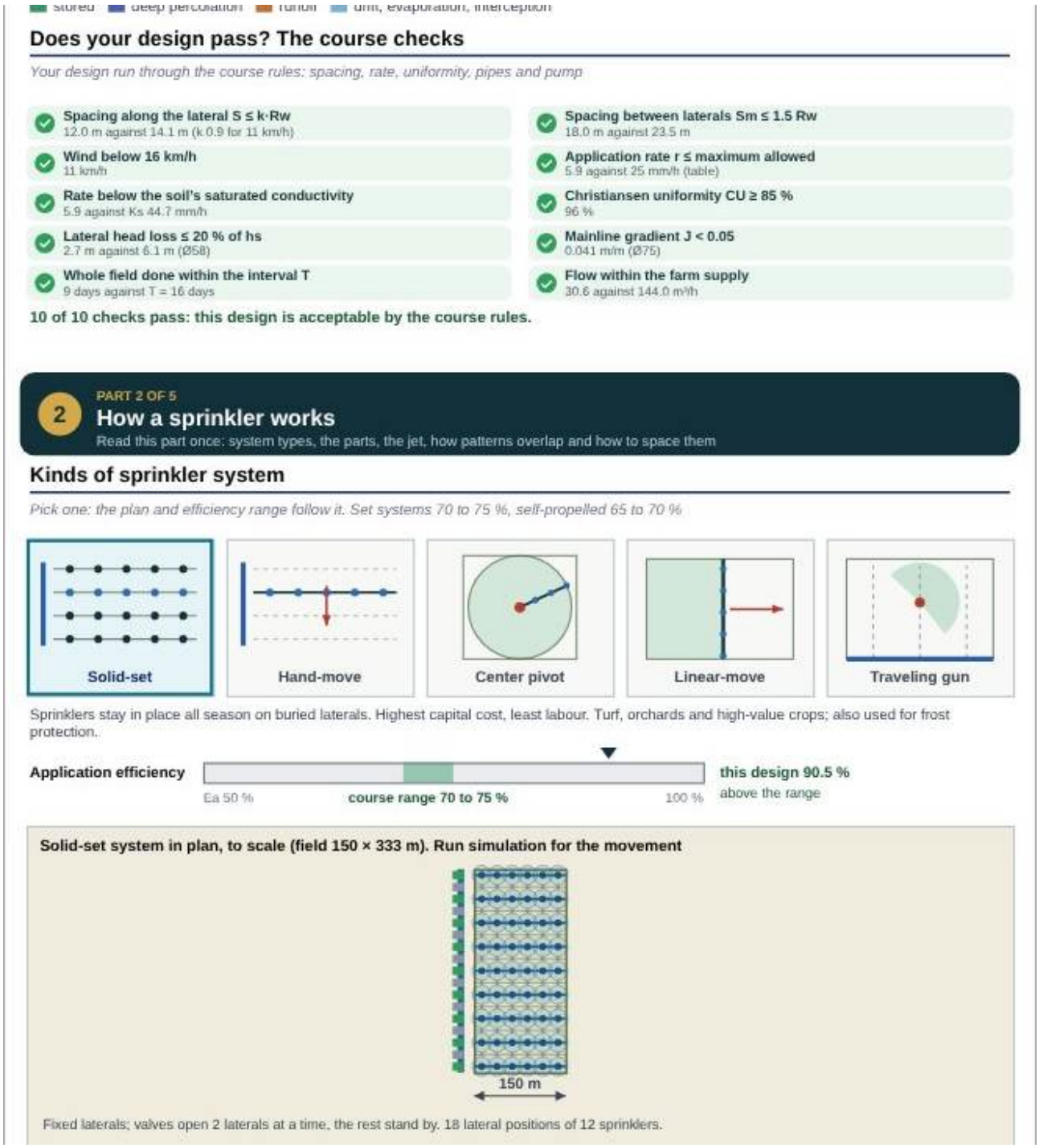


Figure 3.16: Sprinkler design and pipes, Sandy Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

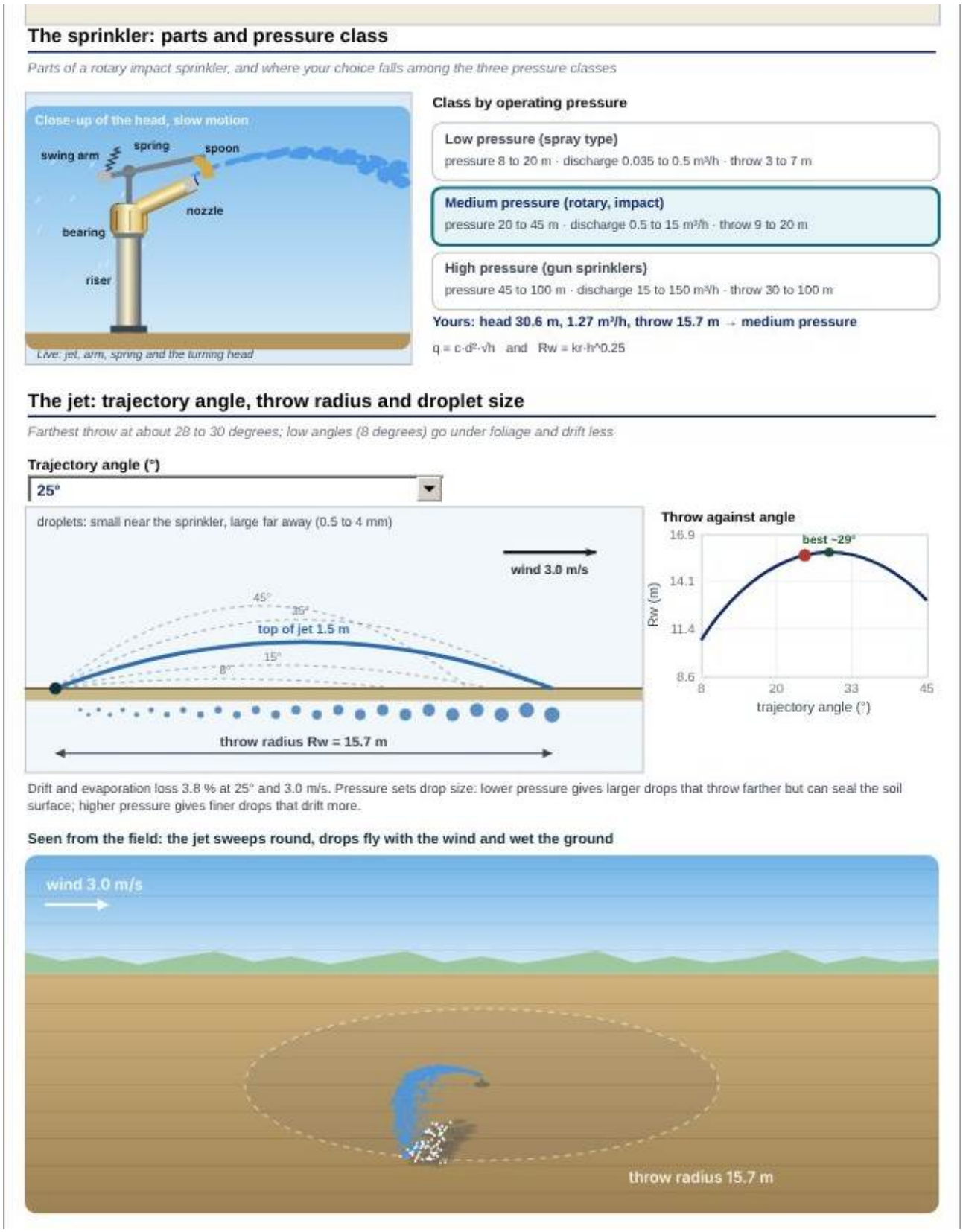


Figure 3.17: Sprinkler design and pipes, Sandy Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

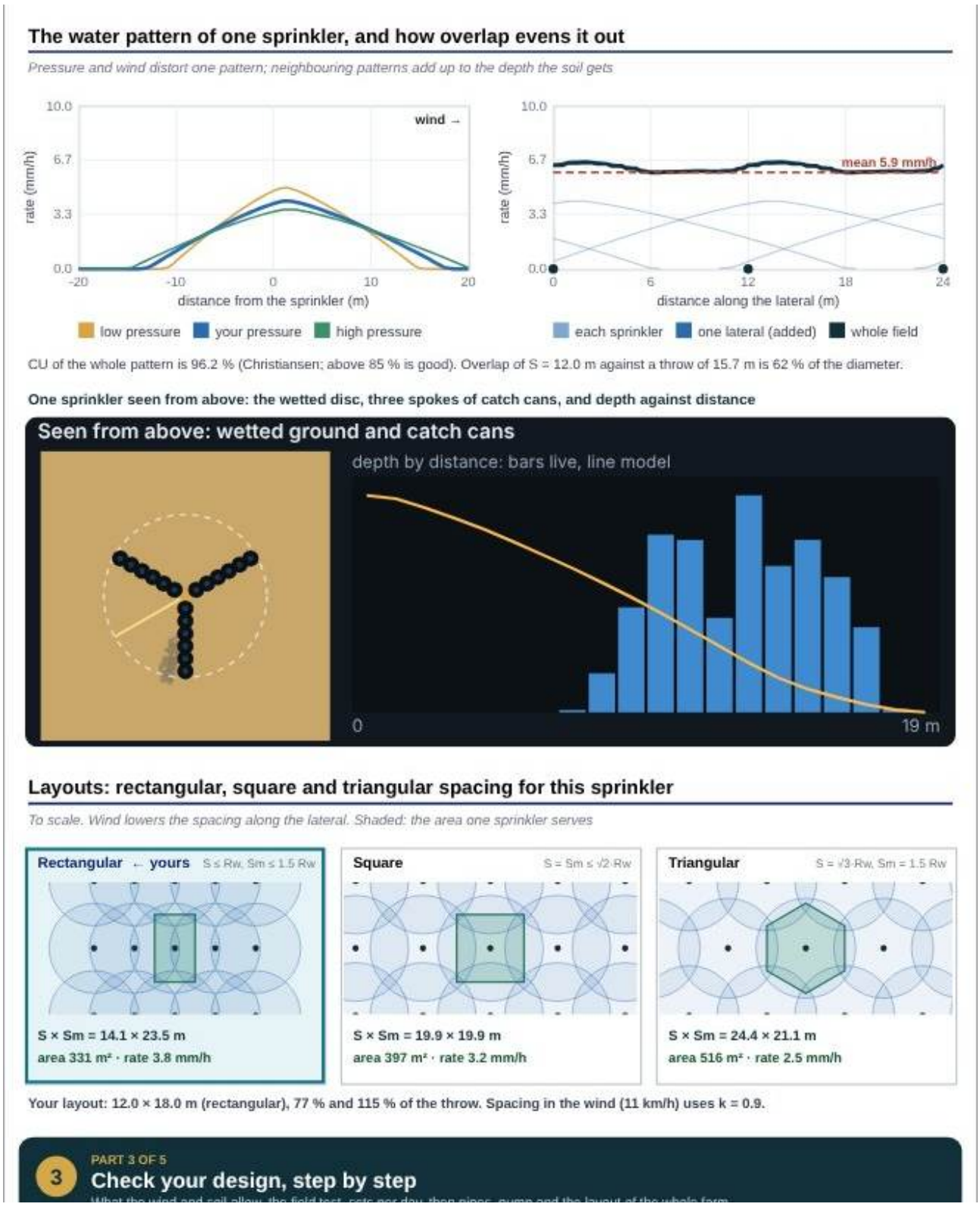


Figure 3.18: Sprinkler design and pipes, Sandy Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

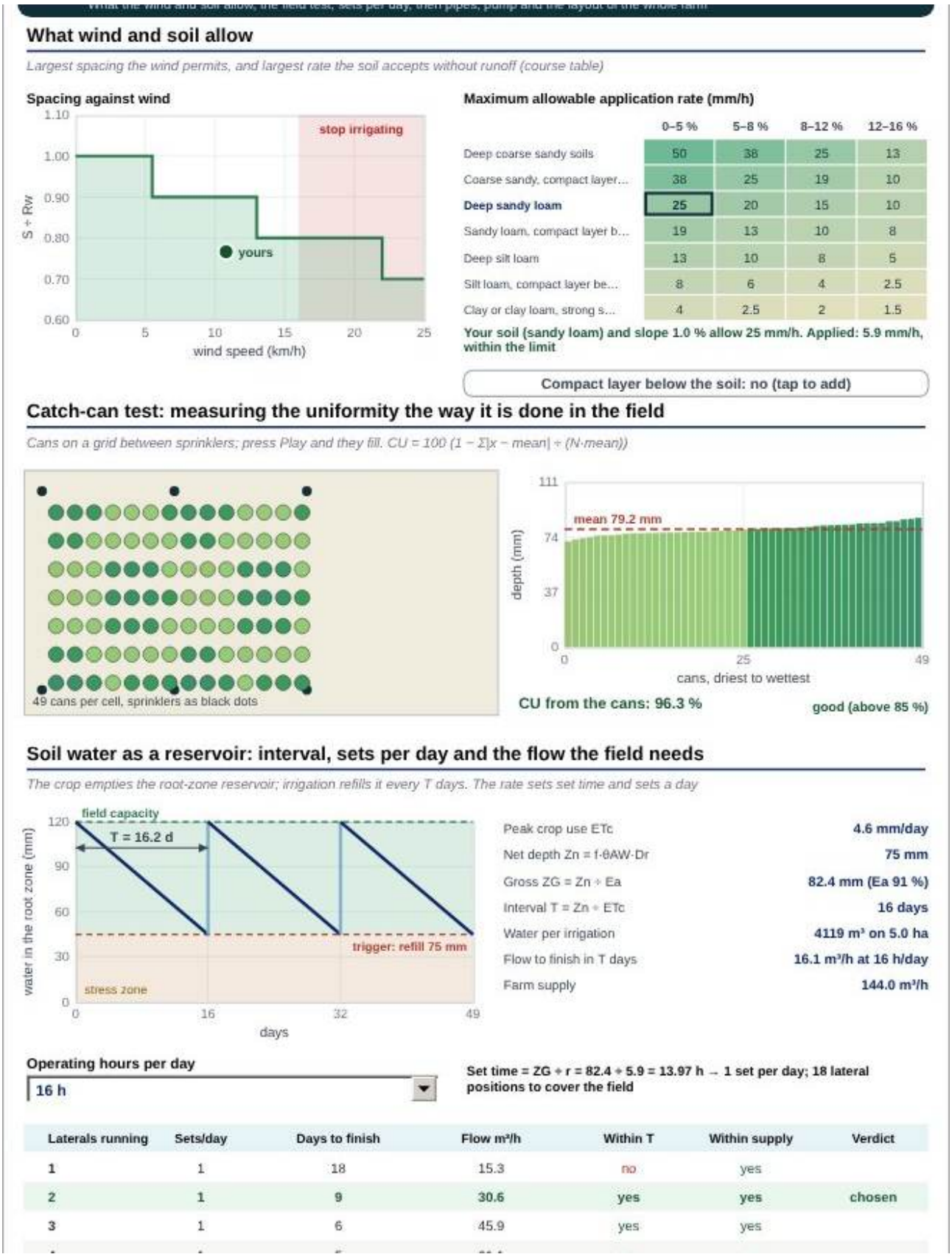


Figure 3.19: Sprinkler design and pipes, Sandy Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

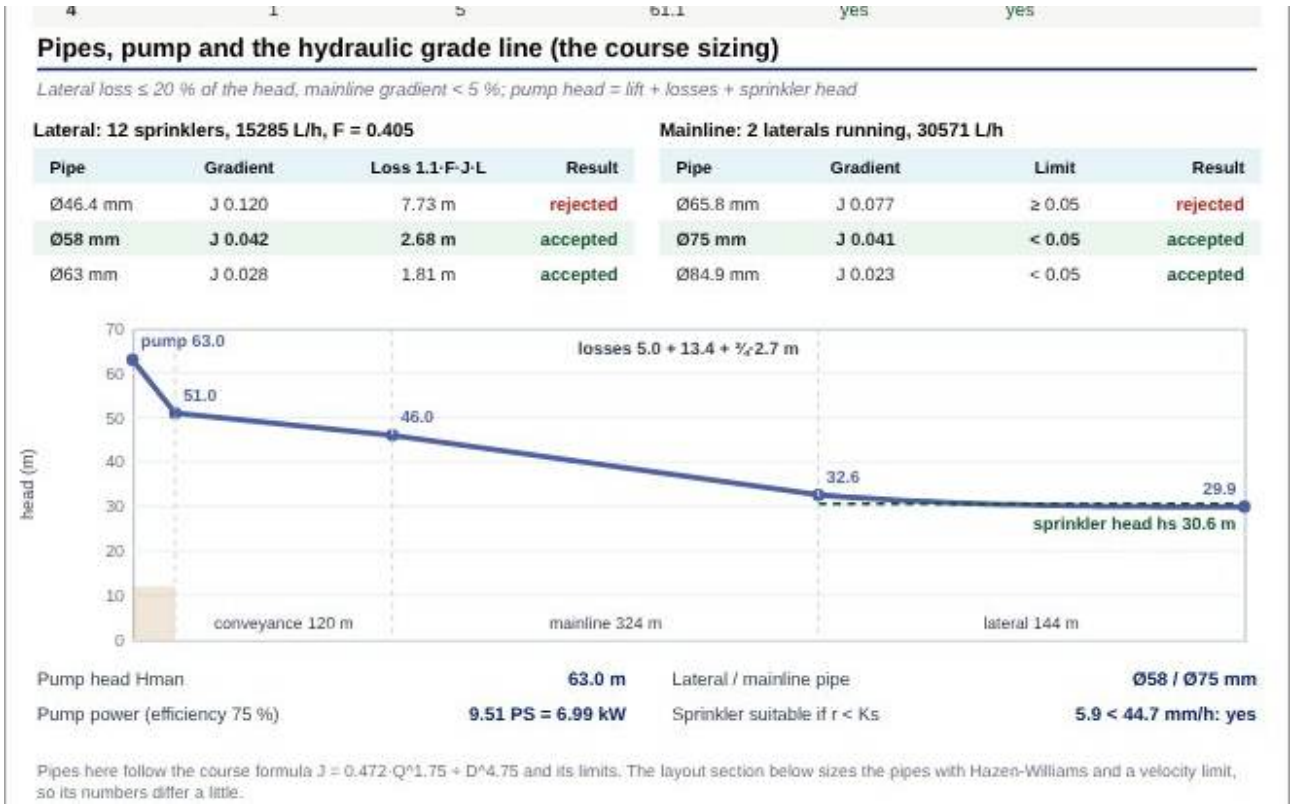
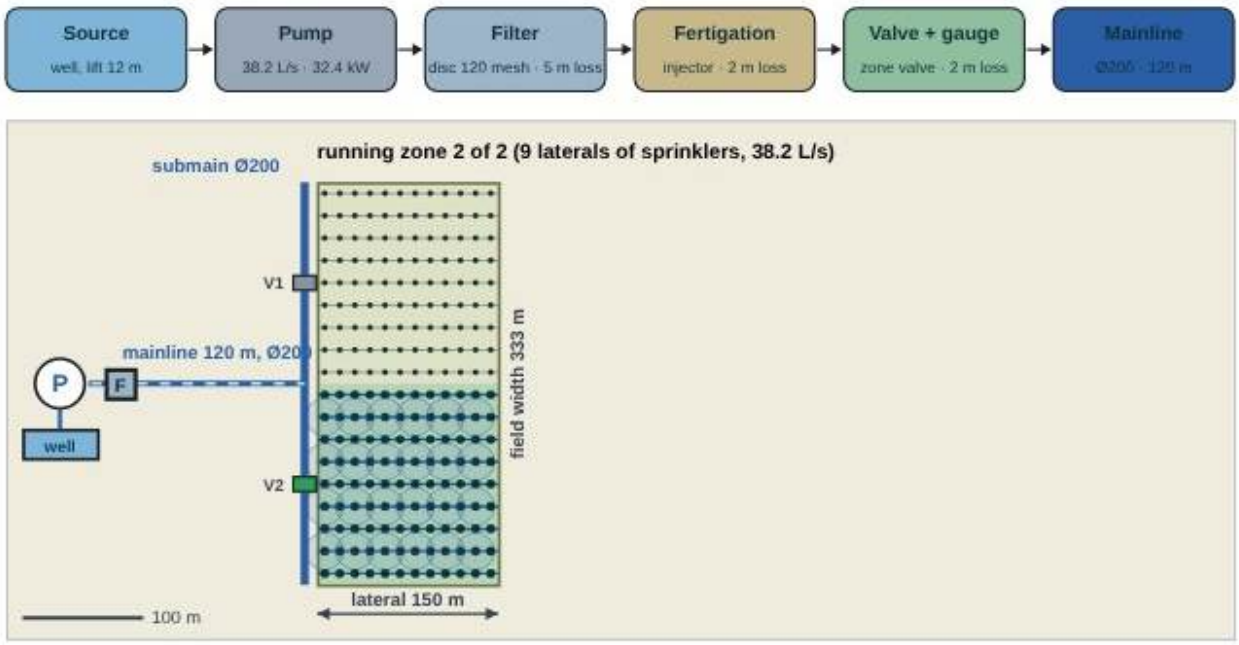
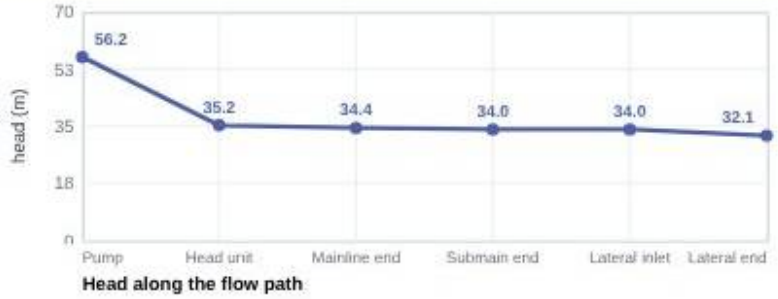


Figure 3.20: Sprinkler design and pipes, Sandy Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

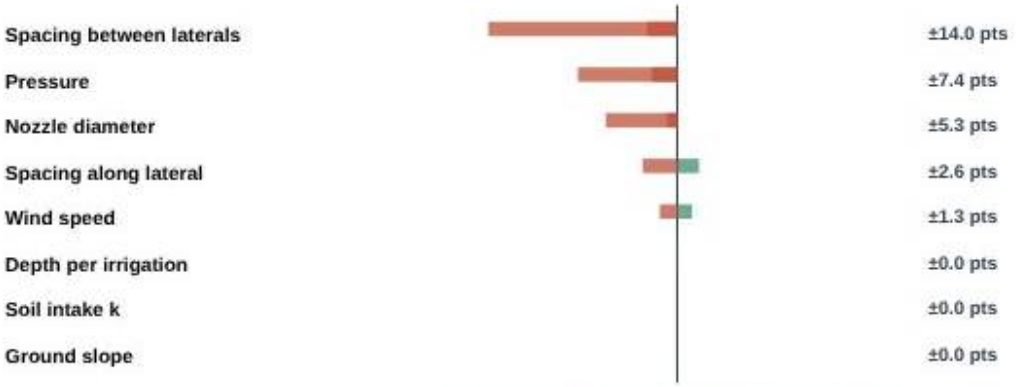
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	838	90.5	99.8	0.0	meets target

Figure 3.21: Sprinkler design and pipes, Sandy Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CU 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

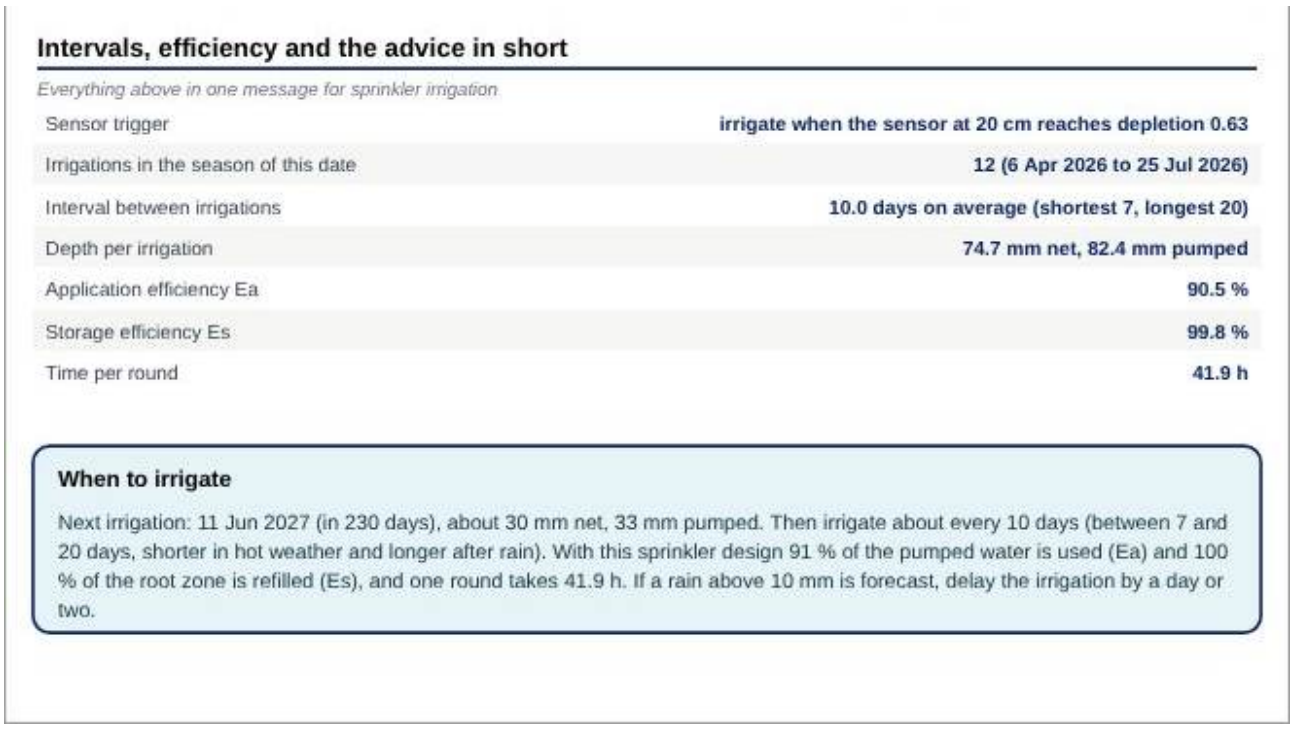
Net irrigation need in a season (DSS)	**358 mm → 5 irrigations of 75 mm**
Gross water per irrigation	**82.4 mm (Ea 91 %)**
Gross water per season	**395 mm = 3945 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 838 min = 41.9 h per round**
Pumping per season	**201 hours · 6511 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**11 Jun 2027**	230 days	30	33	41.9 h	9 days
2	**20 Jun 2027**	239 days	30	33	41.9 h	12 days
3	**2 Jul 2027**	251 days	30	33	41.9 h	9 days
4	**11 Jul 2027**	260 days	30	33	41.9 h	10 days
5	**21 Jul 2027**	270 days	30	33	41.9 h	259 days
6	**5 Apr 2028**	529 days	30	33	41.9 h	11 days
7	**16 Apr 2028**	540 days	30	33	41.9 h	10 days
8	**26 Apr 2028**	550 days	30	33	41.9 h	-

Figure 3.22: Sprinkler design and pipes, Sandy Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.63**
Irrigations in the season of this date	**12 (6 Apr 2026 to 25 Jul 2026)**
Interval between irrigations	**10.0 days on average (shortest 7, longest 20)**
Depth per irrigation	**74.7 mm net, 82.4 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**41.9 h**

When to irrigate

Next irrigation: 11 Jun 2027 (in 230 days), about 30 mm net, 33 mm pumped. Then irrigate about every 10 days (between 7 and 20 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 41.9 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 3.23: Sprinkler design and pipes, Sandy Loam, part 12 of 12: Intervals, efficiency and the advice in short.

3.4 Virtual farm of the surface design (Sandy Loam)

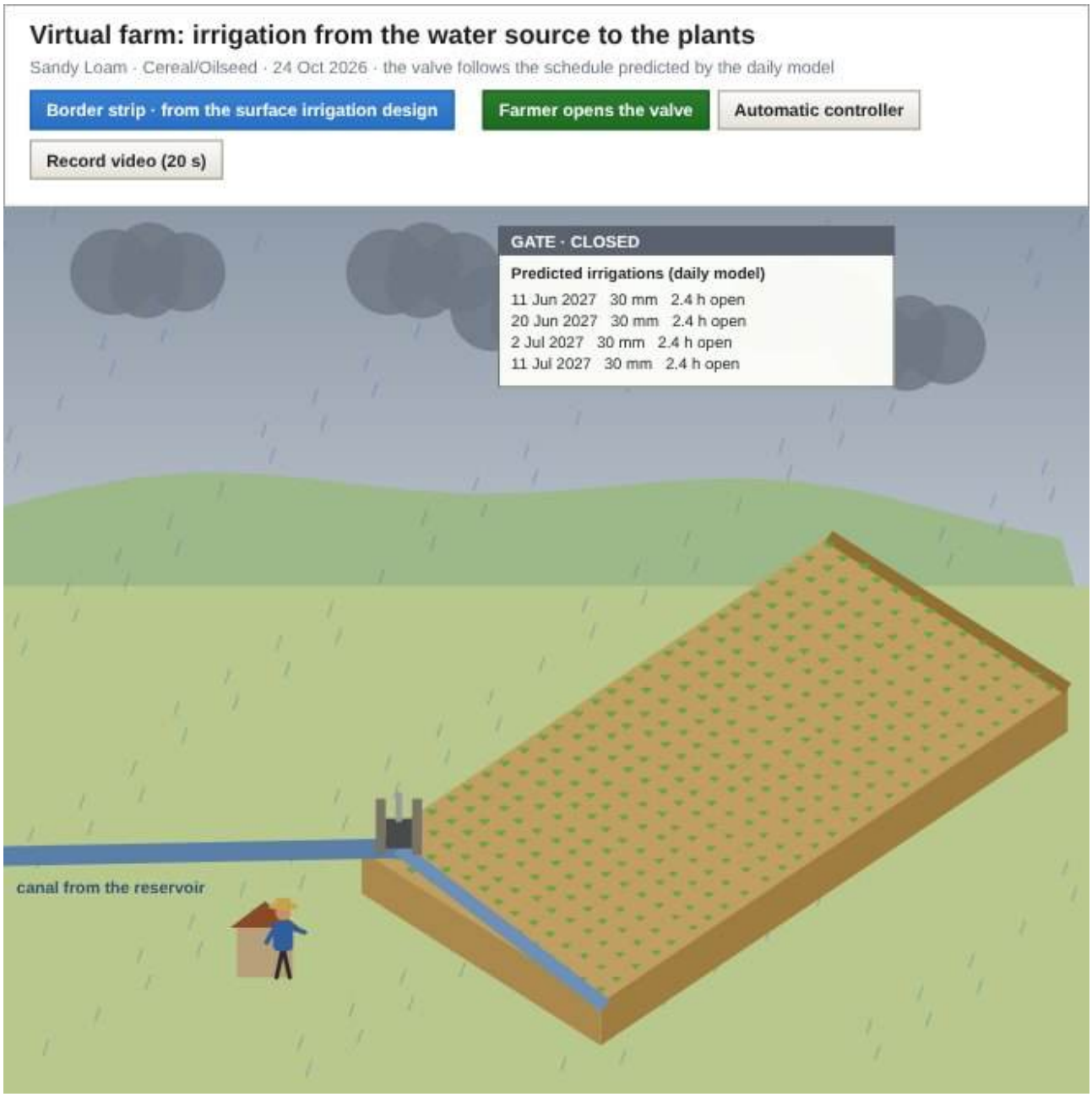


Figure 3.24: Virtual farm, surface design, Sandy Loam, part 1 of 3.

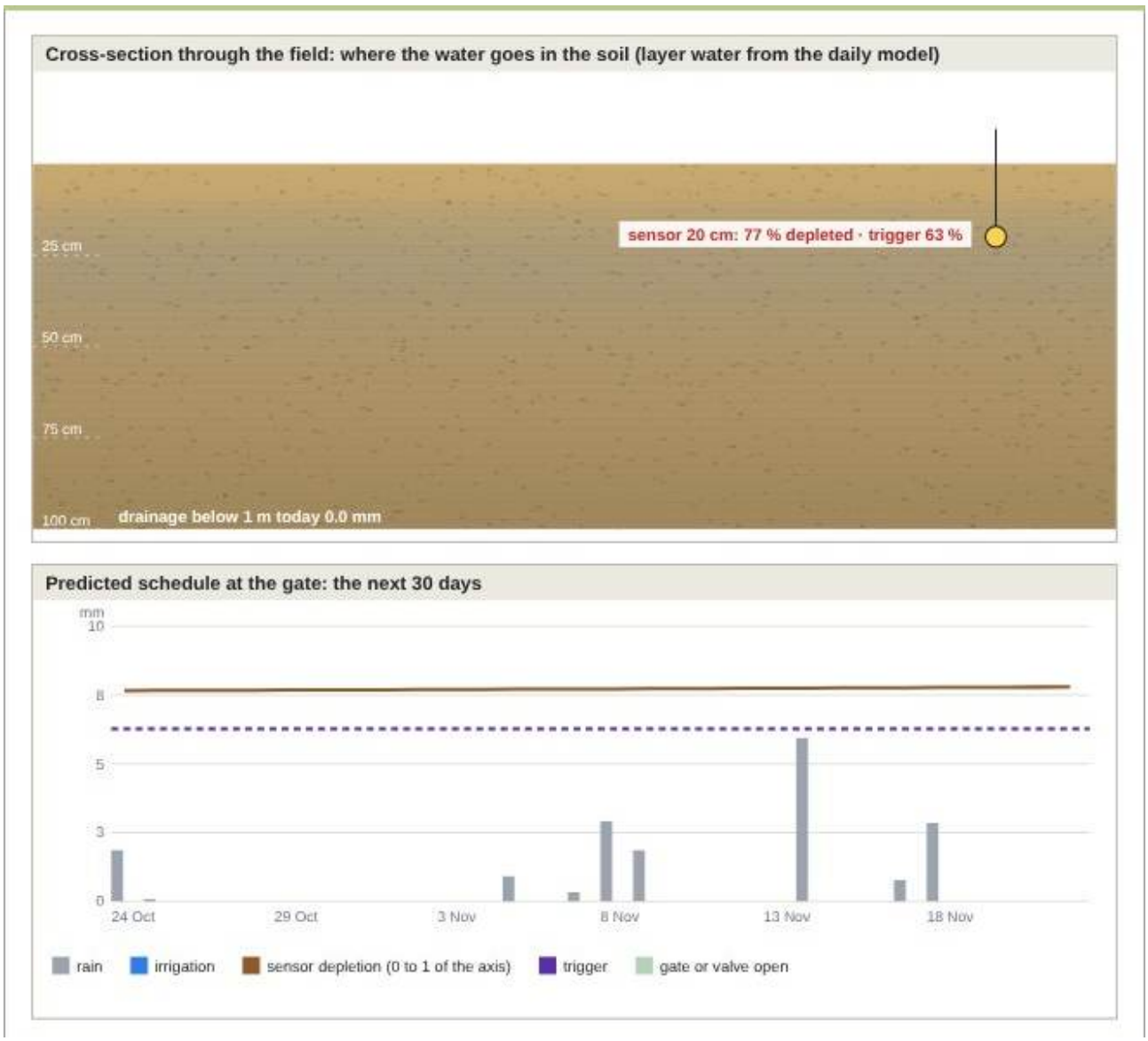


Figure 3.25: Virtual farm, surface design, Sandy Loam, part 2 of 3.

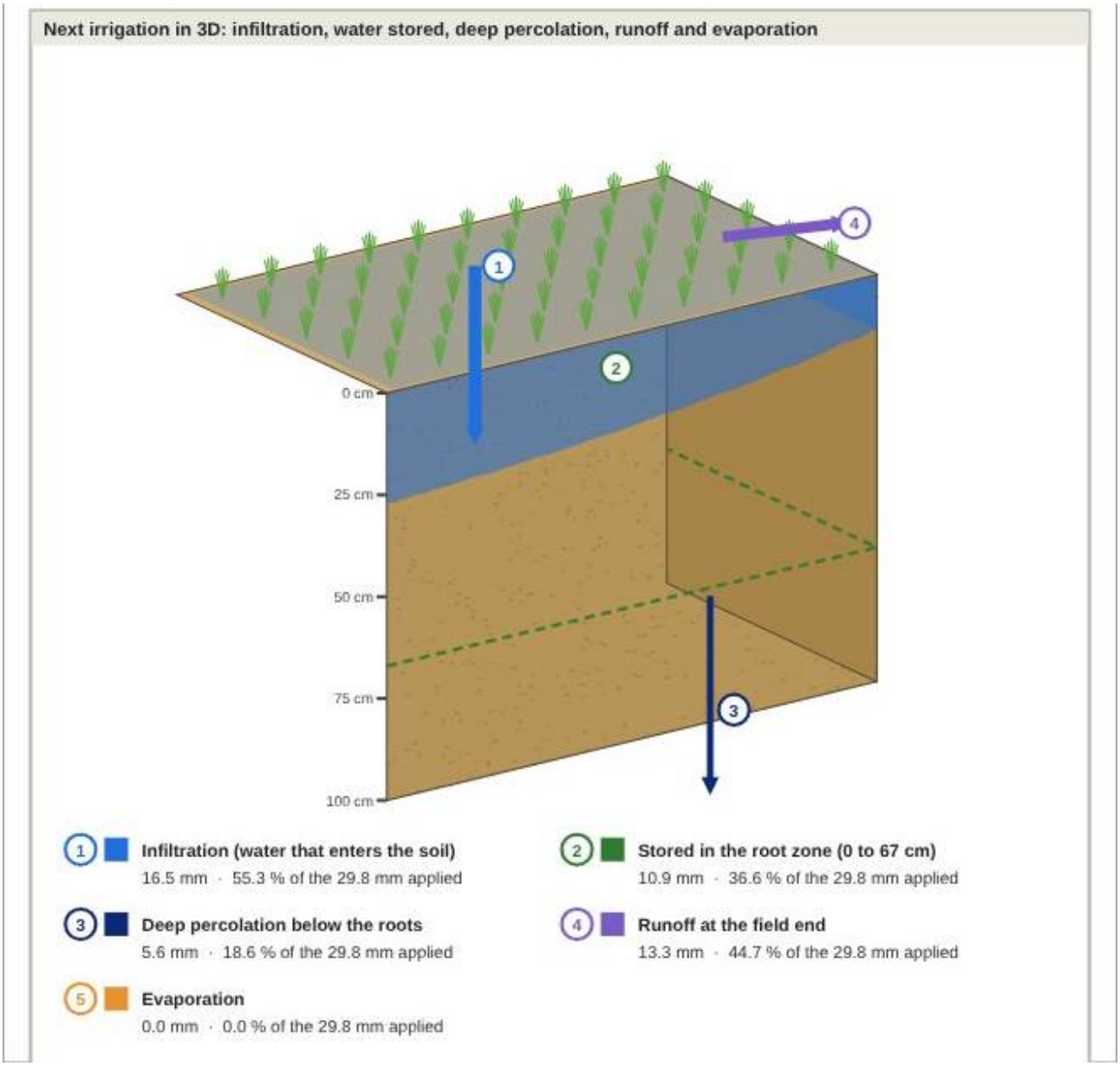


Figure 3.26: Virtual farm, surface design, Sandy Loam, part 3 of 3.

3.5 Virtual farm of the drip design (Sandy Loam)

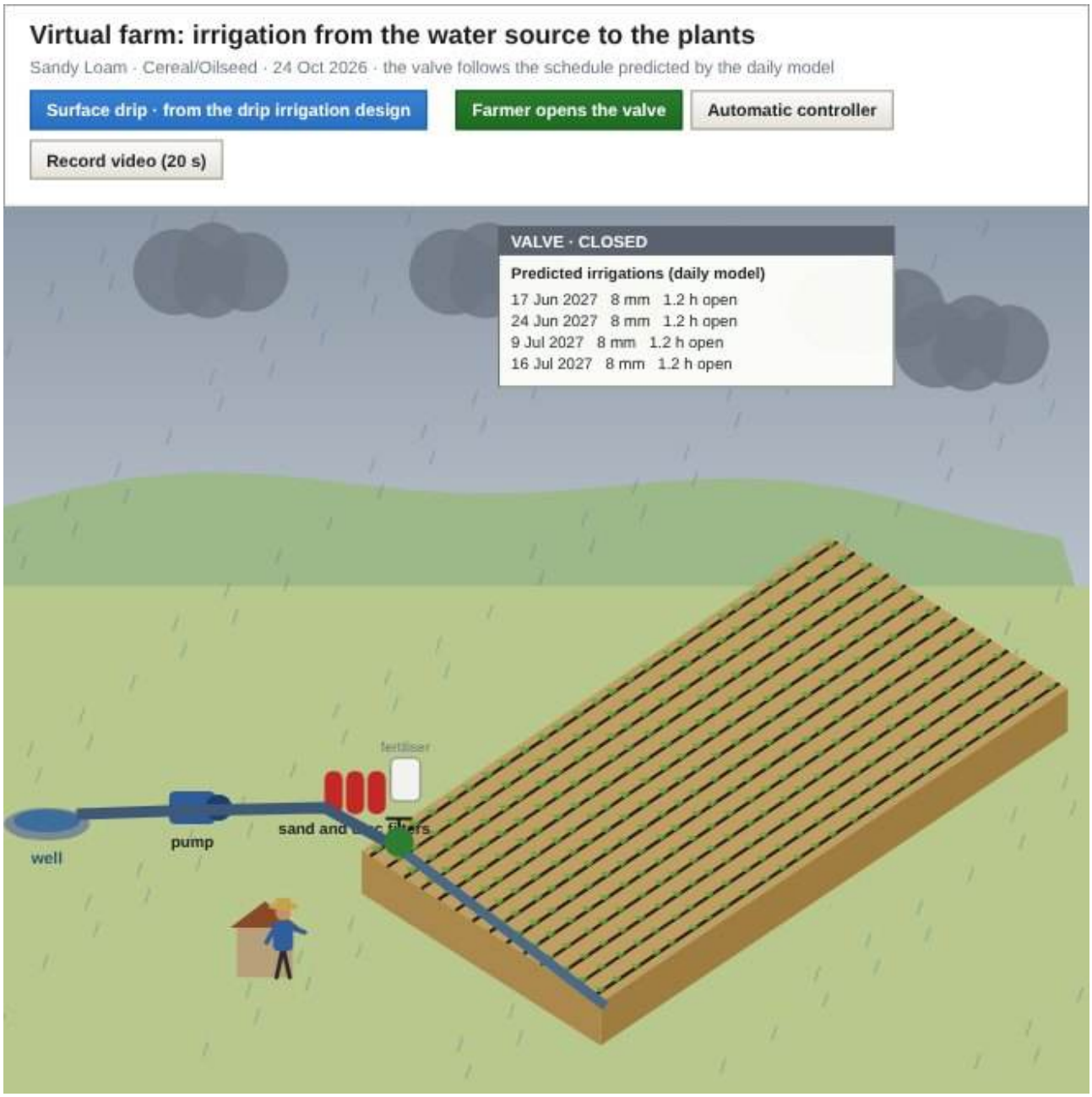


Figure 3.27: Virtual farm, drip design, Sandy Loam, part 1 of 3.

Figure 3.28: Virtual farm, drip design, Sandy Loam, part 2 of 3.

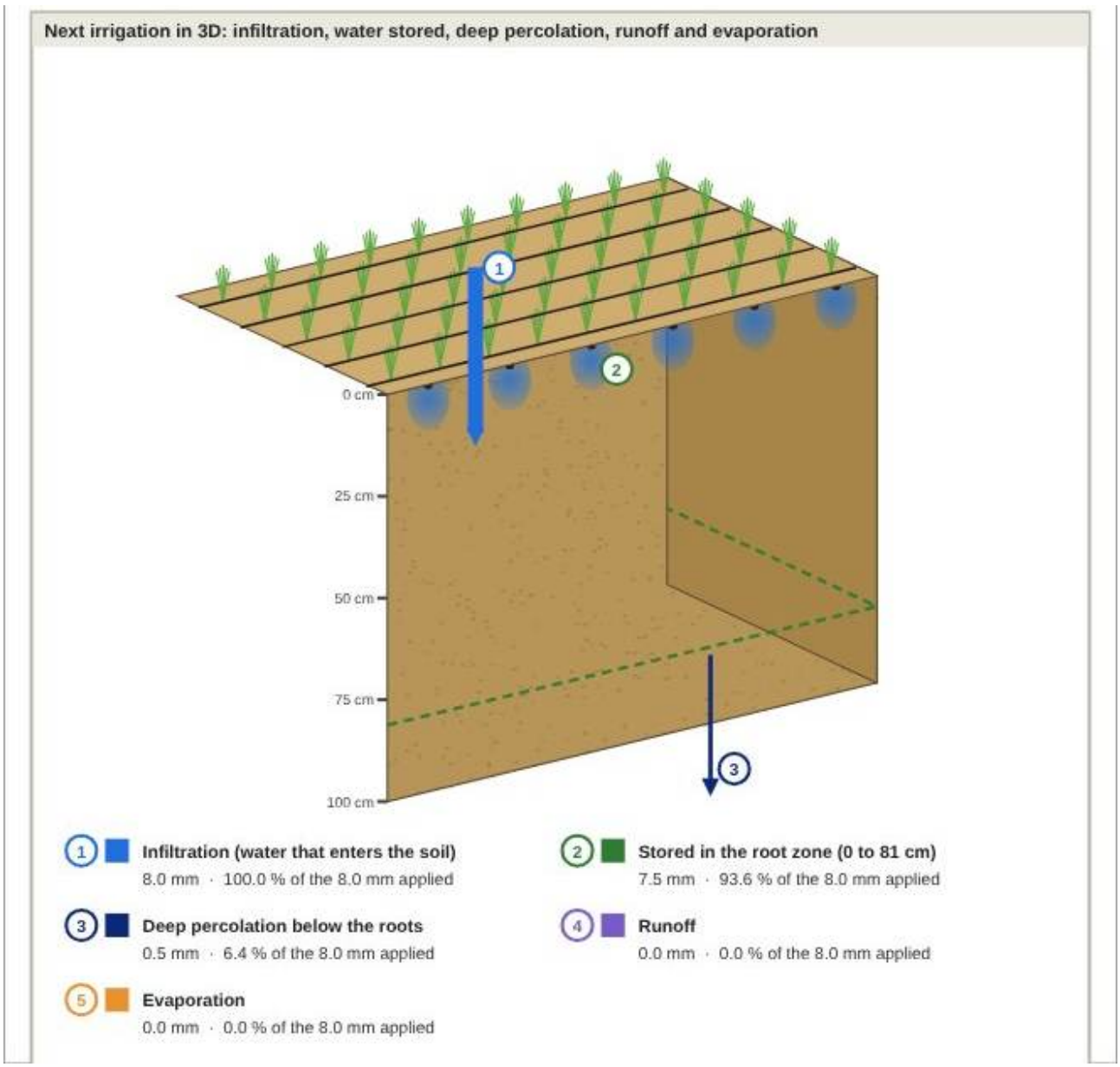


Figure 3.29: Virtual farm, drip design, Sandy Loam, part 3 of 3.

3.6 Virtual farm of the sprinkler design (Sandy Loam)

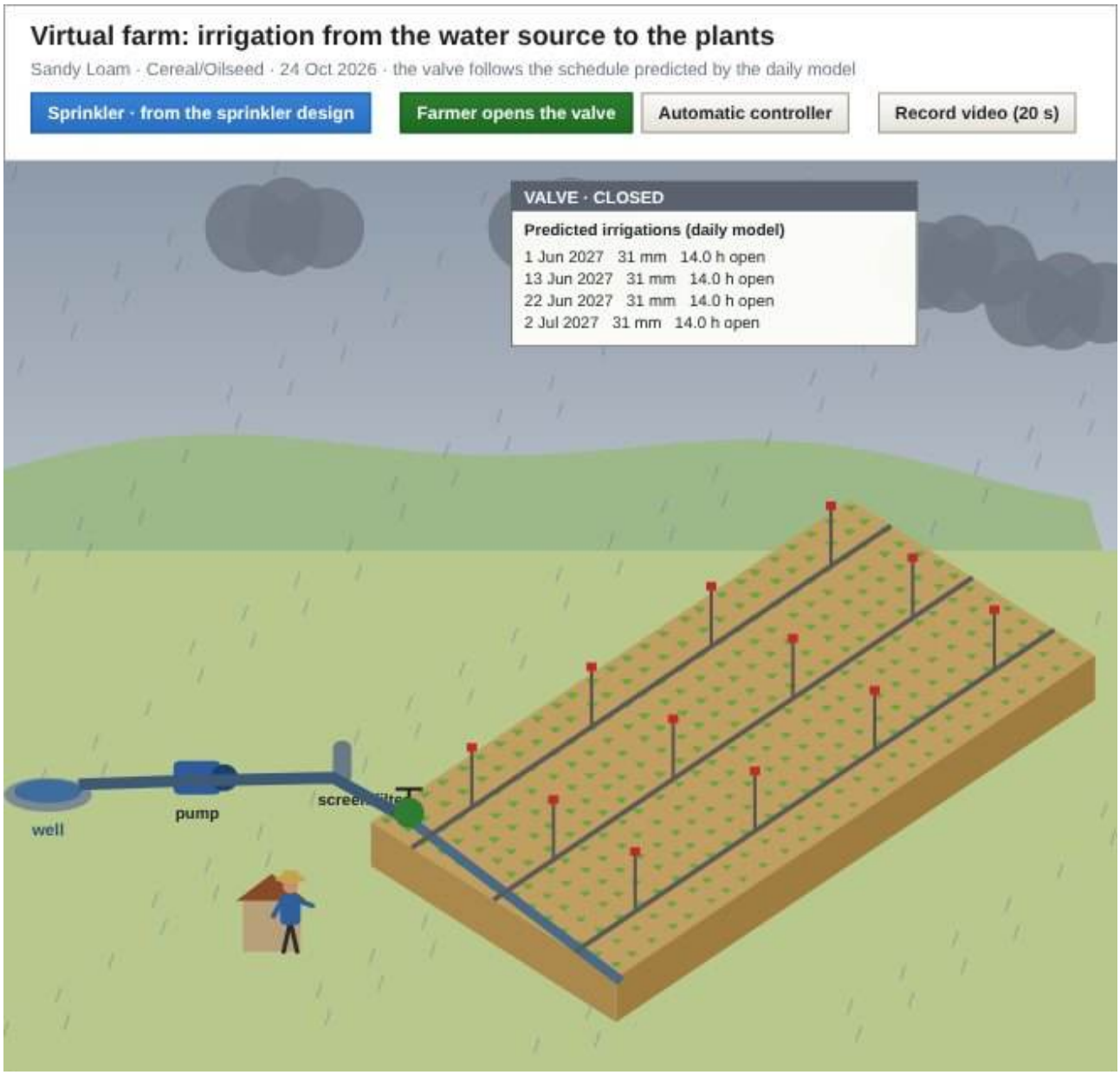


Figure 3.30: Virtual farm, sprinkler design, Sandy Loam, part 1 of 3.

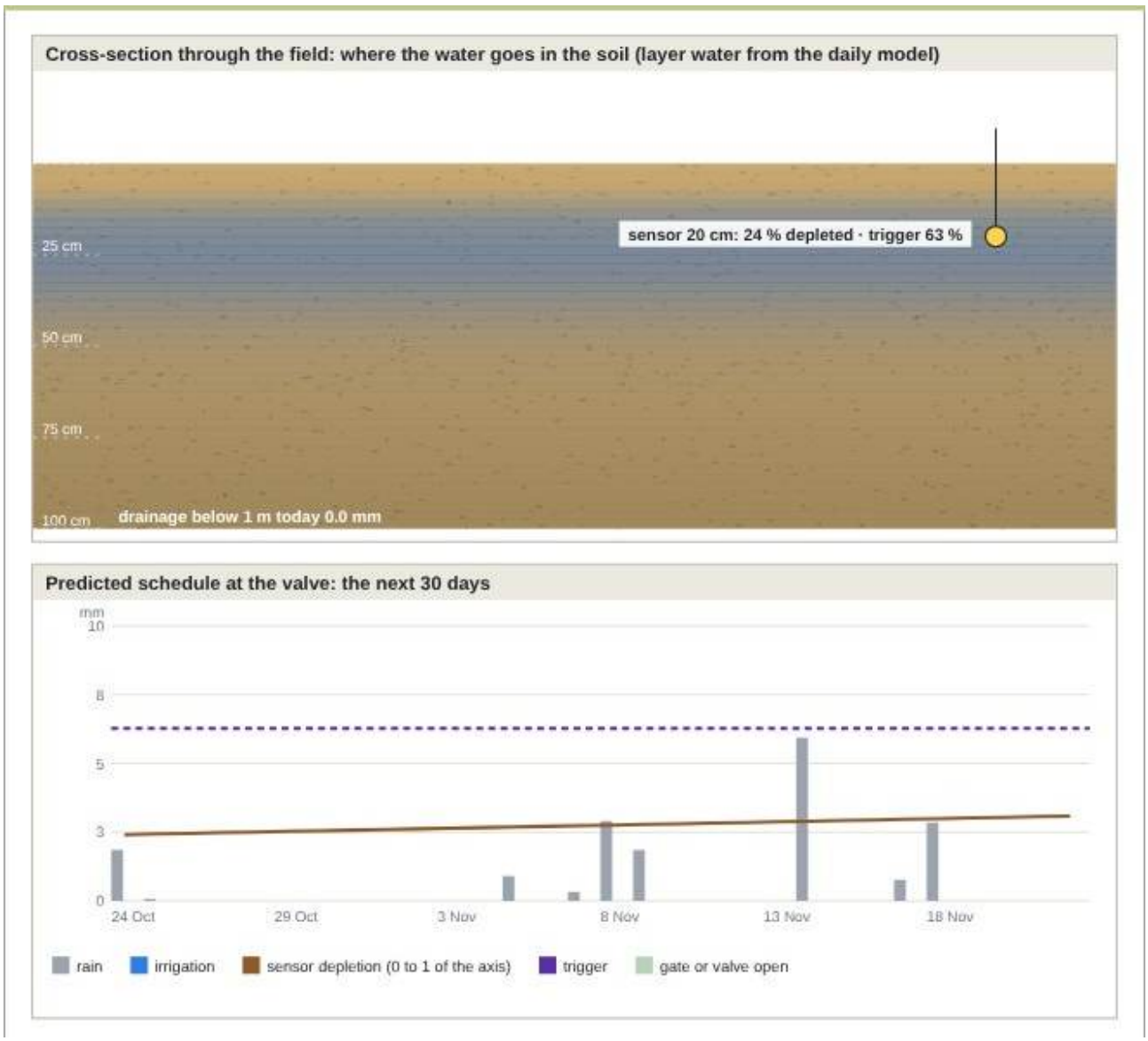


Figure 3.31: Virtual farm, sprinkler design, Sandy Loam, part 2 of 3.

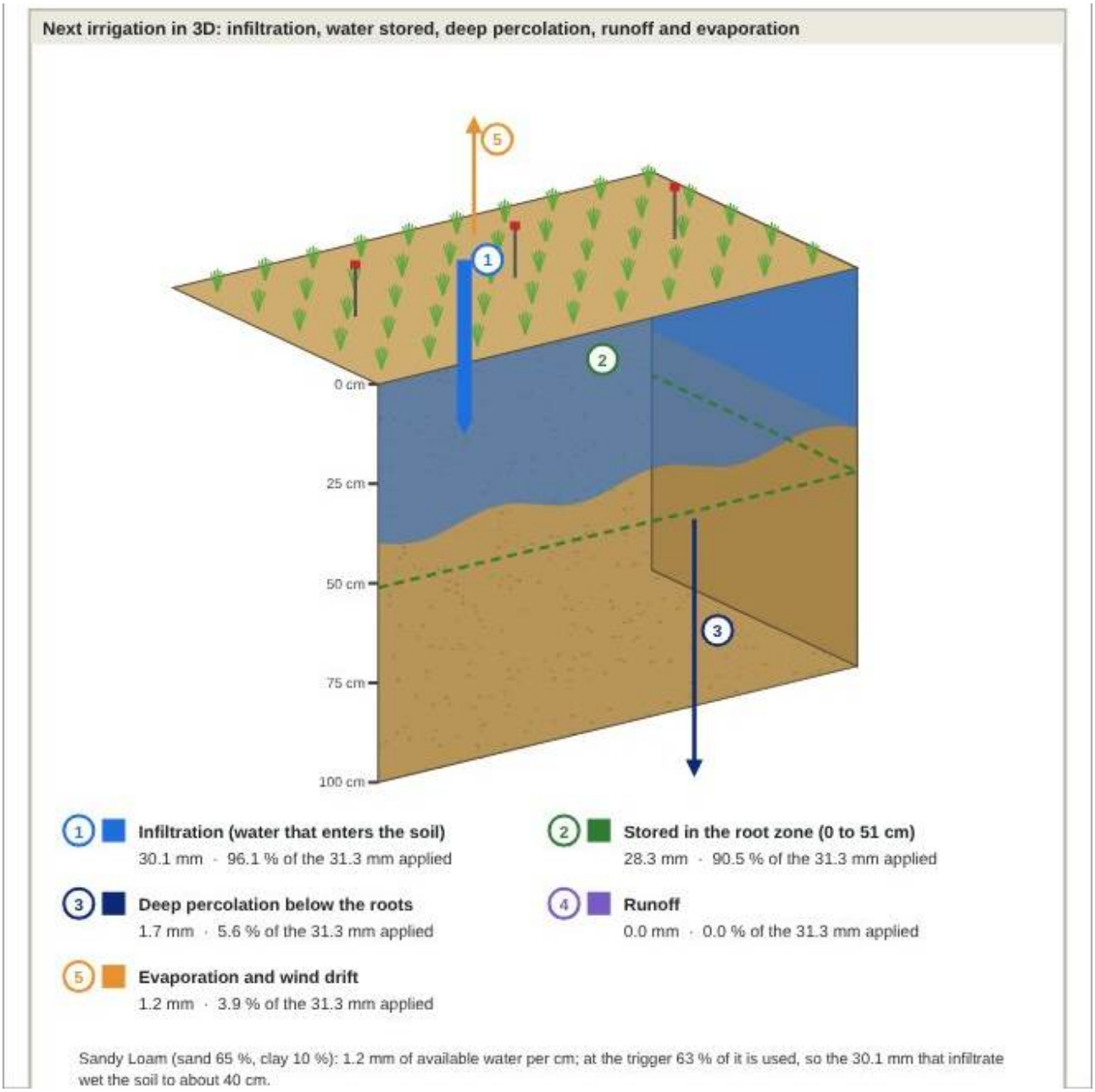


Figure 3.32: Virtual farm, sprinkler design, Sandy Loam, part 3 of 3.

3.7 Sensor and trigger (Sandy Loam)

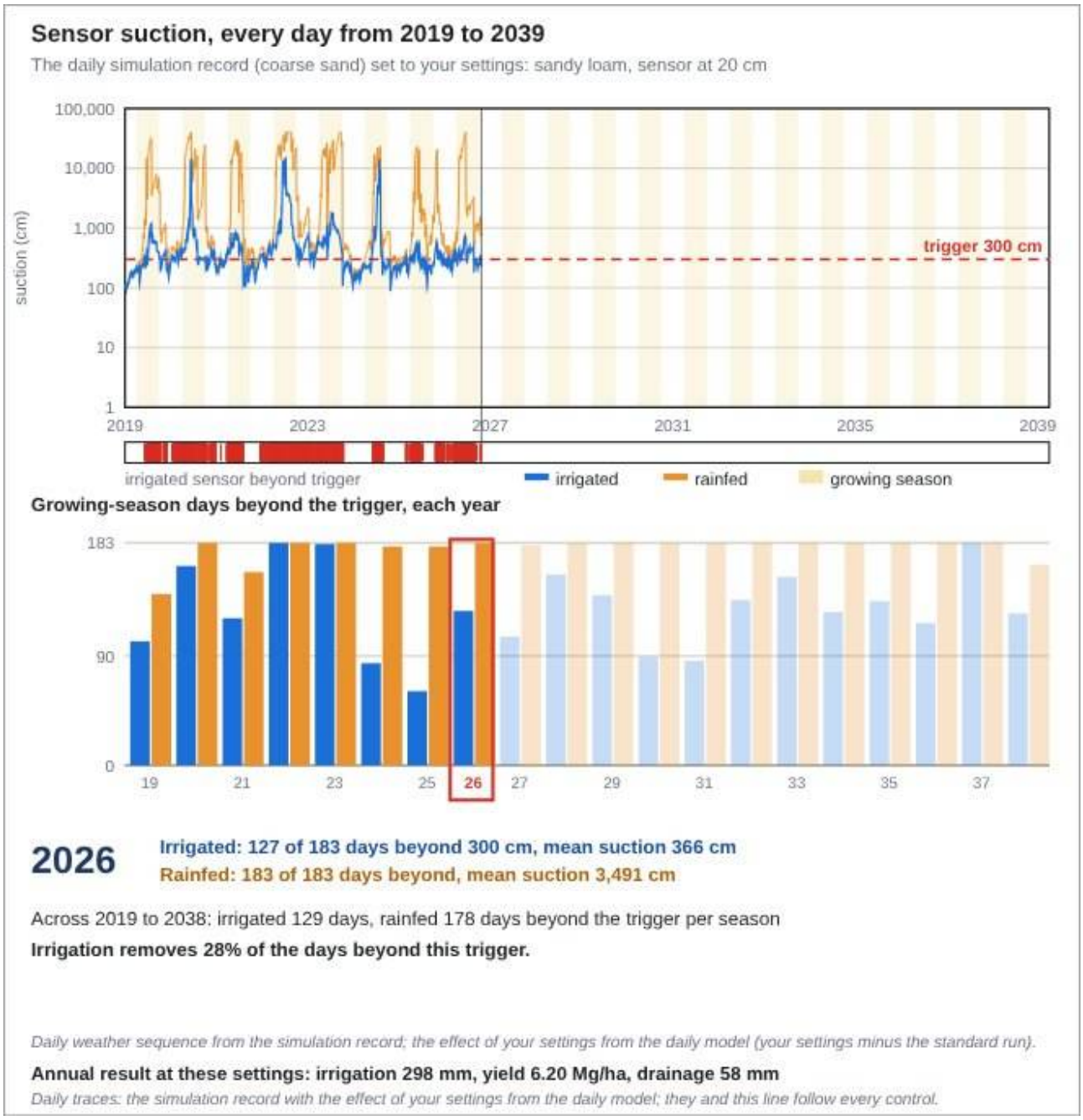


Figure 3.33: Sensor and trigger, Sandy Loam, part 1 of 1.

3.8 Trigger law (Sandy Loam)

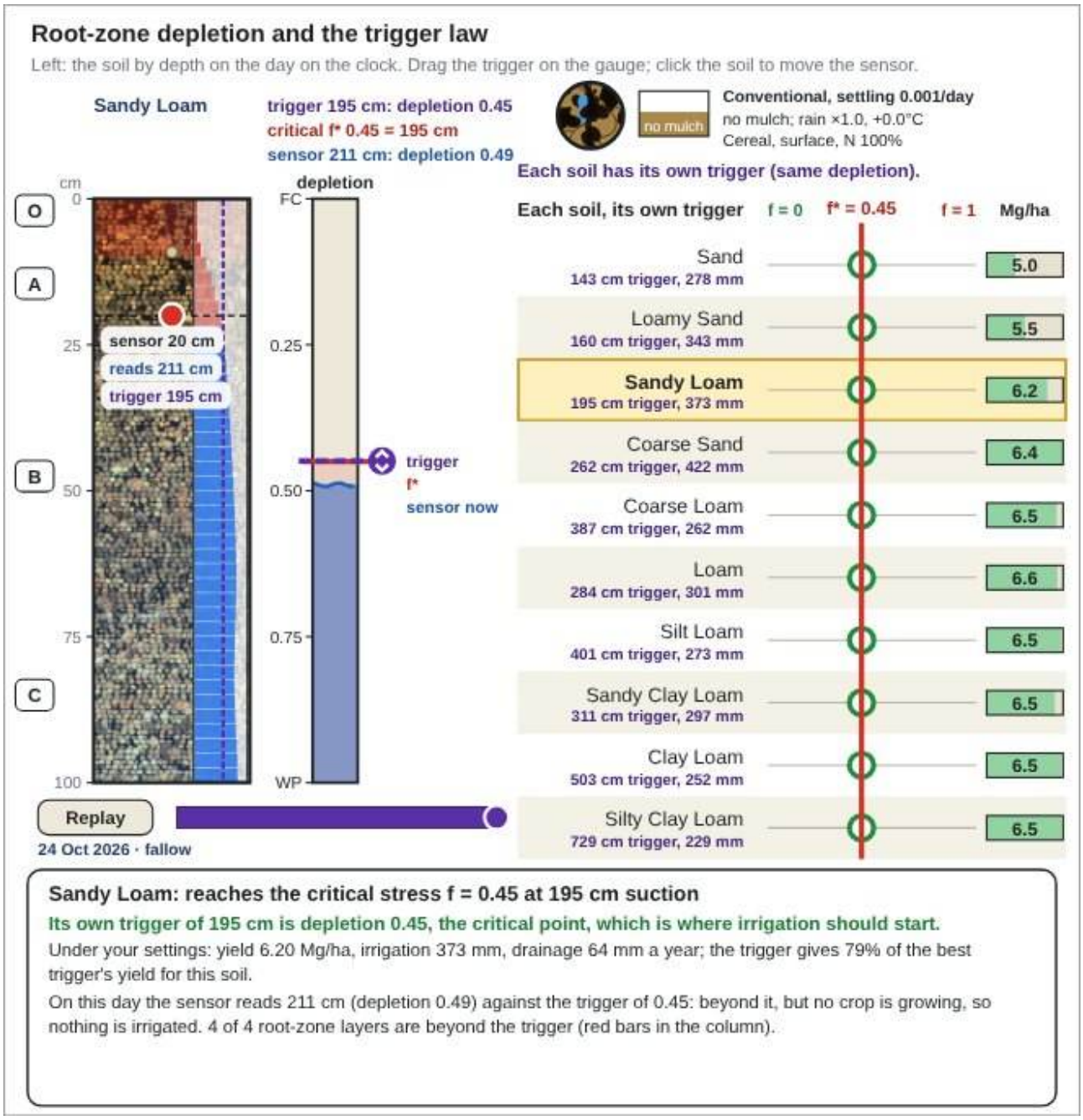


Figure 3.34: Trigger law, Sandy Loam, part 1 of 1.

3.9 Mulch against bare soil (Sandy Loam)

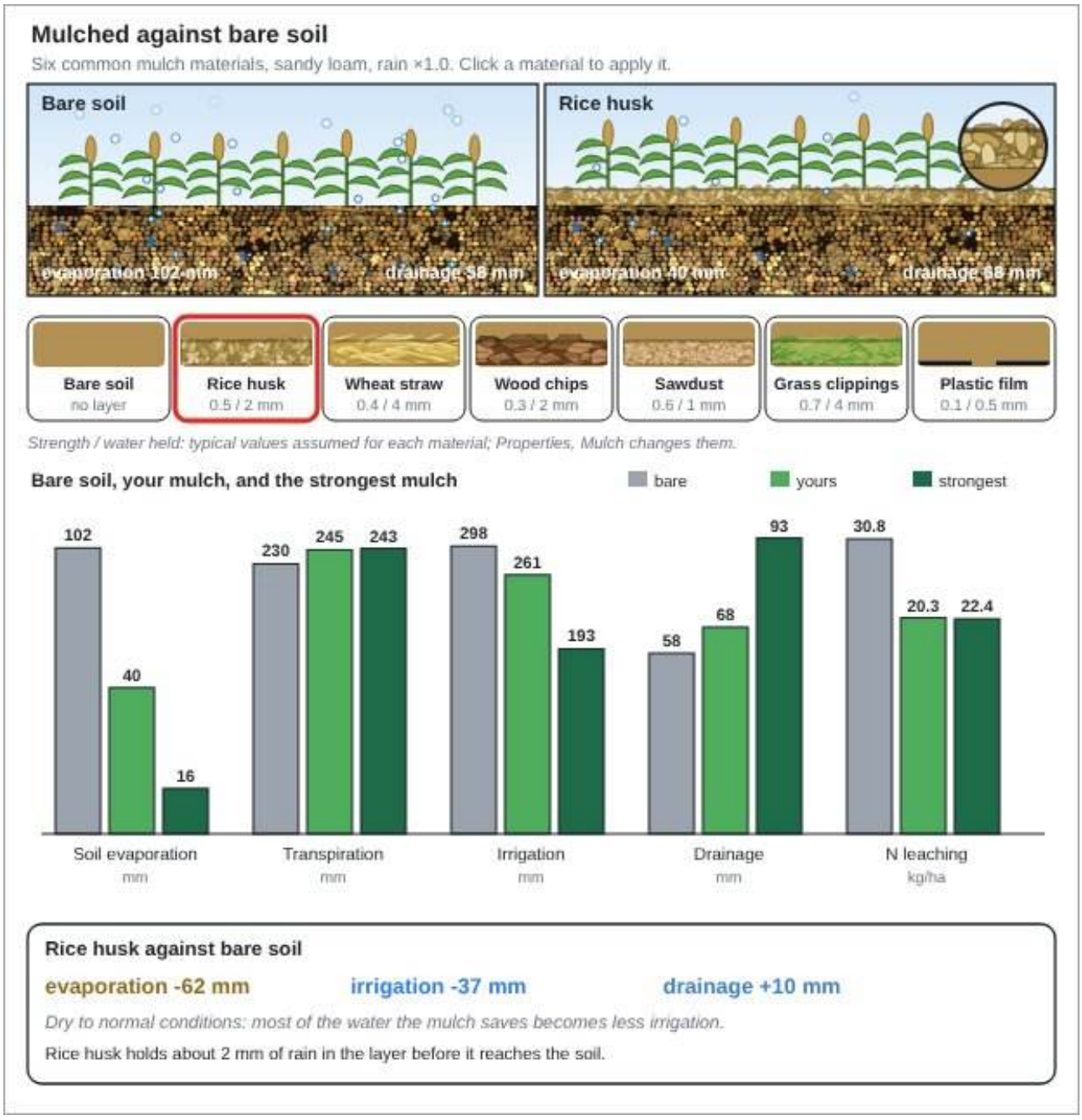


Figure 3.35: Mulch against bare soil, Sandy Loam, part 1 of 1.

3.10 Mulch control (Sandy Loam)

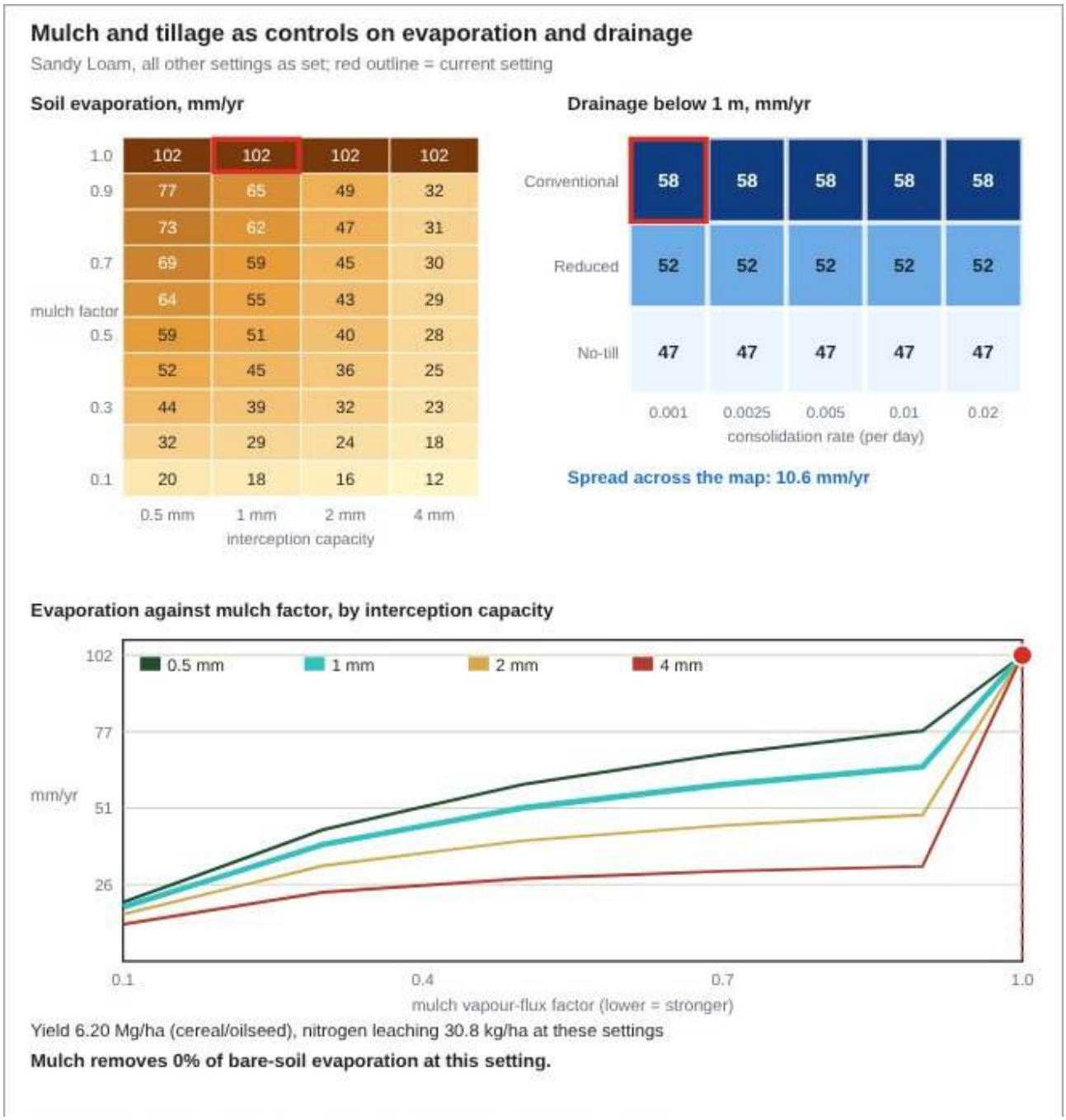


Figure 3.36: Mulch control, Sandy Loam, part 1 of 2.

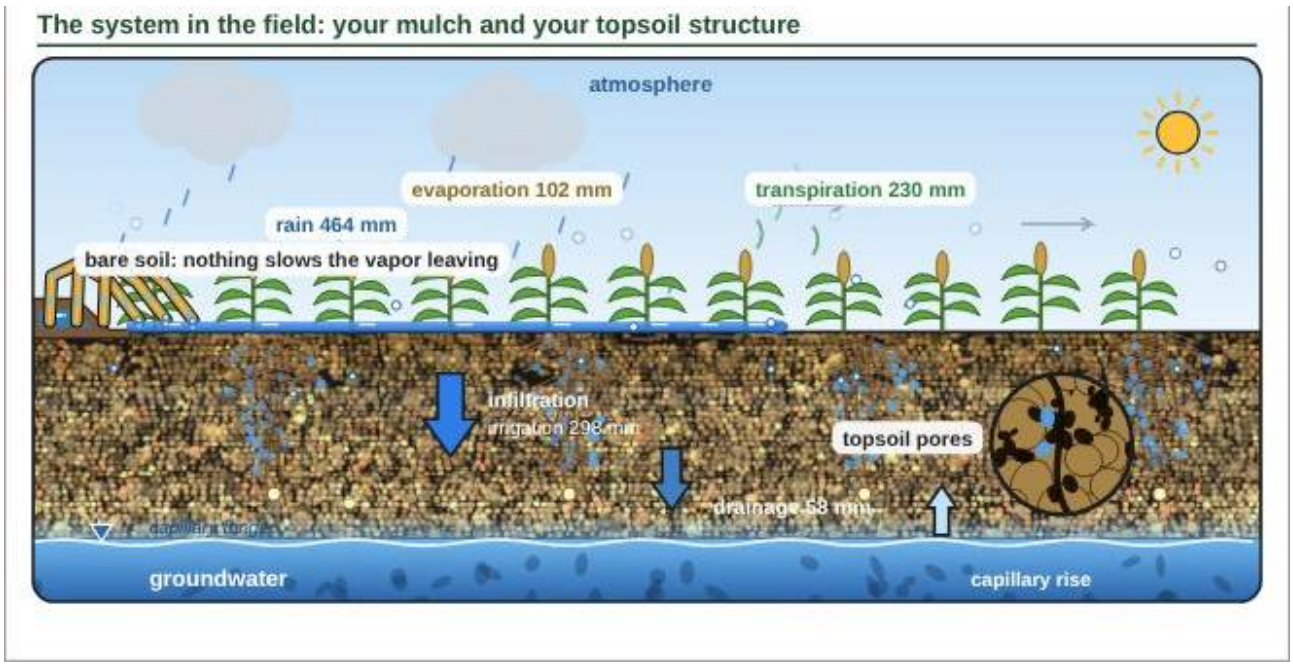


Figure 3.37: Mulch control, Sandy Loam, part 2 of 2.

3.11 Evaporation physics (Sandy Loam)

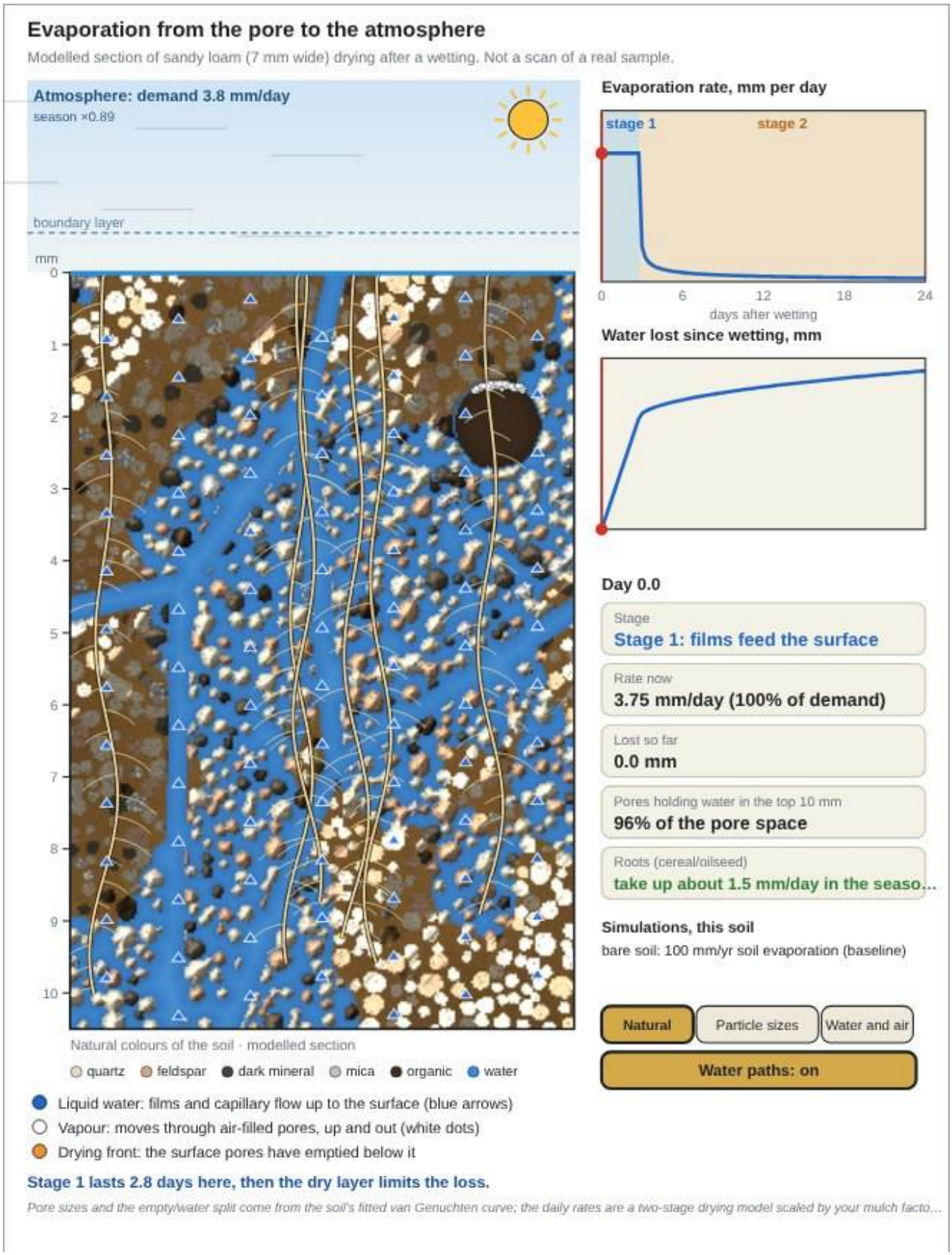


Figure 3.38: Evaporation physics, Sandy Loam, part 1 of 3.

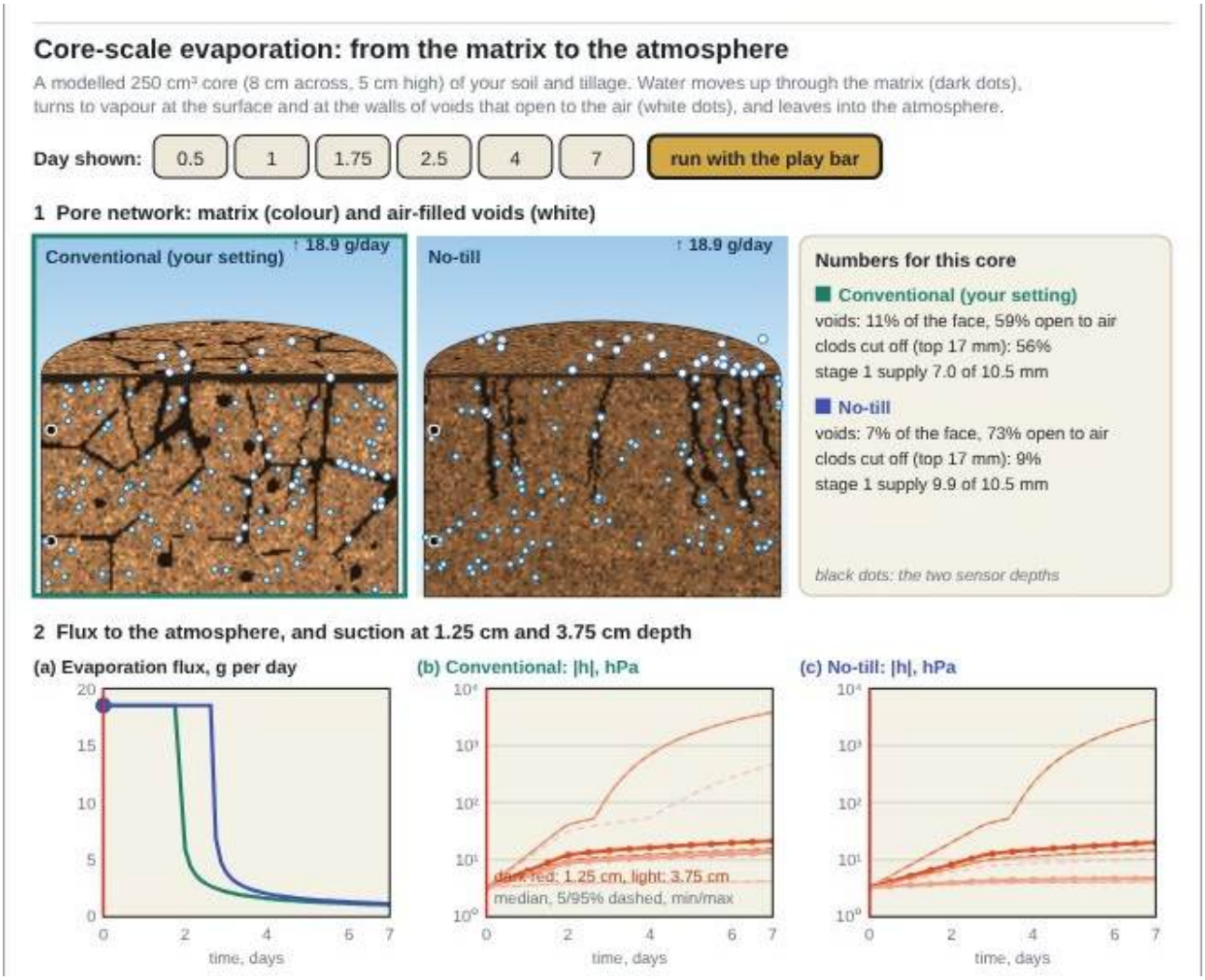


Figure 3.39: Evaporation physics, Sandy Loam, part 2 of 3.

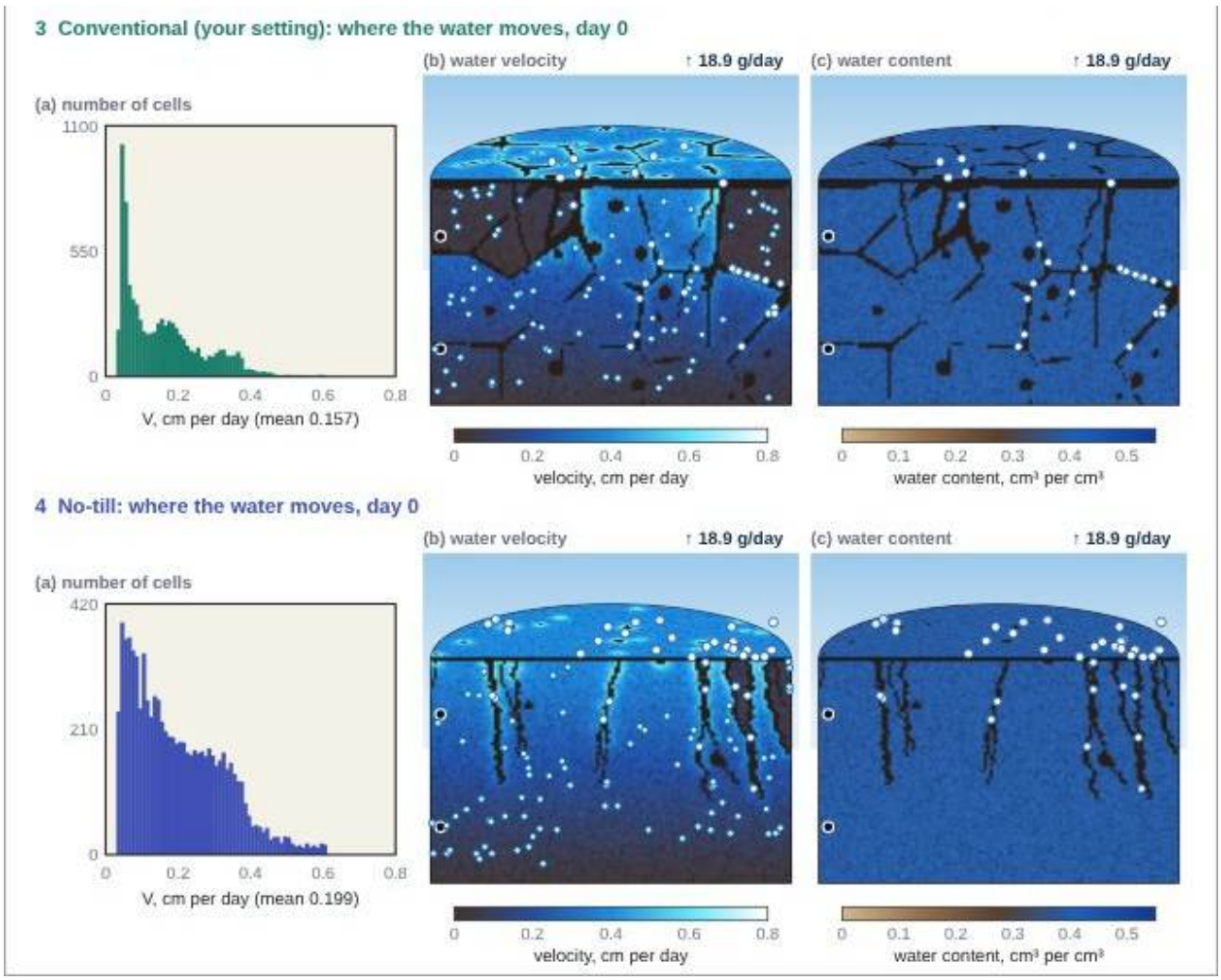


Figure 3.40: Evaporation physics, Sandy Loam, part 3 of 3.

3.12 Fate of saved water (Sandy Loam)

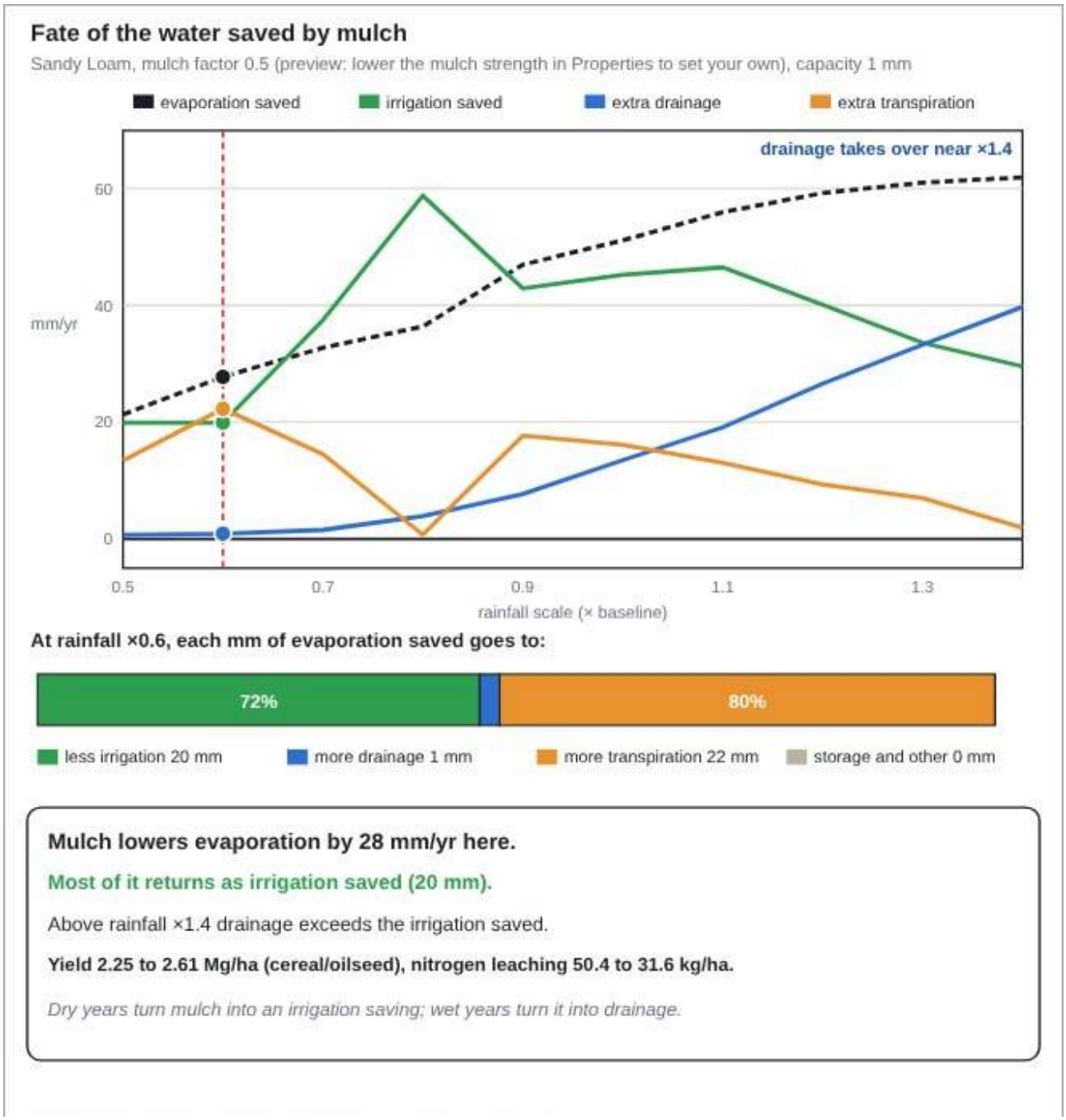


Figure 3.41: Fate of saved water, Sandy Loam, part 1 of 2.

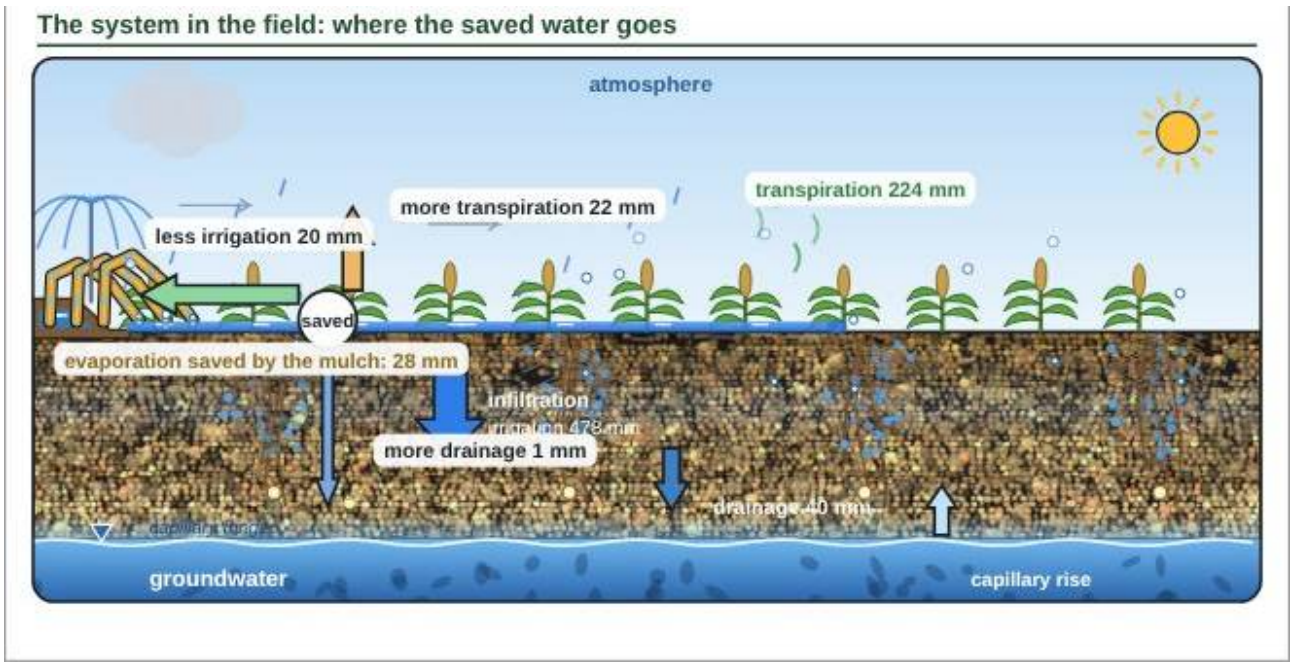


Figure 3.42: Fate of saved water, Sandy Loam, part 2 of 2.

3.13 Regime map (Sandy Loam)

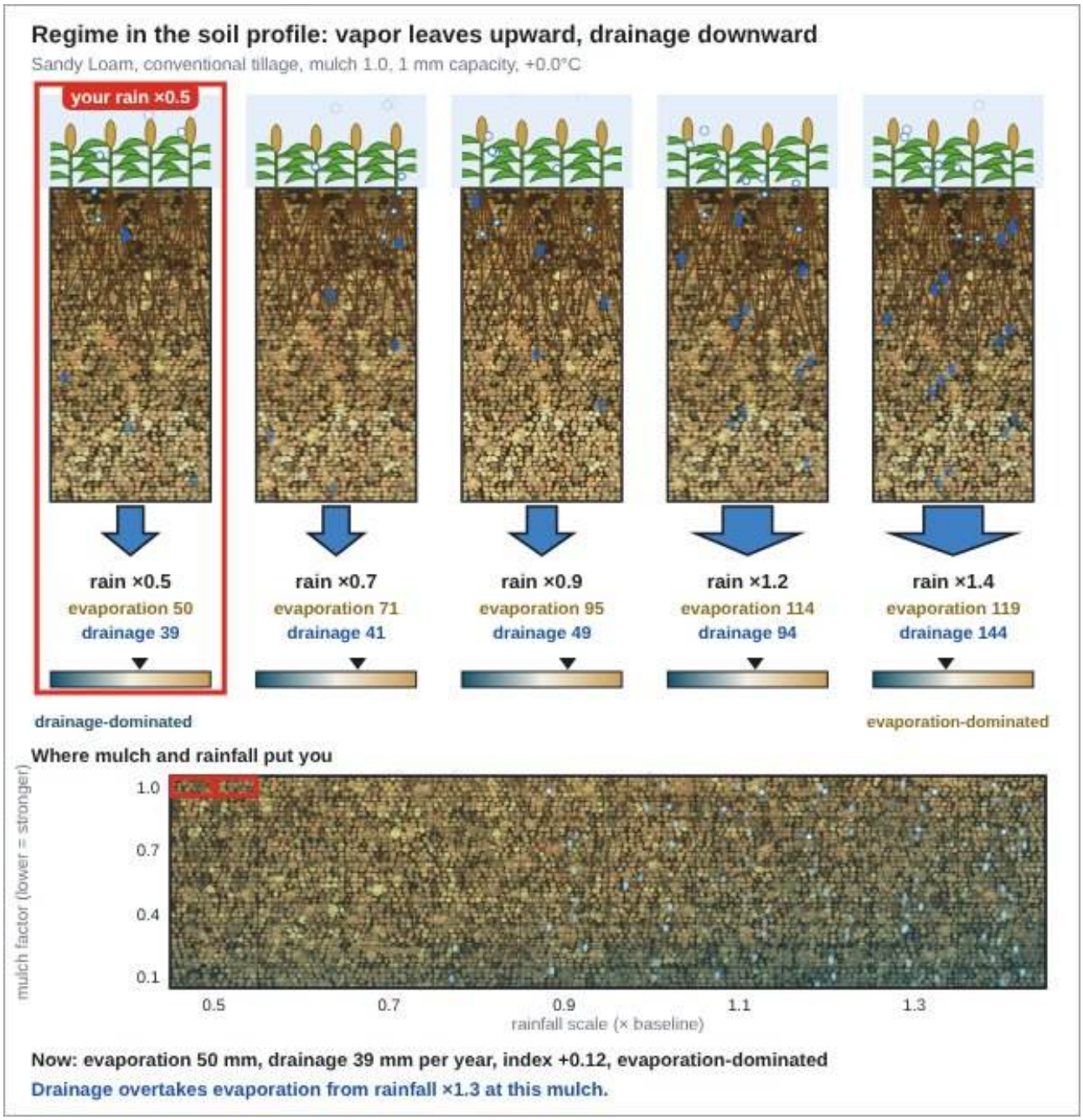


Figure 3.43: Regime map, Sandy Loam, part 1 of 1.

3.14 Tillage and structure (Sandy Loam)

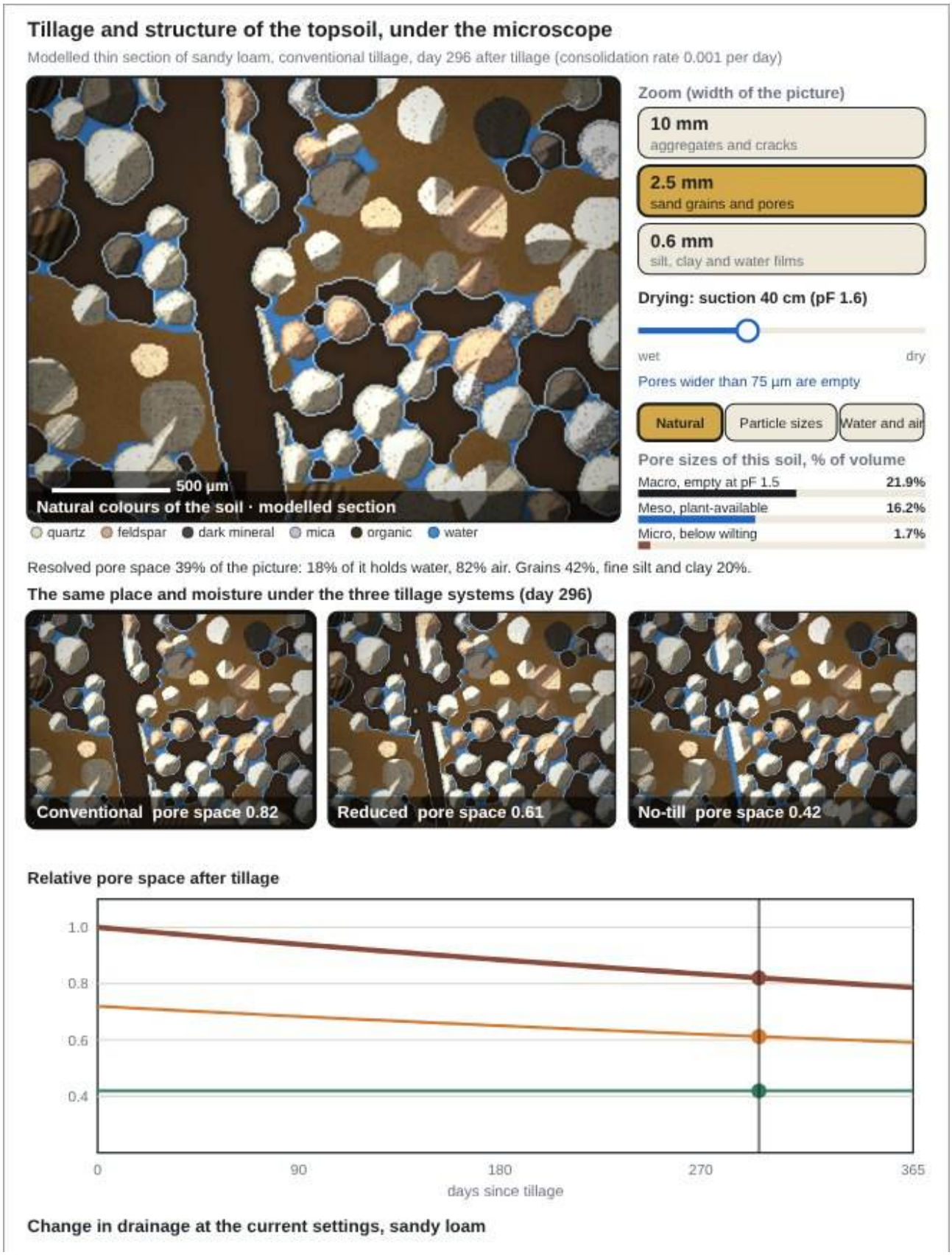


Figure 3.44: Tillage and structure, Sandy Loam, part 1 of 2.

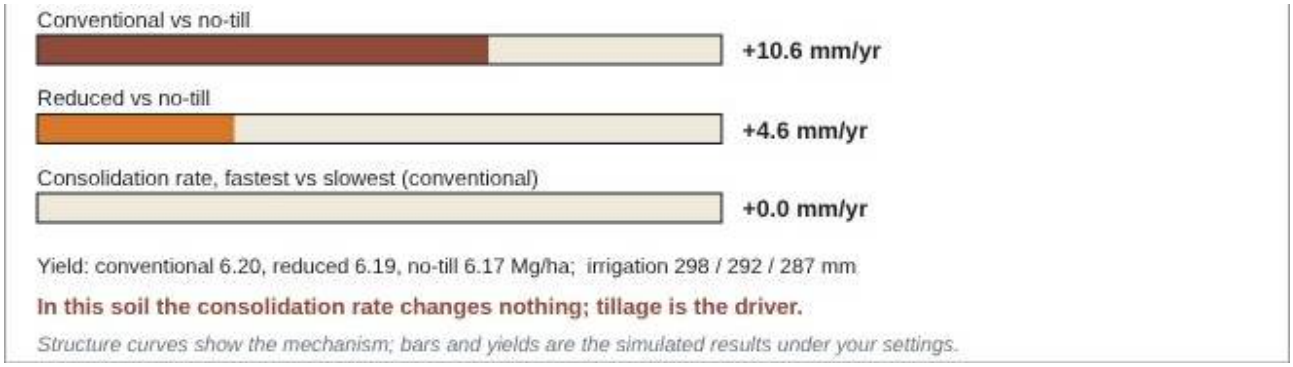


Figure 3.45: Tillage and structure, Sandy Loam, part 2 of 2.

3.15 Soil hydraulic drivers (Sandy Loam)

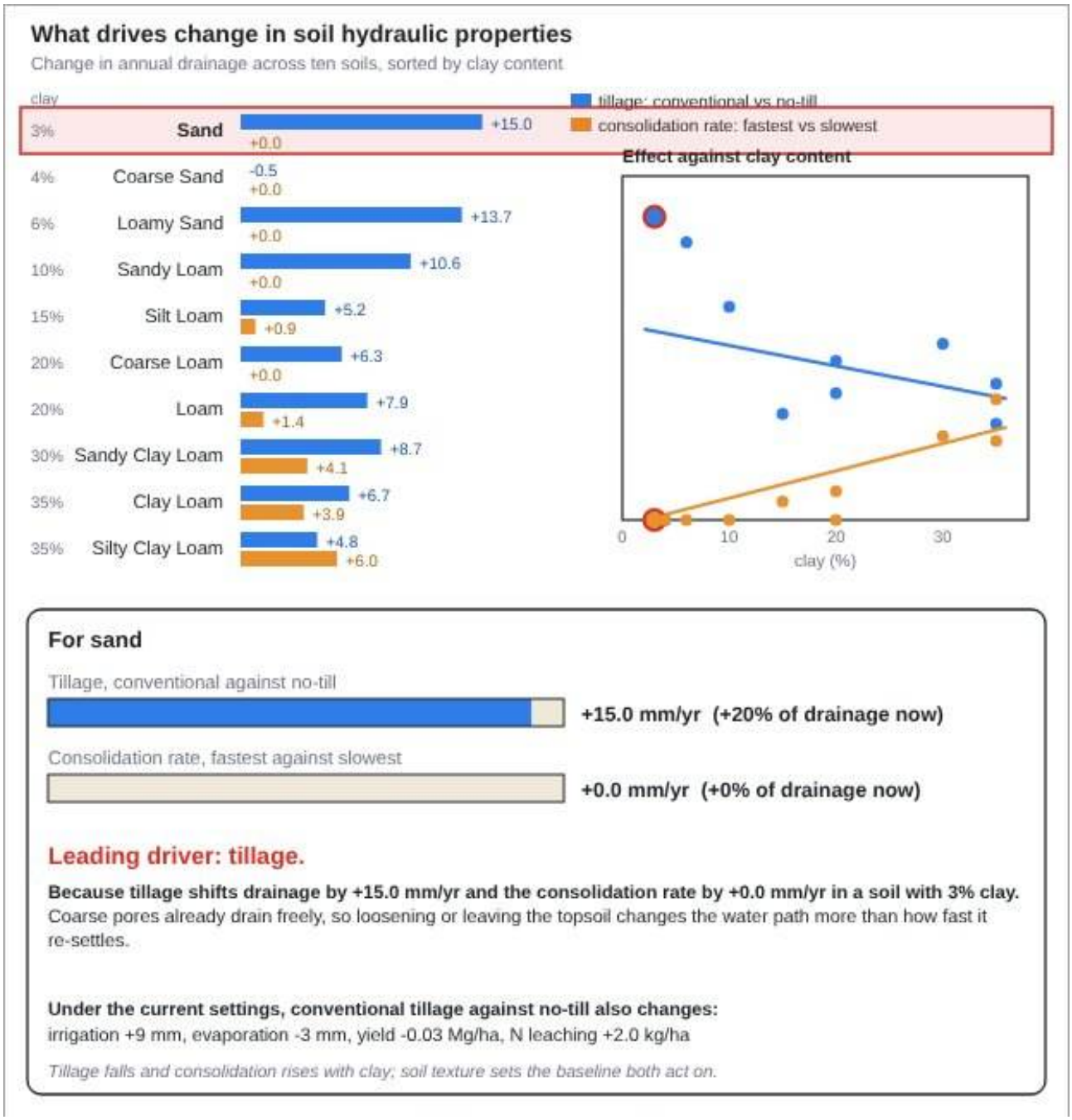


Figure 3.46: Soil hydraulic drivers, Sandy Loam, part 1 of 3.

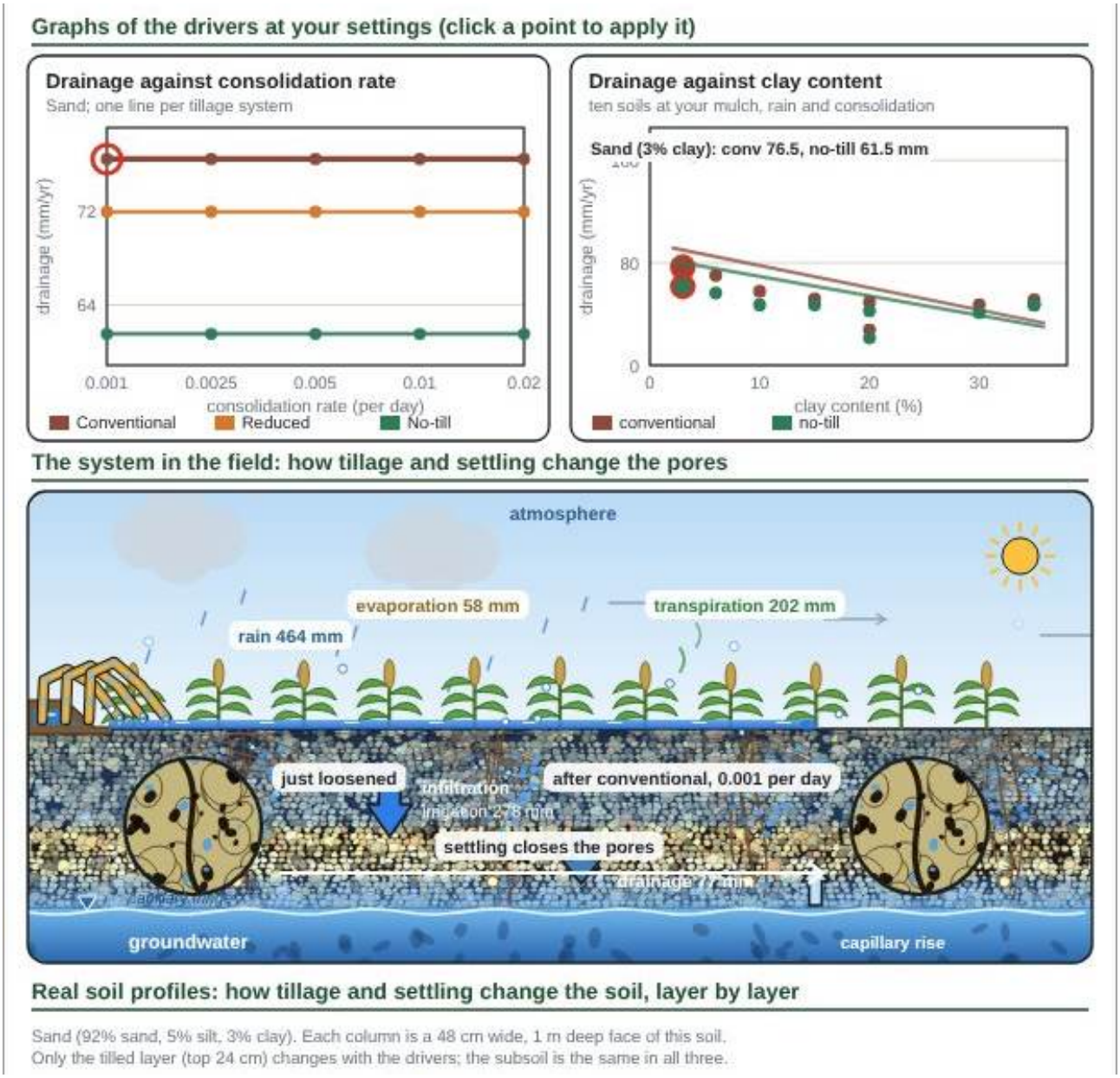


Figure 3.47: Soil hydraulic drivers, Sandy Loam, part 2 of 3.

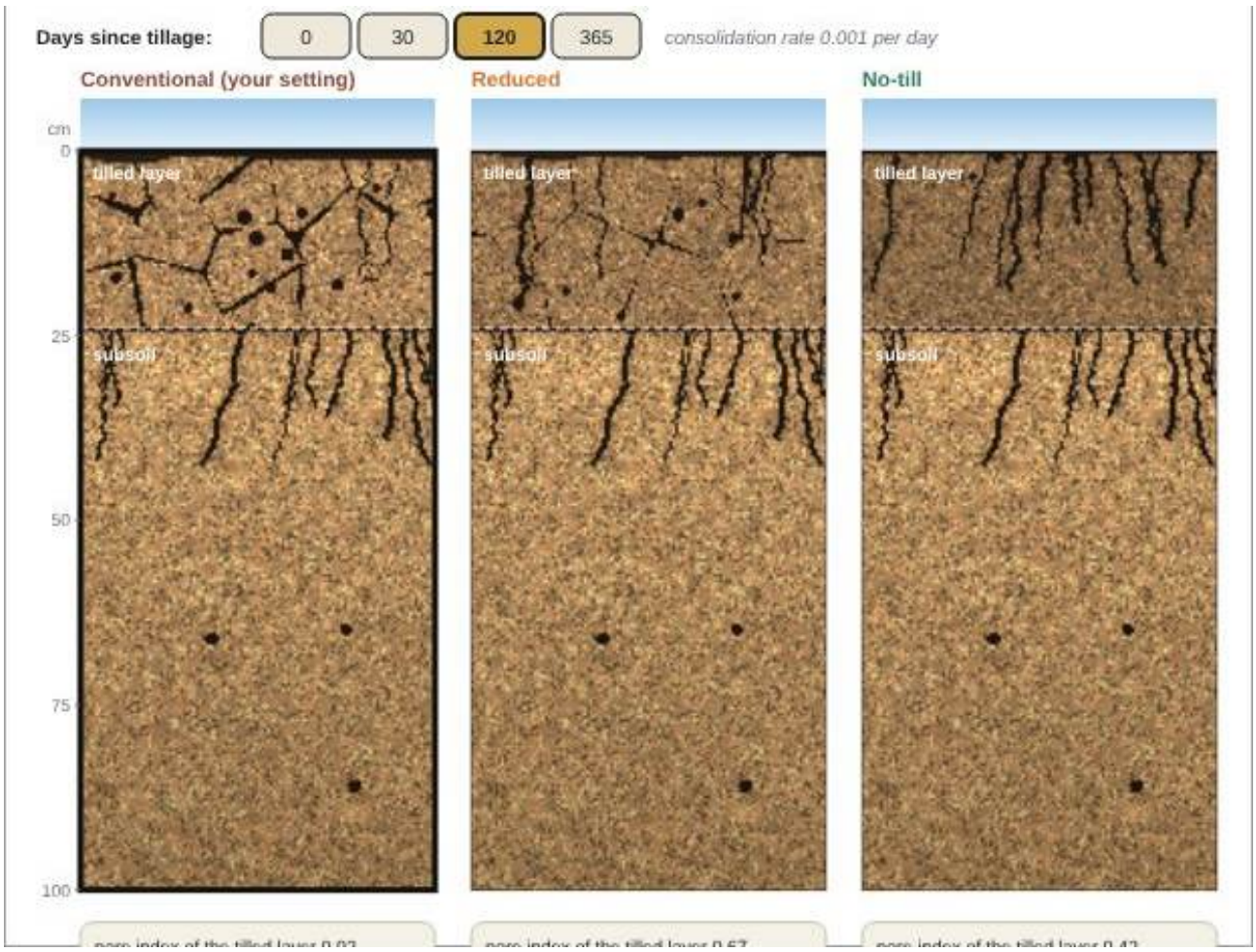


Figure 3.48: Soil hydraulic drivers, Sandy Loam, part 3 of 3.

3.16 Root structure (Sandy Loam)

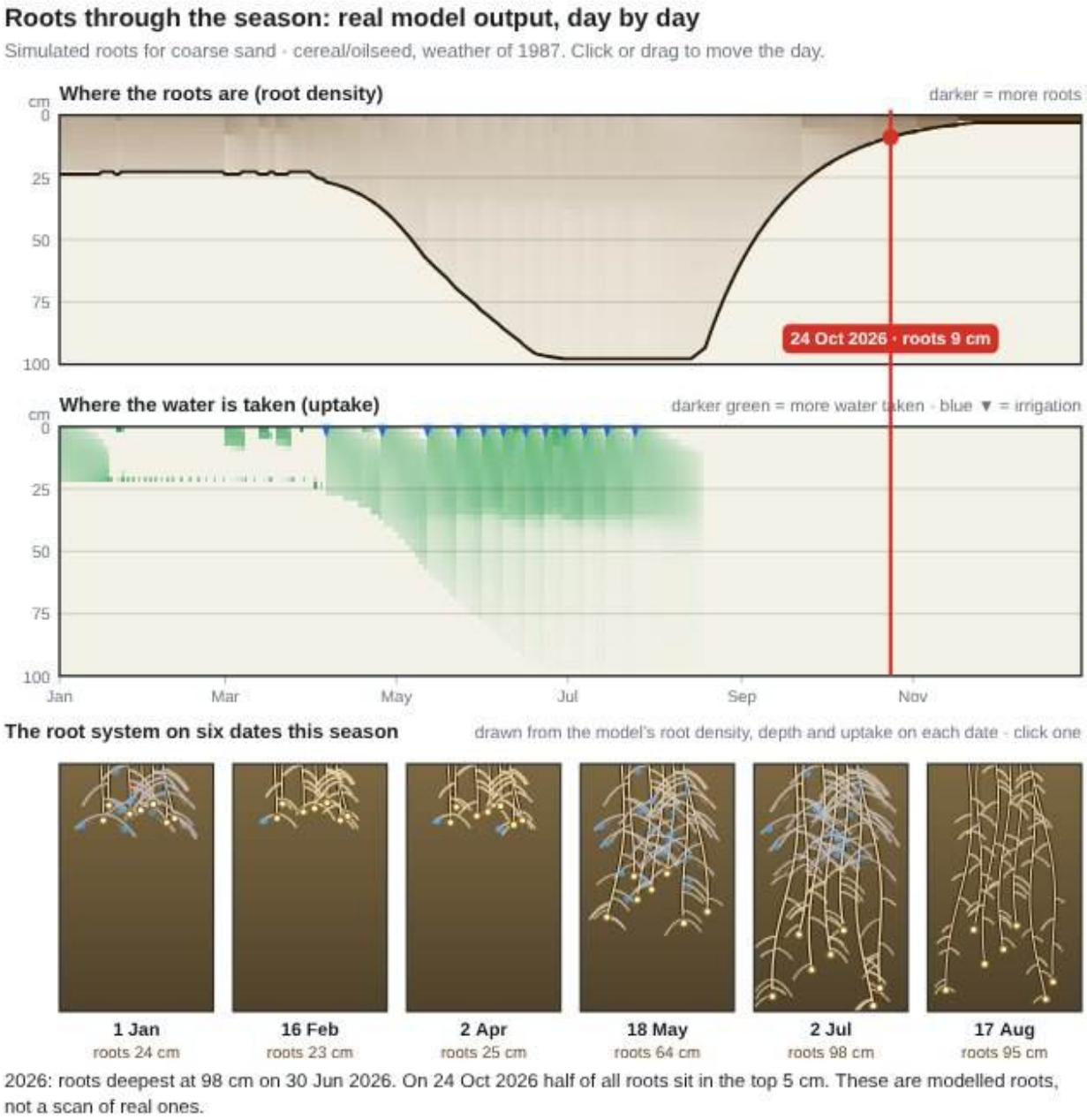


Figure 3.49: Root structure, Sandy Loam, part 1 of 4.

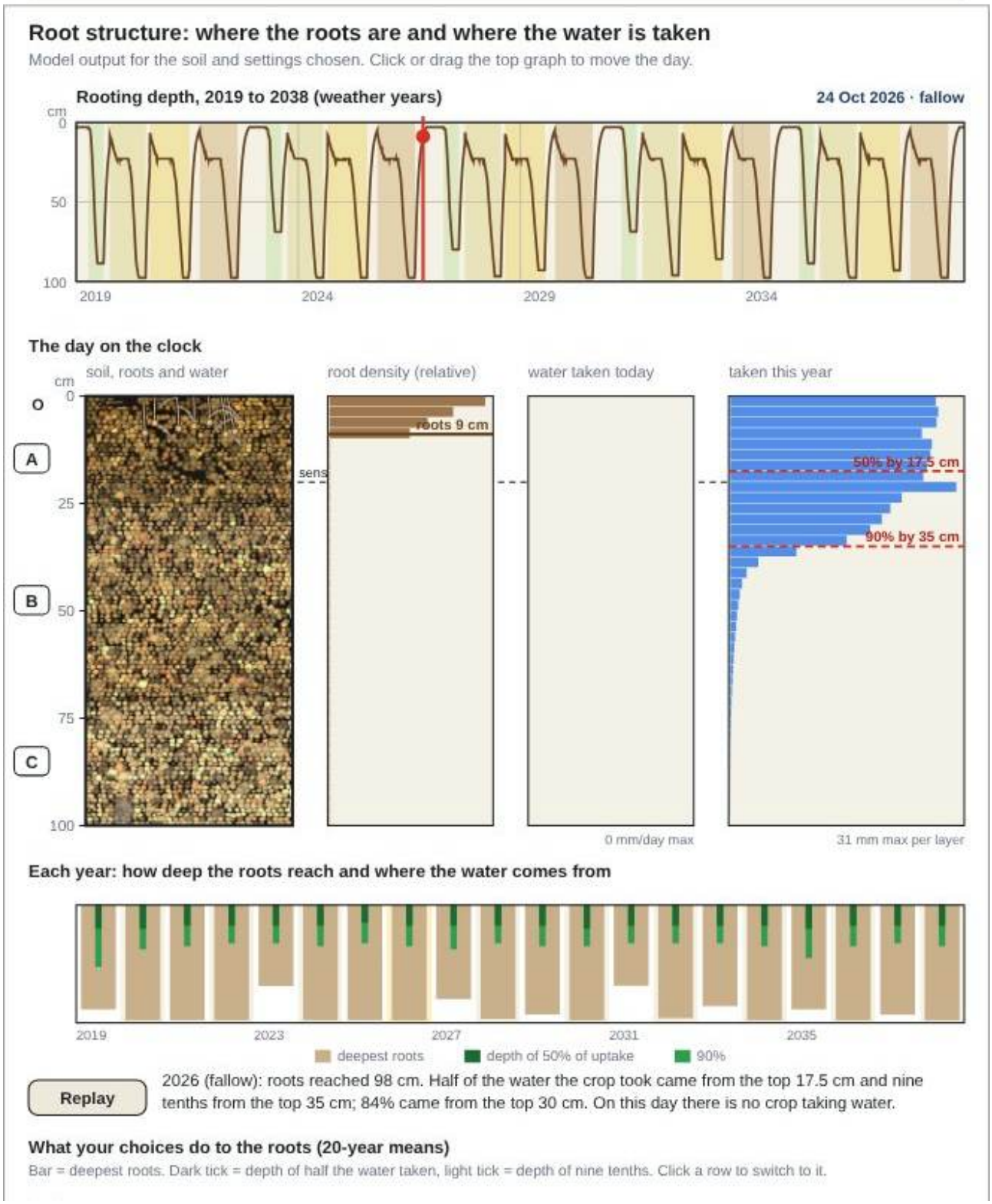


Figure 3.50: Root structure, Sandy Loam, part 2 of 4.

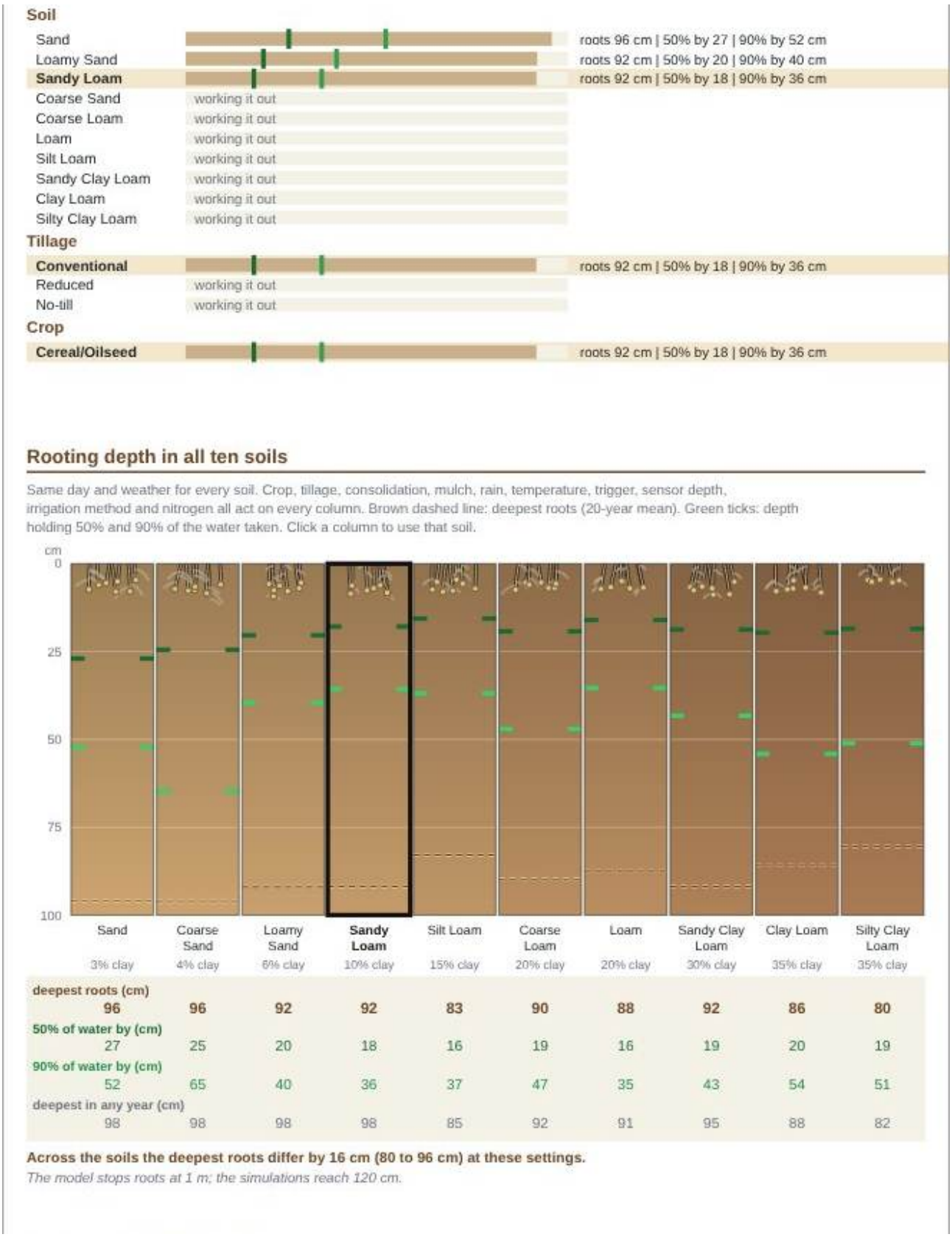


Figure 3.51: Root structure, Sandy Loam, part 3 of 4.

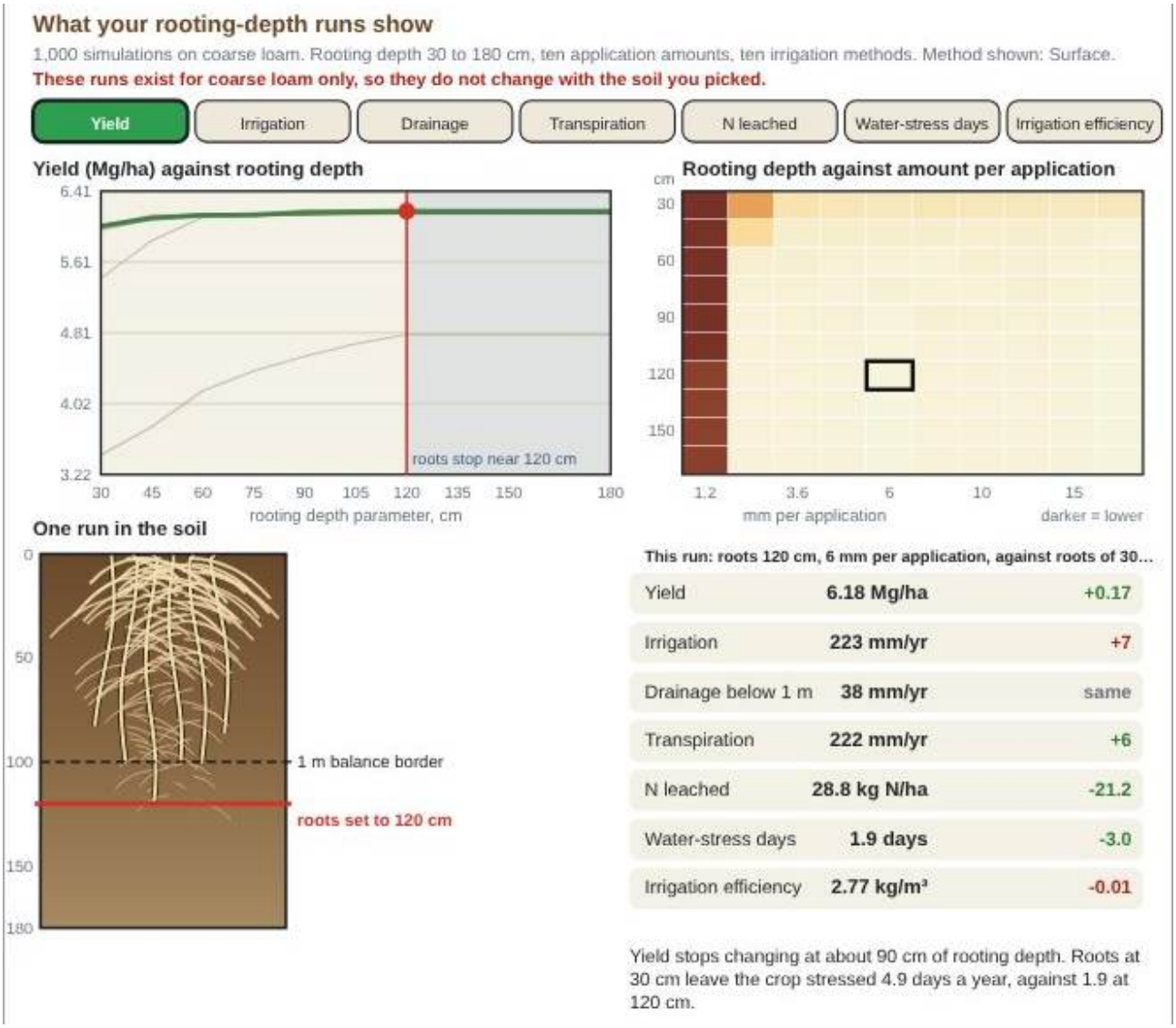


Figure 3.52: Root structure, Sandy Loam, part 4 of 4.

3.17 Field balance (Sandy Loam)

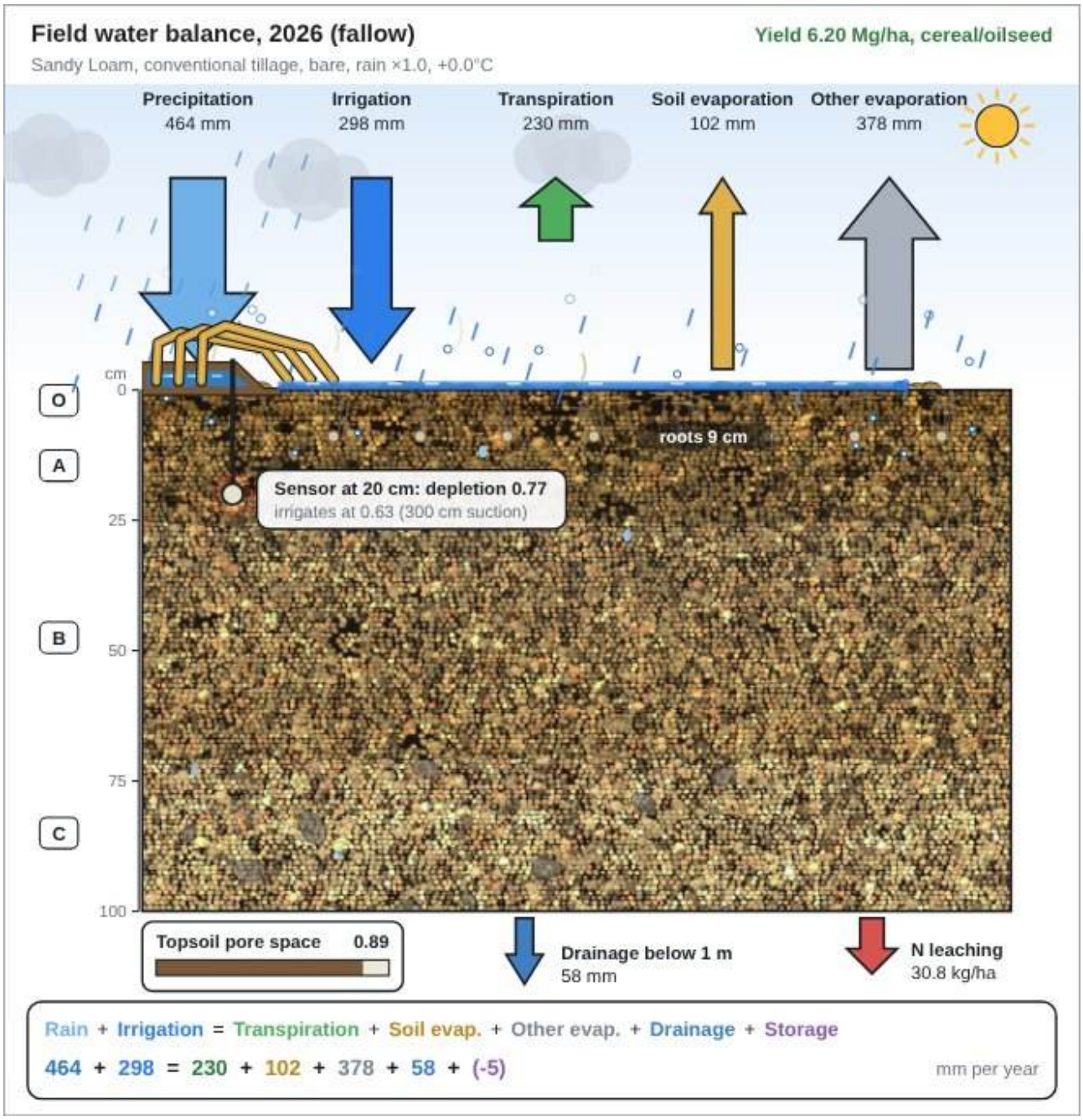


Figure 3.53: Field balance, Sandy Loam, part 1 of 1.

3.18 Soil, plant and air (Sandy Loam)

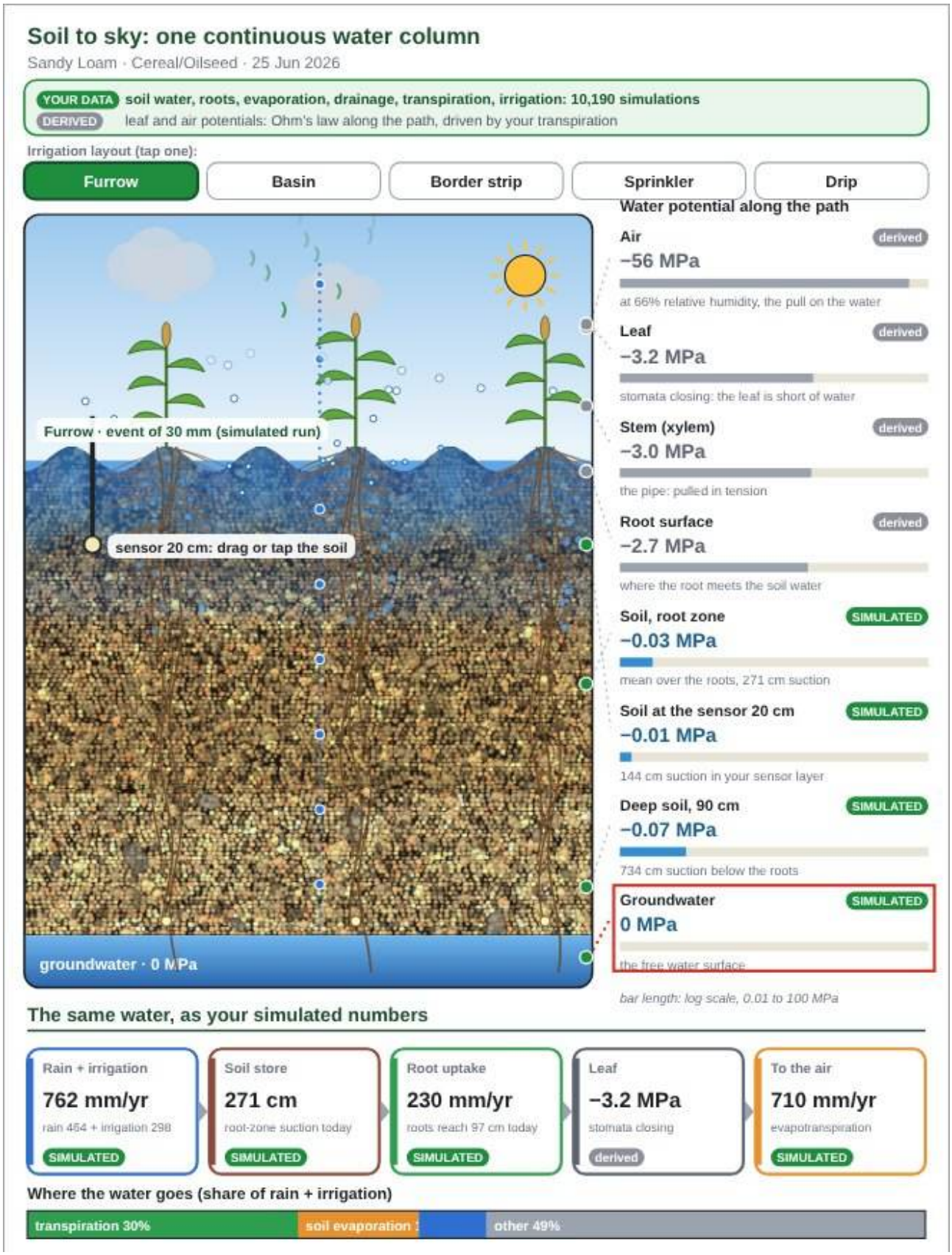


Figure 3.54: Soil, plant and air, Sandy Loam, part 1 of 3.

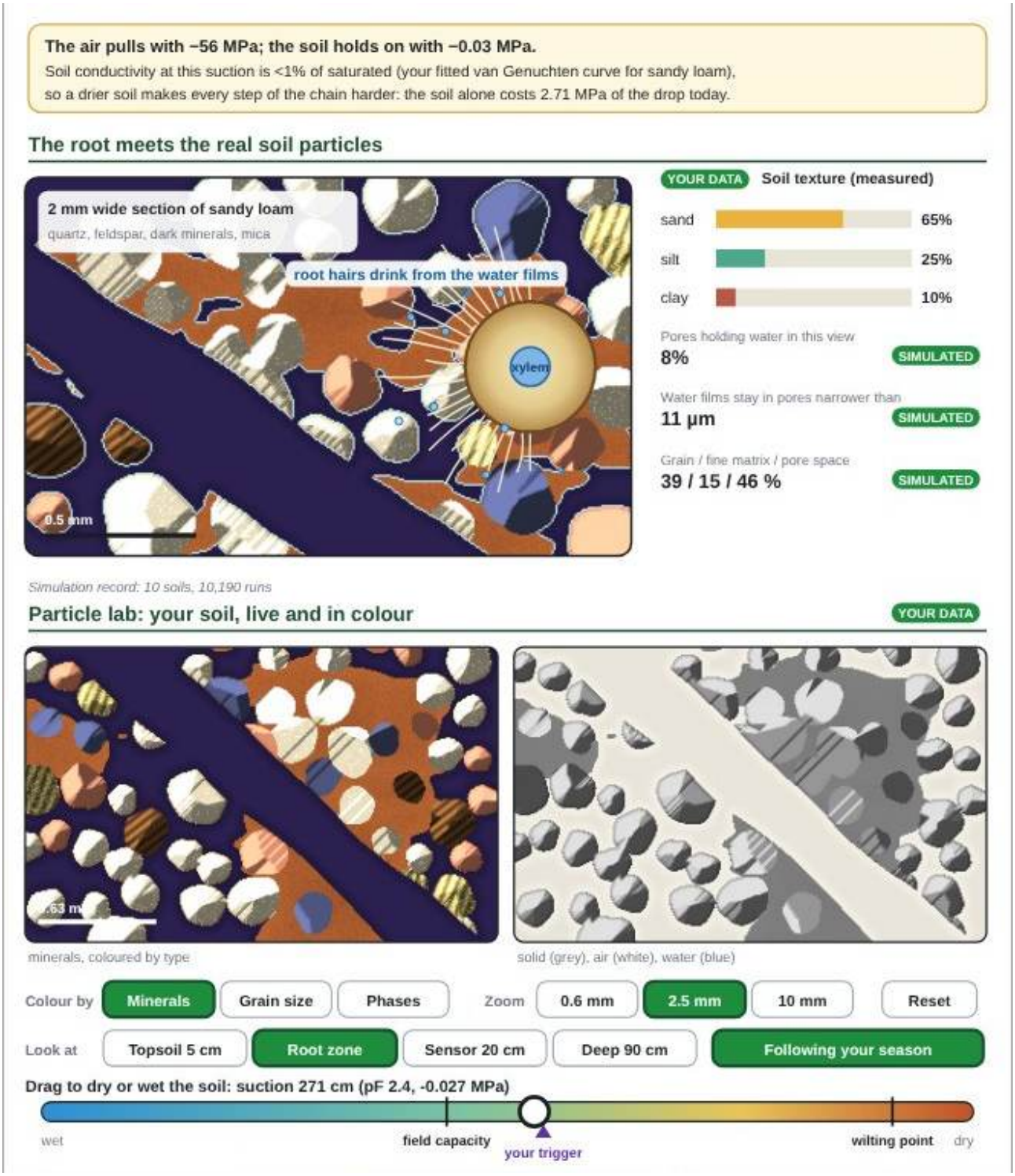


Figure 3.55: Soil, plant and air, Sandy Loam, part 2 of 3.

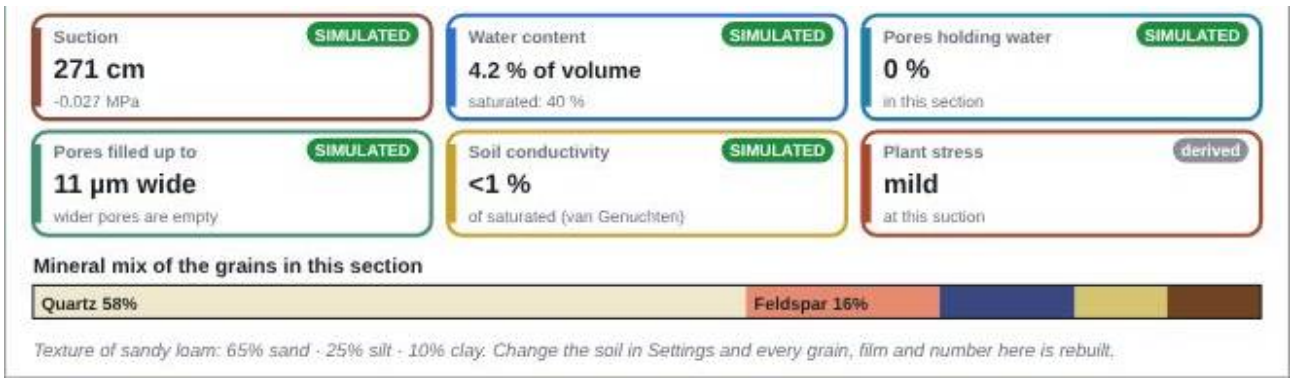


Figure 3.56: Soil, plant and air, Sandy Loam, part 3 of 3.

3.19 Profile evolution (Sandy Loam)

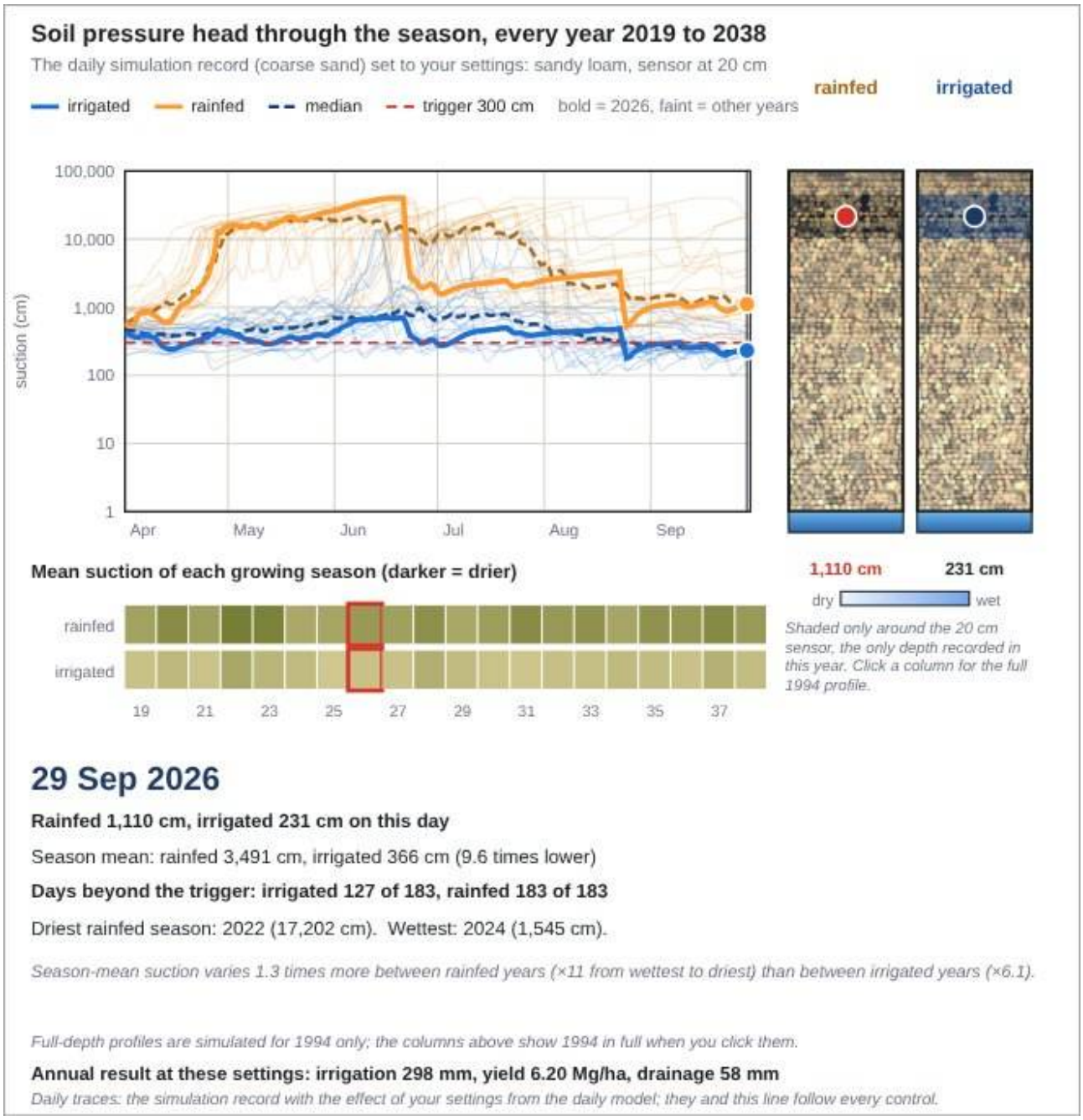


Figure 3.57: Profile evolution, Sandy Loam, part 1 of 1.

3.20 Reliability over 20 seasons (Sandy Loam)

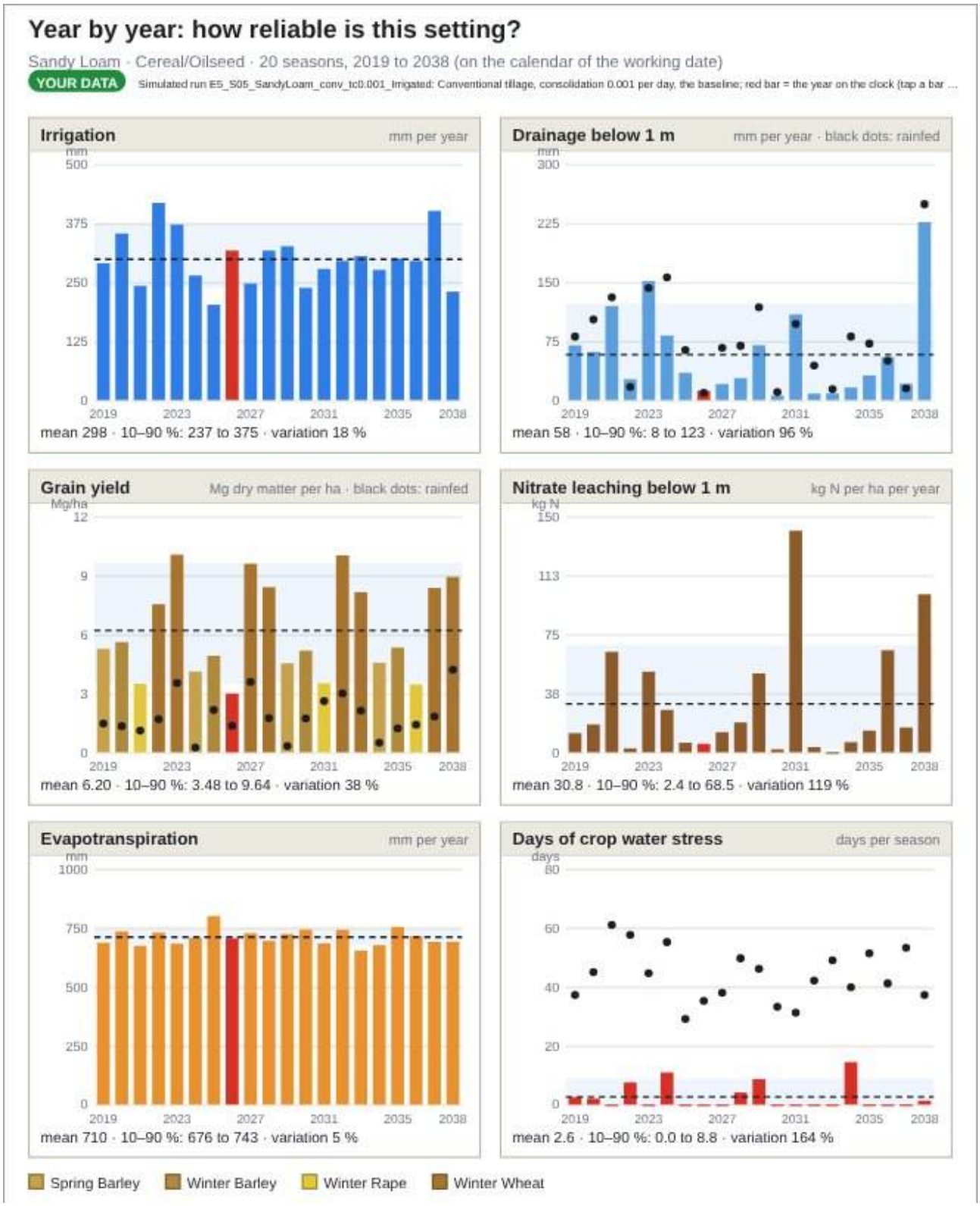


Figure 3.58: Reliability over 20 seasons, Sandy Loam, part 1 of 2.

Reliability of this setting over the 20 seasons — from the yearly values above

Irrigation needed in a dry year (9 years in 10 need less)	**375 mm**	against 298 mm in an average year
Years that need 20 % more irrigation than average	**3 of 20**	plan pump, labour and water rights for these
Years with more than 50 mm drainage below the roots	**9 of 20**	water and nitrate lost to groundwater
Years with a yield below 80 % of the average	**8 of 20**	yield risk at this setting
Nitrate leaching in the worst year	**140.9 kg N/ha**	mean 30.8 kg N/ha

How often a given irrigation is exceeded — the 20 seasons sorted from wettest to driest

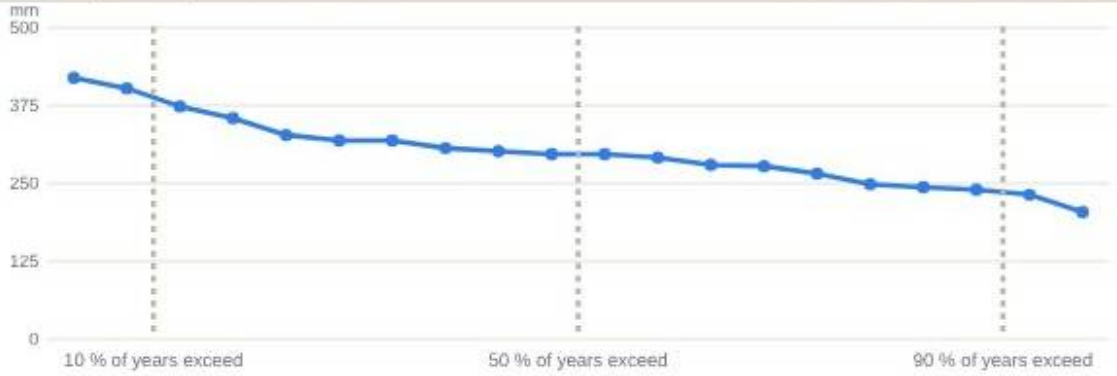


Figure 3.59: Reliability over 20 seasons, Sandy Loam, part 2 of 2.

3.21 Full water balance (Sandy Loam)

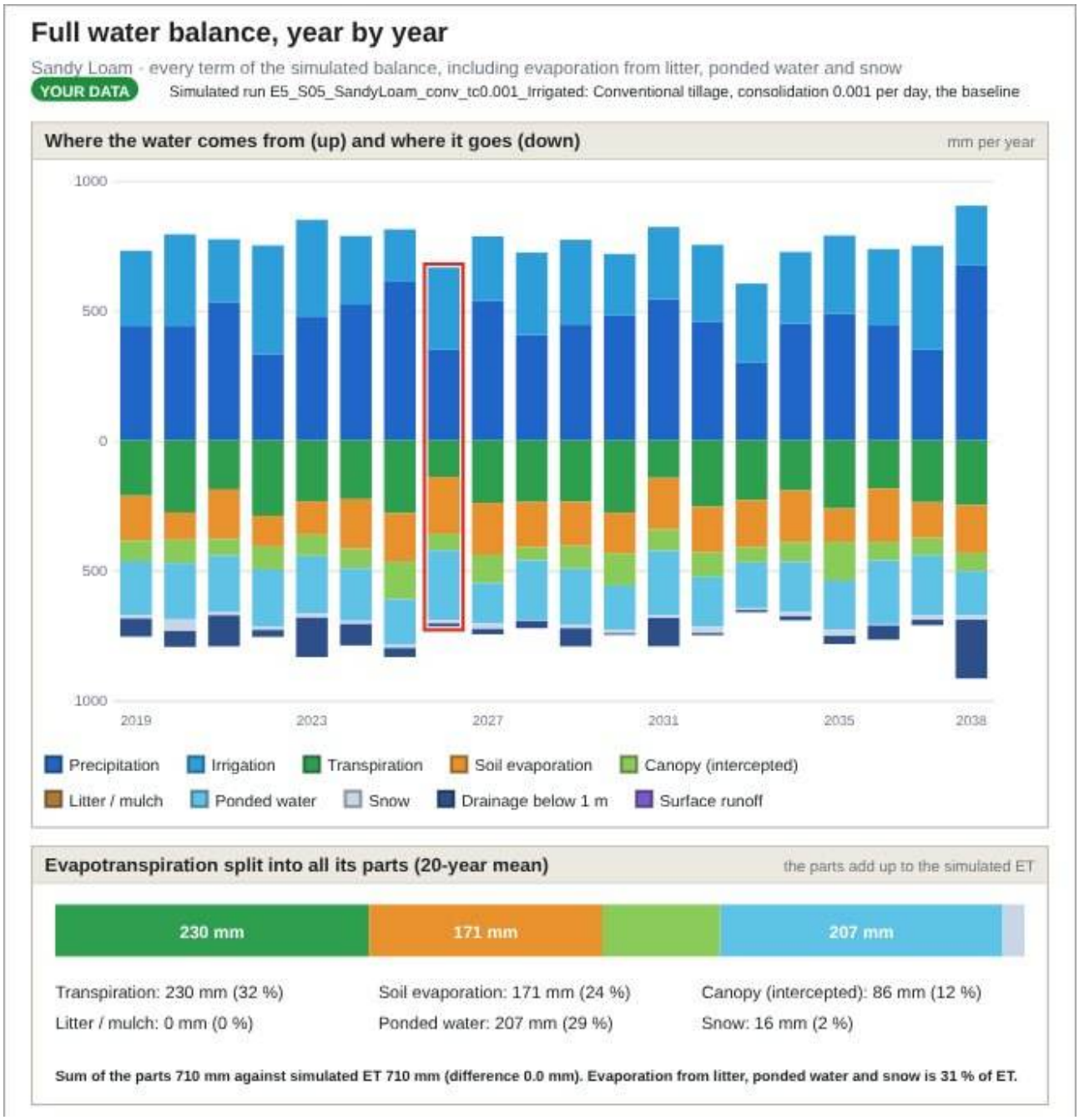


Figure 3.60: Full water balance, Sandy Loam, part 1 of 2.

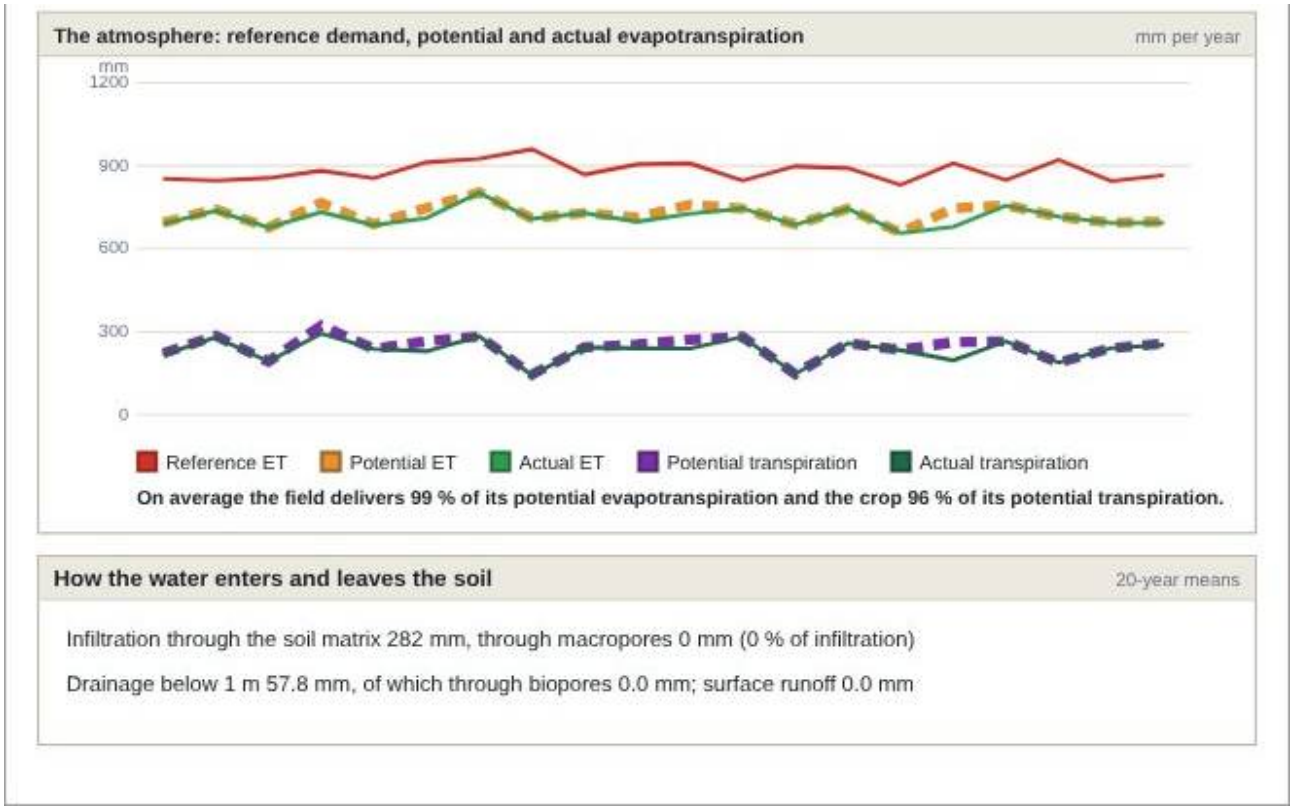


Figure 3.61: Full water balance, Sandy Loam, part 2 of 2.

3.22 Nitrogen budget (Sandy Loam)

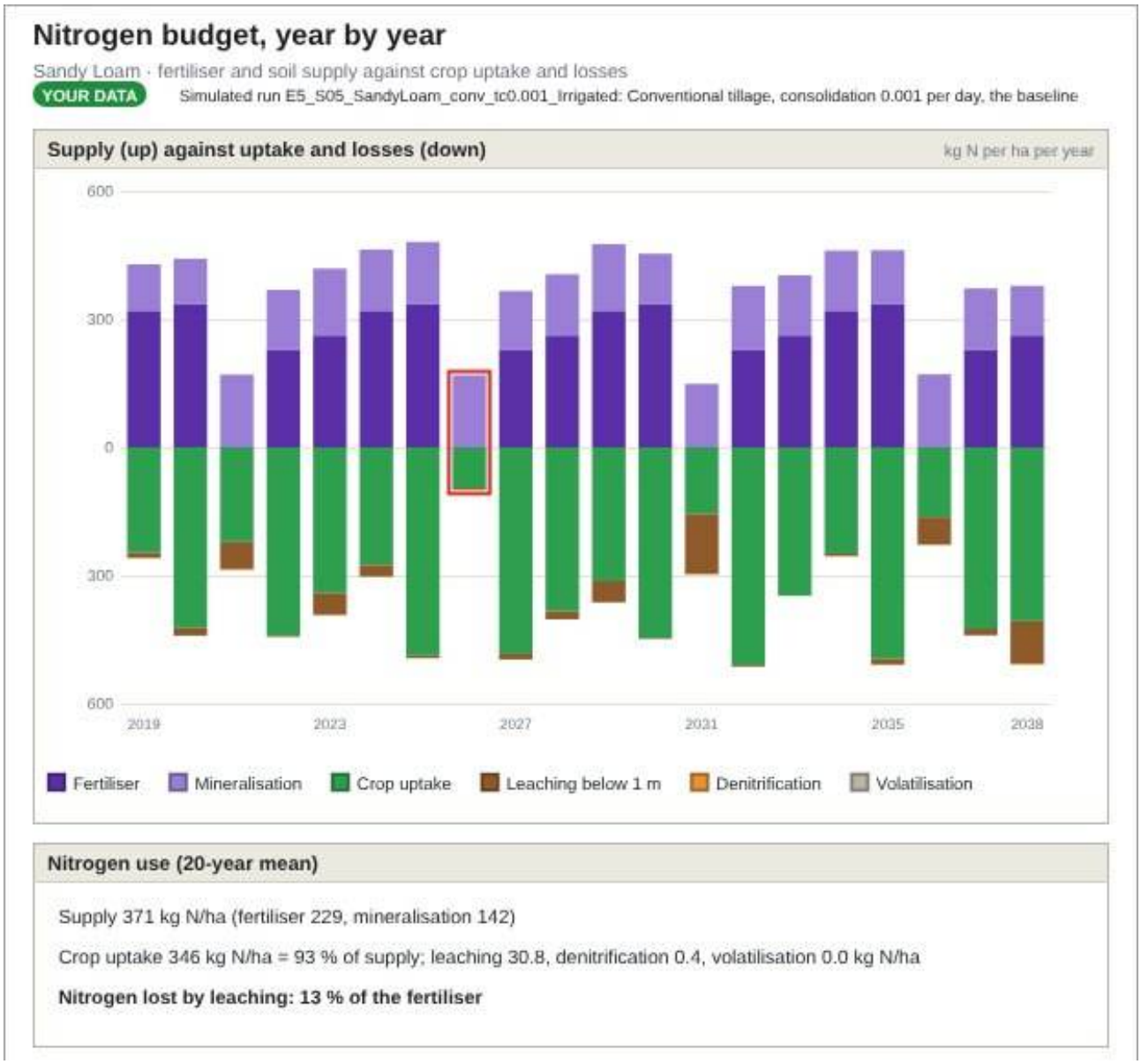


Figure 3.62: Nitrogen budget, Sandy Loam, part 1 of 2.

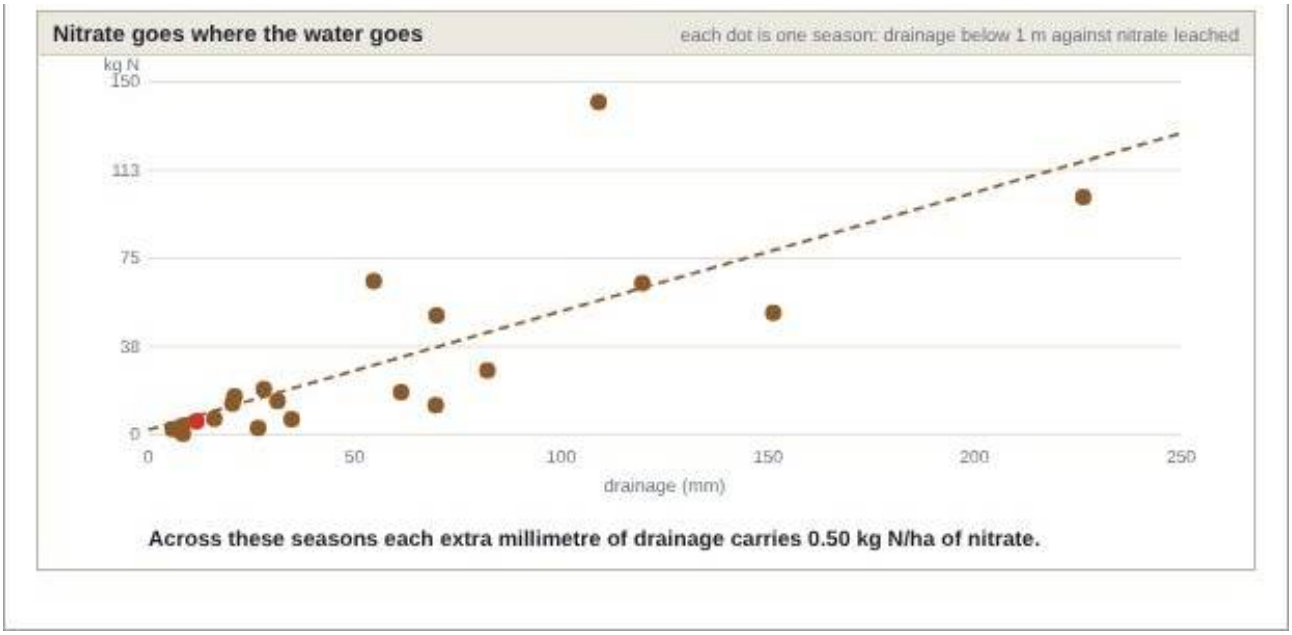


Figure 3.63: Nitrogen budget, Sandy Loam, part 2 of 2.

3.23 Crop water requirement (Sandy Loam)

Figure 3.64: Crop water requirement, Sandy Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**420**	6.5	92 %
Coconut (high)	834	**415**	5.4	100 %
Papaya (high)	814	**414**	6.0	100 %
Oil palm (young)	790	**381**	5.5	100 %
Yam	649	**311**	7.1	97 %
Cotton	571	**297**	4.0	100 %
Avocado (medium)	677	**279**	4.9	100 %
Cabbage	559	**272**	5.7	99 %
Onion (dry)	565	**267**	7.8	91 %
Kiwi (pergola)	626	**259**	3.4	95 %
Guava (high)	651	**255**	3.3	100 %
Potato	508	**242**	6.5	93 %
Mustard	468	**234**	3.3	100 %
Hop (v trellis)	520	**232**	3.0	100 %
Tomato	521	**231**	3.1	96 %
Wheat	469	**230**	3.2	100 %
Groundnut	471	**229**	4.1	99 %
Soybean	473	**229**	3.3	99 %
Rice	573	**228**	4.0	83 %
Walnut (medium)	523	**228**	3.0	100 %
Table grapes (horizontal trellis)	597	**227**	3.0	95 %
Pear (medium)	528	**226**	3.0	100 %
Mango (medium)	620	**223**	2.9	100 %
Maize (grain)	451	**204**	2.7	100 %
Cherry (medium)	503	**202**	2.7	100 %
Pistachio (medium)	503	**193**	2.6	98 %
Chickpea	436	**193**	3.3	99 %
Date palm (medium)	602	**191**	2.5	100 %
Hazelnut (low)	481	**182**	2.4	100 %
Pecan (low)	481	**182**	2.4	100 %
Sweet pepper	475	**180**	3.2	90 %
Sorghum (grain)	429	**178**	2.4	100 %
Cassava	466	**178**	3.7	96 %
Cherimoya (medium)	545	**163**	2.1	100 %
Millet	395	**161**	2.3	100 %

Figure 3.65: Crop water requirement, Sandy Loam, part 2 of 3.

Cowpea (beans)	398	**156**	2.7	98 %
Citrus (medium density, short)	552	**151**	2.0	100 %
Apple (medium)	459	**140**	1.8	100 %
Wine grapes (pergola)	457	**131**	1.7	100 %
Green gram (mung bean)	372	**131**	2.3	98 %
Peach (low)	435	**131**	1.7	100 %
Almond (medium)	435	**130**	1.7	99 %
Apricot (young)	380	**71**	0.9	100 %
Olive (traditional, medium density)	446	**67**	0.9	100 %
Highbush blueberry (low)	338	**57**	0.8	100 %
Persimmon (young, low)	358	**52**	0.7	100 %
Pomegranate (low)	358	**52**	0.7	100 %
Fig tree (young, low)	327	**36**	0.5	100 %

Figure 3.66: Crop water requirement, Sandy Loam, part 3 of 3.

3.24 Feedback loops (Sandy Loam)

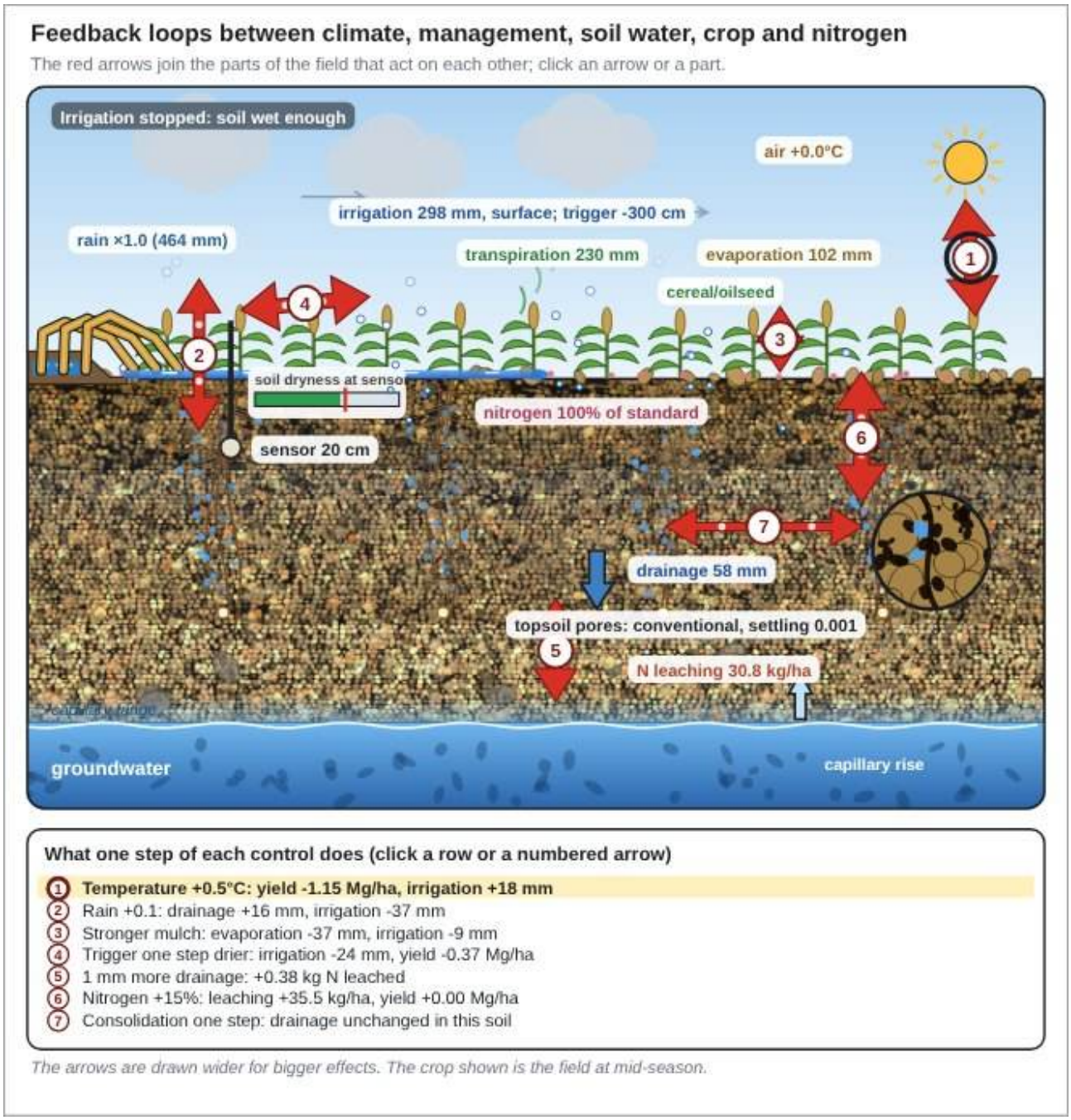


Figure 3.67: Feedback loops, Sandy Loam, part 1 of 1.

Chapter 4

Coarse Sand

4.1 Surface irrigation design (Coarse Sand)

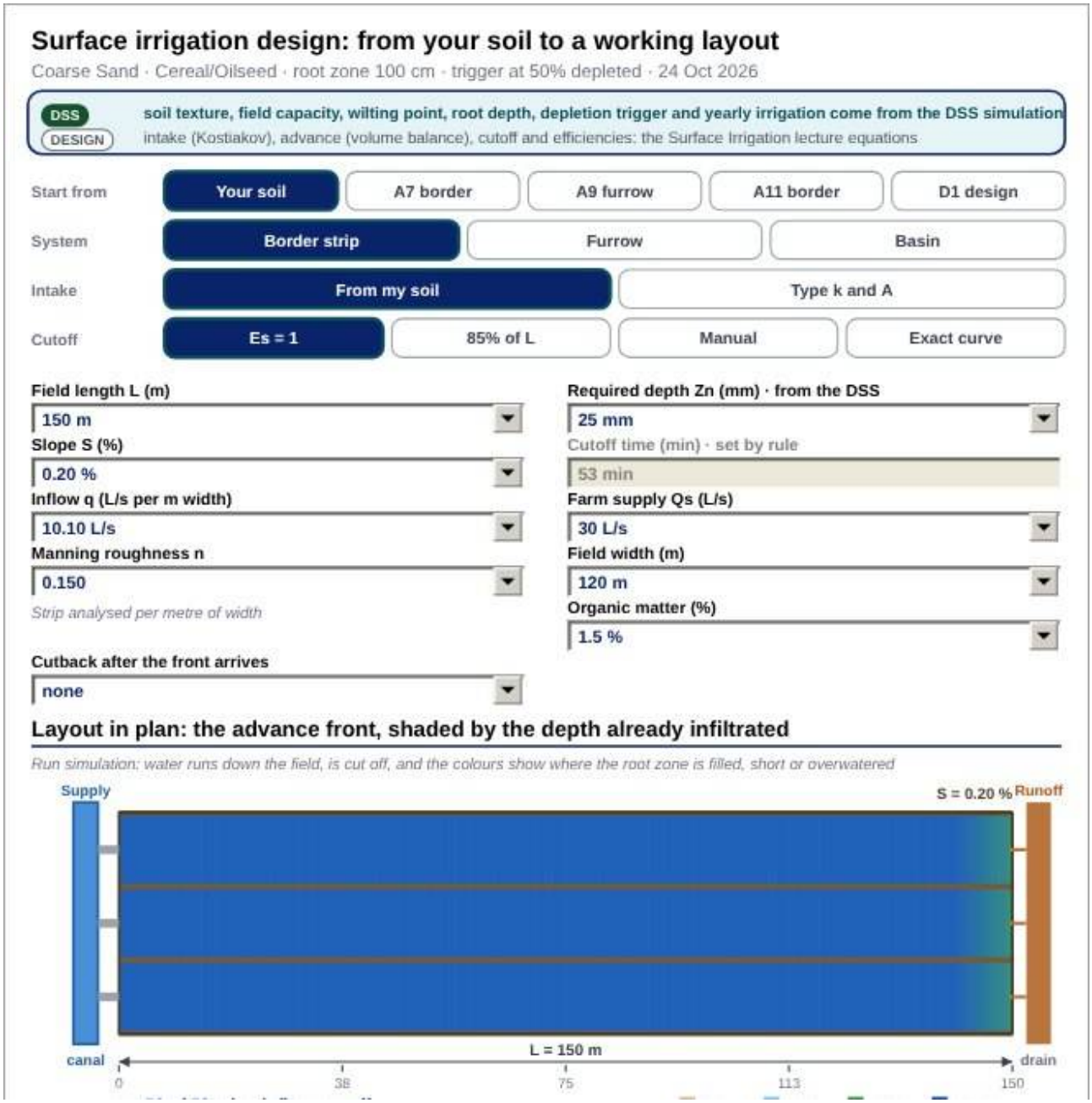


Figure 4.1: Surface irrigation design, Coarse Sand, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

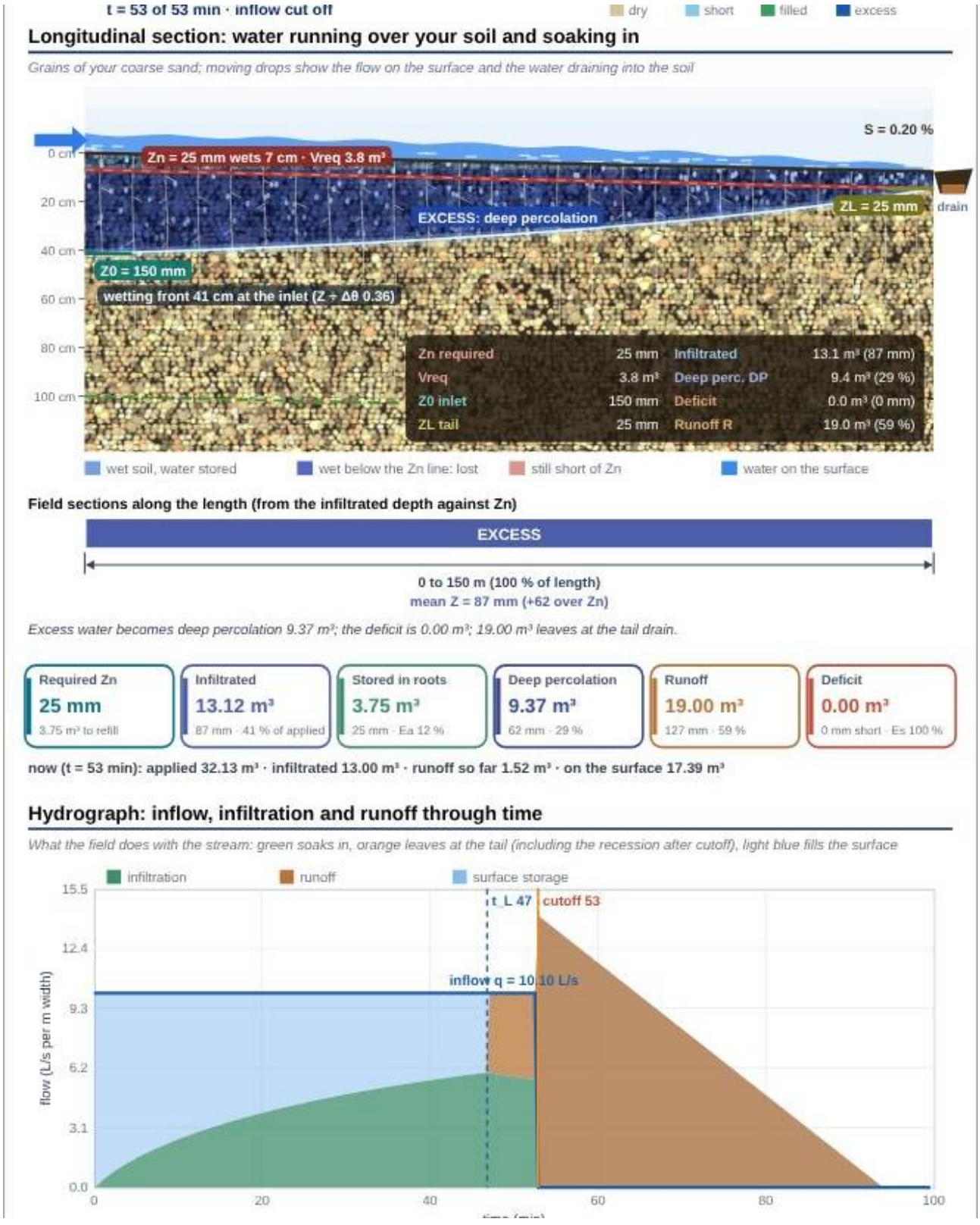


Figure 4.2: Surface irrigation design, Coarse Sand, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

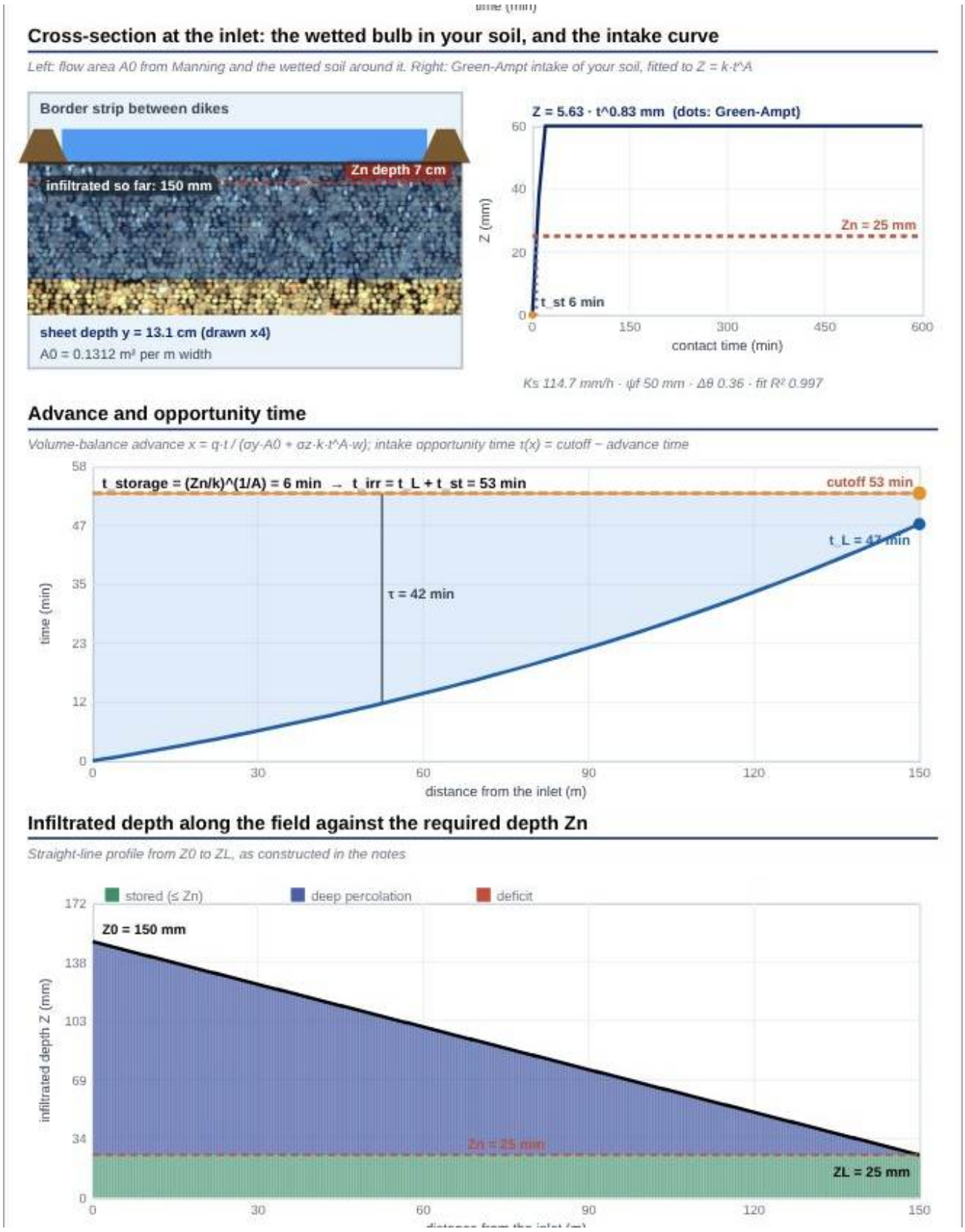


Figure 4.3: Surface irrigation design, Coarse Sand, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

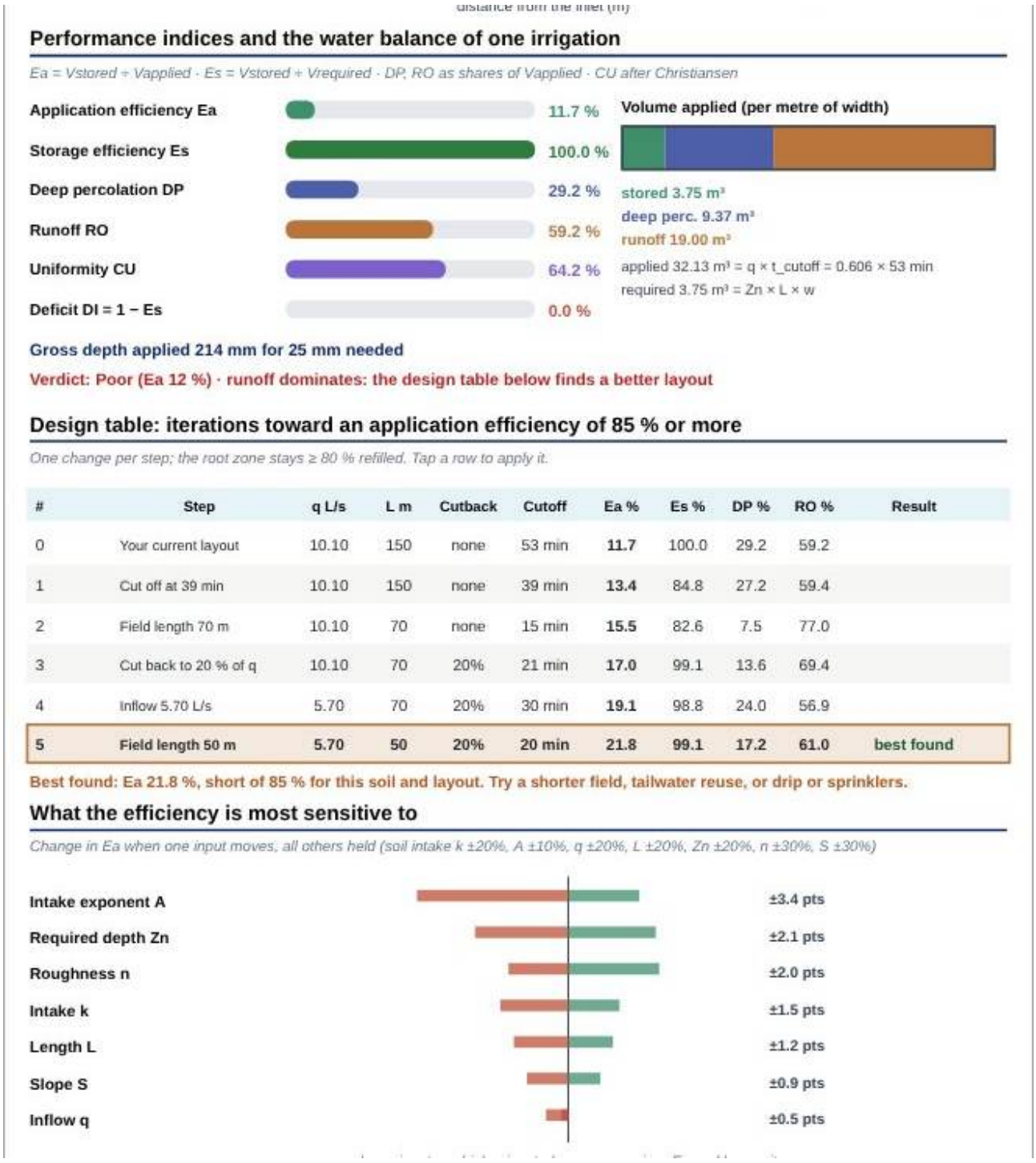


Figure 4.4: Surface irrigation design, Coarse Sand, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

lower input ◄ ► higher input · bars: green raises Ea, red lowers it

What this means for the season and the farm

Change the date or the year in Properties. The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**490 mm → 19.6 irrigations of 25 mm**
Gross water per irrigation	**214 mm (Ea 12 %)**
Gross water per season	**4198 mm = 41978 m³ per hectare**
Units run at once N0M = Qs ÷ q	**3.0 (30 L/s ÷ 10.10 L/s)**
Sets to cover 120 m of width	**40.4 sets × 53 min = 35.7 h per irrigation**
Pumping per season	**700 hours**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	35	300	35.7 h	10 days
2	**26 May 2027**	214 days	35	300	35.7 h	8 days
3	**3 Jun 2027**	222 days	35	300	35.7 h	7 days
4	**10 Jun 2027**	229 days	35	300	35.7 h	7 days
5	**17 Jun 2027**	236 days	35	300	35.7 h	7 days
6	**24 Jun 2027**	243 days	35	300	35.7 h	6 days
7	**30 Jun 2027**	249 days	35	300	35.7 h	12 days
8	**12 Jul 2027**	261 days	35	300	35.7 h	-

Intervals, efficiency and the advice in short

Everything above in one message for surface irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.50**
Irrigations in the season of this date	**14 (6 Apr 2026 to 31 Jul 2026)**
Interval between irrigations	**8.9 days on average (shortest 6, longest 17)**
Depth per irrigation	**25.0 mm net, 214.2 mm pumped**
Application efficiency Ea	**11.7 %**
Storage efficiency Es	**100.0 %**
Time per round	**35.7 h**

When to irrigate

Next irrigation: 16 May 2027 (in 204 days), about 35 mm net, 300 mm pumped. Then irrigate about every 9 days (between 6 and 17 days, shorter in hot weather and longer after rain). With this surface design 12 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 35.7 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 4.5: Surface irrigation design, Coarse Sand, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

4.2 Drip irrigation design (Coarse Sand)

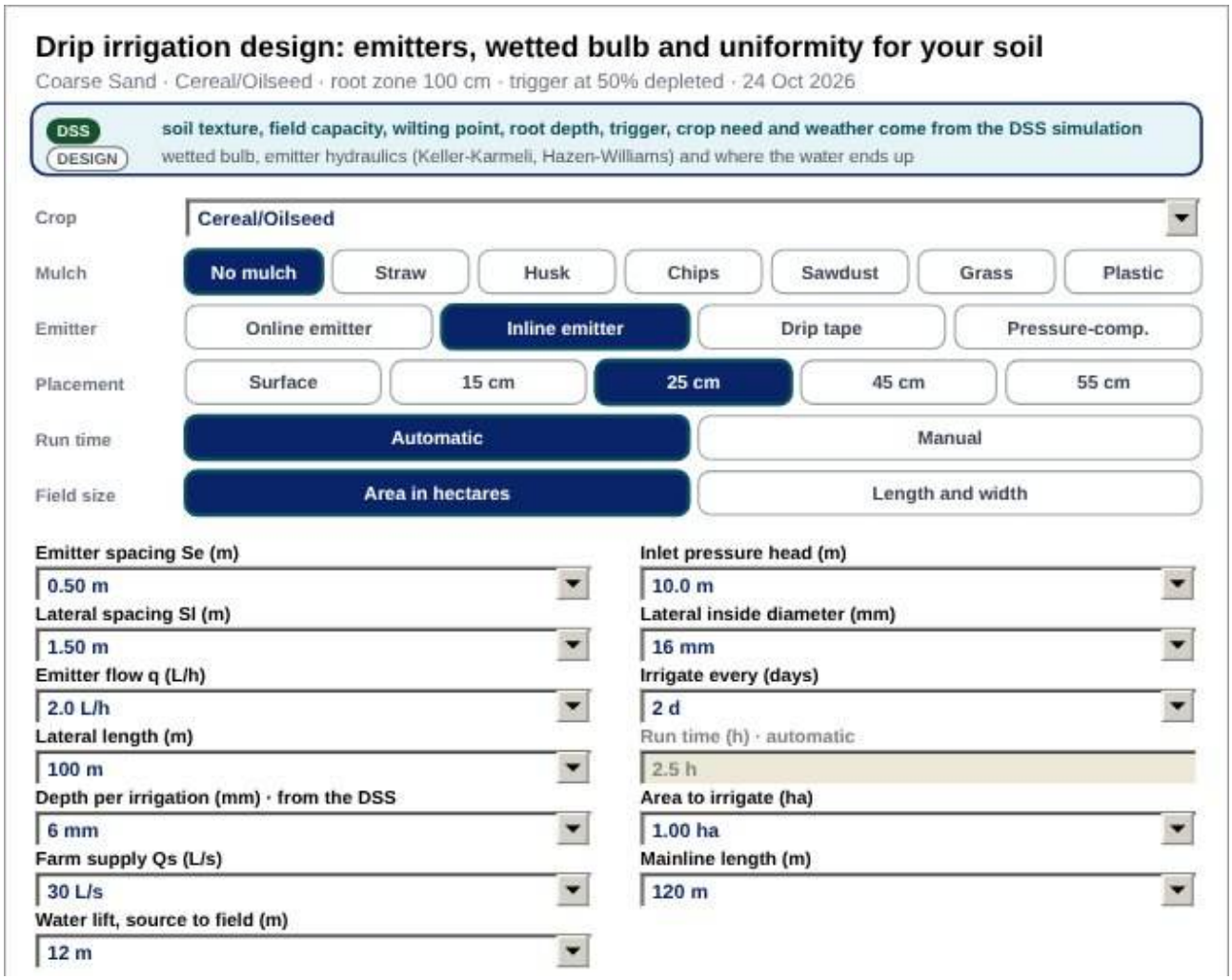


Figure 4.6: Drip irrigation design, Coarse Sand, part 1 of 6.

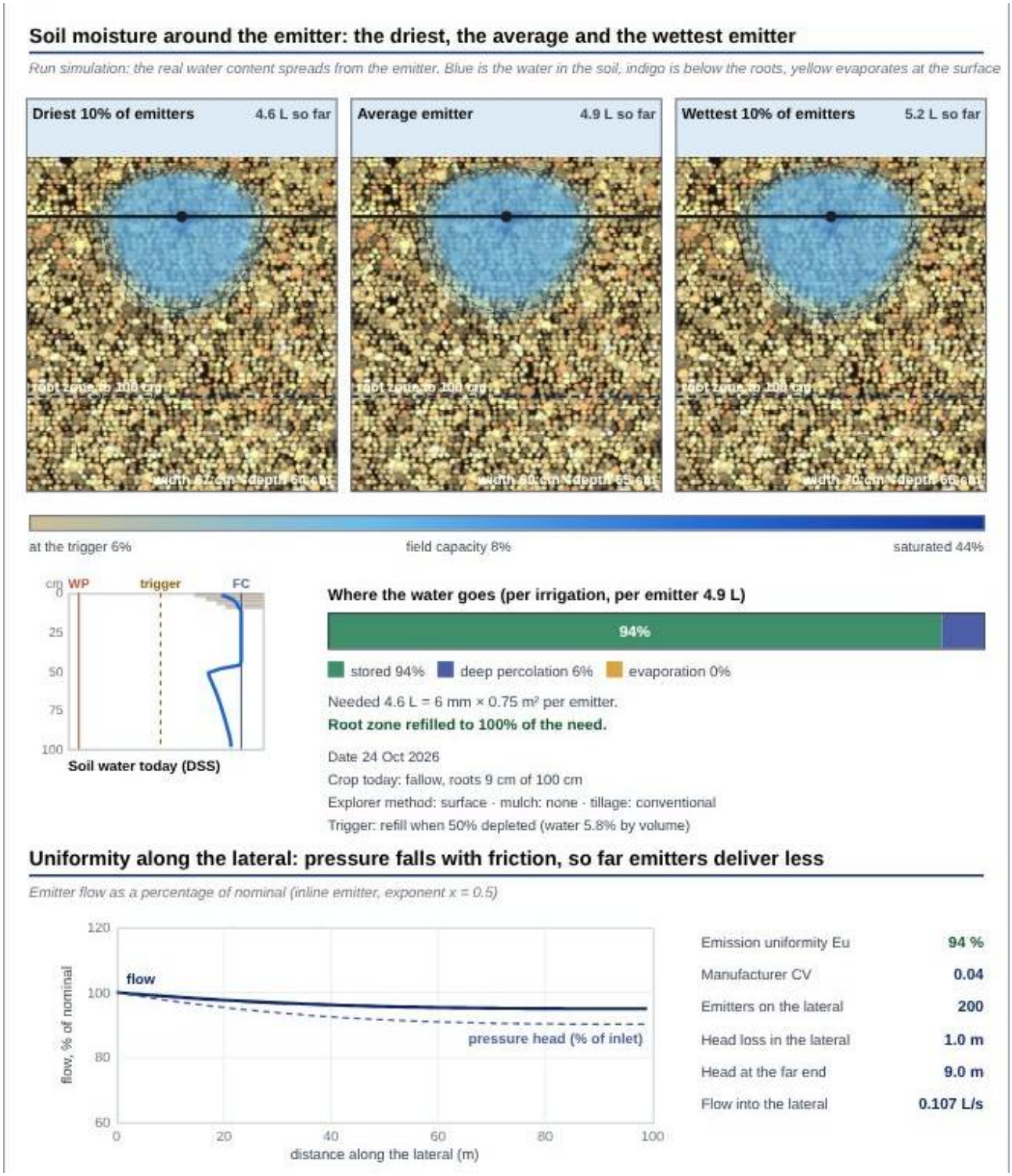


Figure 4.7: Drip irrigation design, Coarse Sand, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

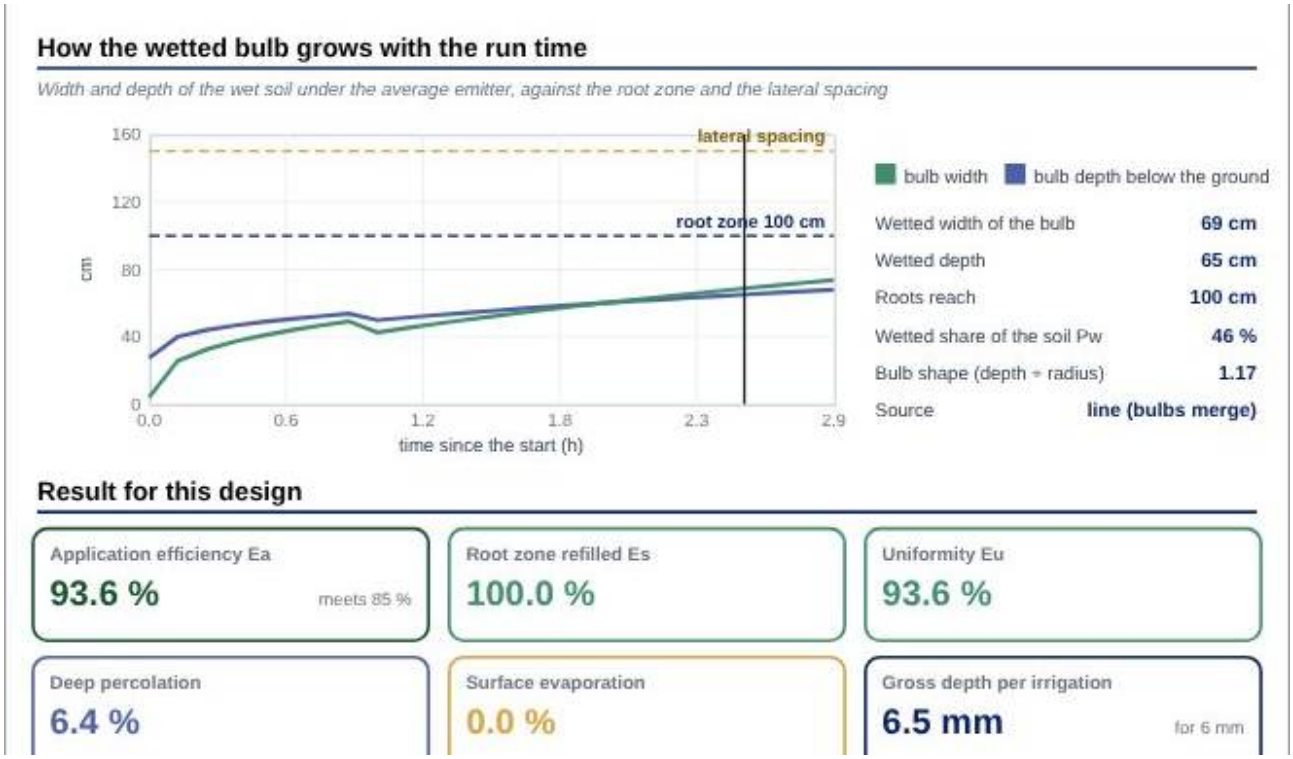
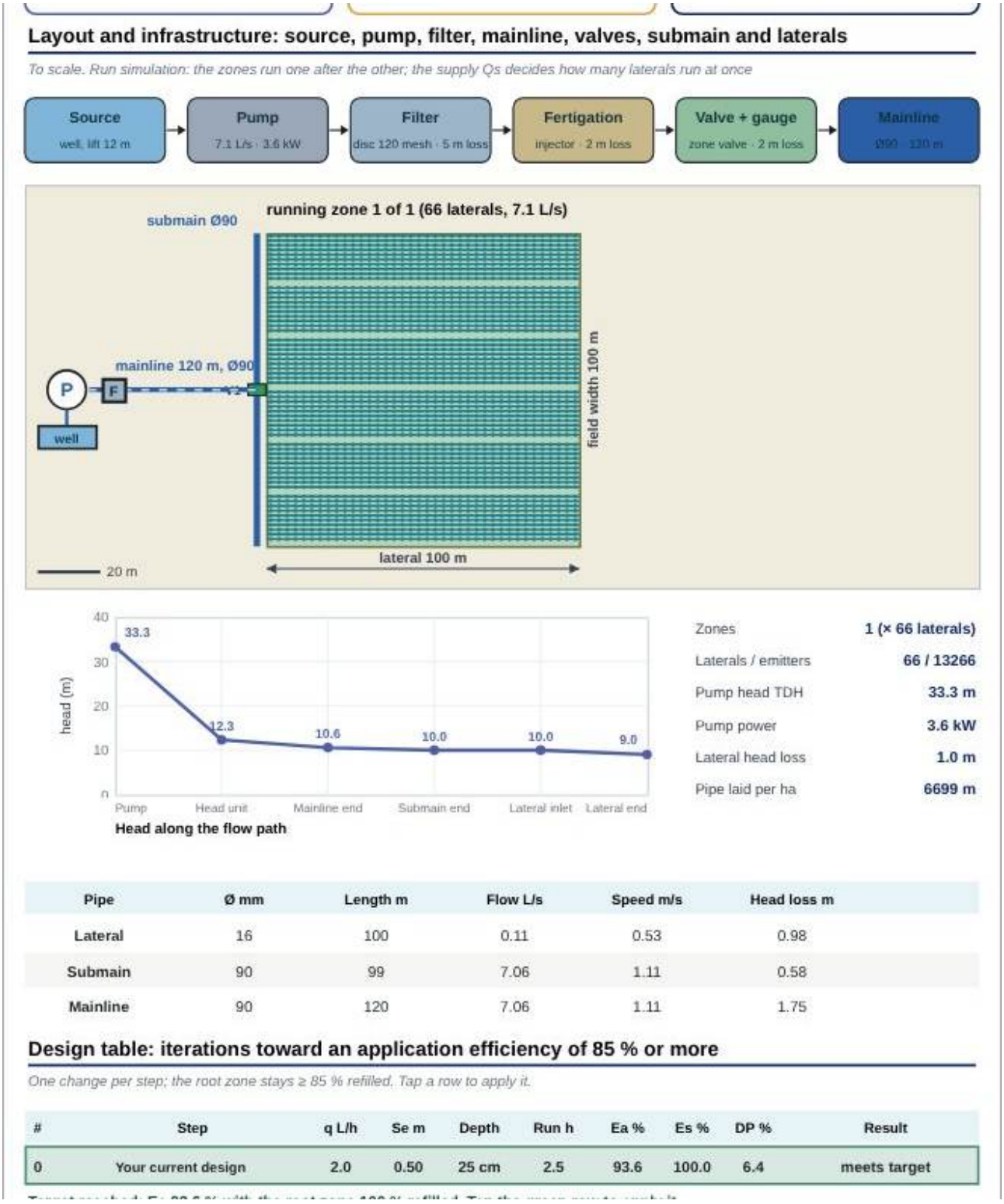


Figure 4.8: Drip irrigation design, Coarse Sand, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	2.5	93.6	100.0	6.4	meets target

Figure 4.9: Drip irrigation design, Coarse Sand, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**490 mm → 80 irrigations of 6 mm**
Gross water per irrigation	**6.5 mm (Ea 94 %)**
Gross water per season	**523 mm = 5234 m³ per hectare**
Peak crop need / interval	**3.1 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 2.5 h = 2.5 h per round**
Pumping per season	**204 hours · 724 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	35	37	2.5 h	10 days
2	**26 May 2027**	214 days	35	37	2.5 h	8 days
3	**3 Jun 2027**	222 days	35	37	2.5 h	7 days
4	**10 Jun 2027**	229 days	35	37	2.5 h	7 days
5	**17 Jun 2027**	236 days	35	37	2.5 h	7 days
6	**24 Jun 2027**	243 days	35	37	2.5 h	6 days
7	**30 Jun 2027**	249 days	35	37	2.5 h	12 days
8	**12 Jul 2027**	261 days	35	37	2.5 h	-

Figure 4.10: Drip irrigation design, Coarse Sand, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.

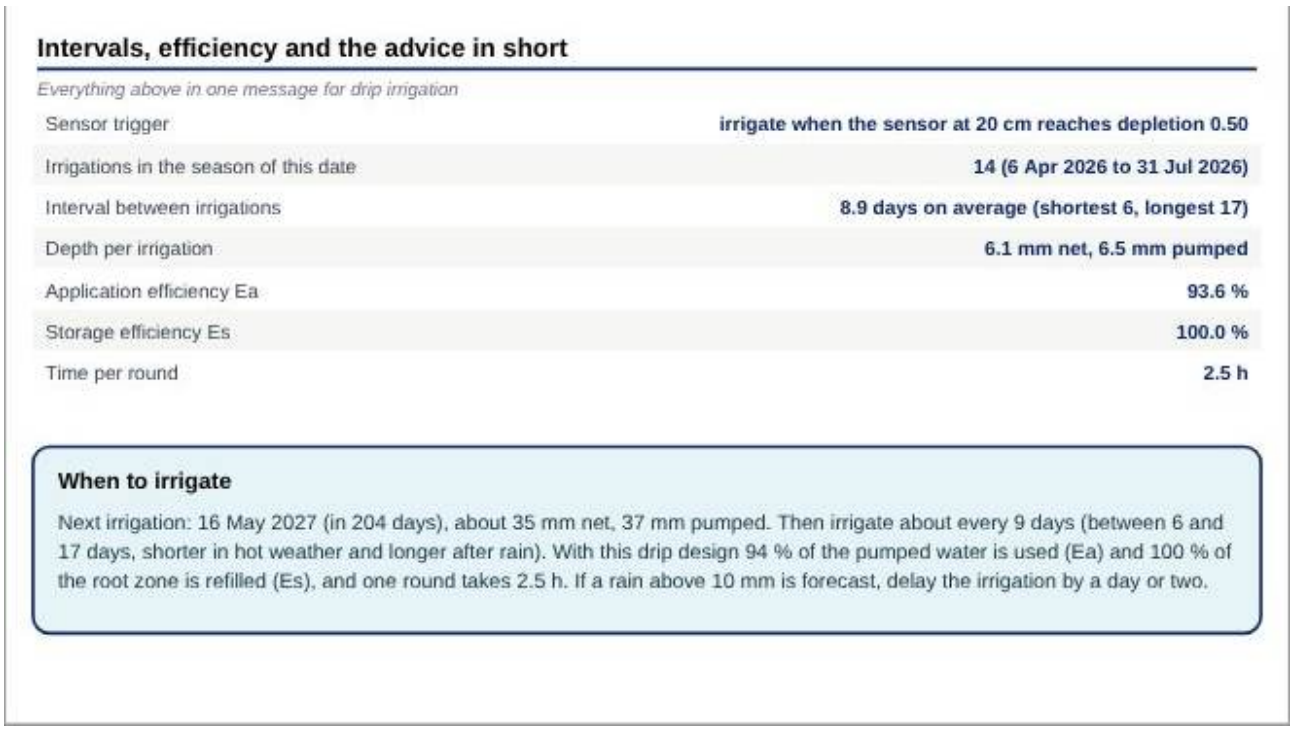


Figure 4.11: Drip irrigation design, Coarse Sand, part 6 of 6: Intervals, efficiency and the advice in short.

4.3 Sprinkler design and pipes (Coarse Sand)

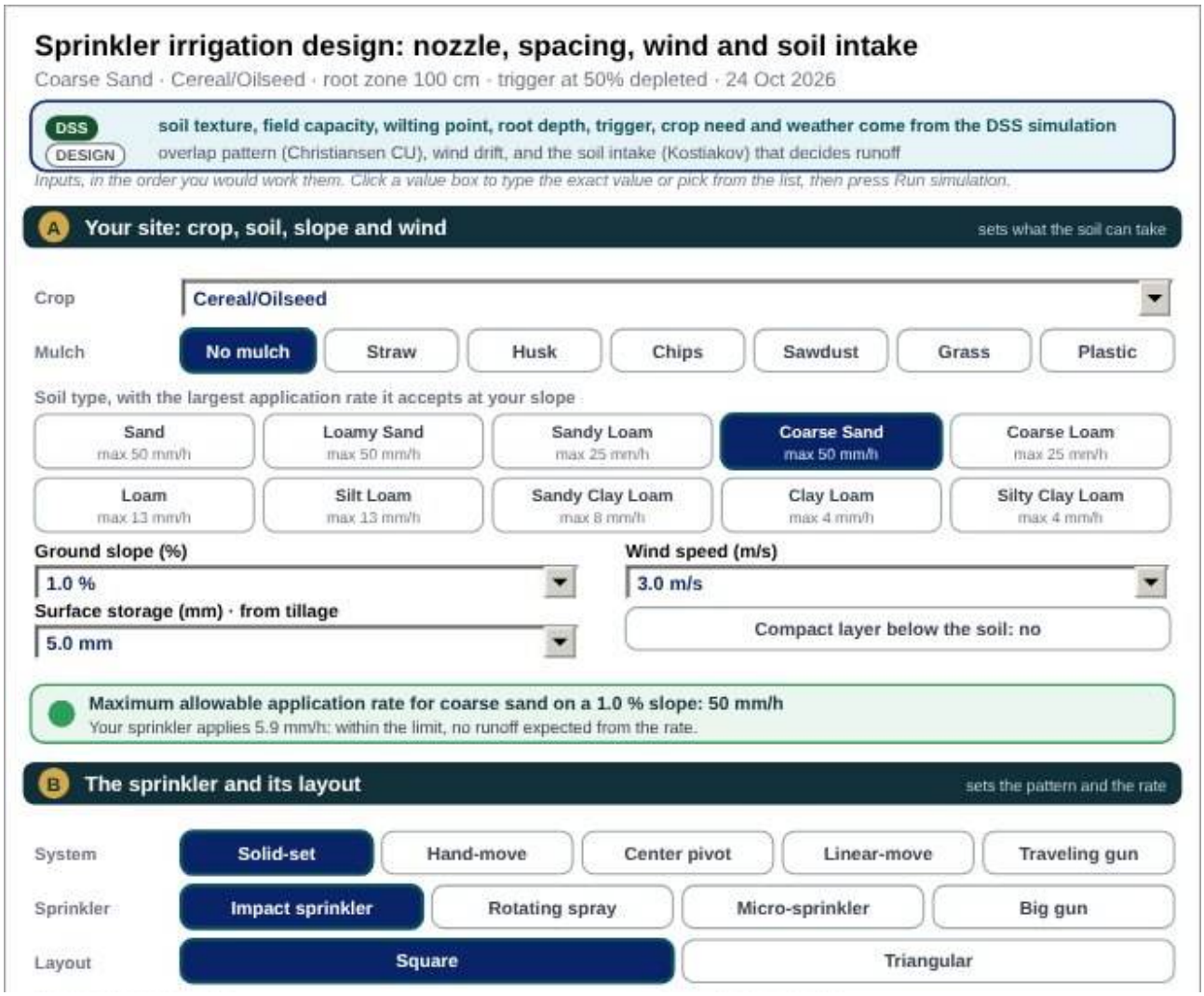


Figure 4.12: Sprinkler design and pipes, Coarse Sand, part 1 of 12.

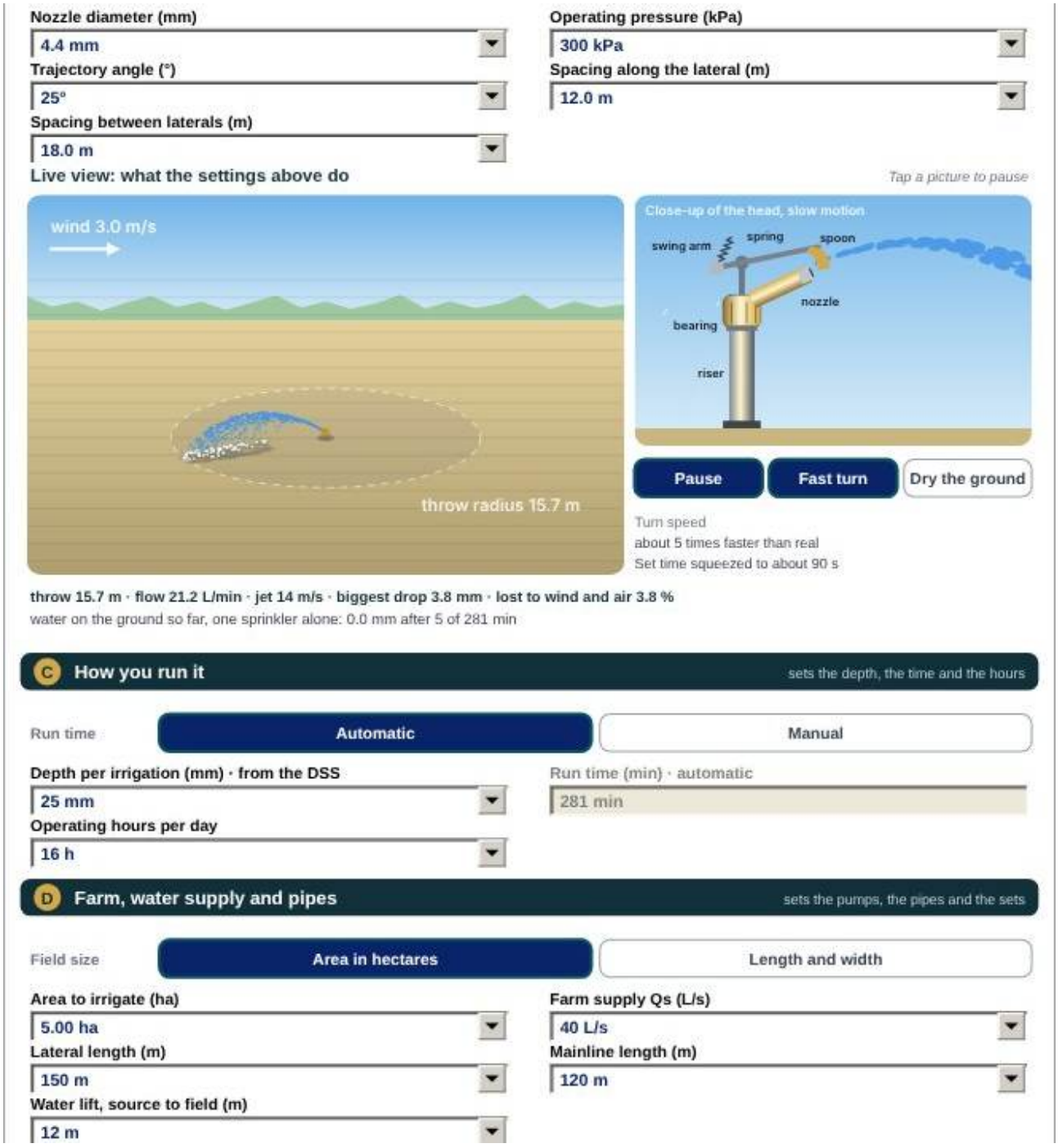


Figure 4.13: Sprinkler design and pipes, Coarse Sand, part 2 of 12.

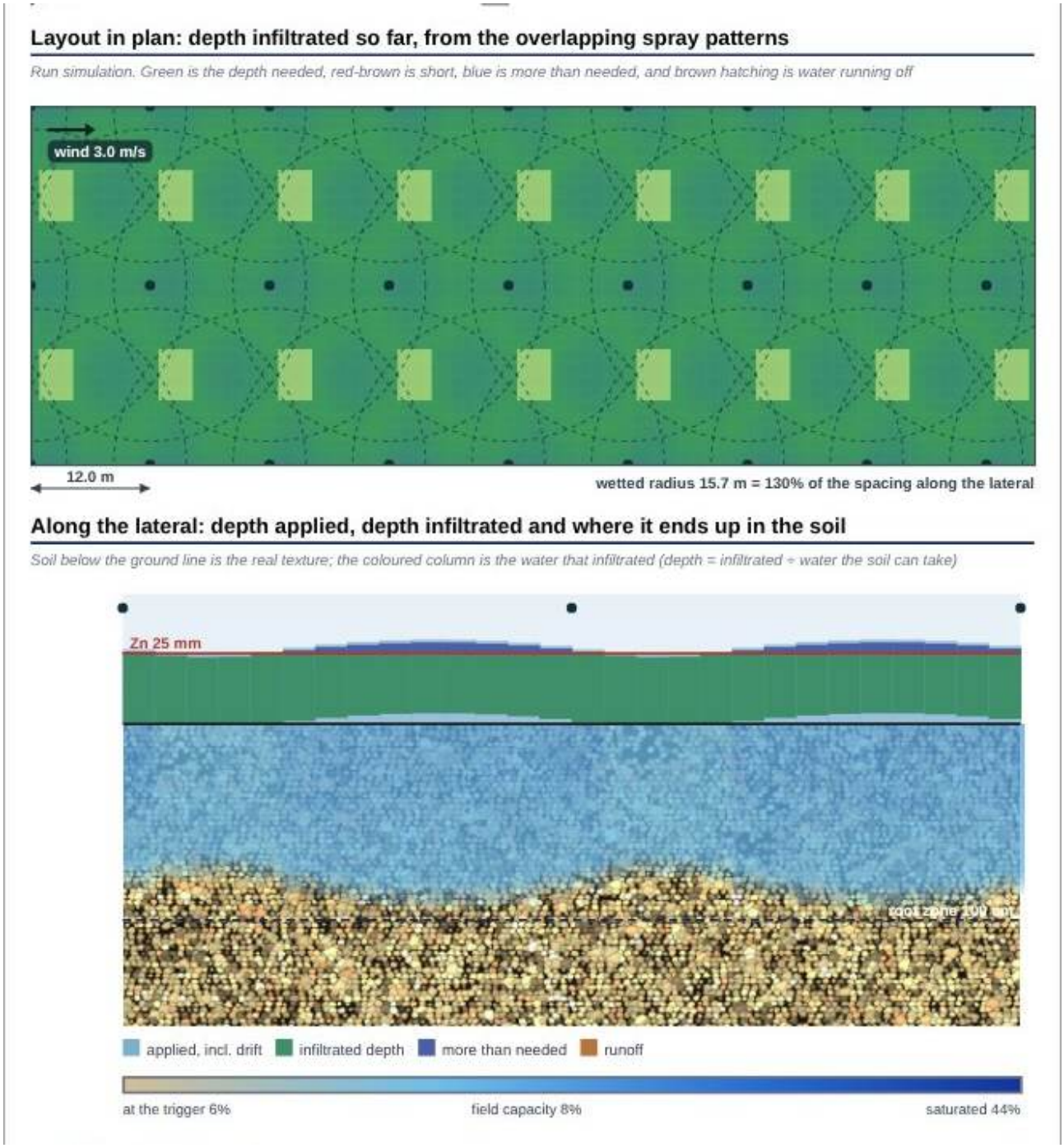


Figure 4.14: Sprinkler design and pipes, Coarse Sand, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

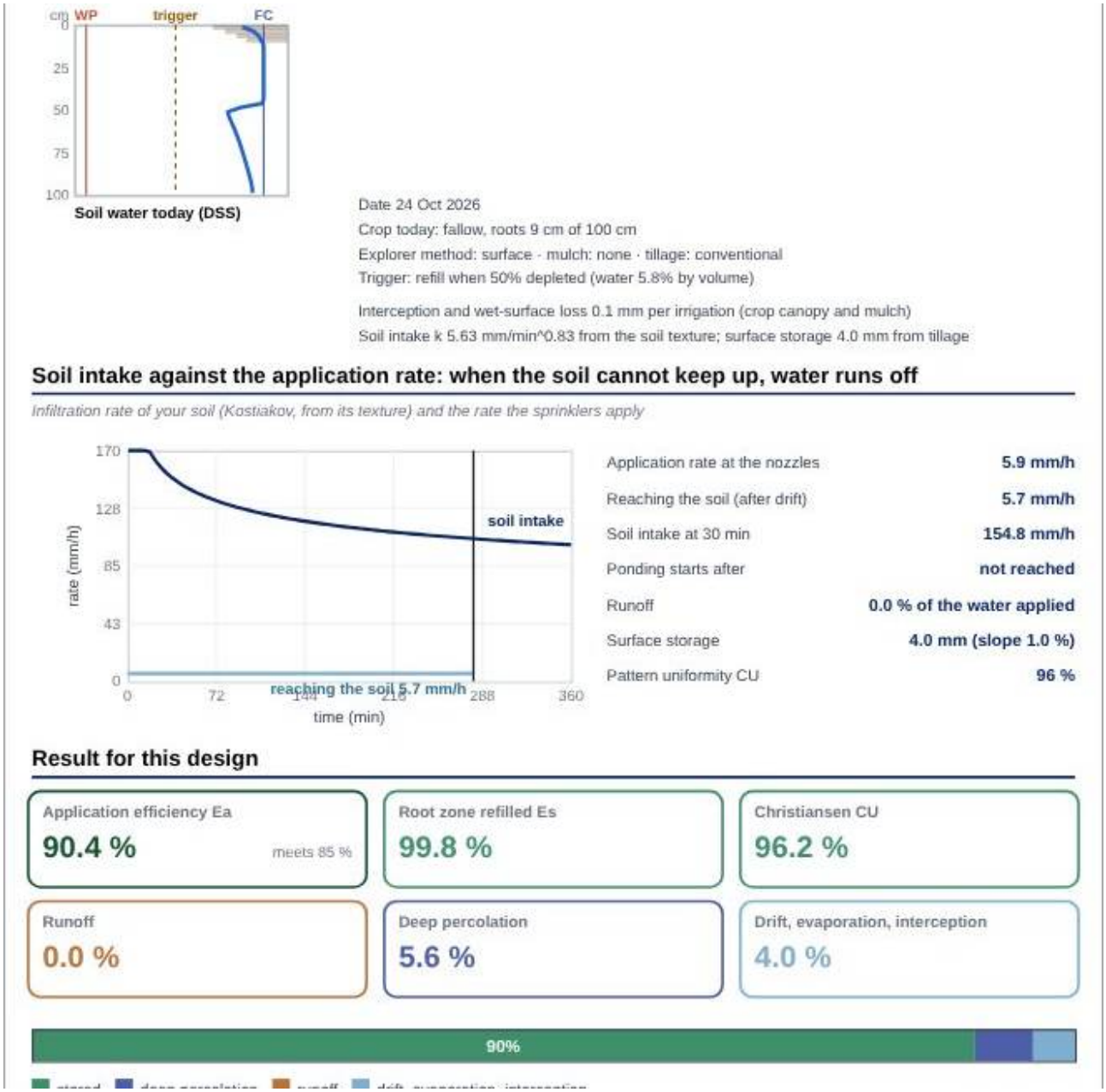


Figure 4.15: Sprinkler design and pipes, Coarse Sand, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

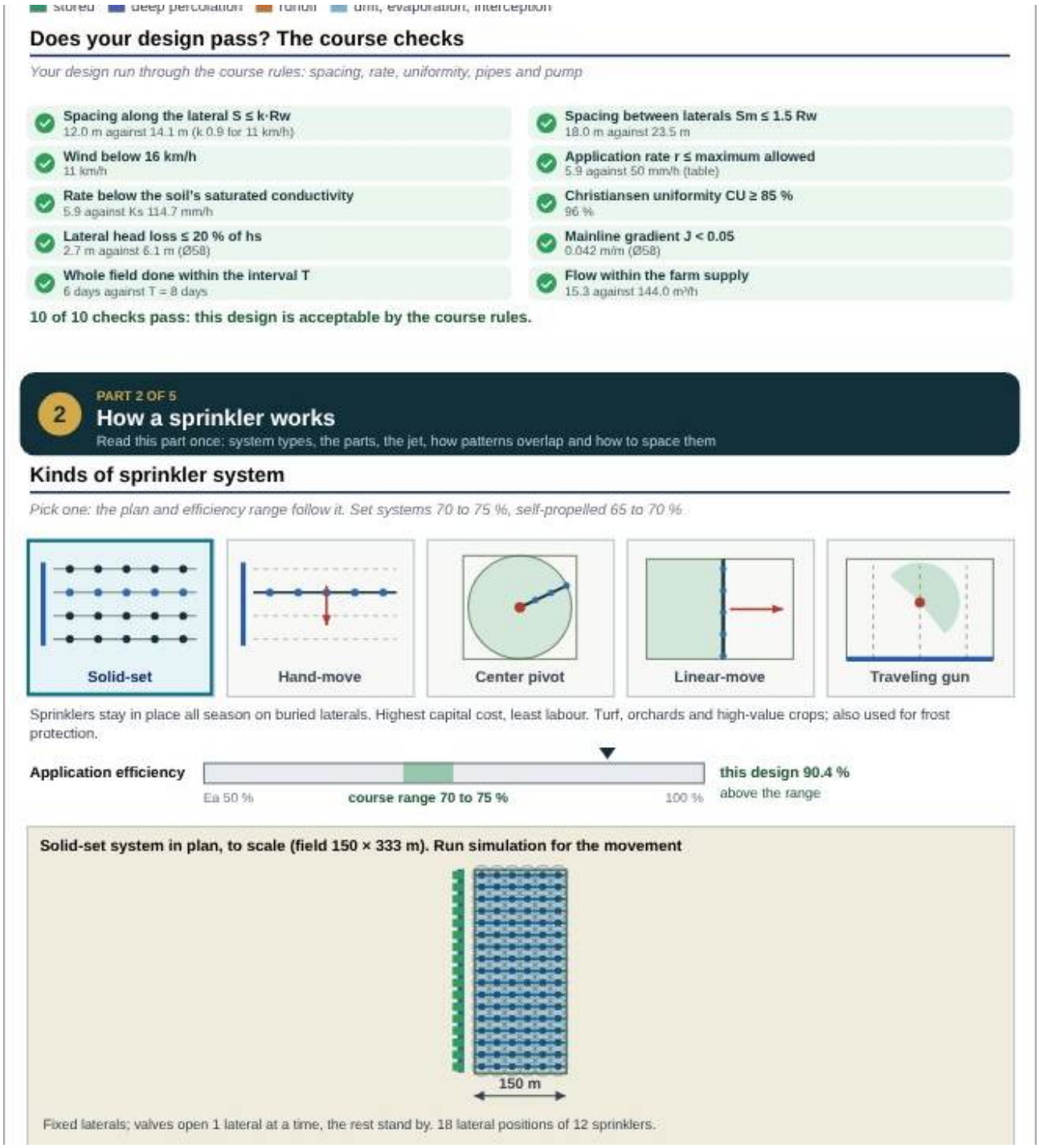


Figure 4.16: Sprinkler design and pipes, Coarse Sand, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

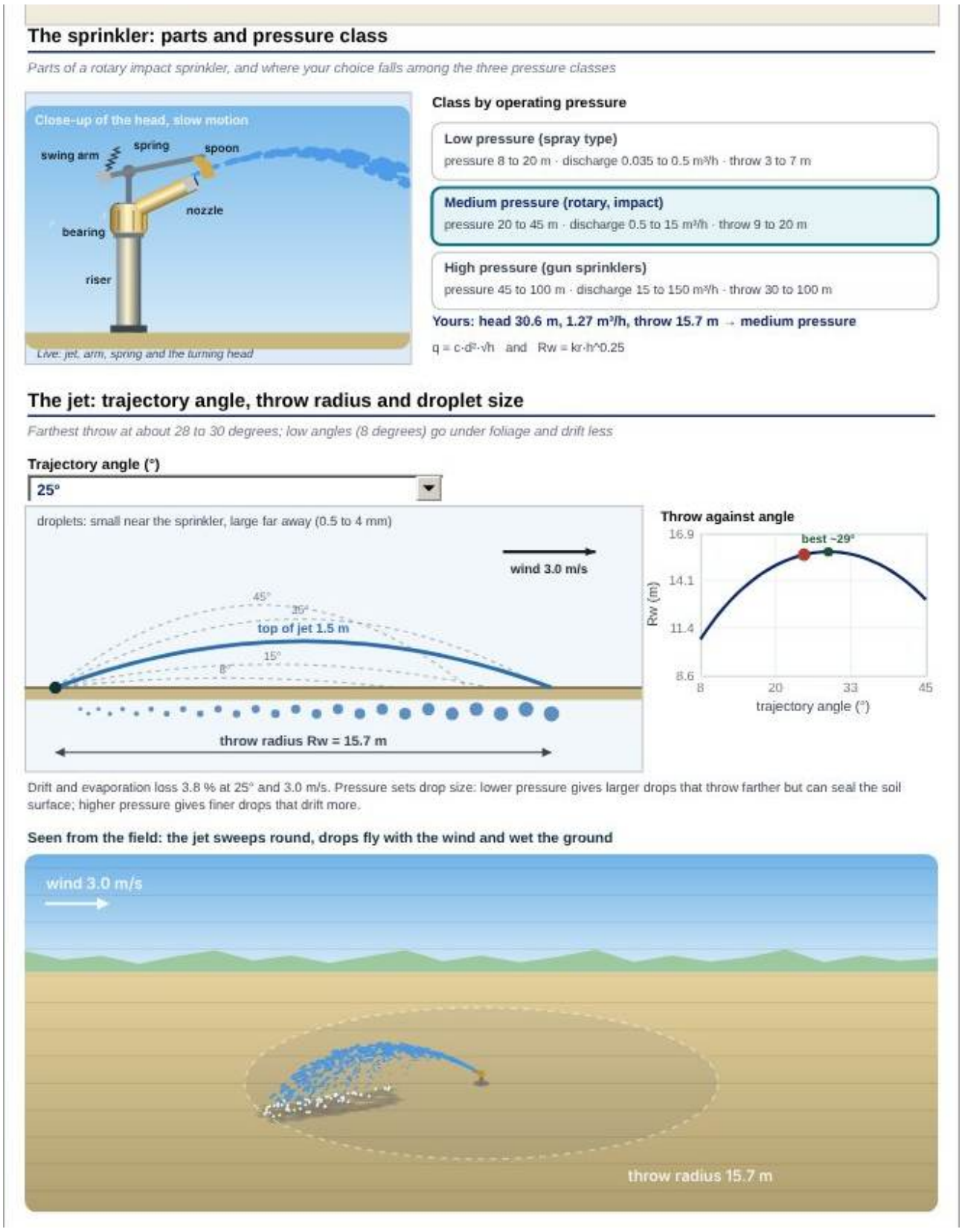


Figure 4.17: Sprinkler design and pipes, Coarse Sand, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

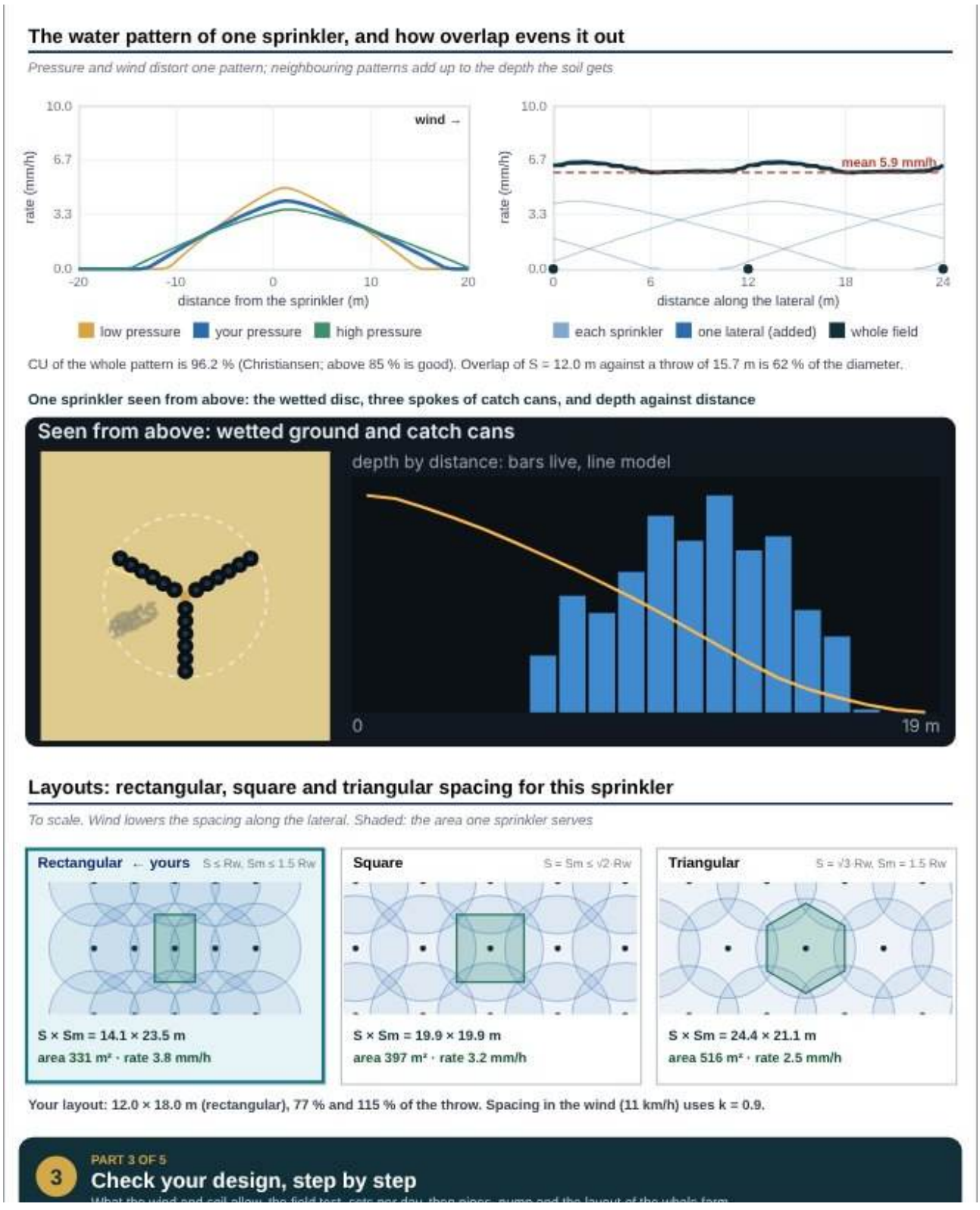


Figure 4.18: Sprinkler design and pipes, Coarse Sand, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

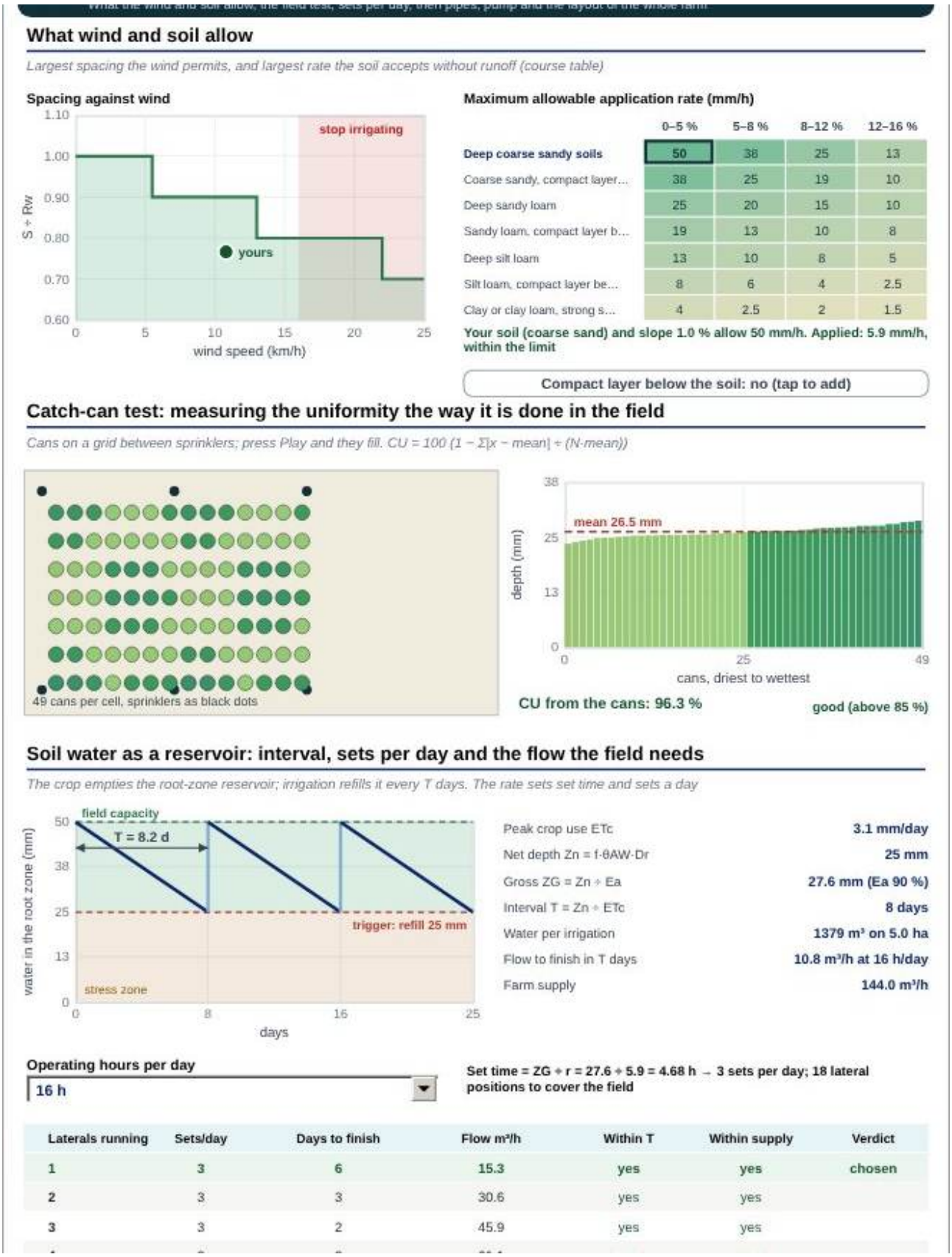


Figure 4.19: Sprinkler design and pipes, Coarse Sand, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

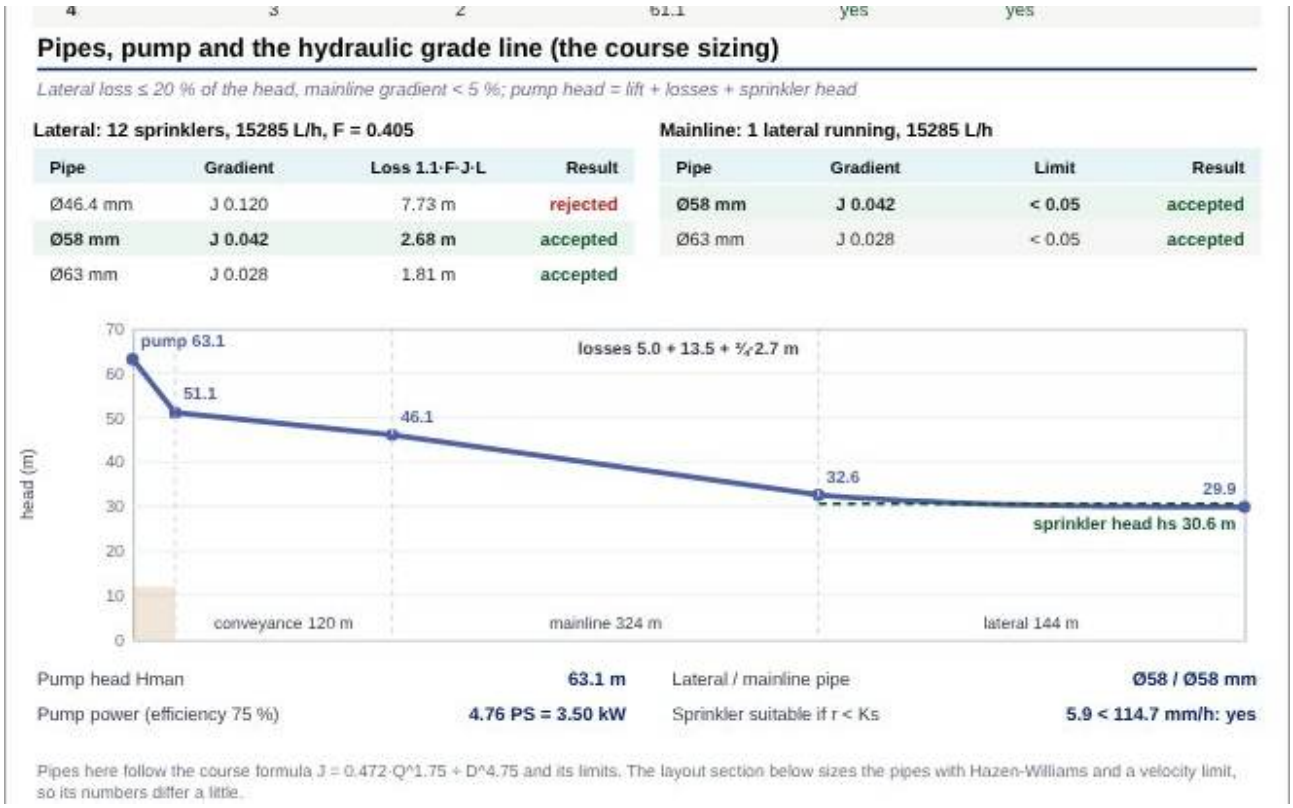
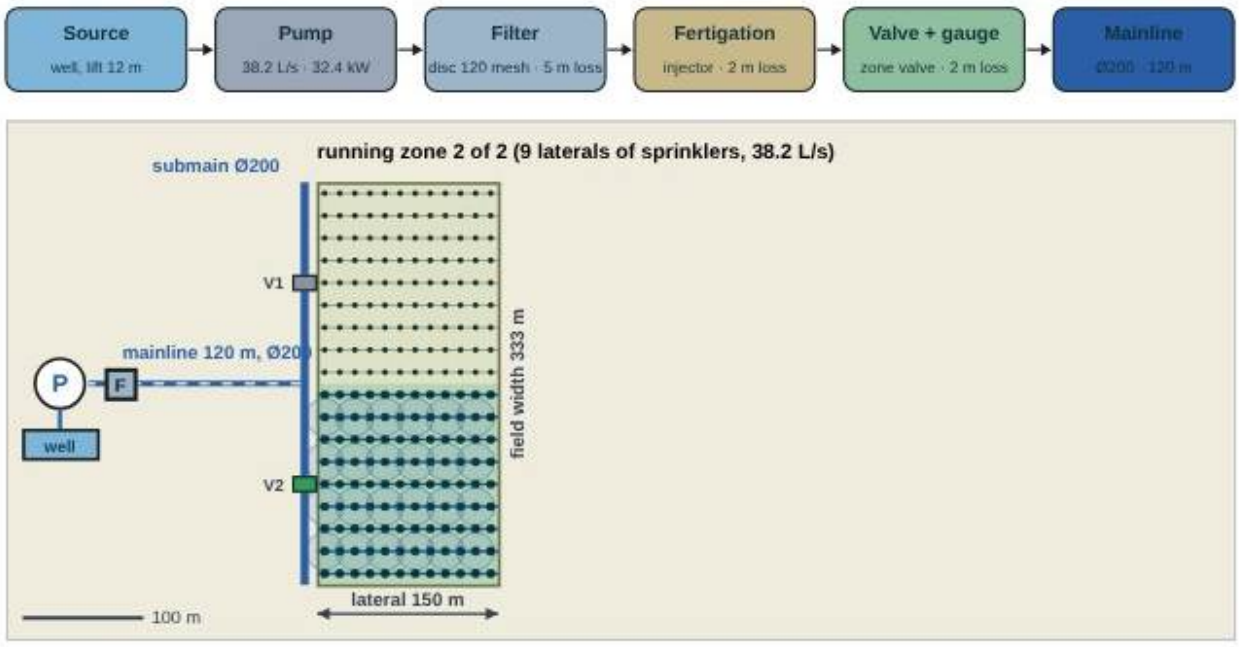
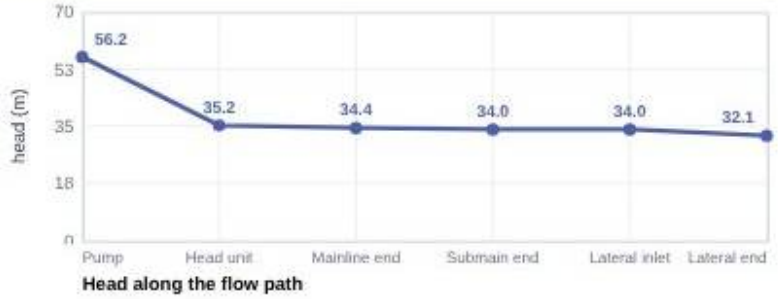


Figure 4.20: Sprinkler design and pipes, Coarse Sand, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4

Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

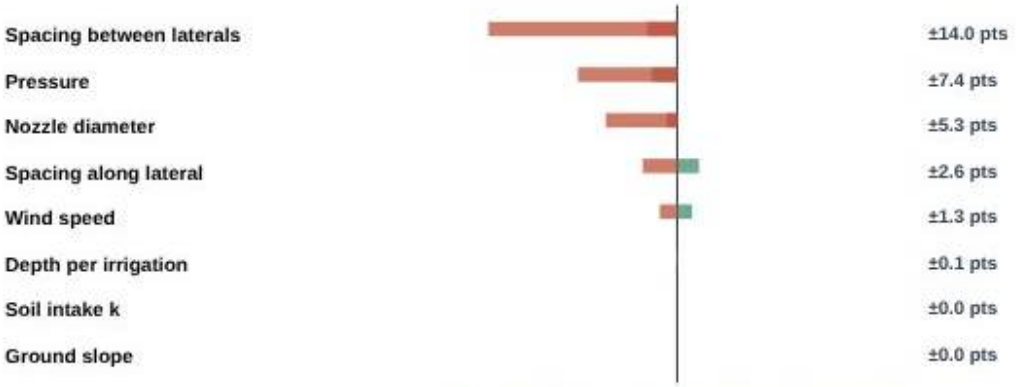
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	281	90.4	99.8	0.0	meets target

Figure 4.21: Sprinkler design and pipes, Coarse Sand, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.4 % with CU 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**490 mm → 20 irrigations of 25 mm**
Gross water per irrigation	**27.6 mm (Ea 90 %)**
Gross water per season	**541 mm = 5408 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 281 min = 14.0 h per round**
Pumping per season	**275 hours · 8924 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	35	39	14.0 h	10 days
2	**26 May 2027**	214 days	35	39	14.0 h	8 days
3	**3 Jun 2027**	222 days	35	39	14.0 h	7 days
4	**10 Jun 2027**	229 days	35	39	14.0 h	7 days
5	**17 Jun 2027**	236 days	35	39	14.0 h	7 days
6	**24 Jun 2027**	243 days	35	39	14.0 h	6 days
7	**30 Jun 2027**	249 days	35	39	14.0 h	12 days
8	**12 Jul 2027**	261 days	35	39	14.0 h	-

Figure 4.22: Sprinkler design and pipes, Coarse Sand, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.

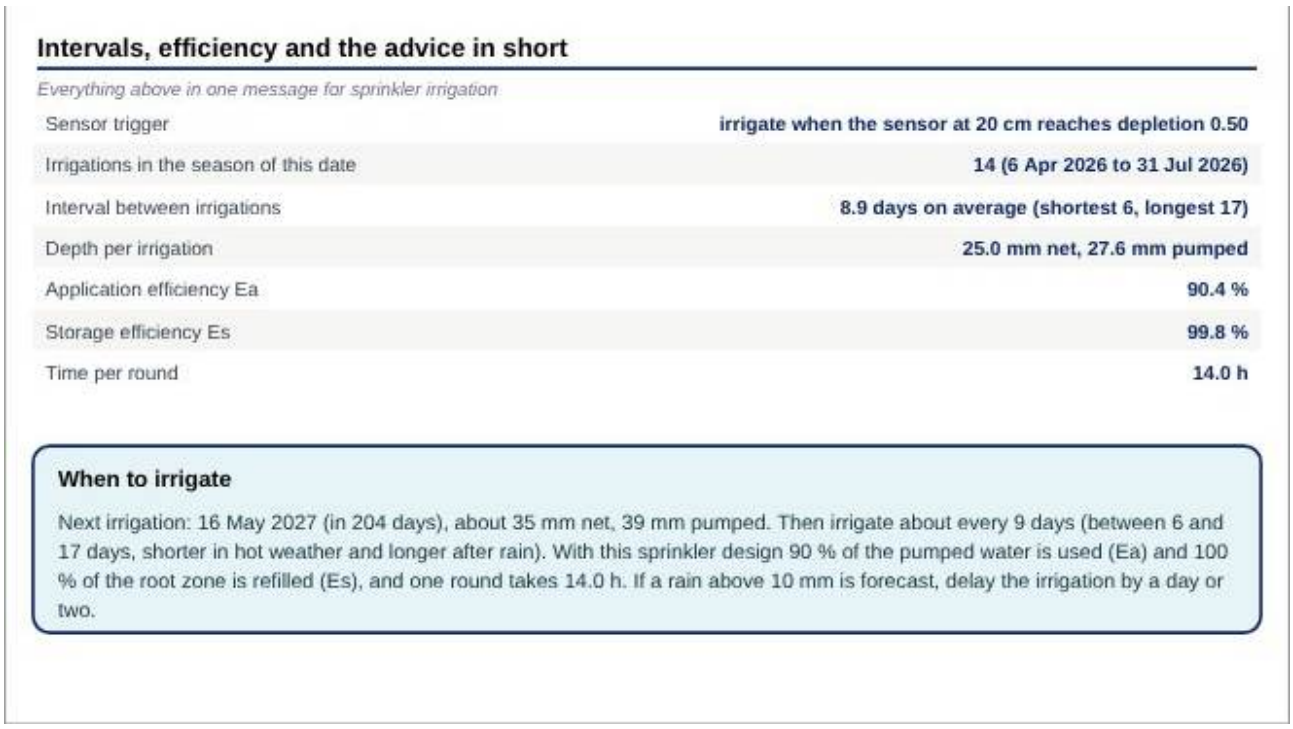


Figure 4.23: Sprinkler design and pipes, Coarse Sand, part 12 of 12: Intervals, efficiency and the advice in short.

4.4 Virtual farm of the surface design (Coarse Sand)

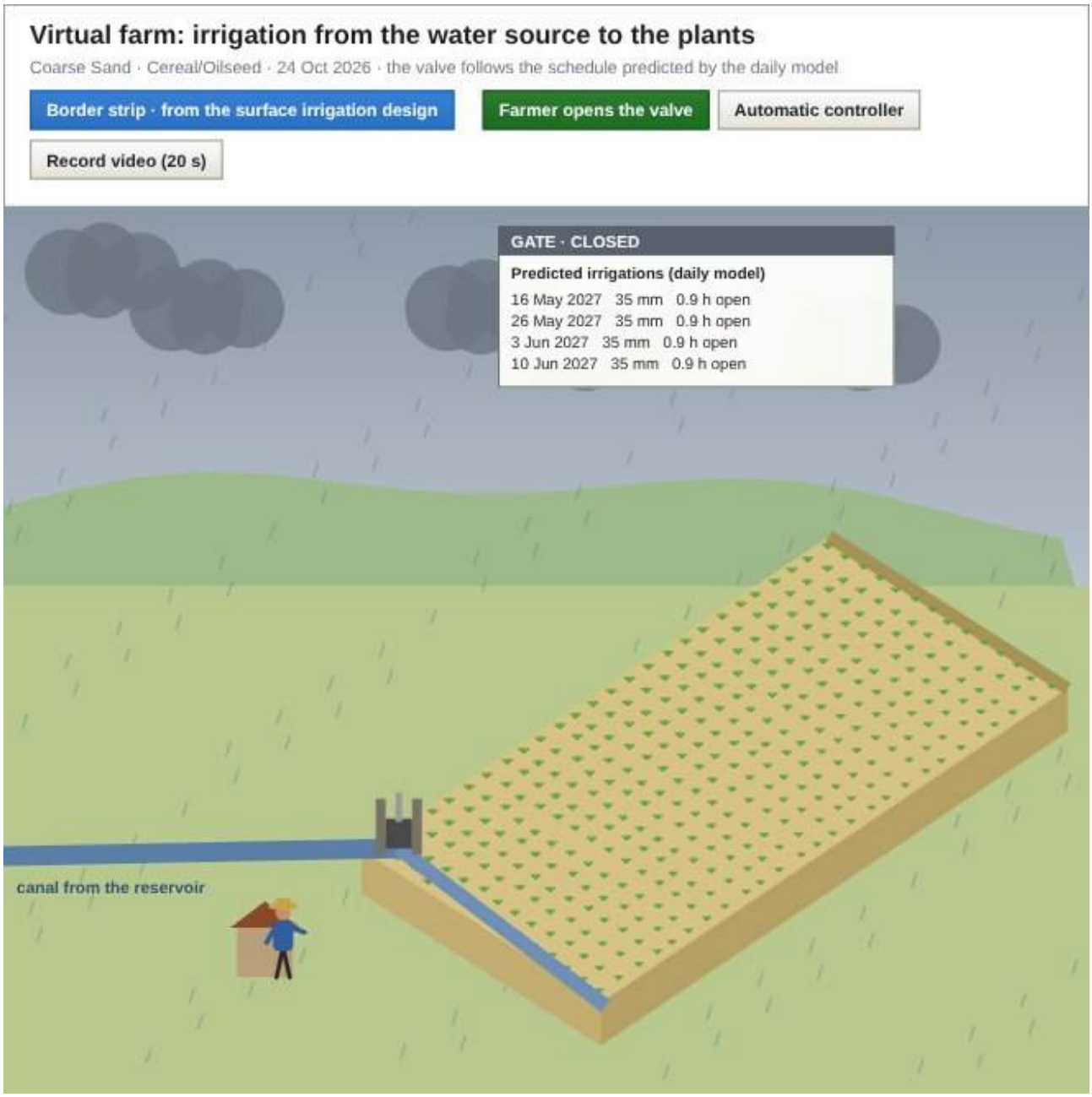


Figure 4.24: Virtual farm, surface design, Coarse Sand, part 1 of 3.

Figure 4.25: Virtual farm, surface design, Coarse Sand, part 2 of 3.

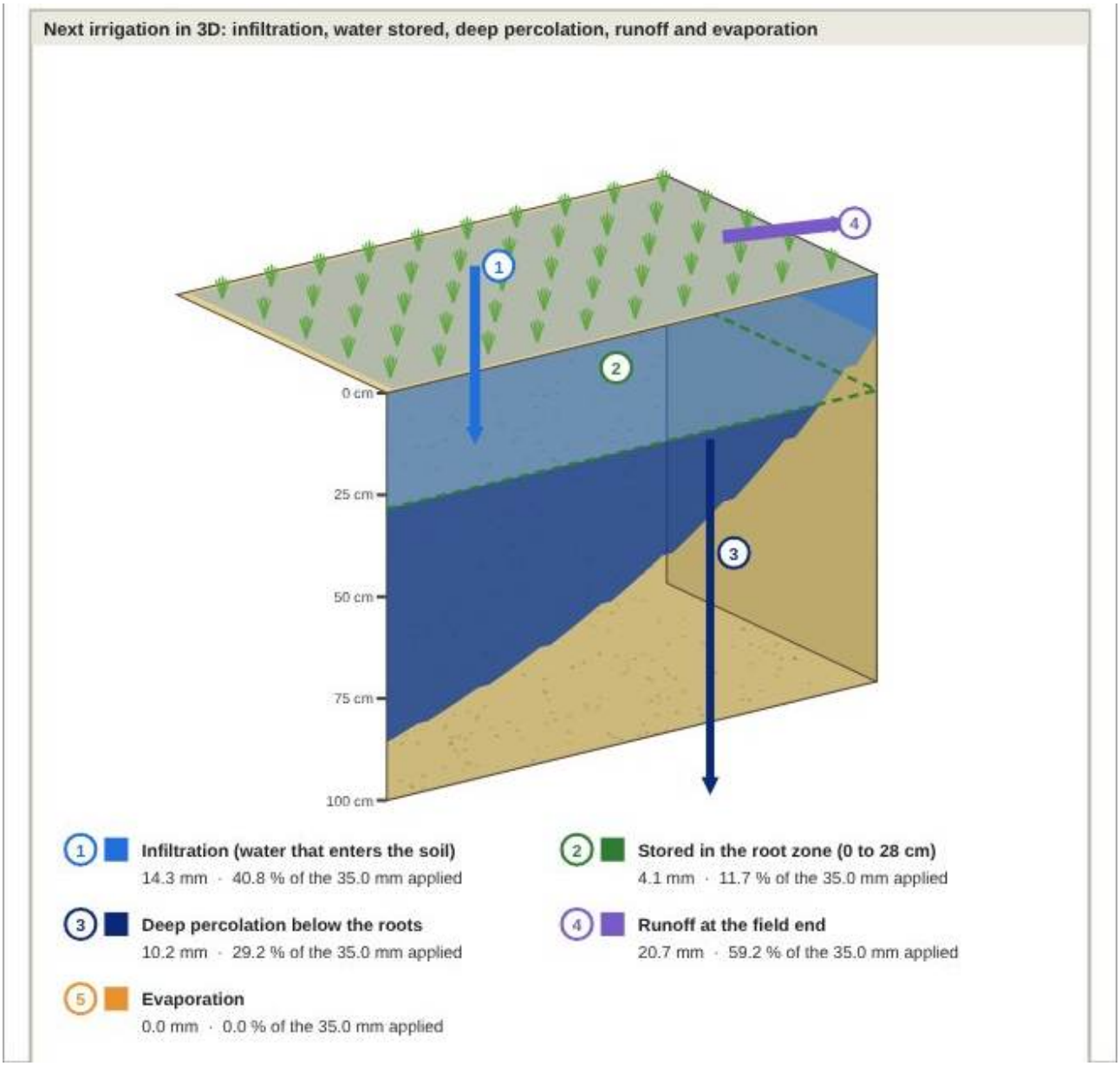


Figure 4.26: Virtual farm, surface design, Coarse Sand, part 3 of 3.

4.5 Virtual farm of the drip design (Coarse Sand)

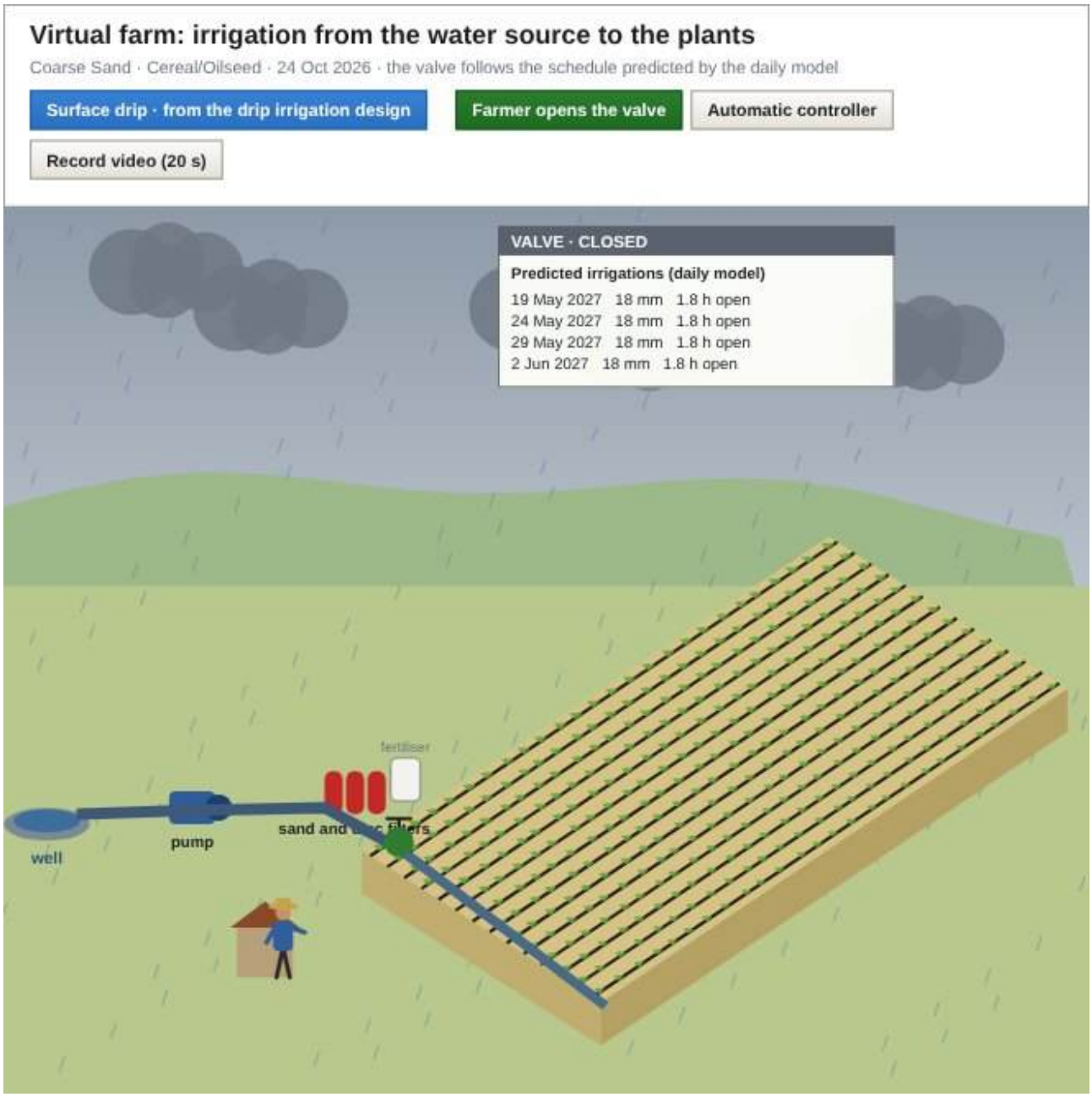


Figure 4.27: Virtual farm, drip design, Coarse Sand, part 1 of 3.

Figure 4.28: Virtual farm, drip design, Coarse Sand, part 2 of 3.

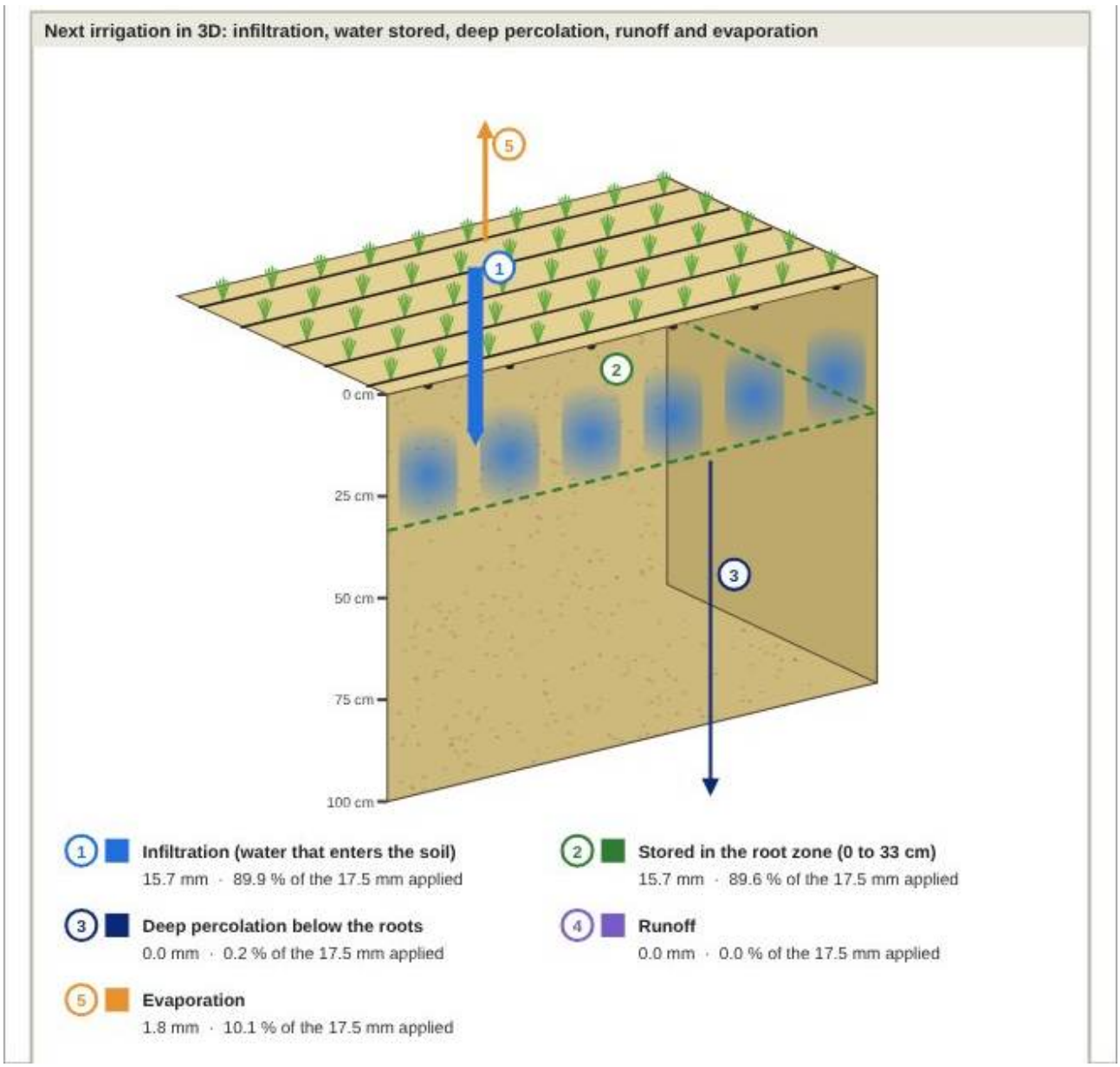


Figure 4.29: Virtual farm, drip design, Coarse Sand, part 3 of 3.

4.6 Virtual farm of the sprinkler design (Coarse Sand)

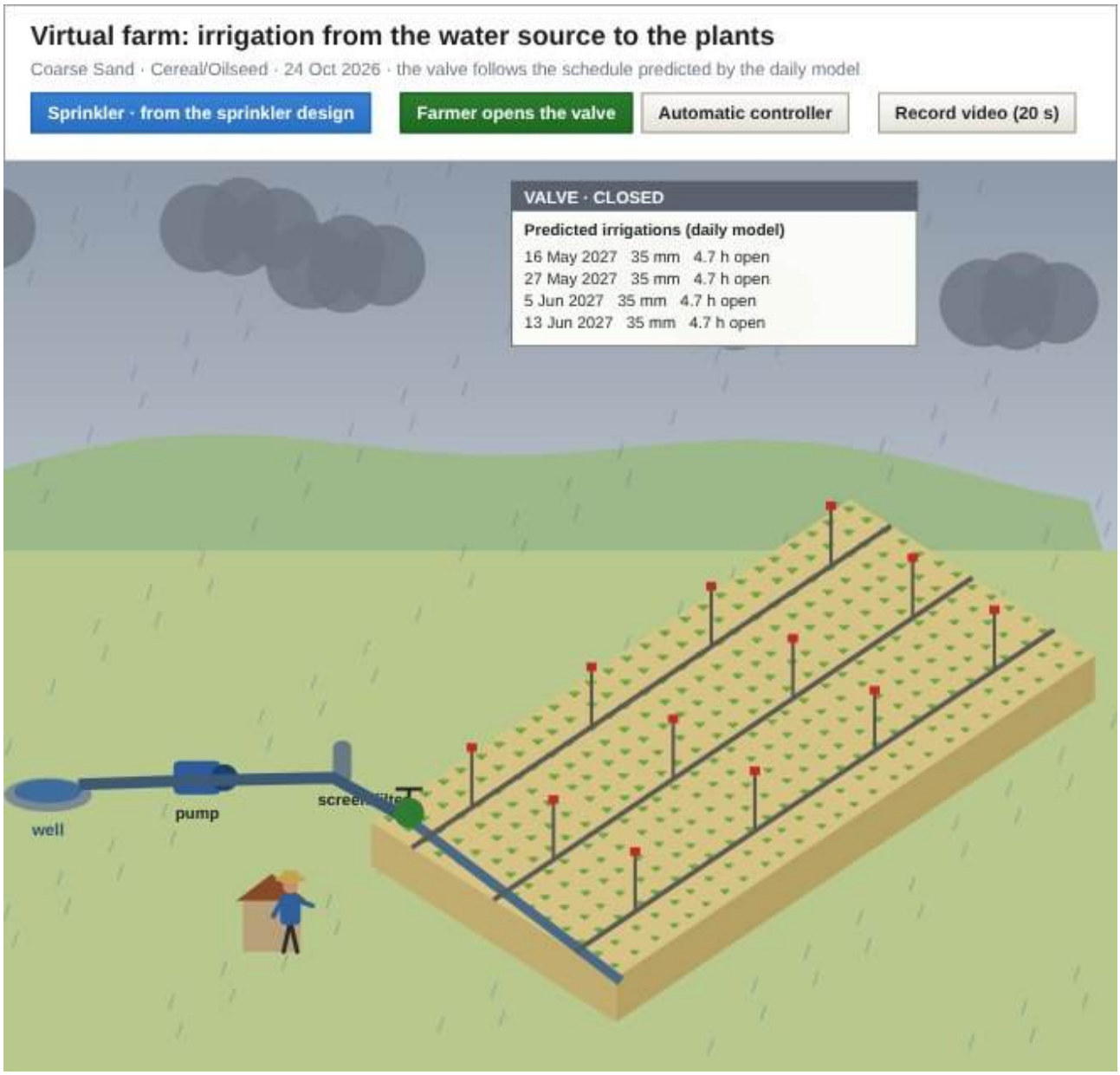


Figure 4.30: Virtual farm, sprinkler design, Coarse Sand, part 1 of 3.

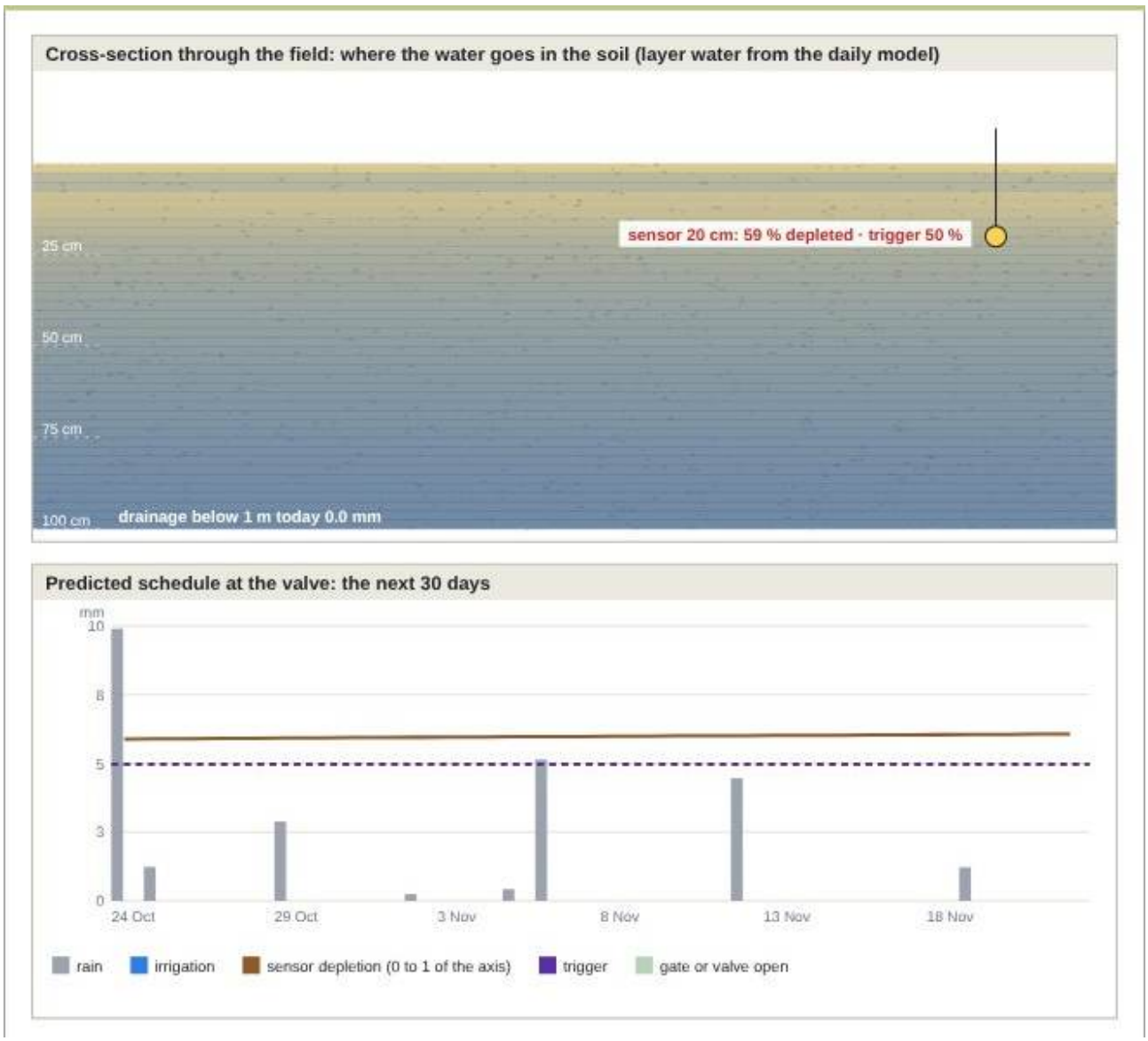


Figure 4.31: Virtual farm, sprinkler design, Coarse Sand, part 2 of 3.

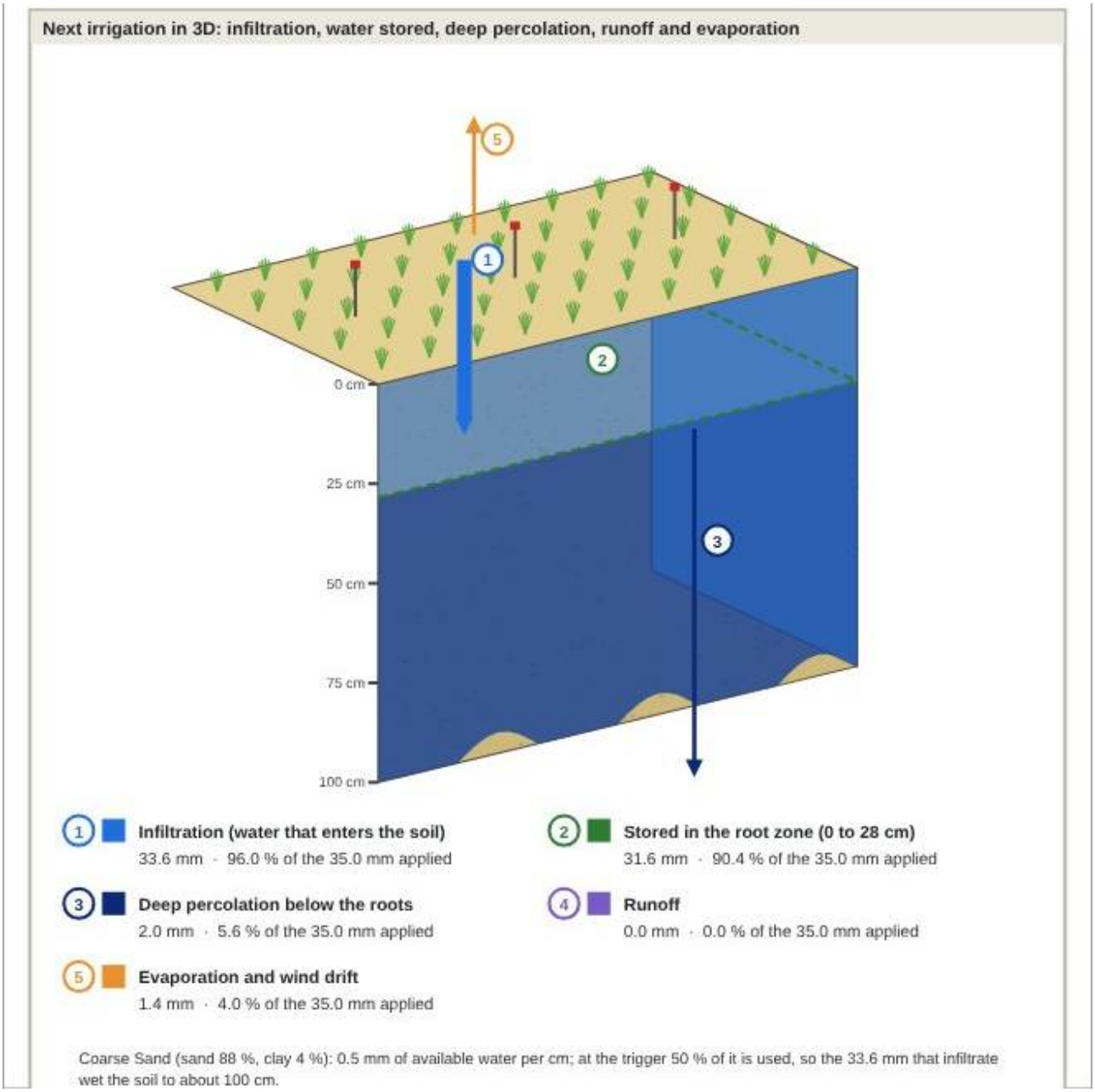


Figure 4.32: Virtual farm, sprinkler design, Coarse Sand, part 3 of 3.

4.7 Sensor and trigger (Coarse Sand)

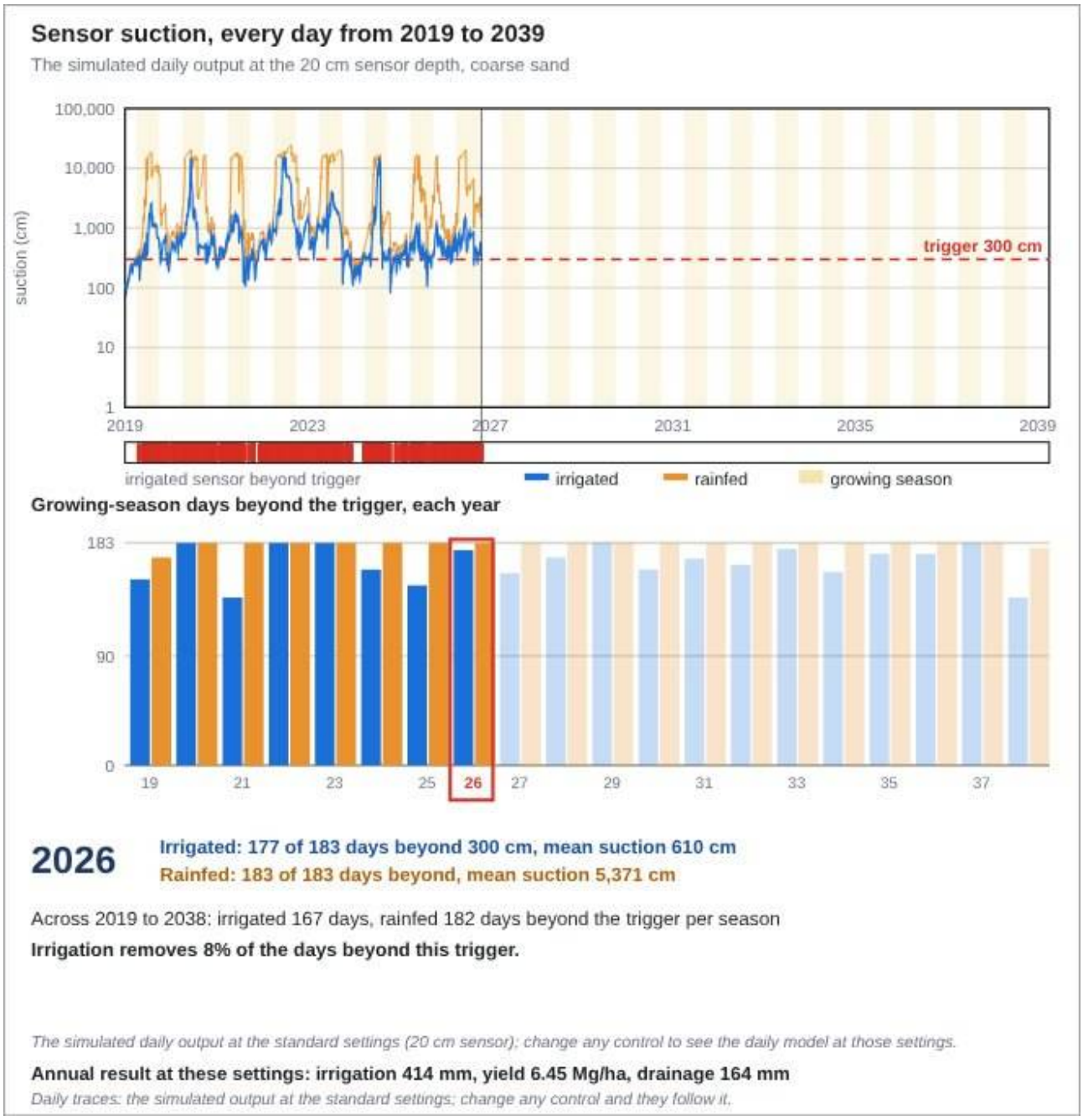


Figure 4.33: Sensor and trigger, Coarse Sand, part 1 of 1.

4.8 Trigger law (Coarse Sand)

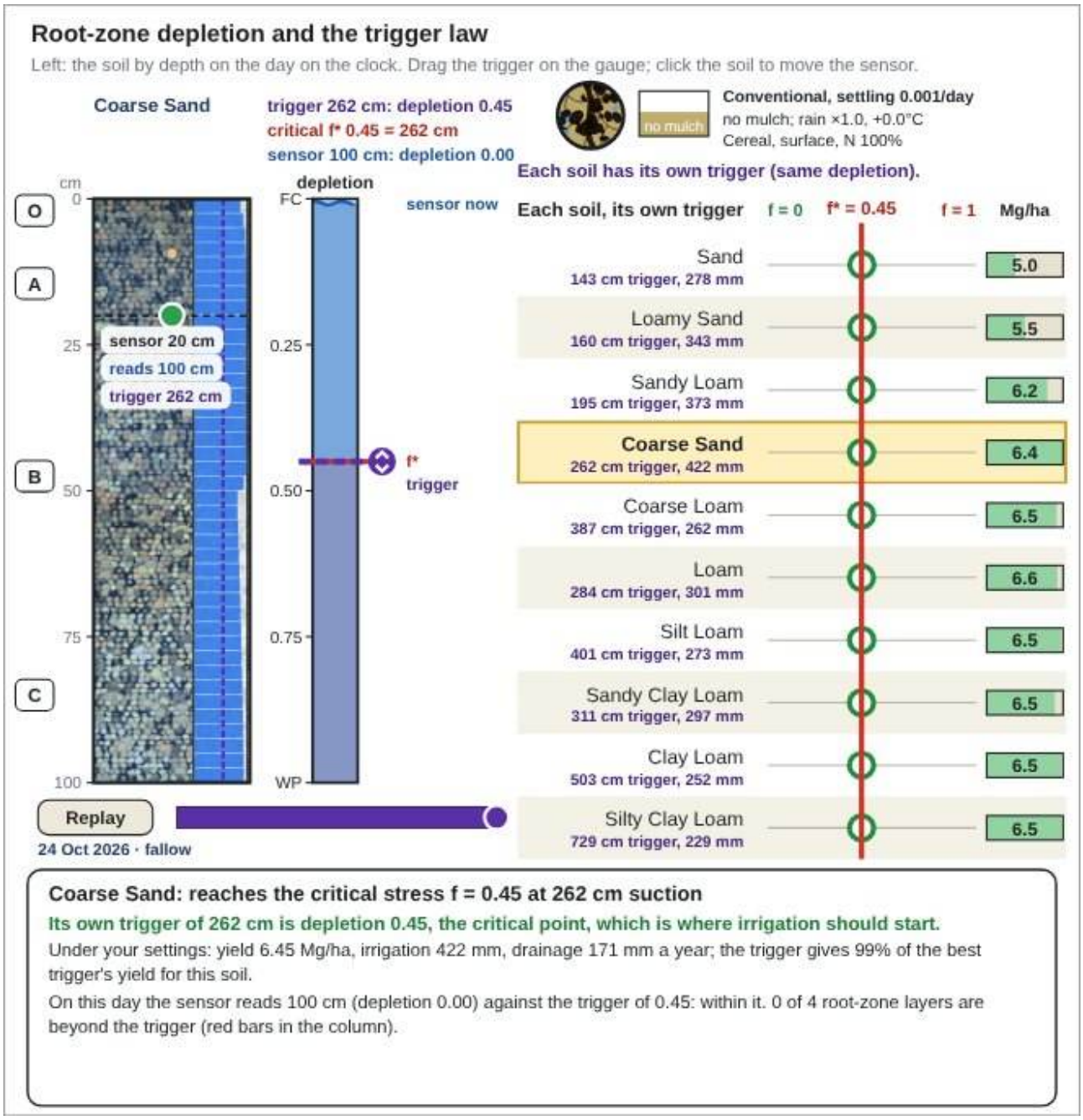


Figure 4.34: Trigger law, Coarse Sand, part 1 of 1.

4.9 Mulch against bare soil (Coarse Sand)

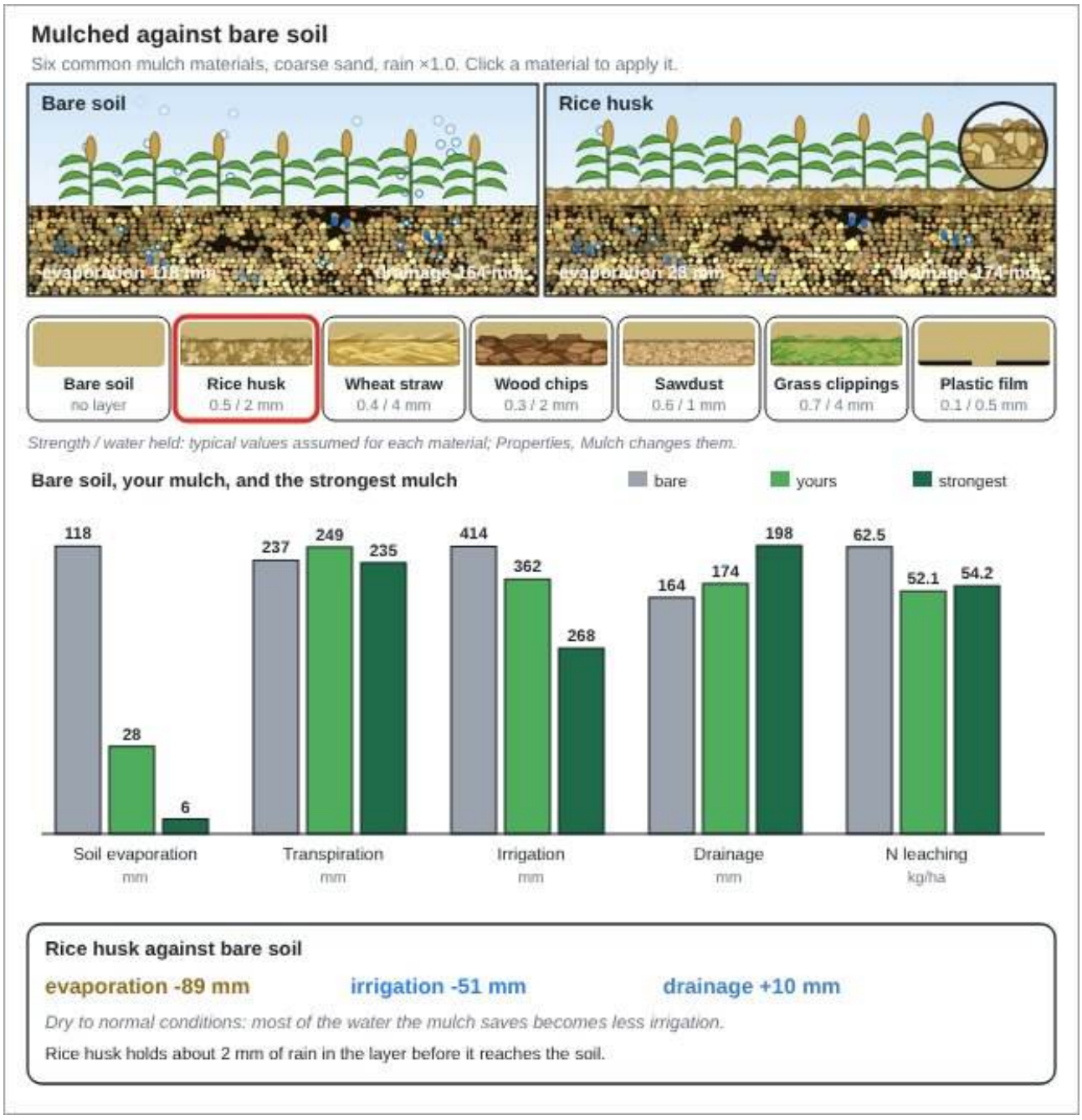


Figure 4.35: Mulch against bare soil, Coarse Sand, part 1 of 1.

4.10 Mulch control (Coarse Sand)

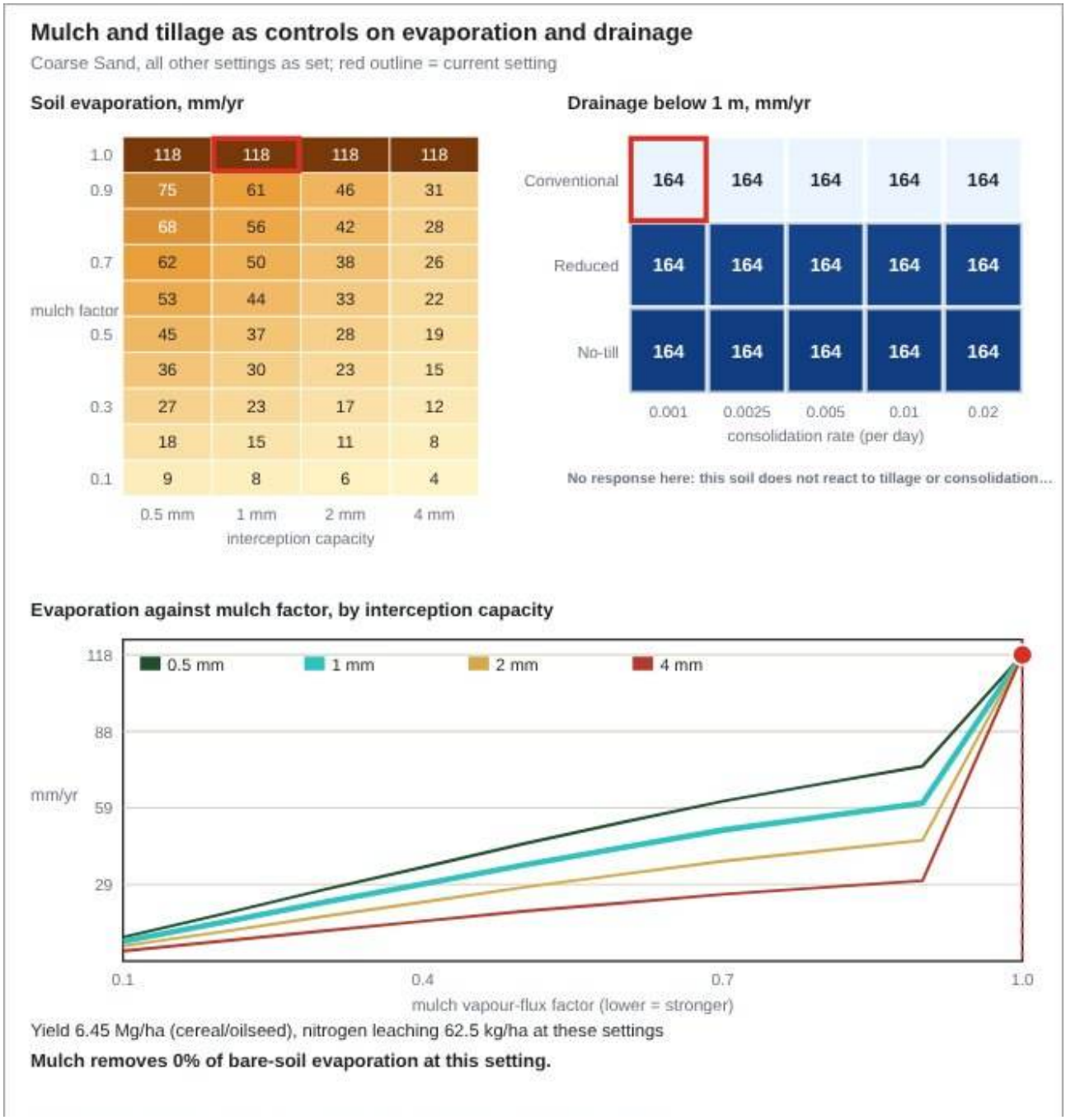


Figure 4.36: Mulch control, Coarse Sand, part 1 of 2.

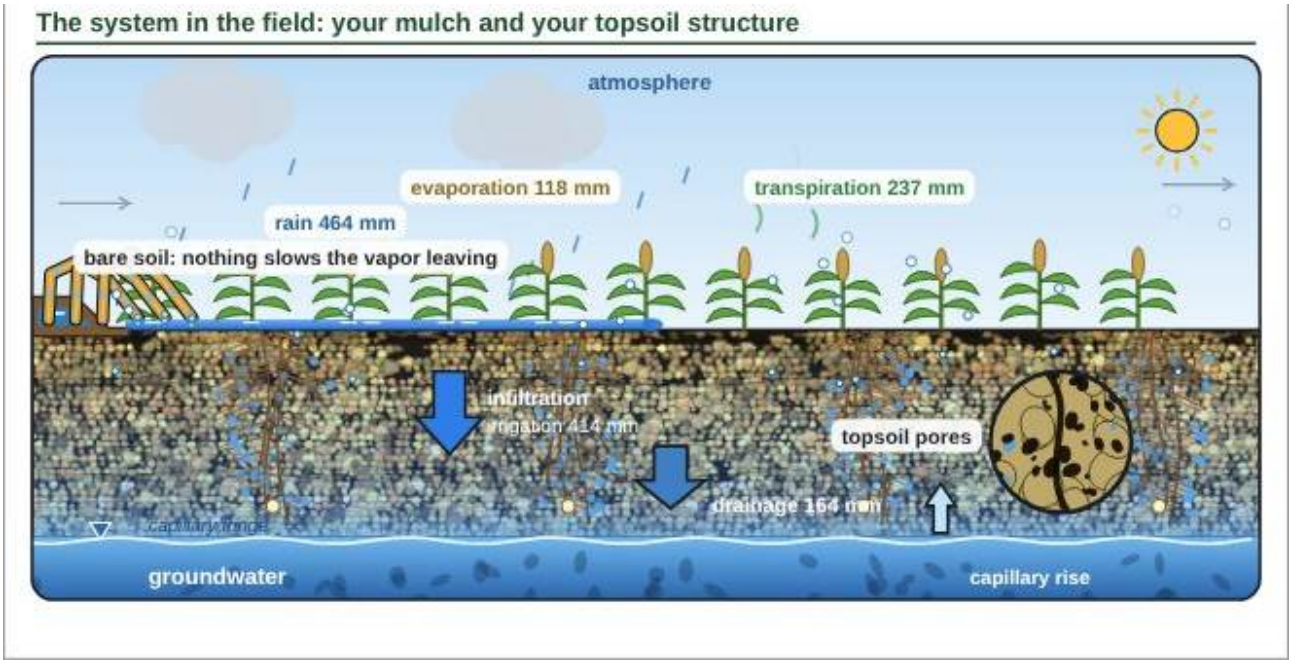


Figure 4.37: Mulch control, Coarse Sand, part 2 of 2.

4.11 Evaporation physics (Coarse Sand)

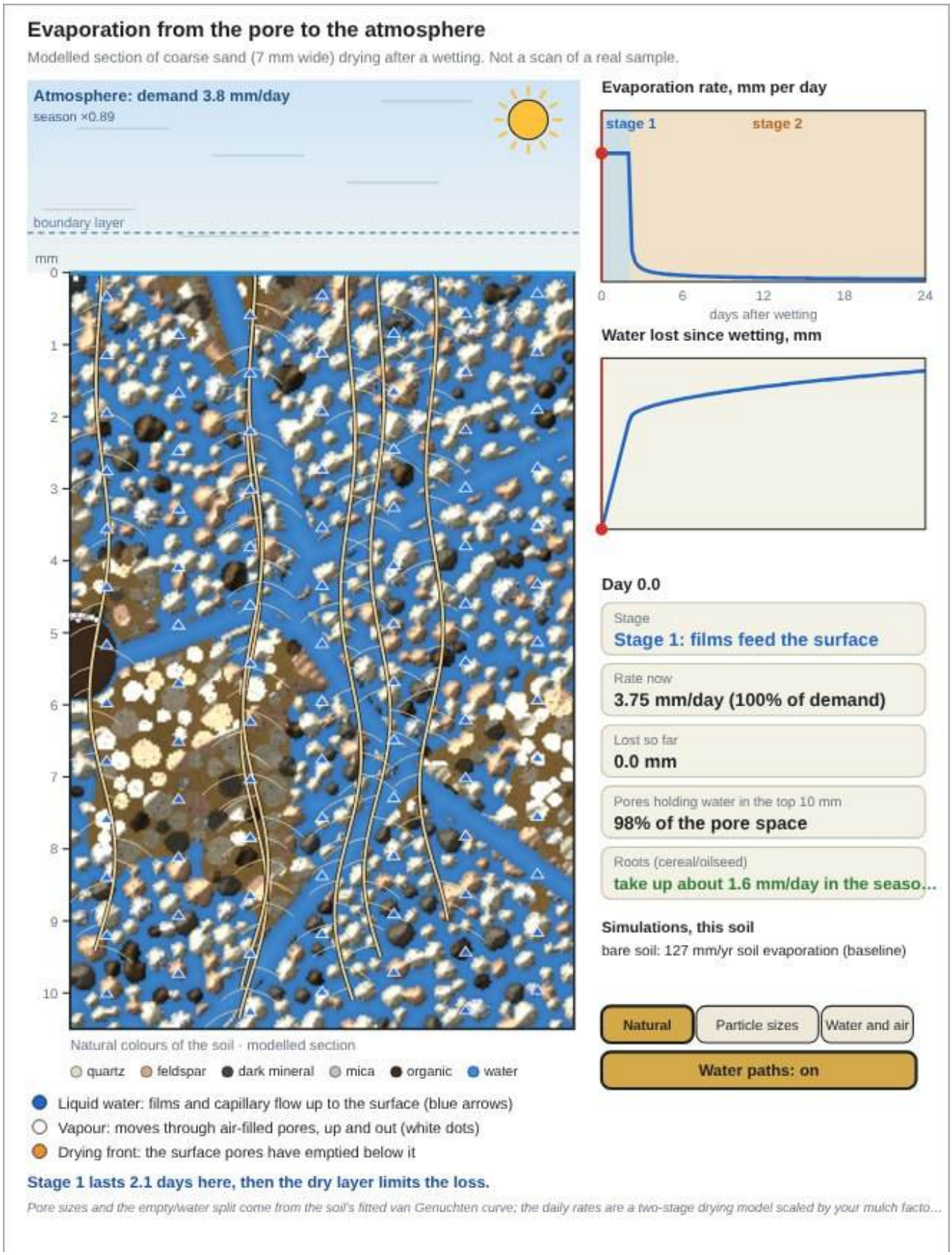


Figure 4.38: Evaporation physics, Coarse Sand, part 1 of 3.

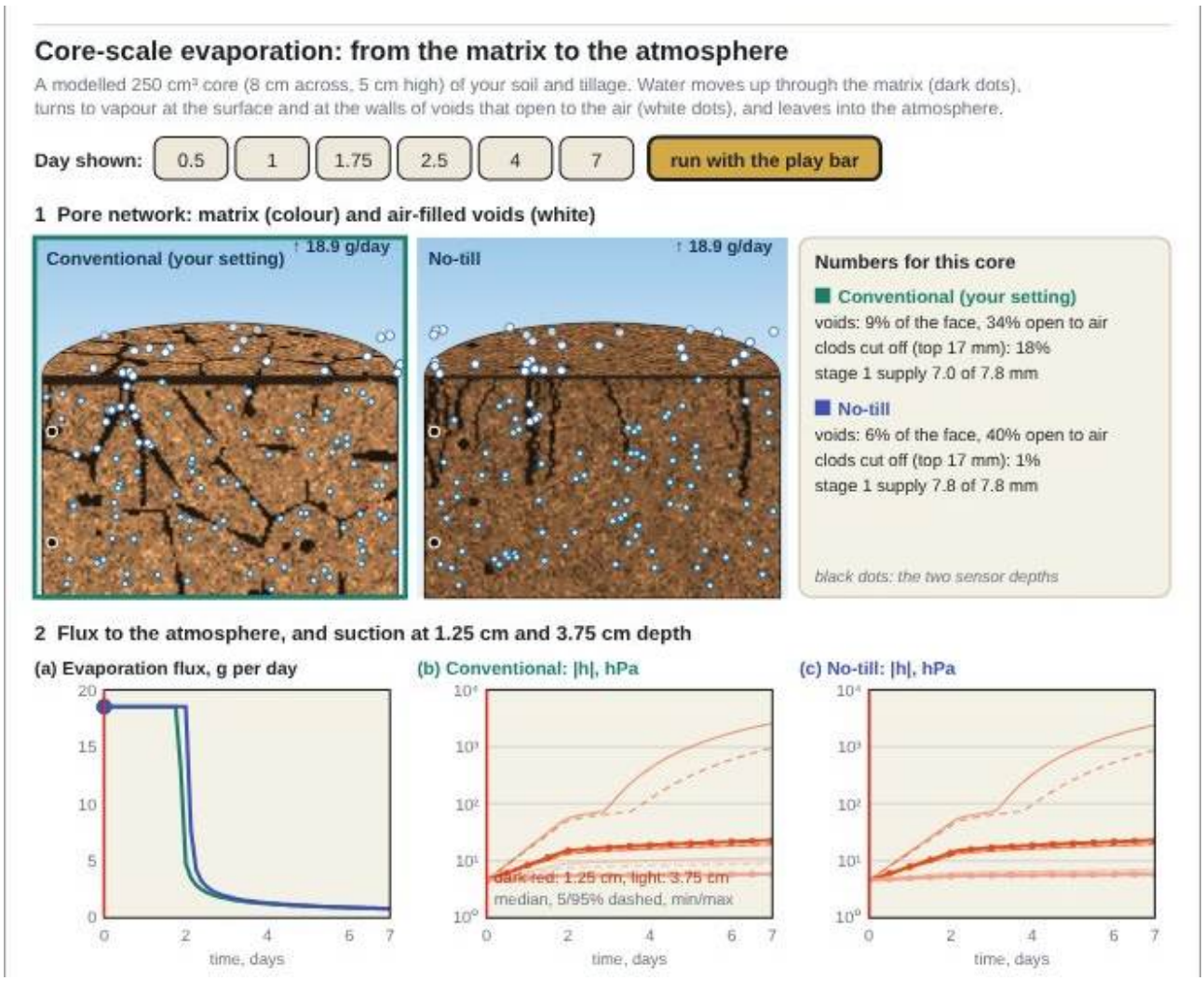


Figure 4.39: Evaporation physics, Coarse Sand, part 2 of 3.

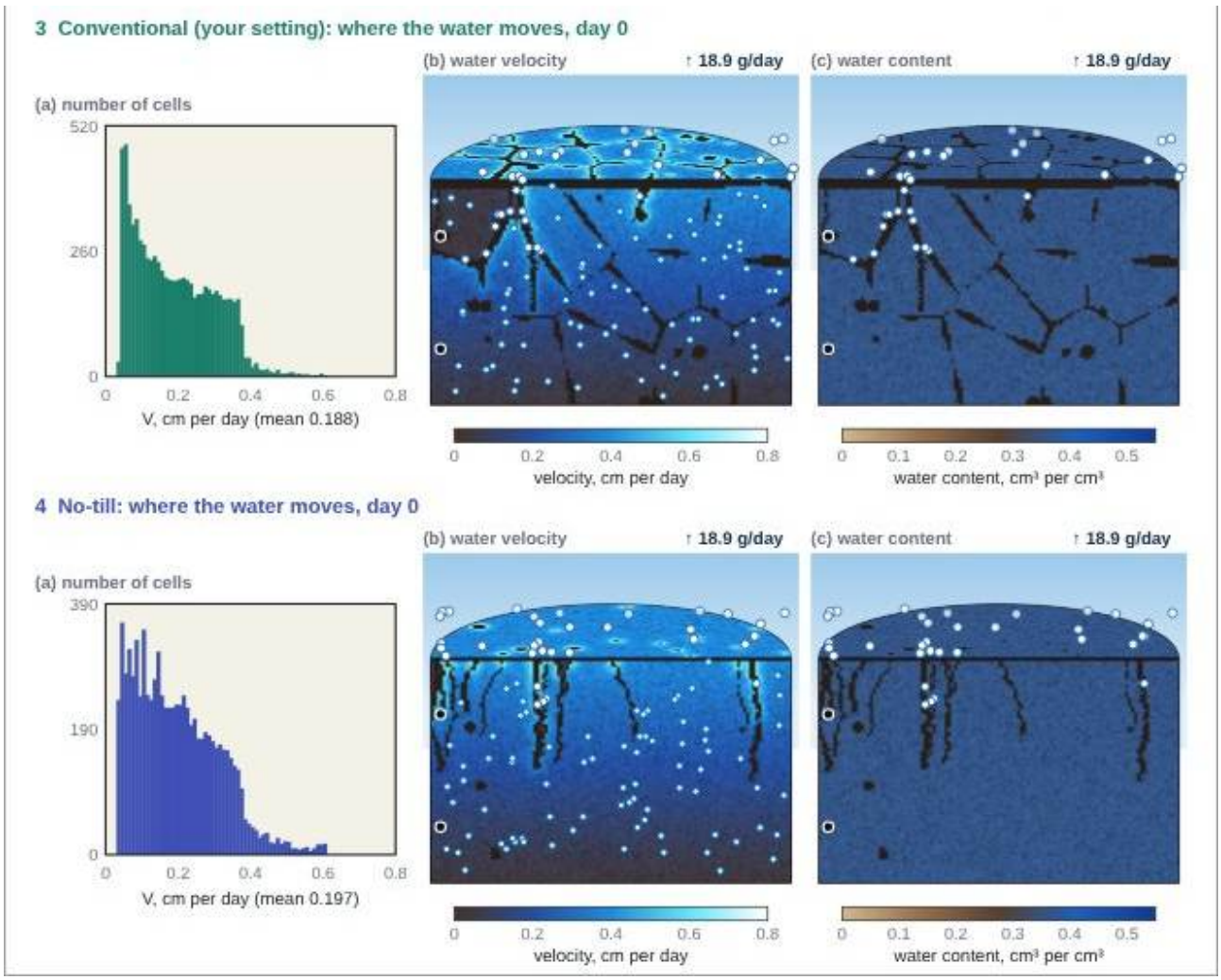


Figure 4.40: Evaporation physics, Coarse Sand, part 3 of 3.

4.12 Fate of saved water (Coarse Sand)

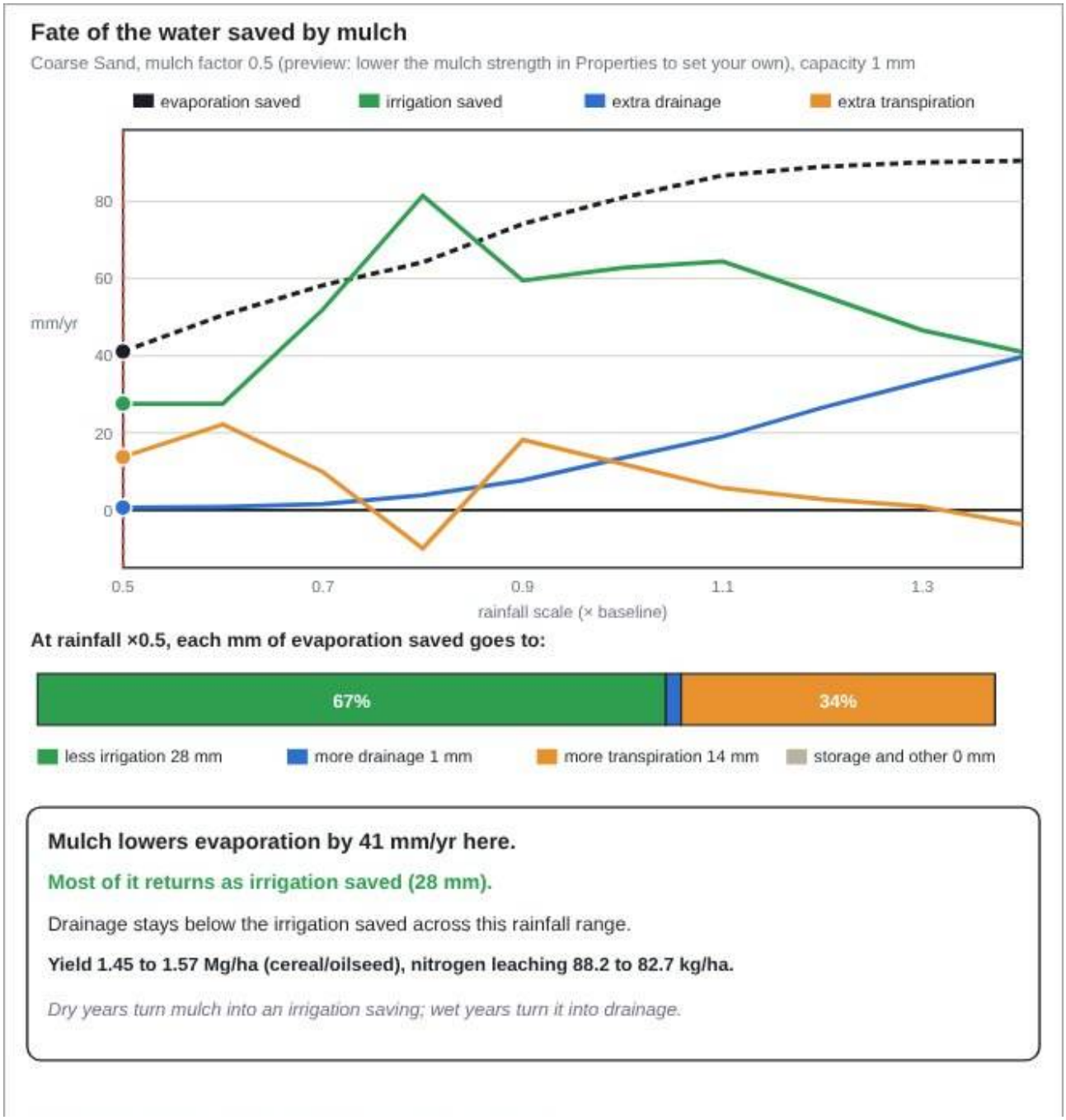


Figure 4.41: Fate of saved water, Coarse Sand, part 1 of 2.

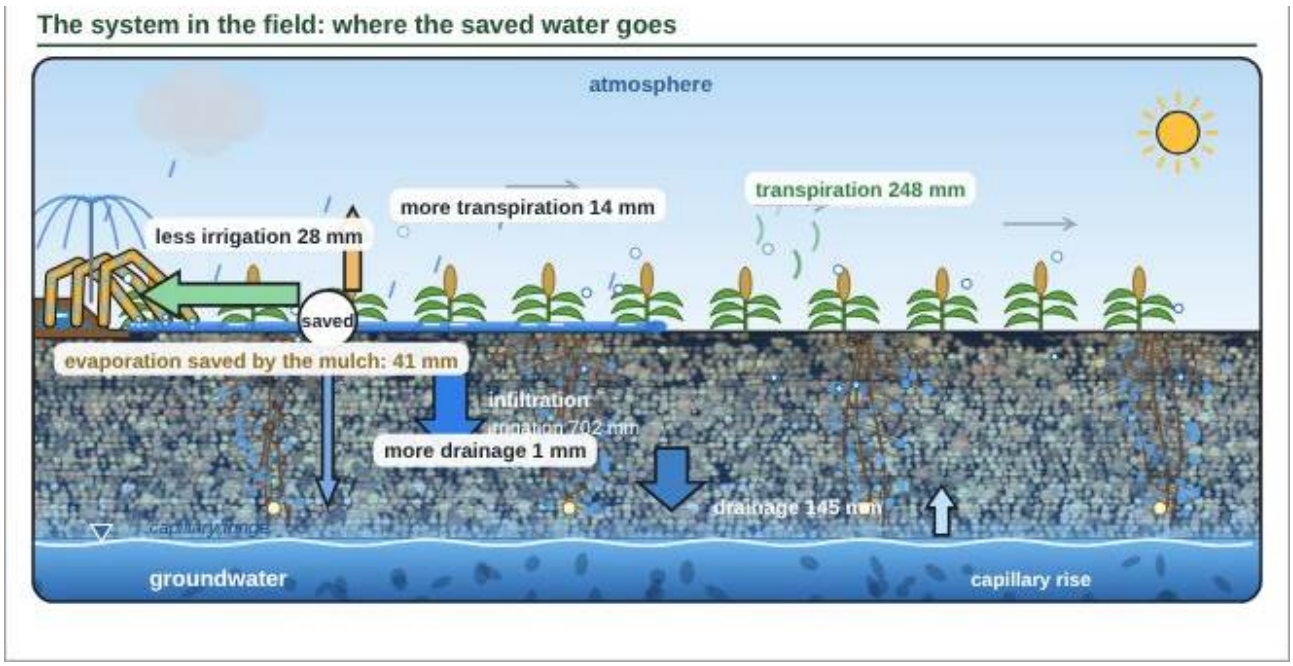


Figure 4.42: Fate of saved water, Coarse Sand, part 2 of 2.

4.13 Regime map (Coarse Sand)

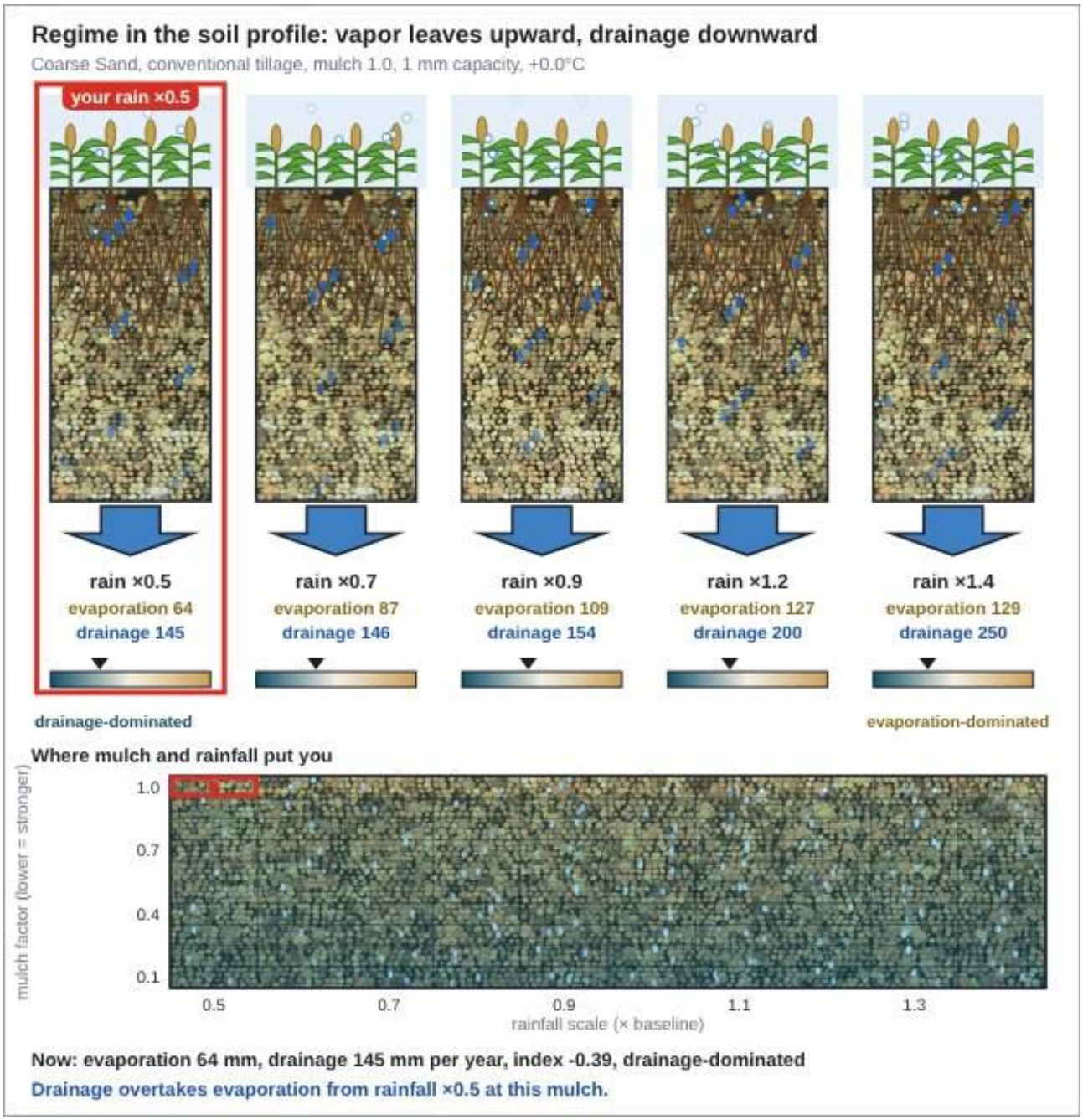


Figure 4.43: Regime map, Coarse Sand, part 1 of 1.

4.14 Tillage and structure (Coarse Sand)

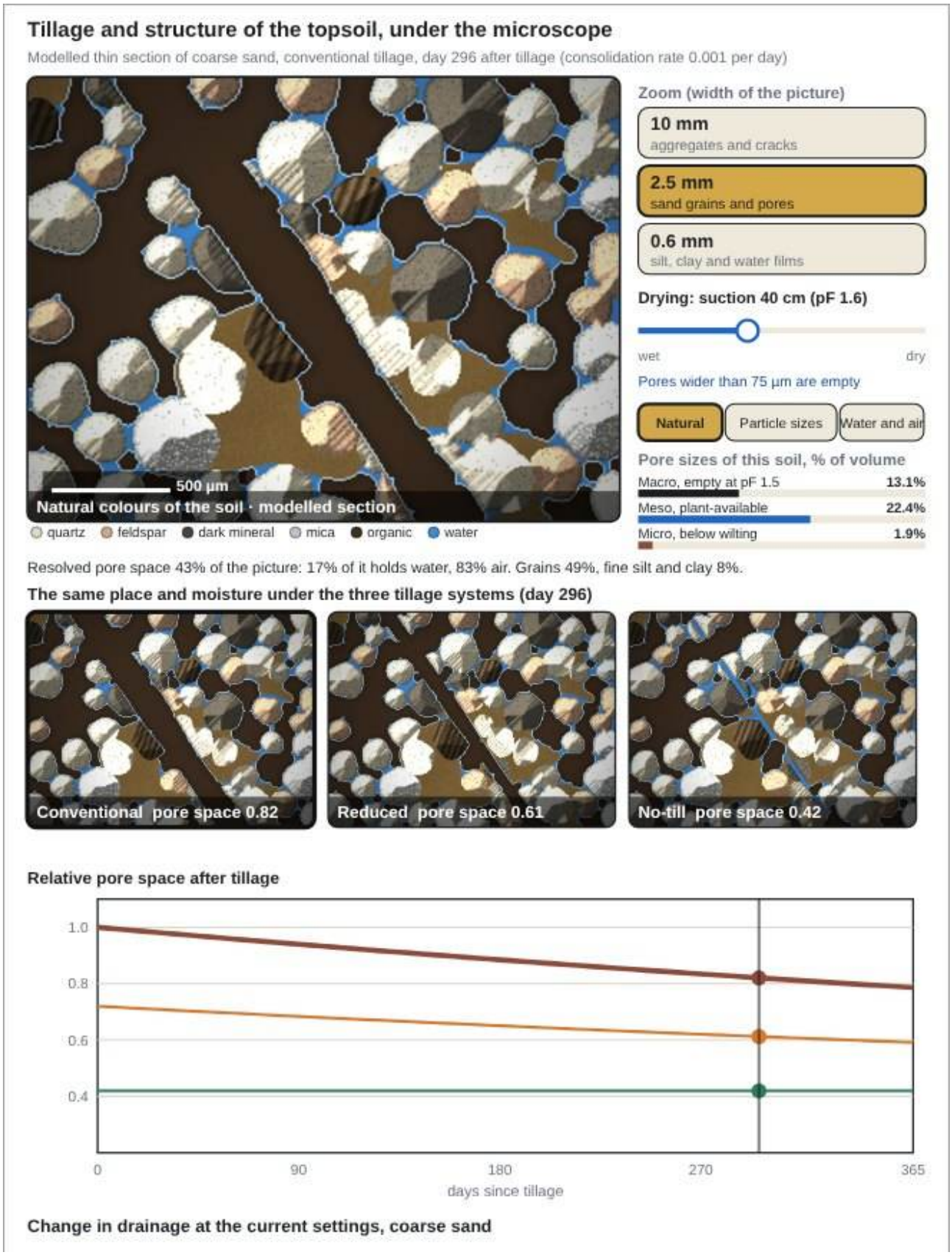


Figure 4.44: Tillage and structure, Coarse Sand, part 1 of 2.

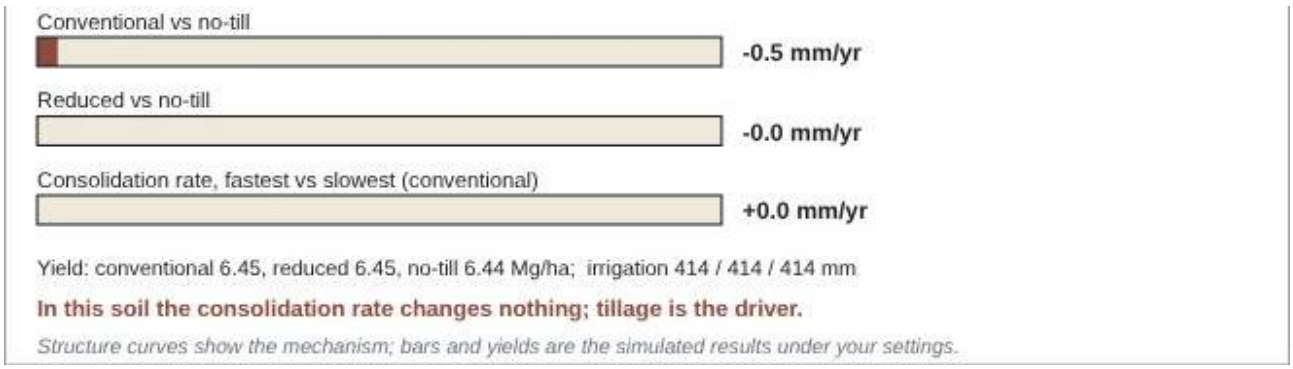


Figure 4.45: Tillage and structure, Coarse Sand, part 2 of 2.

4.15 Soil hydraulic drivers (Coarse Sand)

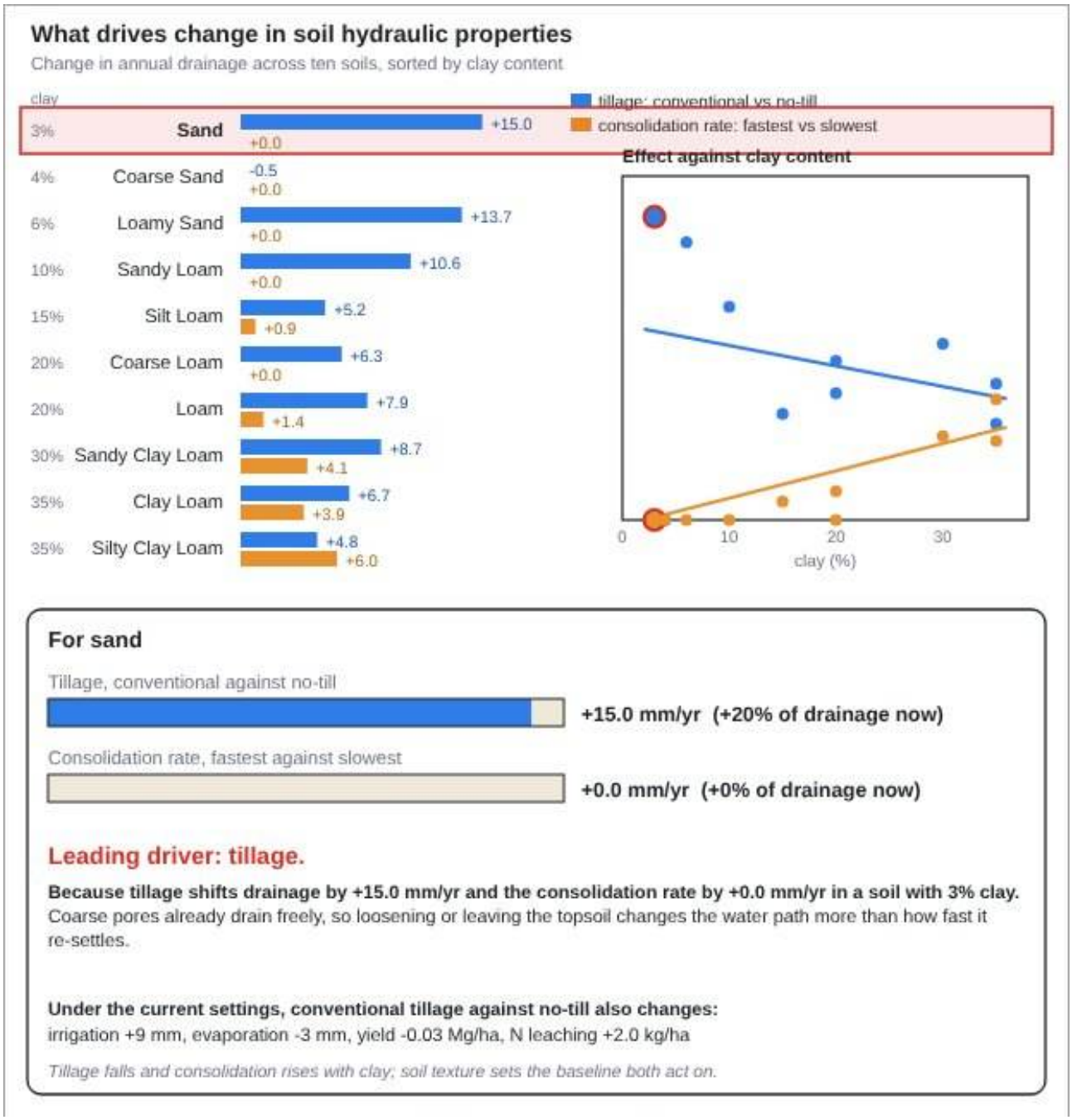


Figure 4.46: Soil hydraulic drivers, Coarse Sand, part 1 of 3.

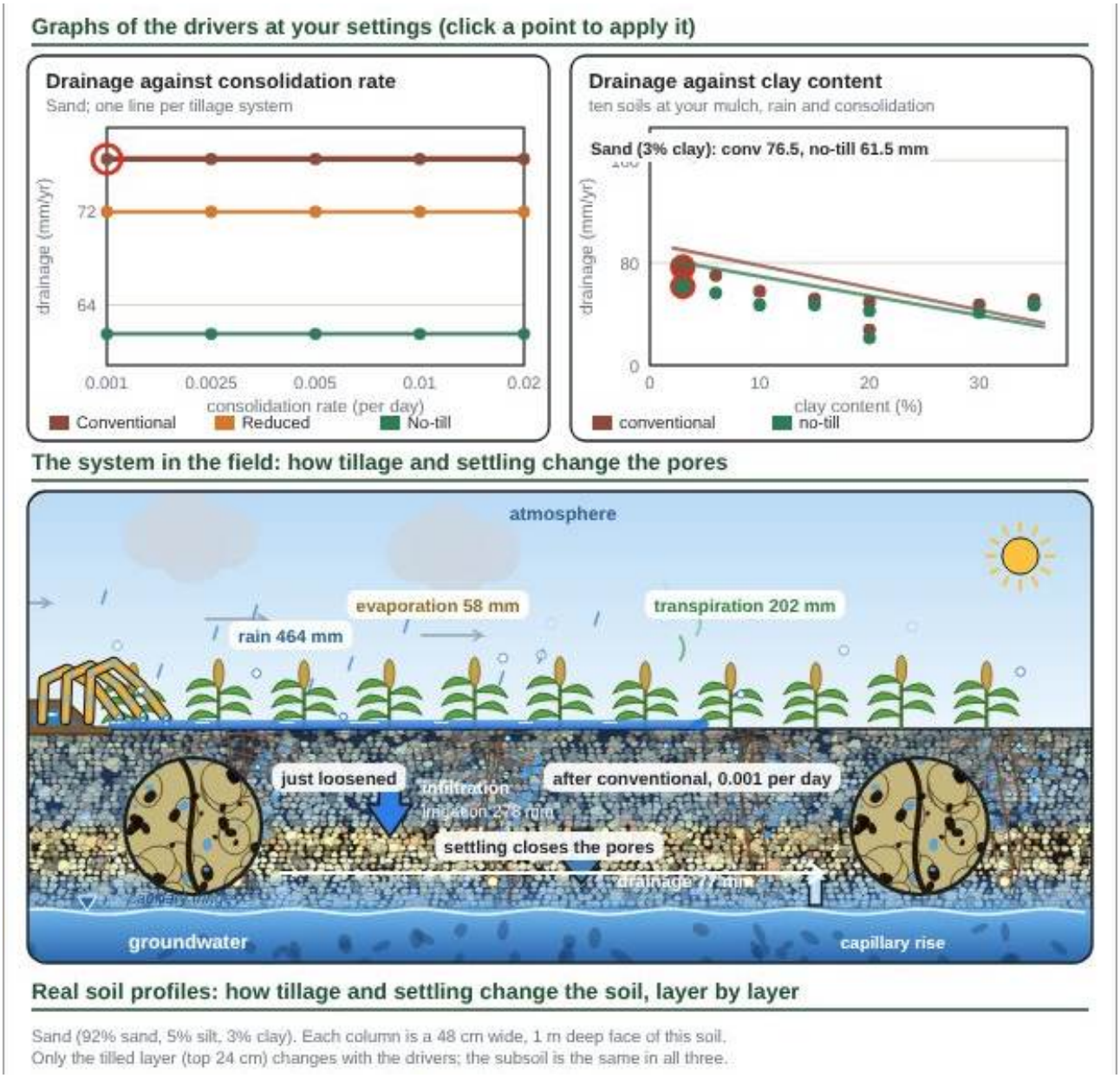


Figure 4.47: Soil hydraulic drivers, Coarse Sand, part 2 of 3.

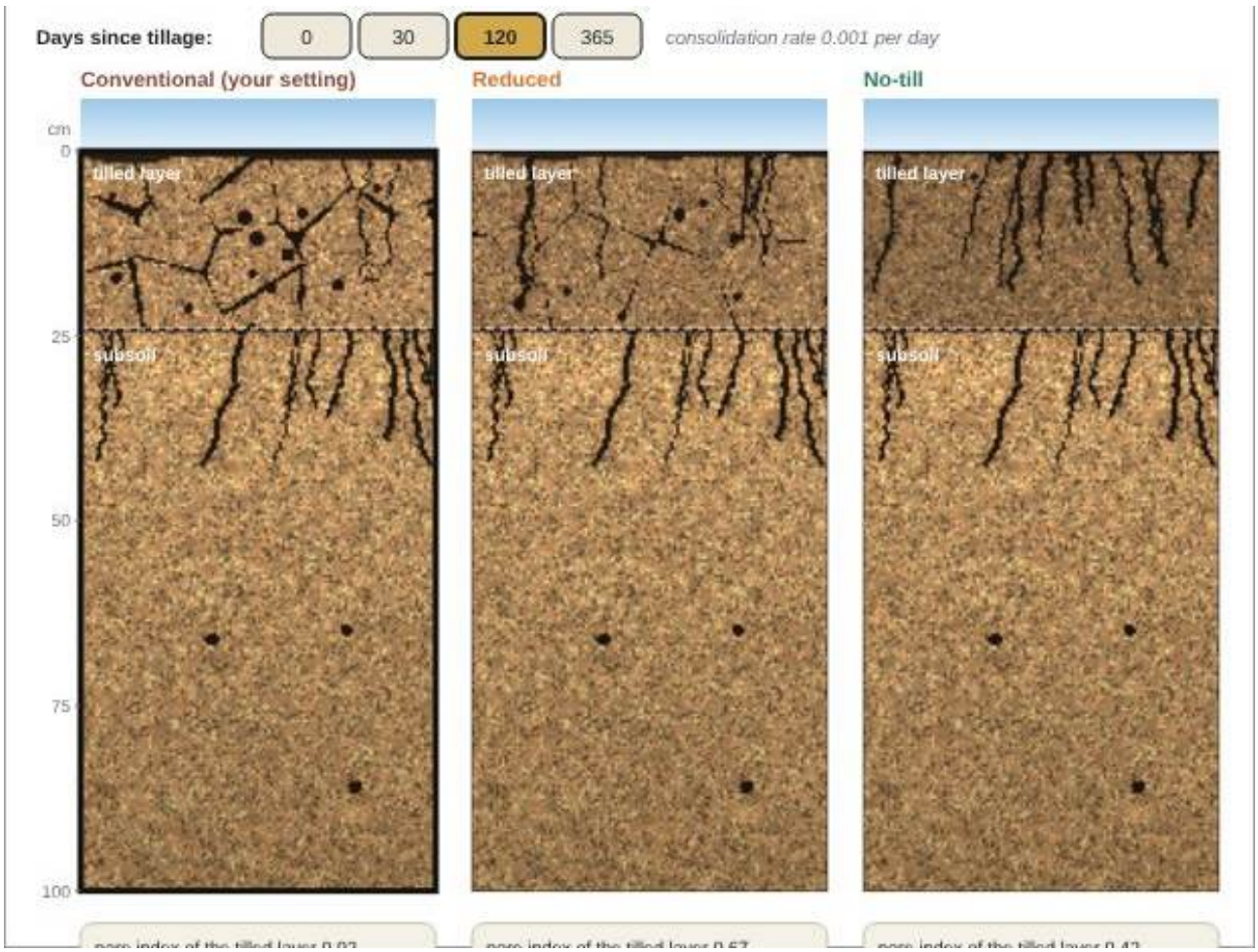


Figure 4.48: Soil hydraulic drivers, Coarse Sand, part 3 of 3.

4.16 Root structure (Coarse Sand)

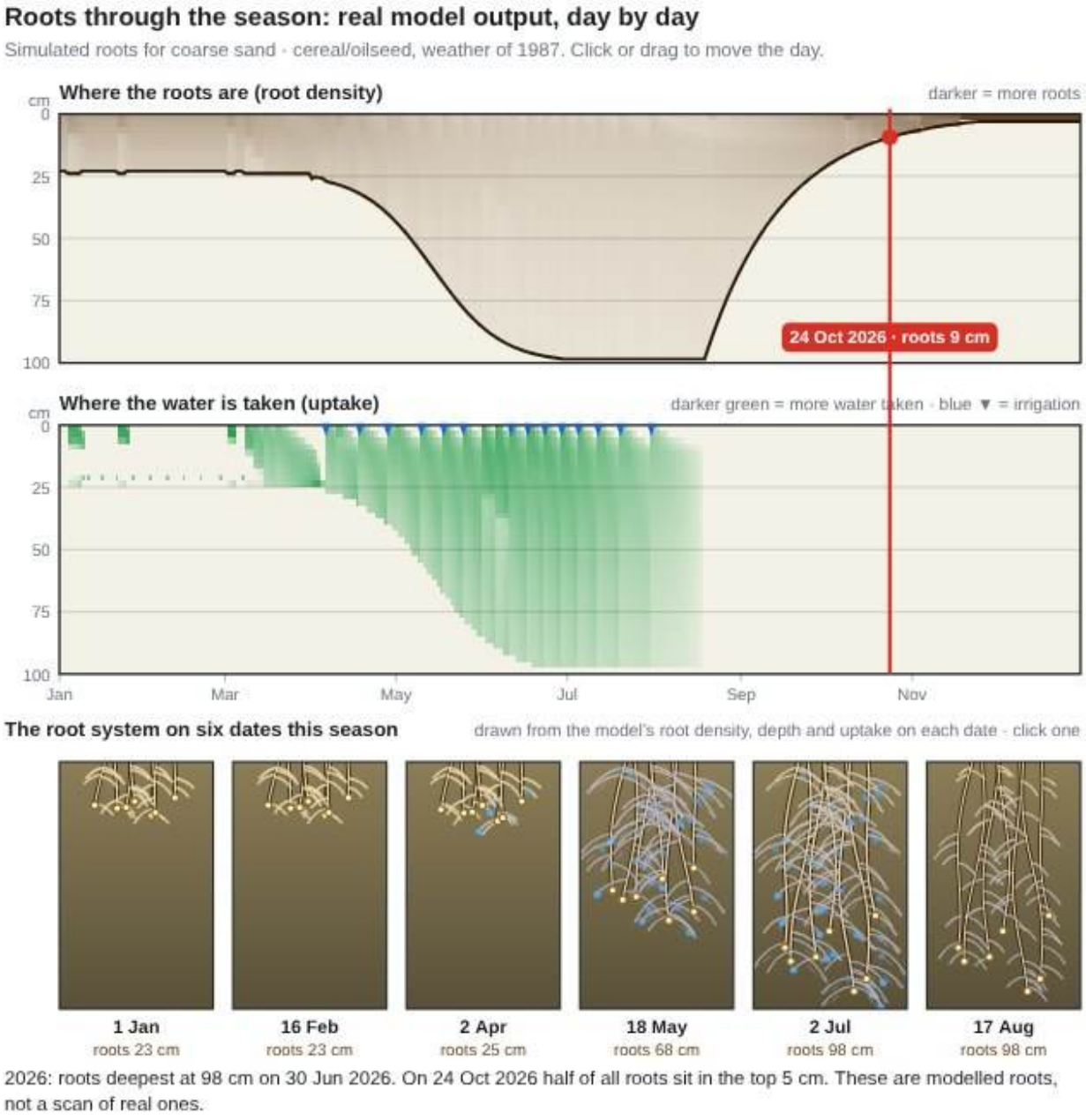


Figure 4.49: Root structure, Coarse Sand, part 1 of 4.

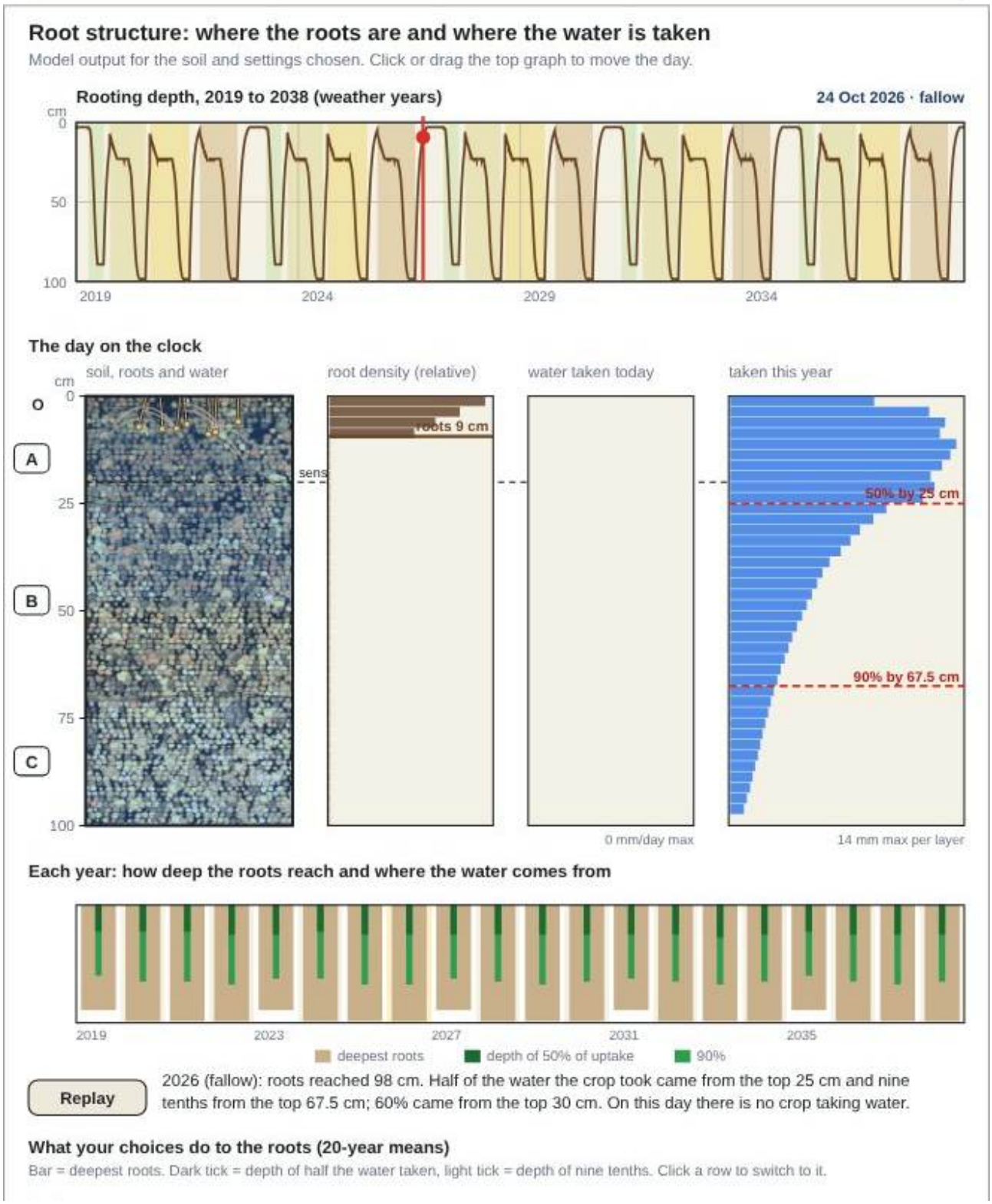


Figure 4.50: Root structure, Coarse Sand, part 2 of 4.

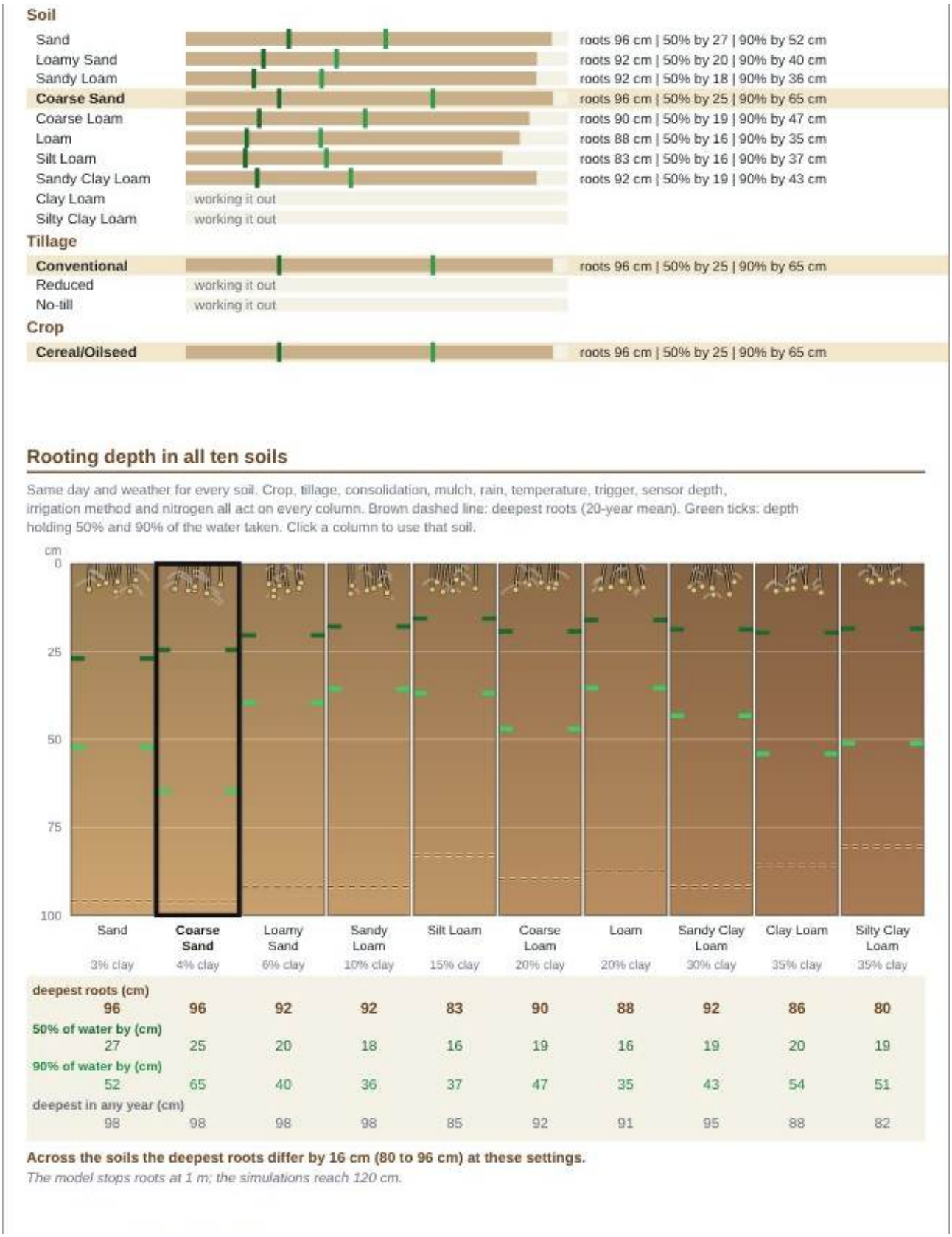


Figure 4.51: Root structure, Coarse Sand, part 3 of 4.

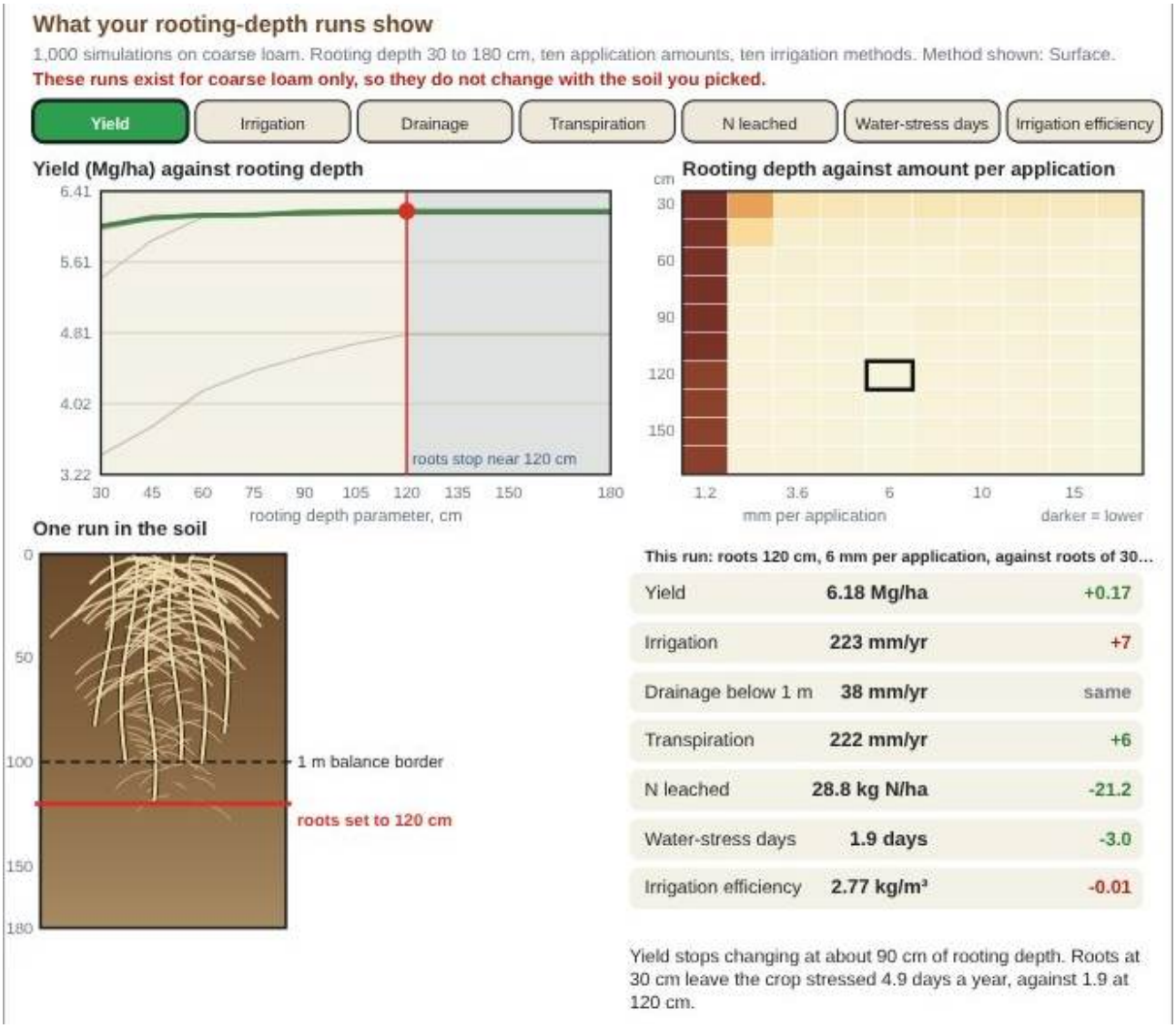


Figure 4.52: Root structure, Coarse Sand, part 4 of 4.

4.17 Field balance (Coarse Sand)

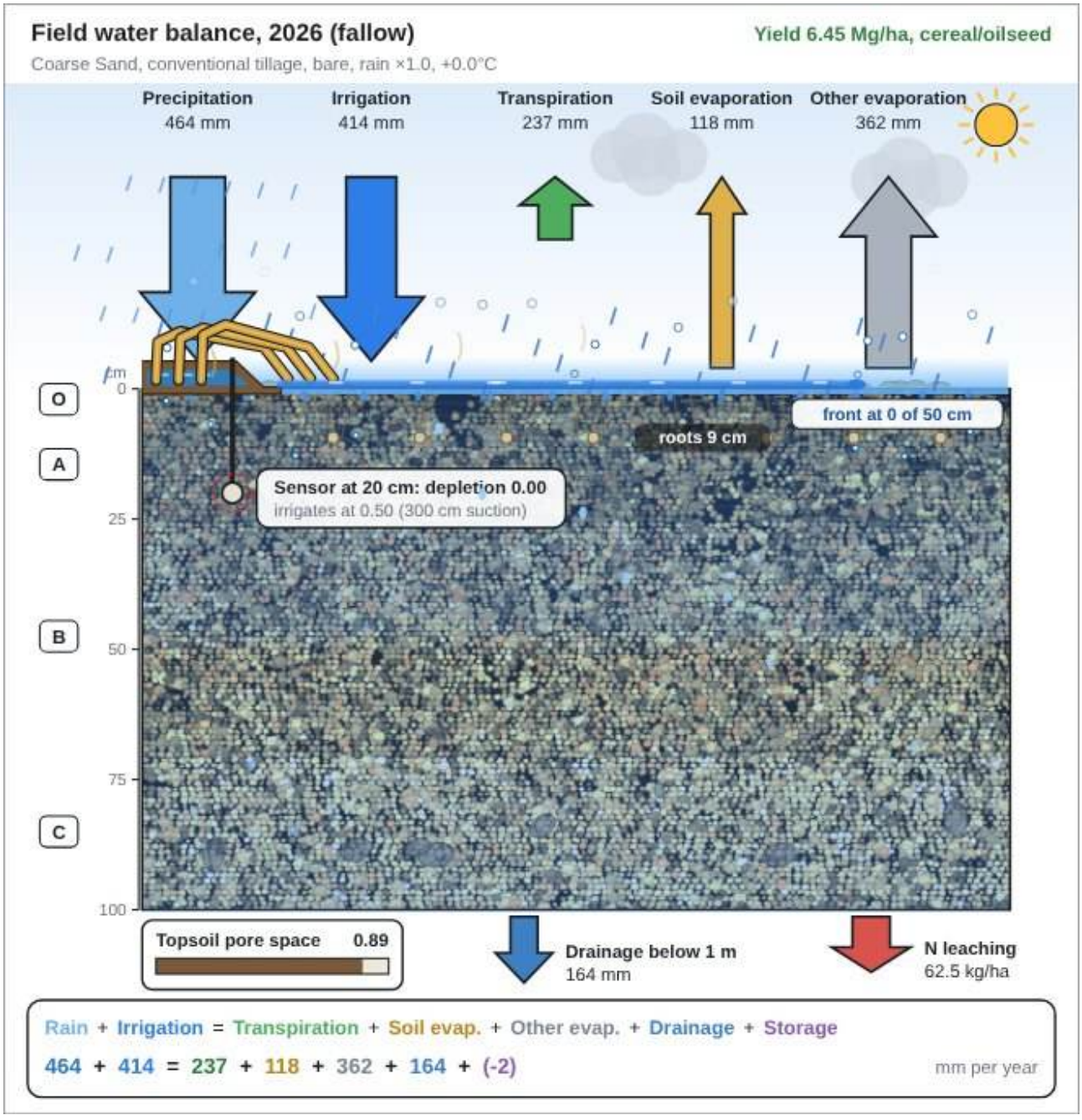


Figure 4.53: Field balance, Coarse Sand, part 1 of 1.

4.18 Soil, plant and air (Coarse Sand)

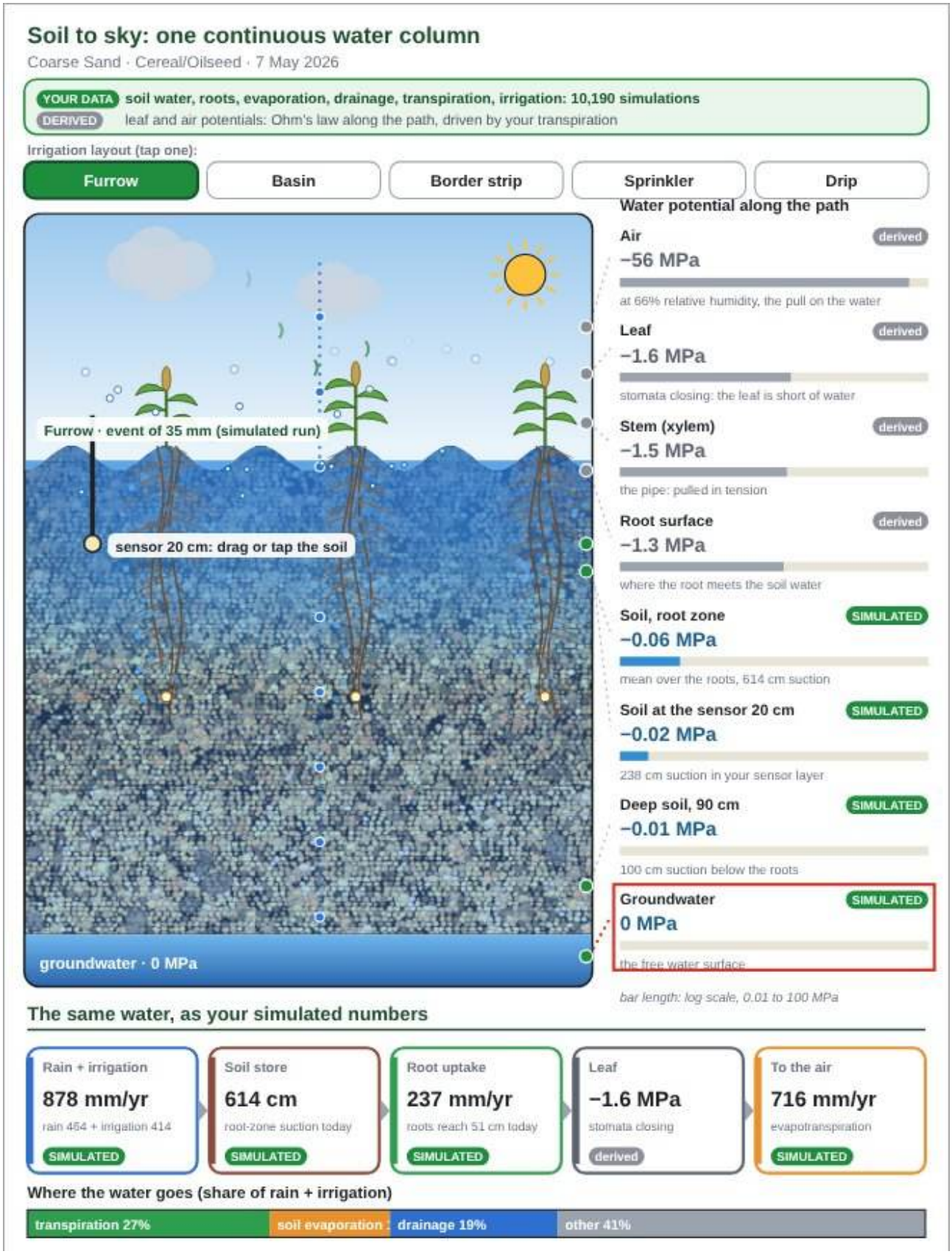


Figure 4.54: Soil, plant and air, Coarse Sand, part 1 of 3.

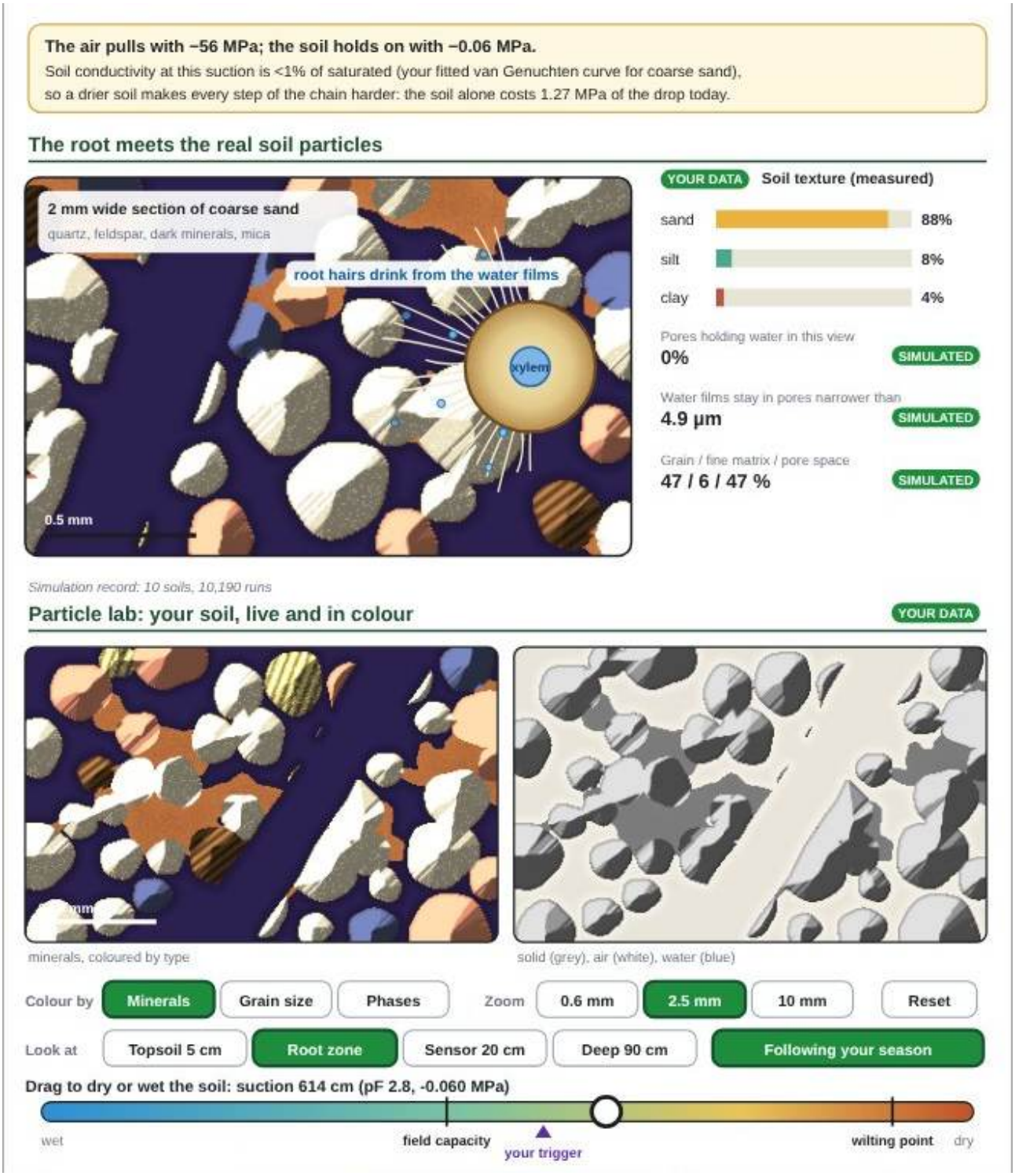


Figure 4.55: Soil, plant and air, Coarse Sand, part 2 of 3.

Figure 4.56: Soil, plant and air, Coarse Sand, part 3 of 3.

4.19 Profile evolution (Coarse Sand)

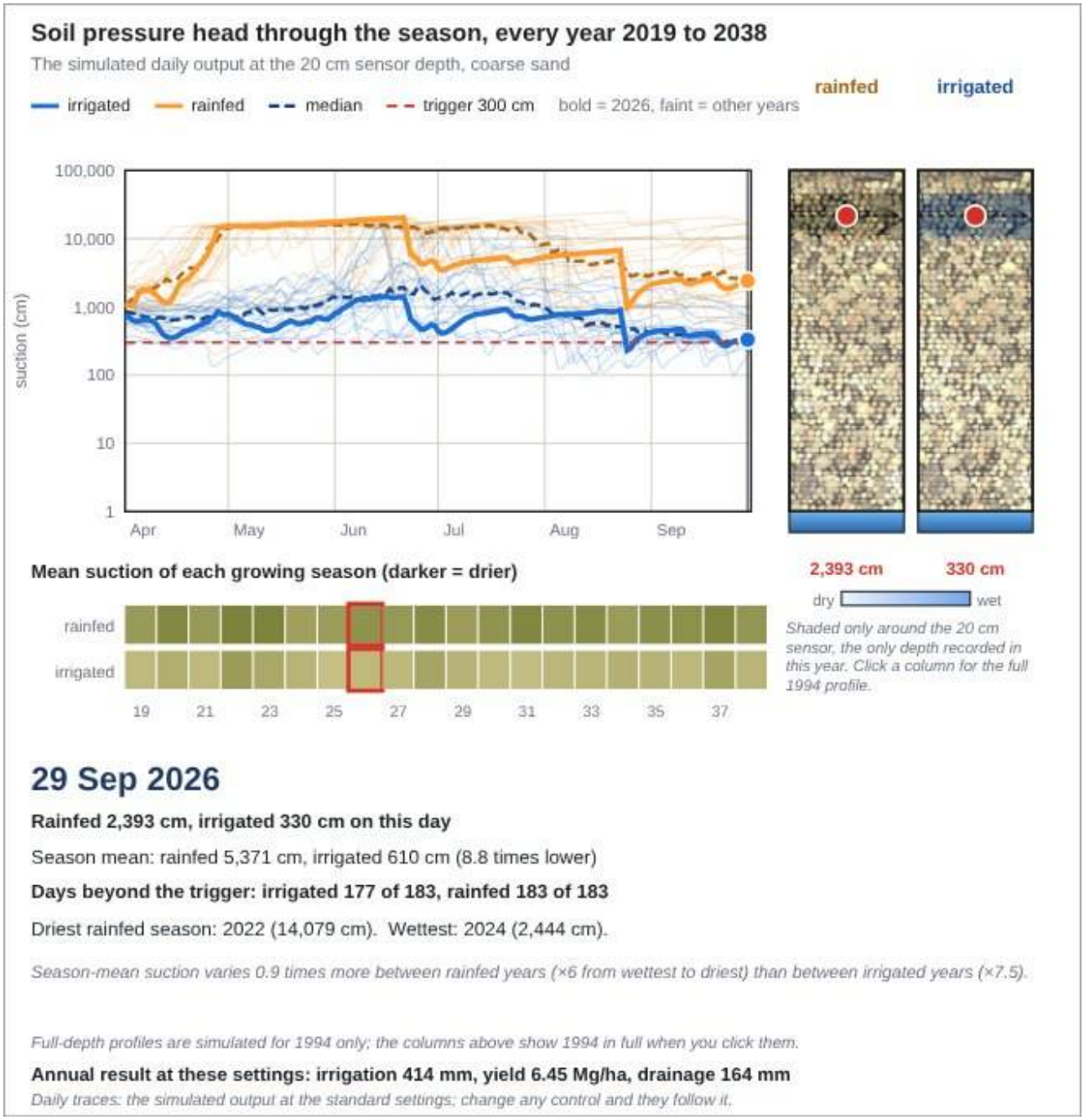


Figure 4.57: Profile evolution, Coarse Sand, part 1 of 1.

4.20 Reliability over 20 seasons (Coarse Sand)

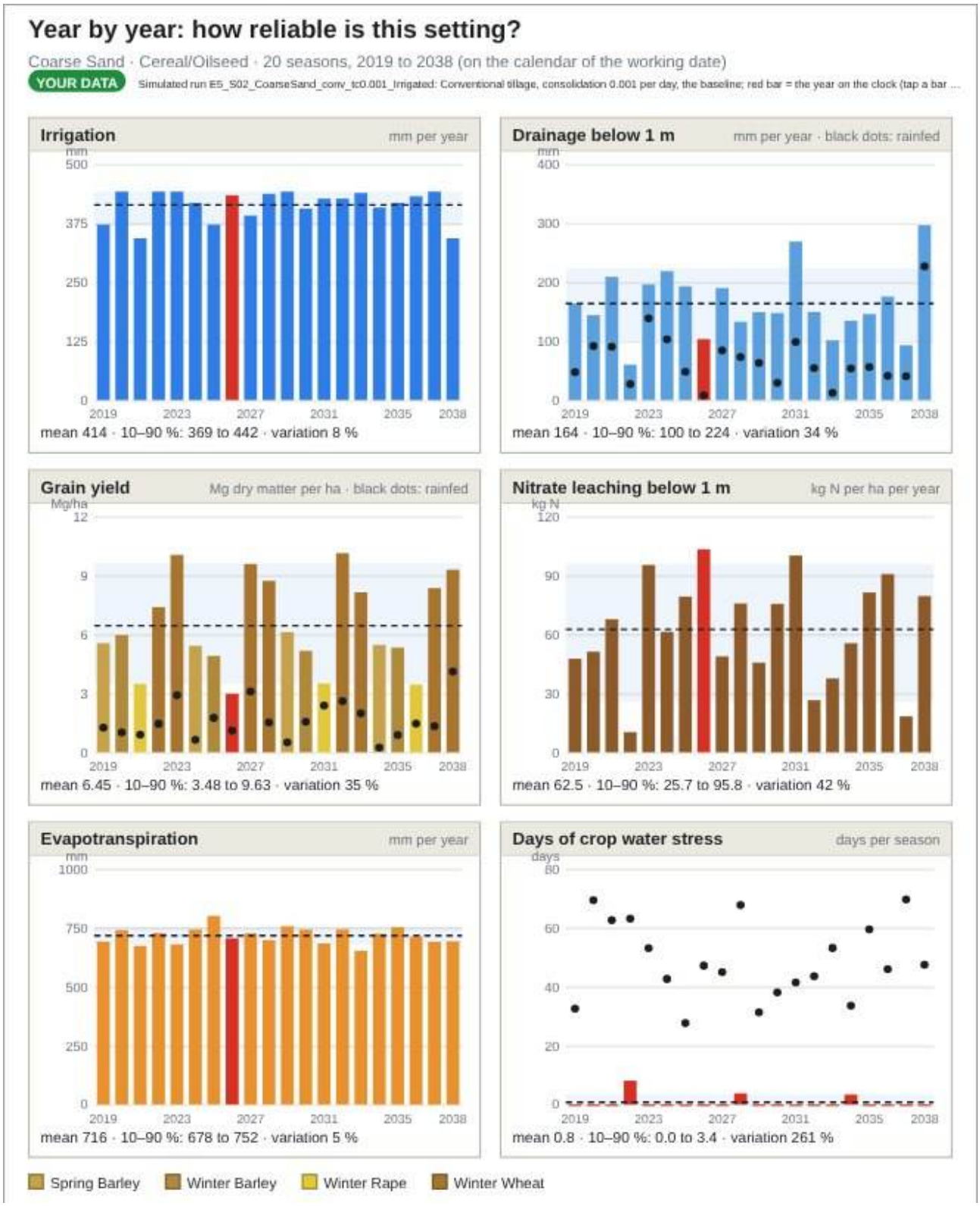


Figure 4.58: Reliability over 20 seasons, Coarse Sand, part 1 of 2.

Figure 4.59: Reliability over 20 seasons, Coarse Sand, part 2 of 2.

4.21 Full water balance (Coarse Sand)

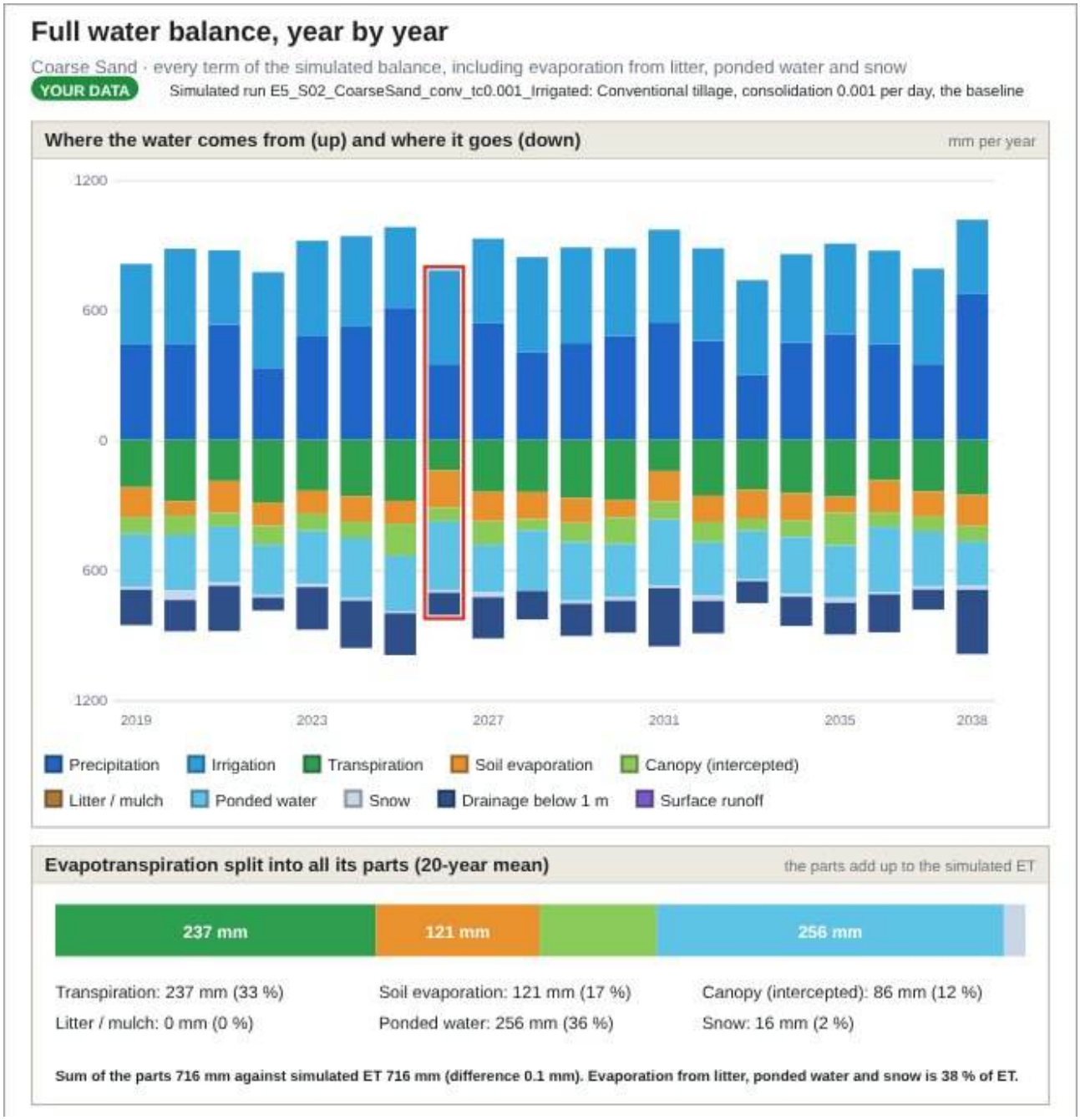


Figure 4.60: Full water balance, Coarse Sand, part 1 of 2.

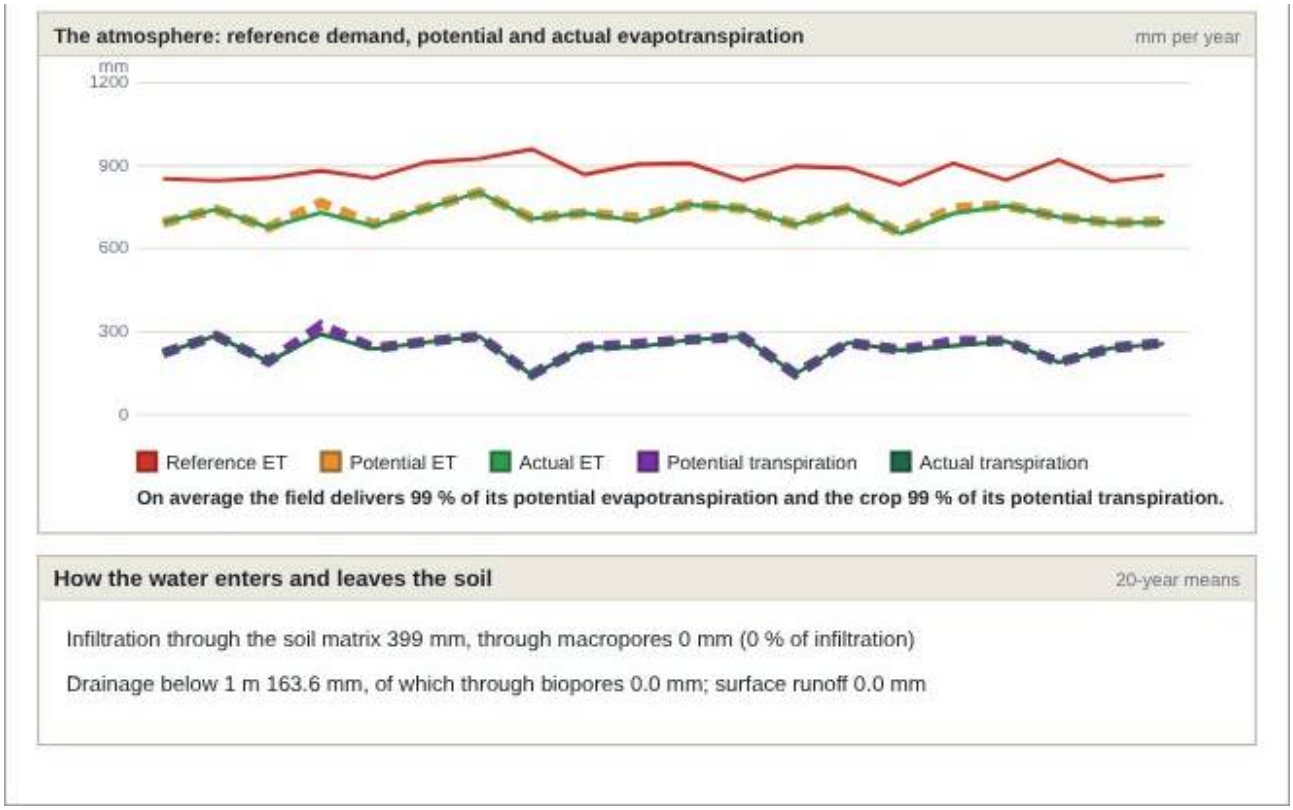


Figure 4.61: Full water balance, Coarse Sand, part 2 of 2.

4.22 Nitrogen budget (Coarse Sand)

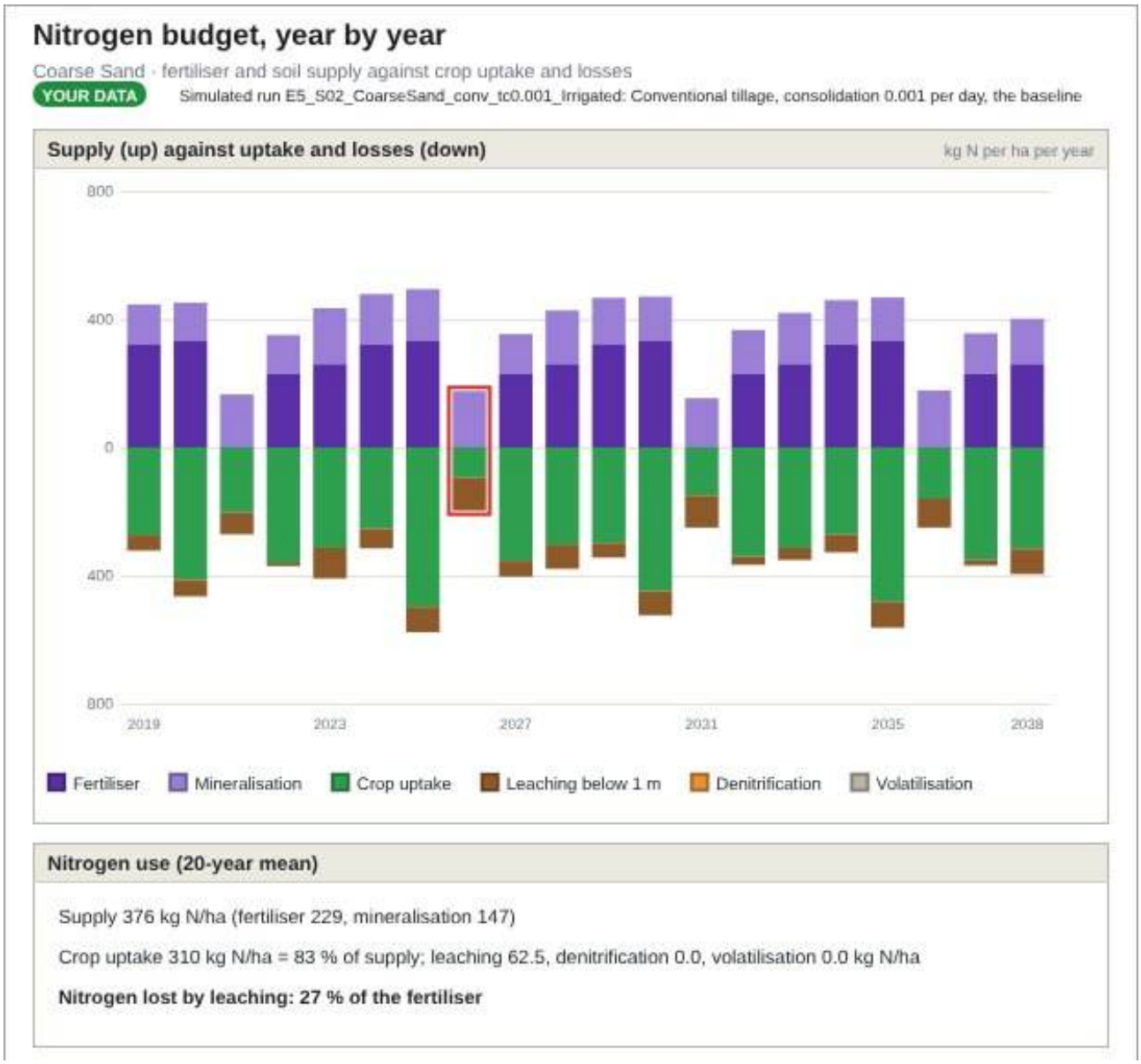


Figure 4.62: Nitrogen budget, Coarse Sand, part 1 of 2.

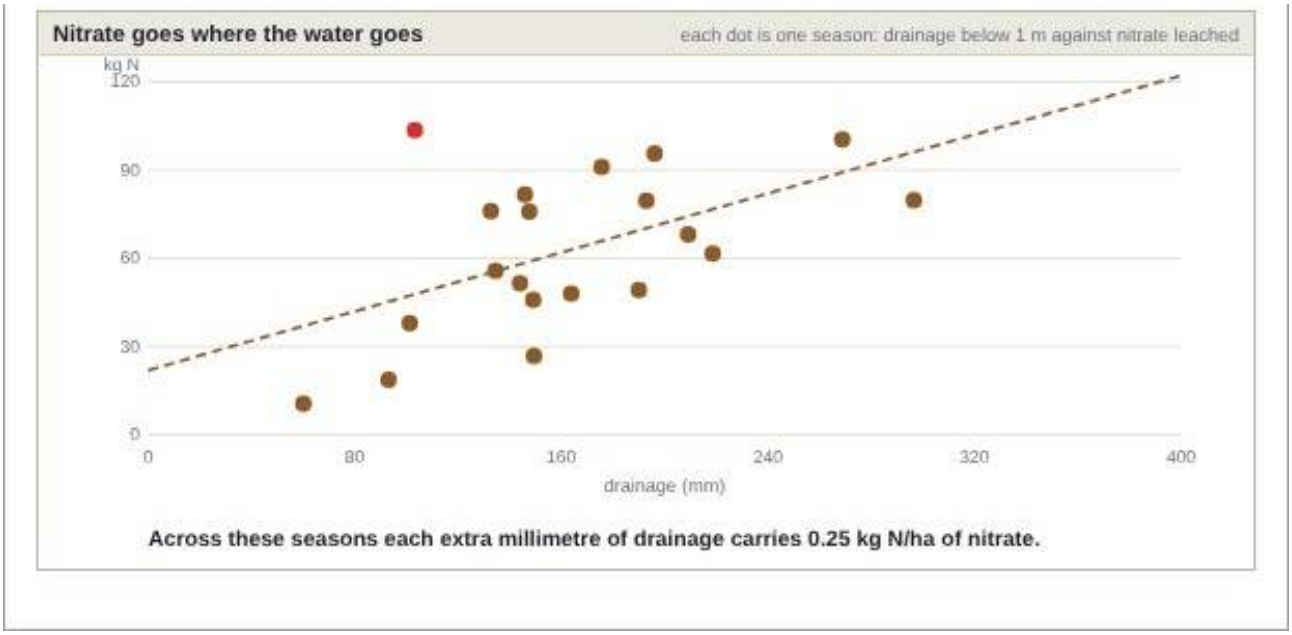


Figure 4.63: Nitrogen budget, Coarse Sand, part 2 of 2.

4.23 Crop water requirement (Coarse Sand)

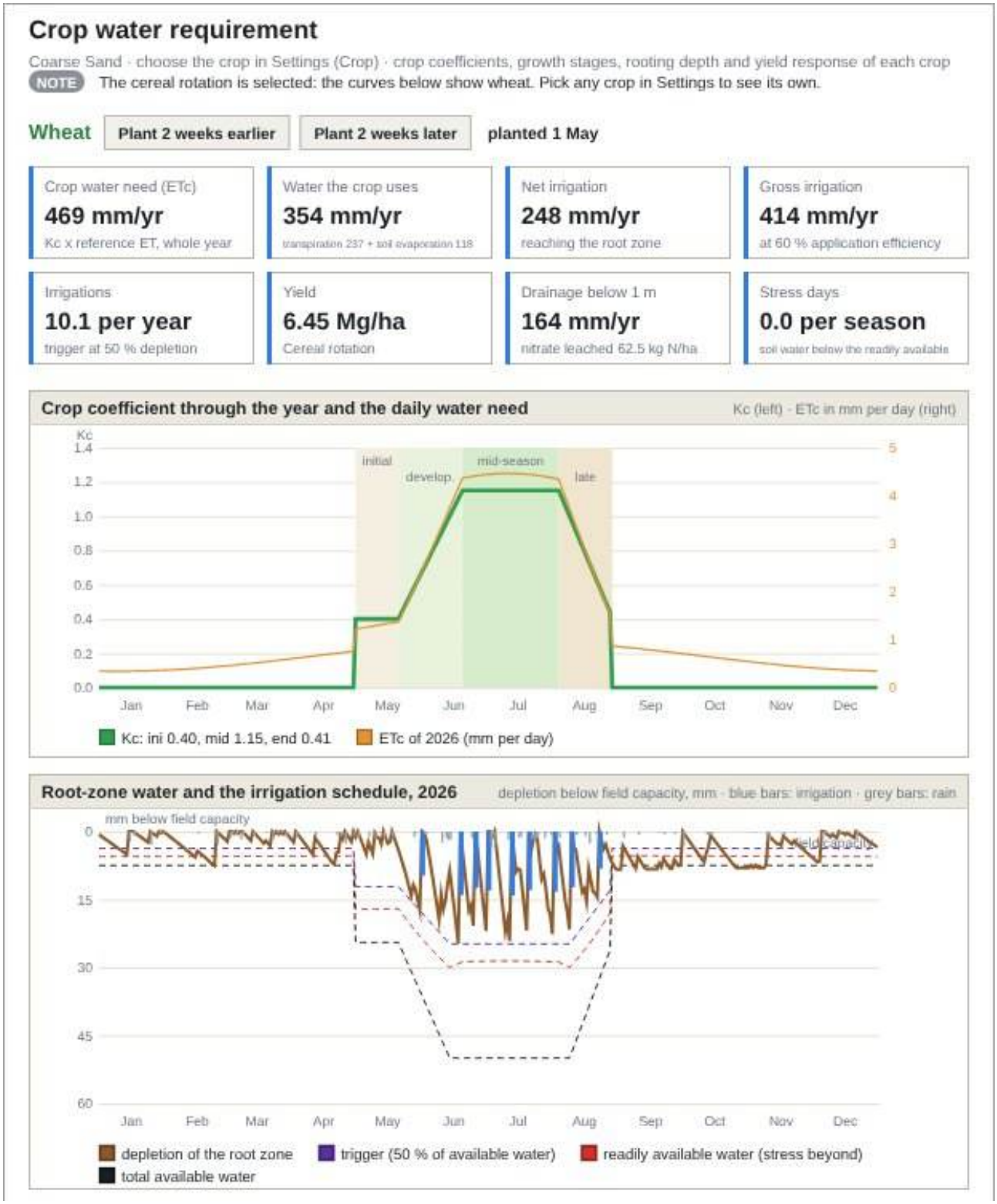


Figure 4.64: Crop water requirement, Coarse Sand, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**688**	31.0	98 %
Coconut (high)	834	**604**	23.1	100 %
Papaya (high)	814	**593**	25.1	100 %
Oil palm (young)	790	**553**	23.5	100 %
Yam	649	**490**	32.0	100 %
Avocado (medium)	677	**440**	22.4	100 %
Kiwi (pergola)	626	**426**	16.7	100 %
Cotton	571	**420**	17.5	100 %
Onion (dry)	565	**417**	34.1	98 %
Cabbage	559	**403**	24.1	100 %
Table grapes (horizontal trellis)	597	**387**	15.2	100 %
Guava (high)	651	**386**	14.9	100 %
Rice	573	**364**	18.4	92 %
Potato	508	**358**	27.3	99 %
Tomato	521	**352**	14.3	100 %
Mango (medium)	620	**347**	13.5	100 %
Hop (v trellis)	520	**335**	13.2	100 %
Pear (medium)	528	**325**	12.8	100 %
Walnut (medium)	523	**324**	12.8	100 %
Date palm (medium)	602	**317**	12.3	100 %
Groundnut	471	**303**	16.1	100 %
Soybean	473	**302**	12.9	100 %
Mustard	468	**298**	12.2	100 %
Wheat	469	**298**	12.2	100 %
Cassava	466	**297**	18.3	100 %
Sweet pepper	475	**293**	15.5	98 %
Cherry (medium)	503	**289**	11.4	100 %

Figure 4.65: Crop water requirement, Coarse Sand, part 2 of 3.

Pistachio (medium)	503		**289**	11.4	100 %
Maize (grain)	451		**276**	11.0	100 %
Hazelnut (low)	481		**271**	10.7	100 %
Pecan (low)	481		**271**	10.7	100 %
Cherimoya (medium)	545		**265**	10.3	100 %
Citrus (medium density, short)	552		**262**	10.2	100 %
Chickpea	436		**260**	13.0	100 %
Sorghum (grain)	429		**247**	10.1	100 %
Apple (medium)	459		**234**	9.2	100 %
Wine grapes (pergola)	457		**222**	8.7	100 %
Cowpea (beans)	398		**217**	11.0	100 %
Millet	395		**212**	8.7	100 %
Peach (low)	435		**209**	8.2	100 %
Almond (medium)	435		**209**	8.2	100 %
Green gram (mung bean)	372		**179**	9.4	100 %
Apricot (young)	380		**145**	5.7	100 %
Olive (traditional, medium density)	446		**135**	5.3	100 %
Persimmon (young, low)	358		**123**	4.8	100 %
Pomegranate (low)	358		**123**	4.8	100 %
Highbush blueberry (low)	338		**114**	5.0	100 %
Fig tree (young, low)	327		**87**	3.4	100 %

Figure 4.66: Crop water requirement, Coarse Sand, part 3 of 3.

4.24 Feedback loops (Coarse Sand)

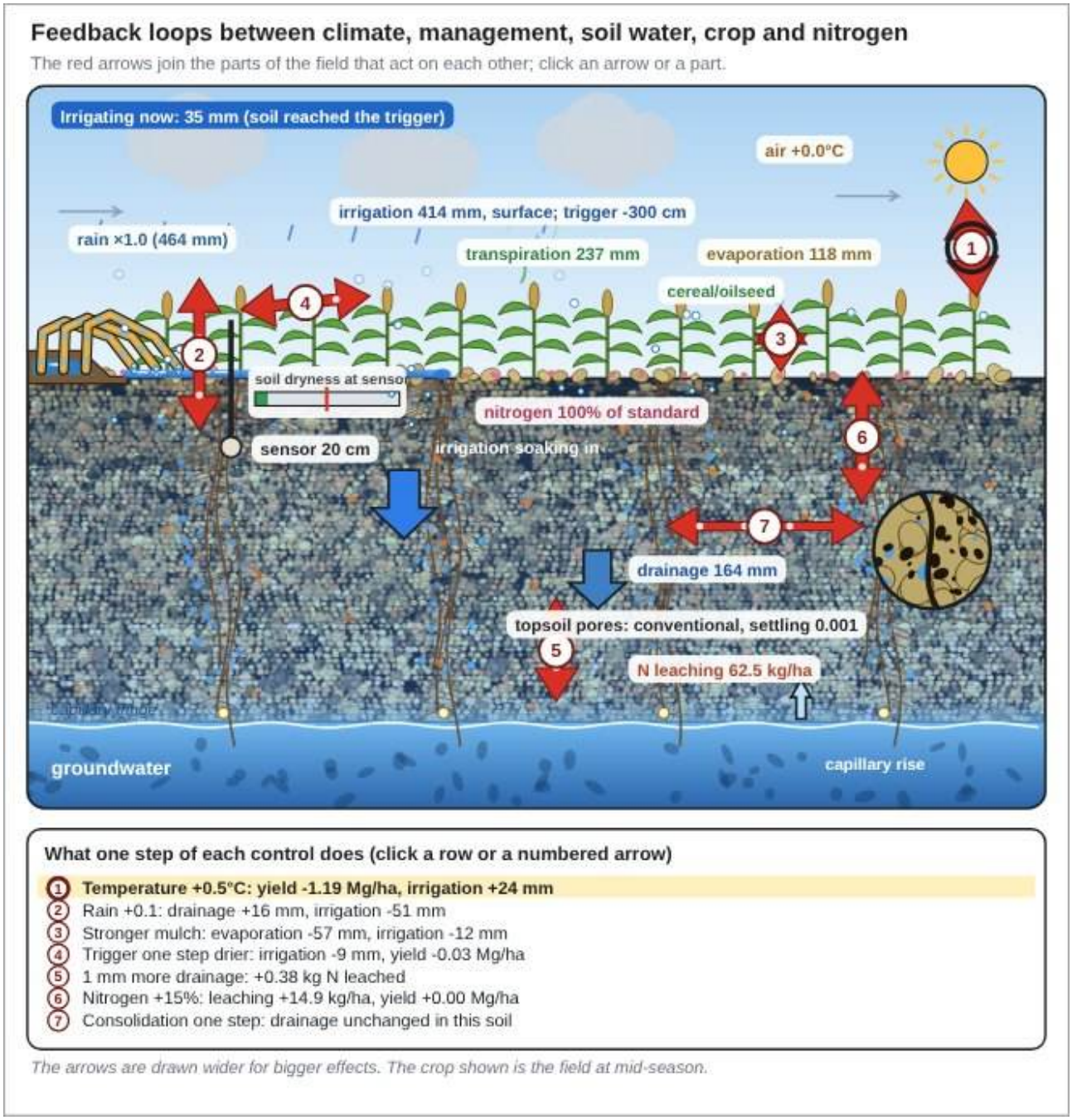


Figure 4.67: Feedback loops, Coarse Sand, part 1 of 1.

Chapter 5

Coarse Loam

5.1 Surface irrigation design (Coarse Loam)

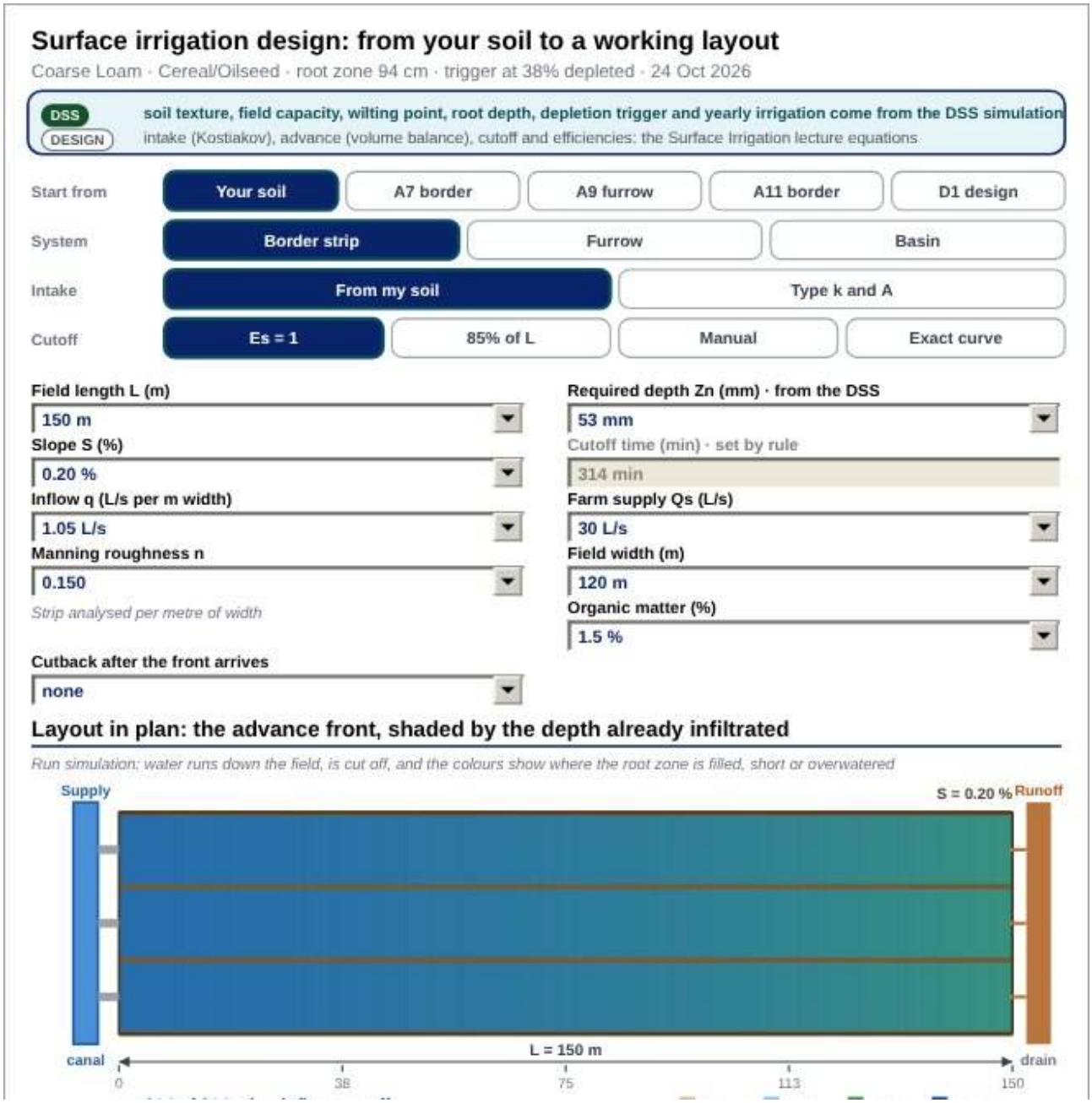


Figure 5.1: Surface irrigation design, Coarse Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

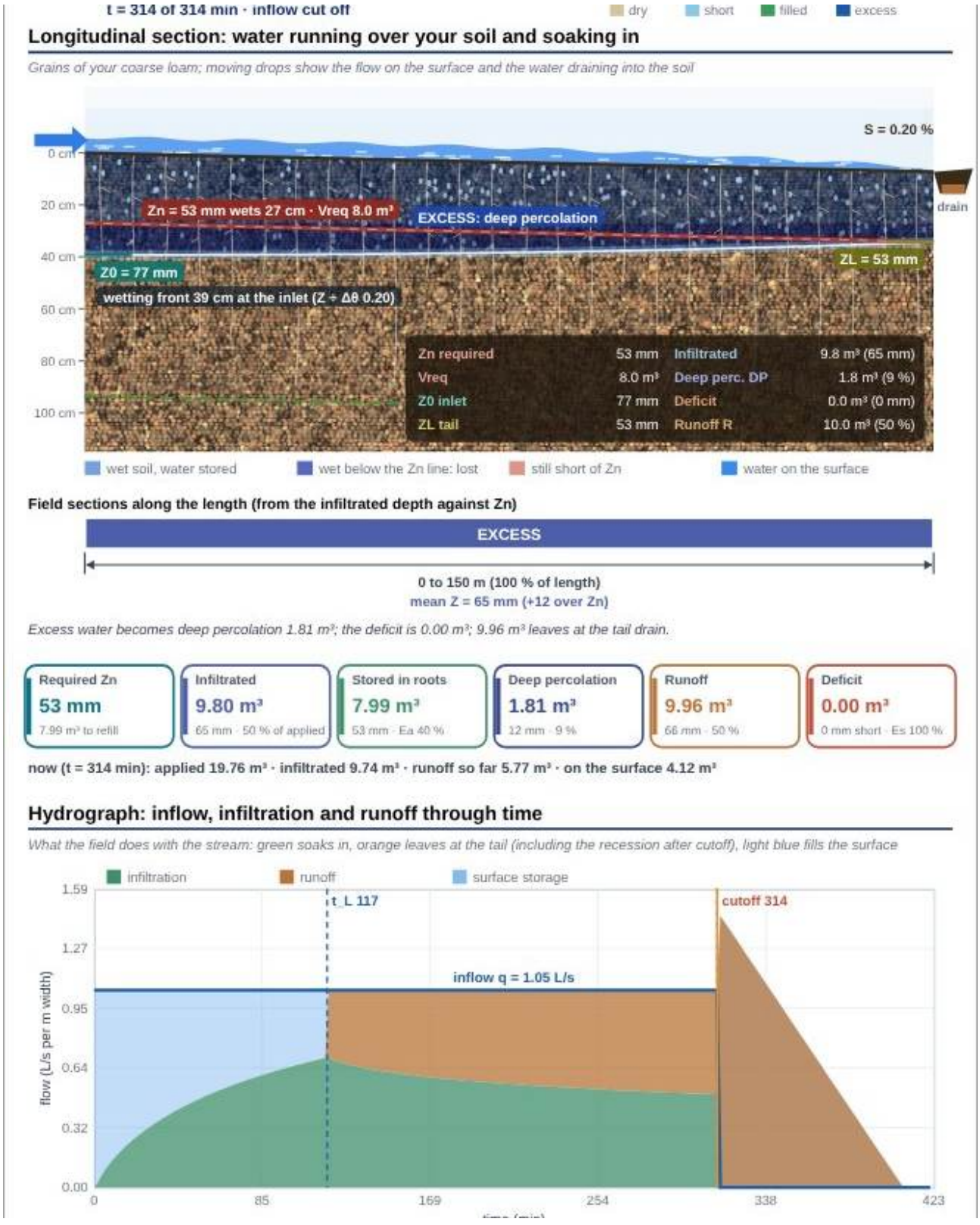


Figure 5.2: Surface irrigation design, Coarse Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

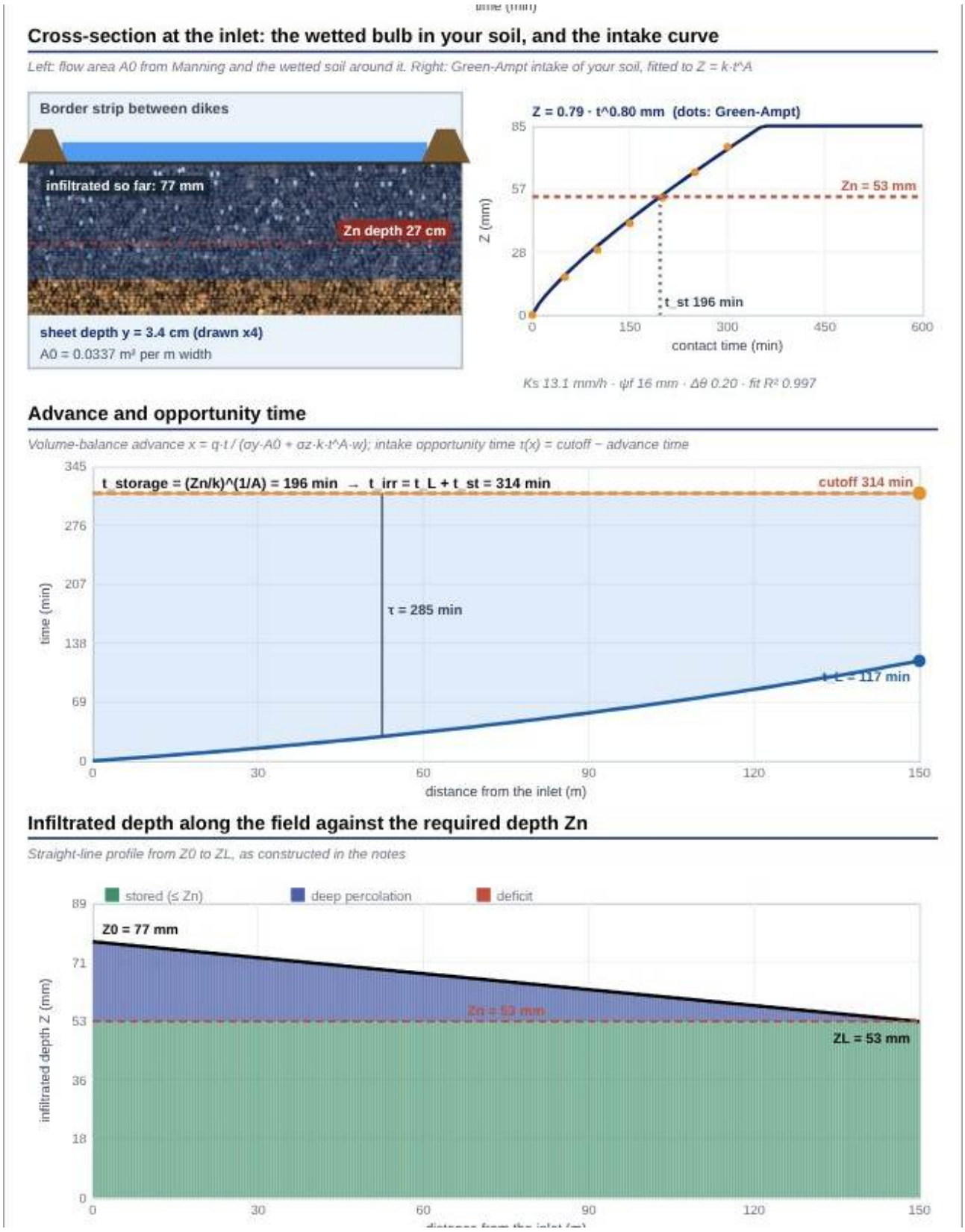


Figure 5.3: Surface irrigation design, Coarse Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

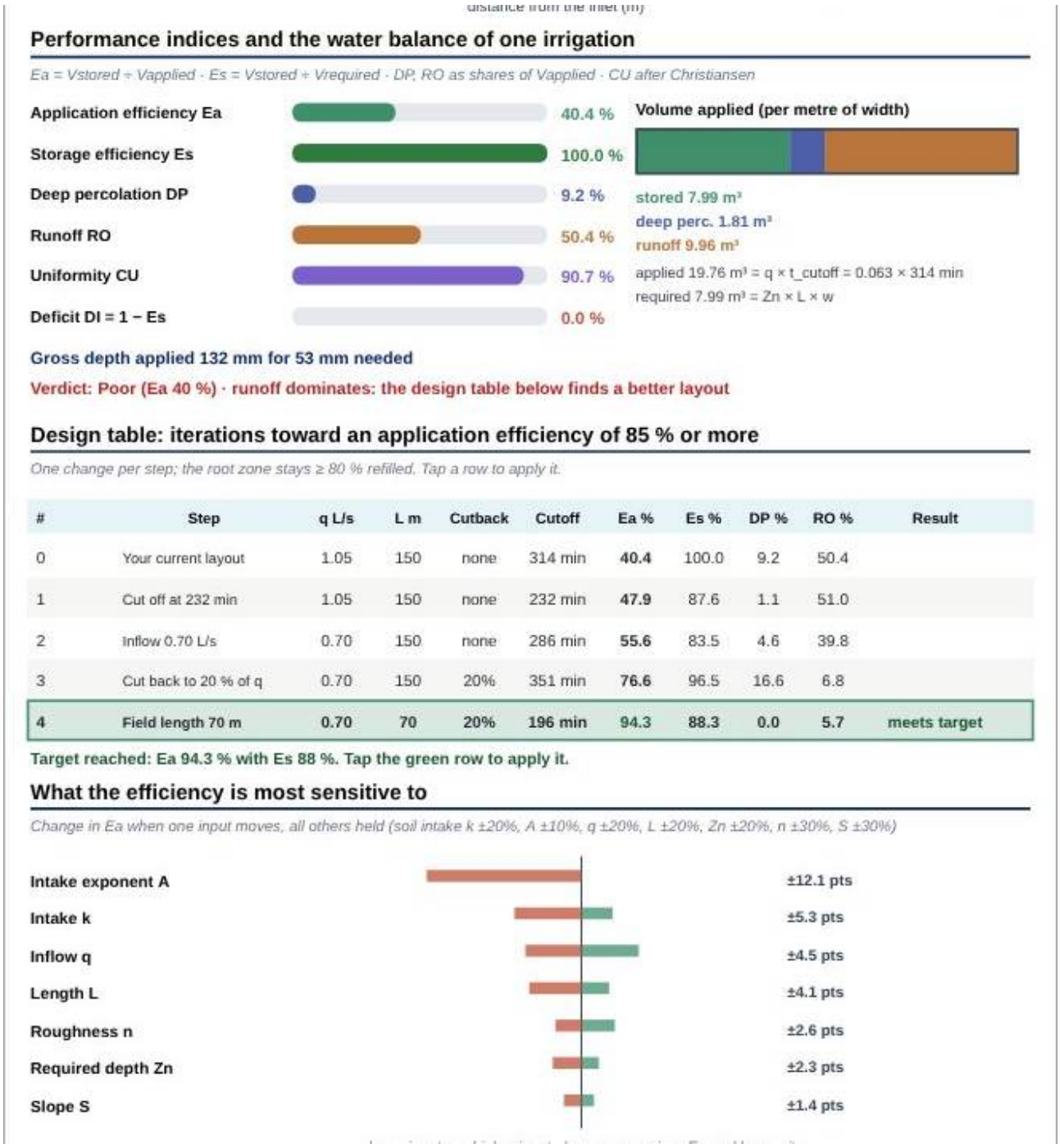


Figure 5.4: Surface irrigation design, Coarse Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

lower input ◄ ► higher input · bars: green raises Ea, red lowers it

What this means for the season and the farm

Change the date or the year in Properties. The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**339 mm → 6.4 irrigations of 53 mm**
Gross water per irrigation	**132 mm (Ea 40 %)**
Gross water per season	**837 mm = 8375 m³ per hectare**
Units run at once N0M = Qs ÷ q	**28.6 (30 L/s ÷ 1.05 L/s)**
Sets to cover 120 m of width	**4.2 sets × 314 min = 22.0 h per irrigation**
Pumping per season	**140 hours**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**22 May 2027**	210 days	28	70	22.0 h	13 days
2	**4 Jun 2027**	223 days	28	70	22.0 h	11 days
3	**15 Jun 2027**	234 days	28	70	22.0 h	9 days
4	**24 Jun 2027**	243 days	28	70	22.0 h	9 days
5	**3 Jul 2027**	252 days	28	70	22.0 h	10 days
6	**13 Jul 2027**	262 days	28	70	22.0 h	16 days
7	**29 Jul 2027**	278 days	28	70	22.0 h	251 days
8	**5 Apr 2028**	529 days	28	70	22.0 h	-

Intervals, efficiency and the advice in short

Everything above in one message for surface irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.38**
Irrigations in the season of this date	**12 (6 Apr 2026 to 24 Jul 2026)**
Interval between irrigations	**9.9 days on average (shortest 1, longest 20)**
Depth per irrigation	**53.3 mm net, 131.7 mm pumped**
Application efficiency Ea	**40.4 %**
Storage efficiency Es	**100.0 %**
Time per round	**22.0 h**

When to irrigate

Next irrigation: 22 May 2027 (in 210 days), about 28 mm net, 70 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this surface design 40 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 22.0 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 5.5: Surface irrigation design, Coarse Loam, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

5.2 Drip irrigation design (Coarse Loam)

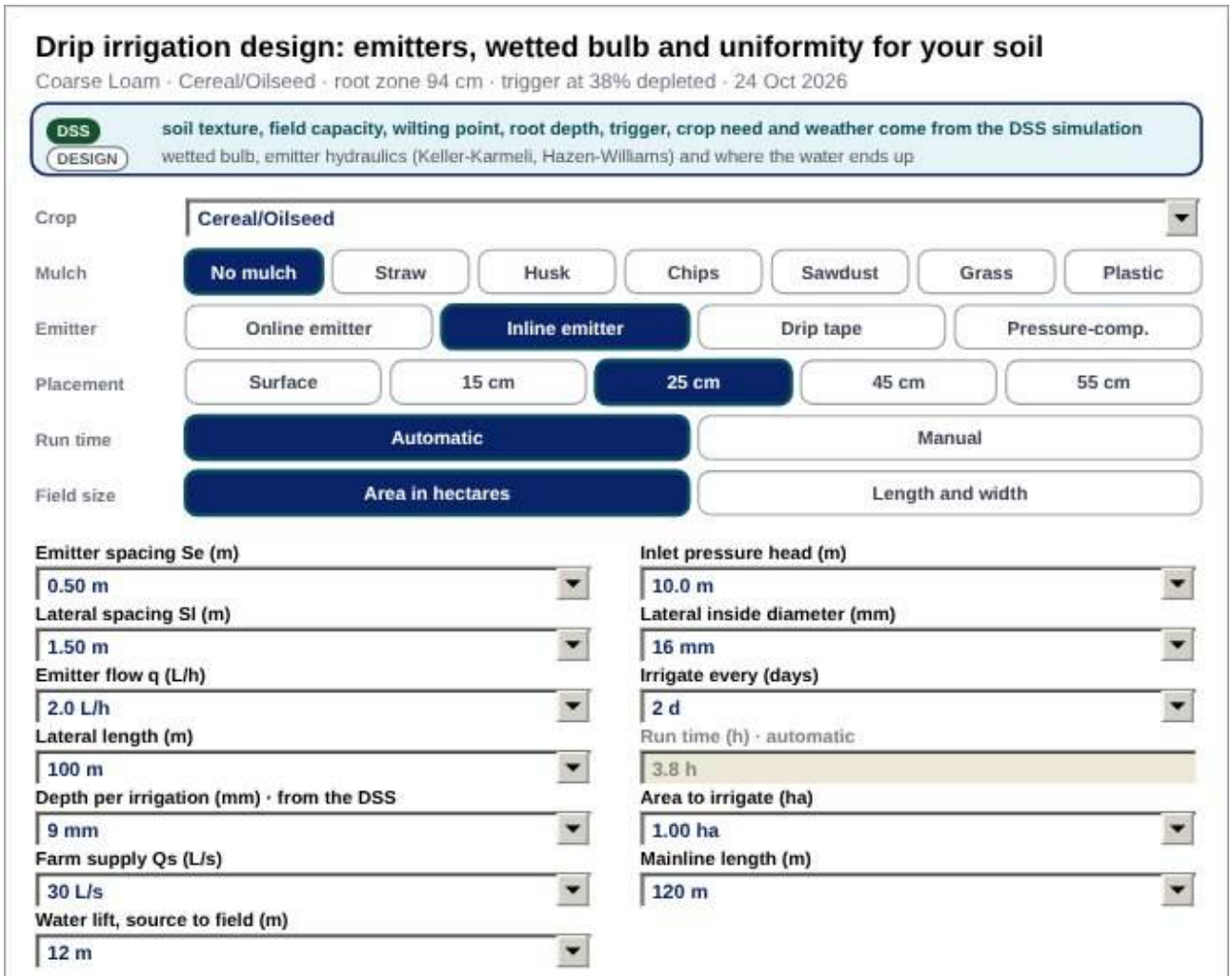


Figure 5.6: Drip irrigation design, Coarse Loam, part 1 of 6.

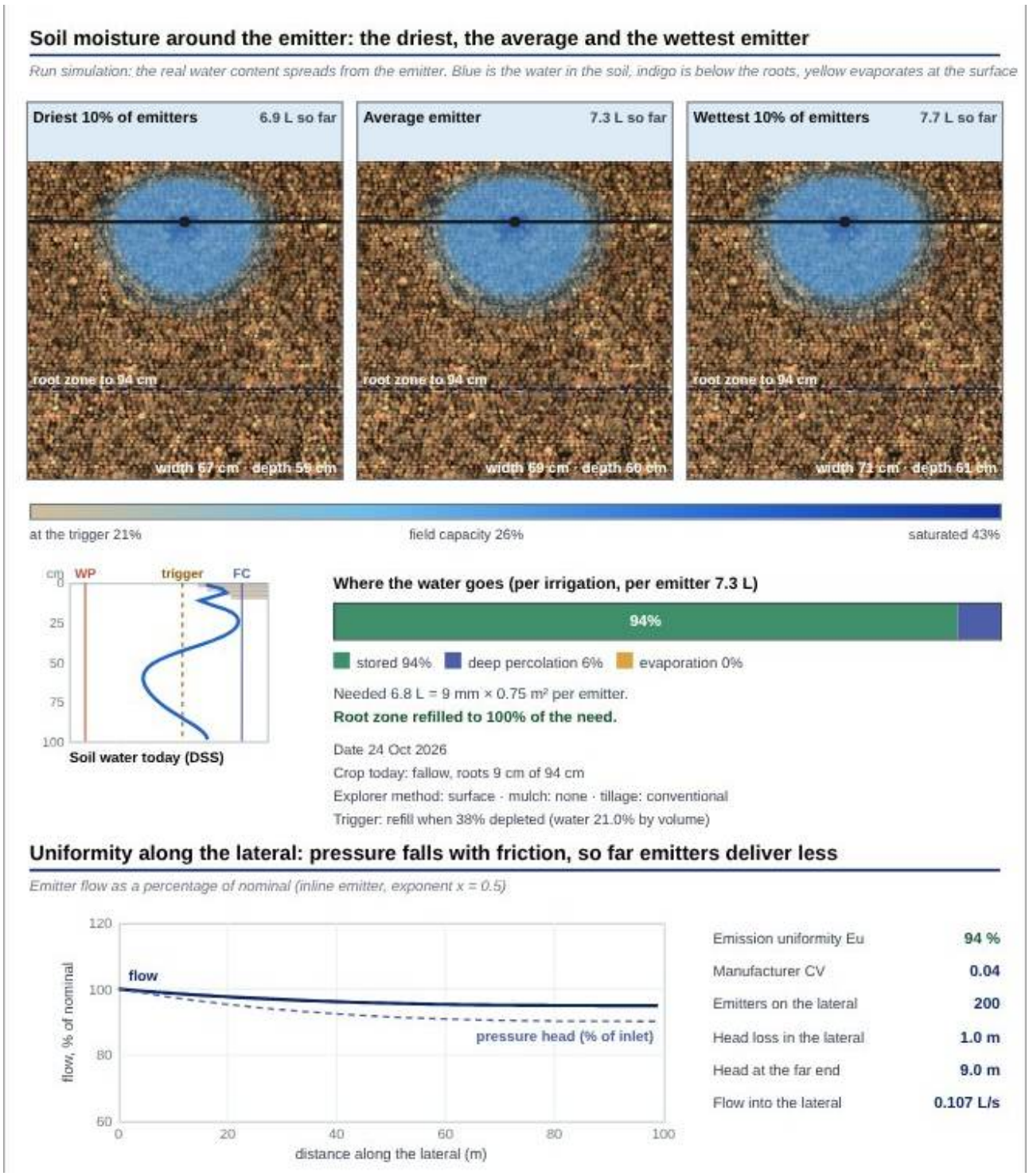


Figure 5.7: Drip irrigation design, Coarse Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

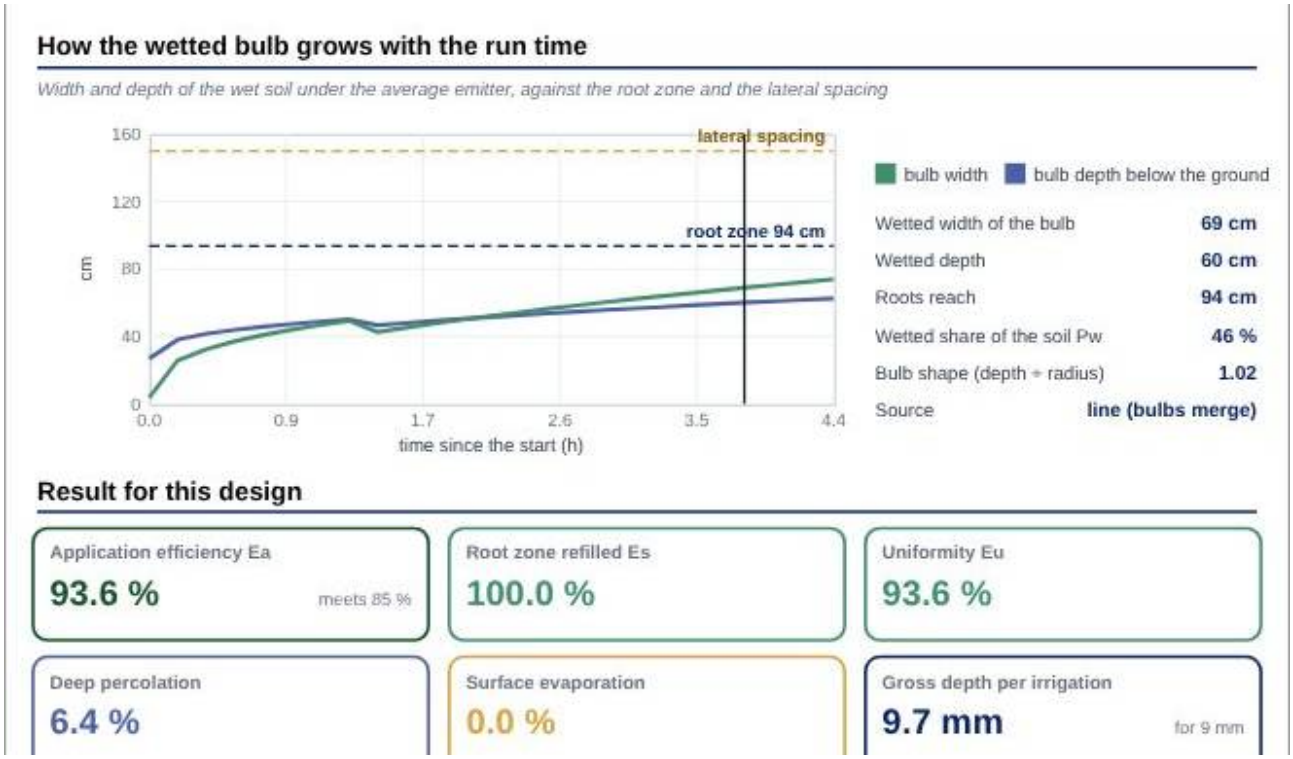
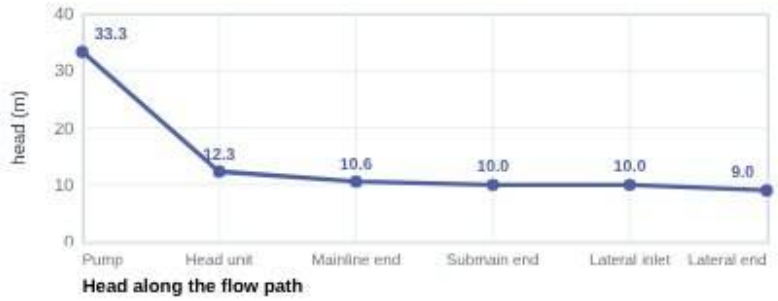


Figure 5.8: Drip irrigation design, Coarse Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.

Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals

To scale. Run simulation: the zones run one after the other; the supply Qs decides how many laterals run at once



Head along the flow path

Zones	**1 (× 66 laterals)**
Laterals / emitters	**66 / 13266**
Pump head TDH	**33.3 m**
Pump power	**3.6 kW**
Lateral head loss	**1.0 m**
Pipe laid per ha	**6699 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.8	93.6	100.0	6.4	meets target

Figure 5.9: Drip irrigation design, Coarse Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

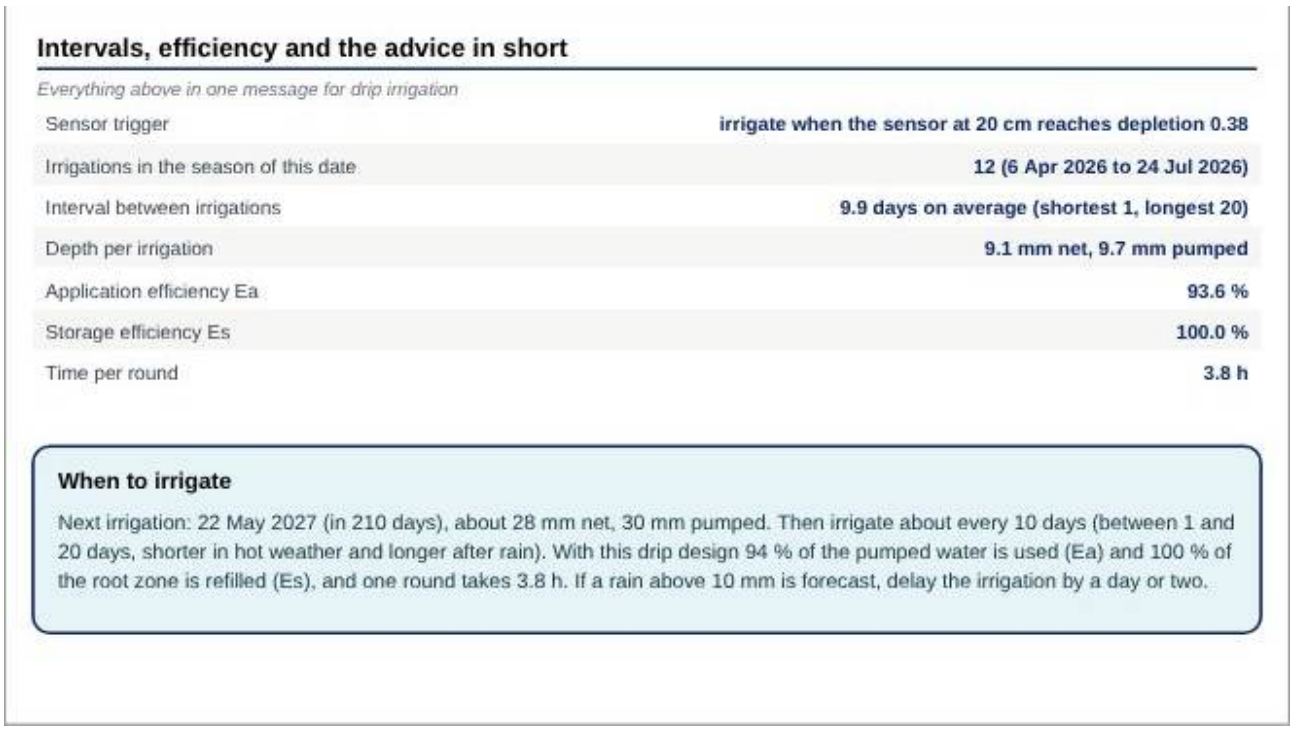
Net irrigation need in a season (DSS)	**339 mm → 37 irrigations of 9 mm**
Gross water per irrigation	**9.7 mm (Ea 94 %)**
Gross water per season	**362 mm = 3618 m³ per hectare**
Peak crop need / interval	**4.6 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.8 h = 3.8 h per round**
Pumping per season	**141 hours · 500 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**22 May 2027**	210 days	28	30	3.8 h	13 days
2	**4 Jun 2027**	223 days	28	30	3.8 h	11 days
3	**15 Jun 2027**	234 days	28	30	3.8 h	9 days
4	**24 Jun 2027**	243 days	28	30	3.8 h	9 days
5	**3 Jul 2027**	252 days	28	30	3.8 h	10 days
6	**13 Jul 2027**	262 days	28	30	3.8 h	16 days
7	**29 Jul 2027**	278 days	28	30	3.8 h	251 days
8	**5 Apr 2028**	529 days	28	30	3.8 h	-

Figure 5.10: Drip irrigation design, Coarse Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.38**
Irrigations in the season of this date	**12 (6 Apr 2026 to 24 Jul 2026)**
Interval between irrigations	**9.9 days on average (shortest 1, longest 20)**
Depth per irrigation	**9.1 mm net, 9.7 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**3.8 h**

When to irrigate

Next irrigation: 22 May 2027 (in 210 days), about 28 mm net, 30 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 3.8 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 5.11: Drip irrigation design, Coarse Loam, part 6 of 6: Intervals, efficiency and the advice in short.

5.3 Sprinkler design and pipes (Coarse Loam)

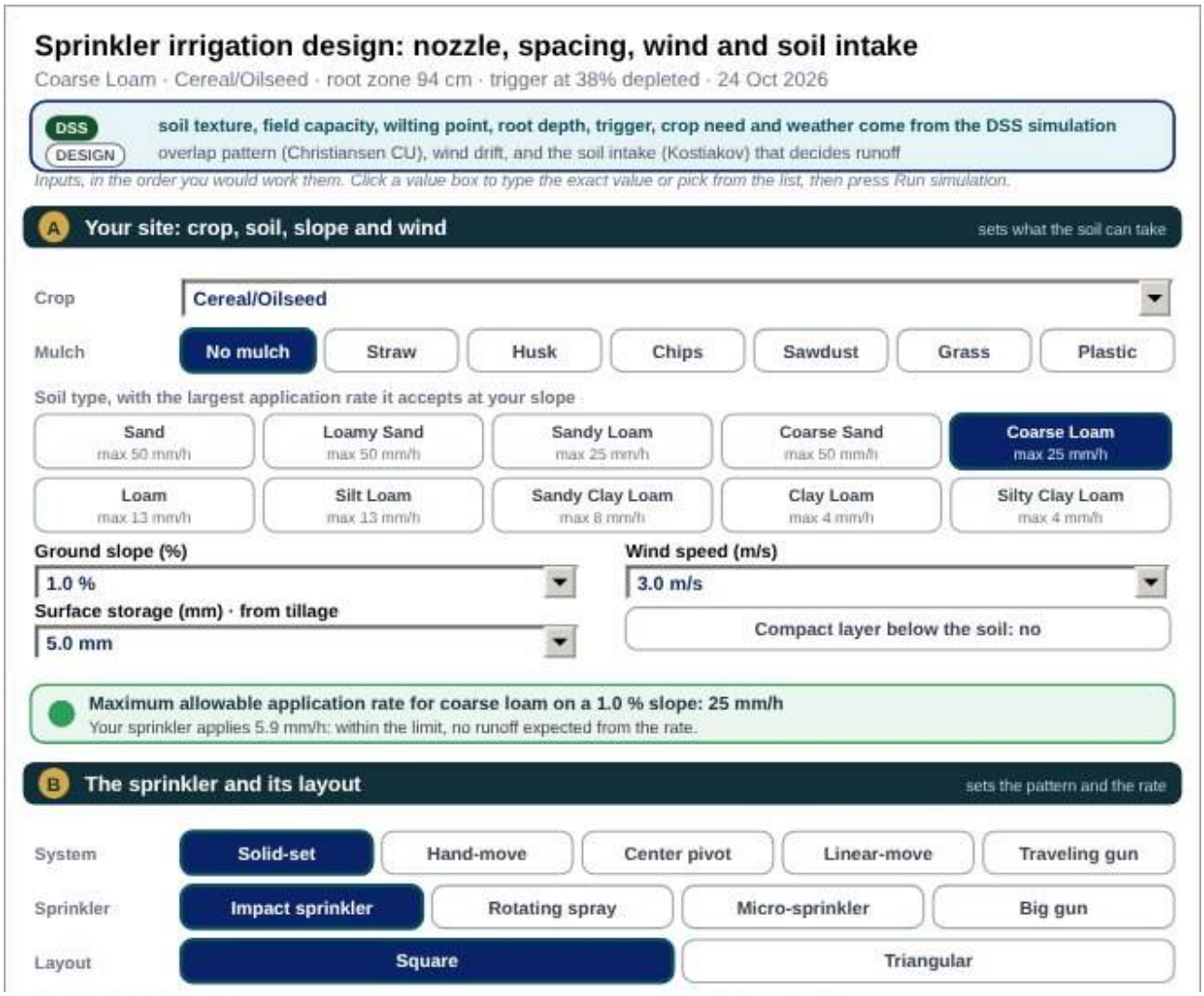


Figure 5.12: Sprinkler design and pipes, Coarse Loam, part 1 of 12.

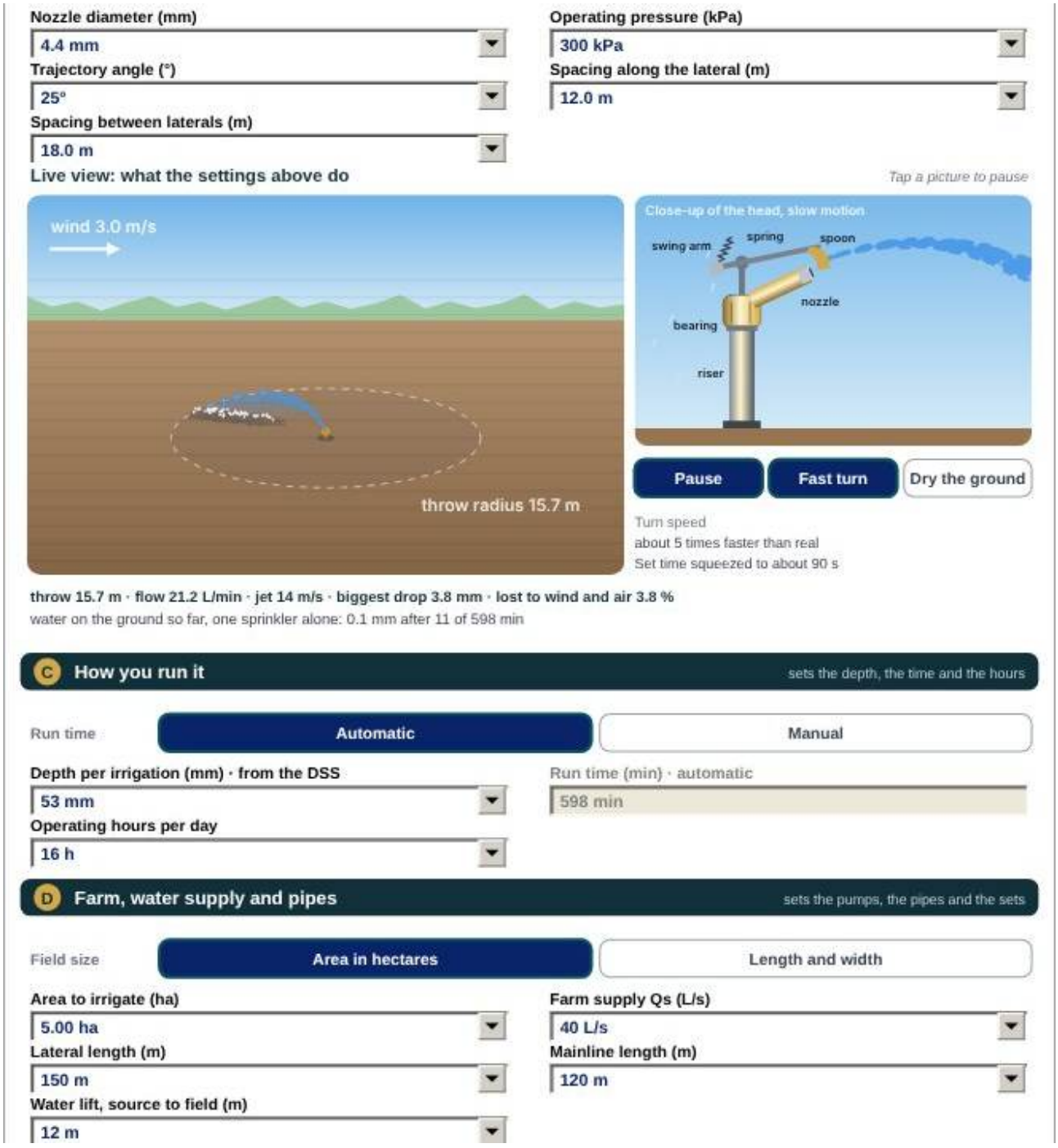


Figure 5.13: Sprinkler design and pipes, Coarse Loam, part 2 of 12.

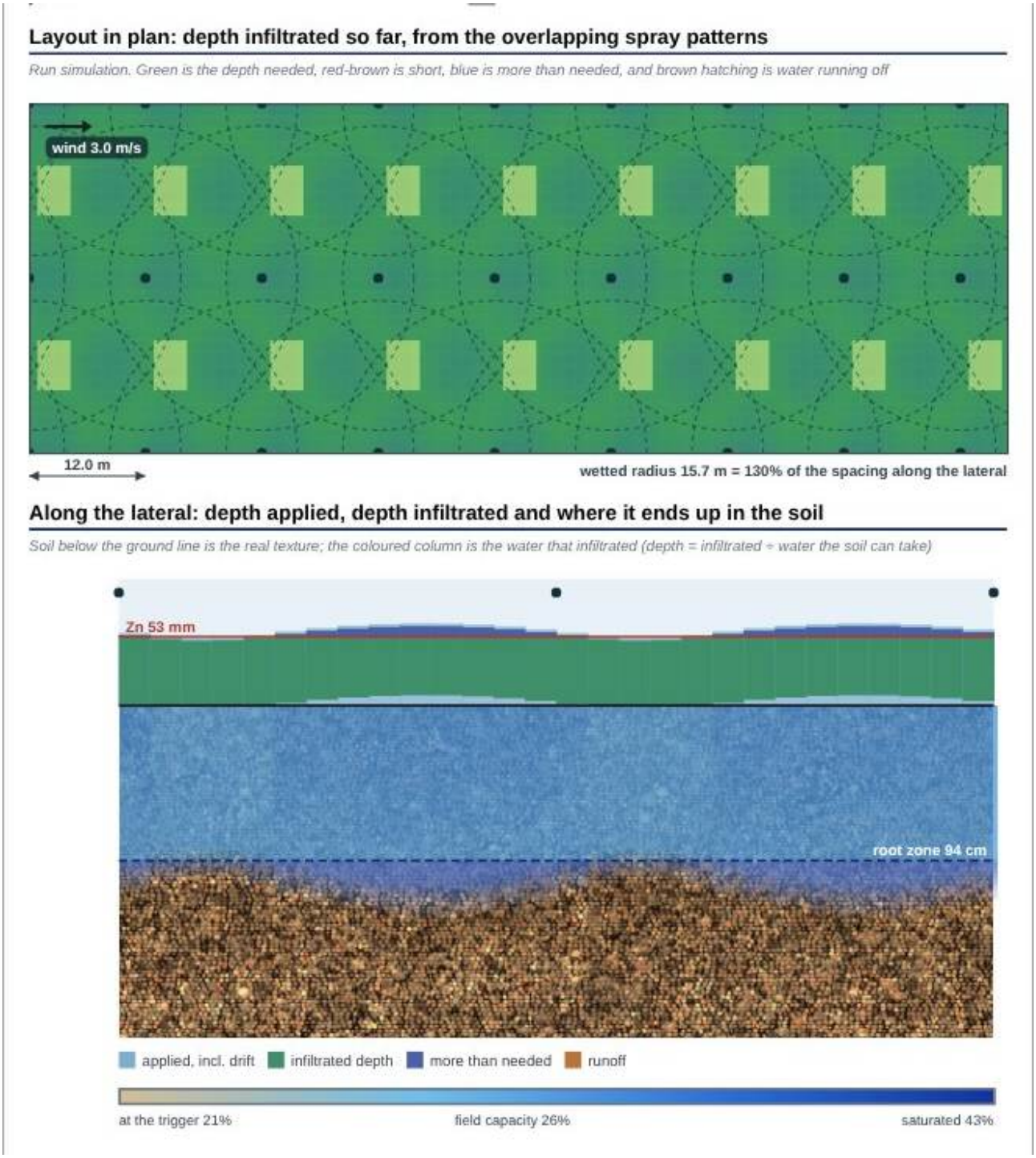


Figure 5.14: Sprinkler design and pipes, Coarse Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

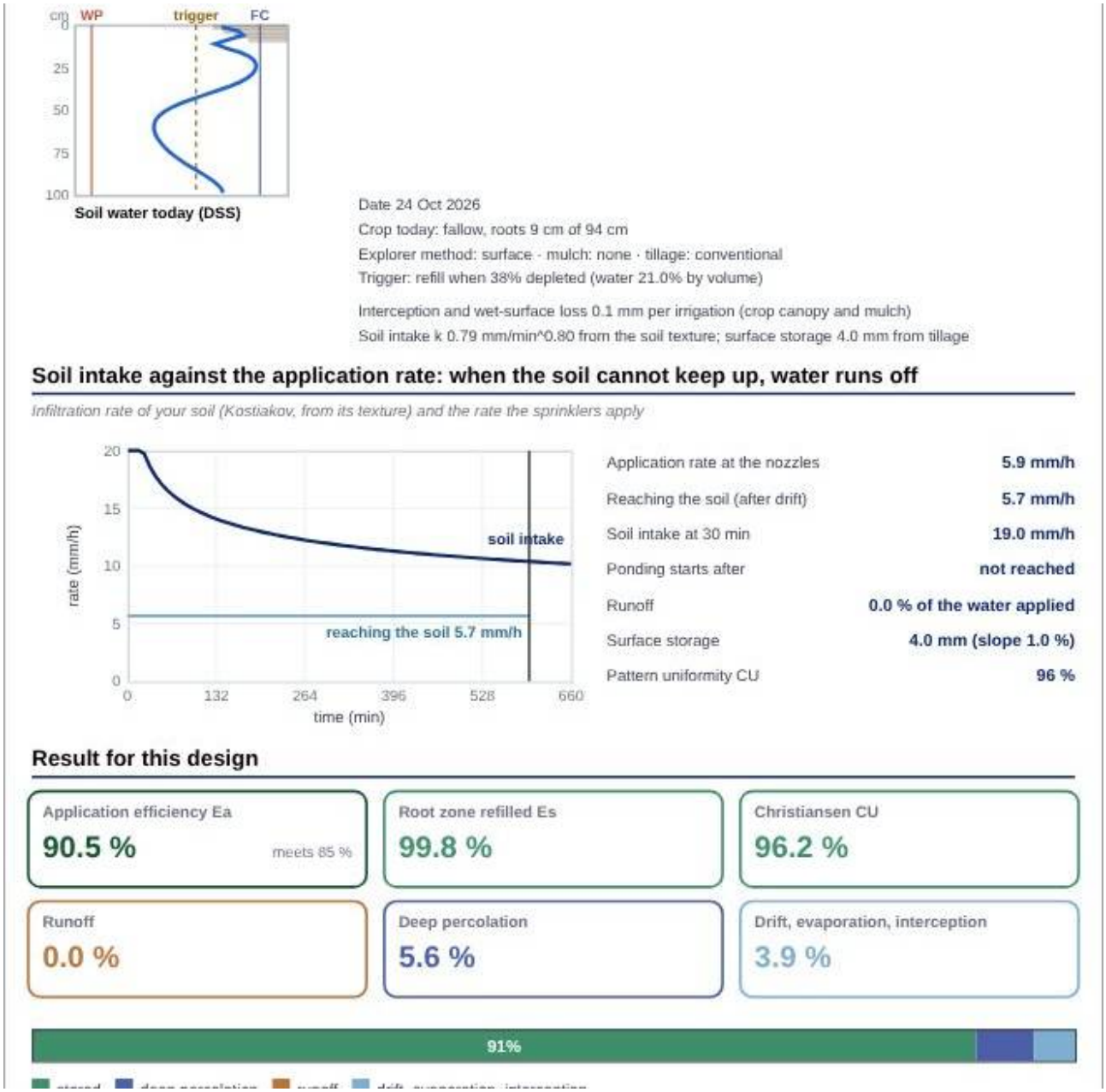


Figure 5.15: Sprinkler design and pipes, Coarse Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

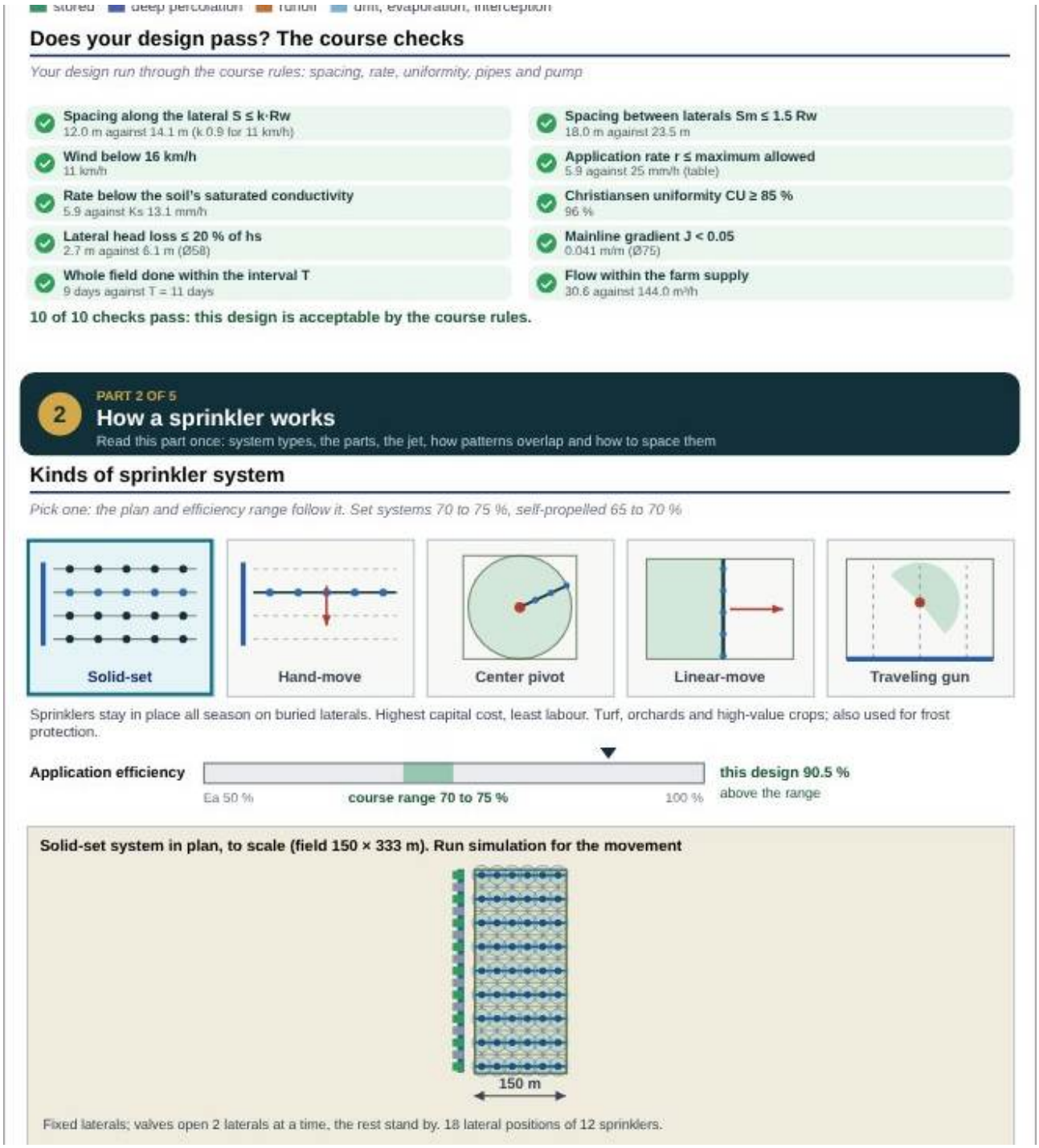


Figure 5.16: Sprinkler design and pipes, Coarse Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

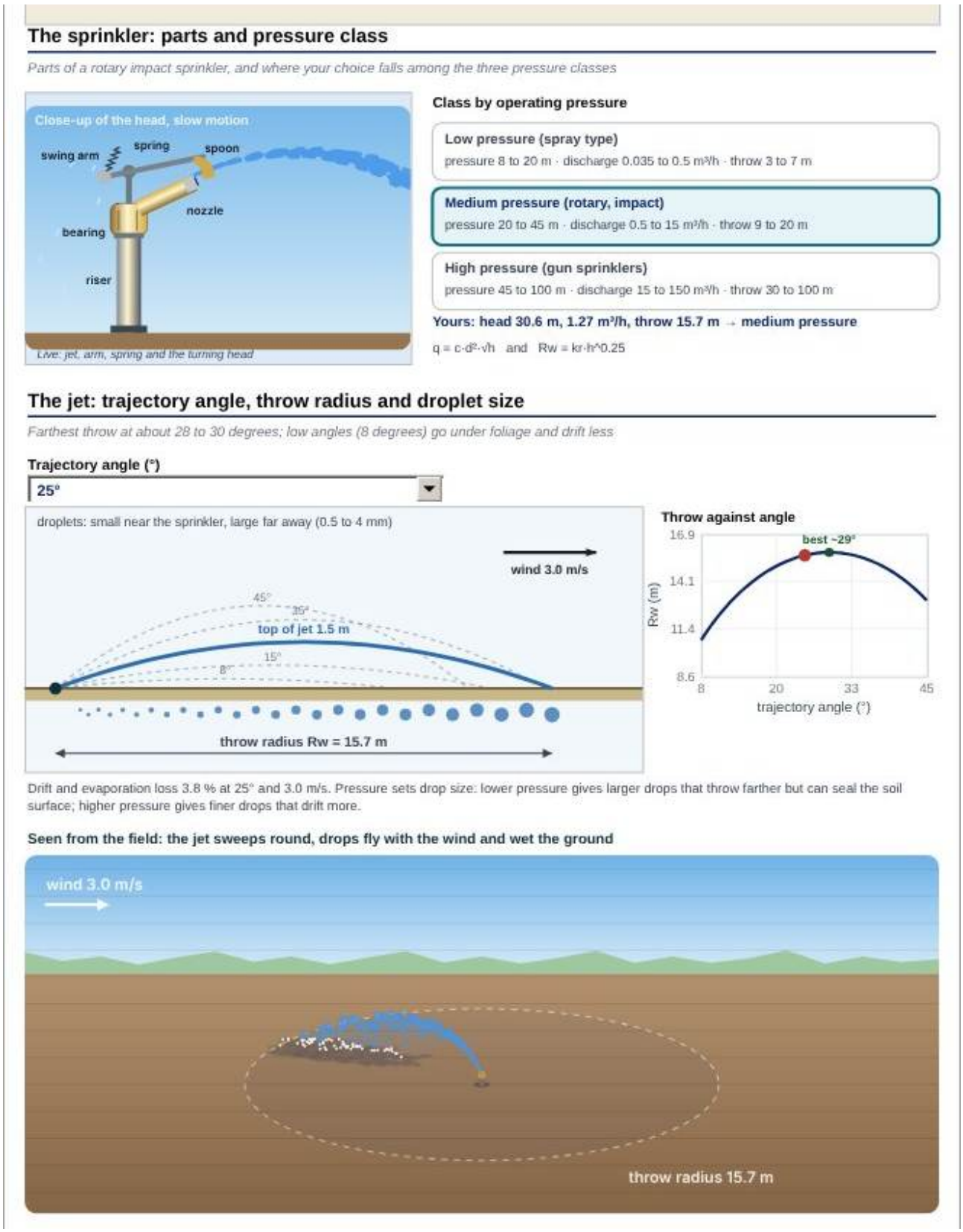


Figure 5.17: Sprinkler design and pipes, Coarse Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

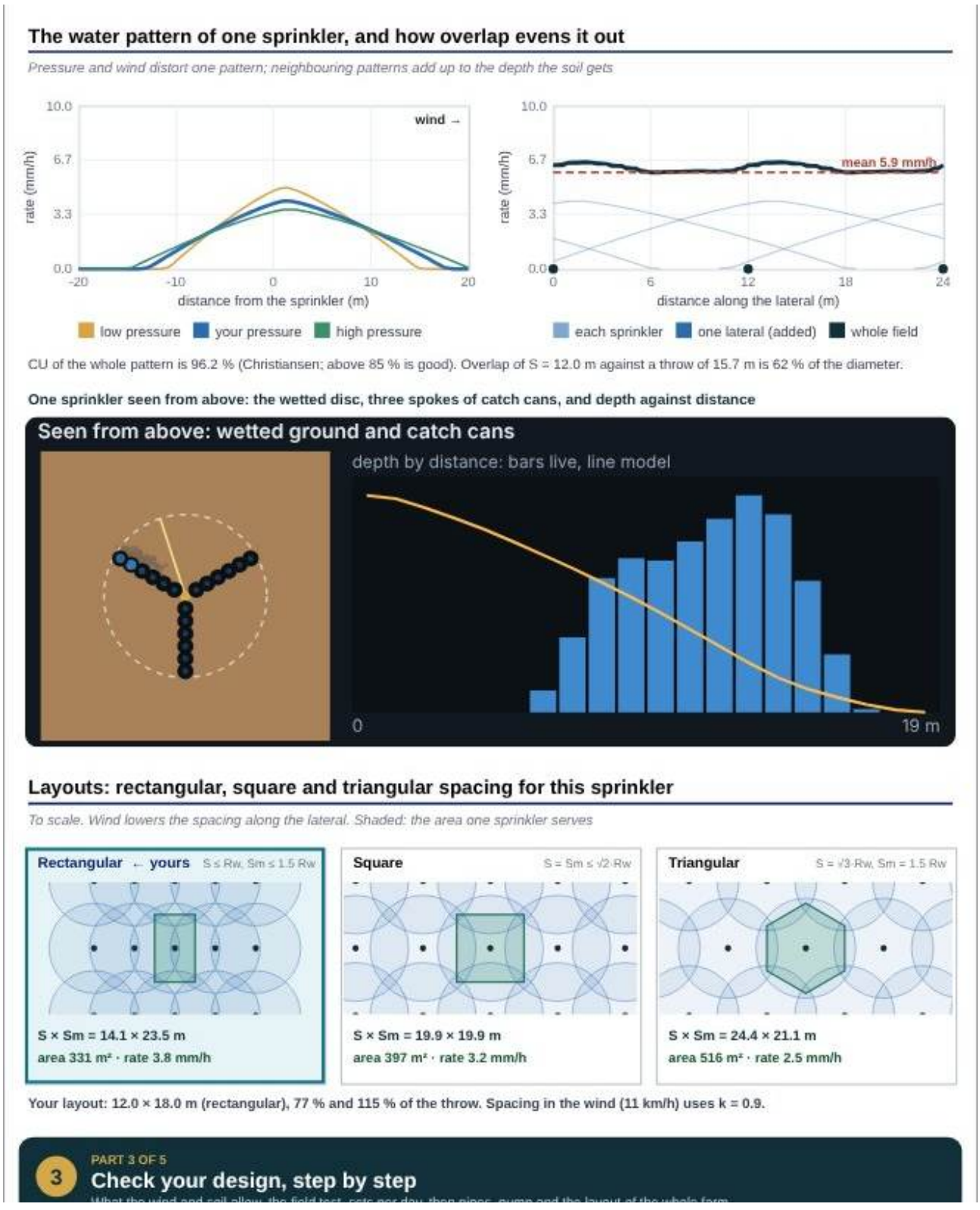


Figure 5.18: Sprinkler design and pipes, Coarse Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

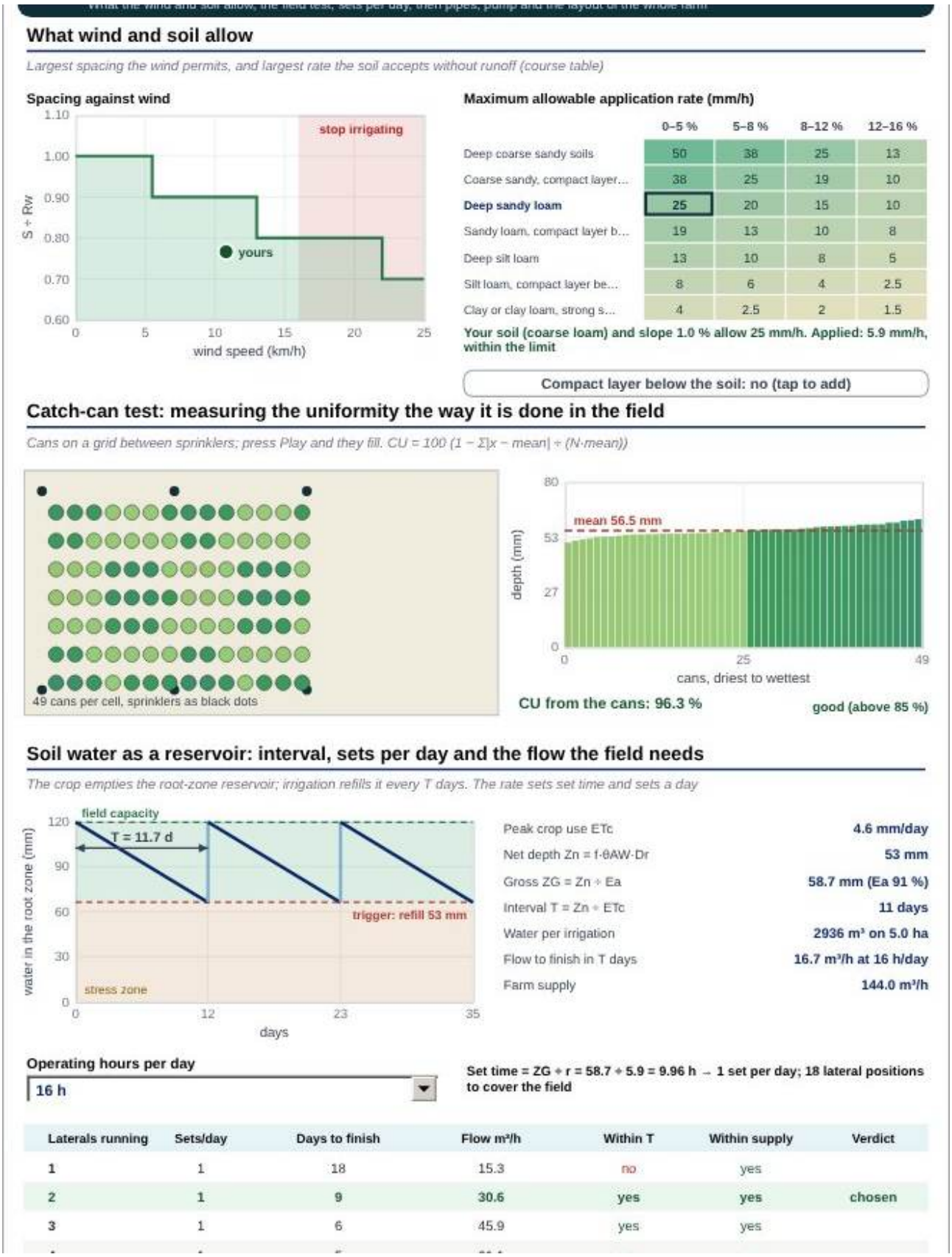


Figure 5.19: Sprinkler design and pipes, Coarse Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

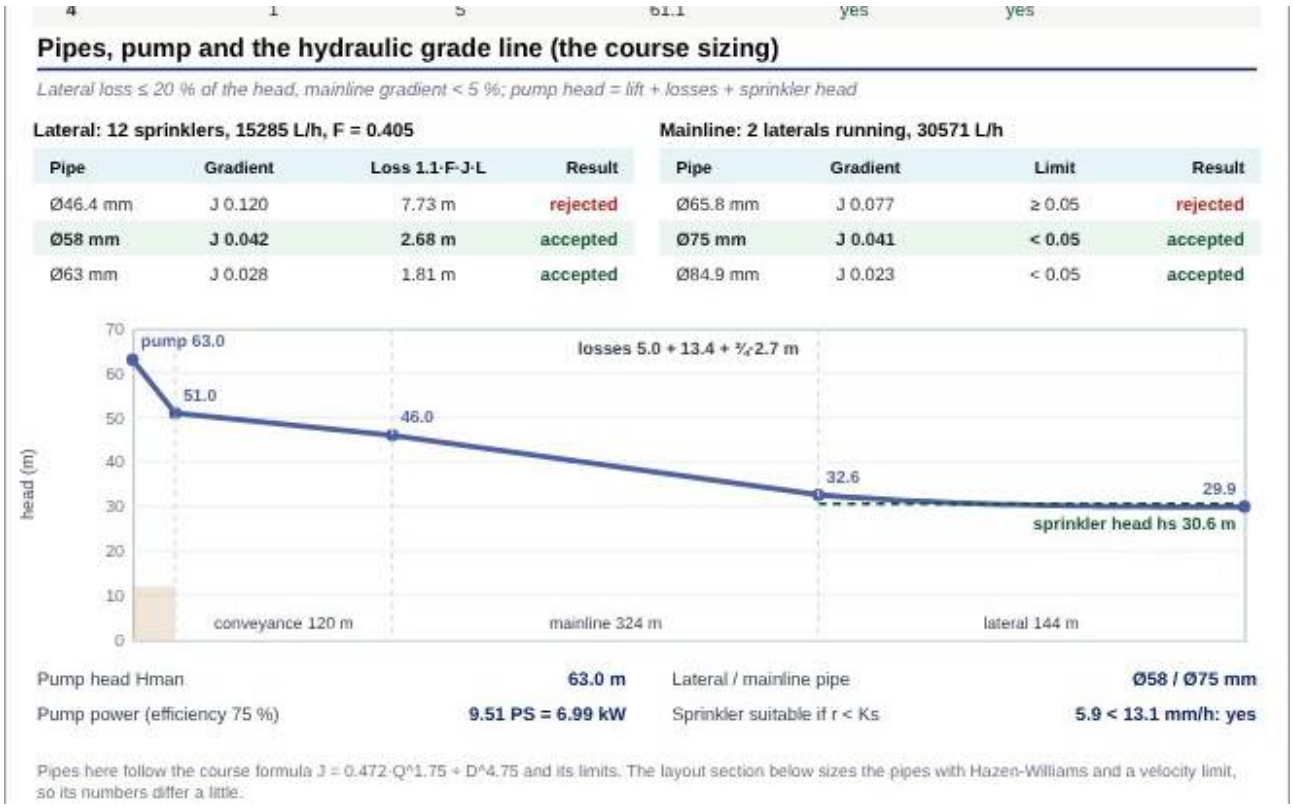
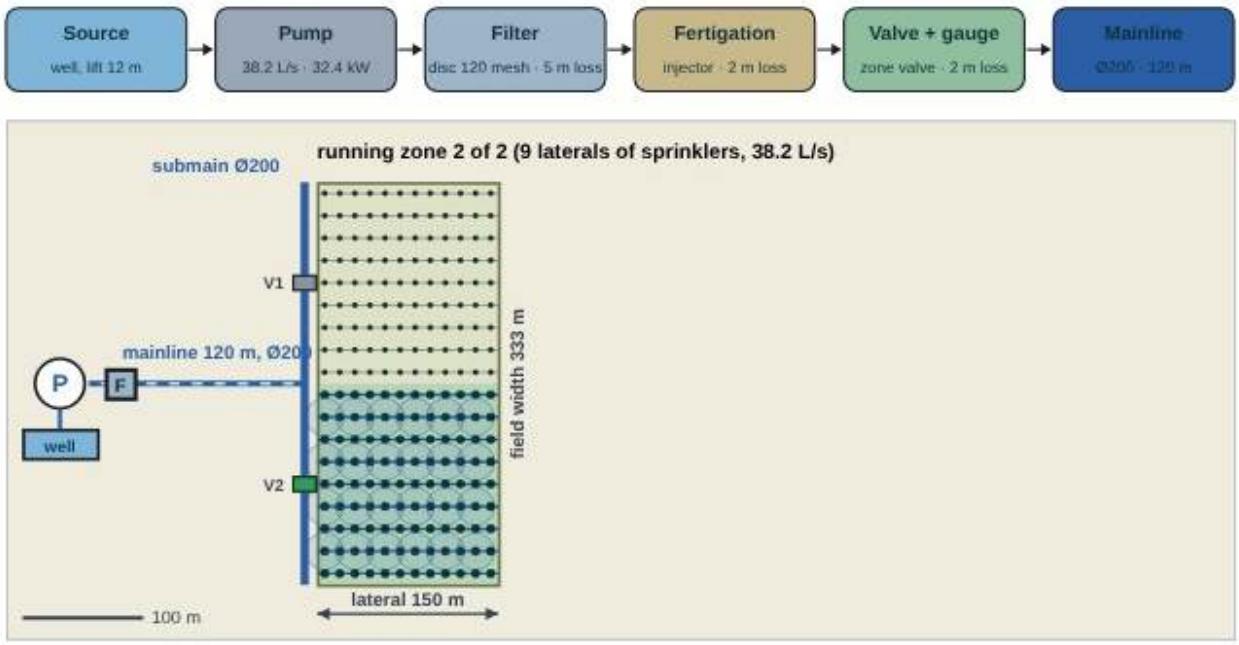
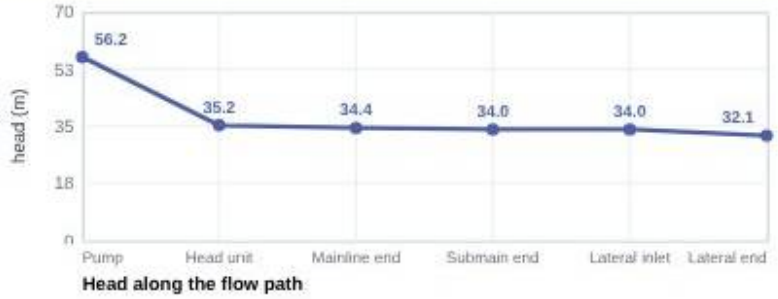


Figure 5.20: Sprinkler design and pipes, Coarse Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4

Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

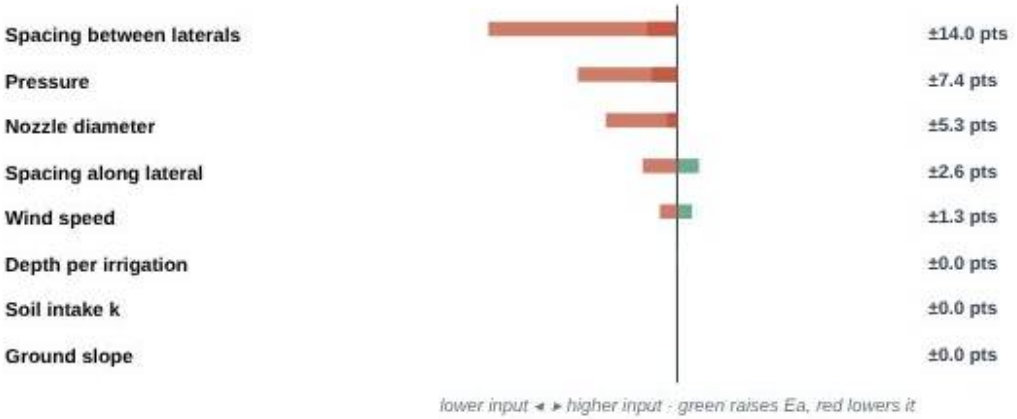
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	598	90.5	99.8	0.0	meets target

Figure 5.21: Sprinkler design and pipes, Coarse Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

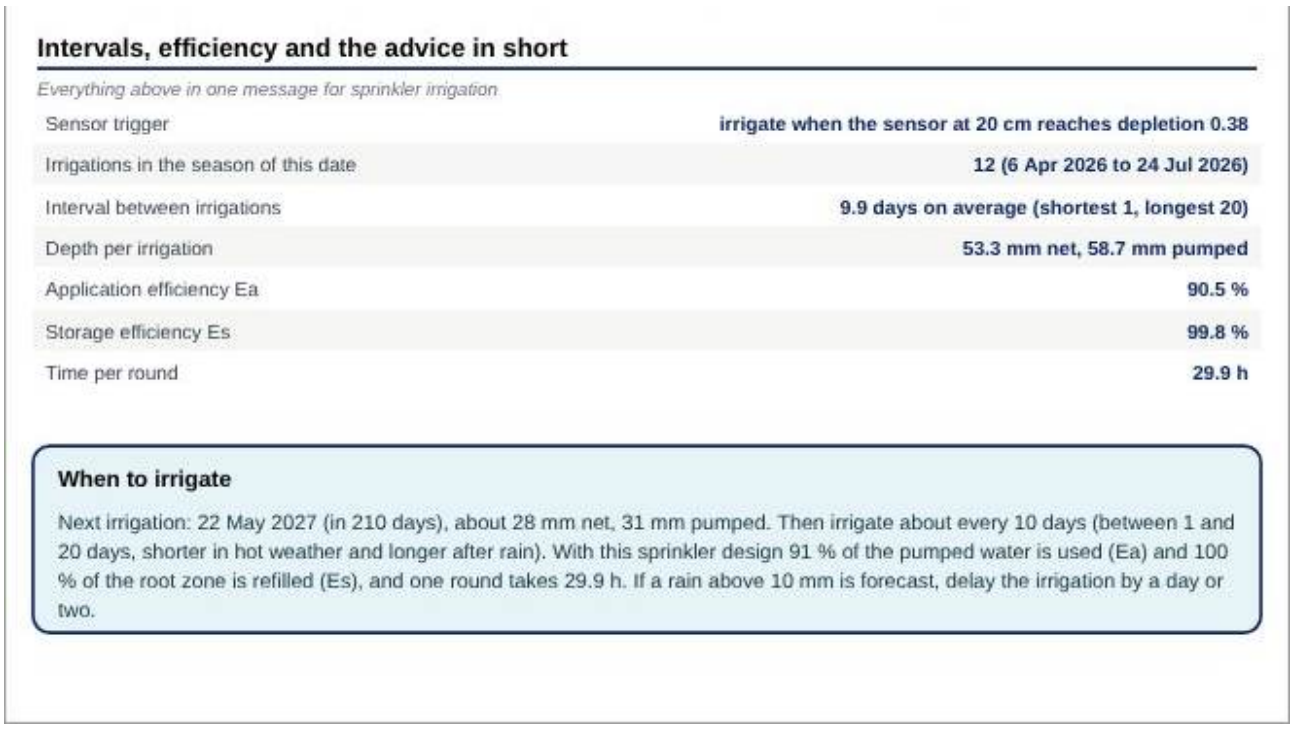
Net irrigation need in a season (DSS)	**339 mm → 6 irrigations of 53 mm**
Gross water per irrigation	**58.7 mm (Ea 91 %)**
Gross water per season	**373 mm = 3734 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 598 min = 29.9 h per round**
Pumping per season	**190 hours · 6162 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**22 May 2027**	210 days	28	31	29.9 h	13 days
2	**4 Jun 2027**	223 days	28	31	29.9 h	11 days
3	**15 Jun 2027**	234 days	28	31	29.9 h	9 days
4	**24 Jun 2027**	243 days	28	31	29.9 h	9 days
5	**3 Jul 2027**	252 days	28	31	29.9 h	10 days
6	**13 Jul 2027**	262 days	28	31	29.9 h	16 days
7	**29 Jul 2027**	278 days	28	31	29.9 h	251 days
8	**5 Apr 2028**	529 days	28	31	29.9 h	-

Figure 5.22: Sprinkler design and pipes, Coarse Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.38**
Irrigations in the season of this date	**12 (6 Apr 2026 to 24 Jul 2026)**
Interval between irrigations	**9.9 days on average (shortest 1, longest 20)**
Depth per irrigation	**53.3 mm net, 58.7 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**29.9 h**

When to irrigate

Next irrigation: 22 May 2027 (in 210 days), about 28 mm net, 31 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 29.9 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 5.23: Sprinkler design and pipes, Coarse Loam, part 12 of 12: Intervals, efficiency and the advice in short.

5.4 Virtual farm of the surface design (Coarse Loam)

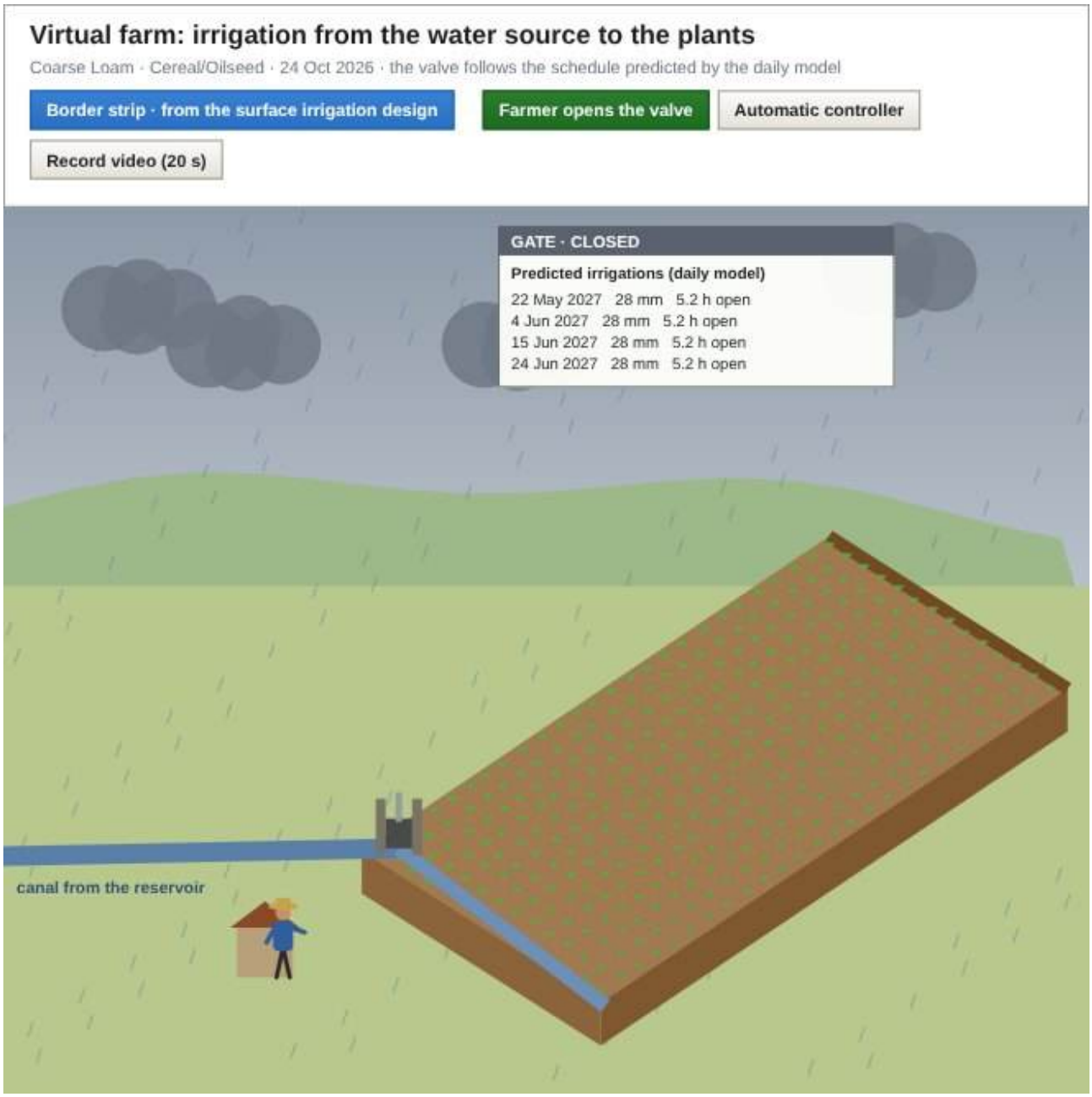


Figure 5.24: Virtual farm, surface design, Coarse Loam, part 1 of 3.

Figure 5.25: Virtual farm, surface design, Coarse Loam, part 2 of 3.

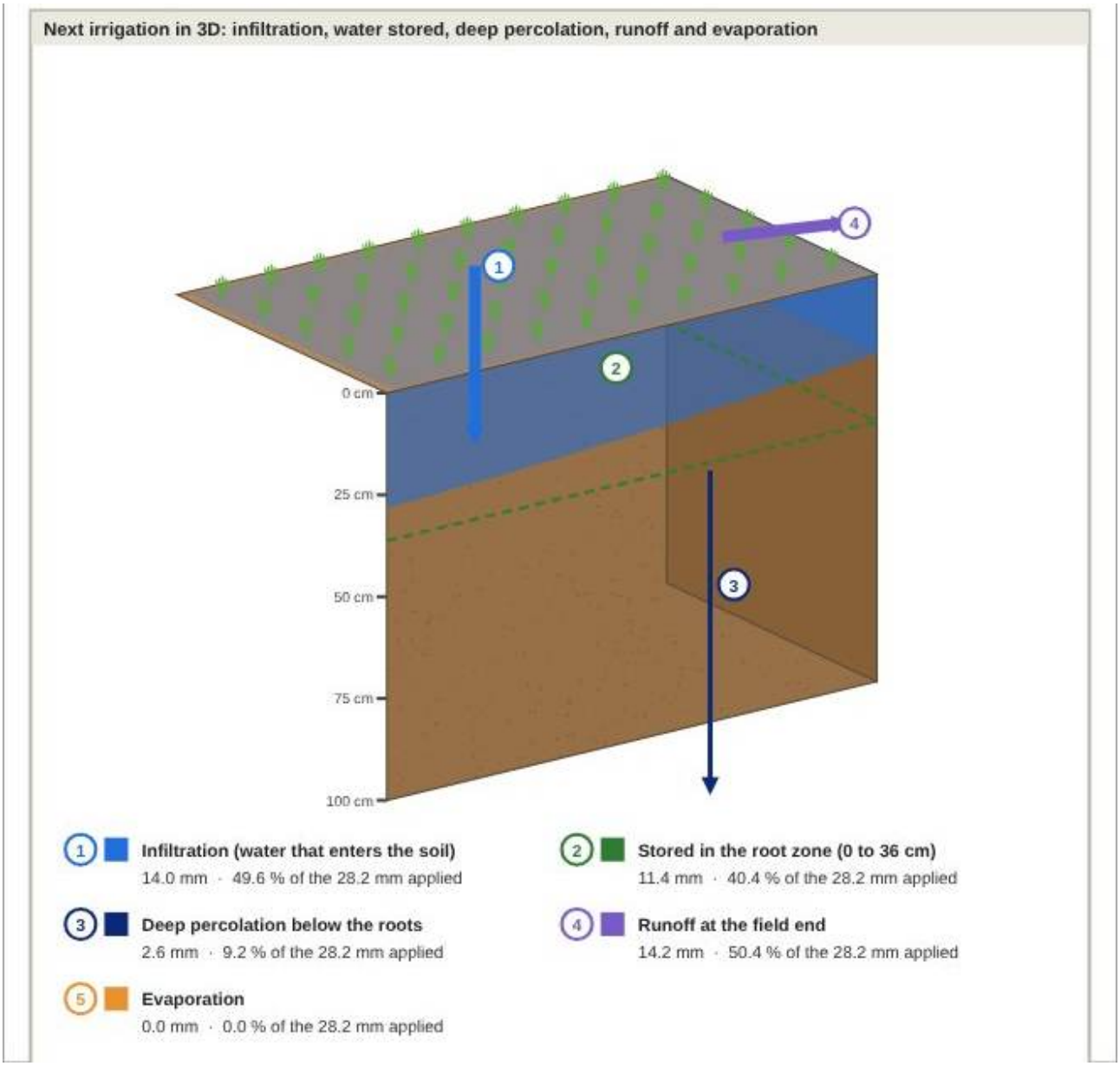


Figure 5.26: Virtual farm, surface design, Coarse Loam, part 3 of 3.

5.5 Virtual farm of the drip design (Coarse Loam)

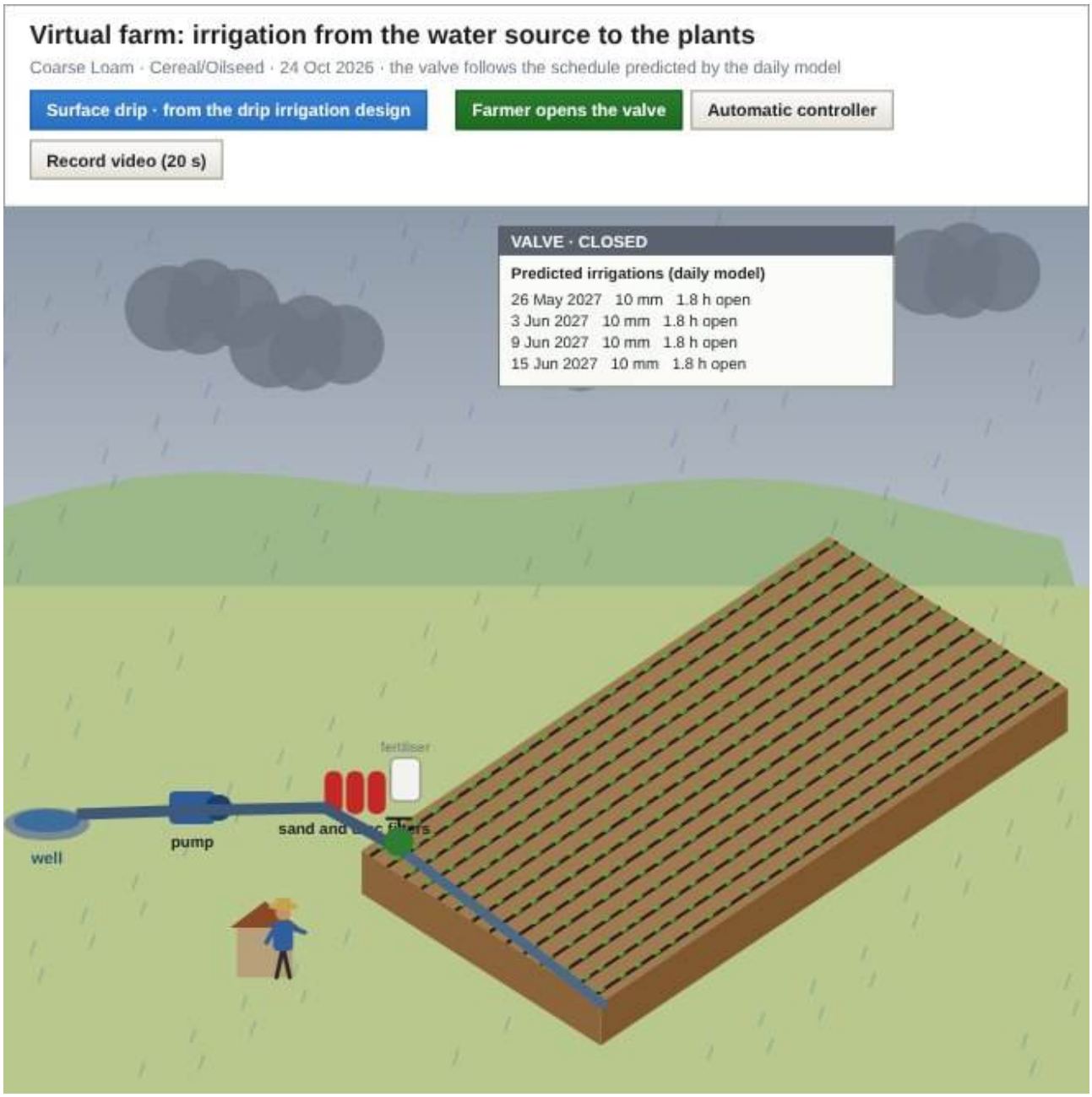


Figure 5.27: Virtual farm, drip design, Coarse Loam, part 1 of 3.

Figure 5.28: Virtual farm, drip design, Coarse Loam, part 2 of 3.

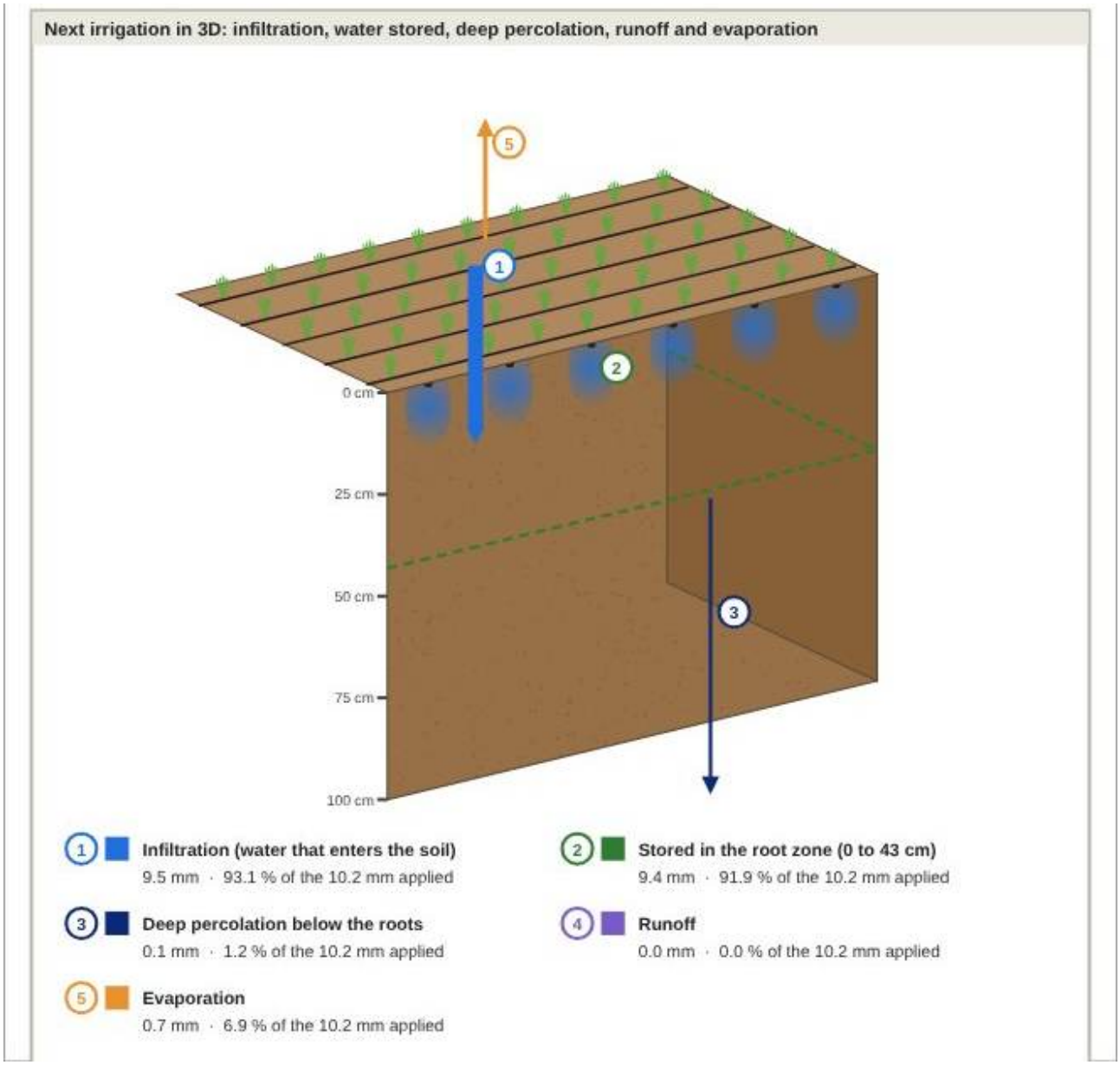


Figure 5.29: Virtual farm, drip design, Coarse Loam, part 3 of 3.

5.6 Virtual farm of the sprinkler design (Coarse Loam)

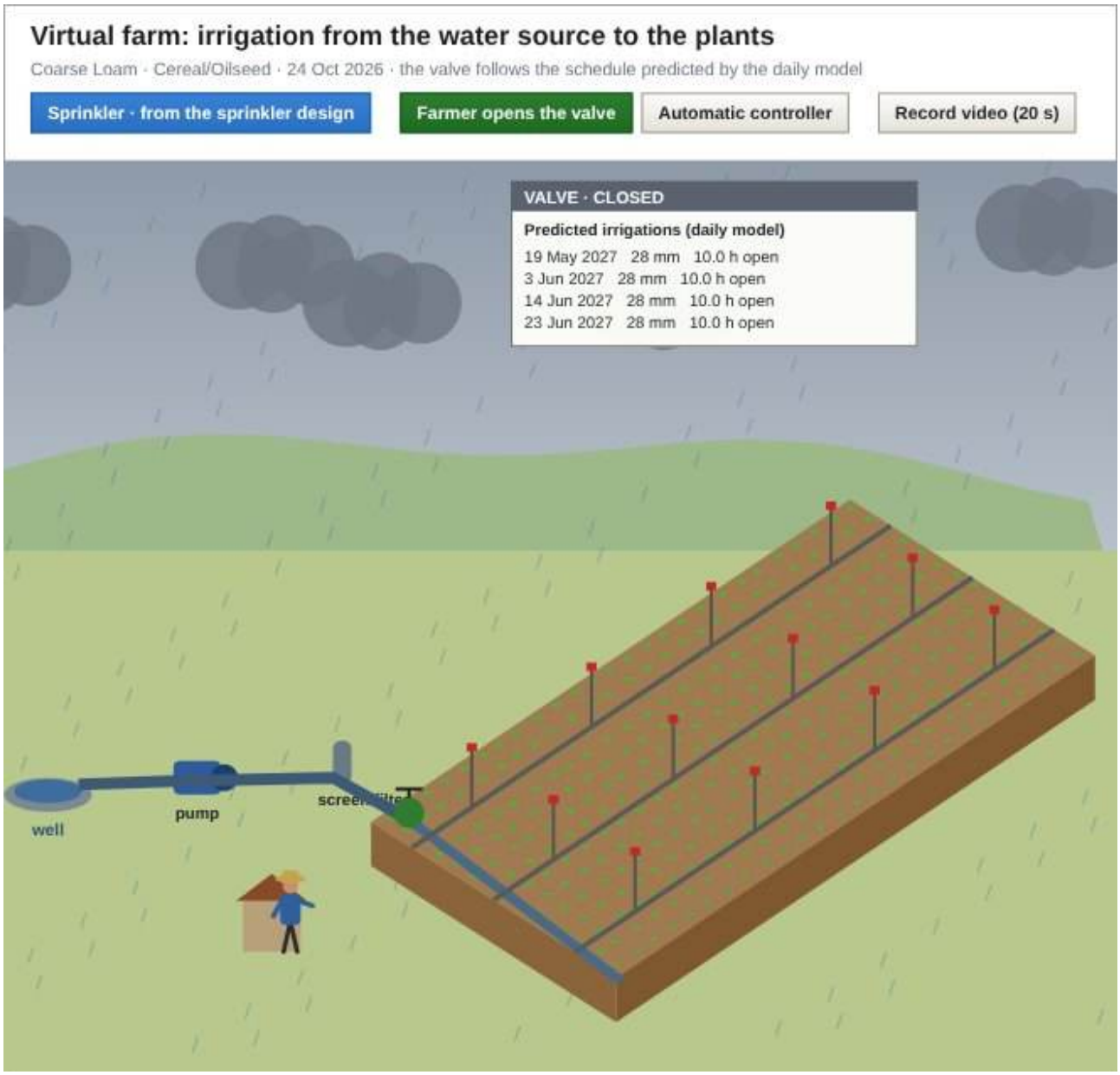


Figure 5.30: Virtual farm, sprinkler design, Coarse Loam, part 1 of 3.

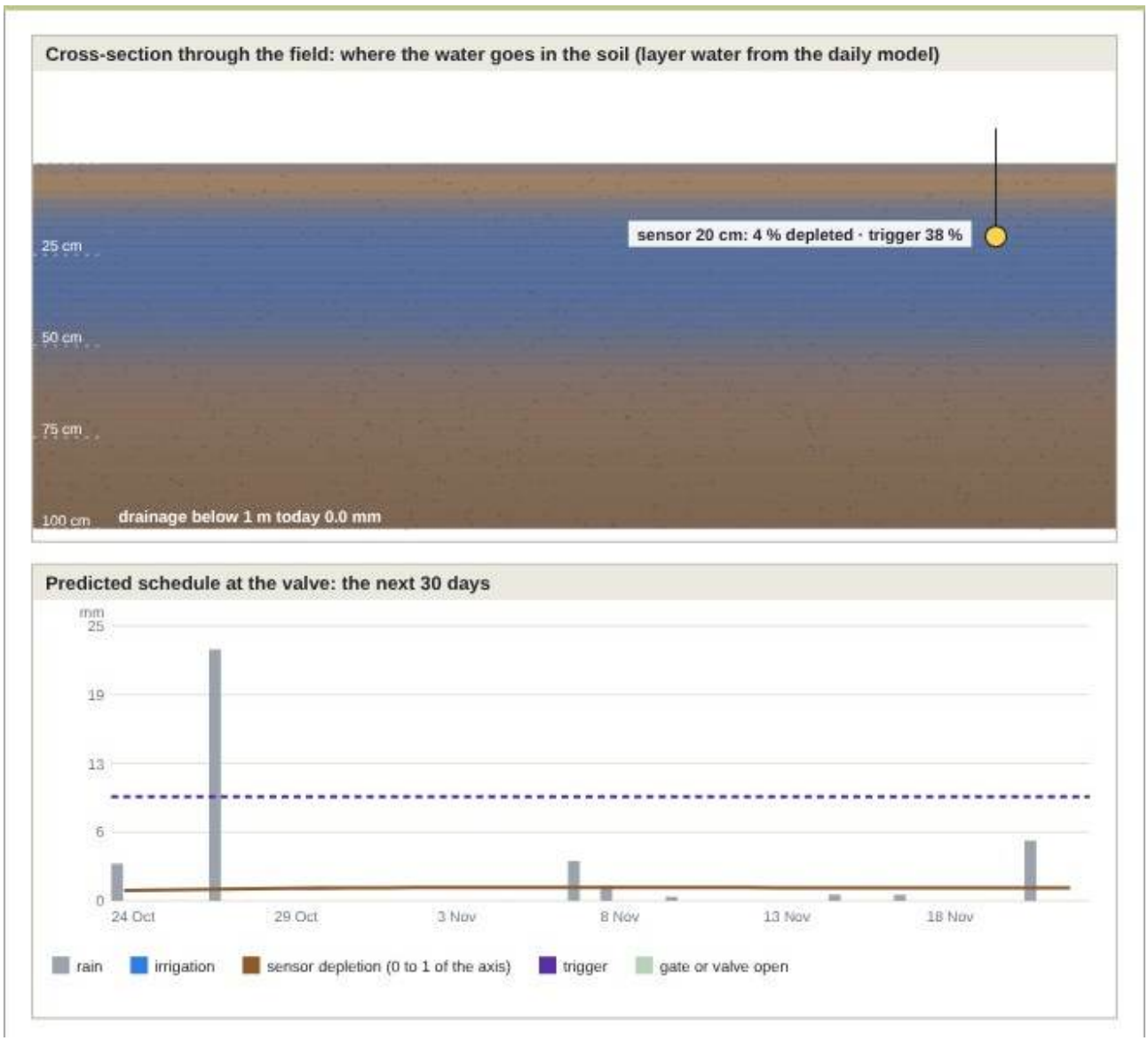


Figure 5.31: Virtual farm, sprinkler design, Coarse Loam, part 2 of 3.

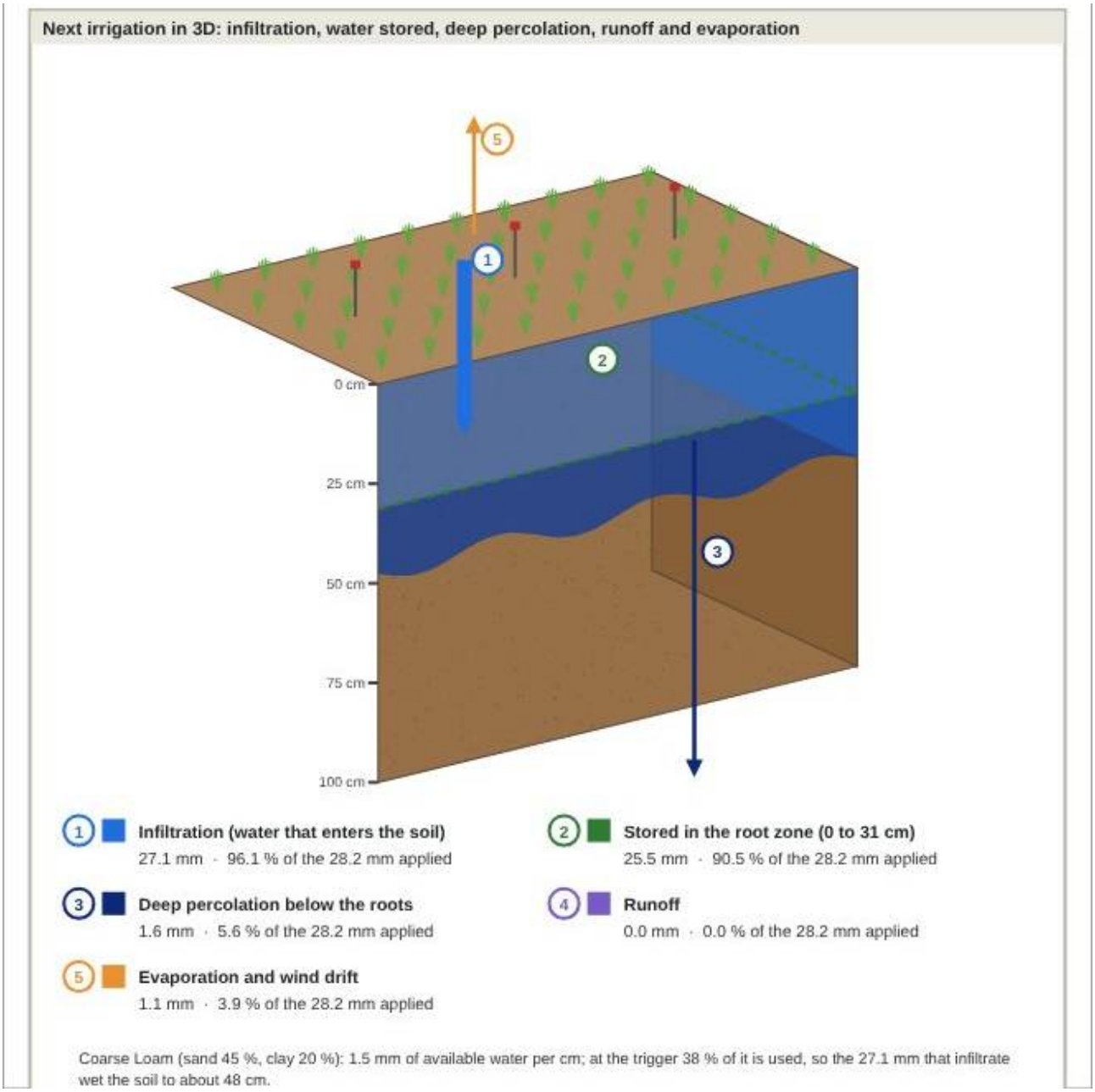


Figure 5.32: Virtual farm, sprinkler design, Coarse Loam, part 3 of 3.

5.7 Sensor and trigger (Coarse Loam)

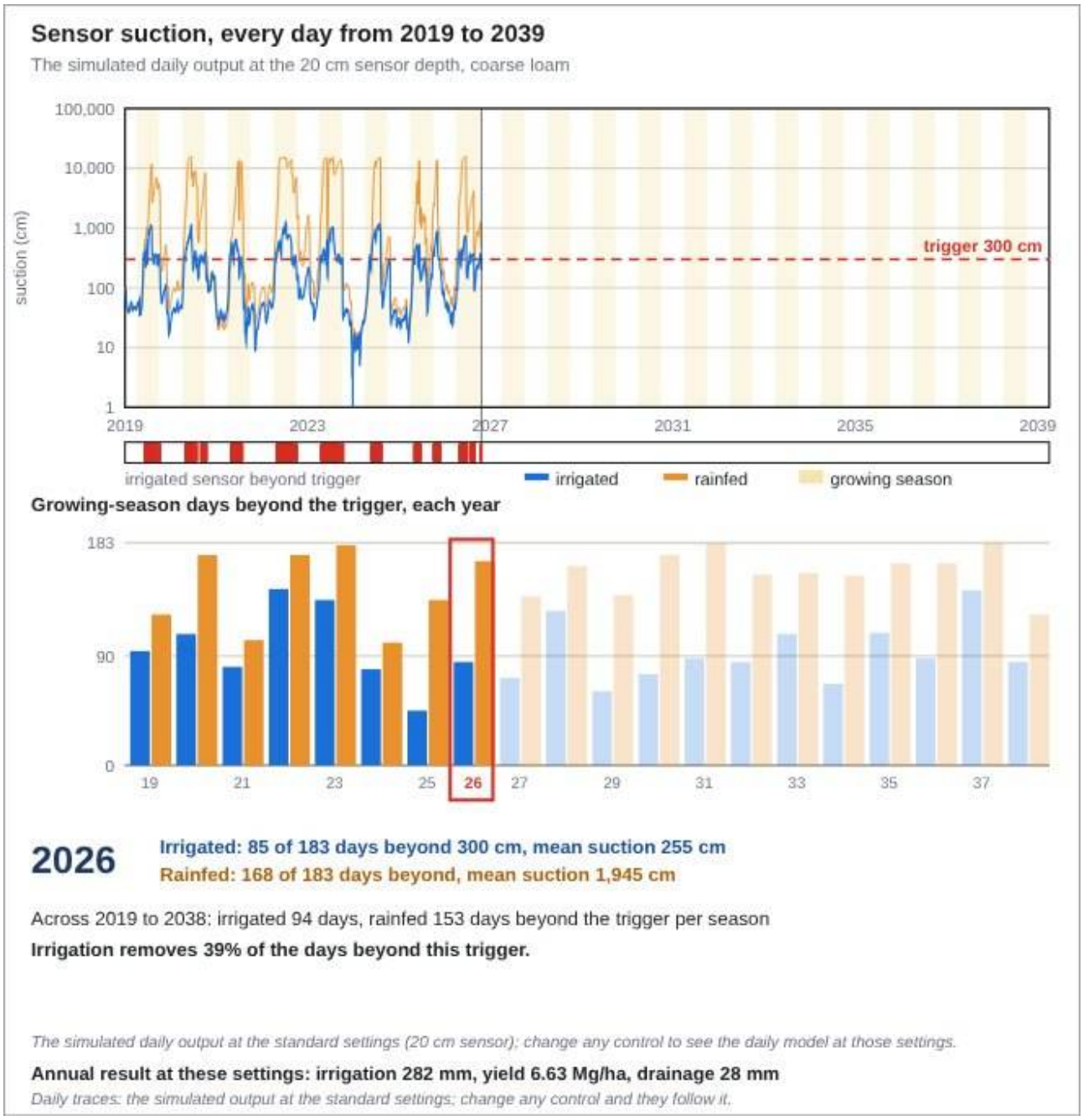


Figure 5.33: Sensor and trigger, Coarse Loam, part 1 of 1.

5.8 Trigger law (Coarse Loam)

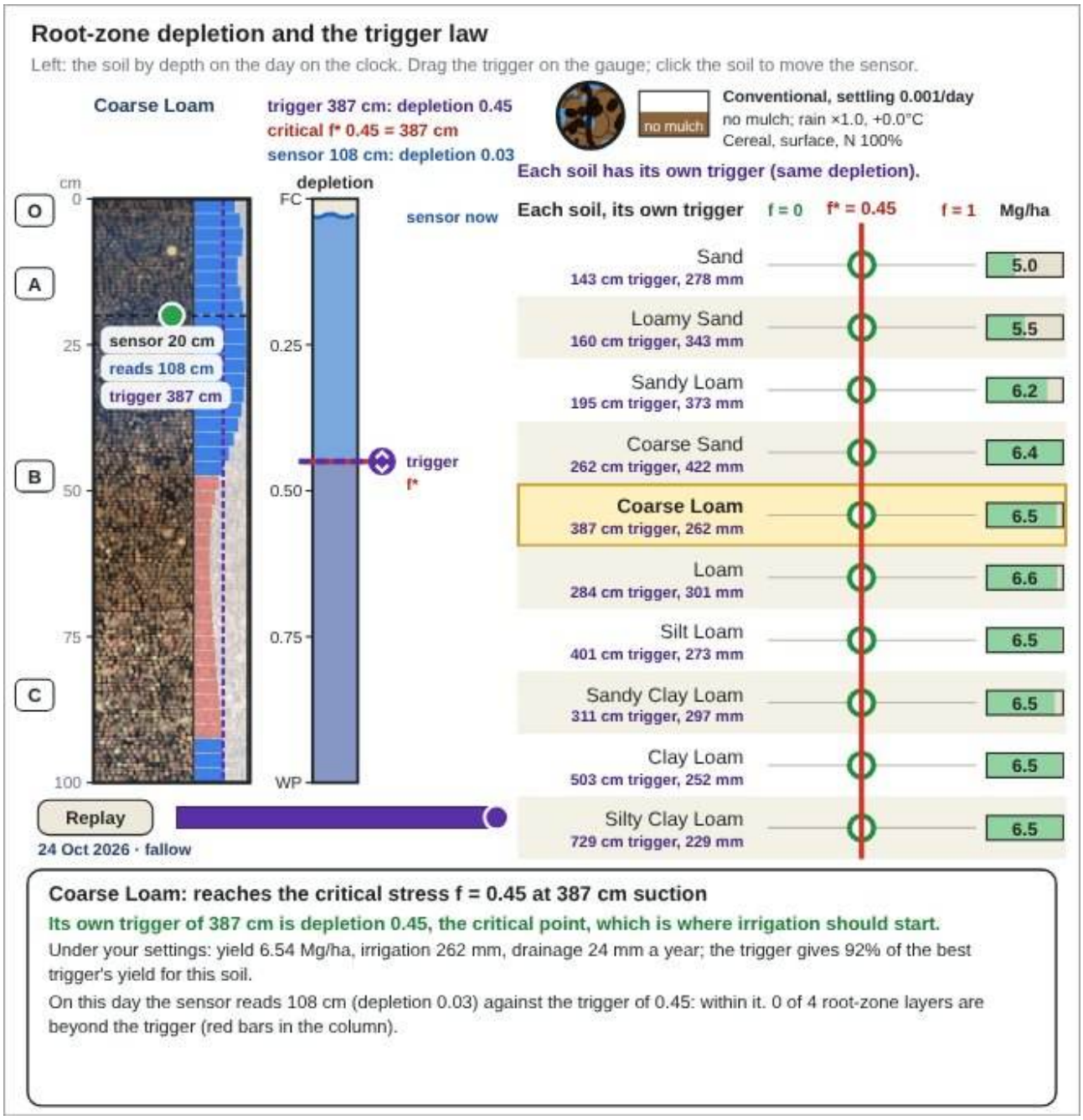


Figure 5.34: Trigger law, Coarse Loam, part 1 of 1.

5.9 Mulch against bare soil (Coarse Loam)

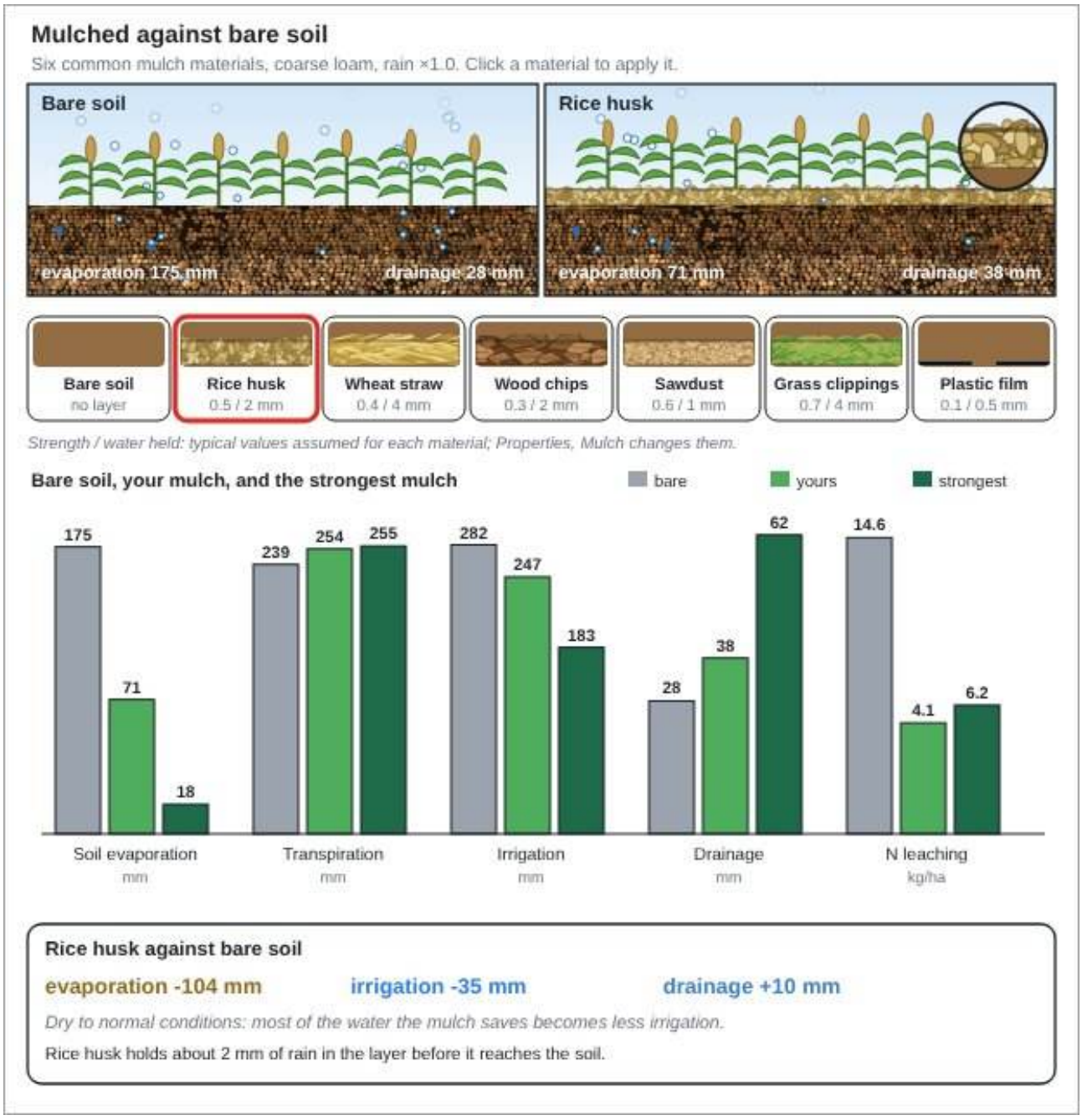


Figure 5.35: Mulch against bare soil, Coarse Loam, part 1 of 1.

5.10 Mulch control (Coarse Loam)

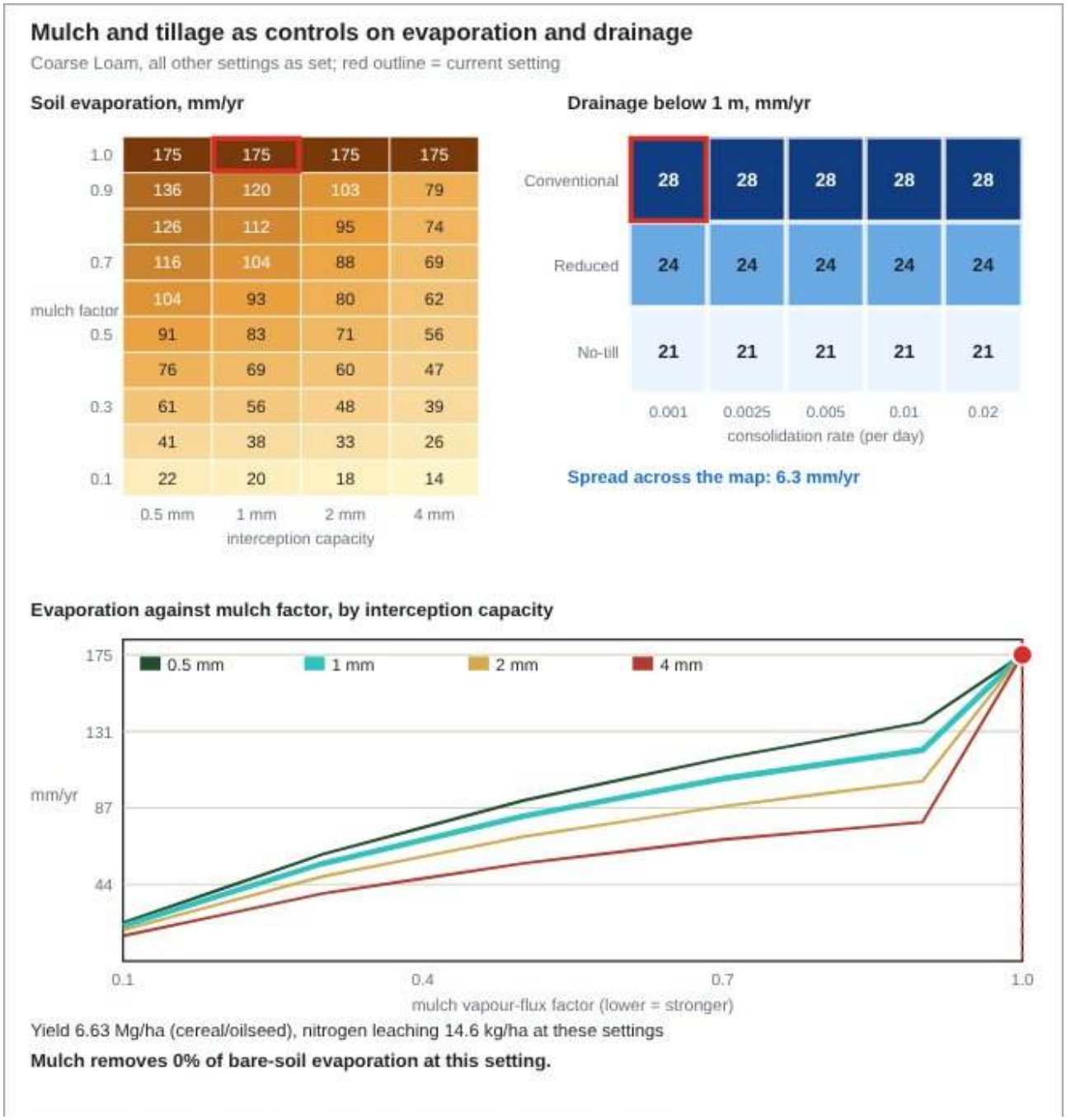


Figure 5.36: Mulch control, Coarse Loam, part 1 of 2.

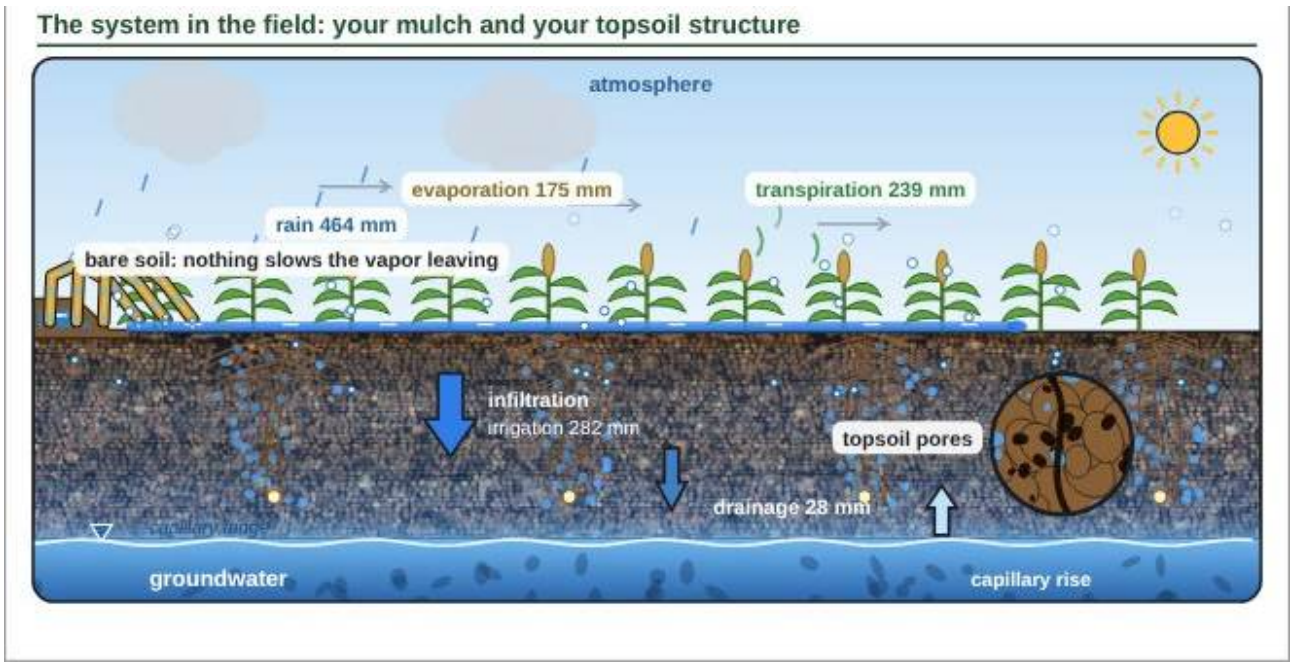


Figure 5.37: Mulch control, Coarse Loam, part 2 of 2.

5.11 Evaporation physics (Coarse Loam)

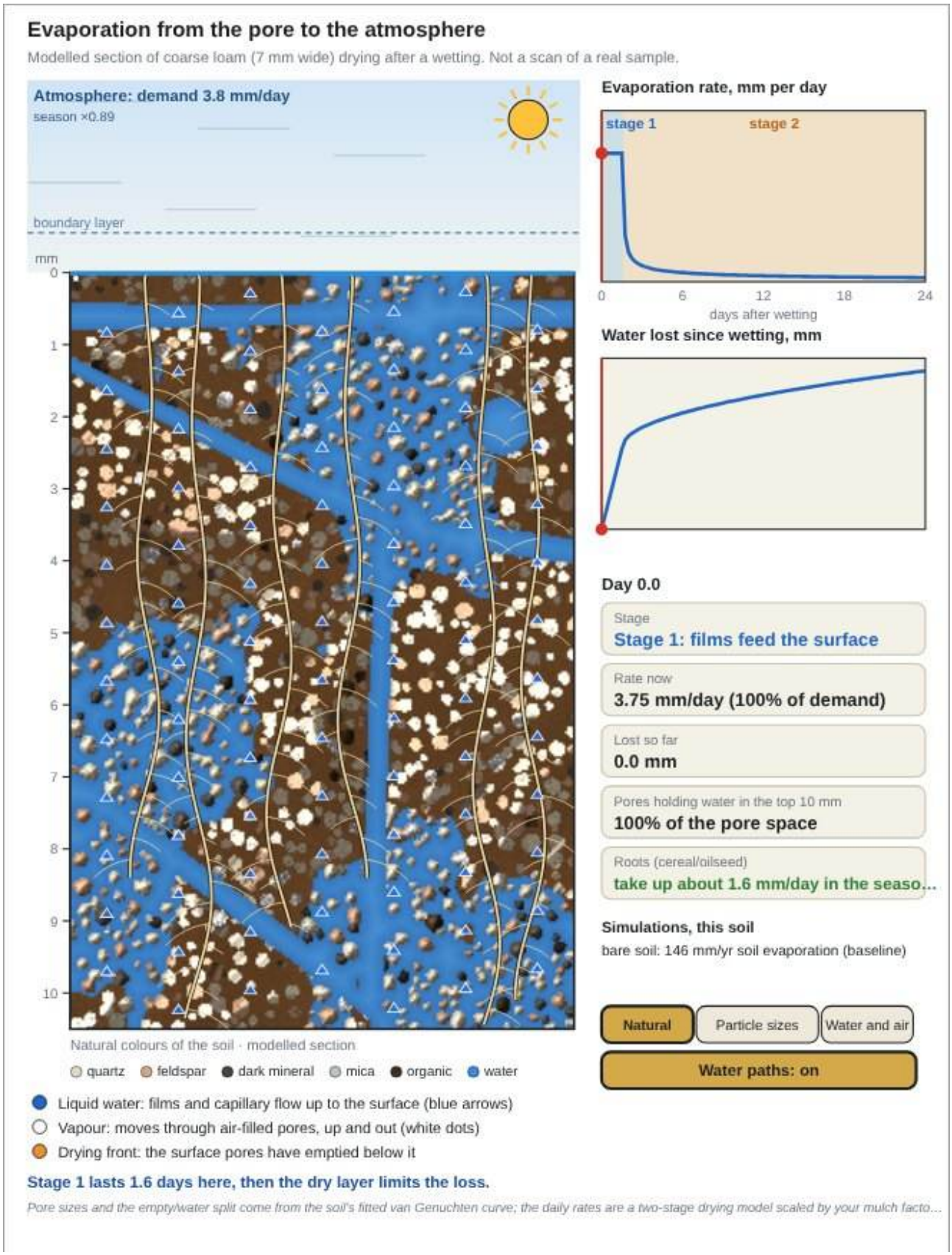


Figure 5.38: Evaporation physics, Coarse Loam, part 1 of 3.

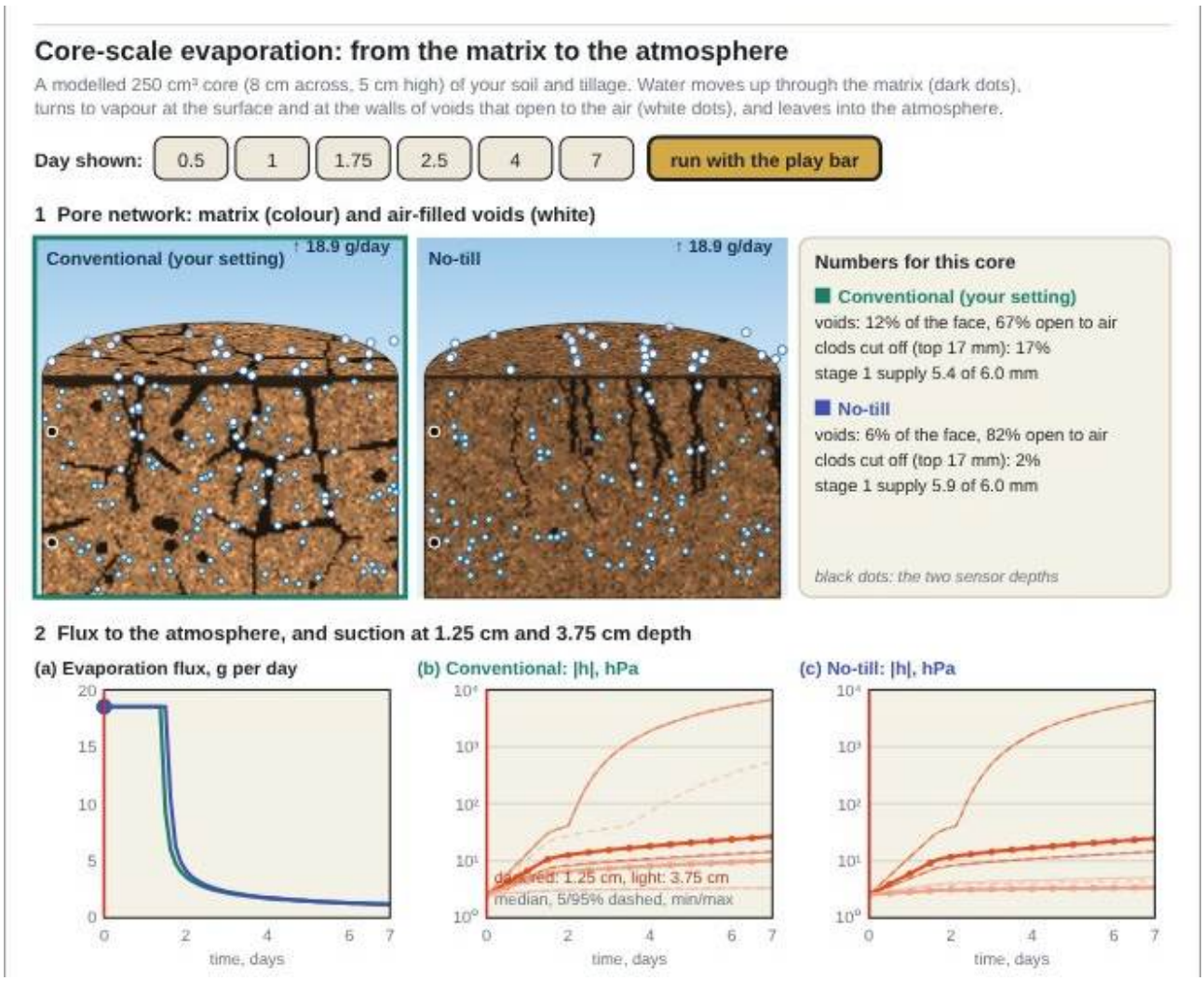


Figure 5.39: Evaporation physics, Coarse Loam, part 2 of 3.

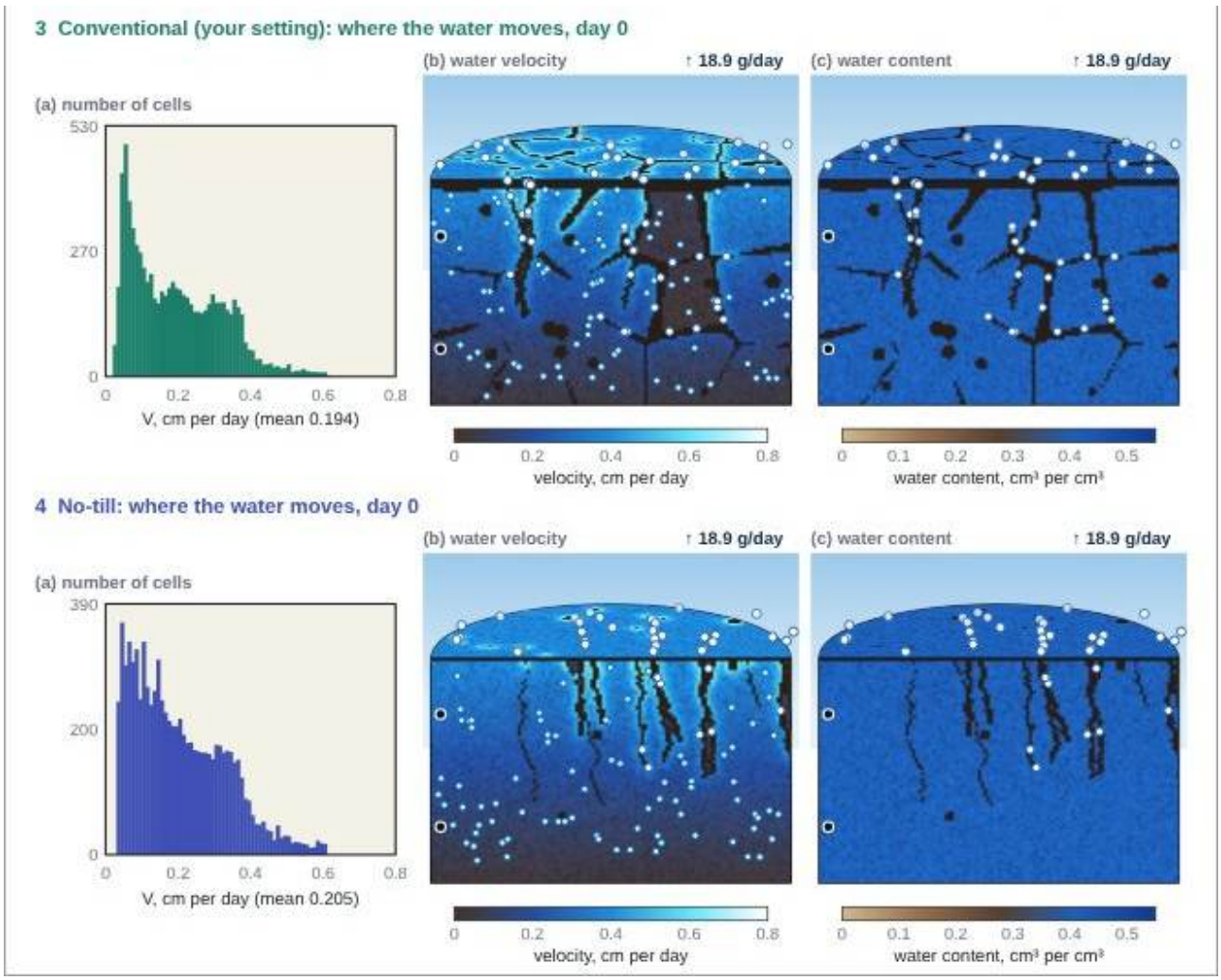


Figure 5.40: Evaporation physics, Coarse Loam, part 3 of 3.

5.12 Fate of saved water (Coarse Loam)

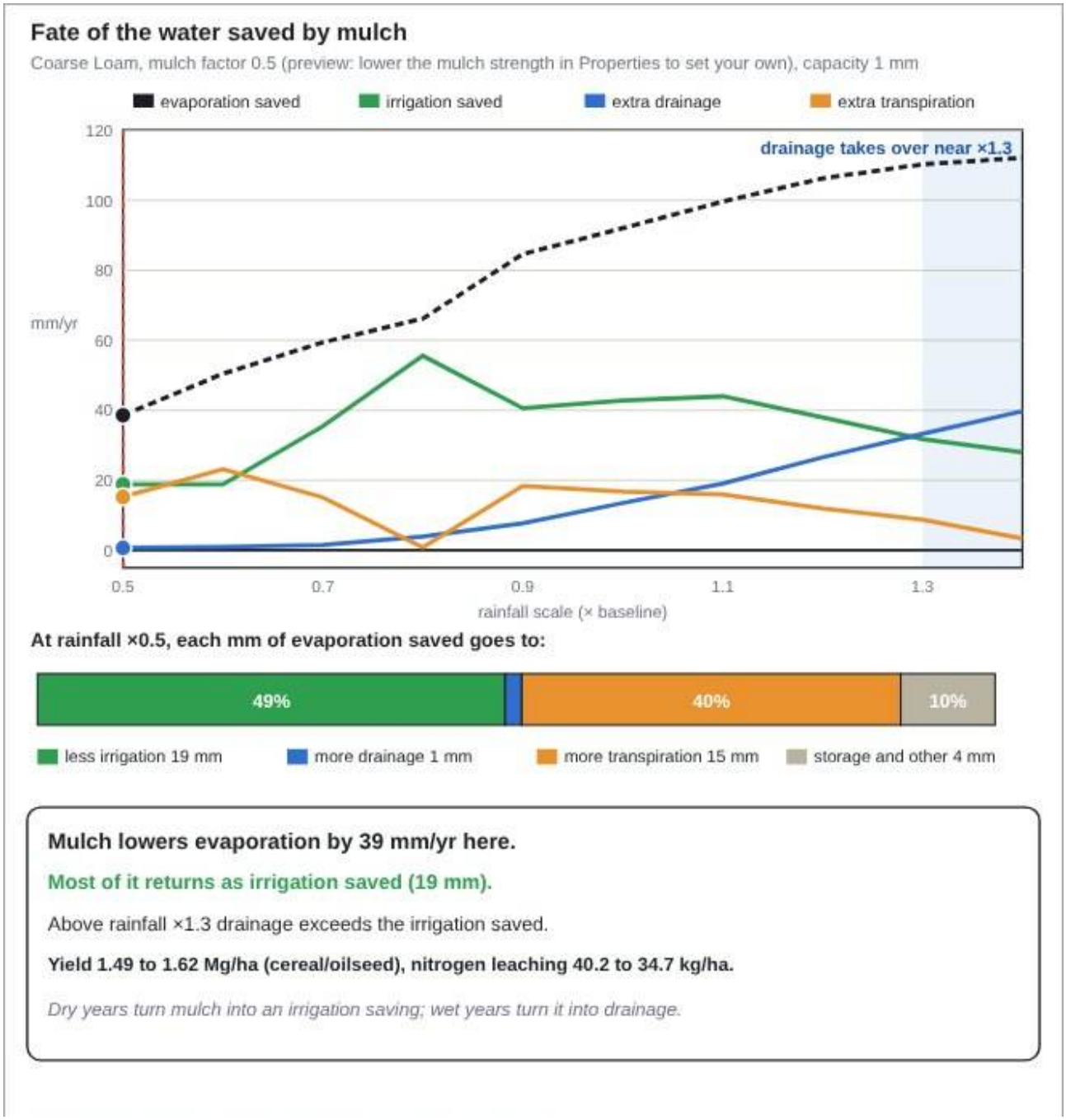


Figure 5.41: Fate of saved water, Coarse Loam, part 1 of 2.

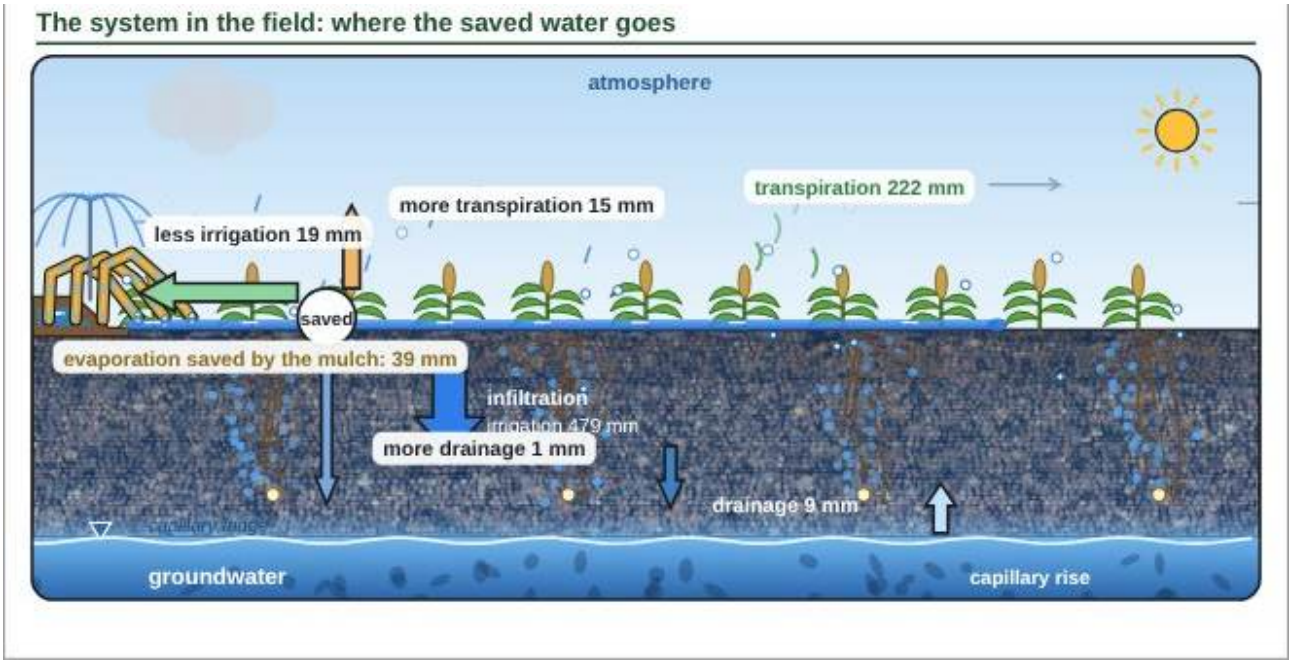


Figure 5.42: Fate of saved water, Coarse Loam, part 2 of 2.

5.13 Regime map (Coarse Loam)

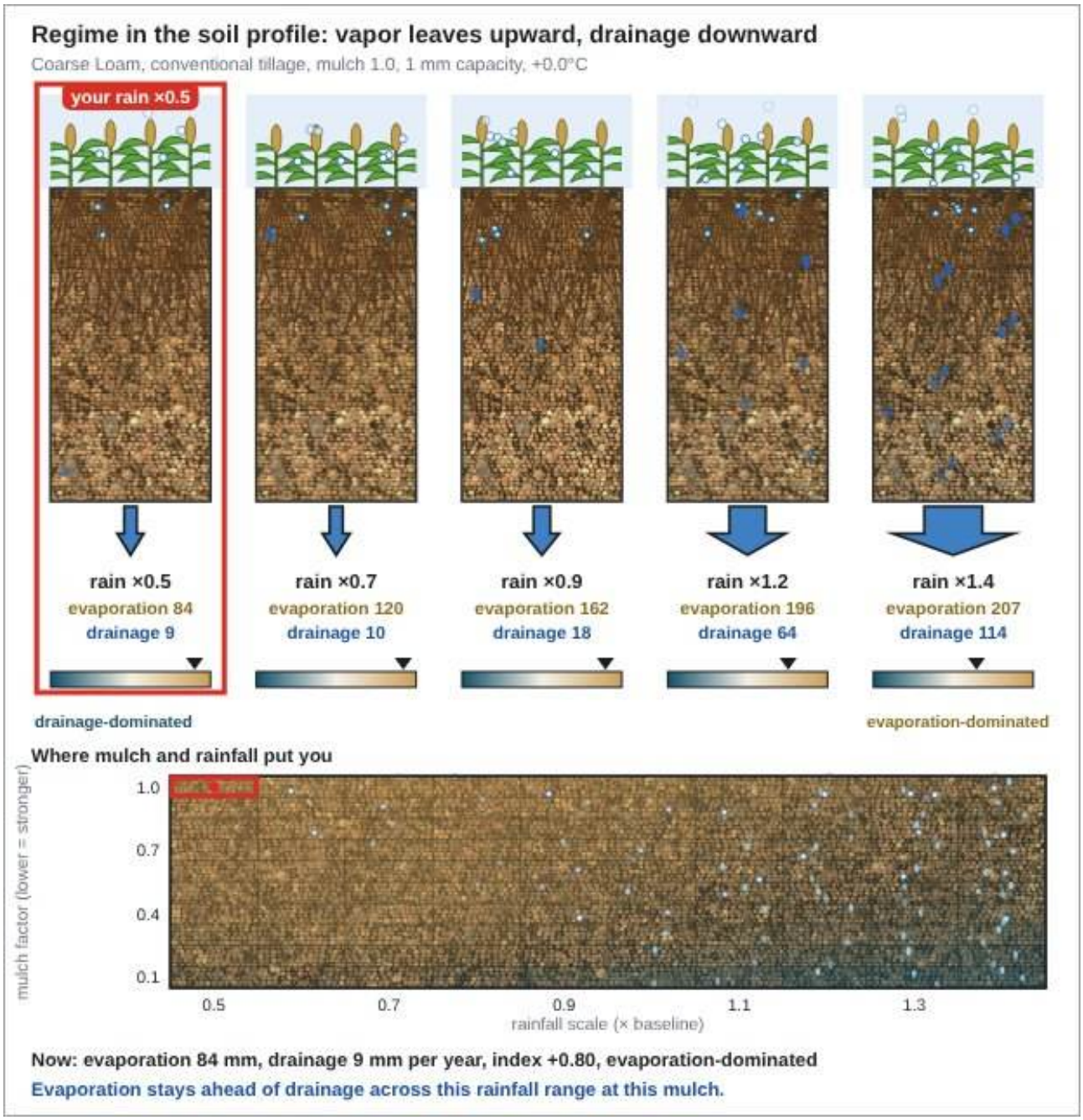


Figure 5.43: Regime map, Coarse Loam, part 1 of 1.

5.14 Tillage and structure (Coarse Loam)

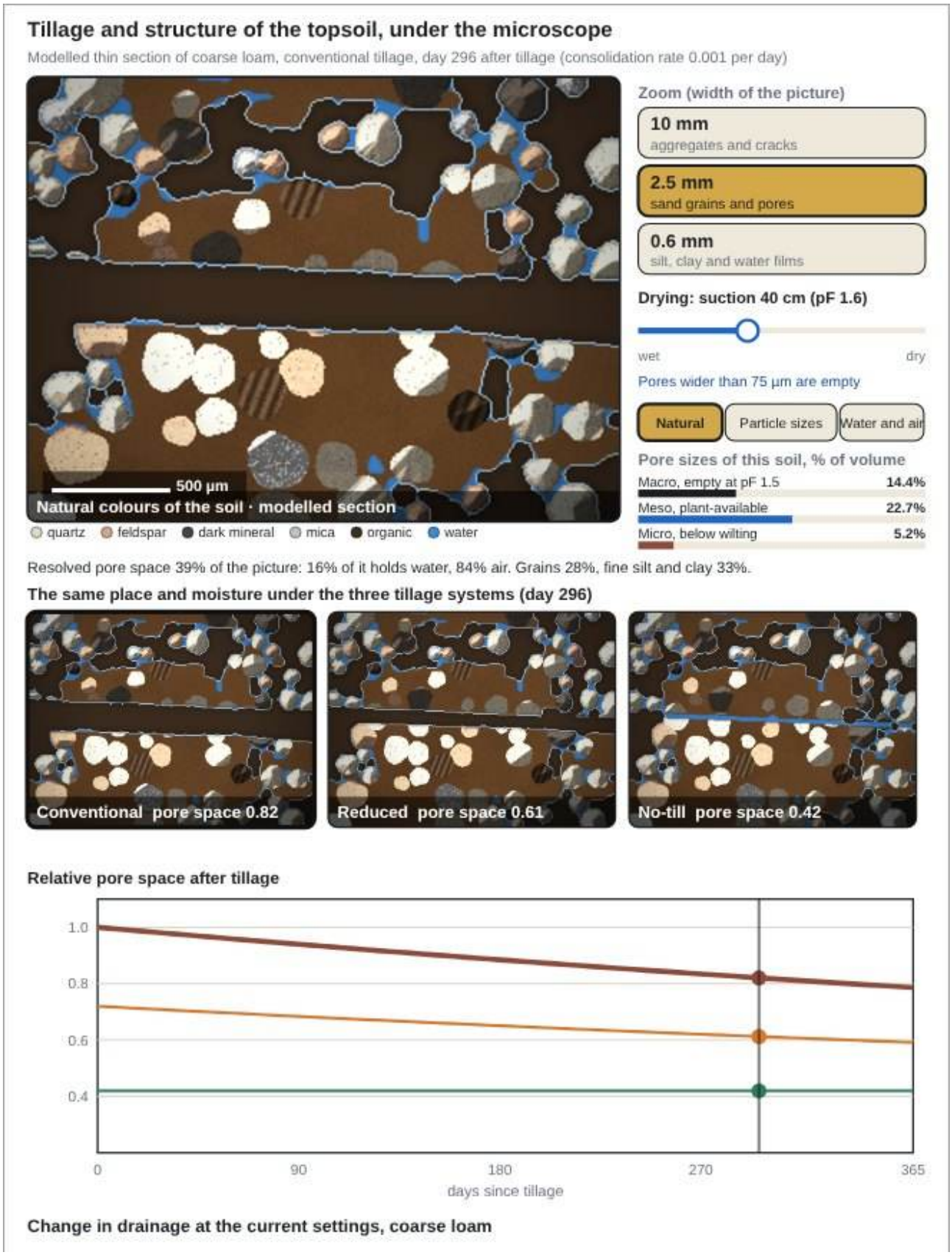


Figure 5.44: Tillage and structure, Coarse Loam, part 1 of 2.

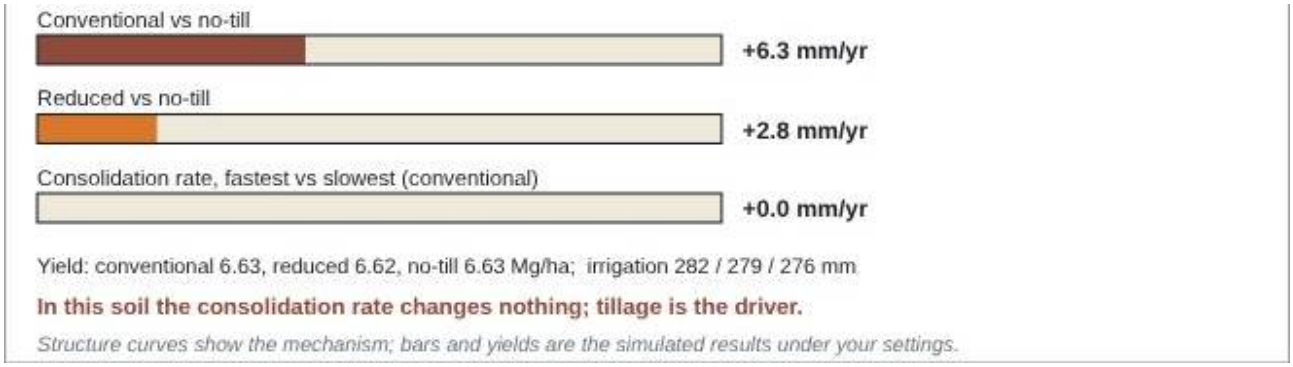


Figure 5.45: Tillage and structure, Coarse Loam, part 2 of 2.

5.15 Soil hydraulic drivers (Coarse Loam)

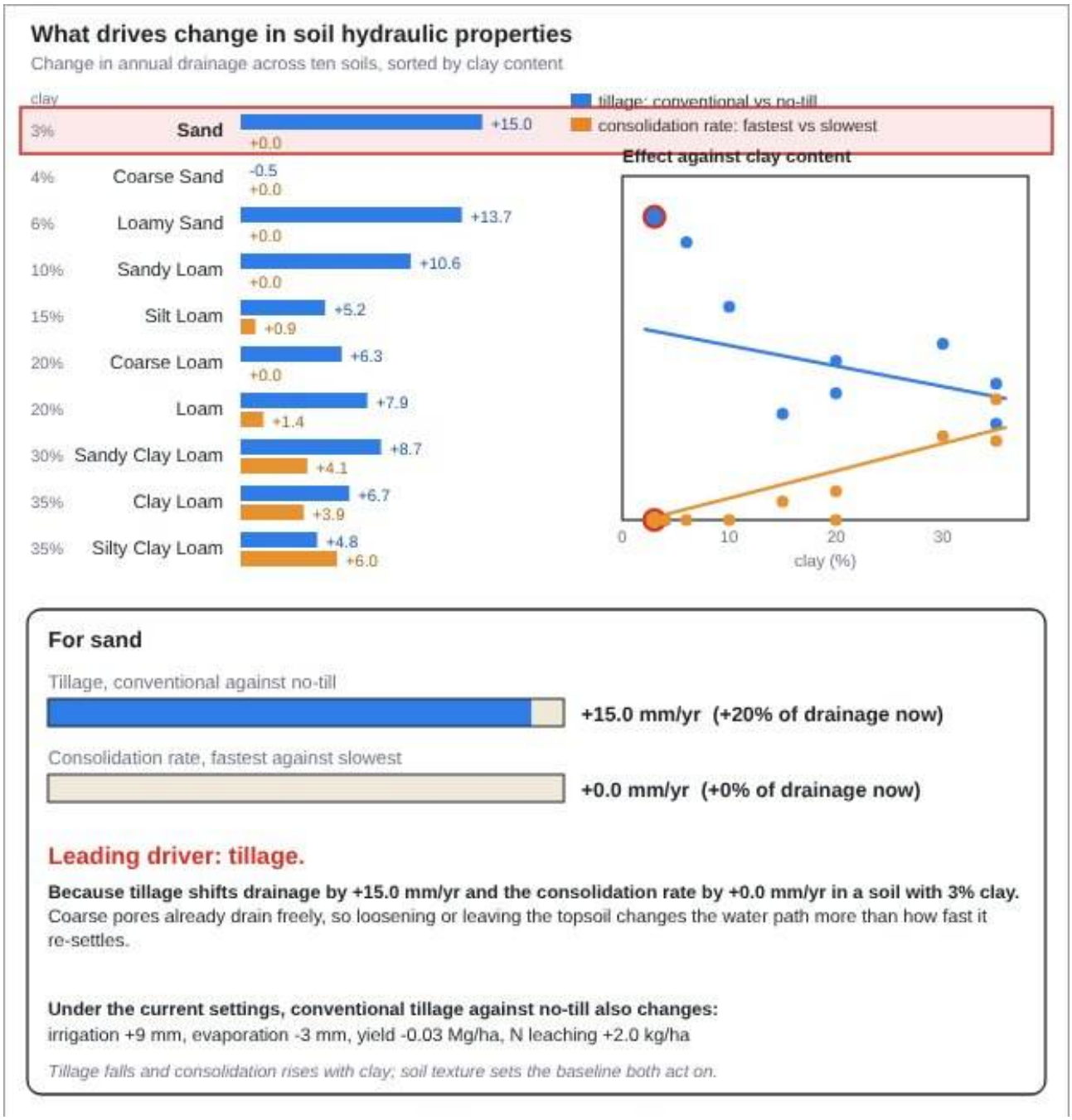


Figure 5.46: Soil hydraulic drivers, Coarse Loam, part 1 of 3.

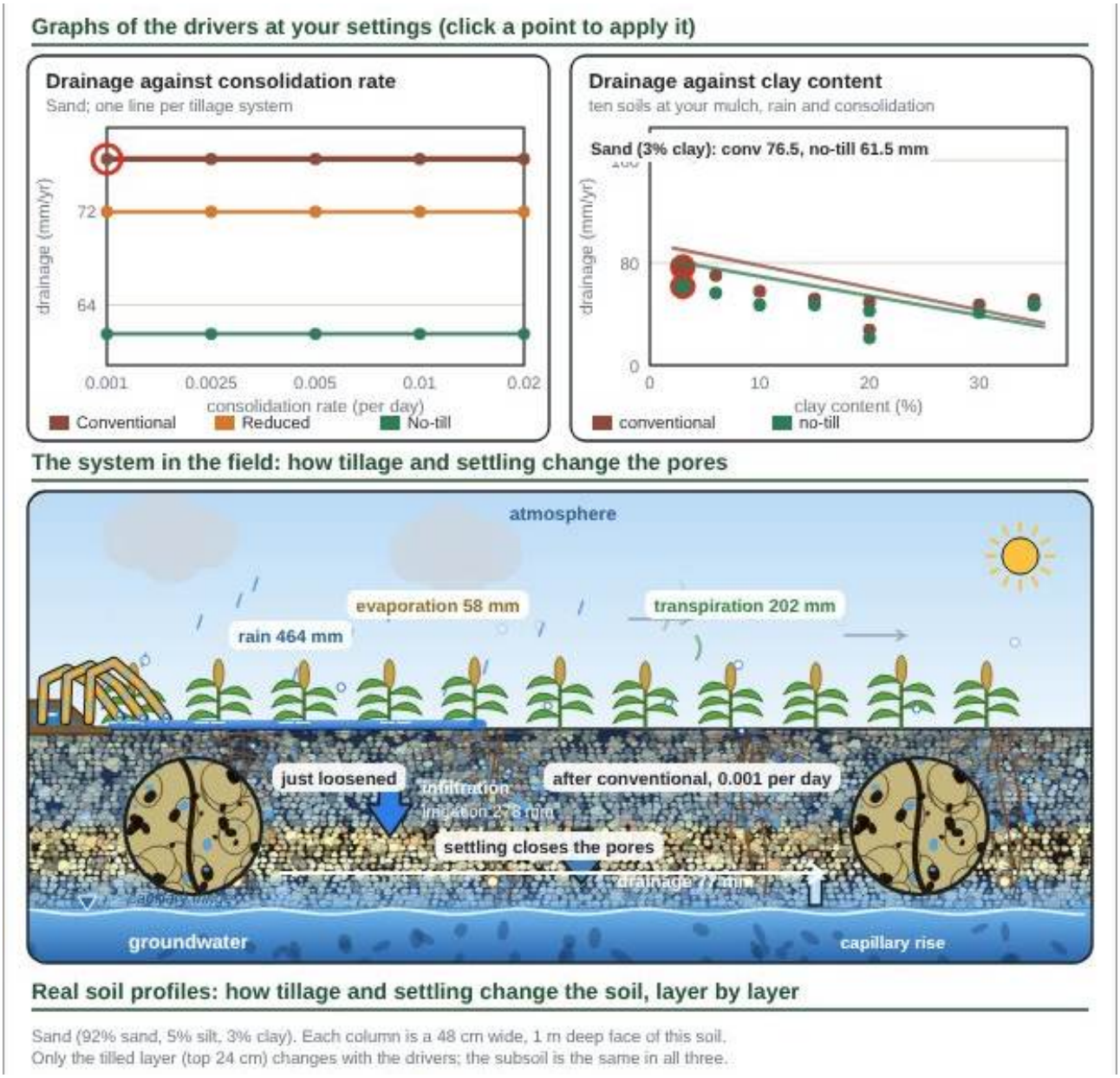


Figure 5.47: Soil hydraulic drivers, Coarse Loam, part 2 of 3.

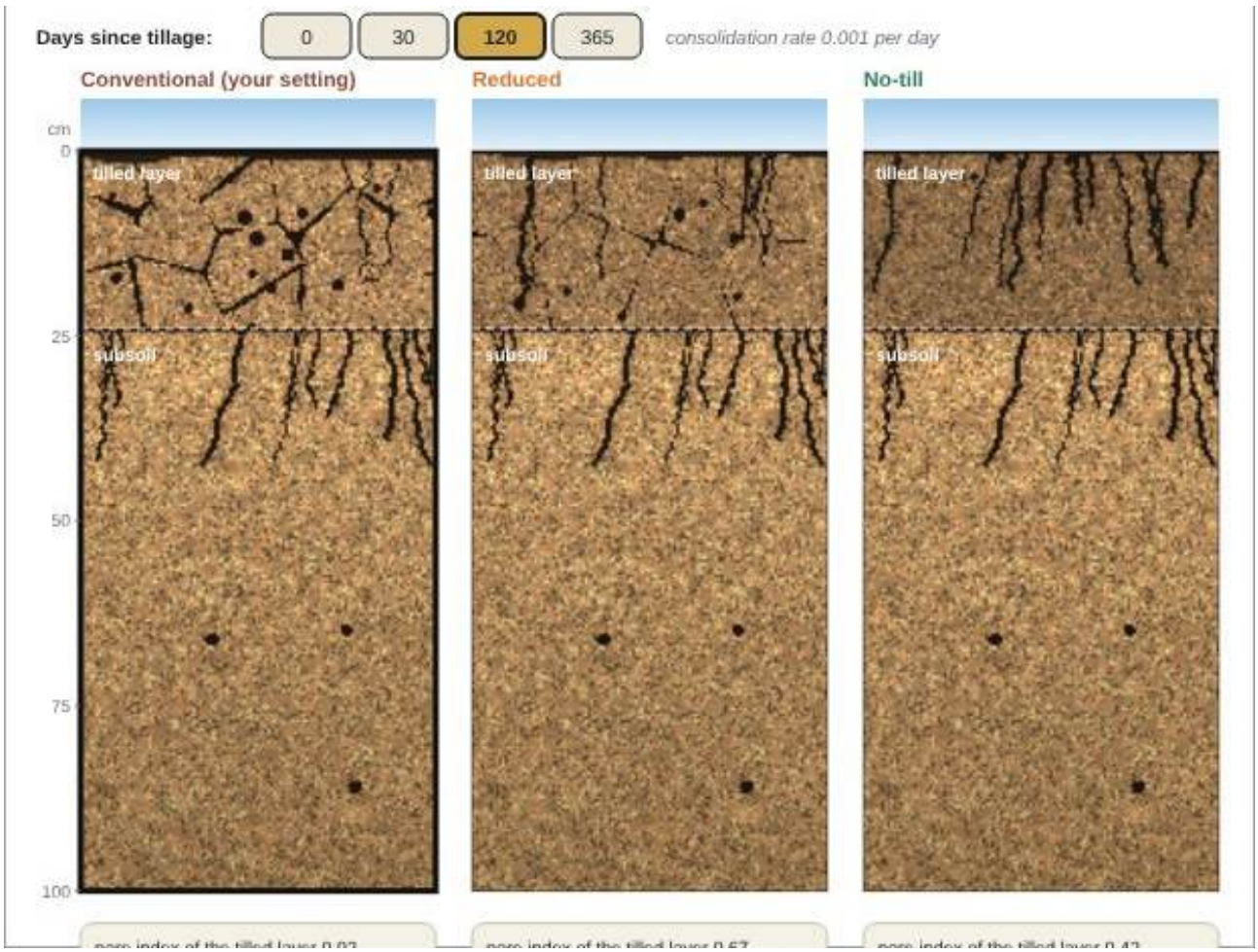


Figure 5.48: Soil hydraulic drivers, Coarse Loam, part 3 of 3.

5.16 Root structure (Coarse Loam)

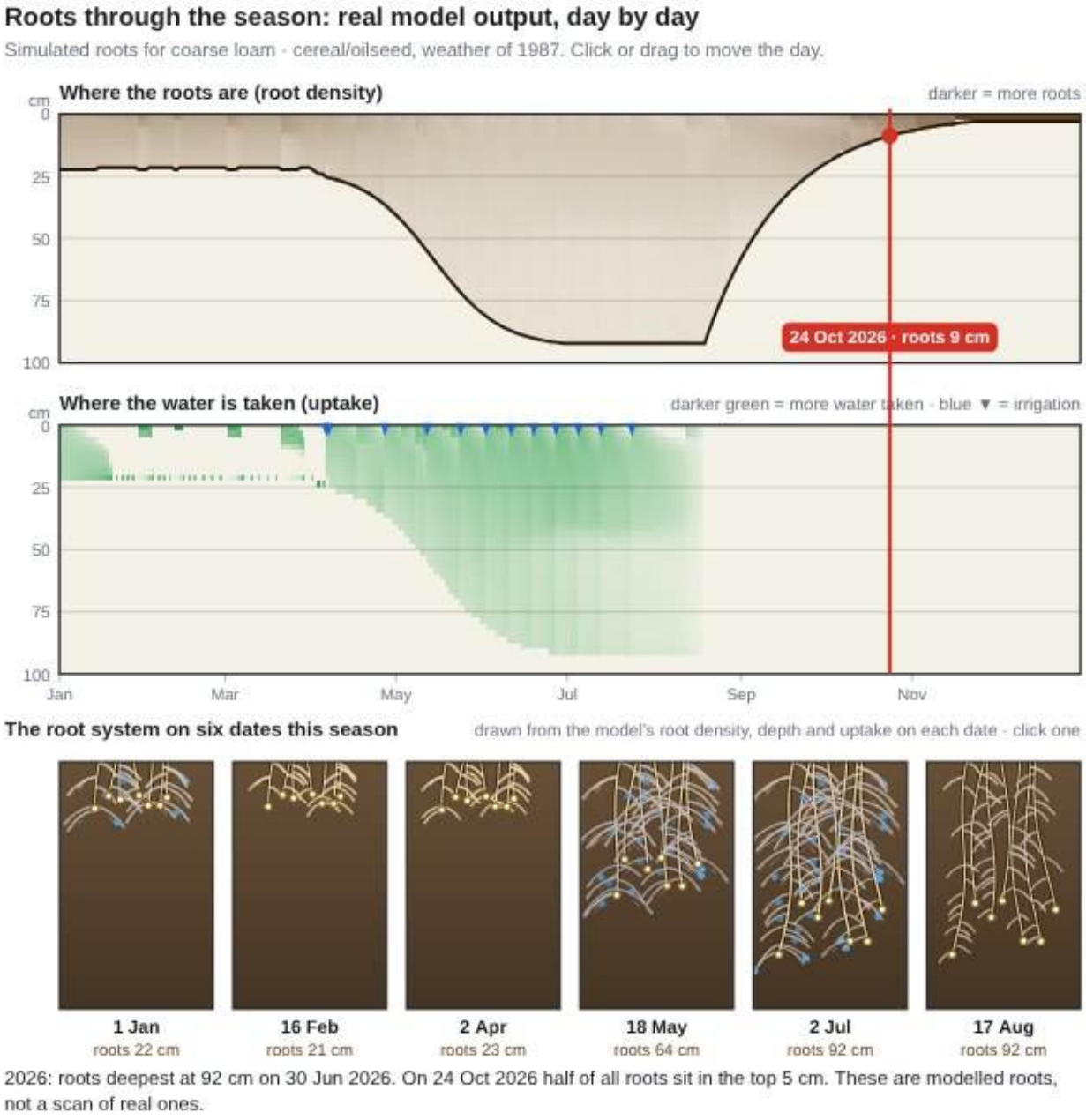


Figure 5.49: Root structure, Coarse Loam, part 1 of 4.

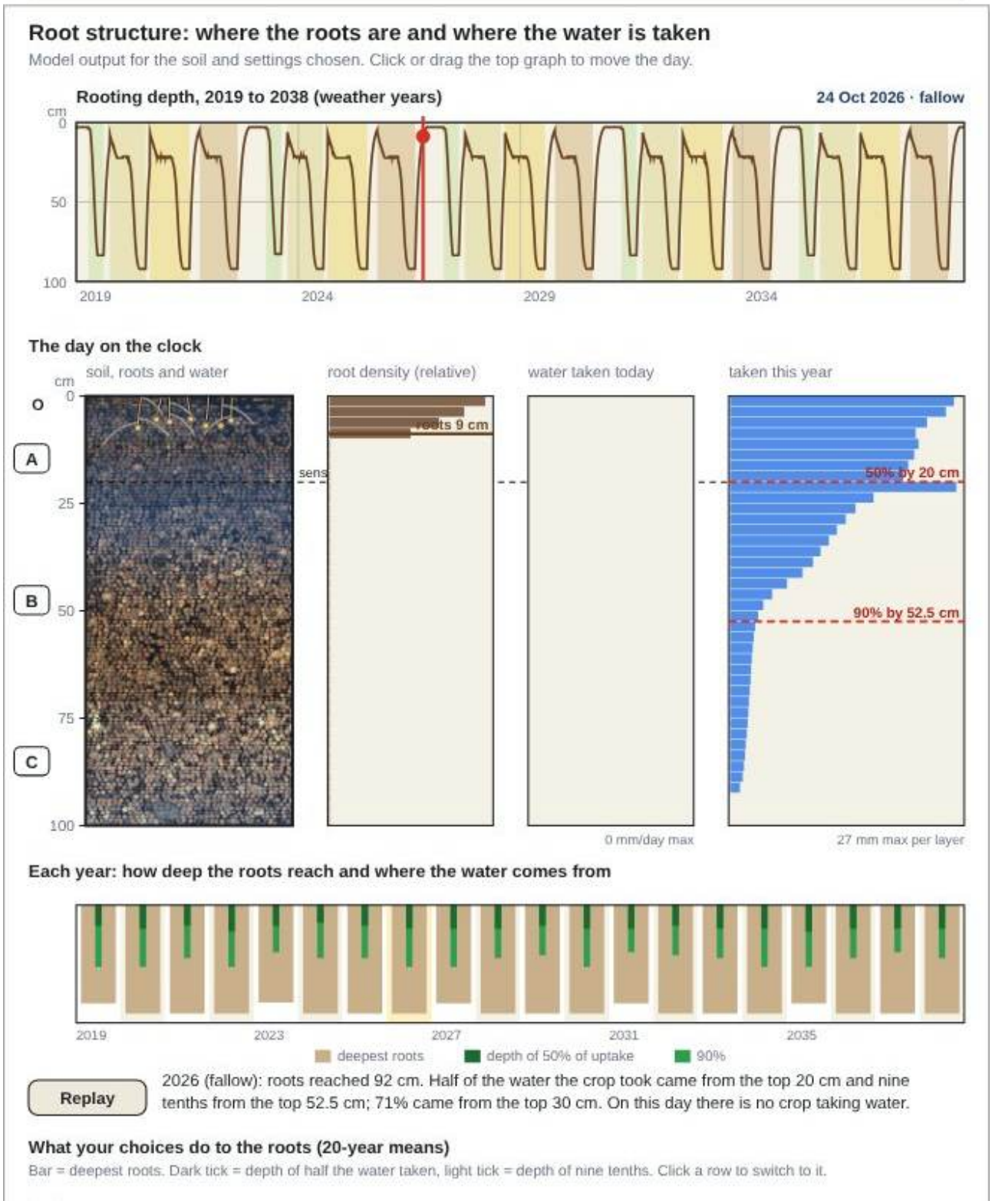


Figure 5.50: Root structure, Coarse Loam, part 2 of 4.

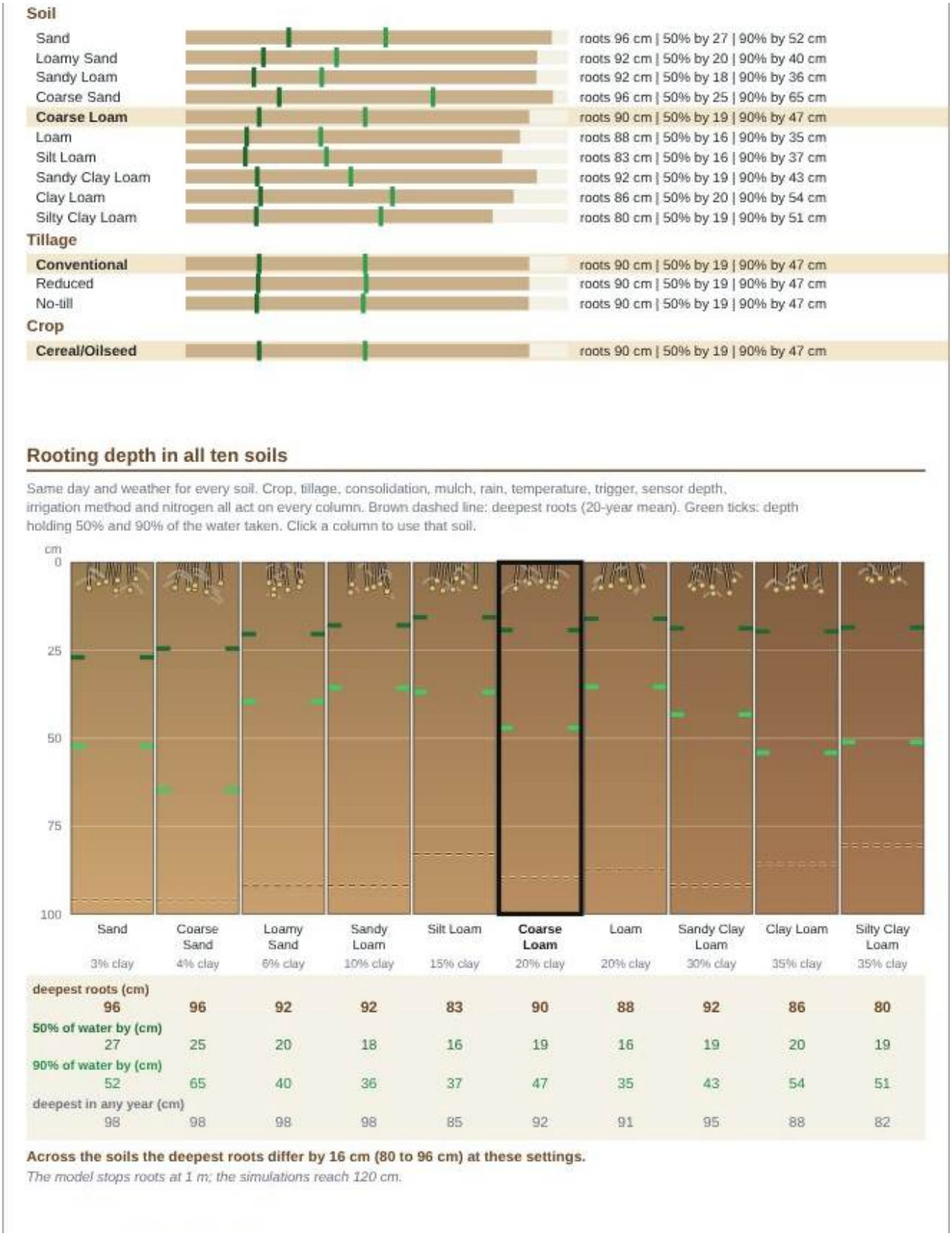


Figure 5.51: Root structure, Coarse Loam, part 3 of 4.

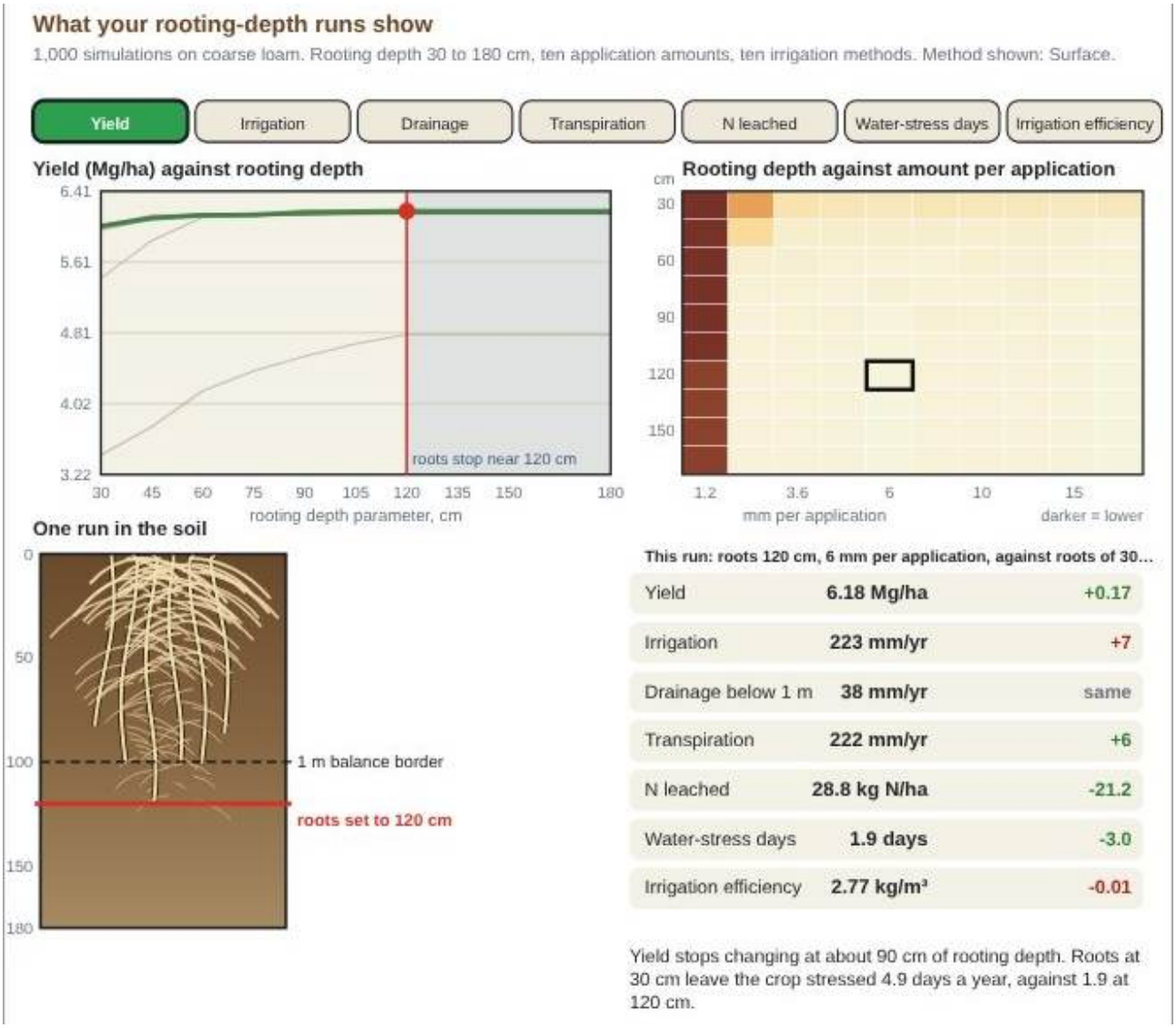


Figure 5.52: Root structure, Coarse Loam, part 4 of 4.

5.17 Field balance (Coarse Loam)

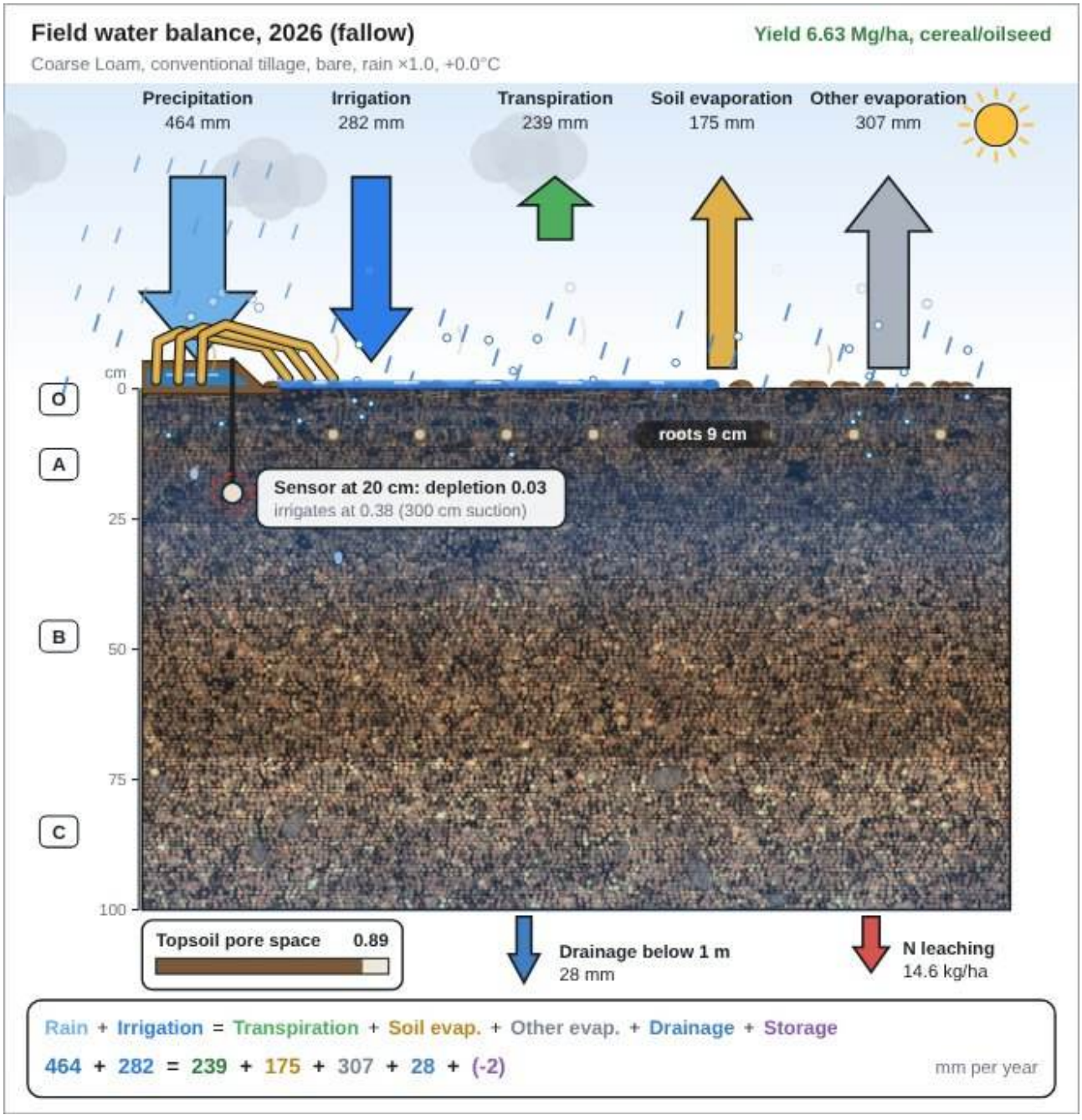


Figure 5.53: Field balance, Coarse Loam, part 1 of 1.

5.18 Soil, plant and air (Coarse Loam)

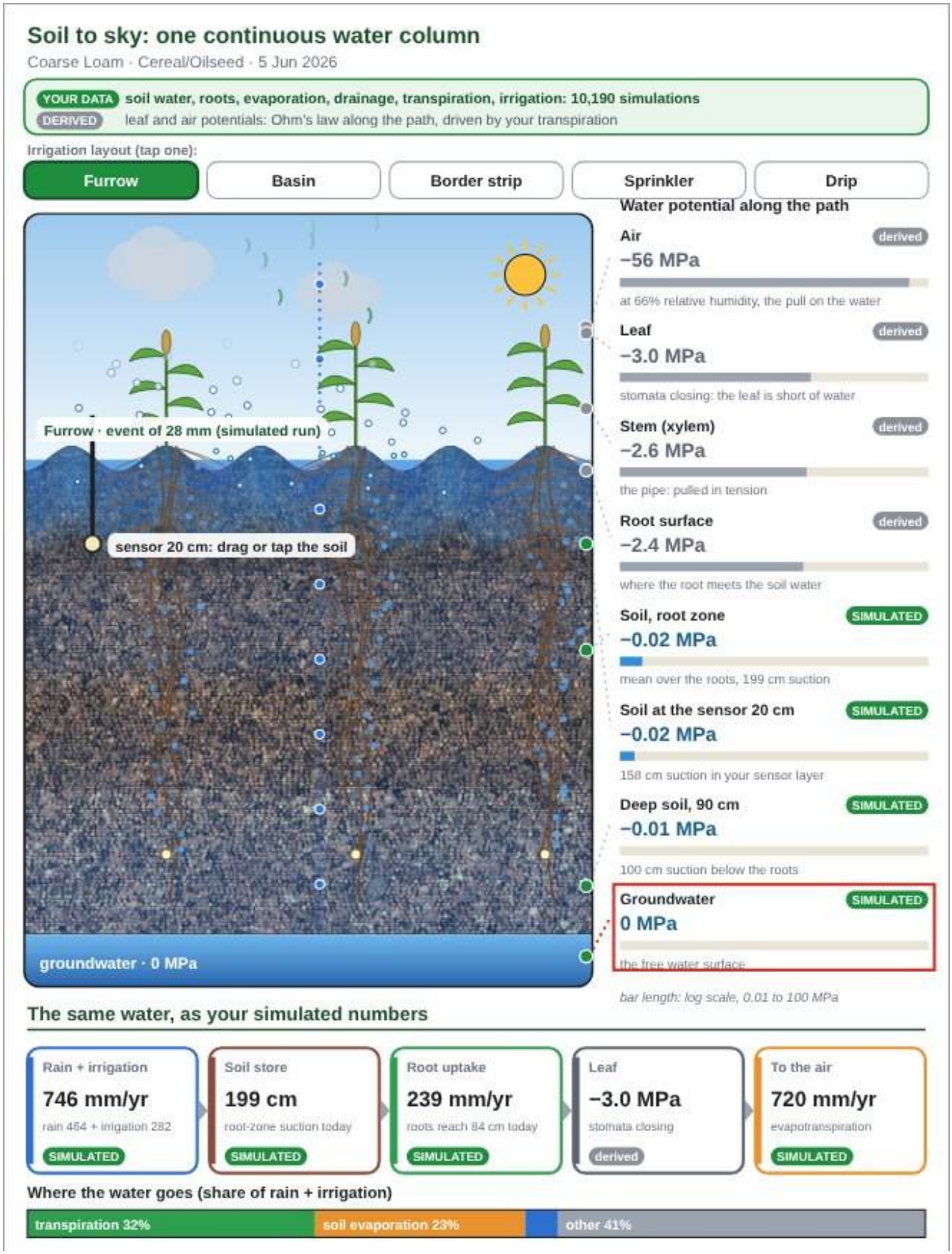


Figure 5.54: Soil, plant and air, Coarse Loam, part 1 of 3.

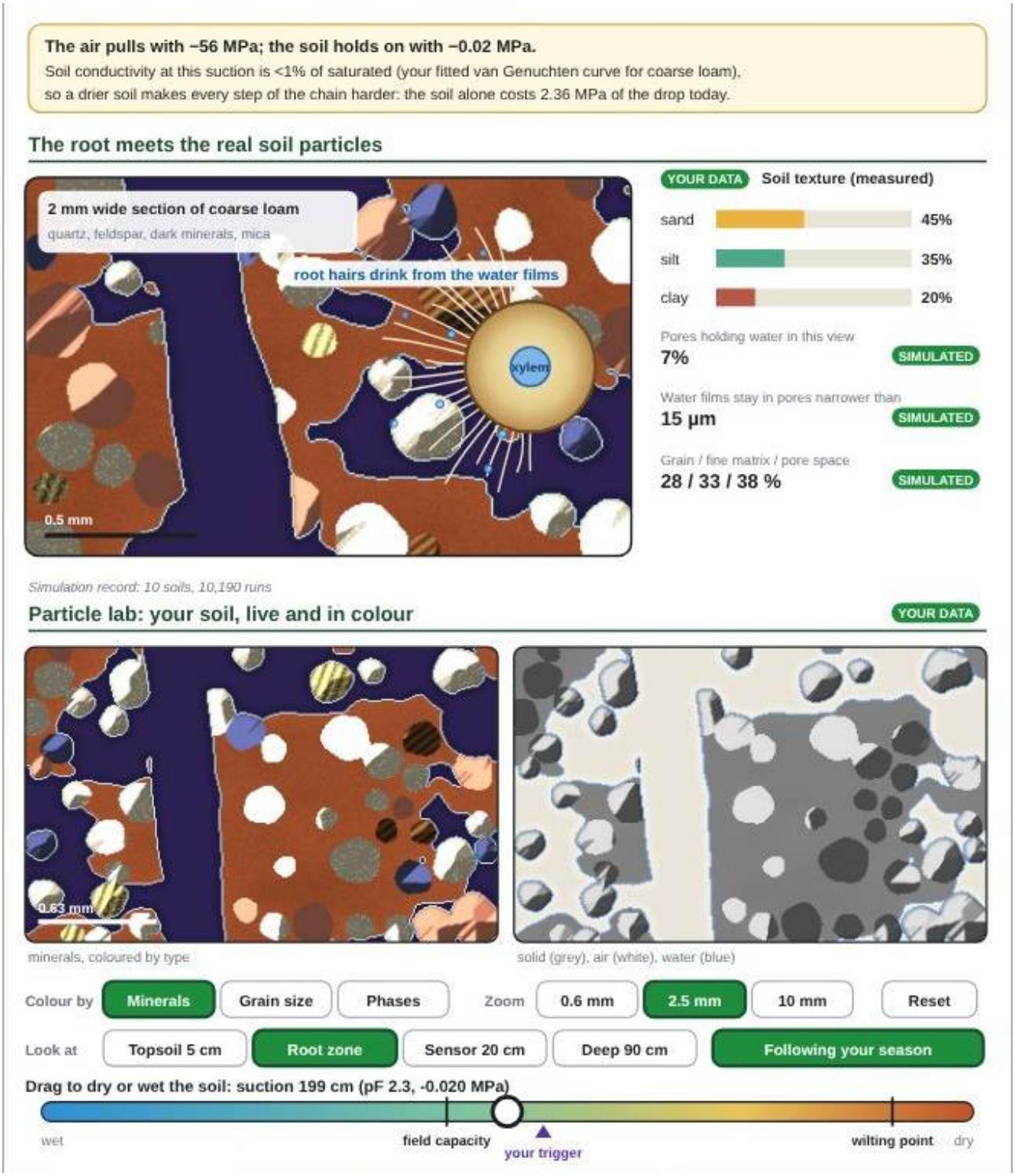


Figure 5.55: Soil, plant and air, Coarse Loam, part 2 of 3.

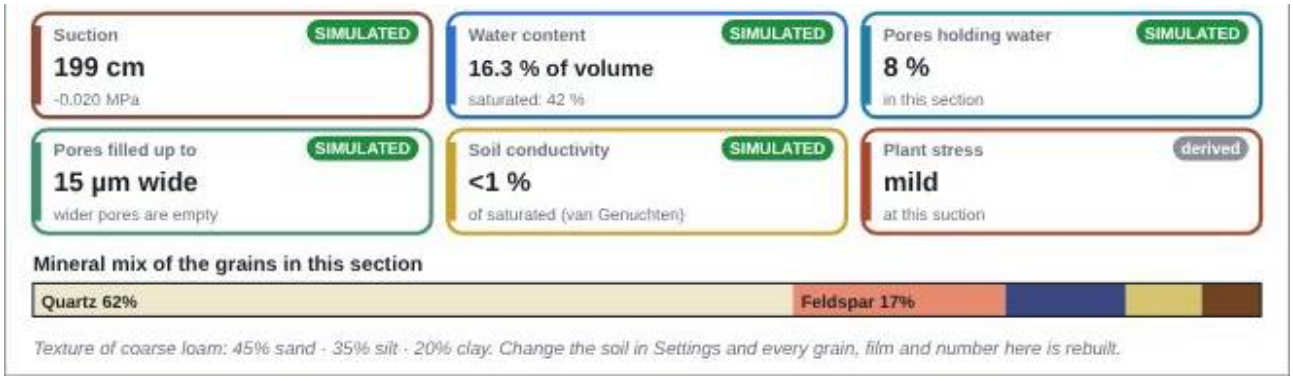


Figure 5.56: Soil, plant and air, Coarse Loam, part 3 of 3.

5.19 Profile evolution (Coarse Loam)

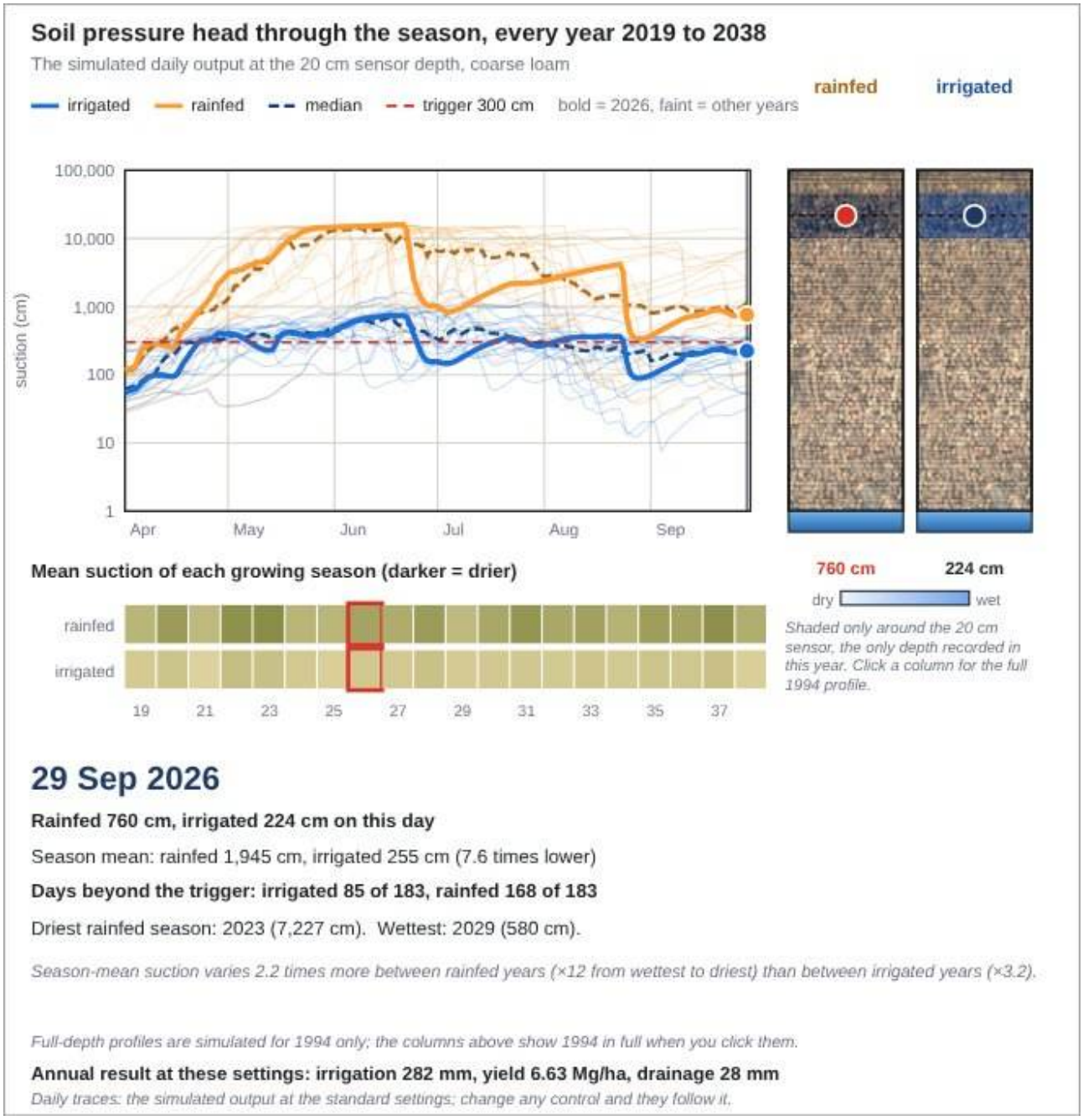


Figure 5.57: Profile evolution, Coarse Loam, part 1 of 1.

5.20 Reliability over 20 seasons (Coarse Loam)

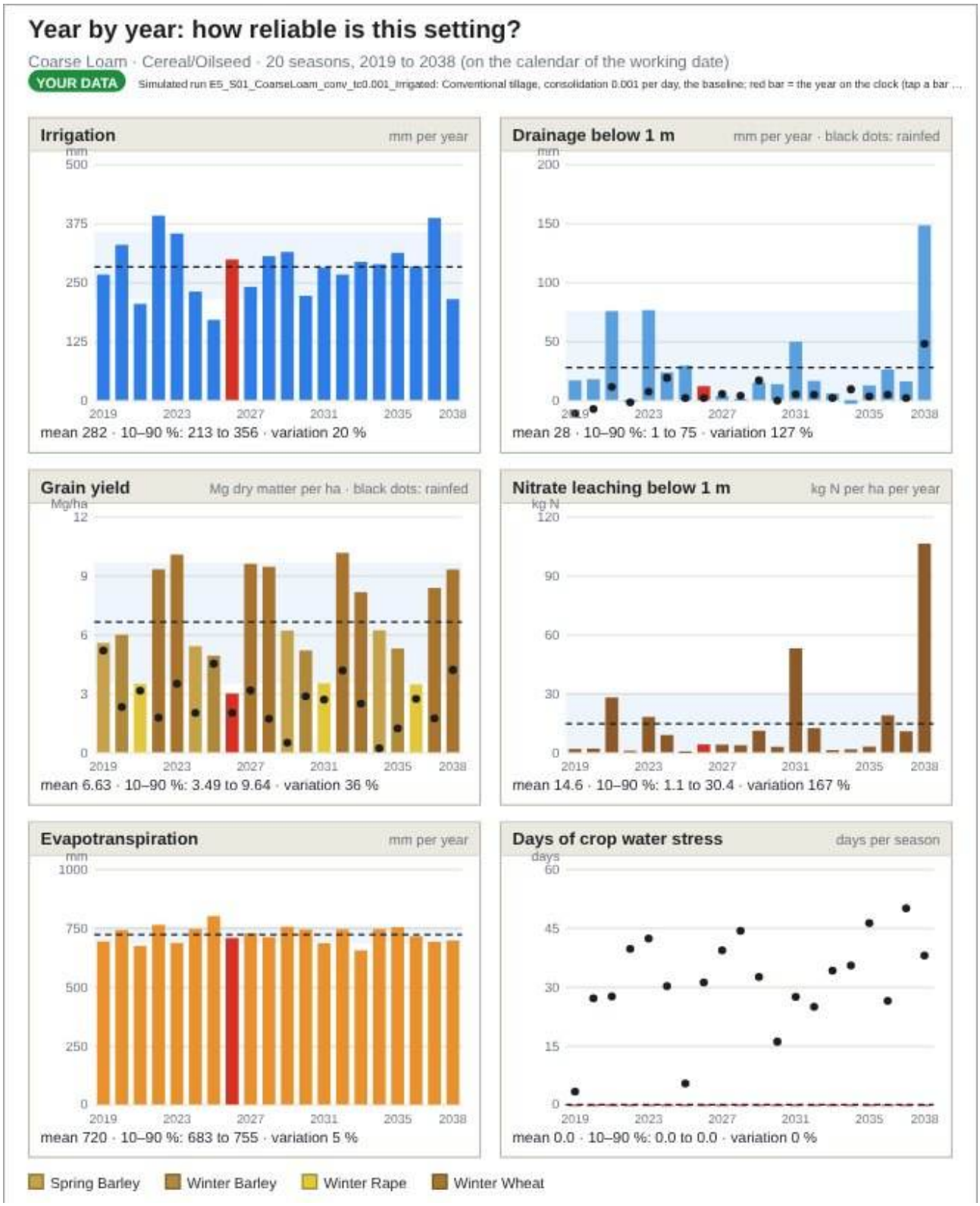


Figure 5.58: Reliability over 20 seasons, Coarse Loam, part 1 of 2.

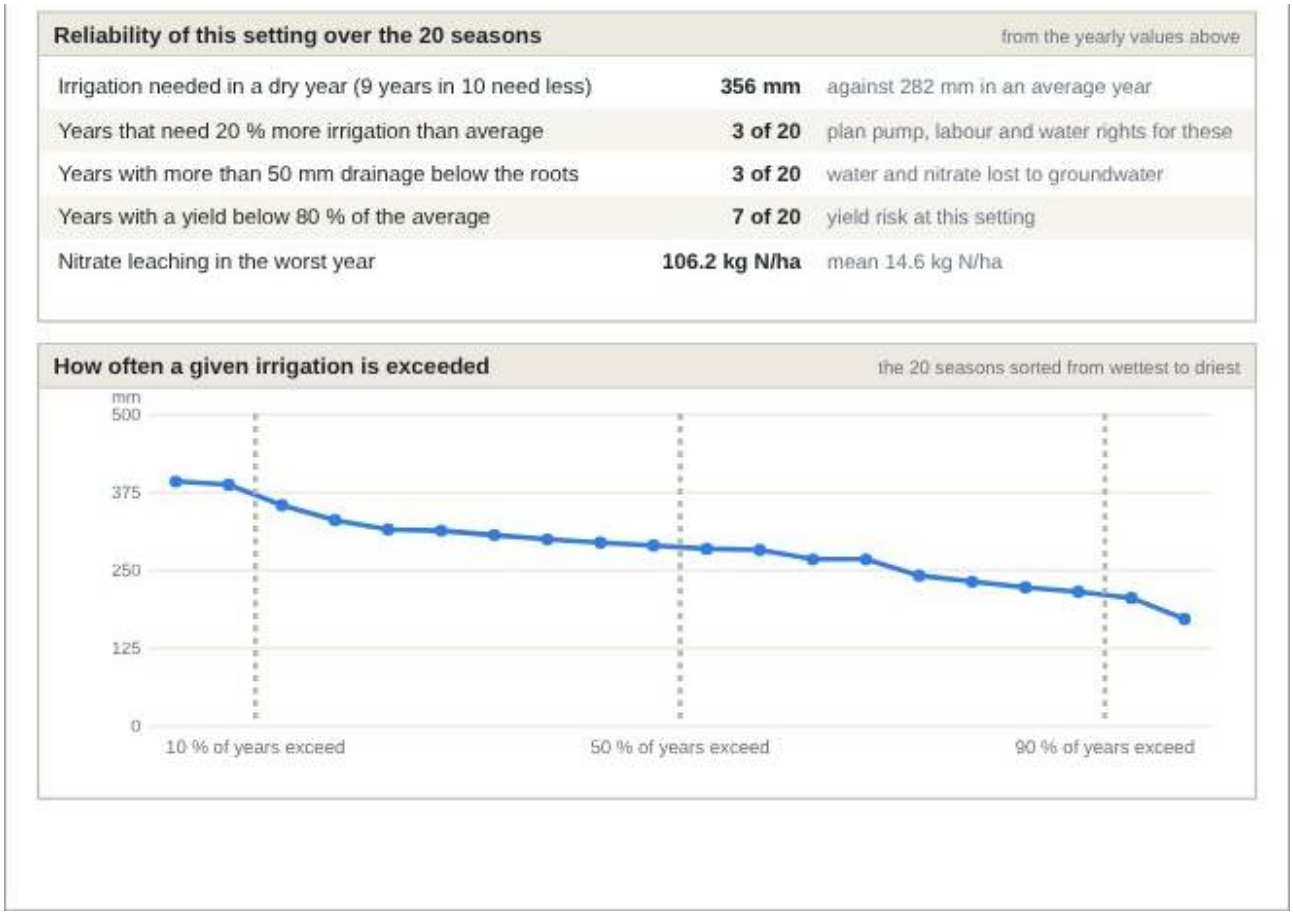


Figure 5.59: Reliability over 20 seasons, Coarse Loam, part 2 of 2.

5.21 Full water balance (Coarse Loam)

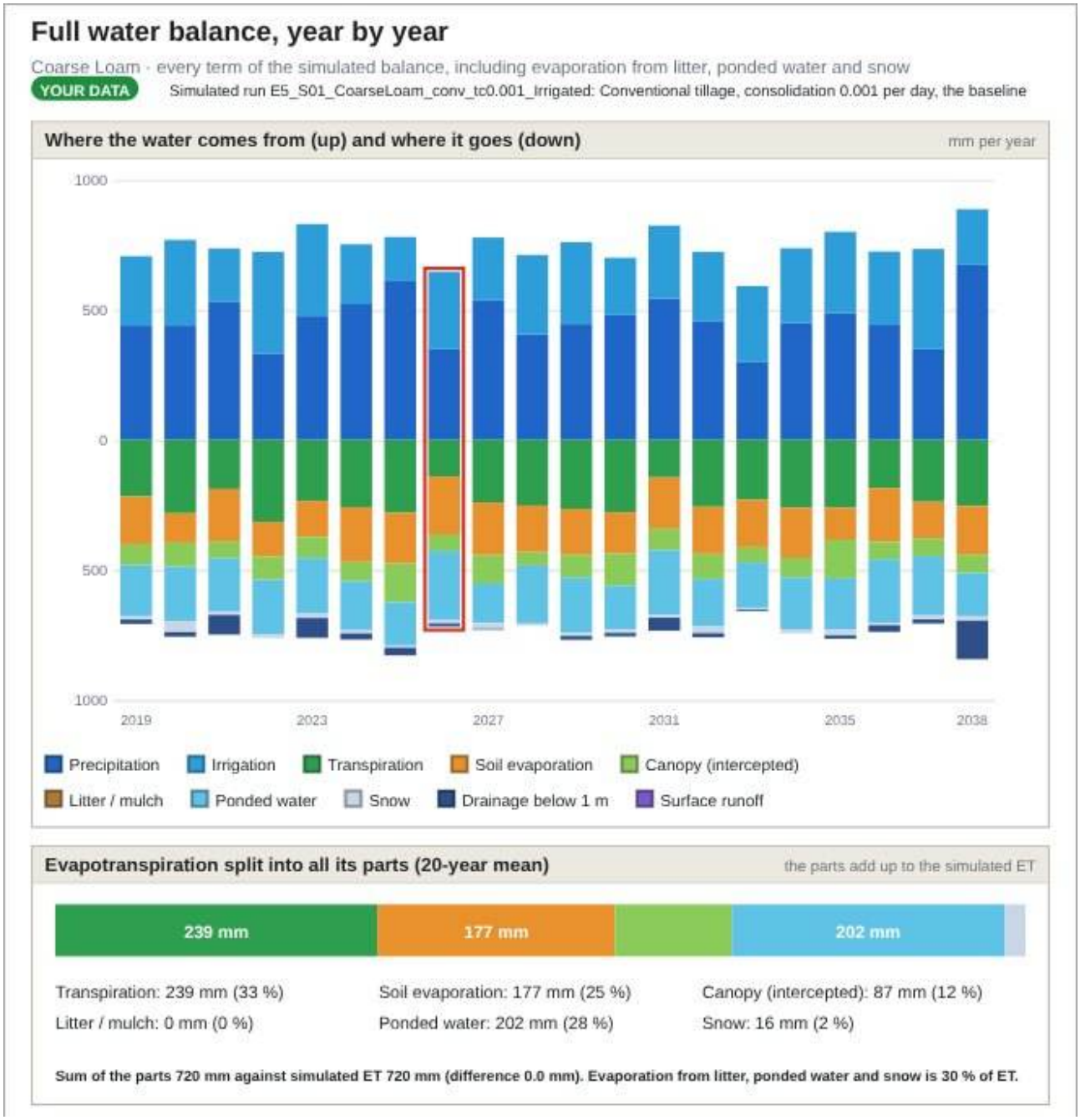


Figure 5.60: Full water balance, Coarse Loam, part 1 of 2.

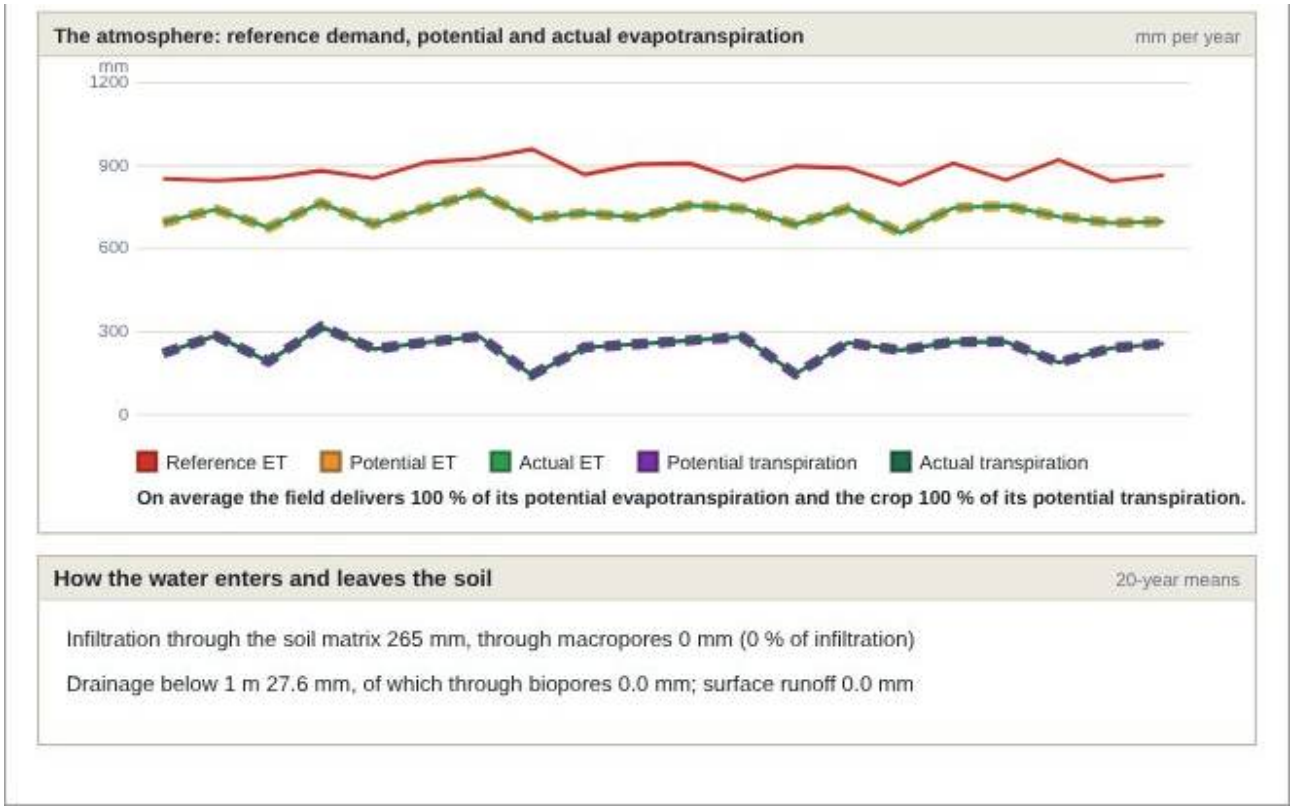


Figure 5.61: Full water balance, Coarse Loam, part 2 of 2.

5.22 Nitrogen budget (Coarse Loam)

Figure 5.62: Nitrogen budget, Coarse Loam, part 1 of 2.

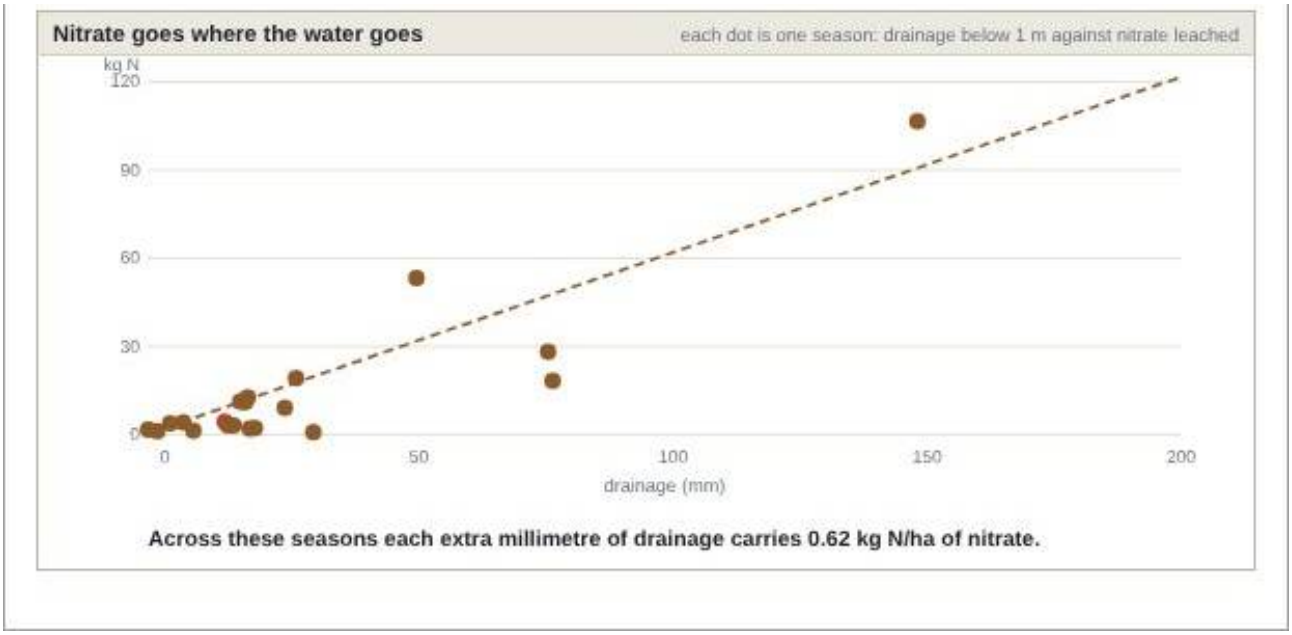


Figure 5.63: Nitrogen budget, Coarse Loam, part 2 of 2.

5.23 Crop water requirement (Coarse Loam)

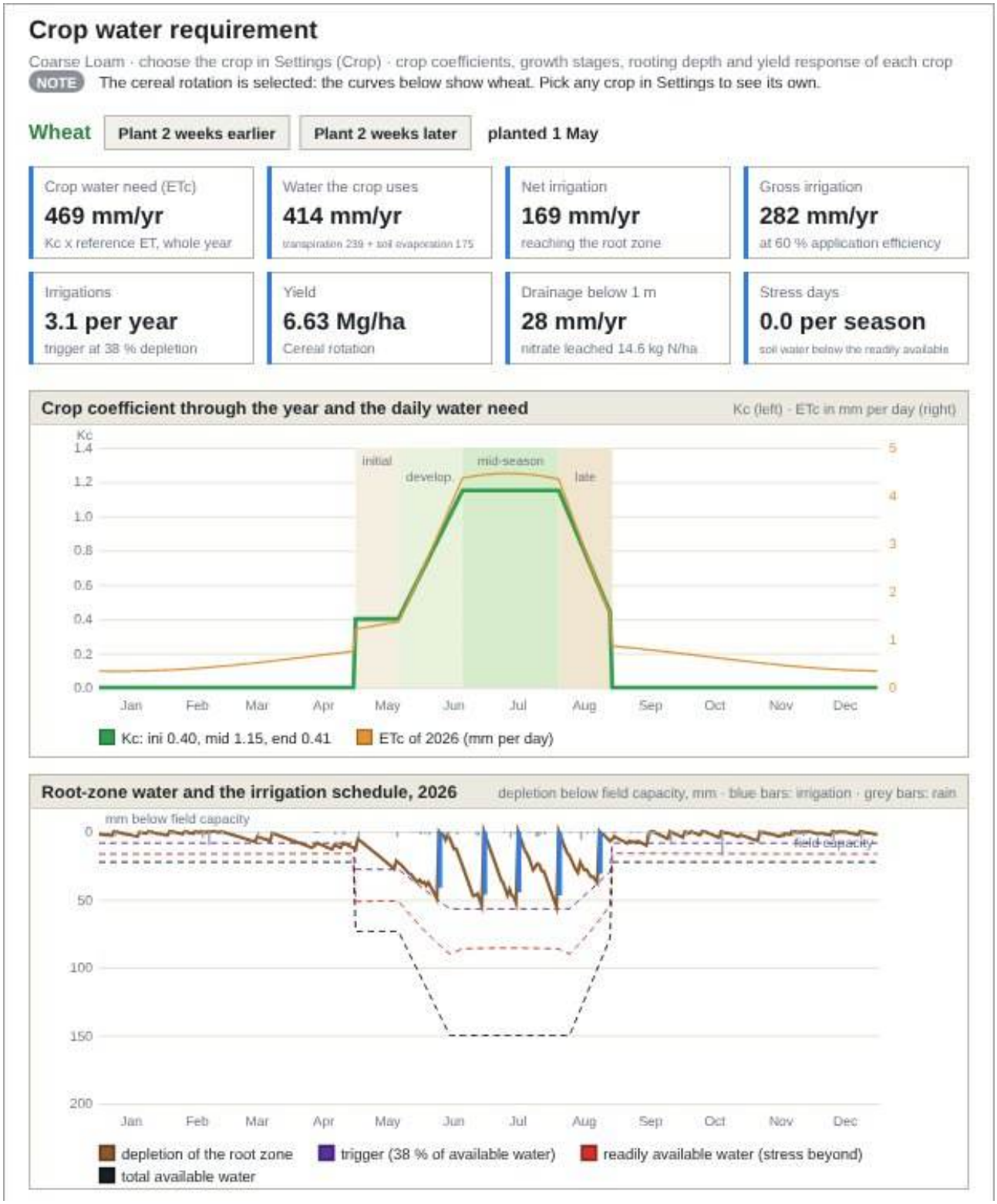


Figure 5.64: Crop water requirement, Coarse Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %
Chickpea	436	**183**	4.2	100 %
Sorghum (grain)	429	**167**	2.9	100 %
Hazelnut (low)	481	**167**	2.9	100 %
Pecan (low)	481	**167**	2.9	100 %
Cherimoya (medium)	545	**161**	2.8	100 %
Citrus (medium density, short)	552	**152**	2.6	100 %

Figure 5.65: Crop water requirement, Coarse Loam, part 2 of 3.

Apple (medium)	459	**149**	2.6	100 %
Cowpea (beans)	398	**148**	3.4	100 %
Millet	395	**144**	2.6	100 %
Wine grapes (pergola)	457	**125**	2.2	100 %
Green gram (mung bean)	372	**123**	2.9	100 %
Peach (low)	435	**119**	2.1	100 %
Almond (medium)	435	**119**	2.1	100 %
Apricot (young)	380	**76**	1.3	100 %
Olive (traditional, medium density)	446	**61**	1.1	100 %
Persimmon (young, low)	358	**58**	1.0	100 %
Pomegranate (low)	358	**58**	1.0	100 %
Highbush blueberry (low)	338	**55**	1.1	100 %
Fig tree (young, low)	327	**24**	0.4	100 %

Figure 5.66: Crop water requirement, Coarse Loam, part 3 of 3.

5.24 Feedback loops (Coarse Loam)

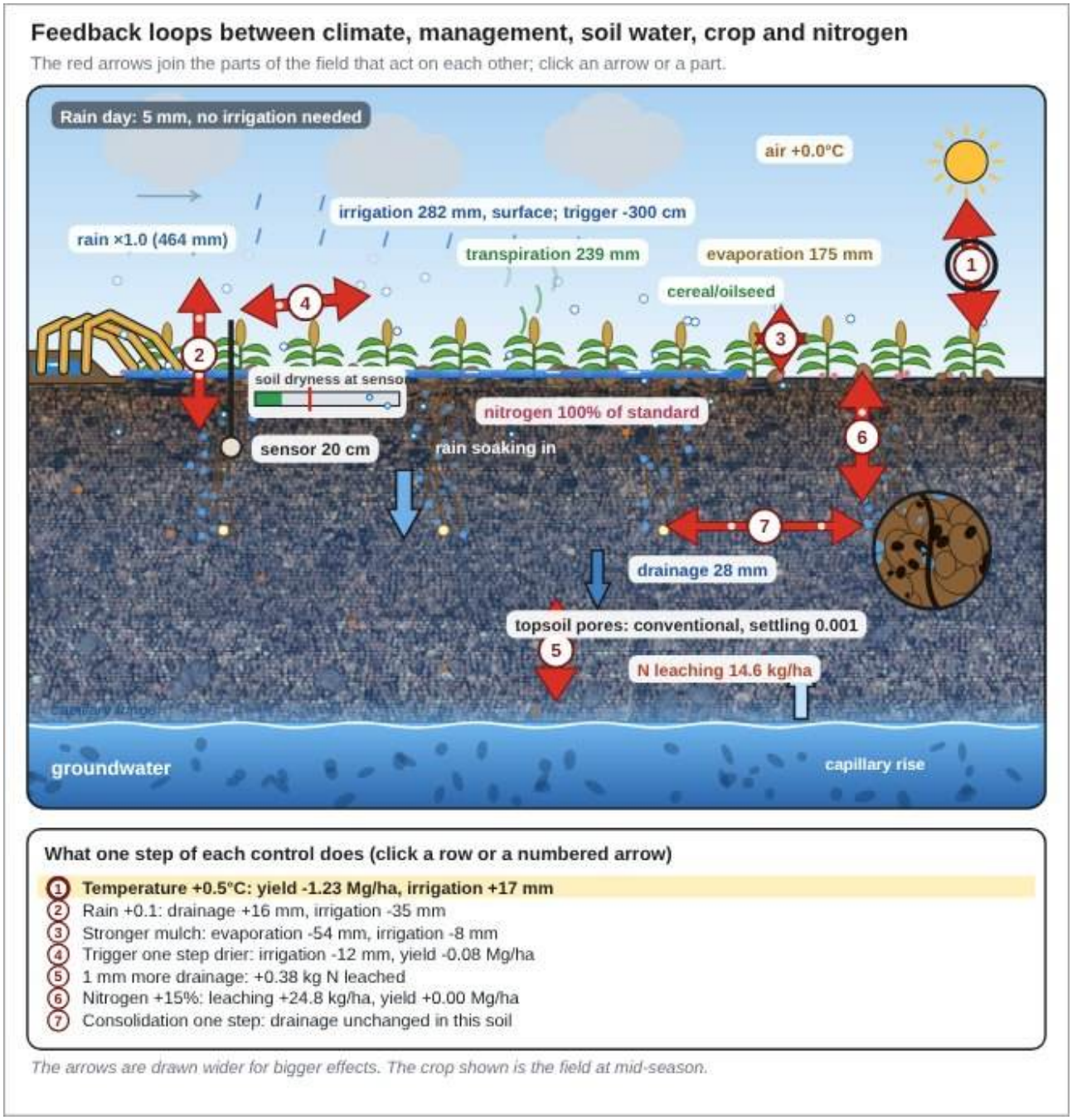


Figure 5.67: Feedback loops, Coarse Loam, part 1 of 1.

Chapter 6

Loam

6.1 Surface irrigation design (Loam)

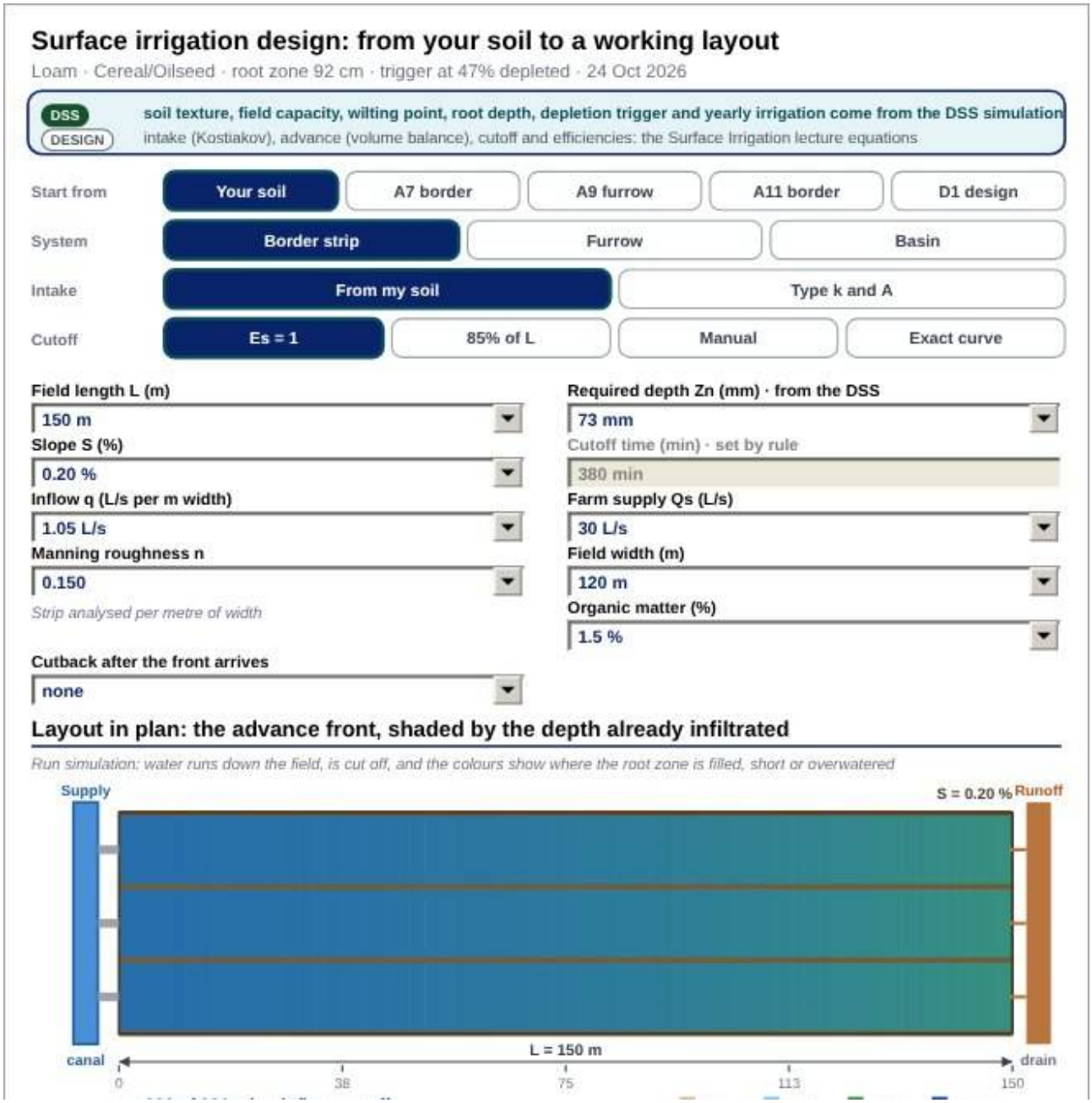


Figure 6.1: Surface irrigation design, Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

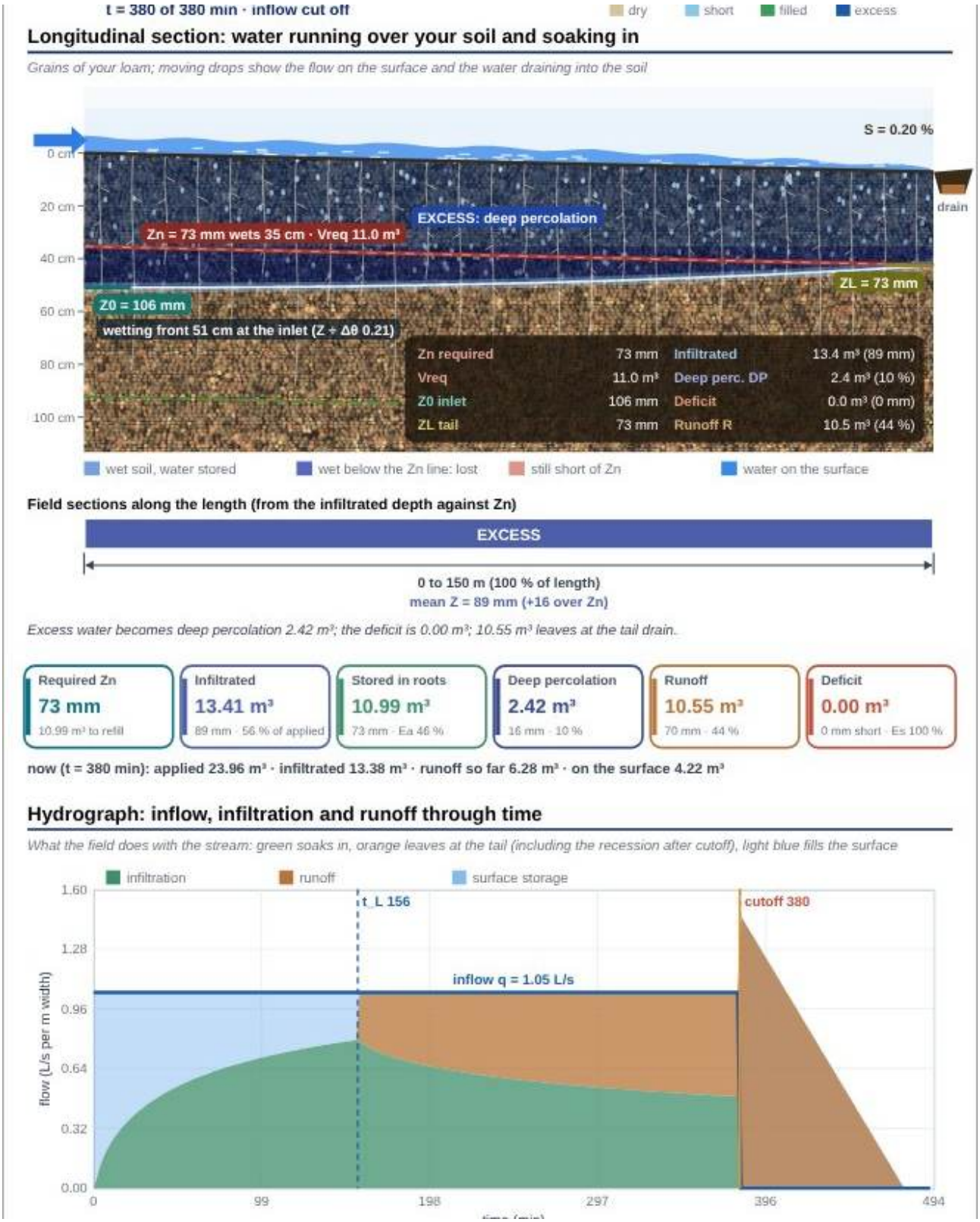


Figure 6.2: Surface irrigation design, Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

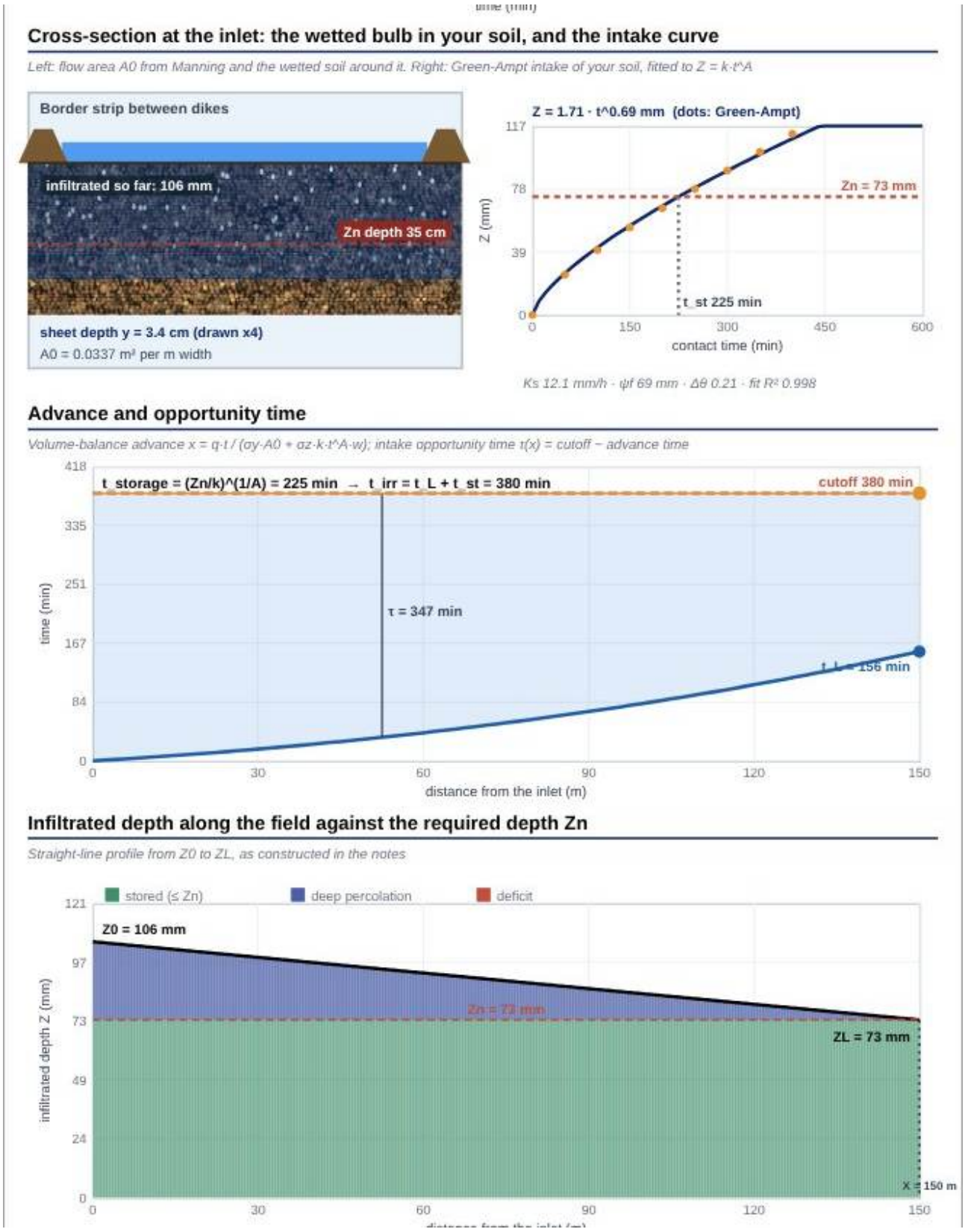


Figure 6.3: Surface irrigation design, Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

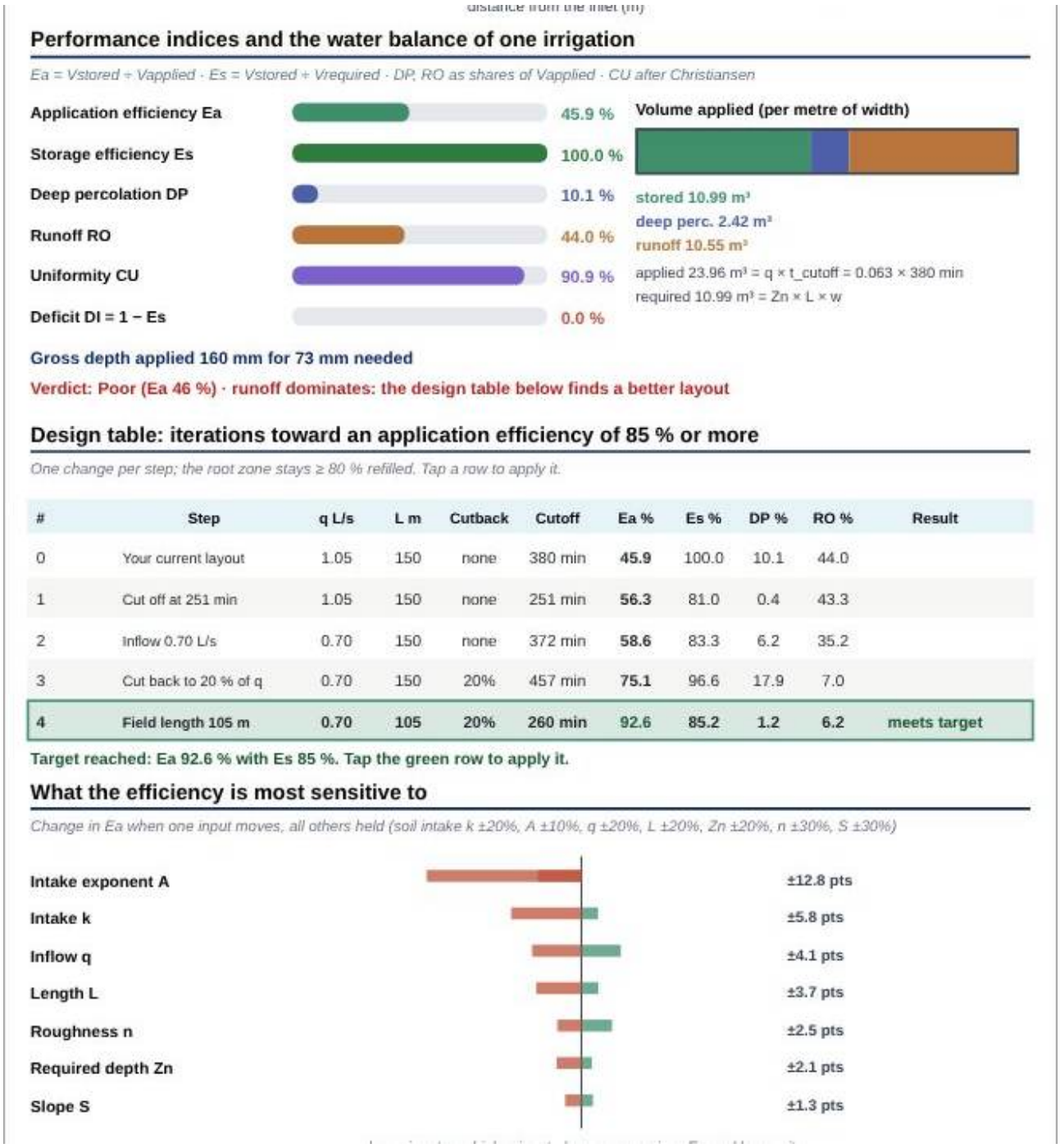


Figure 6.4: Surface irrigation design, Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

Figure 6.5: Surface irrigation design, Loam, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

6.2 Drip irrigation design (Loam)

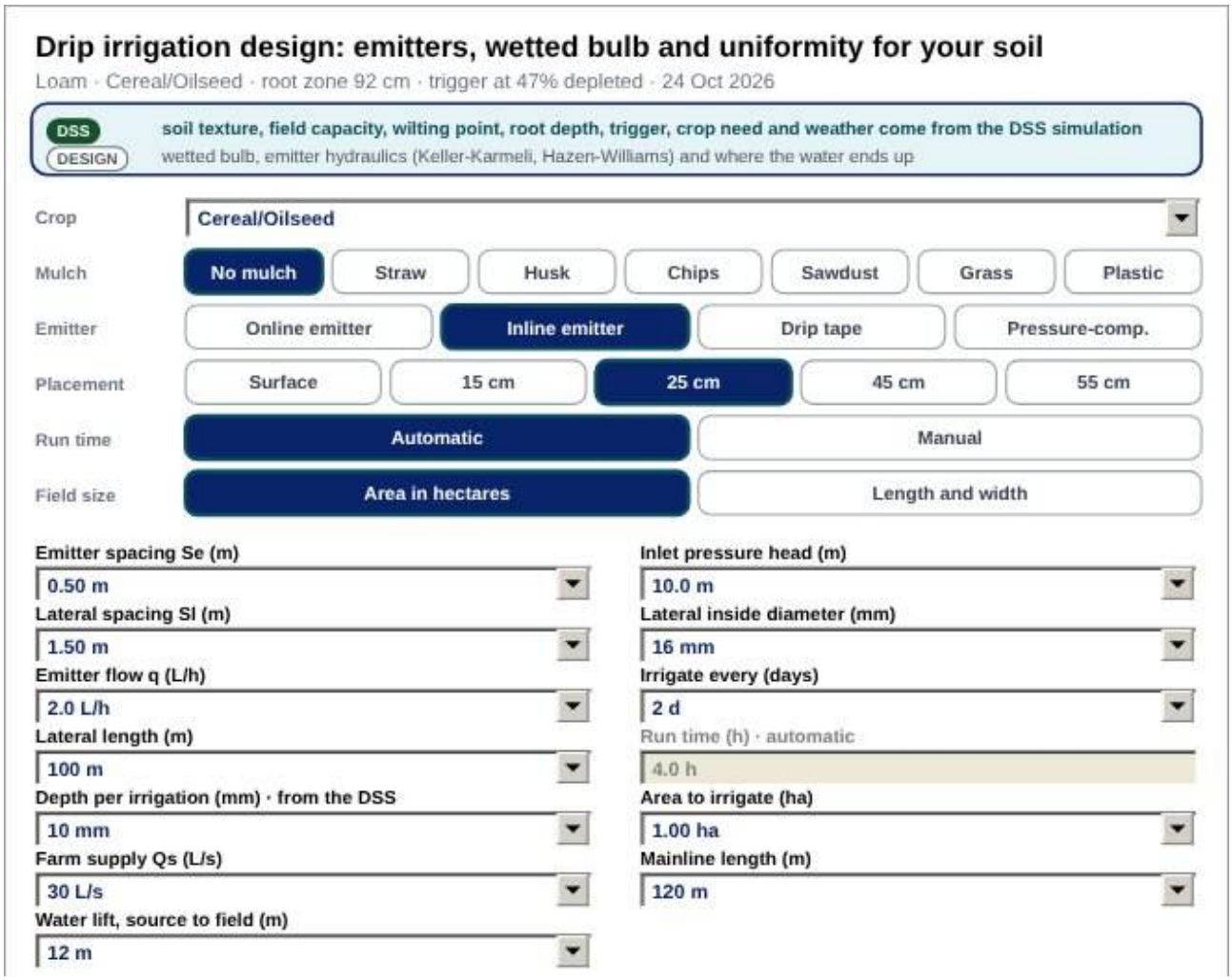


Figure 6.6: Drip irrigation design, Loam, part 1 of 6.

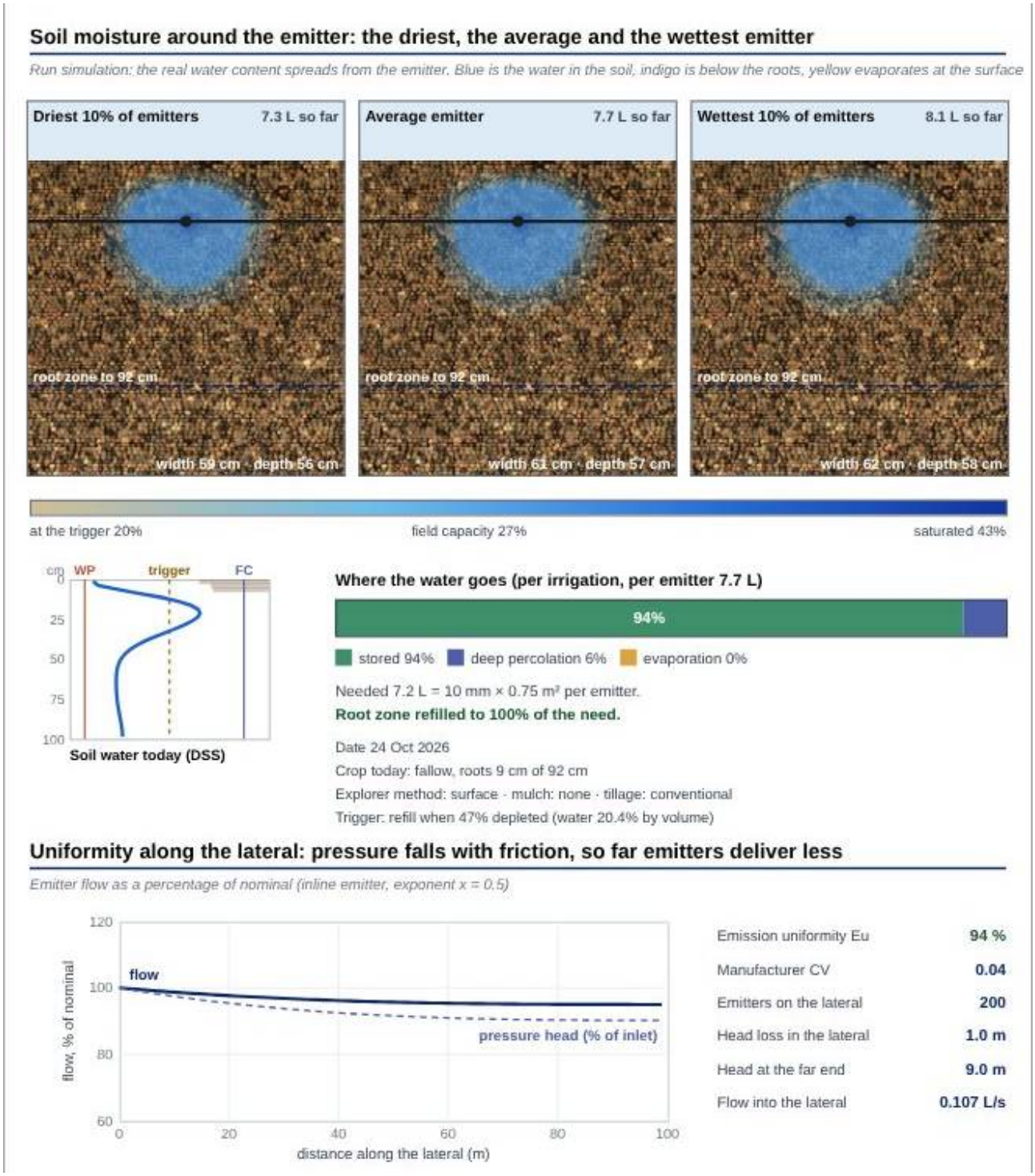
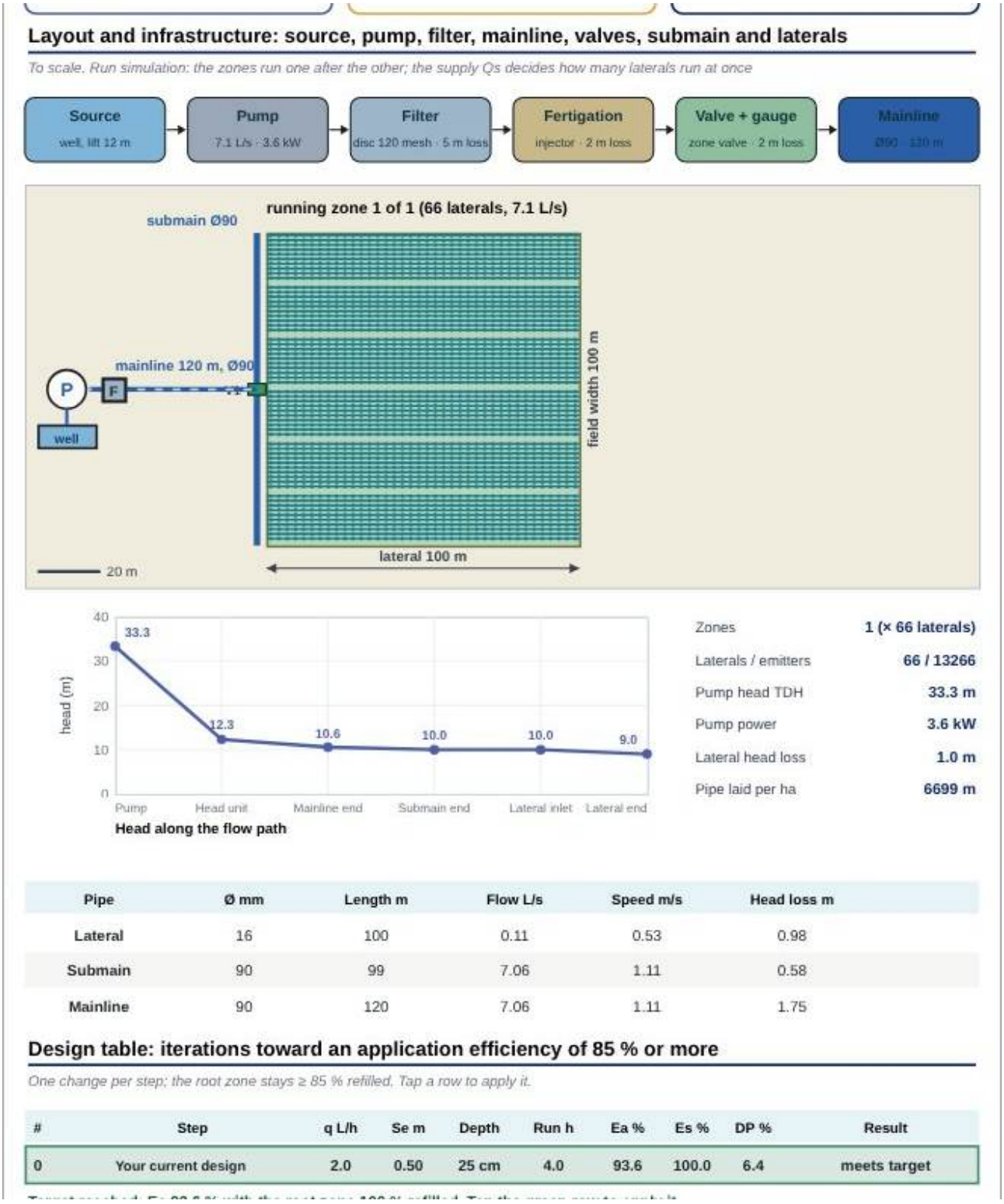


Figure 6.7: Drip irrigation design, Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

Figure 6.8: Drip irrigation design, Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	4.0	93.6	100.0	6.4	meets target

Figure 6.9: Drip irrigation design, Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

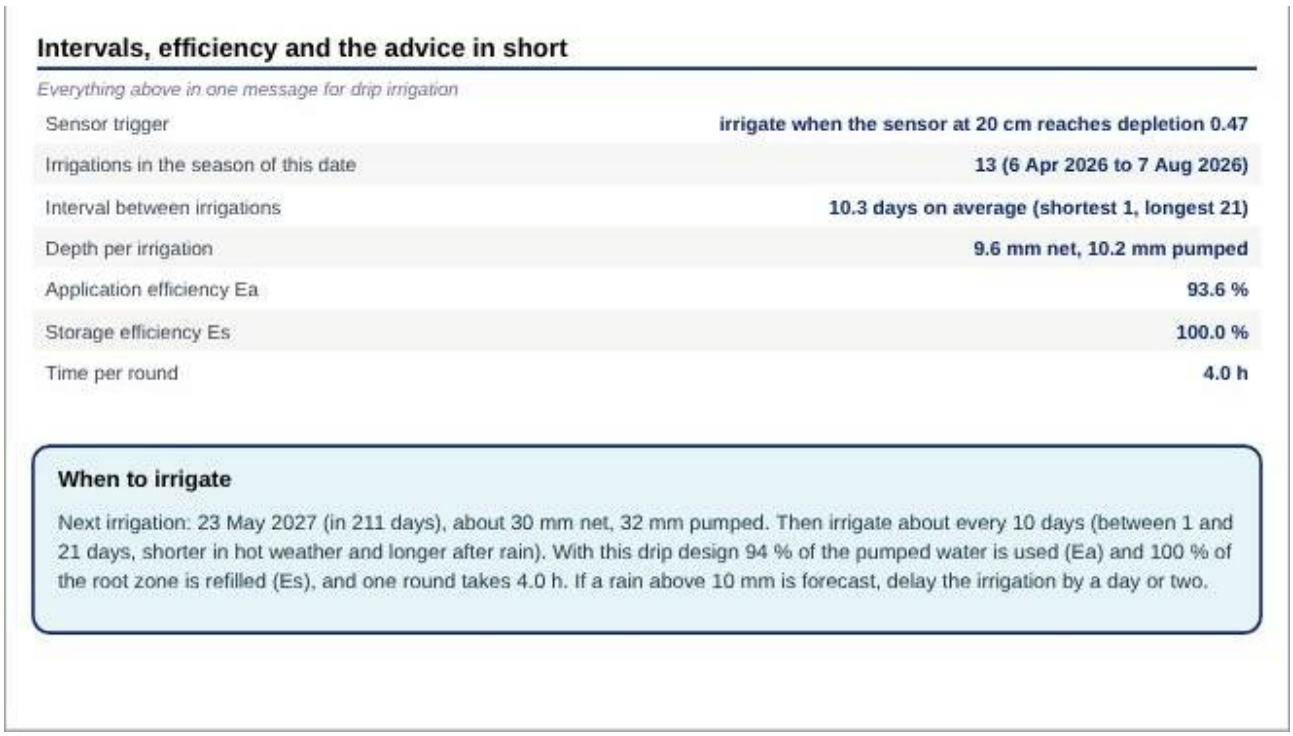
Net irrigation need in a season (DSS)	**391 mm → 41 irrigations of 10 mm**
Gross water per irrigation	**10.2 mm (Ea 94 %)**
Gross water per season	**418 mm = 4176 m³ per hectare**
Peak crop need / interval	**4.8 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 4.0 h = 4.0 h per round**
Pumping per season	**163 hours · 578 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**23 May 2027**	211 days	30	32	4.0 h	16 days
2	**8 Jun 2027**	227 days	30	32	4.0 h	10 days
3	**18 Jun 2027**	237 days	30	32	4.0 h	9 days
4	**27 Jun 2027**	246 days	30	32	4.0 h	8 days
5	**5 Jul 2027**	254 days	30	32	4.0 h	10 days
6	**15 Jul 2027**	264 days	30	32	4.0 h	17 days
7	**1 Aug 2027**	281 days	30	32	4.0 h	248 days
8	**5 Apr 2028**	529 days	30	32	4.0 h	-

Figure 6.10: Drip irrigation design, Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.47**
Irrigations in the season of this date	**13 (6 Apr 2026 to 7 Aug 2026)**
Interval between irrigations	**10.3 days on average (shortest 1, longest 21)**
Depth per irrigation	**9.6 mm net, 10.2 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**4.0 h**

When to irrigate

Next irrigation: 23 May 2027 (in 211 days), about 30 mm net, 32 mm pumped. Then irrigate about every 10 days (between 1 and 21 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 4.0 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 6.11: Drip irrigation design, Loam, part 6 of 6: Intervals, efficiency and the advice in short.

6.3 Sprinkler design and pipes (Loam)

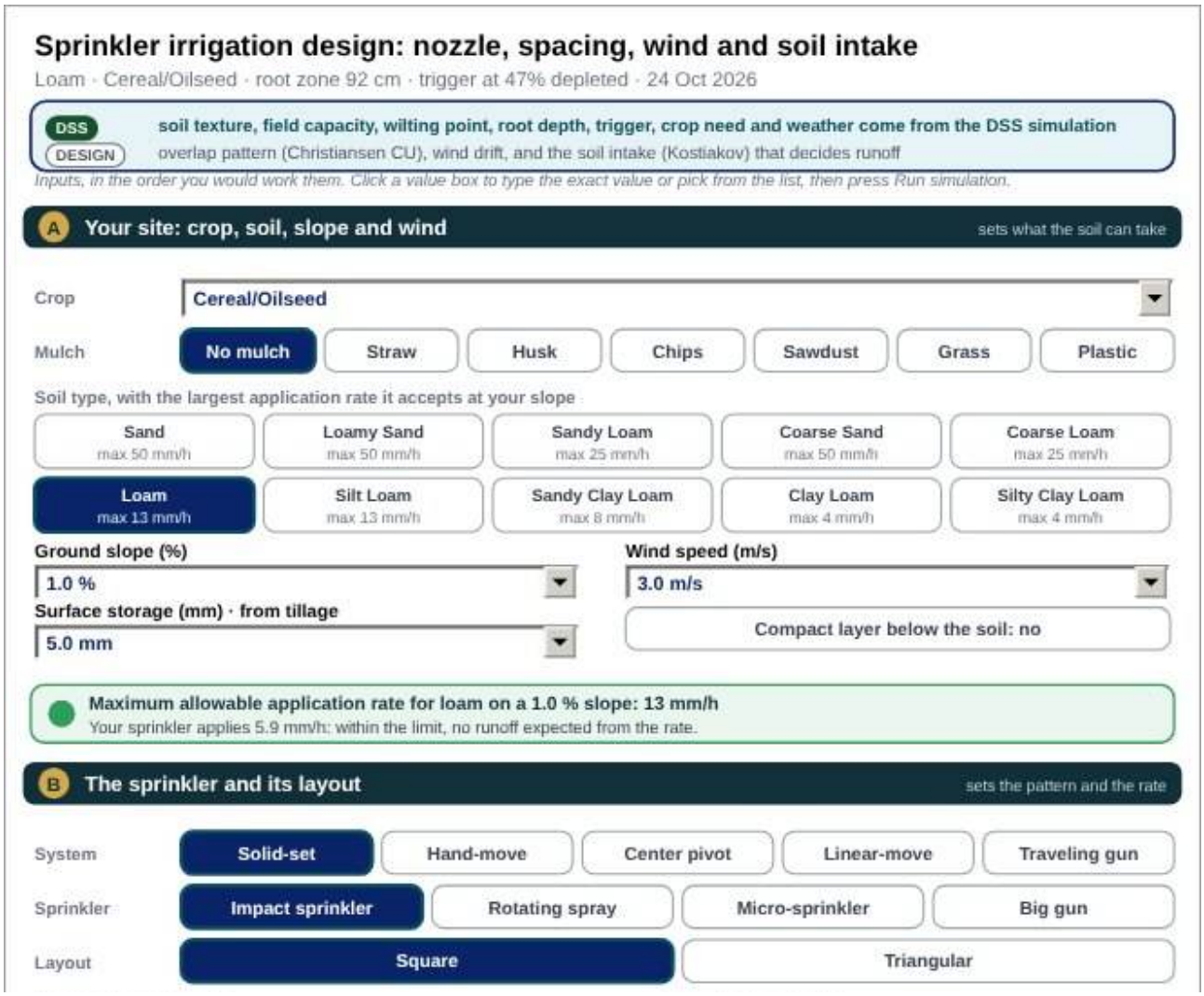


Figure 6.12: Sprinkler design and pipes, Loam, part 1 of 12.

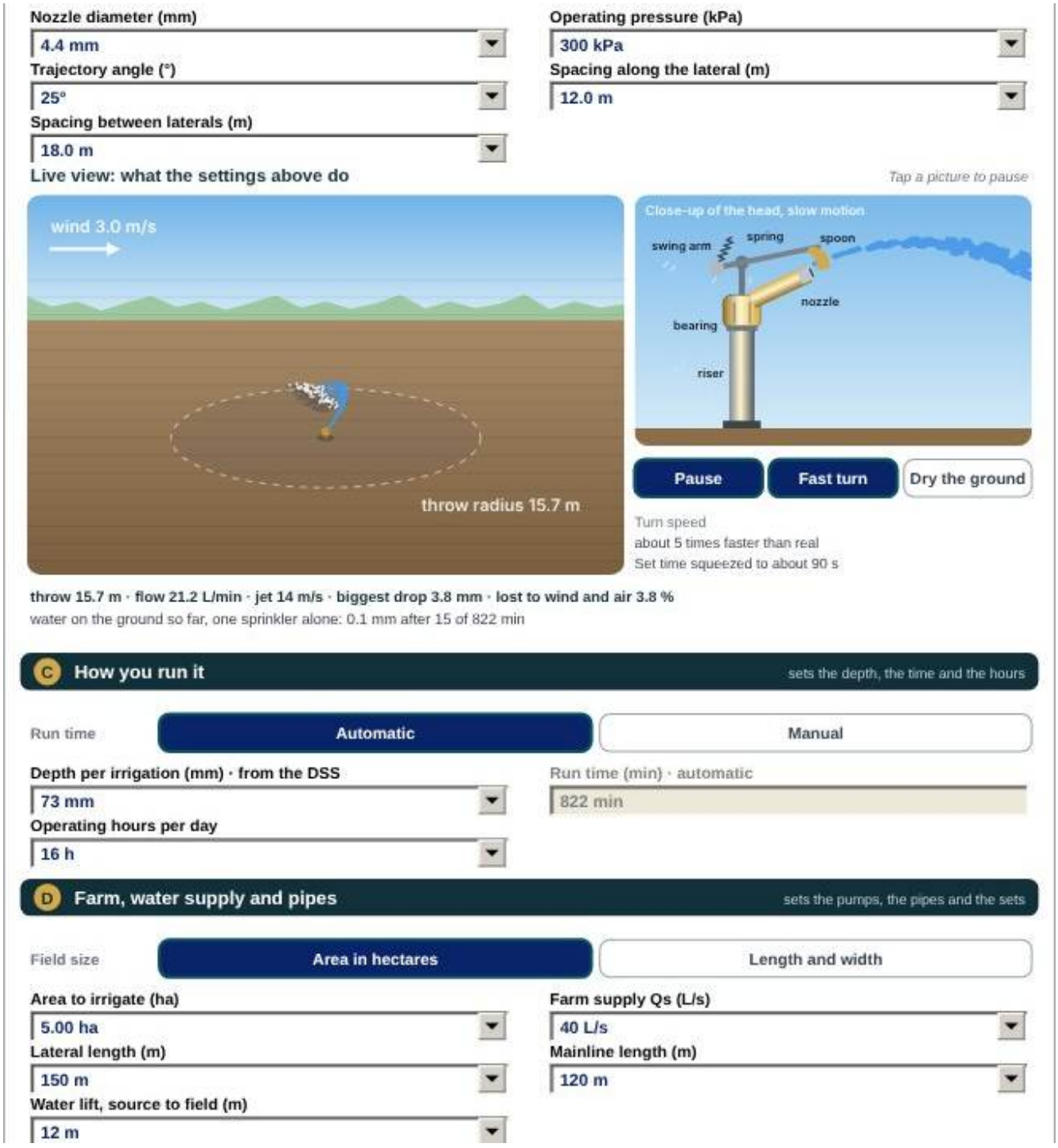


Figure 6.13: Sprinkler design and pipes, Loam, part 2 of 12.

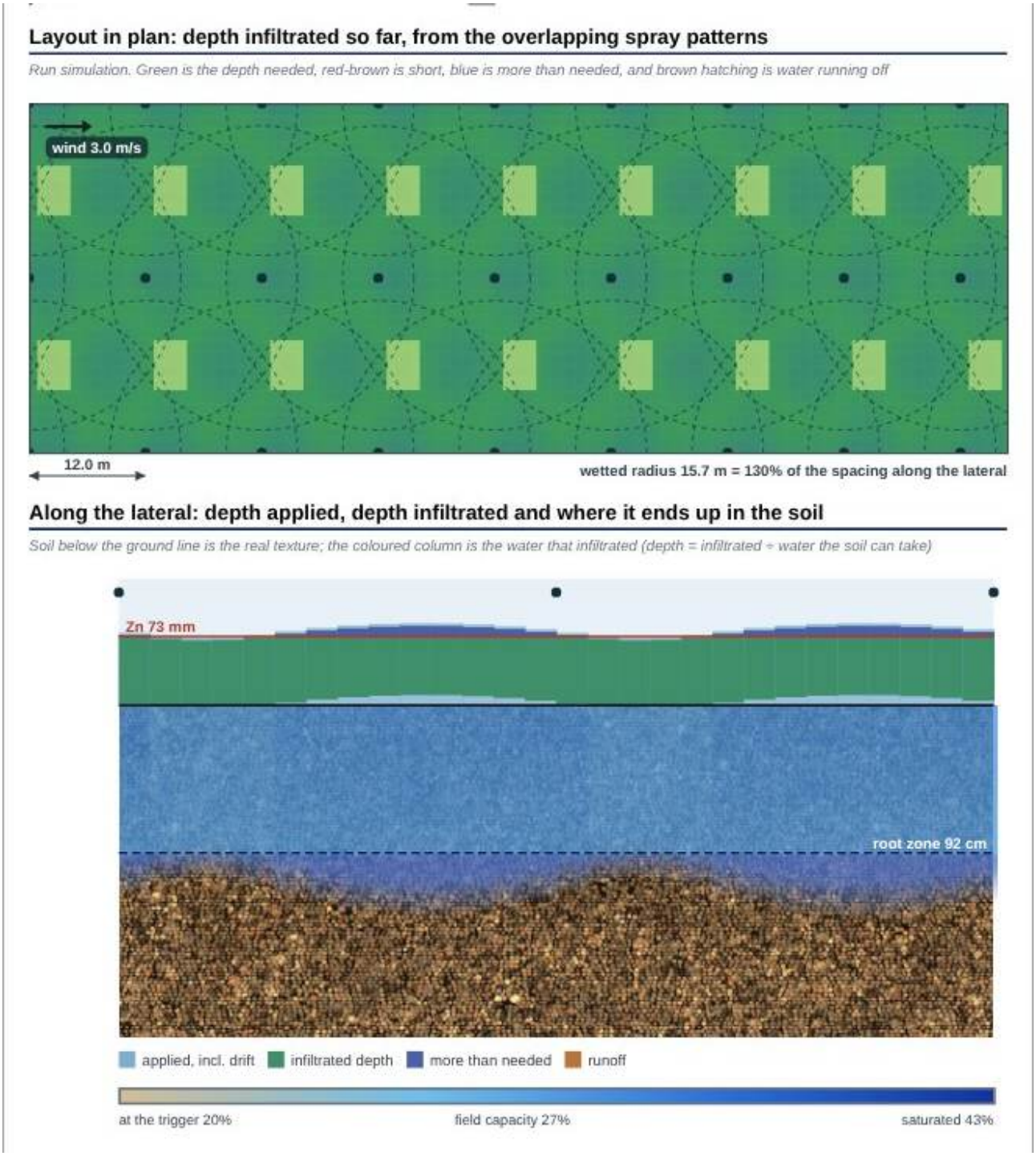


Figure 6.14: Sprinkler design and pipes, Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

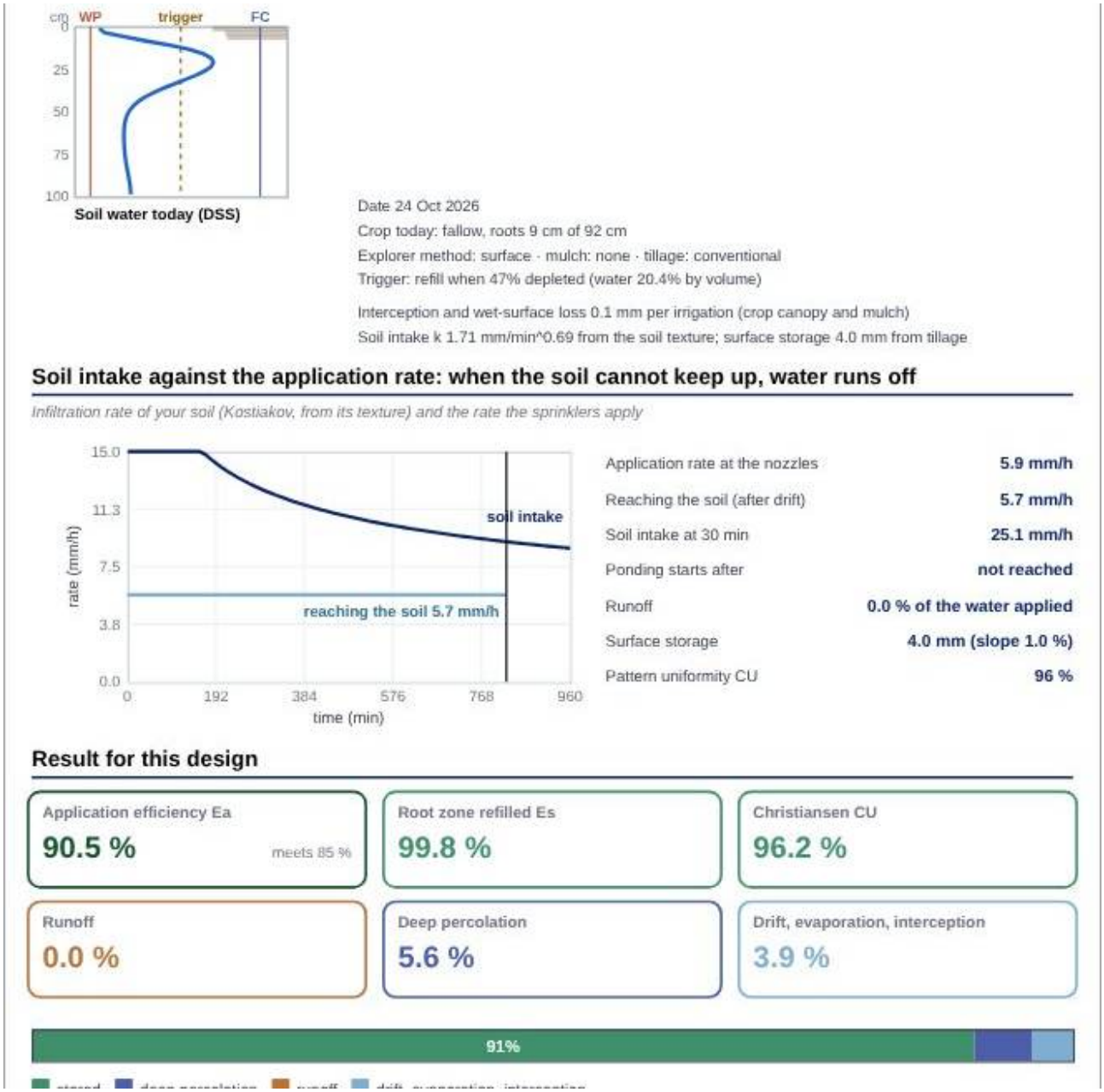


Figure 6.15: Sprinkler design and pipes, Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

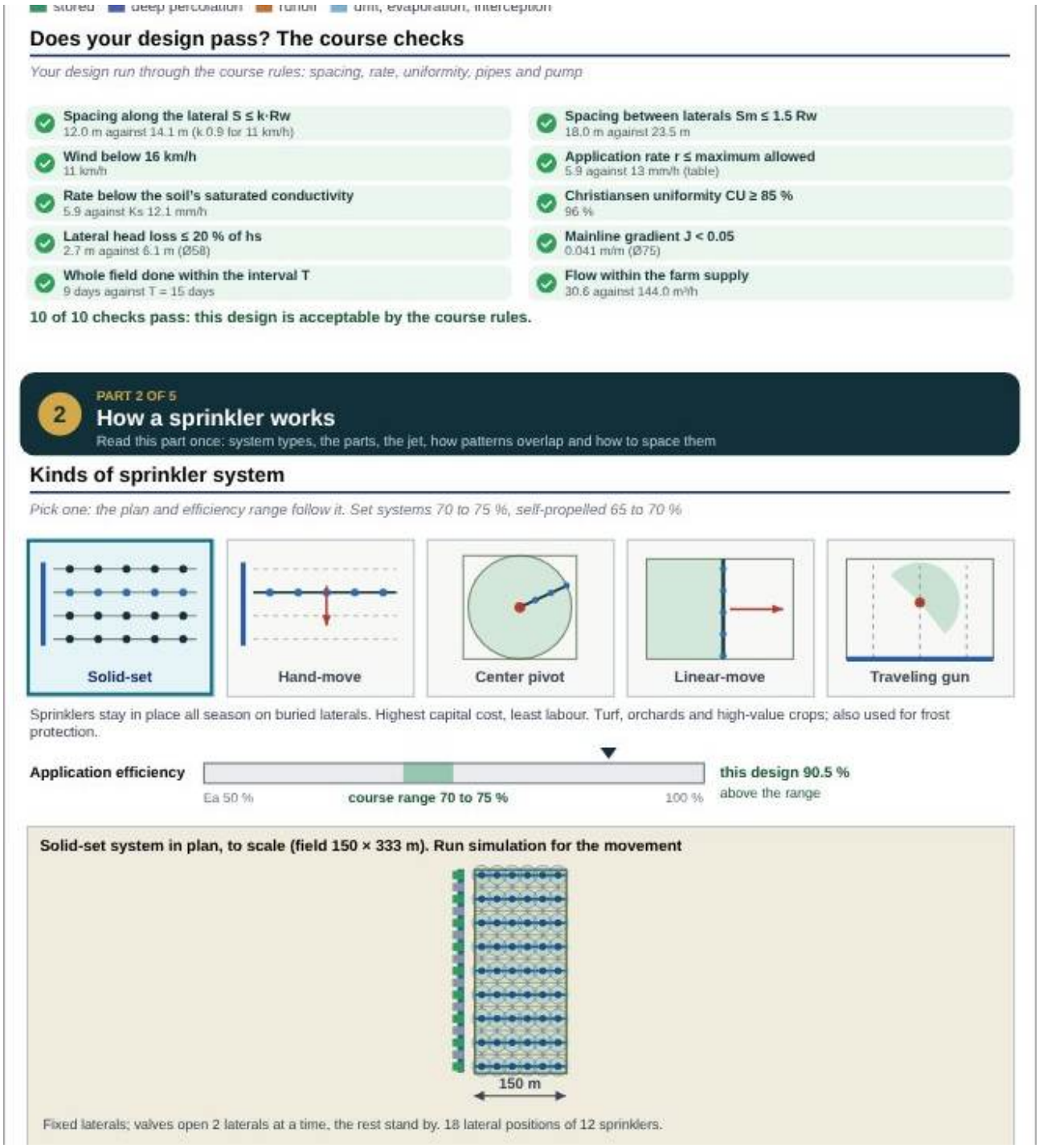
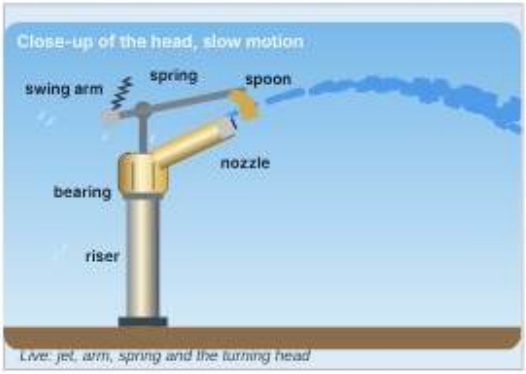


Figure 6.16: Sprinkler design and pipes, Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

The sprinkler: parts and pressure class

Parts of a rotary impact sprinkler, and where your choice falls among the three pressure classes



Class by operating pressure

Low pressure (spray type)
pressure 8 to 20 m · discharge 0.035 to 0.5 m³/h · throw 3 to 7 m

Medium pressure (rotary, impact)
pressure 20 to 45 m · discharge 0.5 to 15 m³/h · throw 9 to 20 m

High pressure (gun sprinklers)
pressure 45 to 100 m · discharge 15 to 150 m³/h · throw 30 to 100 m

Yours: head 30.6 m, 1.27 m³/h, throw 15.7 m → medium pressure

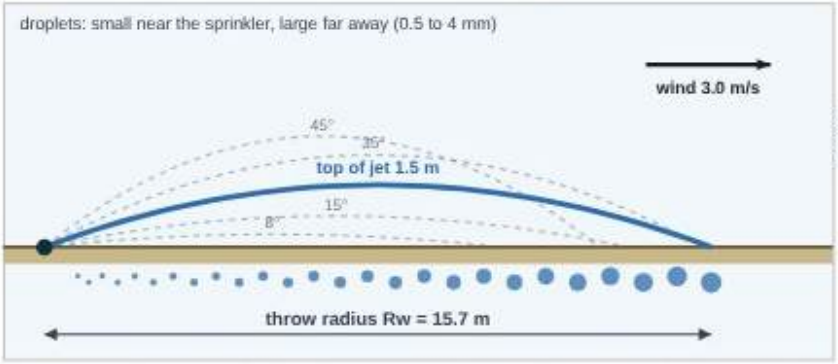
$q = c \cdot d^2 \cdot \sqrt{h}$ and $Rw = kr \cdot h^{0.25}$

The jet: trajectory angle, throw radius and droplet size

Farthest throw at about 28 to 30 degrees; low angles (8 degrees) go under foliage and drift less

Trajectory angle (°)

25°



Drift and evaporation loss 3.8 % at 25° and 3.0 m/s. Pressure sets drop size: lower pressure gives larger drops that throw farther but can seal the soil surface; higher pressure gives finer drops that drift more.

Seen from the field: the jet sweeps round, drops fly with the wind and wet the ground

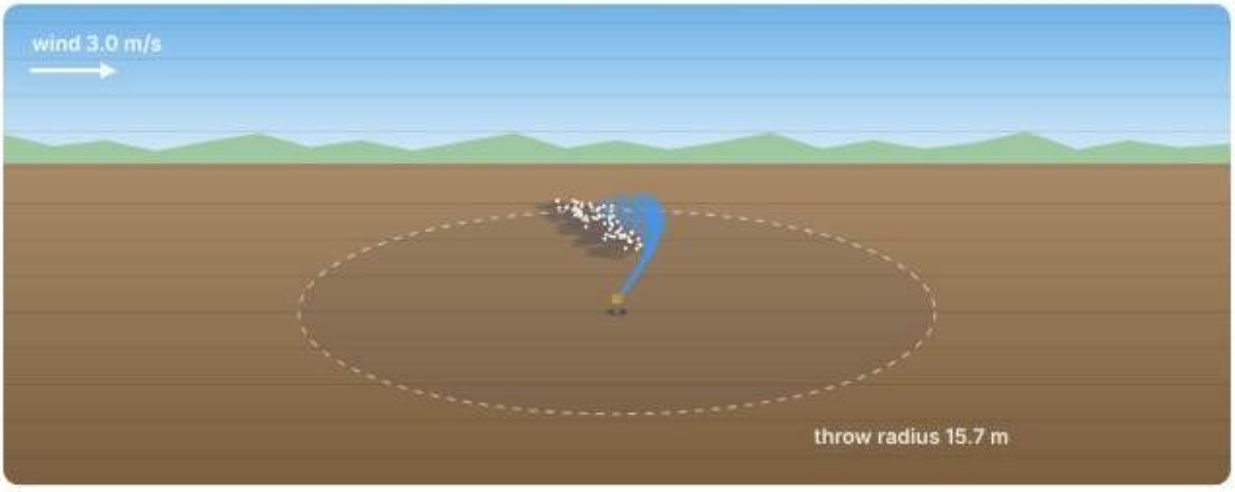


Figure 6.17: Sprinkler design and pipes, Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

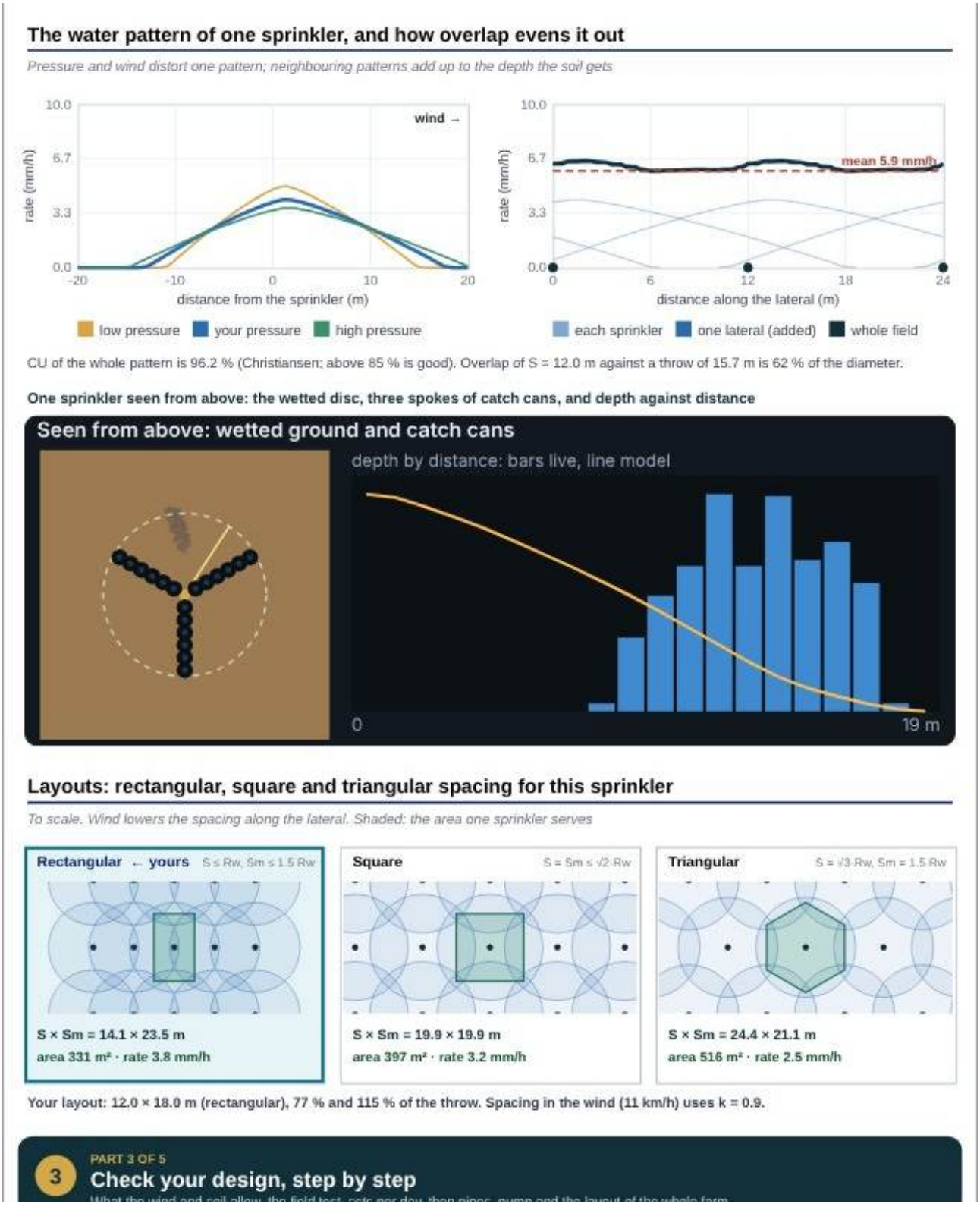


Figure 6.18: Sprinkler design and pipes, Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

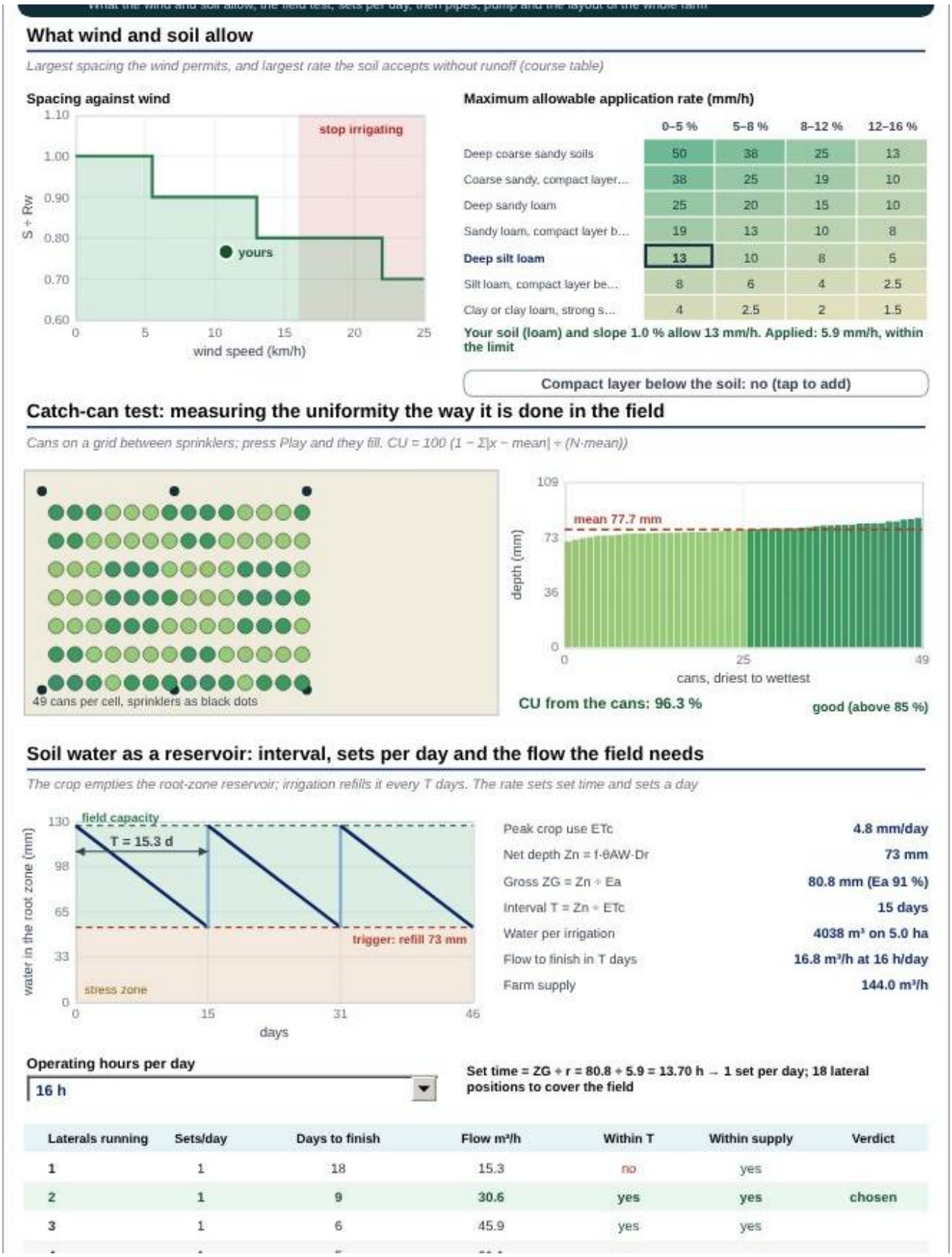


Figure 6.19: Sprinkler design and pipes, Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

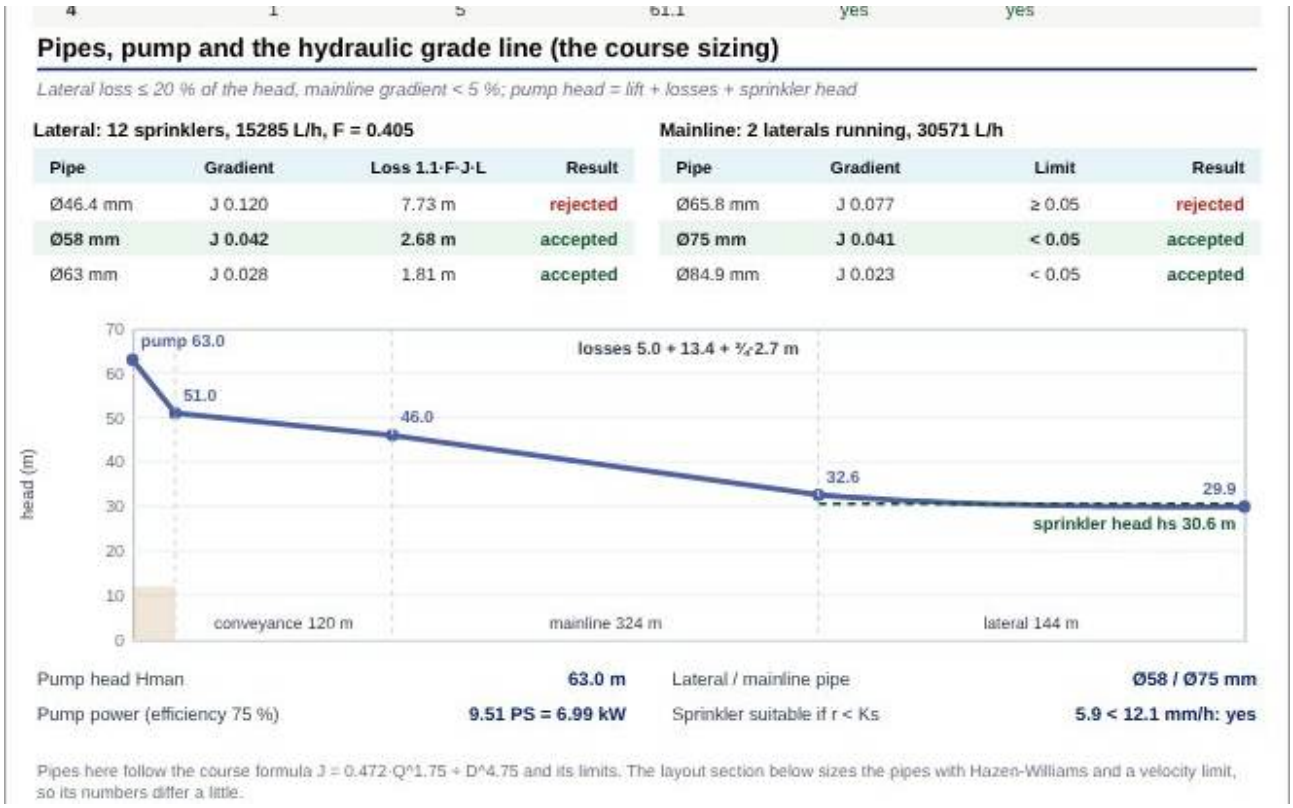
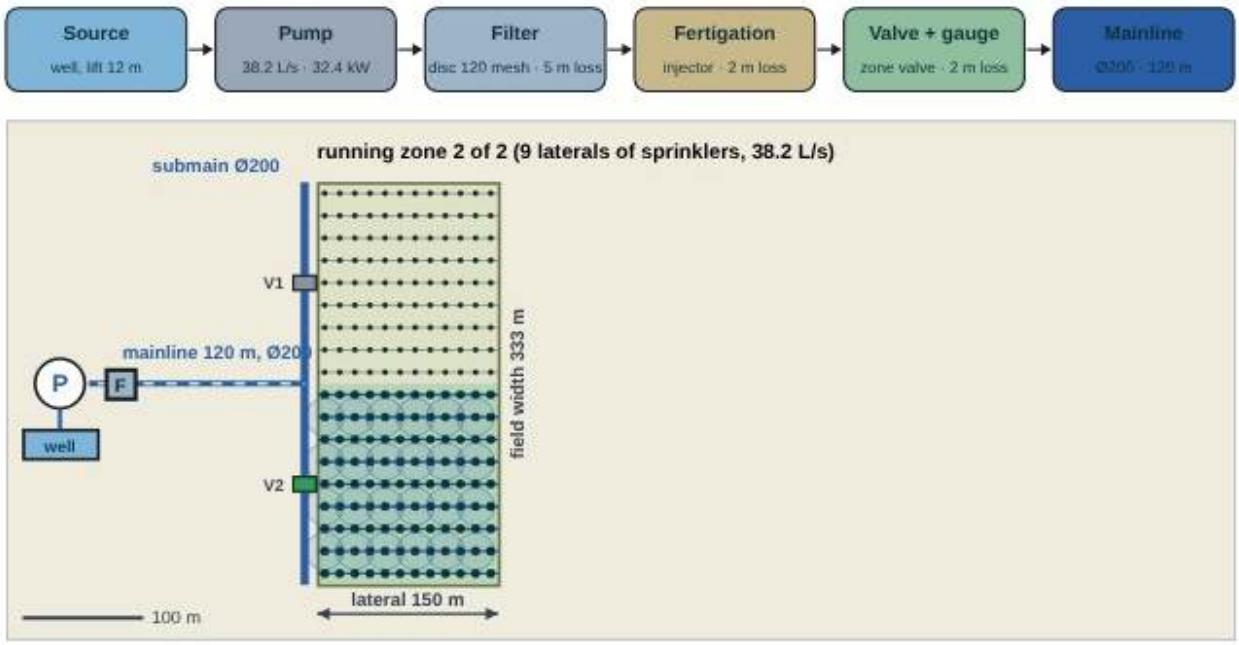
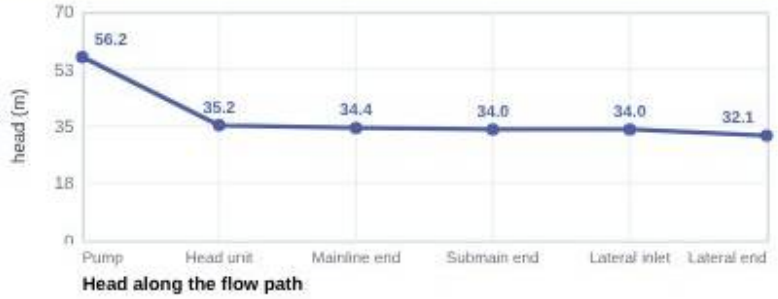


Figure 6.20: Sprinkler design and pipes, Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 **Improve it**

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

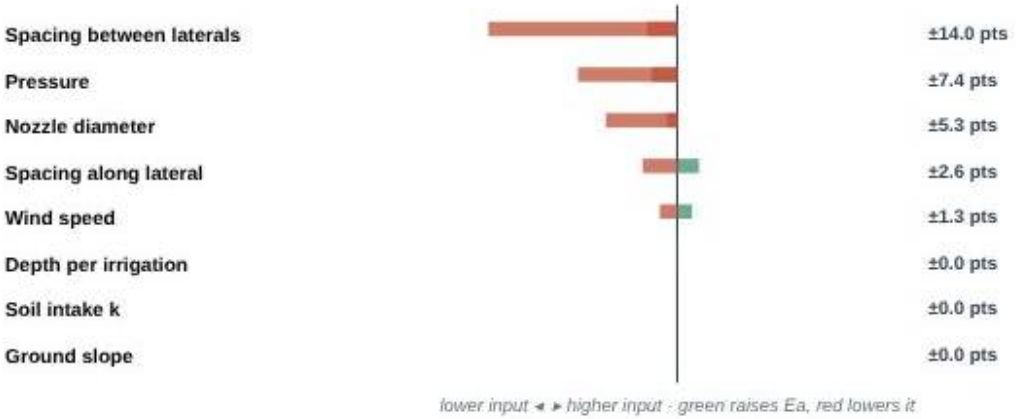
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	822	90.5	99.8	0.0	meets target

Figure 6.21: Sprinkler design and pipes, Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

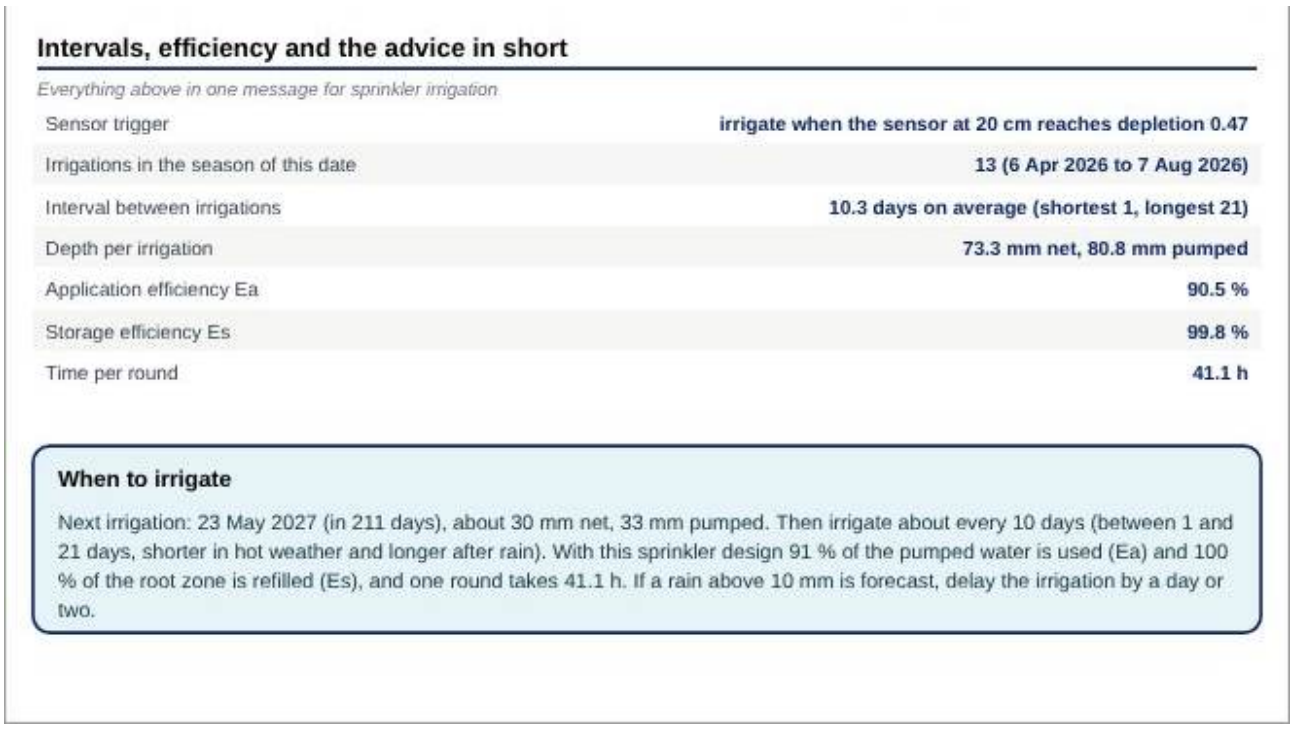
Net irrigation need in a season (DSS)	**391 mm → 5 irrigations of 73 mm**
Gross water per irrigation	**80.8 mm (Ea 91 %)**
Gross water per season	**431 mm = 4308 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 822 min = 41.1 h per round**
Pumping per season	**219 hours · 7110 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**23 May 2027**	211 days	30	33	41.1 h	16 days
2	**8 Jun 2027**	227 days	30	33	41.1 h	10 days
3	**18 Jun 2027**	237 days	30	33	41.1 h	9 days
4	**27 Jun 2027**	246 days	30	33	41.1 h	8 days
5	**5 Jul 2027**	254 days	30	33	41.1 h	10 days
6	**15 Jul 2027**	264 days	30	33	41.1 h	17 days
7	**1 Aug 2027**	281 days	30	33	41.1 h	248 days
8	**5 Apr 2028**	529 days	30	33	41.1 h	-

Figure 6.22: Sprinkler design and pipes, Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.47**
Irrigations in the season of this date	**13 (6 Apr 2026 to 7 Aug 2026)**
Interval between irrigations	**10.3 days on average (shortest 1, longest 21)**
Depth per irrigation	**73.3 mm net, 80.8 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**41.1 h**

When to irrigate

Next irrigation: 23 May 2027 (in 211 days), about 30 mm net, 33 mm pumped. Then irrigate about every 10 days (between 1 and 21 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 41.1 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 6.23: Sprinkler design and pipes, Loam, part 12 of 12: Intervals, efficiency and the advice in short.

6.4 Virtual farm of the surface design (Loam)

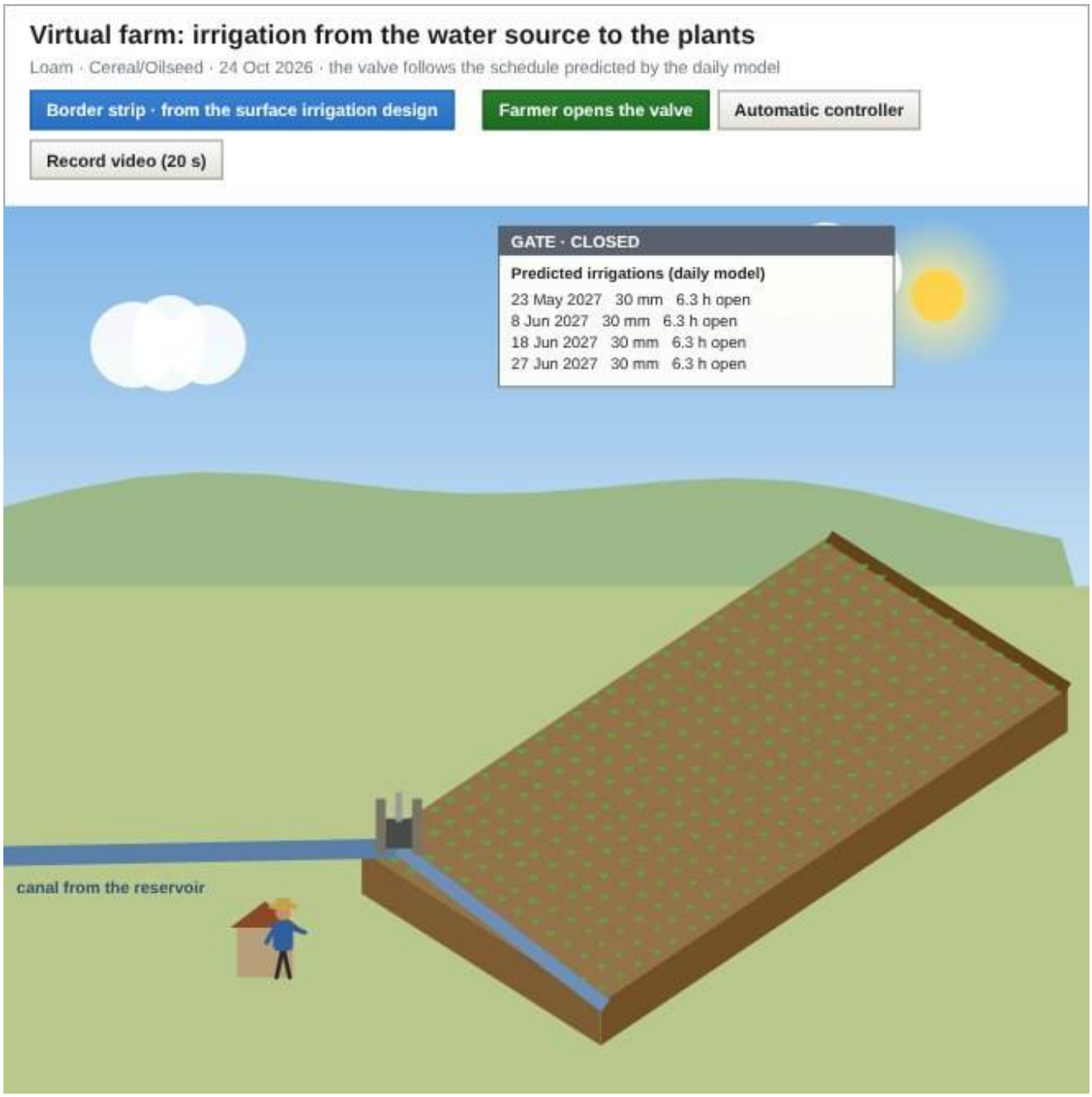


Figure 6.24: Virtual farm, surface design, Loam, part 1 of 3.

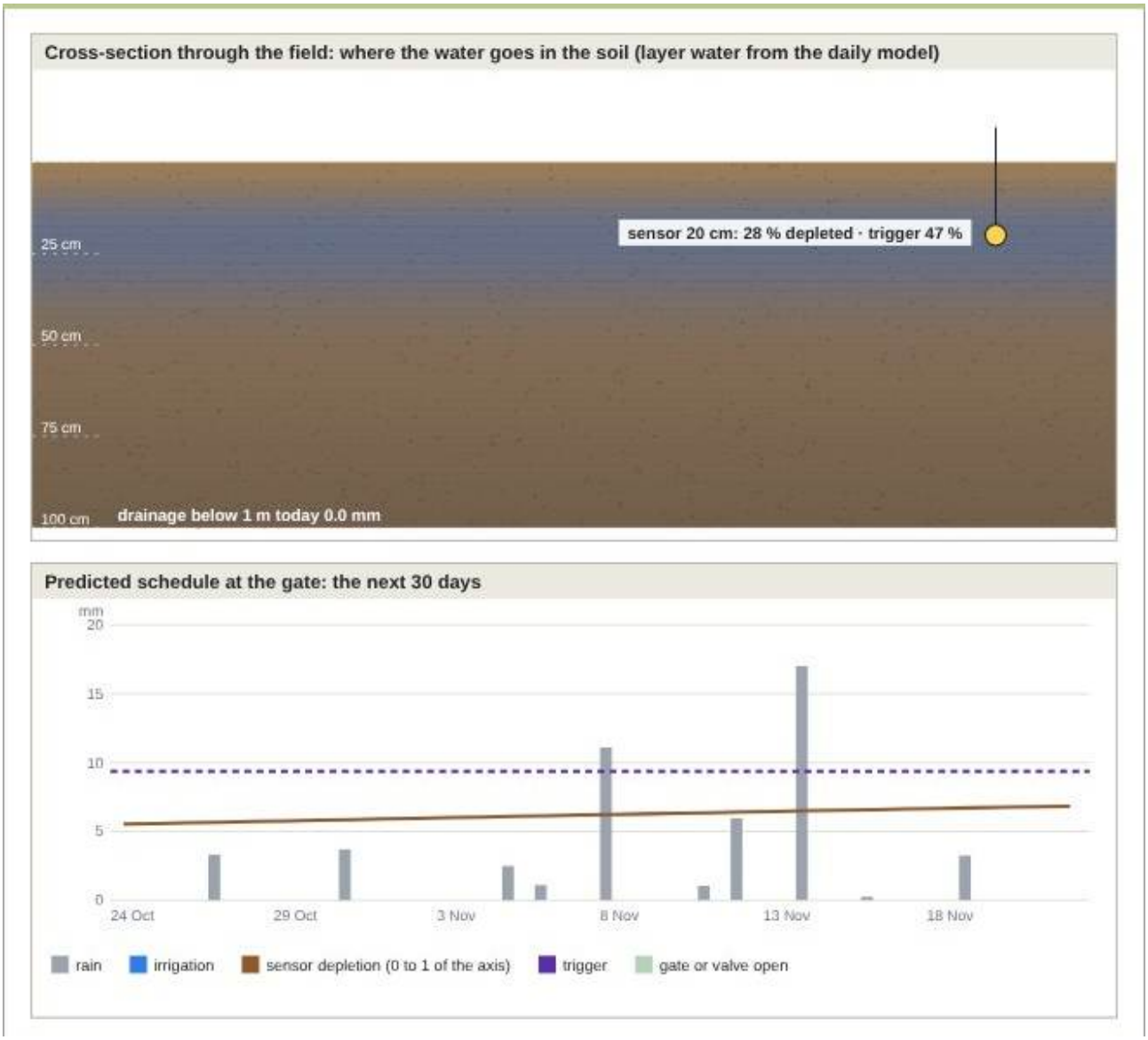


Figure 6.25: Virtual farm, surface design, Loam, part 2 of 3.

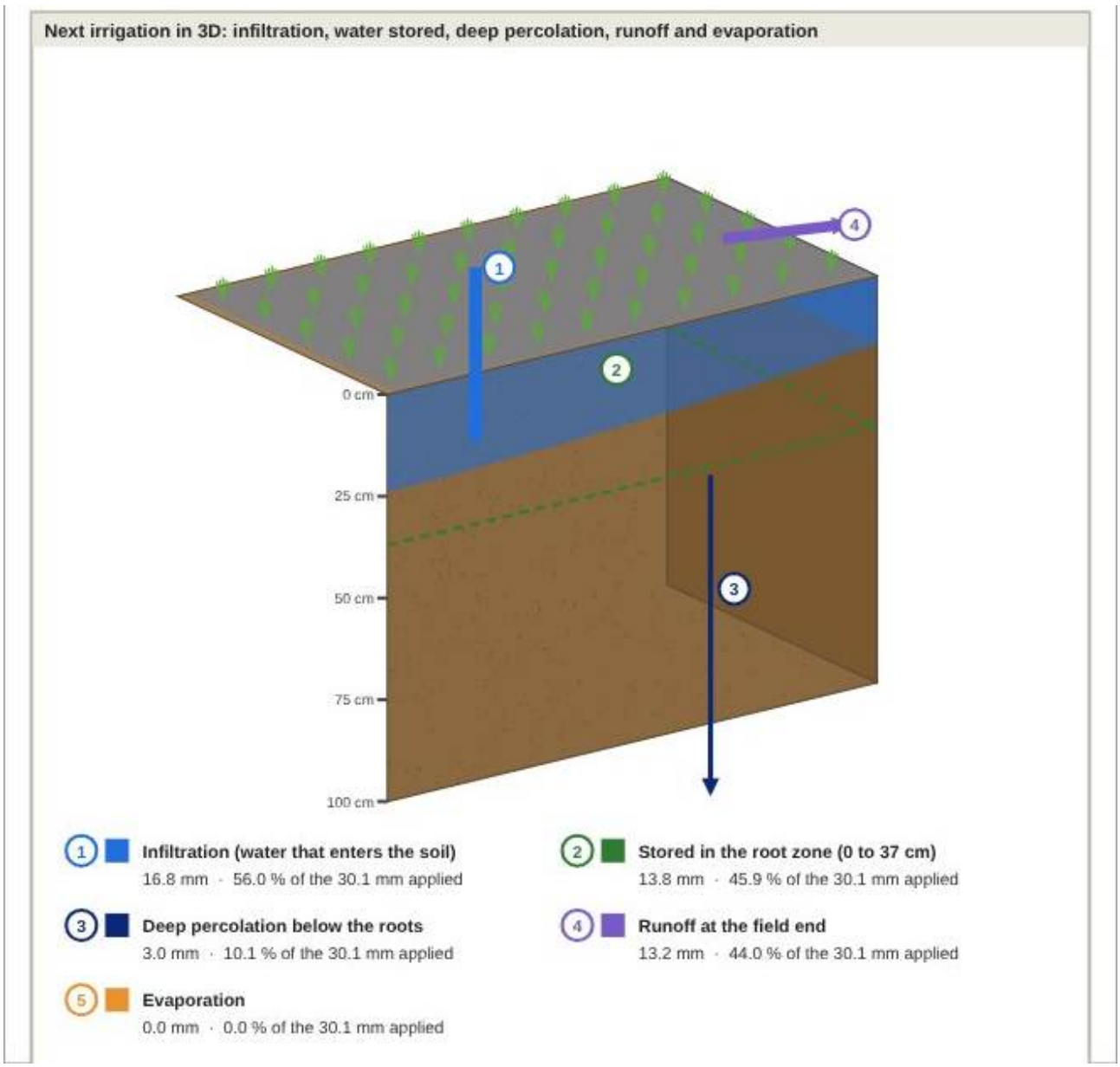


Figure 6.26: Virtual farm, surface design, Loam, part 3 of 3.

6.5 Virtual farm of the drip design (Loam)

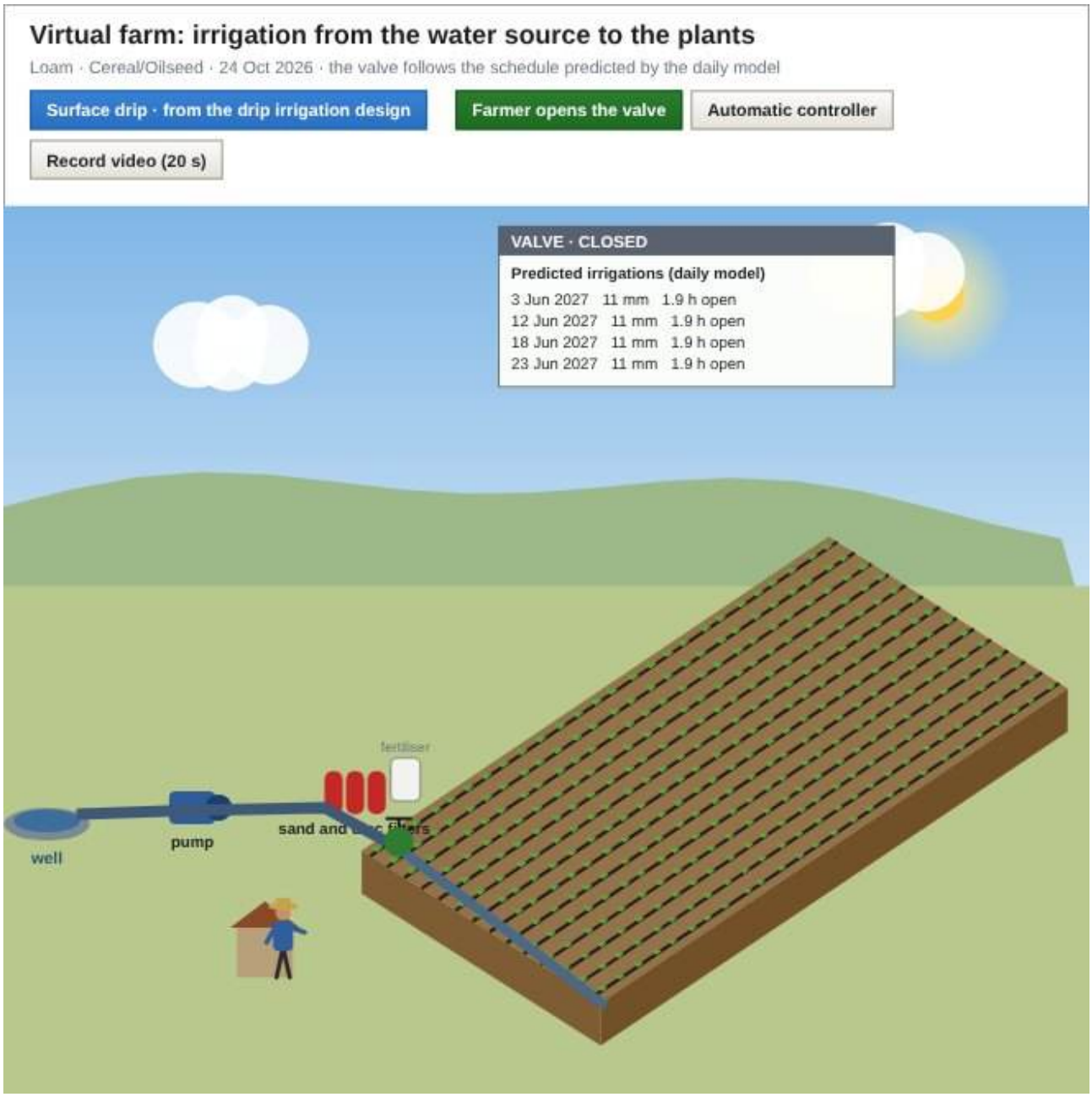


Figure 6.27: Virtual farm, drip design, Loam, part 1 of 3.

Figure 6.28: Virtual farm, drip design, Loam, part 2 of 3.

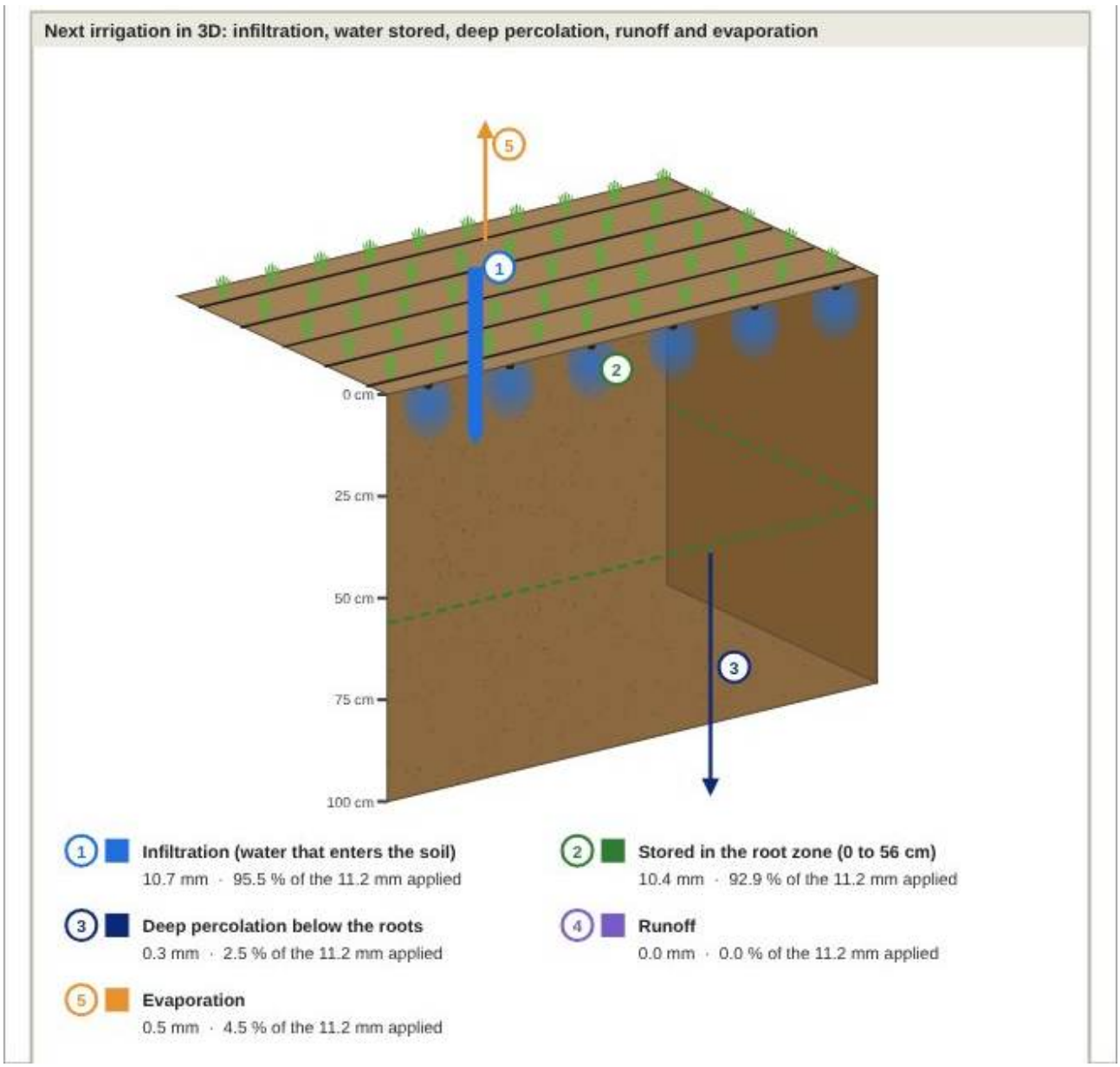


Figure 6.29: Virtual farm, drip design, Loam, part 3 of 3.

6.6 Virtual farm of the sprinkler design (Loam)

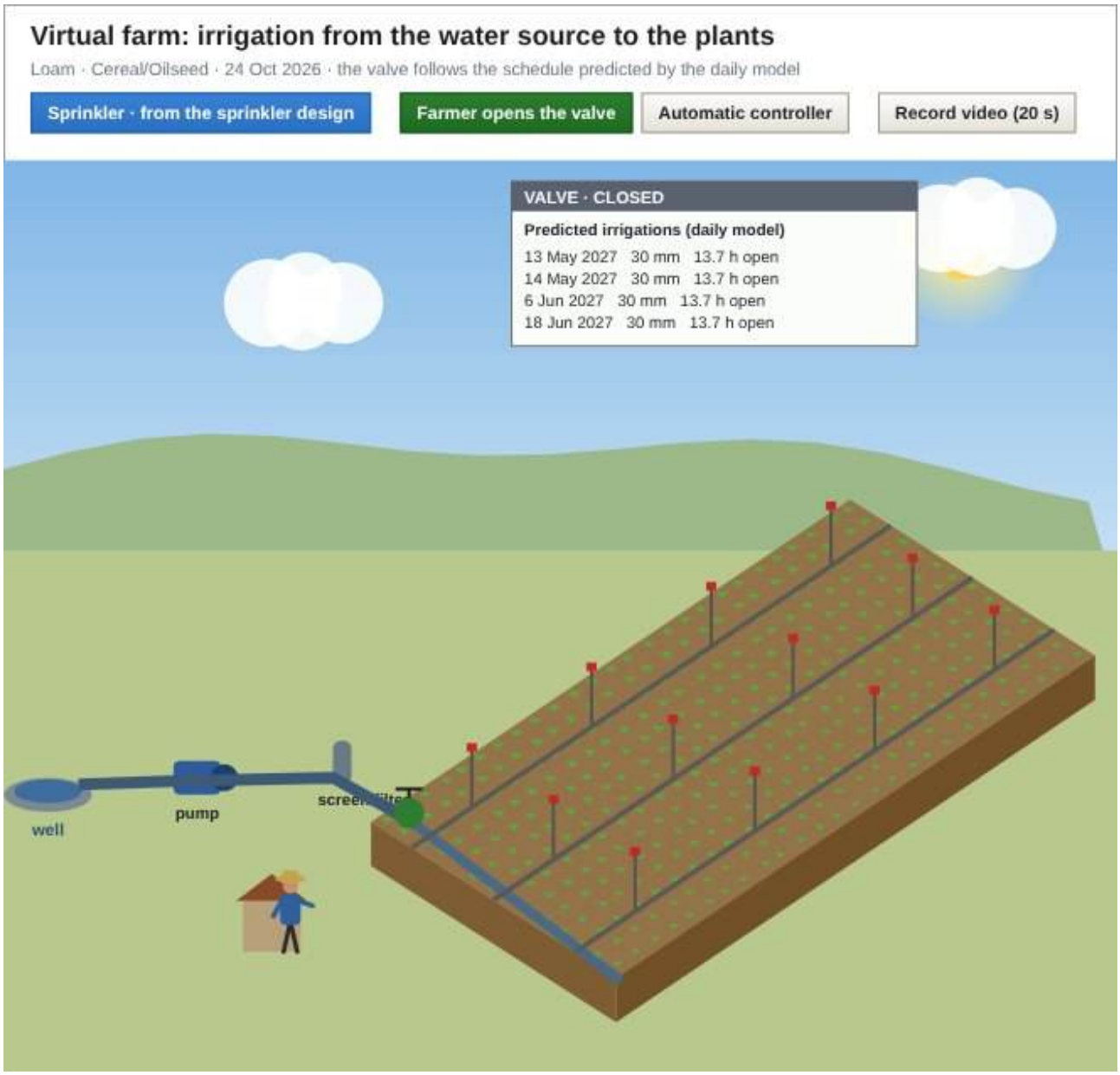


Figure 6.30: Virtual farm, sprinkler design, Loam, part 1 of 3.

Figure 6.31: Virtual farm, sprinkler design, Loam, part 2 of 3.

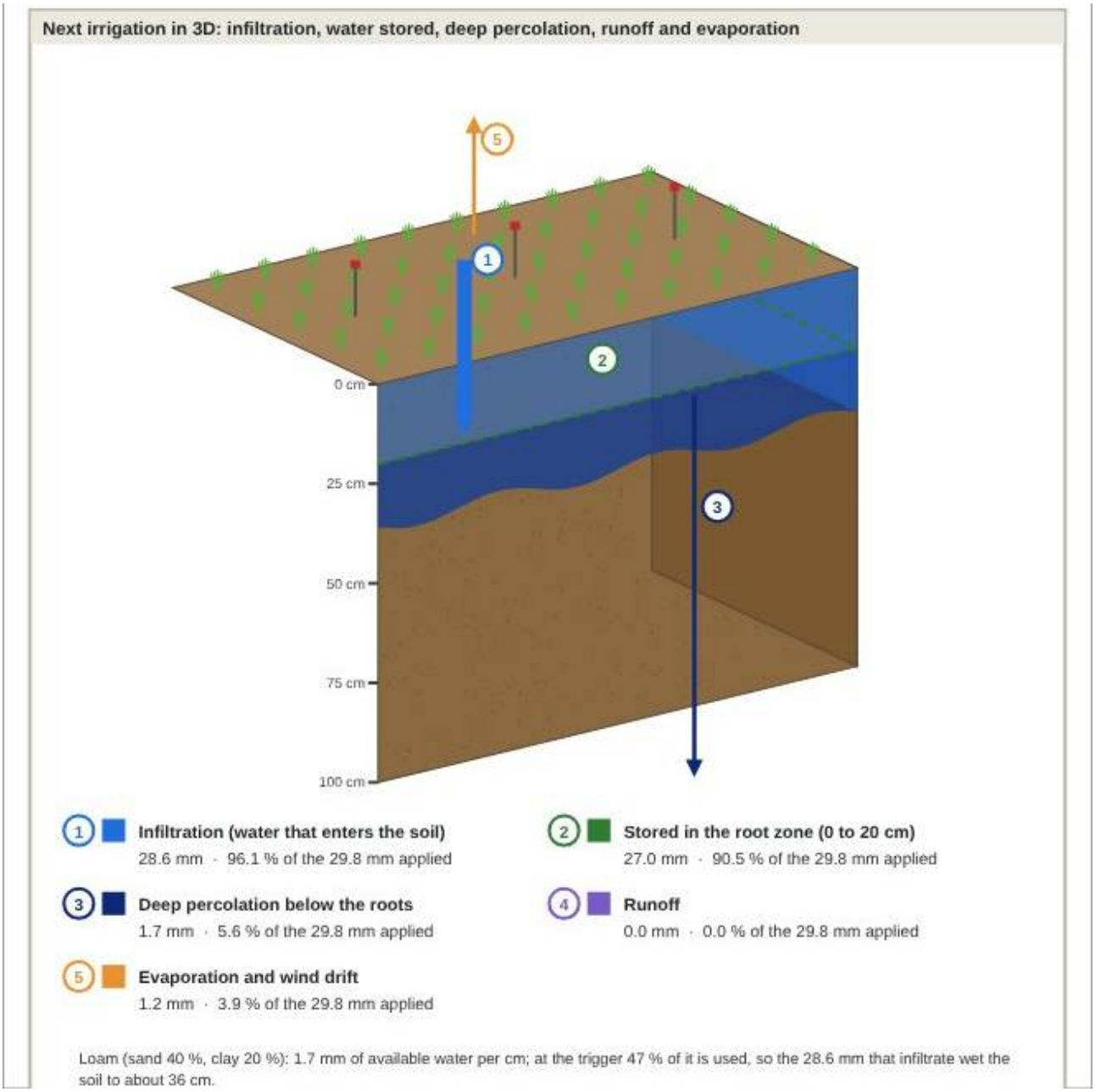


Figure 6.32: Virtual farm, sprinkler design, Loam, part 3 of 3.

6.7 Sensor and trigger (Loam)

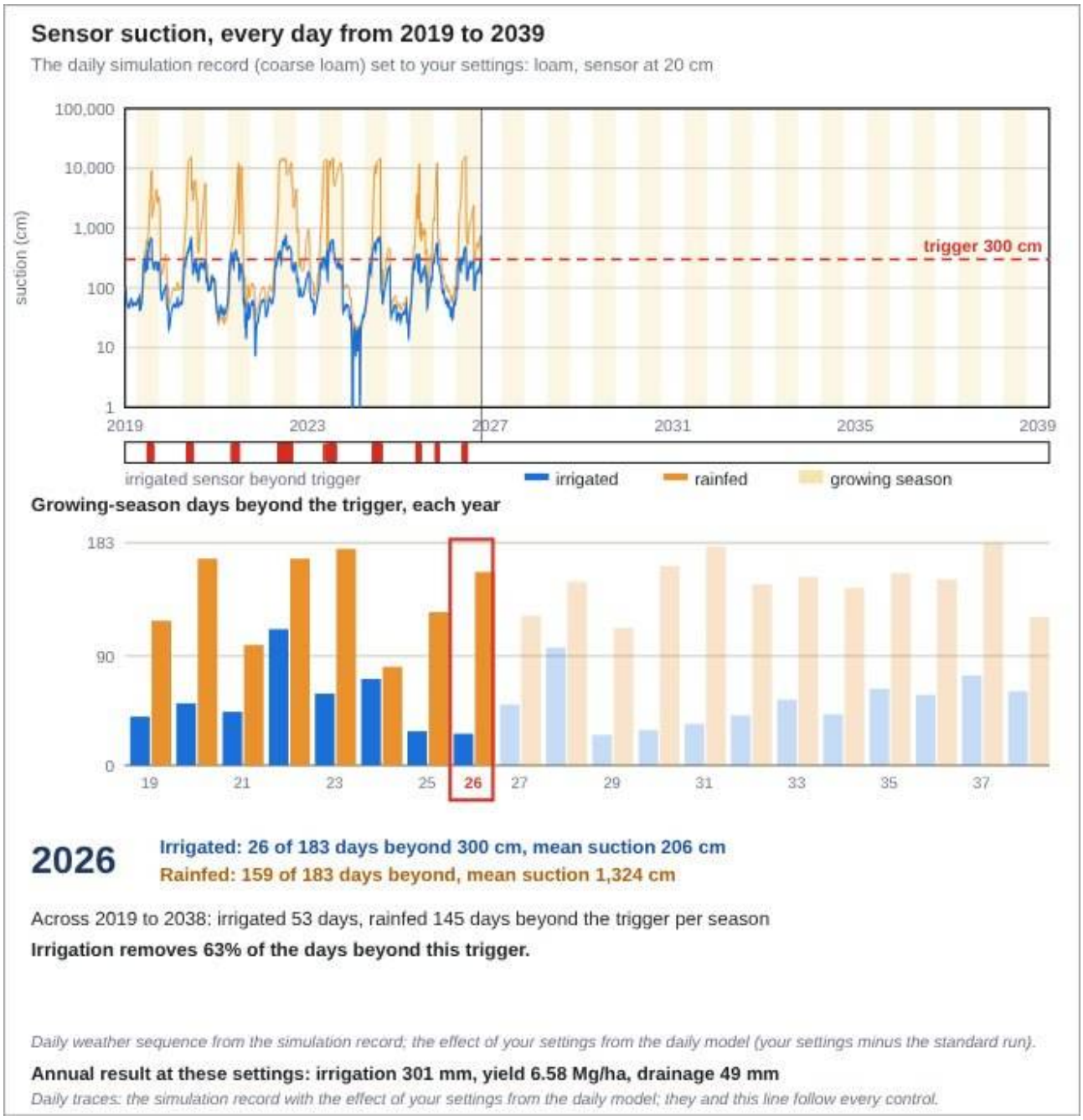


Figure 6.33: Sensor and trigger, Loam, part 1 of 1.

6.8 Trigger law (Loam)

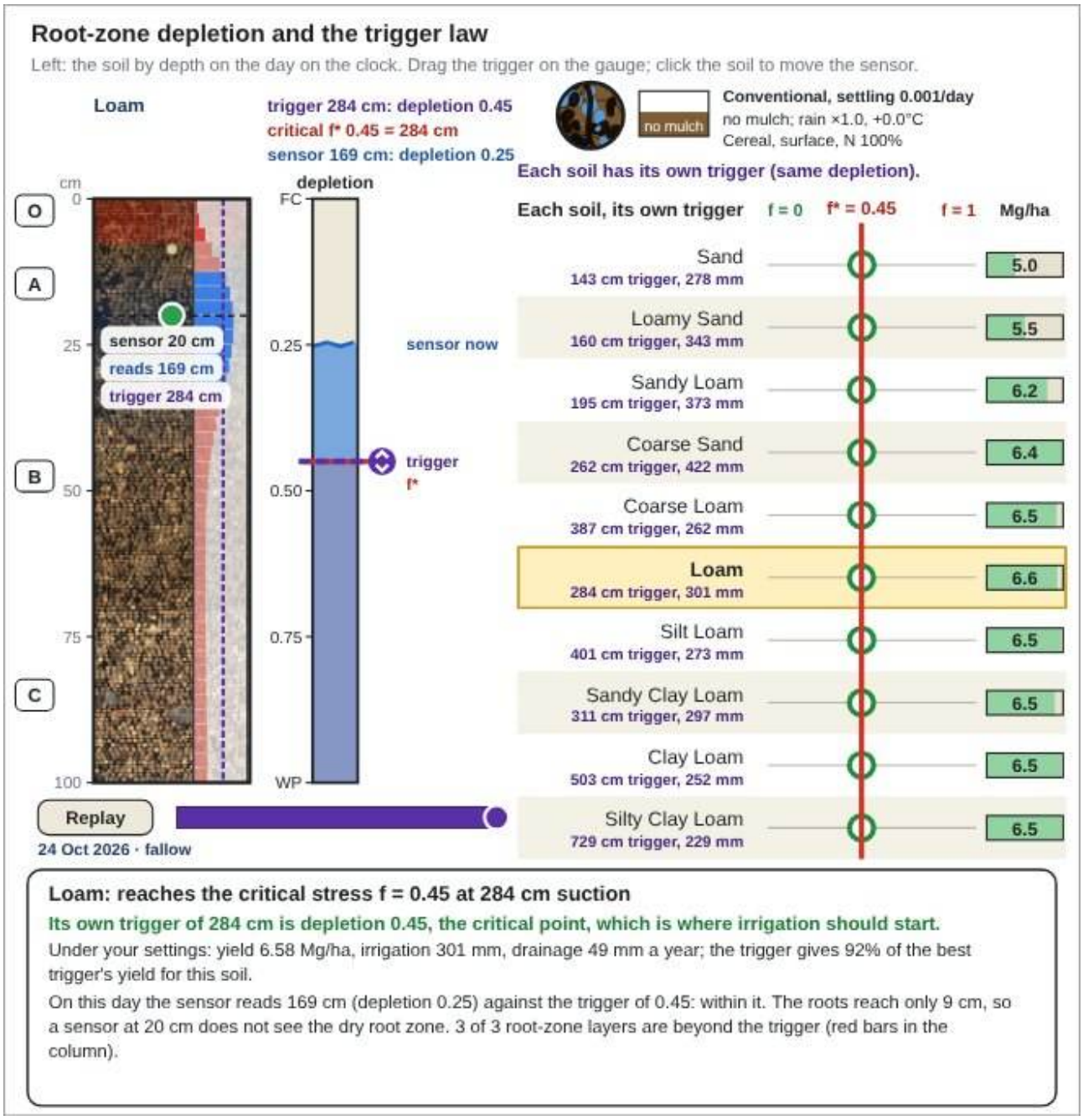


Figure 6.34: Trigger law, Loam, part 1 of 1.

6.9 Mulch against bare soil (Loam)

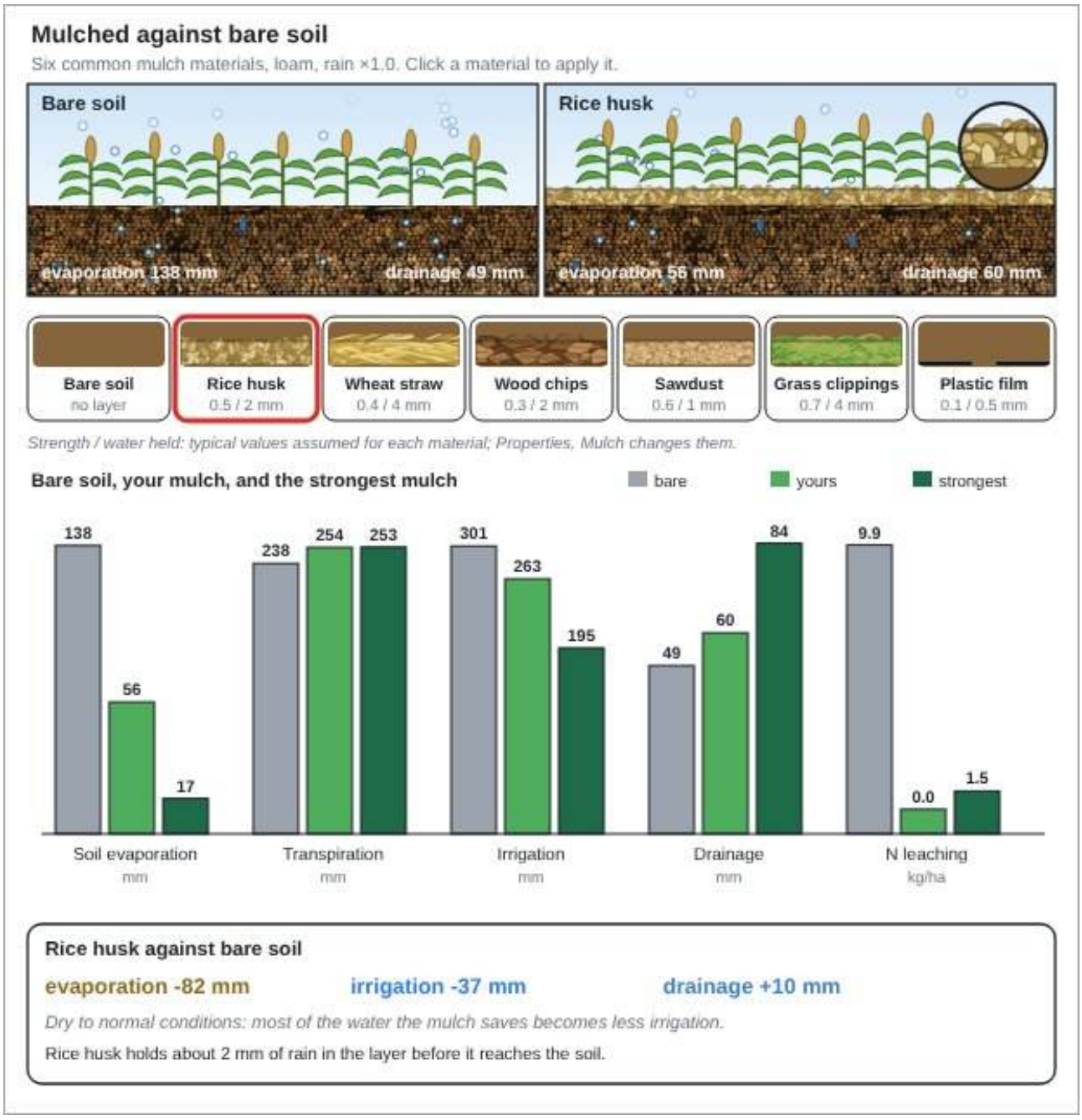


Figure 6.35: Mulch against bare soil, Loam, part 1 of 1.

6.10 Mulch control (Loam)

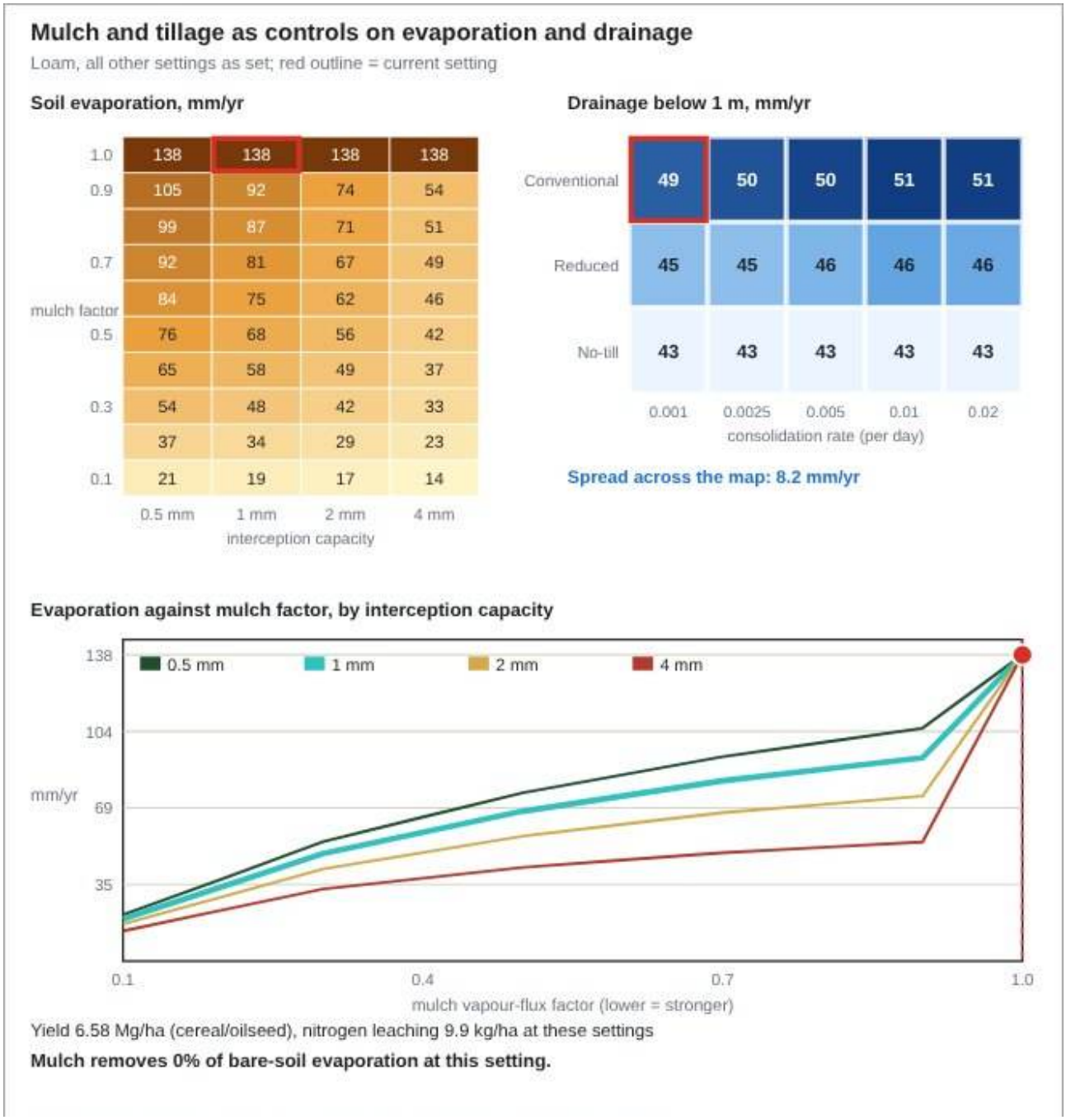


Figure 6.36: Mulch control, Loam, part 1 of 2.

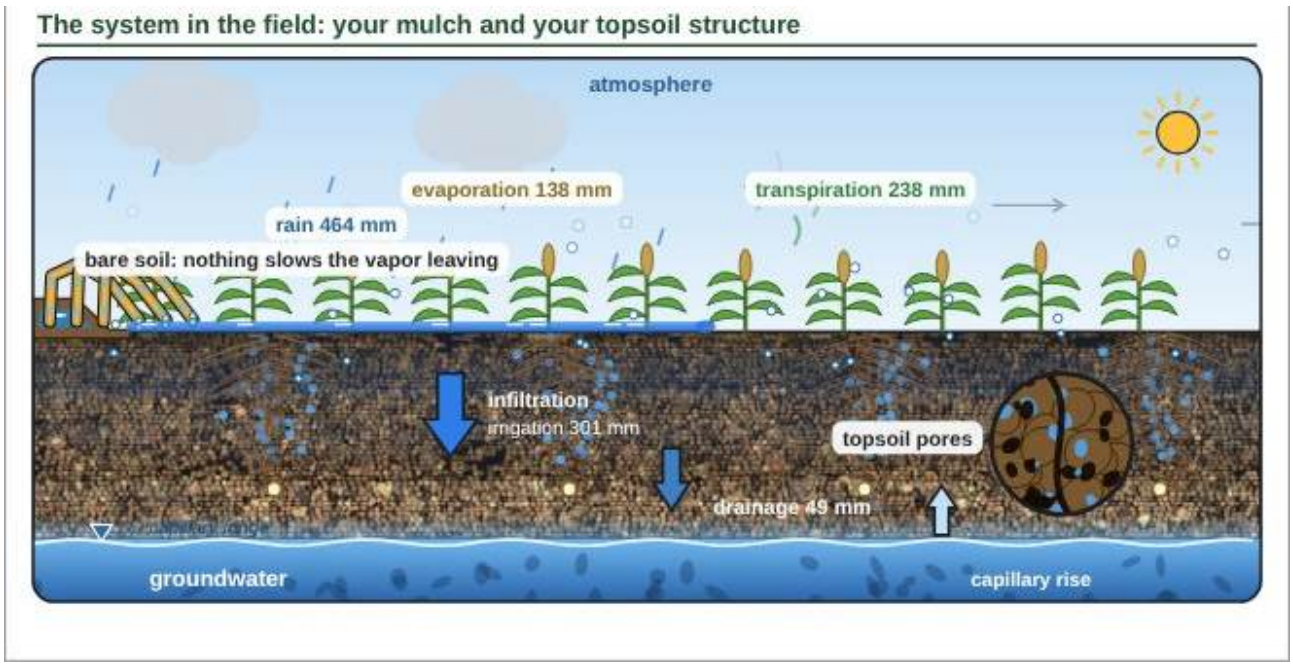


Figure 6.37: Mulch control, Loam, part 2 of 2.

6.11 Evaporation physics (Loam)

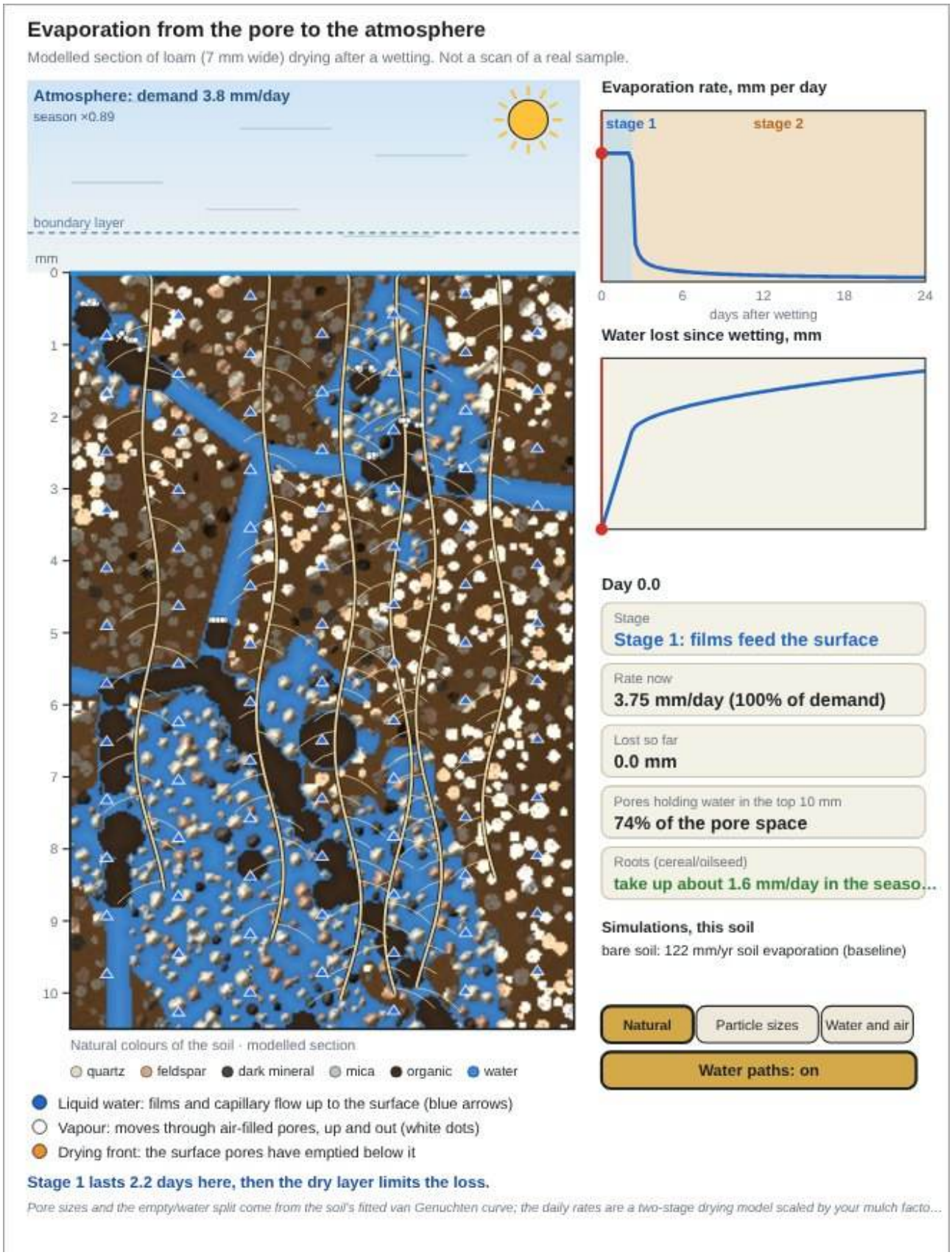


Figure 6.38: Evaporation physics, Loam, part 1 of 3.

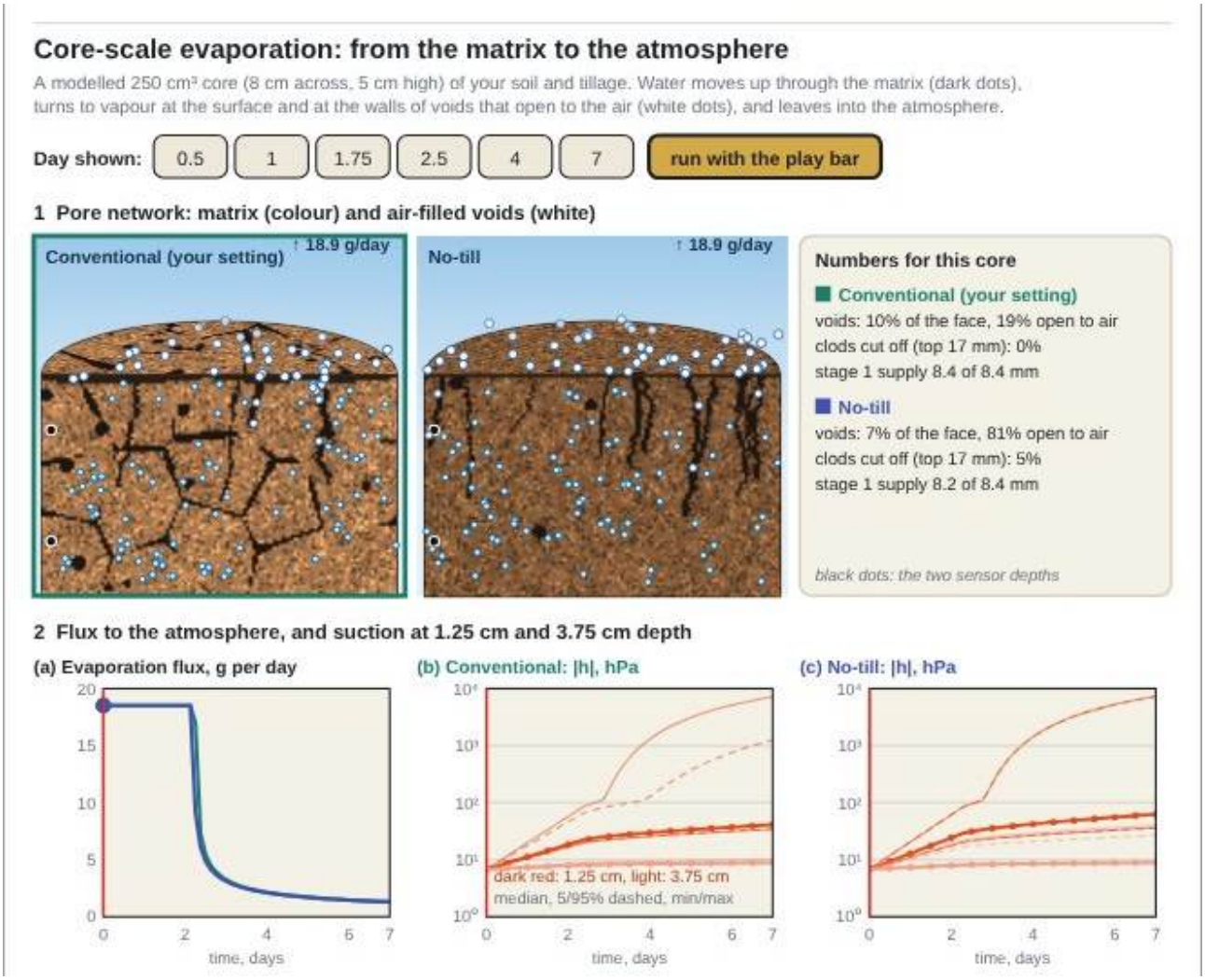


Figure 6.39: Evaporation physics, Loam, part 2 of 3.

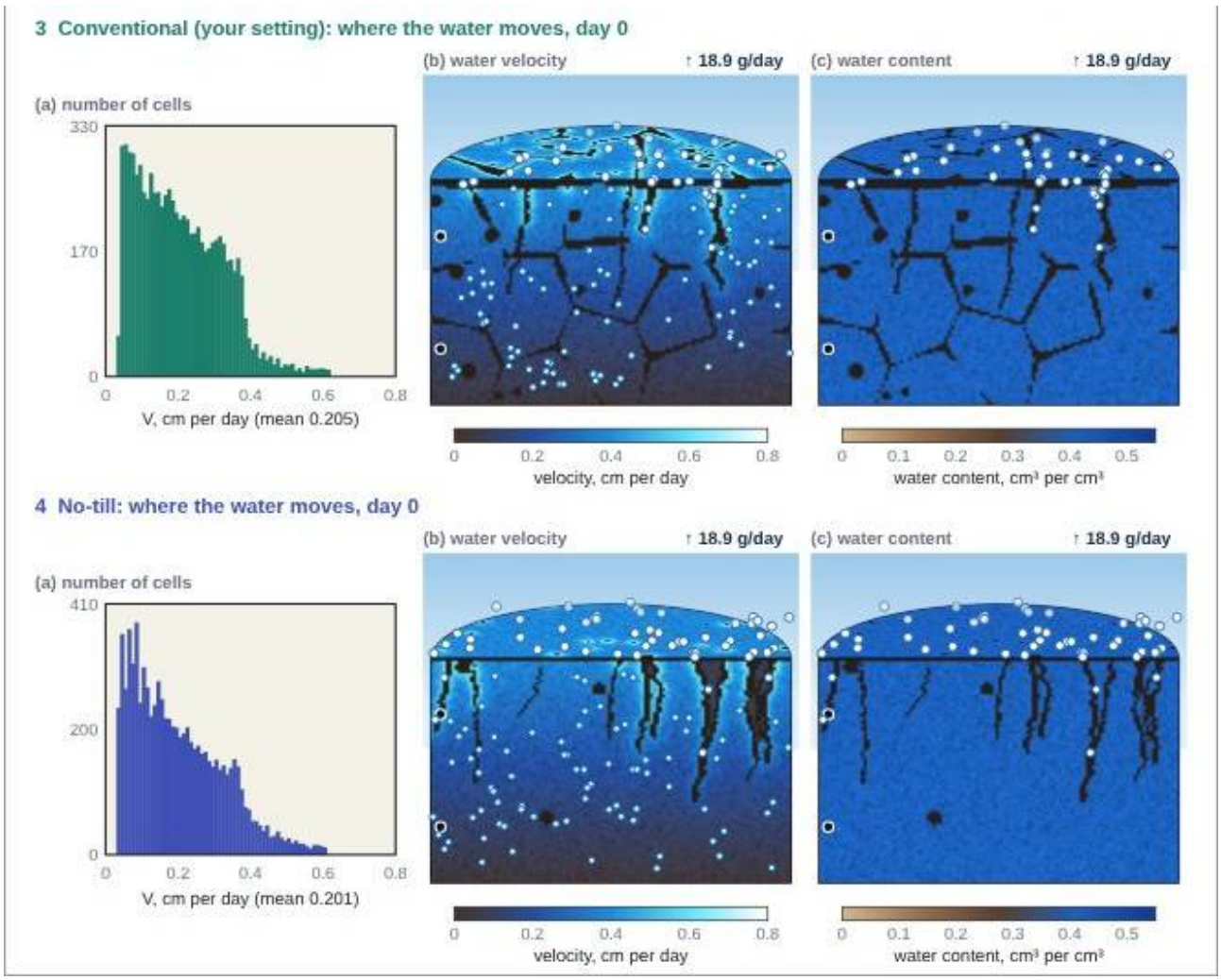


Figure 6.40: Evaporation physics, Loam, part 3 of 3.

6.12 Fate of saved water (Loam)

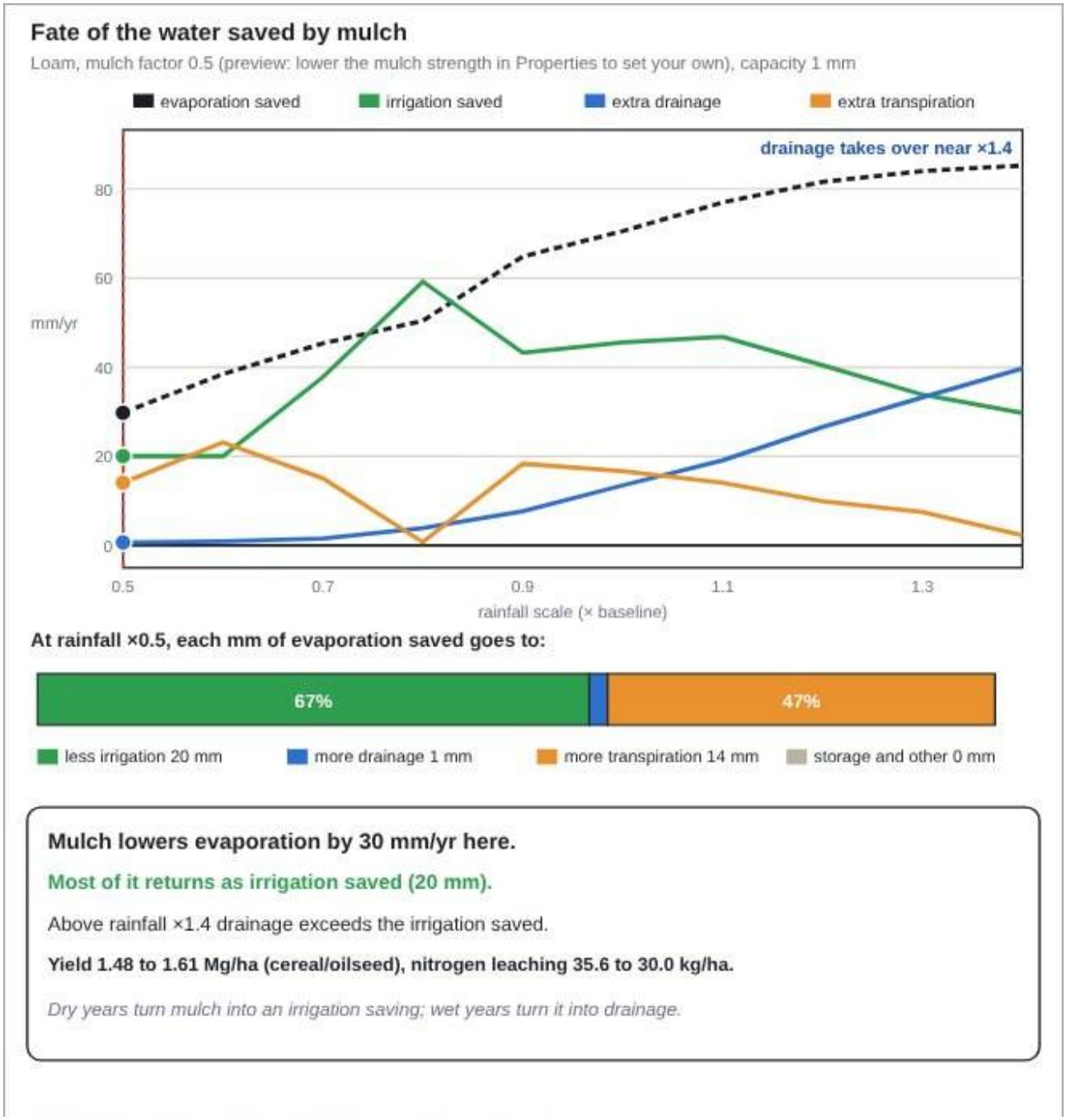


Figure 6.41: Fate of saved water, Loam, part 1 of 2.

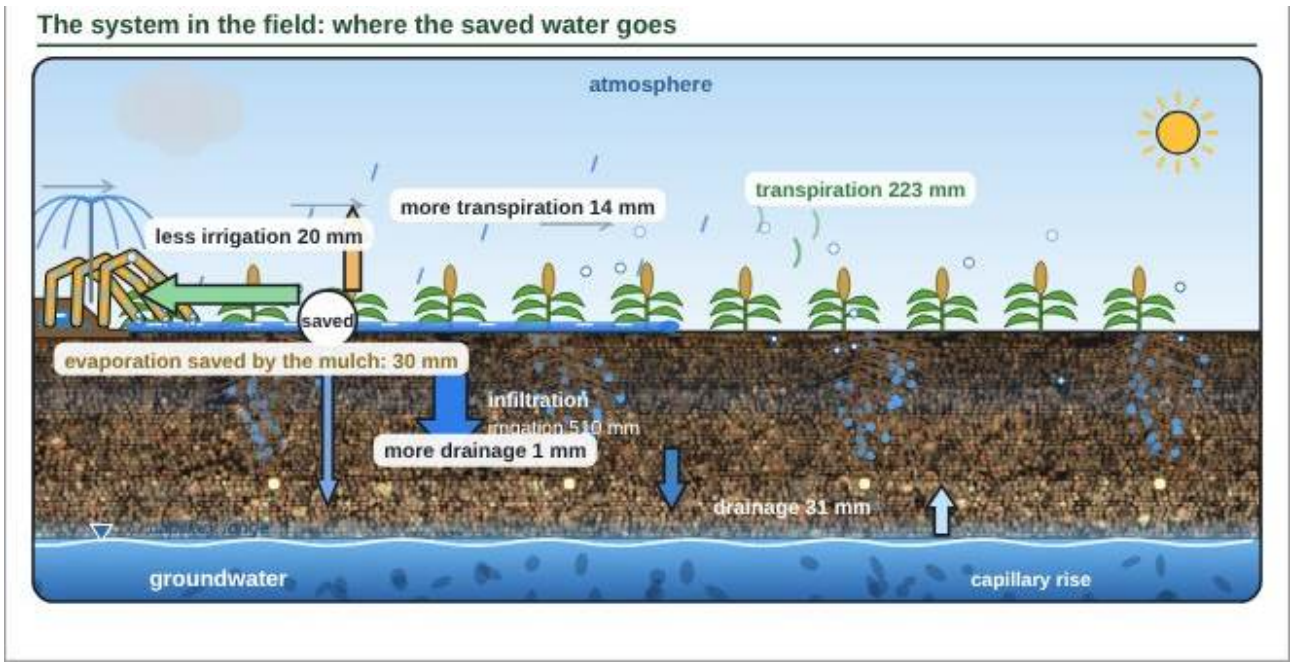


Figure 6.42: Fate of saved water, Loam, part 2 of 2.

6.13 Regime map (Loam)

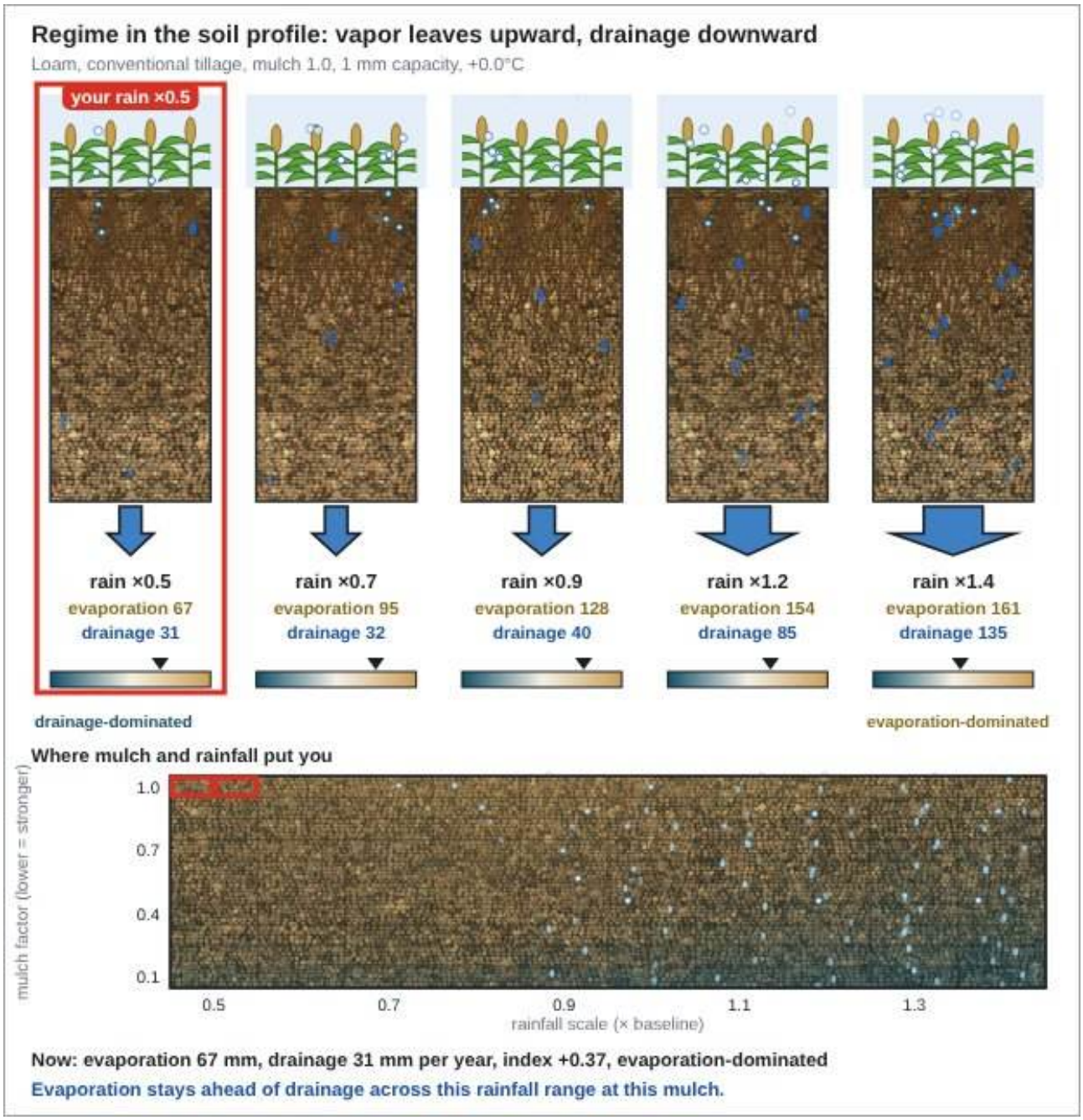


Figure 6.43: Regime map, Loam, part 1 of 1.

6.14 Tillage and structure (Loam)

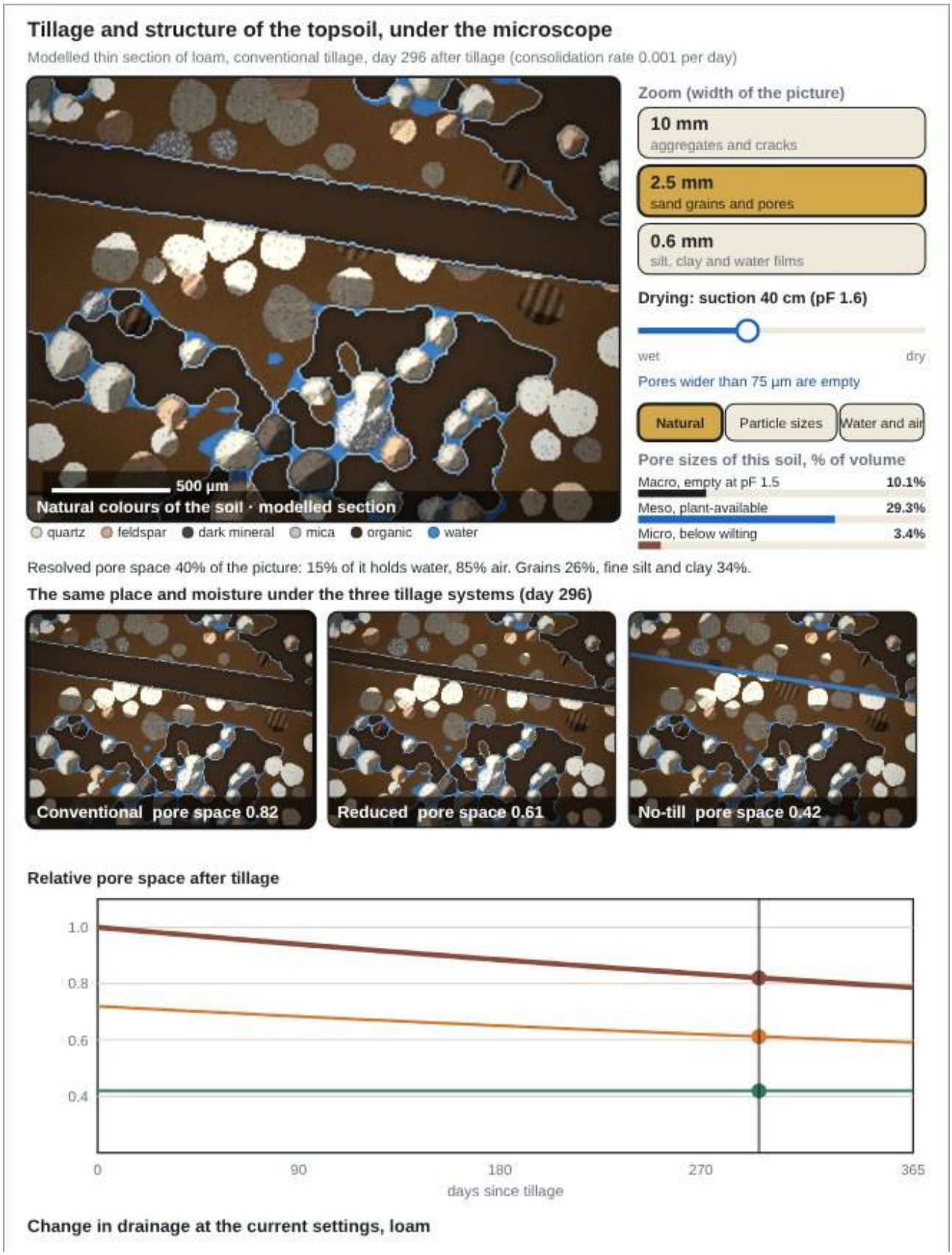


Figure 6.44: Tillage and structure, Loam, part 1 of 2.

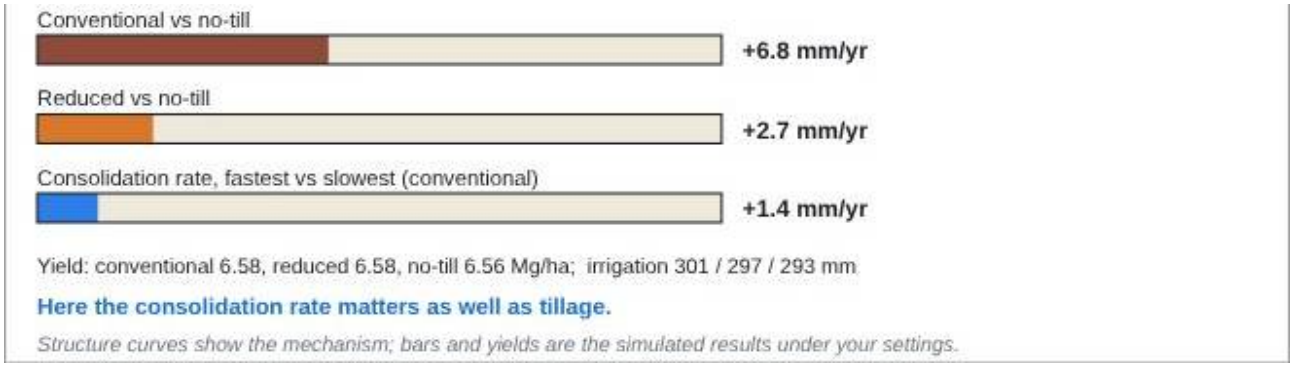


Figure 6.45: Tillage and structure, Loam, part 2 of 2.

6.15 Soil hydraulic drivers (Loam)

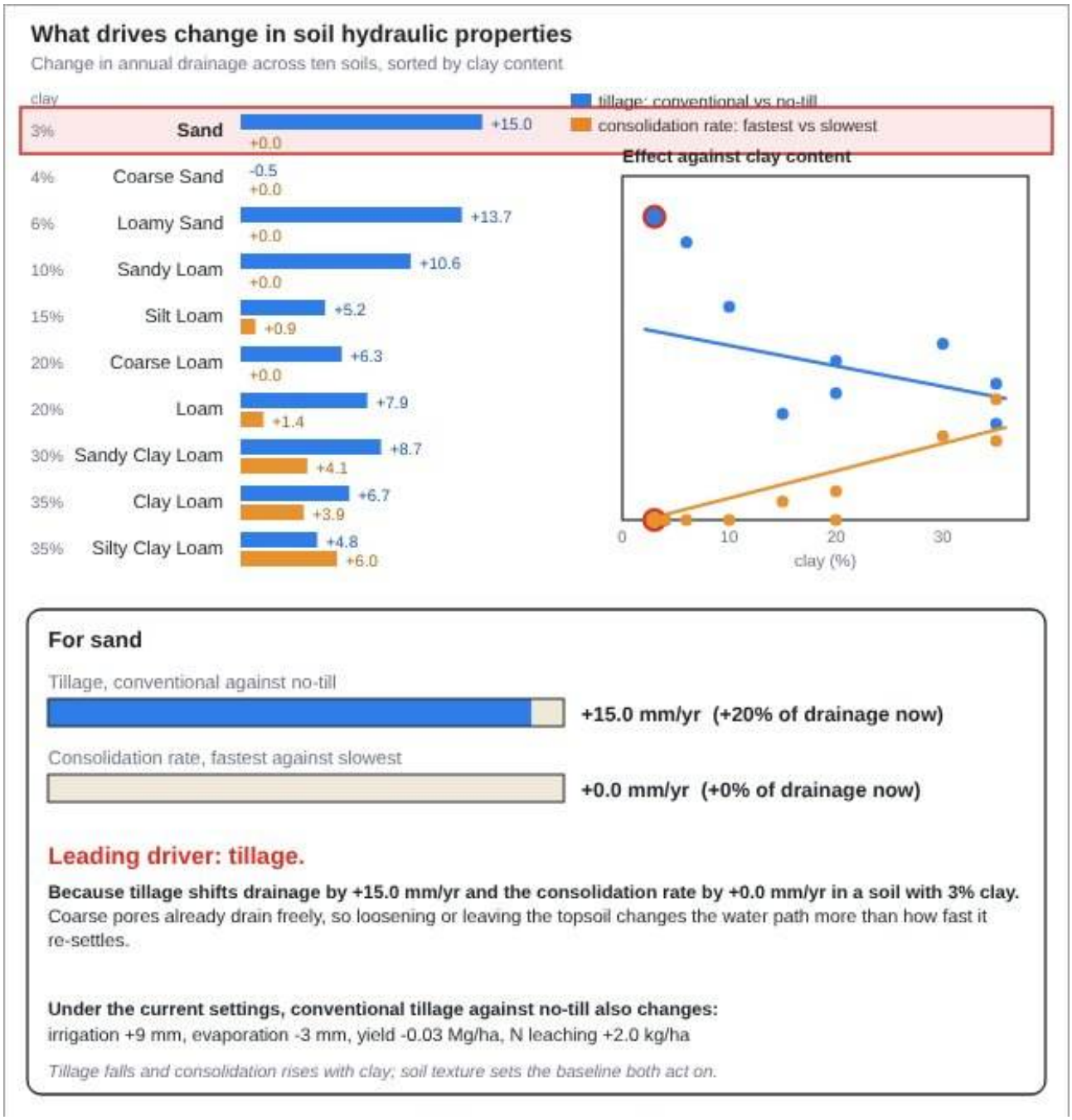


Figure 6.46: Soil hydraulic drivers, Loam, part 1 of 3.

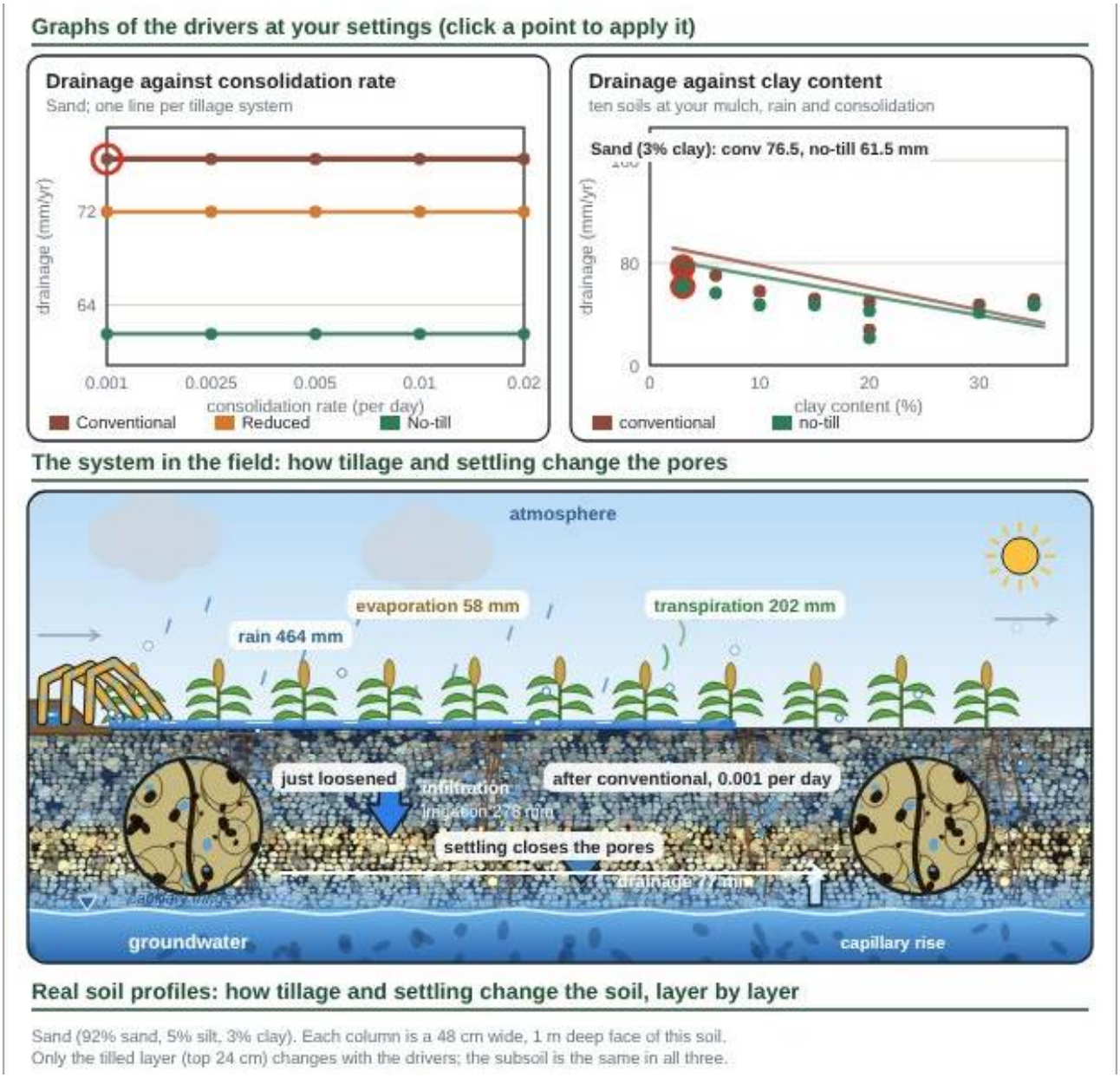


Figure 6.47: Soil hydraulic drivers, Loam, part 2 of 3.

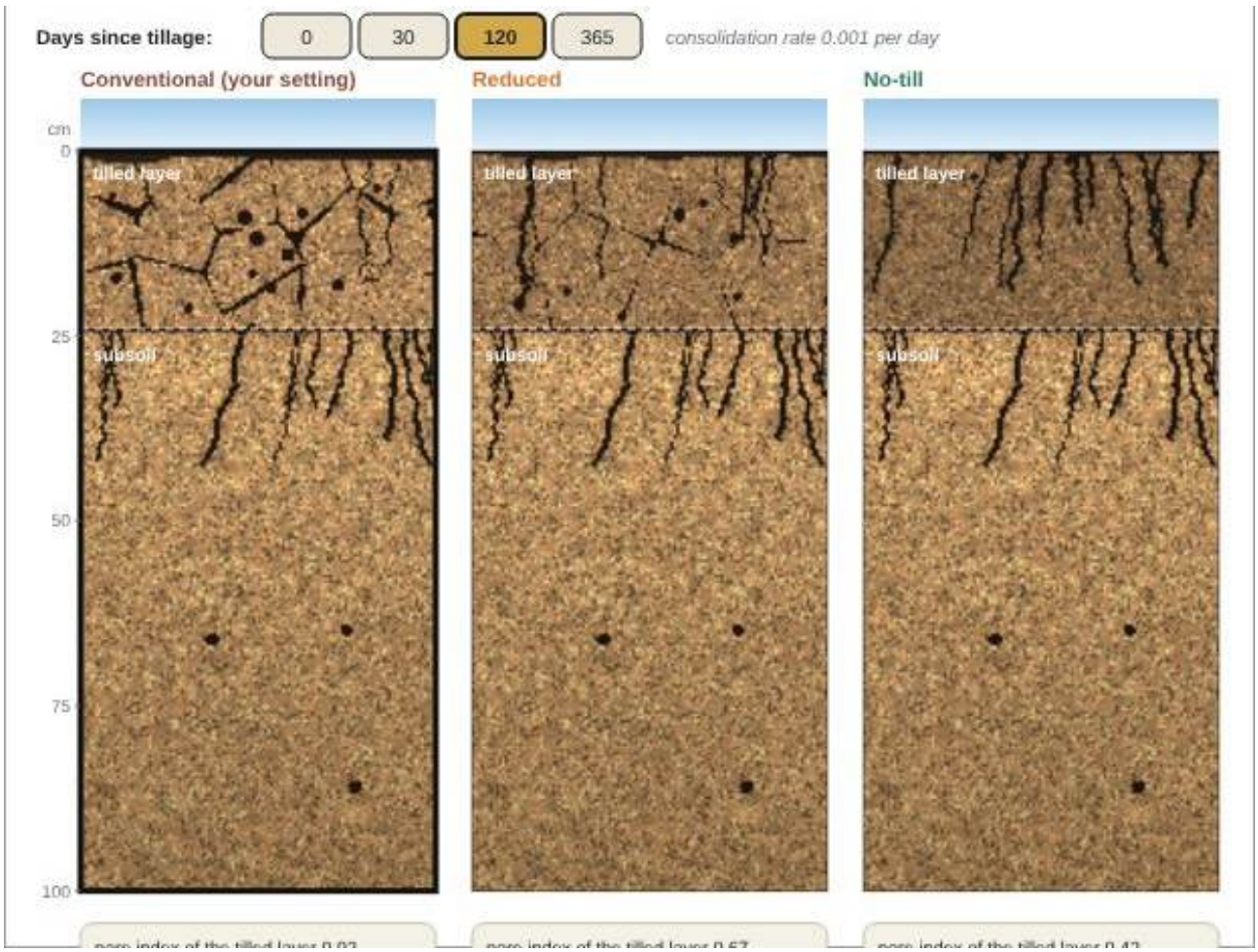


Figure 6.48: Soil hydraulic drivers, Loam, part 3 of 3.

6.16 Root structure (Loam)

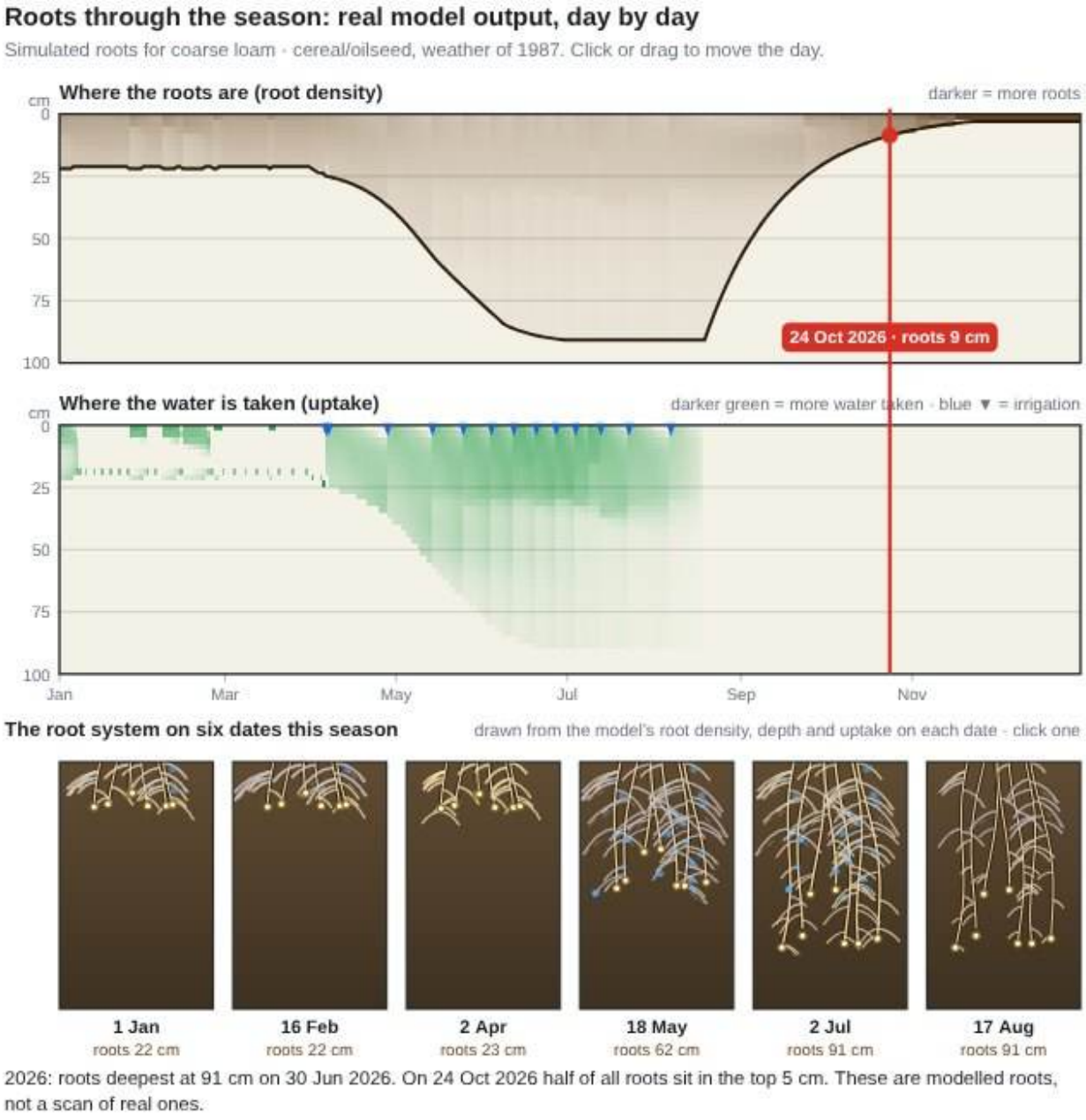


Figure 6.49: Root structure, Loam, part 1 of 4.

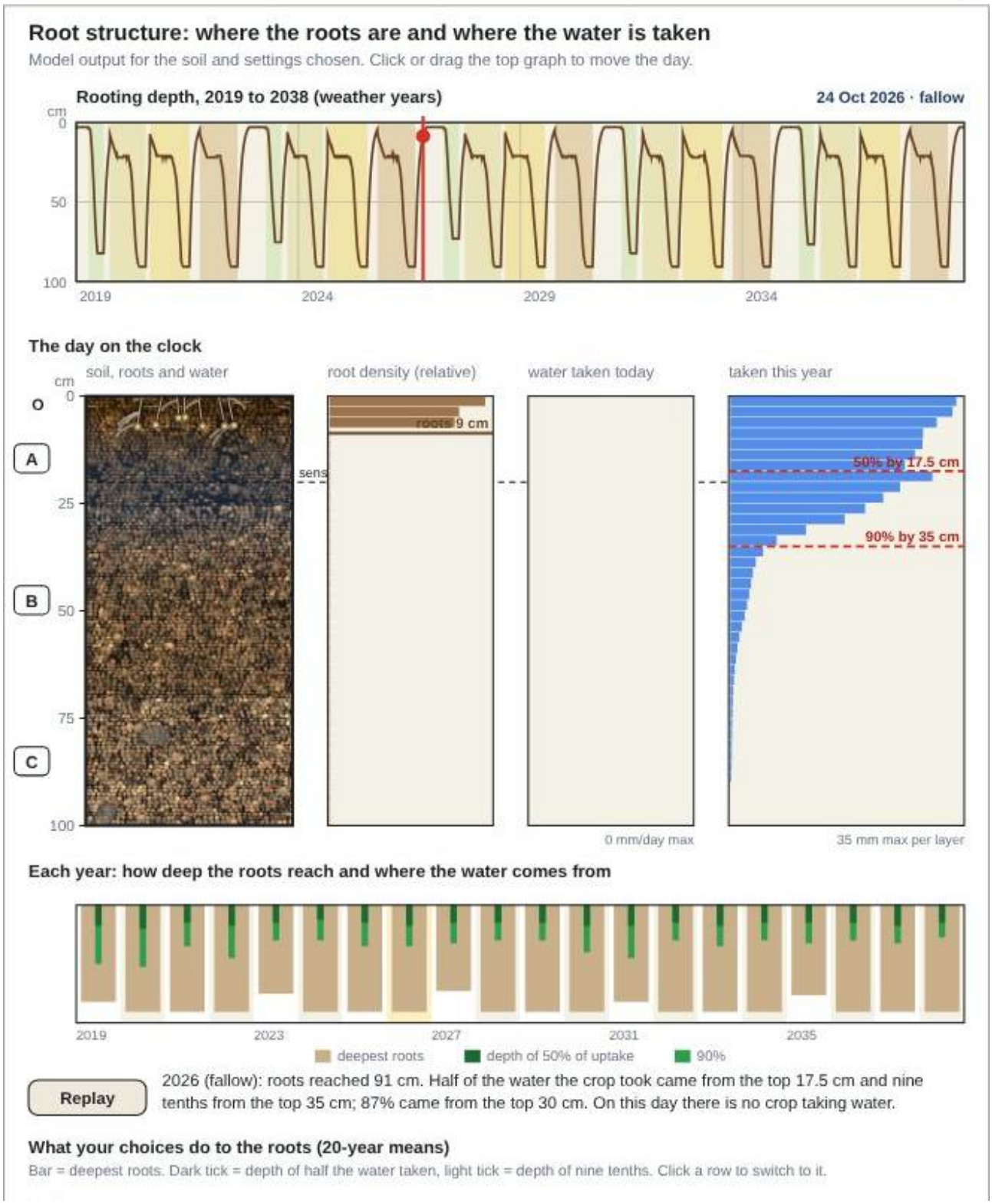


Figure 6.50: Root structure, Loam, part 2 of 4.

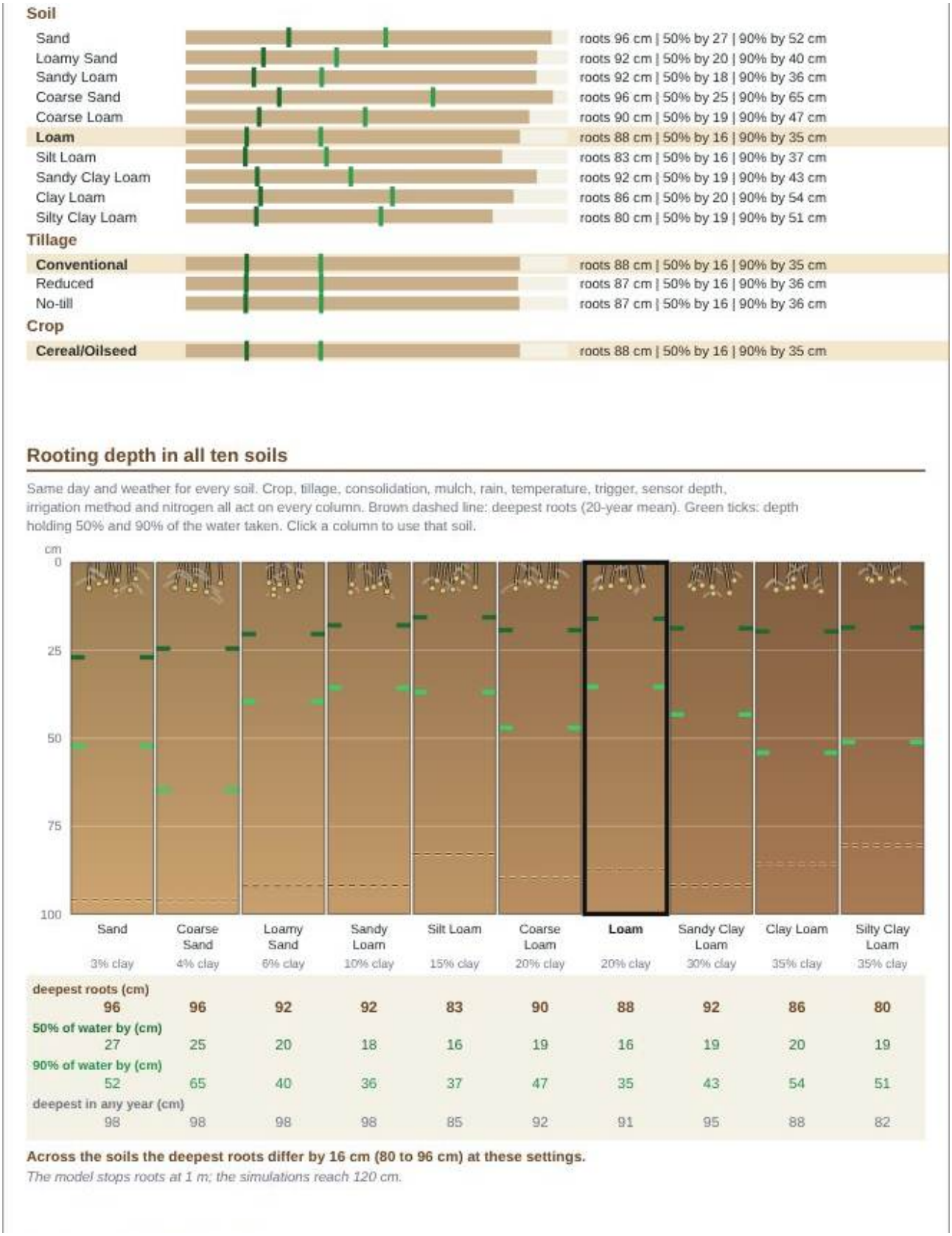


Figure 6.51: Root structure, Loam, part 3 of 4.

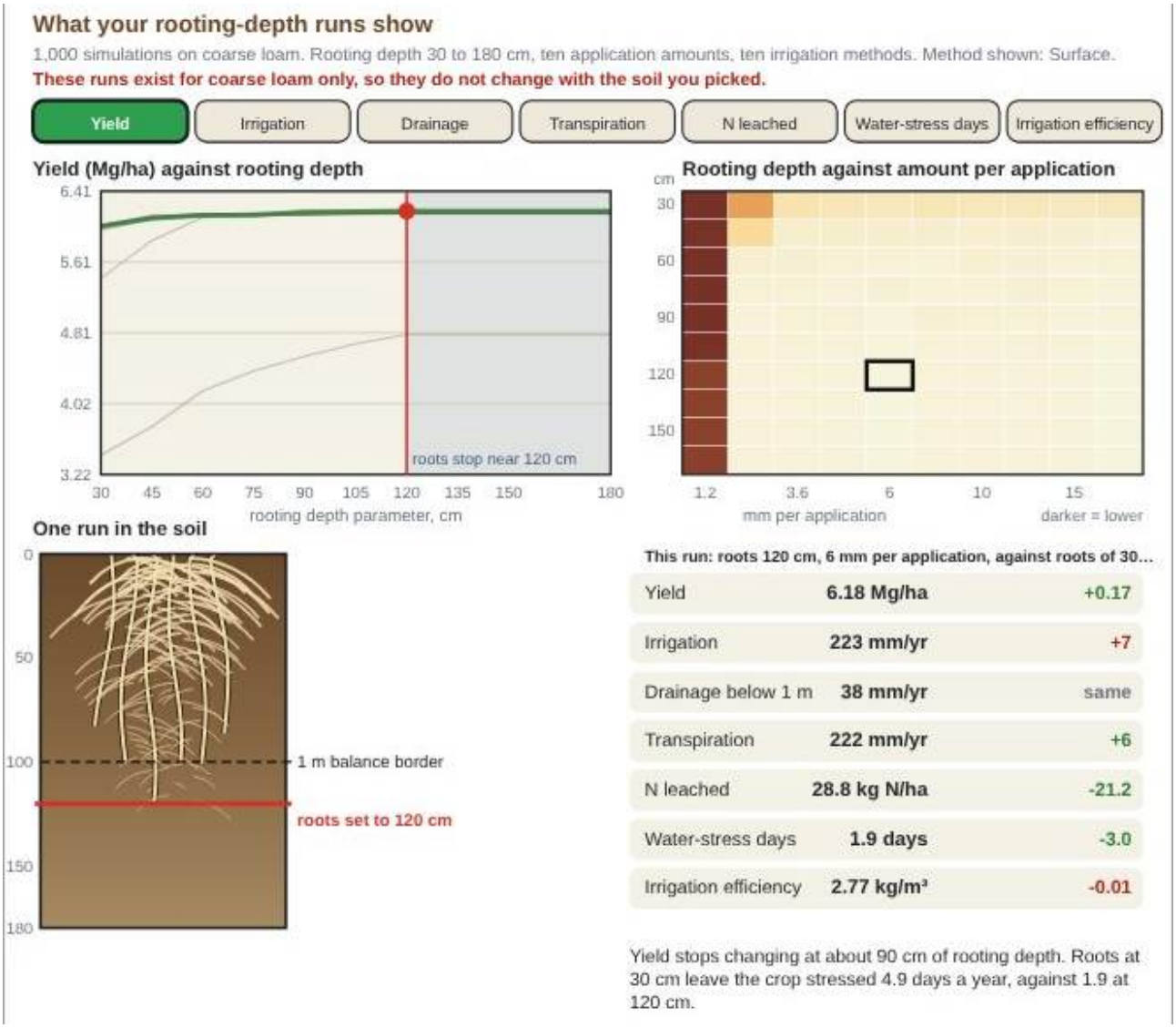


Figure 6.52: Root structure, Loam, part 4 of 4.

6.17 Field balance (Loam)

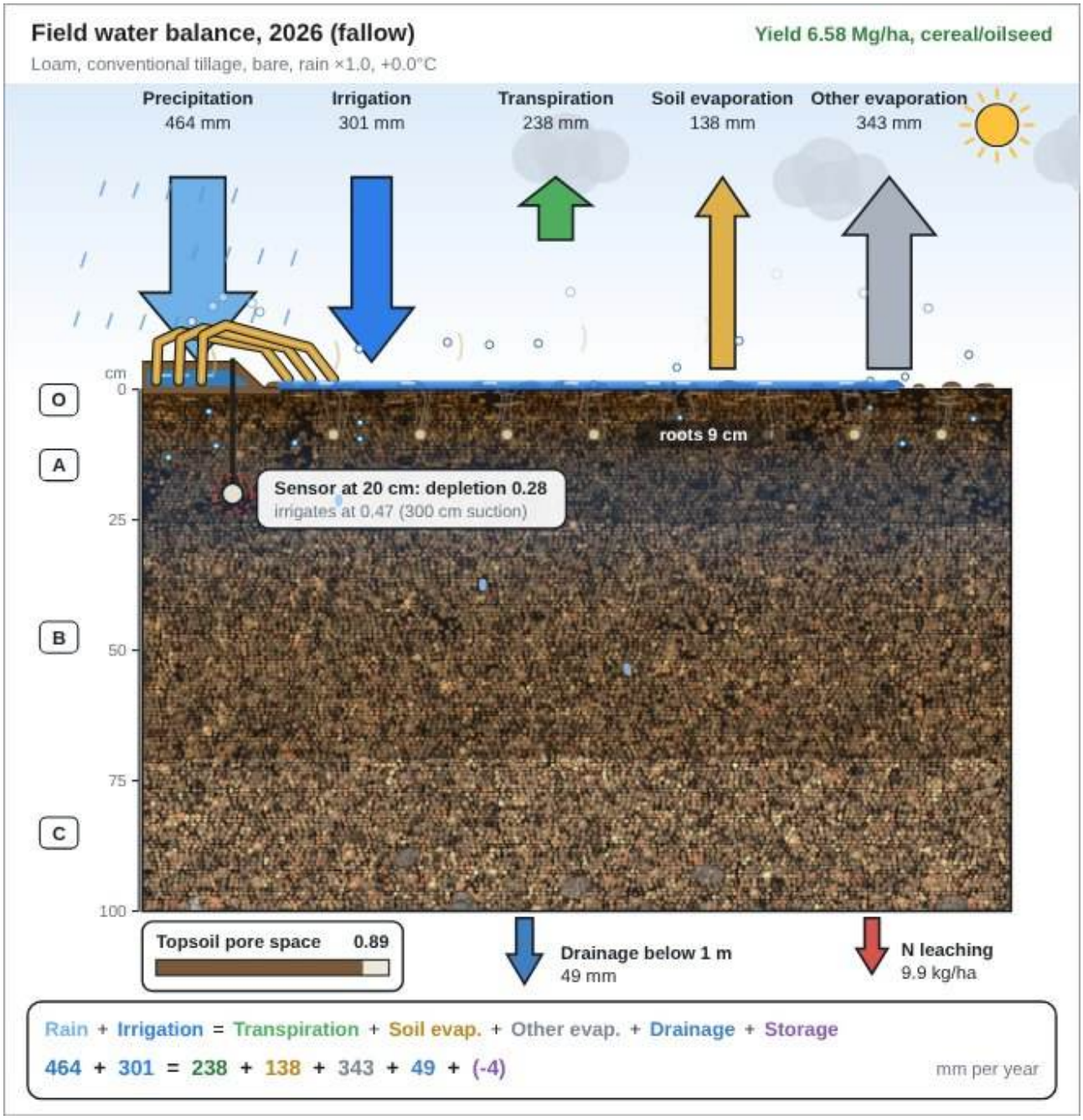


Figure 6.53: Field balance, Loam, part 1 of 1.

6.18 Soil, plant and air (Loam)

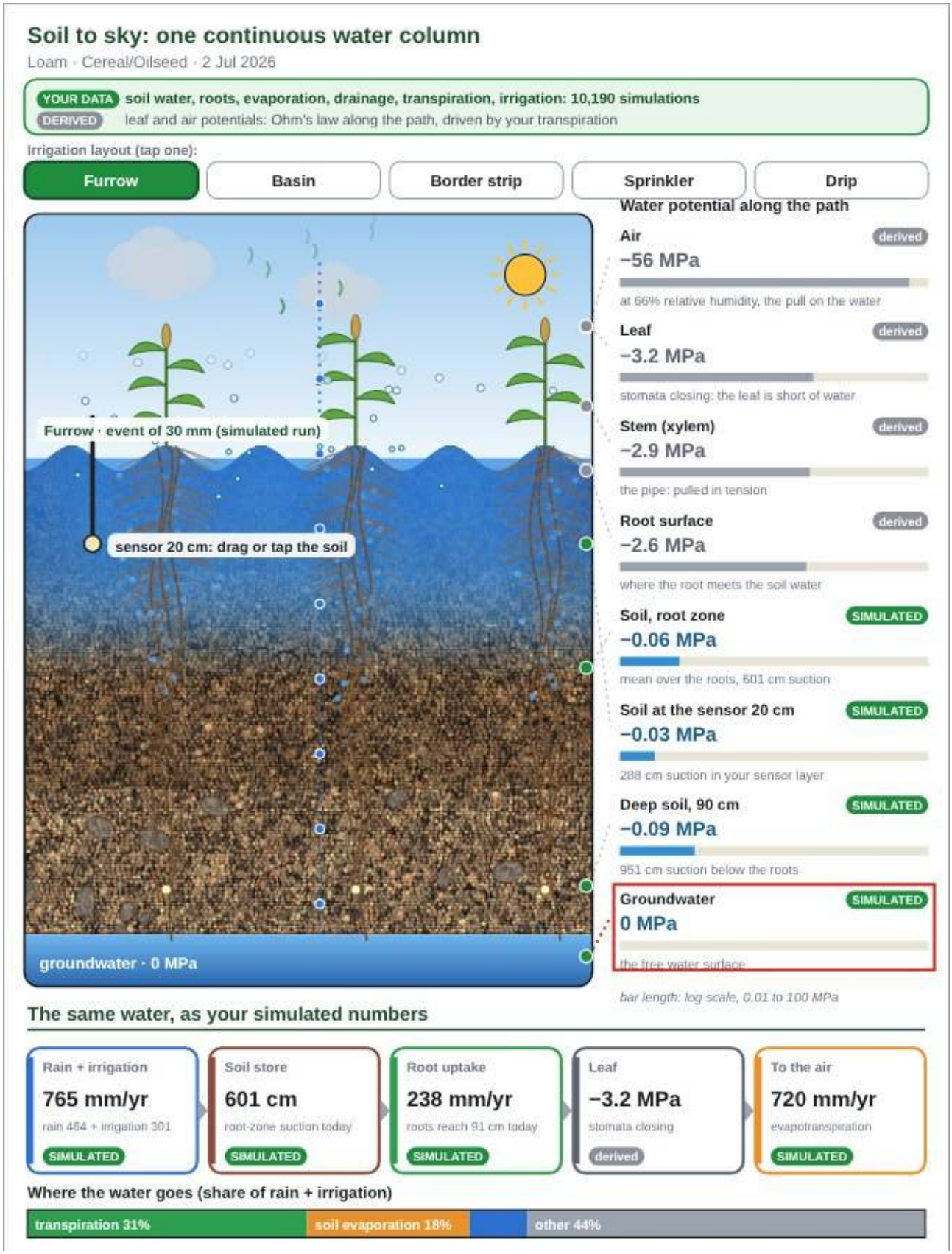


Figure 6.54: Soil, plant and air, Loam, part 1 of 3.

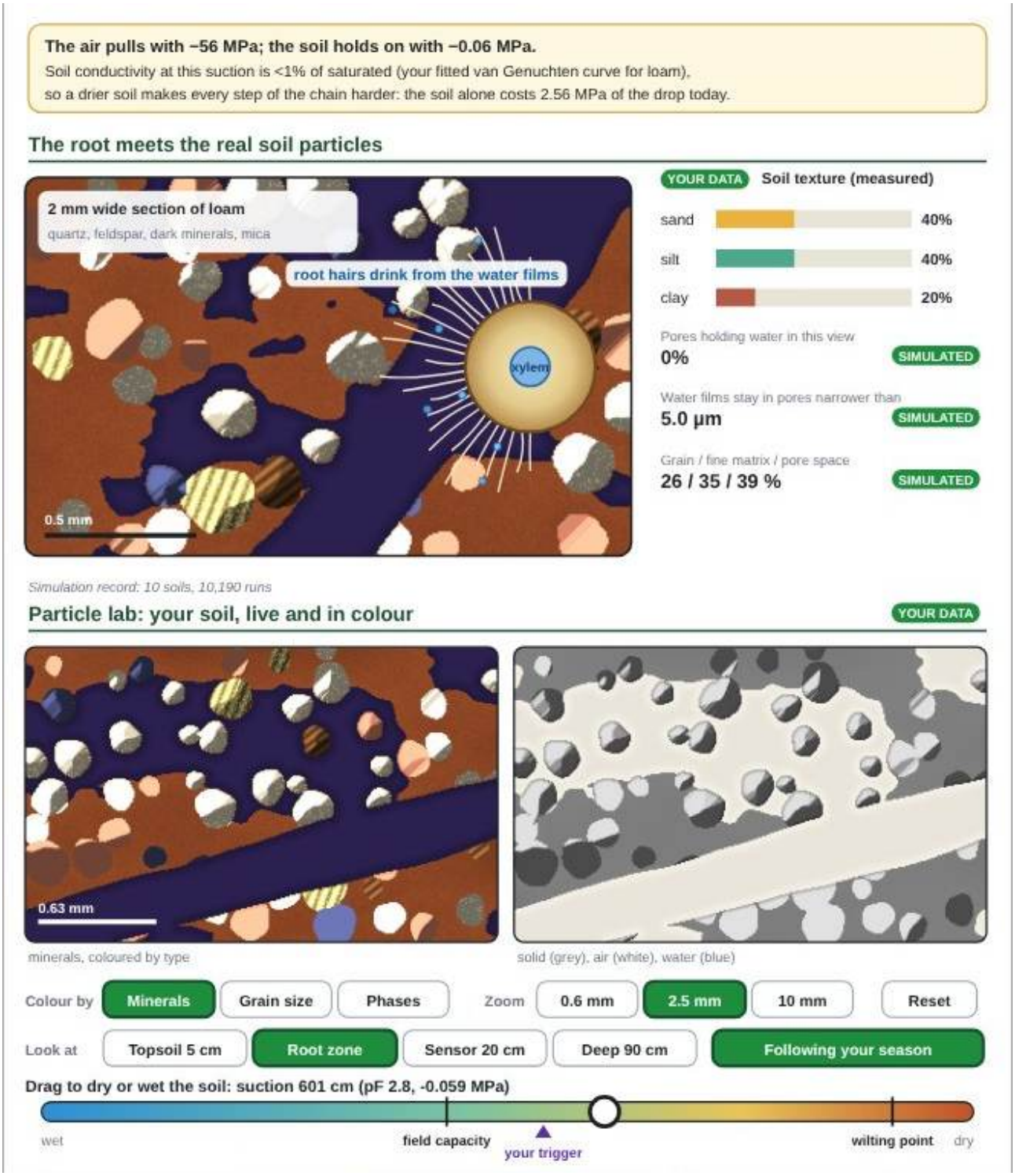


Figure 6.55: Soil, plant and air, Loam, part 2 of 3.

Figure 6.56: Soil, plant and air, Loam, part 3 of 3.

6.19 Profile evolution (Loam)

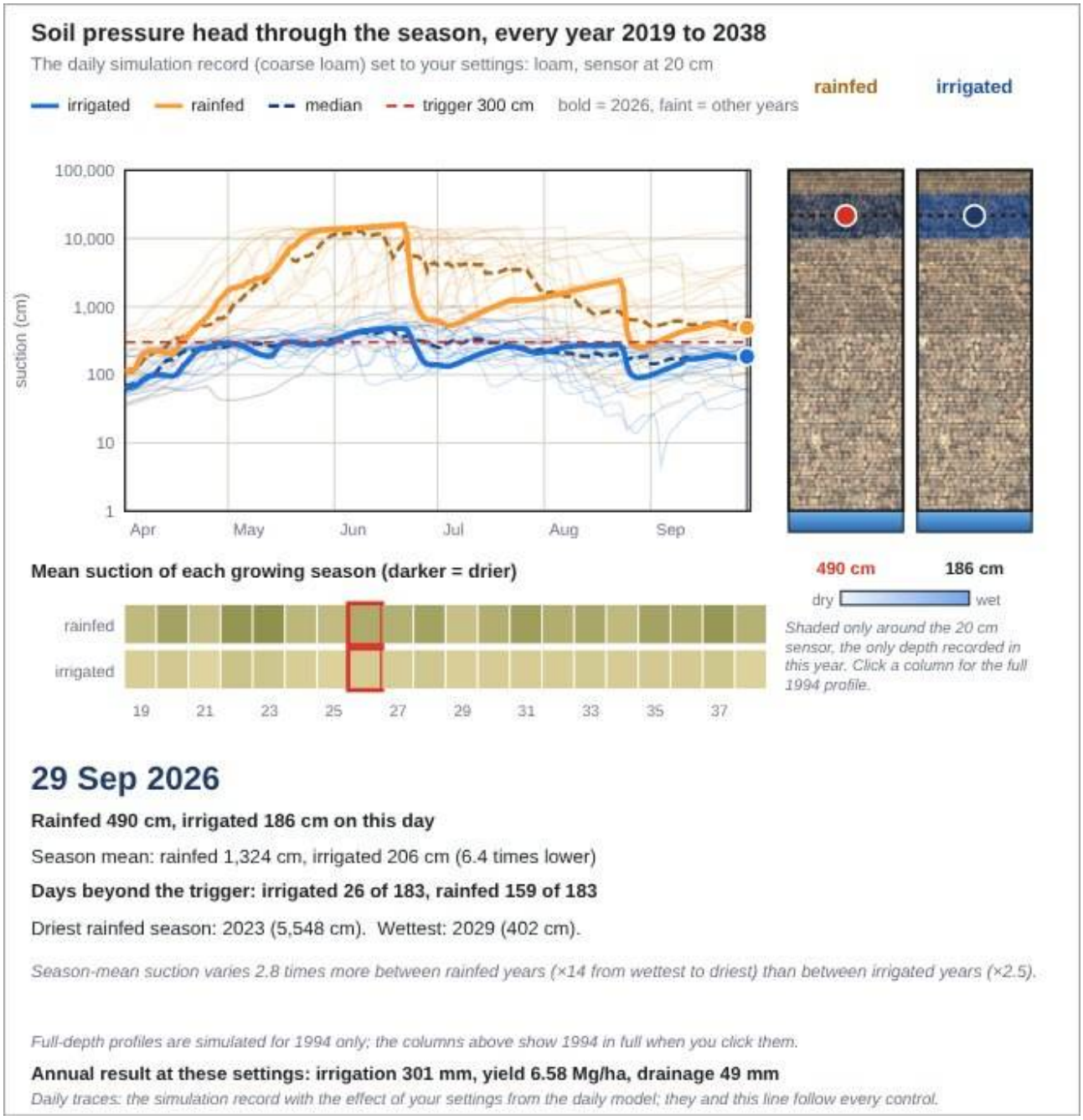


Figure 6.57: Profile evolution, Loam, part 1 of 1.

6.20 Reliability over 20 seasons (Loam)

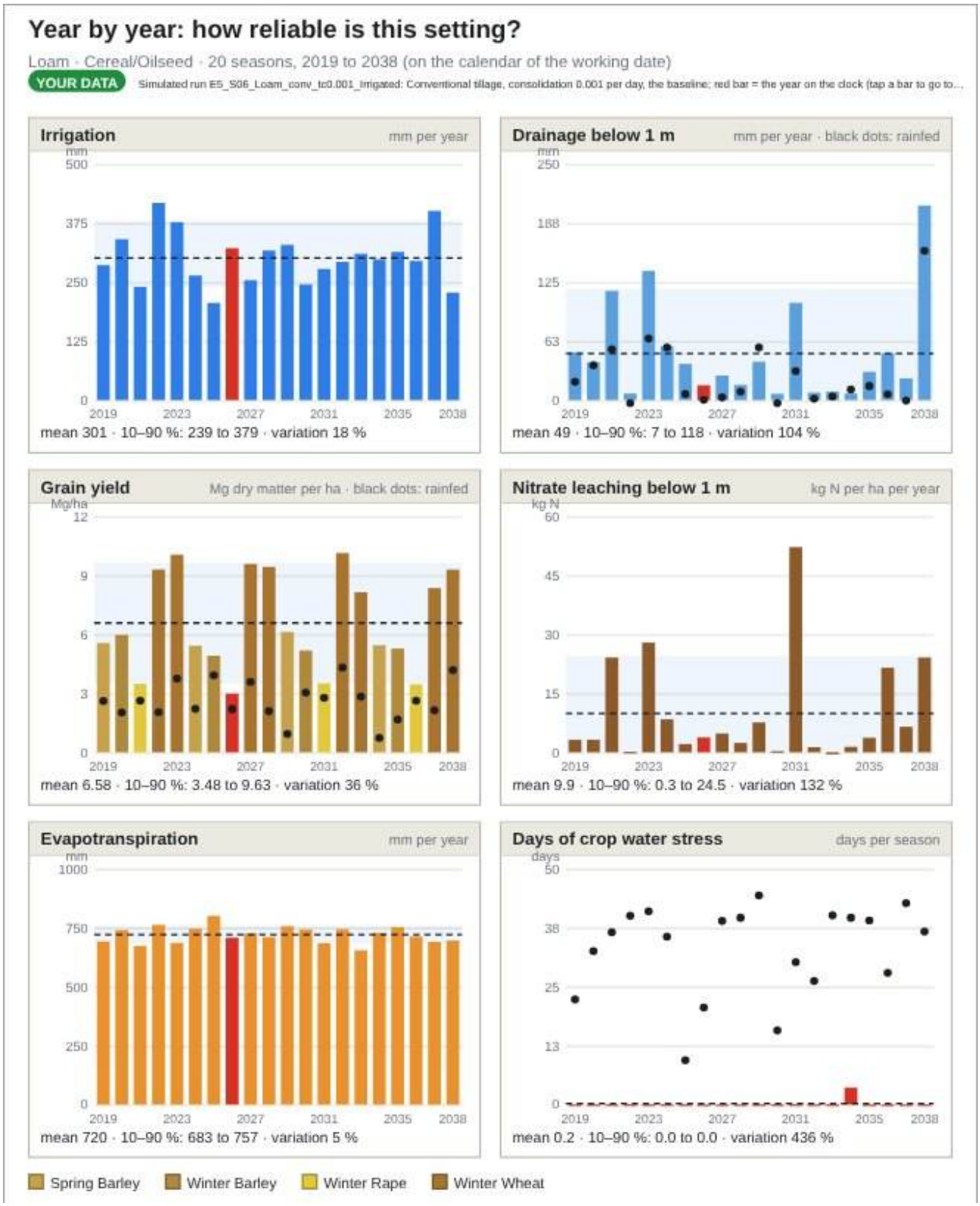


Figure 6.58: Reliability over 20 seasons, Loam, part 1 of 2.

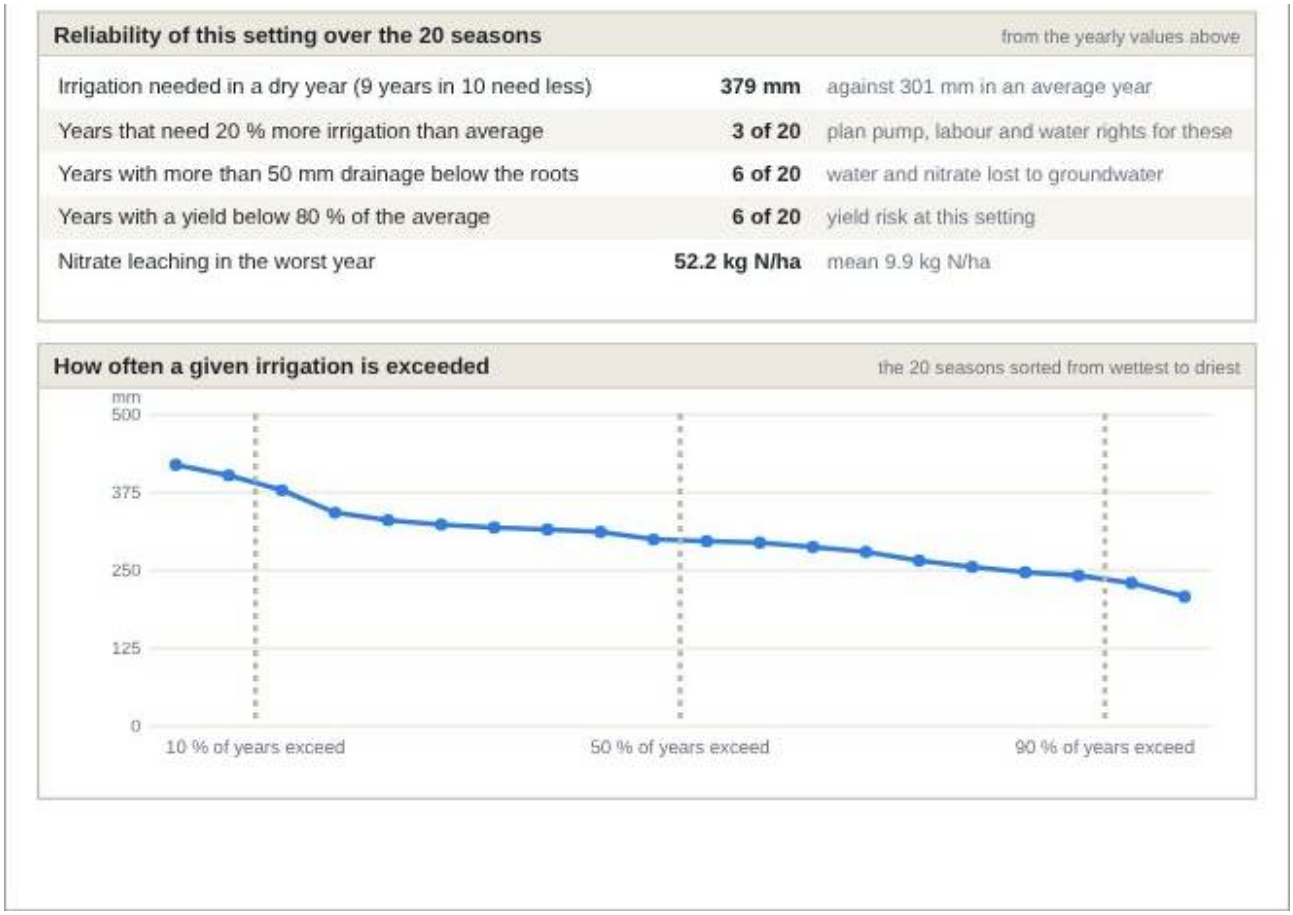


Figure 6.59: Reliability over 20 seasons, Loam, part 2 of 2.

6.21 Full water balance (Loam)

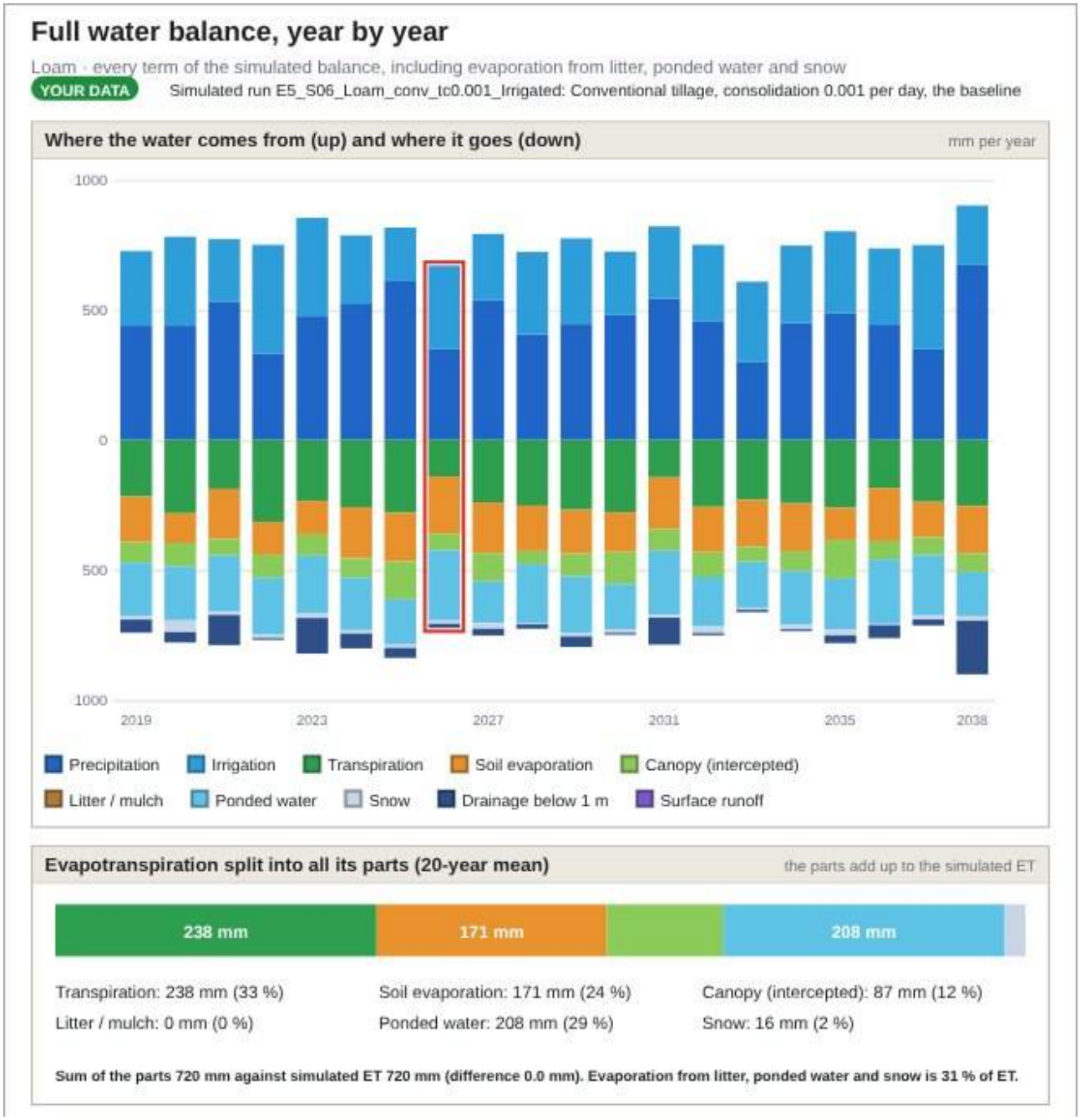


Figure 6.60: Full water balance, Loam, part 1 of 2.

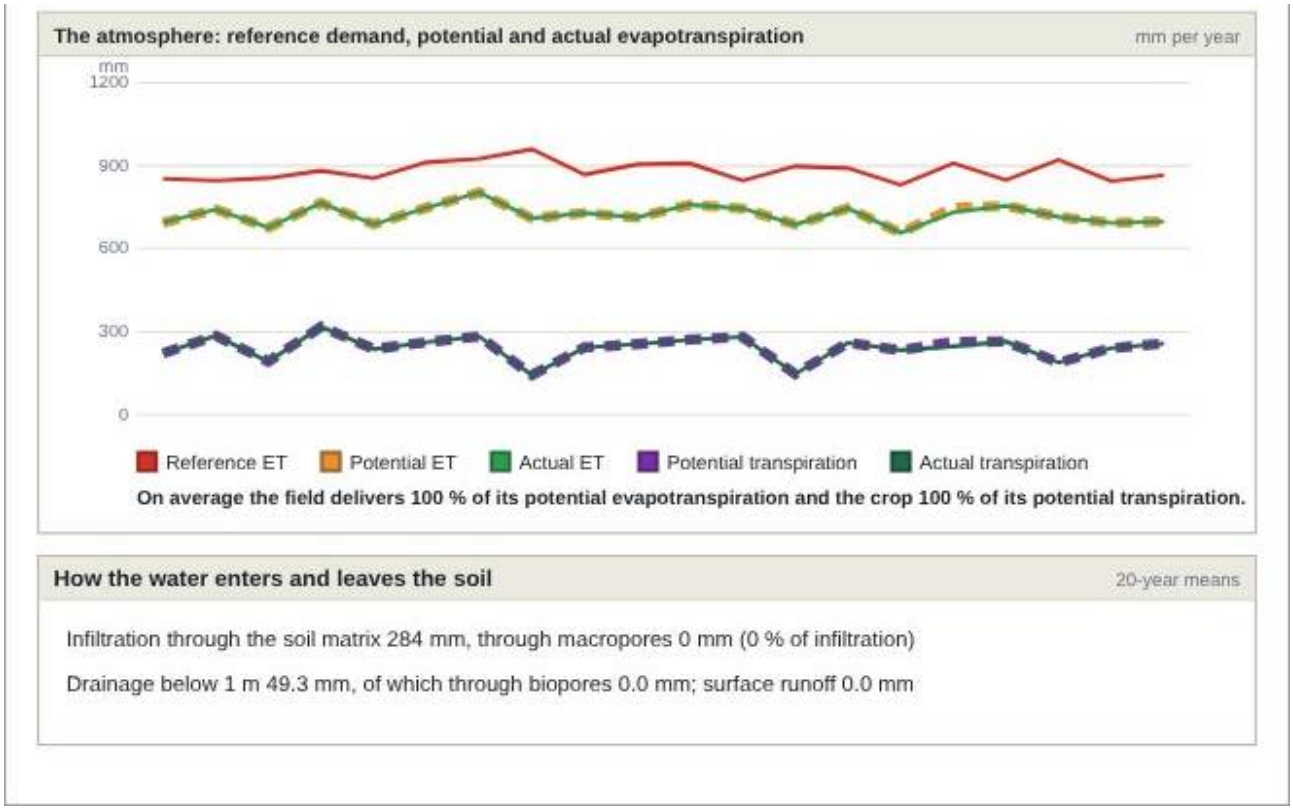


Figure 6.61: Full water balance, Loam, part 2 of 2.

6.22 Nitrogen budget (Loam)

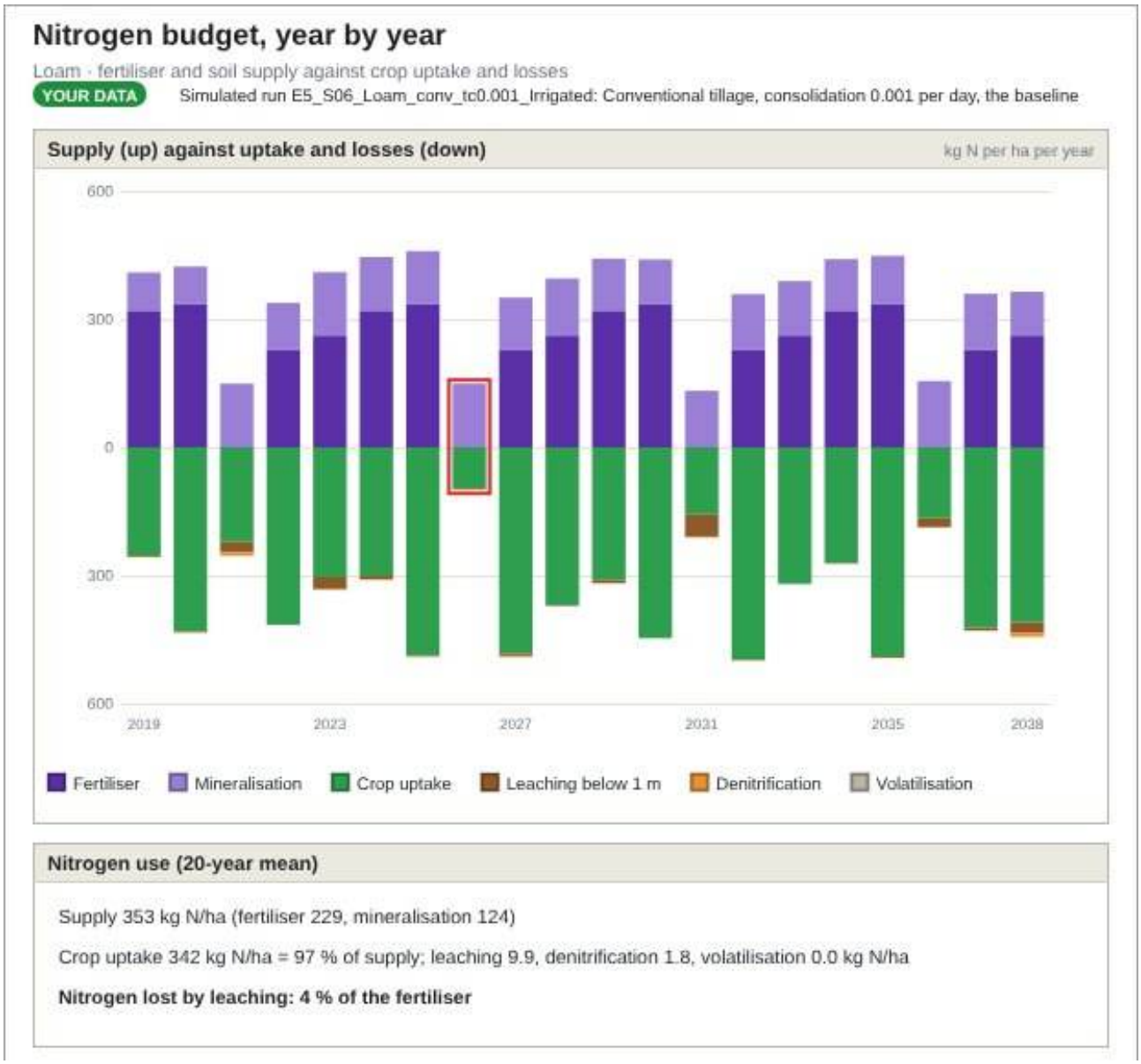


Figure 6.62: Nitrogen budget, Loam, part 1 of 2.

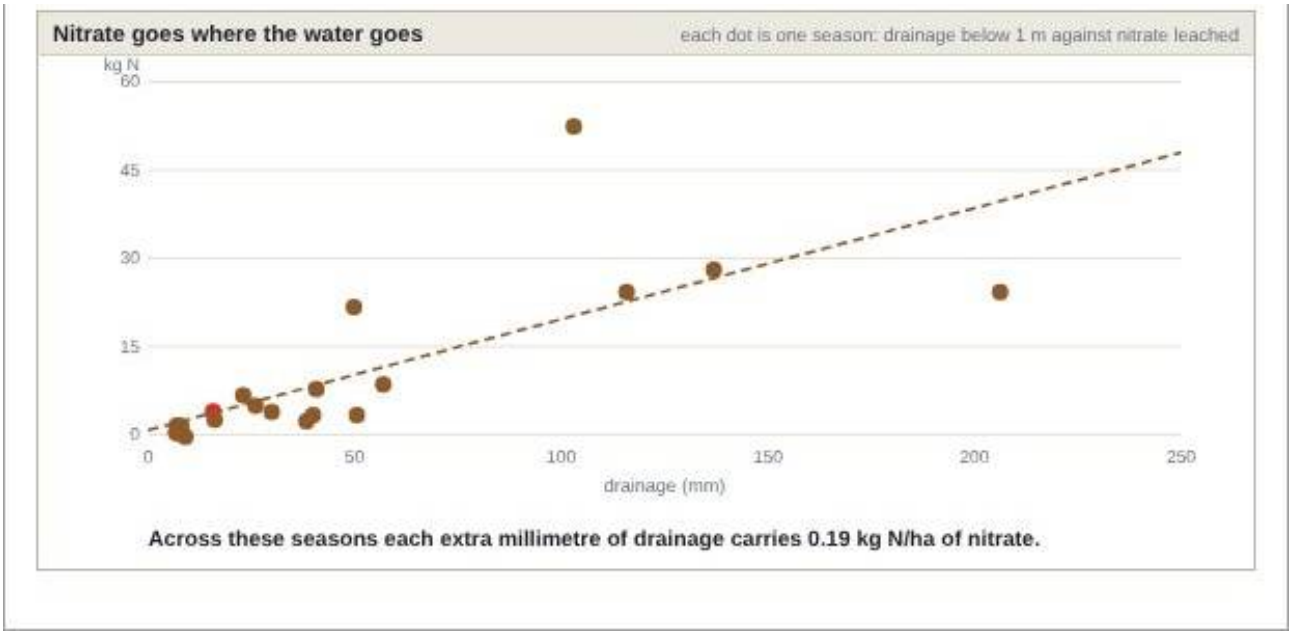


Figure 6.63: Nitrogen budget, Loam, part 2 of 2.

6.23 Crop water requirement (Loam)

Figure 6.64: Crop water requirement, Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**519**	7.5	99 %
Coconut (high)	834	**452**	5.6	100 %
Papaya (high)	814	**444**	6.1	100 %
Oil palm (young)	790	**399**	5.5	100 %
Yam	649	**363**	7.6	100 %
Onion (dry)	565	**325**	8.9	98 %
Kiwi (pergola)	626	**321**	4.0	100 %
Avocado (medium)	677	**306**	5.0	100 %
Cotton	571	**304**	3.9	100 %
Cabbage	559	**297**	5.8	100 %
Rice	573	**285**	4.7	92 %
Table grapes (horizontal trellis)	597	**275**	3.4	100 %
Potato	508	**271**	6.9	99 %
Guava (high)	651	**270**	3.3	100 %
Tomato	521	**259**	3.3	100 %
Hop (v trellis)	520	**249**	3.1	100 %
Mango (medium)	620	**243**	3.0	100 %
Walnut (medium)	523	**232**	2.9	100 %
Soybean	473	**229**	3.1	100 %
Wheat	469	**228**	3.0	100 %
Mustard	468	**227**	3.0	100 %
Pear (medium)	528	**225**	2.8	100 %
Groundnut	471	**223**	3.7	100 %
Date palm (medium)	602	**221**	2.7	100 %
Cherry (medium)	503	**205**	2.6	100 %
Pistachio (medium)	503	**205**	2.6	100 %
Cassava	466	**203**	3.9	100 %

Figure 6.65: Crop water requirement, Loam, part 2 of 3.

Maize (grain)	451		**201**	2.6	100 %
Sweet pepper	475		**200**	3.3	98 %
Chickpea	436		**191**	3.1	100 %
Hazelnut (low)	481		**175**	2.2	100 %
Pecan (low)	481		**175**	2.2	100 %
Sorghum (grain)	429		**168**	2.1	100 %
Cherimoya (medium)	545		**166**	2.1	100 %
Millet	395		**160**	2.2	100 %
Citrus (medium density, short)	552		**157**	2.0	100 %
Apple (medium)	459		**157**	2.0	100 %
Cowpea (beans)	398		**148**	2.3	100 %
Wine grapes (pergola)	457		**130**	1.6	100 %
Green gram (mung bean)	372		**122**	2.1	100 %
Peach (low)	435		**121**	1.5	100 %
Almond (medium)	435		**121**	1.5	100 %
Apricot (young)	380		**76**	0.9	100 %
Highbush blueberry (low)	338		**57**	0.8	100 %
Persimmon (young, low)	358		**54**	0.7	100 %
Pomegranate (low)	358		**54**	0.7	100 %
Olive (traditional, medium density)	446		**49**	0.6	100 %
Fig tree (young, low)	327		**27**	0.3	100 %

Figure 6.66: Crop water requirement, Loam, part 3 of 3.

6.24 Feedback loops (Loam)

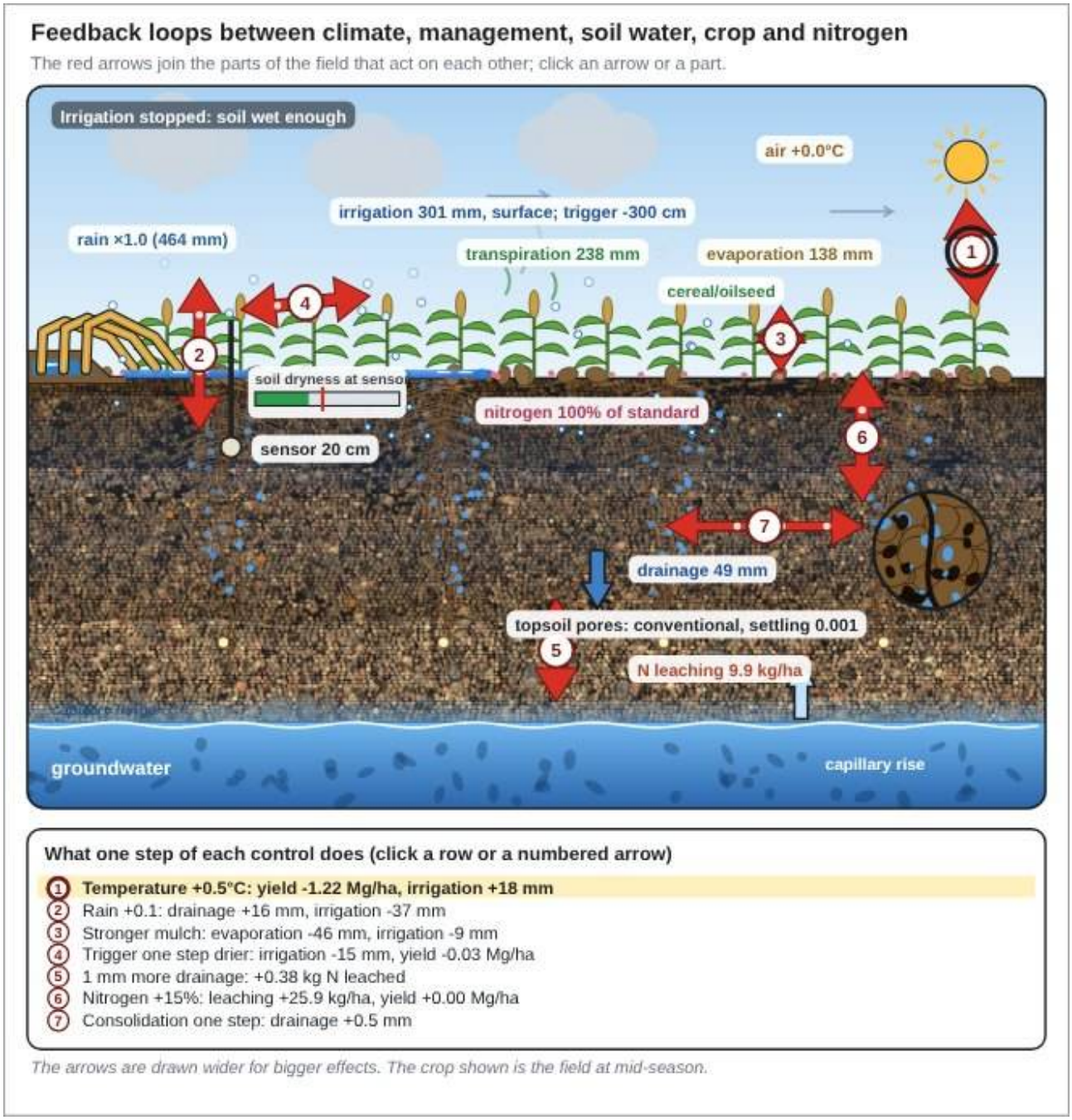


Figure 6.67: Feedback loops, Loam, part 1 of 1.

Chapter 7

Silt Loam

7.1 Surface irrigation design (Silt Loam)

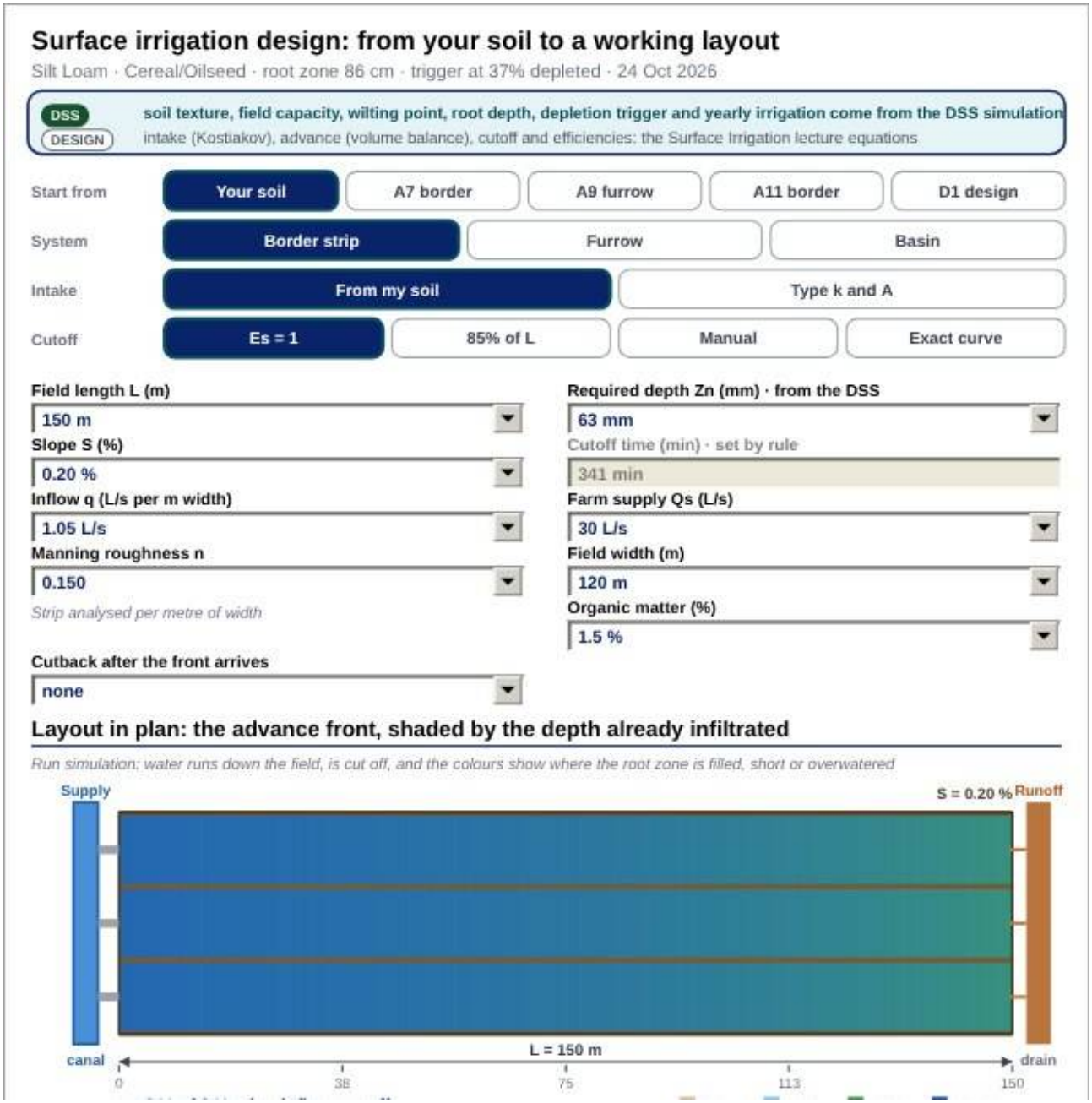


Figure 7.1: Surface irrigation design, Silt Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

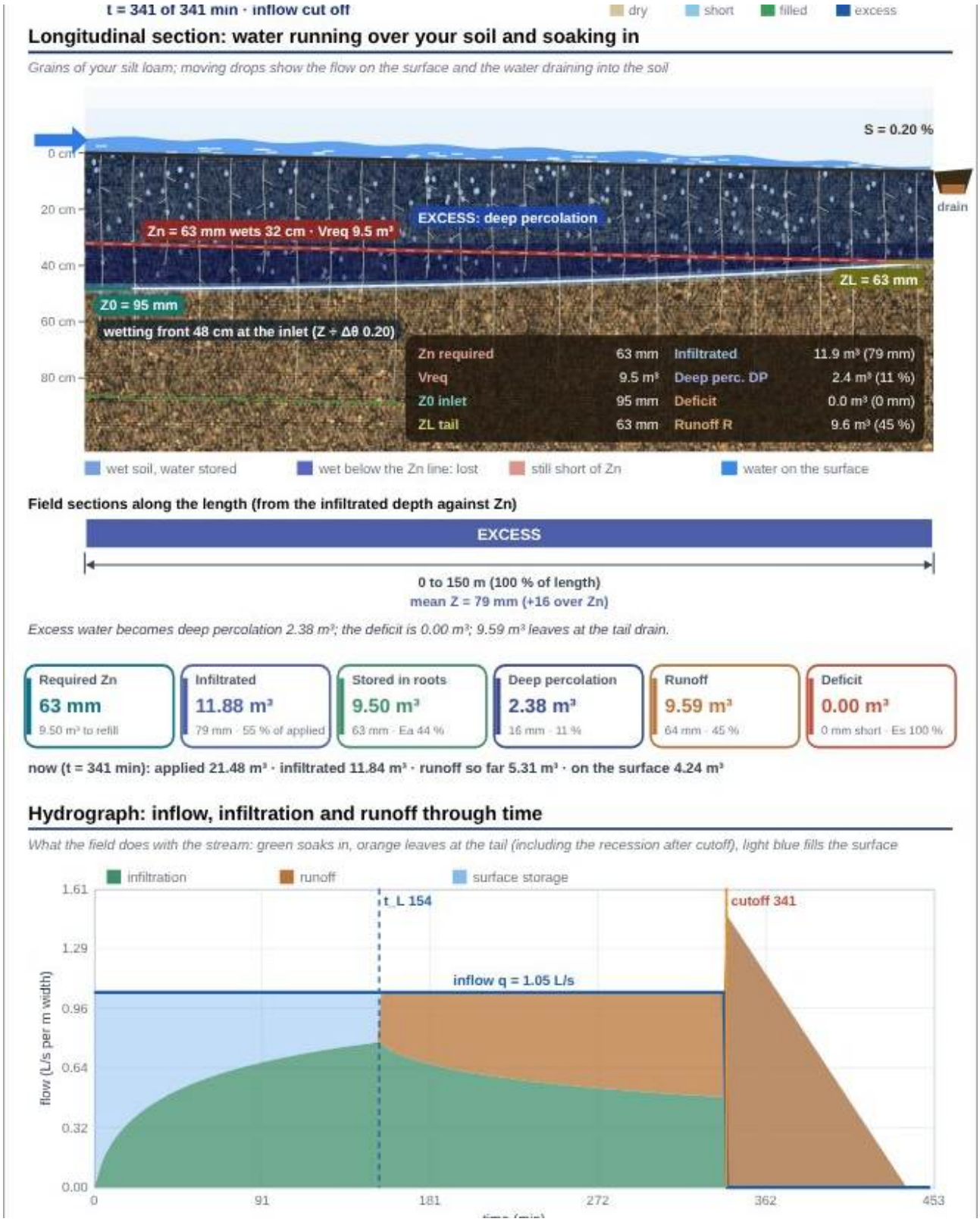


Figure 7.2: Surface irrigation design, Silt Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

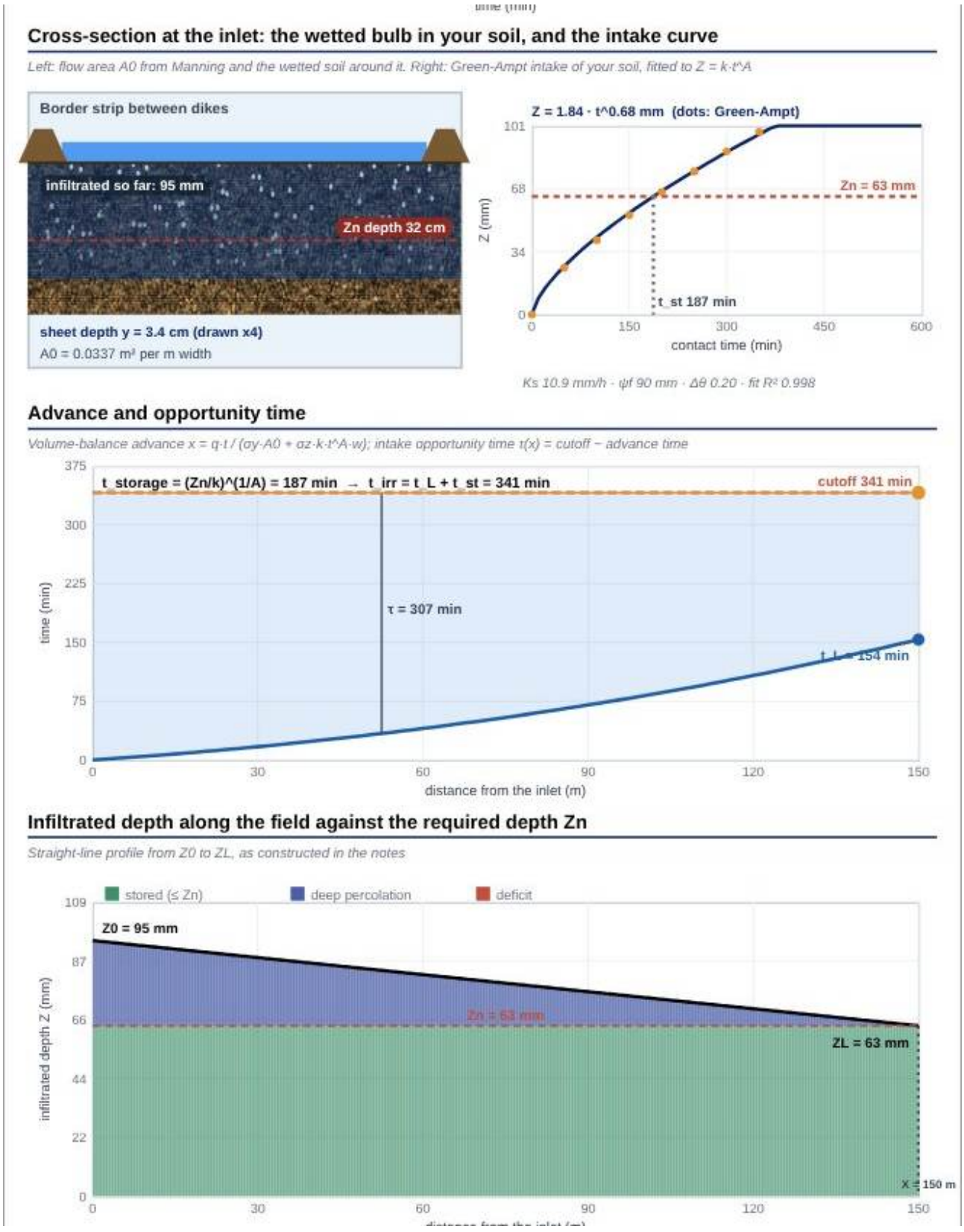


Figure 7.3: Surface irrigation design, Silt Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

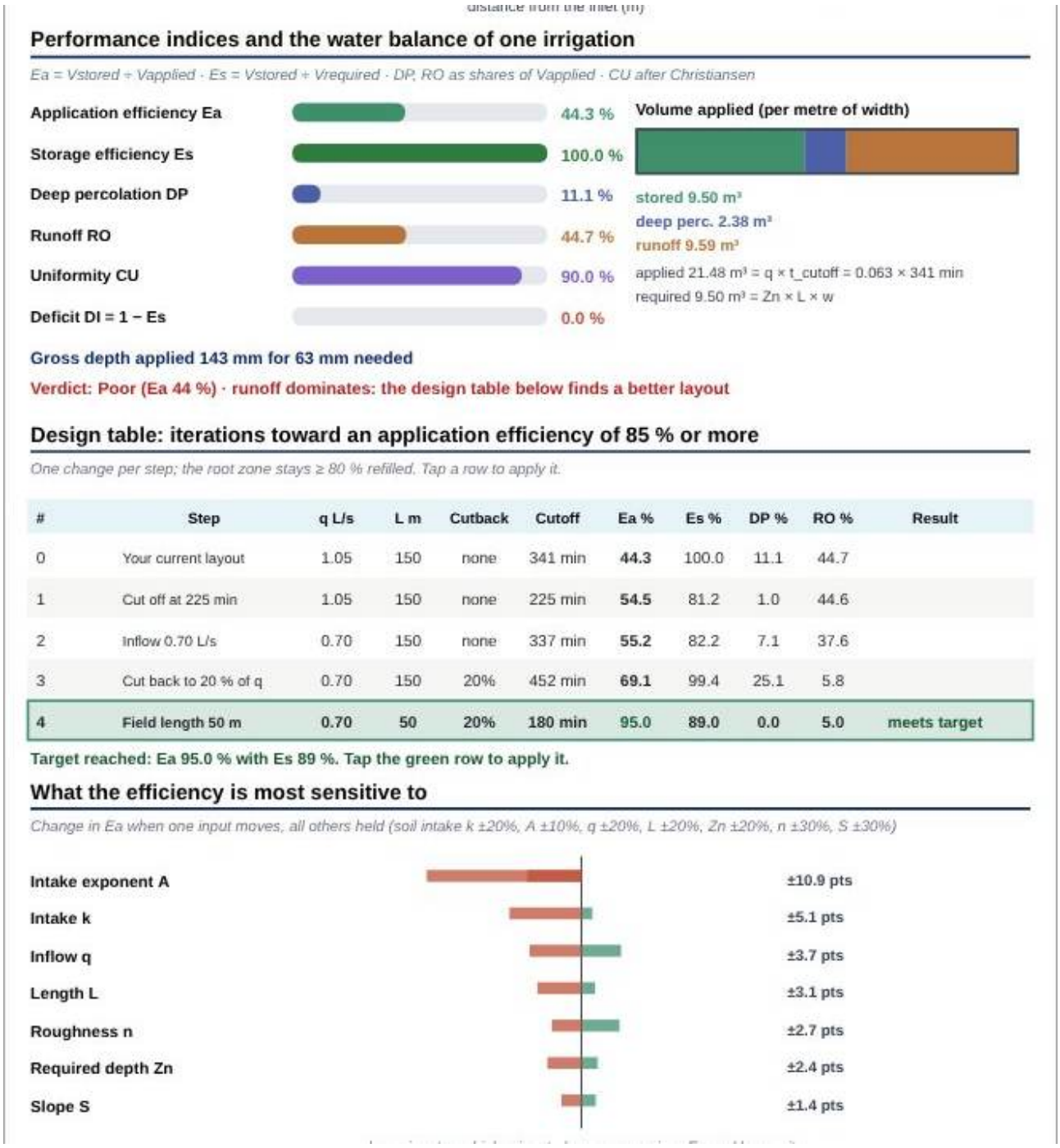


Figure 7.4: Surface irrigation design, Silt Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to.

Figure 7.5: Surface irrigation design, Silt Loam, part 5 of 5: What this means for the season and the farm; Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

7.2 Drip irrigation design (Silt Loam)

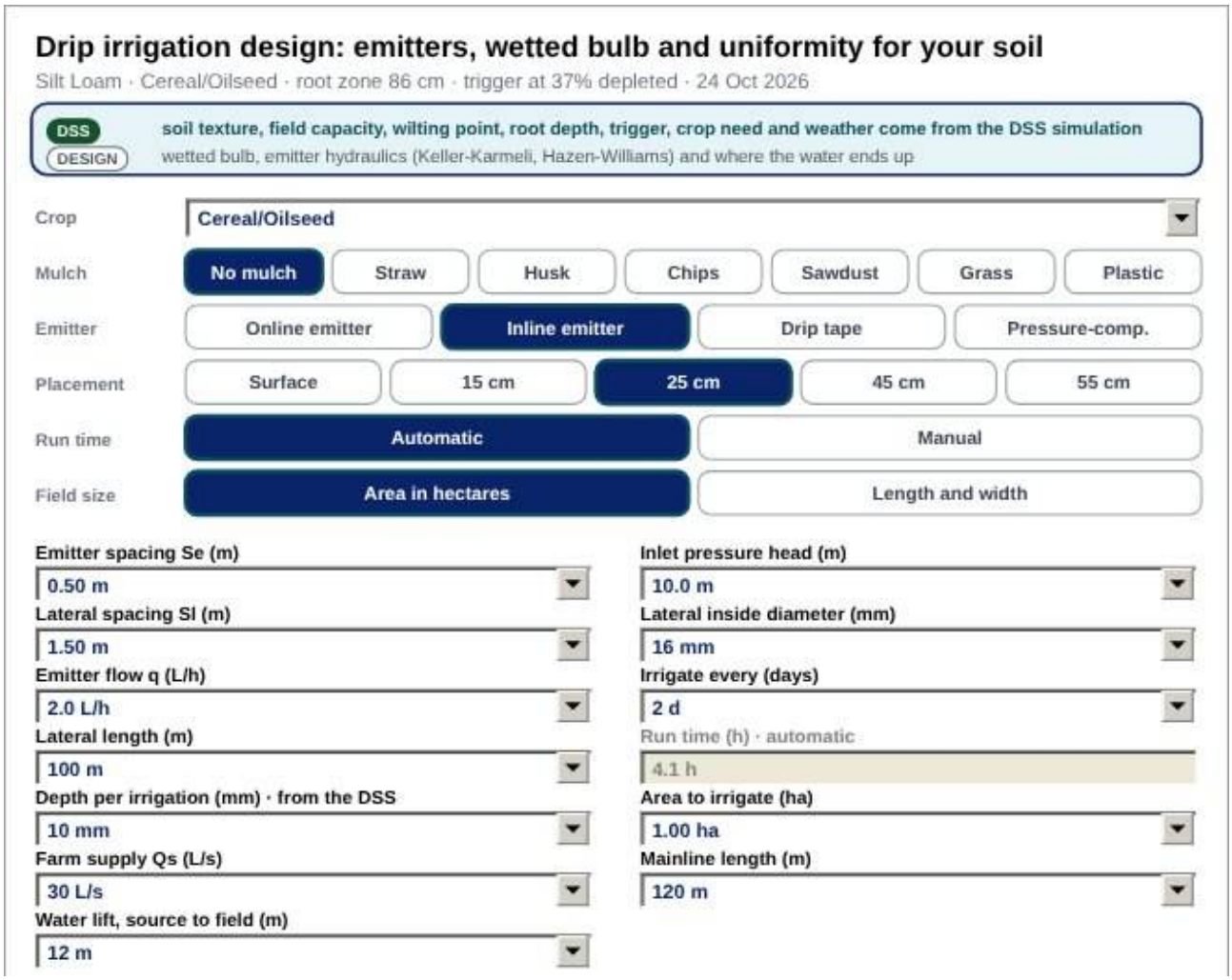


Figure 7.6: Drip irrigation design, Silt Loam, part 1 of 6.

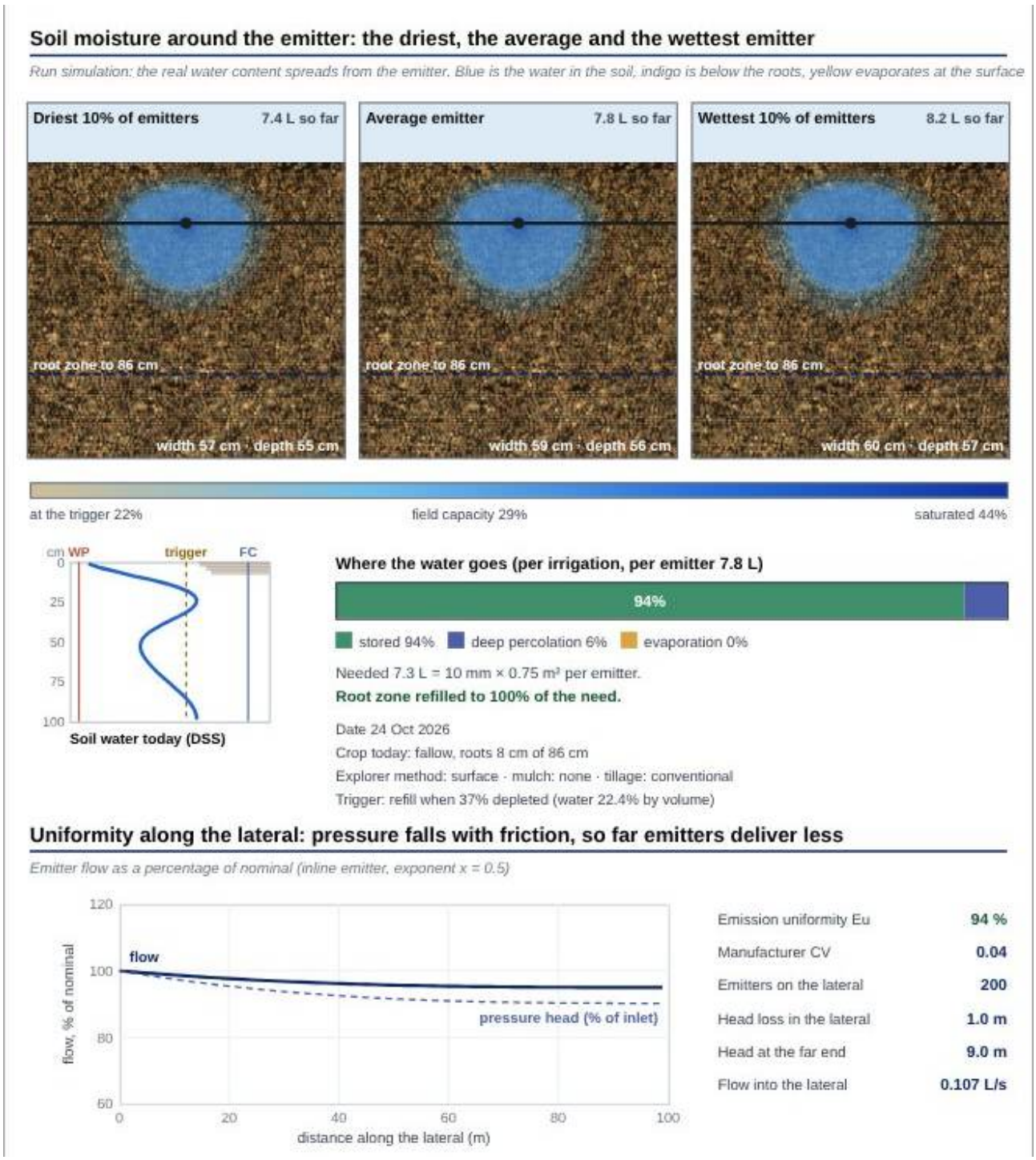
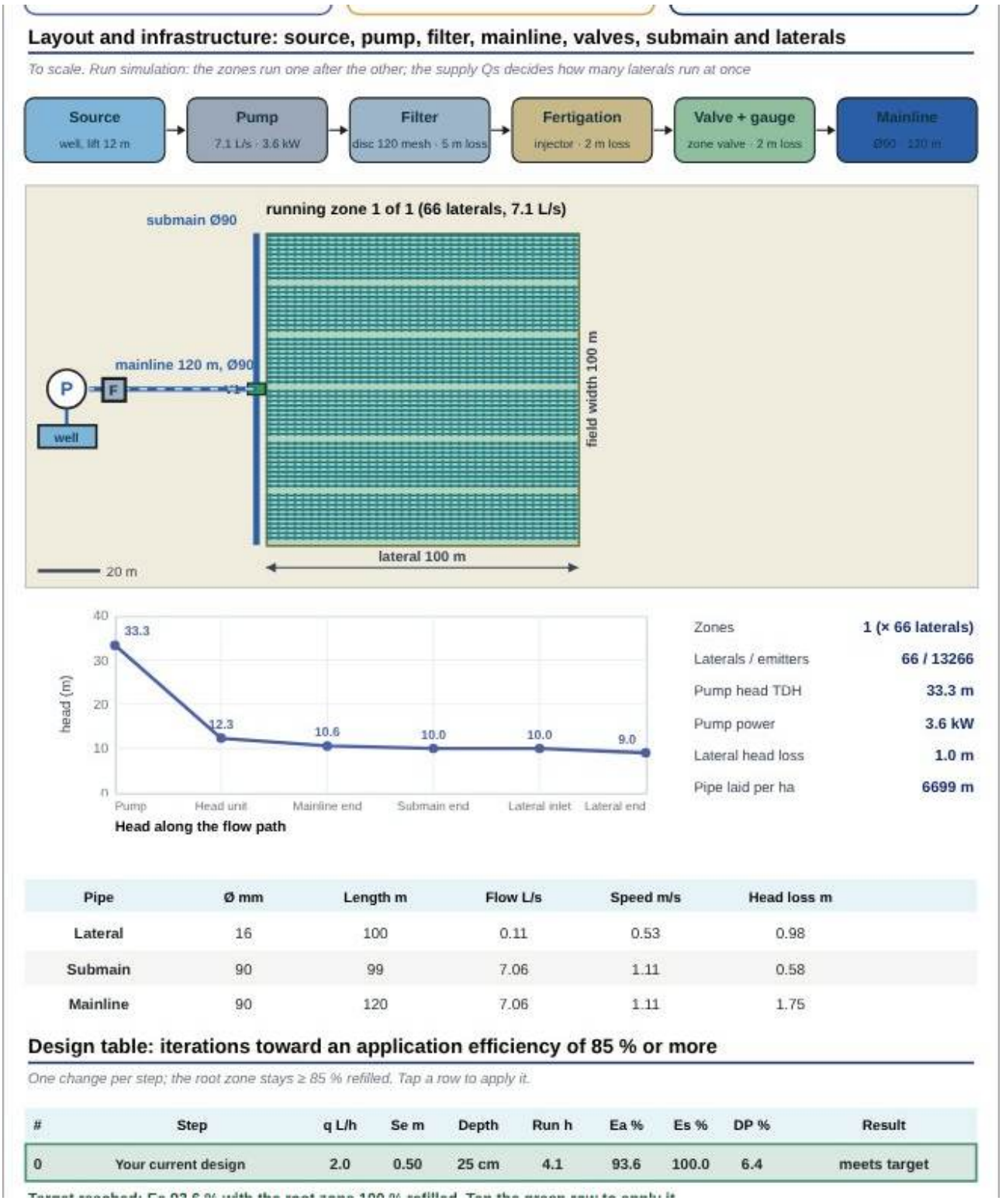


Figure 7.7: Drip irrigation design, Silt Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

Figure 7.8: Drip irrigation design, Silt Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	4.1	93.6	100.0	6.4	meets target

Figure 7.9: Drip irrigation design, Silt Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

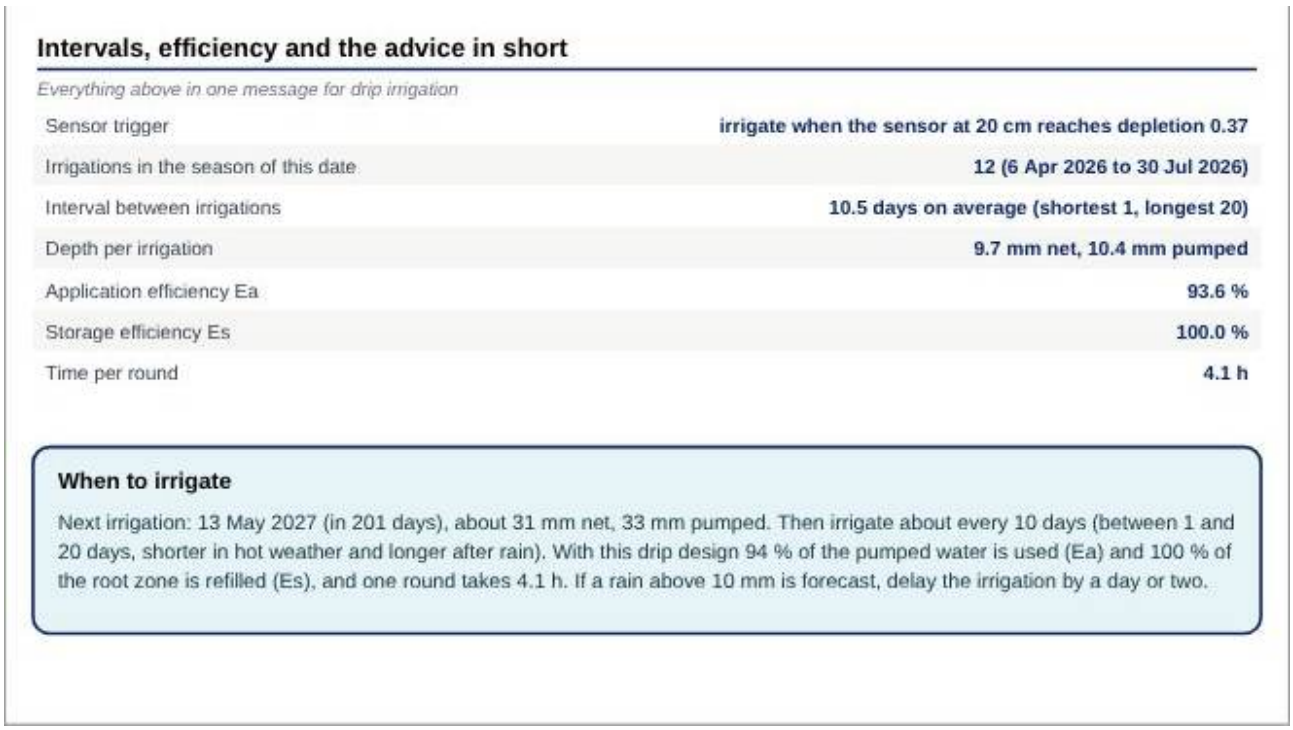
Net irrigation need in a season (DSS)	**366 mm → 38 irrigations of 10 mm**
Gross water per irrigation	**10.4 mm (Ea 94 %)**
Gross water per season	**391 mm = 3911 m³ per hectare**
Peak crop need / interval	**4.9 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 4.1 h = 4.1 h per round**
Pumping per season	**152 hours · 541 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**13 May 2027**	201 days	31	33	4.1 h	22 days
2	**4 Jun 2027**	223 days	31	33	4.1 h	12 days
3	**16 Jun 2027**	235 days	31	33	4.1 h	9 days
4	**25 Jun 2027**	244 days	31	33	4.1 h	8 days
5	**3 Jul 2027**	252 days	31	33	4.1 h	10 days
6	**13 Jul 2027**	262 days	31	33	4.1 h	12 days
7	**25 Jul 2027**	274 days	31	33	4.1 h	255 days
8	**5 Apr 2028**	529 days	31	33	4.1 h	-

Figure 7.10: Drip irrigation design, Silt Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.37**
Irrigations in the season of this date	**12 (6 Apr 2026 to 30 Jul 2026)**
Interval between irrigations	**10.5 days on average (shortest 1, longest 20)**
Depth per irrigation	**9.7 mm net, 10.4 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**4.1 h**

When to irrigate

Next irrigation: 13 May 2027 (in 201 days), about 31 mm net, 33 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 4.1 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 7.11: Drip irrigation design, Silt Loam, part 6 of 6: Intervals, efficiency and the advice in short.

7.3 Sprinkler design and pipes (Silt Loam)

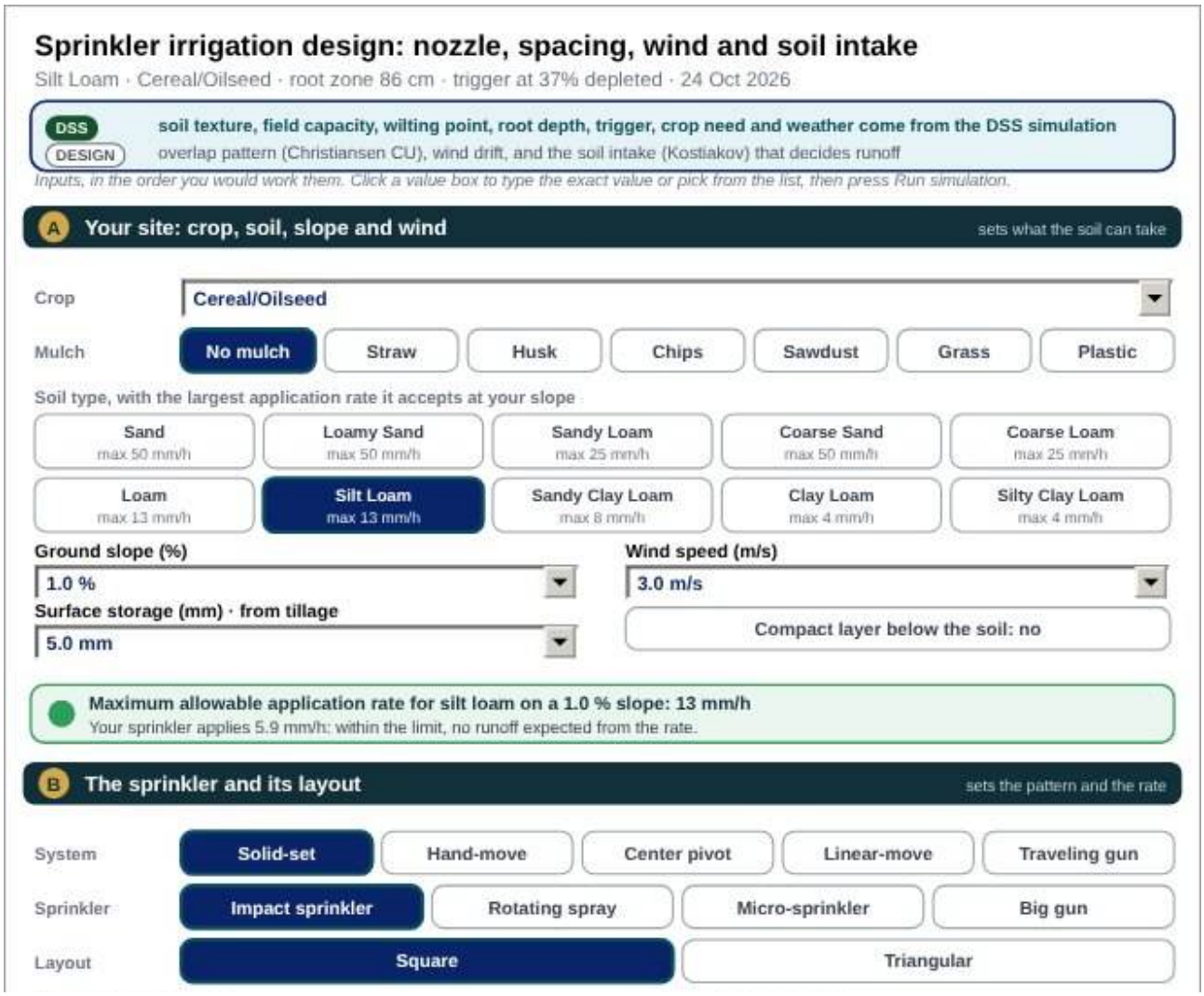


Figure 7.12: Sprinkler design and pipes, Silt Loam, part 1 of 12.

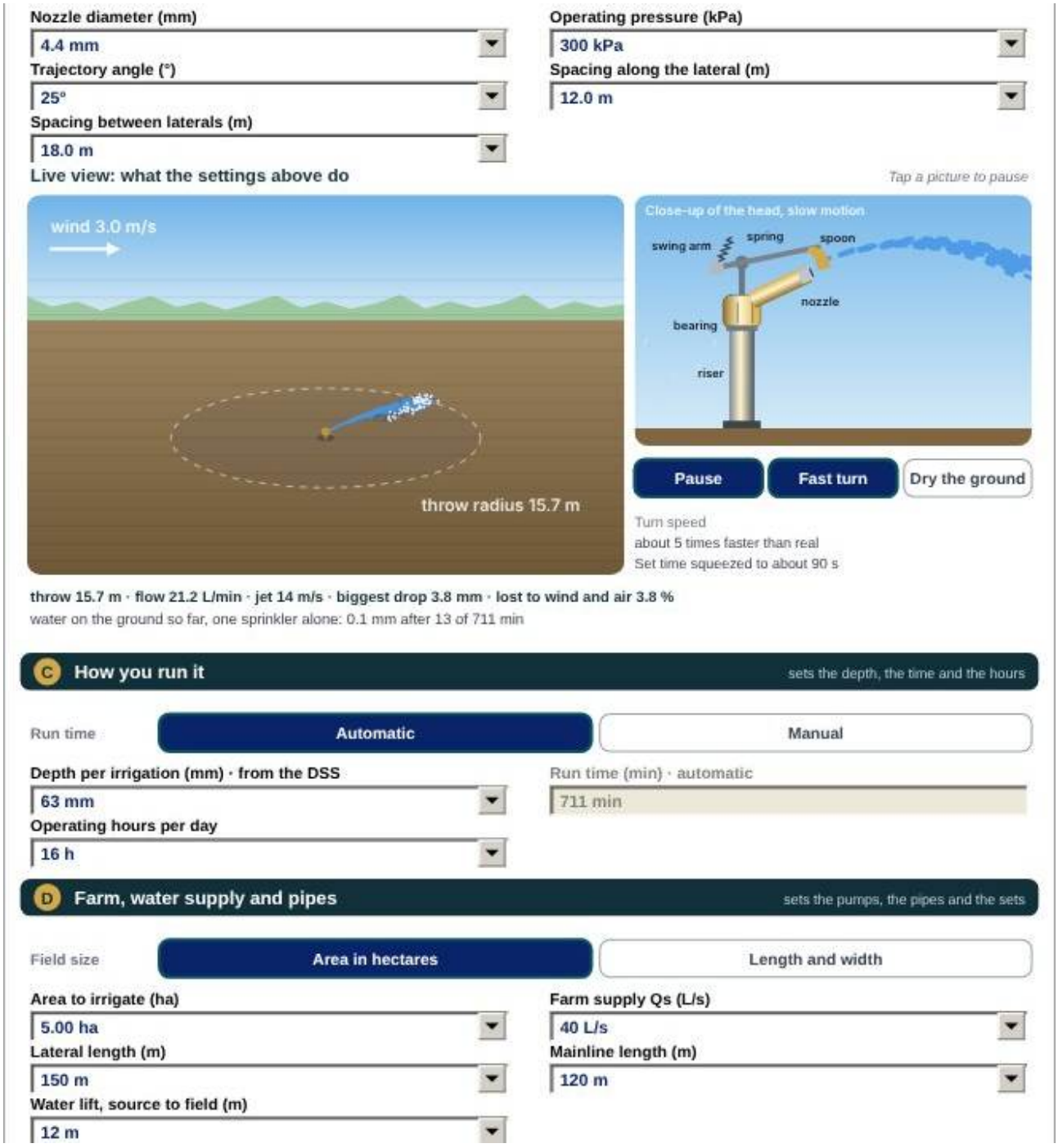


Figure 7.13: Sprinkler design and pipes, Silt Loam, part 2 of 12.

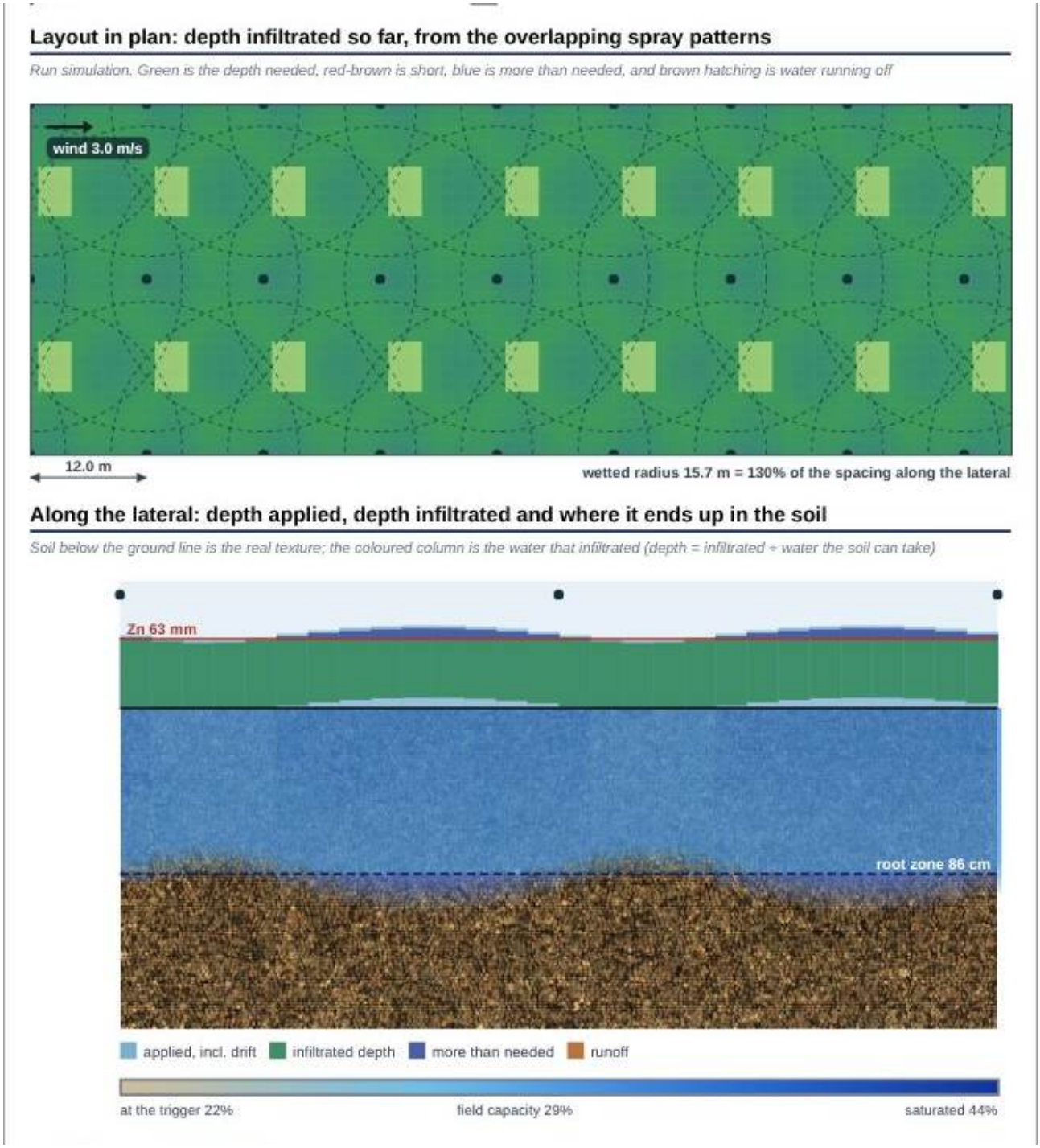


Figure 7.14: Sprinkler design and pipes, Silt Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

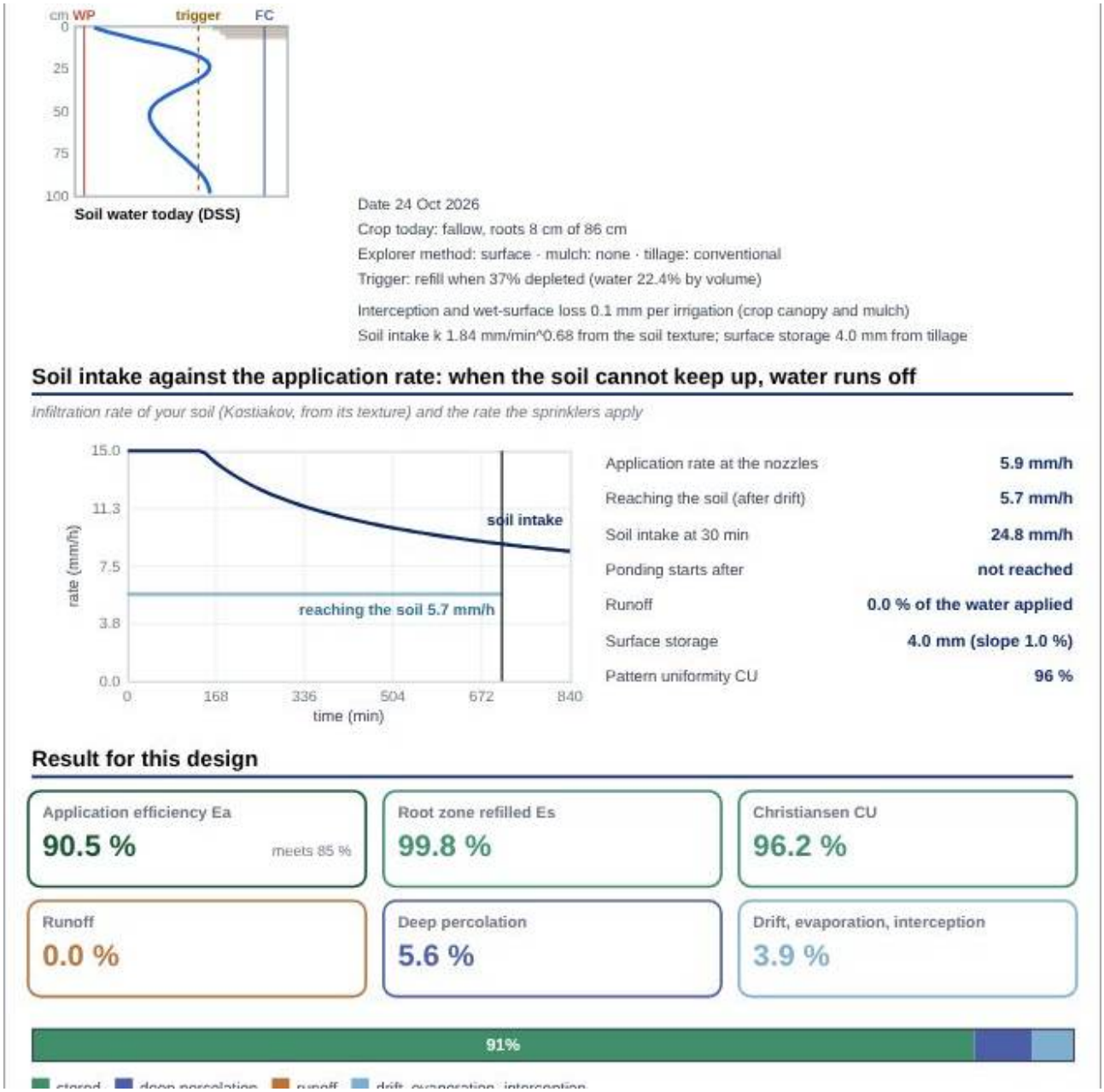


Figure 7.15: Sprinkler design and pipes, Silt Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

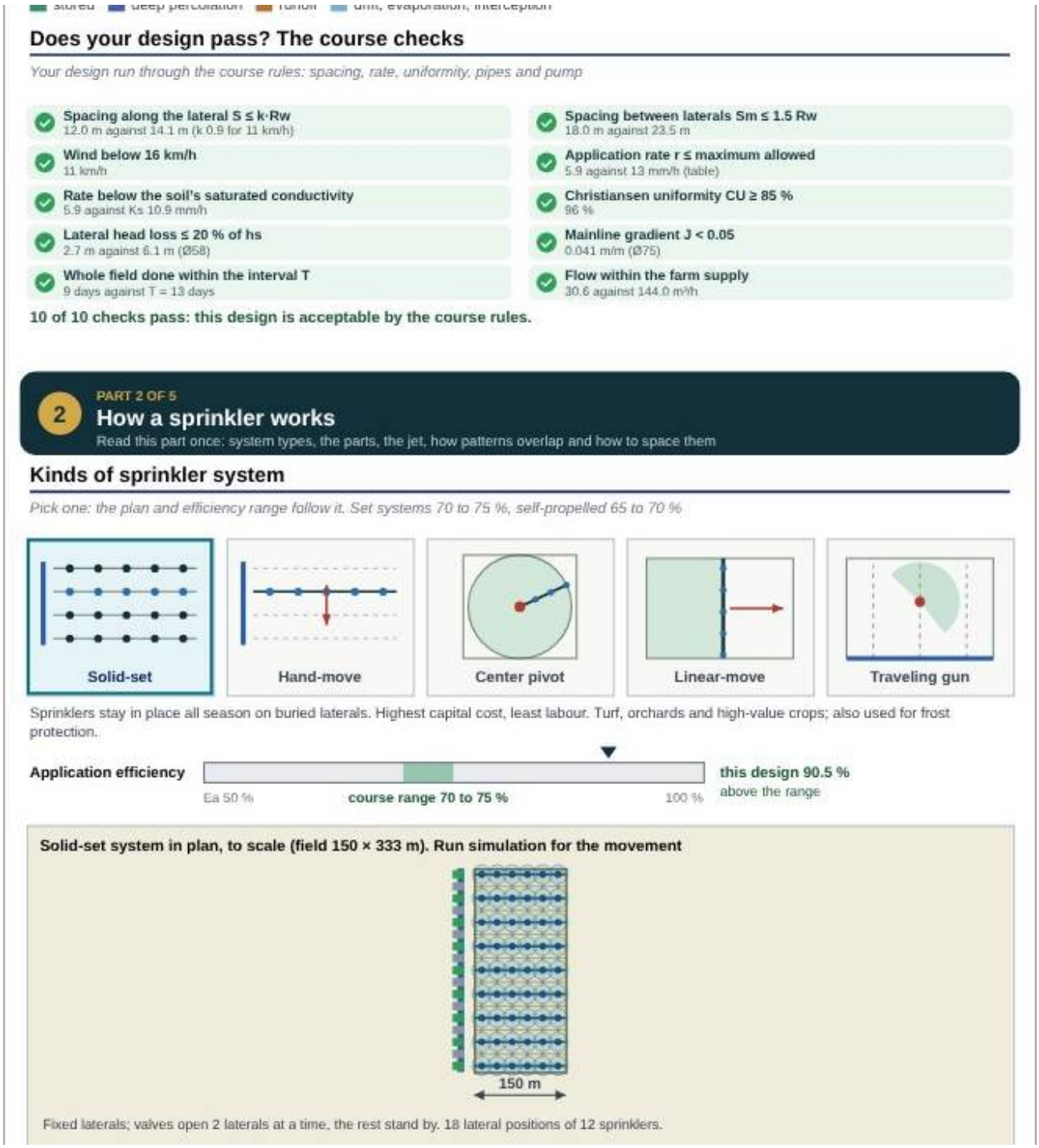


Figure 7.16: Sprinkler design and pipes, Silt Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

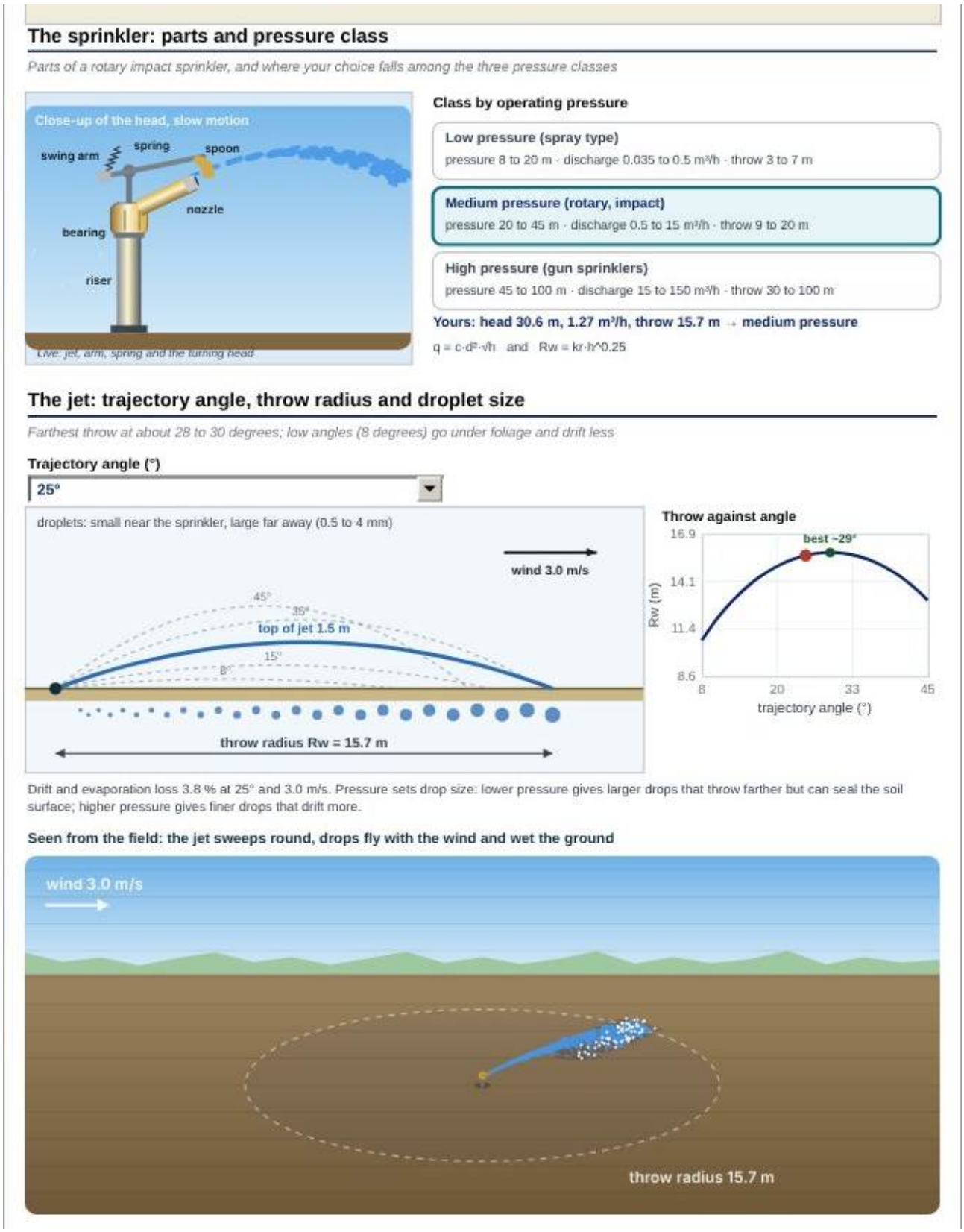


Figure 7.17: Sprinkler design and pipes, Silt Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

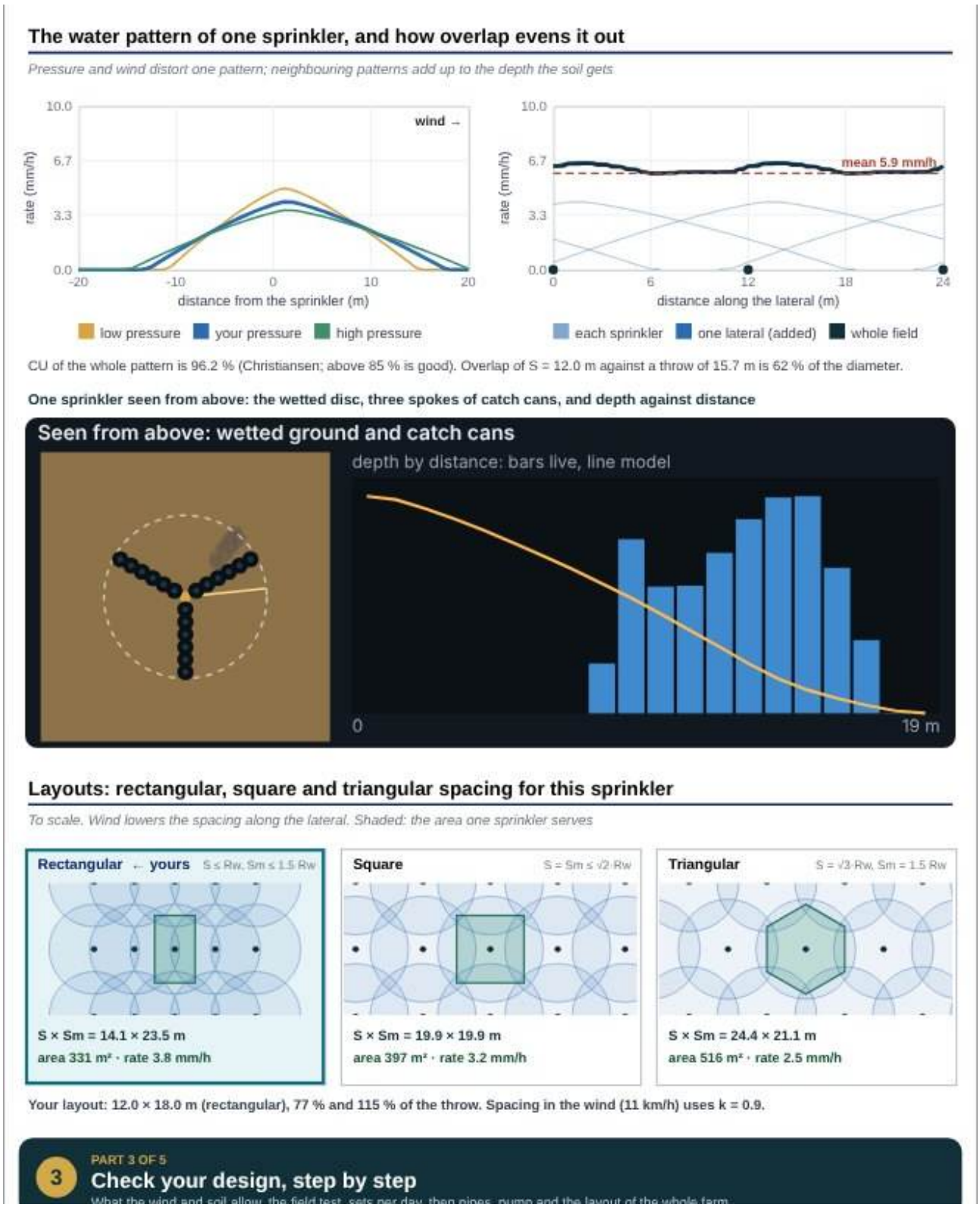


Figure 7.18: Sprinkler design and pipes, Silt Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

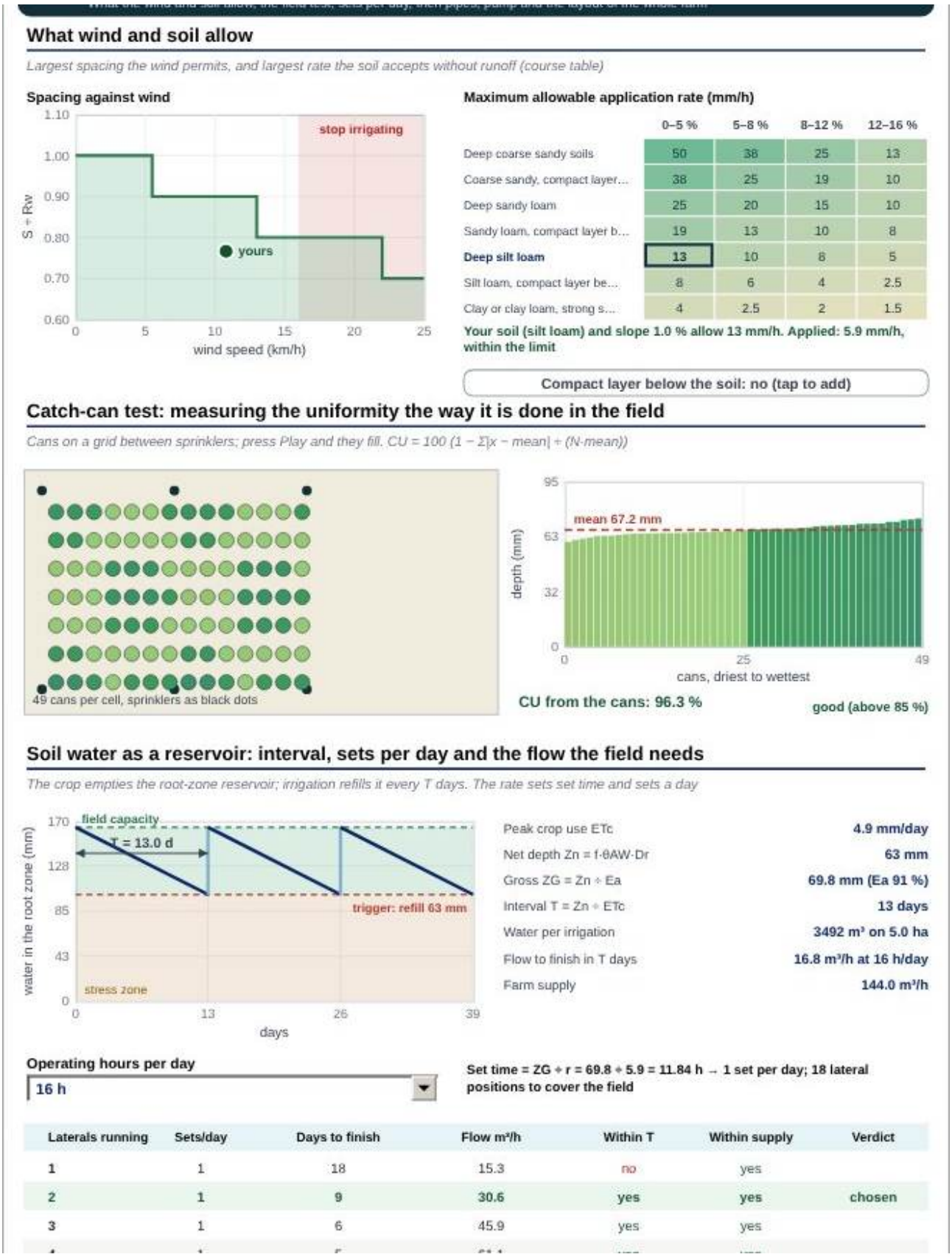


Figure 7.19: Sprinkler design and pipes, Silt Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

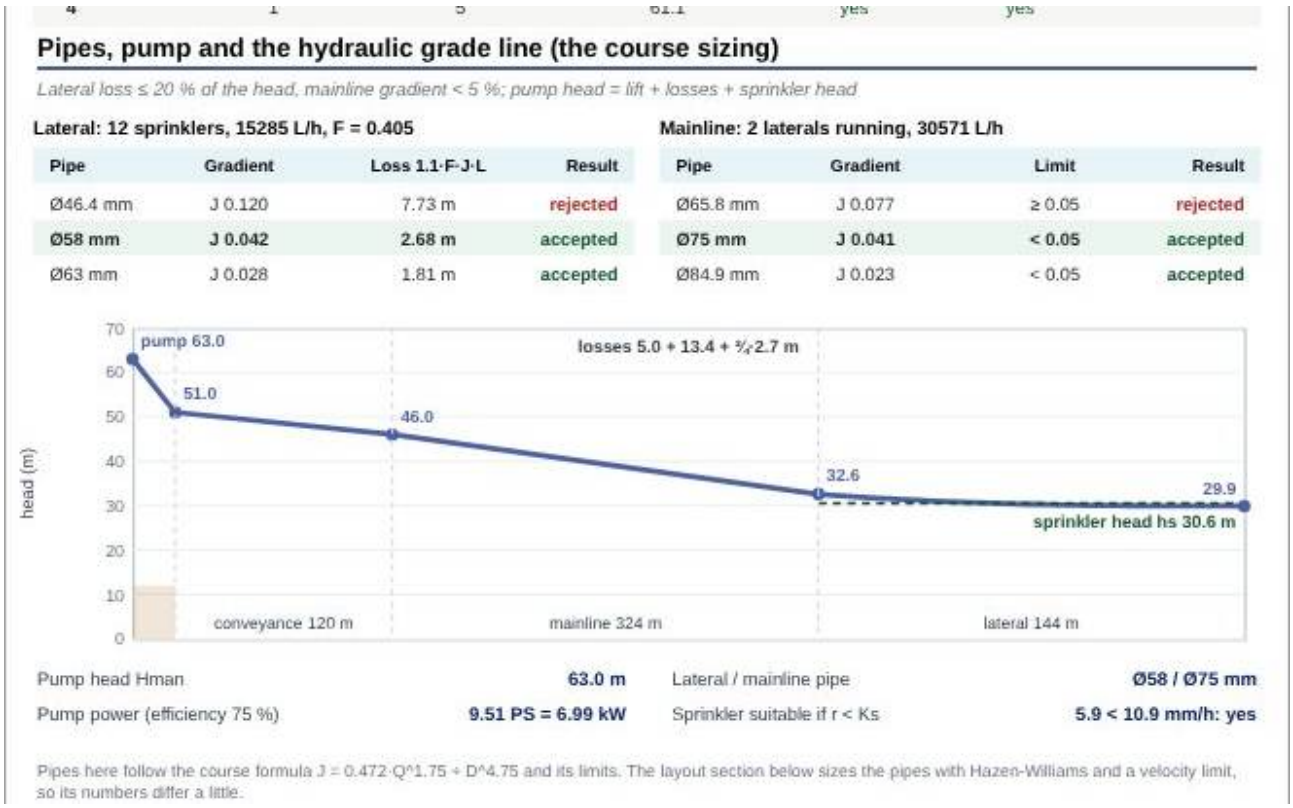
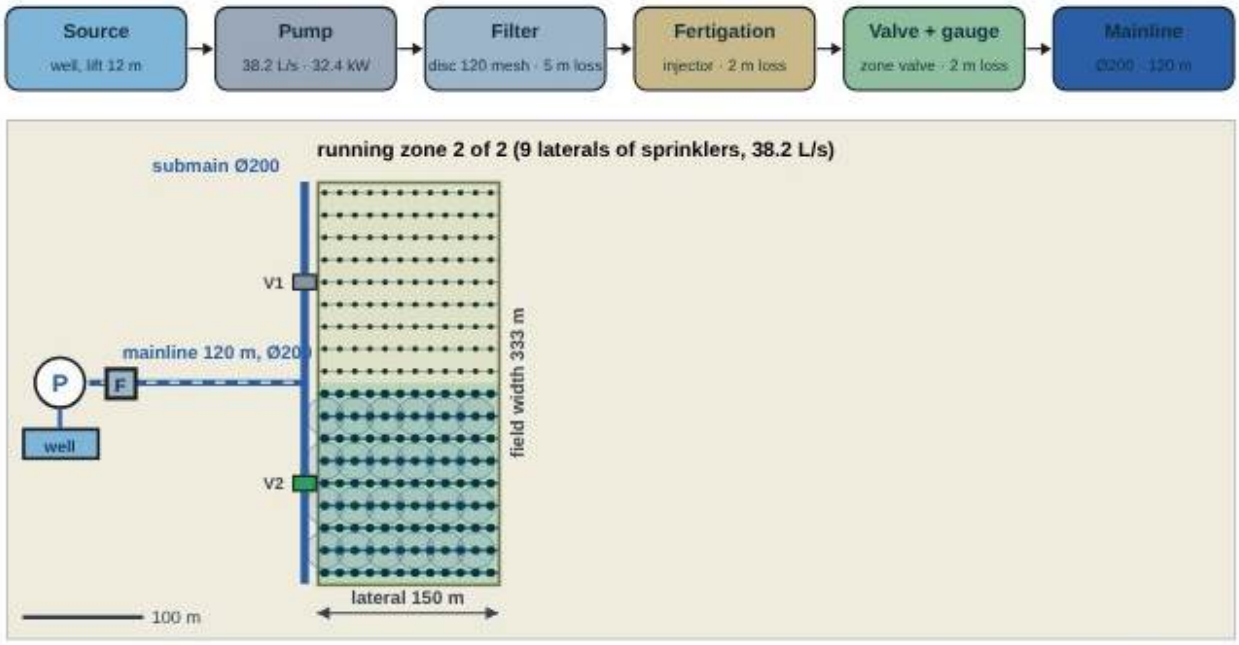
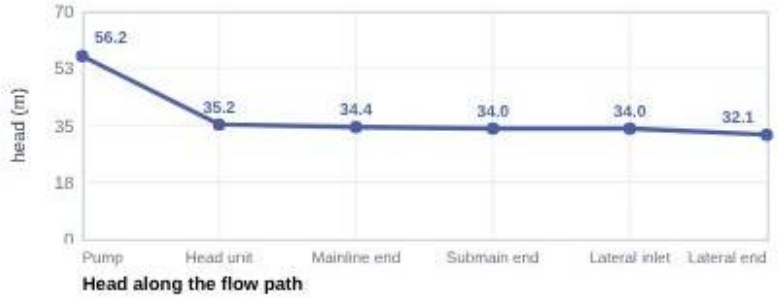


Figure 7.20: Sprinkler design and pipes, Silt Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

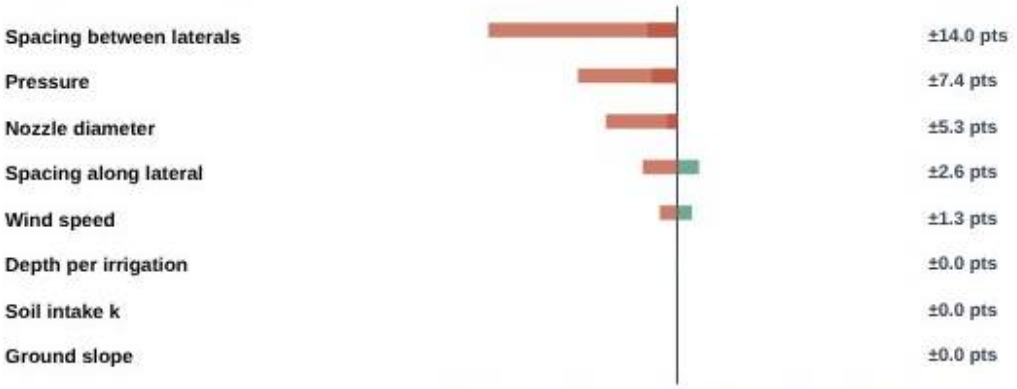
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	711	90.5	99.8	0.0	meets target

Figure 7.21: Sprinkler design and pipes, Silt Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CU 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

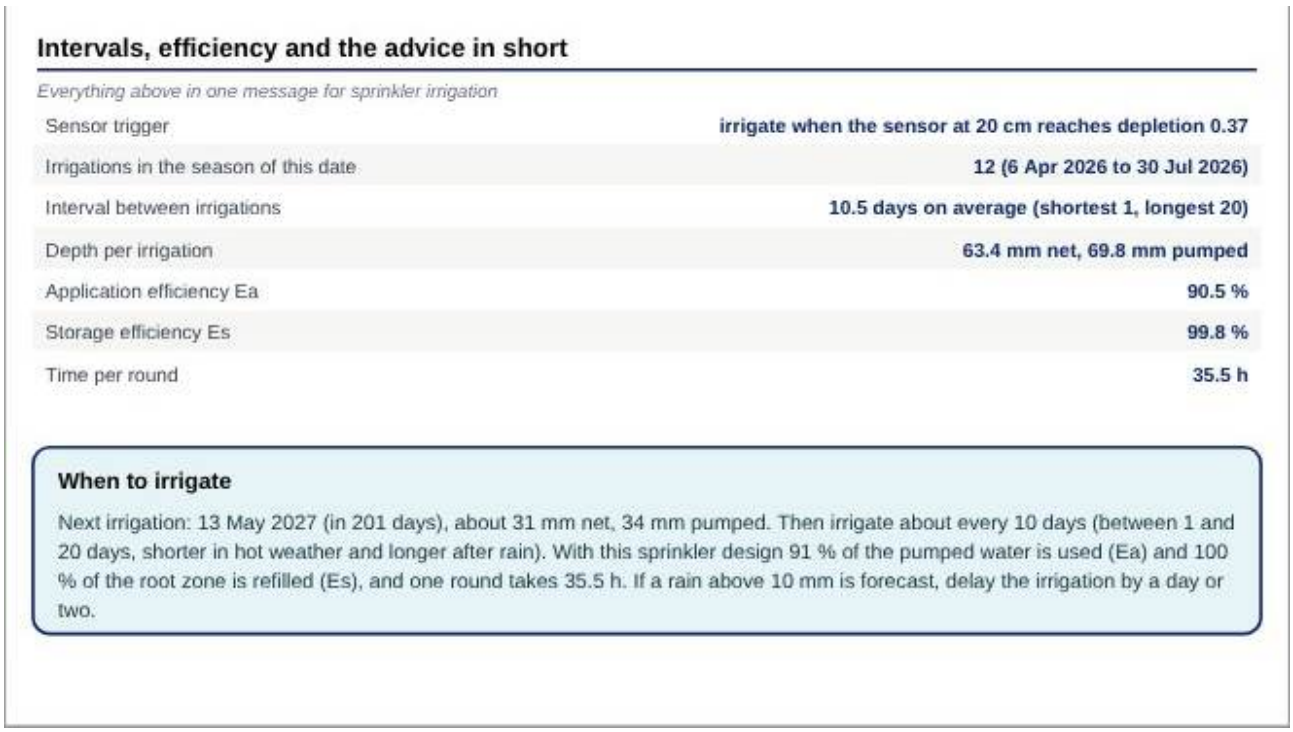
Net irrigation need in a season (DSS)	**366 mm → 6 irrigations of 63 mm**
Gross water per irrigation	**69.8 mm (Ea 91 %)**
Gross water per season	**404 mm = 4036 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 711 min = 35.5 h per round**
Pumping per season	**205 hours · 6661 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**13 May 2027**	201 days	31	34	35.5 h	22 days
2	**4 Jun 2027**	223 days	31	34	35.5 h	12 days
3	**16 Jun 2027**	235 days	31	34	35.5 h	9 days
4	**25 Jun 2027**	244 days	31	34	35.5 h	8 days
5	**3 Jul 2027**	252 days	31	34	35.5 h	10 days
6	**13 Jul 2027**	262 days	31	34	35.5 h	12 days
7	**25 Jul 2027**	274 days	31	34	35.5 h	255 days
8	**5 Apr 2028**	529 days	31	34	35.5 h	-

Figure 7.22: Sprinkler design and pipes, Silt Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.37**
Irrigations in the season of this date	**12 (6 Apr 2026 to 30 Jul 2026)**
Interval between irrigations	**10.5 days on average (shortest 1, longest 20)**
Depth per irrigation	**63.4 mm net, 69.8 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**35.5 h**

When to irrigate

Next irrigation: 13 May 2027 (in 201 days), about 31 mm net, 34 mm pumped. Then irrigate about every 10 days (between 1 and 20 days, shorter in hot weather and longer after rain). With this sprinkler design 91 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 35.5 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 7.23: Sprinkler design and pipes, Silt Loam, part 12 of 12: Intervals, efficiency and the advice in short.

7.4 Virtual farm of the surface design (Silt Loam)

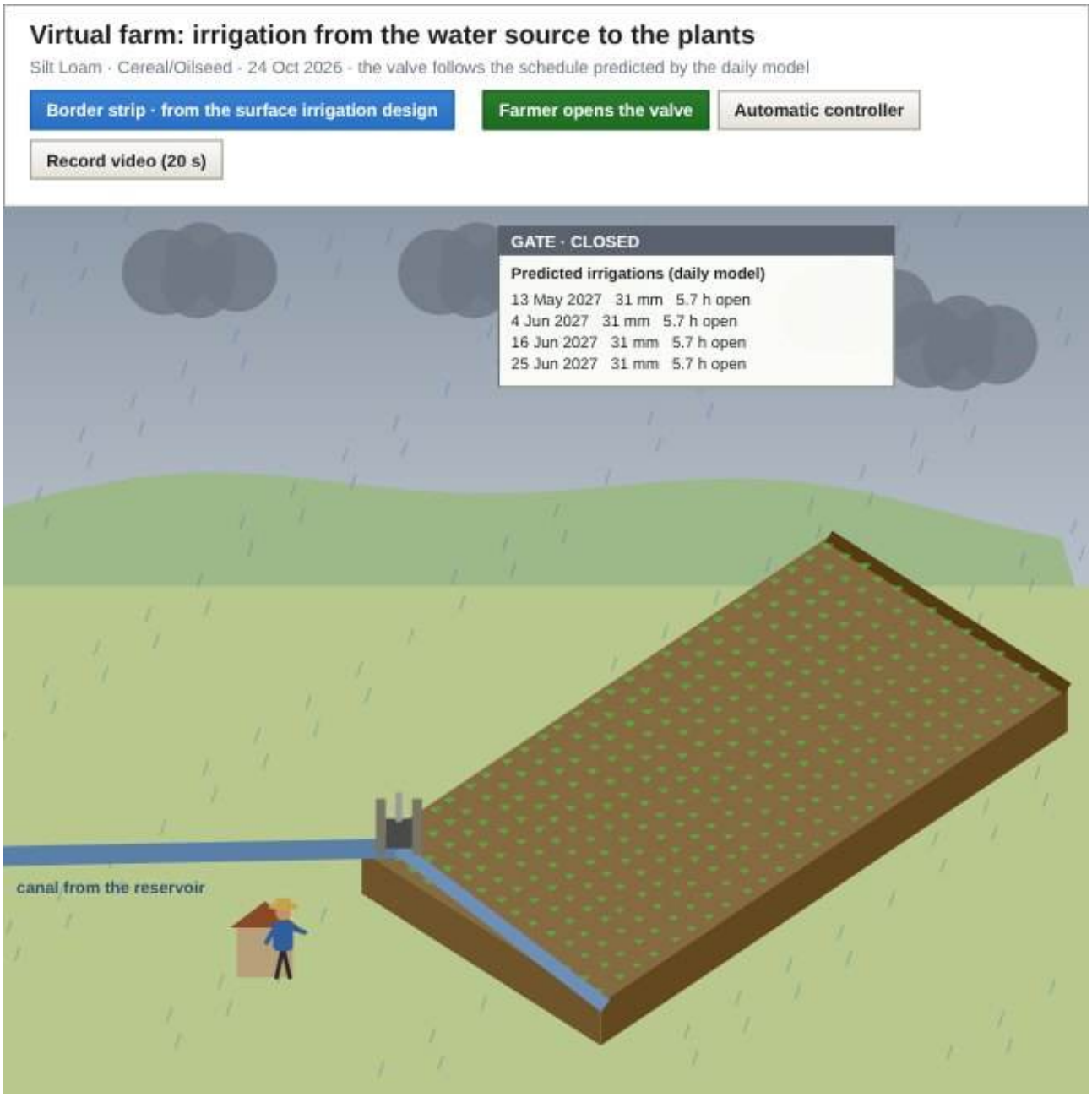


Figure 7.24: Virtual farm, surface design, Silt Loam, part 1 of 3.

Figure 7.25: Virtual farm, surface design, Silt Loam, part 2 of 3.

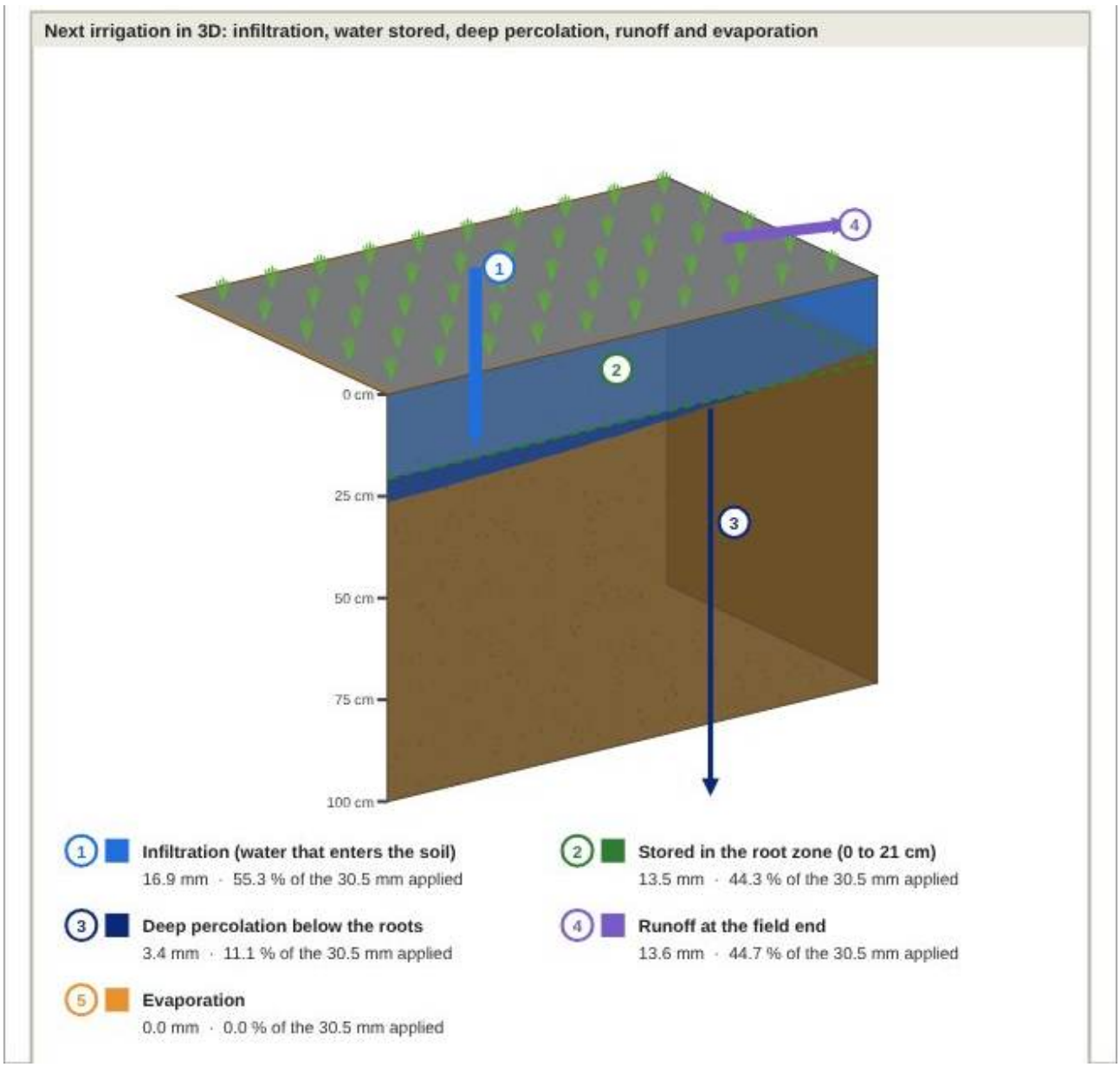


Figure 7.26: Virtual farm, surface design, Silt Loam, part 3 of 3.

7.5 Virtual farm of the drip design (Silt Loam)

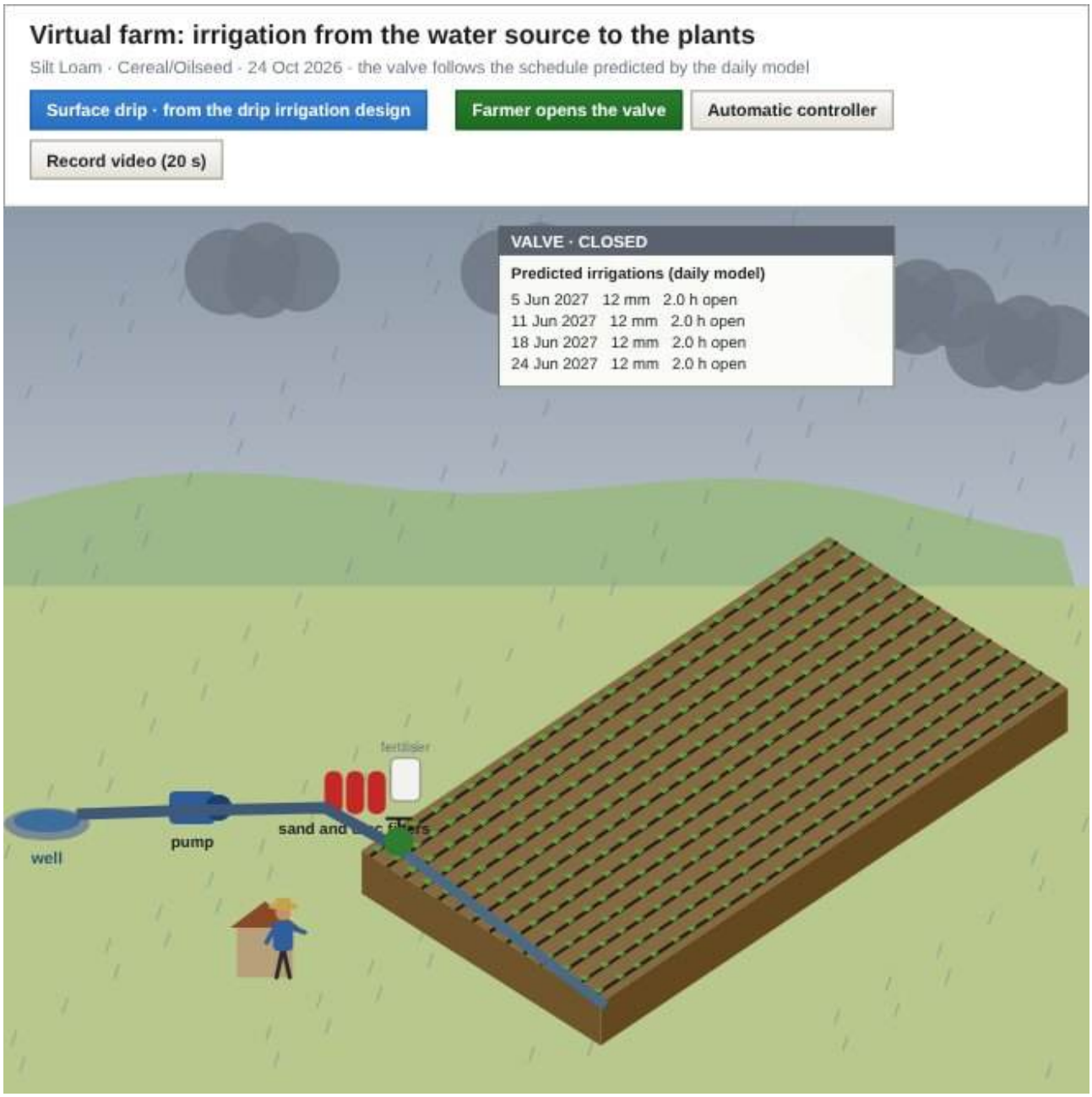


Figure 7.27: Virtual farm, drip design, Silt Loam, part 1 of 3.

Figure 7.28: Virtual farm, drip design, Silt Loam, part 2 of 3.

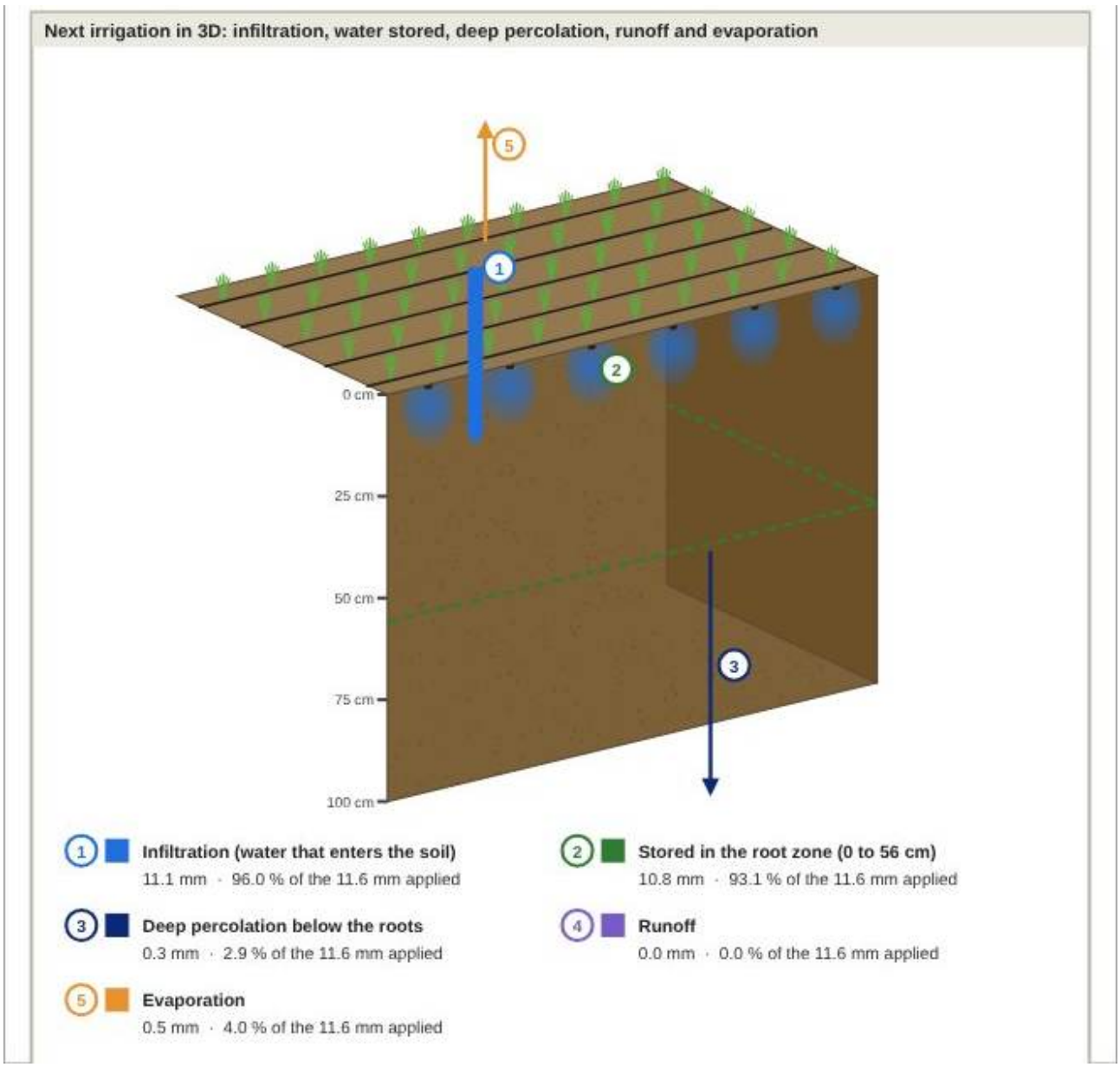


Figure 7.29: Virtual farm, drip design, Silt Loam, part 3 of 3.

7.6 Virtual farm of the sprinkler design (Silt Loam)

Figure 7.30: Virtual farm, sprinkler design, Silt Loam, part 1 of 3.

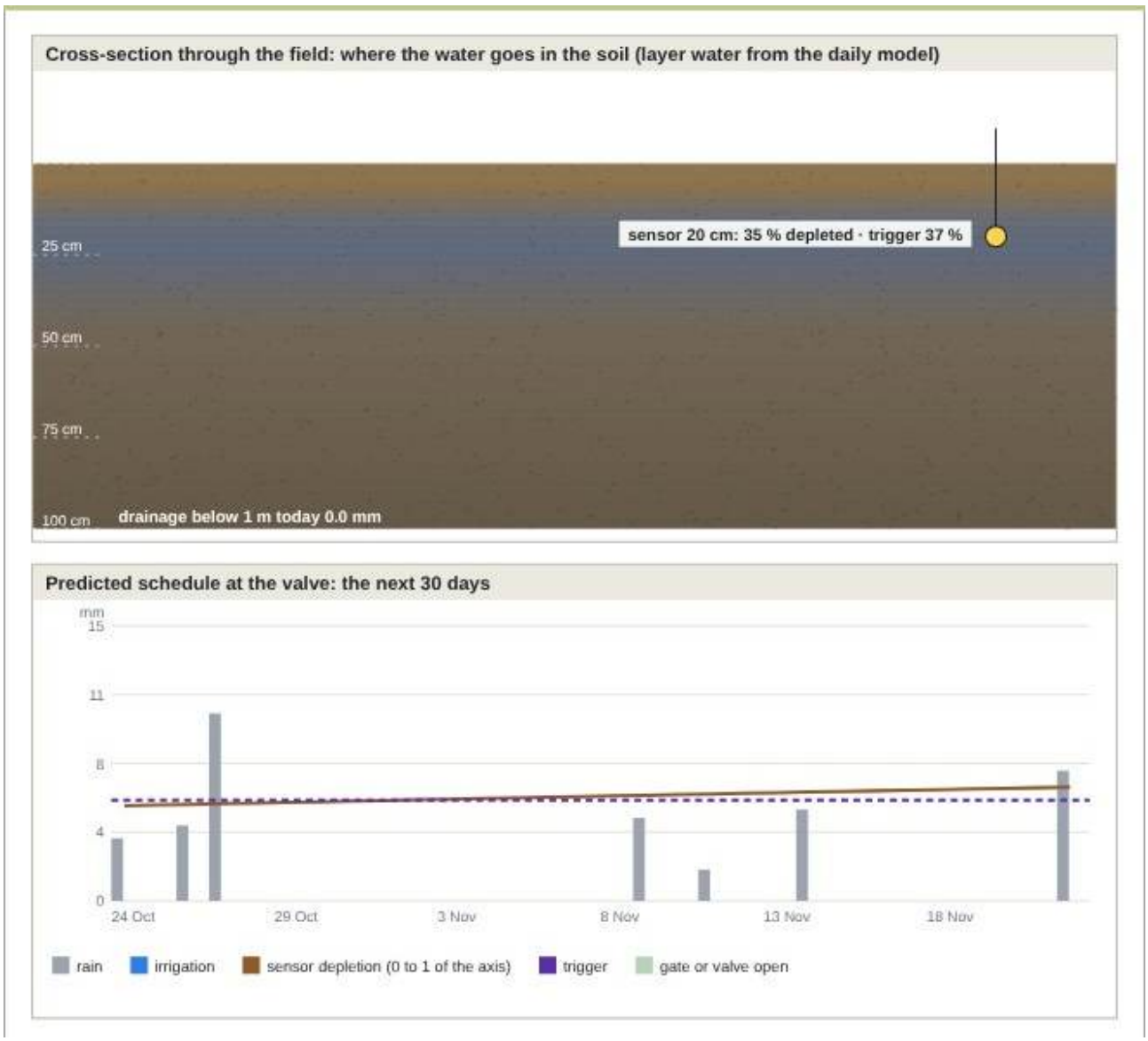


Figure 7.31: Virtual farm, sprinkler design, Silt Loam, part 2 of 3.

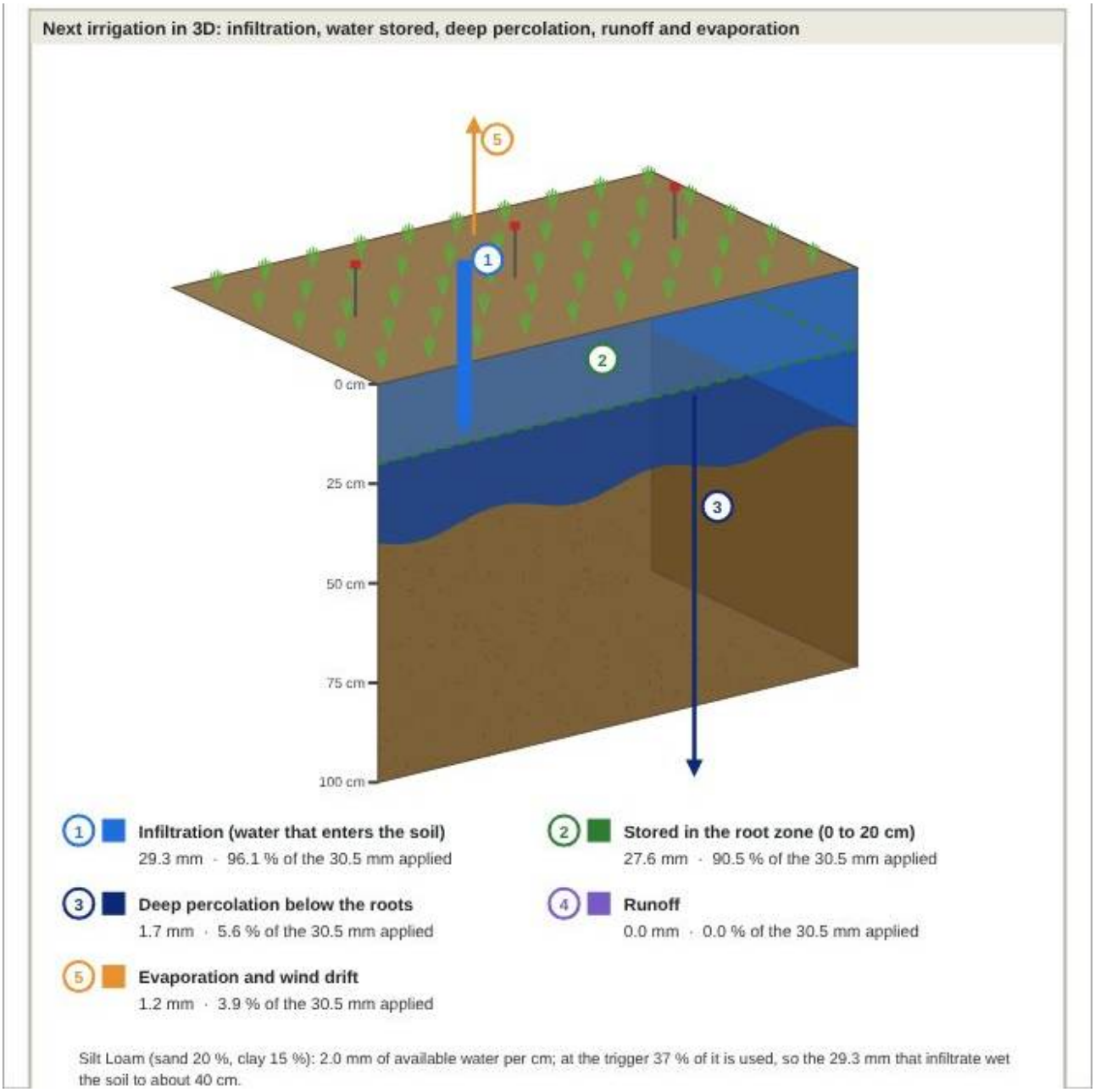


Figure 7.32: Virtual farm, sprinkler design, Silt Loam, part 3 of 3.

7.7 Sensor and trigger (Silt Loam)

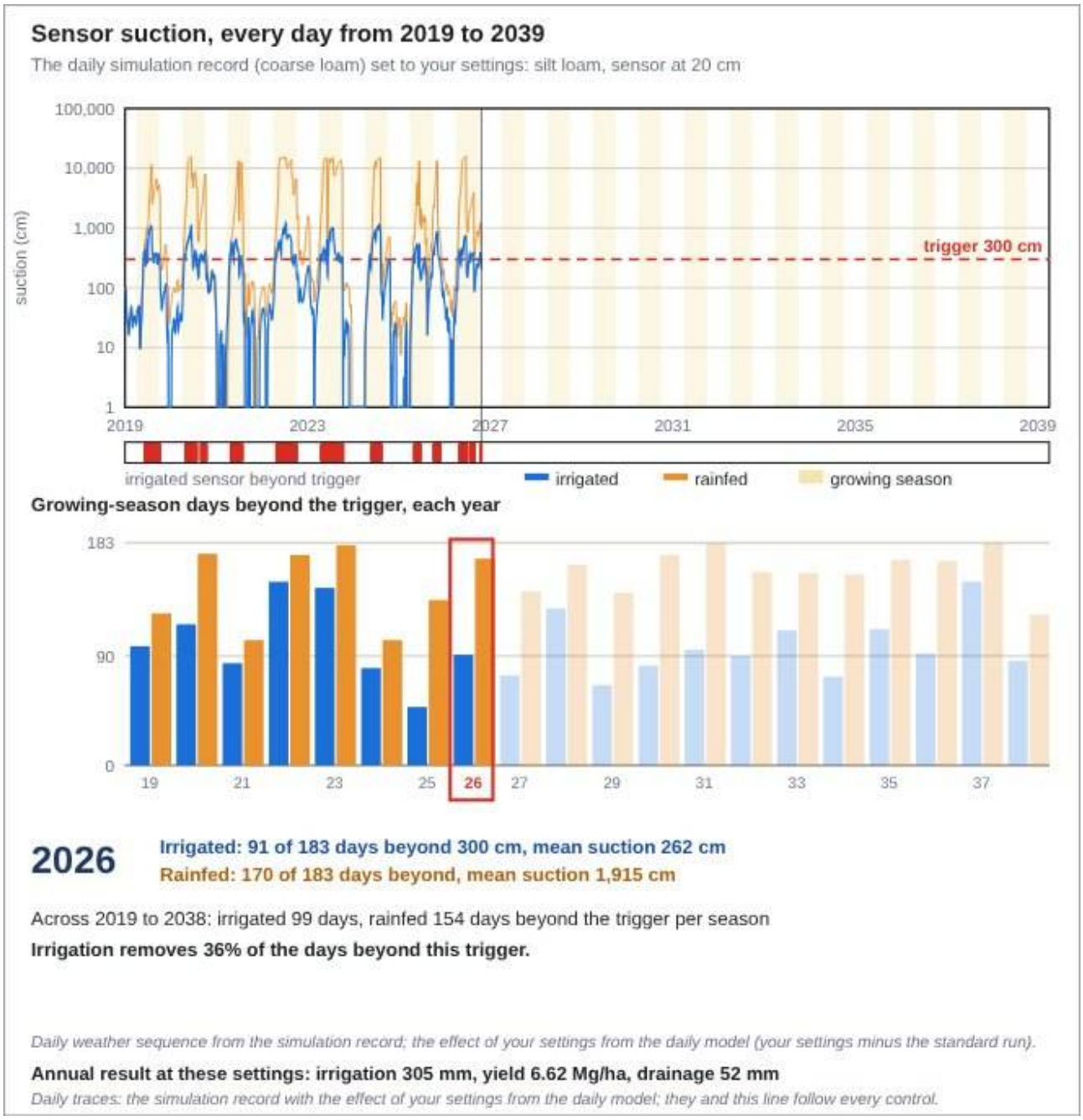


Figure 7.33: Sensor and trigger, Silt Loam, part 1 of 1.

7.8 Trigger law (Silt Loam)

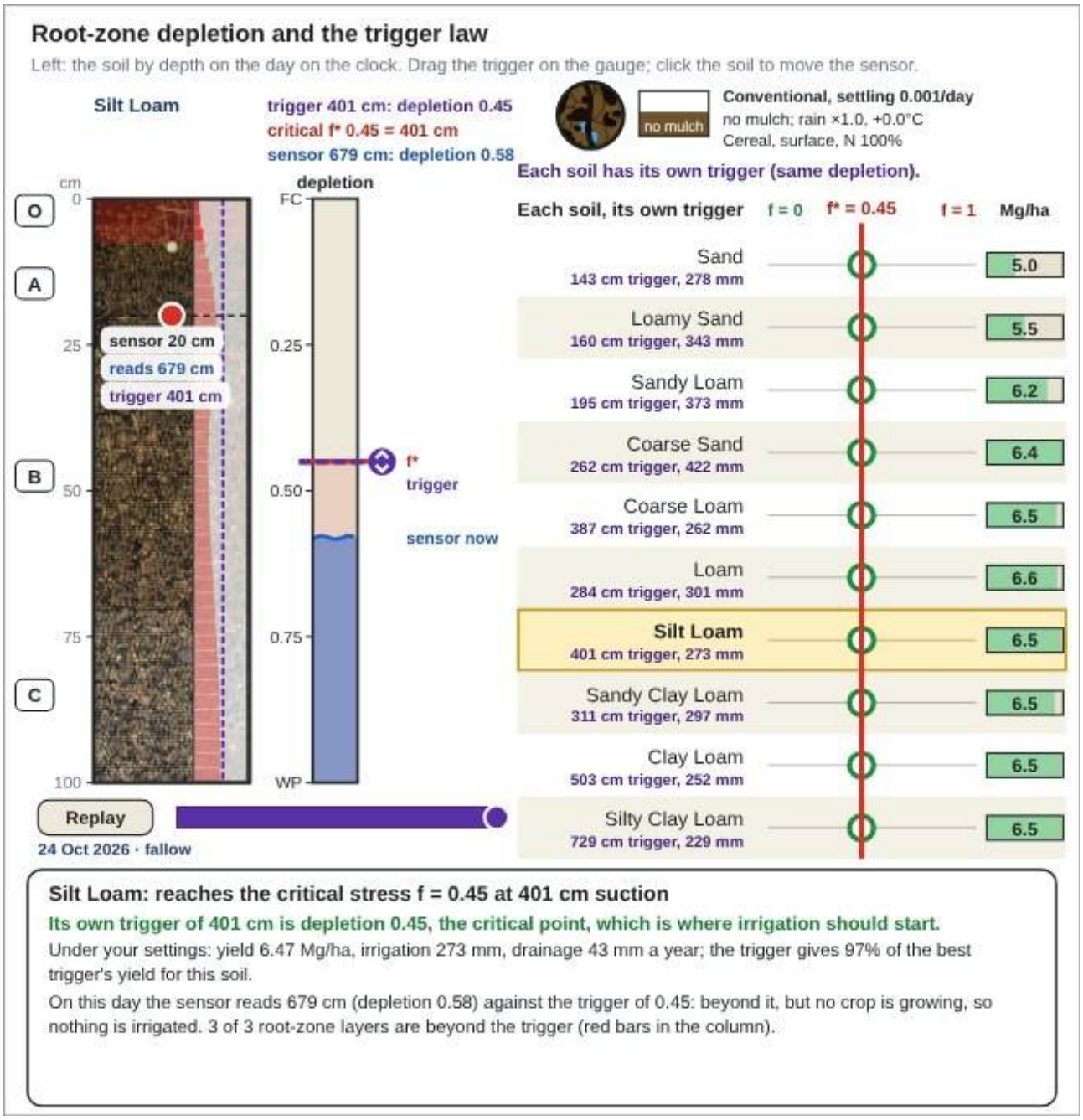


Figure 7.34: Trigger law, Silt Loam, part 1 of 1.

7.9 Mulch against bare soil (Silt Loam)

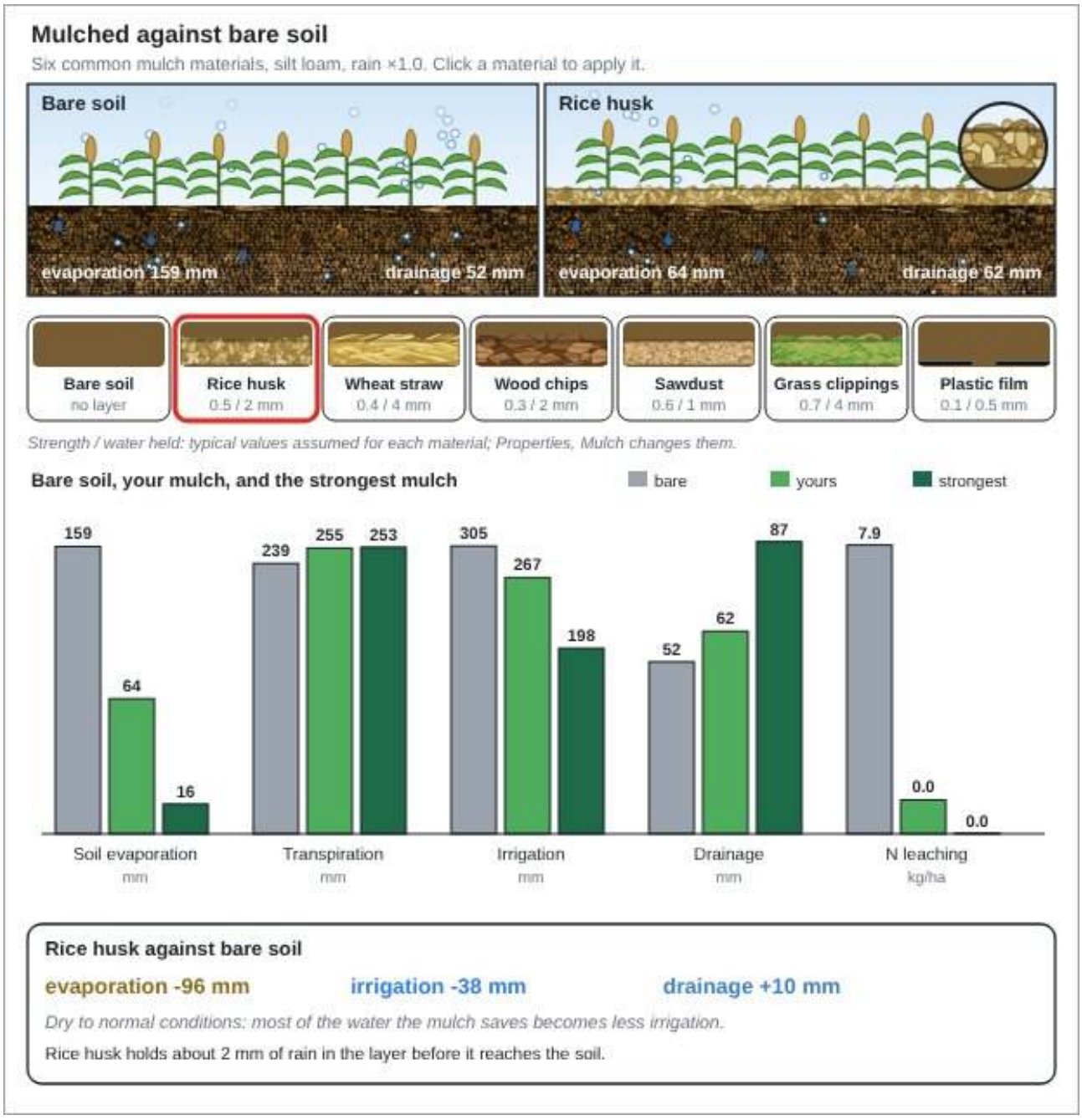


Figure 7.35: Mulch against bare soil, Silt Loam, part 1 of 1.

7.10 Mulch control (Silt Loam)

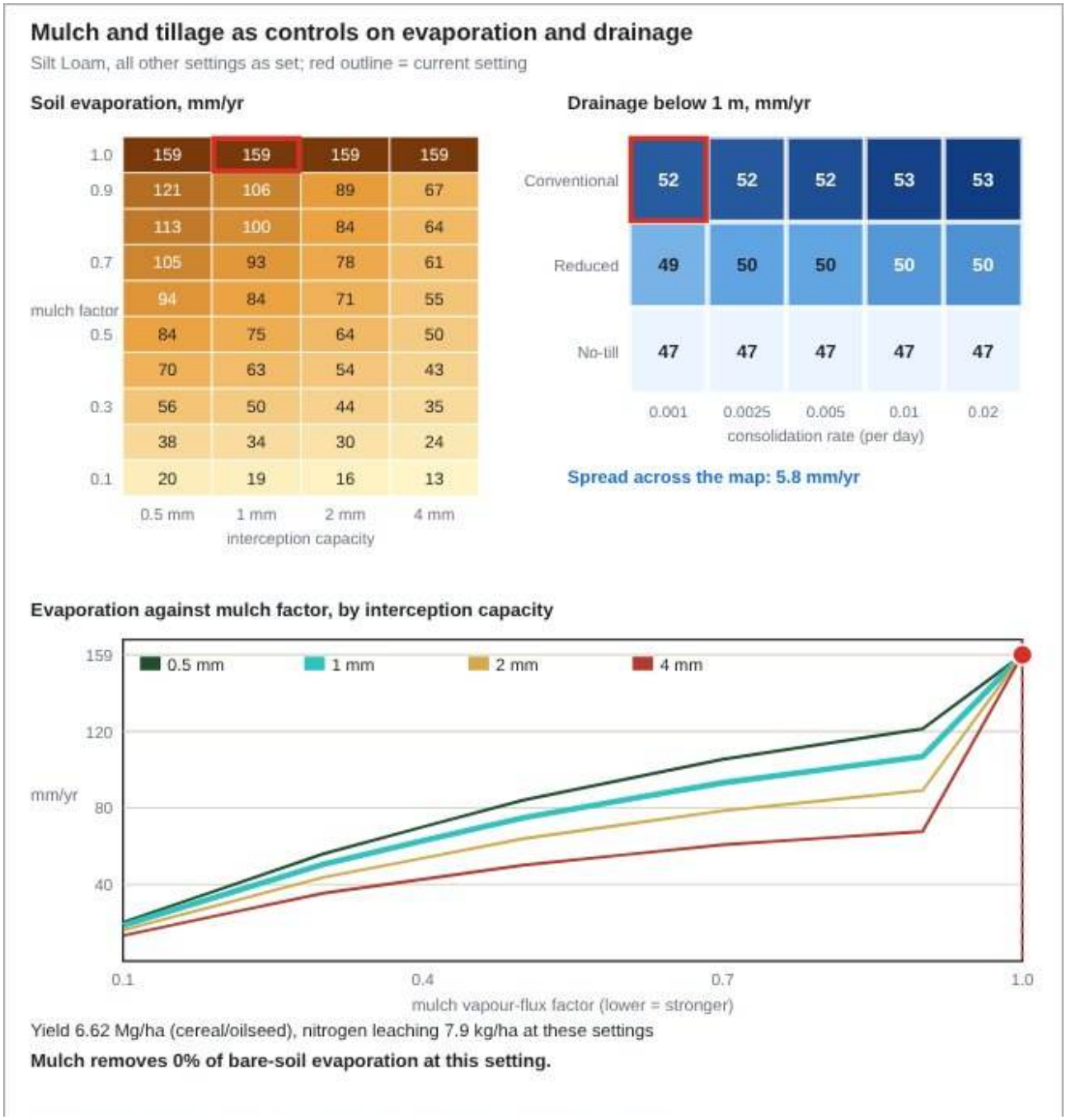


Figure 7.36: Mulch control, Silt Loam, part 1 of 2.

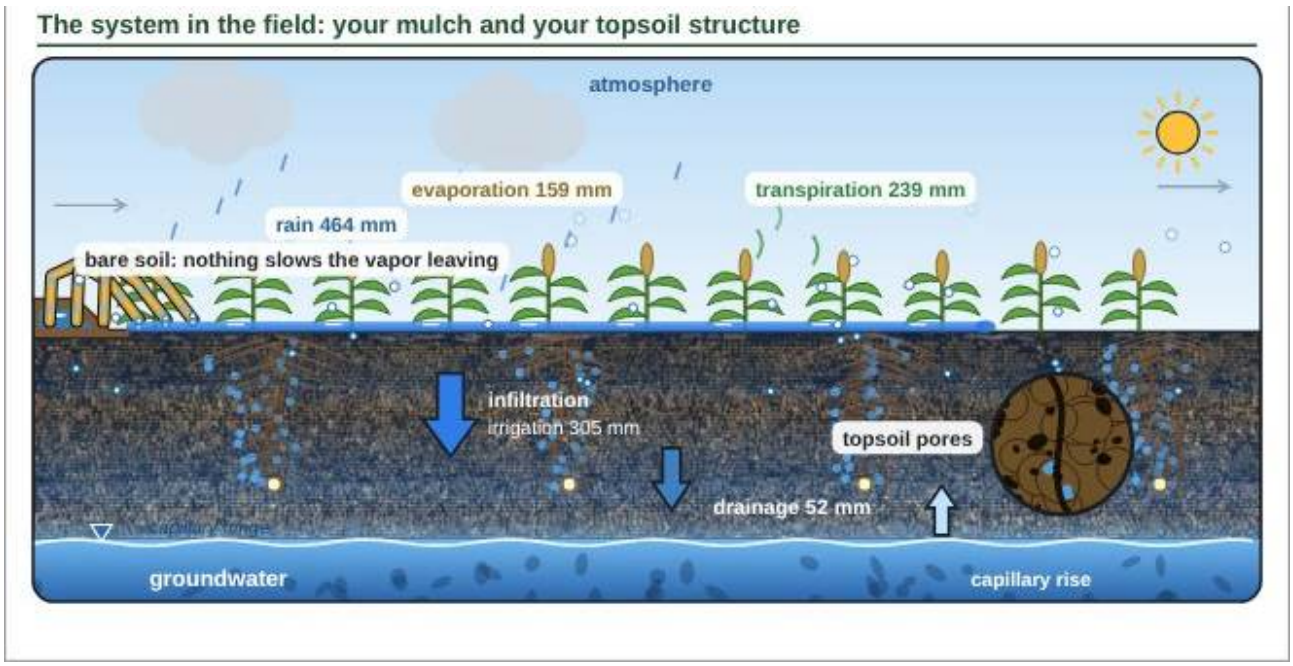


Figure 7.37: Mulch control, Silt Loam, part 2 of 2.

7.11 Evaporation physics (Silt Loam)

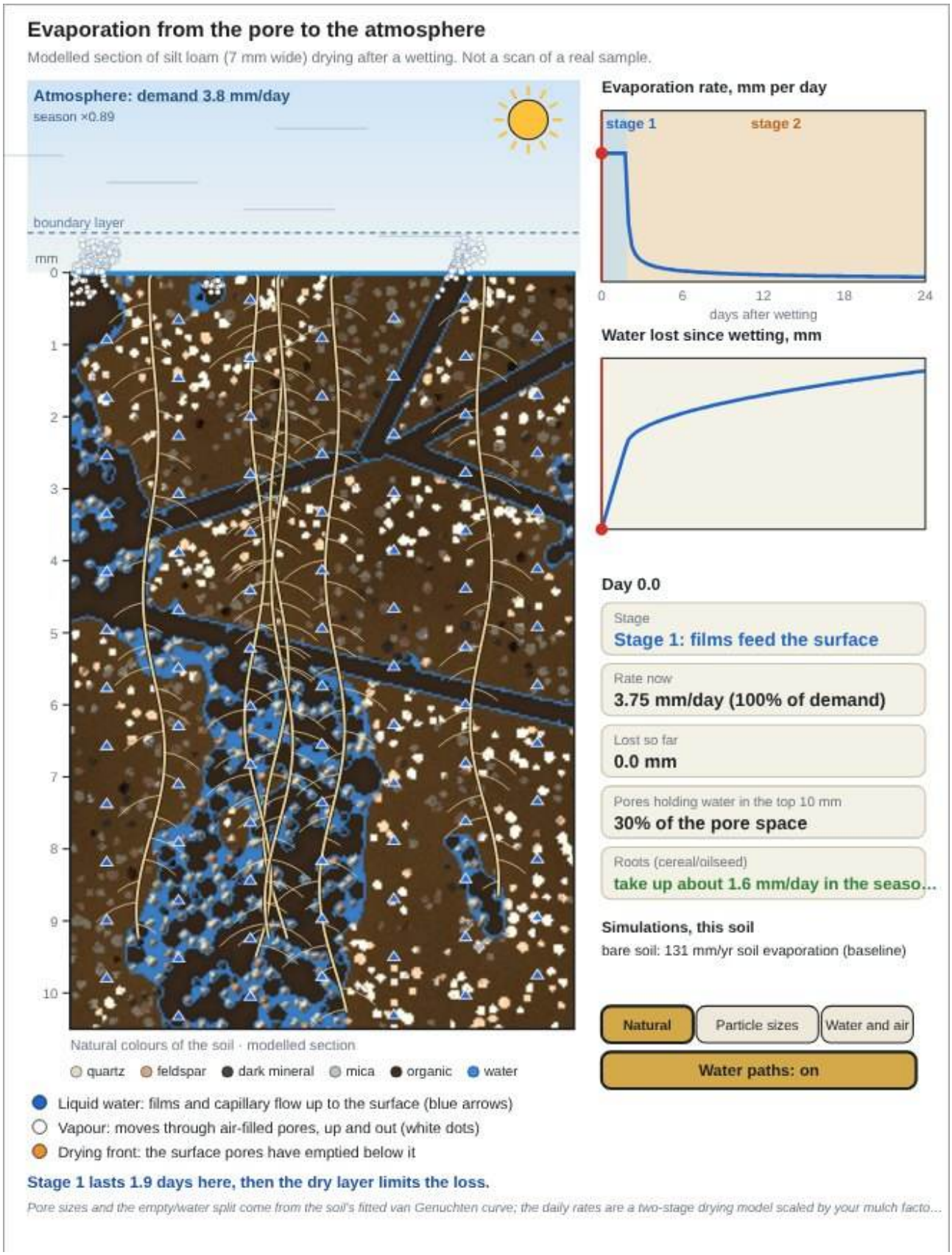


Figure 7.38: Evaporation physics, Silt Loam, part 1 of 3.

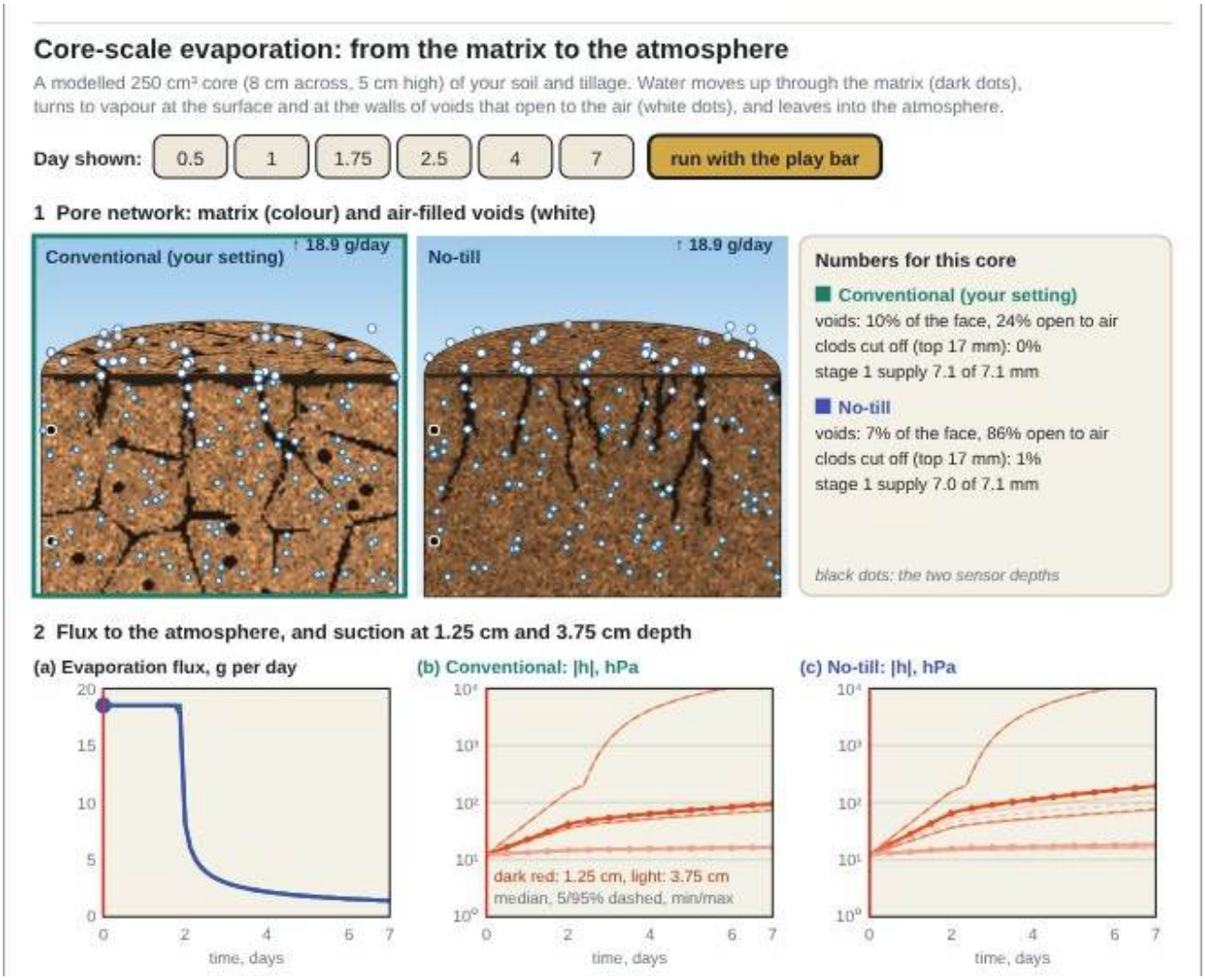


Figure 7.39: Evaporation physics, Silt Loam, part 2 of 3.

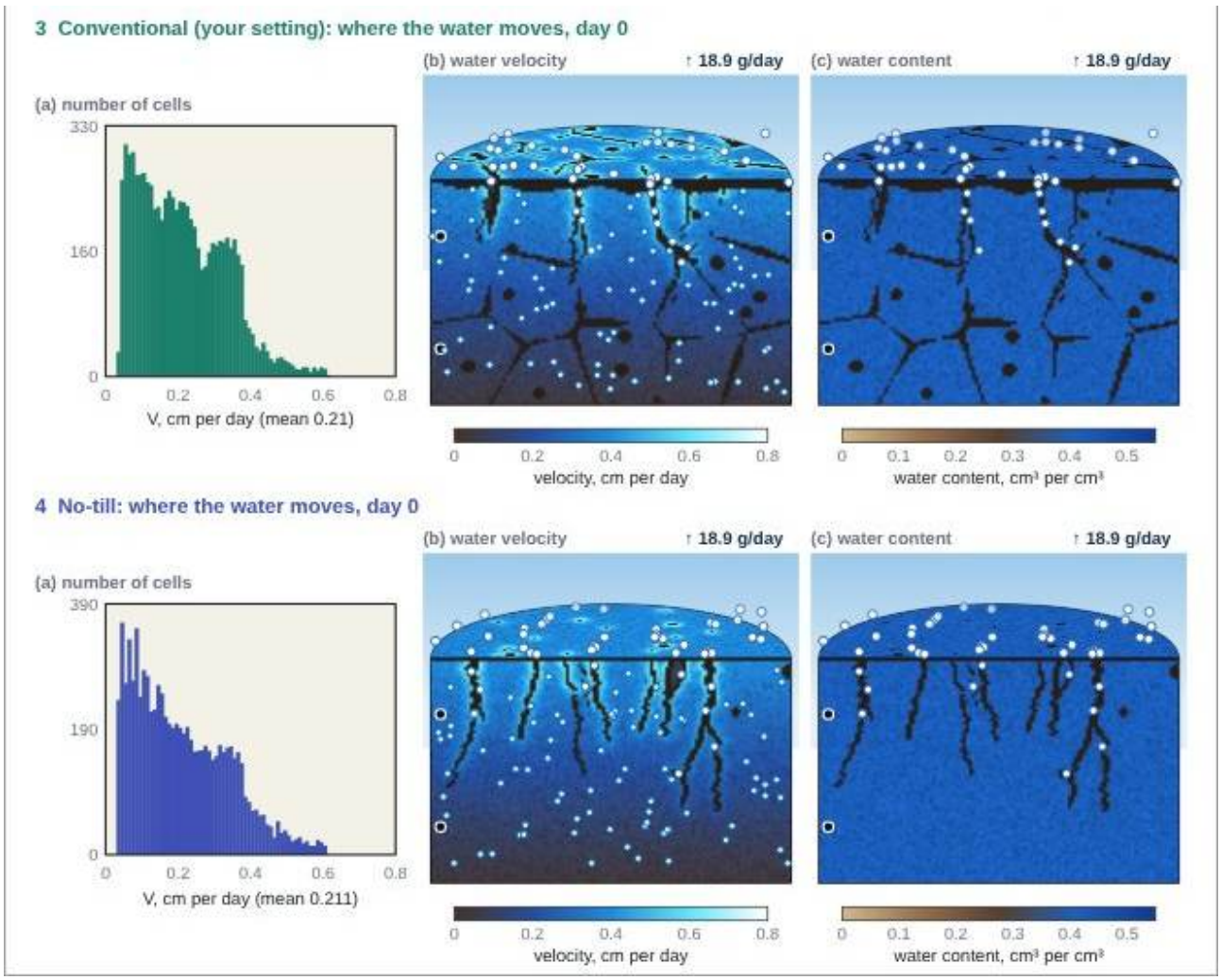


Figure 7.40: Evaporation physics, Silt Loam, part 3 of 3.

7.12 Fate of saved water (Silt Loam)

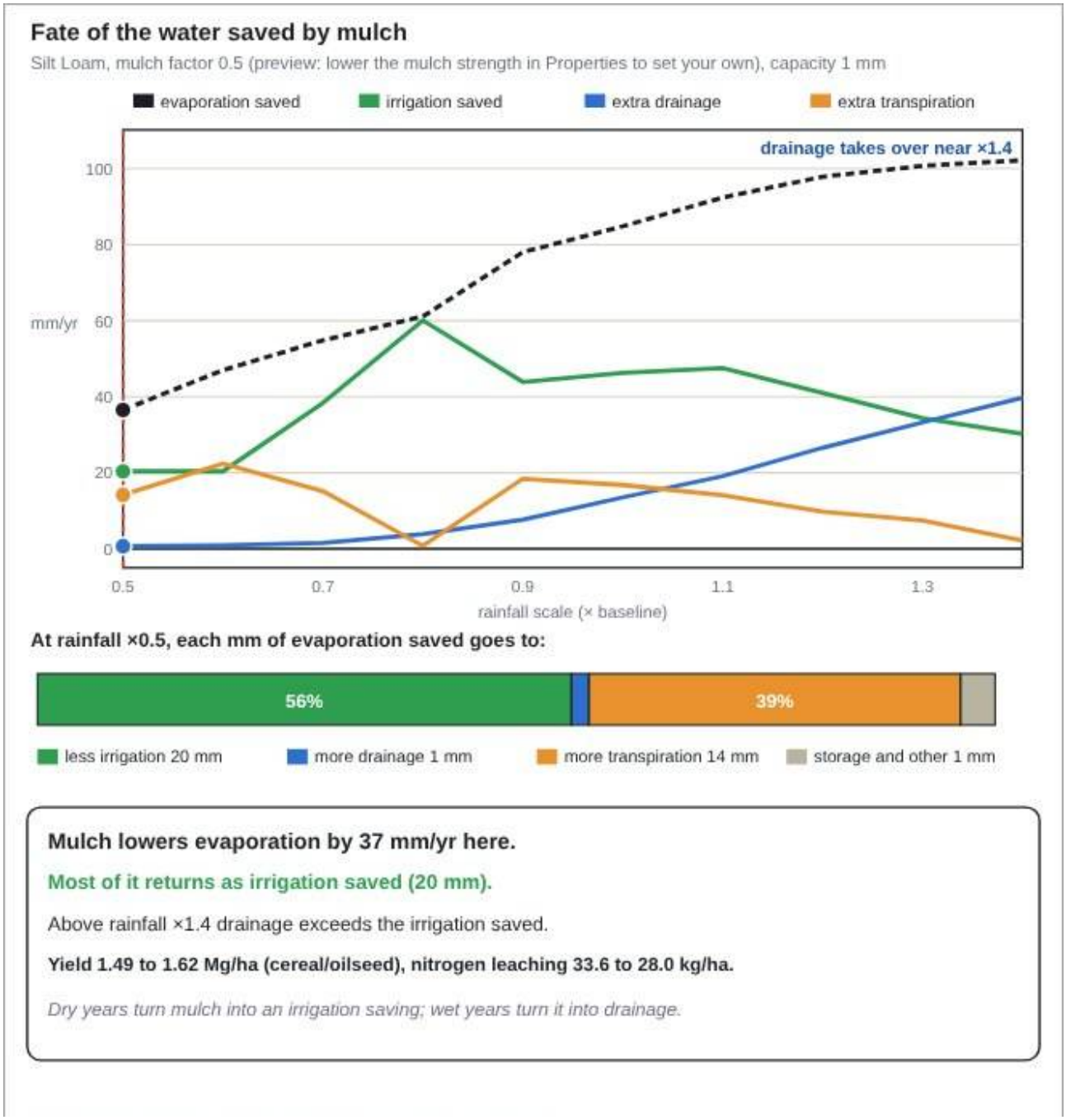


Figure 7.41: Fate of saved water, Silt Loam, part 1 of 2.

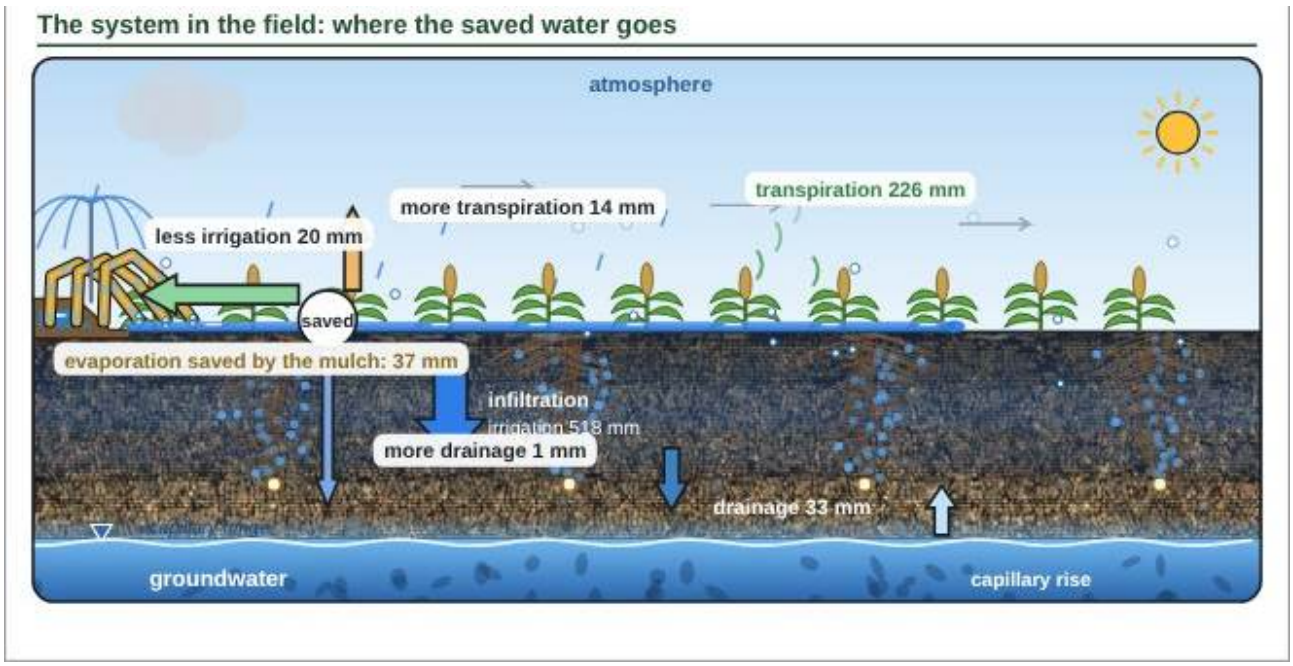


Figure 7.42: Fate of saved water, Silt Loam, part 2 of 2.

7.13 Regime map (Silt Loam)

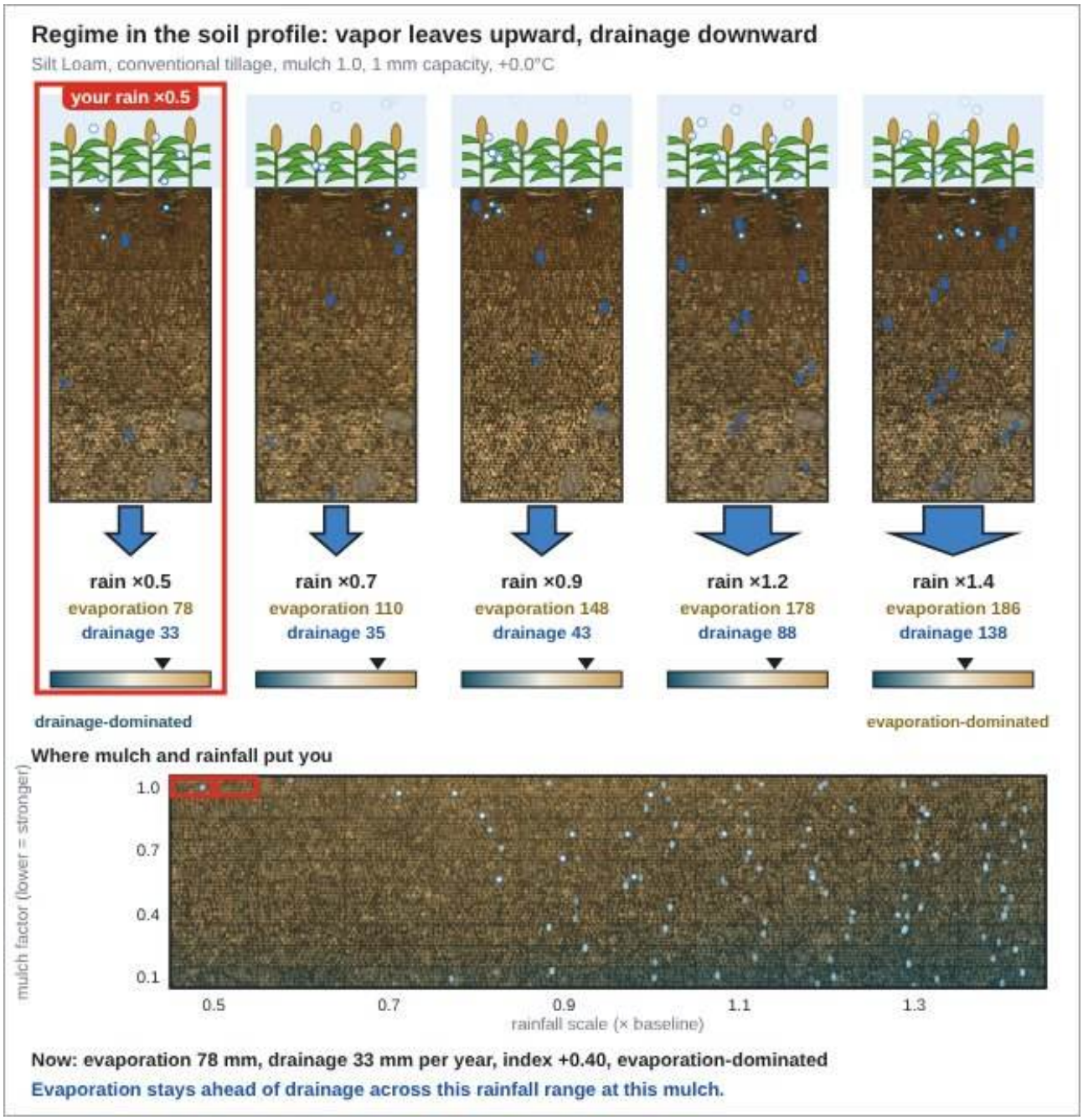


Figure 7.43: Regime map, Silt Loam, part 1 of 1.

7.14 Tillage and structure (Silt Loam)

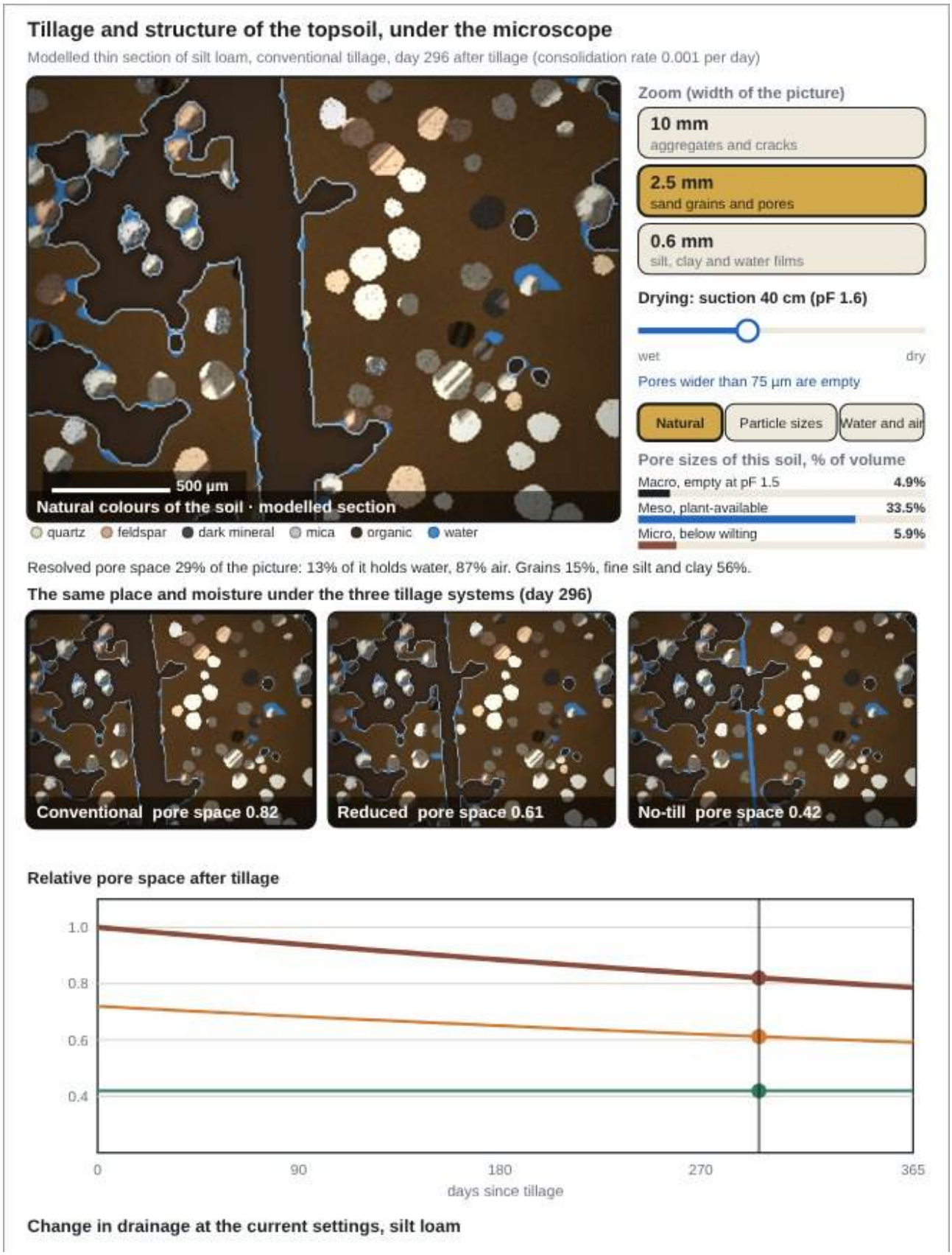


Figure 7.44: Tillage and structure, Silt Loam, part 1 of 2.

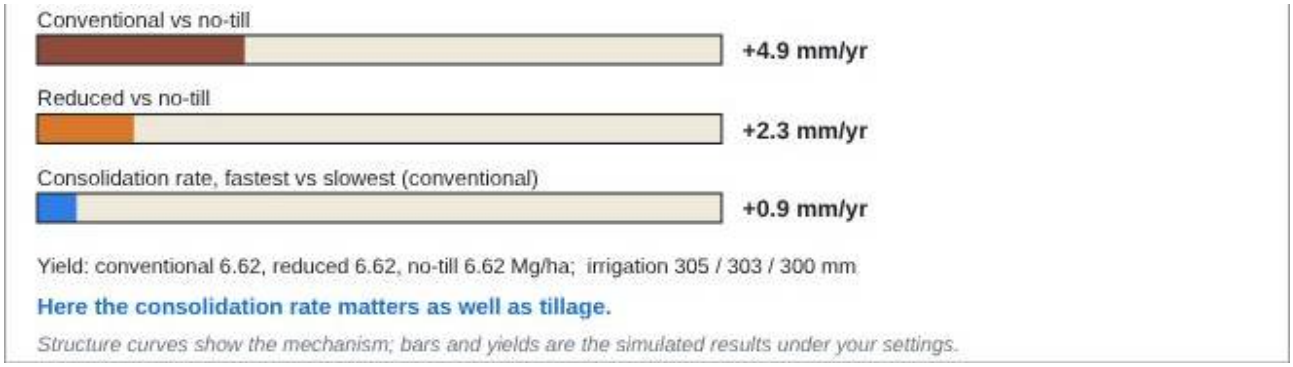


Figure 7.45: Tillage and structure, Silt Loam, part 2 of 2.

7.15 Soil hydraulic drivers (Silt Loam)

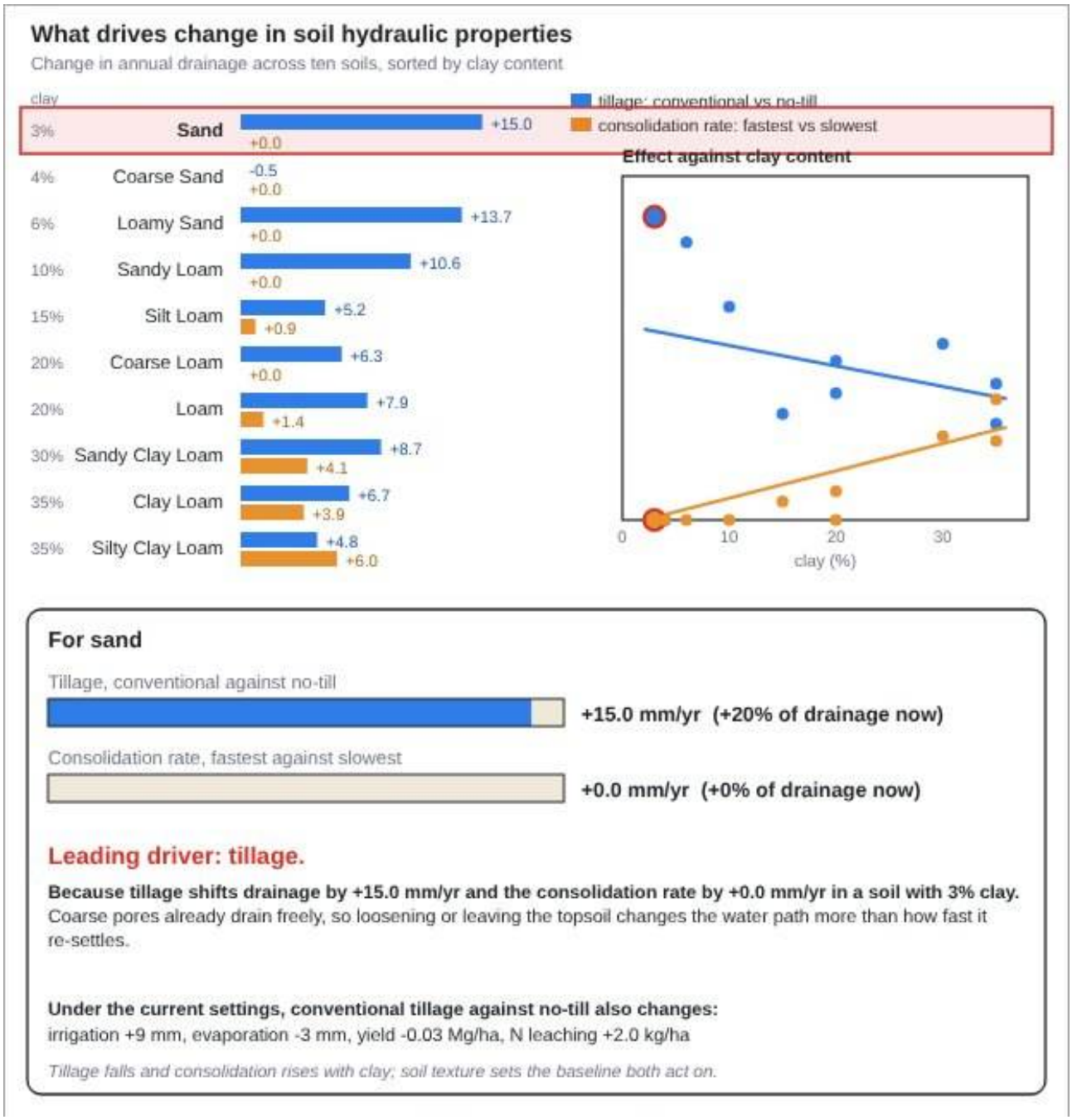


Figure 7.46: Soil hydraulic drivers, Silt Loam, part 1 of 3.

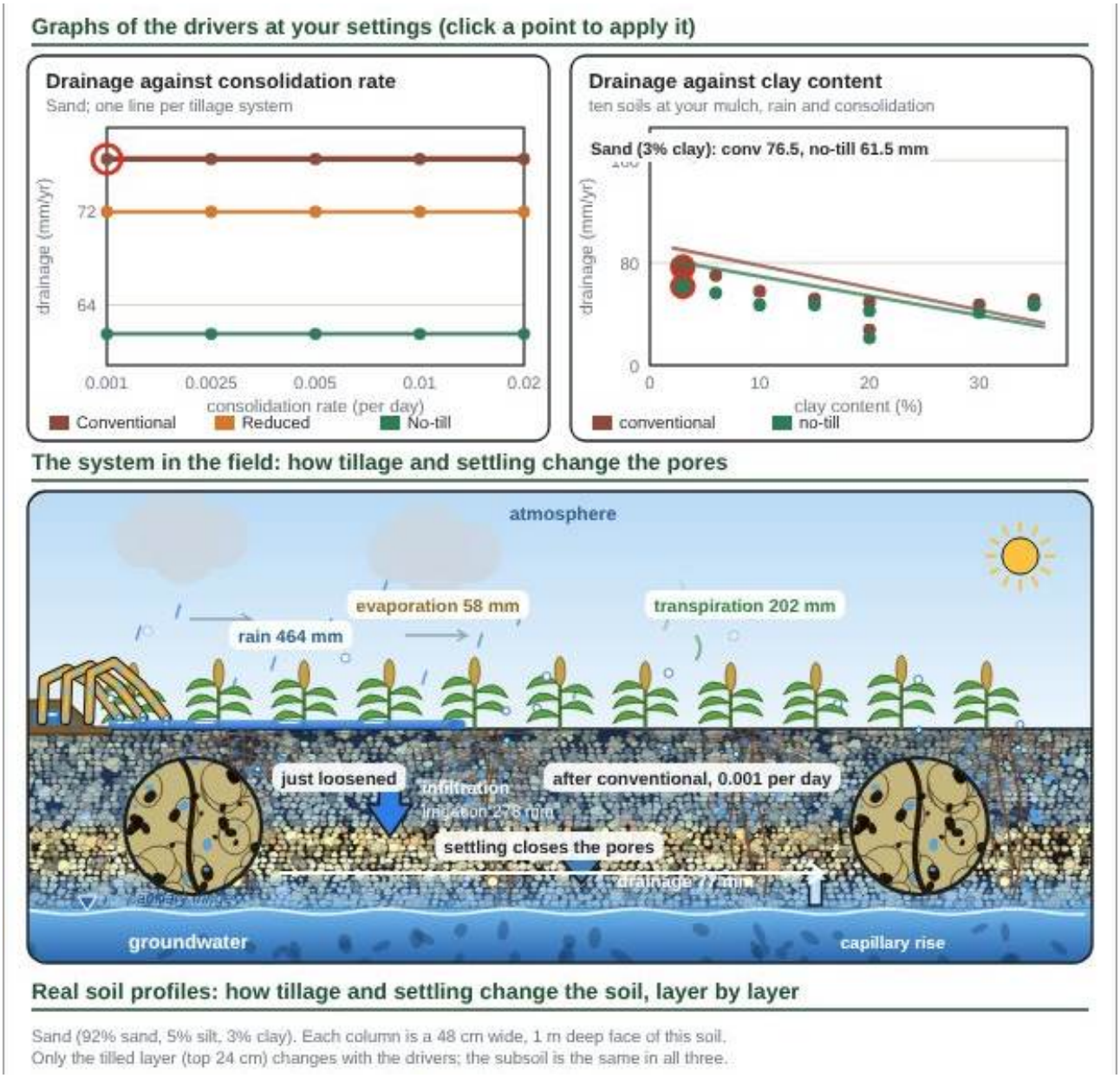


Figure 7.47: Soil hydraulic drivers, Silt Loam, part 2 of 3.

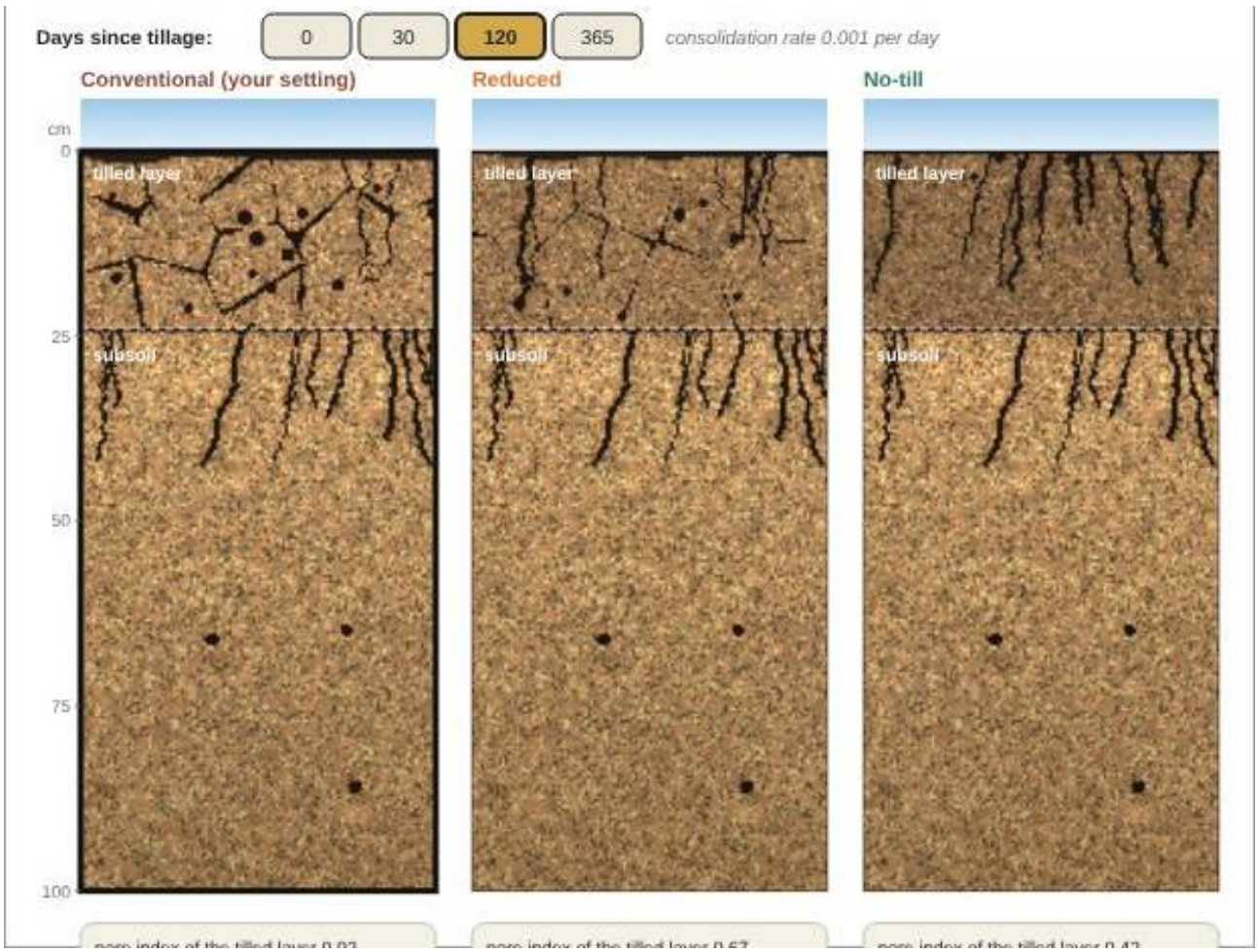


Figure 7.48: Soil hydraulic drivers, Silt Loam, part 3 of 3.

7.16 Root structure (Silt Loam)

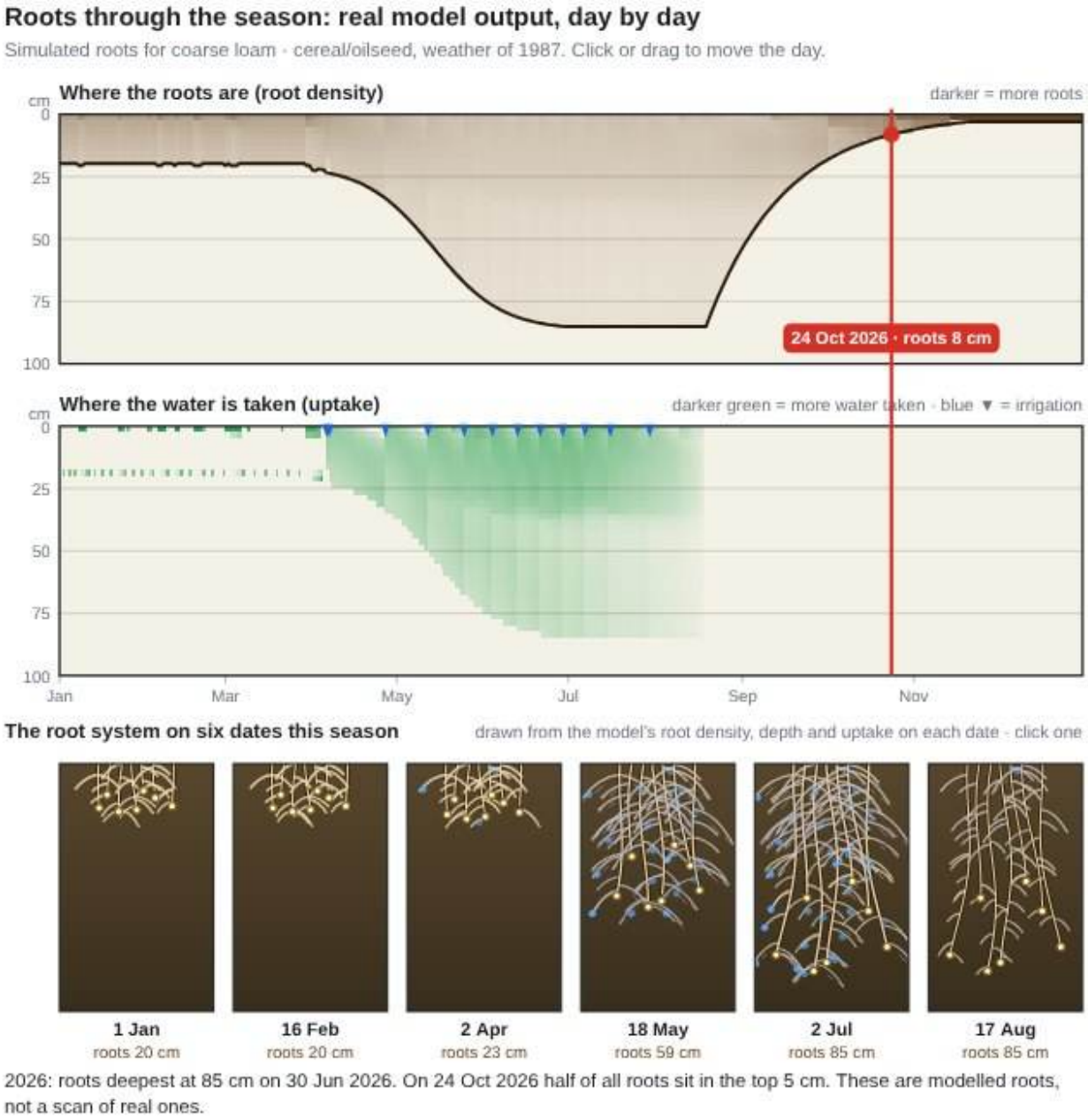


Figure 7.49: Root structure, Silt Loam, part 1 of 4.

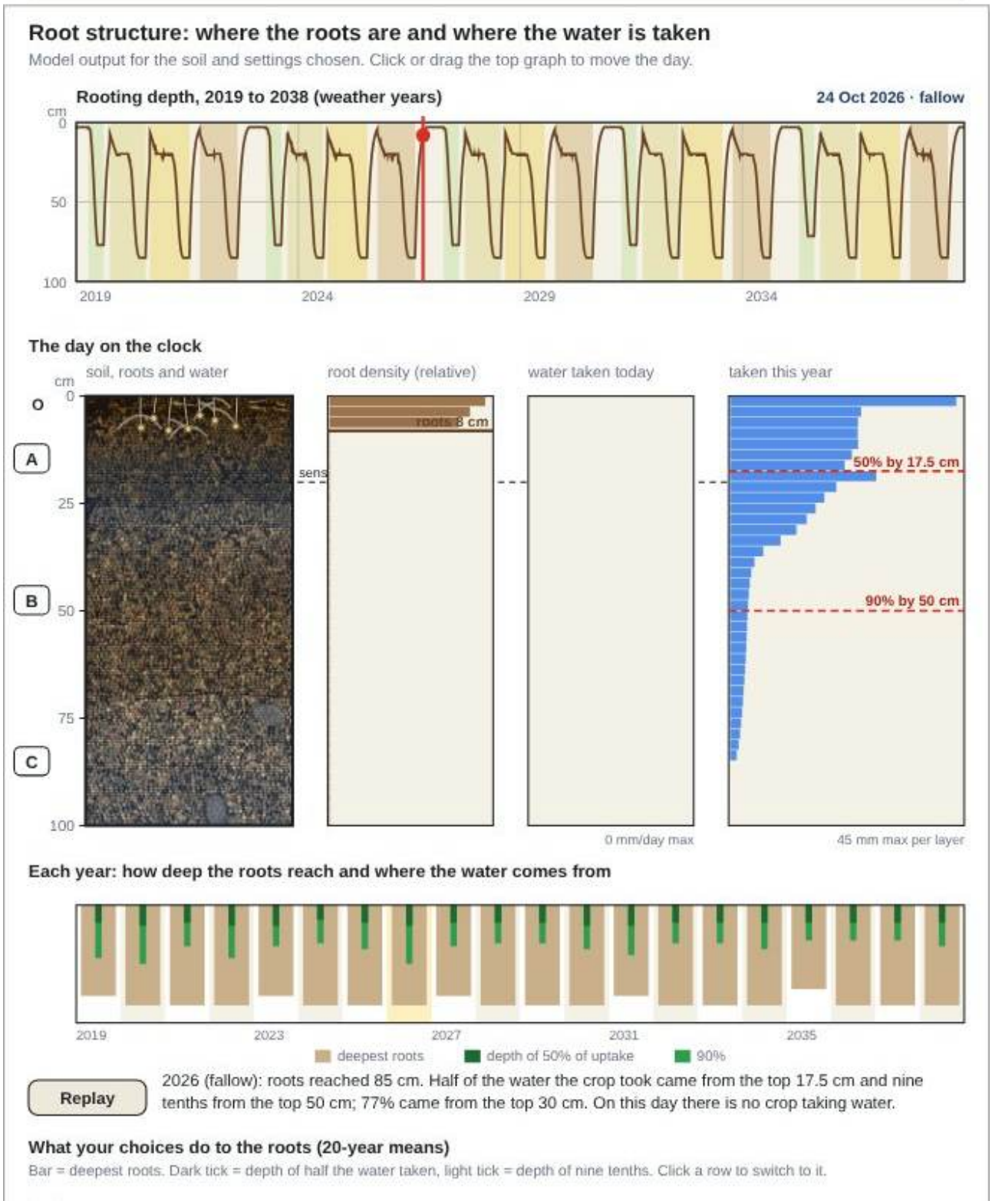


Figure 7.50: Root structure, Silt Loam, part 2 of 4.

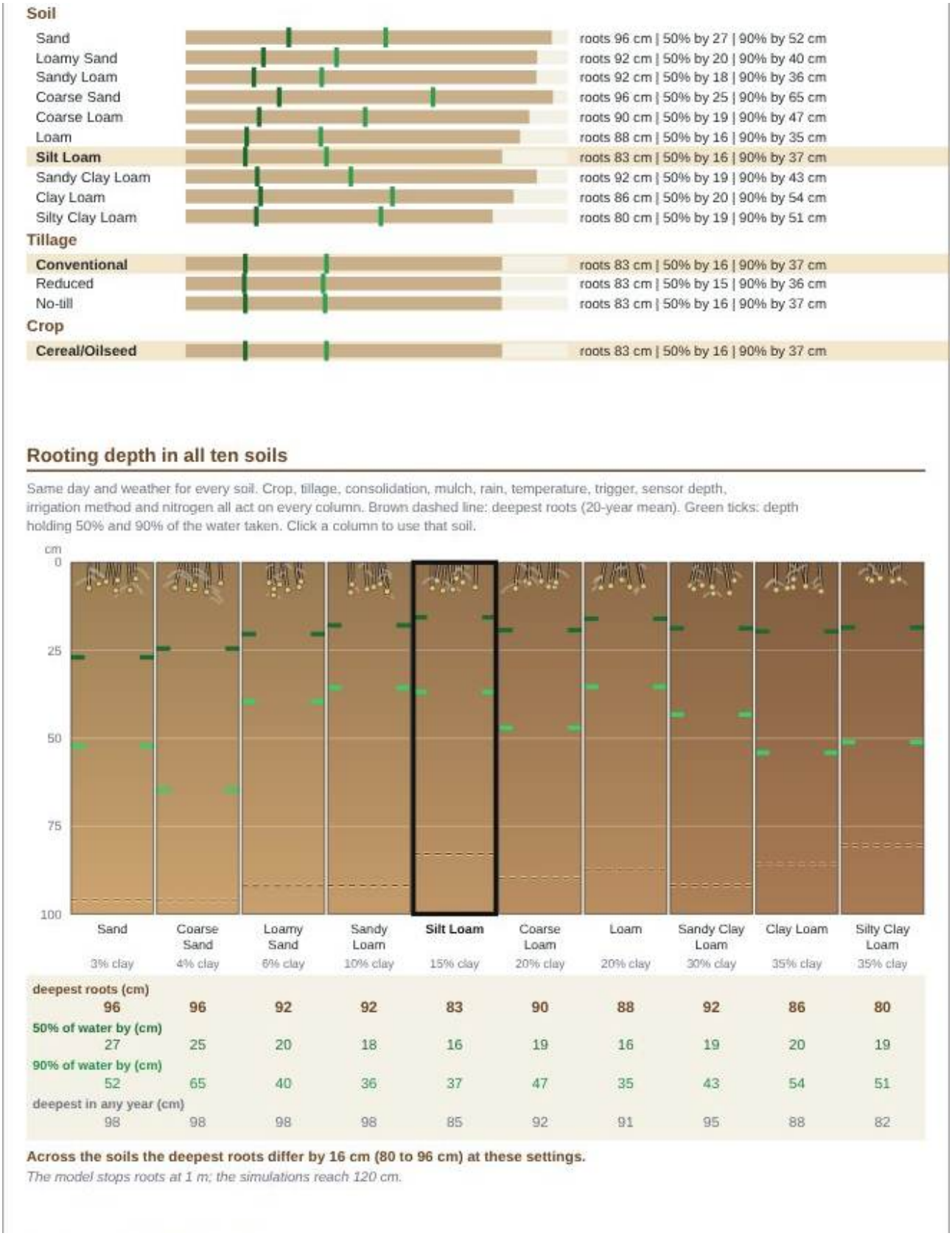


Figure 7.51: Root structure, Silt Loam, part 3 of 4.

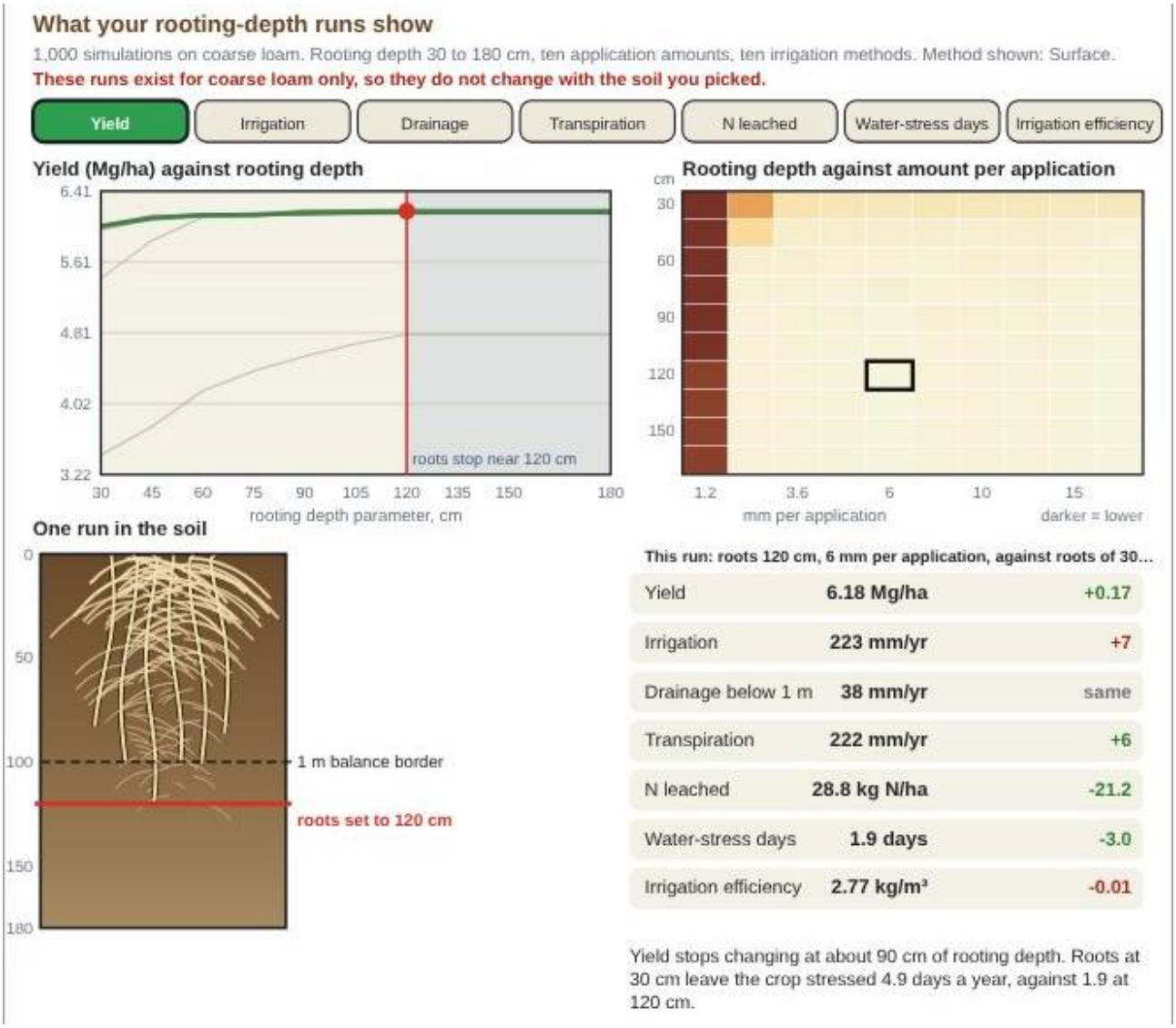


Figure 7.52: Root structure, Silt Loam, part 4 of 4.

7.17 Field balance (Silt Loam)

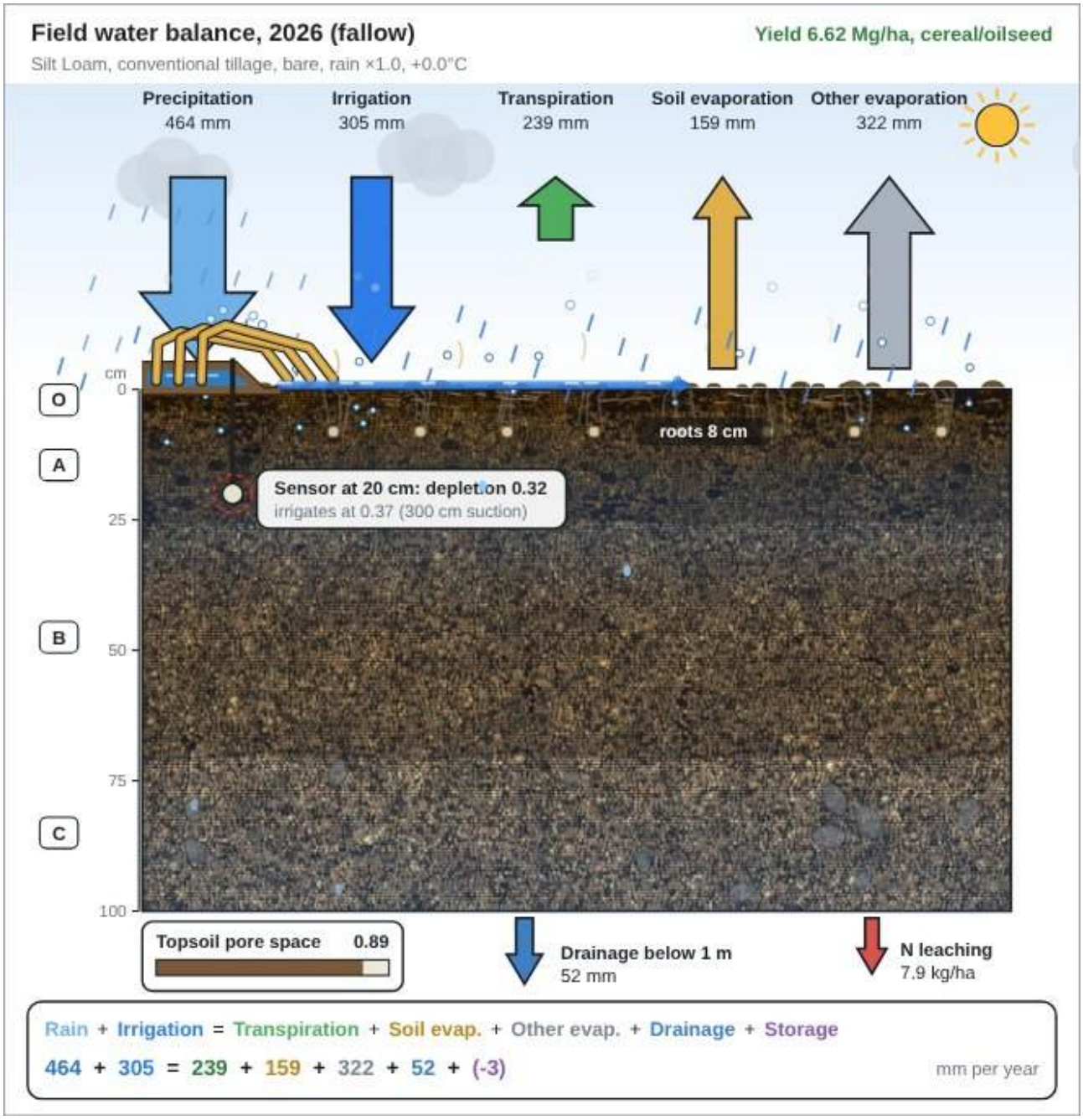


Figure 7.53: Field balance, Silt Loam, part 1 of 1.

7.18 Soil, plant and air (Silt Loam)

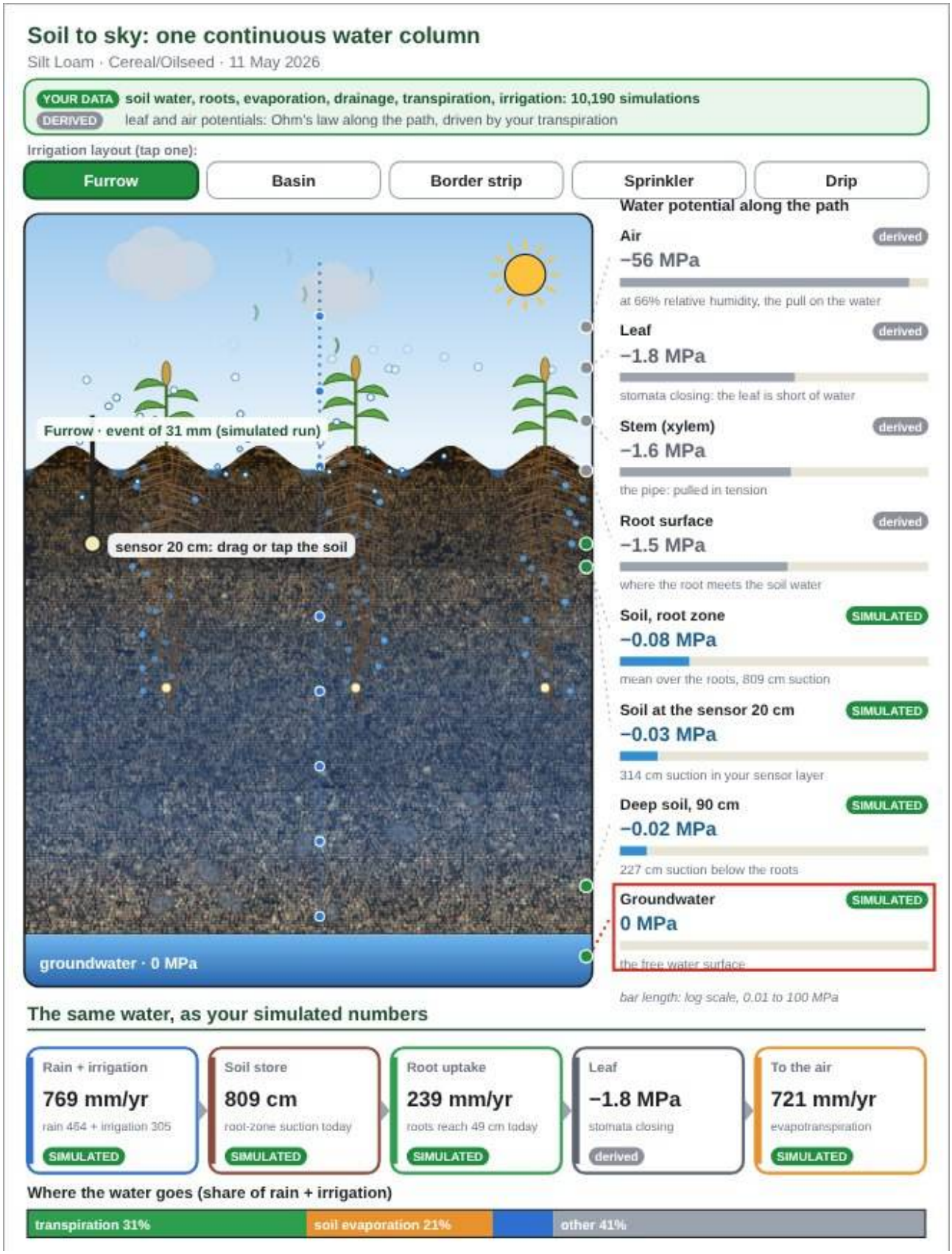


Figure 7.54: Soil, plant and air, Silt Loam, part 1 of 3.

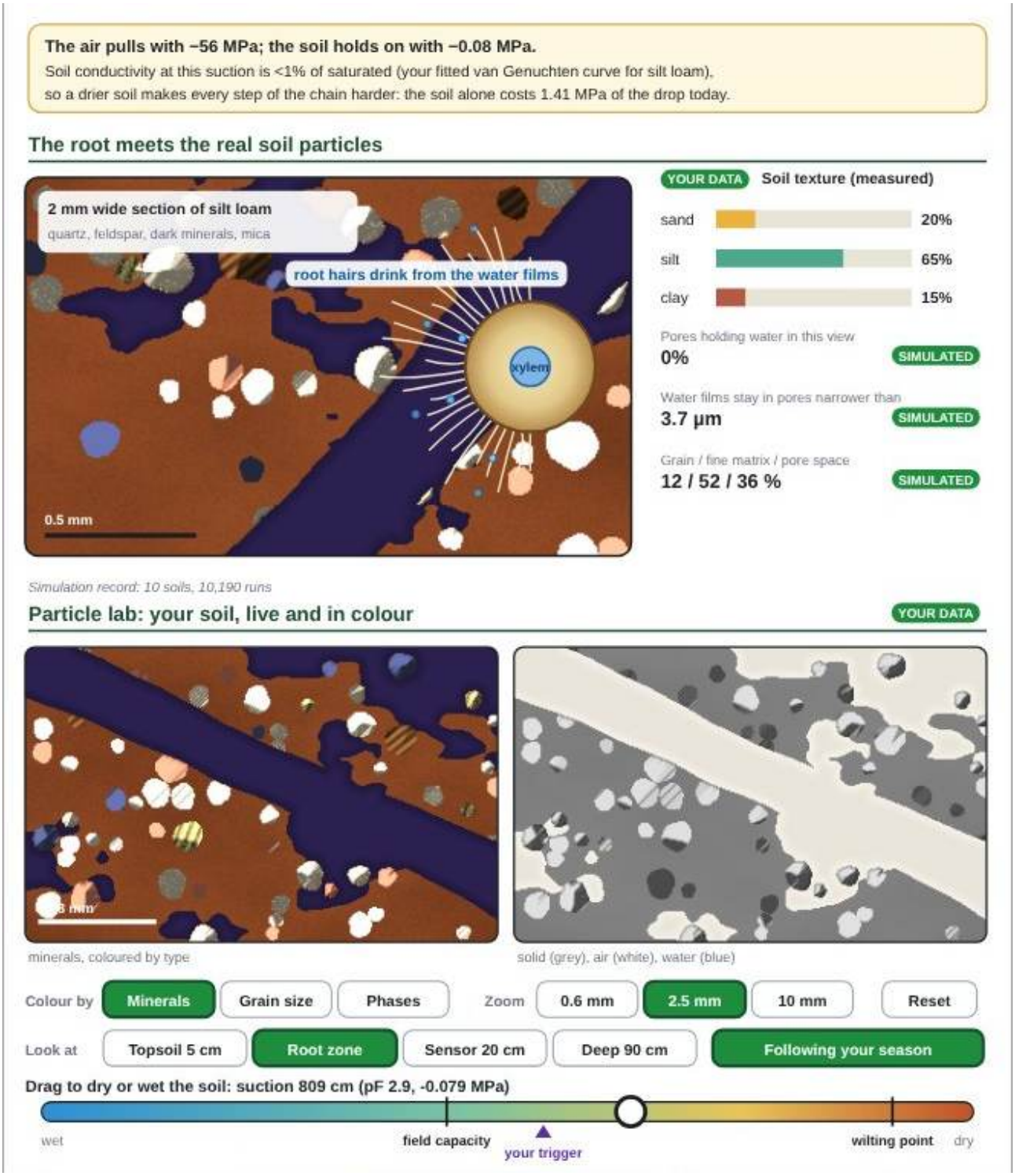


Figure 7.55: Soil, plant and air, Silt Loam, part 2 of 3.

Figure 7.56: Soil, plant and air, Silt Loam, part 3 of 3.

7.19 Profile evolution (Silt Loam)

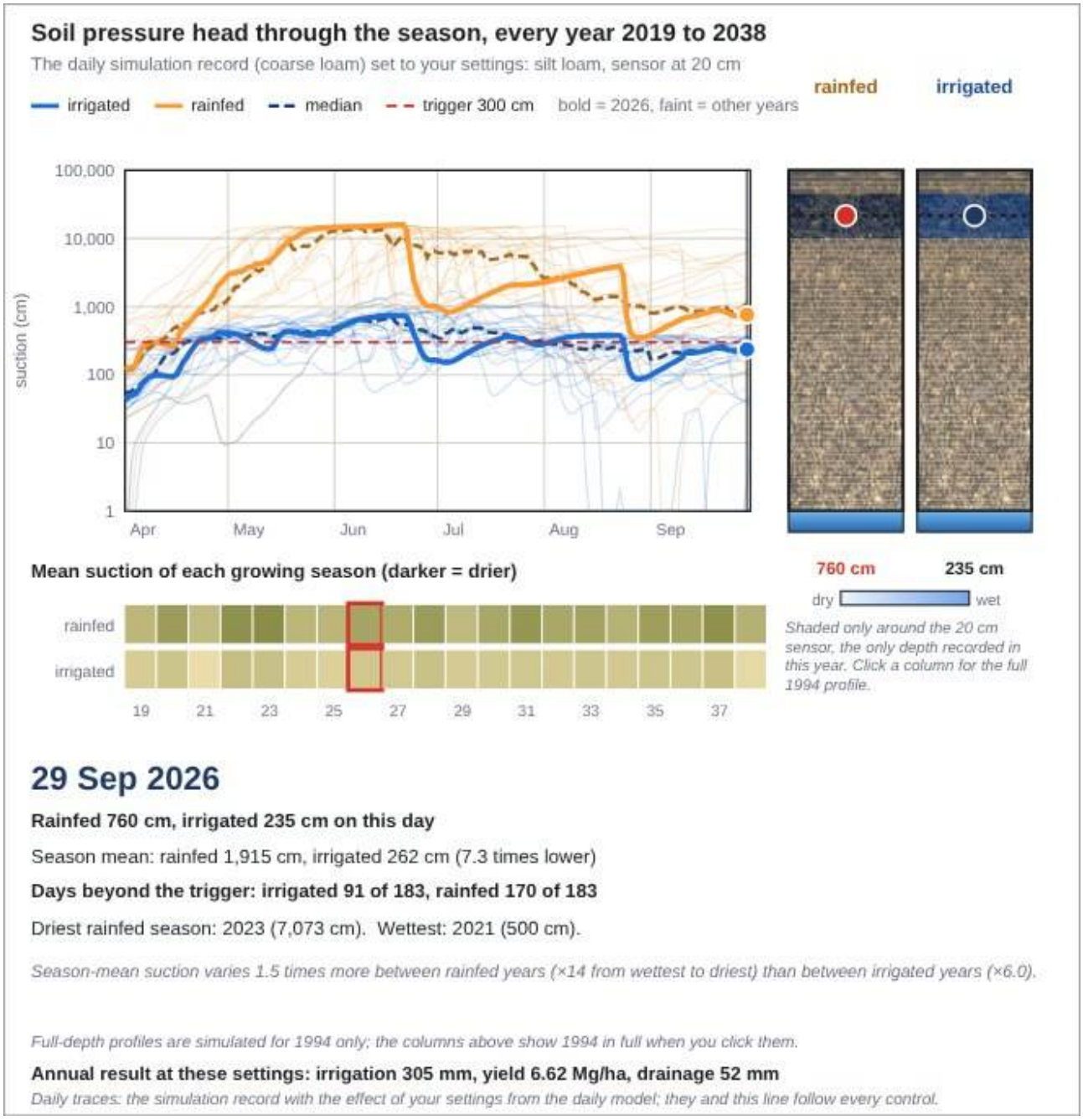


Figure 7.57: Profile evolution, Silt Loam, part 1 of 1.

7.20 Reliability over 20 seasons (Silt Loam)

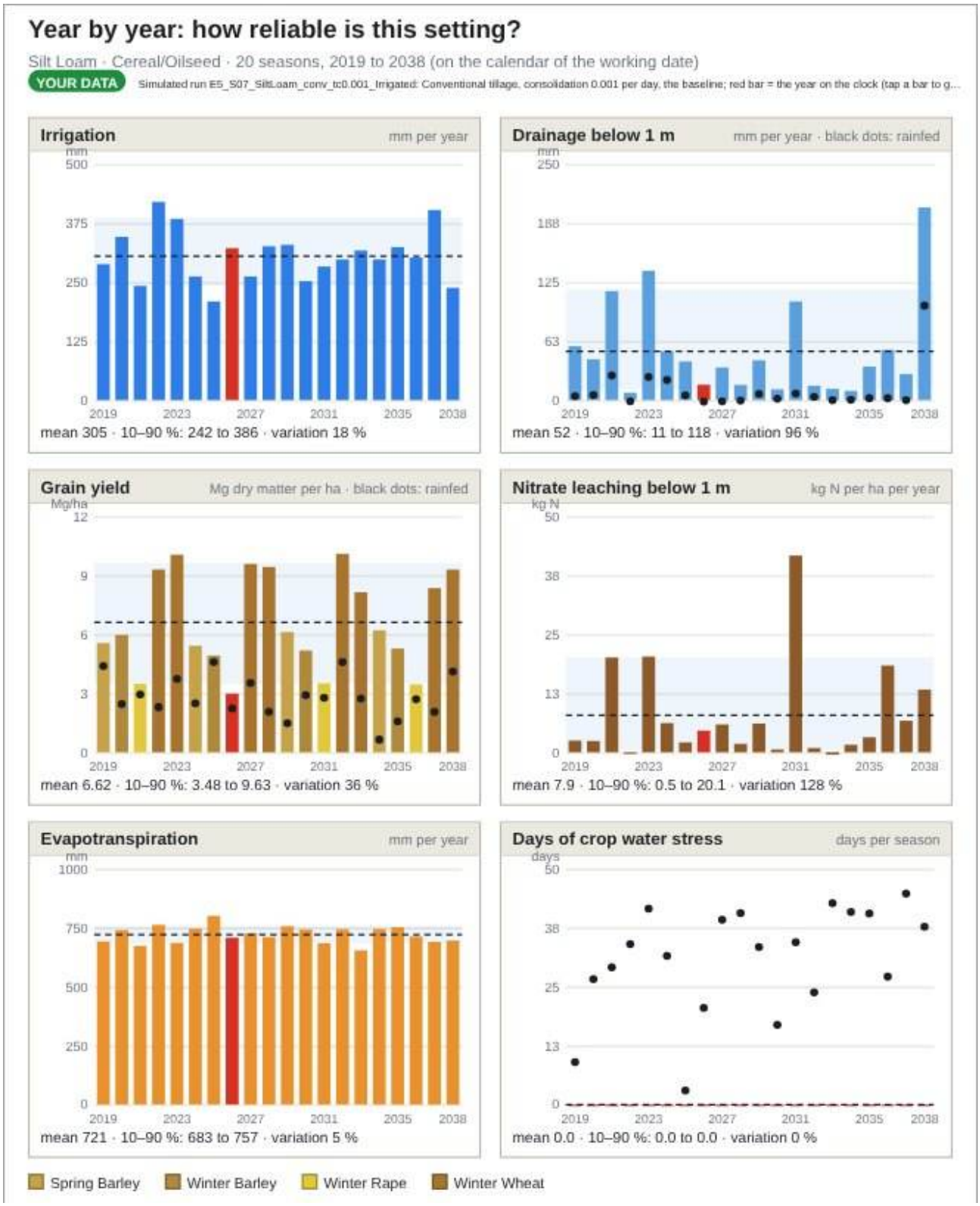


Figure 7.58: Reliability over 20 seasons, Silt Loam, part 1 of 2.

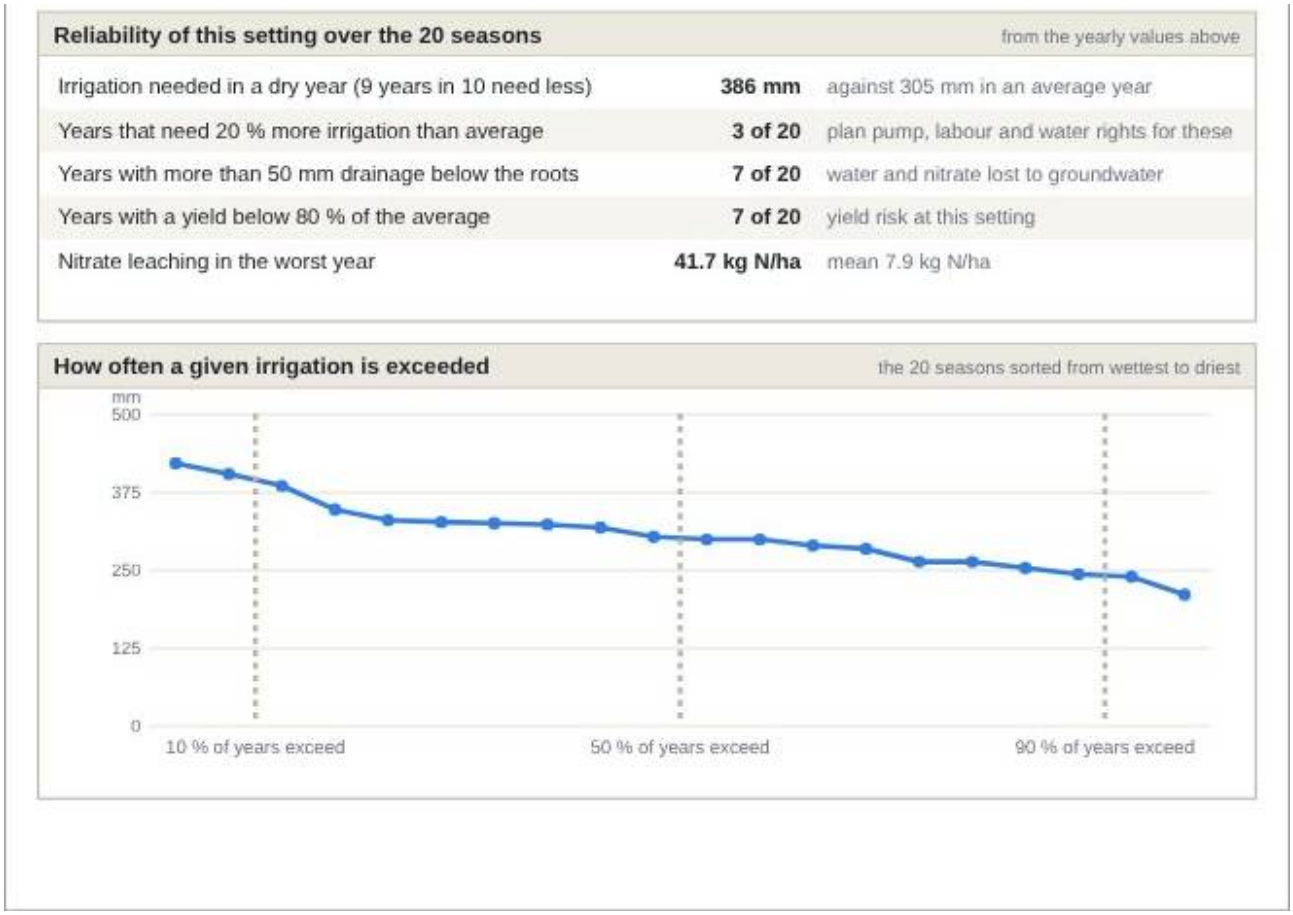


Figure 7.59: Reliability over 20 seasons, Silt Loam, part 2 of 2.

7.21 Full water balance (Silt Loam)

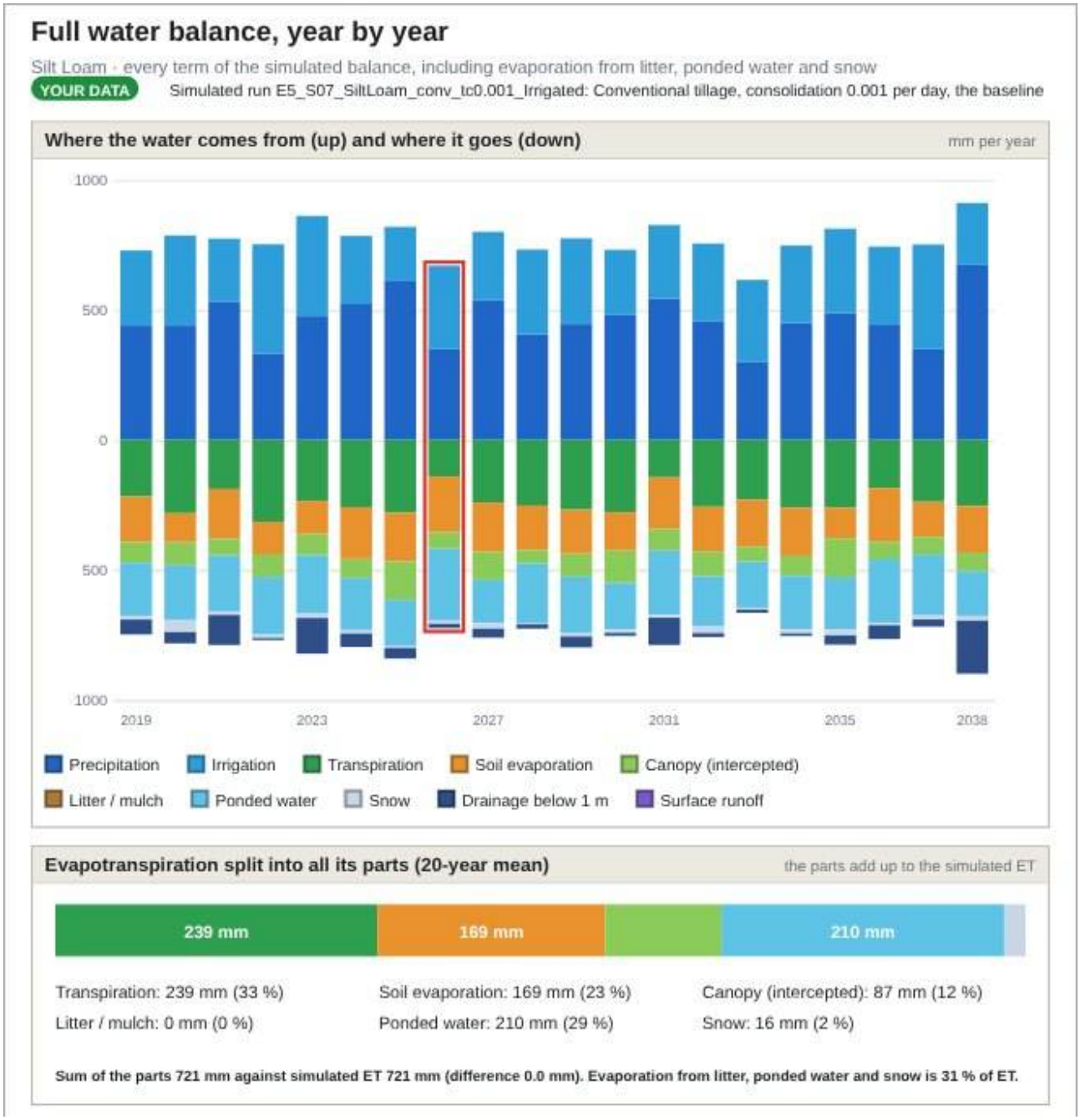


Figure 7.60: Full water balance, Silt Loam, part 1 of 2.

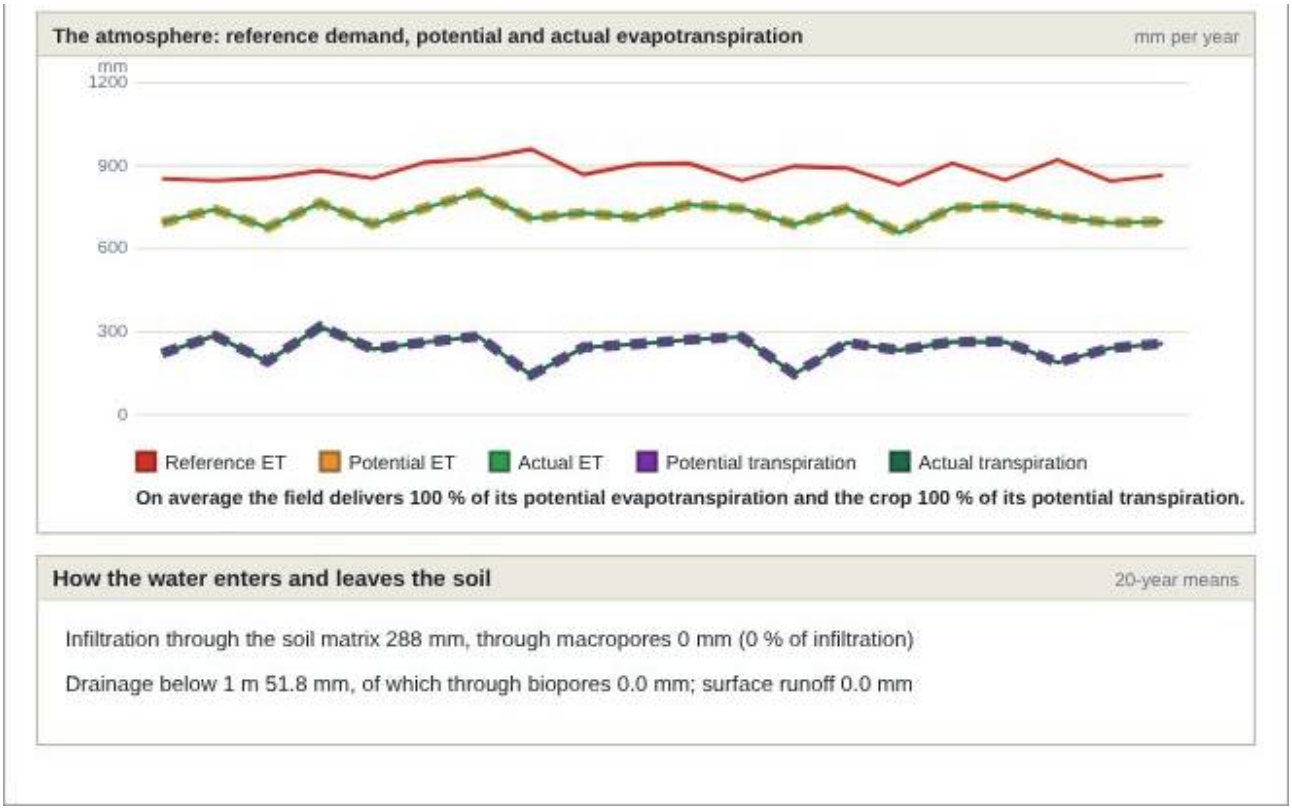


Figure 7.61: Full water balance, Silt Loam, part 2 of 2.

7.22 Nitrogen budget (Silt Loam)

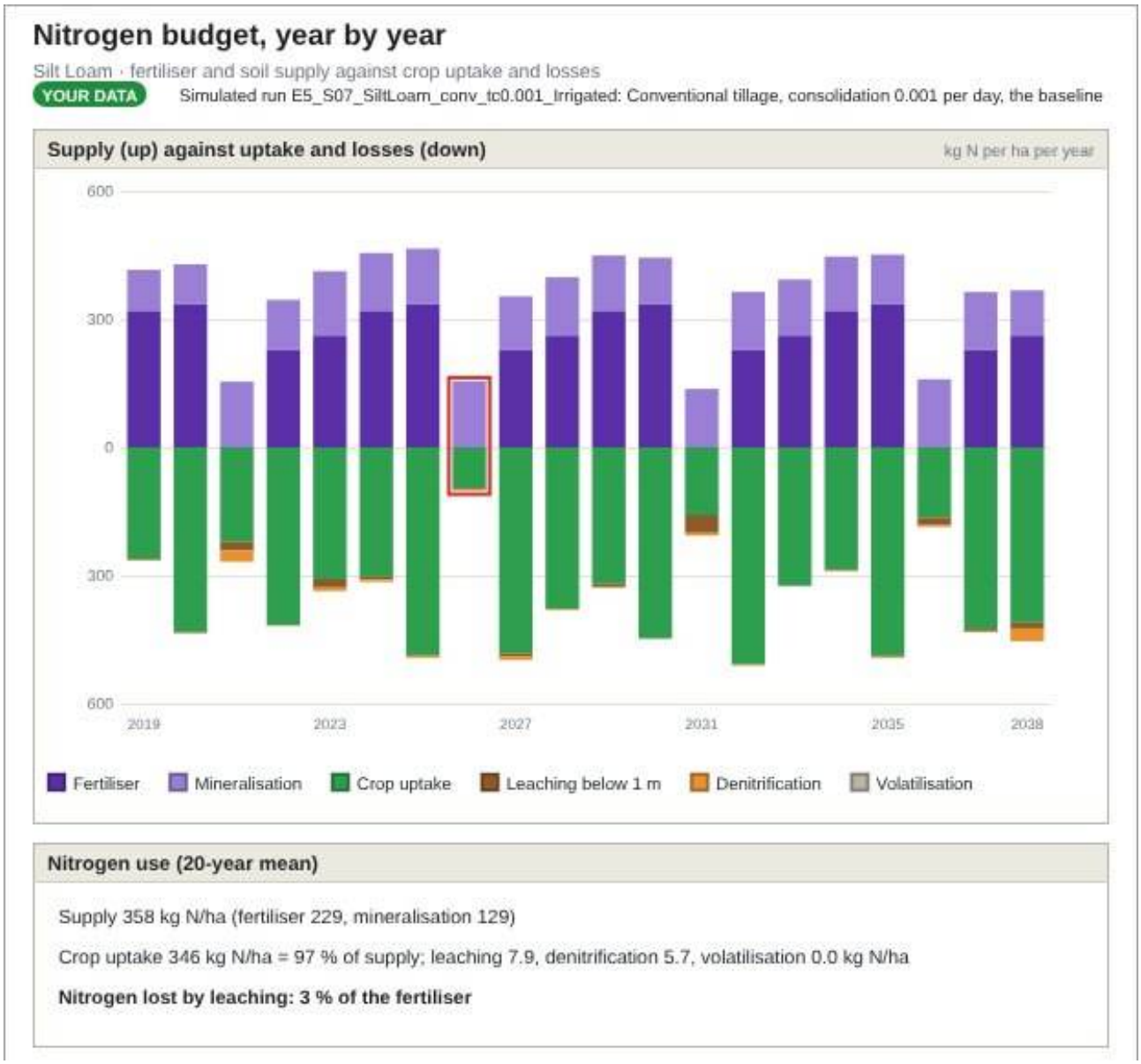


Figure 7.62: Nitrogen budget, Silt Loam, part 1 of 2.

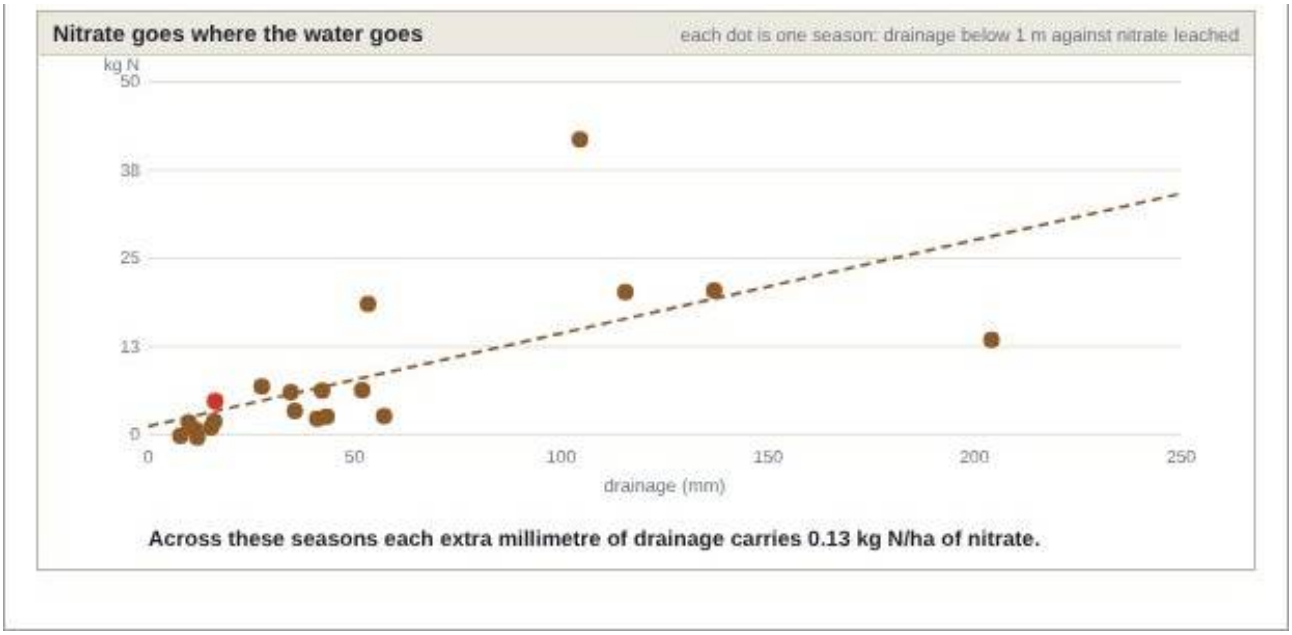


Figure 7.63: Nitrogen budget, Silt Loam, part 2 of 2.

7.23 Crop water requirement (Silt Loam)

Figure 7.64: Crop water requirement, Silt Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**537**	8.4	100 %
Coconut (high)	834	**457**	6.1	100 %
Papaya (high)	814	**442**	6.6	100 %
Oil palm (young)	790	**419**	6.2	100 %
Yam	649	**369**	8.5	100 %
Onion (dry)	565	**332**	9.8	100 %
Rice	573	**324**	5.8	97 %
Cotton	571	**319**	4.4	100 %
Kiwi (pergola)	626	**316**	4.2	100 %
Cabbage	559	**312**	6.6	100 %
Avocado (medium)	677	**305**	5.4	100 %
Guava (high)	651	**290**	3.9	100 %
Table grapes (horizontal trellis)	597	**287**	3.8	100 %
Potato	508	**281**	7.6	100 %
Tomato	521	**268**	3.7	100 %
Hop (v trellis)	520	**258**	3.5	100 %
Walnut (medium)	523	**242**	3.3	100 %
Mango (medium)	620	**240**	3.2	100 %
Pear (medium)	528	**239**	3.2	100 %
Mustard	468	**235**	3.3	100 %
Wheat	469	**233**	3.3	100 %
Groundnut	471	**233**	4.3	100 %
Soybean	473	**232**	3.4	100 %
Date palm (medium)	602	**227**	3.1	100 %
Sweet pepper	475	**225**	4.1	100 %
Cherry (medium)	503	**222**	3.0	100 %
Pistachio (medium)	503	**222**	3.0	100 %

Figure 7.65: Crop water requirement, Silt Loam, part 2 of 3.

Maize (grain)	451		**214**	2.9	100 %
Cassava	466		**209**	4.3	100 %
Chickpea	436		**199**	3.5	100 %
Hazelnut (low)	481		**197**	2.7	100 %
Pecan (low)	481		**197**	2.7	100 %
Cherimoya (medium)	545		**176**	2.4	100 %
Sorghum (grain)	429		**173**	2.4	100 %
Apple (medium)	459		**169**	2.3	100 %
Cowpea (beans)	398		**164**	2.9	100 %
Millet	395		**164**	2.4	100 %
Citrus (medium density, short)	552		**156**	2.1	100 %
Peach (low)	435		**142**	1.9	100 %
Almond (medium)	435		**142**	1.9	100 %
Wine grapes (pergola)	457		**142**	1.9	100 %
Green gram (mung bean)	372		**129**	2.4	100 %
Apricot (young)	380		**92**	1.2	100 %
Olive (traditional, medium density)	446		**71**	1.0	100 %
Highbush blueberry (low)	338		**67**	1.0	100 %
Persimmon (young, low)	358		**67**	0.9	100 %
Pomegranate (low)	358		**67**	0.9	100 %
Fig tree (young, low)	327		**37**	0.5	100 %

Figure 7.66: Crop water requirement, Silt Loam, part 3 of 3.

7.24 Feedback loops (Silt Loam)

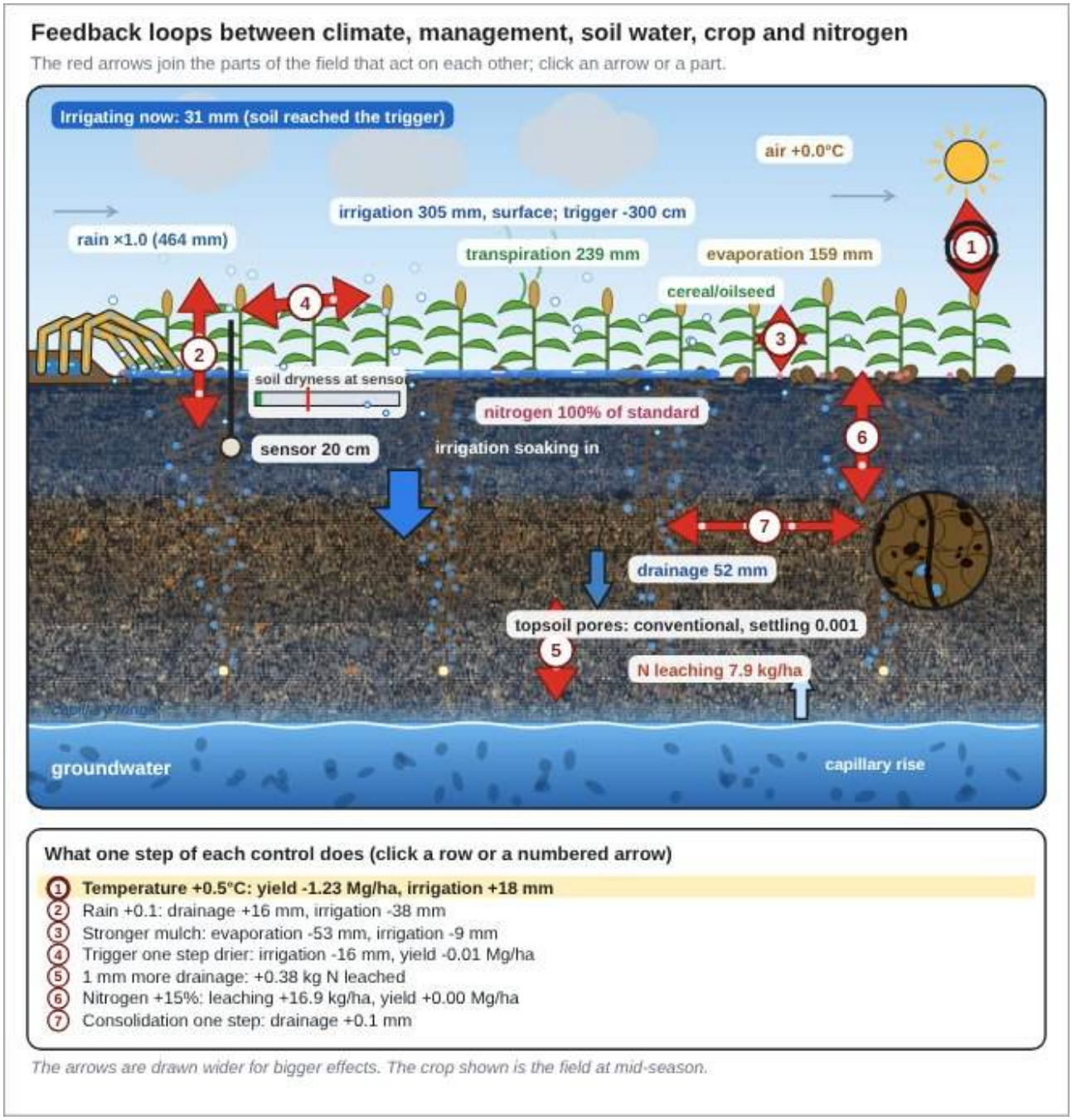


Figure 7.67: Feedback loops, Silt Loam, part 1 of 1.

Chapter 8

Sandy Clay Loam

8.1 Surface irrigation design (Sandy Clay Loam)

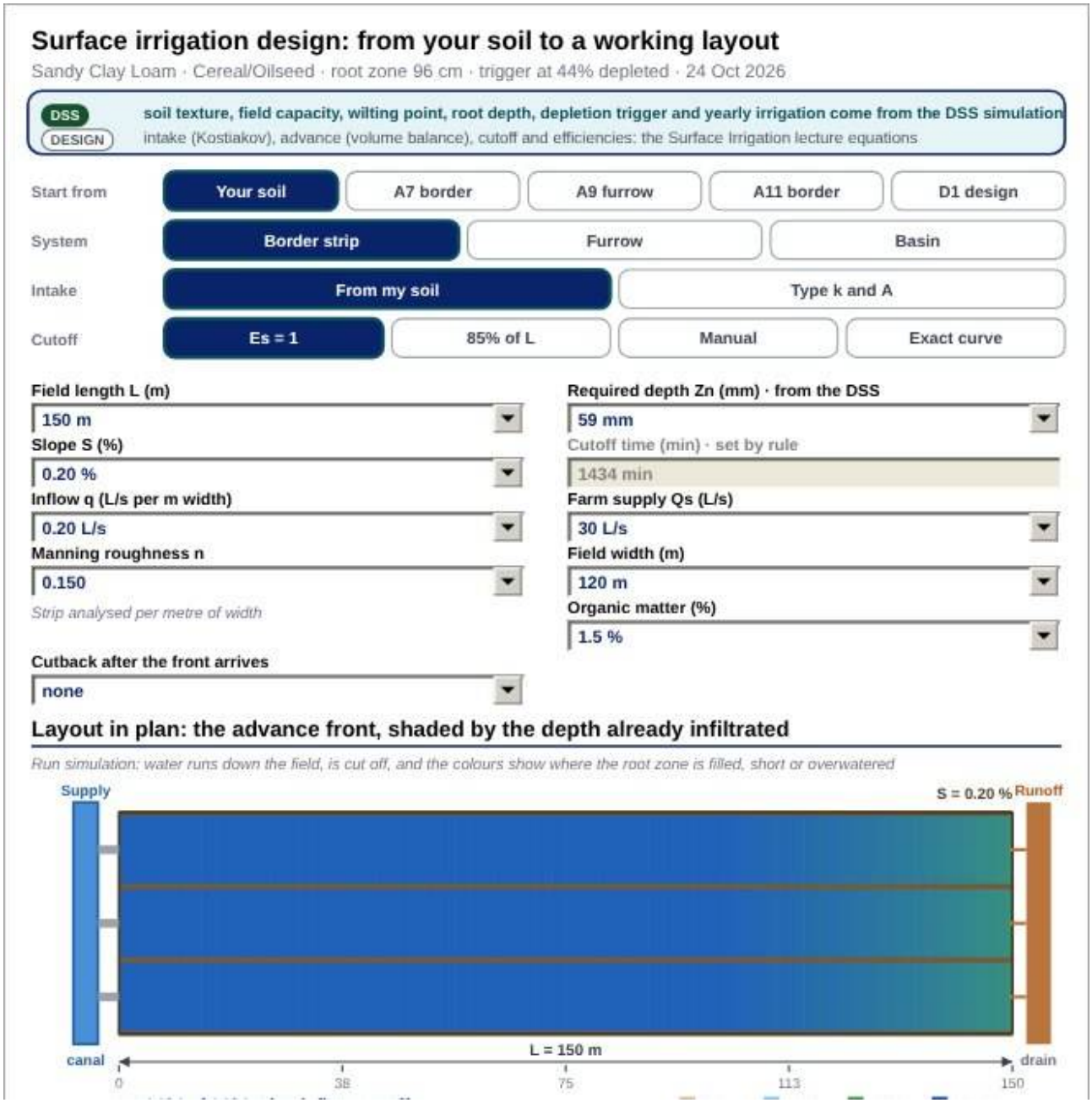


Figure 8.1: Surface irrigation design, Sandy Clay Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

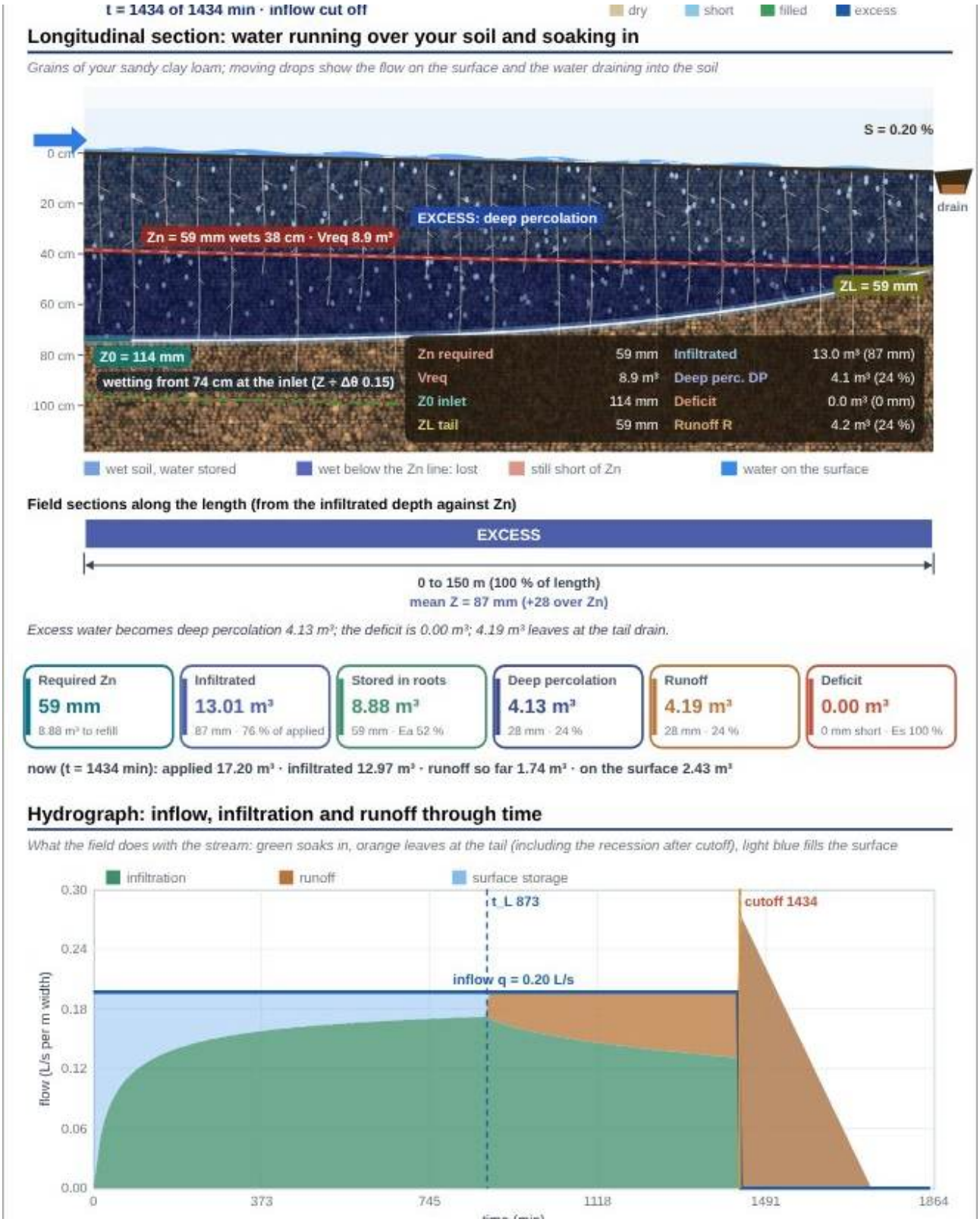


Figure 8.2: Surface irrigation design, Sandy Clay Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

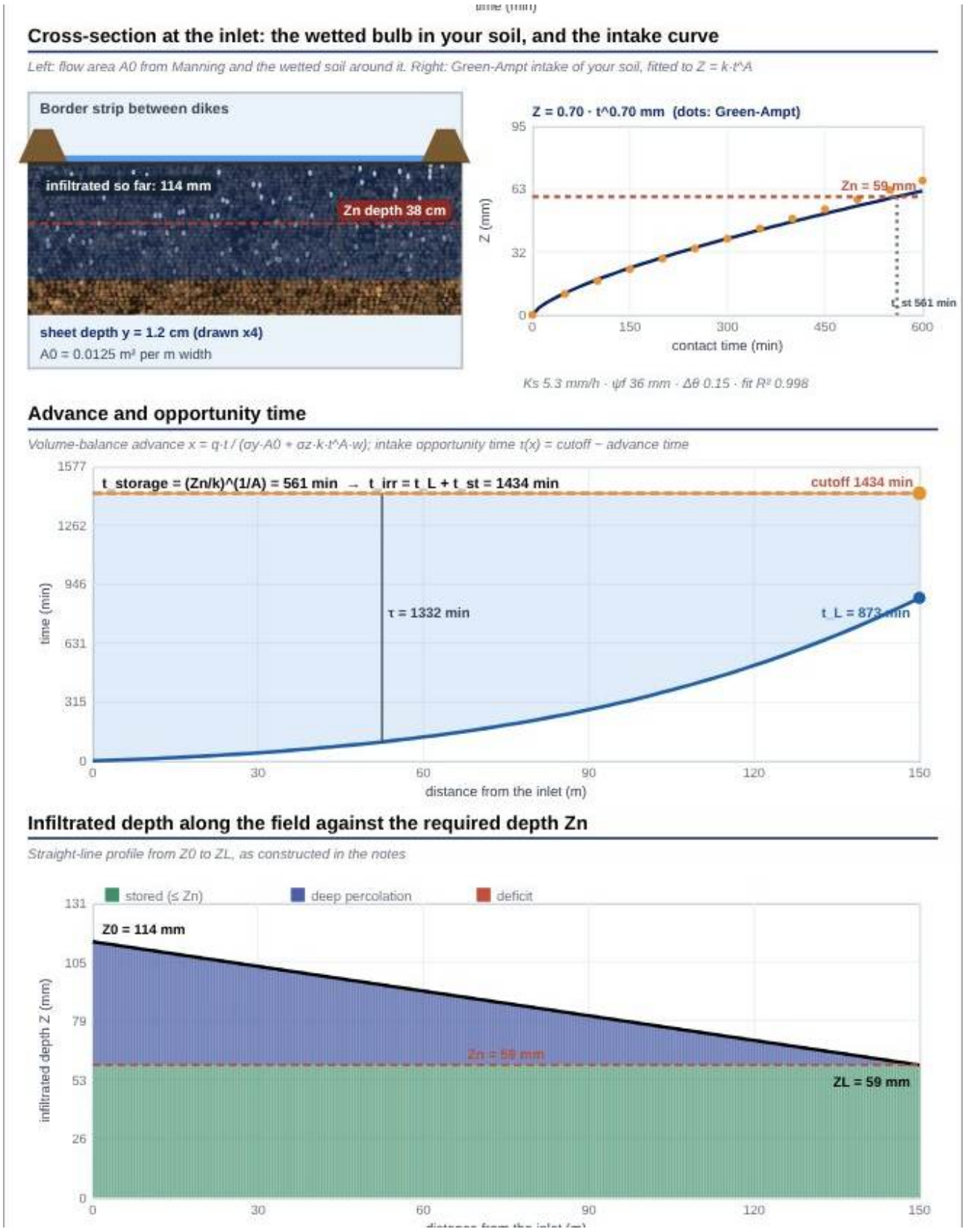


Figure 8.3: Surface irrigation design, Sandy Clay Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

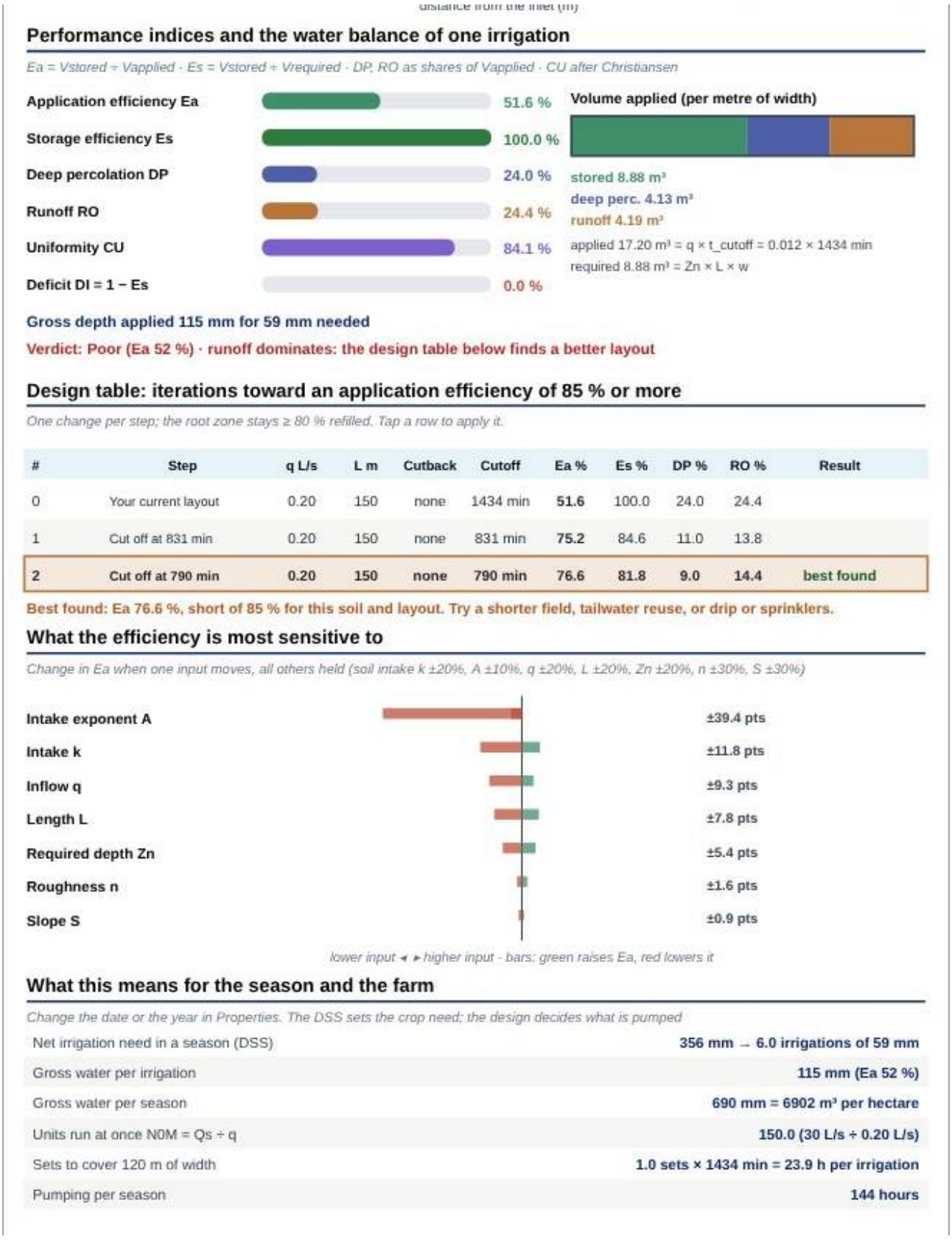
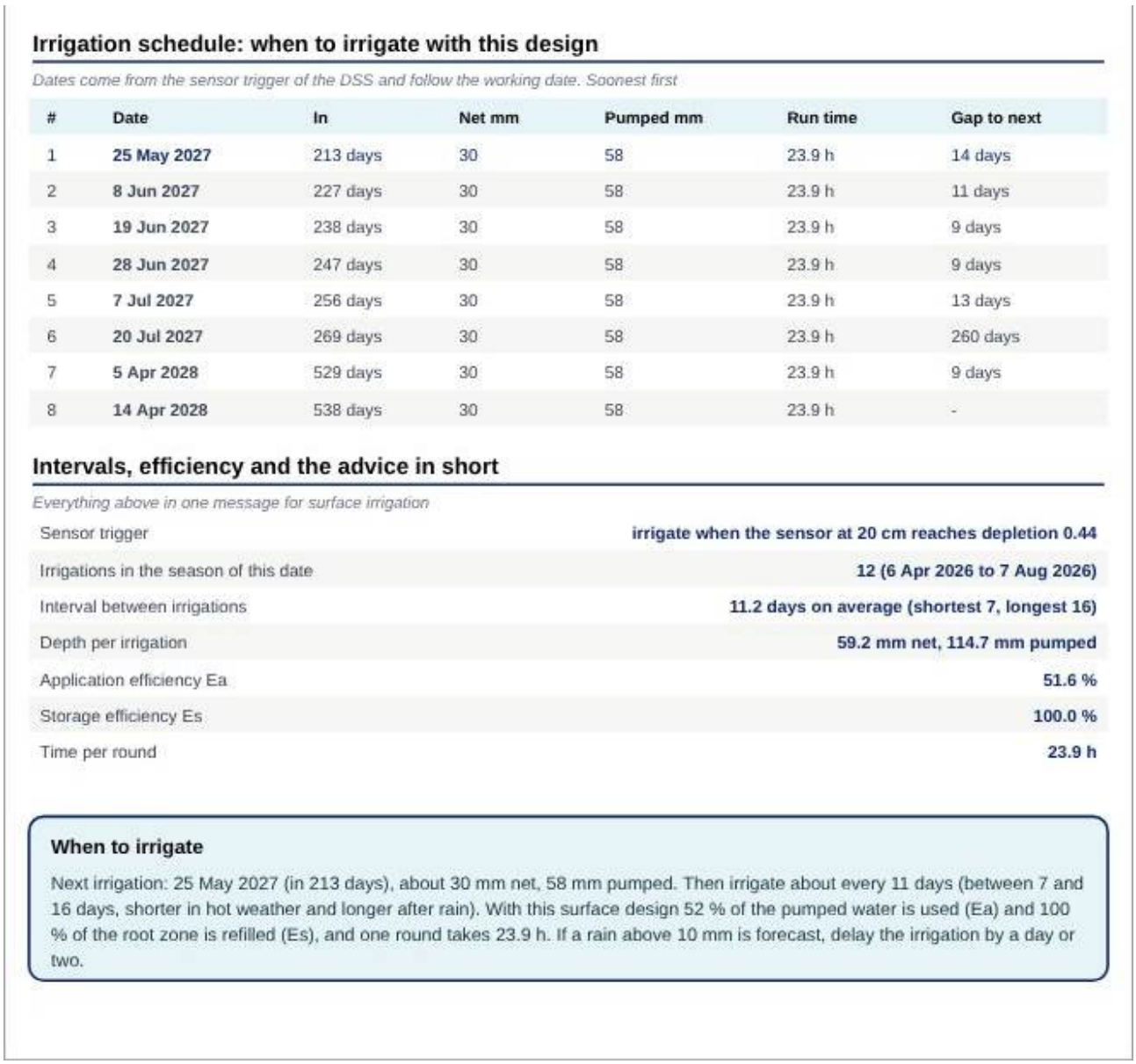


Figure 8.4: Surface irrigation design, Sandy Clay Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to; What this means for the season and the farm.



Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**25 May 2027**	213 days	30	58	23.9 h	14 days
2	**8 Jun 2027**	227 days	30	58	23.9 h	11 days
3	**19 Jun 2027**	238 days	30	58	23.9 h	9 days
4	**28 Jun 2027**	247 days	30	58	23.9 h	9 days
5	**7 Jul 2027**	256 days	30	58	23.9 h	13 days
6	**20 Jul 2027**	269 days	30	58	23.9 h	260 days
7	**5 Apr 2028**	529 days	30	58	23.9 h	9 days
8	**14 Apr 2028**	538 days	30	58	23.9 h	-

Intervals, efficiency and the advice in short

Everything above in one message for surface irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.44**
Irrigations in the season of this date	**12 (6 Apr 2026 to 7 Aug 2026)**
Interval between irrigations	**11.2 days on average (shortest 7, longest 16)**
Depth per irrigation	**59.2 mm net, 114.7 mm pumped**
Application efficiency Ea	**51.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**23.9 h**

When to irrigate

Next irrigation: 25 May 2027 (in 213 days), about 30 mm net, 58 mm pumped. Then irrigate about every 11 days (between 7 and 16 days, shorter in hot weather and longer after rain). With this surface design 52 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 23.9 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 8.5: Surface irrigation design, Sandy Clay Loam, part 5 of 5: Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

8.2 Drip irrigation design (Sandy Clay Loam)

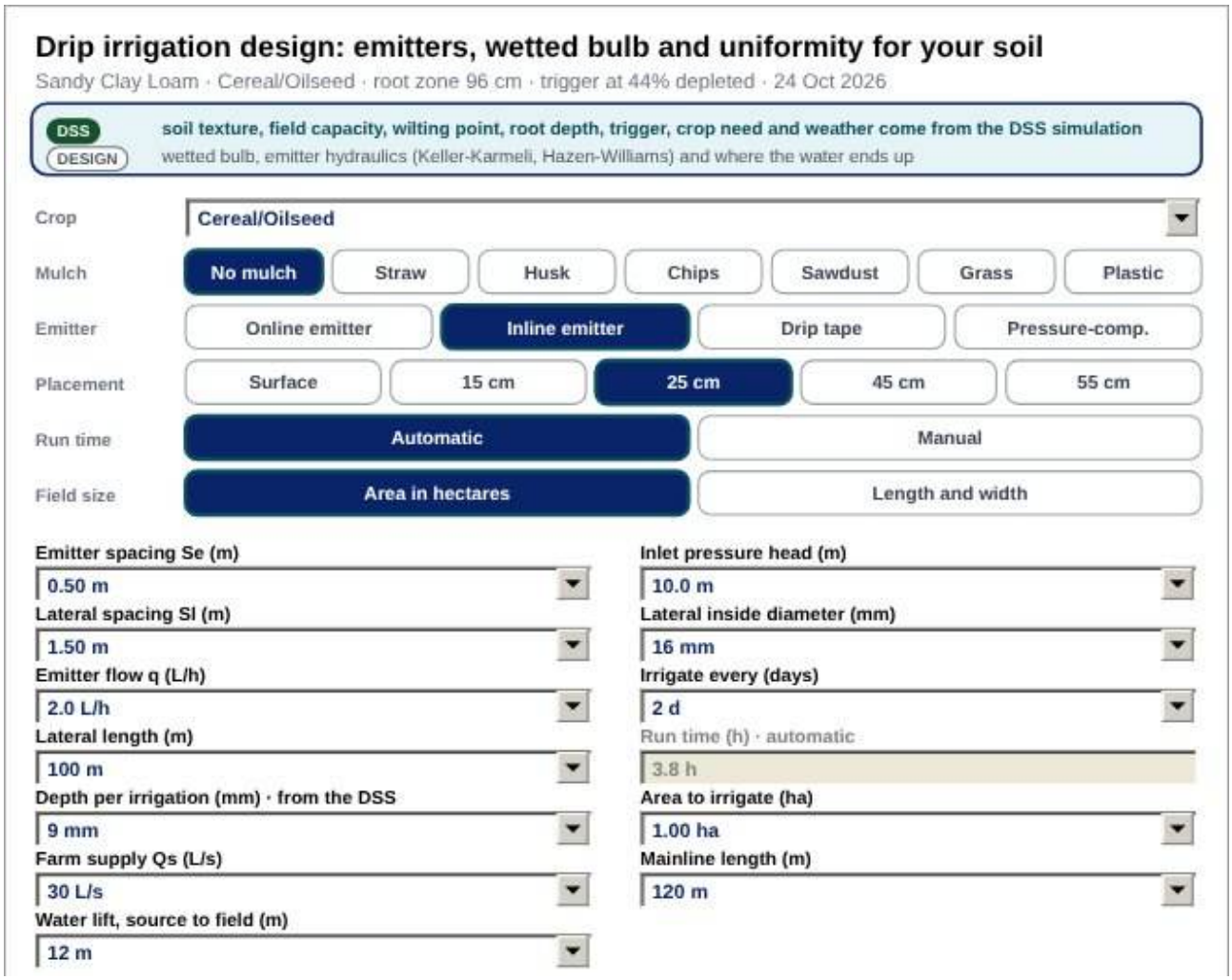


Figure 8.6: Drip irrigation design, Sandy Clay Loam, part 1 of 6.

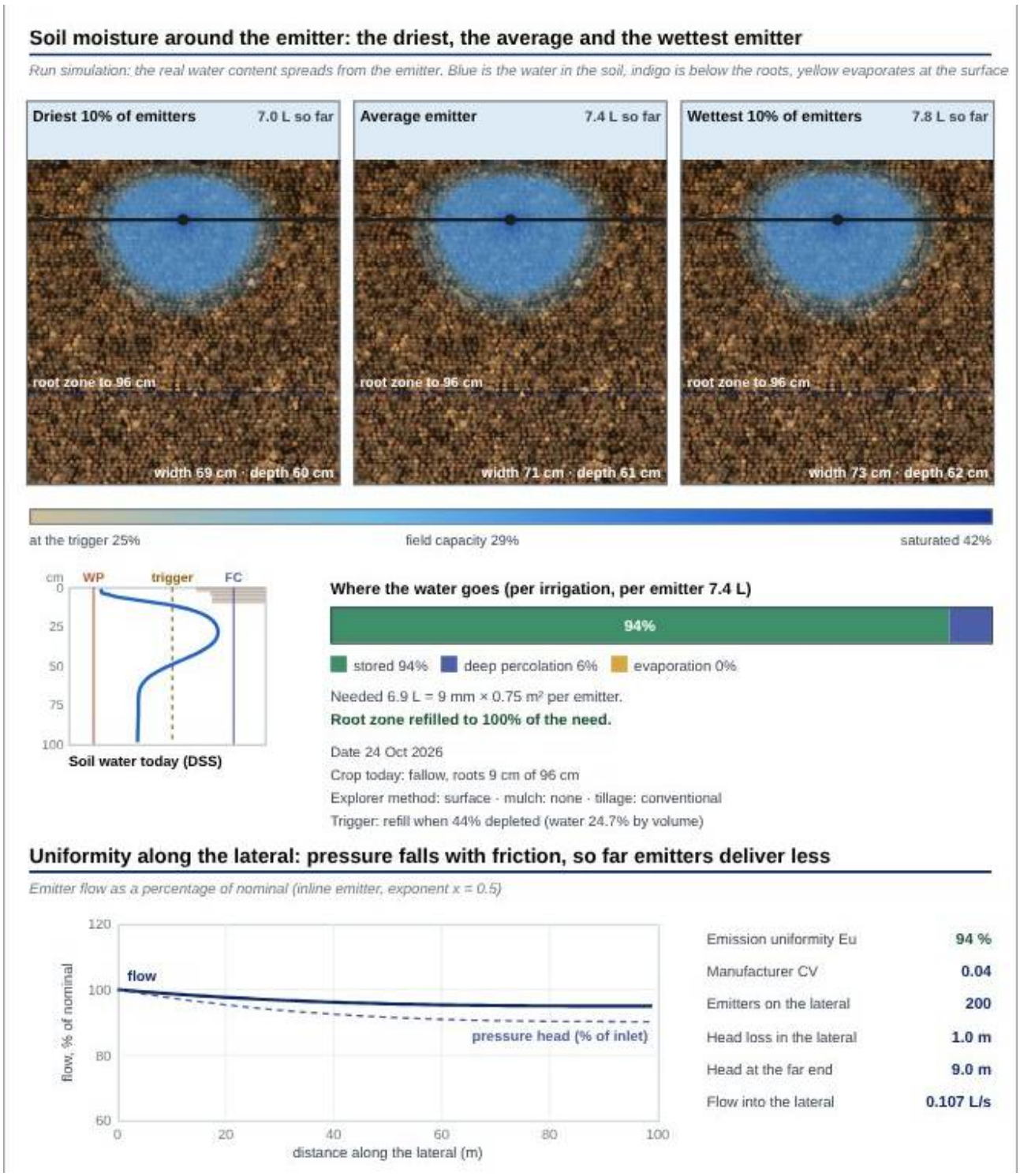


Figure 8.7: Drip irrigation design, Sandy Clay Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

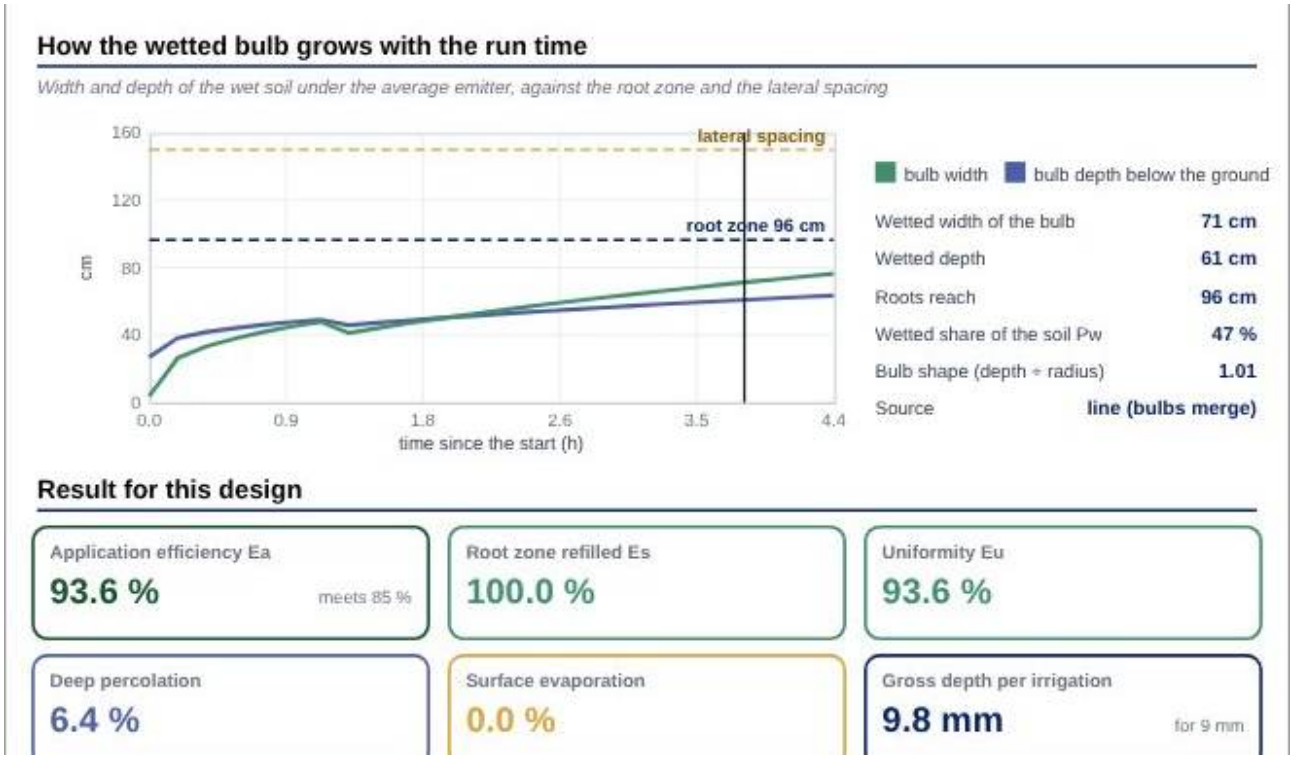
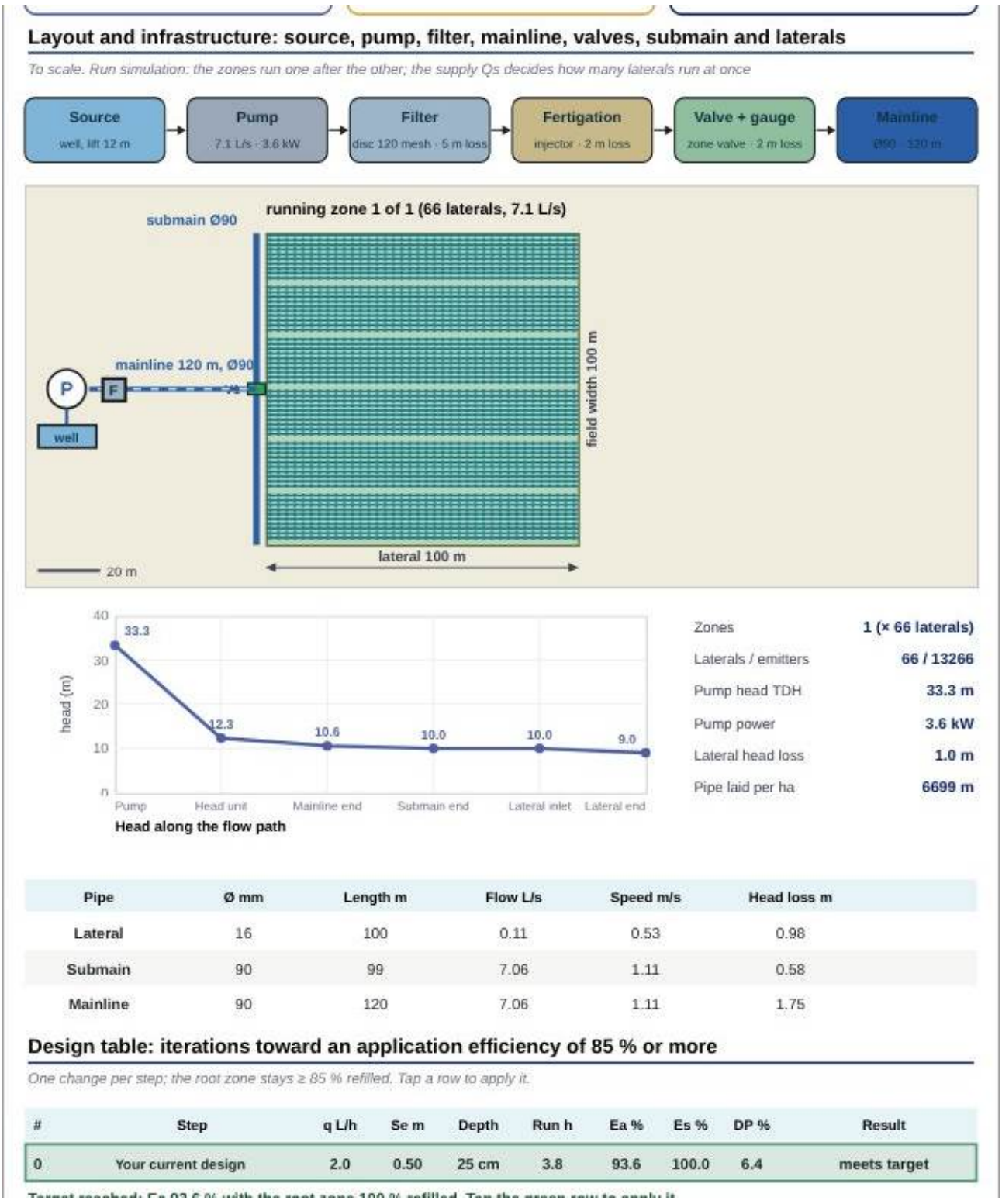


Figure 8.8: Drip irrigation design, Sandy Clay Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.8	93.6	100.0	6.4	meets target

Figure 8.9: Drip irrigation design, Sandy Clay Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

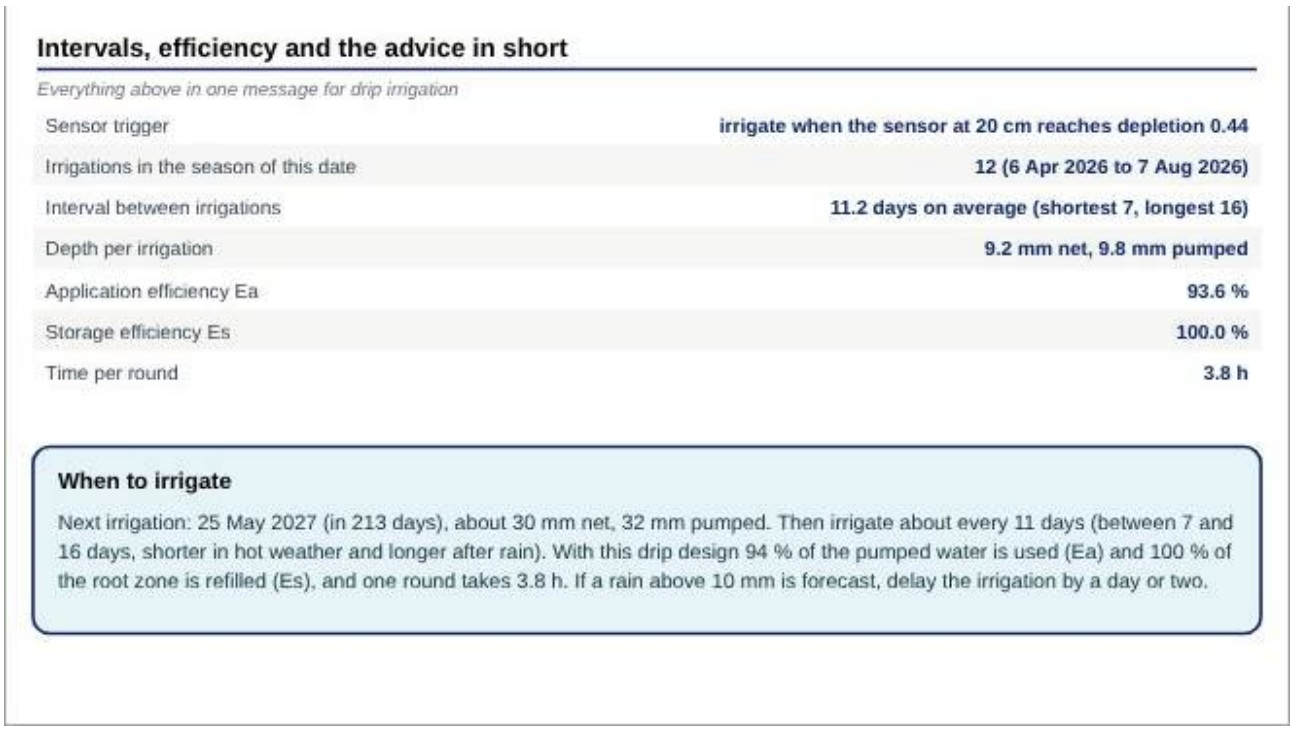
Net irrigation need in a season (DSS)	**356 mm → 39 irrigations of 9 mm**
Gross water per irrigation	**9.8 mm (Ea 94 %)**
Gross water per season	**381 mm = 3805 m³ per hectare**
Peak crop need / interval	**4.6 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.8 h = 3.8 h per round**
Pumping per season	**148 hours · 526 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**25 May 2027**	213 days	30	32	3.8 h	14 days
2	**8 Jun 2027**	227 days	30	32	3.8 h	11 days
3	**19 Jun 2027**	238 days	30	32	3.8 h	9 days
4	**28 Jun 2027**	247 days	30	32	3.8 h	9 days
5	**7 Jul 2027**	256 days	30	32	3.8 h	13 days
6	**20 Jul 2027**	269 days	30	32	3.8 h	260 days
7	**5 Apr 2028**	529 days	30	32	3.8 h	9 days
8	**14 Apr 2028**	538 days	30	32	3.8 h	-

Figure 8.10: Drip irrigation design, Sandy Clay Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.44**
Irrigations in the season of this date	**12 (6 Apr 2026 to 7 Aug 2026)**
Interval between irrigations	**11.2 days on average (shortest 7, longest 16)**
Depth per irrigation	**9.2 mm net, 9.8 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**3.8 h**

When to irrigate

Next irrigation: 25 May 2027 (in 213 days), about 30 mm net, 32 mm pumped. Then irrigate about every 11 days (between 7 and 16 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 3.8 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 8.11: Drip irrigation design, Sandy Clay Loam, part 6 of 6: Intervals, efficiency and the advice in short.

8.3 Sprinkler design and pipes (Sandy Clay Loam)

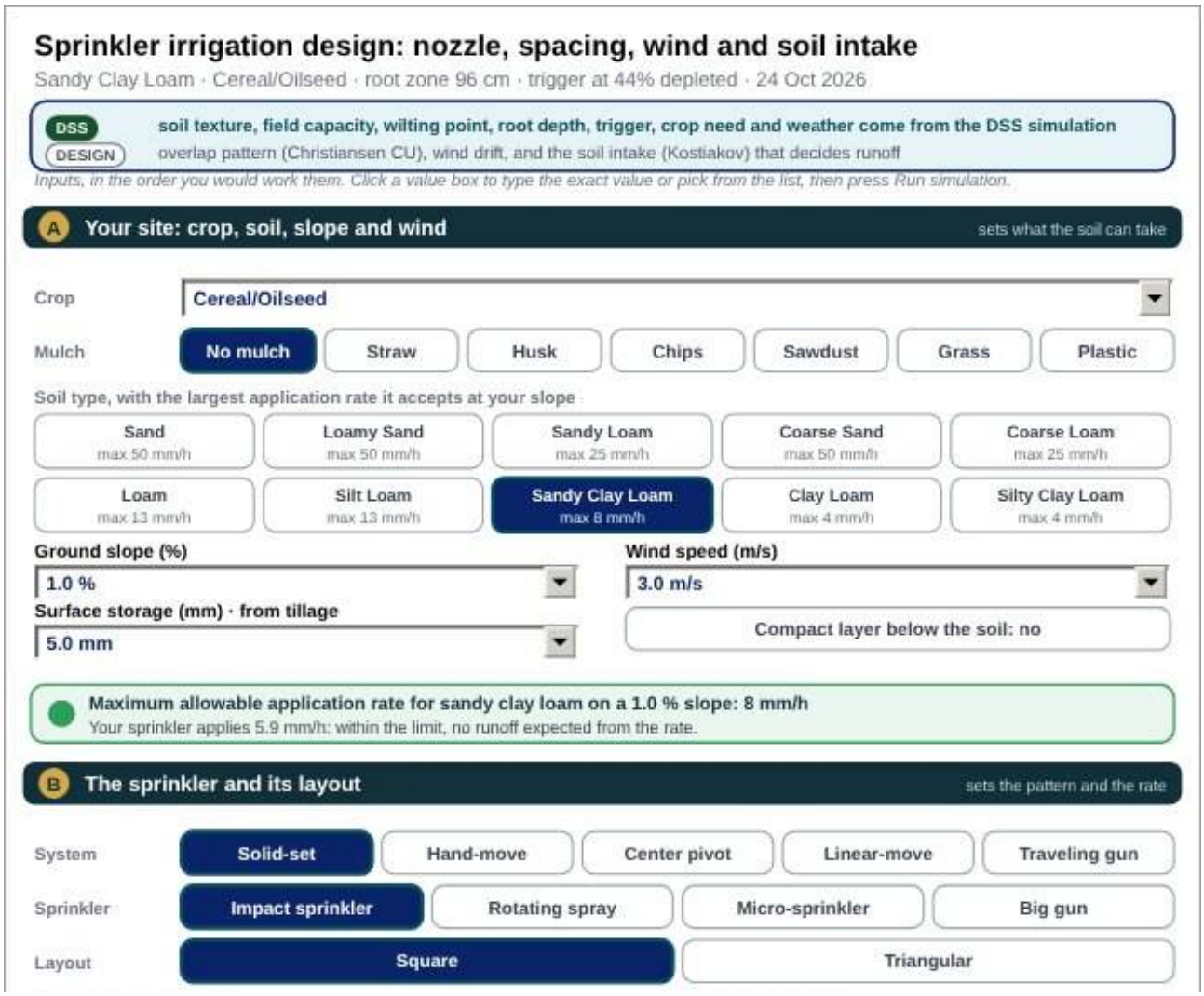


Figure 8.12: Sprinkler design and pipes, Sandy Clay Loam, part 1 of 12.

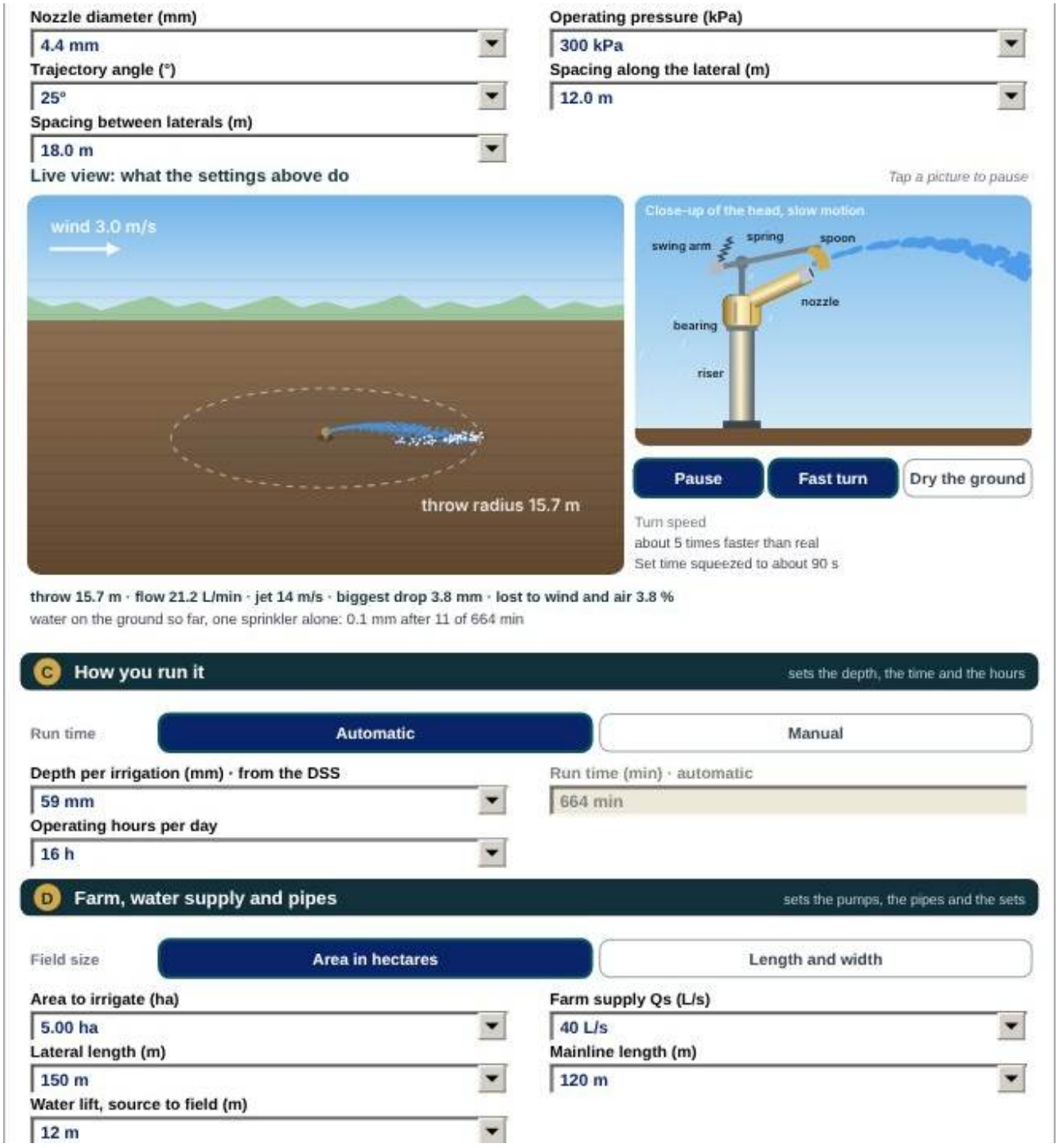


Figure 8.13: Sprinkler design and pipes, Sandy Clay Loam, part 2 of 12.

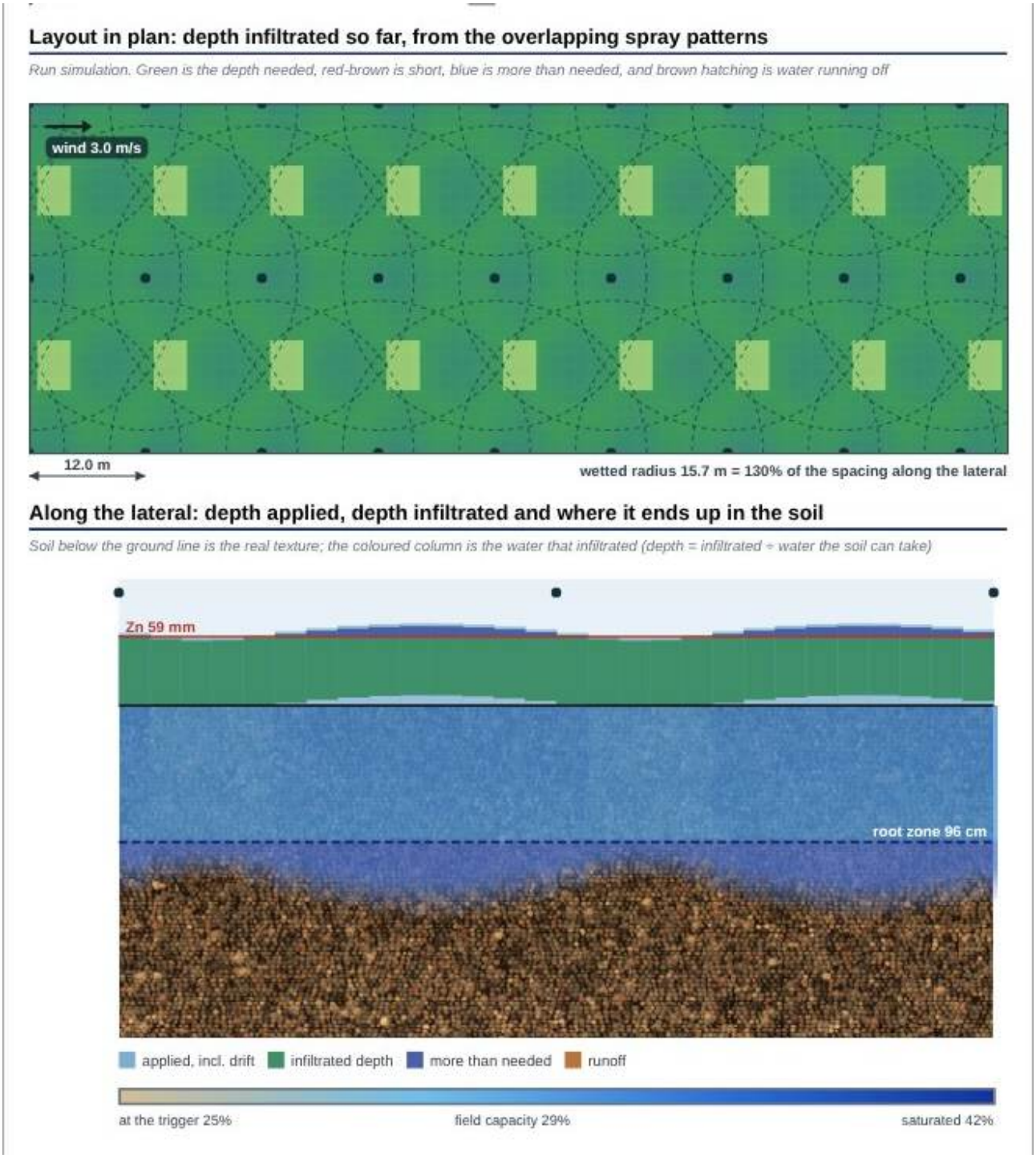


Figure 8.14: Sprinkler design and pipes, Sandy Clay Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

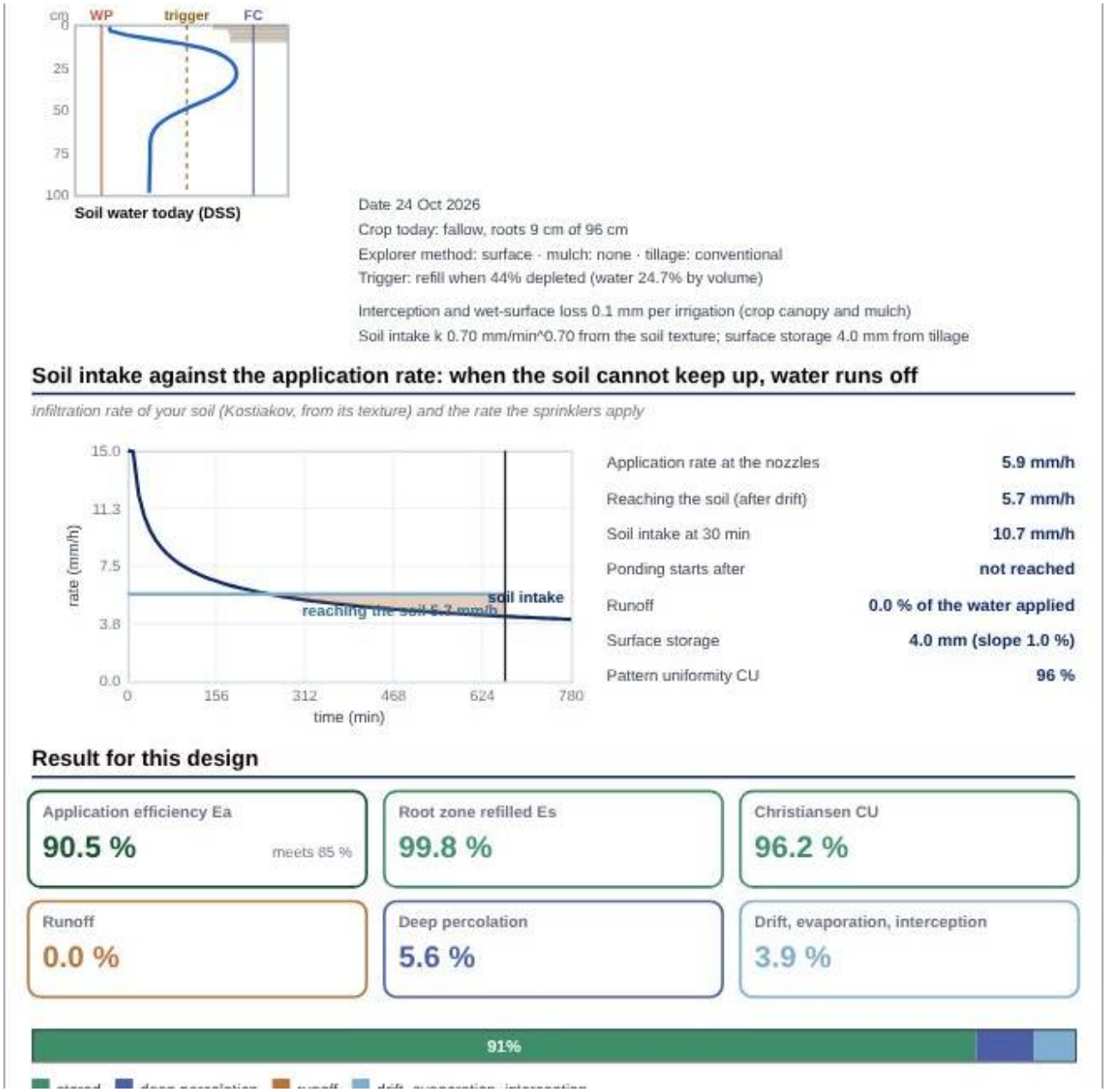


Figure 8.15: Sprinkler design and pipes, Sandy Clay Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

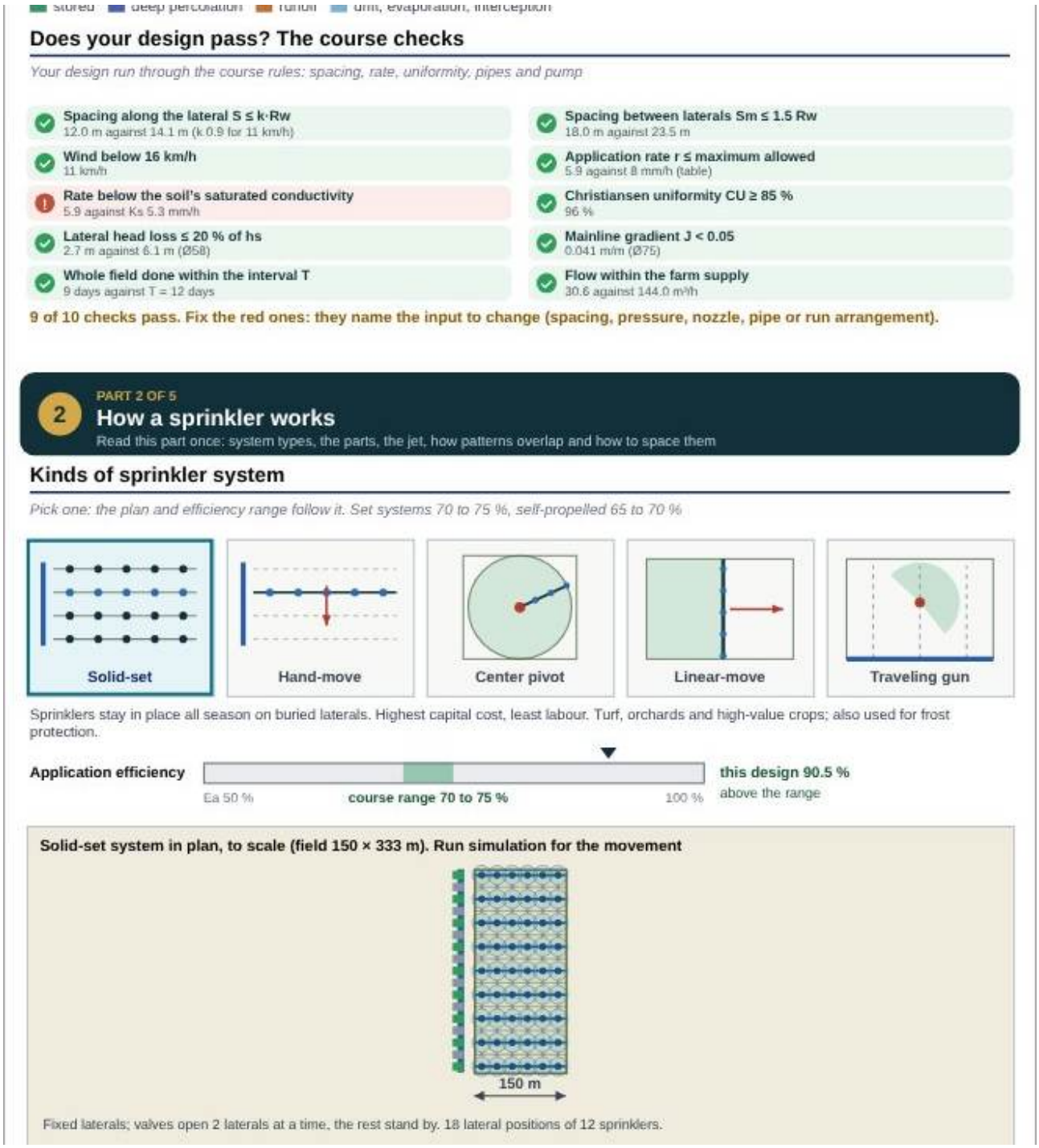


Figure 8.16: Sprinkler design and pipes, Sandy Clay Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

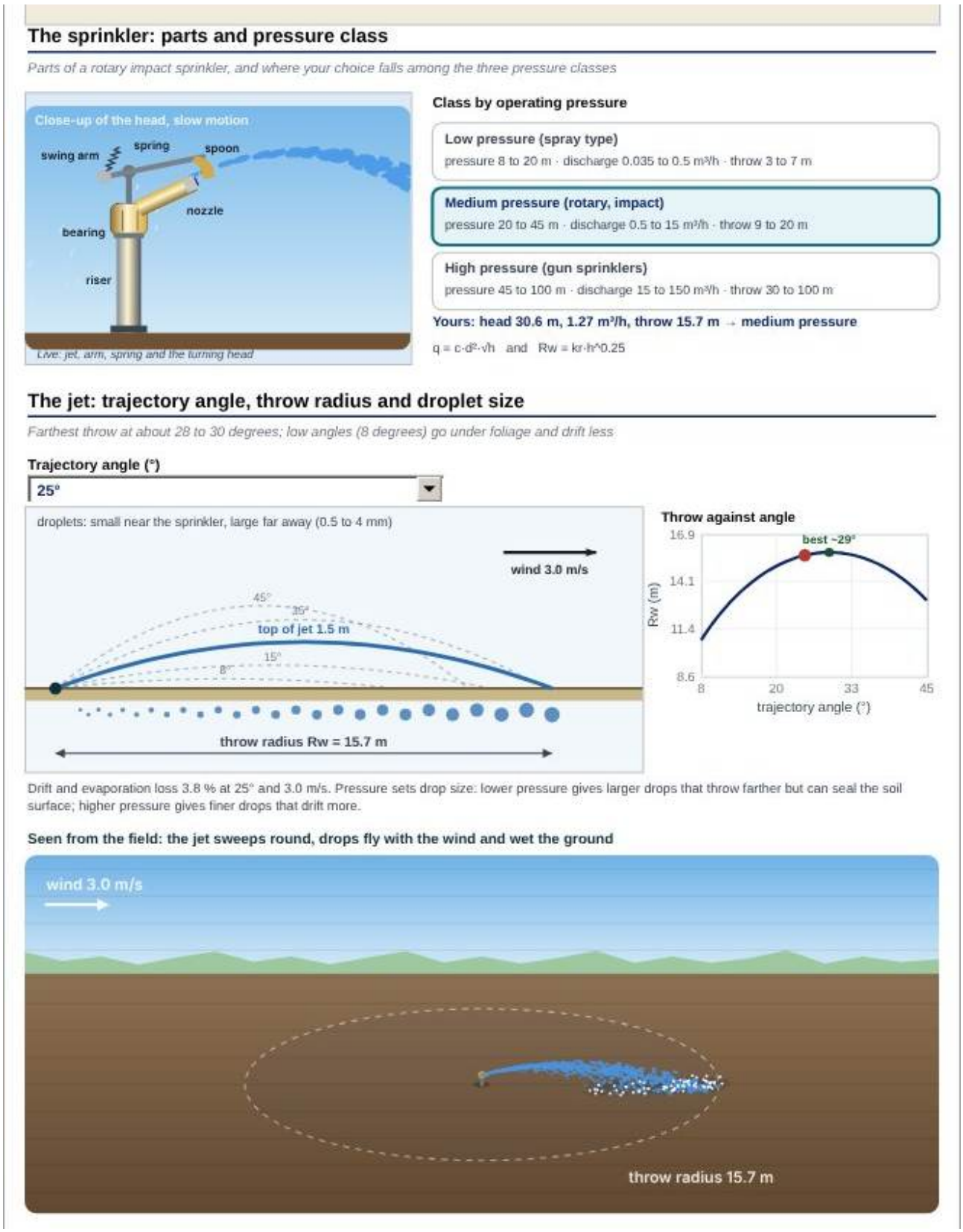


Figure 8.17: Sprinkler design and pipes, Sandy Clay Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

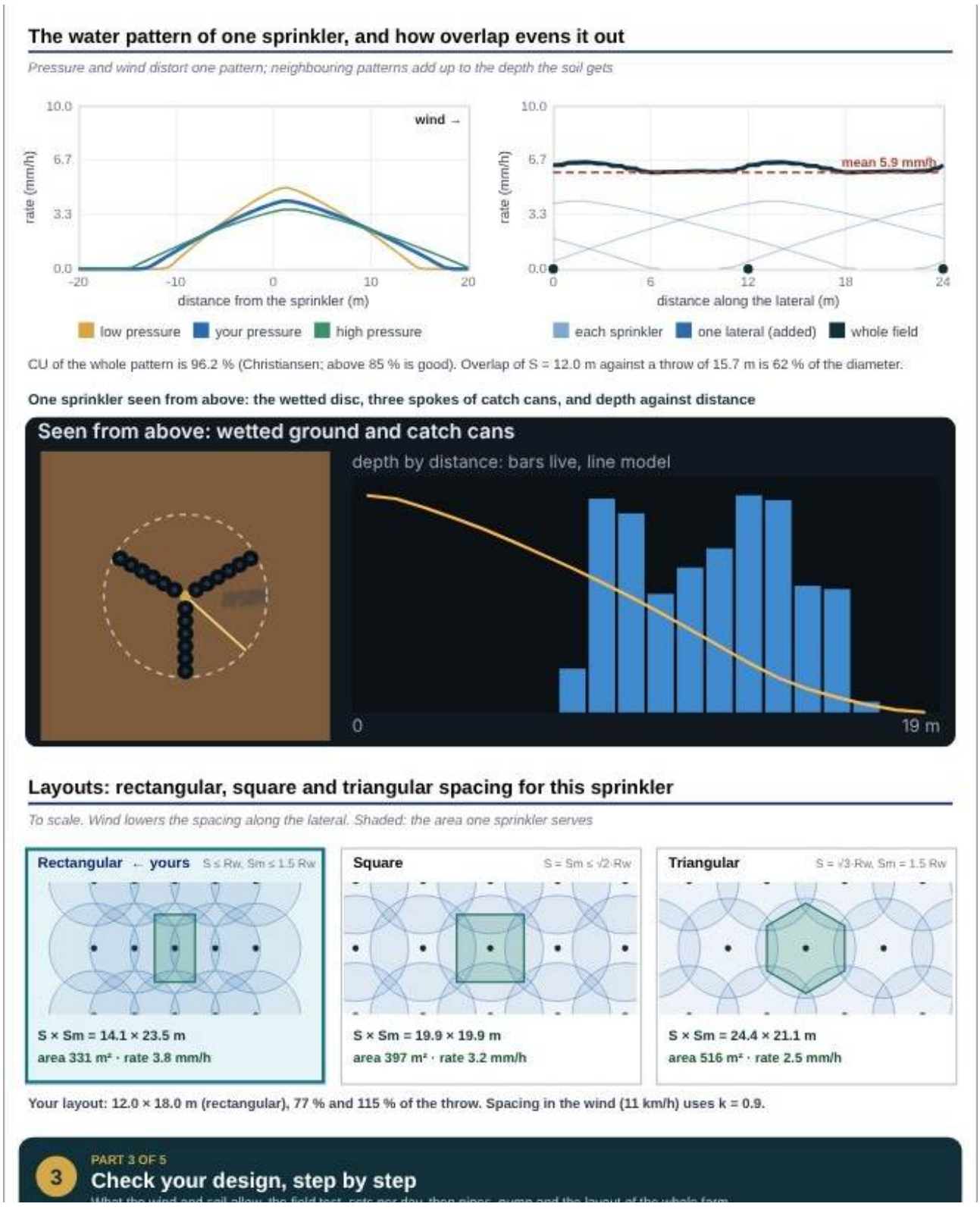


Figure 8.18: Sprinkler design and pipes, Sandy Clay Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

What wind and soil allow

Largest spacing the wind permits, and largest rate the soil accepts without runoff (course table)

Spacing against wind

Maximum allowable application rate (mm/h)

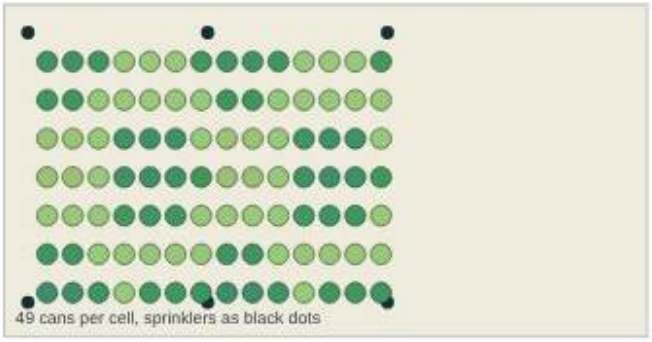
	0–5 %	5–8 %	8–12 %	12–16 %
Deep coarse sandy soils	50	38	25	13
Coarse sandy, compact layer…	38	25	19	10
Deep sandy loam	25	20	15	10
Sandy loam, compact layer b…	19	13	10	8
Deep silt loam	13	10	8	5
Silt loam, compact layer be…	**8**	6	4	2.5
Clay or clay loam, strong s…	4	2.5	2	1.5

Your soil (sandy clay loam) and slope 1.0 % allow 8 mm/h. Applied: 5.9 mm/h, within the limit

Compact layer below the soil: no (tap to add)

Catch-can test: measuring the uniformity the way it is done in the field

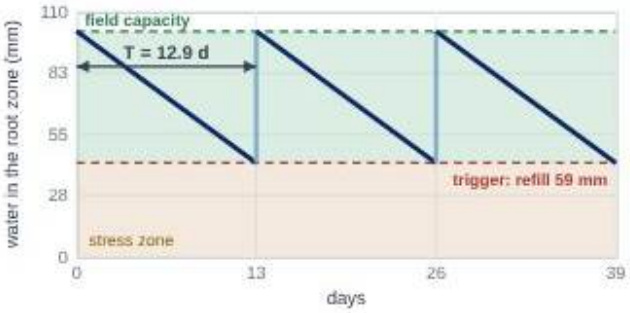
Cans on a grid between sprinklers; press Play and they fill. CU = 100 (1 − Σ|x − mean| ÷ (N·mean))



CU from the cans: 96.3 % — **good (above 85 %)**

Soil water as a reservoir: interval, sets per day and the flow the field needs

The crop empties the root-zone reservoir; irrigation refills it every T days. The rate sets set time and sets a day



Peak crop use ETc	**4.6 mm/day**
Net depth Zn = f·θAW·Dr	**59 mm**
Gross ZG = Zn ÷ Ea	**65.2 mm (Ea 91 %)**
Interval T = Zn ÷ ETc	**12 days**
Water per irrigation	**3262 m³ on 5.0 ha**
Flow to finish in T days	**17.0 m³/h at 16 h/day**
Farm supply	**144.0 m³/h**

Operating hours per day

16 h

Set time = ZG ÷ r = 65.2 ÷ 5.9 = 11.07 h → 1 set per day; 18 lateral positions to cover the field

Laterals running	Sets/day	Days to finish	Flow m³/h	Within T	Within supply	Verdict
1	1	18	15.3	no	yes	
2	**1**	**9**	**30.6**	**yes**	**yes**	**chosen**
3	1	6	45.9	yes	yes	

Figure 8.19: Sprinkler design and pipes, Sandy Clay Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

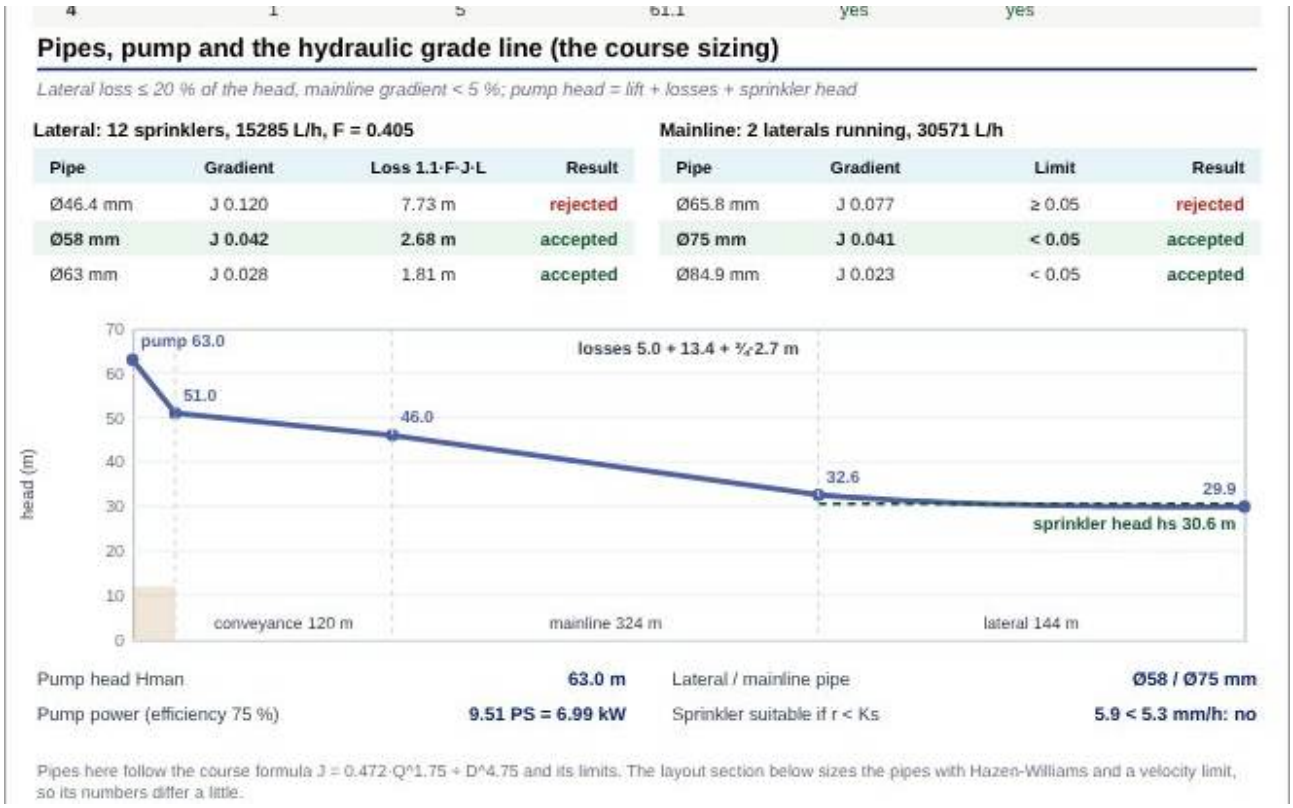
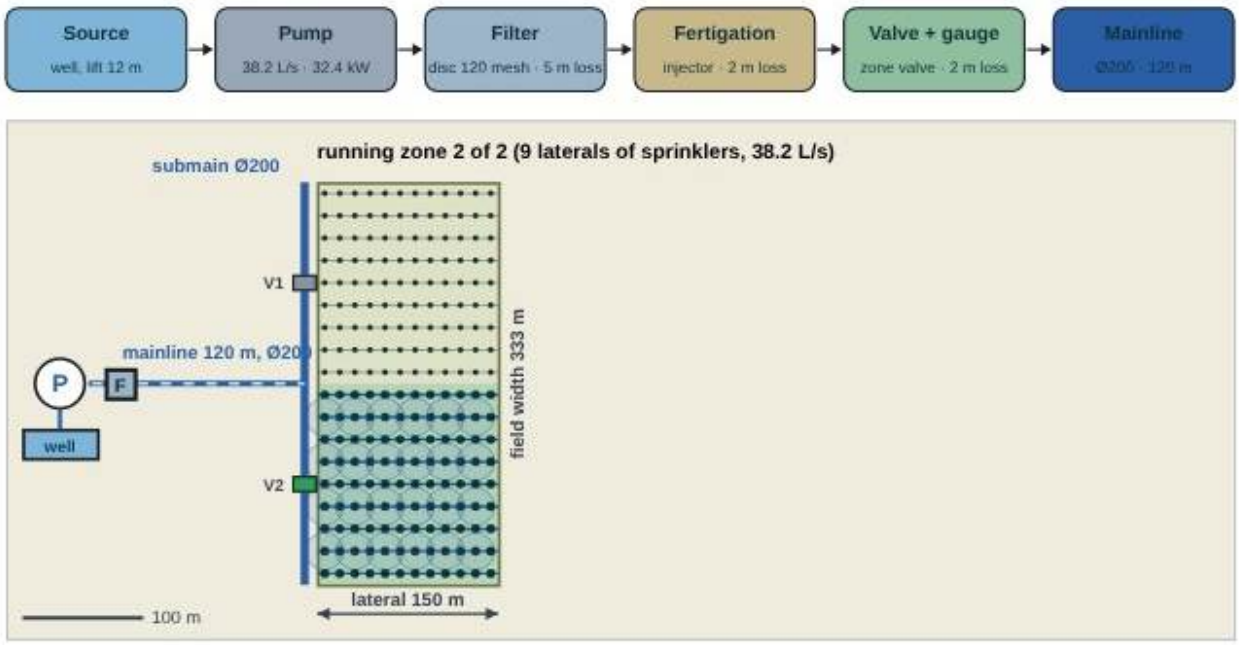
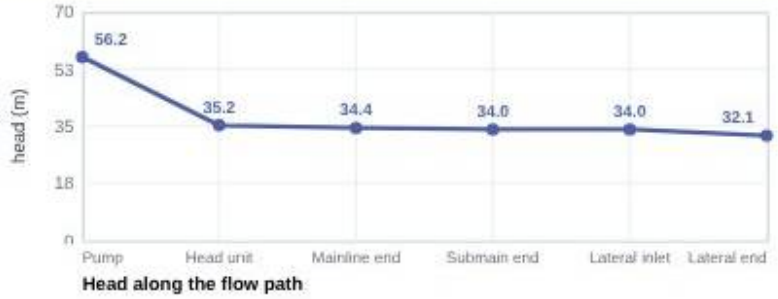


Figure 8.20: Sprinkler design and pipes, Sandy Clay Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4

Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

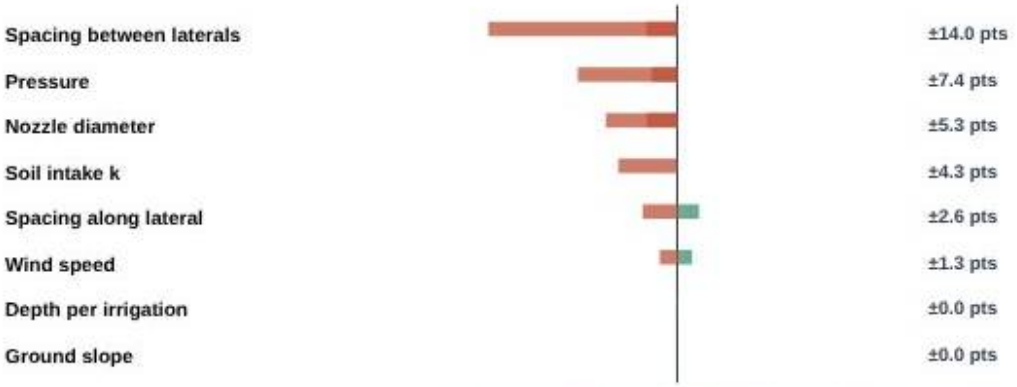
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	664	90.5	99.8	0.0	meets target

Figure 8.21: Sprinkler design and pipes, Sandy Clay Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**356 mm → 6 irrigations of 59 mm**
Gross water per irrigation	**65.2 mm (Ea 91 %)**
Gross water per season	**393 mm = 3927 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 664 min = 33.2 h per round**
Pumping per season	**200 hours · 6481 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**25 May 2027**	213 days	30	33	33.2 h	14 days
2	**8 Jun 2027**	227 days	30	33	33.2 h	11 days
3	**19 Jun 2027**	238 days	30	33	33.2 h	9 days
4	**28 Jun 2027**	247 days	30	33	33.2 h	9 days
5	**7 Jul 2027**	256 days	30	33	33.2 h	13 days
6	**20 Jul 2027**	269 days	30	33	33.2 h	260 days
7	**5 Apr 2028**	529 days	30	33	33.2 h	9 days
8	**14 Apr 2028**	538 days	30	33	33.2 h	-

Figure 8.22: Sprinkler design and pipes, Sandy Clay Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.

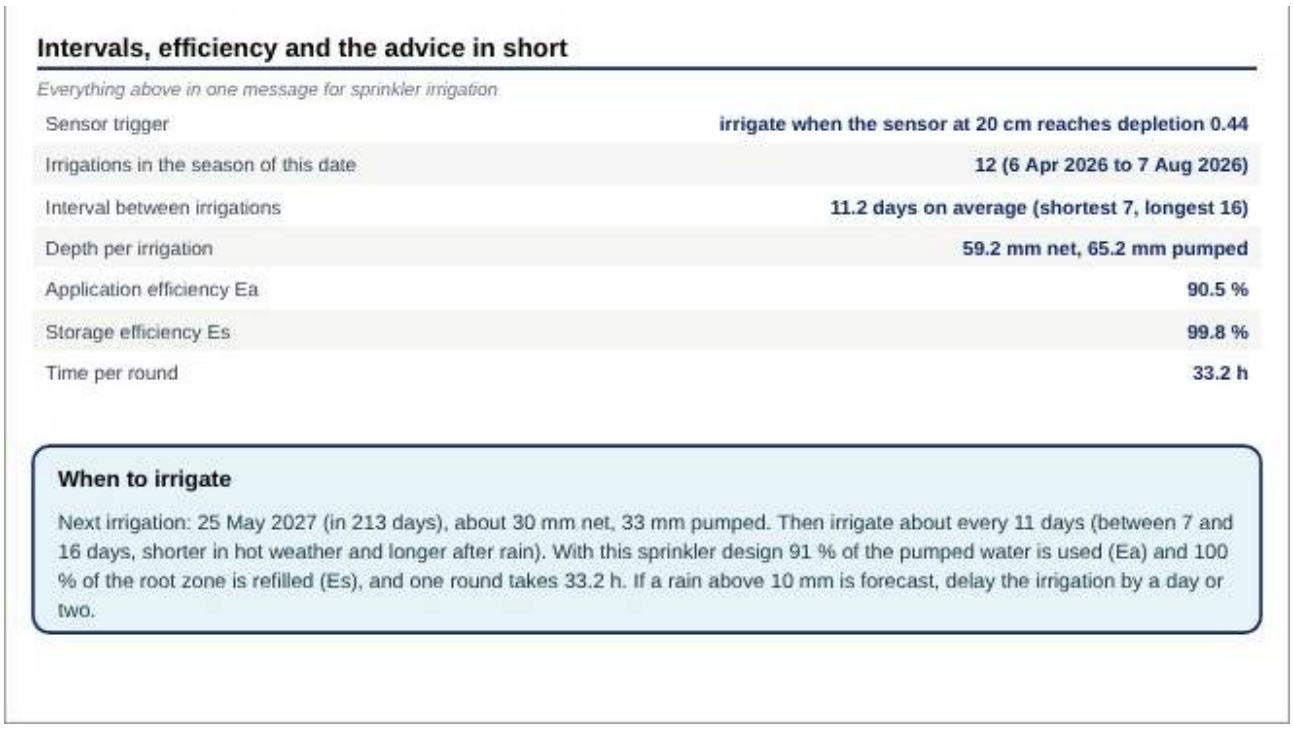


Figure 8.23: Sprinkler design and pipes, Sandy Clay Loam, part 12 of 12: Intervals, efficiency and the advice in short.

8.4 Virtual farm of the surface design (Sandy Clay Loam)

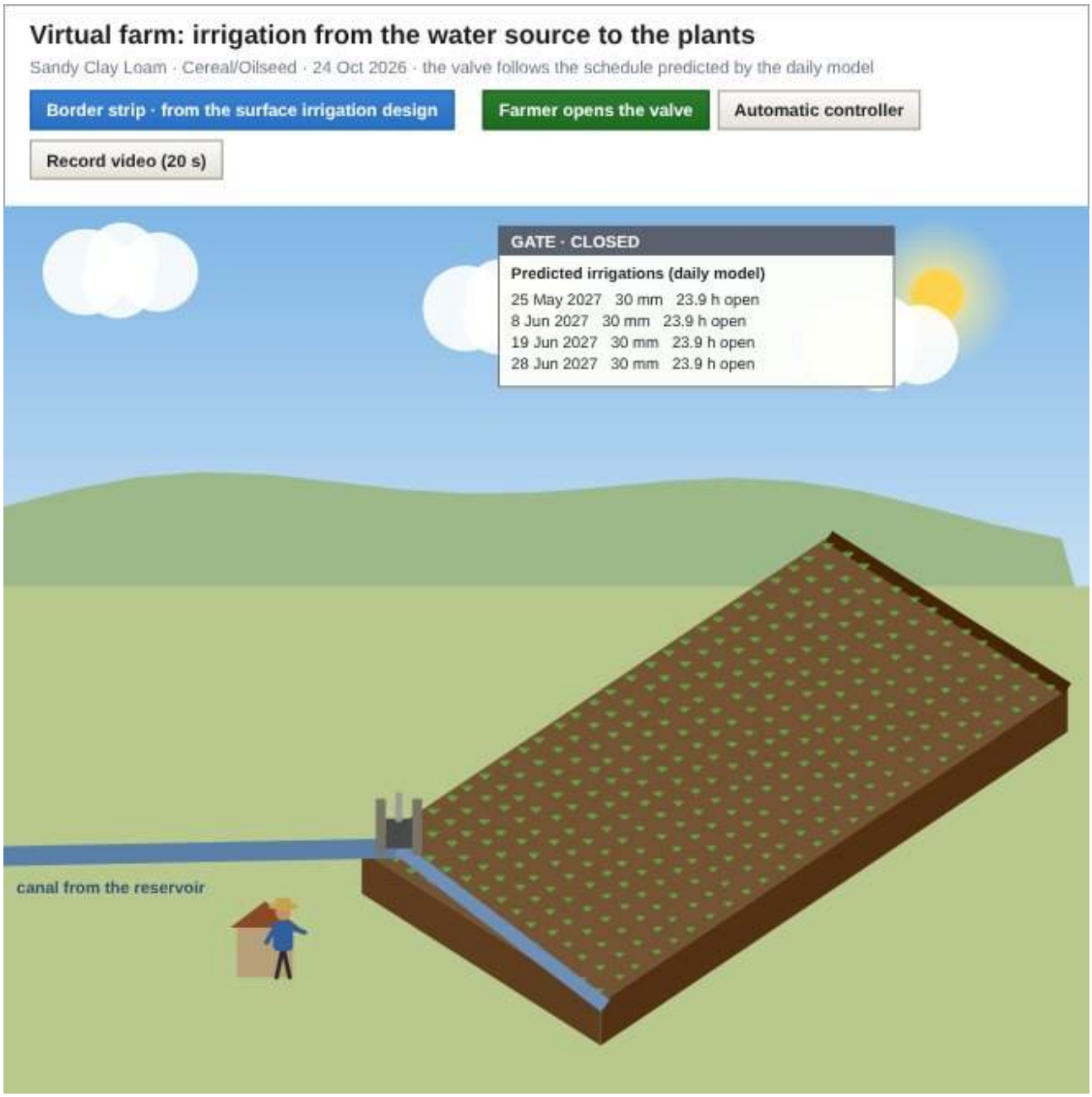


Figure 8.24: Virtual farm, surface design, Sandy Clay Loam, part 1 of 3.

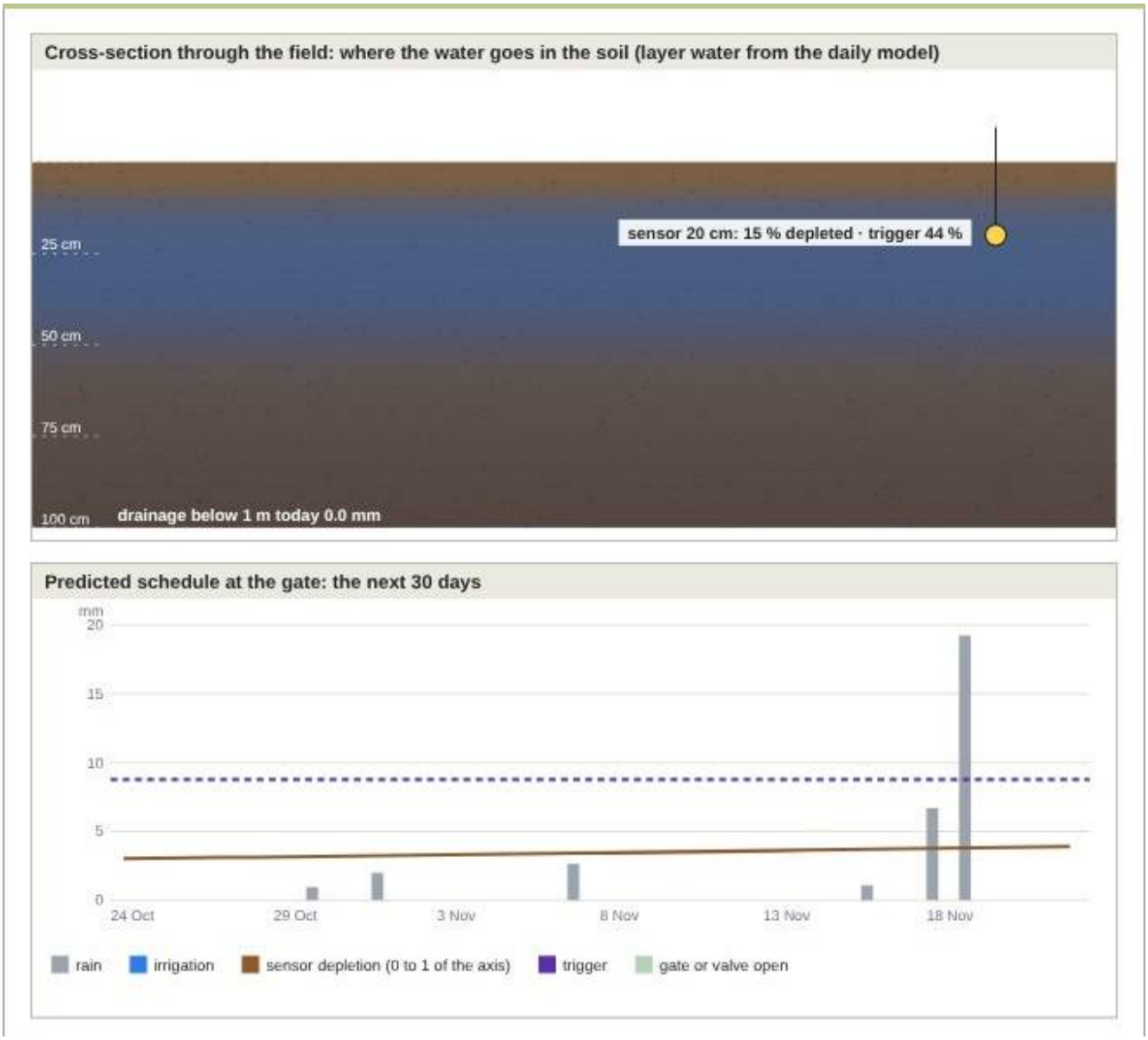


Figure 8.25: Virtual farm, surface design, Sandy Clay Loam, part 2 of 3.

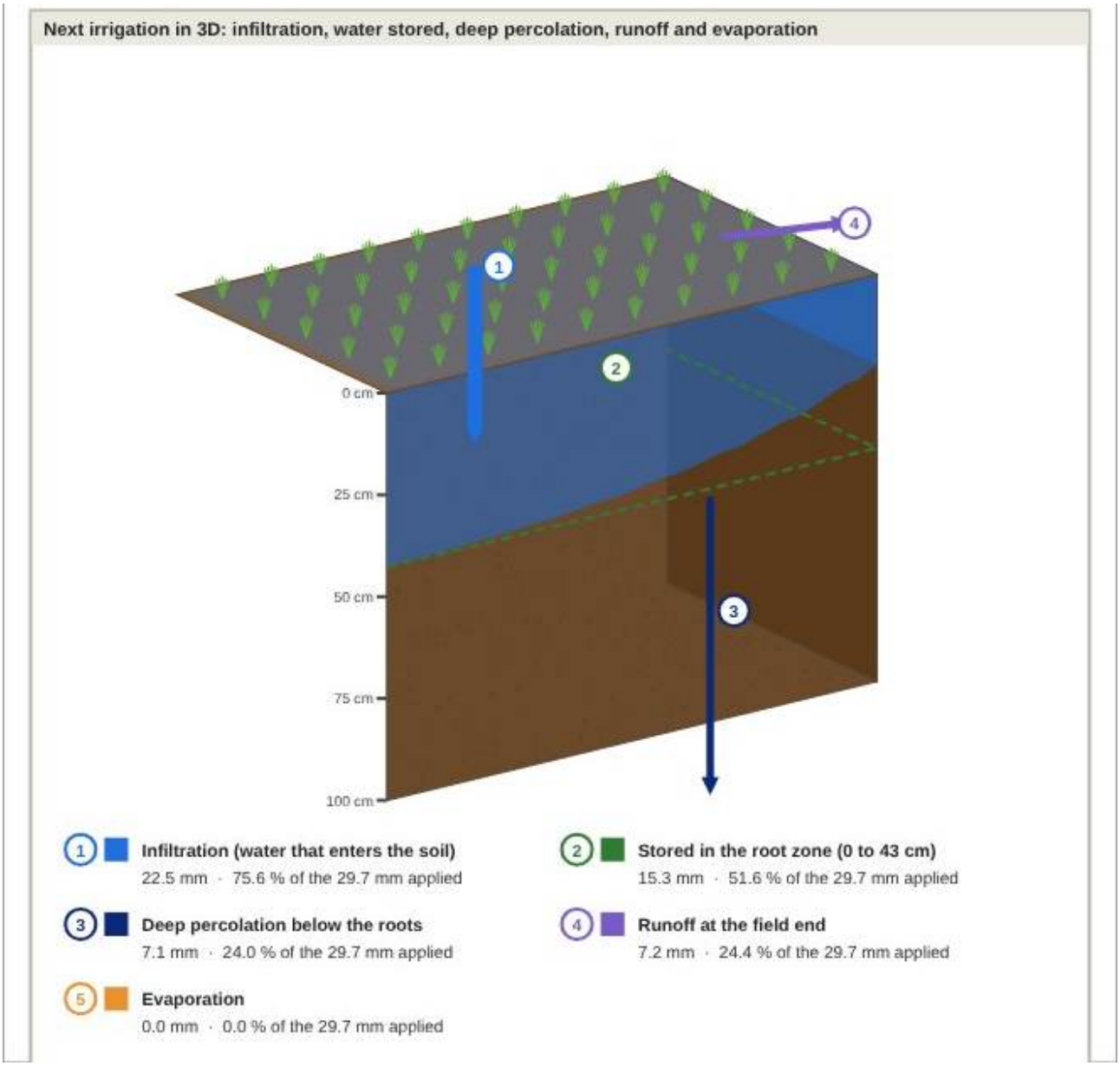


Figure 8.26: Virtual farm, surface design, Sandy Clay Loam, part 3 of 3.

8.5 Virtual farm of the drip design (Sandy Clay Loam)

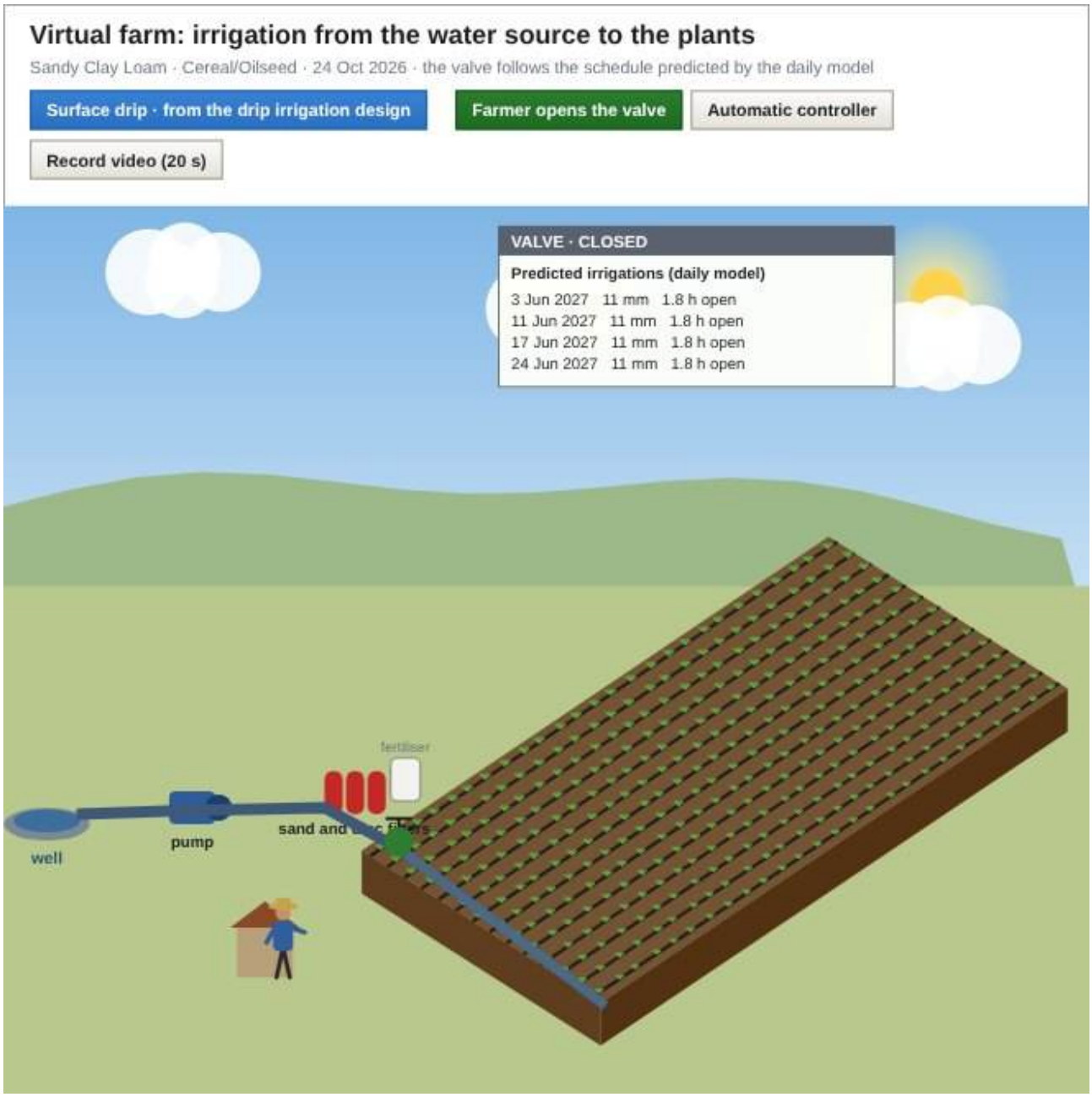


Figure 8.27: Virtual farm, drip design, Sandy Clay Loam, part 1 of 3.

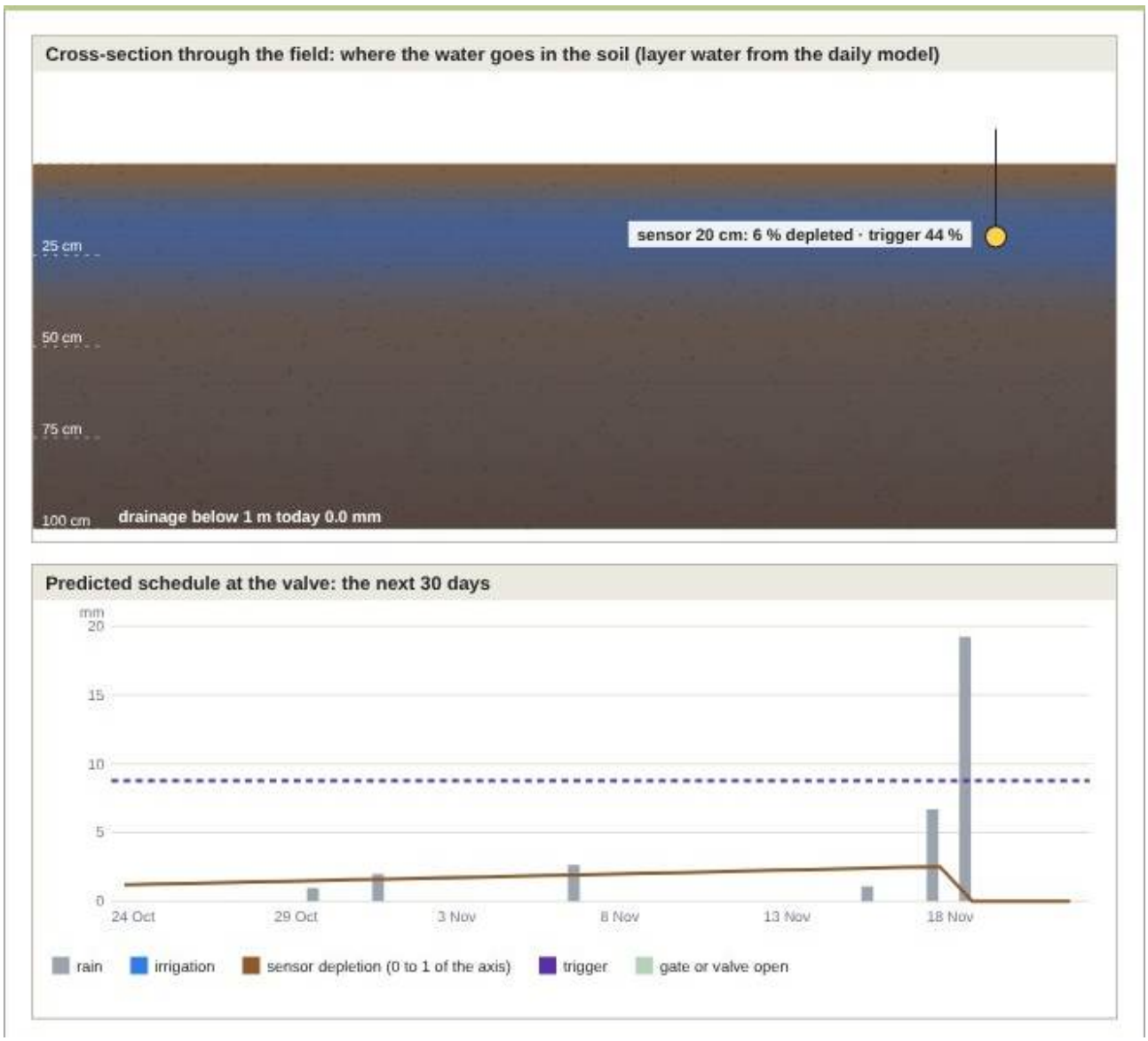


Figure 8.28: Virtual farm, drip design, Sandy Clay Loam, part 2 of 3.

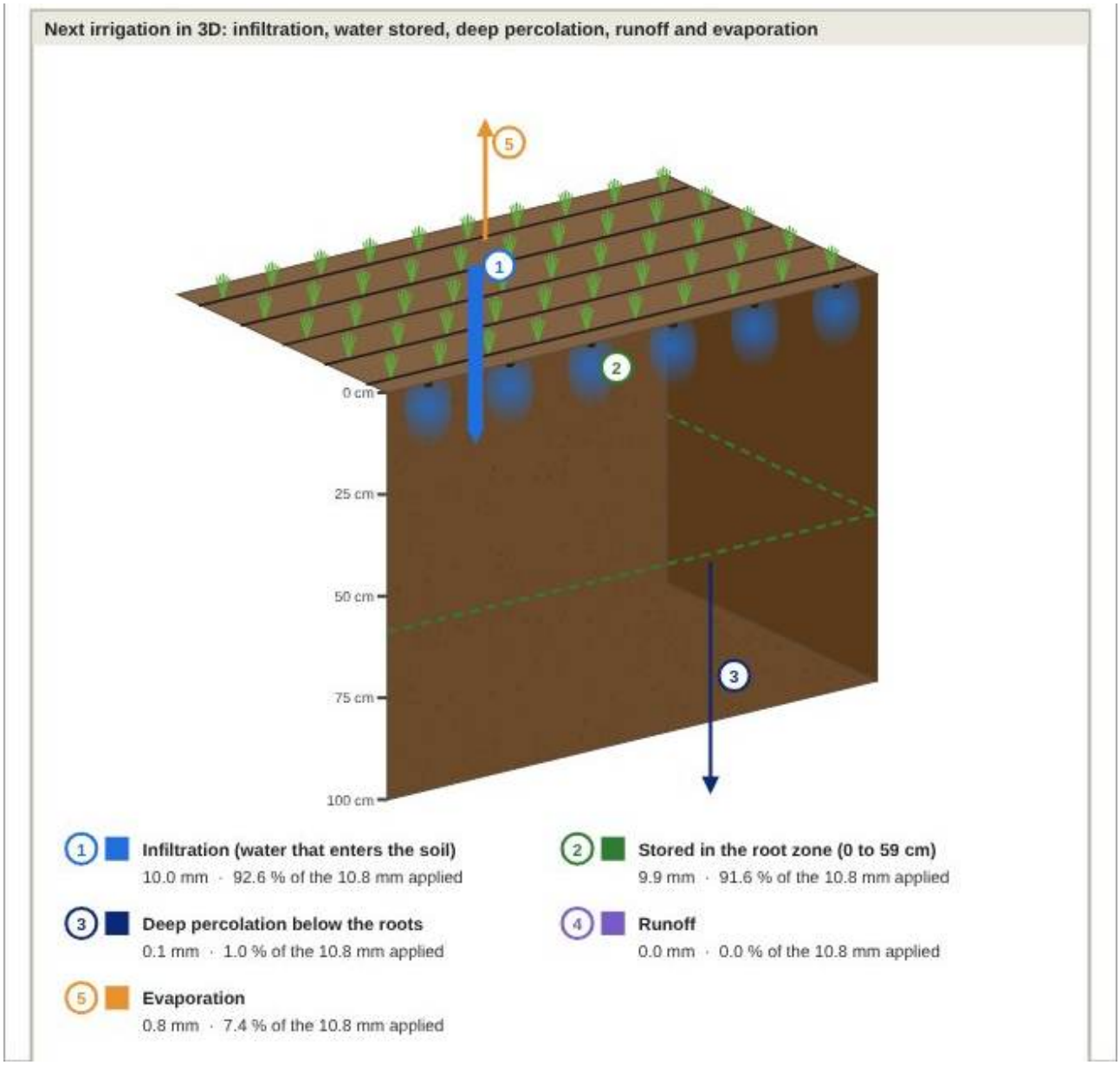


Figure 8.29: Virtual farm, drip design, Sandy Clay Loam, part 3 of 3.

8.6 Virtual farm of the sprinkler design (Sandy Clay Loam)

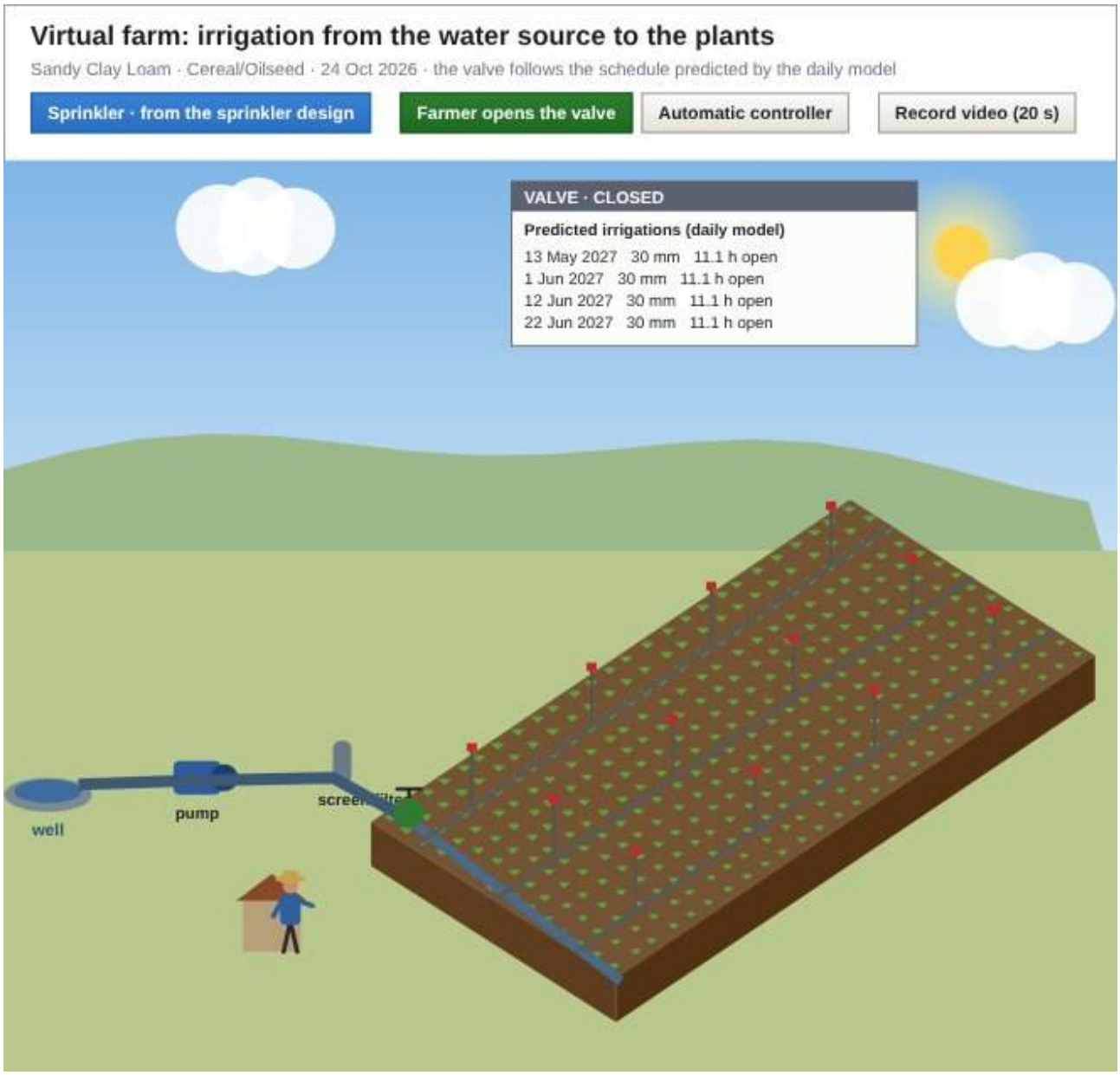


Figure 8.30: Virtual farm, sprinkler design, Sandy Clay Loam, part 1 of 3.

Figure 8.31: Virtual farm, sprinkler design, Sandy Clay Loam, part 2 of 3.

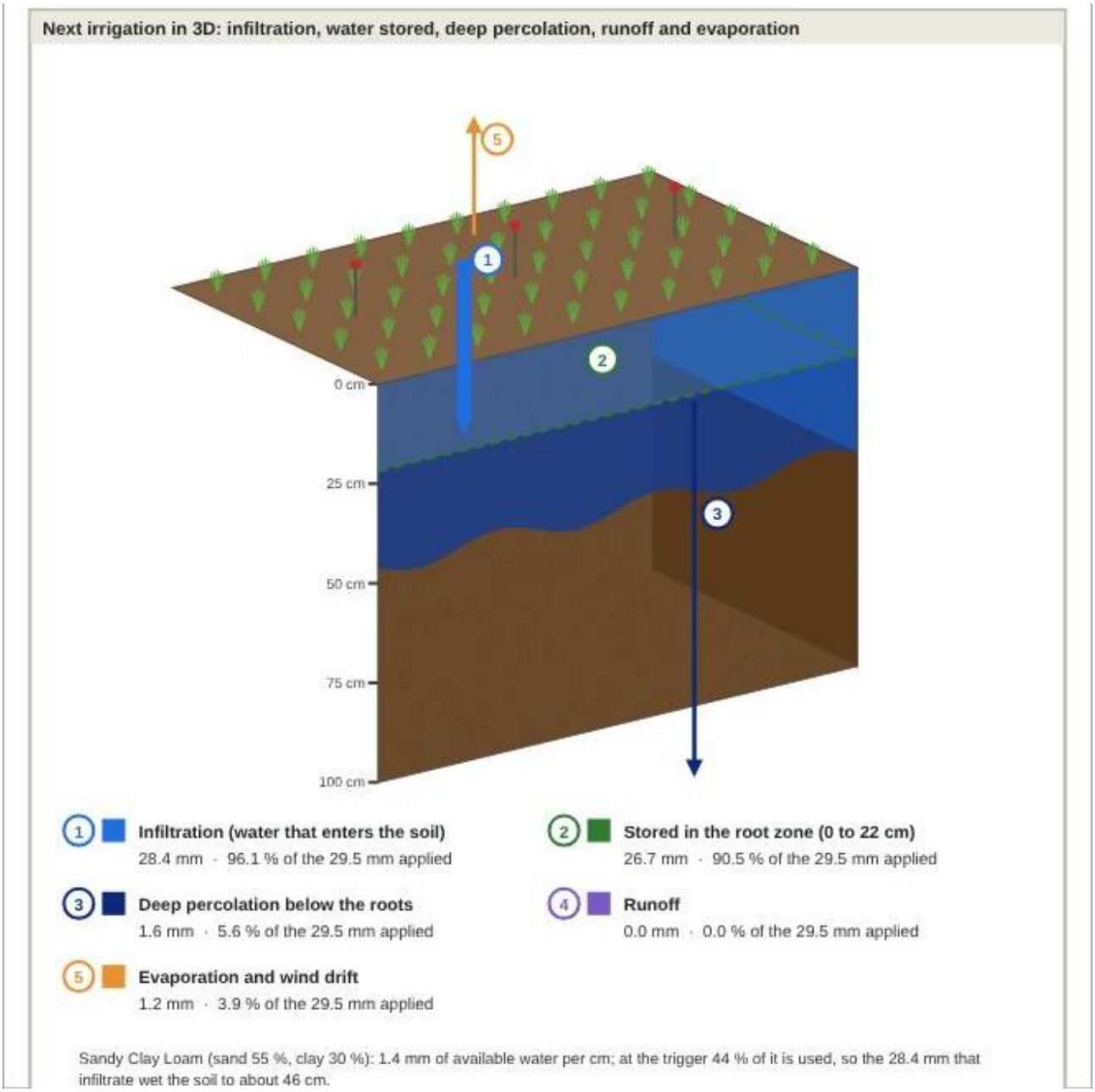


Figure 8.32: Virtual farm, sprinkler design, Sandy Clay Loam, part 3 of 3.

8.7 Sensor and trigger (Sandy Clay Loam)

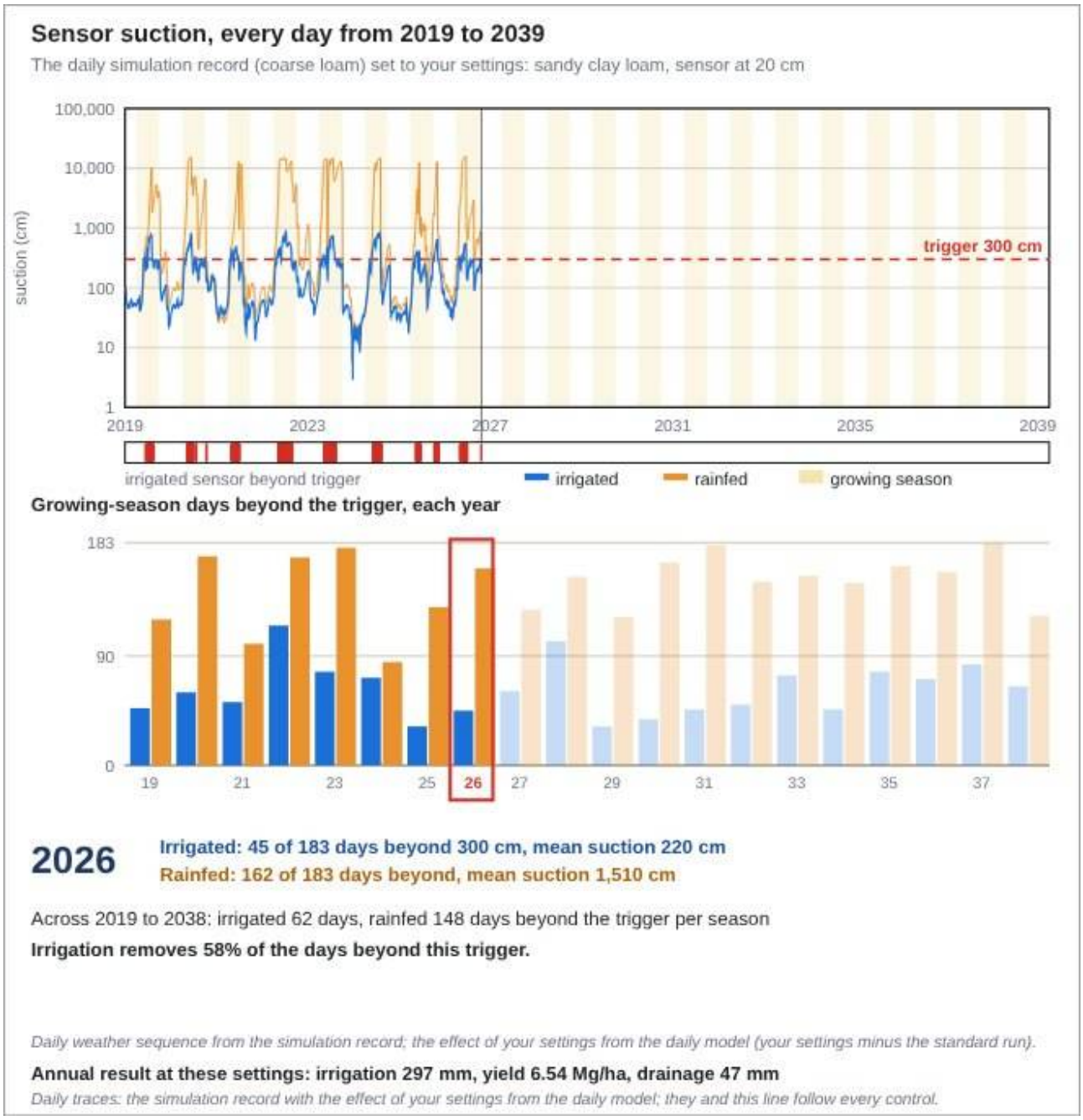


Figure 8.33: Sensor and trigger, Sandy Clay Loam, part 1 of 1.

8.8 Trigger law (Sandy Clay Loam)

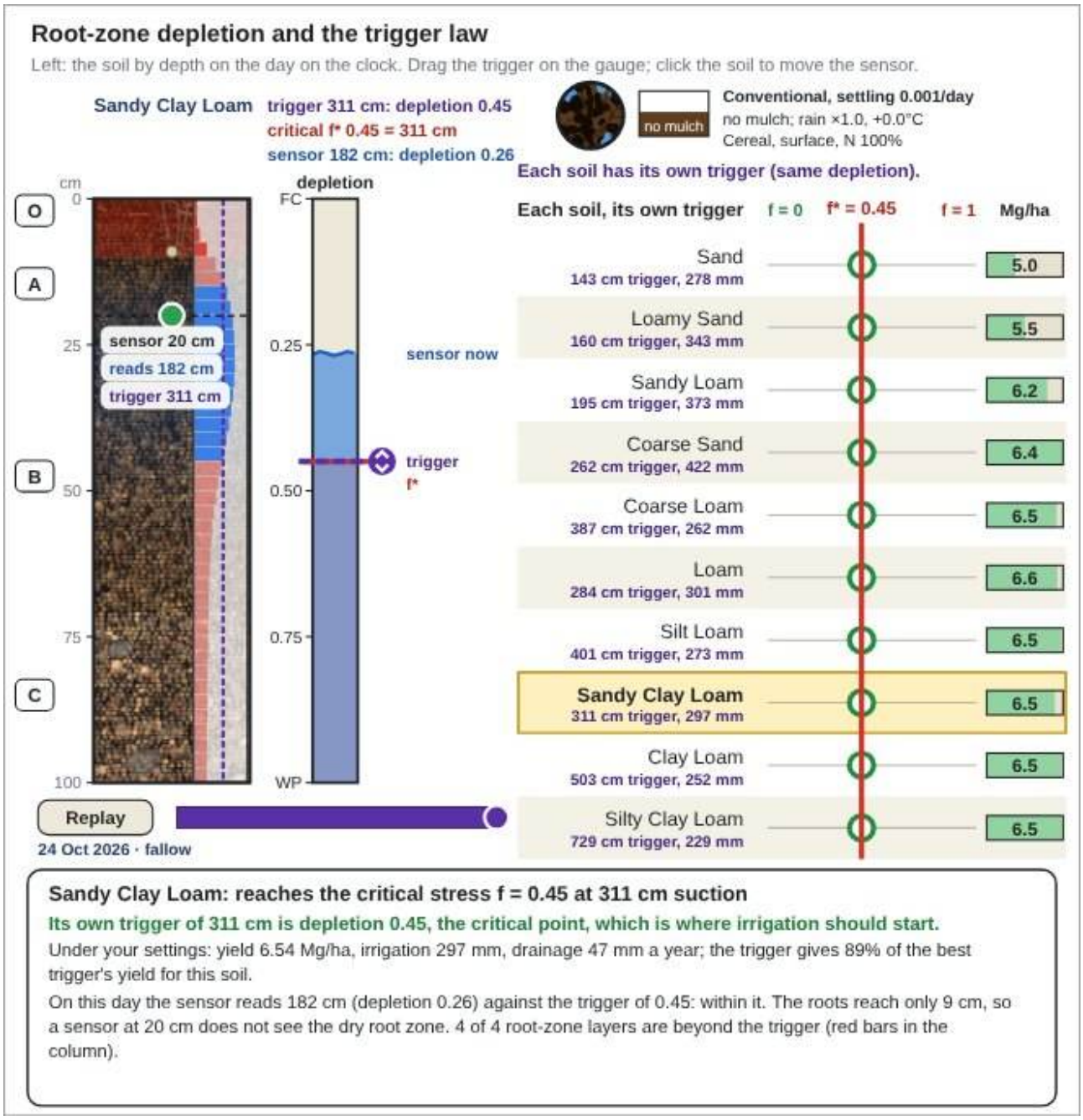


Figure 8.34: Trigger law, Sandy Clay Loam, part 1 of 1.

8.9 Mulch against bare soil (Sandy Clay Loam)

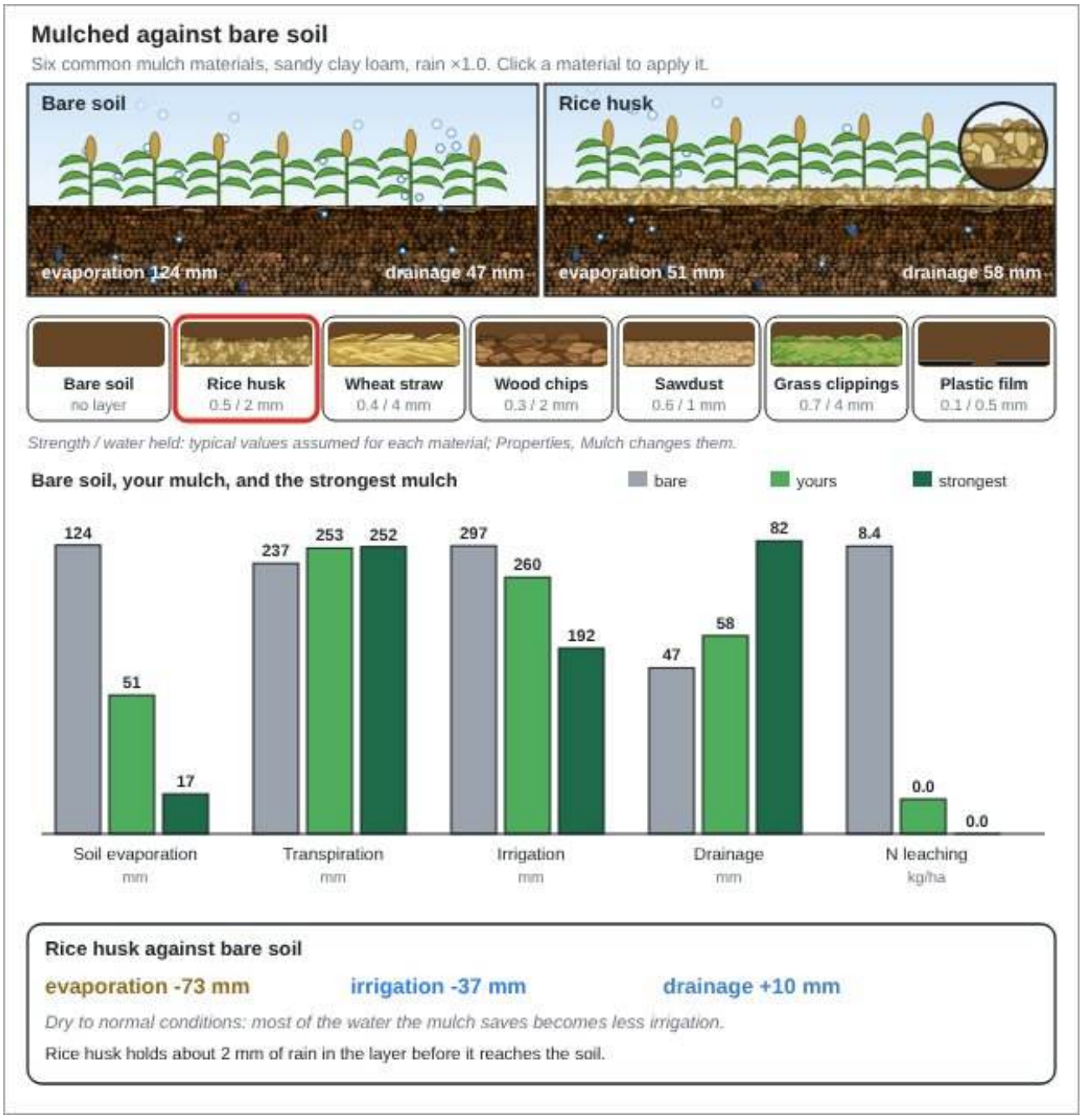


Figure 8.35: Mulch against bare soil, Sandy Clay Loam, part 1 of 1.

8.10 Mulch control (Sandy Clay Loam)

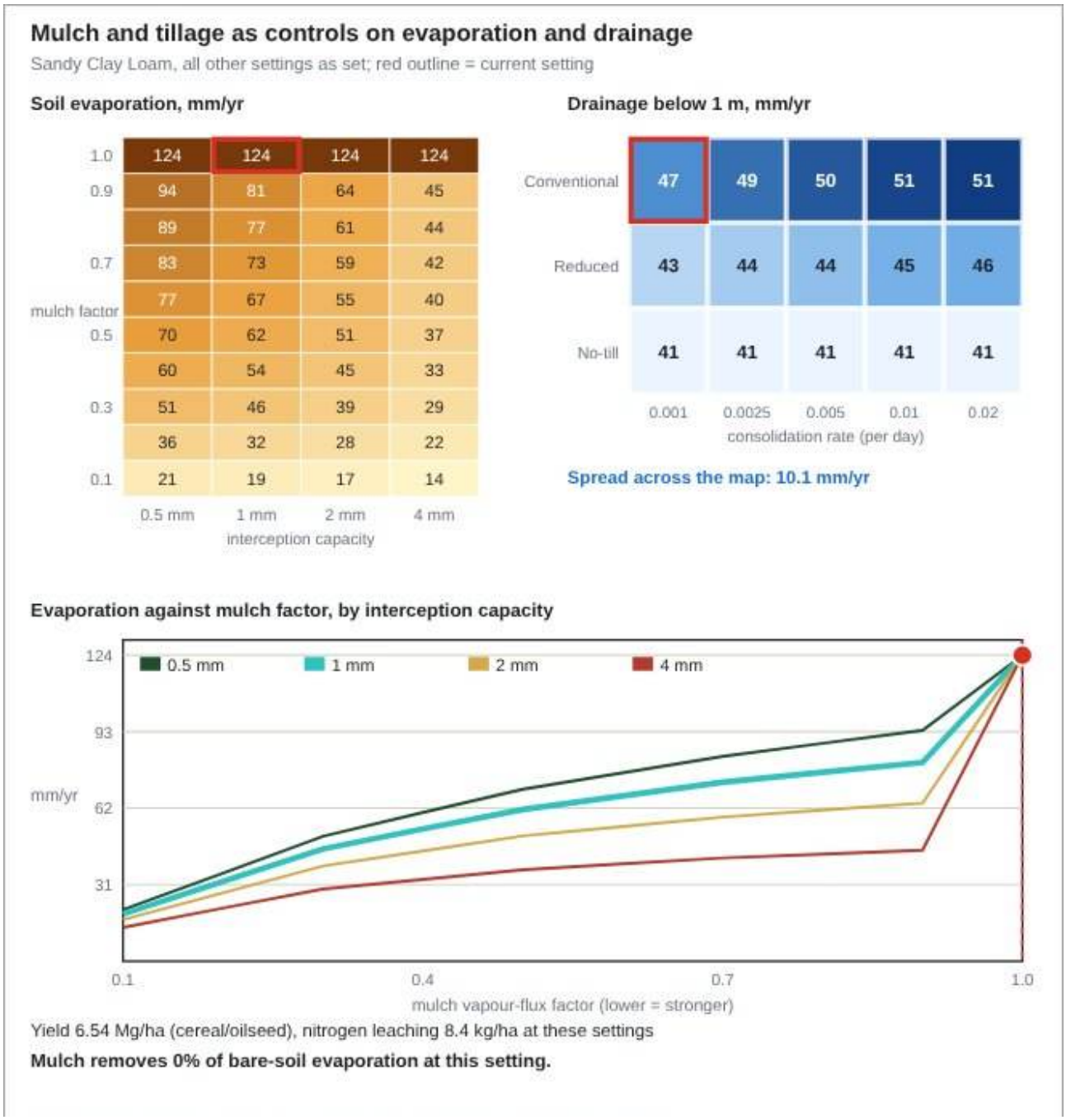


Figure 8.36: Mulch control, Sandy Clay Loam, part 1 of 2.

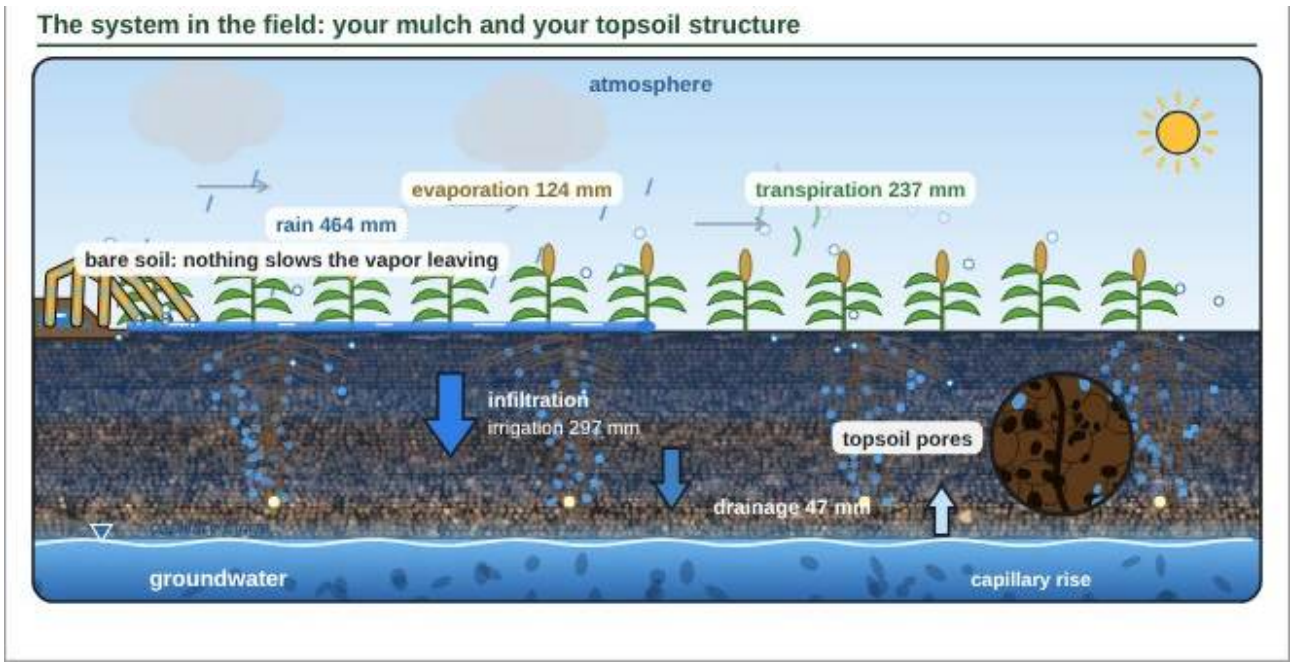


Figure 8.37: Mulch control, Sandy Clay Loam, part 2 of 2.

8.11 Evaporation physics (Sandy Clay Loam)

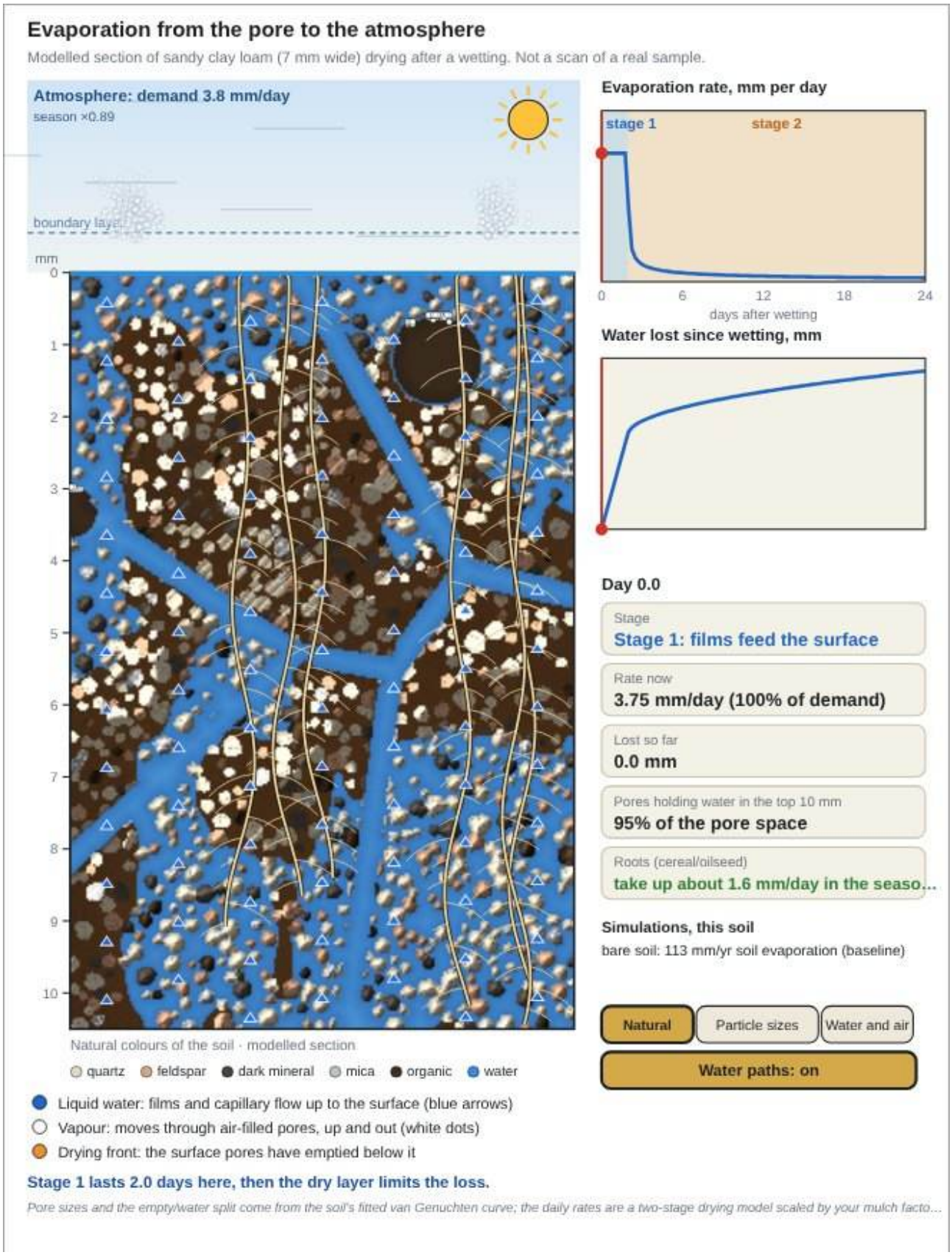


Figure 8.38: Evaporation physics, Sandy Clay Loam, part 1 of 3.

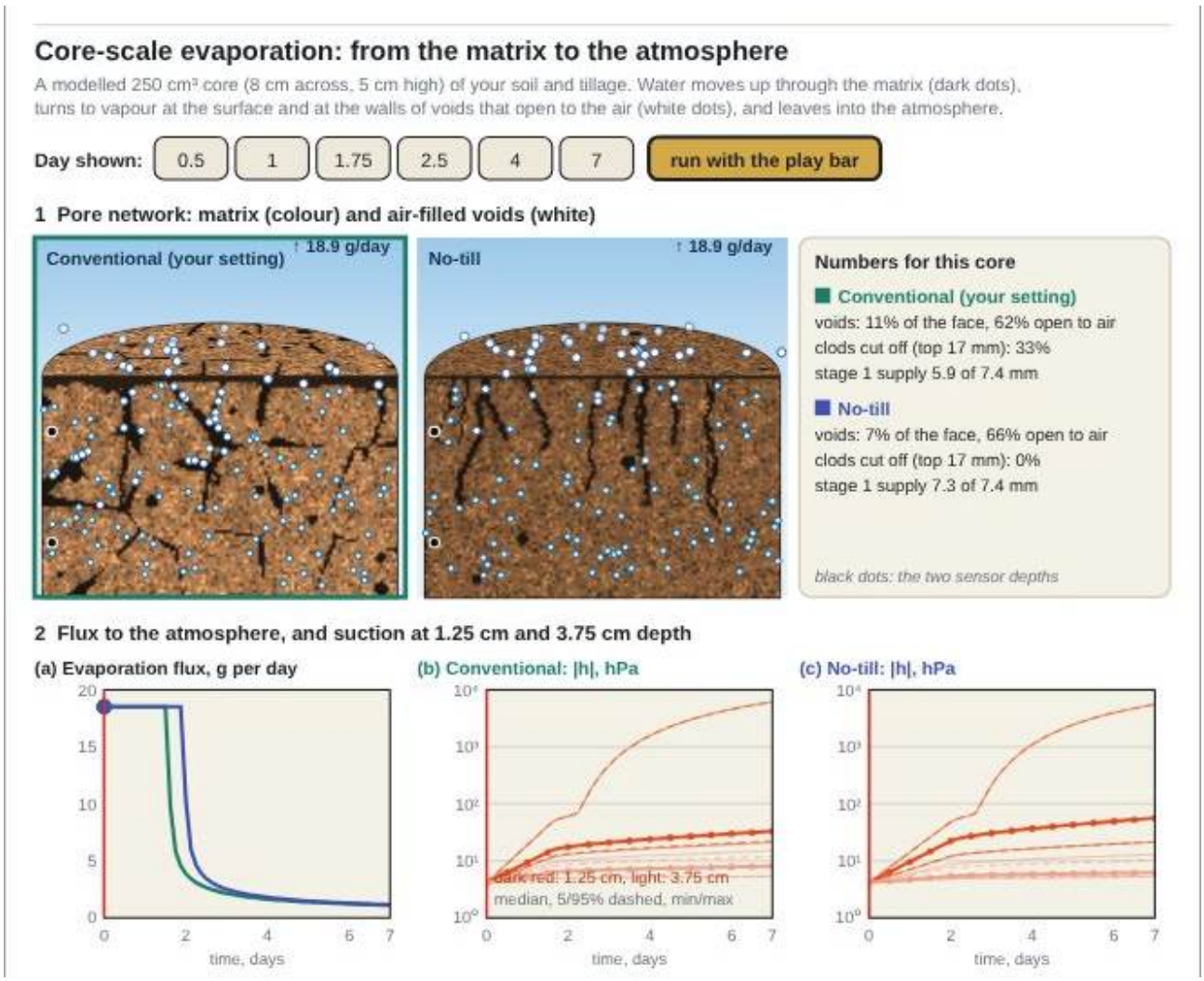


Figure 8.39: Evaporation physics, Sandy Clay Loam, part 2 of 3.

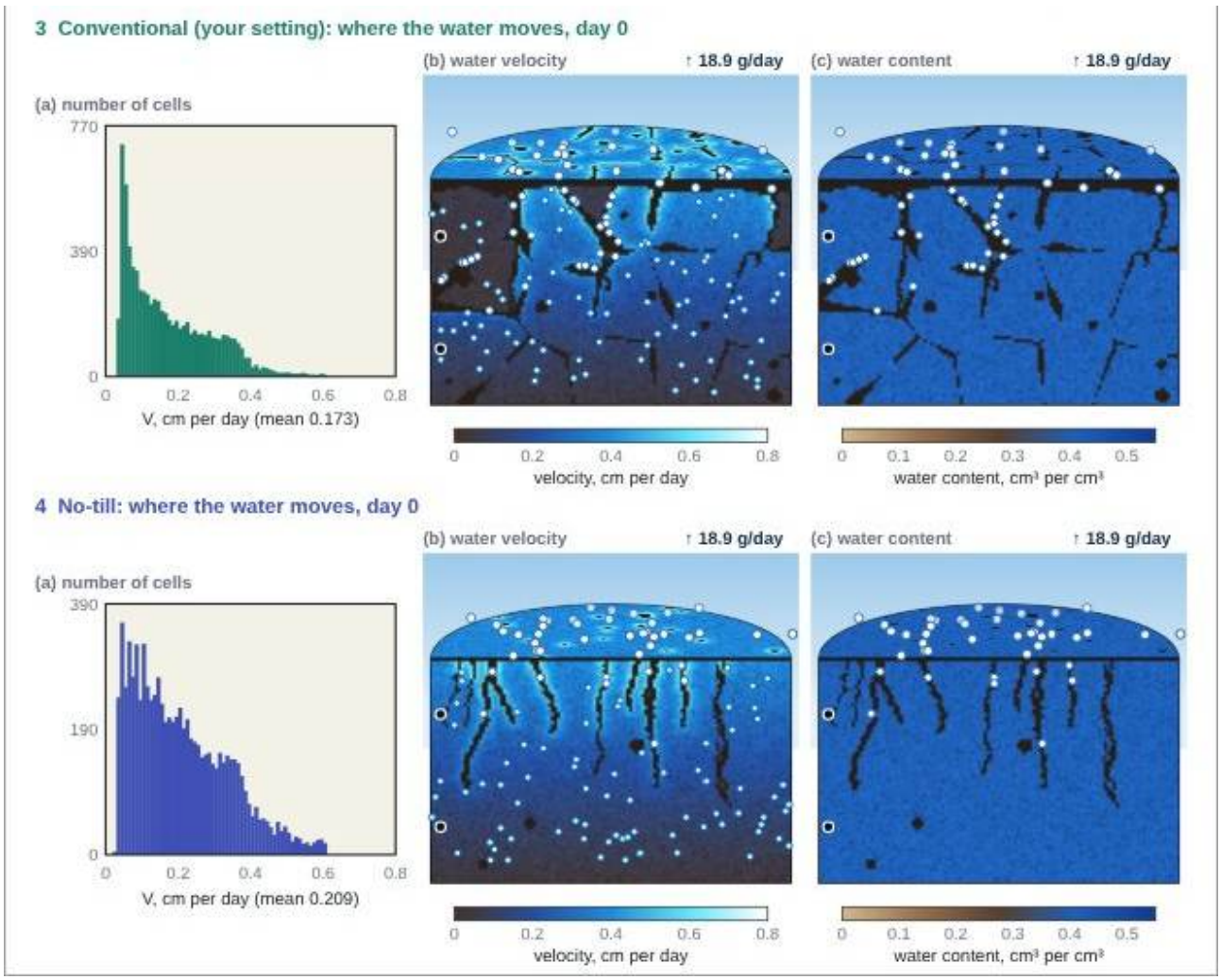


Figure 8.40: Evaporation physics, Sandy Clay Loam, part 3 of 3.

8.12 Fate of saved water (Sandy Clay Loam)

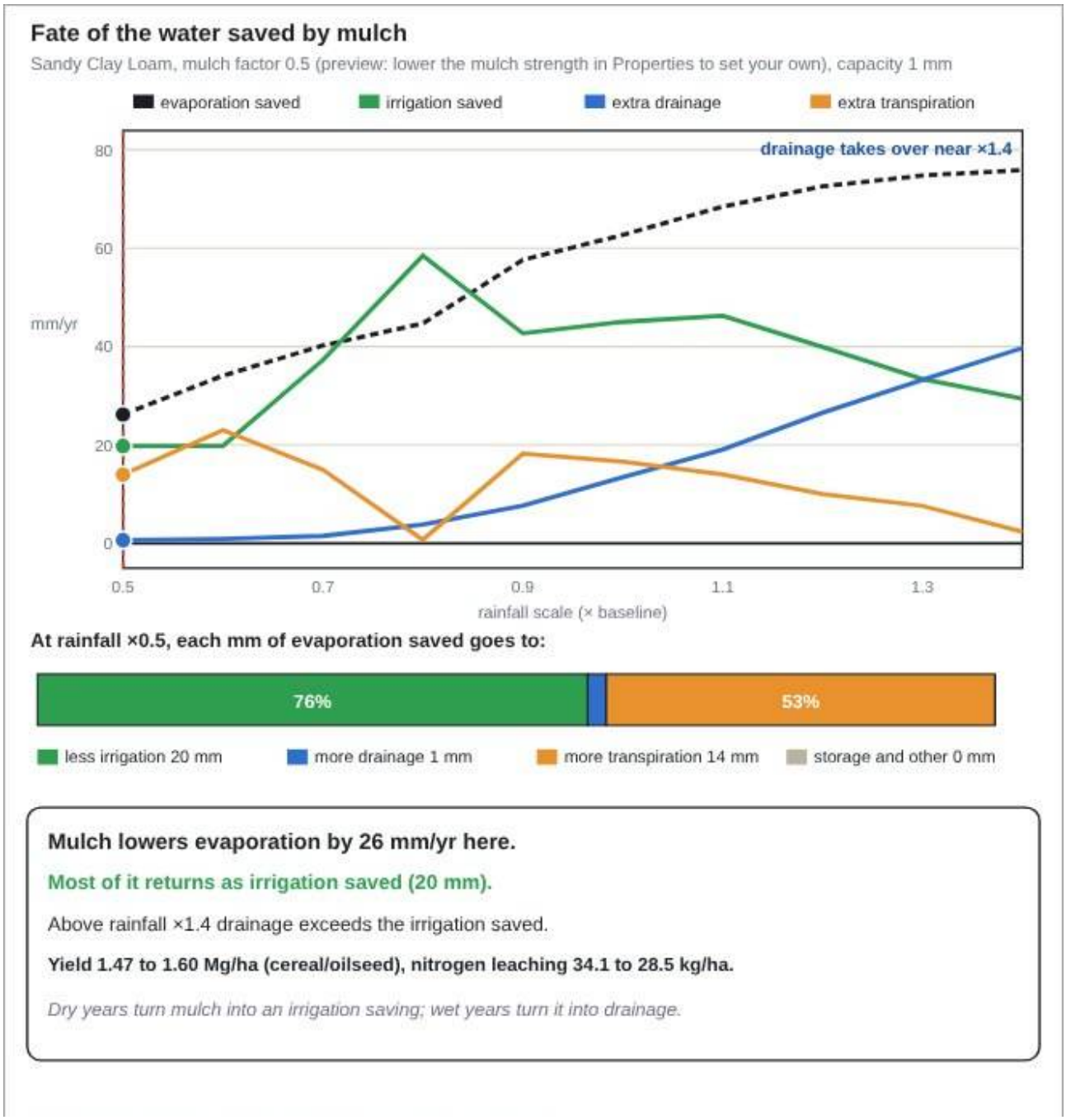


Figure 8.41: Fate of saved water, Sandy Clay Loam, part 1 of 2.

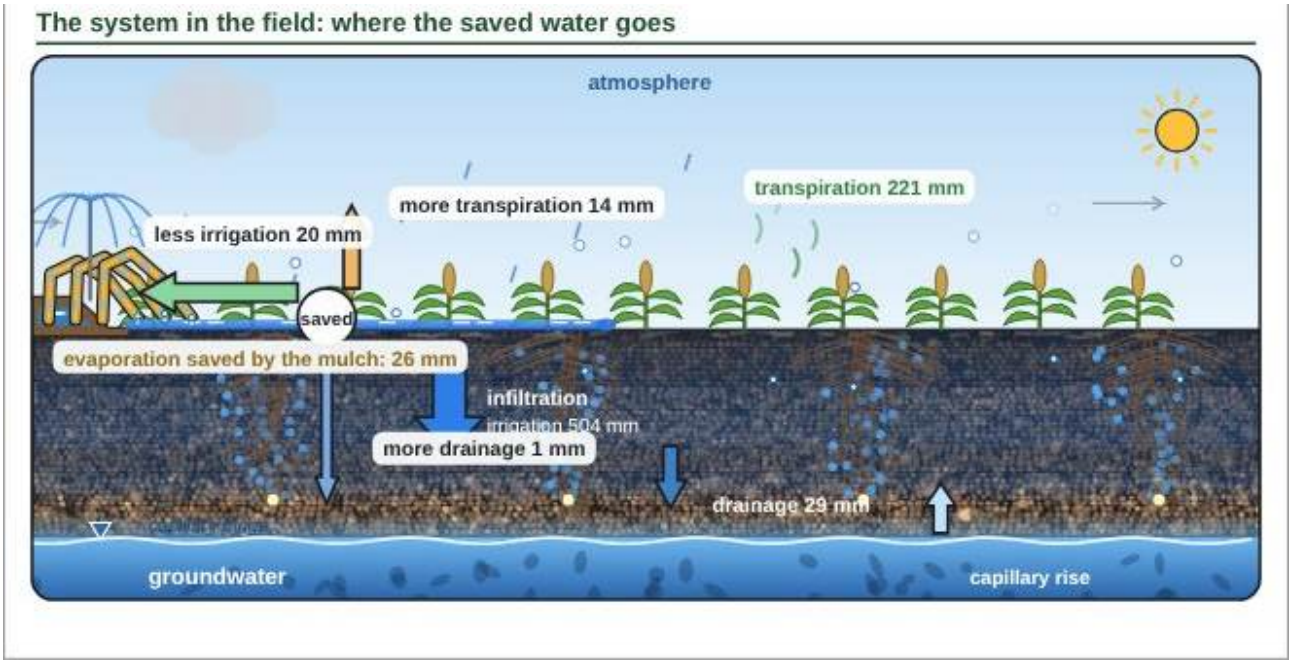


Figure 8.42: Fate of saved water, Sandy Clay Loam, part 2 of 2.

8.13 Regime map (Sandy Clay Loam)

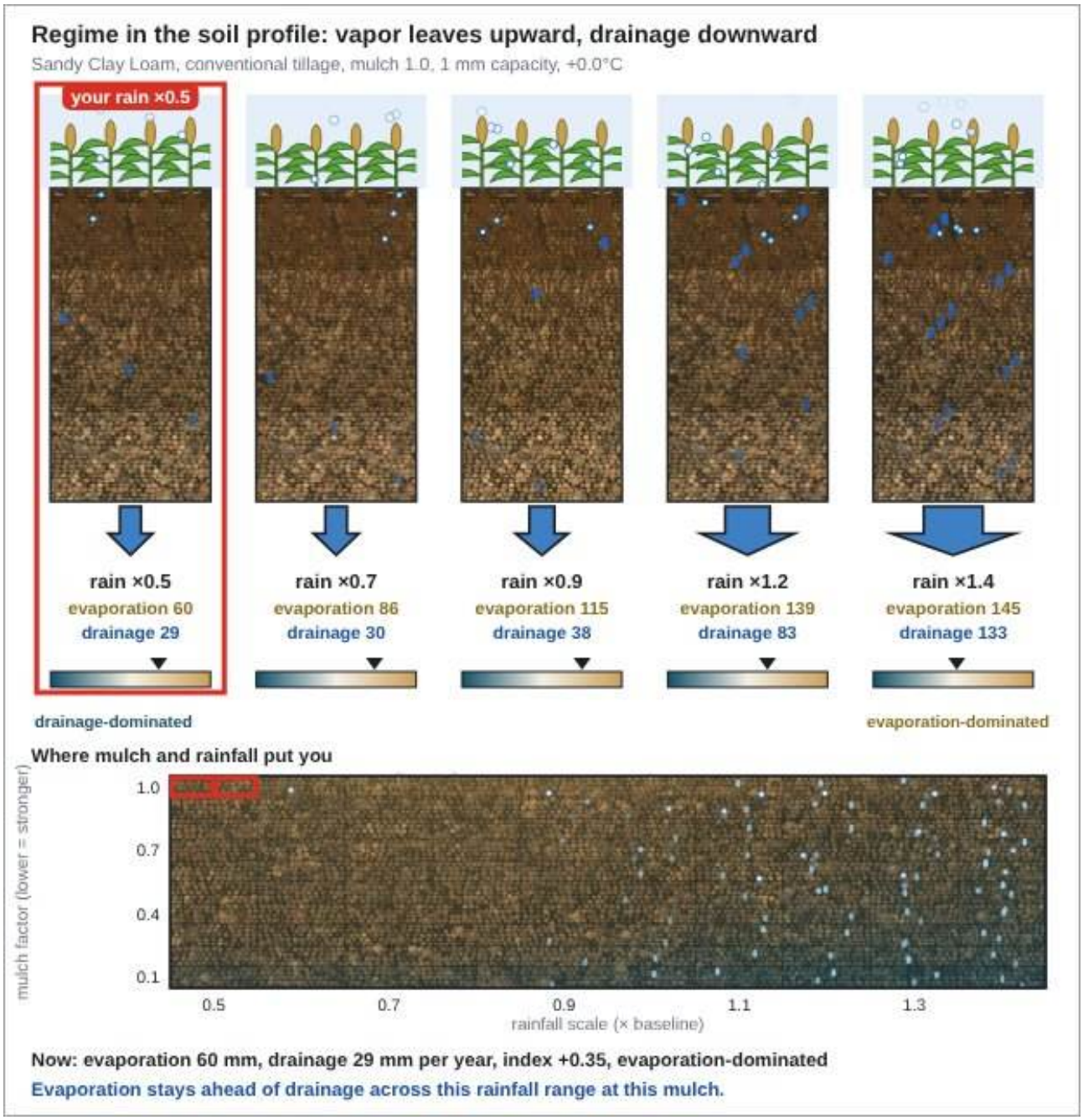


Figure 8.43: Regime map, Sandy Clay Loam, part 1 of 1.

8.14 Tillage and structure (Sandy Clay Loam)

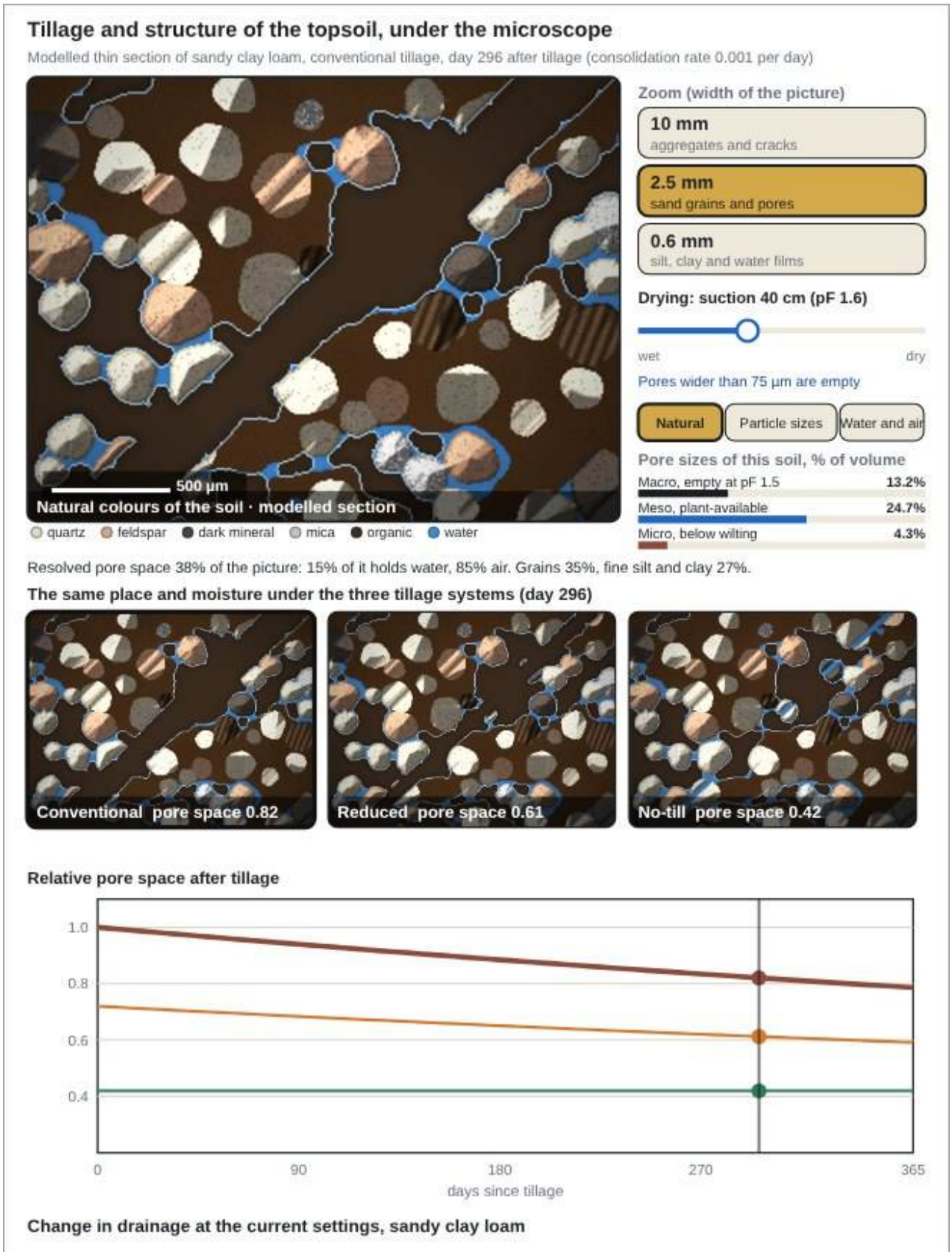


Figure 8.44: Tillage and structure, Sandy Clay Loam, part 1 of 2.

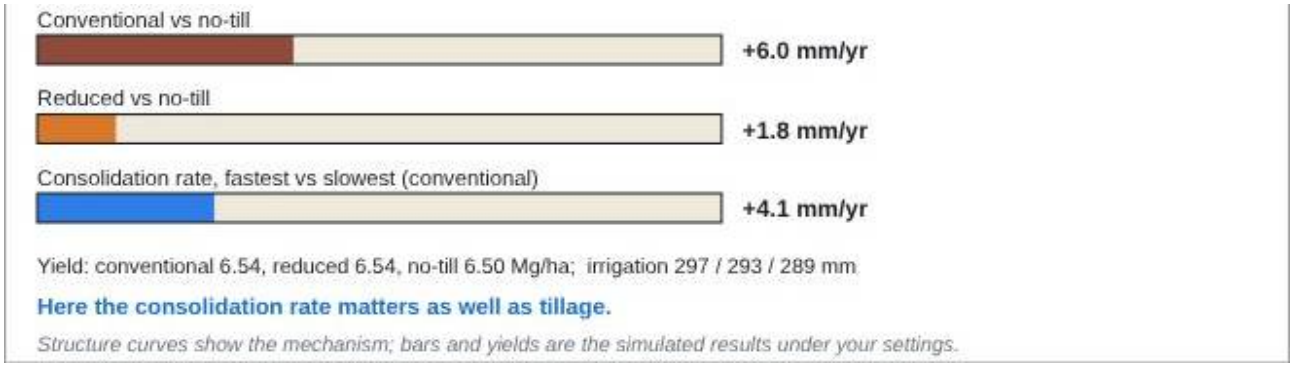


Figure 8.45: Tillage and structure, Sandy Clay Loam, part 2 of 2.

8.15 Soil hydraulic drivers (Sandy Clay Loam)

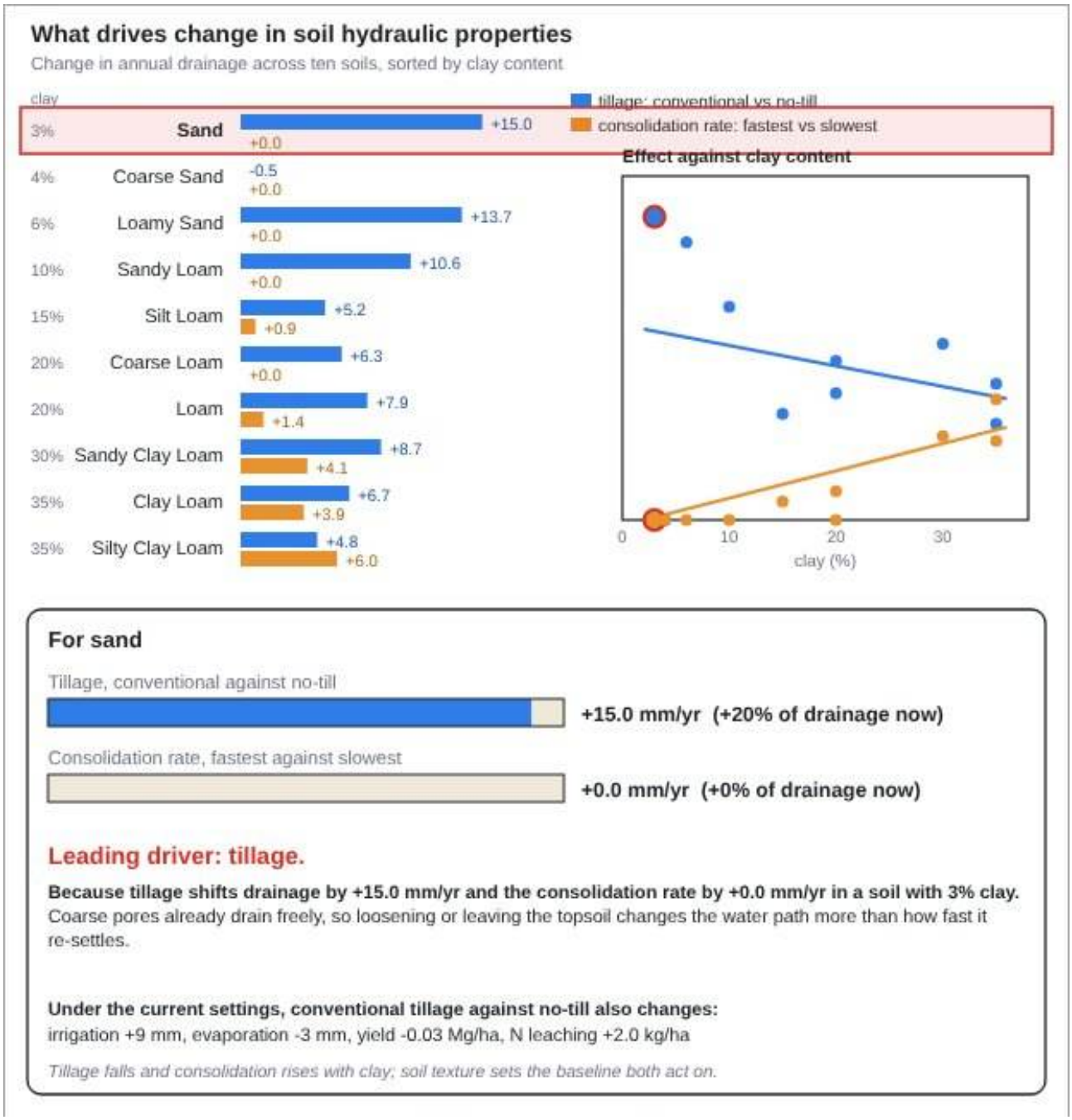


Figure 8.46: Soil hydraulic drivers, Sandy Clay Loam, part 1 of 3.

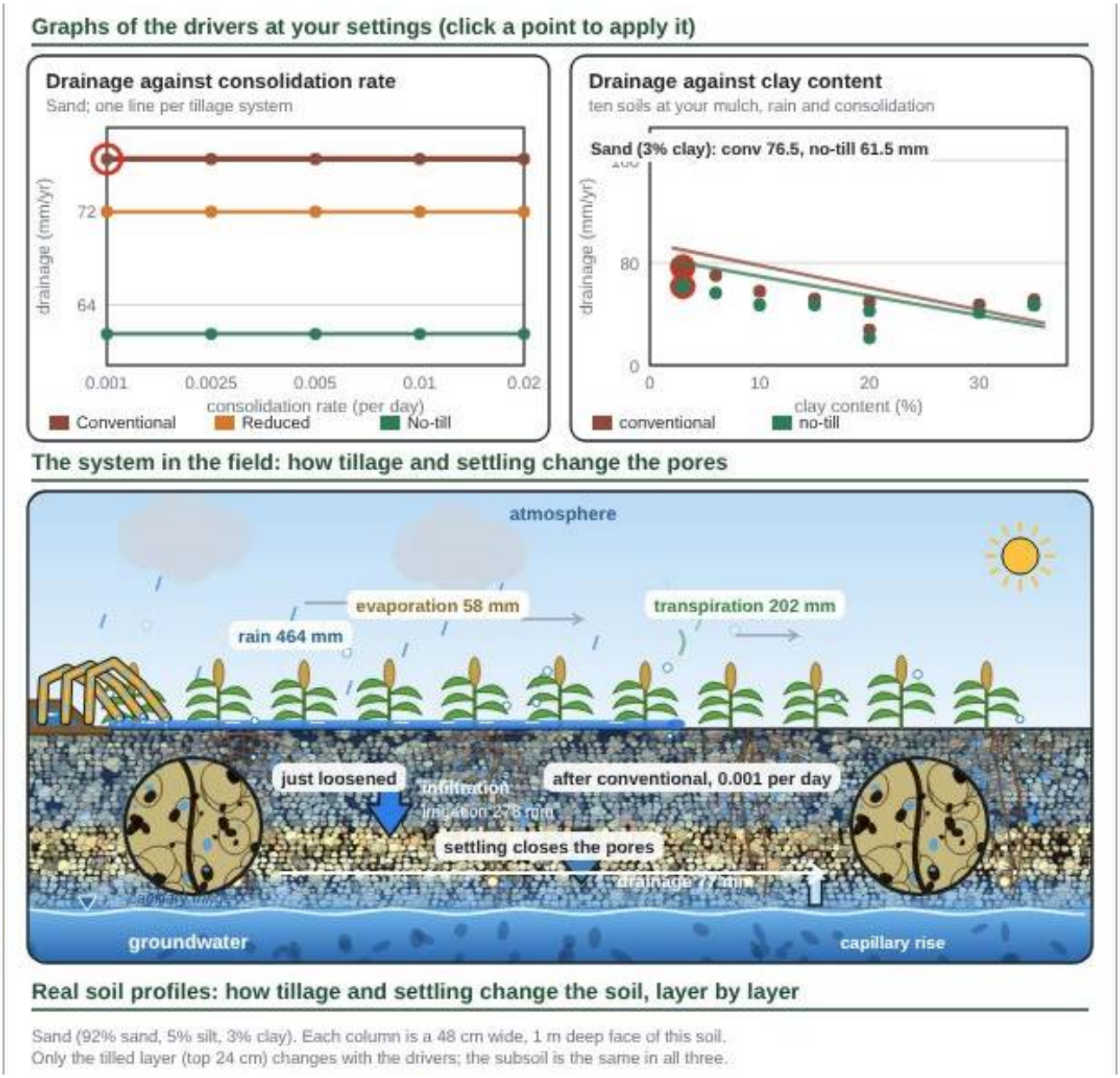


Figure 8.47: Soil hydraulic drivers, Sandy Clay Loam, part 2 of 3.

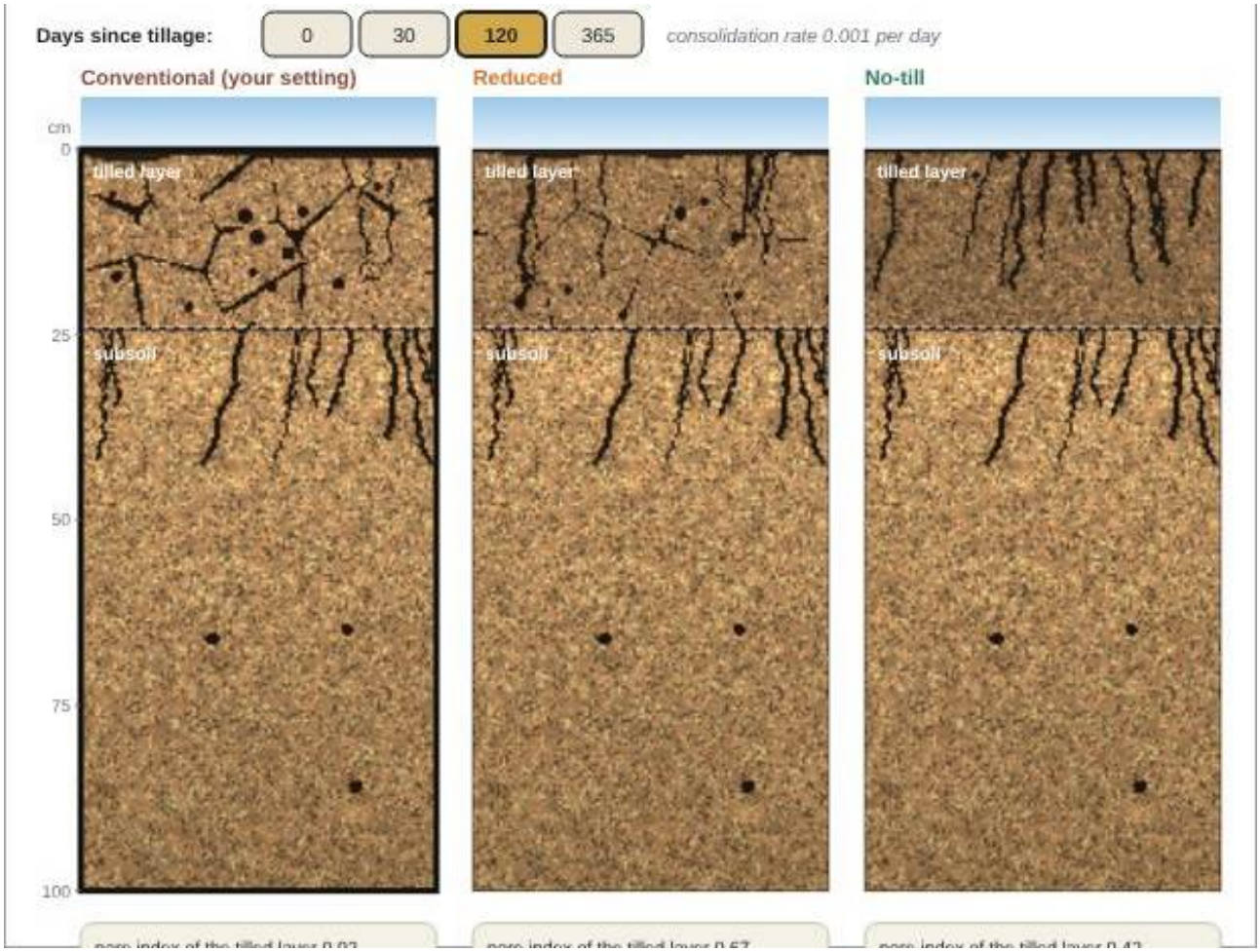


Figure 8.48: Soil hydraulic drivers, Sandy Clay Loam, part 3 of 3.

8.16 Root structure (Sandy Clay Loam)

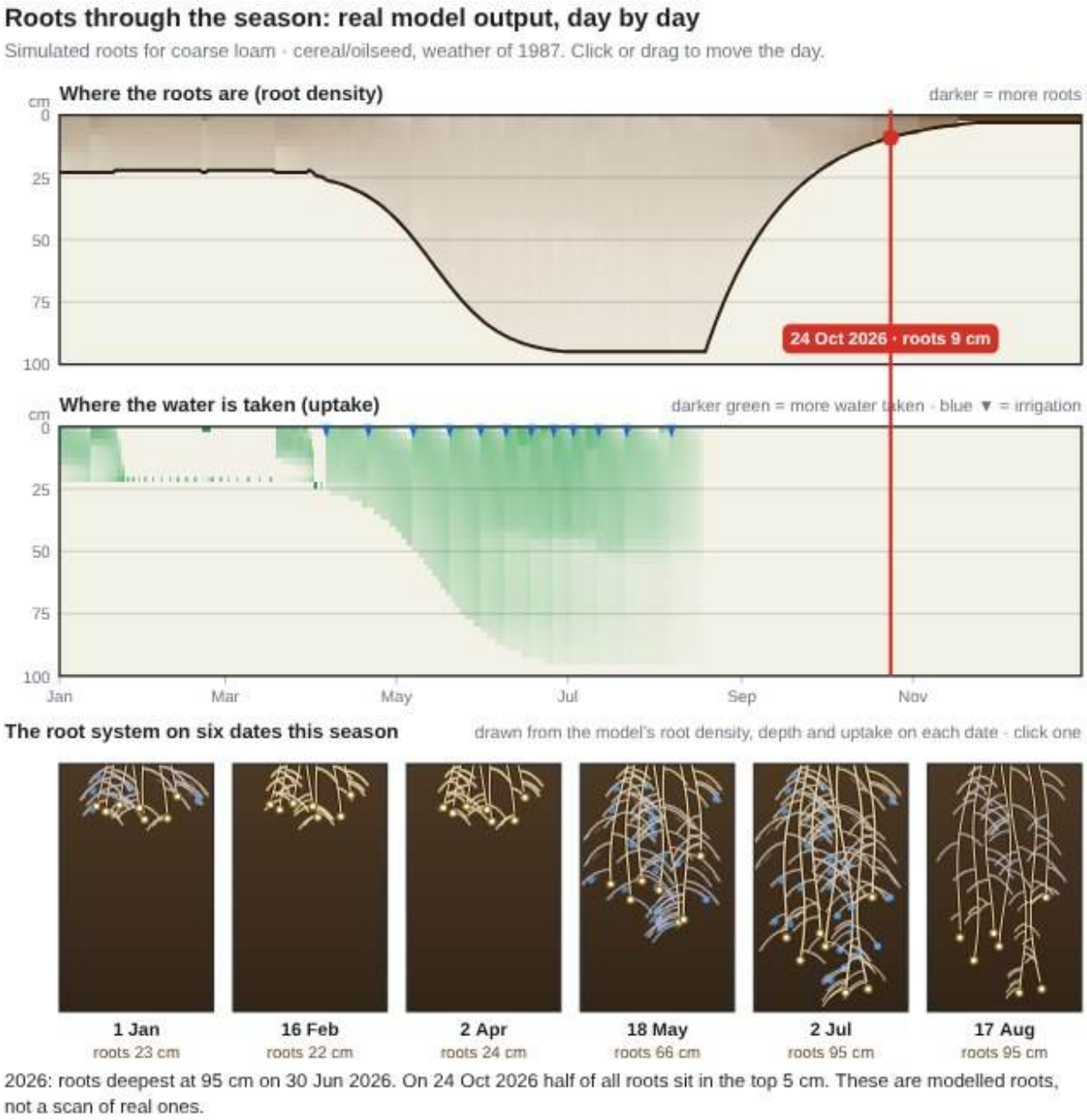


Figure 8.49: Root structure, Sandy Clay Loam, part 1 of 4.

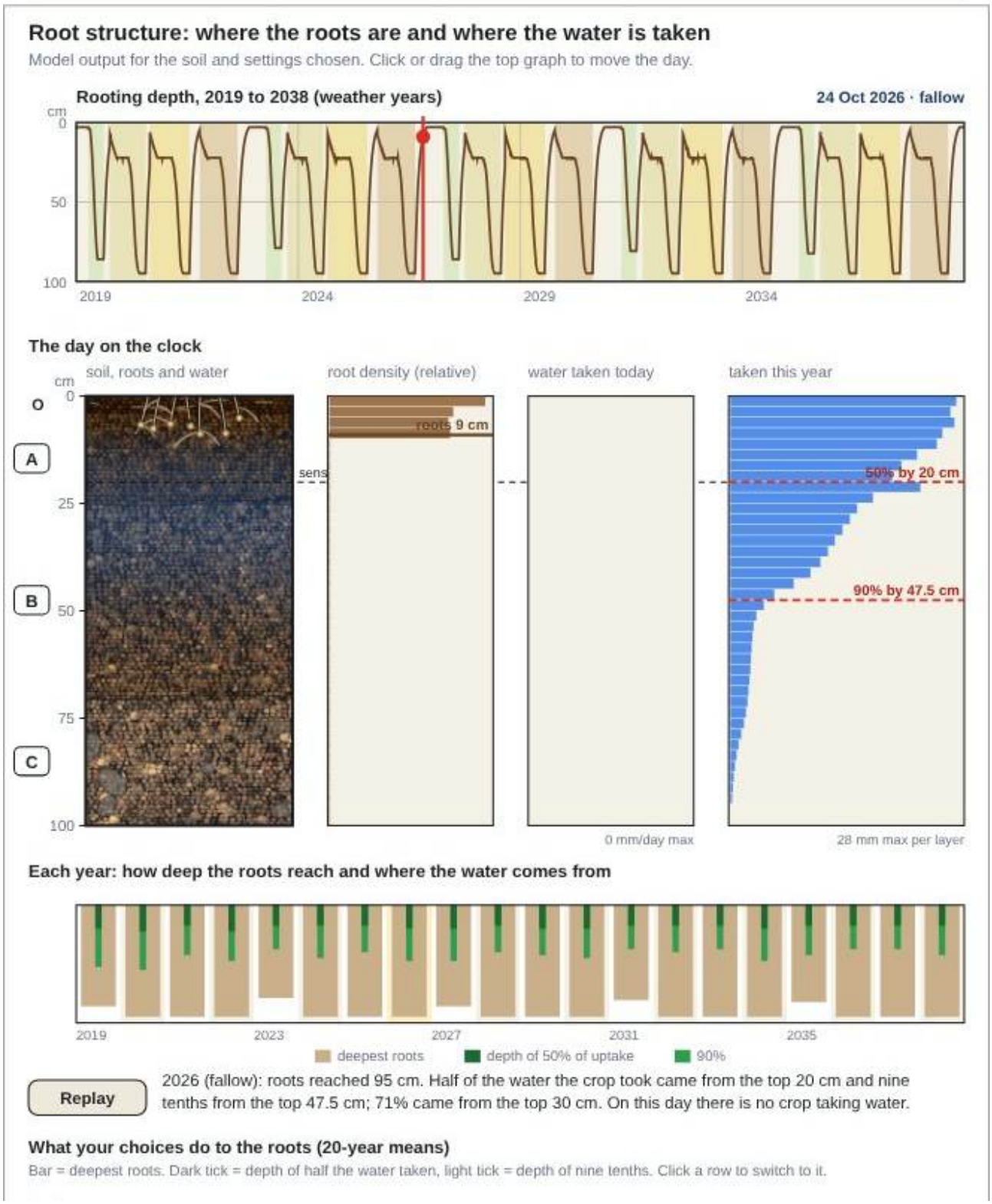


Figure 8.50: Root structure, Sandy Clay Loam, part 2 of 4.

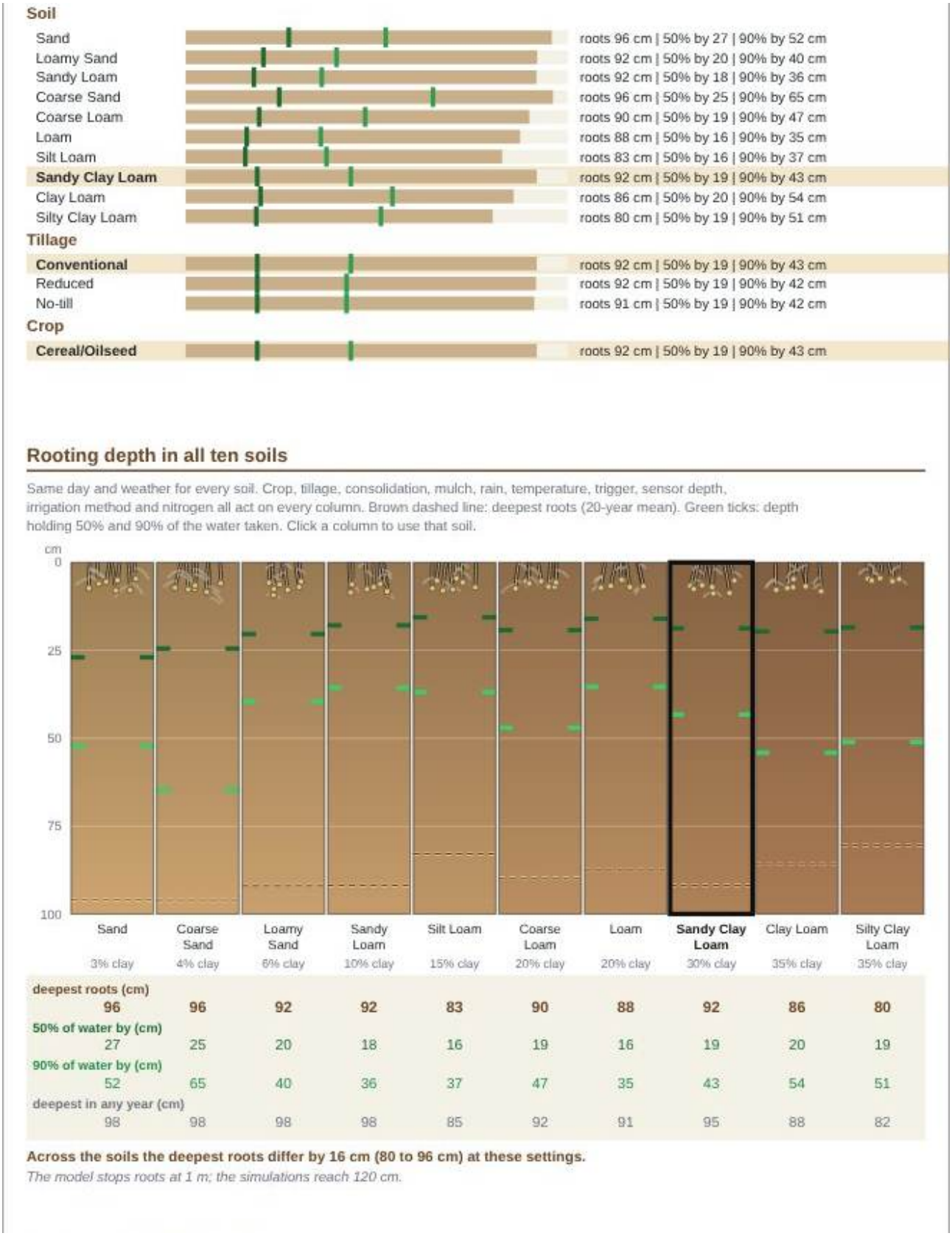


Figure 8.51: Root structure, Sandy Clay Loam, part 3 of 4.

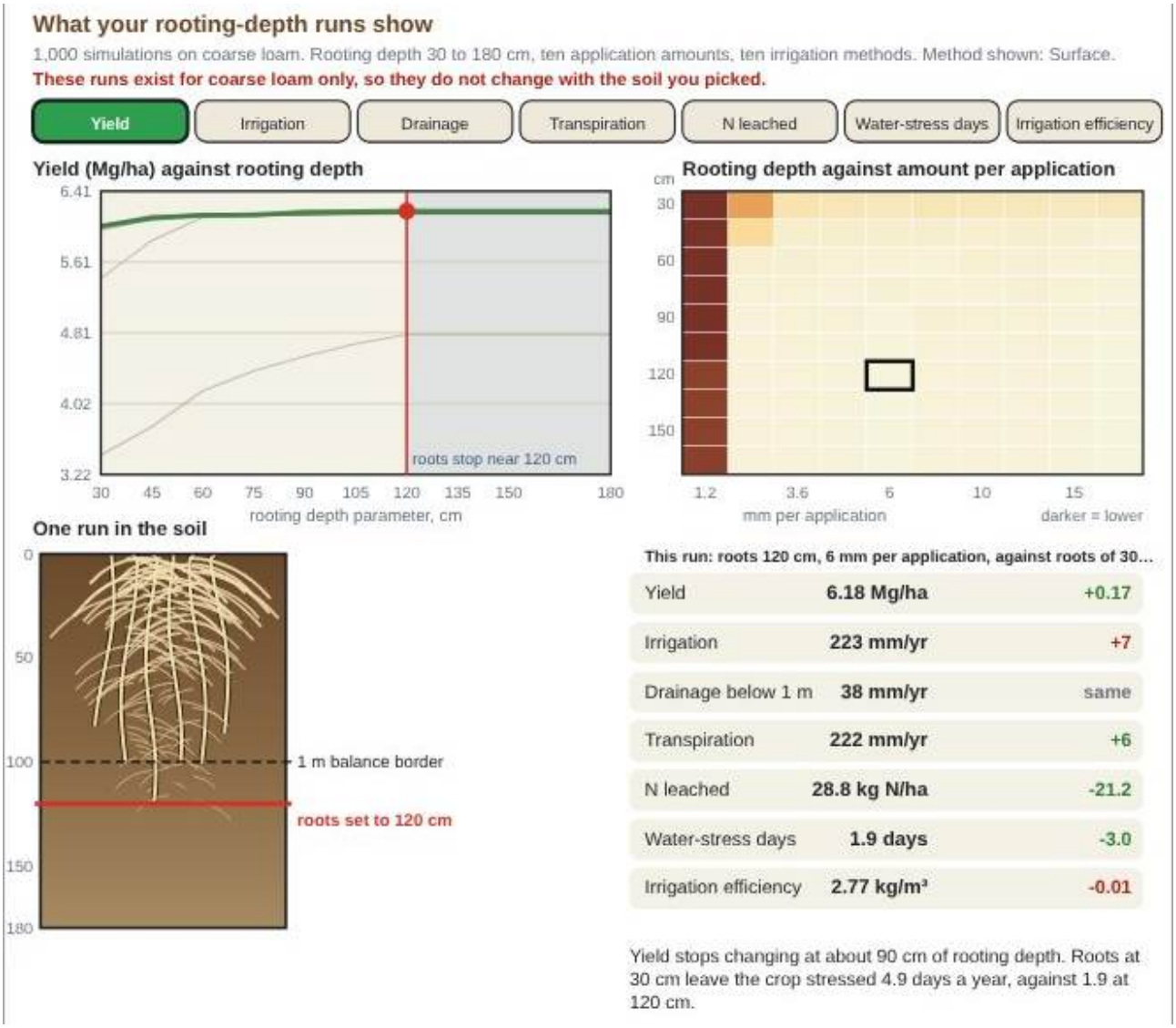


Figure 8.52: Root structure, Sandy Clay Loam, part 4 of 4.

8.17 Field balance (Sandy Clay Loam)

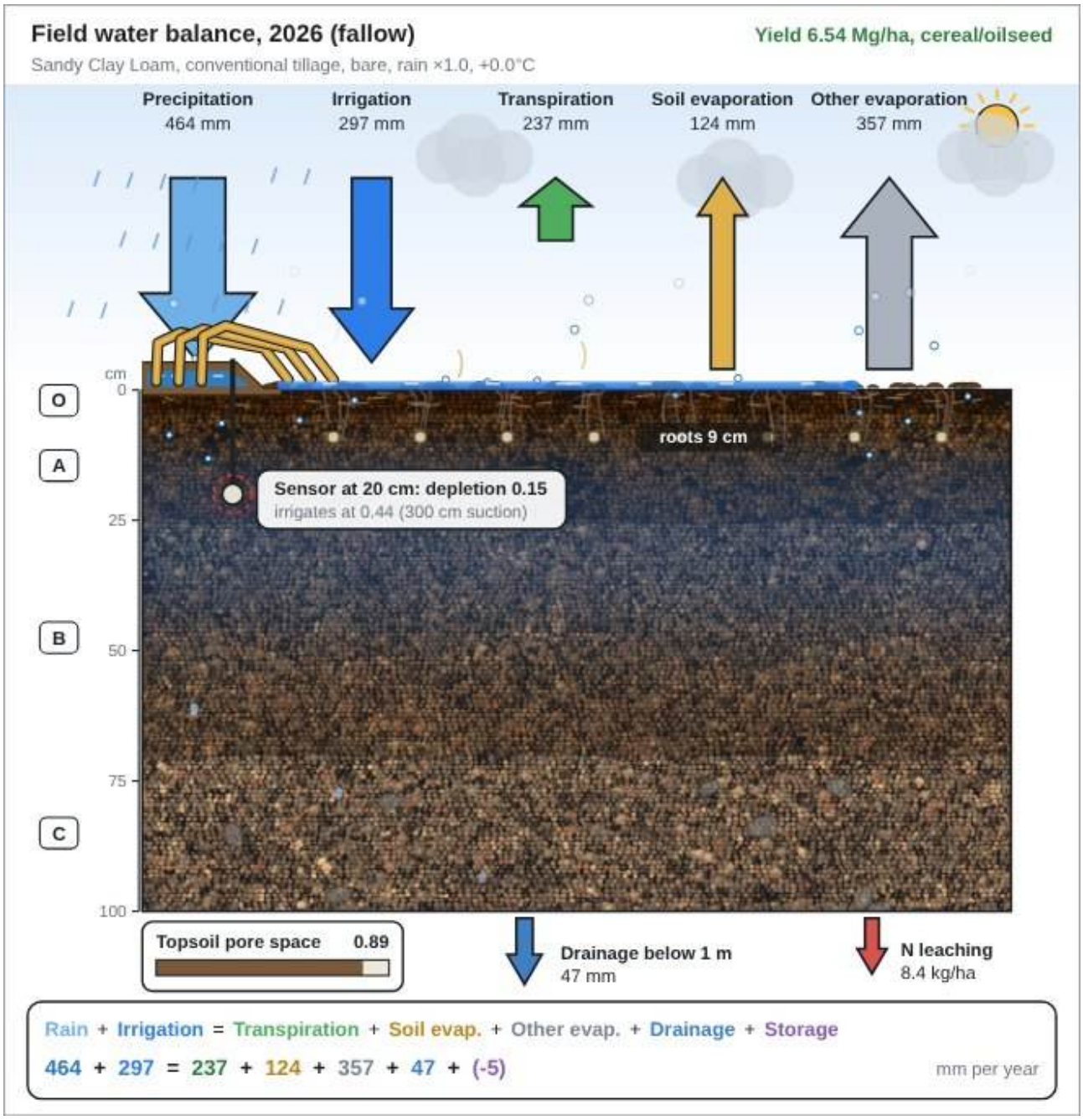


Figure 8.53: Field balance, Sandy Clay Loam, part 1 of 1.

8.18 Soil, plant and air (Sandy Clay Loam)

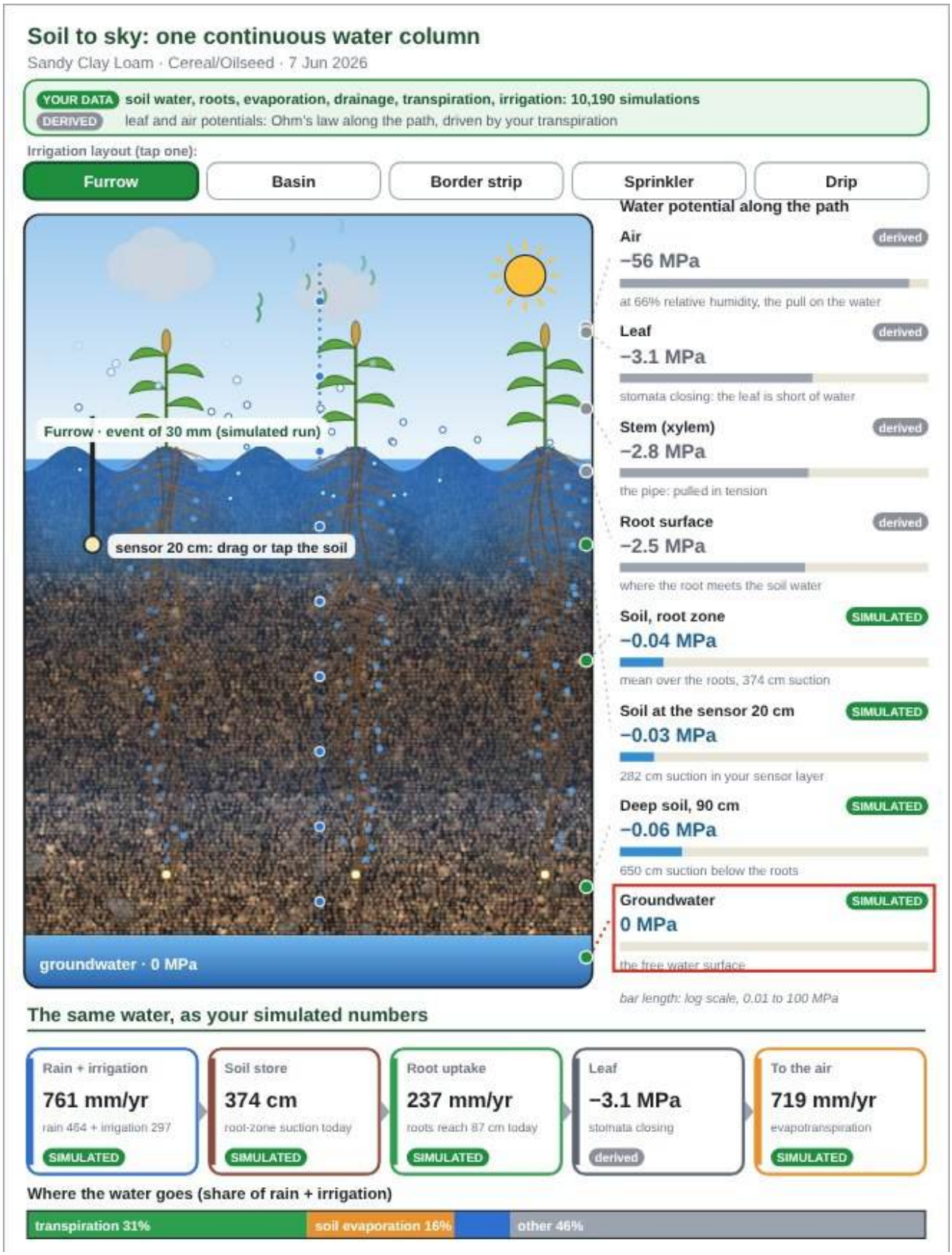


Figure 8.54: Soil, plant and air, Sandy Clay Loam, part 1 of 3.

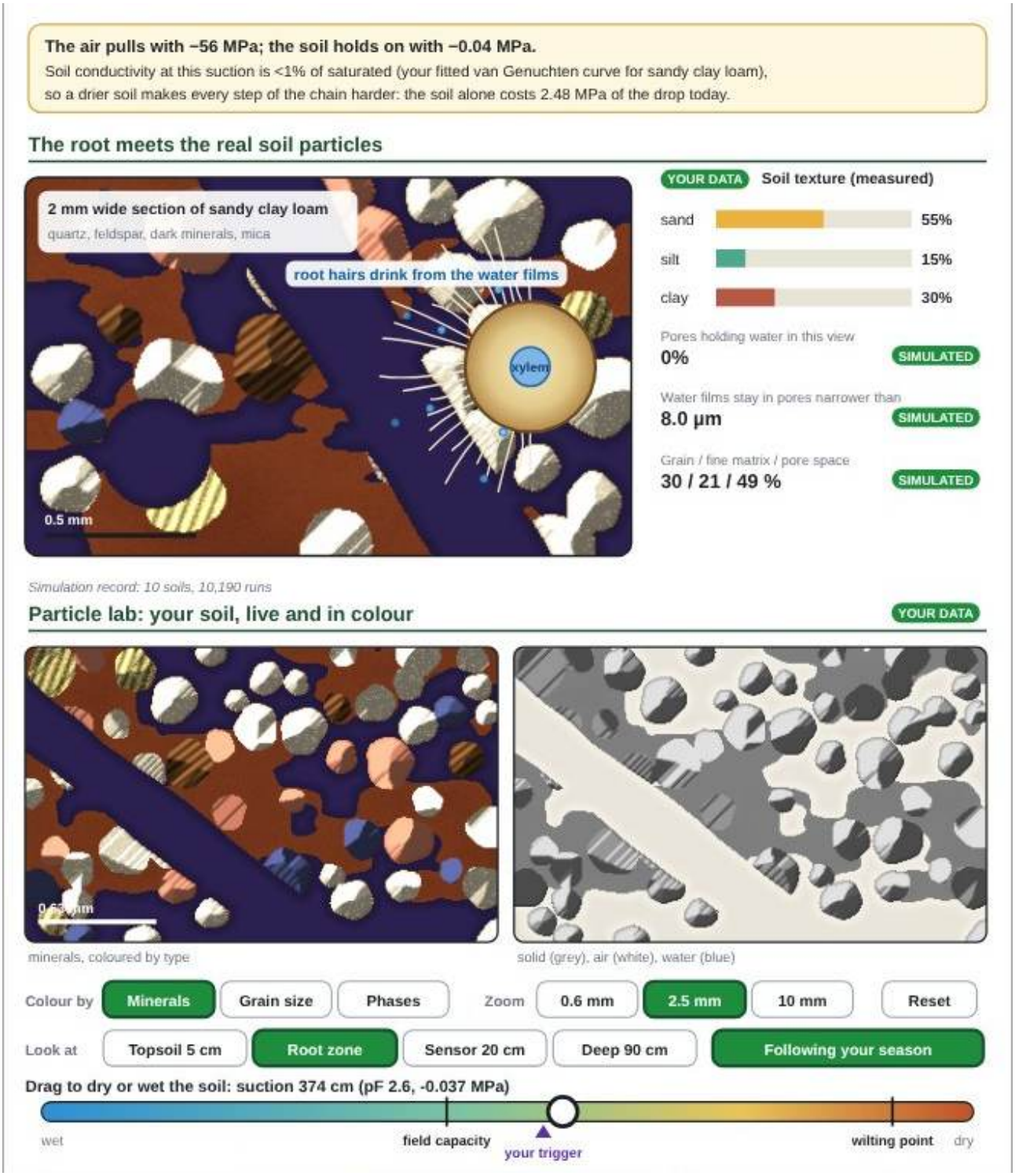


Figure 8.55: Soil, plant and air, Sandy Clay Loam, part 2 of 3.

Figure 8.56: Soil, plant and air, Sandy Clay Loam, part 3 of 3.

8.19 Profile evolution (Sandy Clay Loam)

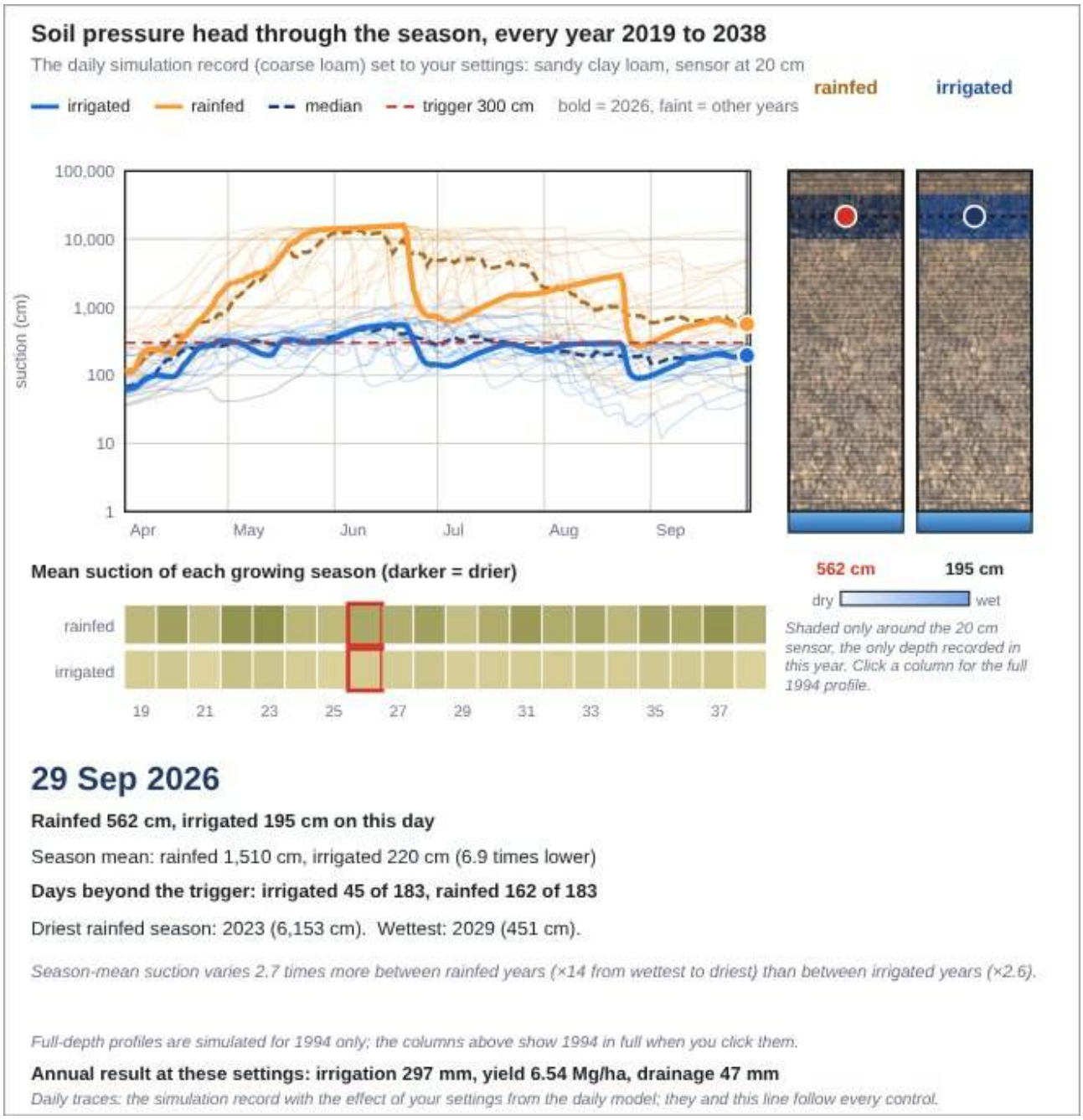


Figure 8.57: Profile evolution, Sandy Clay Loam, part 1 of 1.

8.20 Reliability over 20 seasons (Sandy Clay Loam)

Figure 8.58: Reliability over 20 seasons, Sandy Clay Loam, part 1 of 2.

Reliability of this setting over the 20 seasons — from the yearly values above

Irrigation needed in a dry year (9 years in 10 need less)	**373 mm**	against 297 mm in an average year
Years that need 20 % more irrigation than average	**3 of 20**	plan pump, labour and water rights for these
Years with more than 50 mm drainage below the roots	**5 of 20**	water and nitrate lost to groundwater
Years with a yield below 80 % of the average	**7 of 20**	yield risk at this setting
Nitrate leaching in the worst year	**44.4 kg N/ha**	mean 8.4 kg N/ha

How often a given irrigation is exceeded — the 20 seasons sorted from wettest to driest

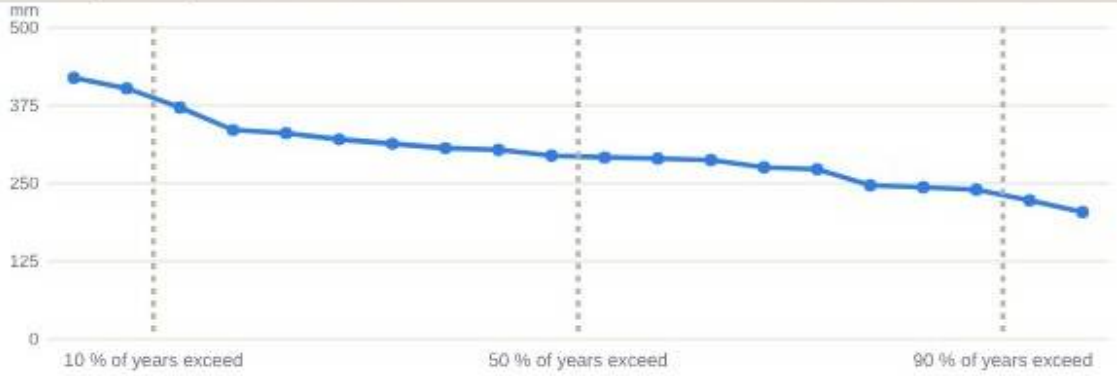


Figure 8.59: Reliability over 20 seasons, Sandy Clay Loam, part 2 of 2.

8.21 Full water balance (Sandy Clay Loam)

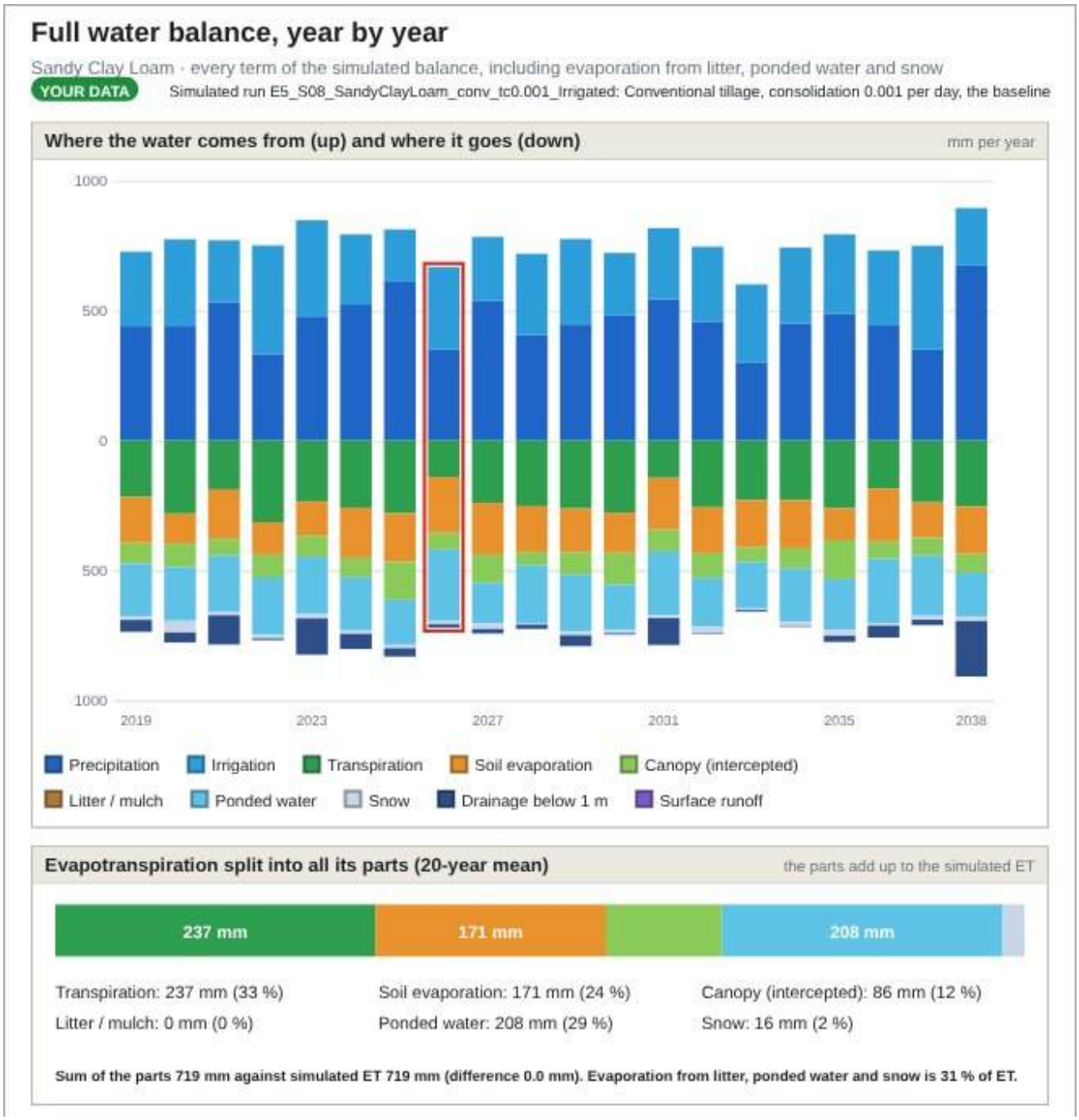


Figure 8.60: Full water balance, Sandy Clay Loam, part 1 of 2.

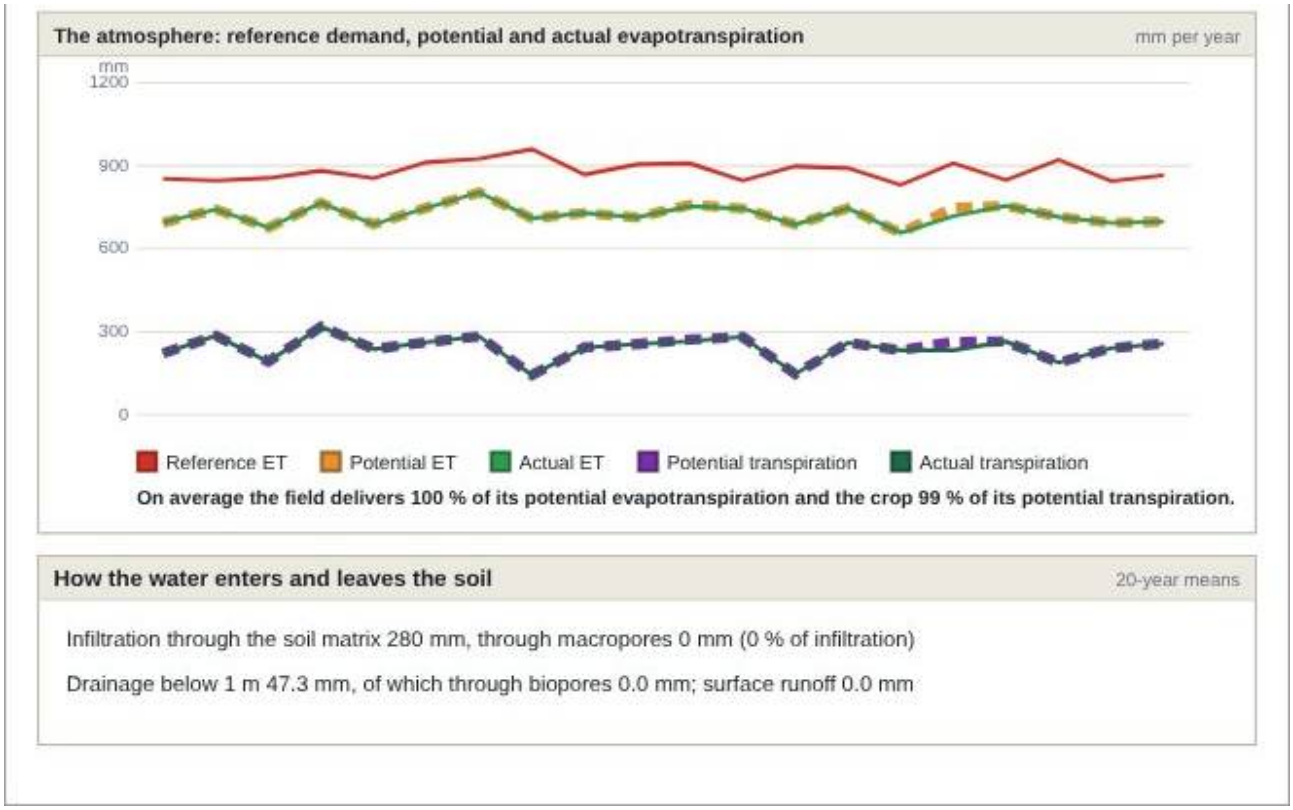


Figure 8.61: Full water balance, Sandy Clay Loam, part 2 of 2.

8.22 Nitrogen budget (Sandy Clay Loam)

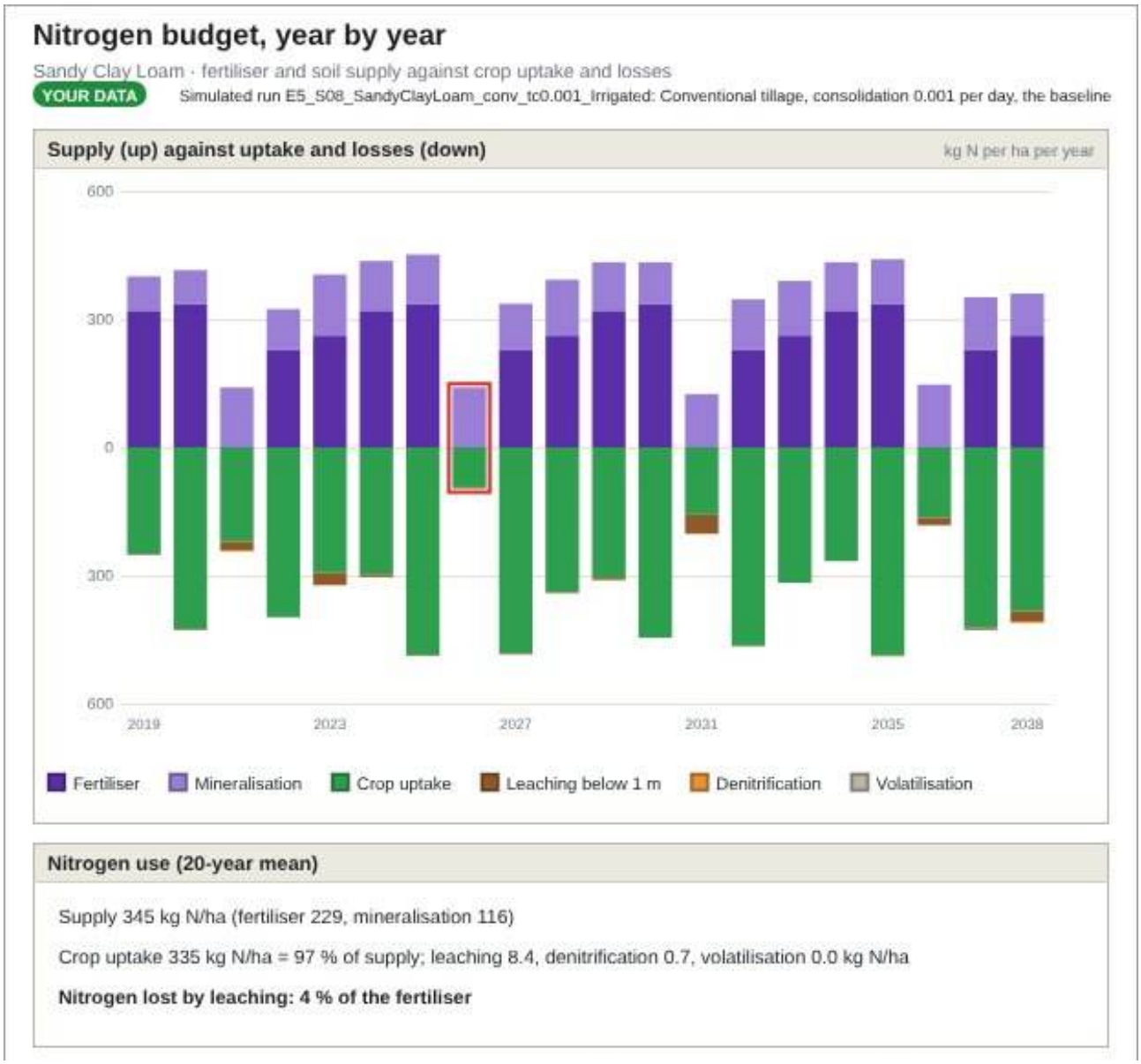


Figure 8.62: Nitrogen budget, Sandy Clay Loam, part 1 of 2.

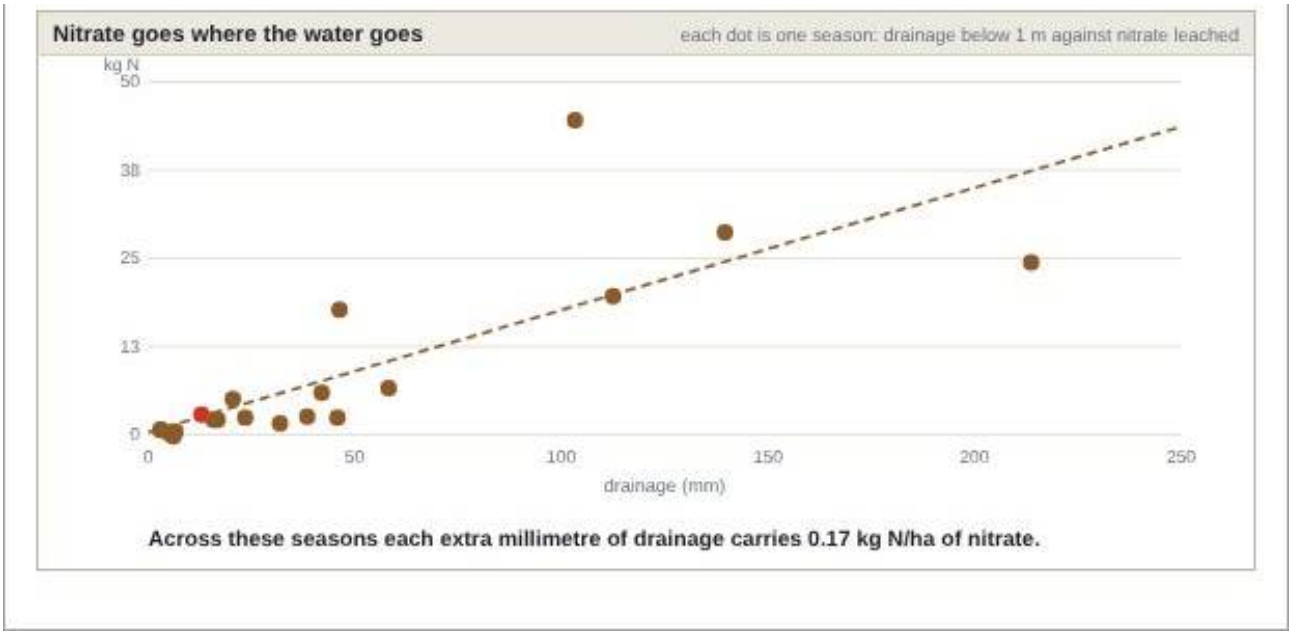


Figure 8.63: Nitrogen budget, Sandy Clay Loam, part 2 of 2.

8.23 Crop water requirement (Sandy Clay Loam)

Figure 8.64: Crop water requirement, Sandy Clay Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**510**	9.5	99 %
Coconut (high)	834	**435**	7.0	100 %
Papaya (high)	814	**423**	7.5	100 %
Oil palm (young)	790	**396**	7.0	100 %
Yam	649	**337**	9.3	100 %
Onion (dry)	565	**310**	10.9	99 %
Avocado (medium)	677	**296**	6.3	100 %
Kiwi (pergola)	626	**294**	4.7	100 %
Cotton	571	**294**	4.9	100 %
Rice	573	**293**	6.2	95 %
Cabbage	559	**283**	7.2	100 %
Table grapes (horizontal trellis)	597	**264**	4.2	100 %
Potato	508	**264**	8.6	100 %
Guava (high)	651	**261**	4.2	100 %
Tomato	521	**253**	4.2	100 %
Hop (v trellis)	520	**233**	3.7	100 %
Mango (medium)	620	**230**	3.7	100 %
Walnut (medium)	523	**229**	3.7	100 %
Groundnut	471	**224**	5.0	100 %
Soybean	473	**223**	3.9	100 %
Mustard	468	**219**	3.7	100 %
Wheat	469	**219**	3.7	100 %
Pear (medium)	528	**218**	3.5	100 %
Date palm (medium)	602	**213**	3.4	100 %
Sweet pepper	475	**207**	4.5	99 %
Maize (grain)	451	**204**	3.3	100 %
Cherry (medium)	503	**199**	3.2	100 %

Figure 8.65: Crop water requirement, Sandy Clay Loam, part 2 of 3.

Pistachio (medium)	503		**199**	3.2	100 %
Cassava	466		**195**	4.9	100 %
Chickpea	436		**188**	3.9	100 %
Hazelnut (low)	481		**183**	2.9	100 %
Pecan (low)	481		**183**	2.9	100 %
Sorghum (grain)	429		**174**	2.9	100 %
Cherimoya (medium)	545		**167**	2.7	100 %
Citrus (medium density, short)	552		**163**	2.6	100 %
Cowpea (beans)	398		**157**	3.3	100 %
Millet	395		**153**	2.6	100 %
Apple (medium)	459		**146**	2.4	100 %
Wine grapes (pergola)	457		**137**	2.2	100 %
Green gram (mung bean)	372		**127**	2.8	100 %
Peach (low)	435		**127**	2.0	100 %
Almond (medium)	435		**127**	2.0	100 %
Apricot (young)	380		**77**	1.2	100 %
Olive (traditional, medium density)	446		**66**	1.1	100 %
Persimmon (young, low)	358		**66**	1.1	100 %
Pomegranate (low)	358		**66**	1.1	100 %
Highbush blueberry (low)	338		**63**	1.1	100 %
Fig tree (young, low)	327		**30**	0.5	100 %

Figure 8.66: Crop water requirement, Sandy Clay Loam, part 3 of 3.

8.24 Feedback loops (Sandy Clay Loam)

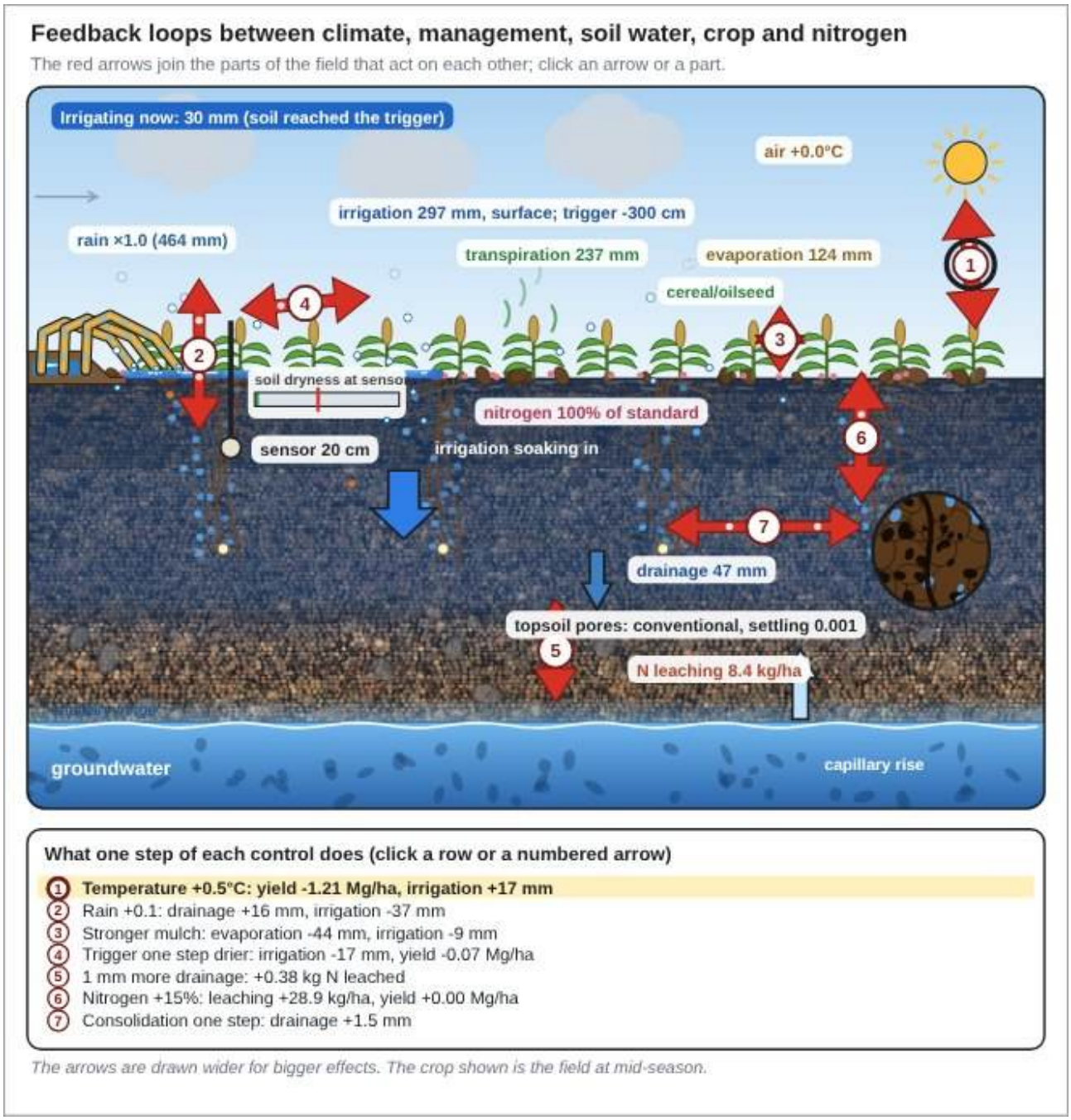


Figure 8.67: Feedback loops, Sandy Clay Loam, part 1 of 1.

Chapter 9

Clay Loam

9.1 Surface irrigation design (Clay Loam)

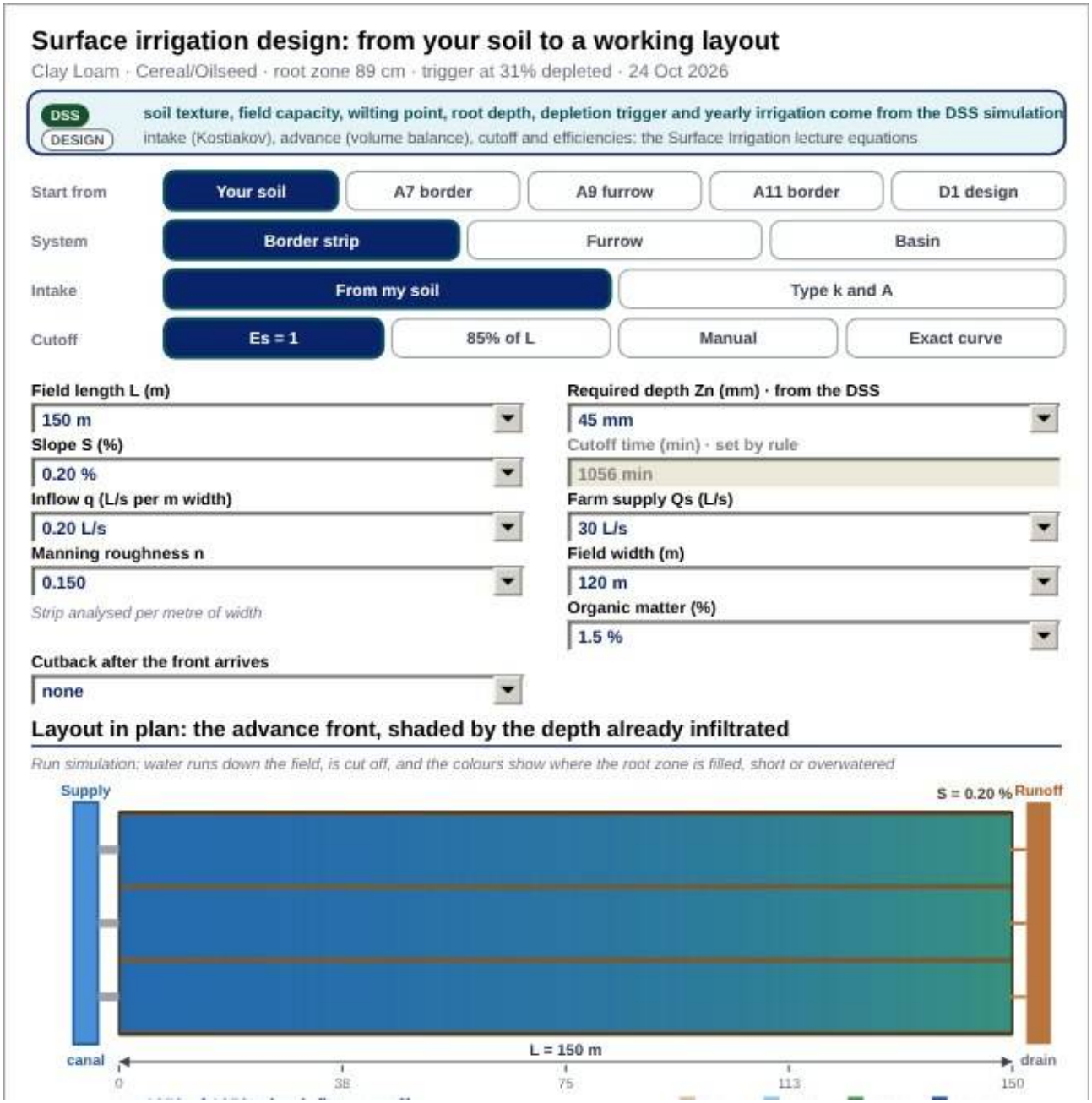


Figure 9.1: Surface irrigation design, Clay Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

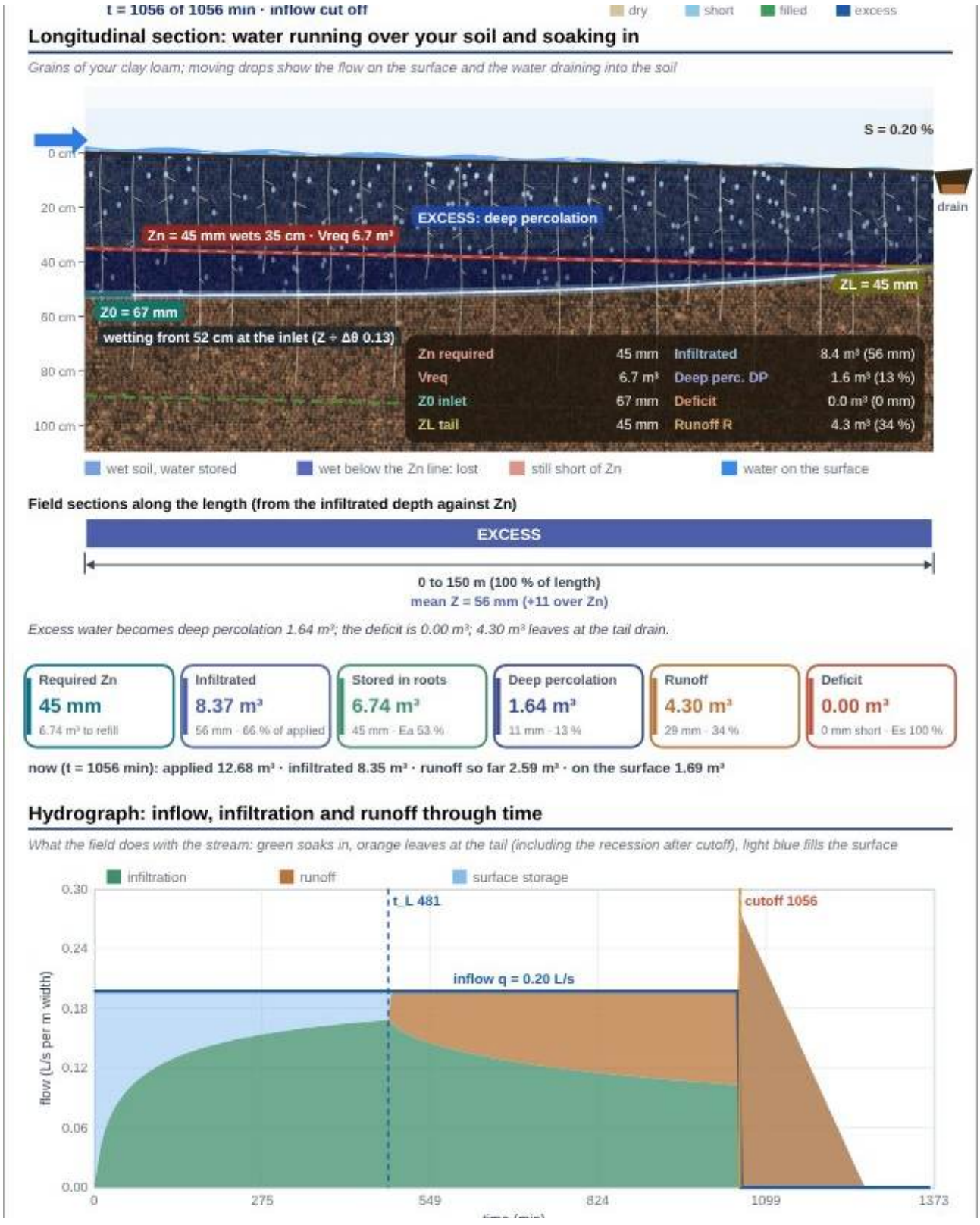


Figure 9.2: Surface irrigation design, Clay Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

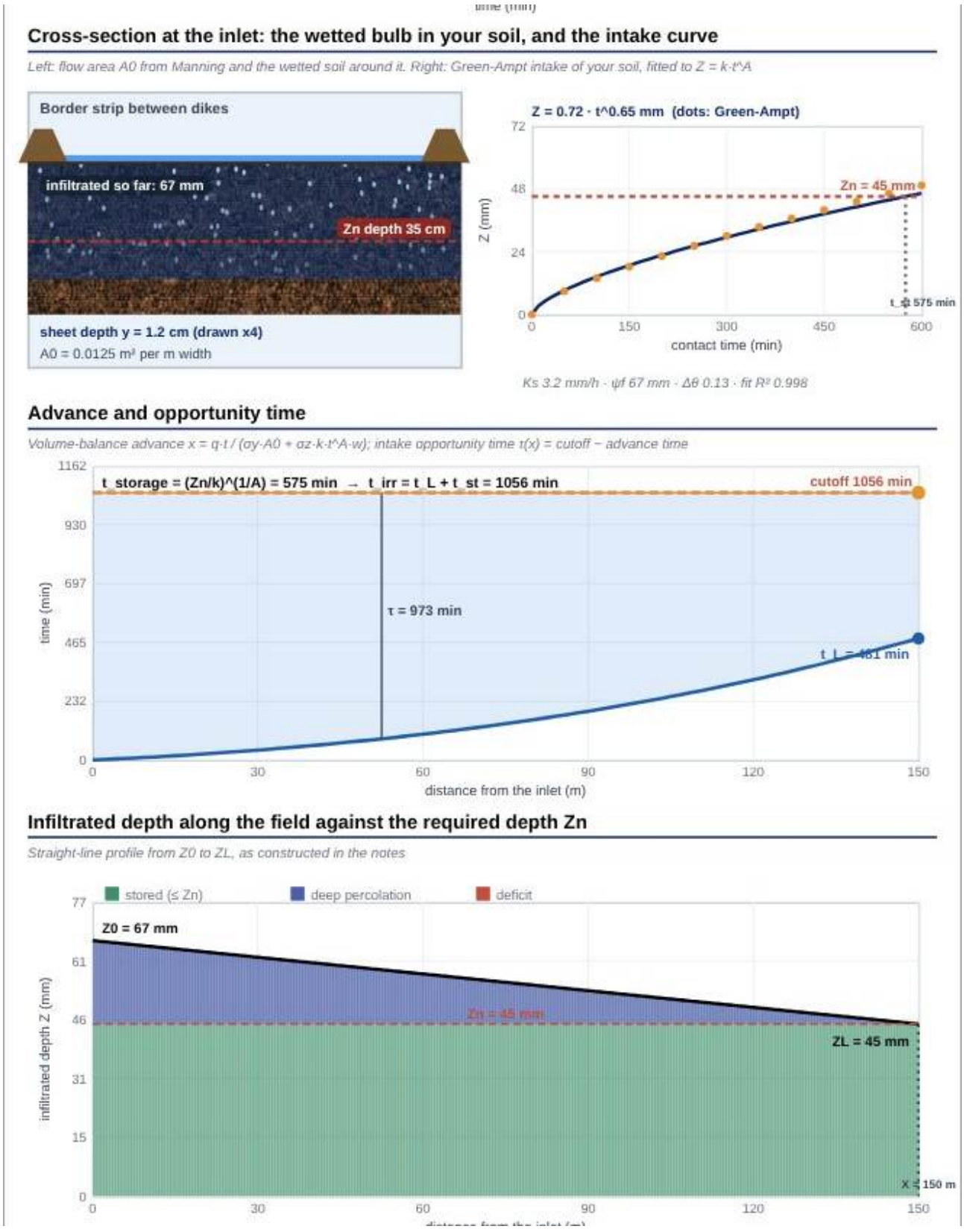


Figure 9.3: Surface irrigation design, Clay Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

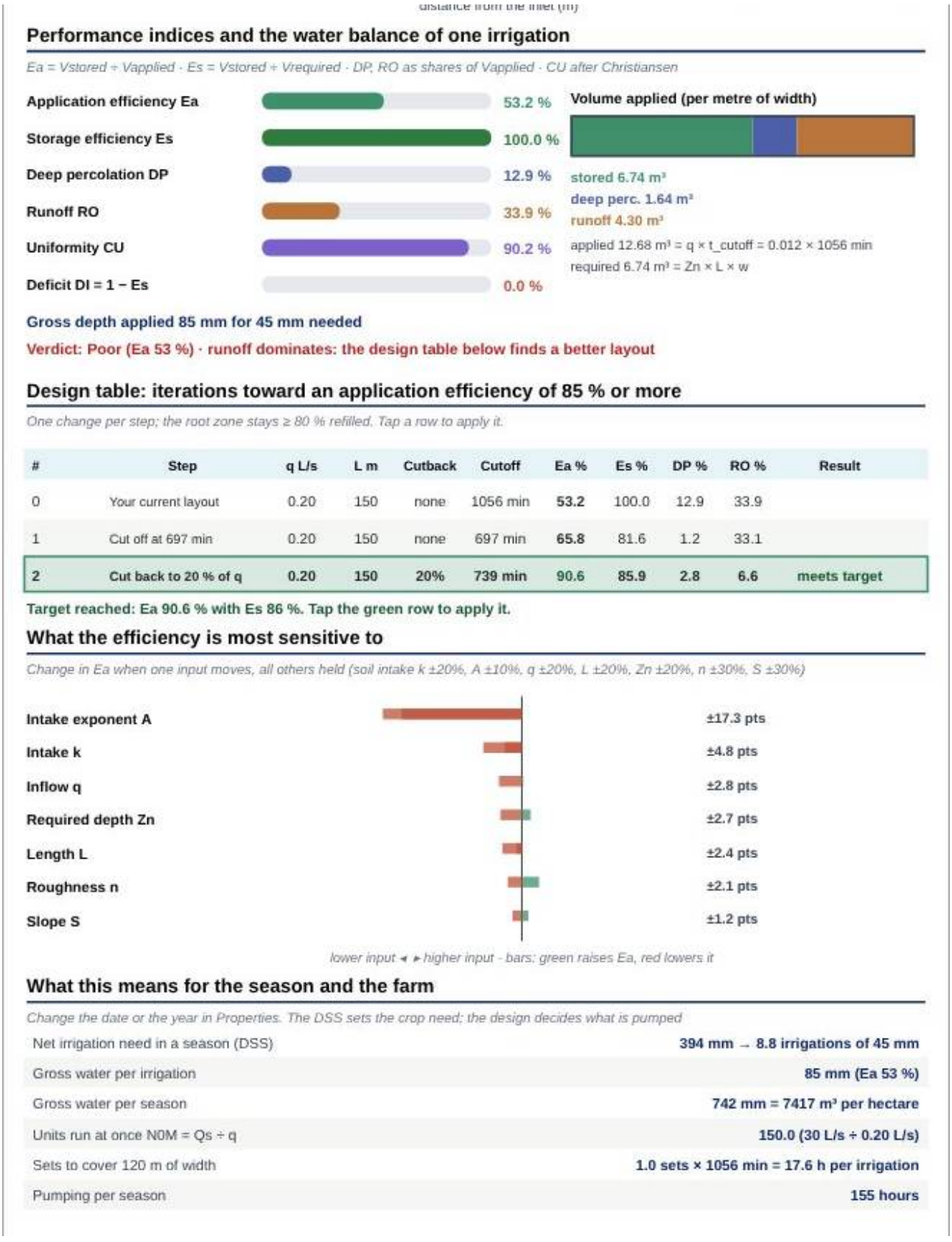


Figure 9.4: Surface irrigation design, Clay Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to; What this means for the season and the farm.

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	30	57	17.6 h	13 days
2	**29 May 2027**	217 days	30	57	17.6 h	10 days
3	**8 Jun 2027**	227 days	30	57	17.6 h	9 days
4	**17 Jun 2027**	236 days	30	57	17.6 h	12 days
5	**29 Jun 2027**	248 days	30	57	17.6 h	8 days
6	**7 Jul 2027**	256 days	30	57	17.6 h	10 days
7	**17 Jul 2027**	266 days	30	57	17.6 h	15 days
8	**1 Aug 2027**	281 days	30	57	17.6 h	-

Intervals, efficiency and the advice in short

Everything above in one message for surface irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.31**
Irrigations in the season of this date	**13 (6 Apr 2026 to 26 Jul 2026)**
Interval between irrigations	**9.3 days on average (shortest 1, longest 16)**
Depth per irrigation	**44.9 mm net, 84.5 mm pumped**
Application efficiency Ea	**53.2 %**
Storage efficiency Es	**100.0 %**
Time per round	**17.6 h**

When to irrigate

Next irrigation: 16 May 2027 (in 204 days), about 30 mm net, 57 mm pumped. Then irrigate about every 9 days (between 1 and 16 days, shorter in hot weather and longer after rain). With this surface design 53 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 17.6 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 9.5: Surface irrigation design, Clay Loam, part 5 of 5: Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

9.2 Drip irrigation design (Clay Loam)

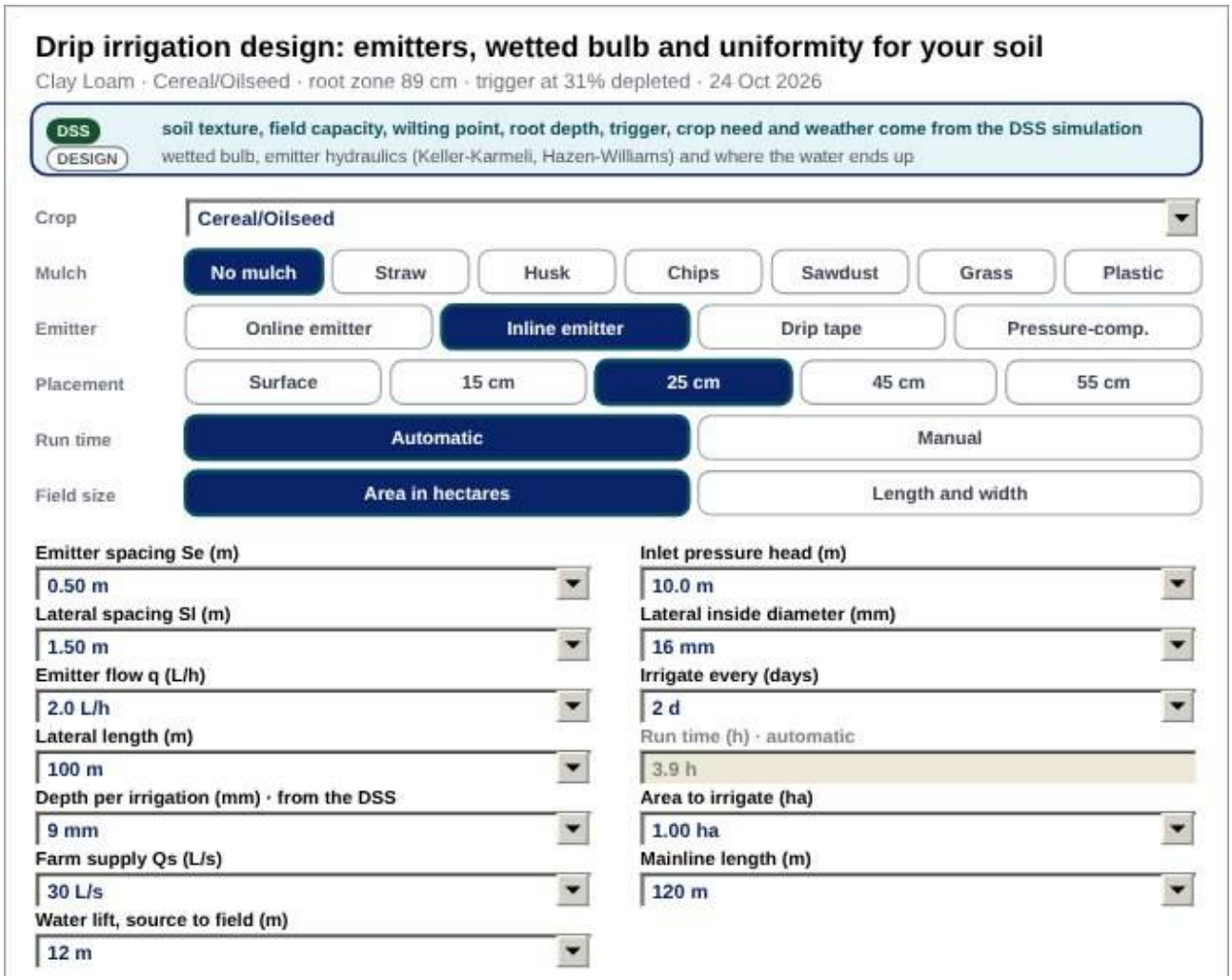


Figure 9.6: Drip irrigation design, Clay Loam, part 1 of 6.

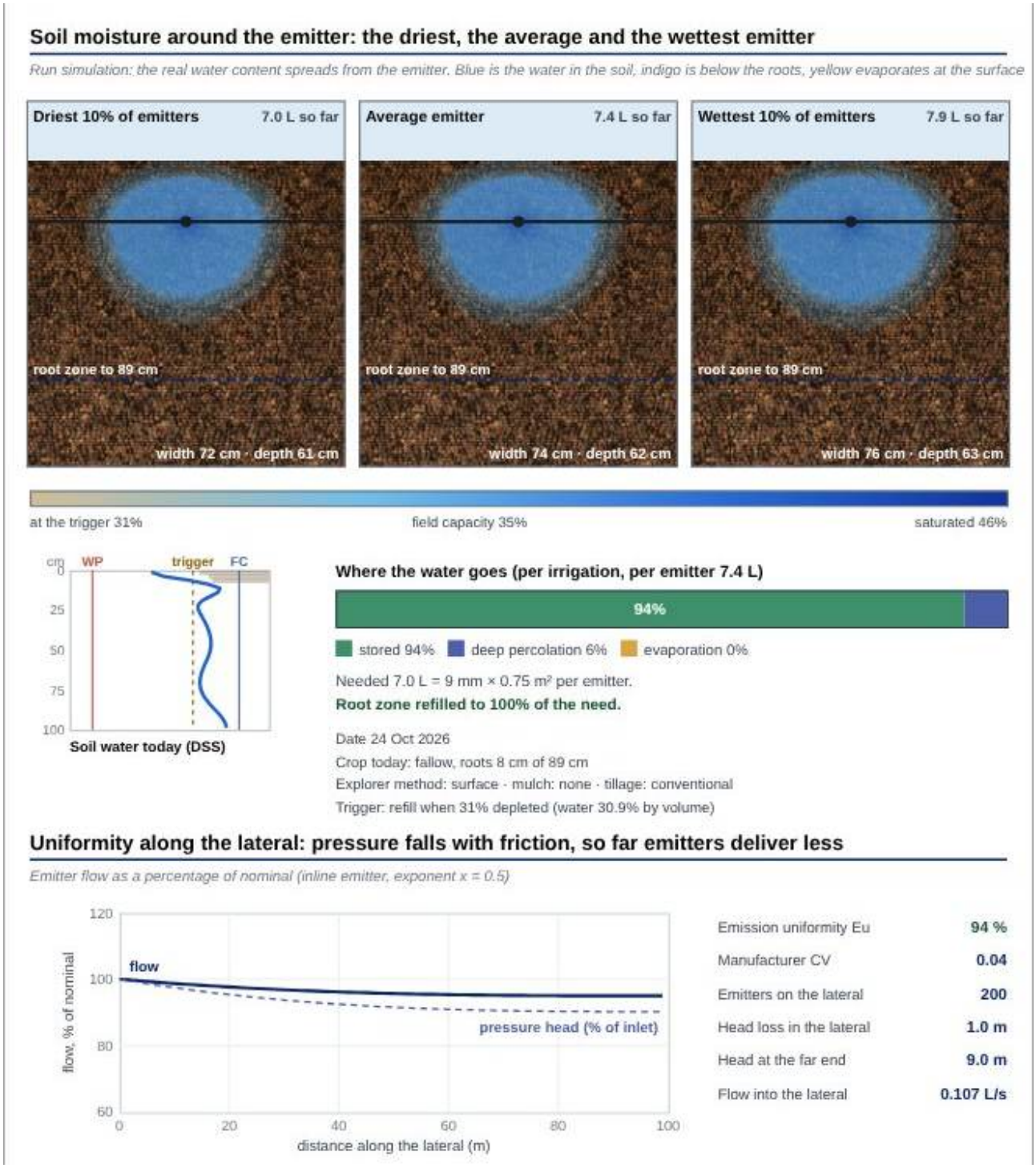


Figure 9.7: Drip irrigation design, Clay Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

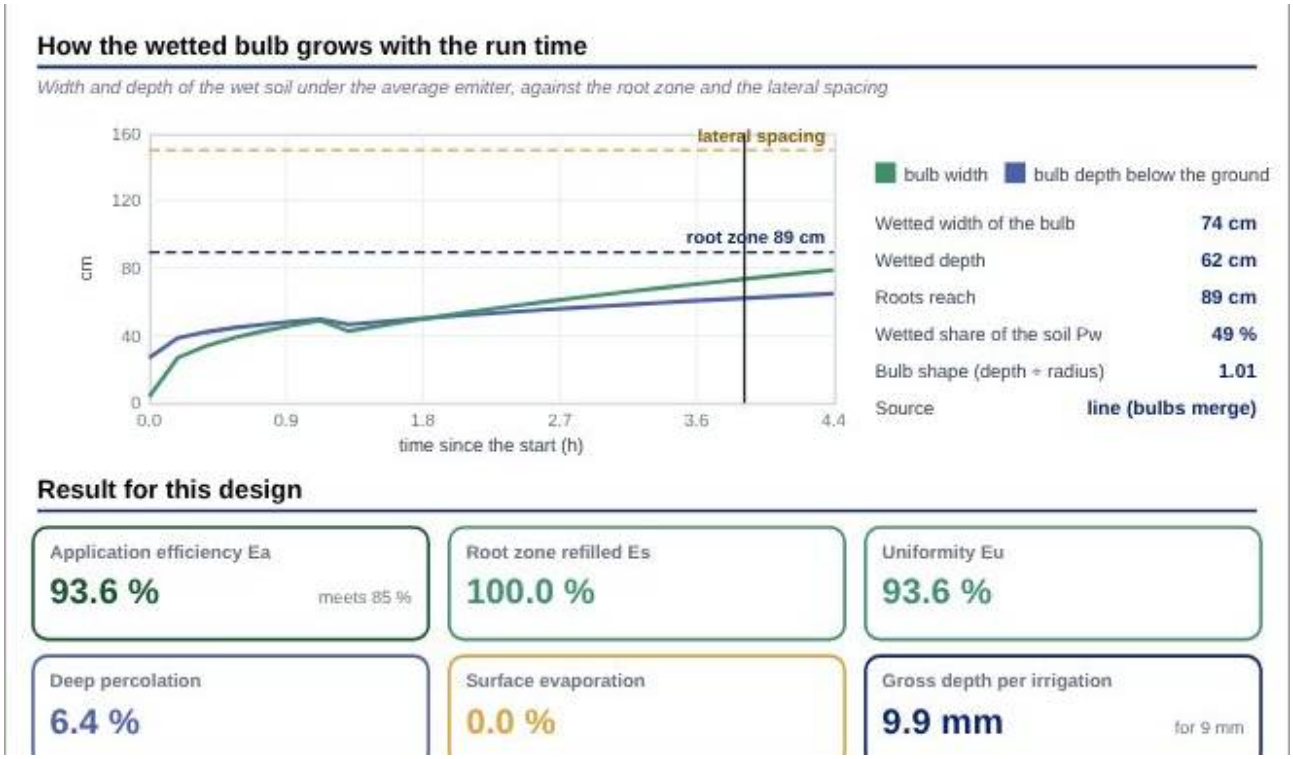
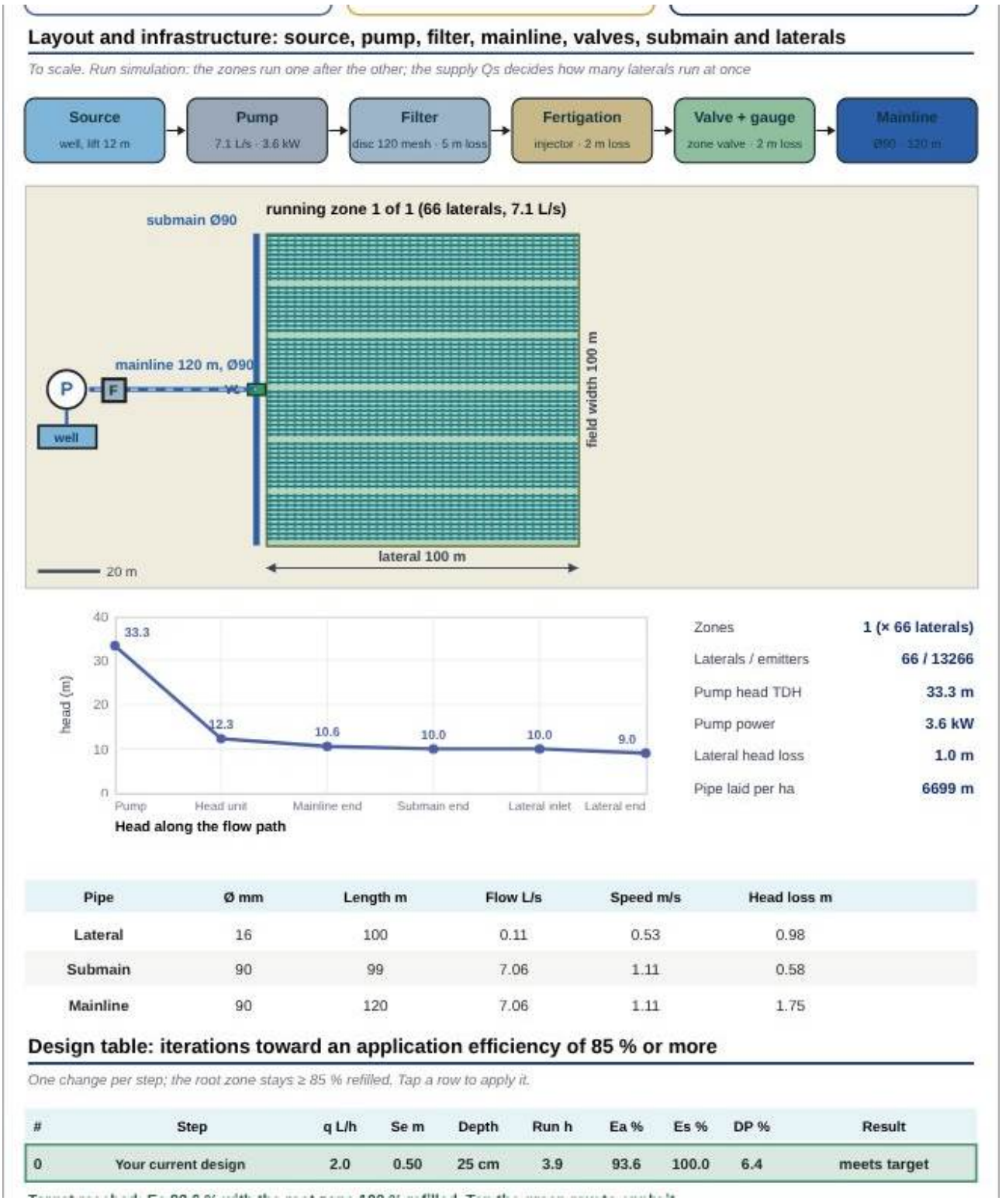


Figure 9.8: Drip irrigation design, Clay Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the root zone stays ≥ 85 % refilled. Tap a row to apply it.

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.9	93.6	100.0	6.4	meets target

Figure 9.9: Drip irrigation design, Clay Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**394 mm → 42 irrigations of 9 mm**
Gross water per irrigation	**9.9 mm (Ea 94 %)**
Gross water per season	**421 mm = 4212 m³ per hectare**
Peak crop need / interval	**4.6 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.9 h = 3.9 h per round**
Pumping per season	**164 hours · 583 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	30	32	3.9 h	13 days
2	**29 May 2027**	217 days	30	32	3.9 h	10 days
3	**8 Jun 2027**	227 days	30	32	3.9 h	9 days
4	**17 Jun 2027**	236 days	30	32	3.9 h	12 days
5	**29 Jun 2027**	248 days	30	32	3.9 h	8 days
6	**7 Jul 2027**	256 days	30	32	3.9 h	10 days
7	**17 Jul 2027**	266 days	30	32	3.9 h	15 days
8	**1 Aug 2027**	281 days	30	32	3.9 h	-

Figure 9.10: Drip irrigation design, Clay Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.

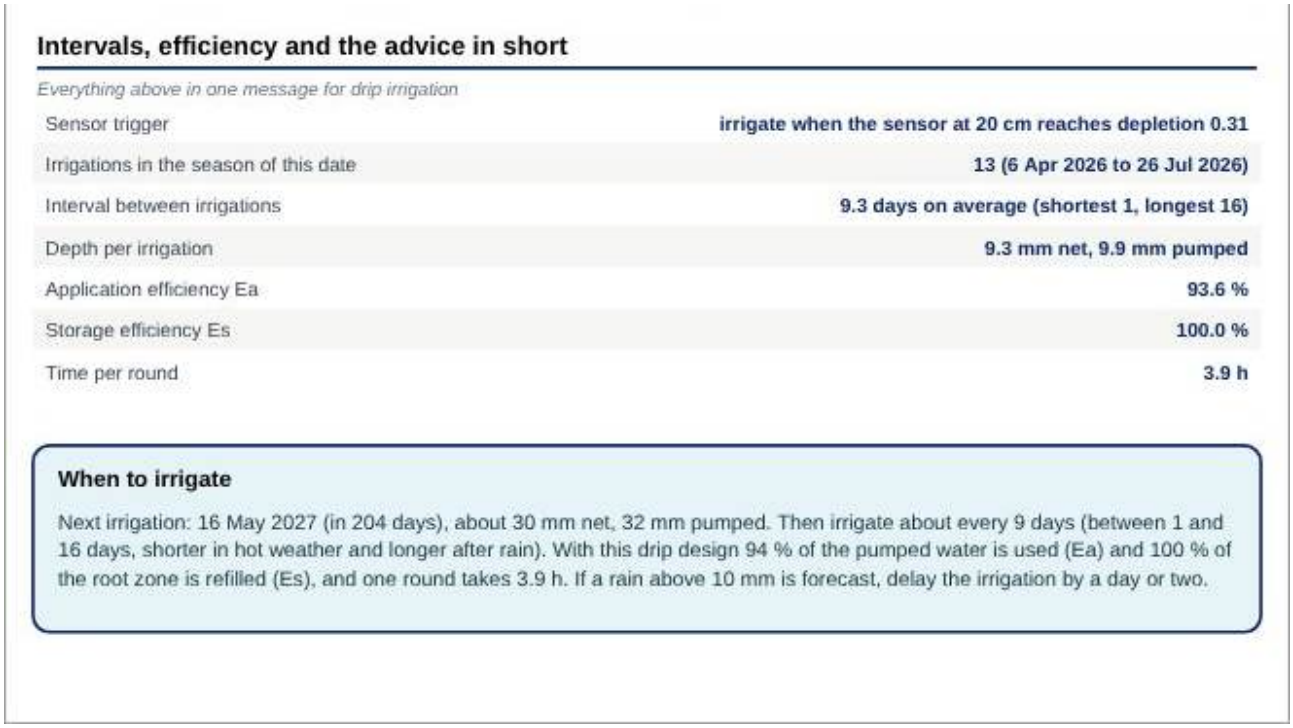


Figure 9.11: Drip irrigation design, Clay Loam, part 6 of 6: Intervals, efficiency and the advice in short.

9.3 Sprinkler design and pipes (Clay Loam)

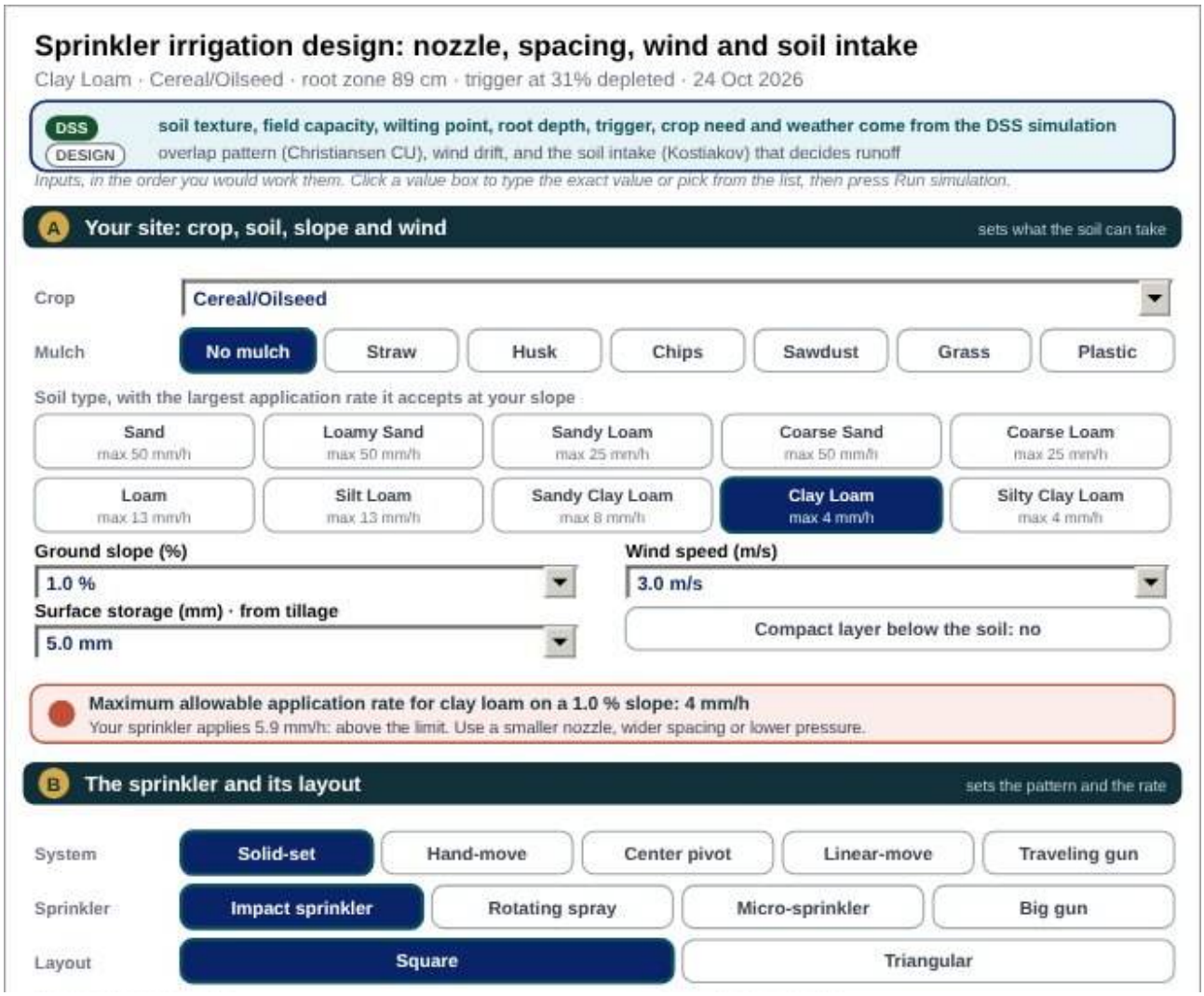


Figure 9.12: Sprinkler design and pipes, Clay Loam, part 1 of 12.

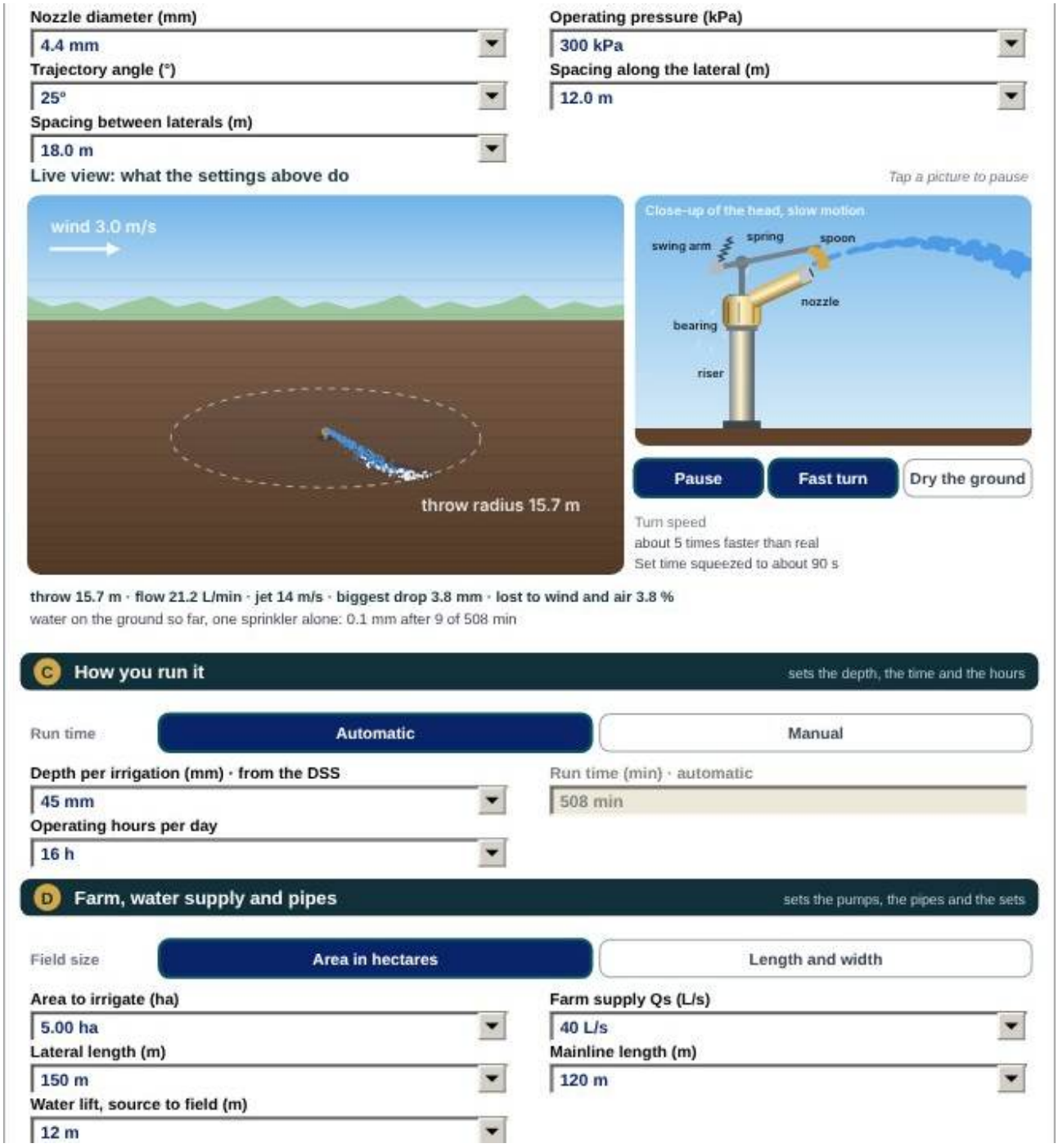


Figure 9.13: Sprinkler design and pipes, Clay Loam, part 2 of 12.

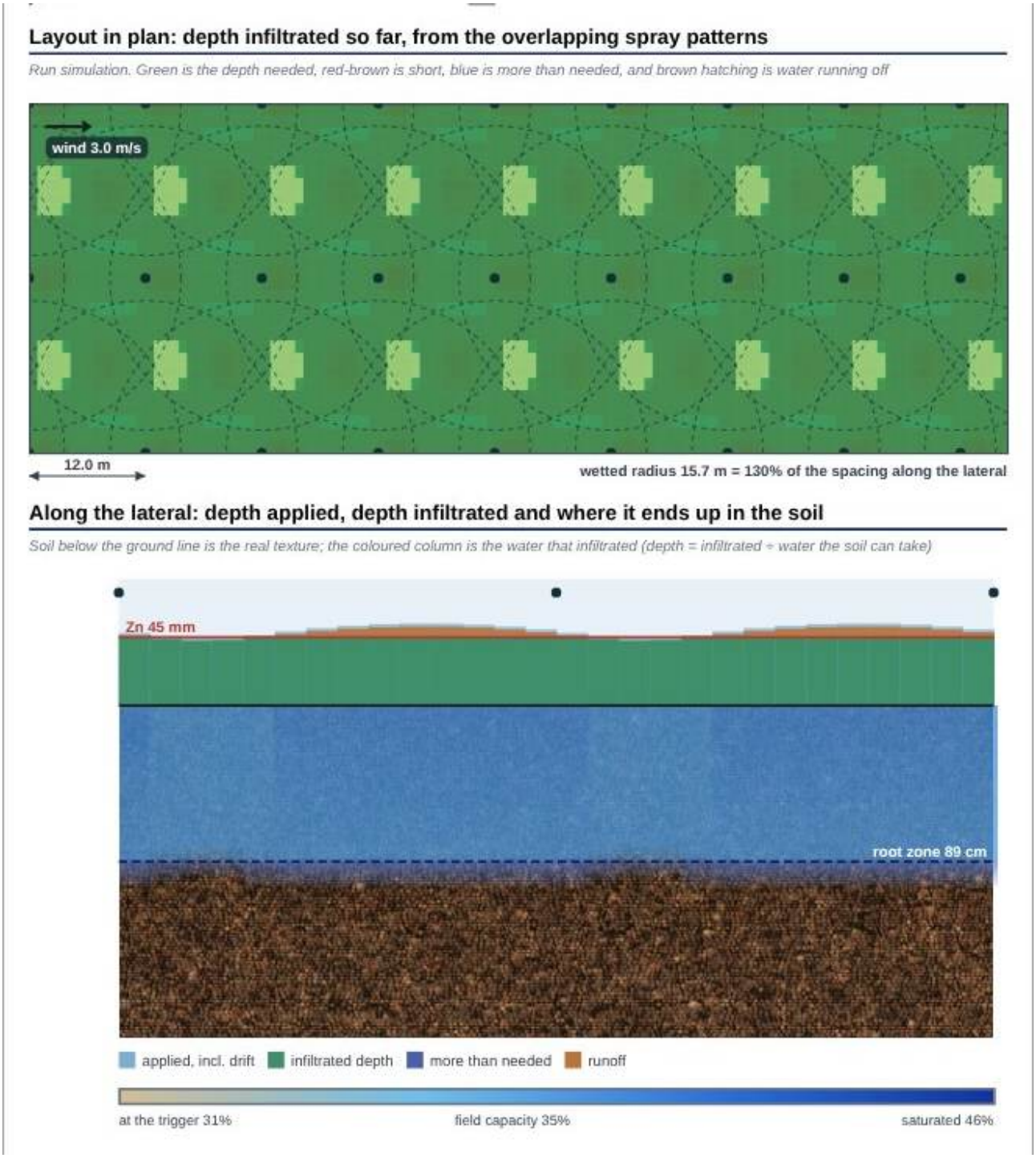


Figure 9.14: Sprinkler design and pipes, Clay Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

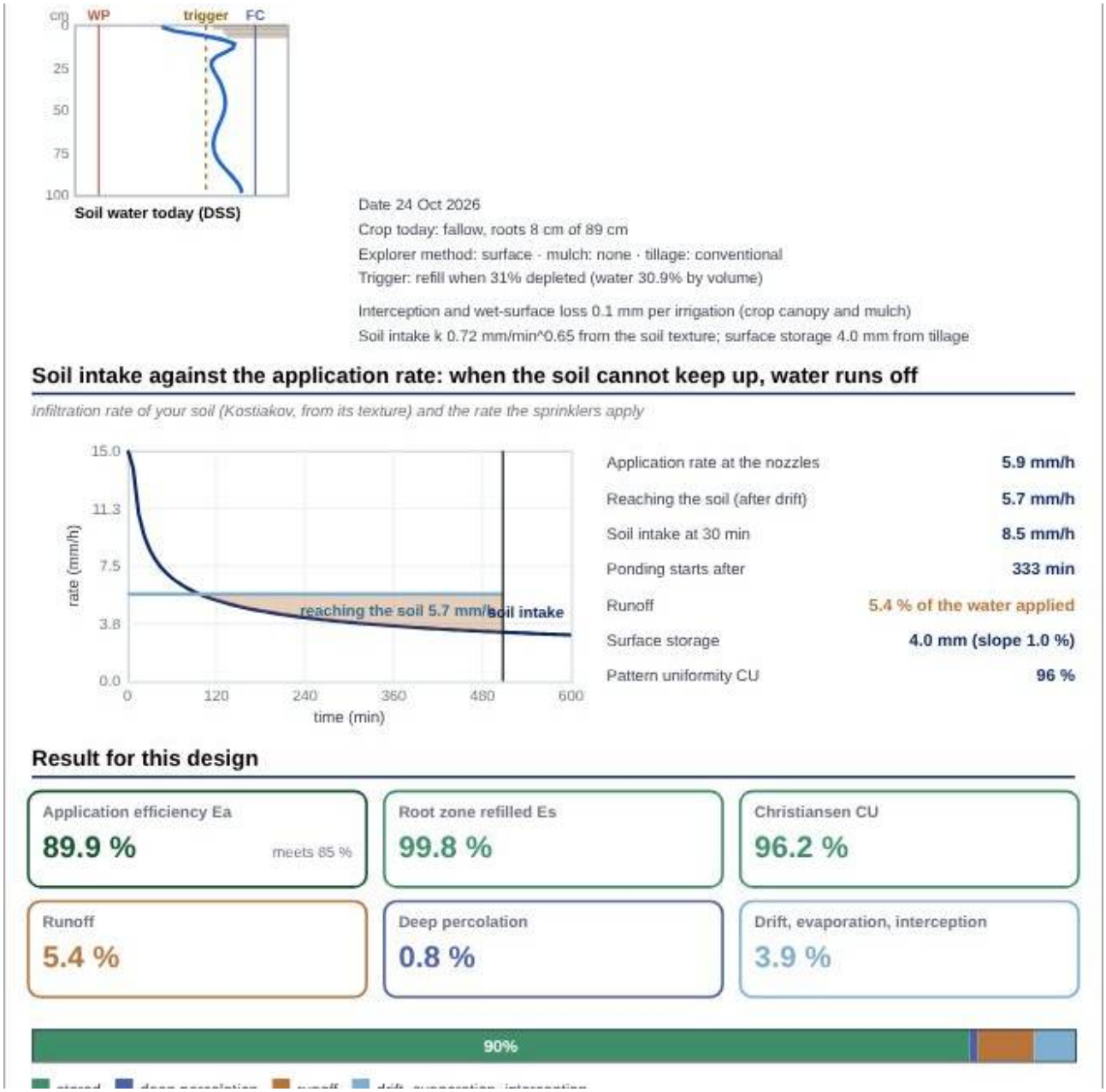


Figure 9.15: Sprinkler design and pipes, Clay Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

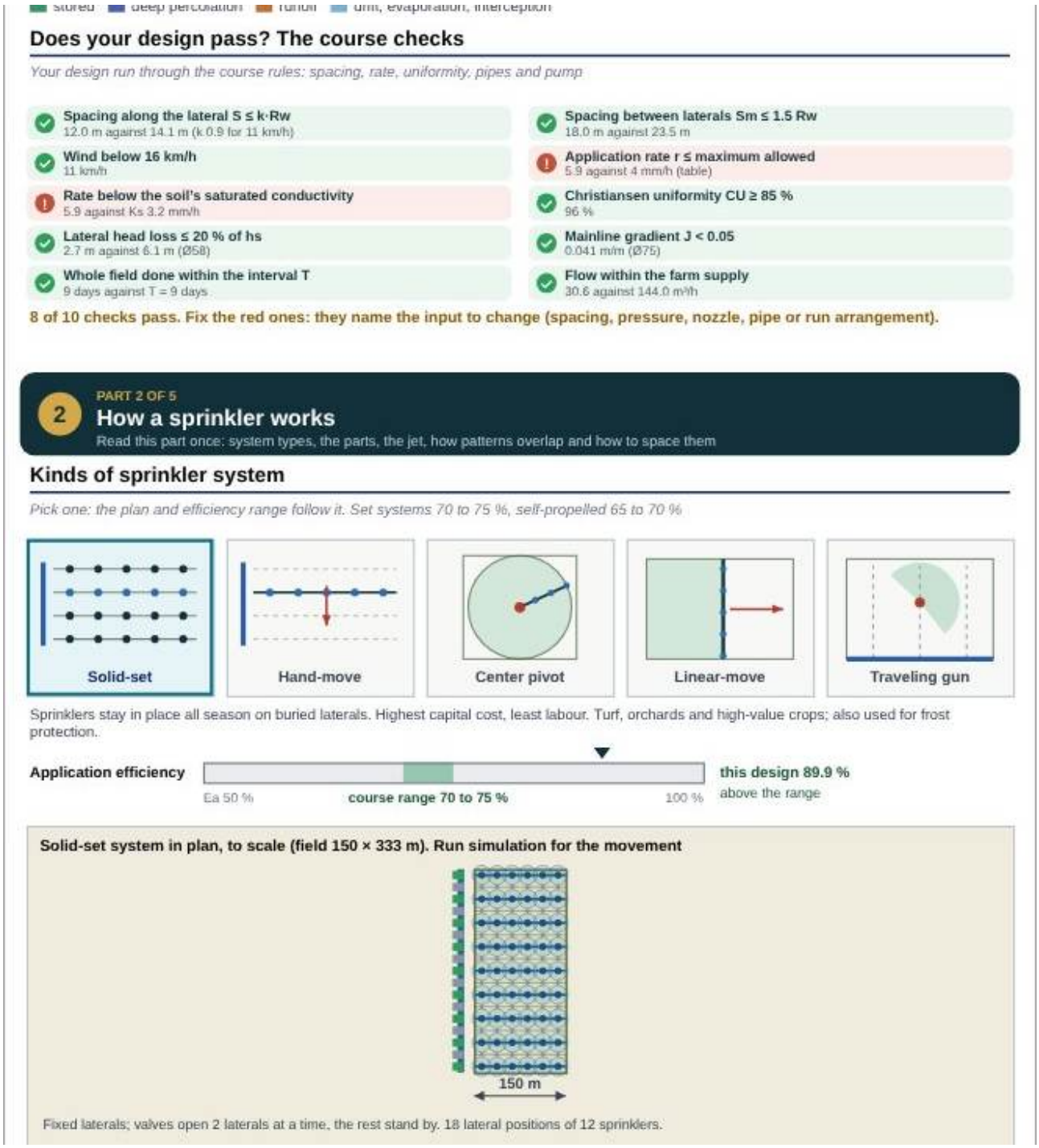


Figure 9.16: Sprinkler design and pipes, Clay Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

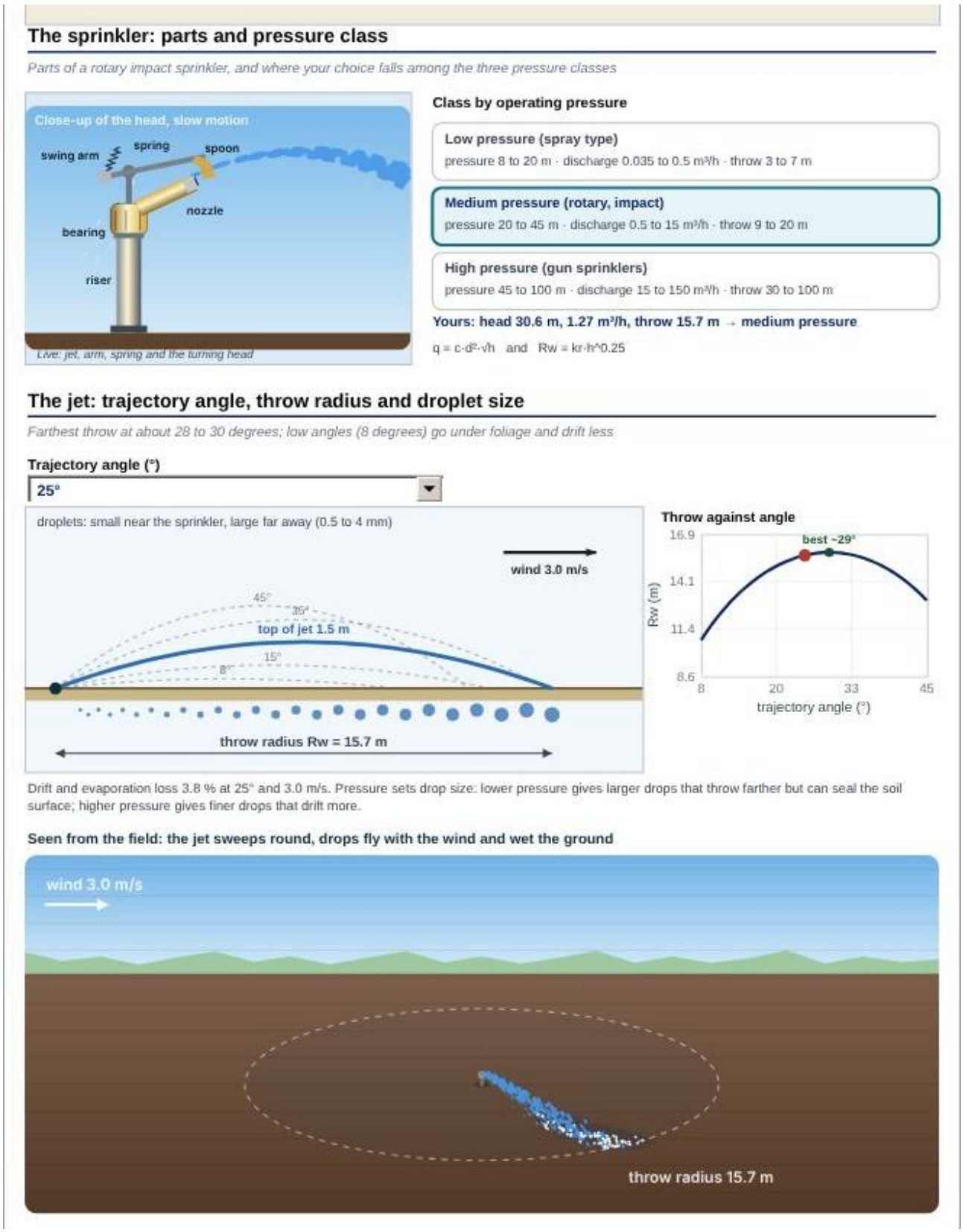


Figure 9.17: Sprinkler design and pipes, Clay Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

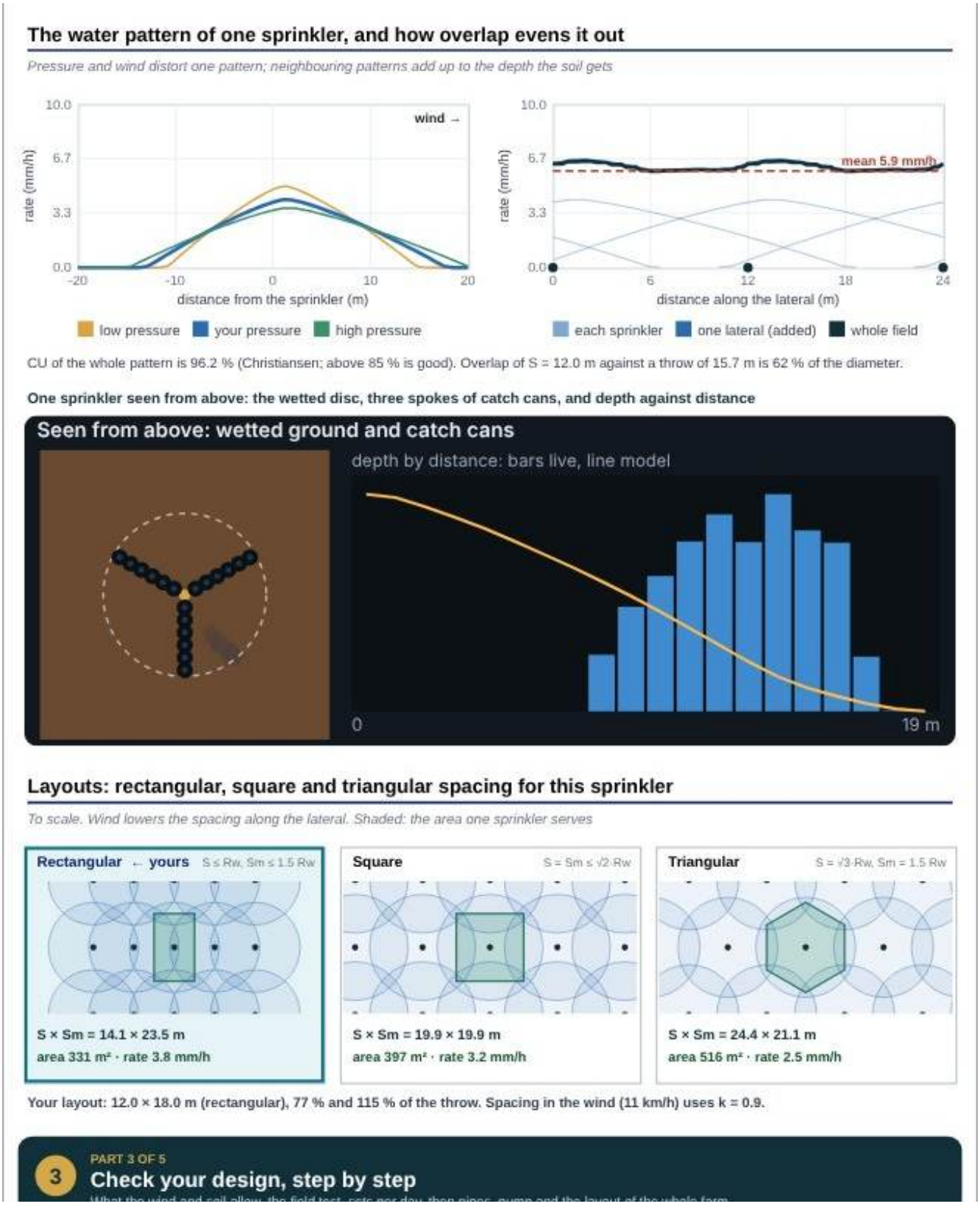


Figure 9.18: Sprinkler design and pipes, Clay Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

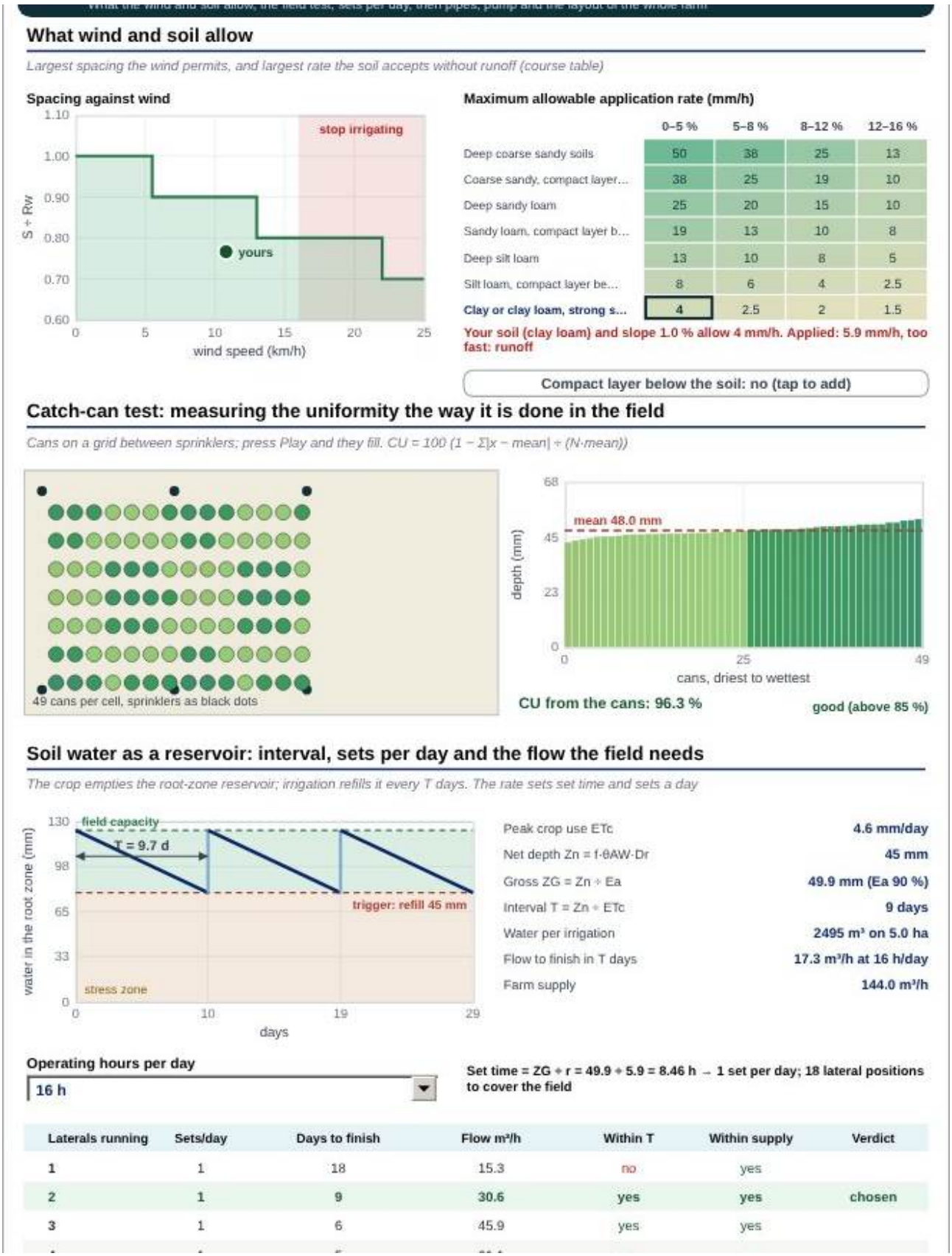


Figure 9.19: Sprinkler design and pipes, Clay Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

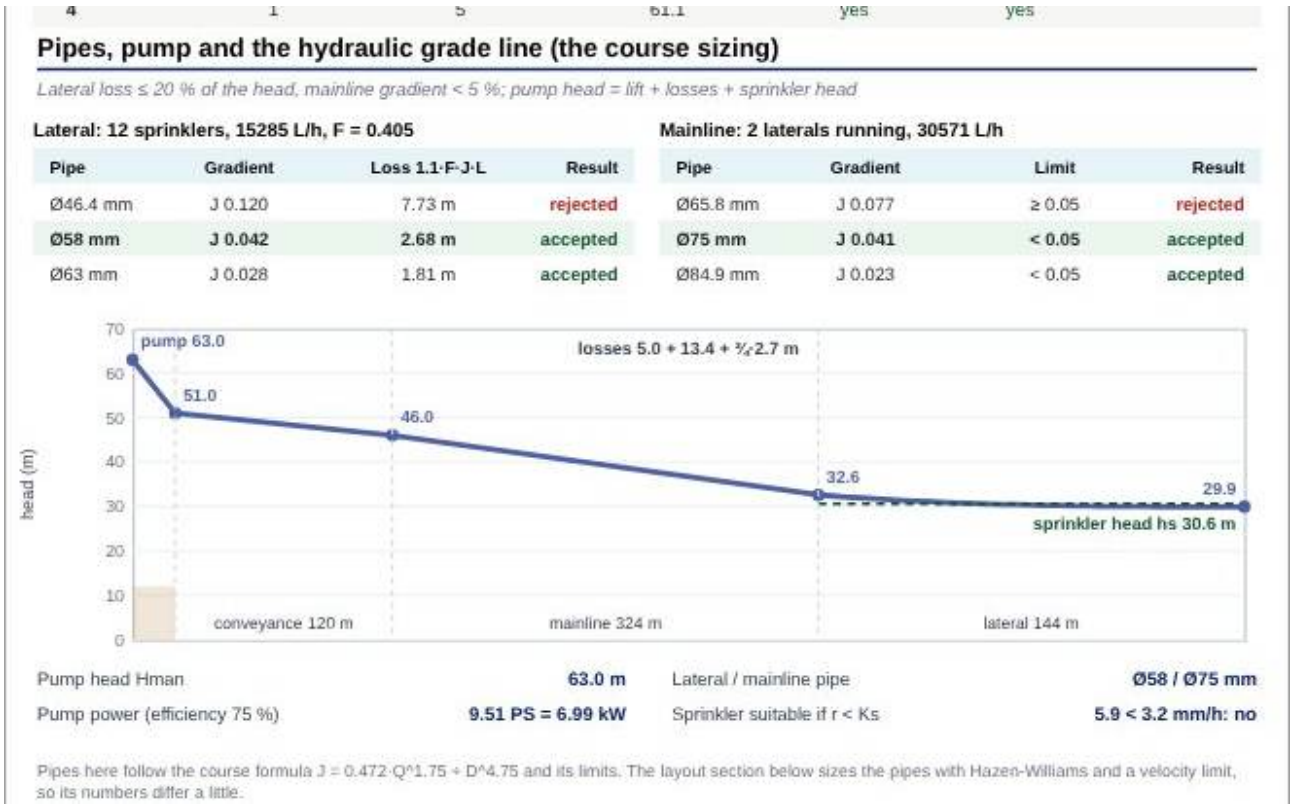
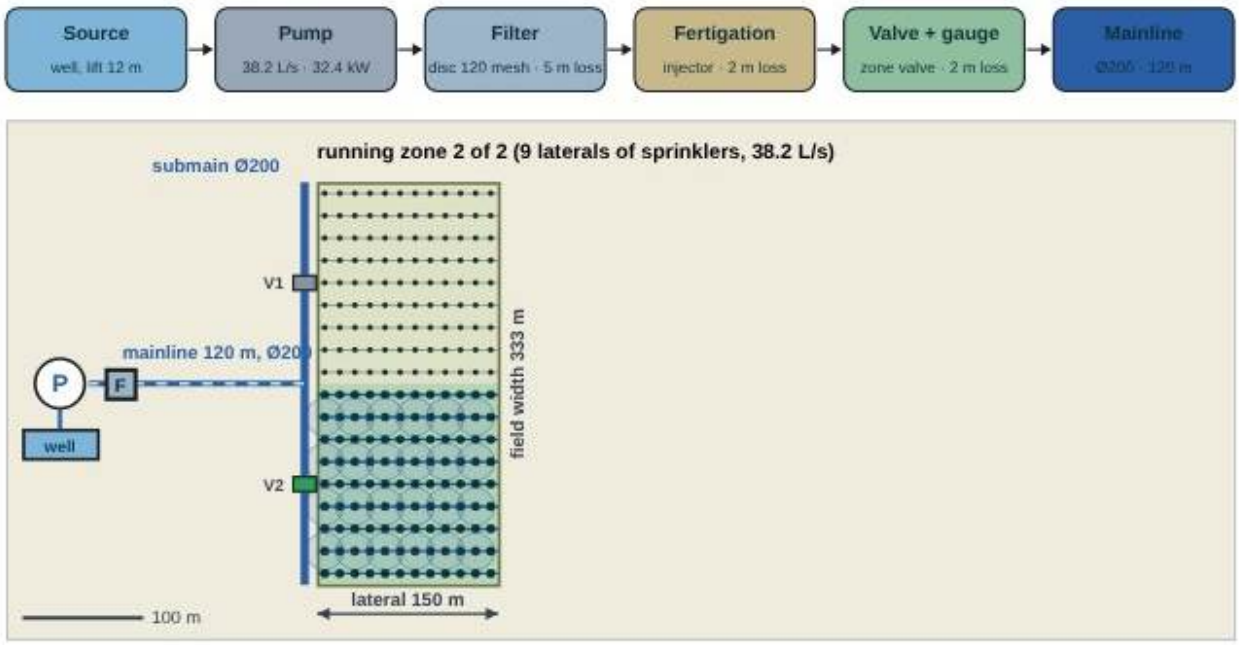
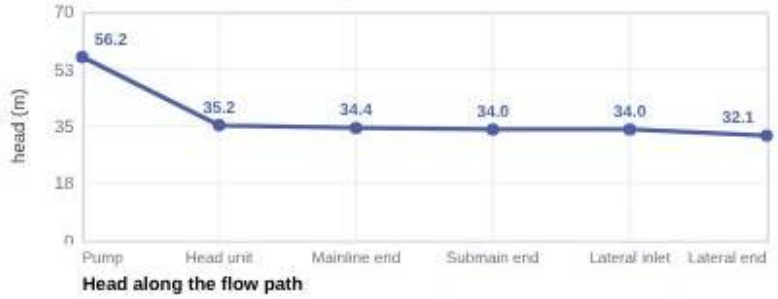


Figure 9.20: Sprinkler design and pipes, Clay Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 Improve it

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

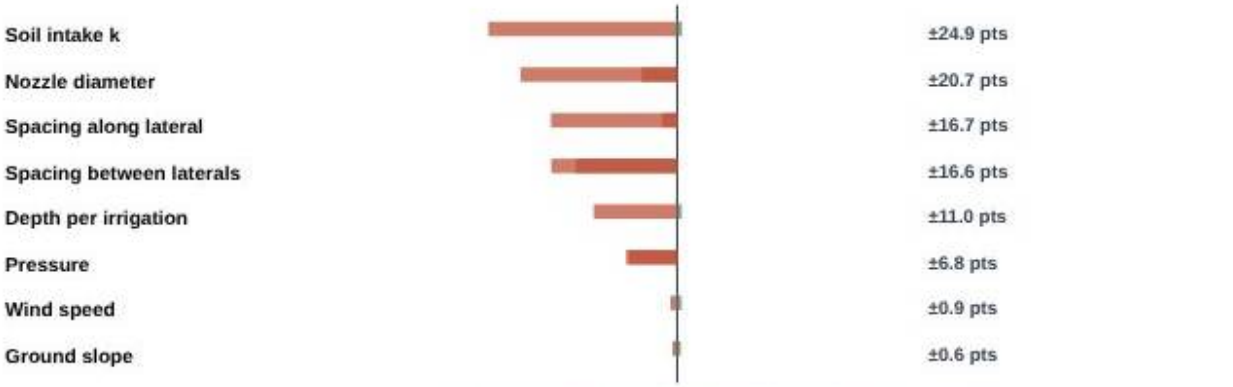
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	508	89.9	99.8	5.4	meets target

Figure 9.21: Sprinkler design and pipes, Clay Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 89.9 % with CO 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

5 PART 5 OF 5

What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

Net irrigation need in a season (DSS)	**394 mm → 9 irrigations of 45 mm**
Gross water per irrigation	**49.9 mm (Ea 90 %)**
Gross water per season	**438 mm = 4380 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 508 min = 25.4 h per round**
Pumping per season	**223 hours · 7228 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**16 May 2027**	204 days	30	34	25.4 h	13 days
2	**29 May 2027**	217 days	30	34	25.4 h	10 days
3	**8 Jun 2027**	227 days	30	34	25.4 h	9 days
4	**17 Jun 2027**	236 days	30	34	25.4 h	12 days
5	**29 Jun 2027**	248 days	30	34	25.4 h	8 days
6	**7 Jul 2027**	256 days	30	34	25.4 h	10 days
7	**17 Jul 2027**	266 days	30	34	25.4 h	15 days
8	**1 Aug 2027**	281 days	30	34	25.4 h	-

Figure 9.22: Sprinkler design and pipes, Clay Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.

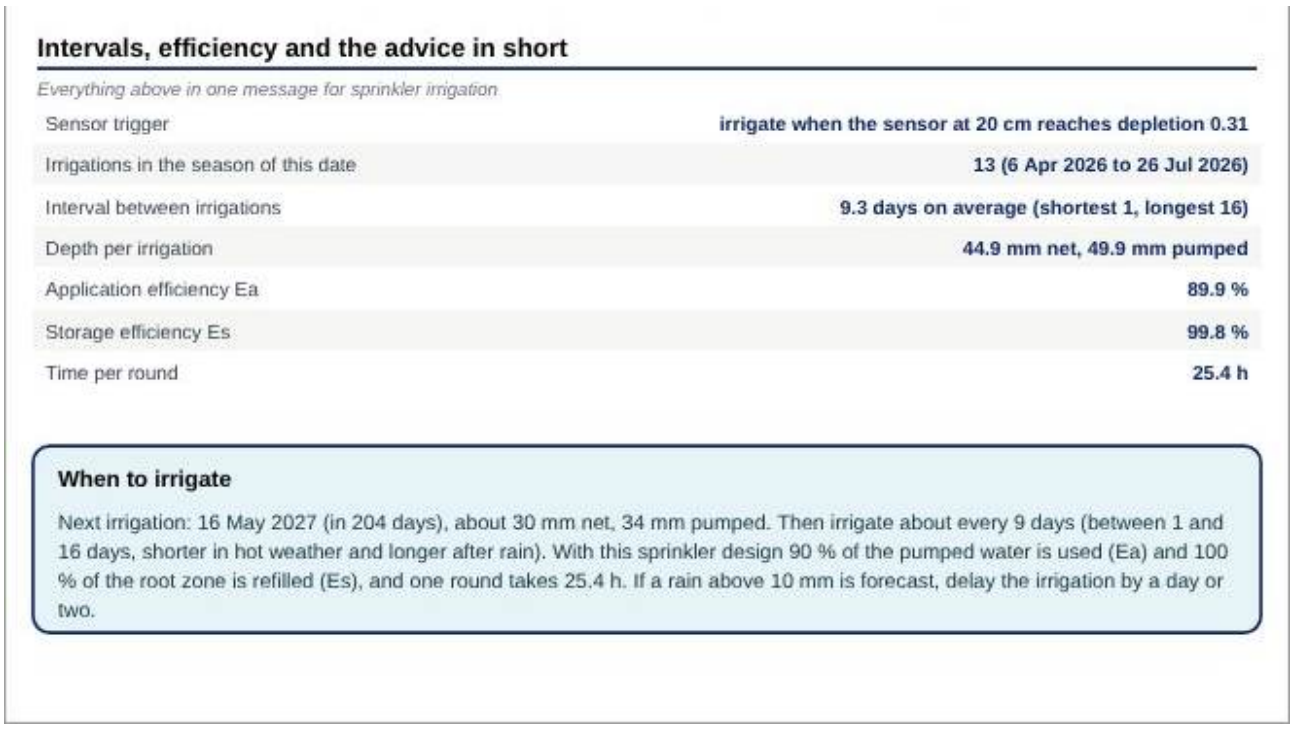


Figure 9.23: Sprinkler design and pipes, Clay Loam, part 12 of 12: Intervals, efficiency and the advice in short.

9.4 Virtual farm of the surface design (Clay Loam)

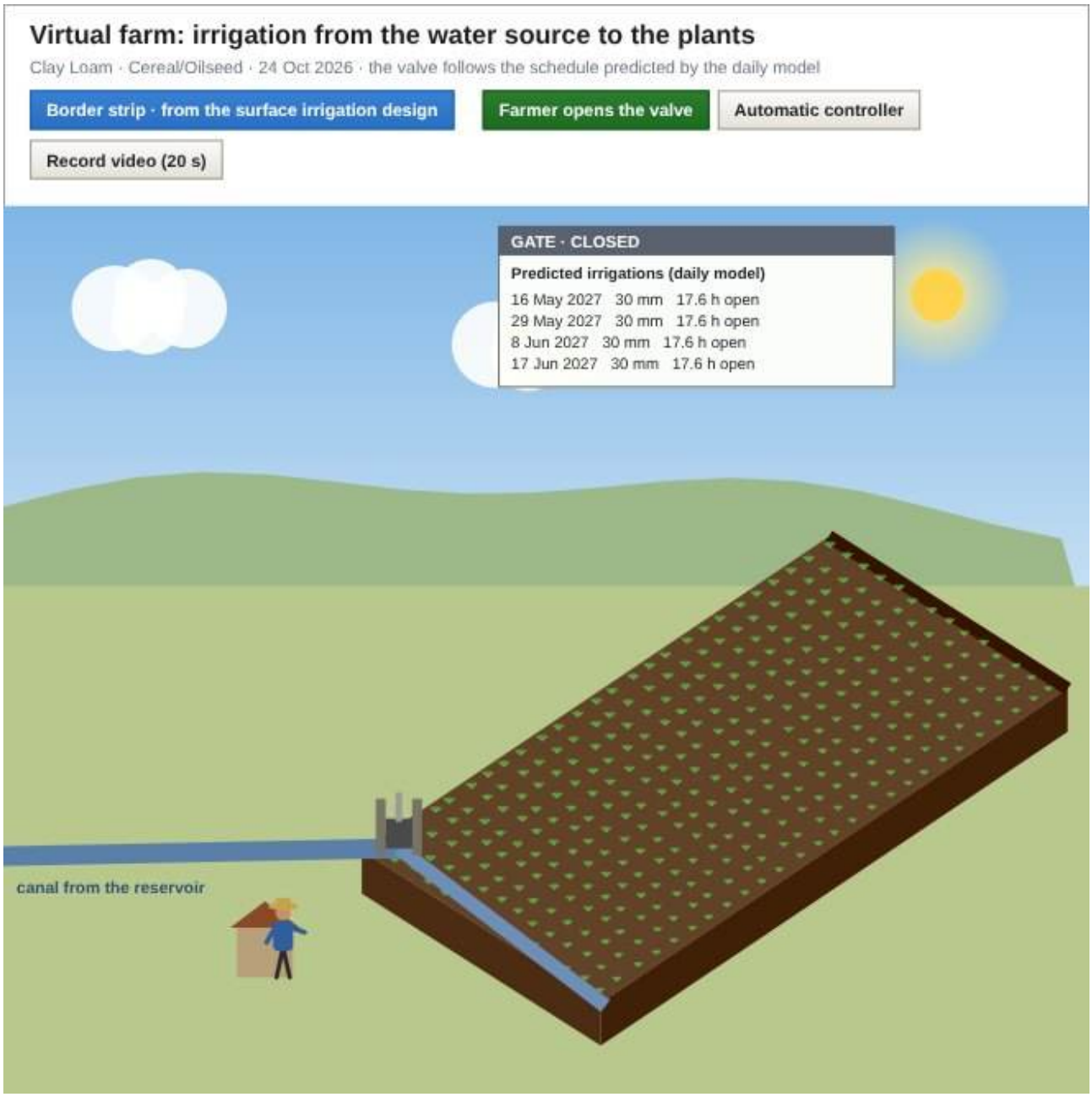


Figure 9.24: Virtual farm, surface design, Clay Loam, part 1 of 3.

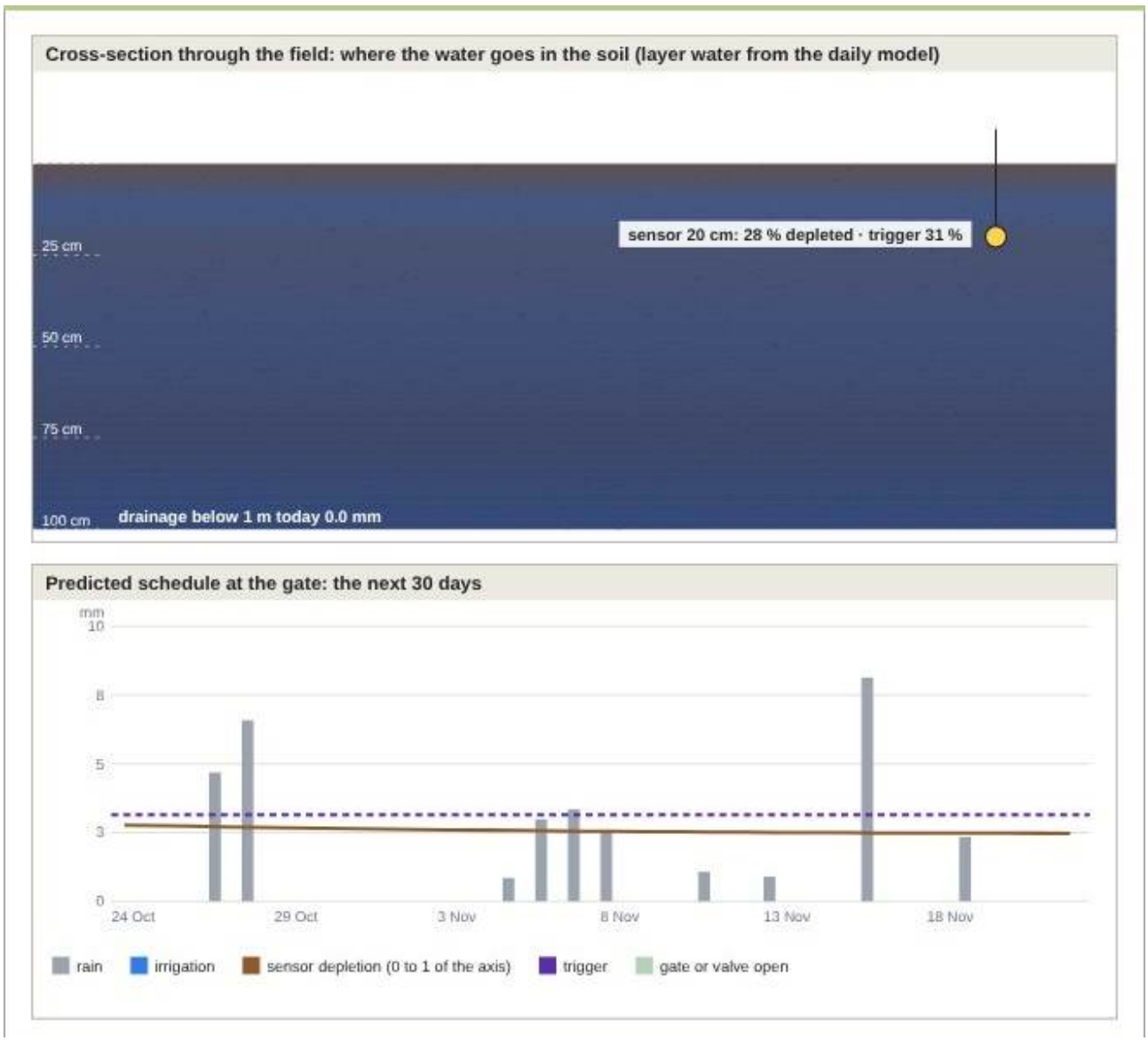


Figure 9.25: Virtual farm, surface design, Clay Loam, part 2 of 3.

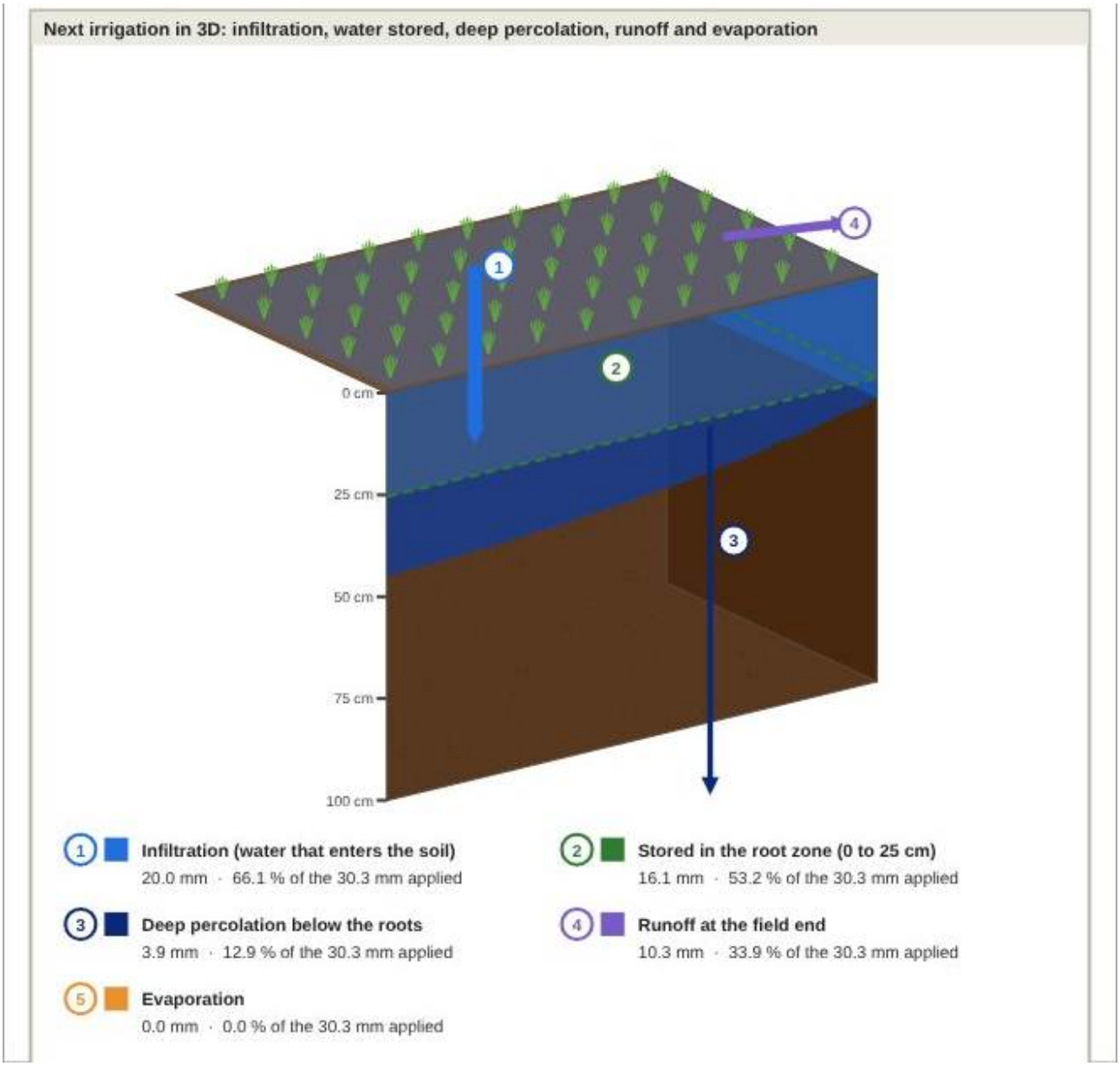


Figure 9.26: Virtual farm, surface design, Clay Loam, part 3 of 3.

9.5 Virtual farm of the drip design (Clay Loam)

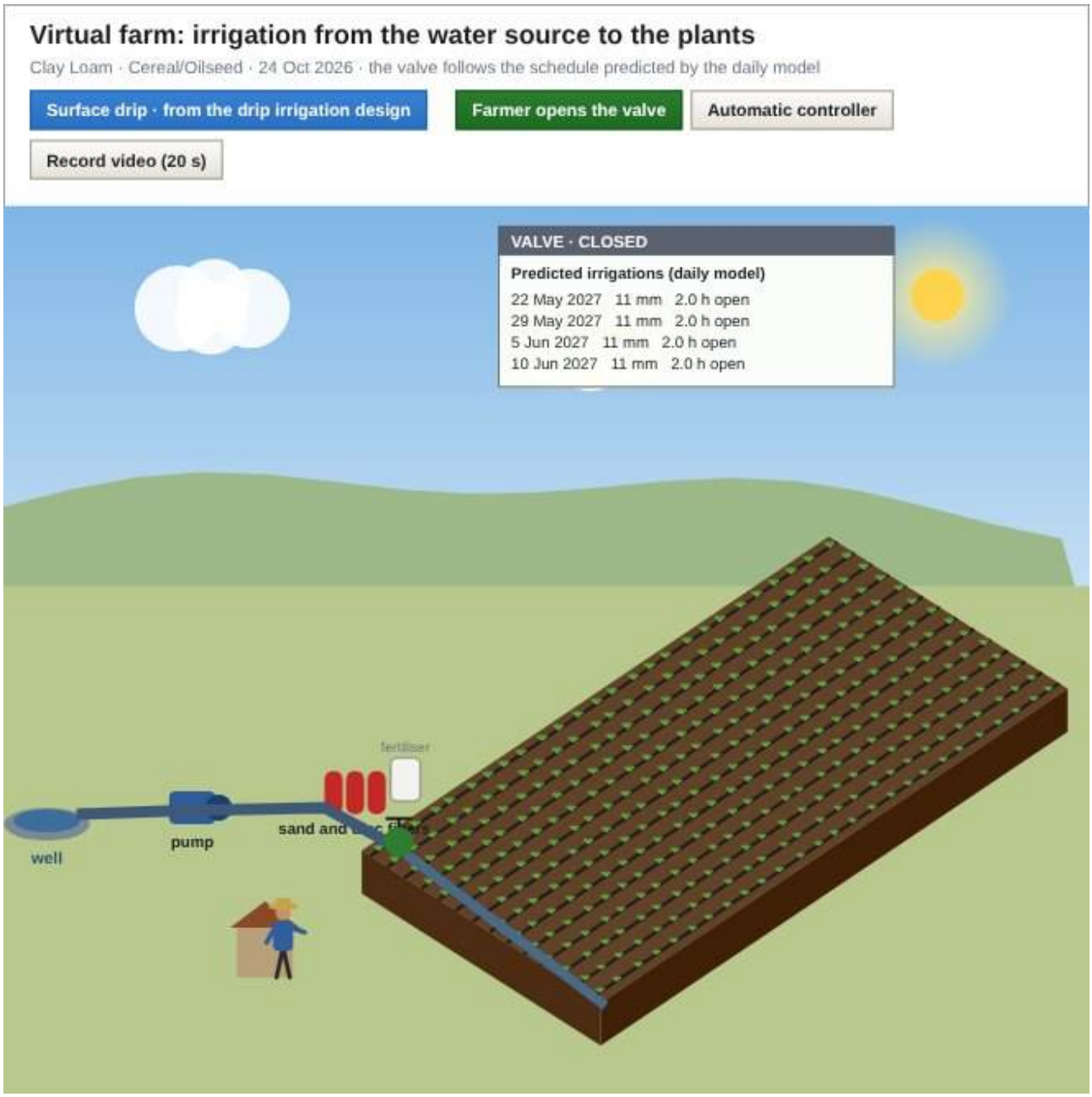


Figure 9.27: Virtual farm, drip design, Clay Loam, part 1 of 3.

Figure 9.28: Virtual farm, drip design, Clay Loam, part 2 of 3.

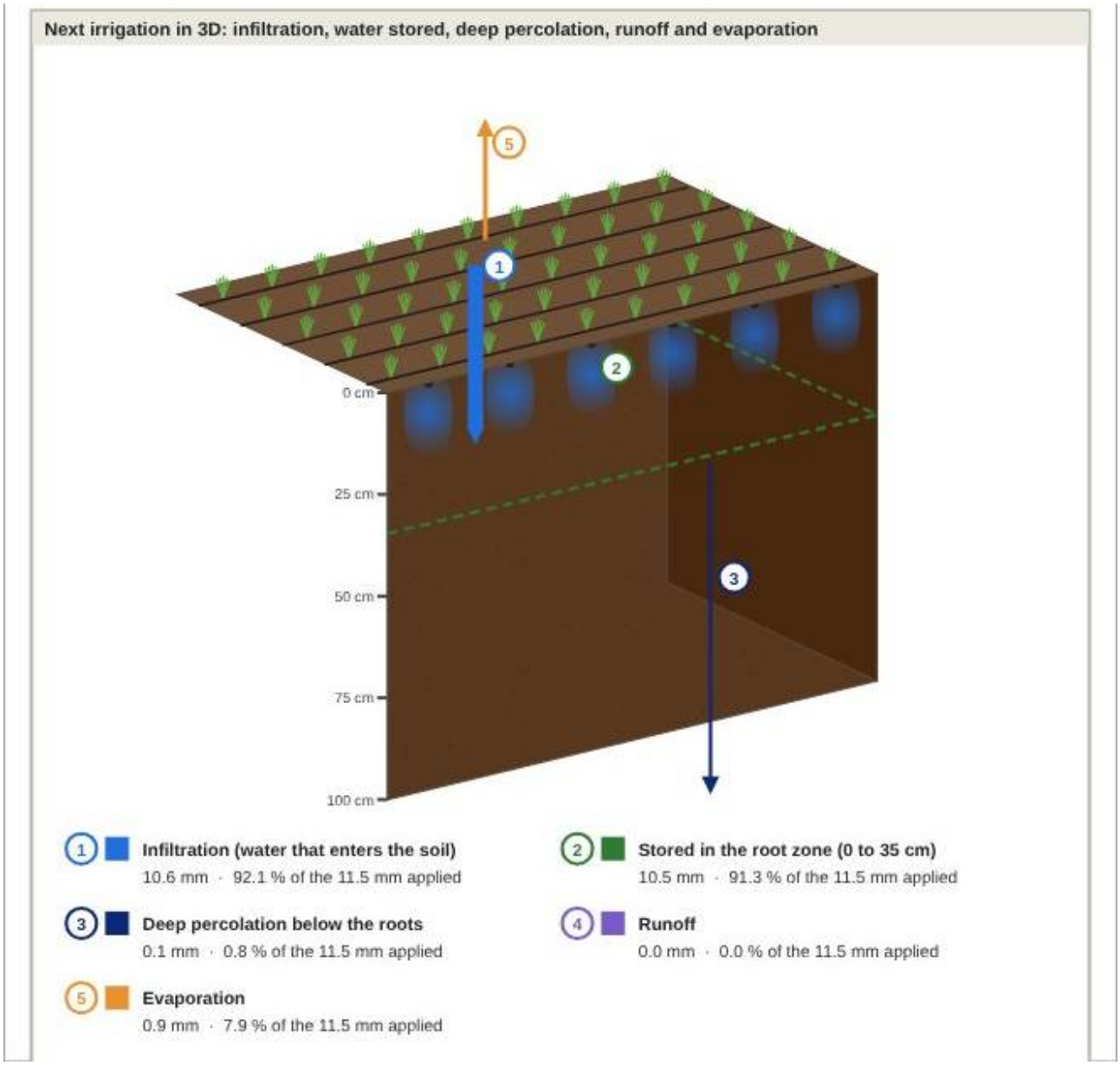


Figure 9.29: Virtual farm, drip design, Clay Loam, part 3 of 3.

9.6 Virtual farm of the sprinkler design (Clay Loam)

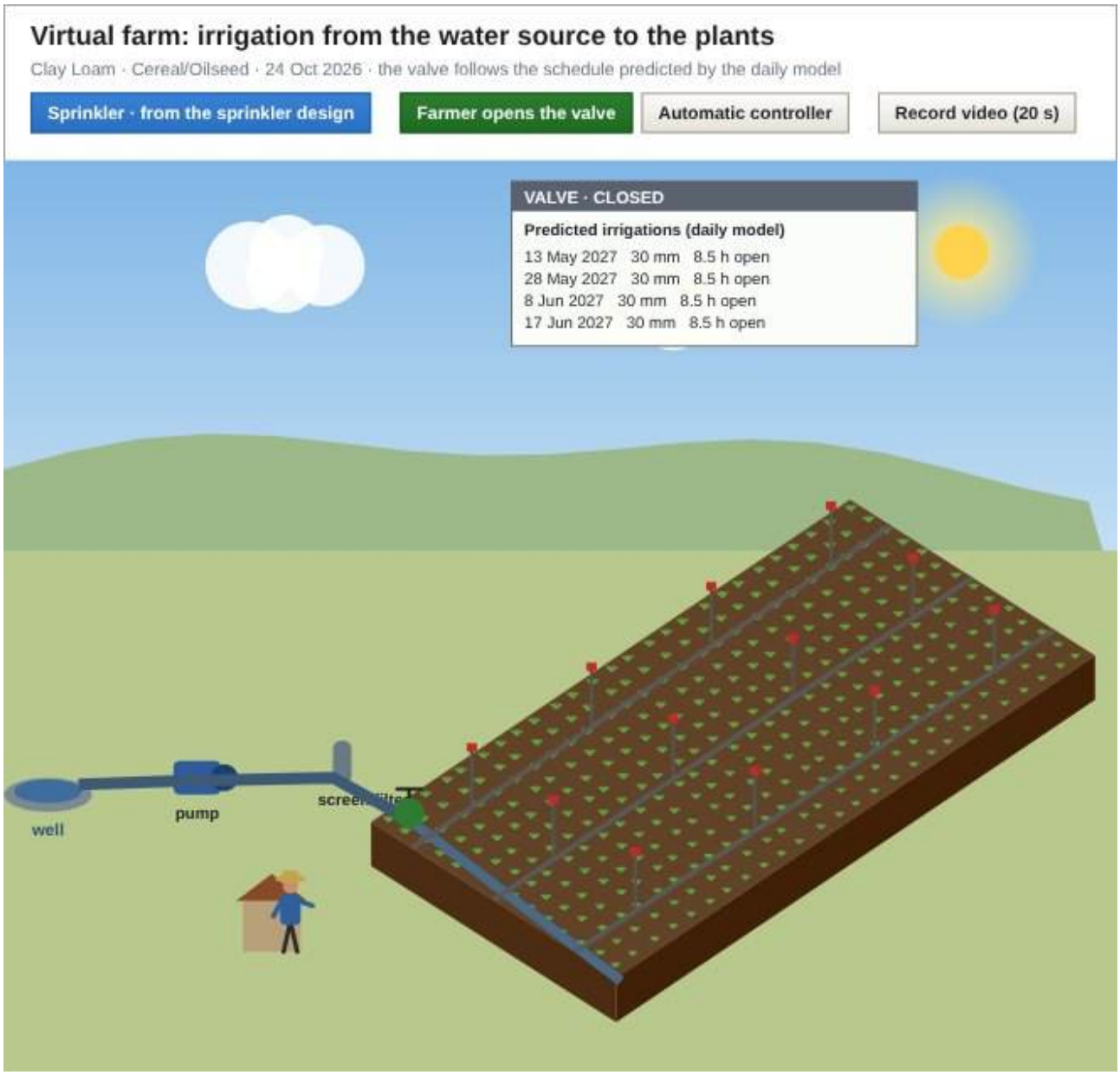


Figure 9.30: Virtual farm, sprinkler design, Clay Loam, part 1 of 3.

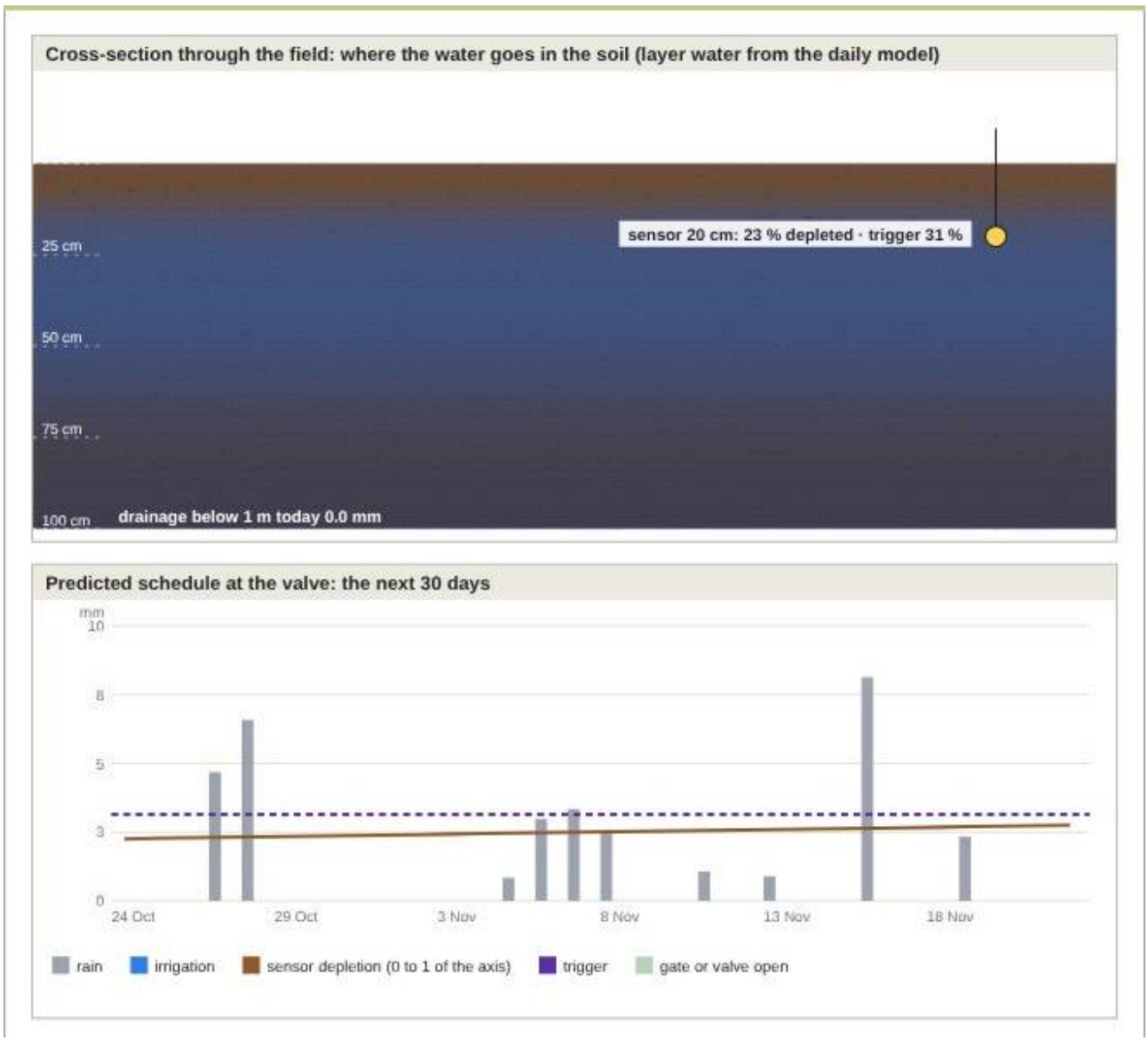


Figure 9.31: Virtual farm, sprinkler design, Clay Loam, part 2 of 3.

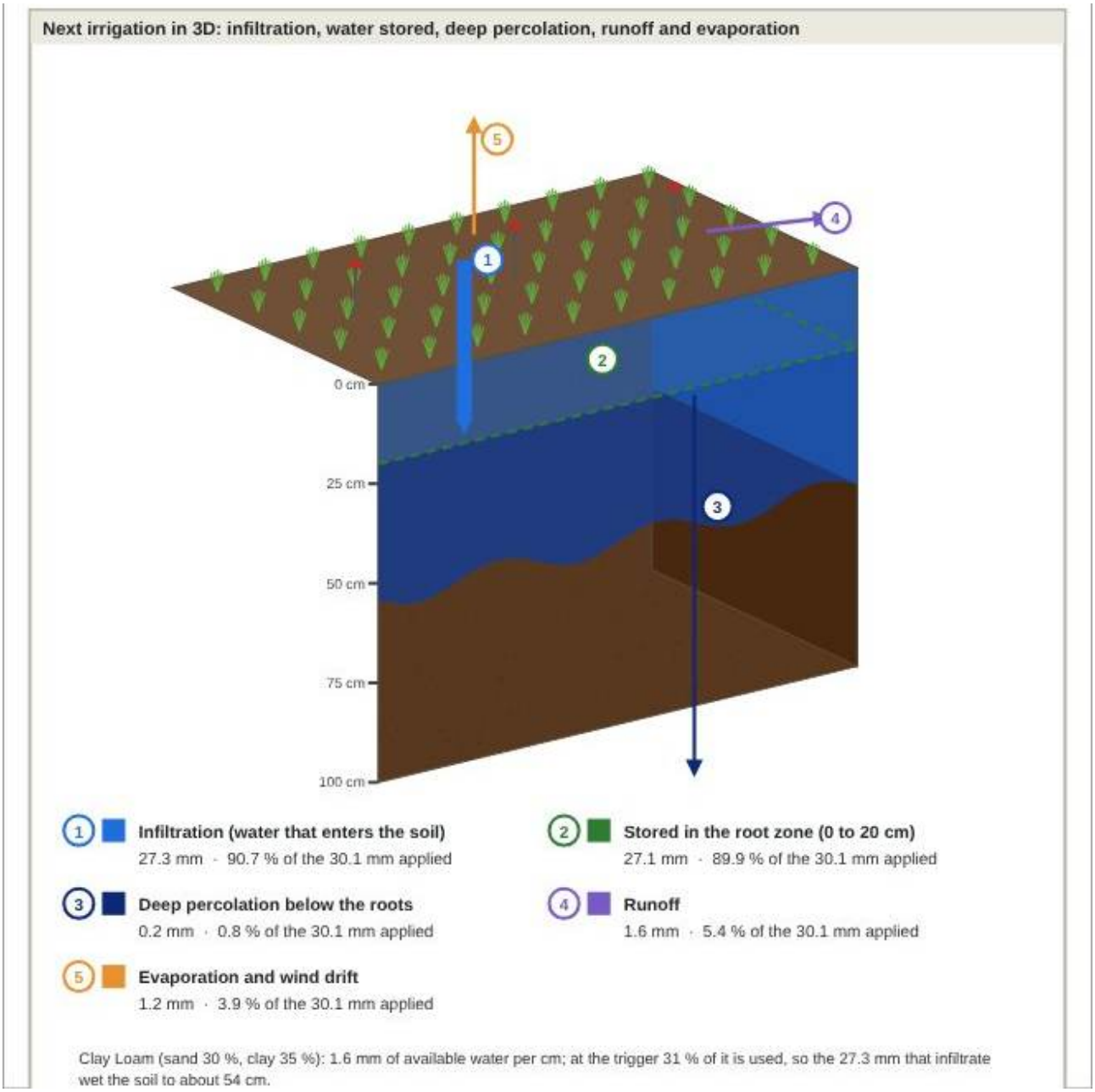


Figure 9.32: Virtual farm, sprinkler design, Clay Loam, part 3 of 3.

9.7 Sensor and trigger (Clay Loam)

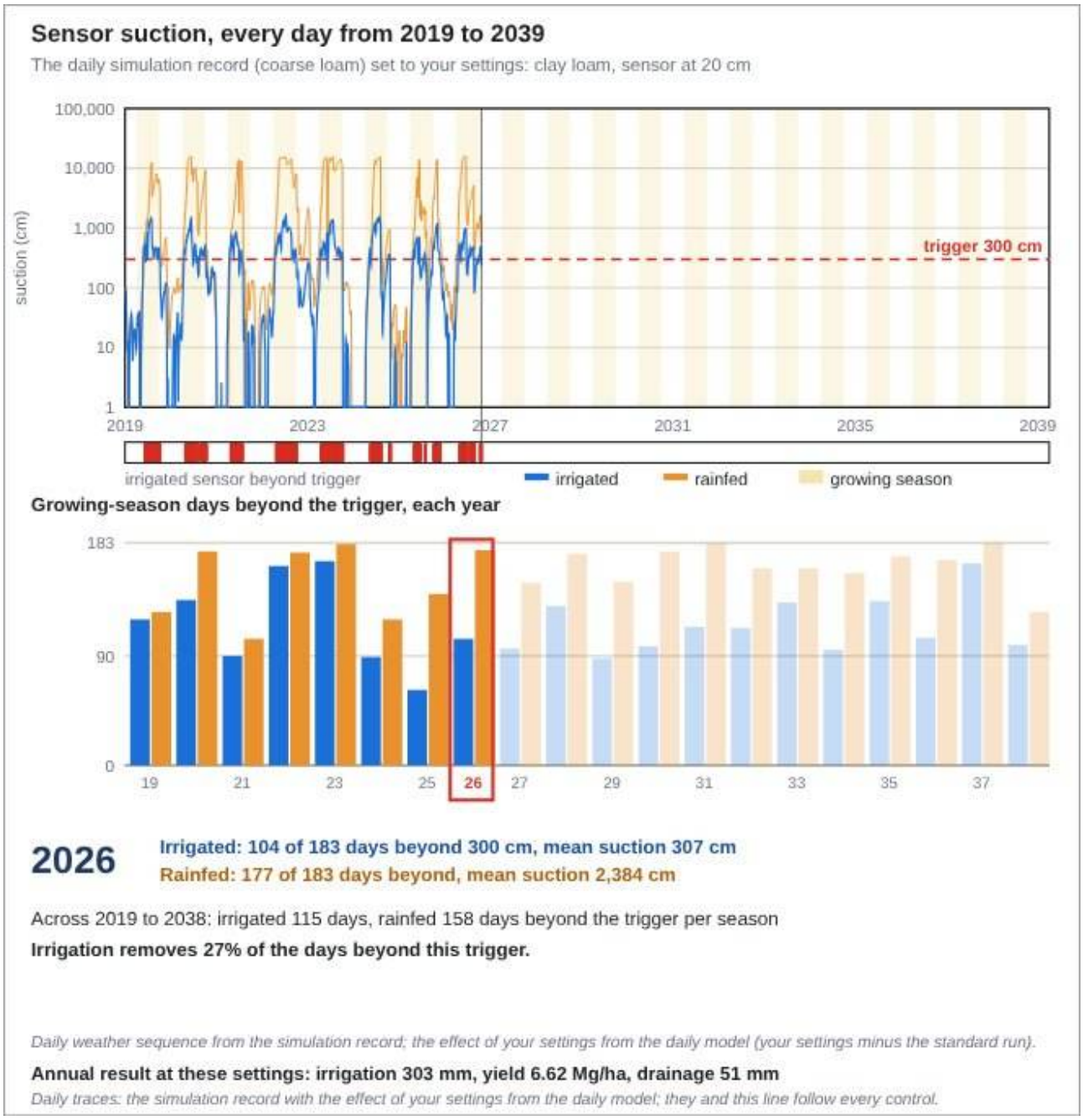


Figure 9.33: Sensor and trigger, Clay Loam, part 1 of 1.

9.8 Trigger law (Clay Loam)

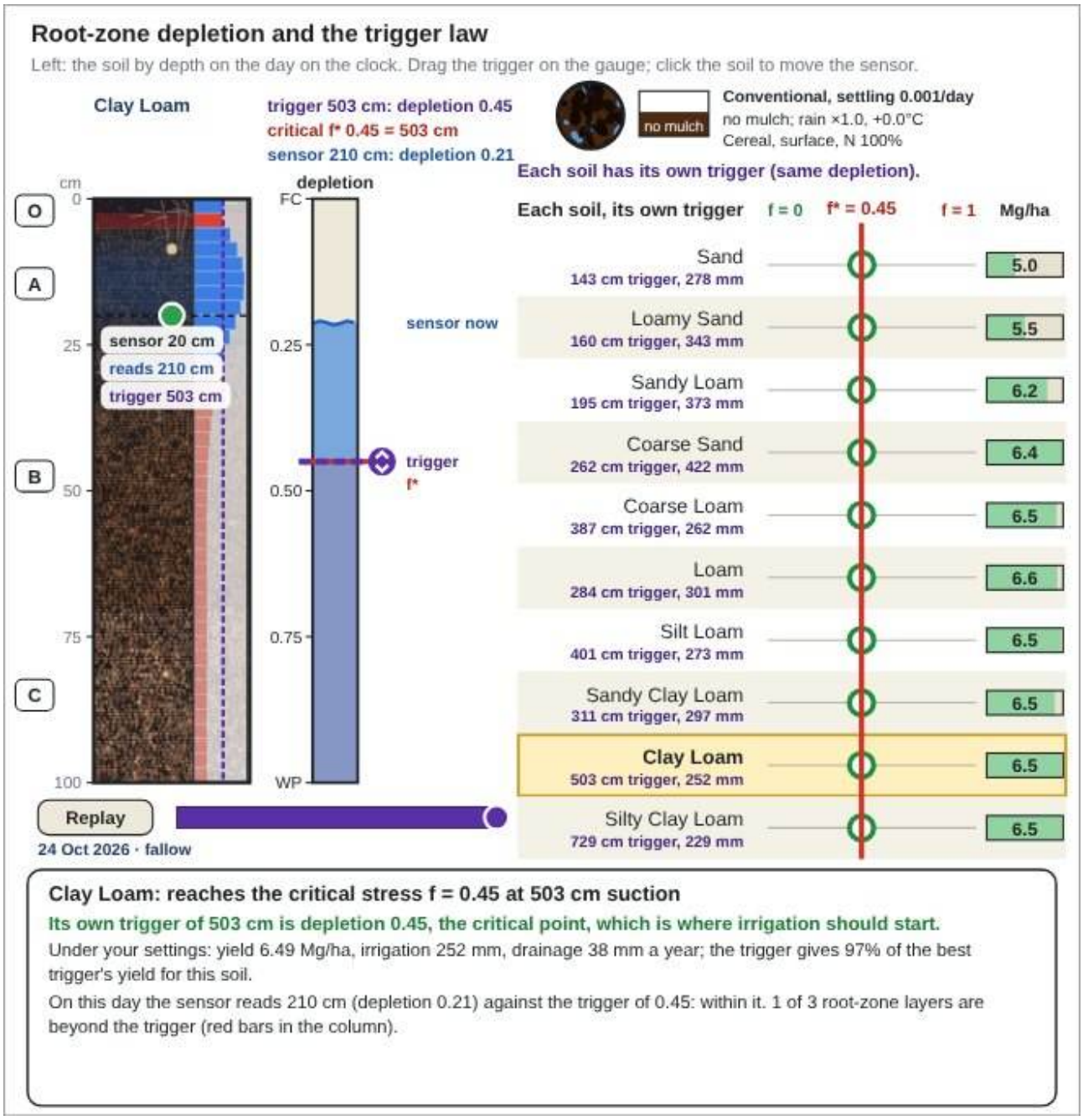


Figure 9.34: Trigger law, Clay Loam, part 1 of 1.

9.9 Mulch against bare soil (Clay Loam)

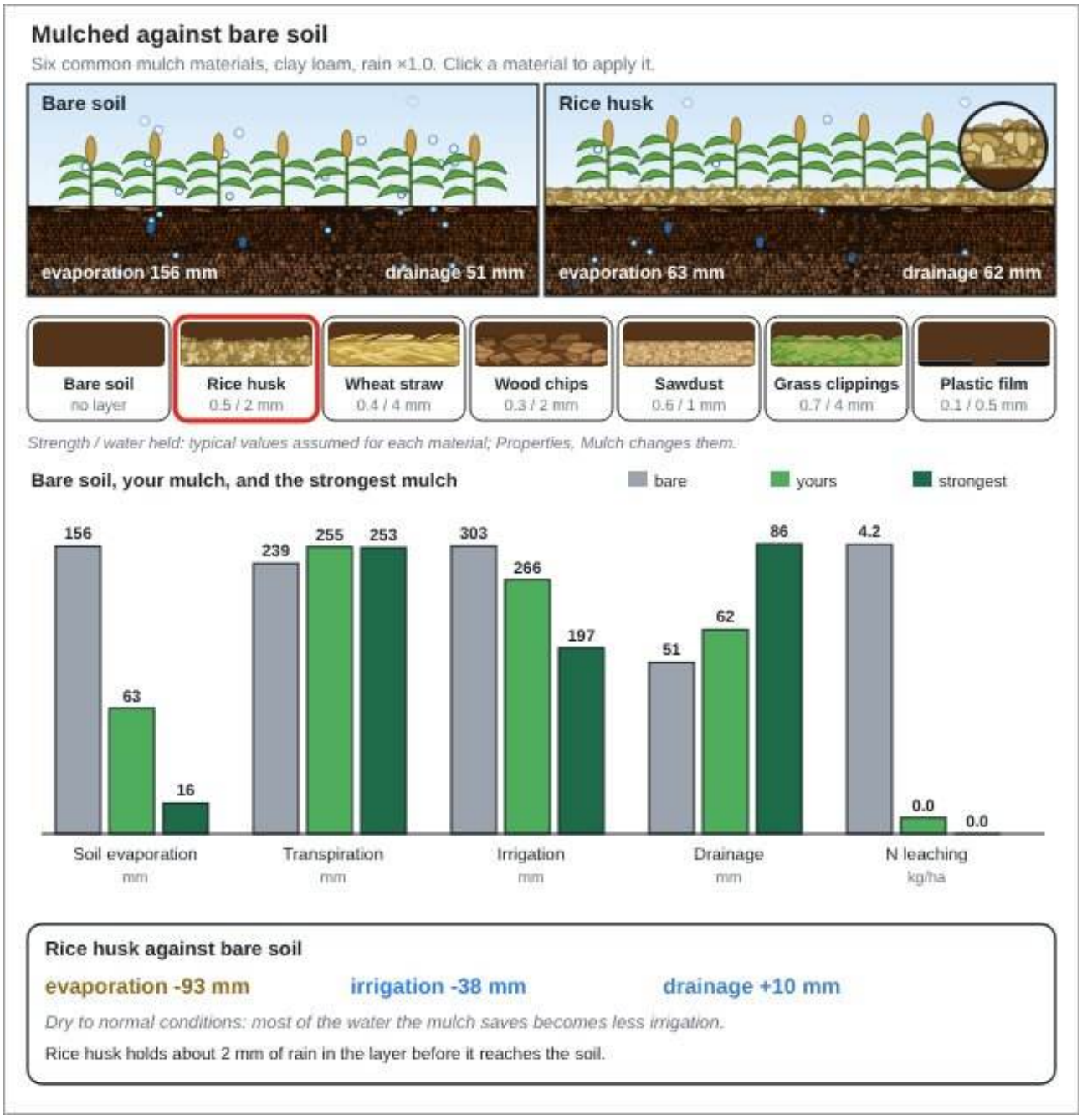


Figure 9.35: Mulch against bare soil, Clay Loam, part 1 of 1.

9.10 Mulch control (Clay Loam)

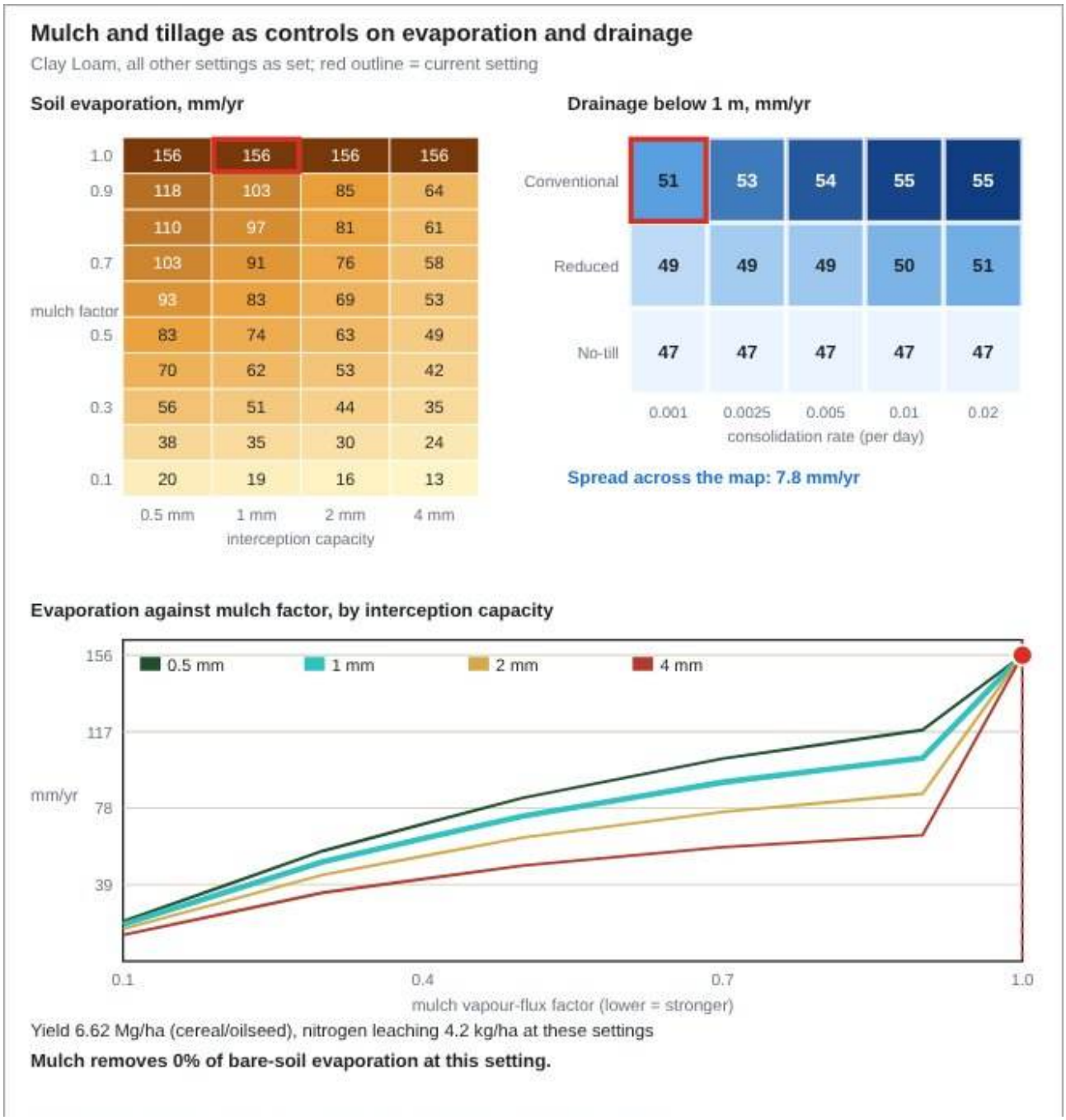


Figure 9.36: Mulch control, Clay Loam, part 1 of 2.

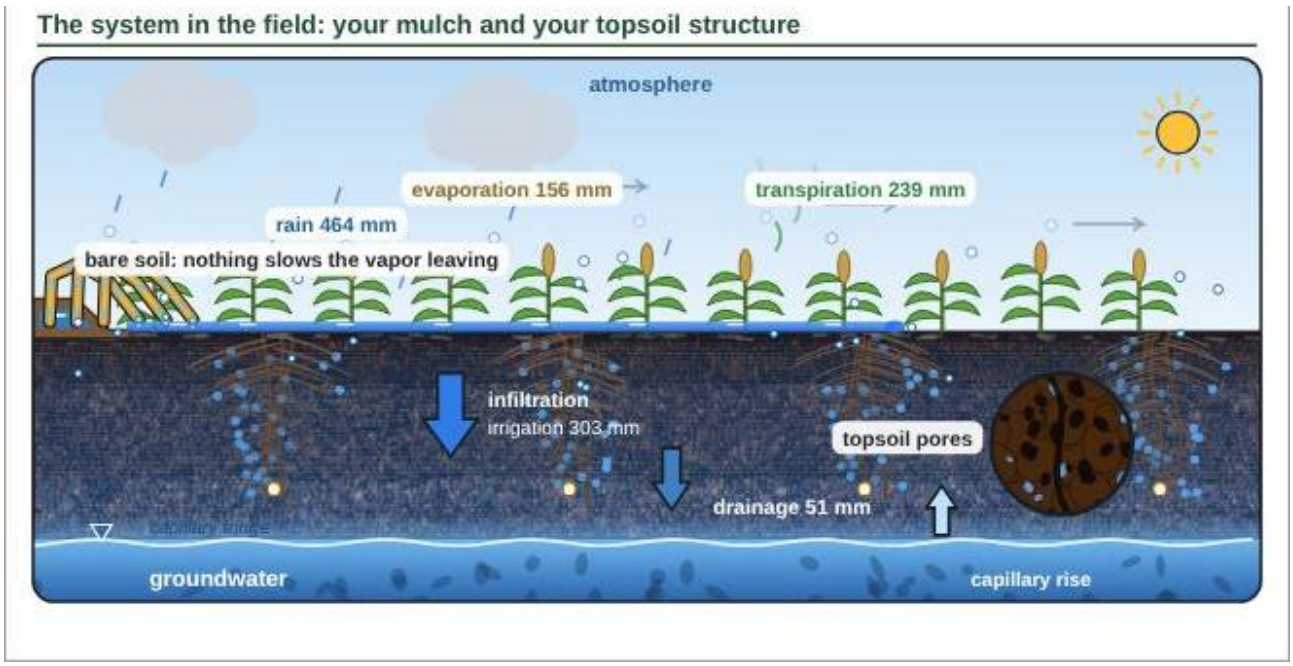


Figure 9.37: Mulch control, Clay Loam, part 2 of 2.

9.11 Evaporation physics (Clay Loam)

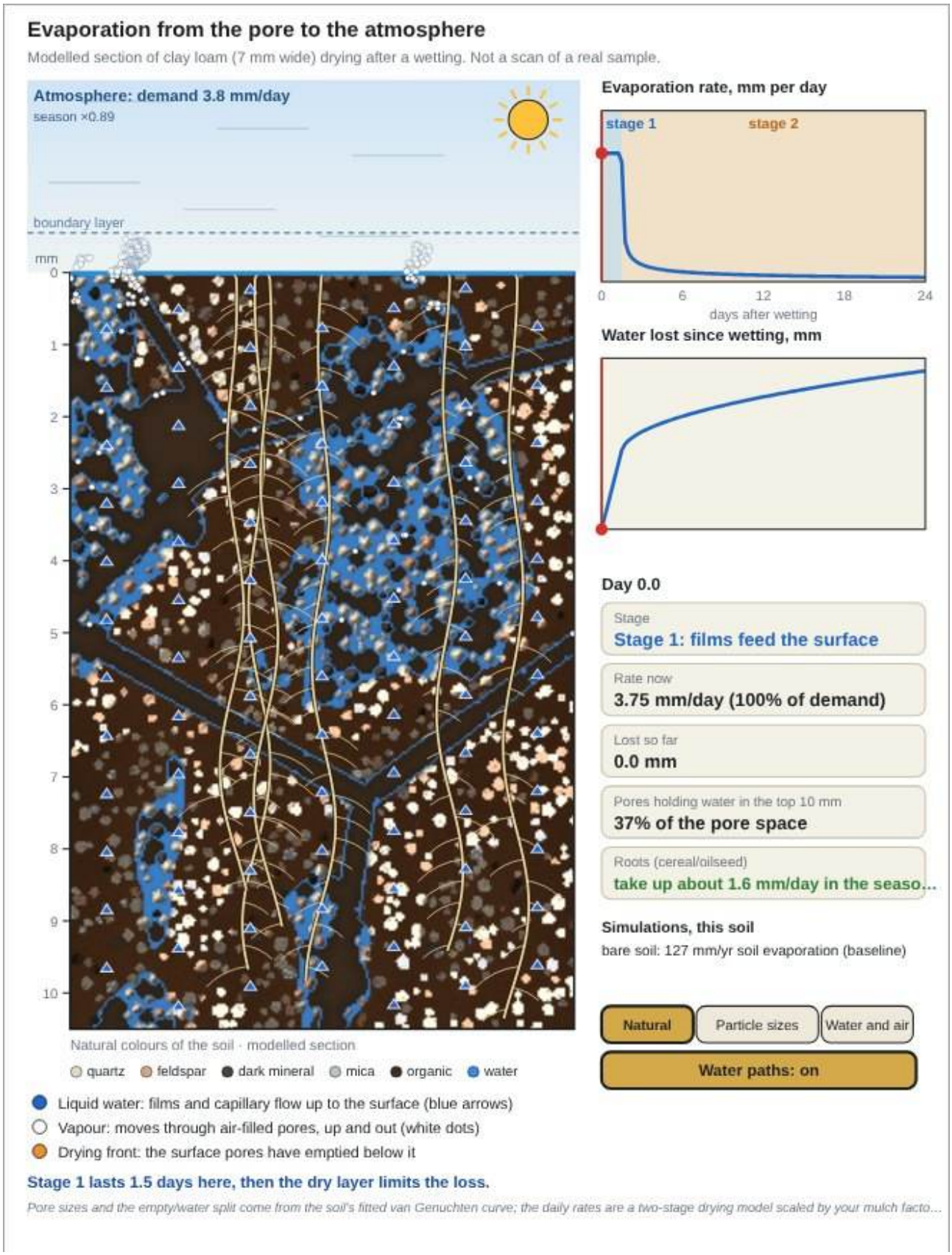


Figure 9.38: Evaporation physics, Clay Loam, part 1 of 3.

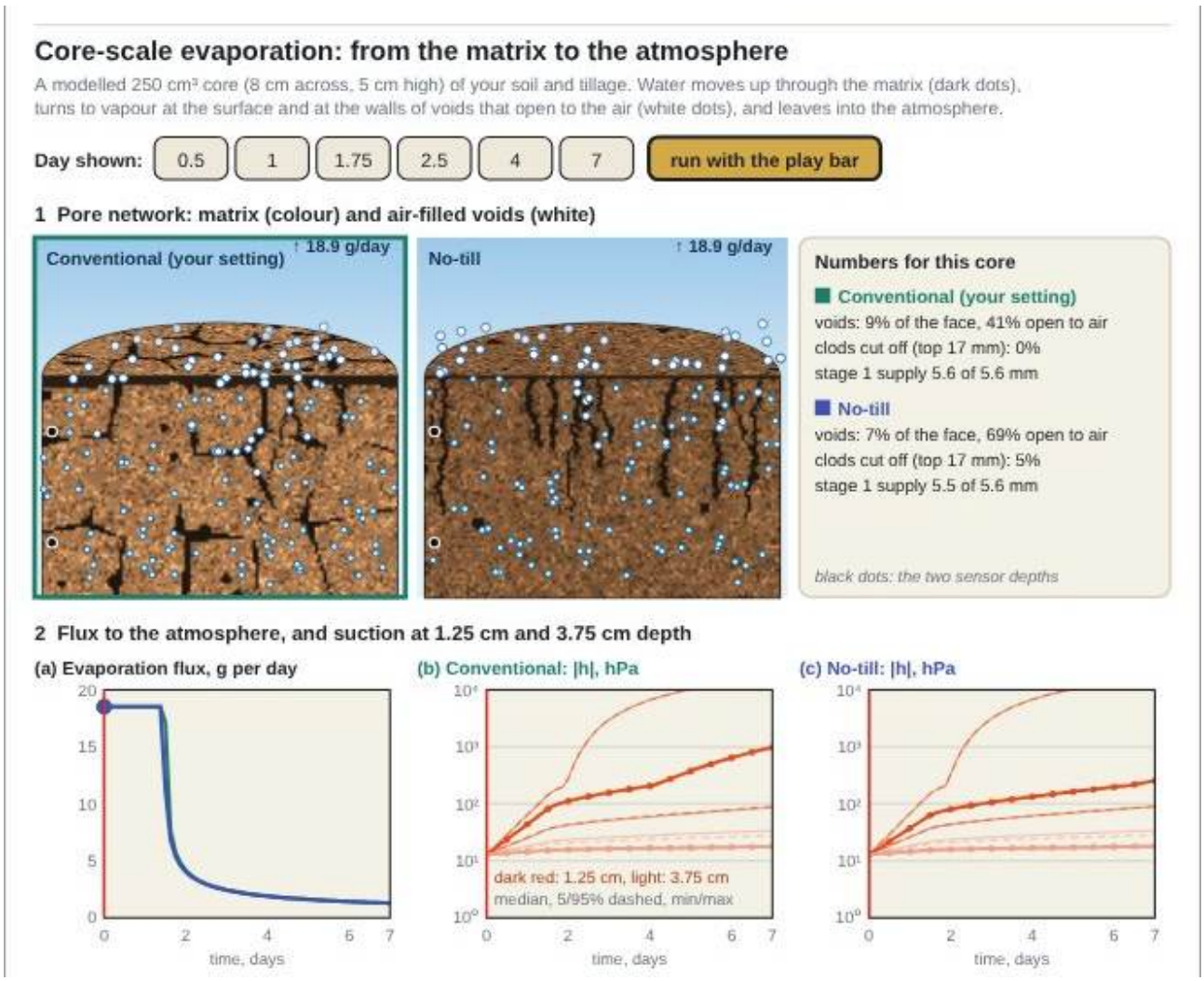


Figure 9.39: Evaporation physics, Clay Loam, part 2 of 3.

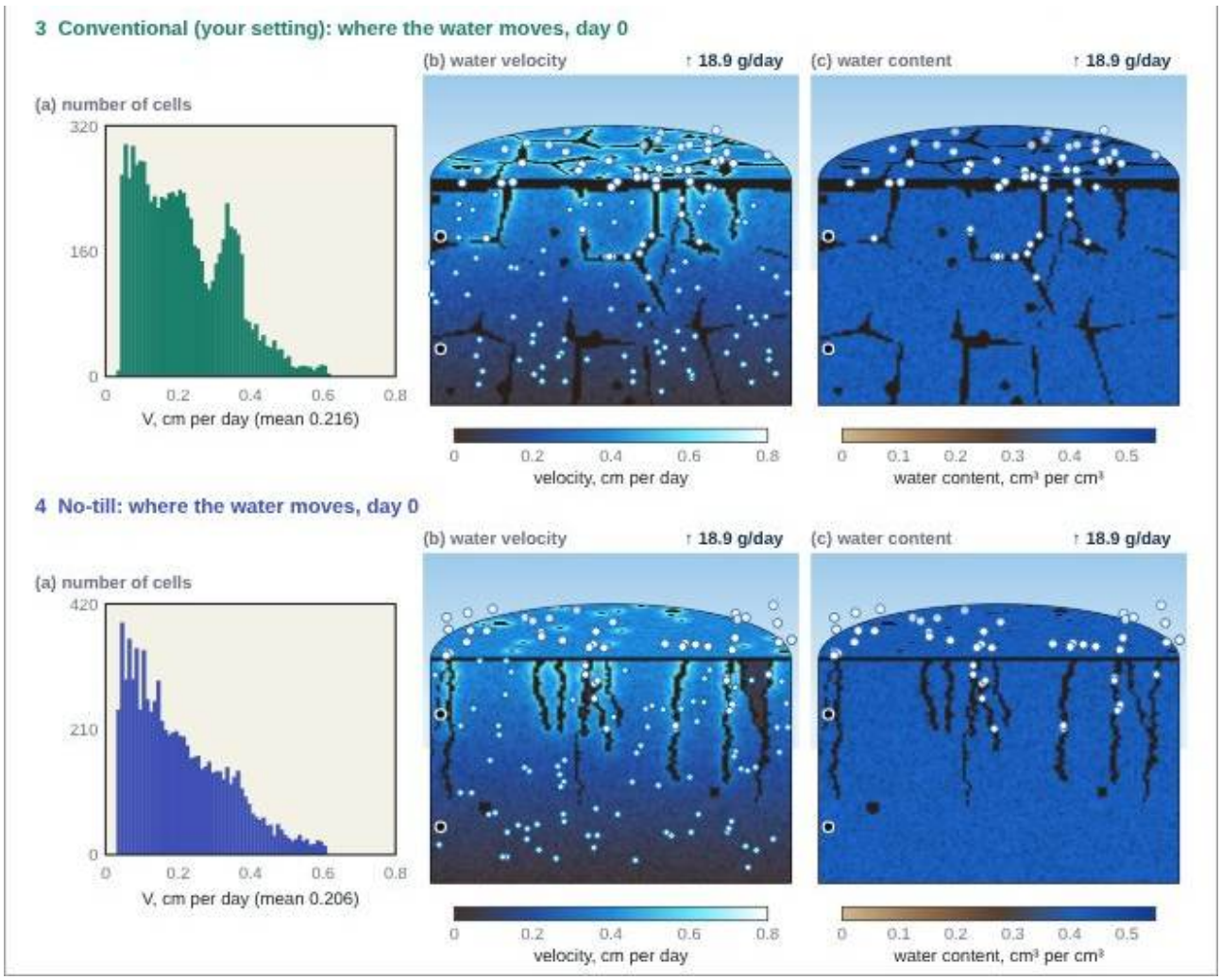


Figure 9.40: Evaporation physics, Clay Loam, part 3 of 3.

9.12 Fate of saved water (Clay Loam)

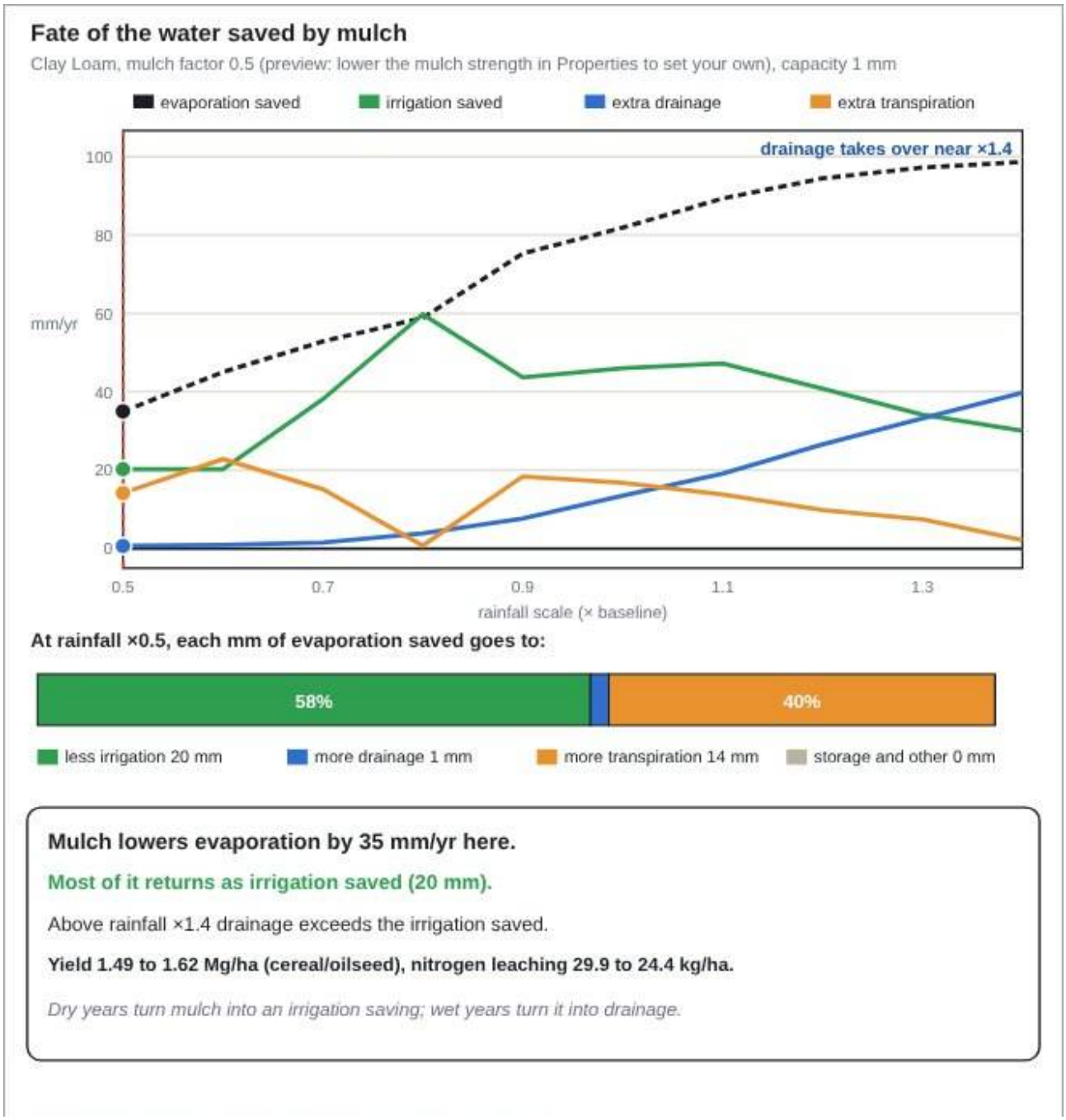


Figure 9.41: Fate of saved water, Clay Loam, part 1 of 2.

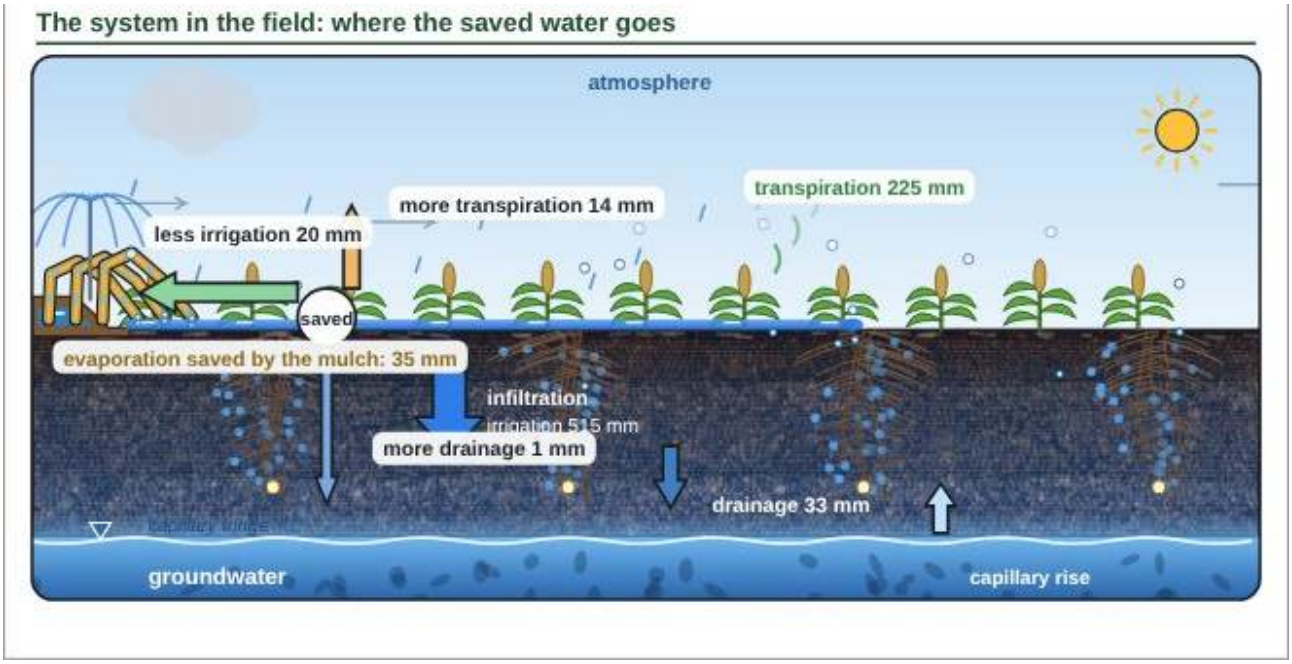


Figure 9.42: Fate of saved water, Clay Loam, part 2 of 2.

9.13 Regime map (Clay Loam)

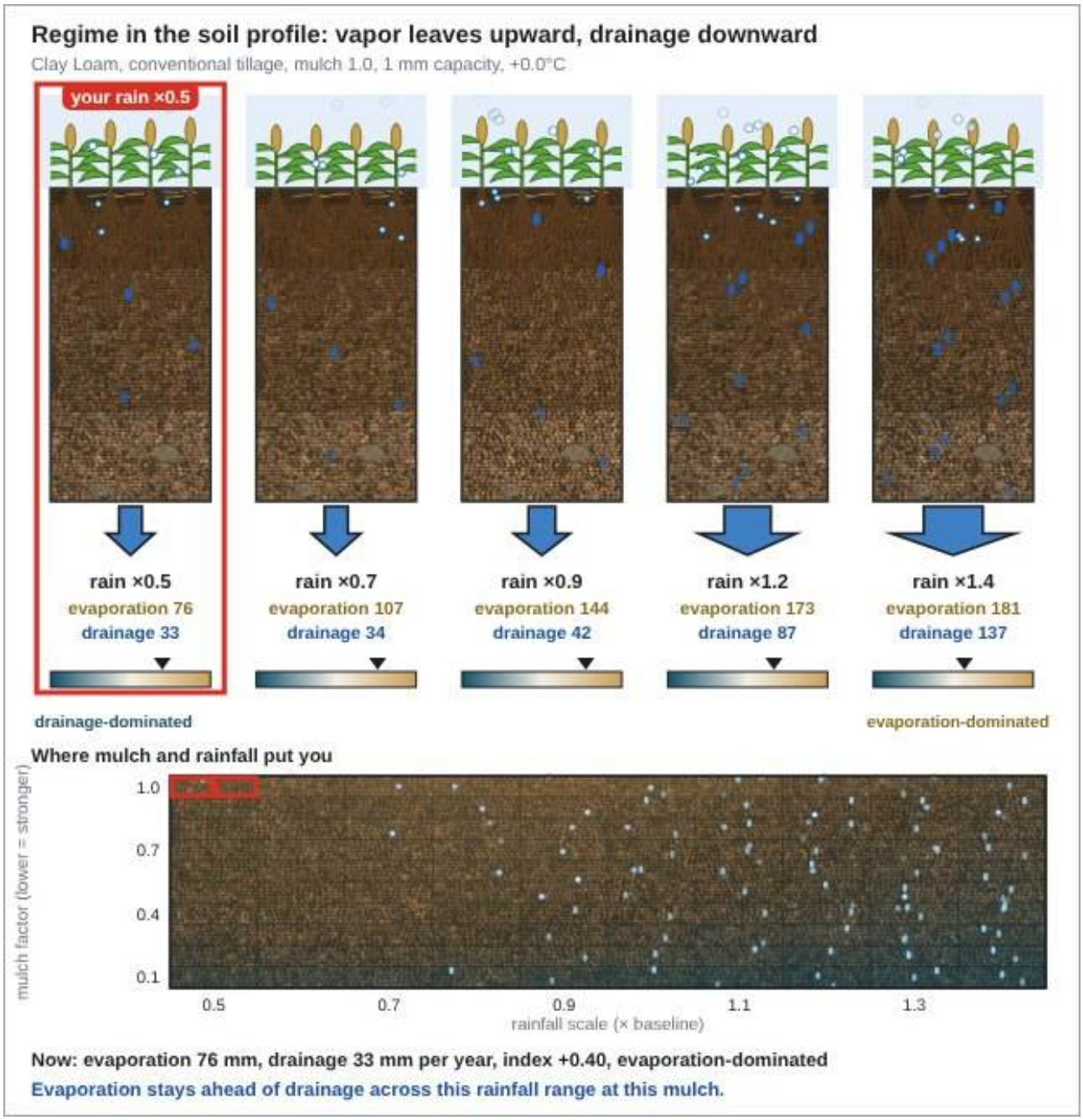


Figure 9.43: Regime map, Clay Loam, part 1 of 1.

9.14 Tillage and structure (Clay Loam)

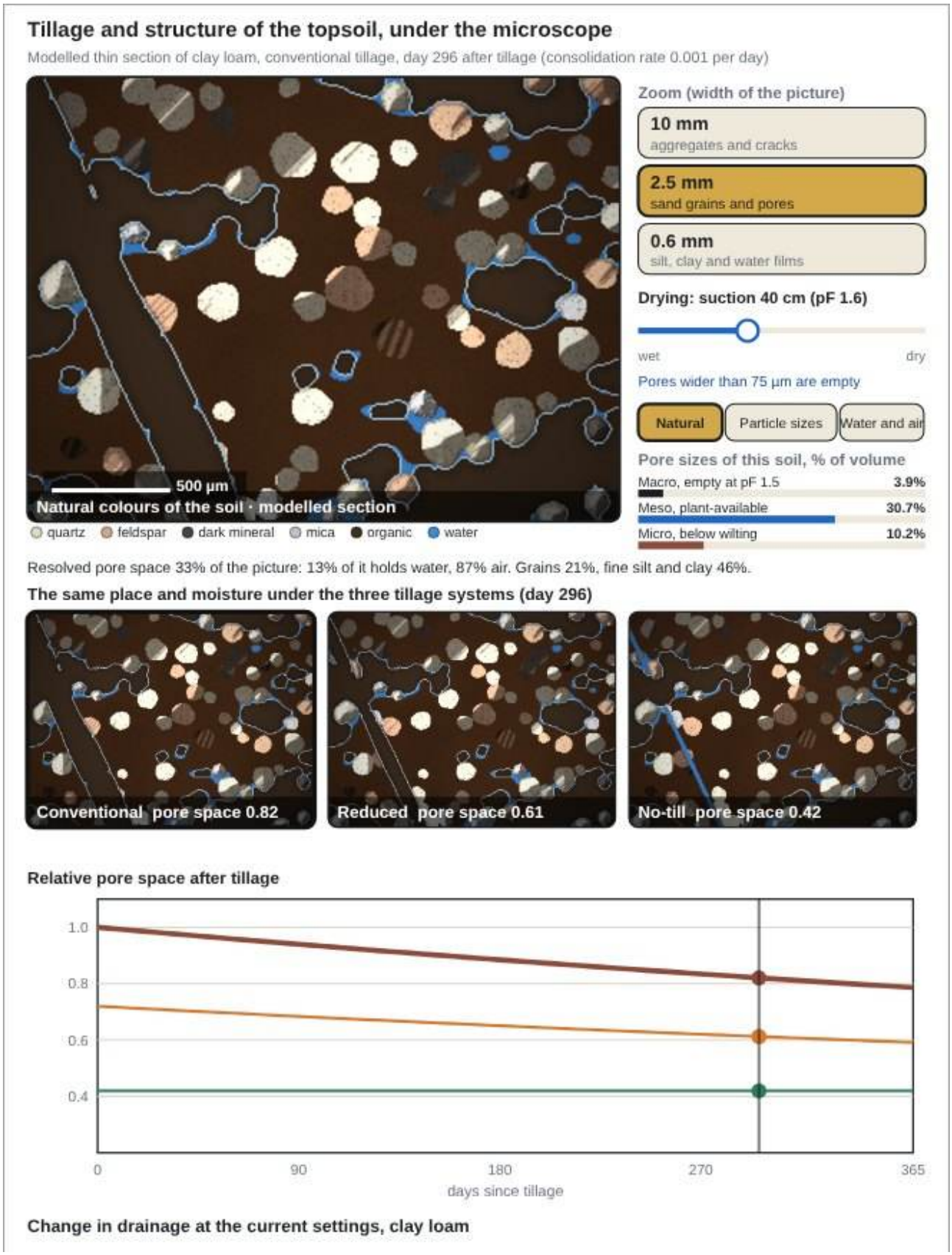


Figure 9.44: Tillage and structure, Clay Loam, part 1 of 2.

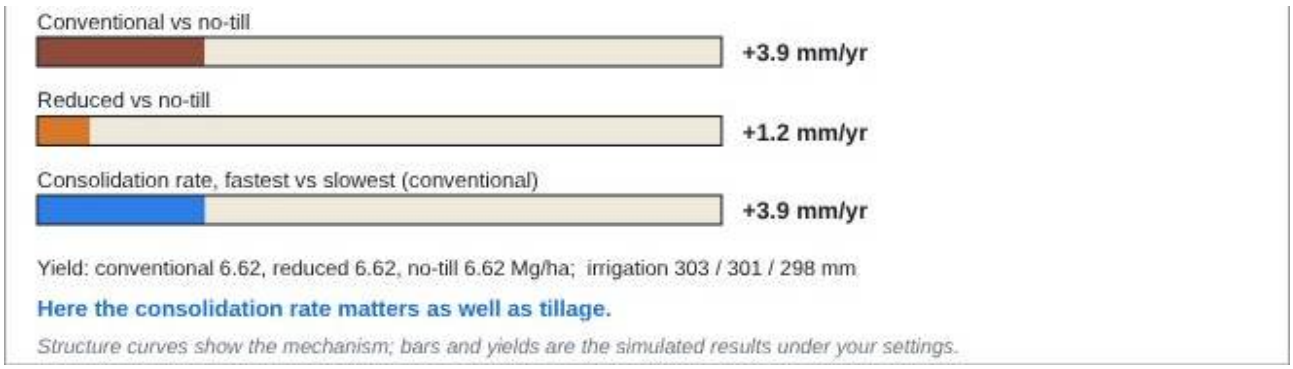


Figure 9.45: Tillage and structure, Clay Loam, part 2 of 2.

9.15 Soil hydraulic drivers (Clay Loam)

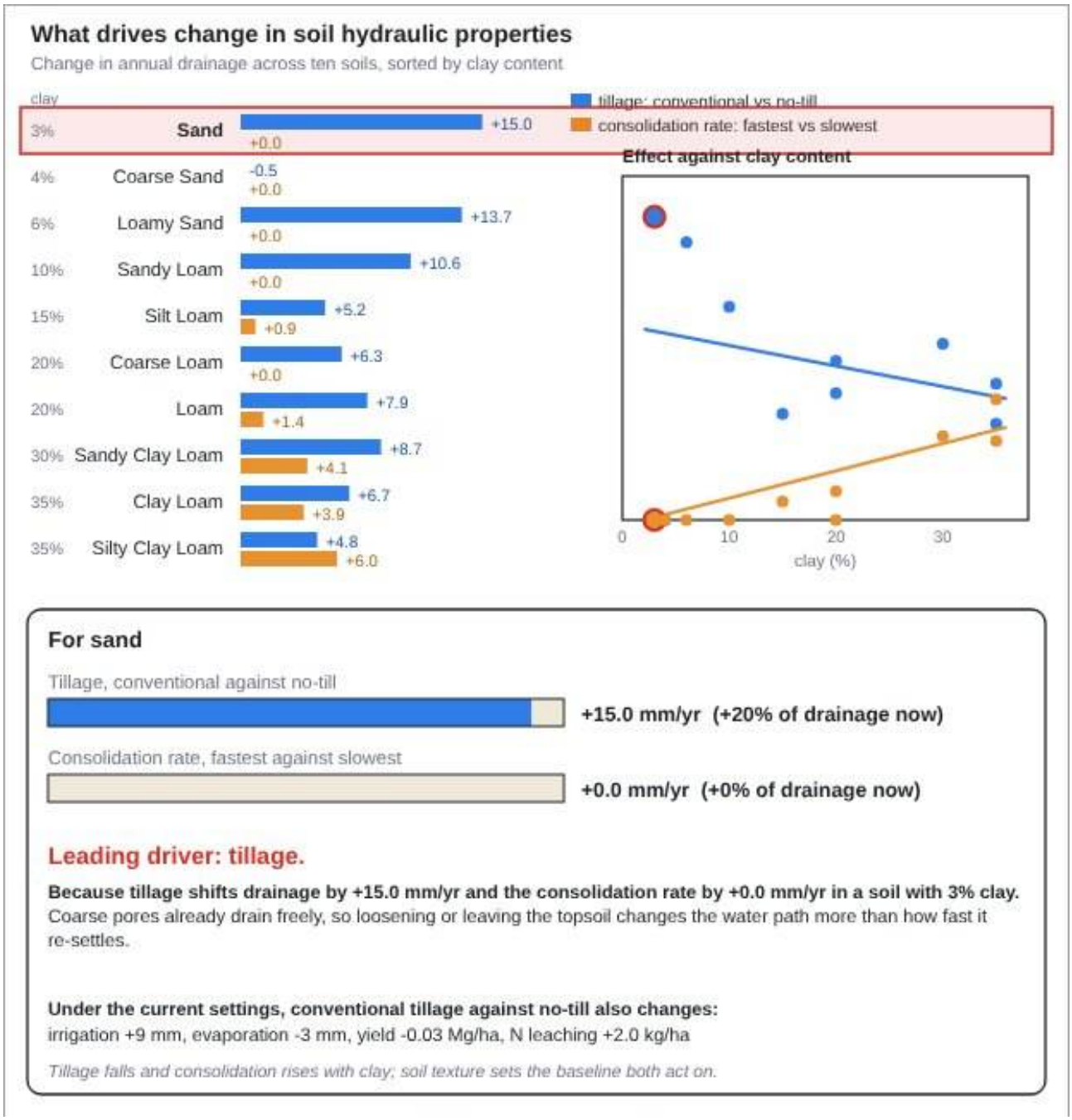


Figure 9.46: Soil hydraulic drivers, Clay Loam, part 1 of 3.

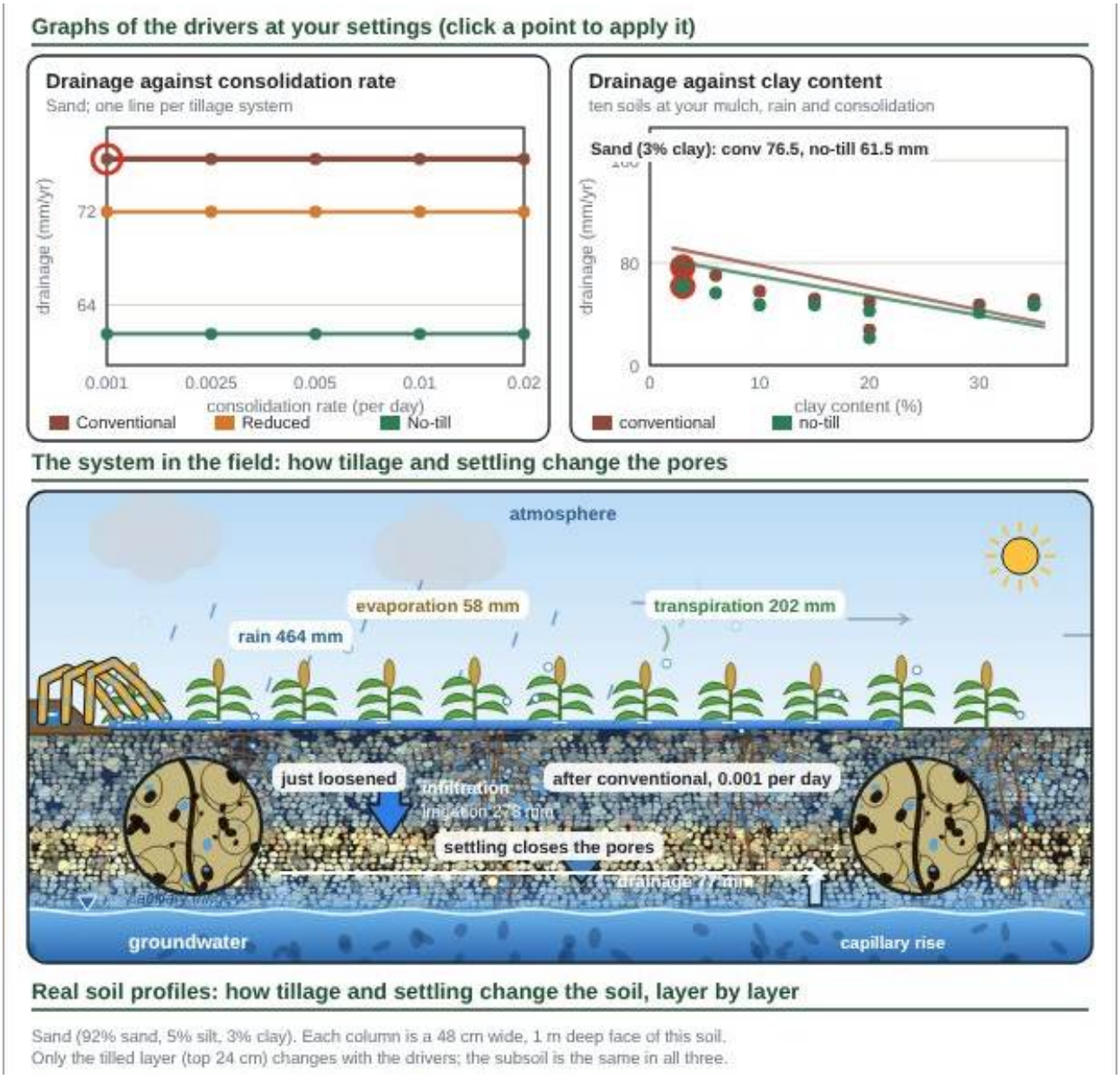


Figure 9.47: Soil hydraulic drivers, Clay Loam, part 2 of 3.

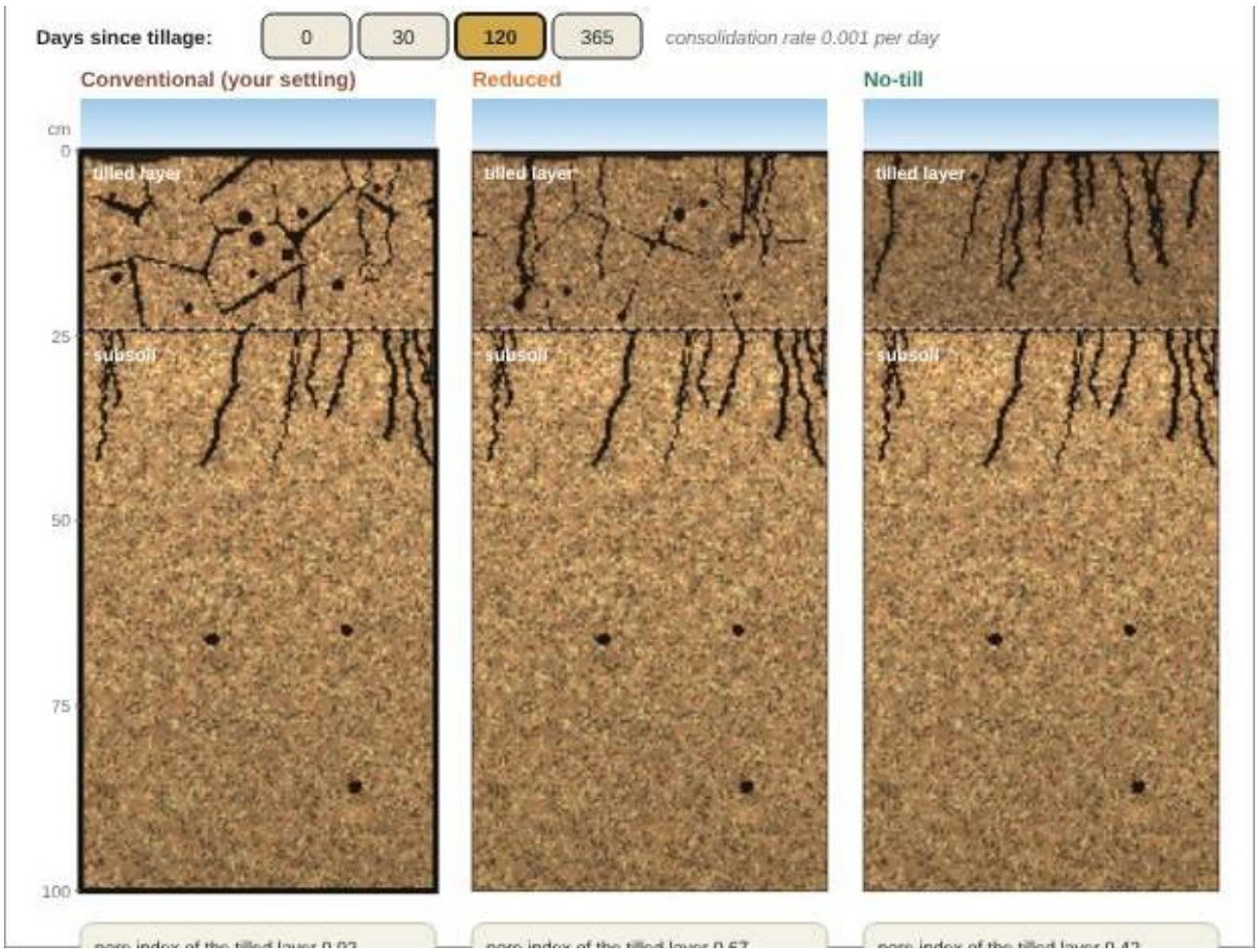


Figure 9.48: Soil hydraulic drivers, Clay Loam, part 3 of 3.

9.16 Root structure (Clay Loam)

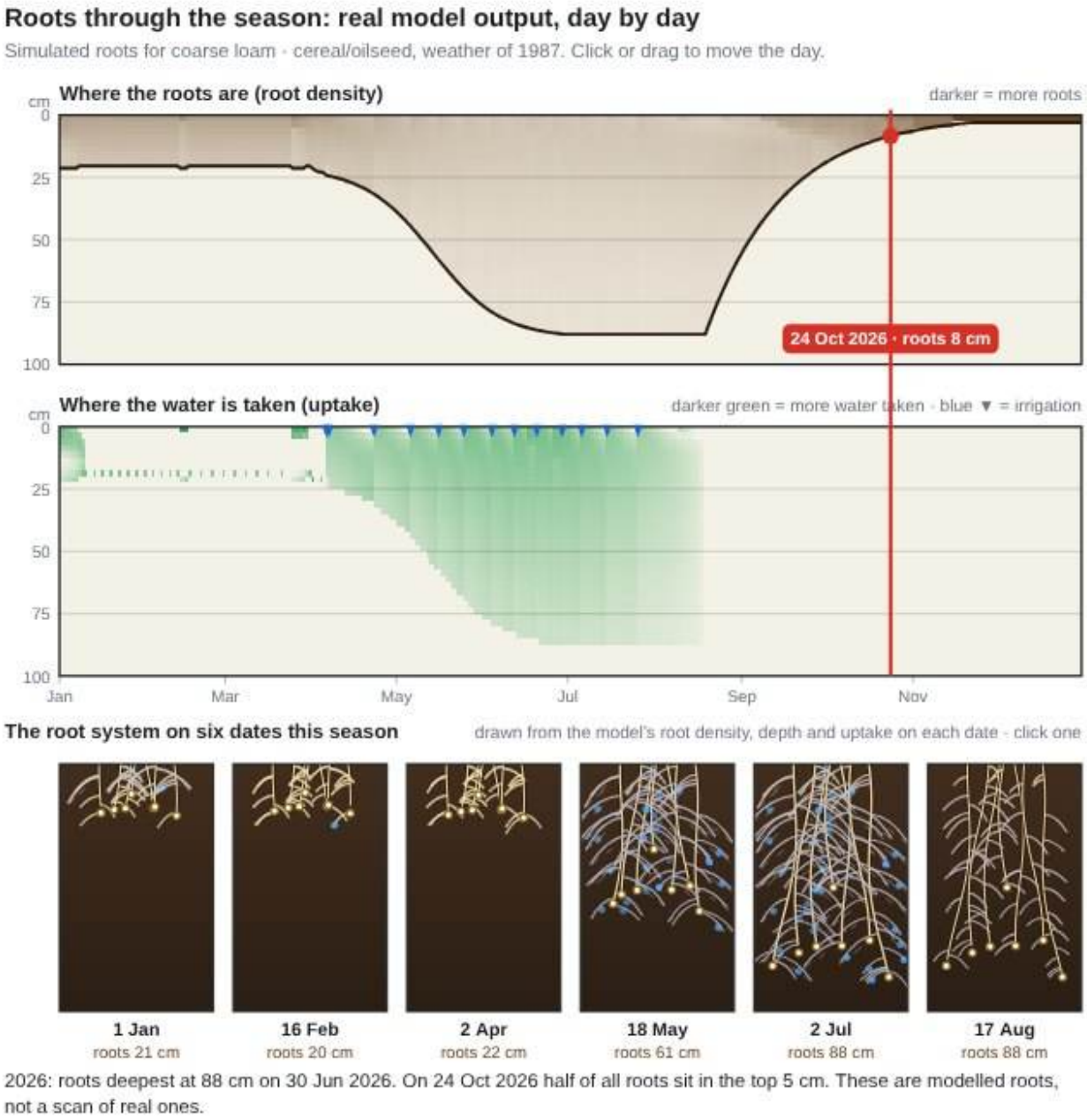


Figure 9.49: Root structure, Clay Loam, part 1 of 4.

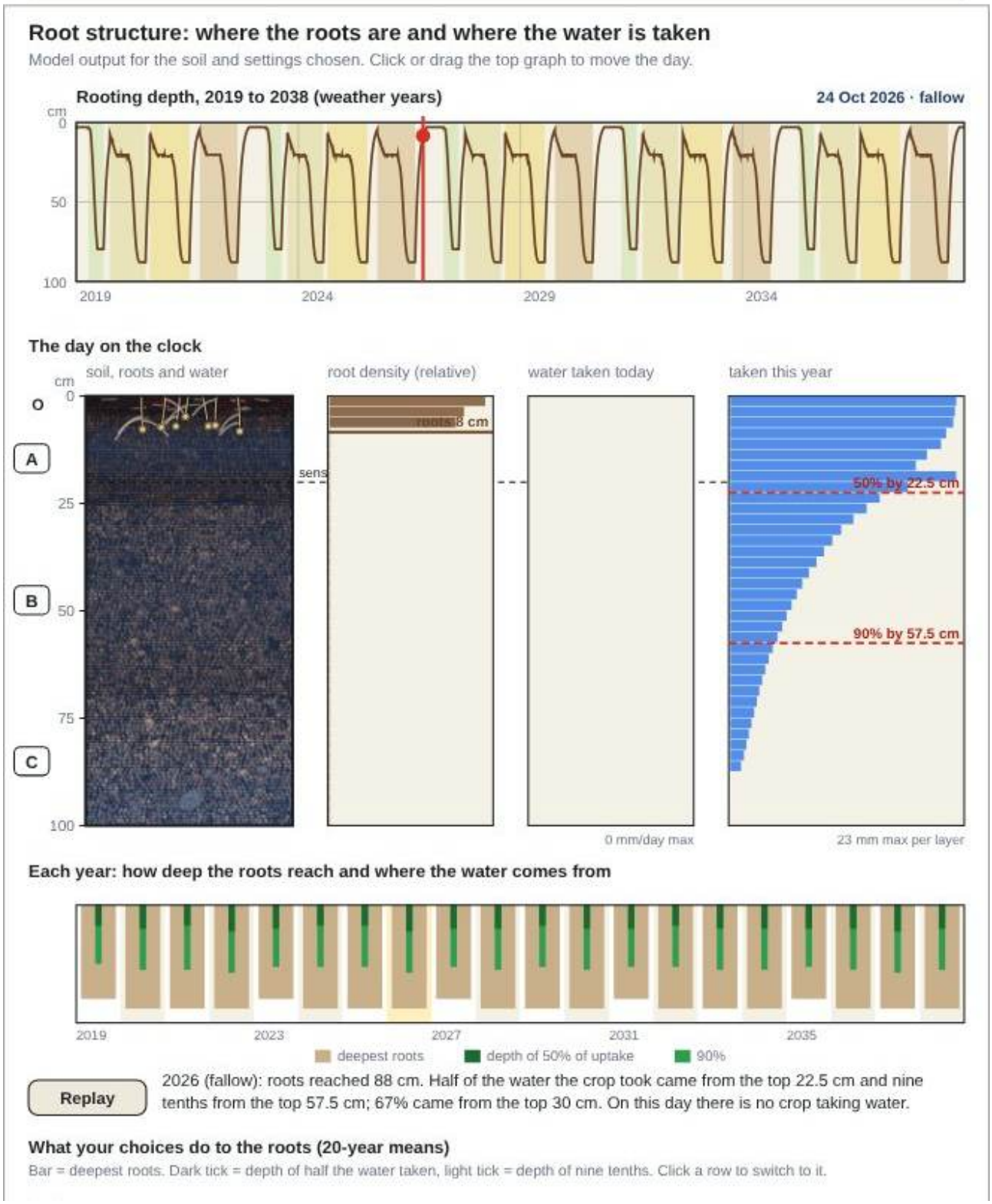


Figure 9.50: Root structure, Clay Loam, part 2 of 4.

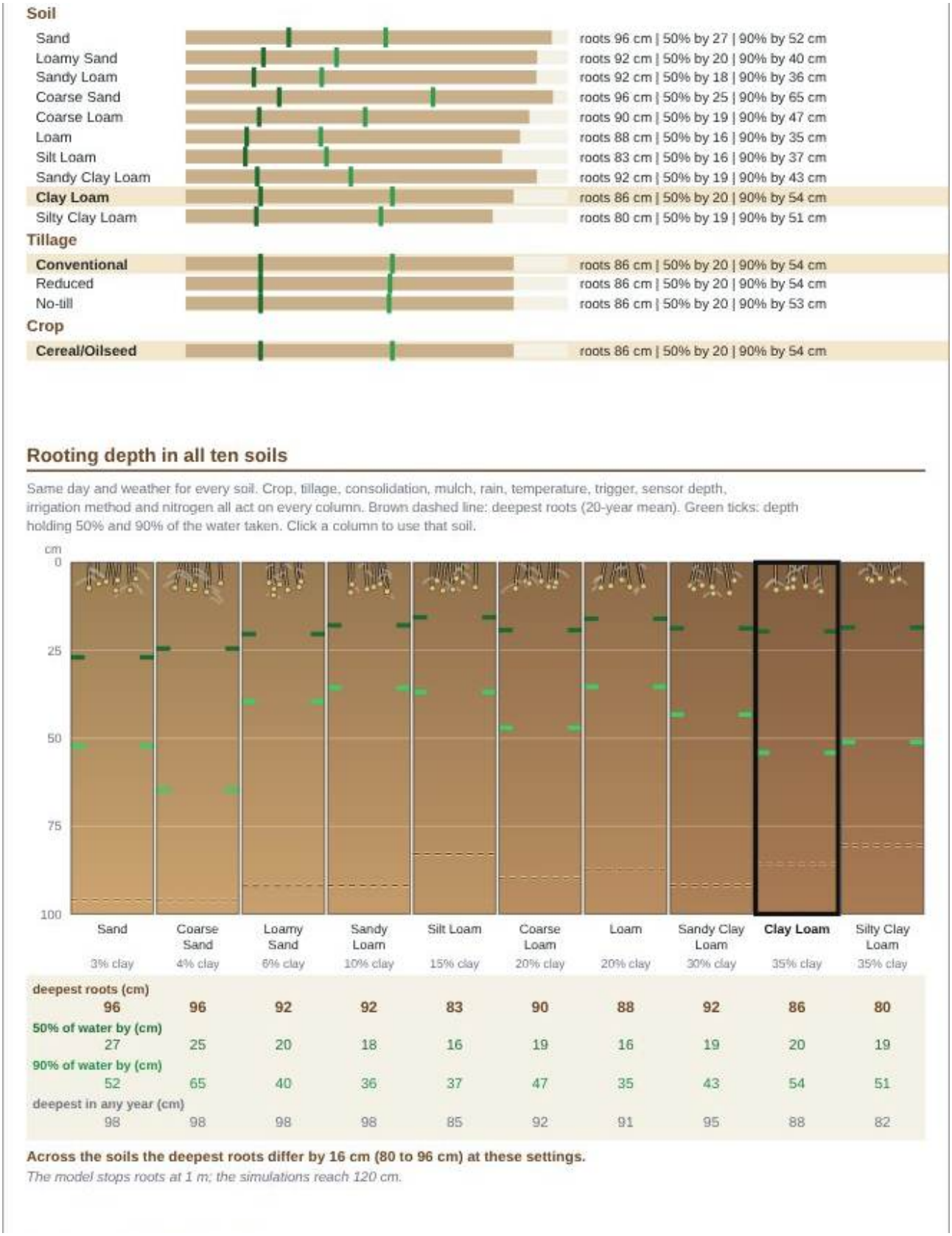


Figure 9.51: Root structure, Clay Loam, part 3 of 4.

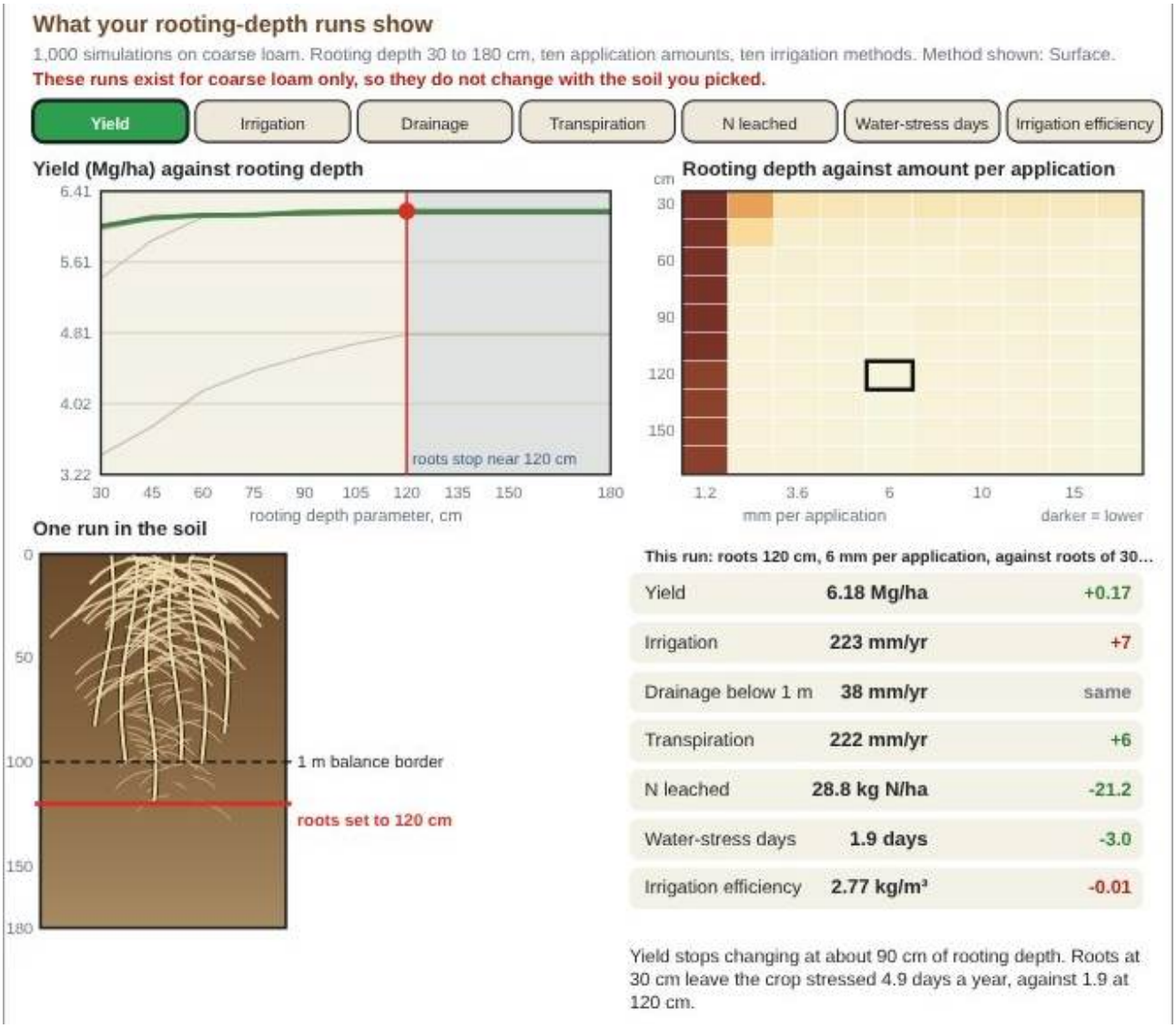


Figure 9.52: Root structure, Clay Loam, part 4 of 4.

9.17 Field balance (Clay Loam)

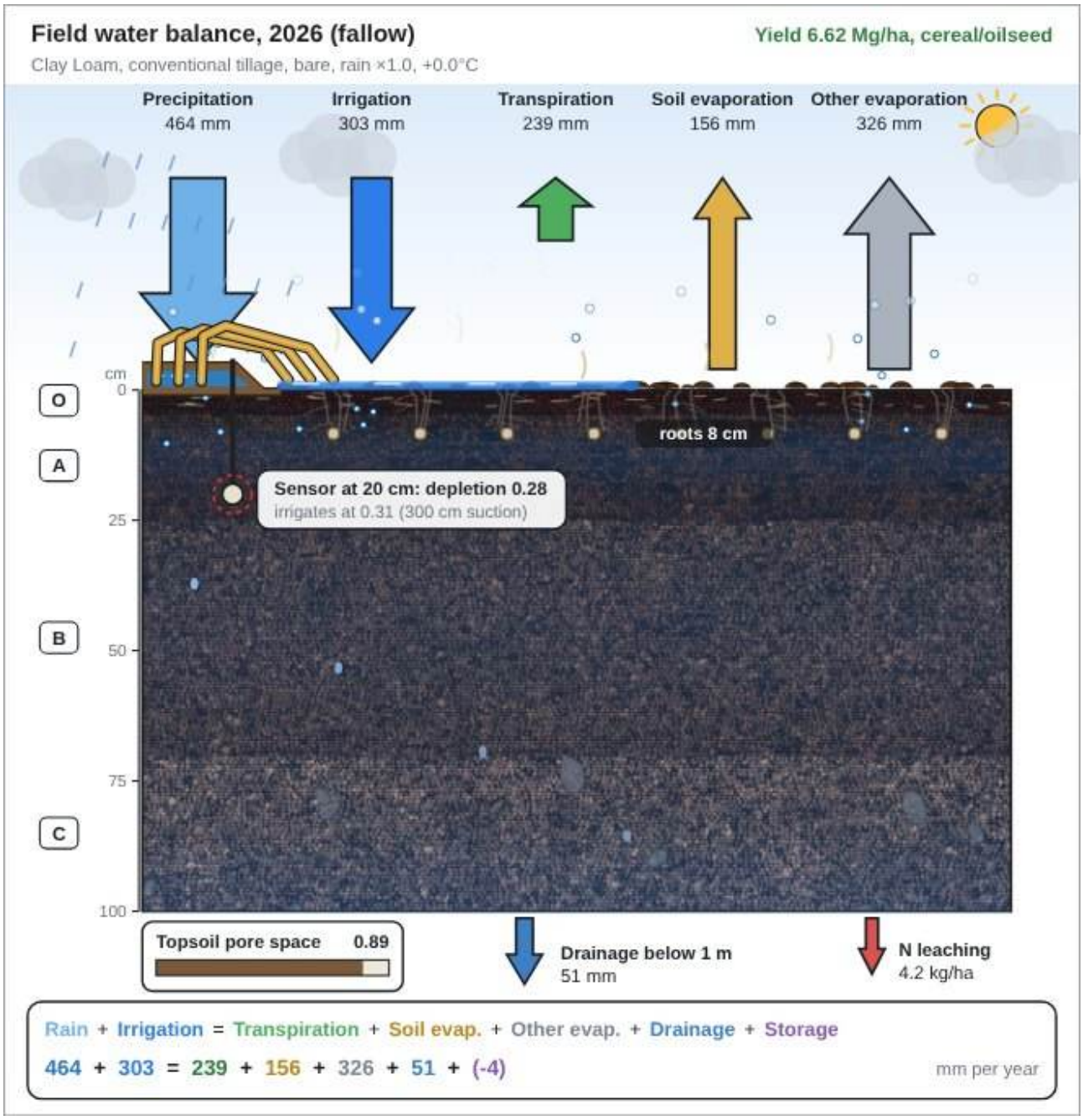


Figure 9.53: Field balance, Clay Loam, part 1 of 1.

9.18 Soil, plant and air (Clay Loam)

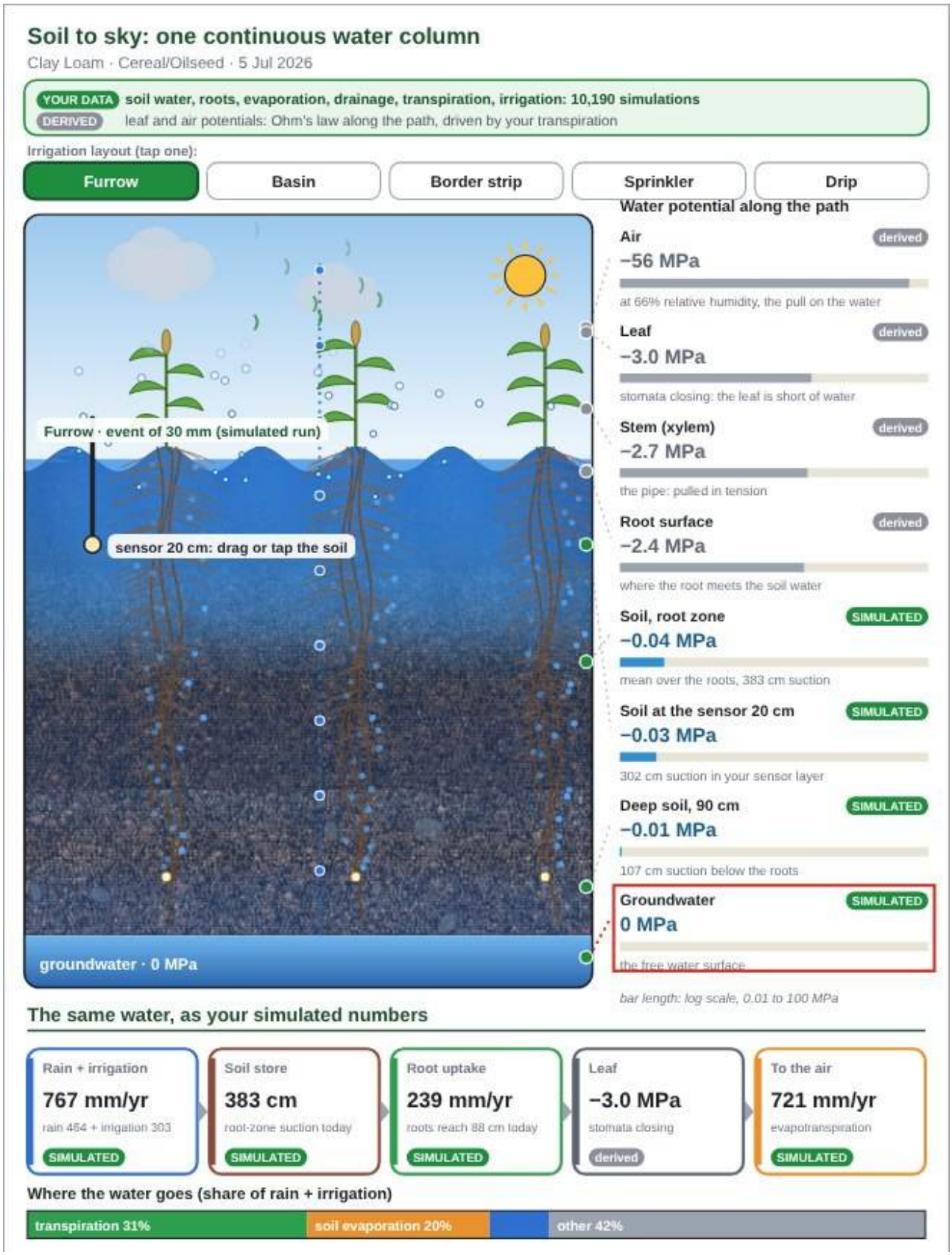


Figure 9.54: Soil, plant and air, Clay Loam, part 1 of 3.

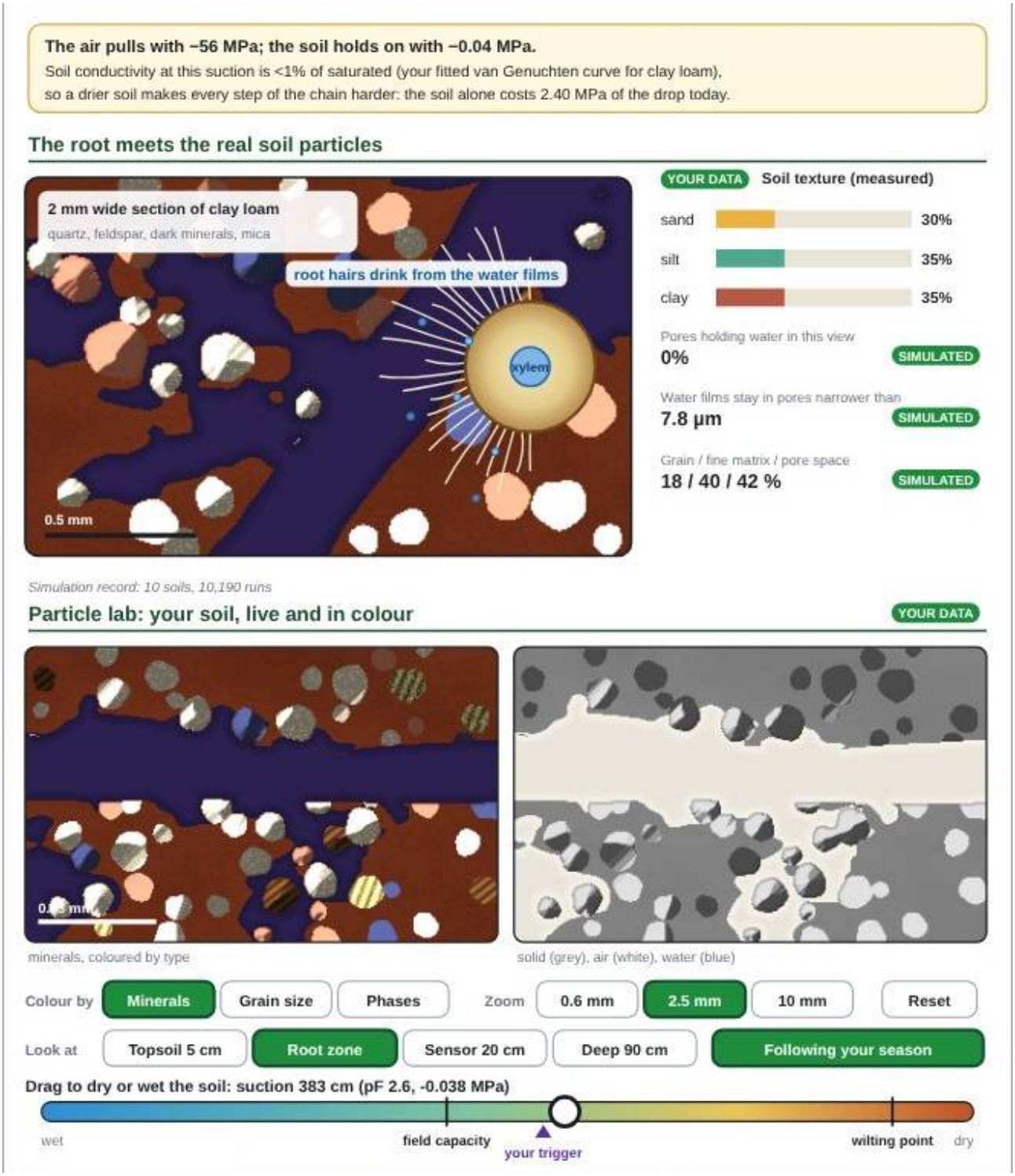


Figure 9.55: Soil, plant and air, Clay Loam, part 2 of 3.

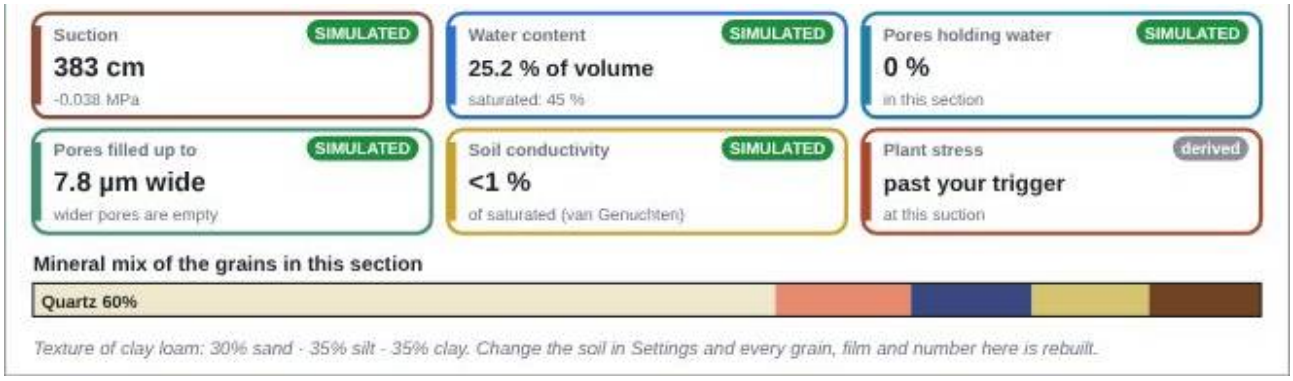


Figure 9.56: Soil, plant and air, Clay Loam, part 3 of 3.

9.19 Profile evolution (Clay Loam)

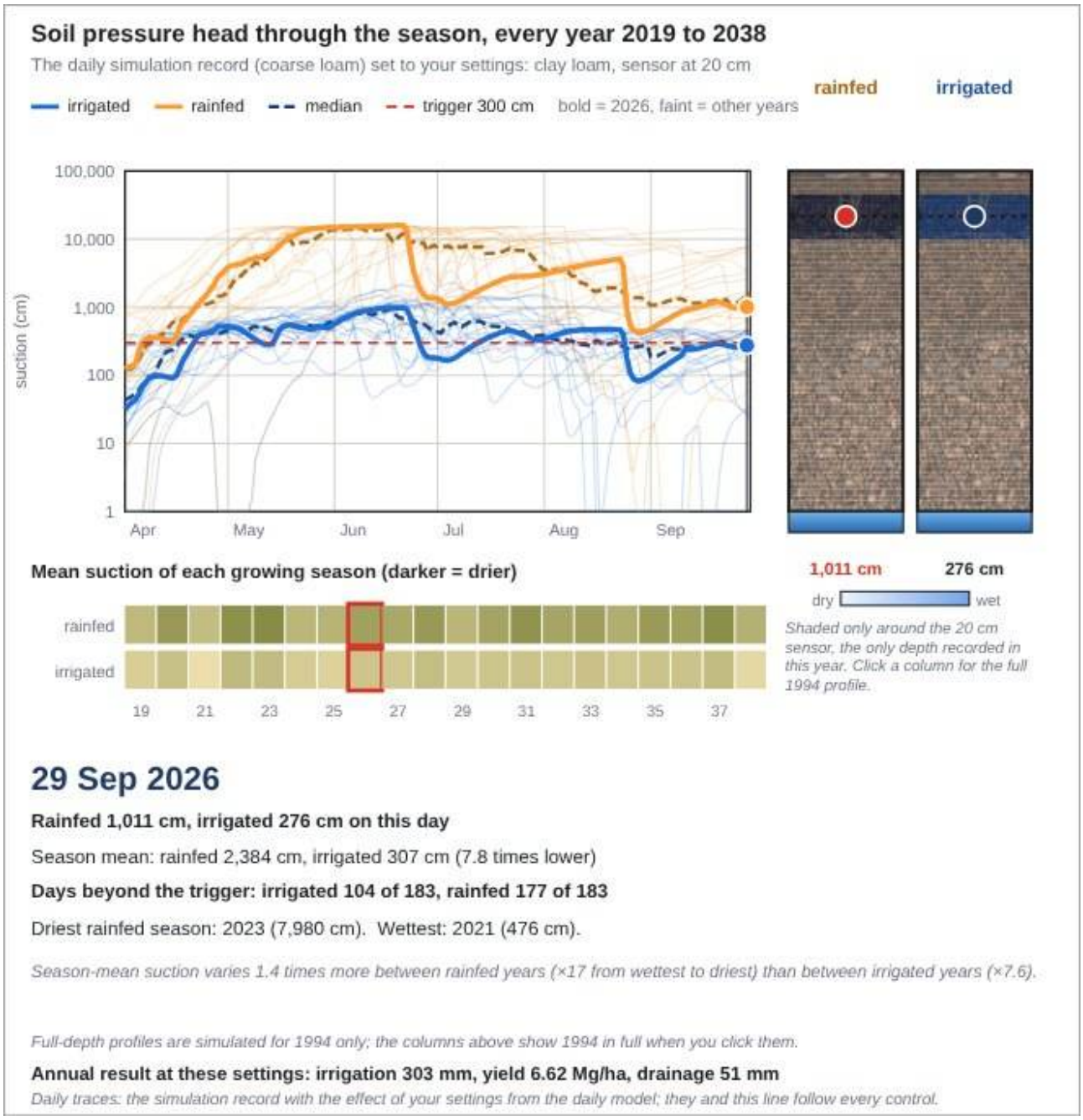


Figure 9.57: Profile evolution, Clay Loam, part 1 of 1.

9.20 Reliability over 20 seasons (Clay Loam)

Figure 9.58: Reliability over 20 seasons, Clay Loam, part 1 of 2.

Reliability of this setting over the 20 seasons — from the yearly values above

Irrigation needed in a dry year (9 years in 10 need less)	**381 mm**	against 303 mm in an average year
Years that need 20 % more irrigation than average	**3 of 20**	plan pump, labour and water rights for these
Years with more than 50 mm drainage below the roots	**5 of 20**	water and nitrate lost to groundwater
Years with a yield below 80 % of the average	**7 of 20**	yield risk at this setting
Nitrate leaching in the worst year	**25.7 kg N/ha**	mean 4.2 kg N/ha

How often a given irrigation is exceeded — the 20 seasons sorted from wettest to driest

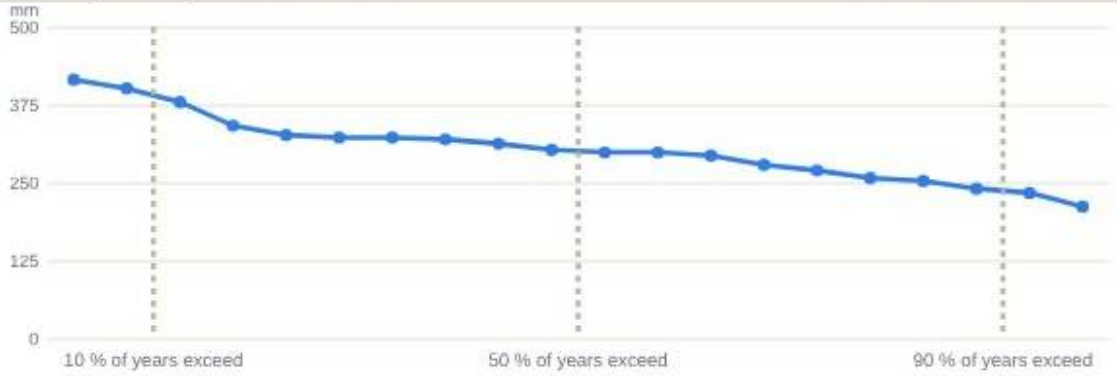


Figure 9.59: Reliability over 20 seasons, Clay Loam, part 2 of 2.

9.21 Full water balance (Clay Loam)

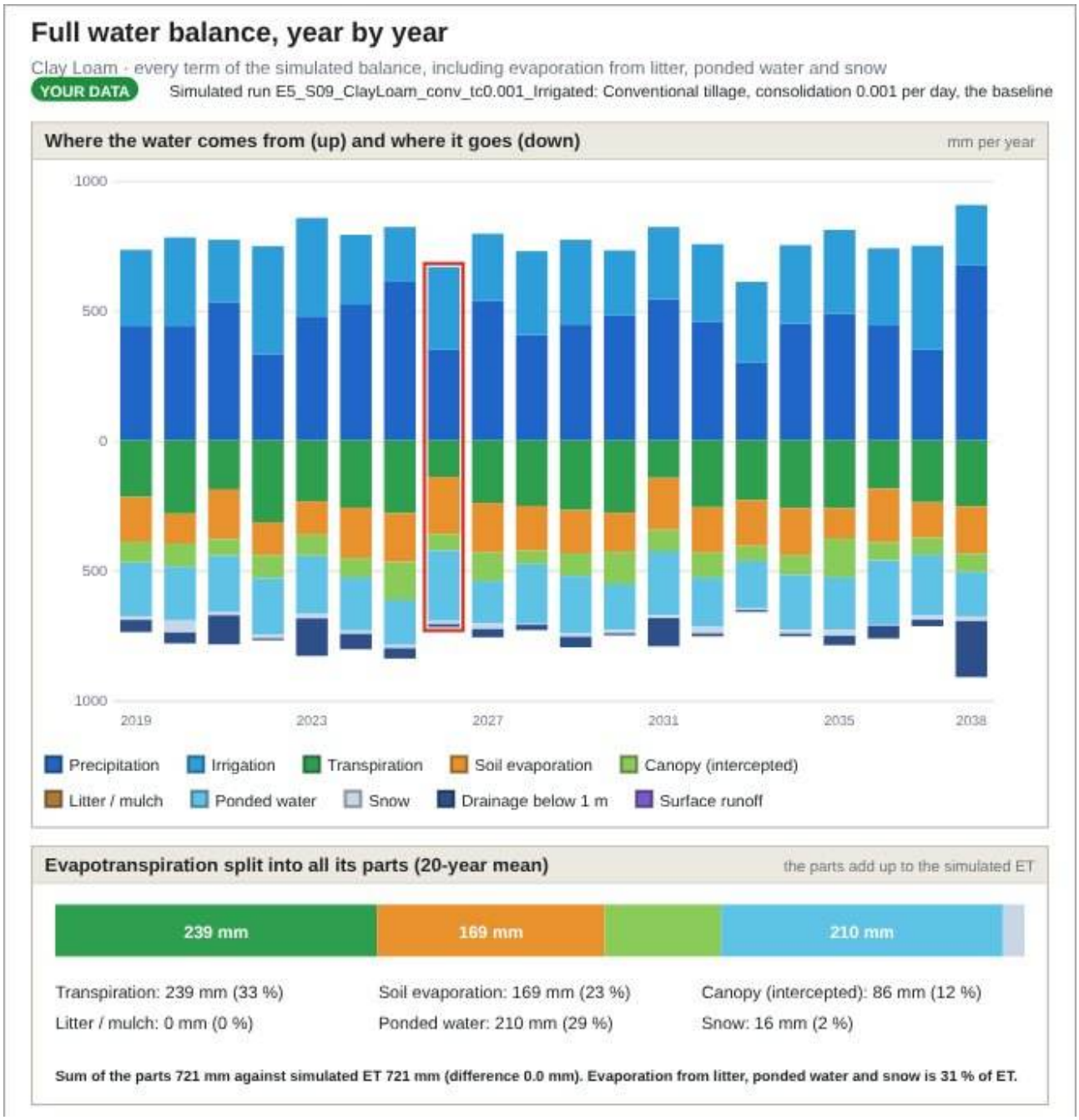


Figure 9.60: Full water balance, Clay Loam, part 1 of 2.

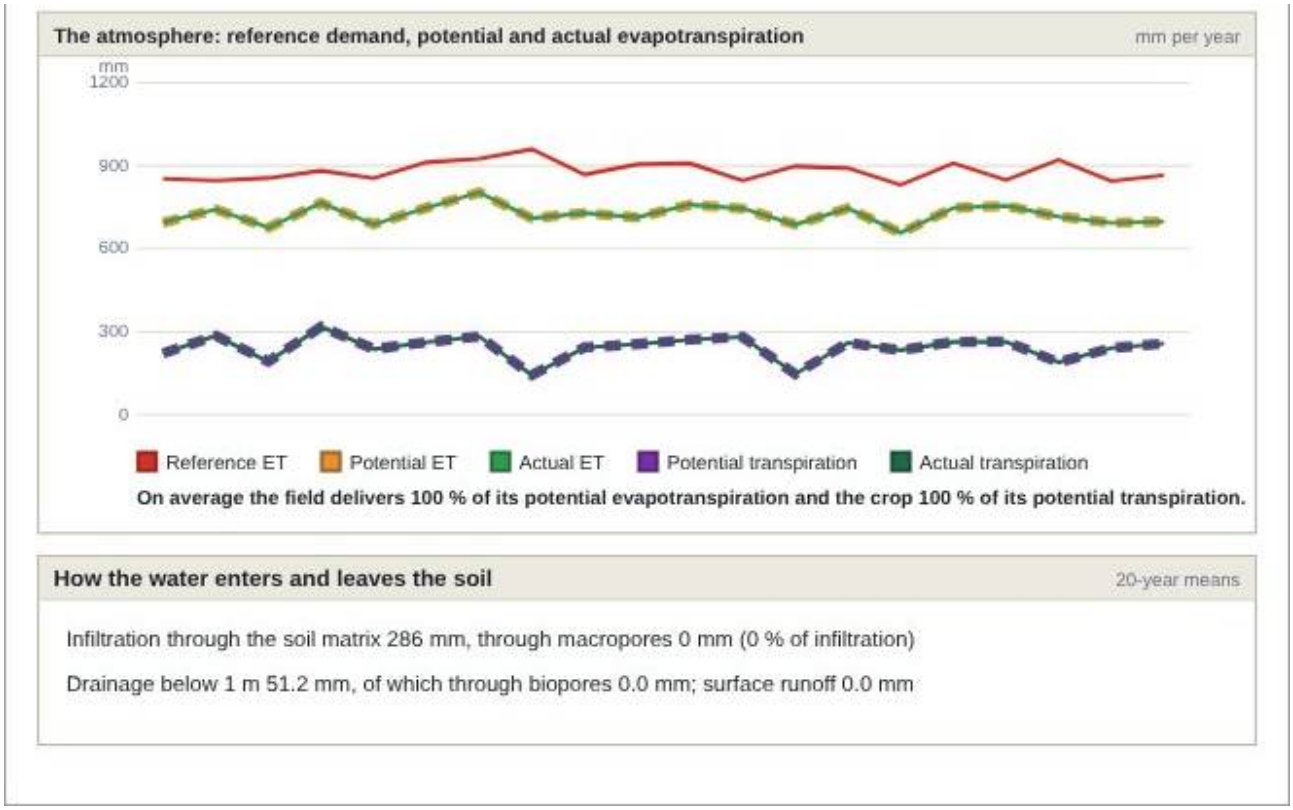


Figure 9.61: Full water balance, Clay Loam, part 2 of 2.

9.22 Nitrogen budget (Clay Loam)

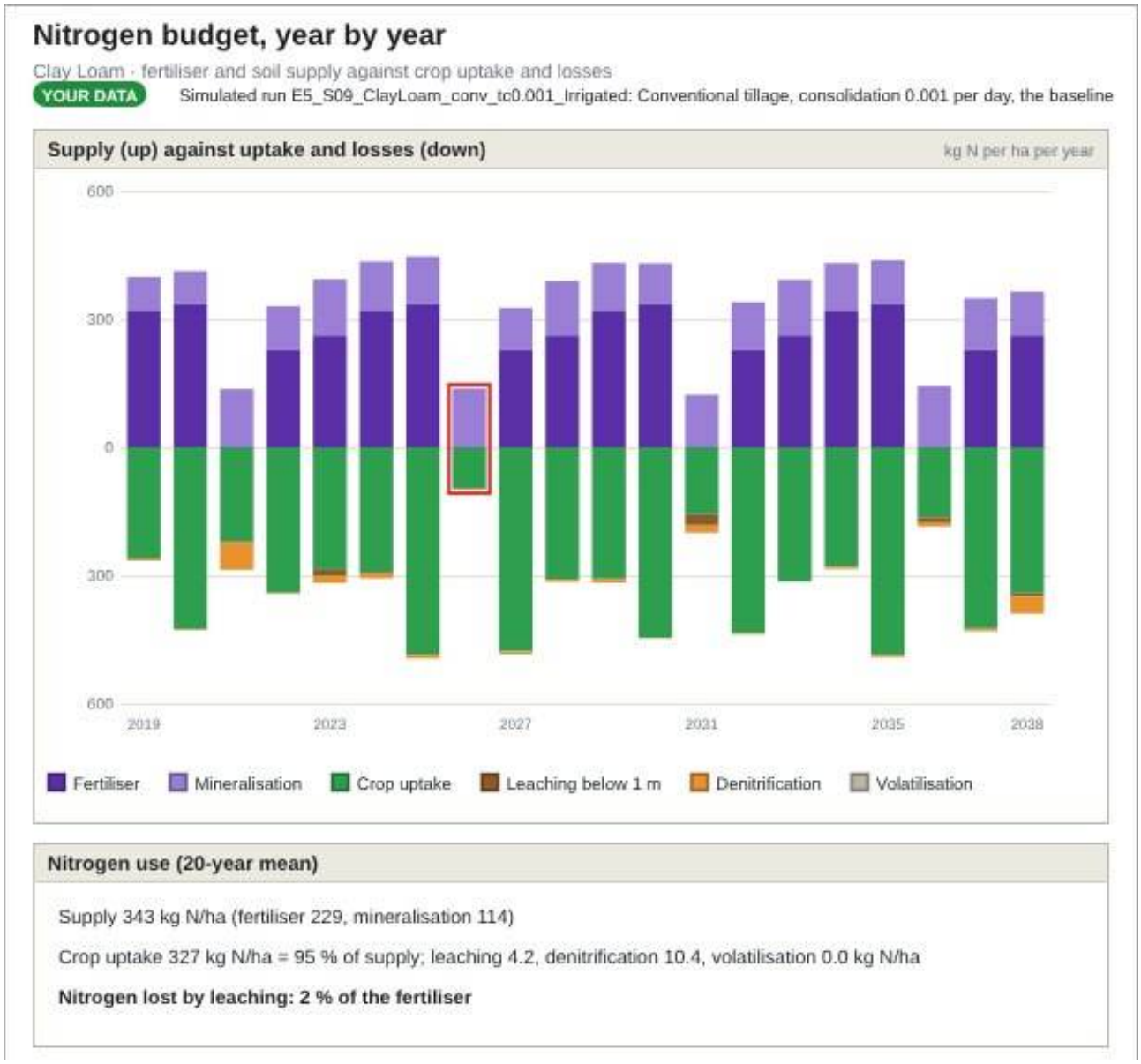


Figure 9.62: Nitrogen budget, Clay Loam, part 1 of 2.

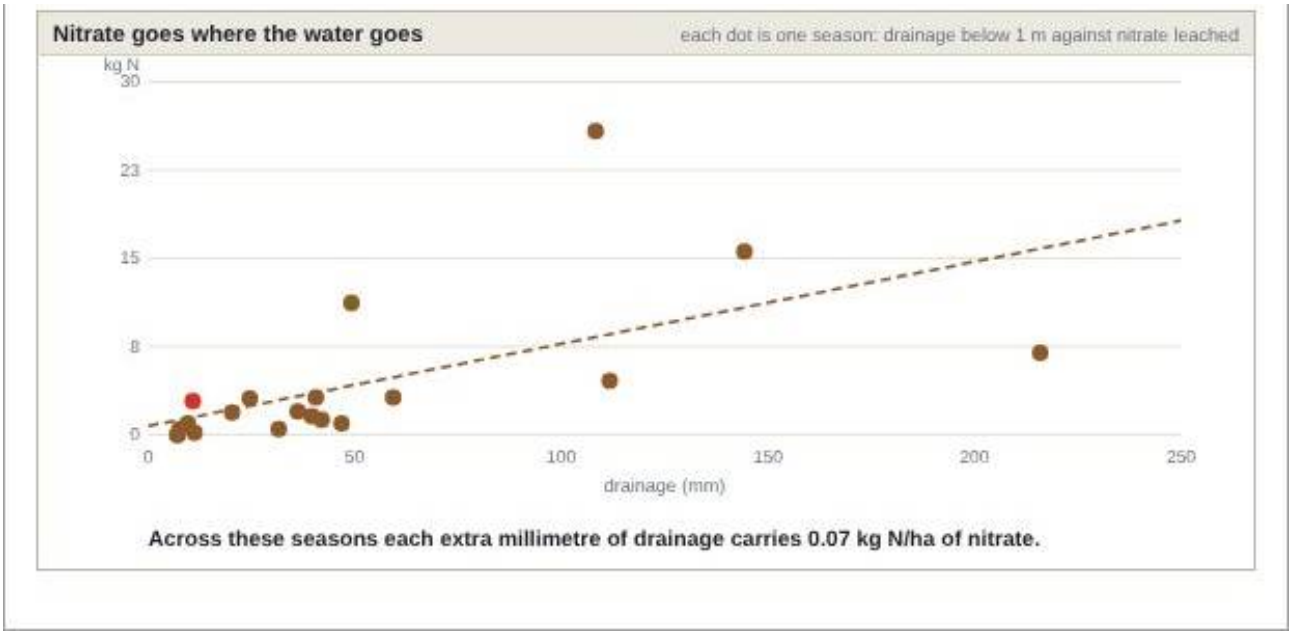


Figure 9.63: Nitrogen budget, Clay Loam, part 2 of 2.

9.23 Crop water requirement (Clay Loam)

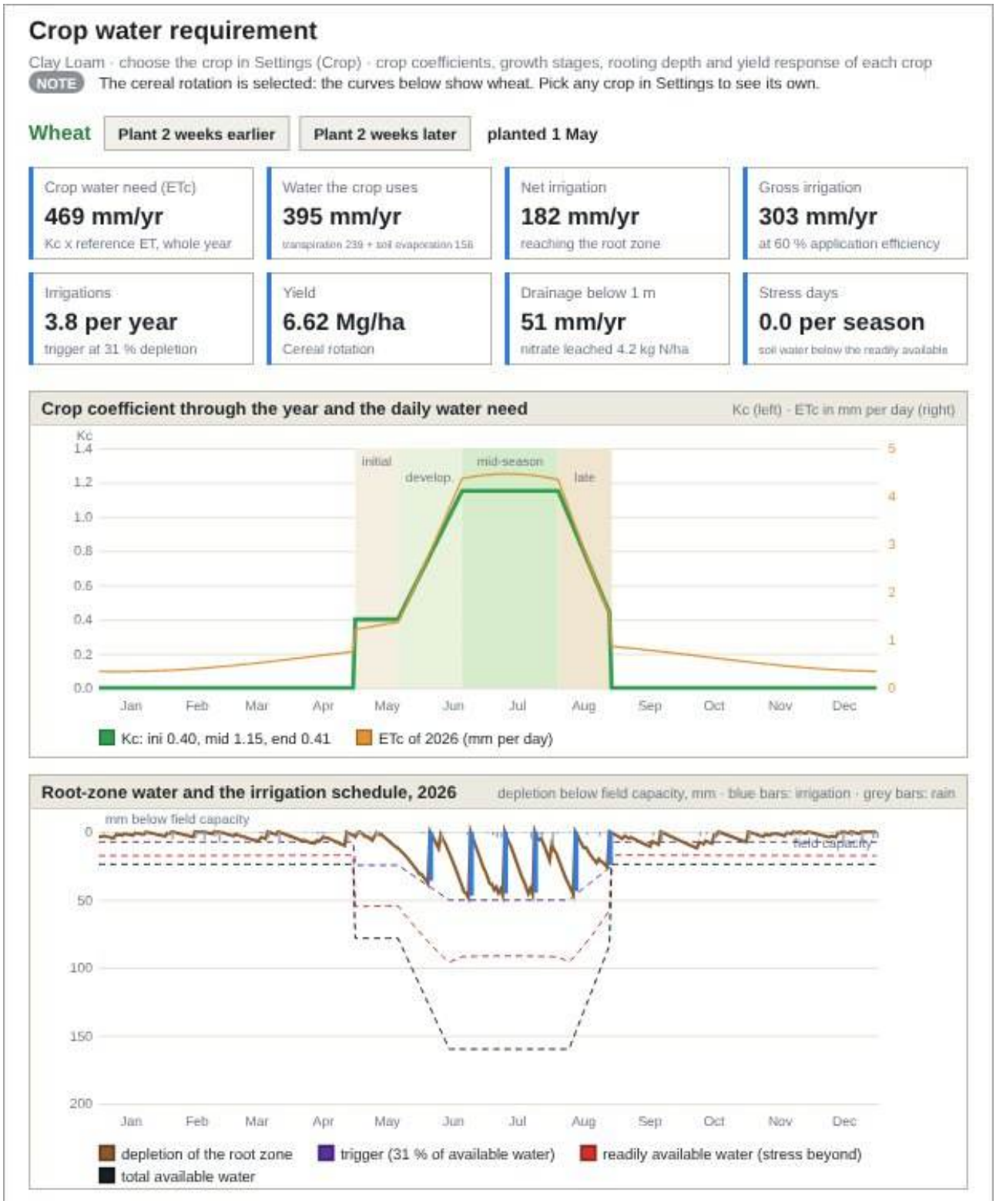


Figure 9.64: Crop water requirement, Clay Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**535**	12.1	100 %
Papaya (high)	814	**452**	9.7	100 %
Coconut (high)	834	**440**	8.5	100 %
Oil palm (young)	790	**409**	8.8	100 %
Yam	649	**357**	11.9	100 %
Onion (dry)	565	**326**	13.7	100 %
Rice	573	**319**	8.1	99 %
Avocado (medium)	677	**315**	8.1	100 %
Kiwi (pergola)	626	**315**	6.2	100 %
Cotton	571	**314**	6.5	100 %
Cabbage	559	**304**	9.4	100 %
Guava (high)	651	**283**	5.5	100 %
Table grapes (horizontal trellis)	597	**274**	5.3	100 %
Potato	508	**273**	10.7	100 %
Tomato	521	**263**	5.3	100 %
Mango (medium)	620	**255**	5.0	100 %
Hop (v trellis)	520	**246**	4.9	100 %
Walnut (medium)	523	**237**	4.7	100 %
Pear (medium)	528	**235**	4.6	100 %
Soybean	473	**235**	5.1	100 %
Groundnut	471	**234**	6.3	100 %
Wheat	469	**230**	4.8	100 %
Date palm (medium)	602	**230**	4.5	100 %
Mustard	468	**228**	4.7	100 %
Sweet pepper	475	**226**	5.9	100 %
Cherry (medium)	503	**213**	4.2	100 %
Pistachio (medium)	503	**213**	4.2	100 %

Figure 9.65: Crop water requirement, Clay Loam, part 2 of 3.

Maize (grain)	451		**210**	4.1	100 %
Cassava	466		**207**	6.3	100 %
Hazelnut (low)	481		**197**	3.9	100 %
Pecan (low)	481		**197**	3.9	100 %
Chickpea	436		**195**	5.0	100 %
Cherimoya (medium)	545		**182**	3.5	100 %
Sorghum (grain)	429		**178**	3.5	100 %
Apple (medium)	459		**169**	3.3	100 %
Citrus (medium density, short)	552		**165**	3.2	100 %
Cowpea (beans)	398		**161**	4.1	100 %
Millet	395		**154**	3.2	100 %
Wine grapes (pergola)	457		**151**	2.9	100 %
Peach (low)	435		**145**	2.8	100 %
Almond (medium)	435		**145**	2.8	100 %
Green gram (mung bean)	372		**137**	3.7	100 %
Apricot (young)	380		**89**	1.7	100 %
Olive (traditional, medium density)	446		**72**	1.4	100 %
Highbush blueberry (low)	338		**72**	1.6	100 %
Persimmon (young, low)	358		**70**	1.4	100 %
Pomegranate (low)	358		**70**	1.4	100 %
Fig tree (young, low)	327		**39**	0.8	100 %

Figure 9.66: Crop water requirement, Clay Loam, part 3 of 3.

9.24 Feedback loops (Clay Loam)

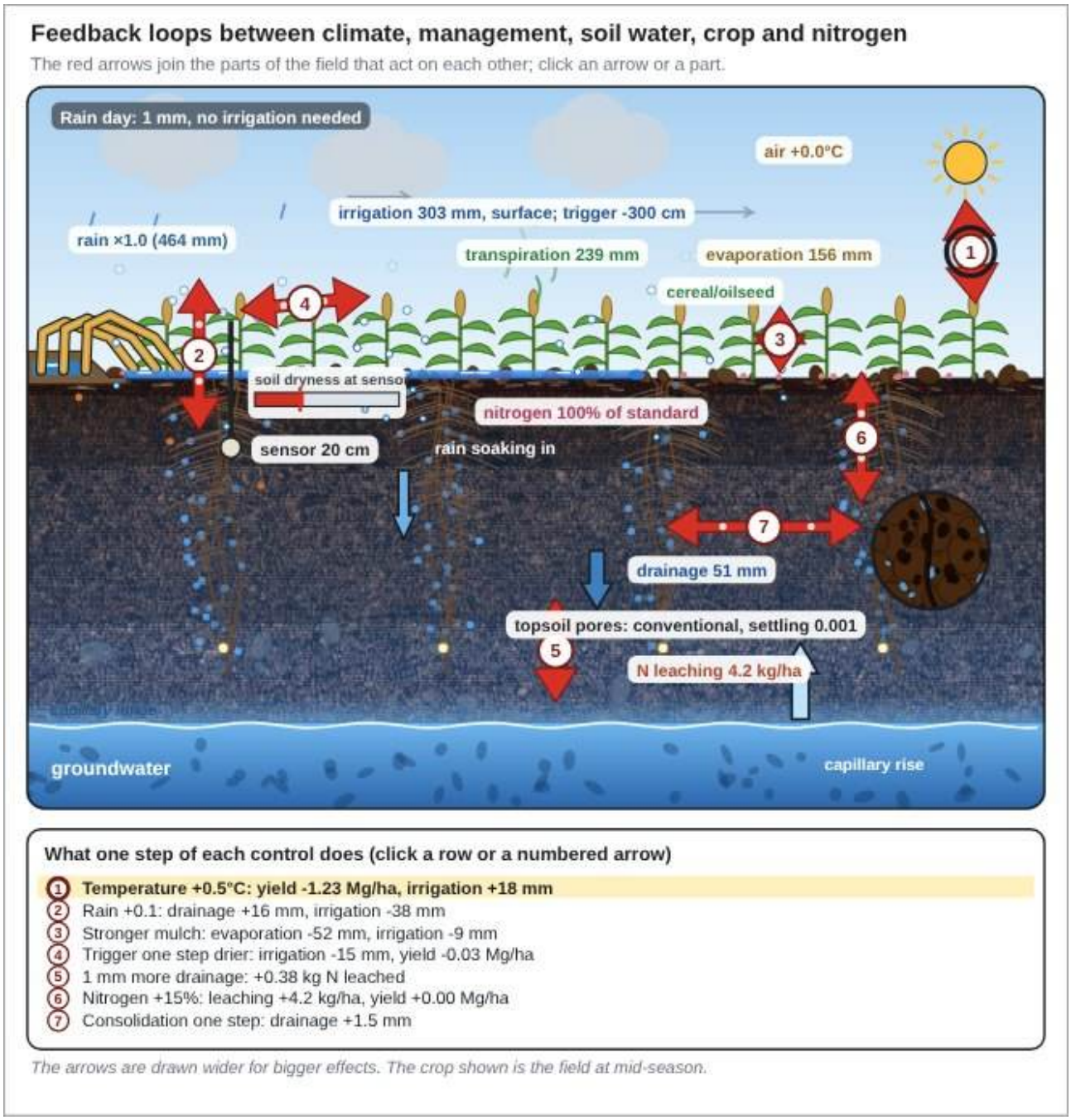


Figure 9.67: Feedback loops, Clay Loam, part 1 of 1.

Chapter 10

Silty Clay Loam

10.1 Surface irrigation design (Silty Clay Loam)

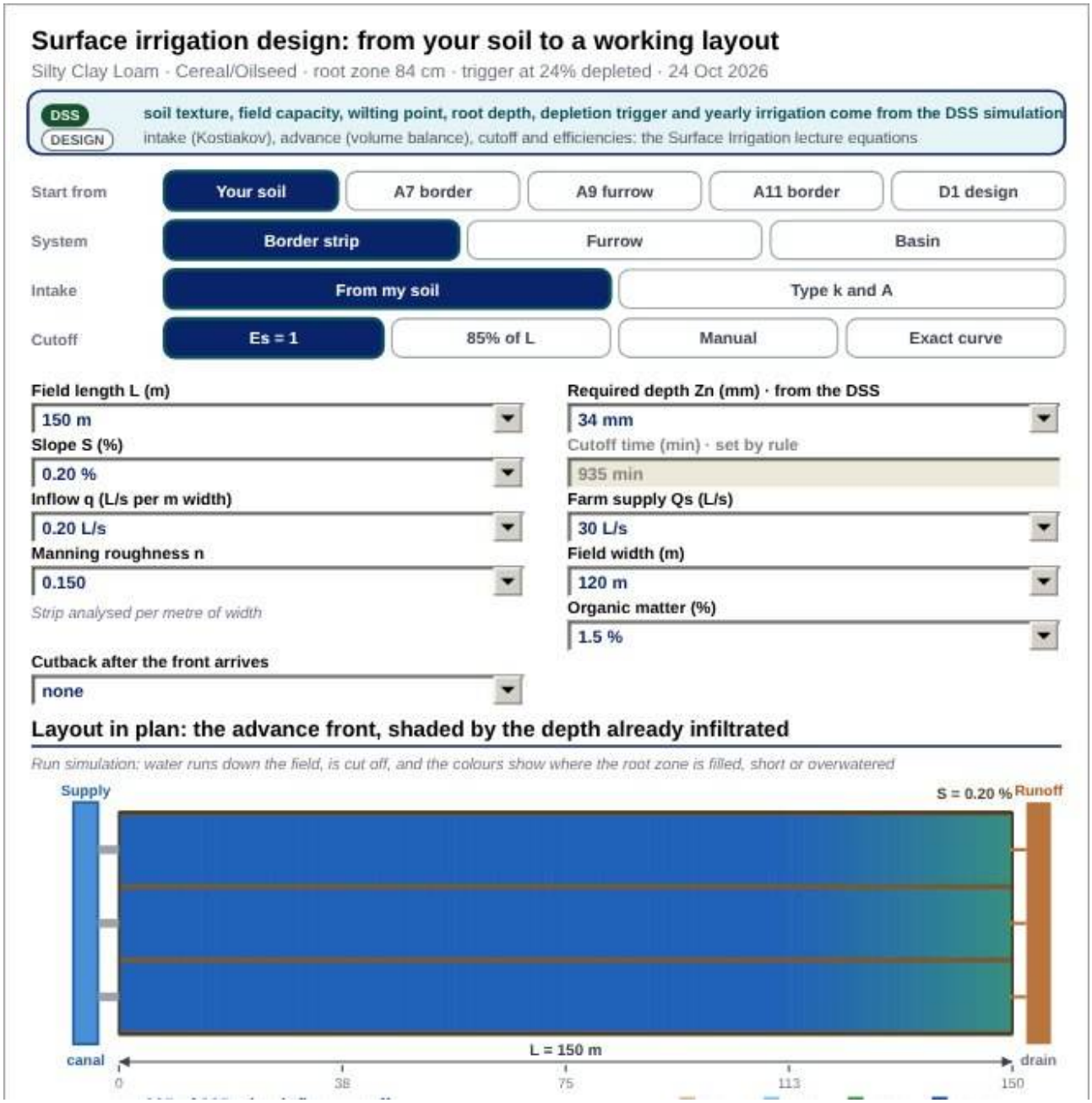


Figure 10.1: Surface irrigation design, Silty Clay Loam, part 1 of 5: Layout in plan: the advance front, shaded by the depth already infiltrated.

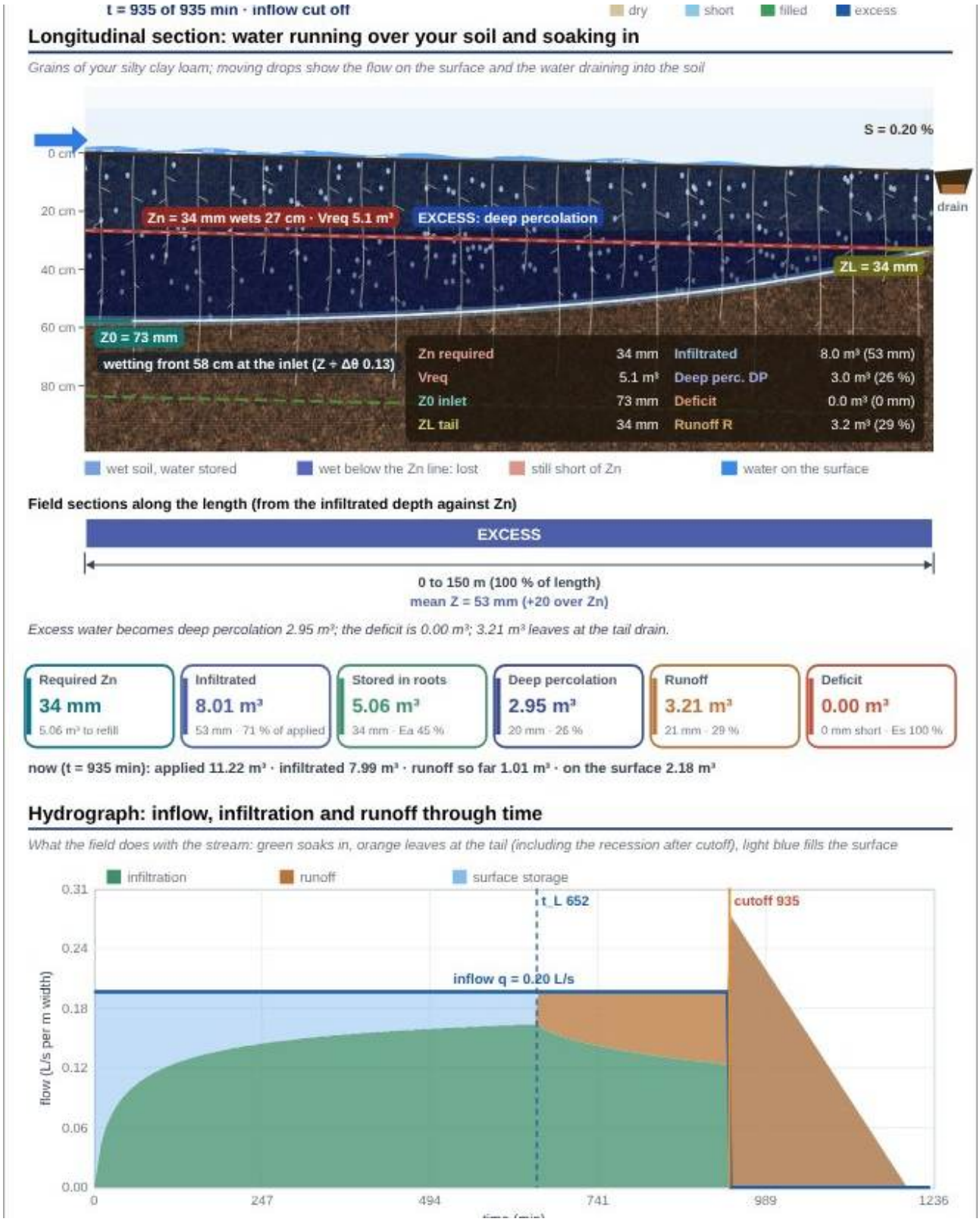


Figure 10.2: Surface irrigation design, Silty Clay Loam, part 2 of 5: Longitudinal section: water running over your soil and soaking in; Hydrograph: inflow, infiltration and runoff through time.

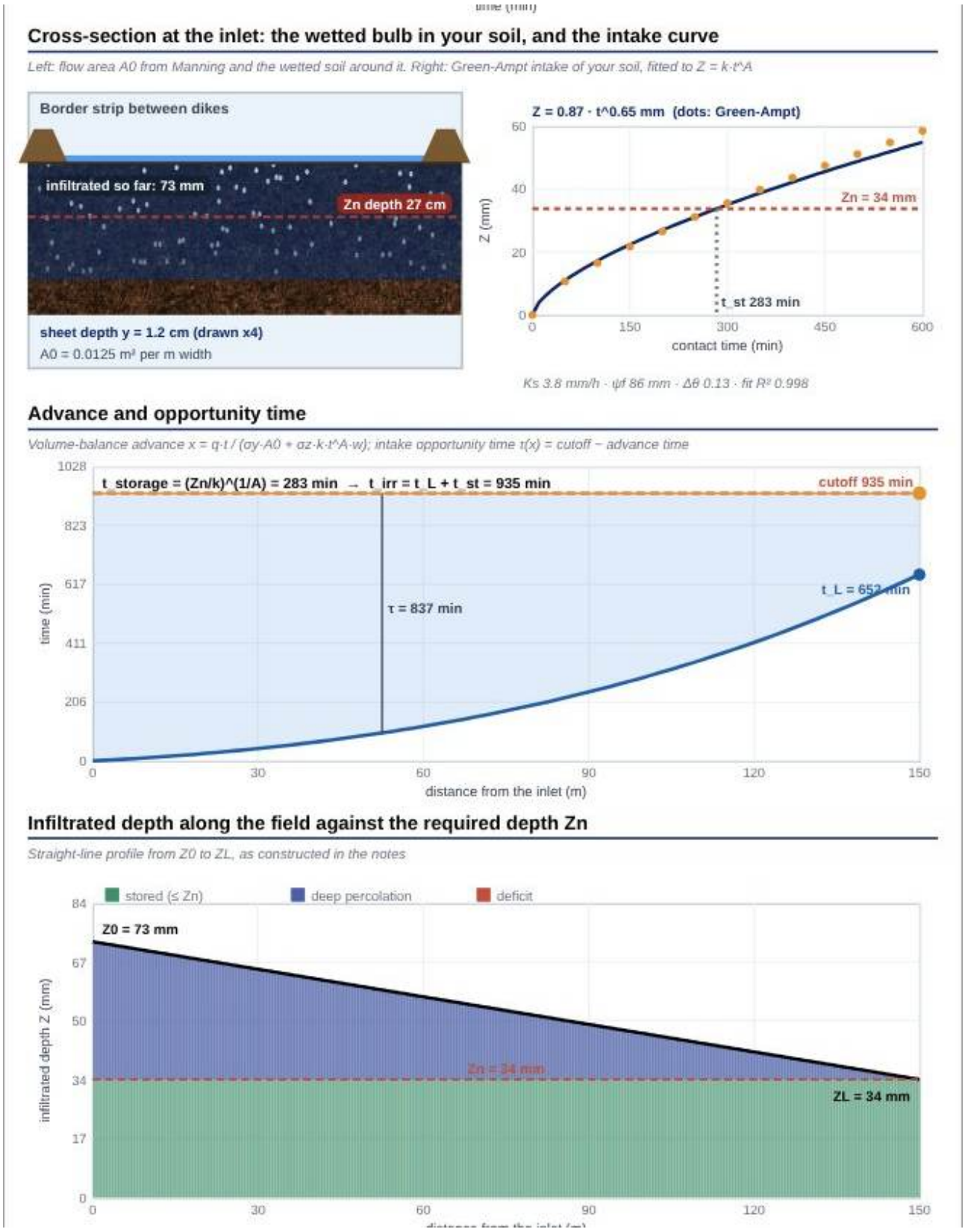


Figure 10.3: Surface irrigation design, Silty Clay Loam, part 3 of 5: Cross-section at the inlet: the wetted bulb in your soil, and the intake curve; Advance and opportunity time; Infiltrated depth along the field against the required depth Zn.

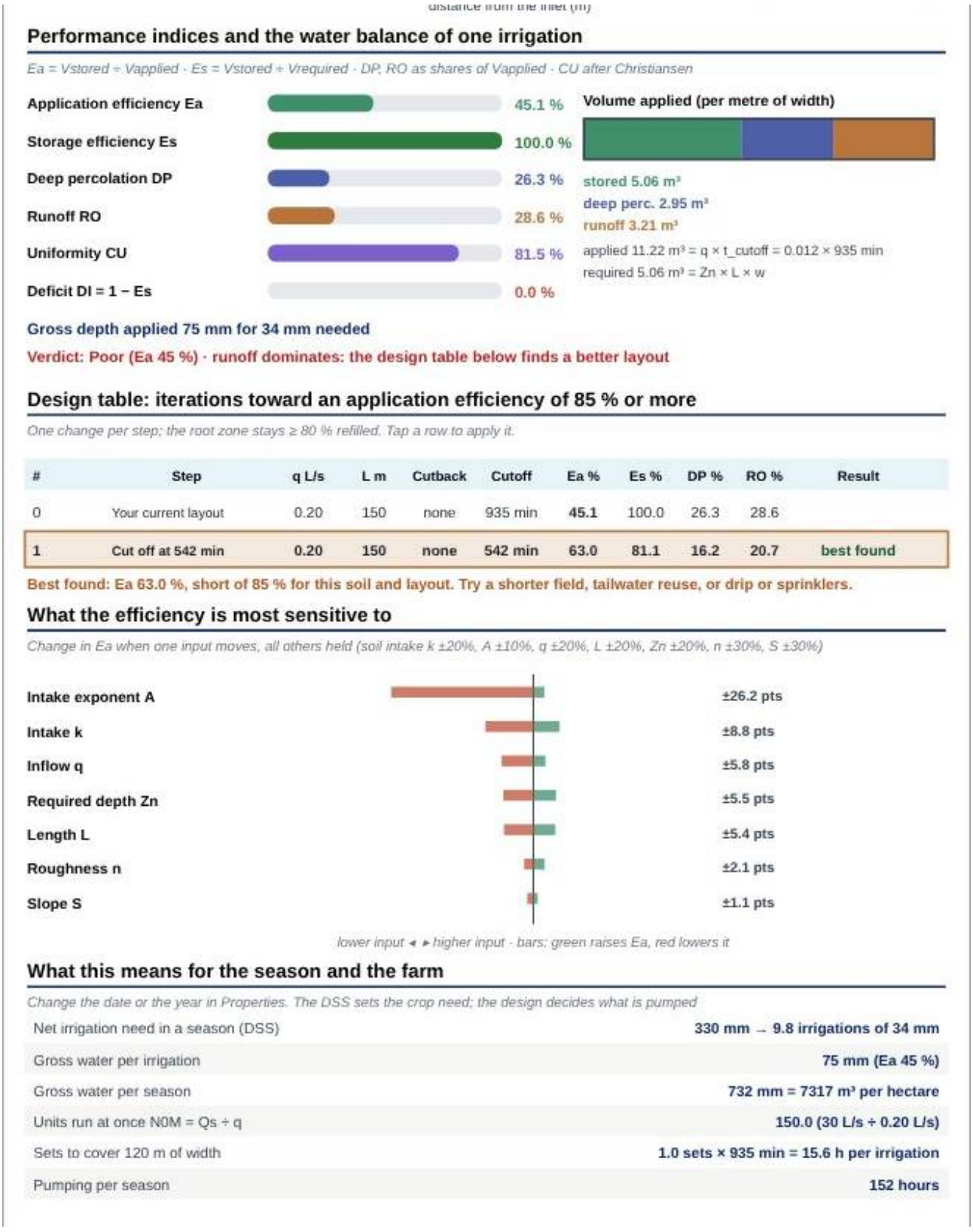
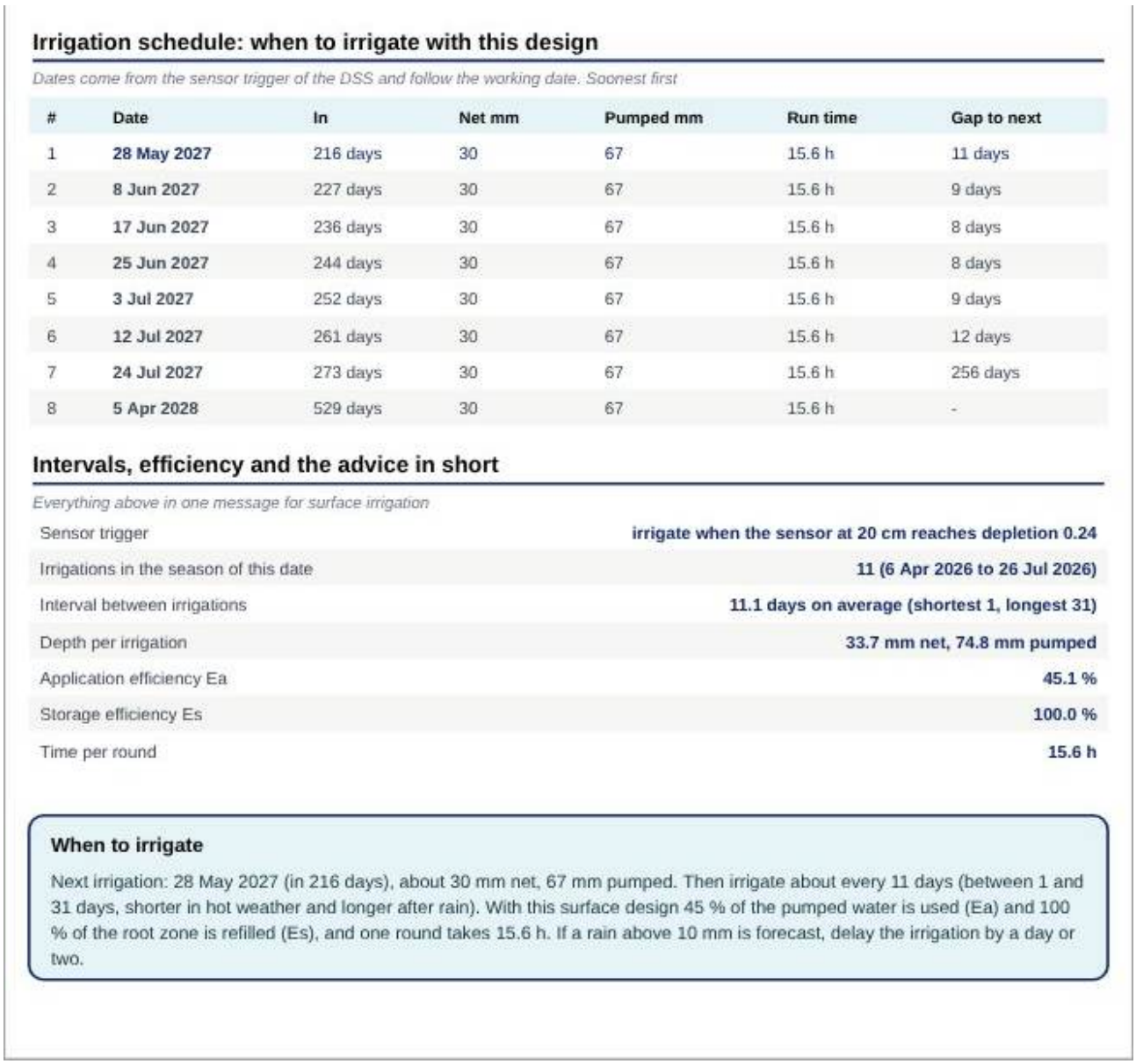


Figure 10.4: Surface irrigation design, Silty Clay Loam, part 4 of 5: Performance indices and the water balance of one irrigation; Design table: iterations toward an application efficiency of 85 % or more; What the efficiency is most sensitive to; What this means for the season and the farm.



Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**28 May 2027**	216 days	30	67	15.6 h	11 days
2	**8 Jun 2027**	227 days	30	67	15.6 h	9 days
3	**17 Jun 2027**	236 days	30	67	15.6 h	8 days
4	**25 Jun 2027**	244 days	30	67	15.6 h	8 days
5	**3 Jul 2027**	252 days	30	67	15.6 h	9 days
6	**12 Jul 2027**	261 days	30	67	15.6 h	12 days
7	**24 Jul 2027**	273 days	30	67	15.6 h	256 days
8	**5 Apr 2028**	529 days	30	67	15.6 h	-

Intervals, efficiency and the advice in short

Everything above in one message for surface irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.24**
Irrigations in the season of this date	**11 (6 Apr 2026 to 26 Jul 2026)**
Interval between irrigations	**11.1 days on average (shortest 1, longest 31)**
Depth per irrigation	**33.7 mm net, 74.8 mm pumped**
Application efficiency Ea	**45.1 %**
Storage efficiency Es	**100.0 %**
Time per round	**15.6 h**

When to irrigate

Next irrigation: 28 May 2027 (in 216 days), about 30 mm net, 67 mm pumped. Then irrigate about every 11 days (between 1 and 31 days, shorter in hot weather and longer after rain). With this surface design 45 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 15.6 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 10.5: Surface irrigation design, Silty Clay Loam, part 5 of 5: Irrigation schedule: when to irrigate with this design; Intervals, efficiency and the advice in short.

10.2 Drip irrigation design (Silty Clay Loam)

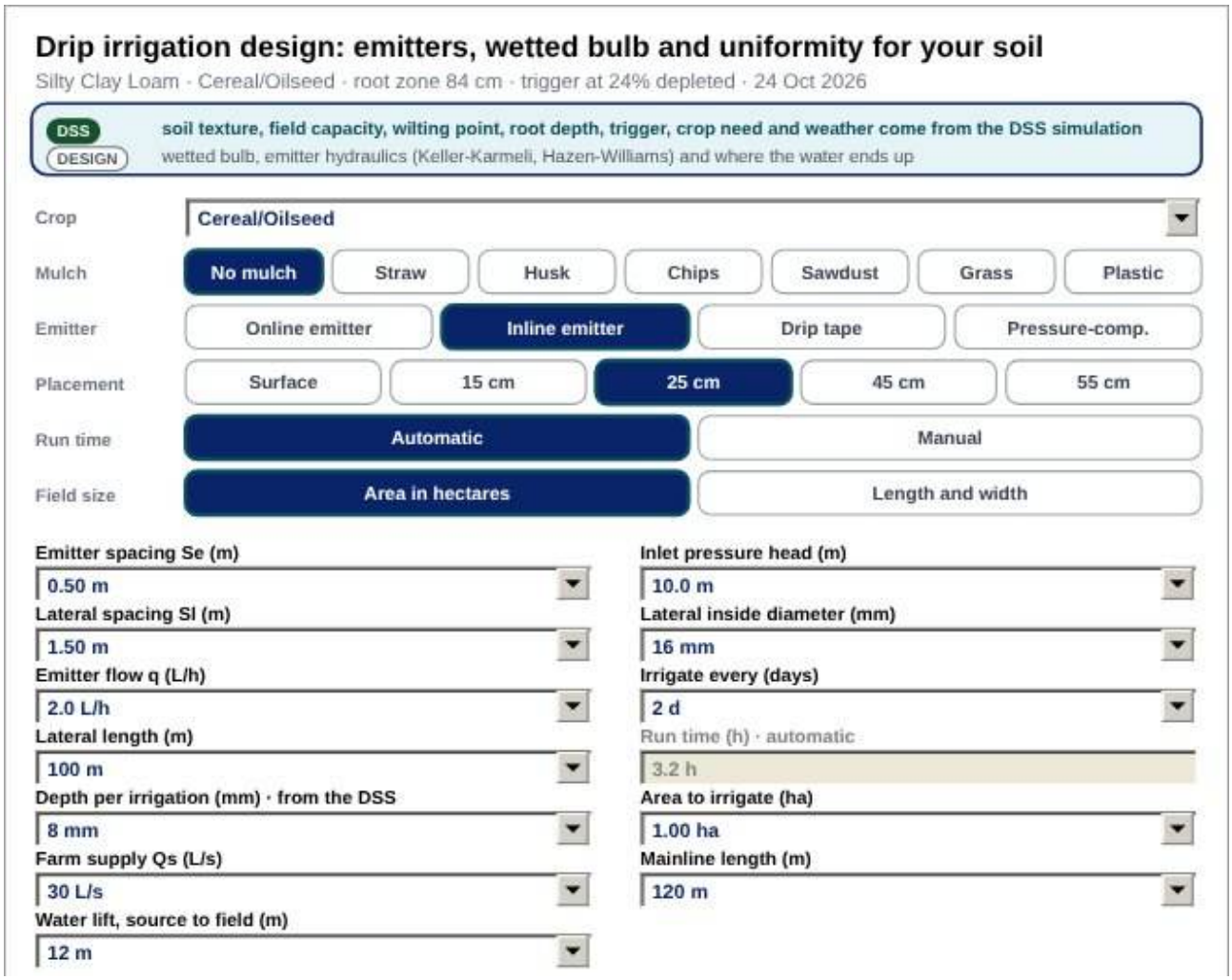


Figure 10.6: Drip irrigation design, Silty Clay Loam, part 1 of 6.

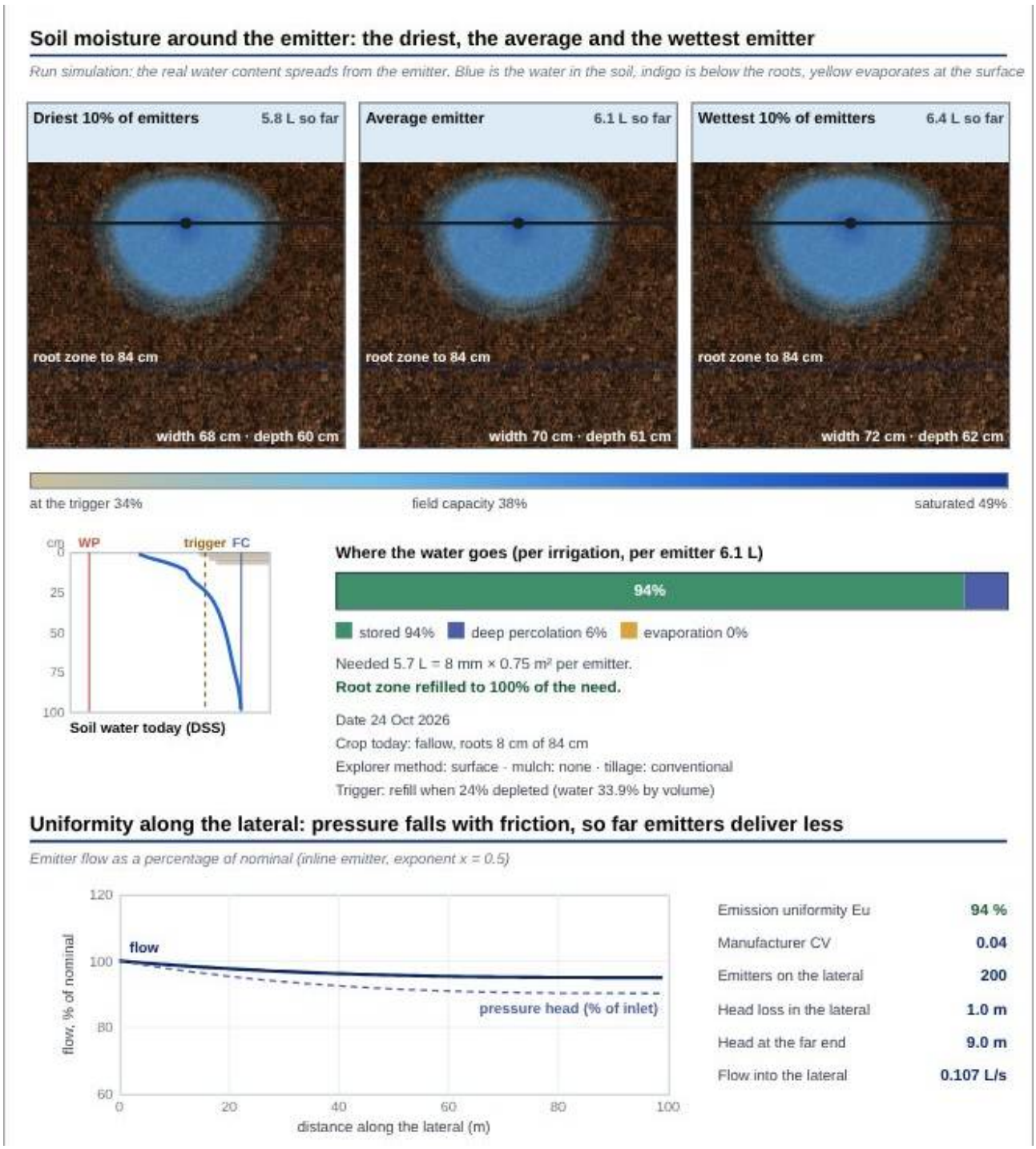
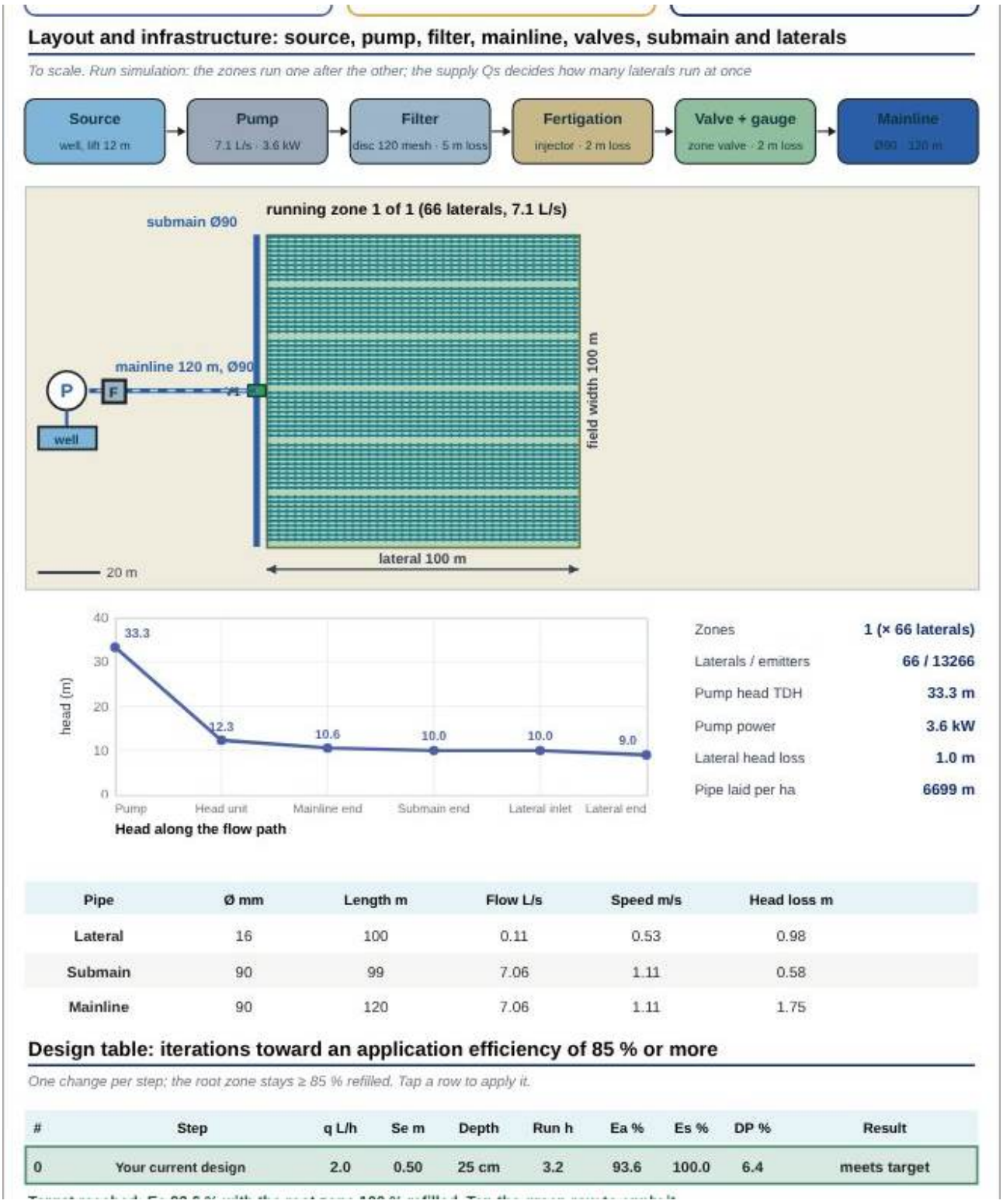


Figure 10.7: Drip irrigation design, Silty Clay Loam, part 2 of 6: Soil moisture around the emitter: the driest, the average and the wettest emitter; Uniformity along the lateral: pressure falls with friction, so far emitters deliver less.

Figure 10.8: Drip irrigation design, Silty Clay Loam, part 3 of 6: How the wetted bulb grows with the run time; Result for this design.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

#	Step	q L/h	Se m	Depth	Run h	Ea %	Es %	DP %	Result
0	Your current design	2.0	0.50	25 cm	3.2	93.6	100.0	6.4	meets target

Figure 10.9: Drip irrigation design, Silty Clay Loam, part 4 of 6: Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 93.6 % with the root zone 100 % refilled. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves ±25–30 %, all others held

lower input ◂ ▸ higher input · green raises Ea, red lowers it

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

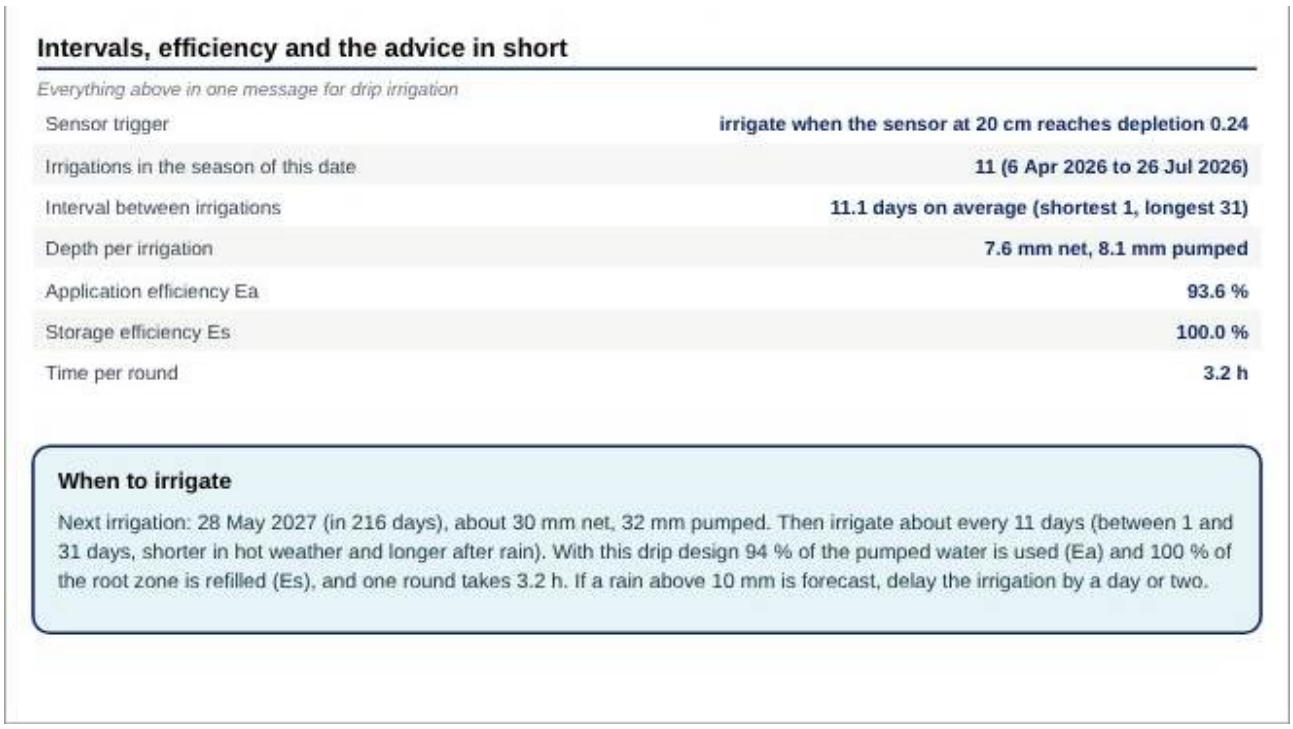
Net irrigation need in a season (DSS)	**330 mm → 43 irrigations of 8 mm**
Gross water per irrigation	**8.1 mm (Ea 94 %)**
Gross water per season	**352 mm = 3523 m³ per hectare**
Peak crop need / interval	**3.8 mm/day × 02 days**
System flow per hectare	**7.41 L/s/ha (13333 emitters/ha, 6667 m of lateral/ha)**
Sets for 1.0 ha with 30 L/s	**1 sets × 3.2 h = 3.2 h per round**
Pumping per season	**137 hours · 487 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**28 May 2027**	216 days	30	32	3.2 h	11 days
2	**8 Jun 2027**	227 days	30	32	3.2 h	9 days
3	**17 Jun 2027**	236 days	30	32	3.2 h	8 days
4	**25 Jun 2027**	244 days	30	32	3.2 h	8 days
5	**3 Jul 2027**	252 days	30	32	3.2 h	9 days
6	**12 Jul 2027**	261 days	30	32	3.2 h	12 days
7	**24 Jul 2027**	273 days	30	32	3.2 h	256 days
8	**5 Apr 2028**	529 days	30	32	3.2 h	-

Figure 10.10: Drip irrigation design, Silty Clay Loam, part 5 of 6: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for drip irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.24**
Irrigations in the season of this date	**11 (6 Apr 2026 to 26 Jul 2026)**
Interval between irrigations	**11.1 days on average (shortest 1, longest 31)**
Depth per irrigation	**7.6 mm net, 8.1 mm pumped**
Application efficiency Ea	**93.6 %**
Storage efficiency Es	**100.0 %**
Time per round	**3.2 h**

When to irrigate

Next irrigation: 28 May 2027 (in 216 days), about 30 mm net, 32 mm pumped. Then irrigate about every 11 days (between 1 and 31 days, shorter in hot weather and longer after rain). With this drip design 94 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 3.2 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 10.11: Drip irrigation design, Silty Clay Loam, part 6 of 6: Intervals, efficiency and the advice in short.

10.3 Sprinkler design and pipes (Silty Clay Loam)

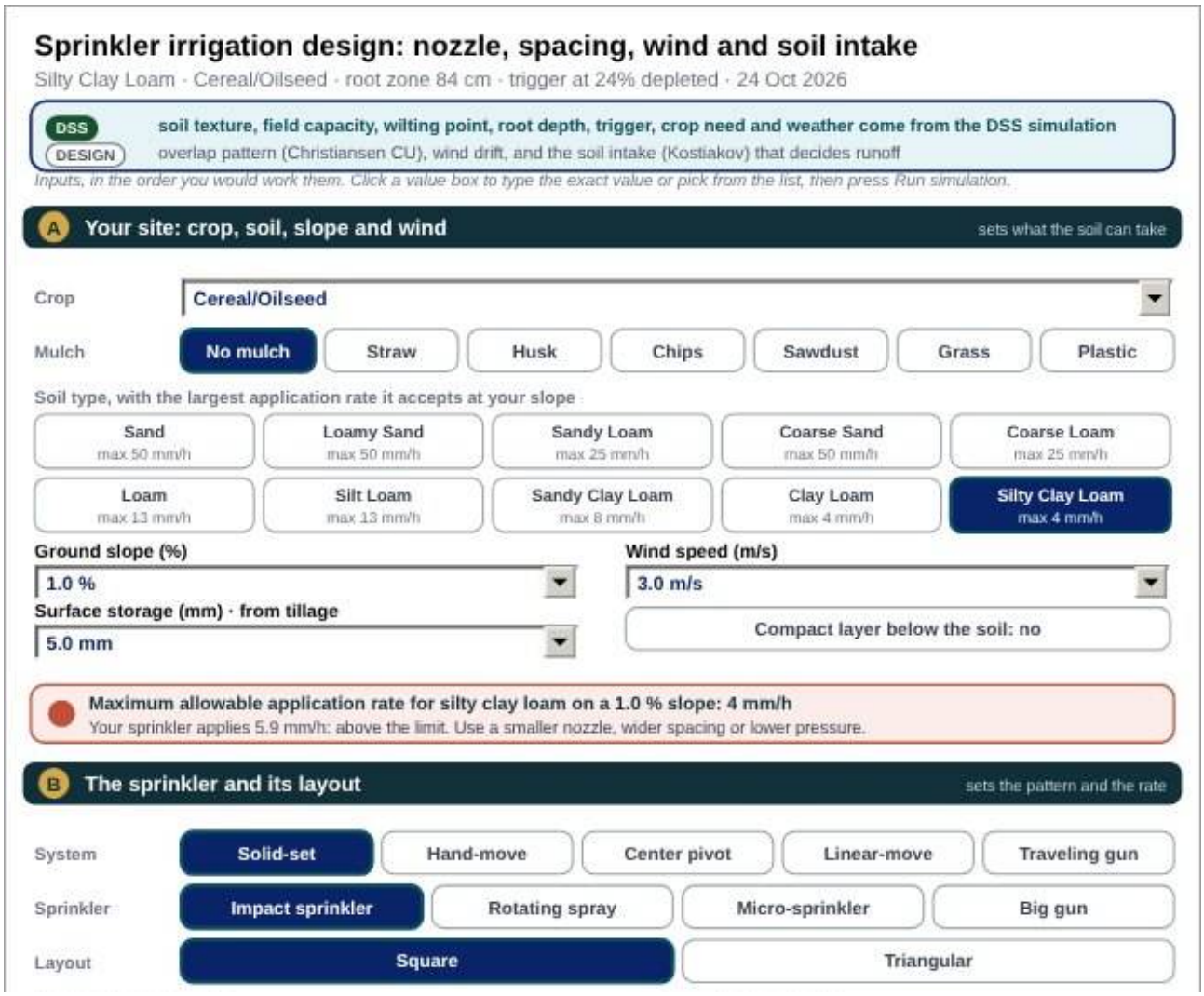


Figure 10.12: Sprinkler design and pipes, Silty Clay Loam, part 1 of 12.

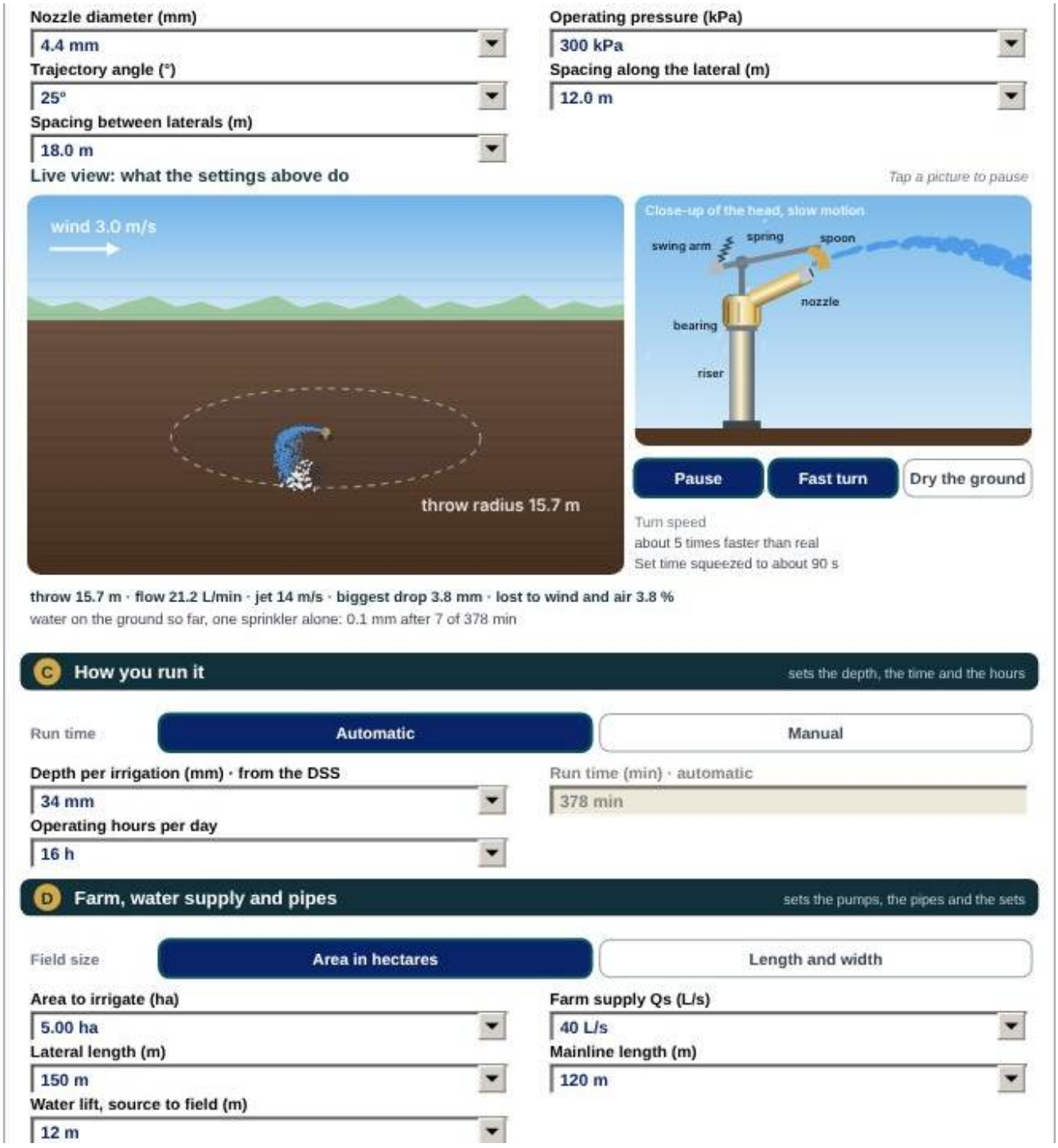


Figure 10.13: Sprinkler design and pipes, Silty Clay Loam, part 2 of 12.

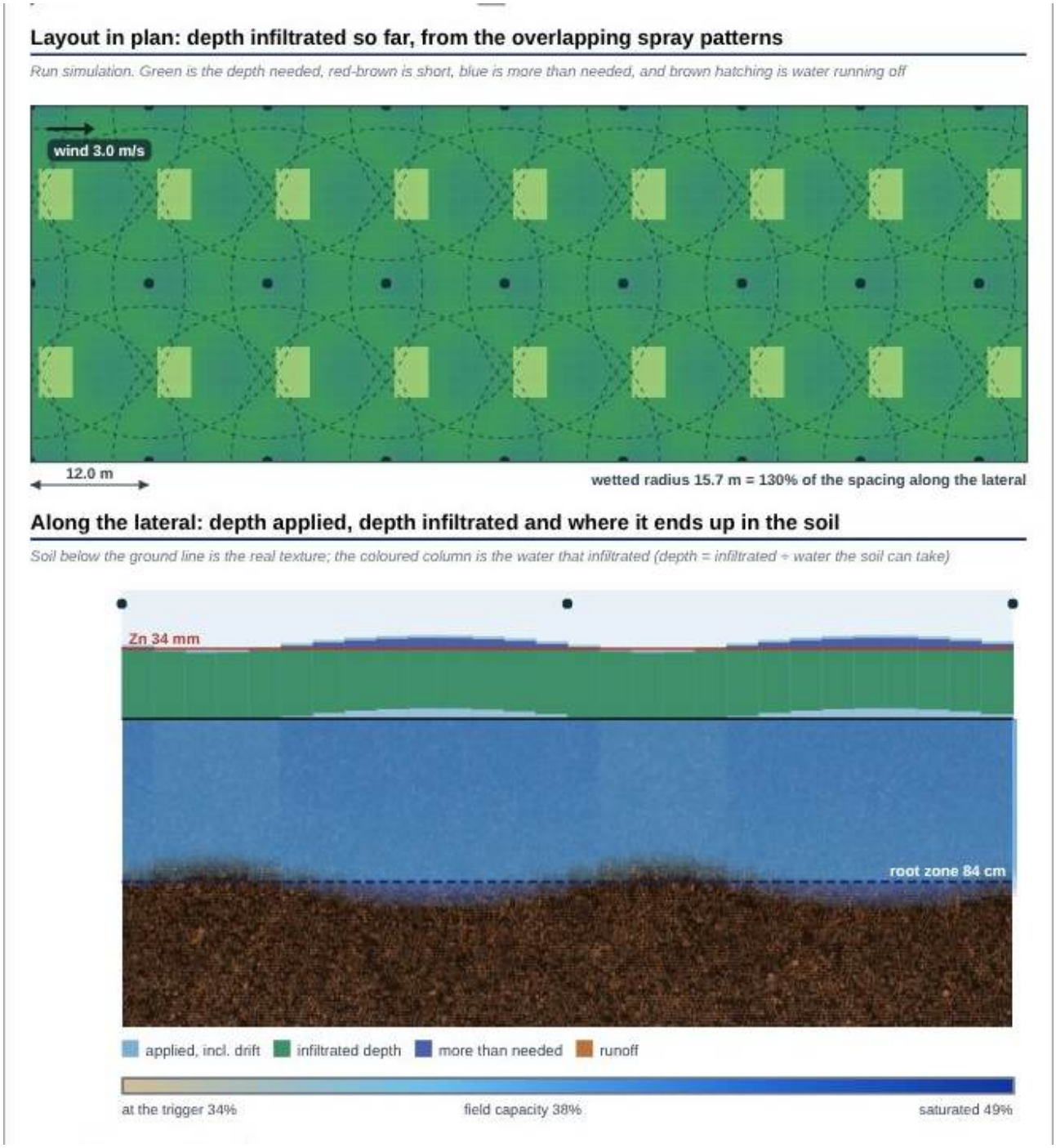


Figure 10.14: Sprinkler design and pipes, Silty Clay Loam, part 3 of 12: Layout in plan: depth infiltrated so far, from the overlapping spray patterns; Along the lateral: depth applied, depth infiltrated and where it ends up in the soil.

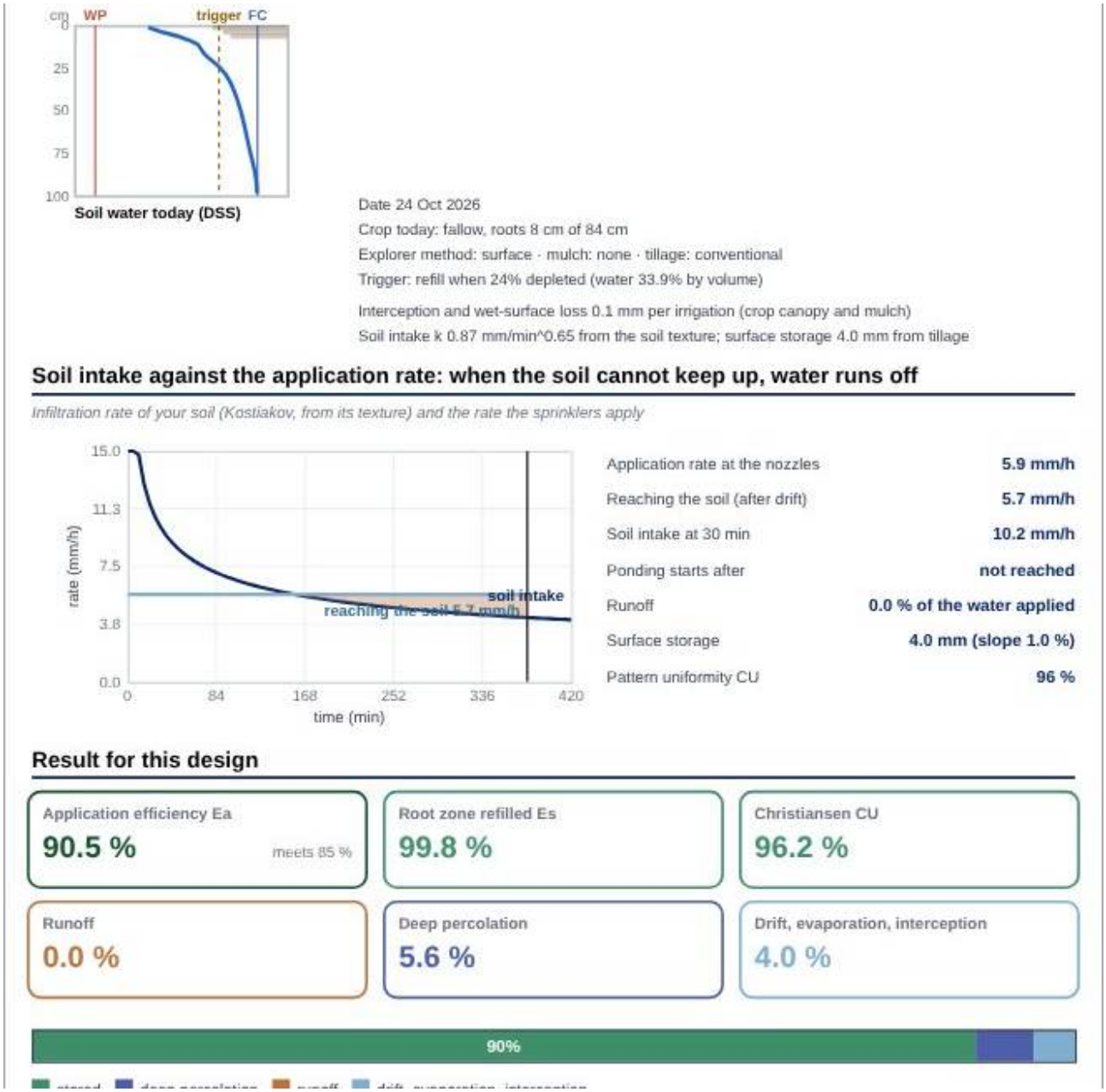


Figure 10.15: Sprinkler design and pipes, Silty Clay Loam, part 4 of 12: Soil intake against the application rate: when the soil cannot keep up, water runs off; Result for this design.

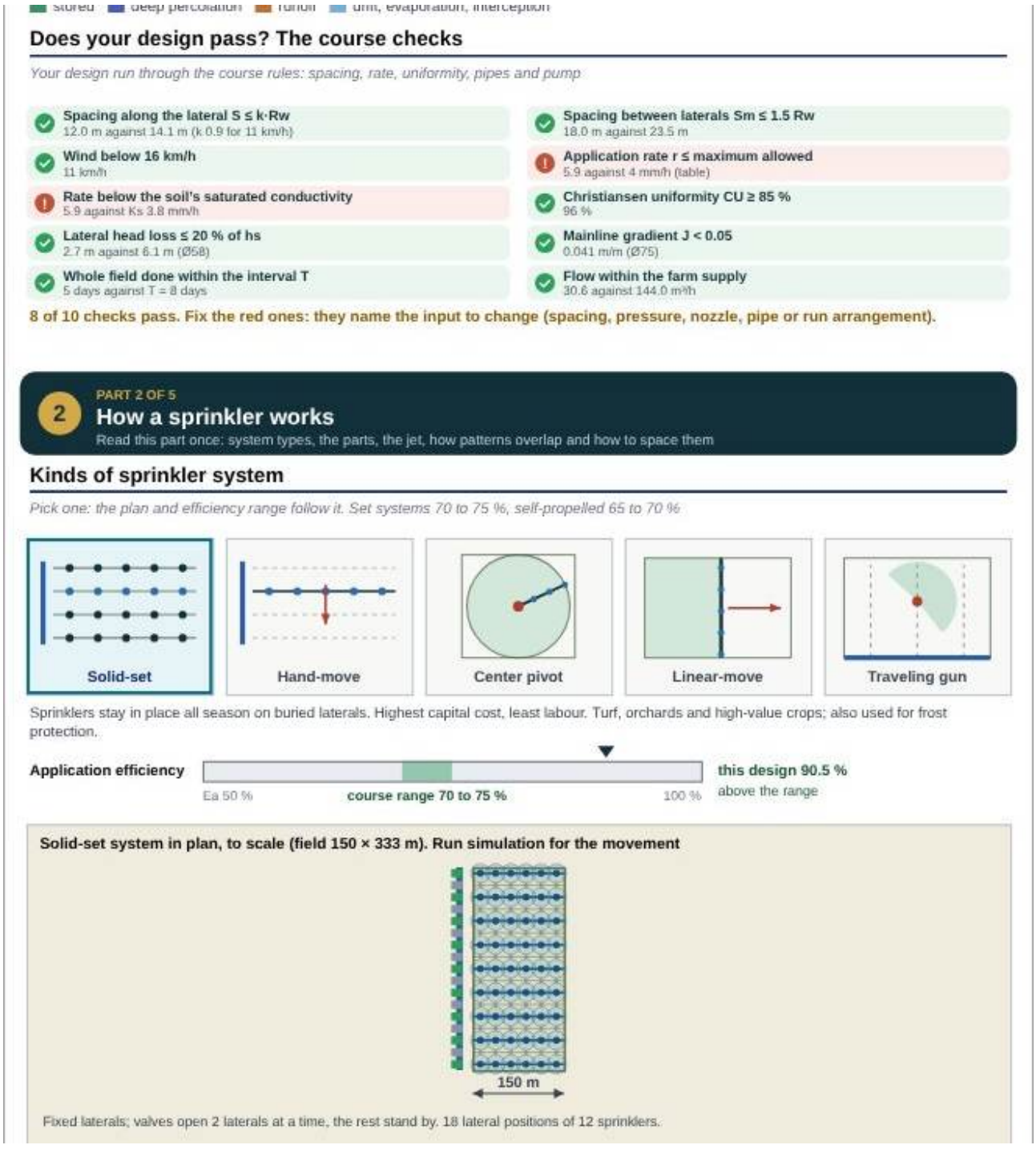
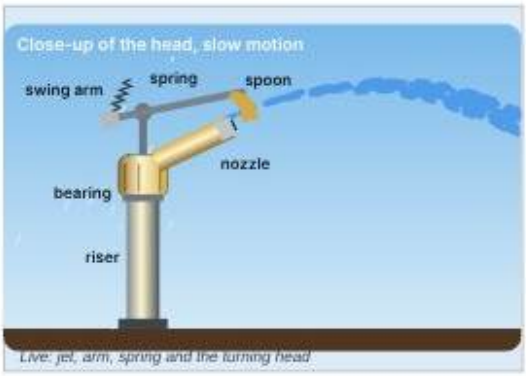


Figure 10.16: Sprinkler design and pipes, Silty Clay Loam, part 5 of 12: Does your design pass? The course checks; Kinds of sprinkler system.

The sprinkler: parts and pressure class

Parts of a rotary impact sprinkler, and where your choice falls among the three pressure classes



Class by operating pressure

Low pressure (spray type)
pressure 8 to 20 m · discharge 0.035 to 0.5 m³/h · throw 3 to 7 m

Medium pressure (rotary, impact)
pressure 20 to 45 m · discharge 0.5 to 15 m³/h · throw 9 to 20 m

High pressure (gun sprinklers)
pressure 45 to 100 m · discharge 15 to 150 m³/h · throw 30 to 100 m

Yours: head 30.6 m, 1.27 m³/h, throw 15.7 m → medium pressure

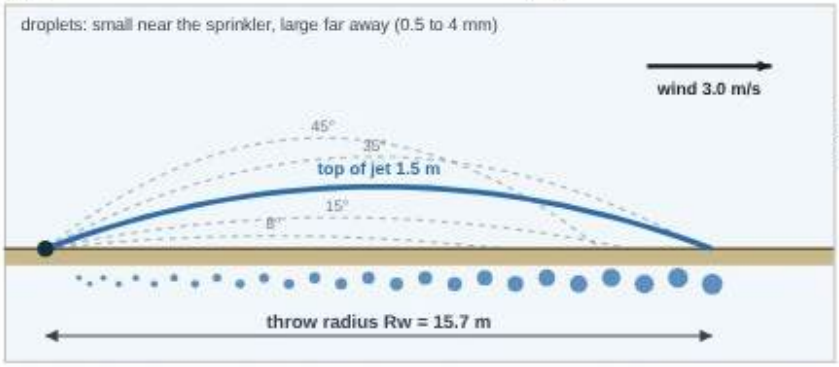
$q = c \cdot d^2 \cdot \sqrt{h}$ and $Rw = kr \cdot h^{0.25}$

The jet: trajectory angle, throw radius and droplet size

Farthest throw at about 28 to 30 degrees; low angles (8 degrees) go under foliage and drift less

Trajectory angle (°)

25°



Throw against angle

Drift and evaporation loss 3.8 % at 25° and 3.0 m/s. Pressure sets drop size: lower pressure gives larger drops that throw farther but can seal the soil surface; higher pressure gives finer drops that drift more.

Seen from the field: the jet sweeps round, drops fly with the wind and wet the ground

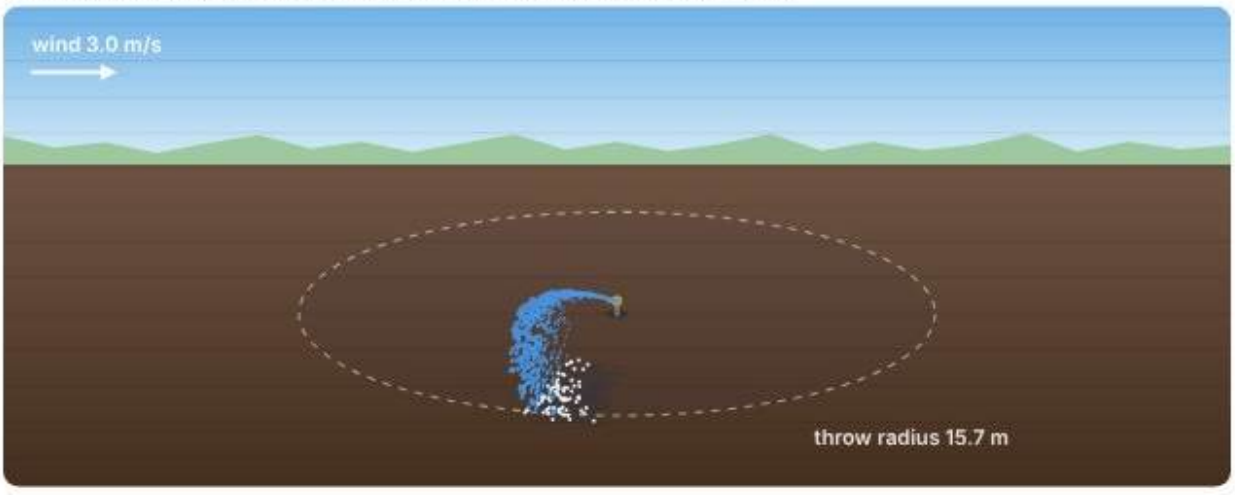


Figure 10.17: Sprinkler design and pipes, Silty Clay Loam, part 6 of 12: The sprinkler: parts and pressure class; The jet: trajectory angle, throw radius and droplet size.

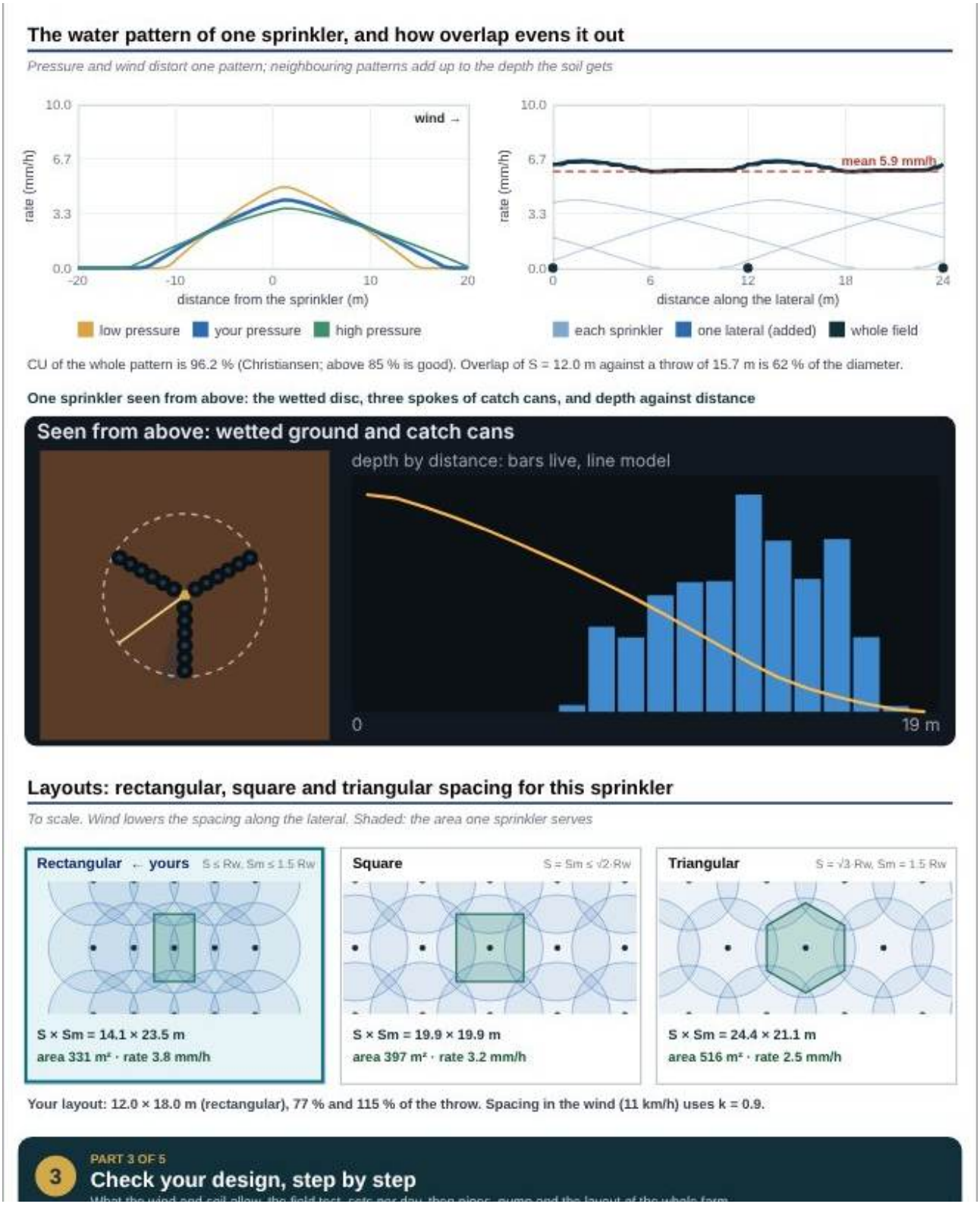


Figure 10.18: Sprinkler design and pipes, Silty Clay Loam, part 7 of 12: The water pattern of one sprinkler, and how overlap evens it out; Layouts: rectangular, square and triangular spacing for this sprinkler.

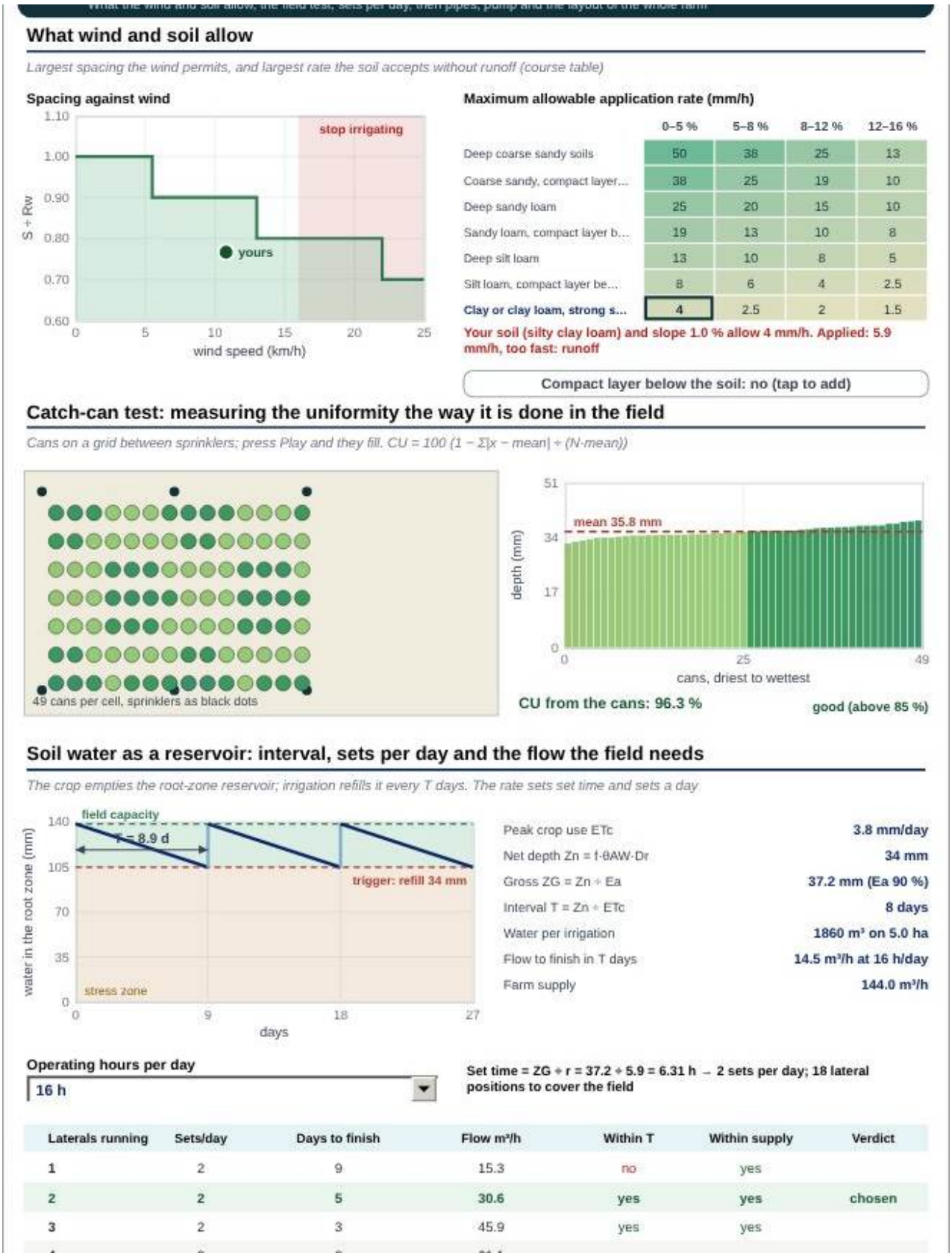


Figure 10.19: Sprinkler design and pipes, Silty Clay Loam, part 8 of 12: What wind and soil allow; Catch-can test: measuring the uniformity the way it is done in the field; Soil water as a reservoir: interval, sets per day and the flow the field needs.

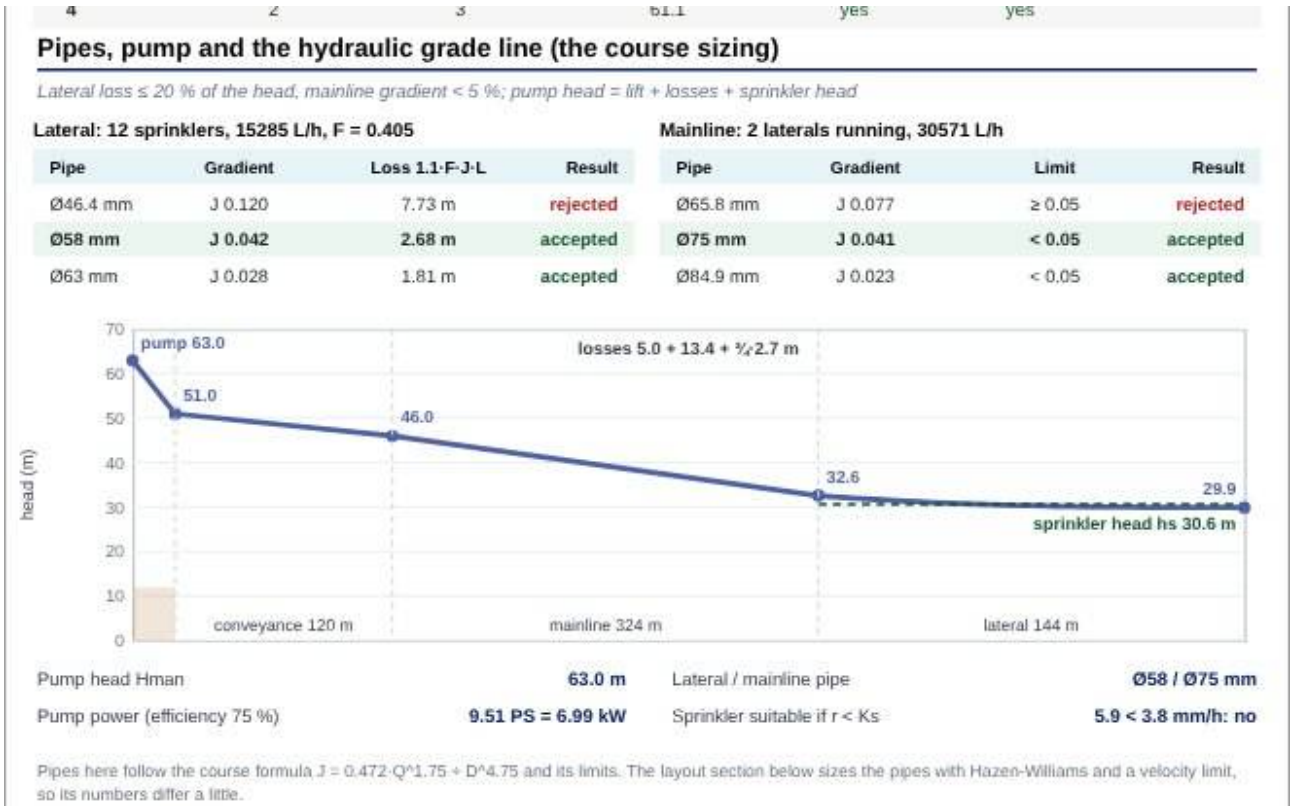
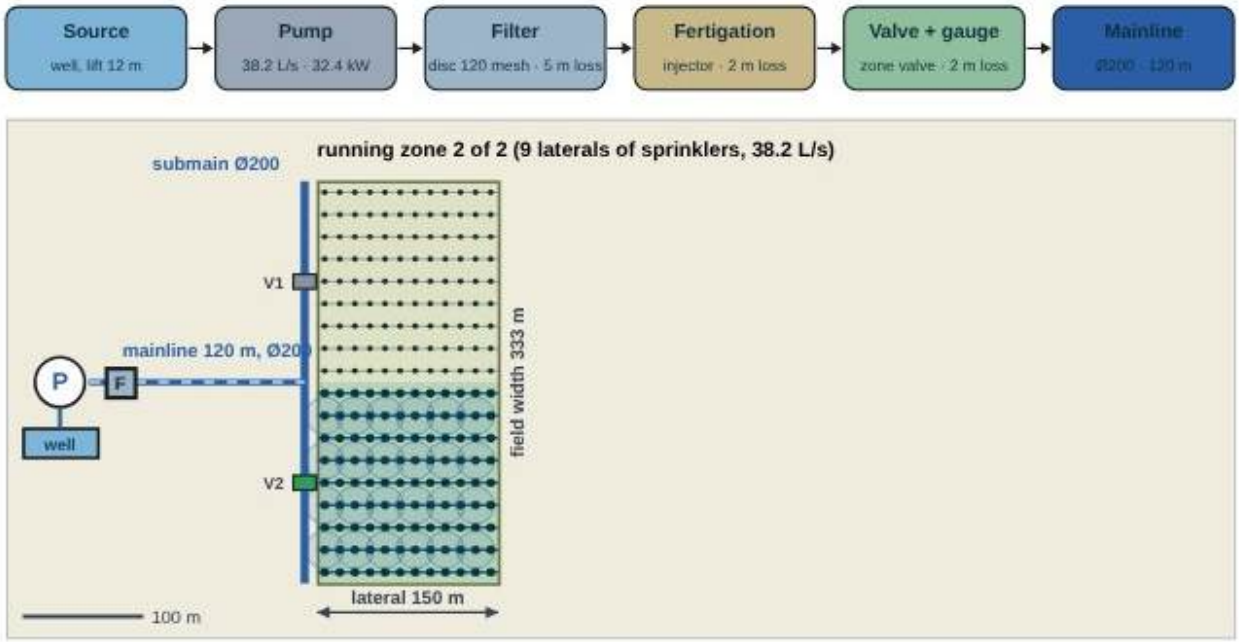
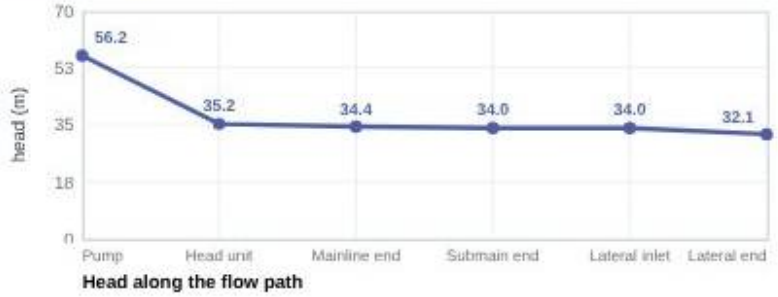


Figure 10.20: Sprinkler design and pipes, Silty Clay Loam, part 9 of 12: Pipes, pump and the hydraulic grade line (the course sizing).

Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

To scale, with the wetted circles of the running zone. Run simulation: the sets run one after the other





Head along the flow path

Sets	**2 (× 9 laterals)**
Laterals / sprinklers	**18 / 216**
Pump head TDH	**56.2 m**
Pump power	**32.4 kW**
Lateral head loss	**1.9 m (≤ 20%)**
Pipe laid per ha	**572 m**

Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	63	150	4.25	1.36	1.90
Submain	200	162	38.21	1.22	0.44
Mainline	200	120	38.21	1.22	0.82

PART 4 OF 5

4 **Improve it**

The search below changes one input at a time toward 85 % efficiency; the bars show what matters most

Design table: iterations toward an application efficiency of 85 % or more

One change per step; the driest quarter must still get the depth. Tap a row to apply it.

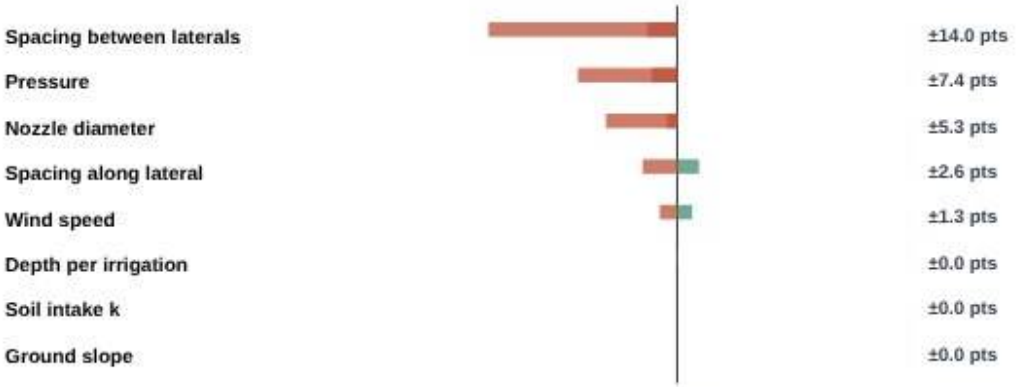
#	Step	kPa	Nozzle	Ss×Sl m	Run min	Ea %	Es %	RO %	Result
0	Your current design	300	4.4	12×18	378	90.5	99.8	0.0	meets target

Figure 10.21: Sprinkler design and pipes, Silty Clay Loam, part 10 of 12: Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals; Design table: iterations toward an application efficiency of 85 % or more.

Target reached: Ea 90.5 % with CU 96 %. Tap the green row to apply it.

What the efficiency is most sensitive to

Change in Ea when one input moves, all others held



lower input ◂ ▸ higher input · green raises Ea, red lowers it

PART 5 OF 5

5 What it means for the season

Water, hours and energy for a whole year of irrigating

What this means for the season and the farm

The DSS sets the crop need; the design decides what is pumped

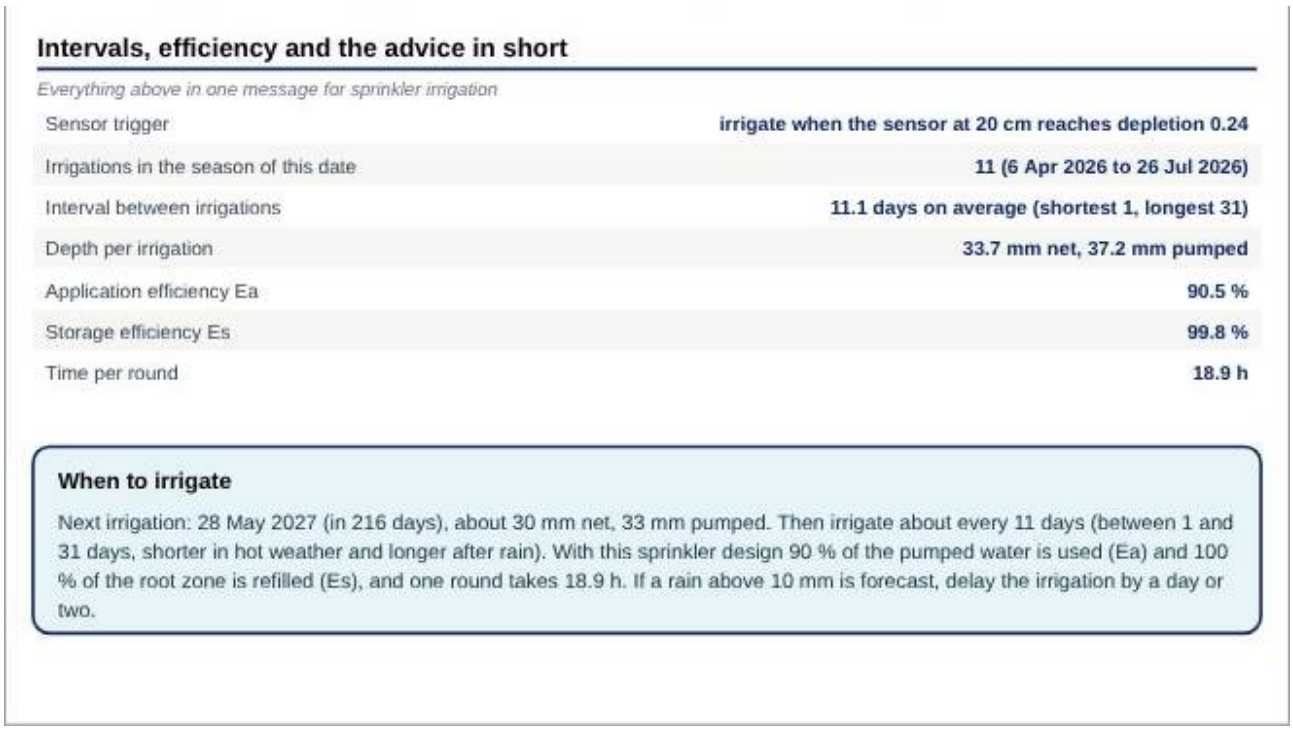
Net irrigation need in a season (DSS)	**330 mm → 10 irrigations of 34 mm**
Gross water per irrigation	**37.2 mm (Ea 90 %)**
Gross water per season	**364 mm = 3638 m³ per hectare**
One sprinkler	**21.2 L/min at 300 kPa, reaches 15.7 m**
Sprinklers for 5.0 ha	**231 (81.7 L/s if all run)**
Sets with 40 L/s	**3 sets × 378 min = 18.9 h per round**
Pumping per season	**185 hours · 6005 kWh**

Irrigation schedule: when to irrigate with this design

Dates come from the sensor trigger of the DSS and follow the working date. Soonest first

#	Date	In	Net mm	Pumped mm	Run time	Gap to next
1	**28 May 2027**	216 days	30	33	18.9 h	11 days
2	**8 Jun 2027**	227 days	30	33	18.9 h	9 days
3	**17 Jun 2027**	236 days	30	33	18.9 h	8 days
4	**25 Jun 2027**	244 days	30	33	18.9 h	8 days
5	**3 Jul 2027**	252 days	30	33	18.9 h	9 days
6	**12 Jul 2027**	261 days	30	33	18.9 h	12 days
7	**24 Jul 2027**	273 days	30	33	18.9 h	256 days
8	**5 Apr 2028**	529 days	30	33	18.9 h	-

Figure 10.22: Sprinkler design and pipes, Silty Clay Loam, part 11 of 12: What the efficiency is most sensitive to; What this means for the season and the farm; Irrigation schedule: when to irrigate with this design.



Intervals, efficiency and the advice in short

Everything above in one message for sprinkler irrigation

Sensor trigger	**irrigate when the sensor at 20 cm reaches depletion 0.24**
Irrigations in the season of this date	**11 (6 Apr 2026 to 26 Jul 2026)**
Interval between irrigations	**11.1 days on average (shortest 1, longest 31)**
Depth per irrigation	**33.7 mm net, 37.2 mm pumped**
Application efficiency Ea	**90.5 %**
Storage efficiency Es	**99.8 %**
Time per round	**18.9 h**

When to irrigate

Next irrigation: 28 May 2027 (in 216 days), about 30 mm net, 33 mm pumped. Then irrigate about every 11 days (between 1 and 31 days, shorter in hot weather and longer after rain). With this sprinkler design 90 % of the pumped water is used (Ea) and 100 % of the root zone is refilled (Es), and one round takes 18.9 h. If a rain above 10 mm is forecast, delay the irrigation by a day or two.

Figure 10.23: Sprinkler design and pipes, Silty Clay Loam, part 12 of 12: Intervals, efficiency and the advice in short.

10.4 Virtual farm of the surface design (Silty Clay Loam)

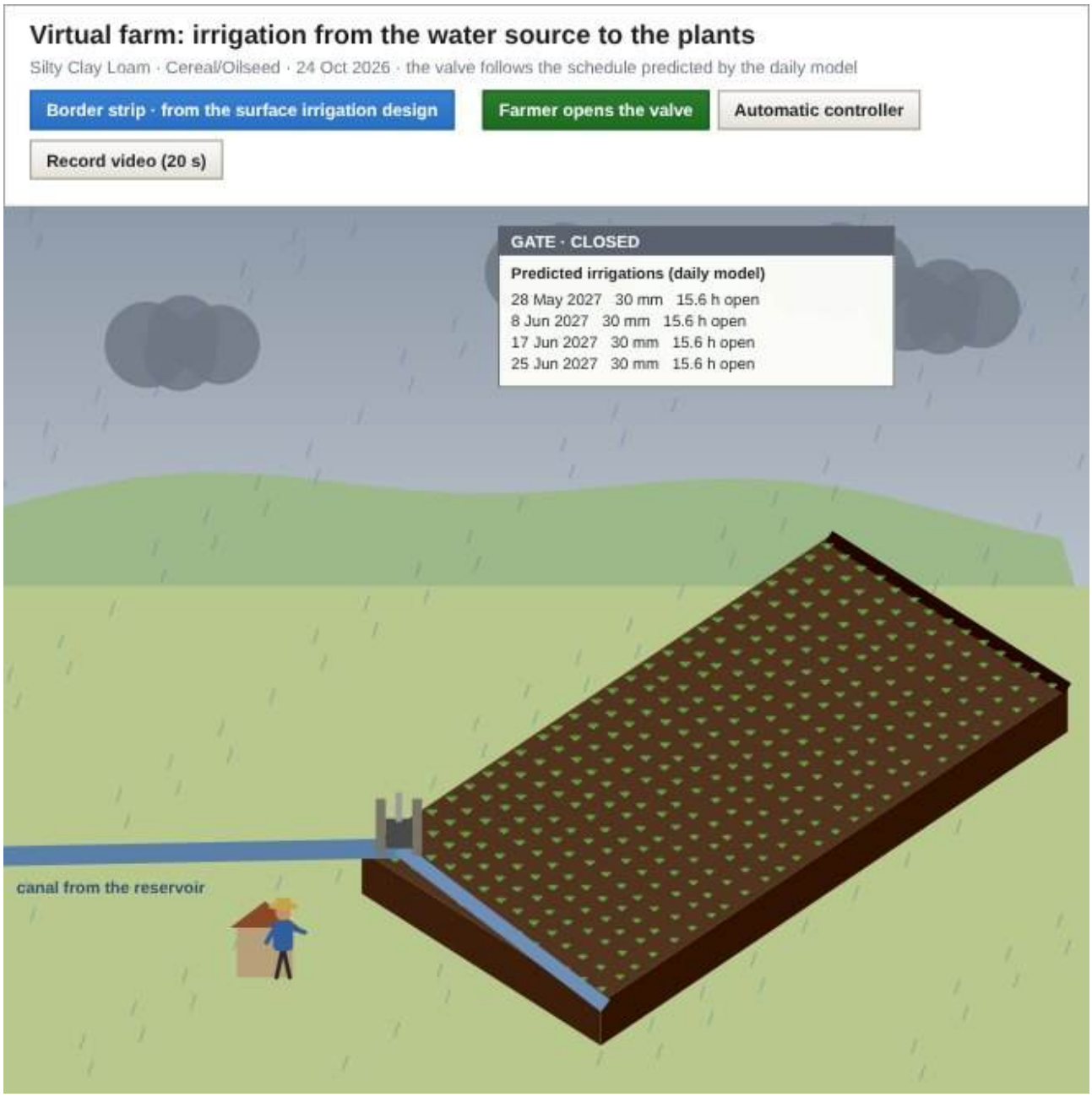


Figure 10.24: Virtual farm, surface design, Silty Clay Loam, part 1 of 3.

Figure 10.25: Virtual farm, surface design, Silty Clay Loam, part 2 of 3.

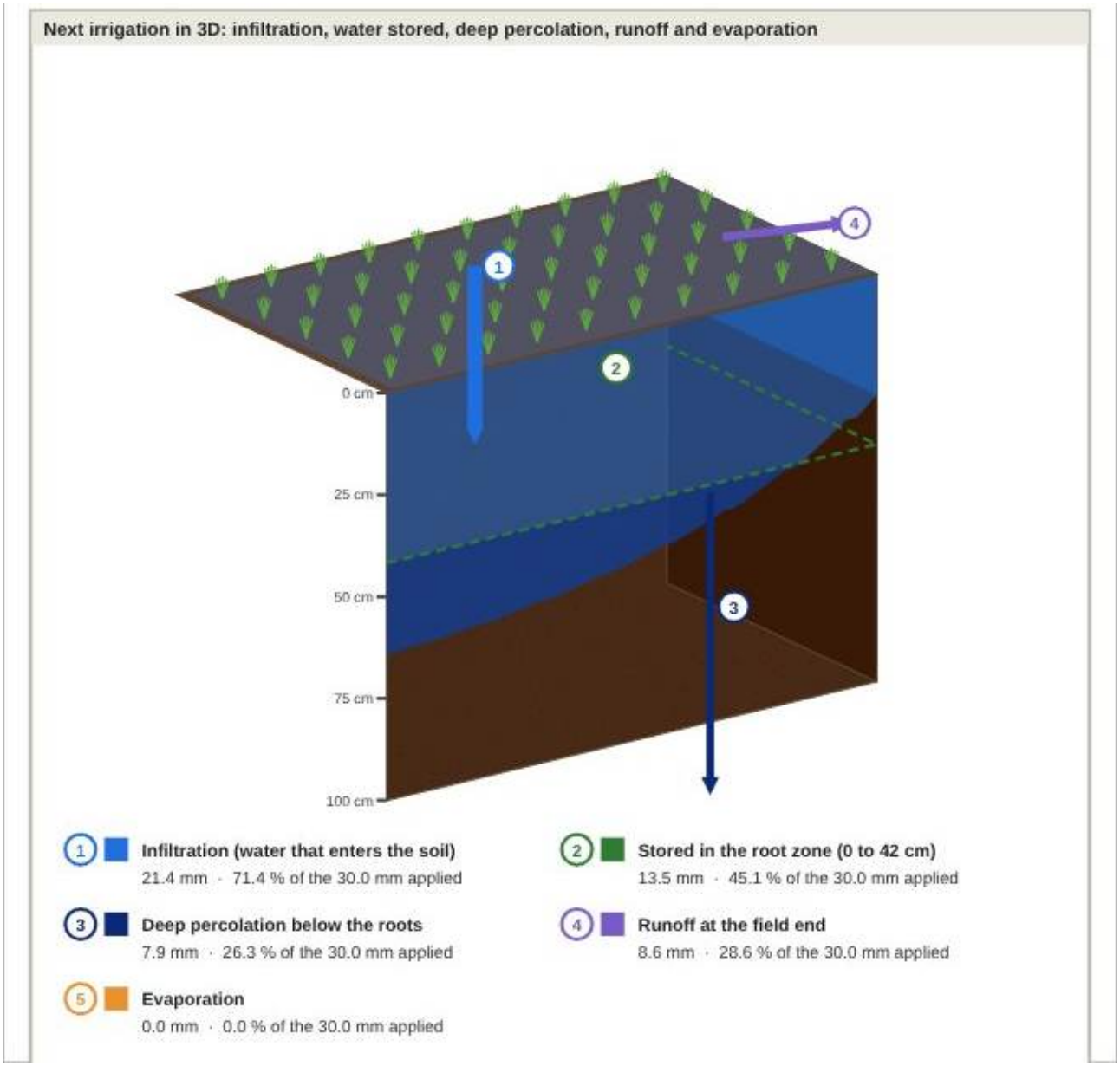


Figure 10.26: Virtual farm, surface design, Silty Clay Loam, part 3 of 3.

10.5 Virtual farm of the drip design (Silty Clay Loam)

Figure 10.27: Virtual farm, drip design, Silty Clay Loam, part 1 of 3.

Figure 10.28: Virtual farm, drip design, Silty Clay Loam, part 2 of 3.

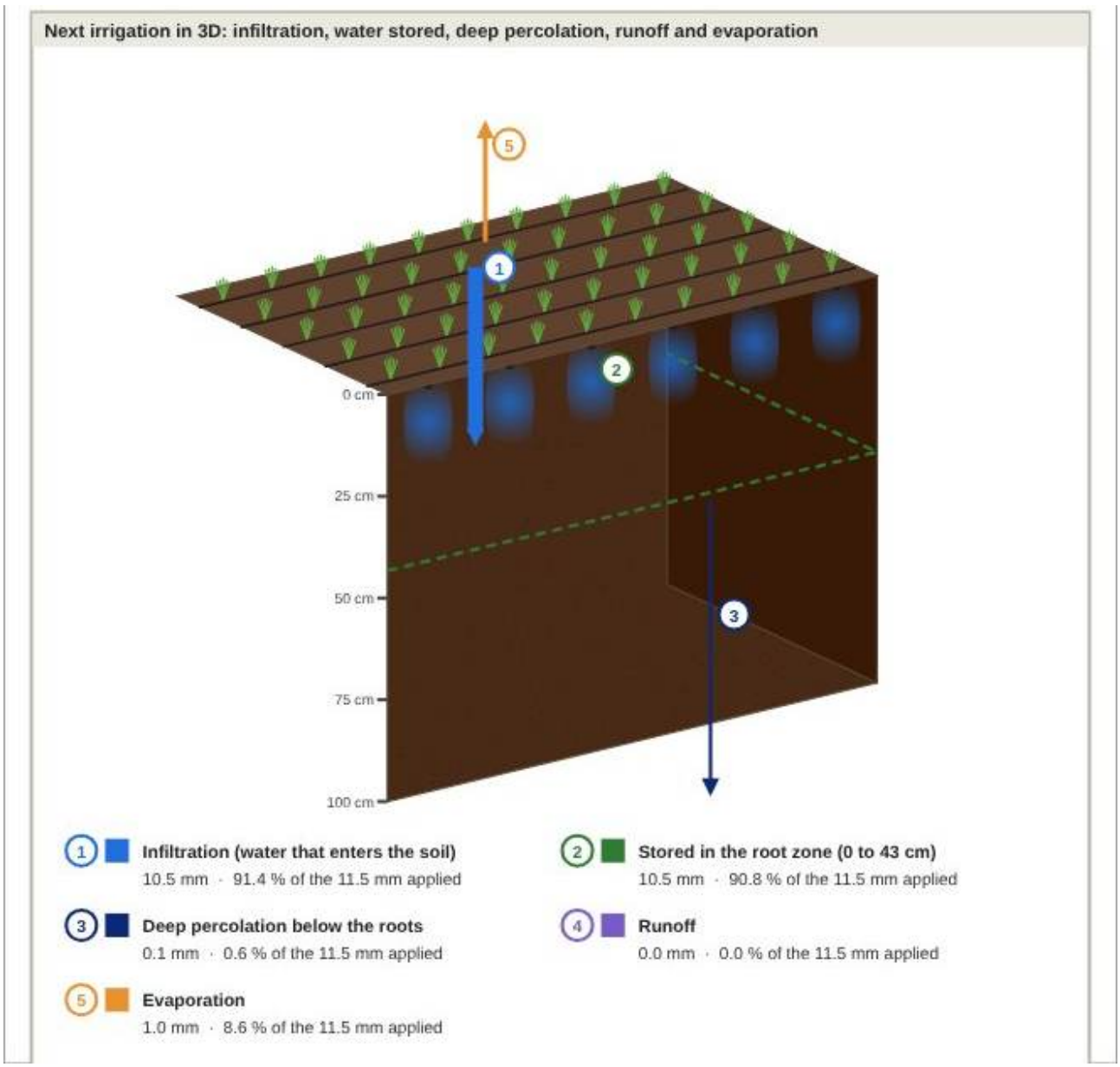


Figure 10.29: Virtual farm, drip design, Silty Clay Loam, part 3 of 3.

10.6 Virtual farm of the sprinkler design (Silty Clay Loam)

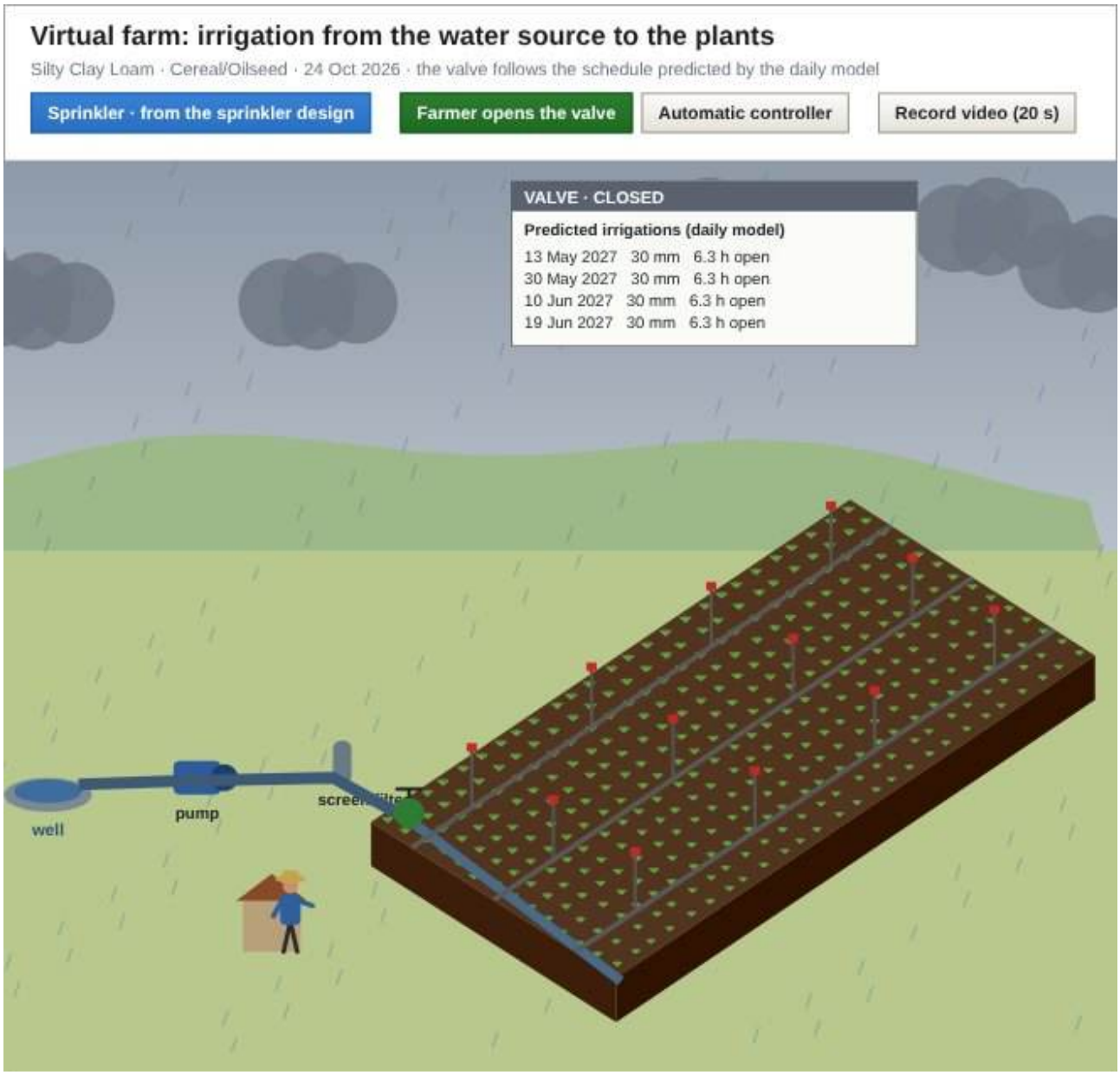


Figure 10.30: Virtual farm, sprinkler design, Silty Clay Loam, part 1 of 3.

Figure 10.31: Virtual farm, sprinkler design, Silty Clay Loam, part 2 of 3.

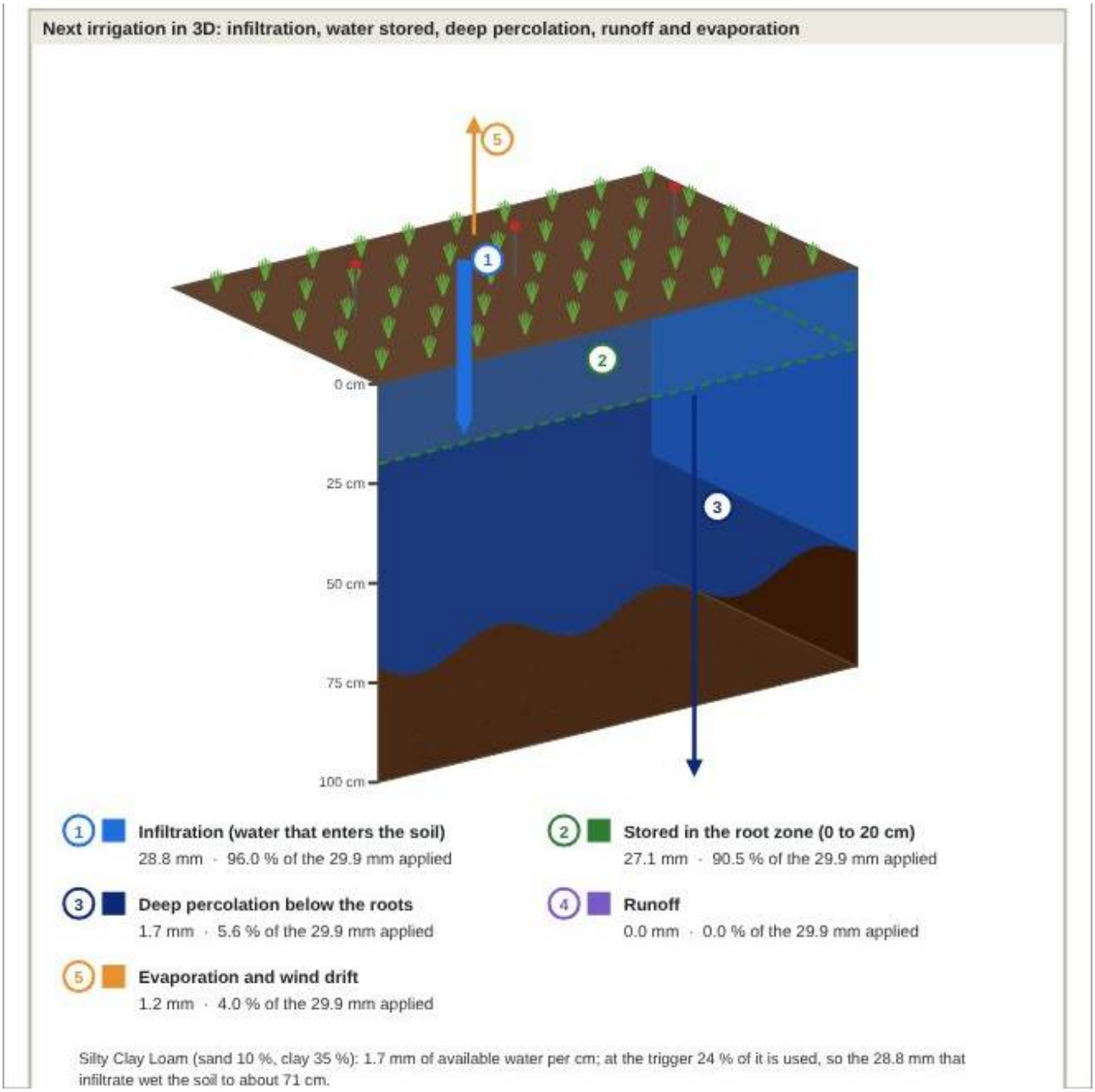


Figure 10.32: Virtual farm, sprinkler design, Silty Clay Loam, part 3 of 3.

10.7 Sensor and trigger (Silty Clay Loam)

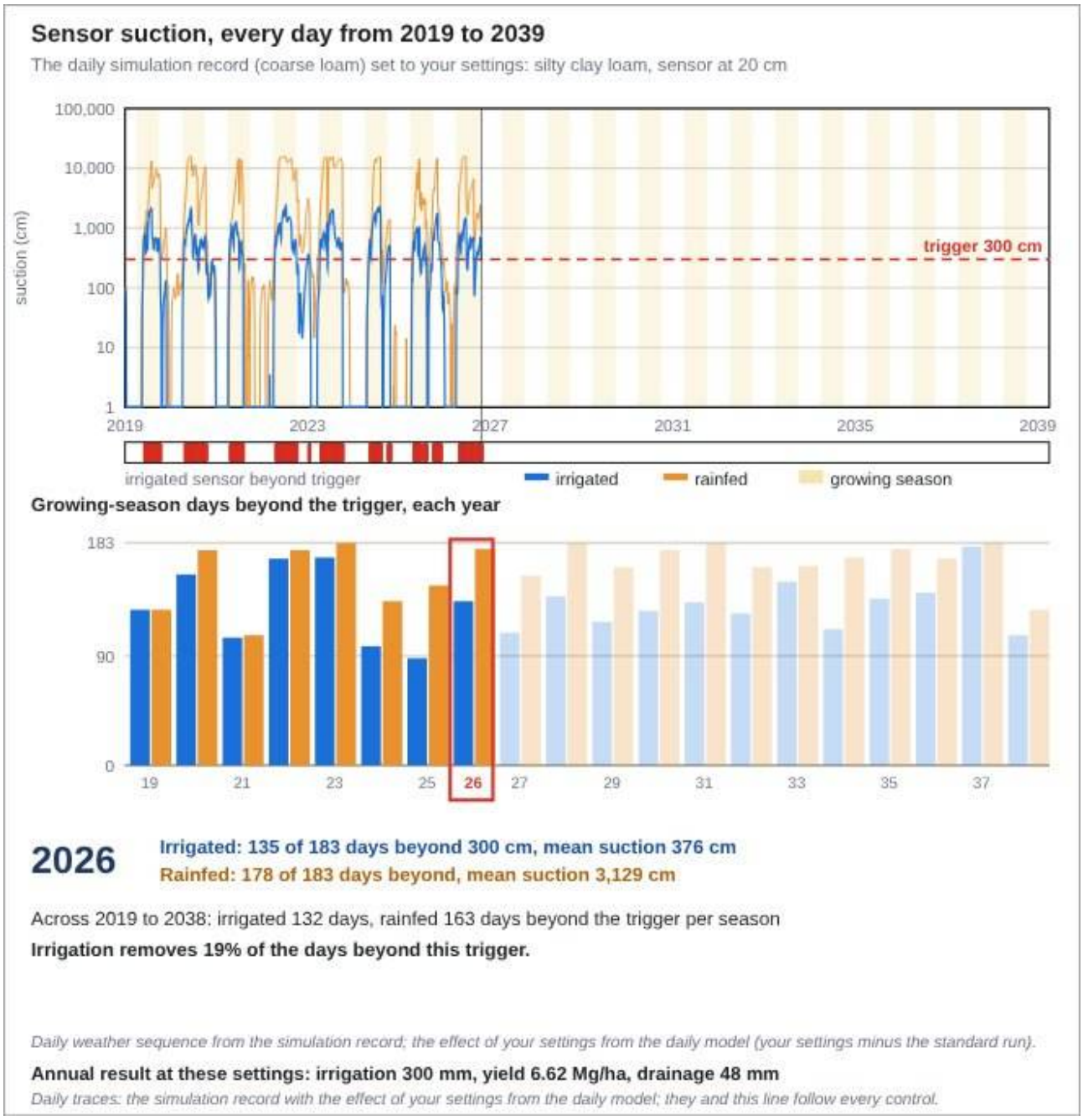


Figure 10.33: Sensor and trigger, Silty Clay Loam, part 1 of 1.

10.8 Trigger law (Silty Clay Loam)

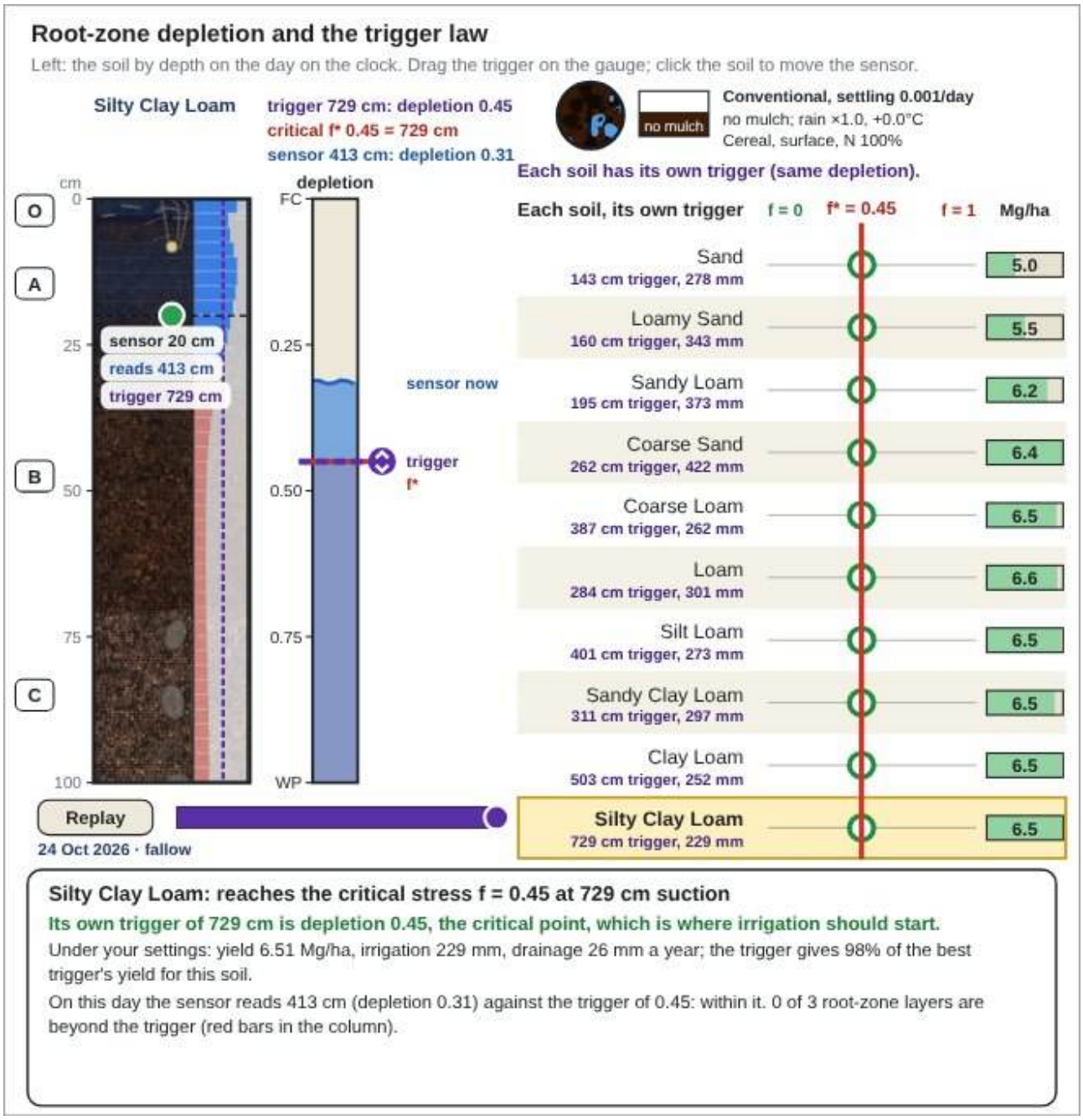


Figure 10.34: Trigger law, Silty Clay Loam, part 1 of 1.

10.9 Mulch against bare soil (Silty Clay Loam)

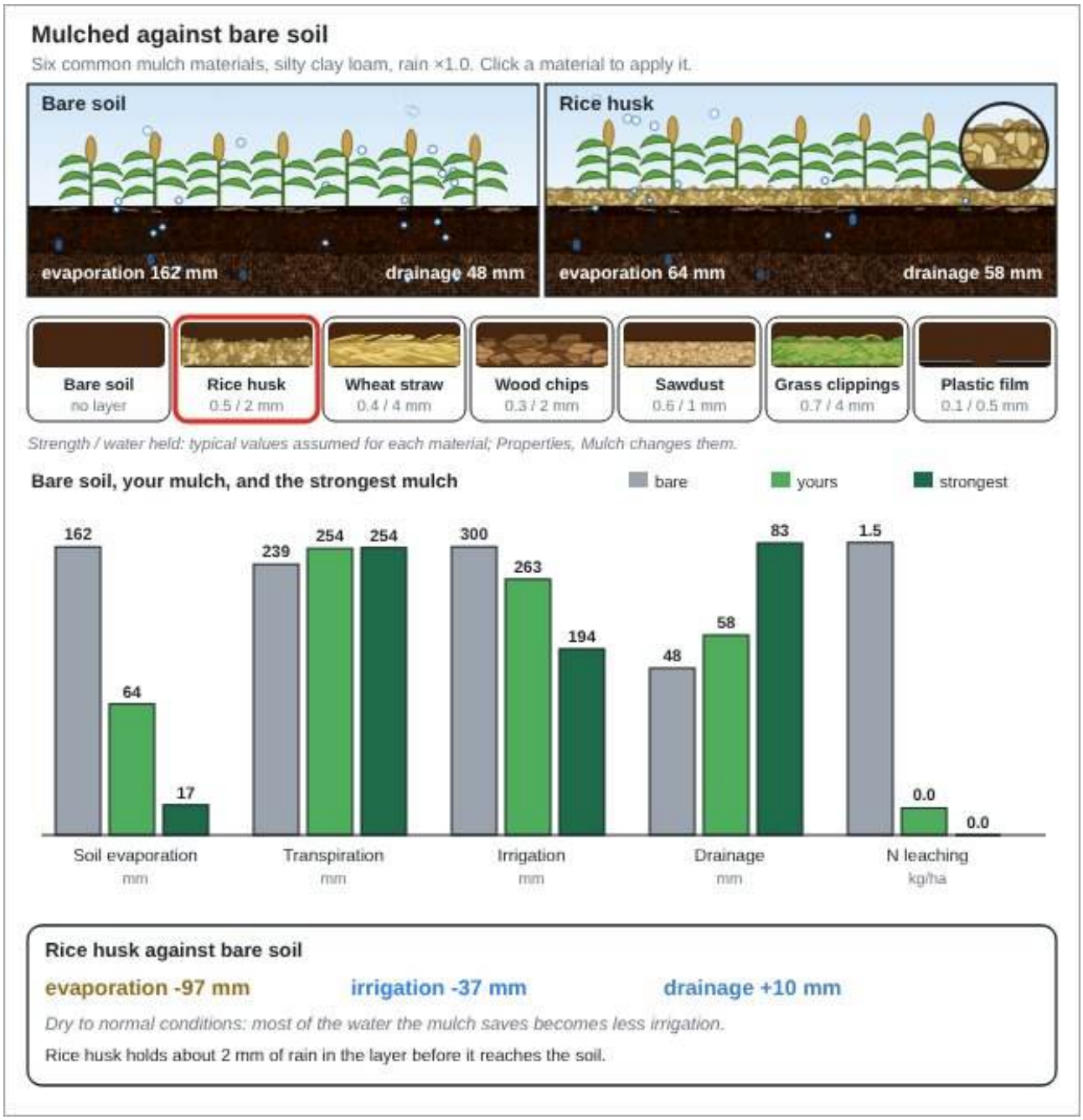


Figure 10.35: Mulch against bare soil, Silty Clay Loam, part 1 of 1.

10.10 Mulch control (Silty Clay Loam)

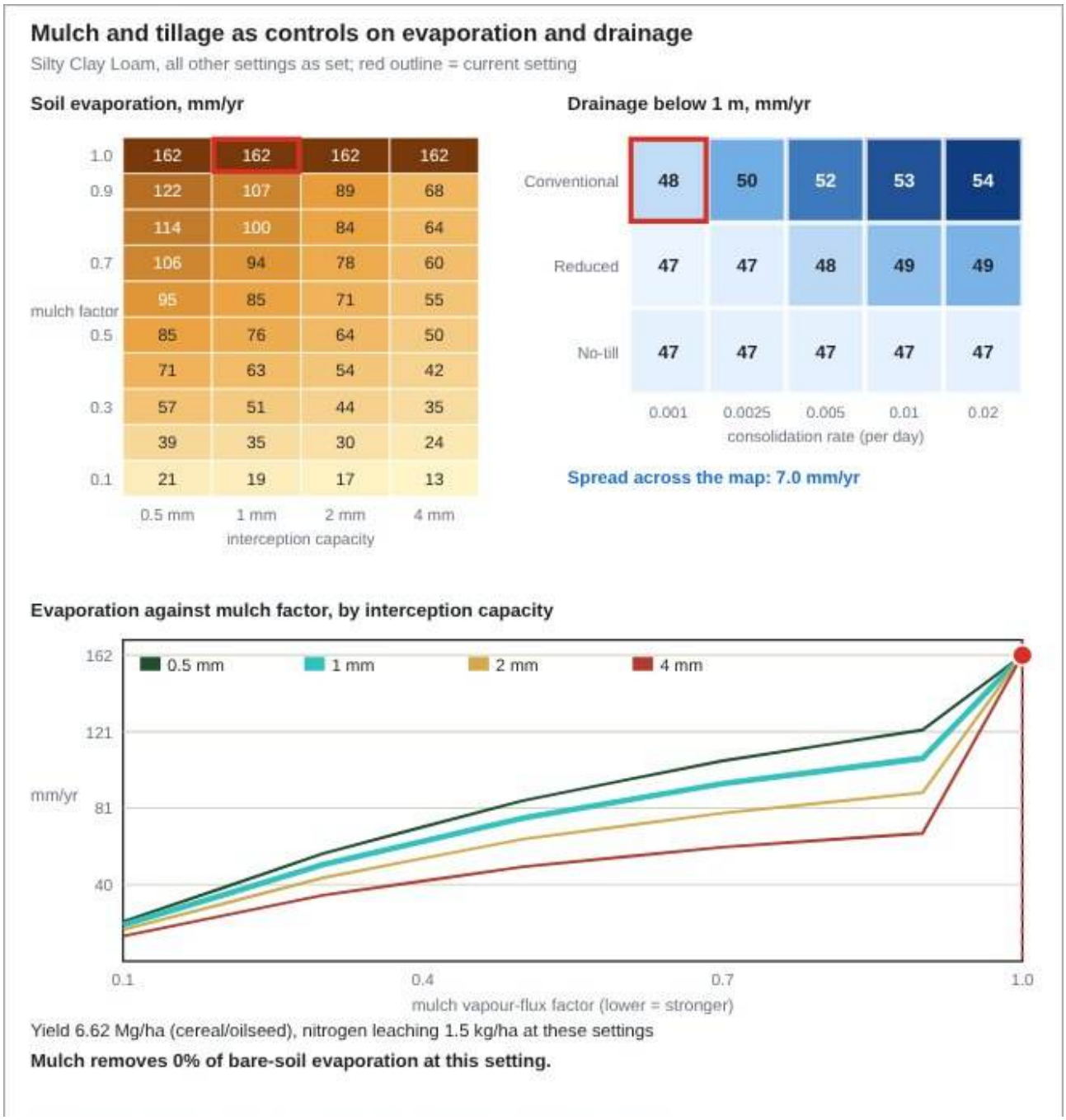


Figure 10.36: Mulch control, Silty Clay Loam, part 1 of 2.

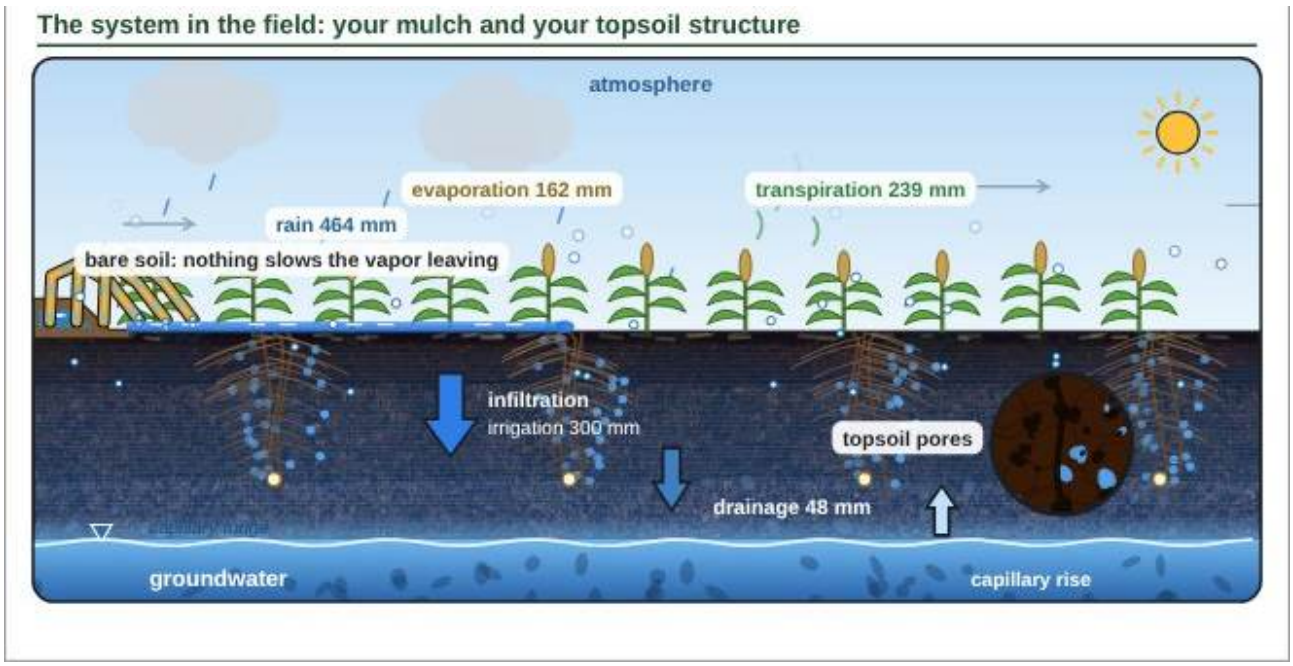


Figure 10.37: Mulch control, Silty Clay Loam, part 2 of 2.

10.11 Evaporation physics (Silty Clay Loam)

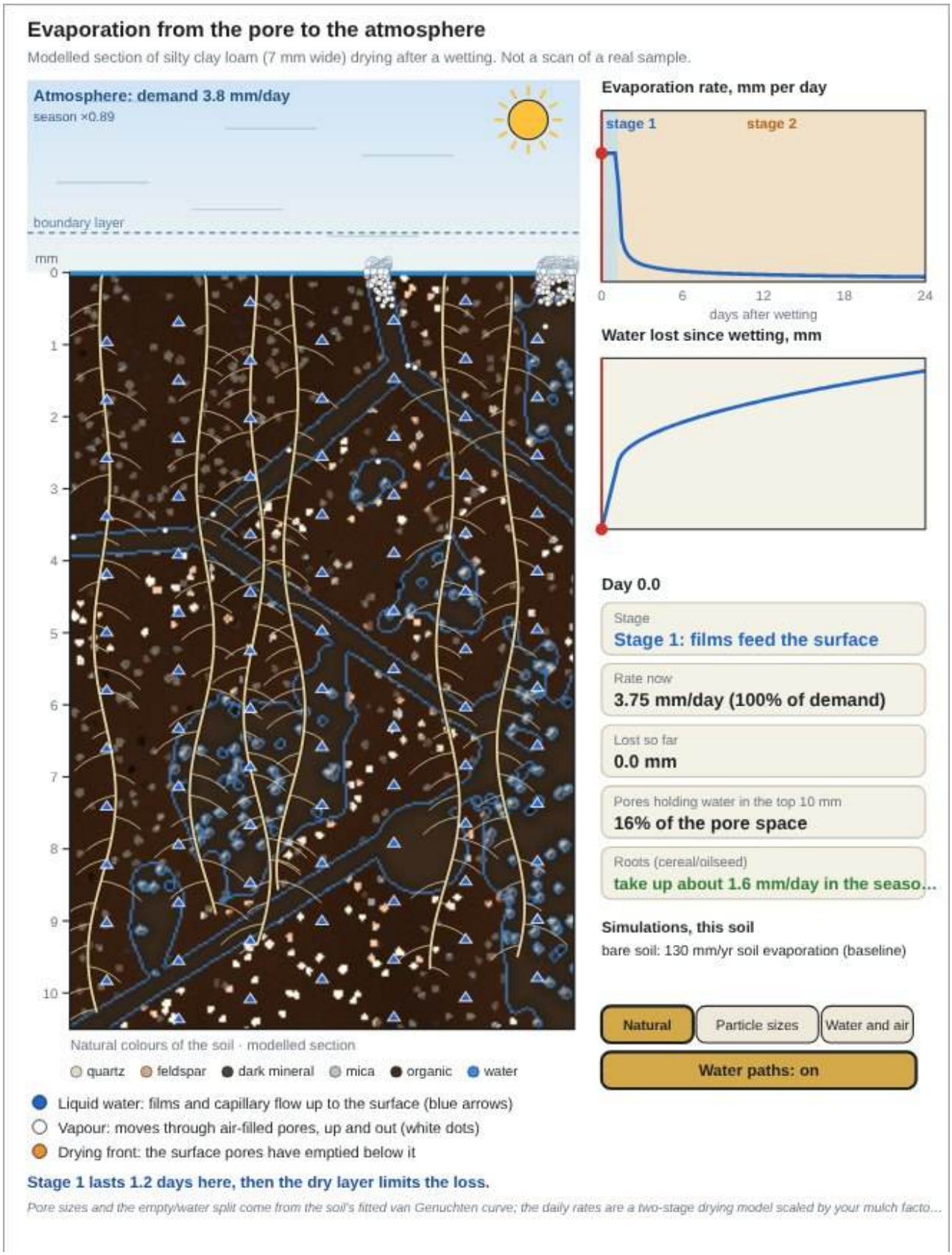


Figure 10.38: Evaporation physics, Silty Clay Loam, part 1 of 3.

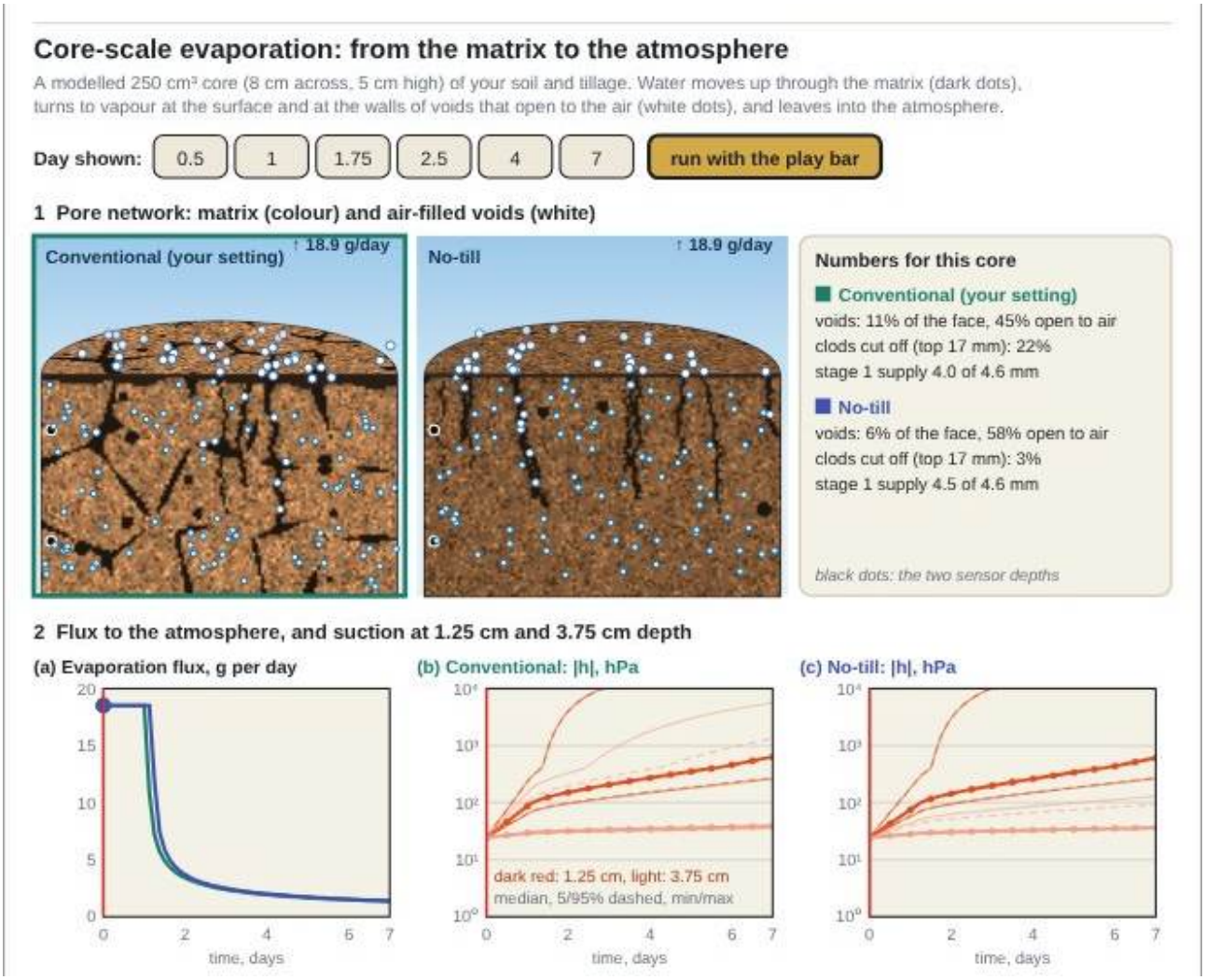


Figure 10.39: Evaporation physics, Silty Clay Loam, part 2 of 3.

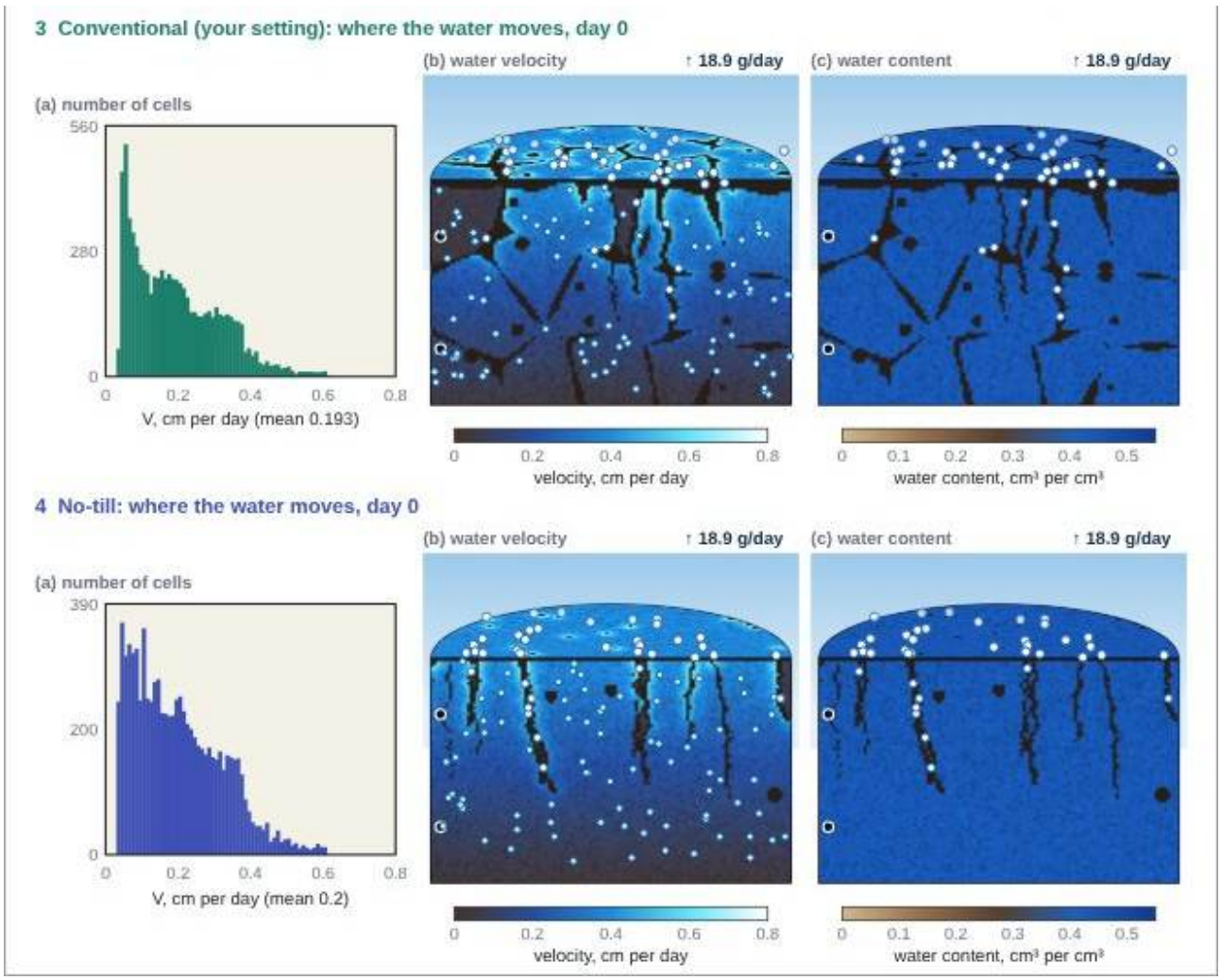


Figure 10.40: Evaporation physics, Silty Clay Loam, part 3 of 3.

10.12 Fate of saved water (Silty Clay Loam)

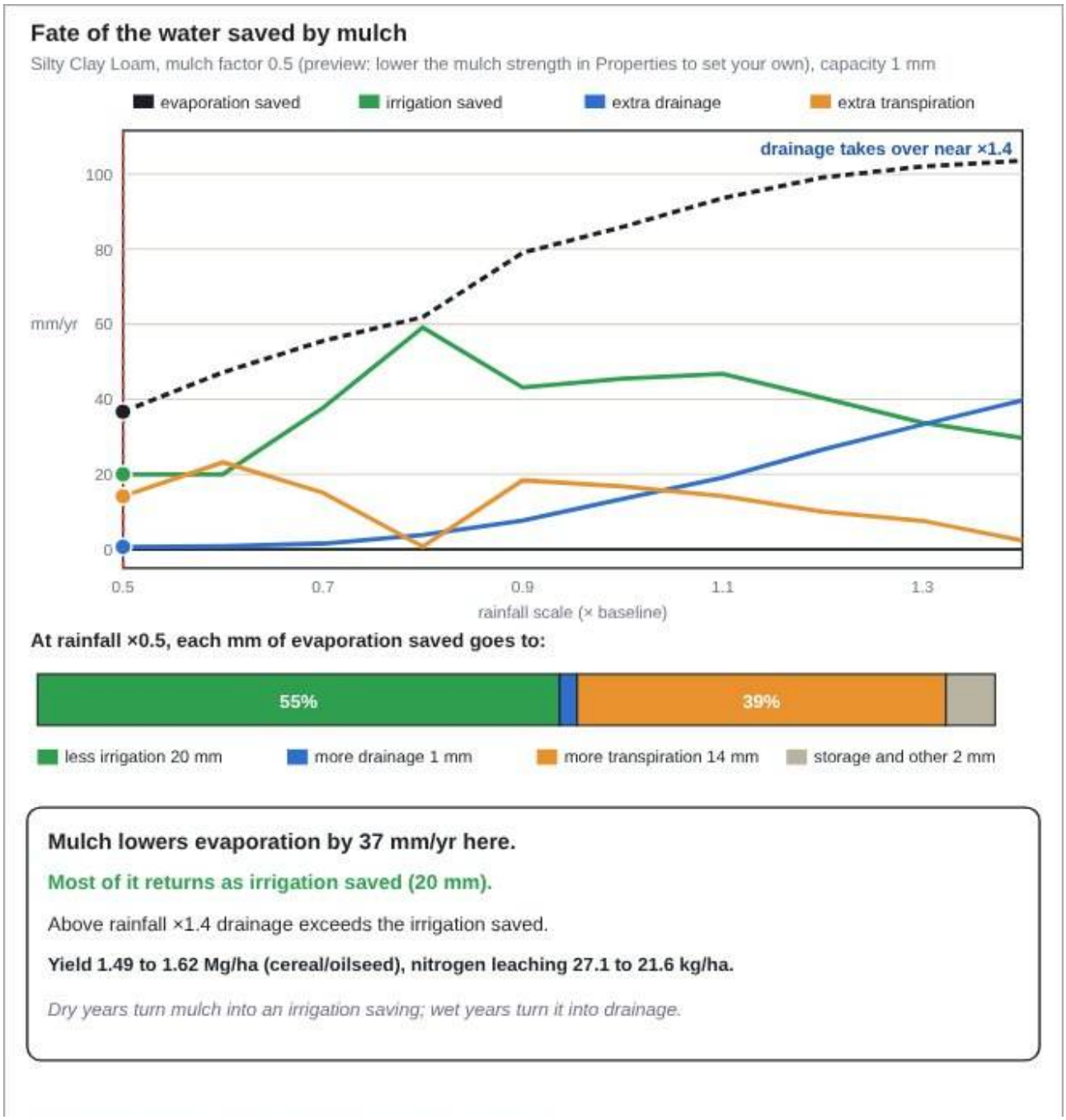


Figure 10.41: Fate of saved water, Silty Clay Loam, part 1 of 2.

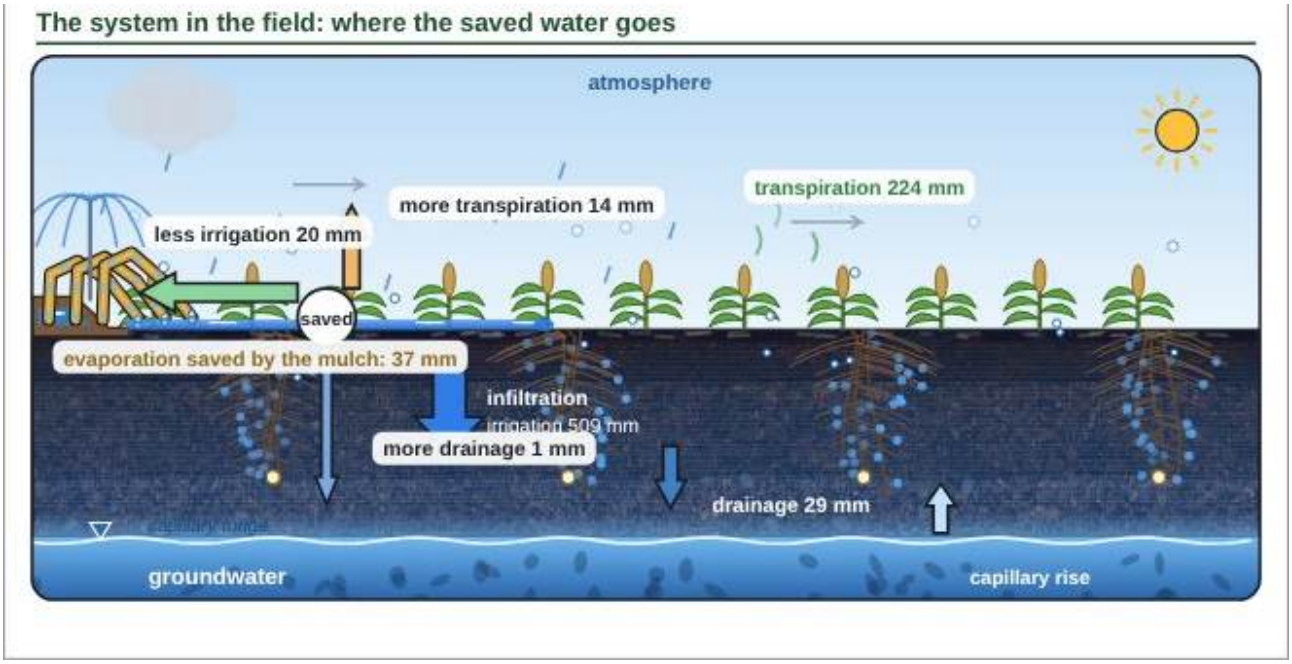


Figure 10.42: Fate of saved water, Silty Clay Loam, part 2 of 2.

10.13 Regime map (Silty Clay Loam)

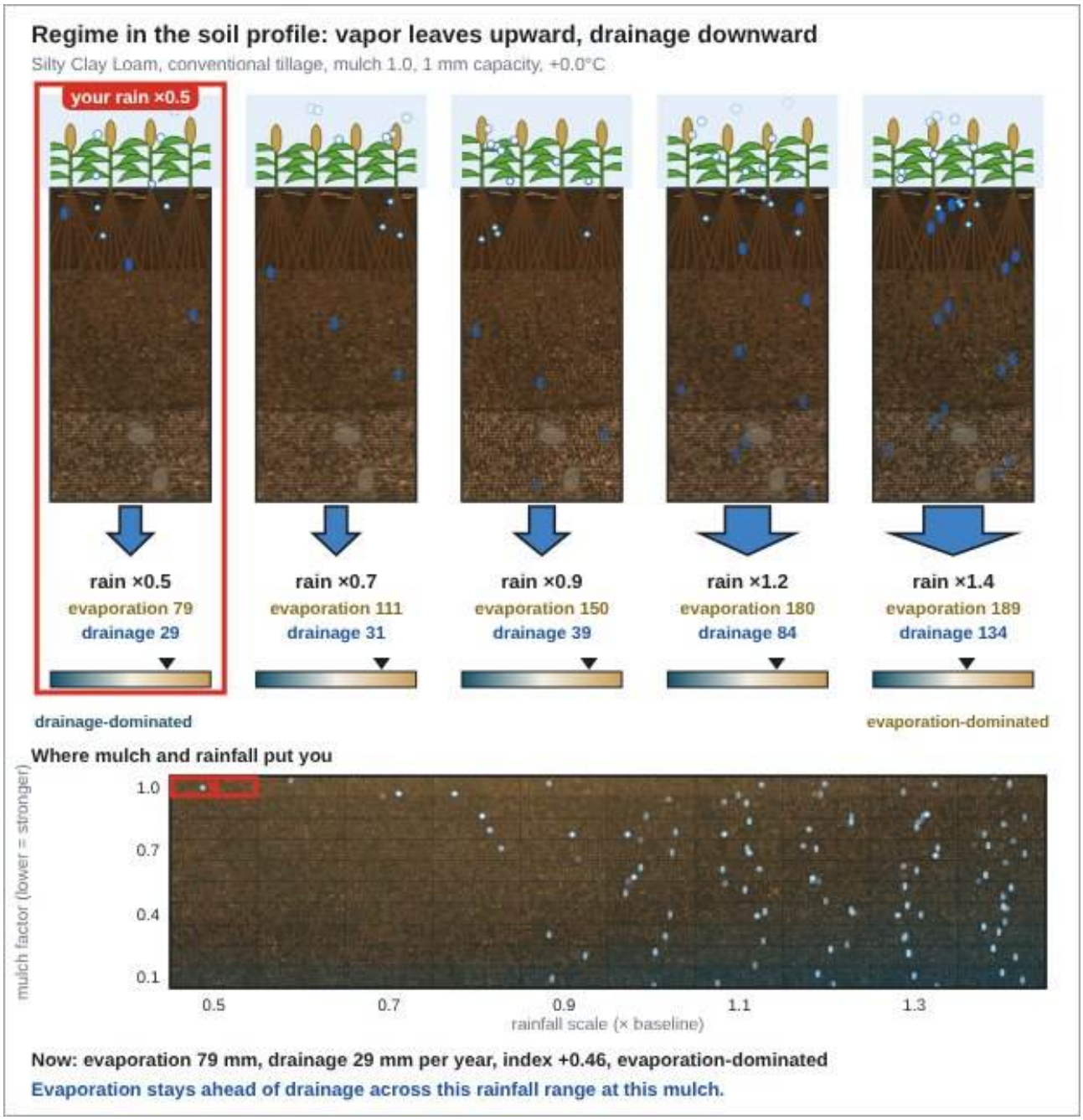


Figure 10.43: Regime map, Silty Clay Loam, part 1 of 1.

10.14 Tillage and structure (Silty Clay Loam)

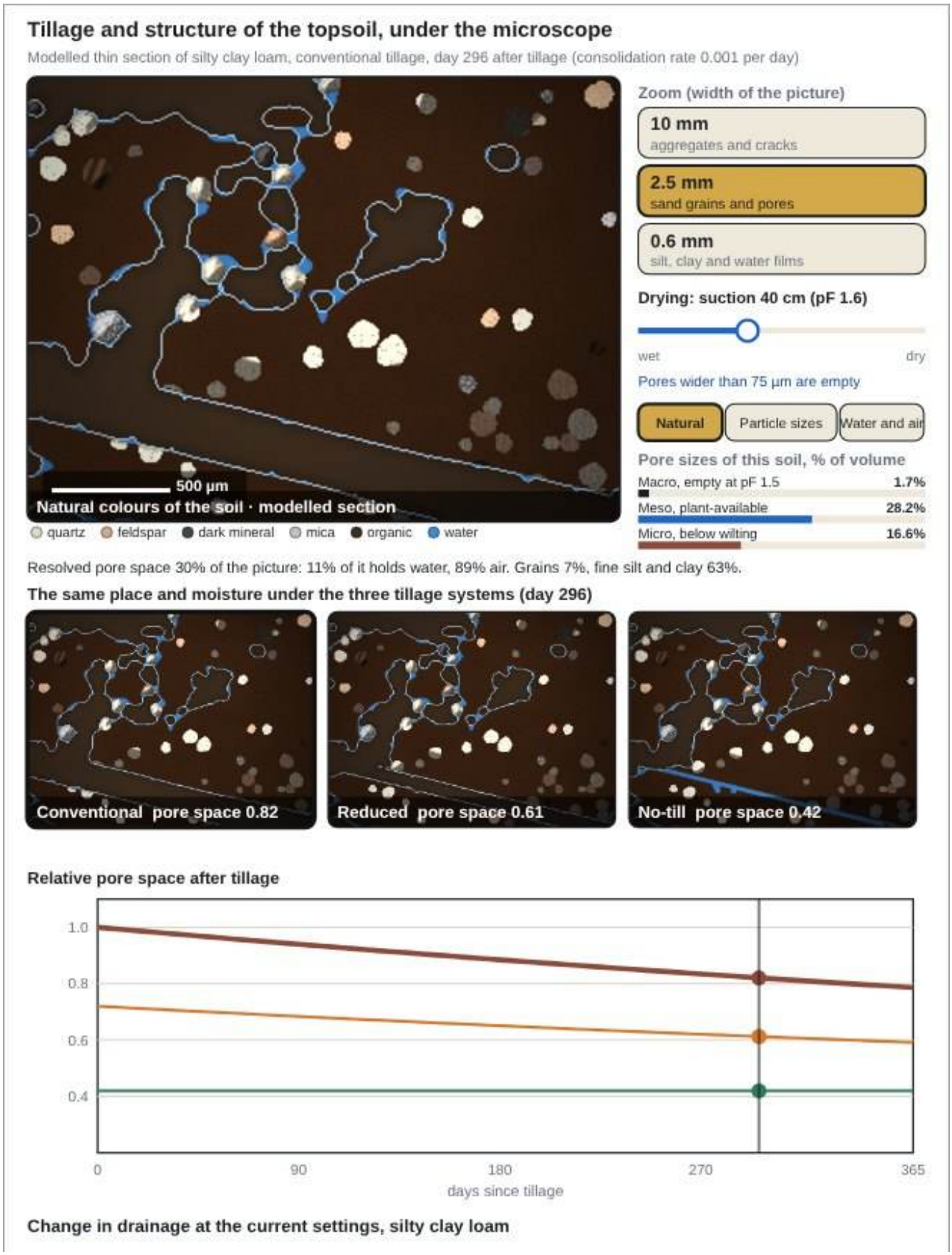


Figure 10.44: Tillage and structure, Silty Clay Loam, part 1 of 2.

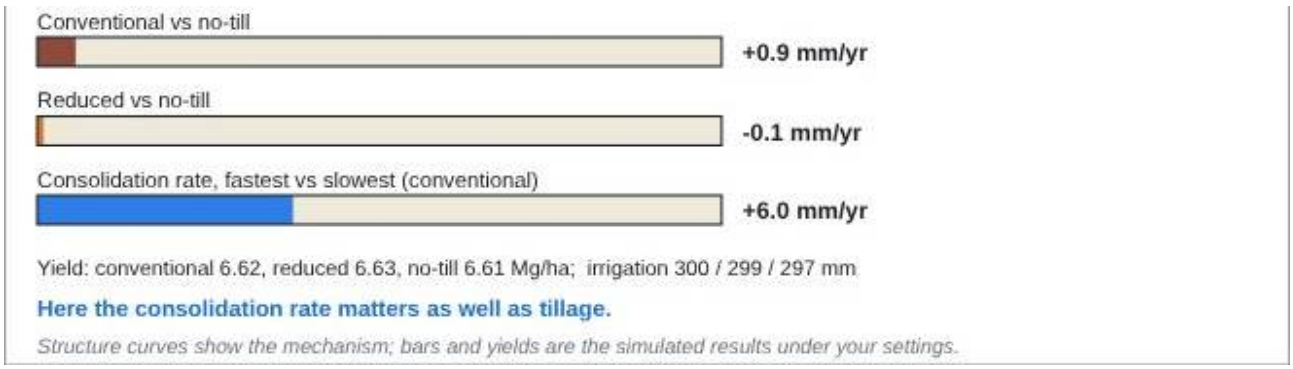


Figure 10.45: Tillage and structure, Silty Clay Loam, part 2 of 2.

10.15 Soil hydraulic drivers (Silty Clay Loam)

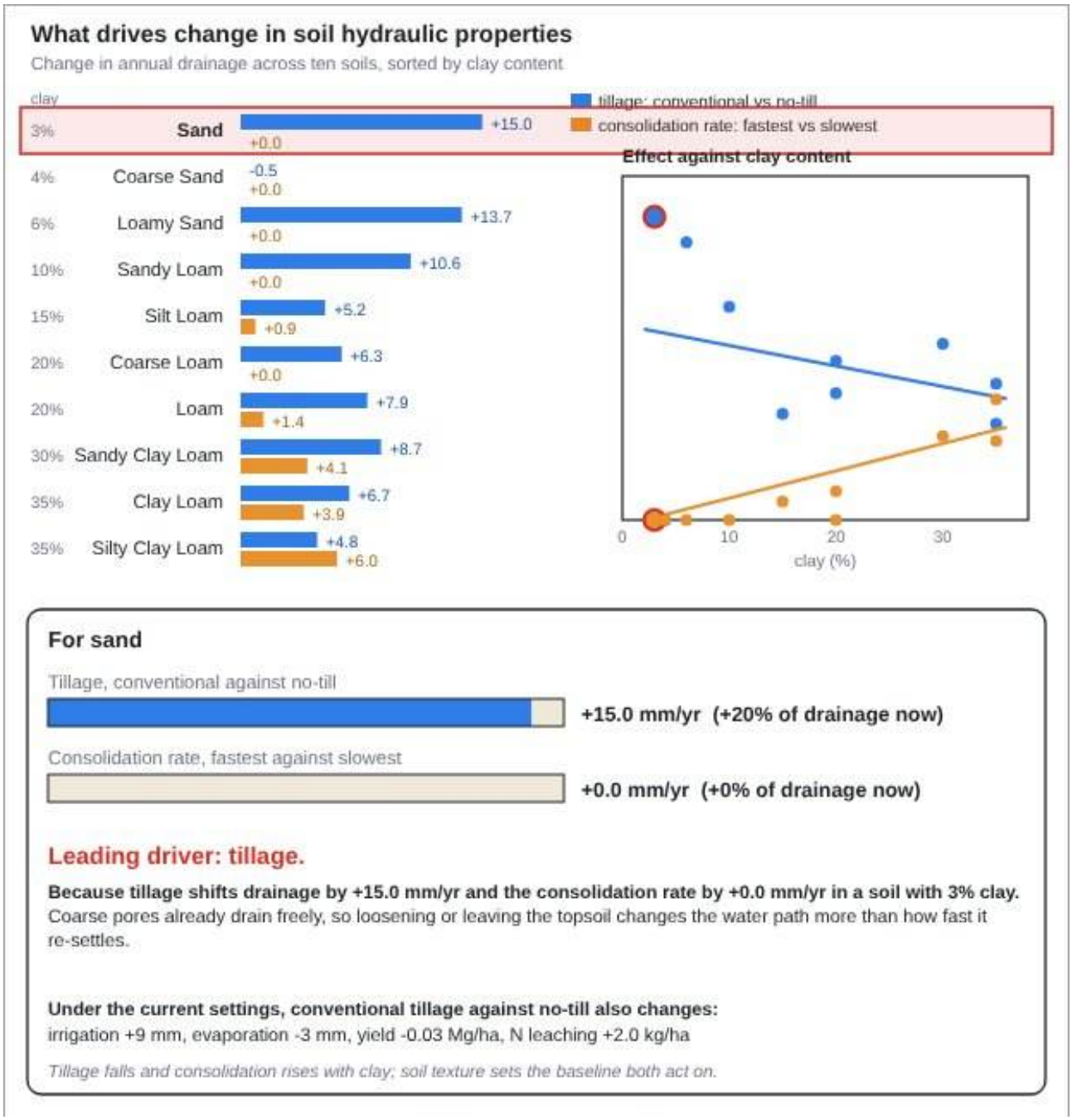


Figure 10.46: Soil hydraulic drivers, Silty Clay Loam, part 1 of 3.

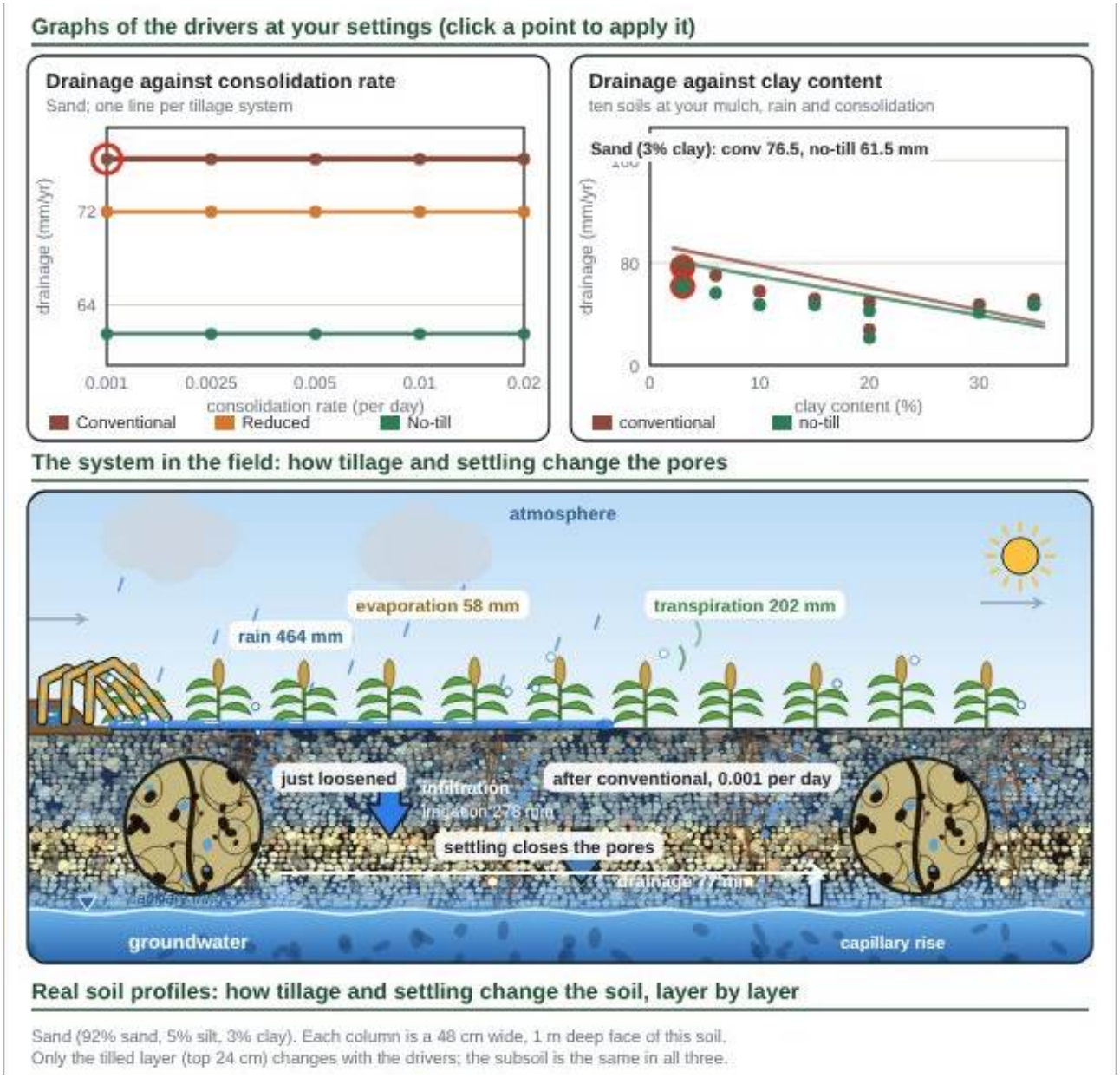


Figure 10.47: Soil hydraulic drivers, Silty Clay Loam, part 2 of 3.

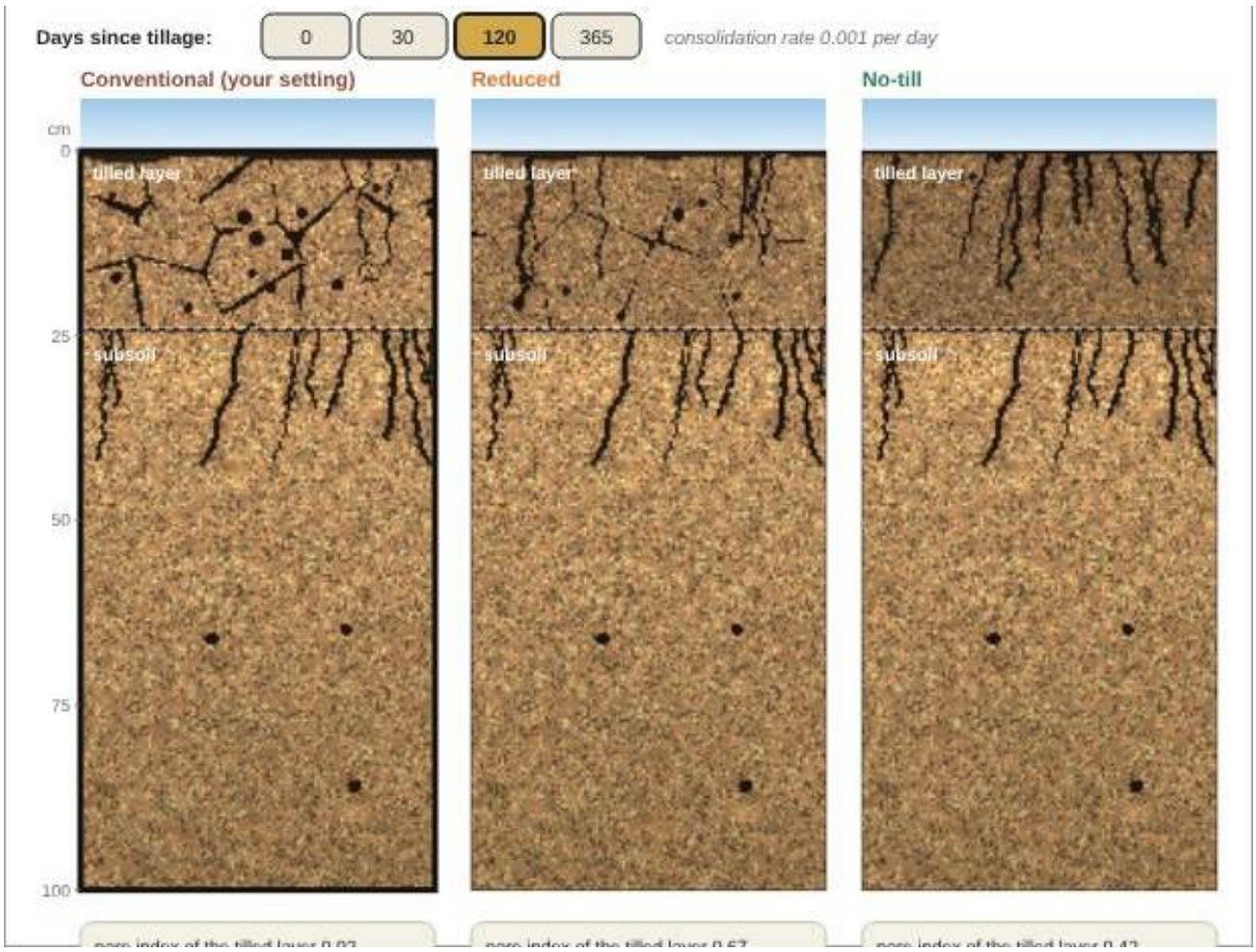


Figure 10.48: Soil hydraulic drivers, Silty Clay Loam, part 3 of 3.

10.16 Root structure (Silty Clay Loam)

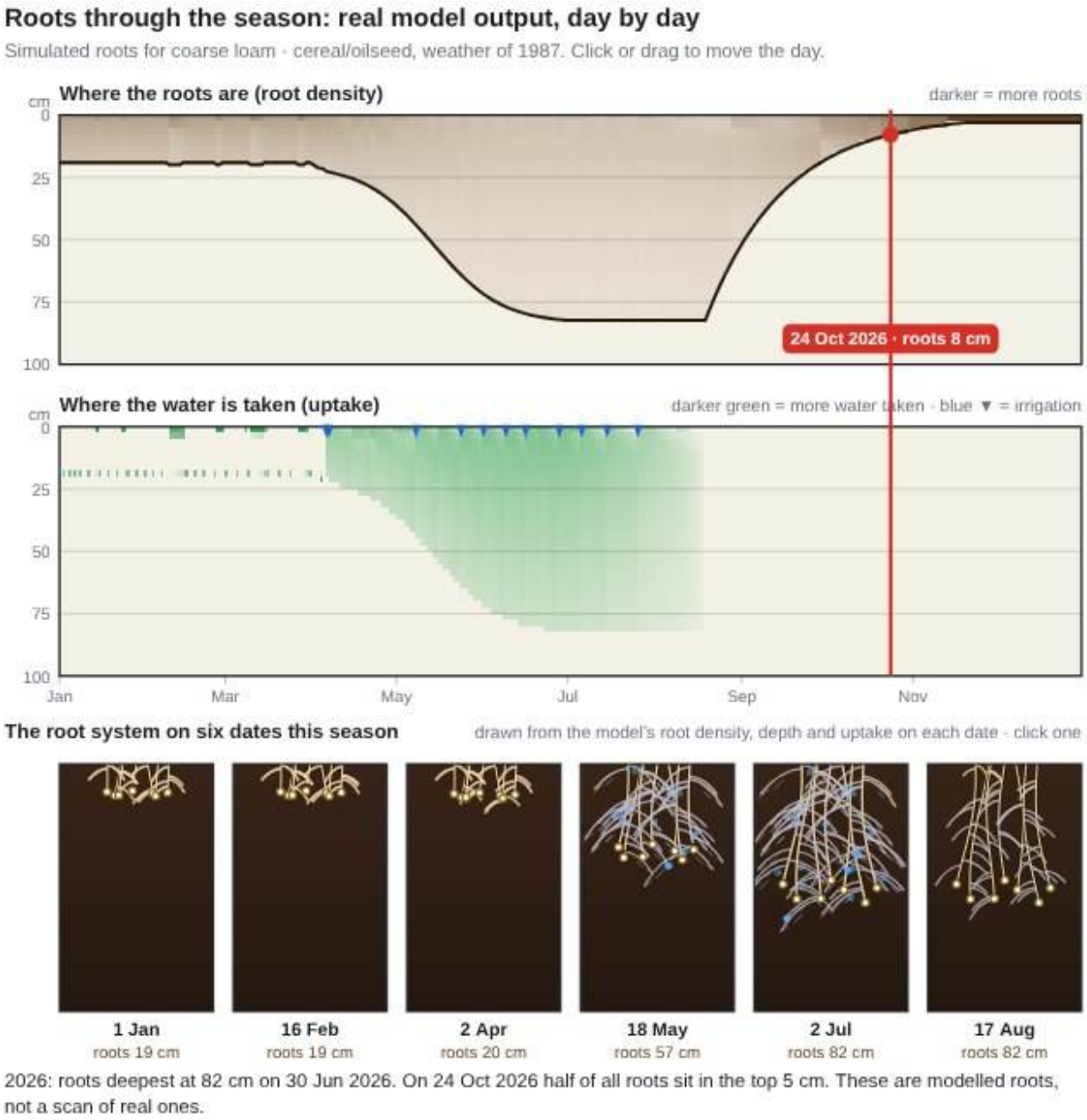


Figure 10.49: Root structure, Silty Clay Loam, part 1 of 4.

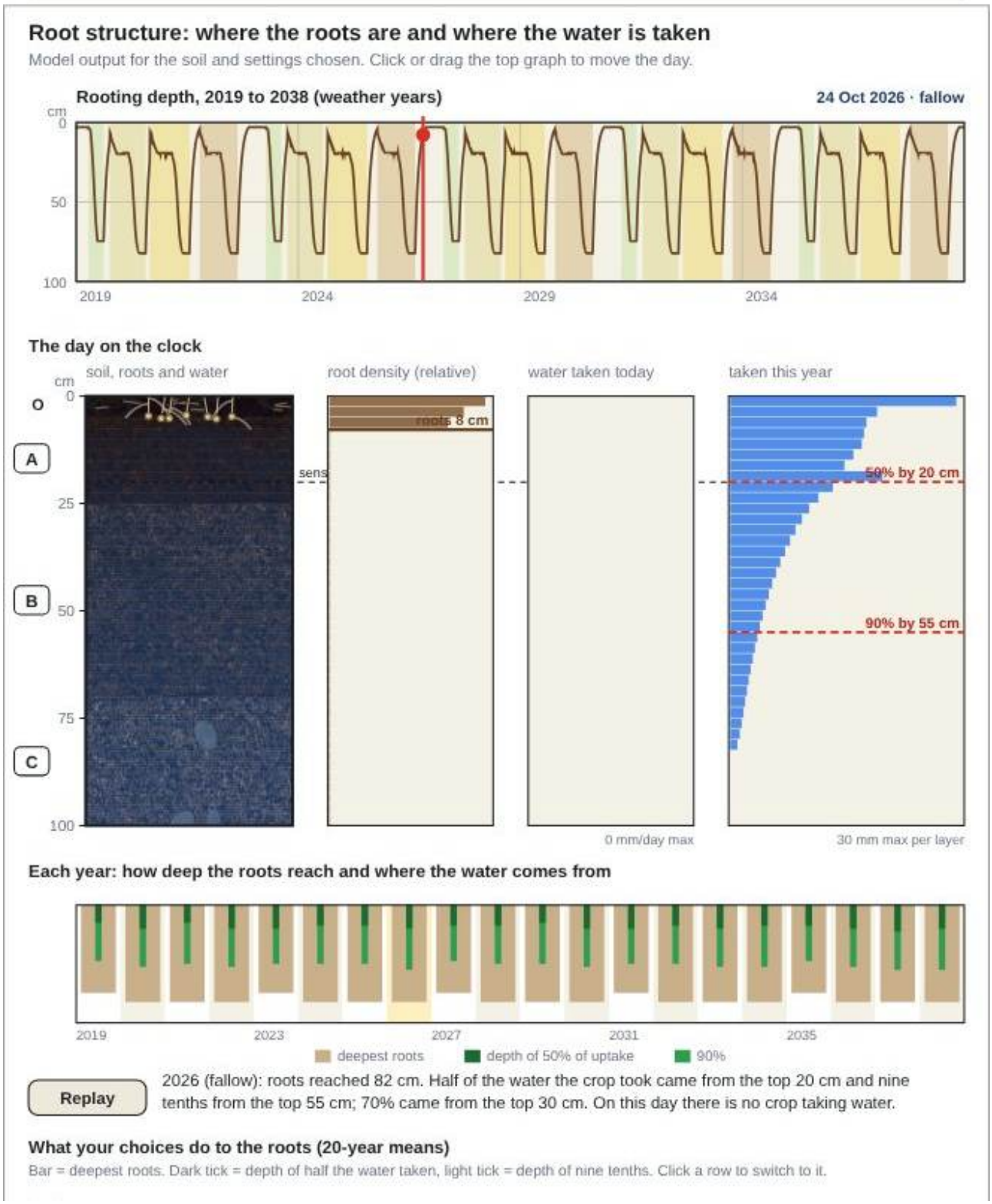


Figure 10.50: Root structure, Silty Clay Loam, part 2 of 4.

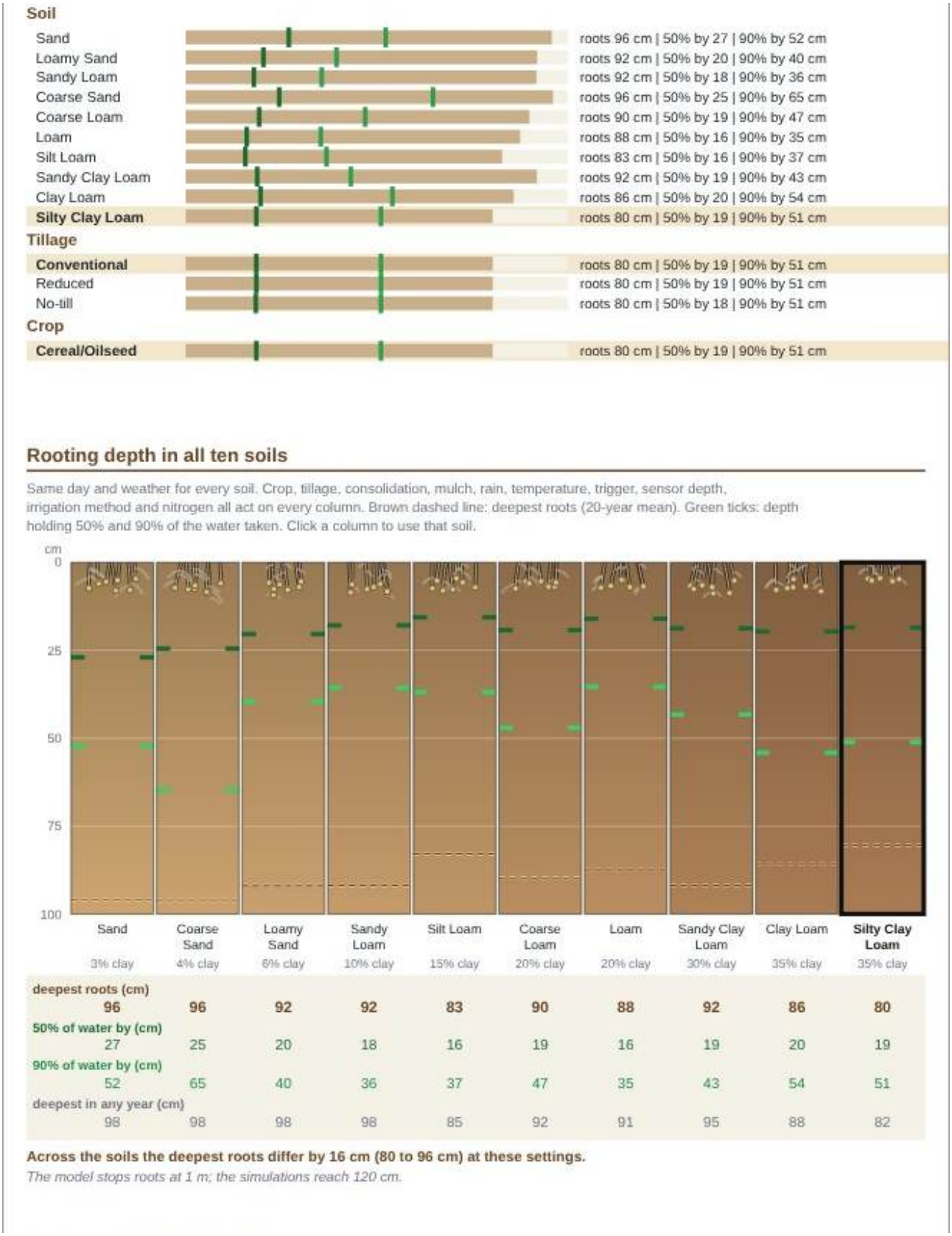


Figure 10.51: Root structure, Silty Clay Loam, part 3 of 4.

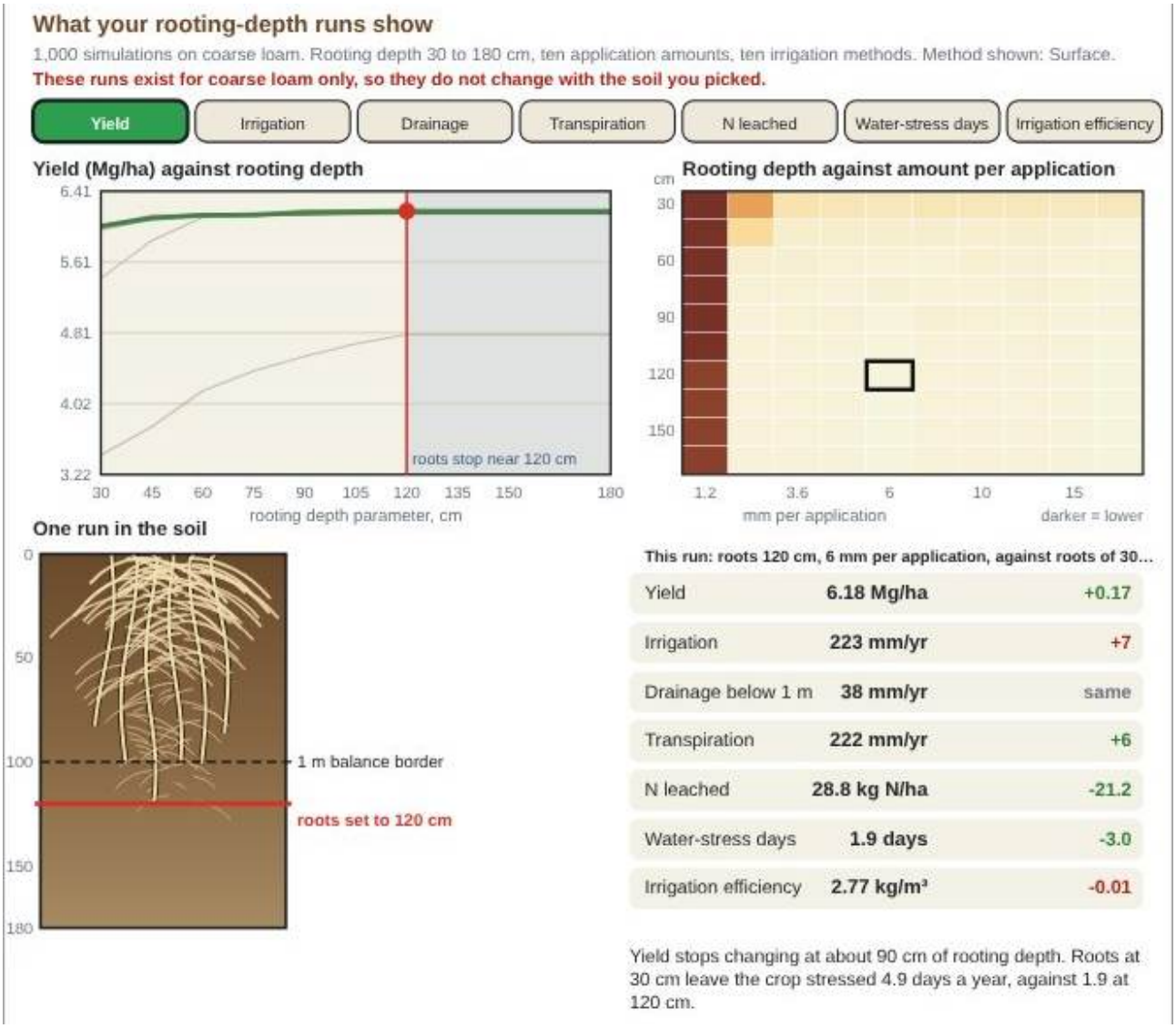


Figure 10.52: Root structure, Silty Clay Loam, part 4 of 4.

10.17 Field balance (Silty Clay Loam)

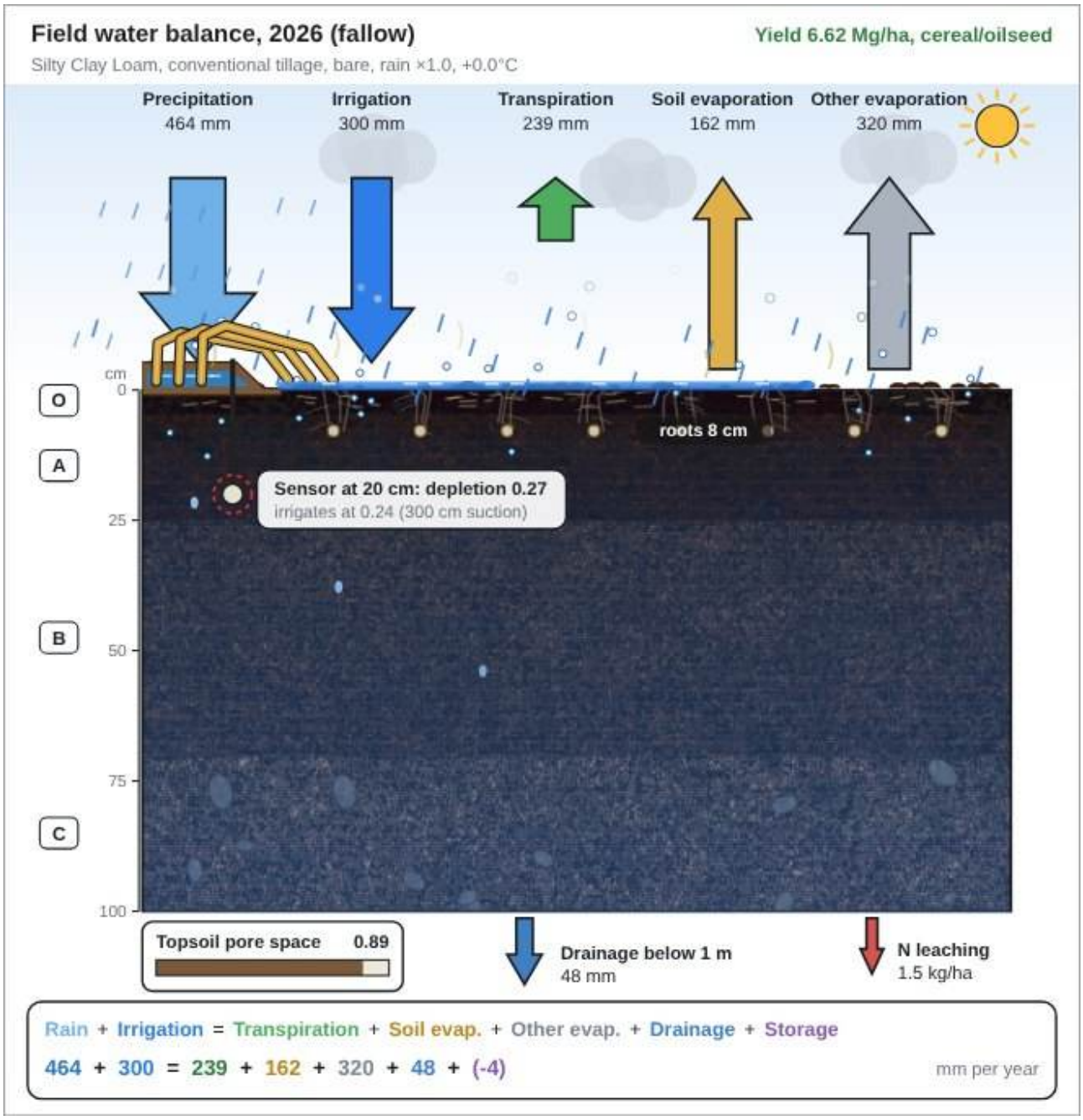


Figure 10.53: Field balance, Silty Clay Loam, part 1 of 1.

10.18 Soil, plant and air (Silty Clay Loam)

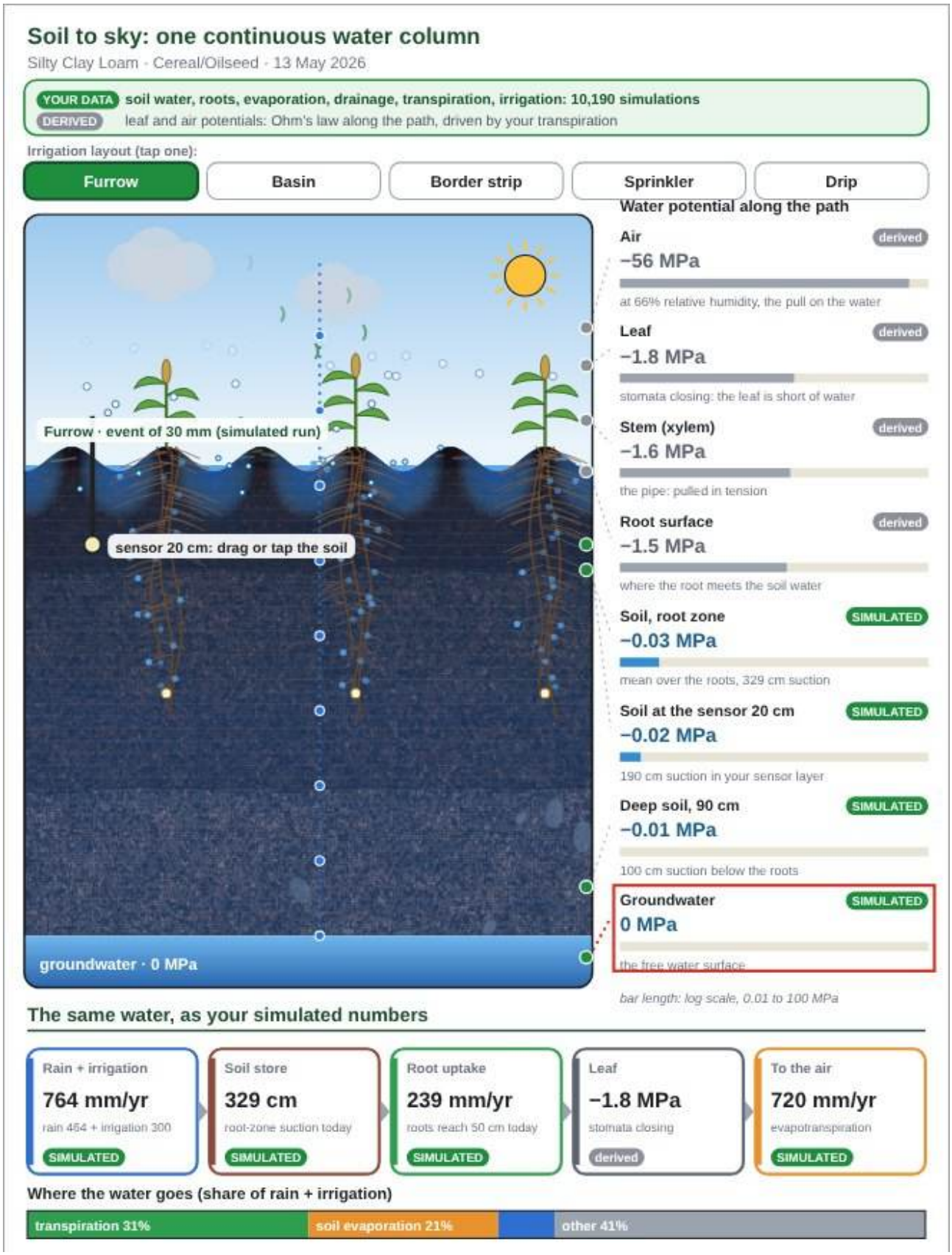


Figure 10.54: Soil, plant and air, Silty Clay Loam, part 1 of 3.

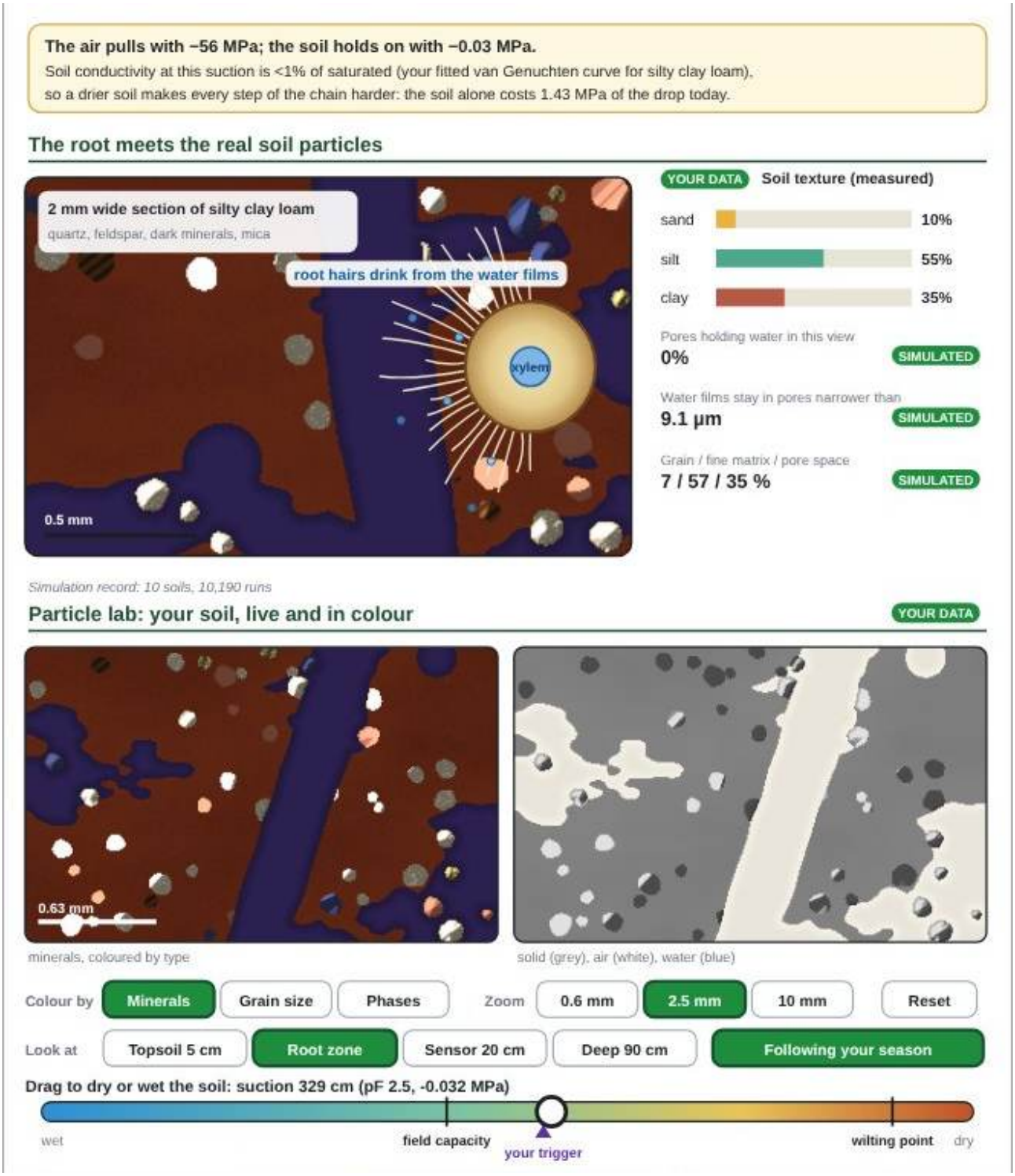


Figure 10.55: Soil, plant and air, Silty Clay Loam, part 2 of 3.

Figure 10.56: Soil, plant and air, Silty Clay Loam, part 3 of 3.

10.19 Profile evolution (Silty Clay Loam)

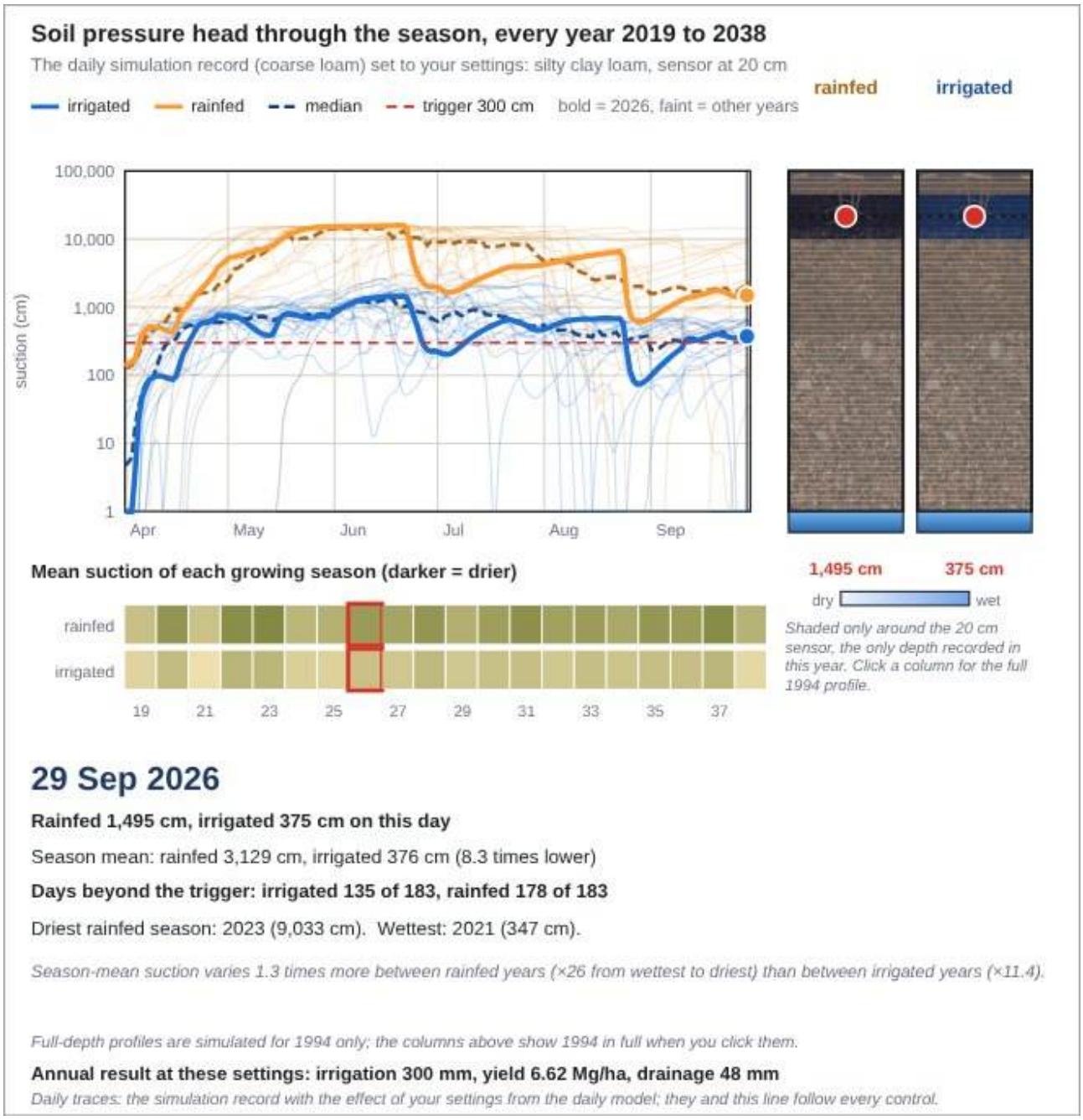


Figure 10.57: Profile evolution, Silty Clay Loam, part 1 of 1.

10.20 Reliability over 20 seasons (Silty Clay Loam)

Figure 10.58: Reliability over 20 seasons, Silty Clay Loam, part 1 of 2.

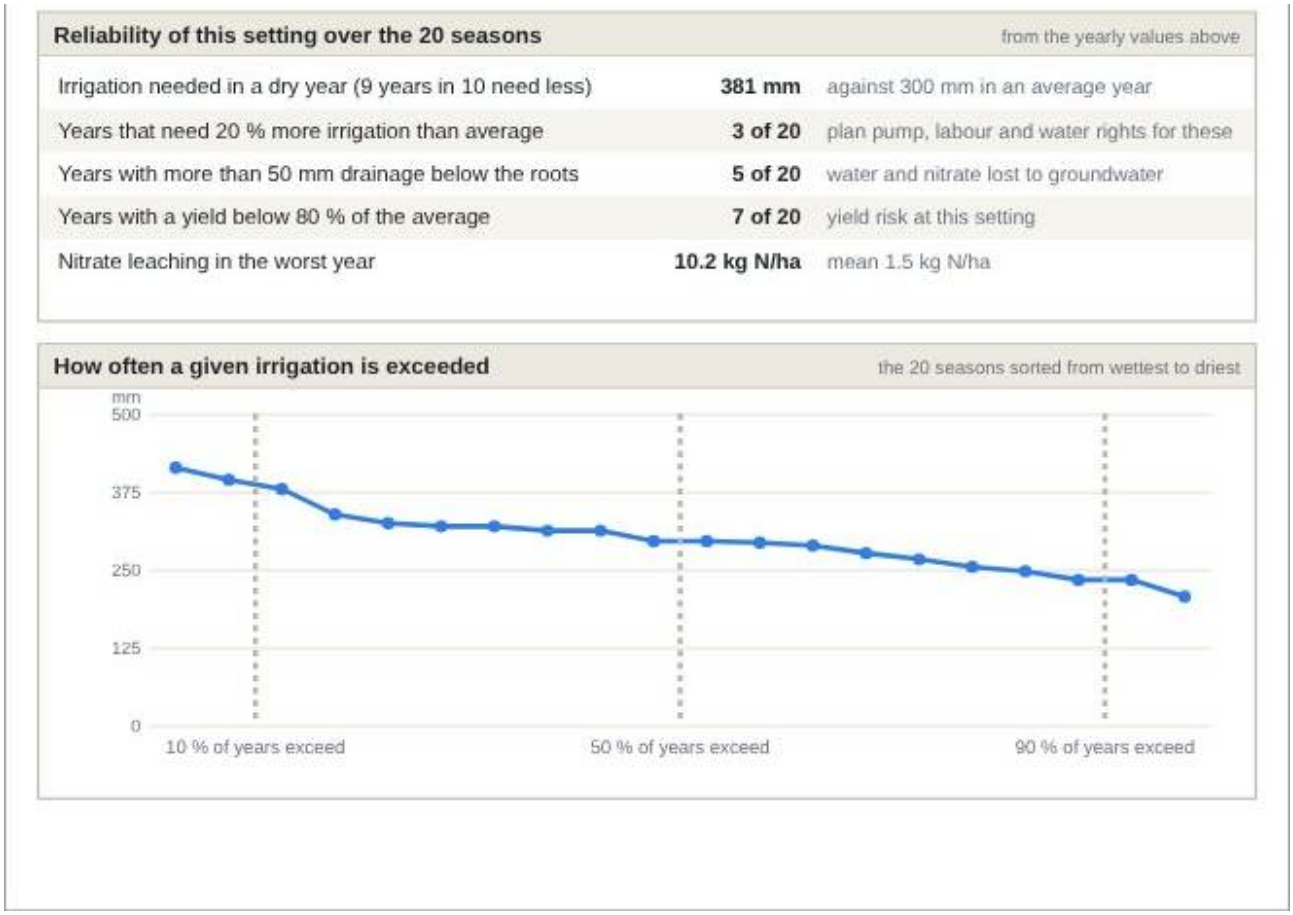


Figure 10.59: Reliability over 20 seasons, Silty Clay Loam, part 2 of 2.

10.21 Full water balance (Silty Clay Loam)

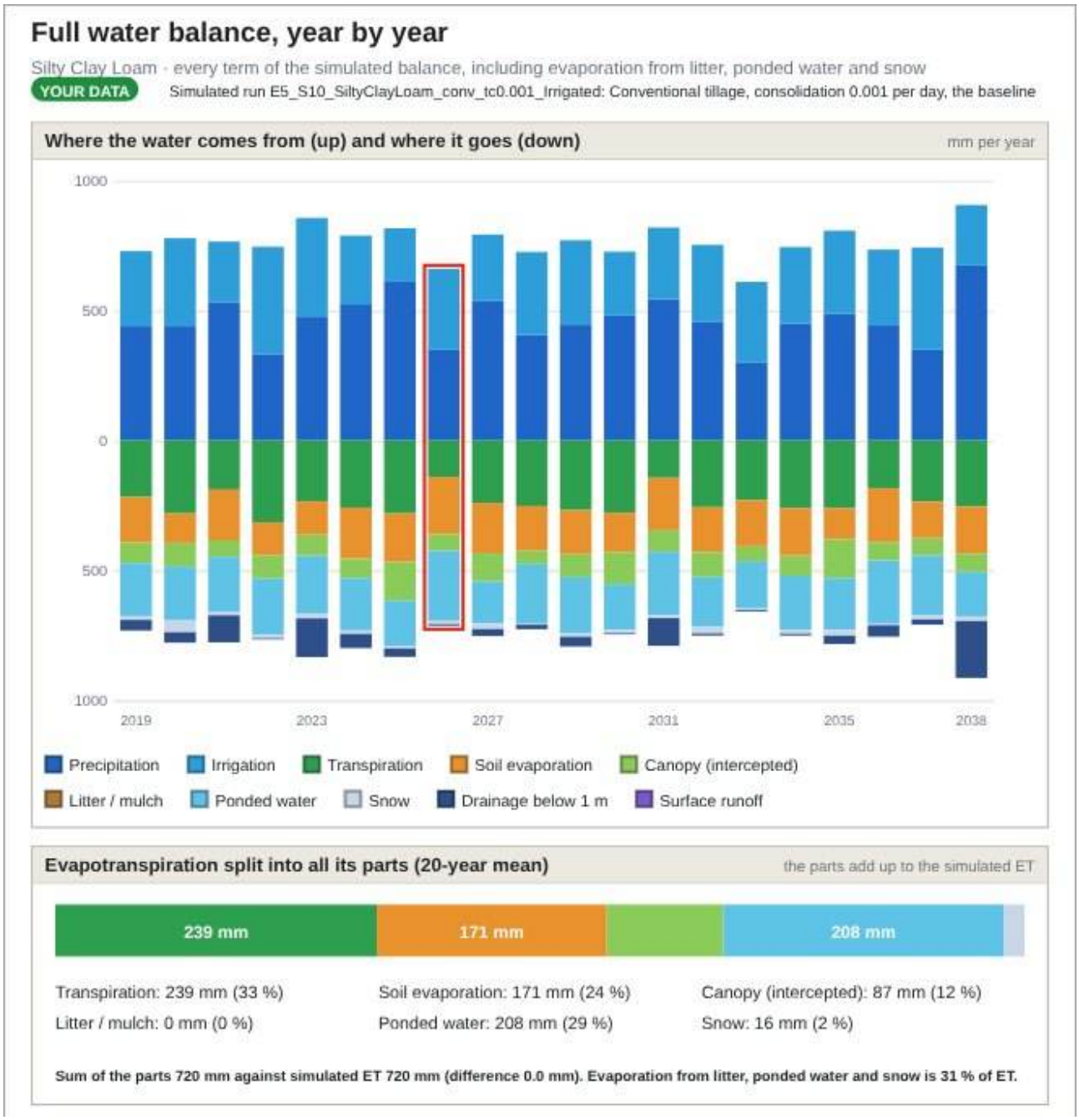


Figure 10.60: Full water balance, Silty Clay Loam, part 1 of 2.

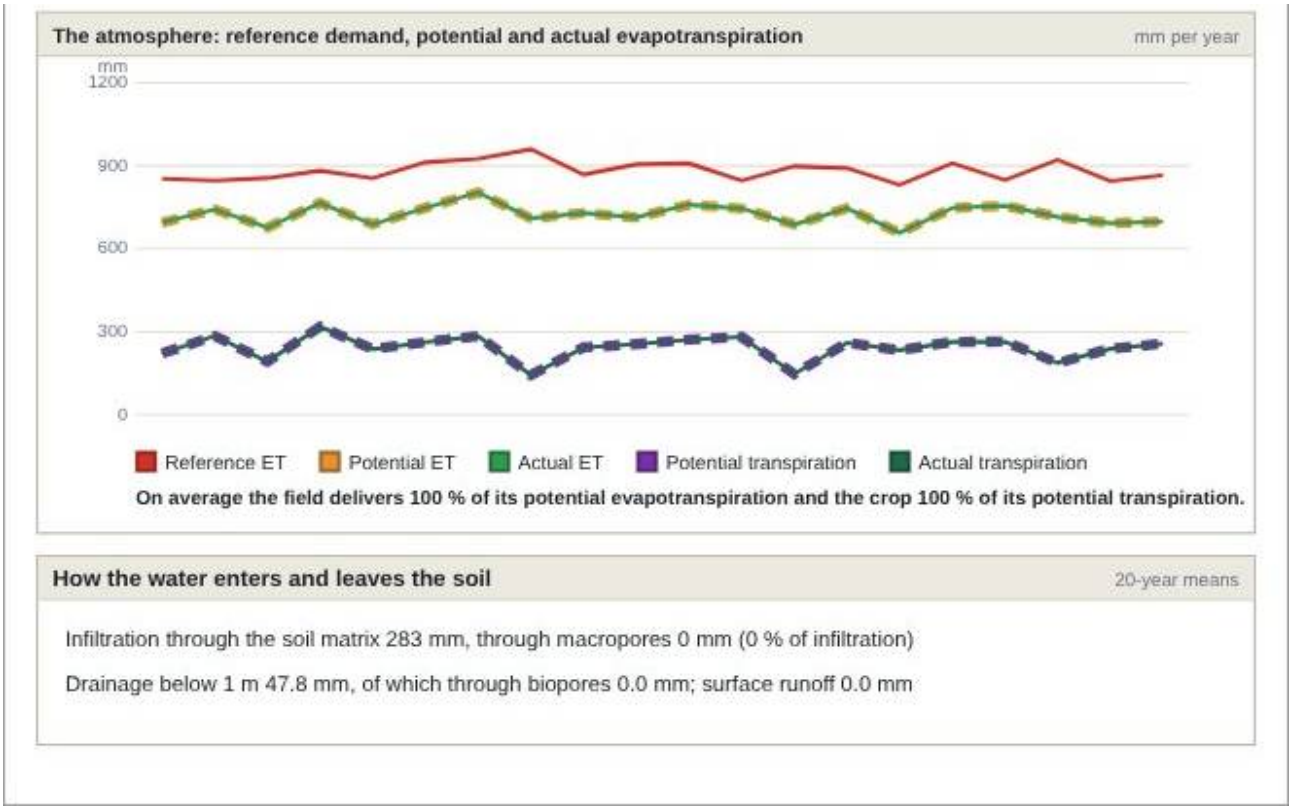


Figure 10.61: Full water balance, Silty Clay Loam, part 2 of 2.

10.22 Nitrogen budget (Silty Clay Loam)

Figure 10.62: Nitrogen budget, Silty Clay Loam, part 1 of 2.

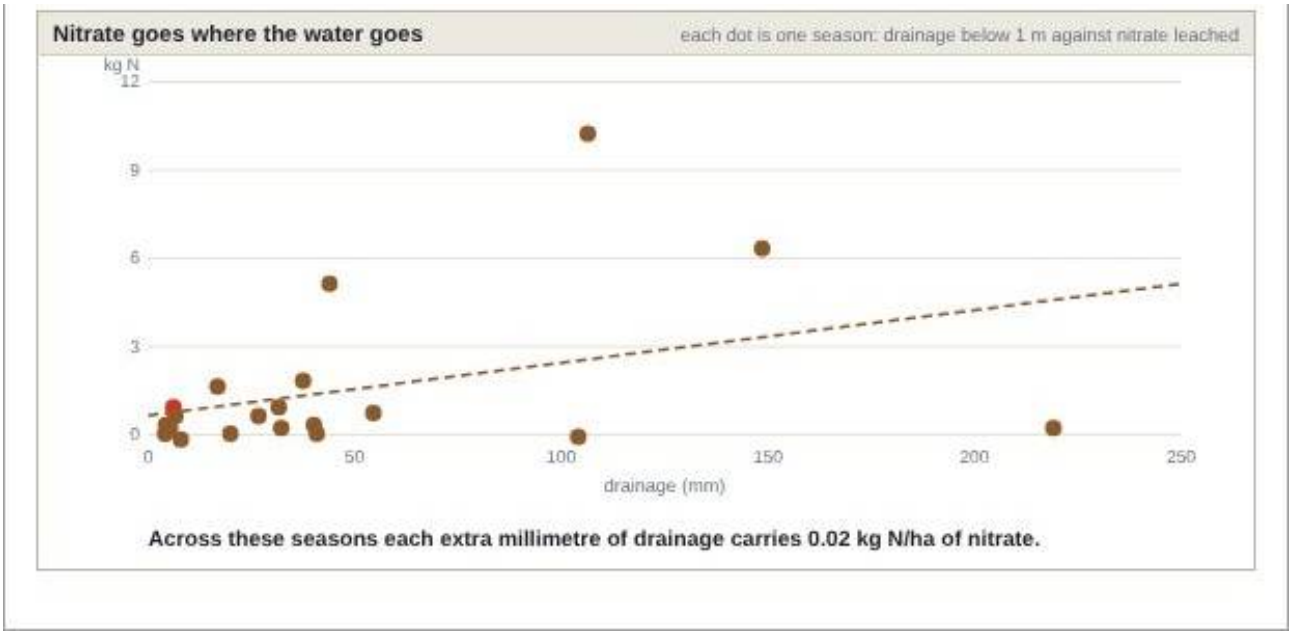


Figure 10.63: Nitrogen budget, Silty Clay Loam, part 2 of 2.

10.23 Crop water requirement (Silty Clay Loam)

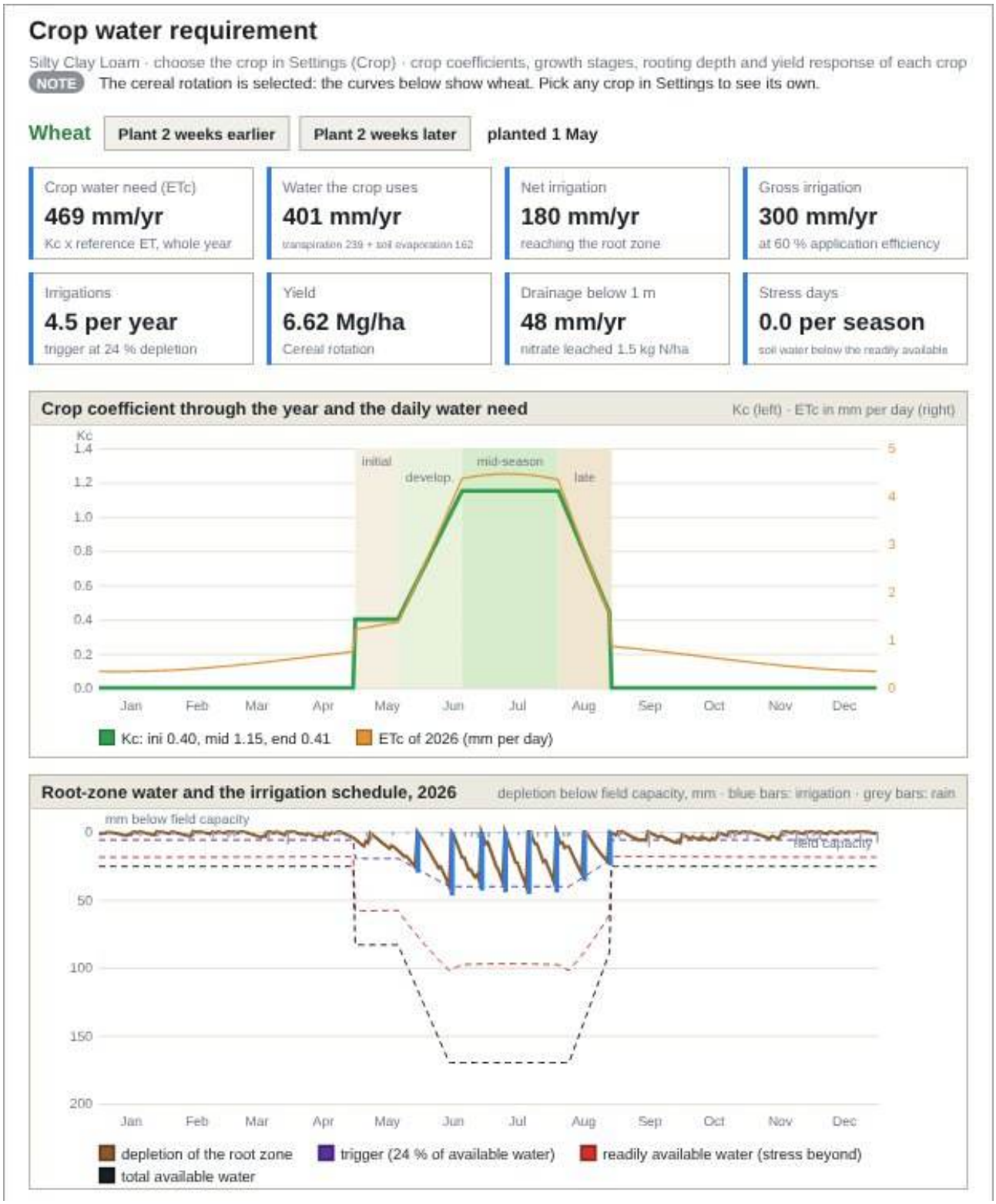


Figure 10.64: Crop water requirement, Silty Clay Loam, part 1 of 3.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**510**	14.3	100 %
Papaya (high)	814	**420**	11.2	100 %
Coconut (high)	834	**418**	10.0	100 %
Oil palm (young)	790	**383**	10.2	100 %
Yam	649	**339**	14.0	100 %
Onion (dry)	565	**307**	15.8	100 %
Rice	573	**302**	9.4	100 %
Avocado (medium)	677	**298**	9.5	100 %
Kiwi (pergola)	626	**292**	7.0	100 %
Cotton	571	**289**	7.4	100 %
Cabbage	559	**281**	10.6	100 %
Guava (high)	651	**278**	6.7	100 %
Table grapes (horizontal trellis)	597	**267**	6.4	100 %
Potato	508	**260**	12.5	100 %
Tomato	521	**255**	6.5	100 %
Hop (v trellis)	520	**240**	5.8	100 %
Mango (medium)	620	**239**	5.7	100 %
Walnut (medium)	523	**224**	5.5	100 %
Pear (medium)	528	**222**	5.4	100 %
Groundnut	471	**221**	7.4	100 %
Date palm (medium)	602	**221**	5.3	100 %
Mustard	468	**217**	5.5	100 %
Sweet pepper	475	**215**	7.0	100 %
Soybean	473	**215**	5.7	100 %
Wheat	469	**214**	5.4	100 %
Cherry (medium)	503	**202**	4.9	100 %
Pistachio (medium)	503	**202**	4.9	100 %

Figure 10.65: Crop water requirement, Silty Clay Loam, part 2 of 3.

Cassava	466		**201**	7.6	100 %
Maize (grain)	451		**198**	4.9	100 %
Chickpea	436		**183**	5.7	100 %
Cherimoya (medium)	545		**179**	4.3	100 %
Hazelnut (low)	481		**178**	4.3	100 %
Pecan (low)	481		**178**	4.3	100 %
Sorghum (grain)	429		**170**	4.2	100 %
Citrus (medium density, short)	552		**167**	4.0	100 %
Cowpea (beans)	398		**153**	4.8	100 %
Apple (medium)	459		**152**	3.7	100 %
Millet	395		**152**	3.8	100 %
Wine grapes (pergola)	457		**144**	3.5	100 %
Peach (low)	435		**132**	3.2	100 %
Almond (medium)	435		**132**	3.2	100 %
Green gram (mung bean)	372		**131**	4.3	100 %
Apricot (young)	380		**88**	2.1	100 %
Olive (traditional, medium density)	446		**80**	1.9	100 %
Highbush blueberry (low)	338		**72**	1.9	100 %
Persimmon (young, low)	358		**70**	1.7	100 %
Pomegranate (low)	358		**70**	1.7	100 %
Fig tree (young, low)	327		**47**	1.1	100 %

Figure 10.66: Crop water requirement, Silty Clay Loam, part 3 of 3.

10.24 Feedback loops (Silty Clay Loam)

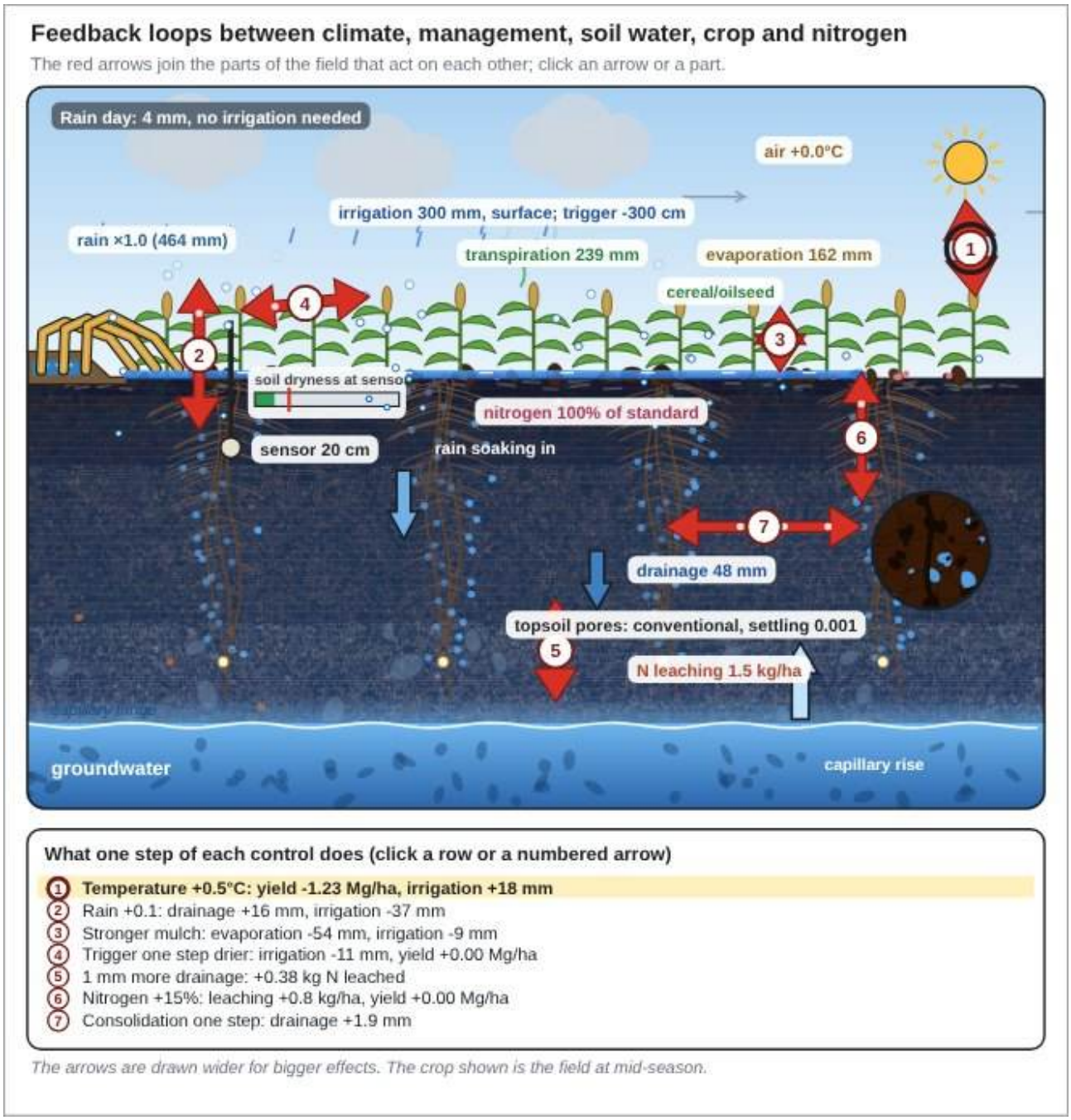


Figure 10.67: Feedback loops, Silty Clay Loam, part 1 of 1.

Chapter 11

The crop water requirement of every crop

11.1 Almond

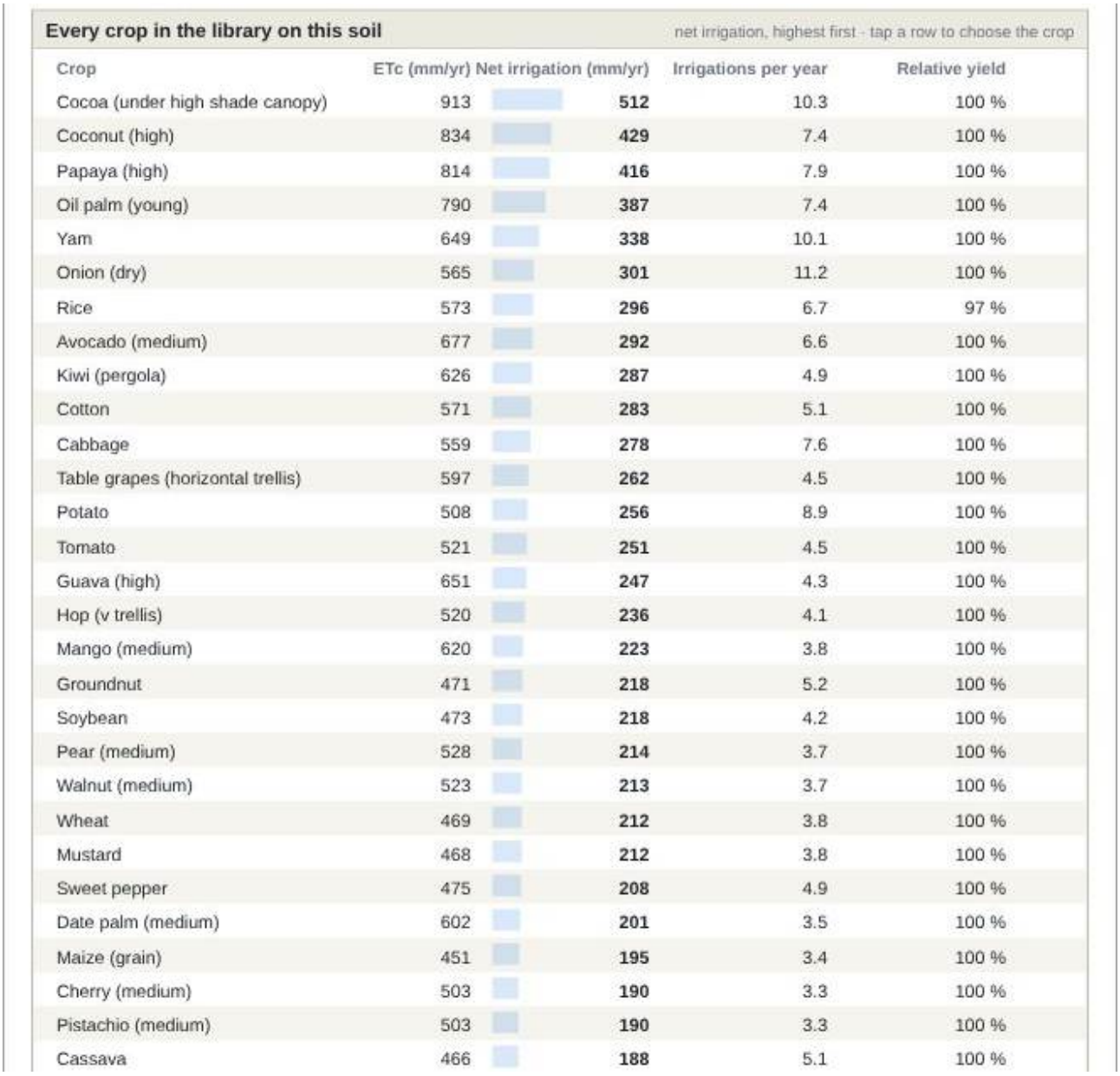
Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.2: Crop water requirement, Almond, part 2.

11.2 Apple

Figure 11.3: Crop water requirement, Apple, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.4: Crop water requirement, Apple, part 2.

11.3 Apricot

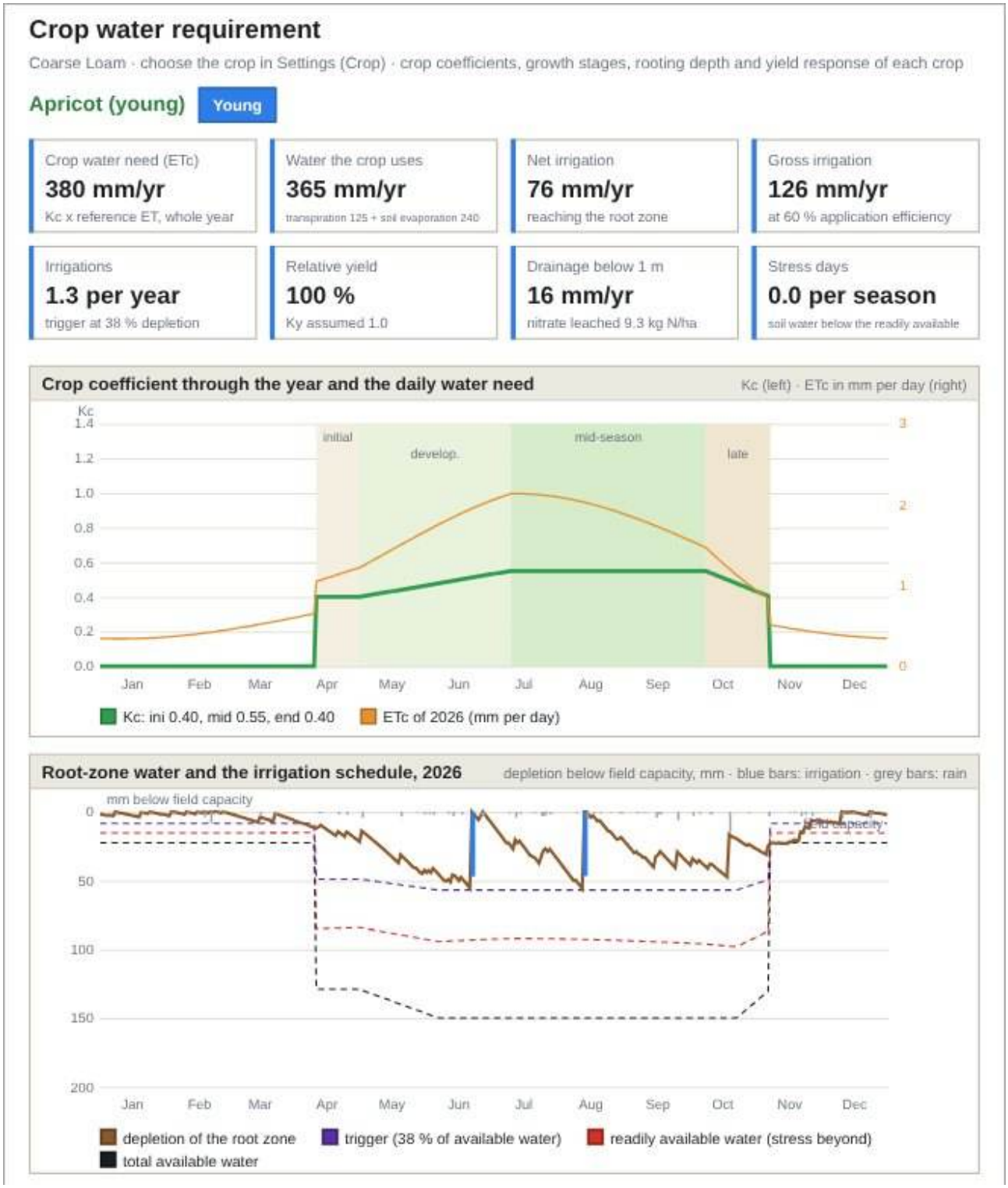
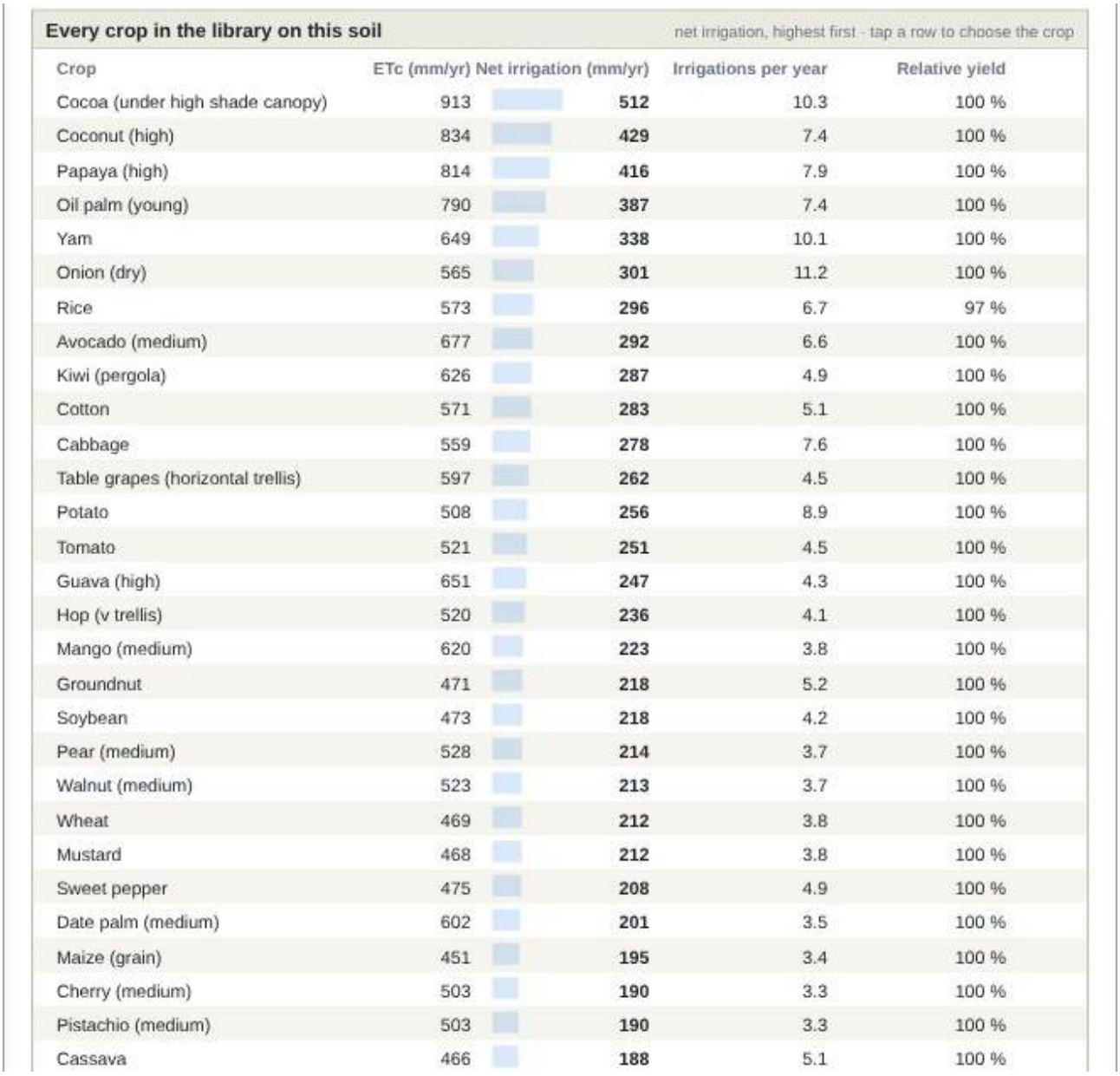


Figure 11.5: Crop water requirement, Apricot, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.6: Crop water requirement, Apricot, part 2.

11.4 Avocado

Figure 11.7: Crop water requirement, Avocado, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.8: Crop water requirement, Avocado, part 2.

11.5 Highbush blueberry

Figure 11.9: Crop water requirement, Highbush blueberry, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.10: Crop water requirement, Highbush blueberry, part 2.

11.6 Cabbage

Figure 11.11: Crop water requirement, Cabbage, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.12: Crop water requirement, Cabbage, part 2.

11.7 Cassava

Figure 11.13: Crop water requirement, Cassava, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %
Chickpea	436	**183**	4.2	100 %
Sorghum (grain)	429	**167**	2.9	100 %
Hazelnut (low)	481	**167**	2.9	100 %

Figure 11.14: Crop water requirement, Cassava, part 2.

11.8 Cereal/Oilseed (simulated rotation)

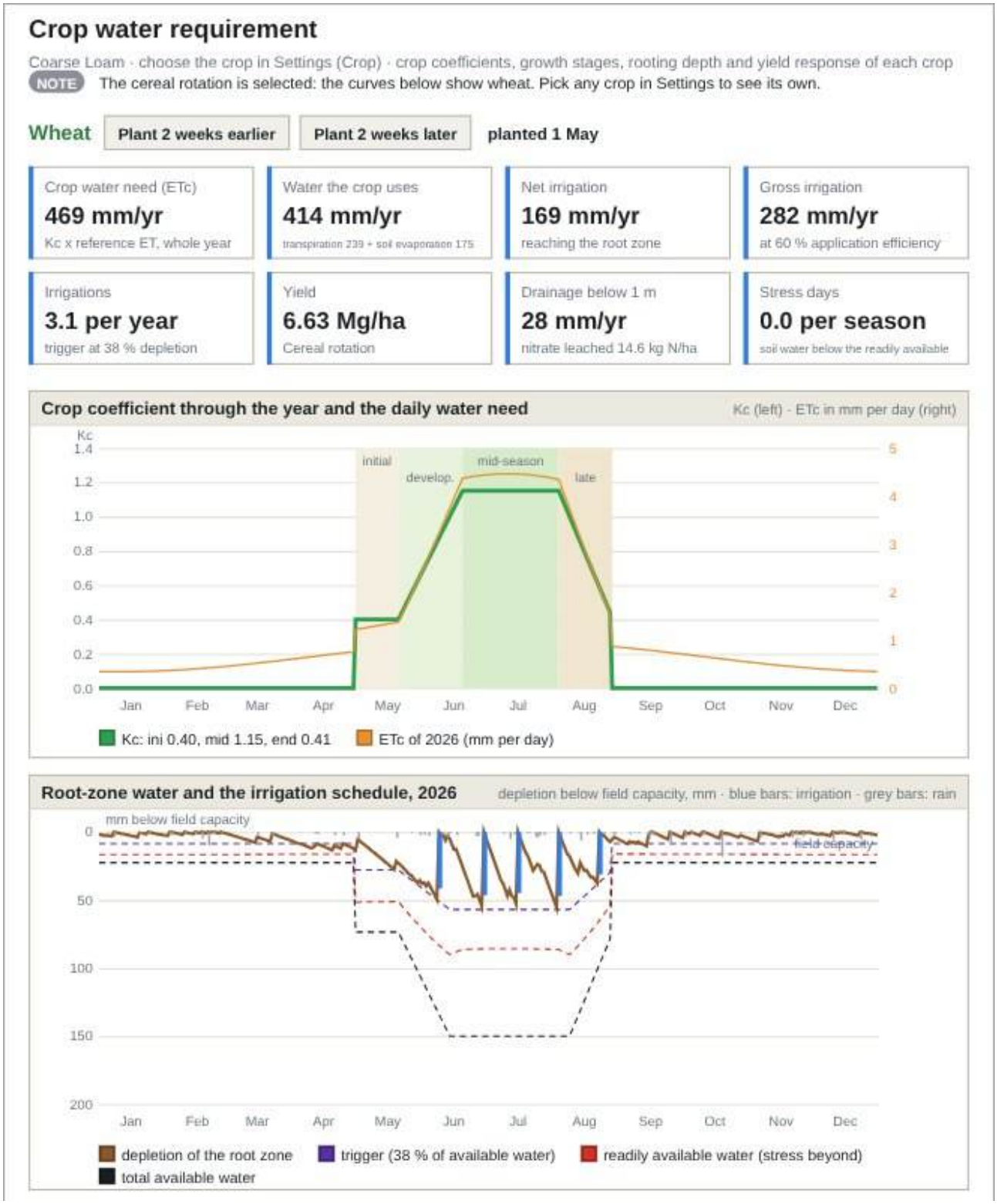


Figure 11.15: Crop water requirement, Cereal/Oilseed (simulated rotation), part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %
Chickpea	436	**183**	4.2	100 %
Sorghum (grain)	429	**167**	2.9	100 %
Hazelnut (low)	481	**167**	2.9	100 %
Pecan (low)	481	**167**	2.9	100 %
Cherimoya (medium)	545	**161**	2.8	100 %
Citrus (medium density, short)	552	**152**	2.6	100 %

Figure 11.16: Crop water requirement, Cereal/Oilseed (simulated rotation), part 2.

11.9 Cherimoya

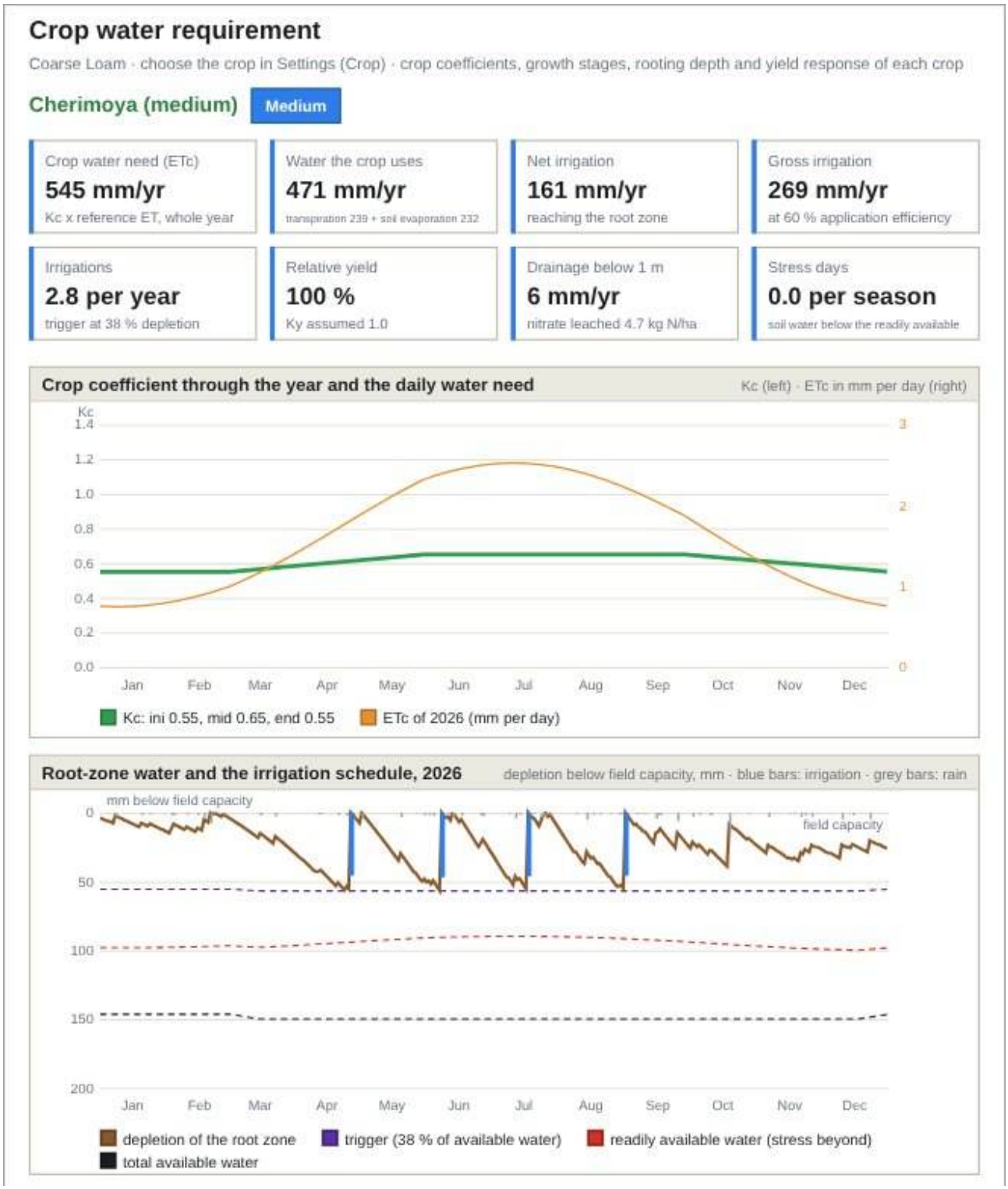
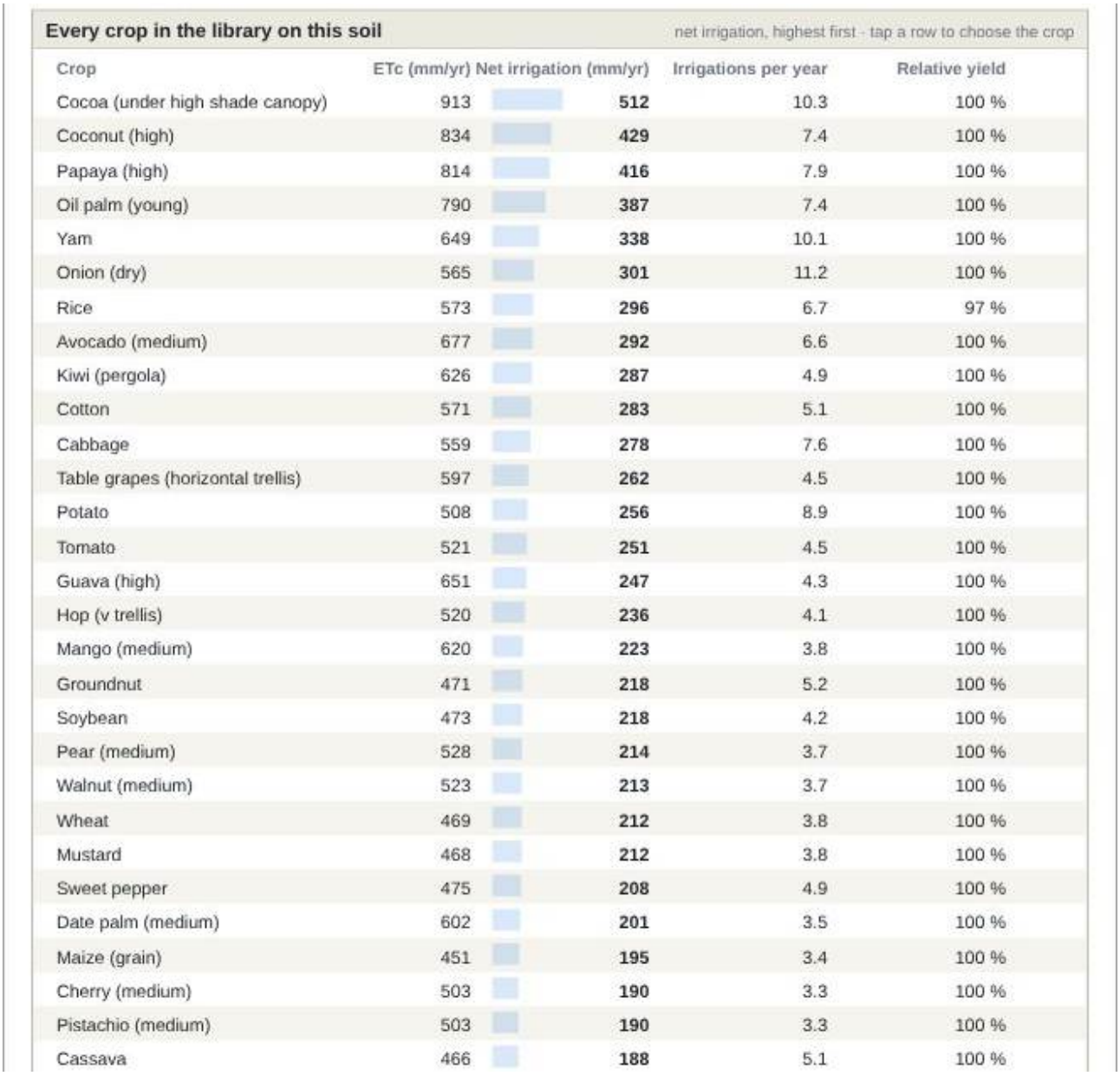


Figure 11.17: Crop water requirement, Cherimoya, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
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Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.18: Crop water requirement, Cherimoya, part 2.

11.10 Cherry

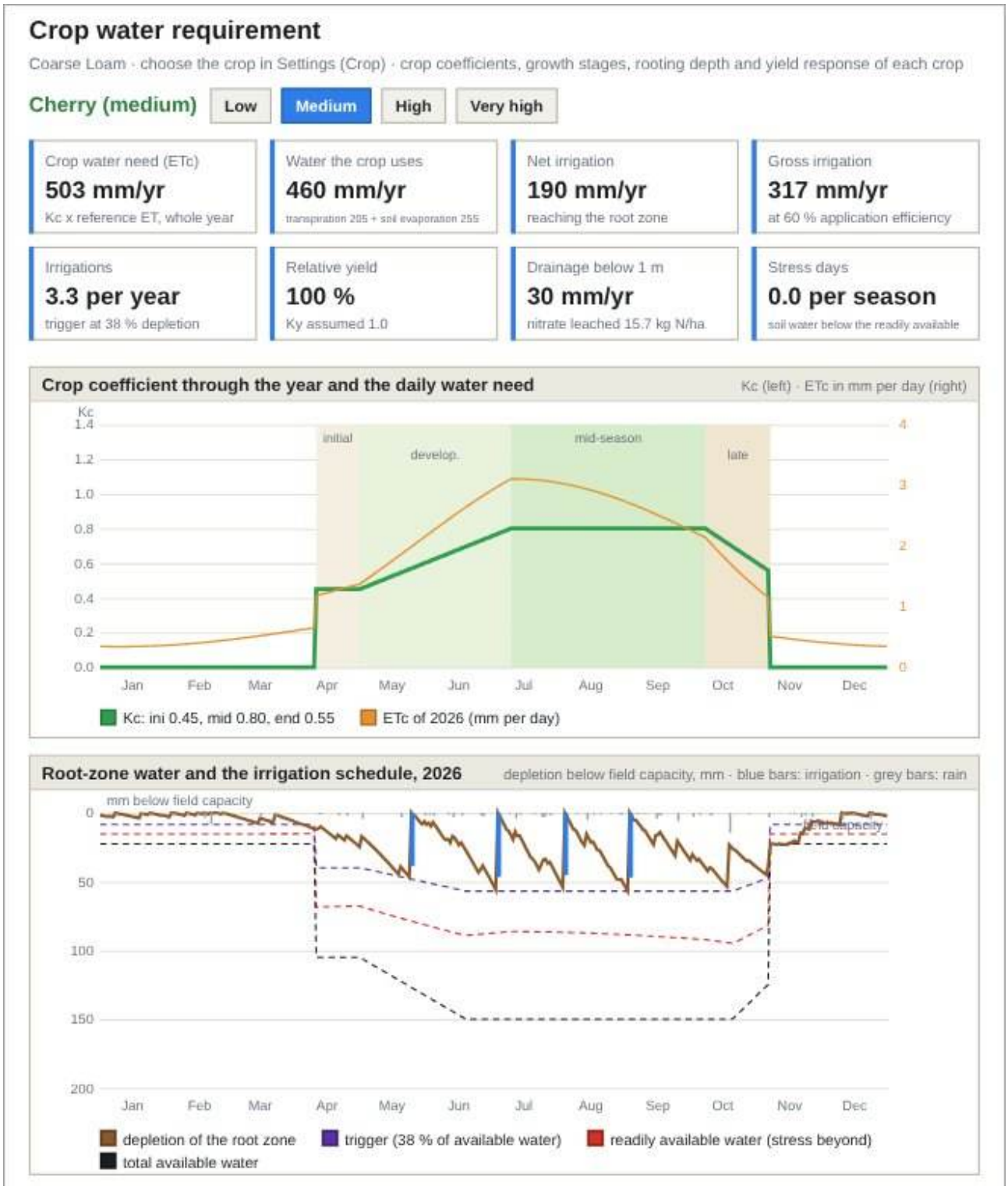
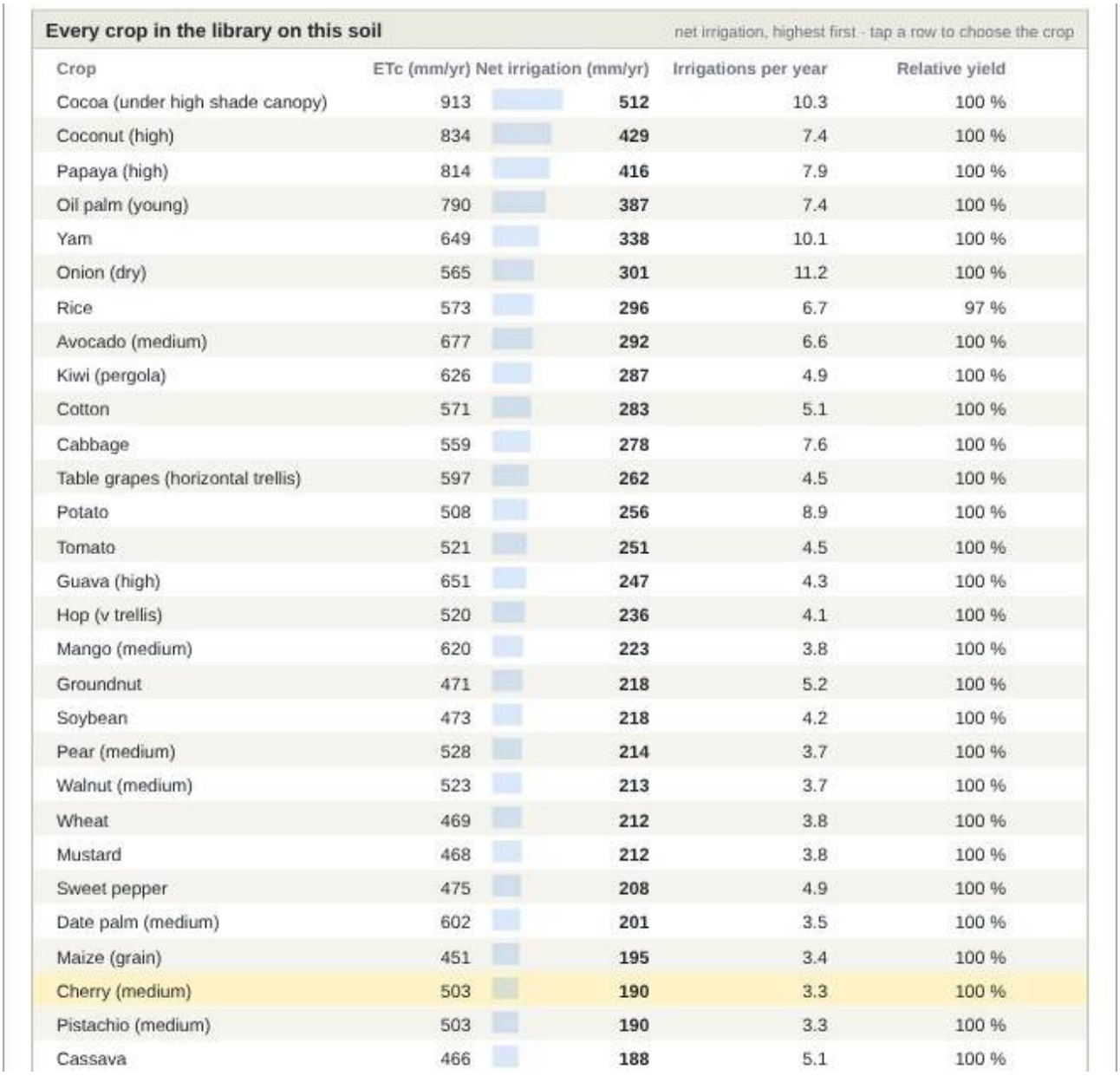


Figure 11.19: Crop water requirement, Cherry, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.20: Crop water requirement, Cherry, part 2.

11.11 Chickpea

Figure 11.21: Crop water requirement, Chickpea, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
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Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %
Chickpea	436	**183**	4.2	100 %
Sorghum (grain)	429	**167**	2.9	100 %
Hazelnut (low)	481	**167**	2.9	100 %

Figure 11.22: Crop water requirement, Chickpea, part 2.

11.12 Citrus

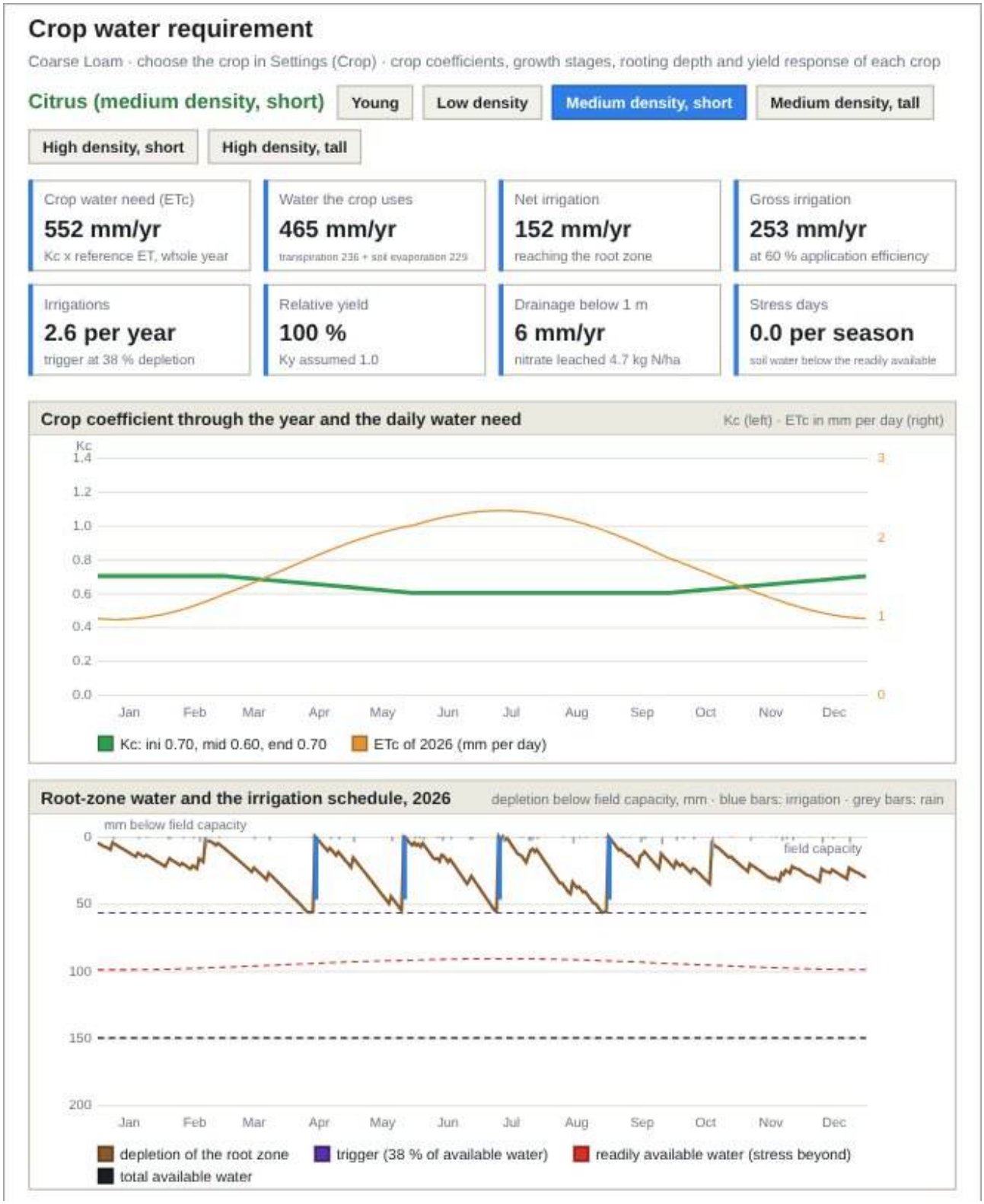
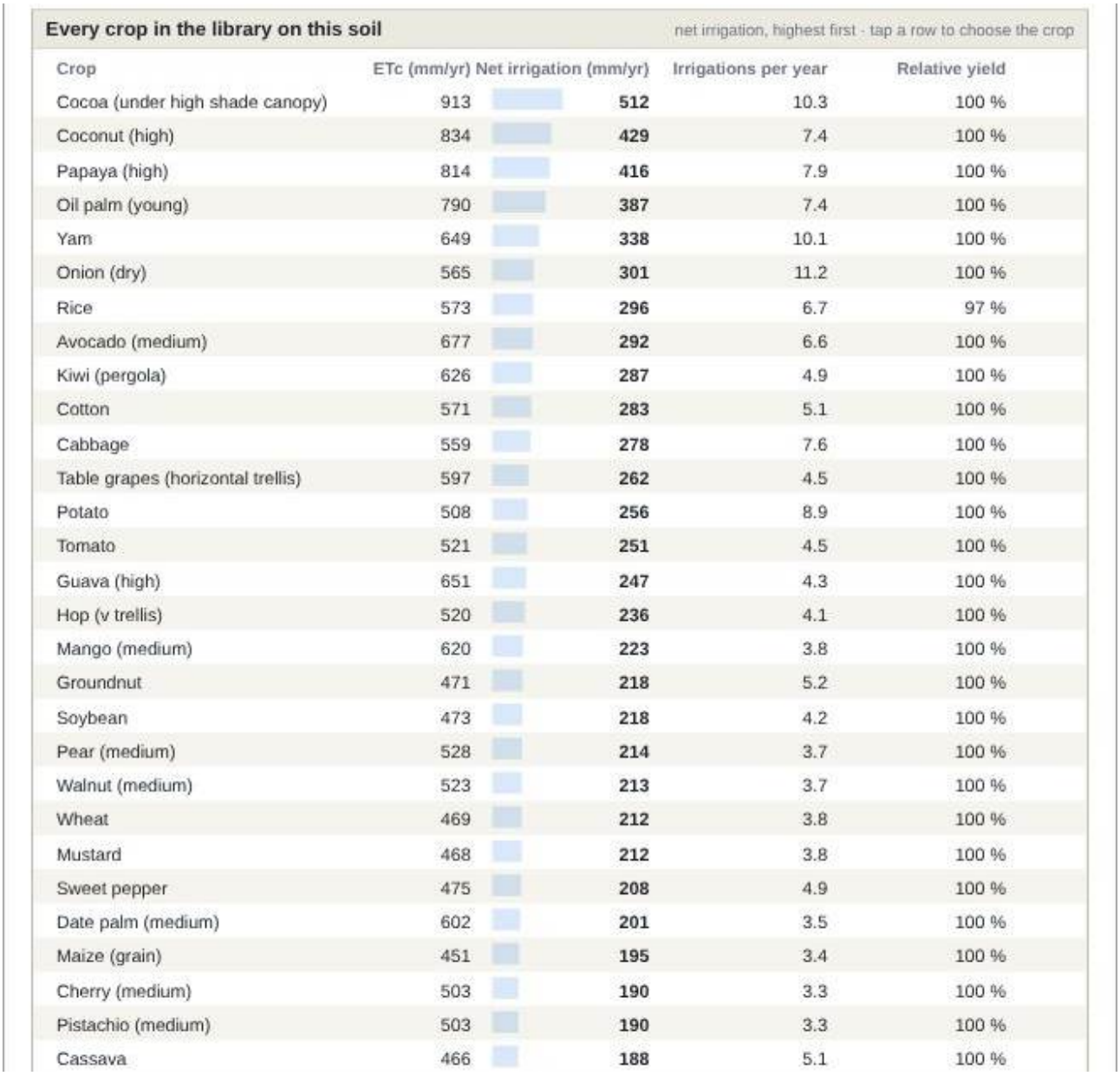


Figure 11.23: Crop water requirement, Citrus, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
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Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.24: Crop water requirement, Citrus, part 2.

11.13 Cocoa

Figure 11.25: Crop water requirement, Cocoa, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
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Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
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Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.26: Crop water requirement, Cocoa, part 2.

11.14 Coconut

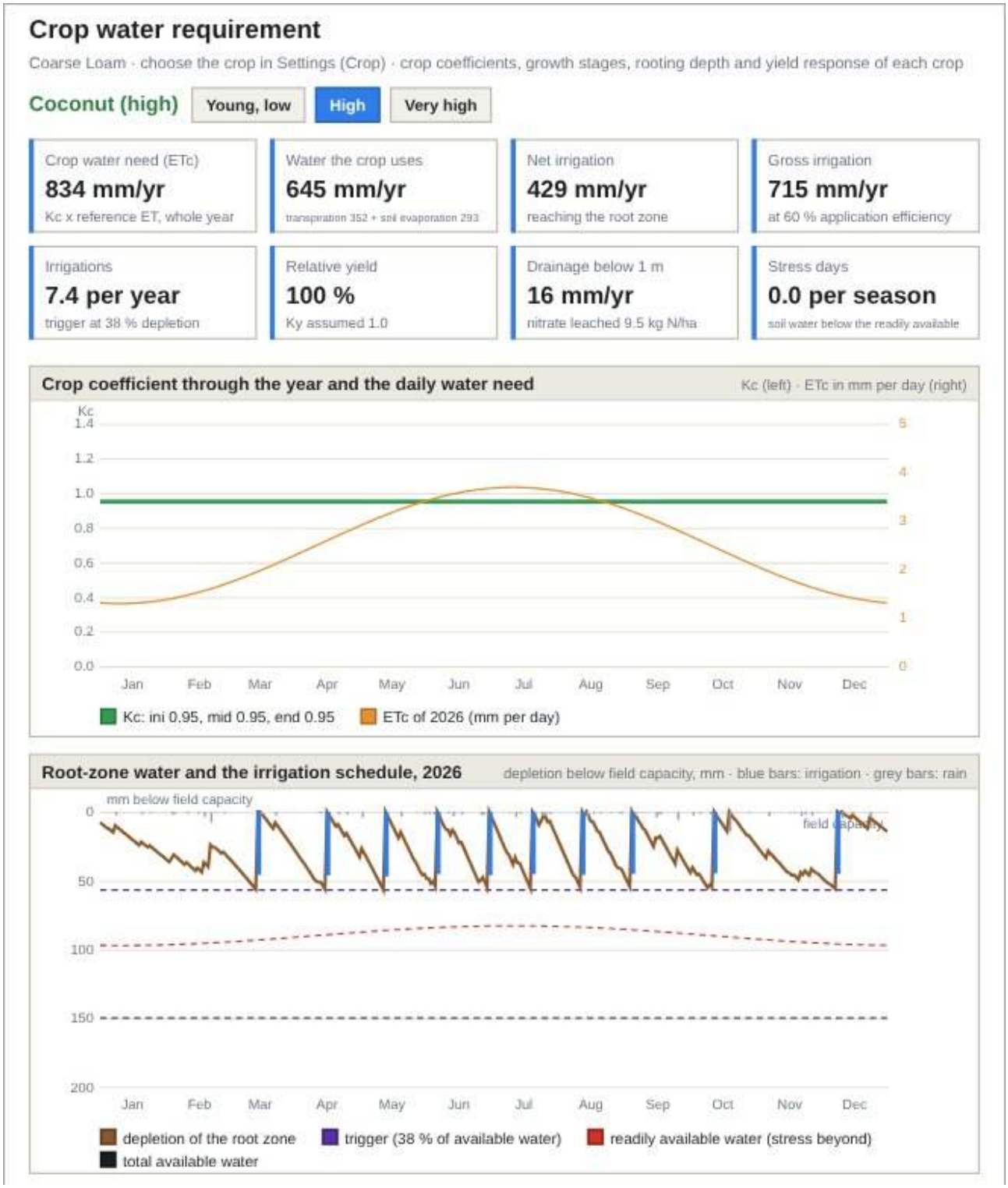

Figure 11.27: Crop water requirement, Coconut, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
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Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.28: Crop water requirement, Coconut, part 2.

11.15 Cotton

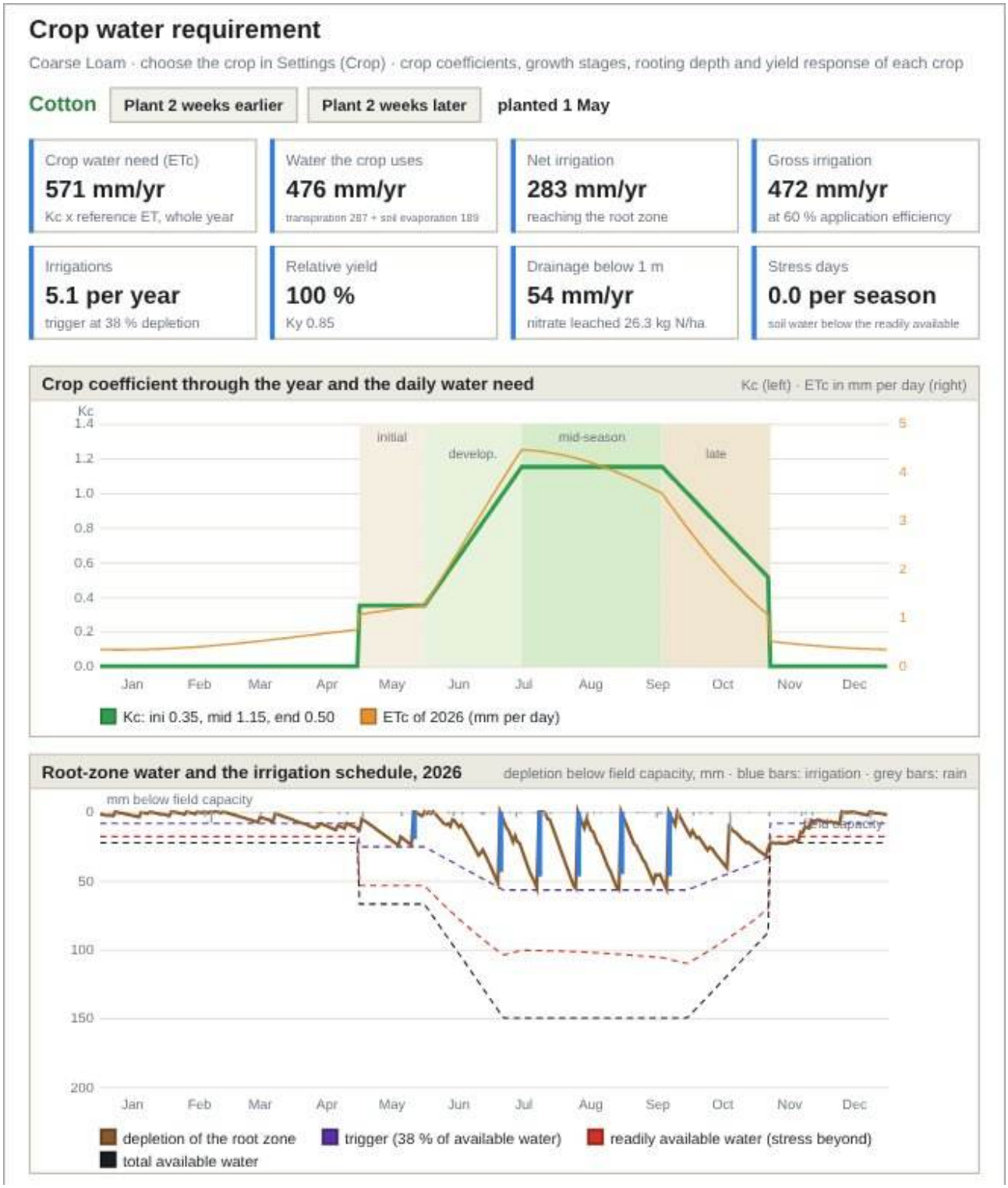

Figure 11.29: Crop water requirement, Cotton, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
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Yam	649	**338**	10.1	100 %
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Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
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Soybean	473	**218**	4.2	100 %
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Walnut (medium)	523	**213**	3.7	100 %
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Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.30: Crop water requirement, Cotton, part 2.

11.16 Cowpea (beans)

Figure 11.31: Crop water requirement, Cowpea (beans), part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
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Soybean	473	**218**	4.2	100 %
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Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.32: Crop water requirement, Cowpea (beans), part 2.

11.17 Date palm

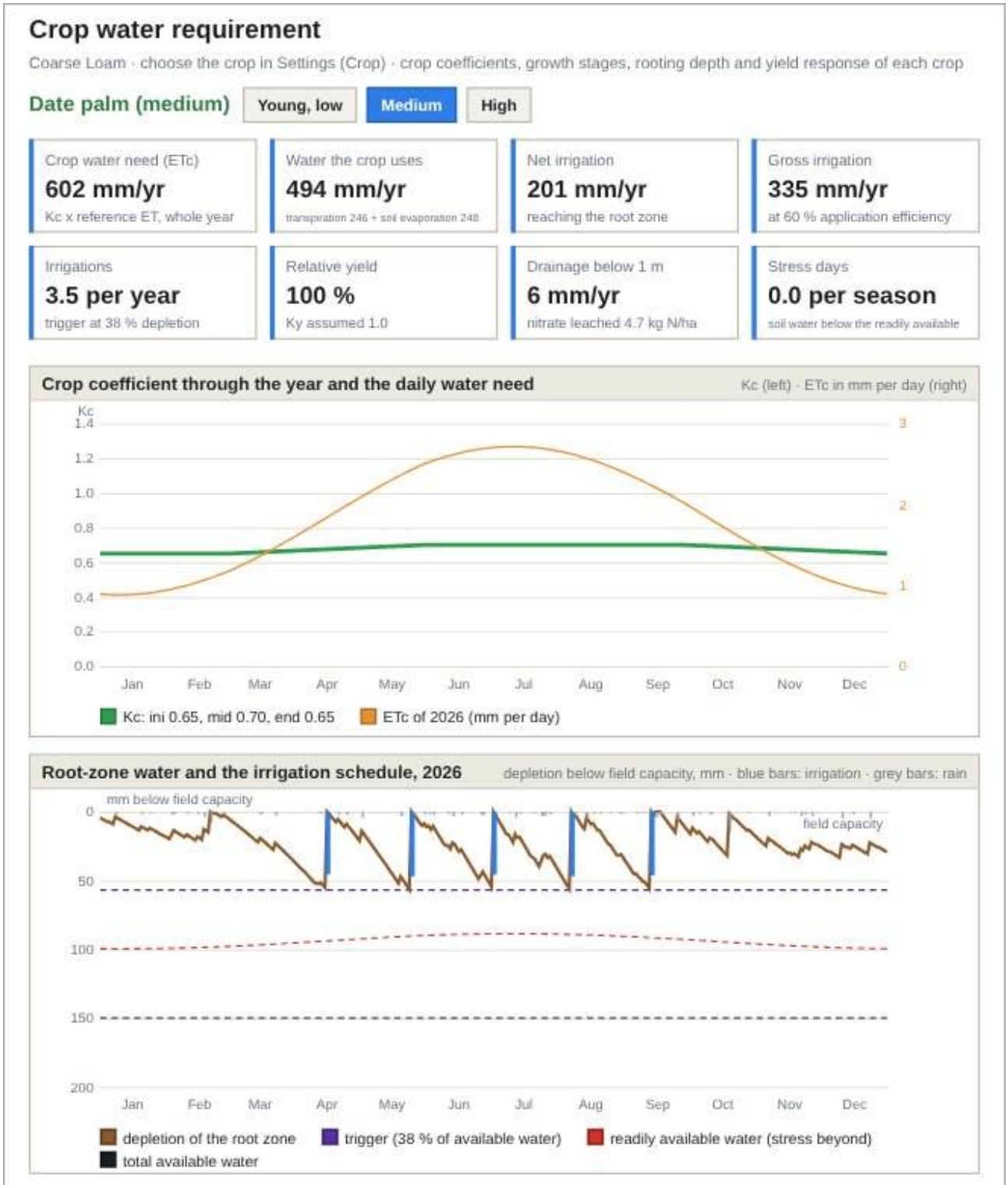

Figure 11.33: Crop water requirement, Date palm, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
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Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
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Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
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Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.34: Crop water requirement, Date palm, part 2.

11.18 Fig tree

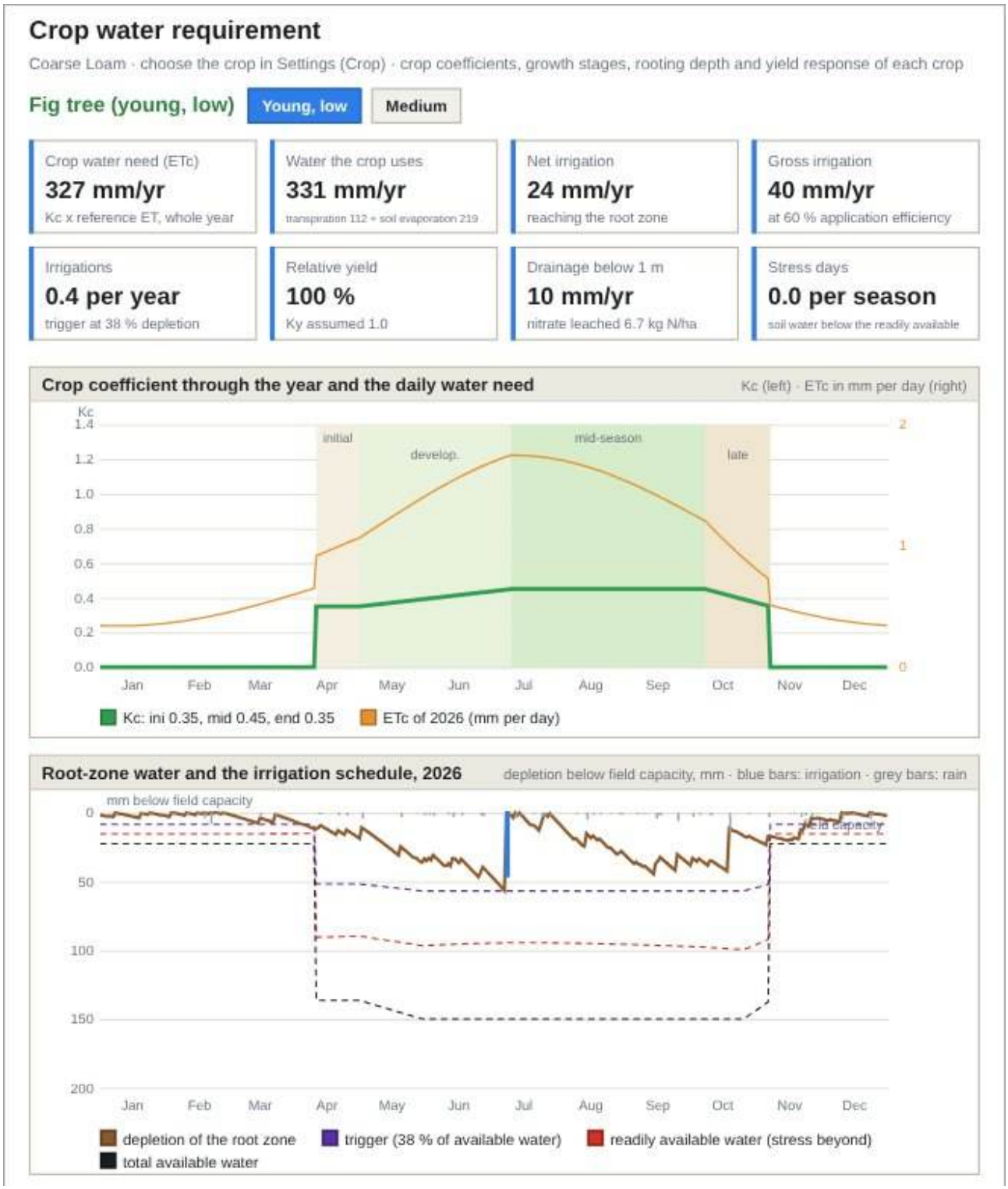
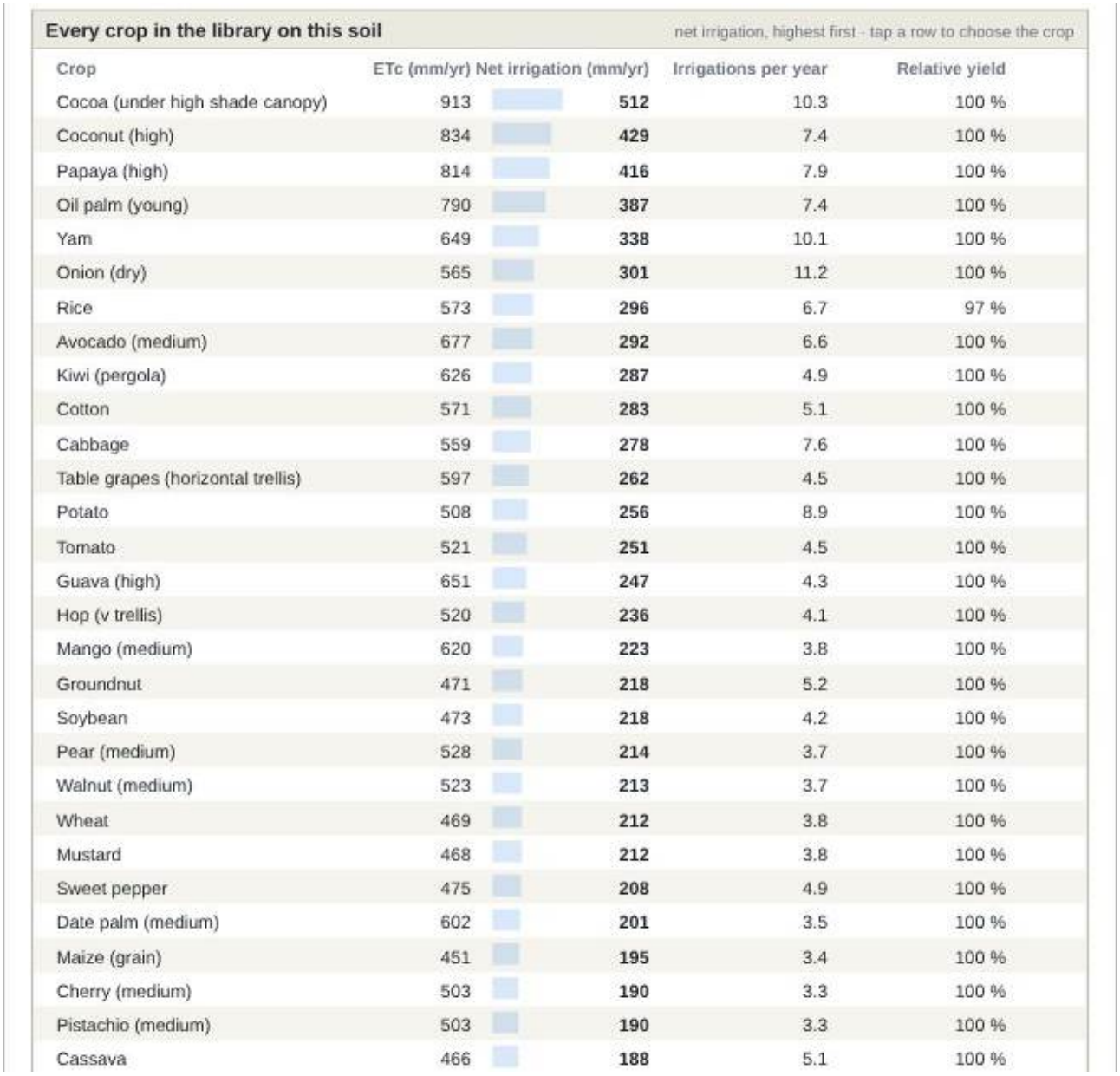


Figure 11.35: Crop water requirement, Fig tree, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.36: Crop water requirement, Fig tree, part 2.

11.19 Green gram (mung bean)

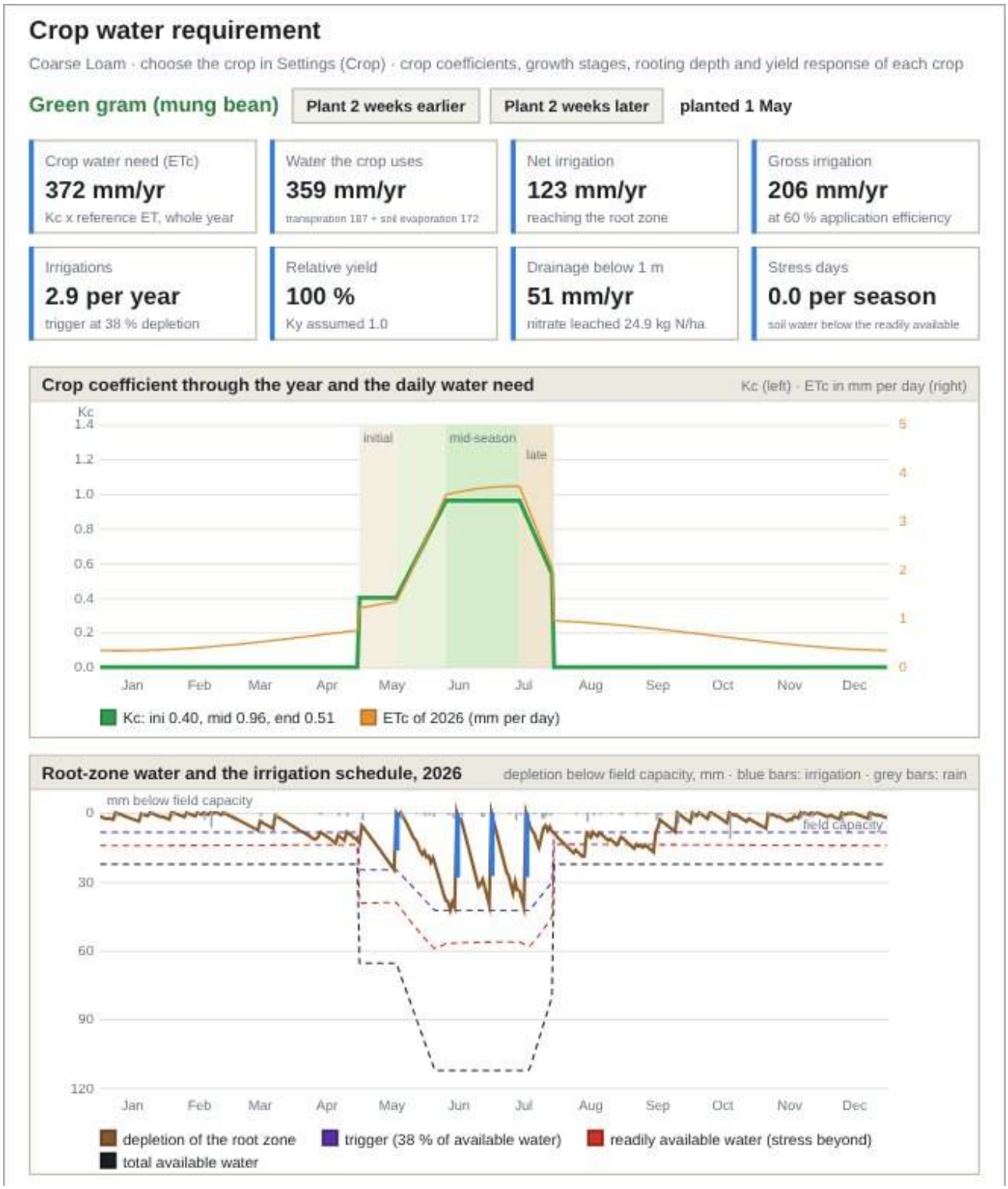
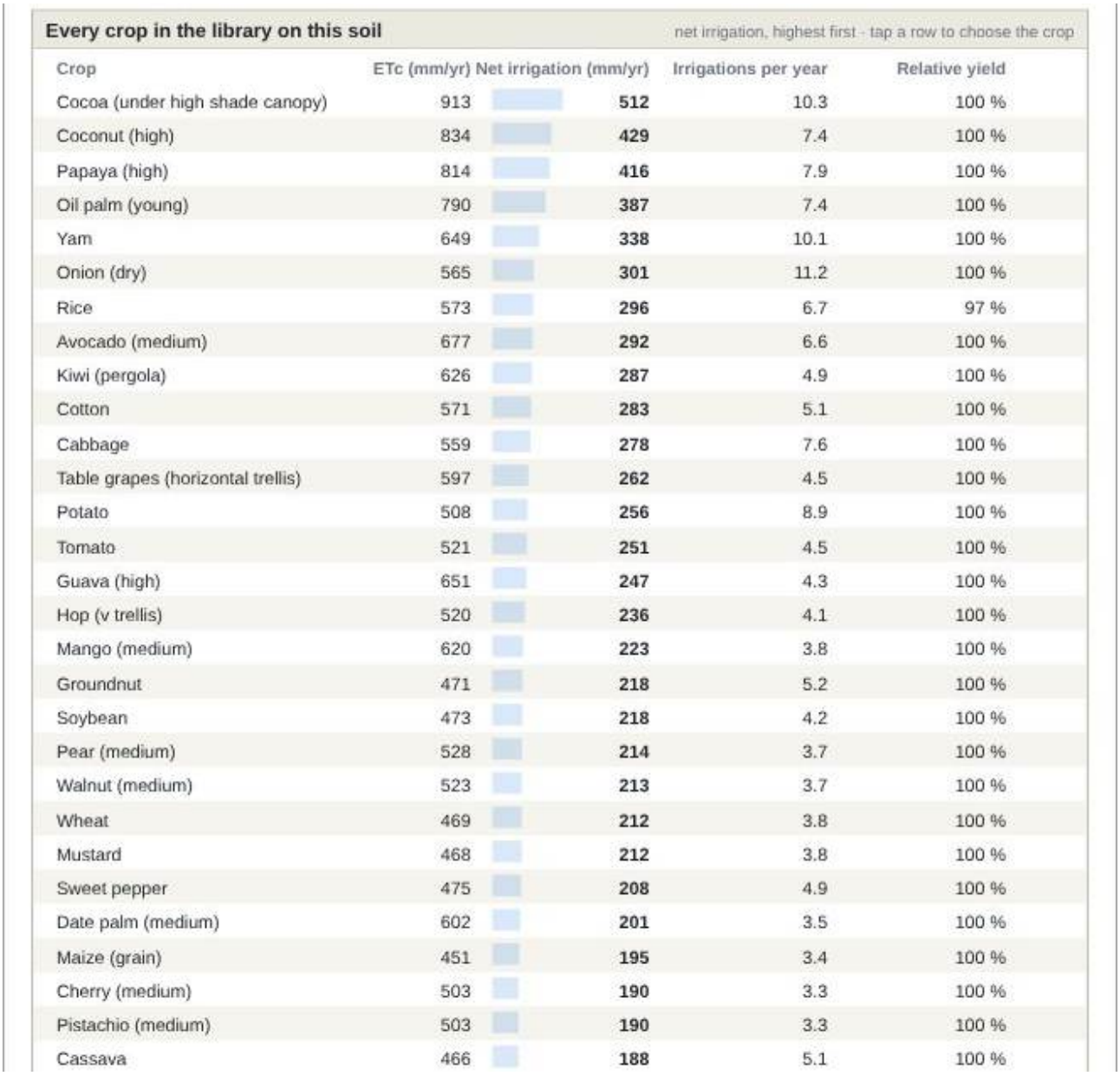


Figure 11.37: Crop water requirement, Green gram (mung bean), part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
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Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
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Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.38: Crop water requirement, Green gram (mung bean), part 2.

11.20 Groundnut

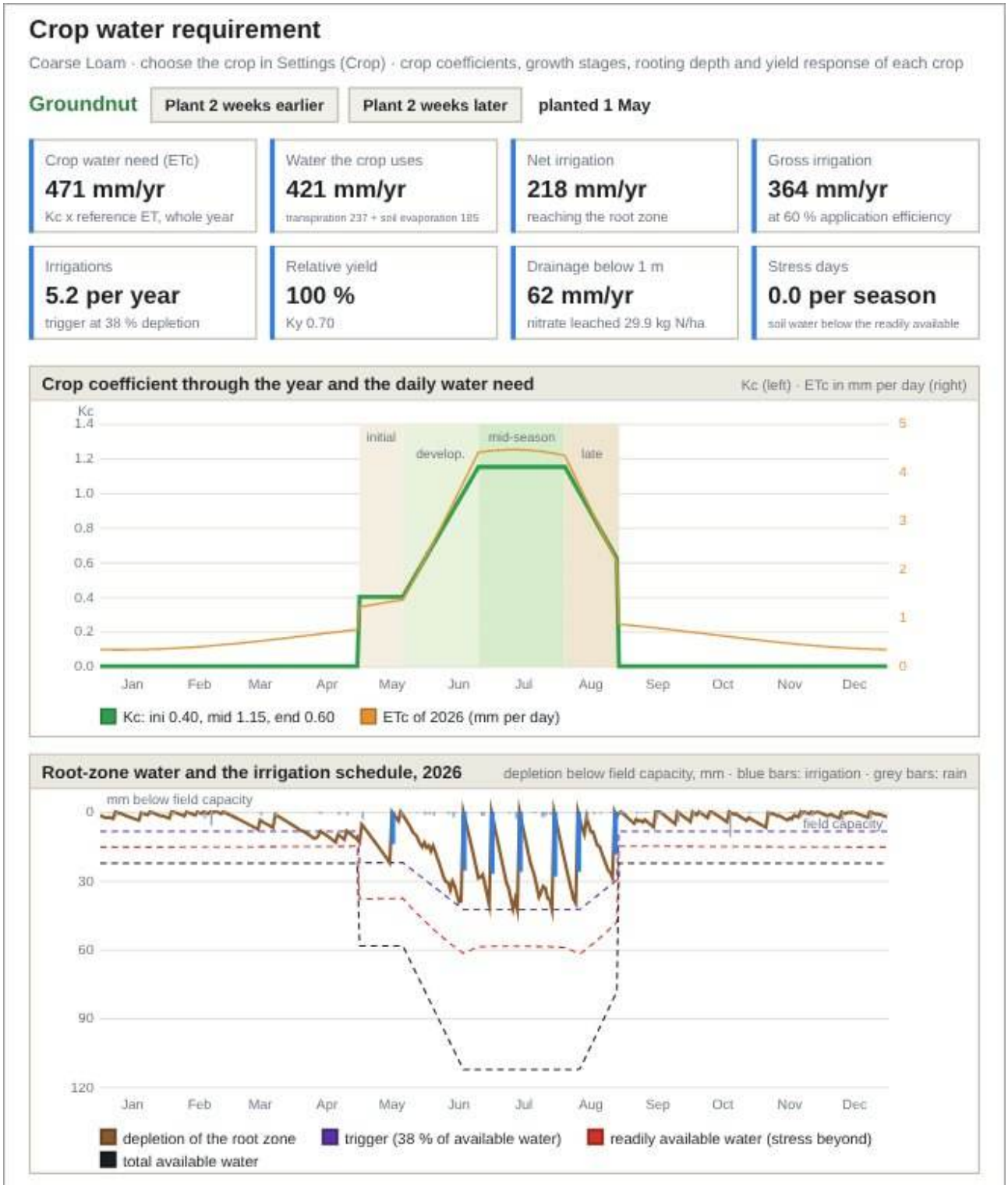
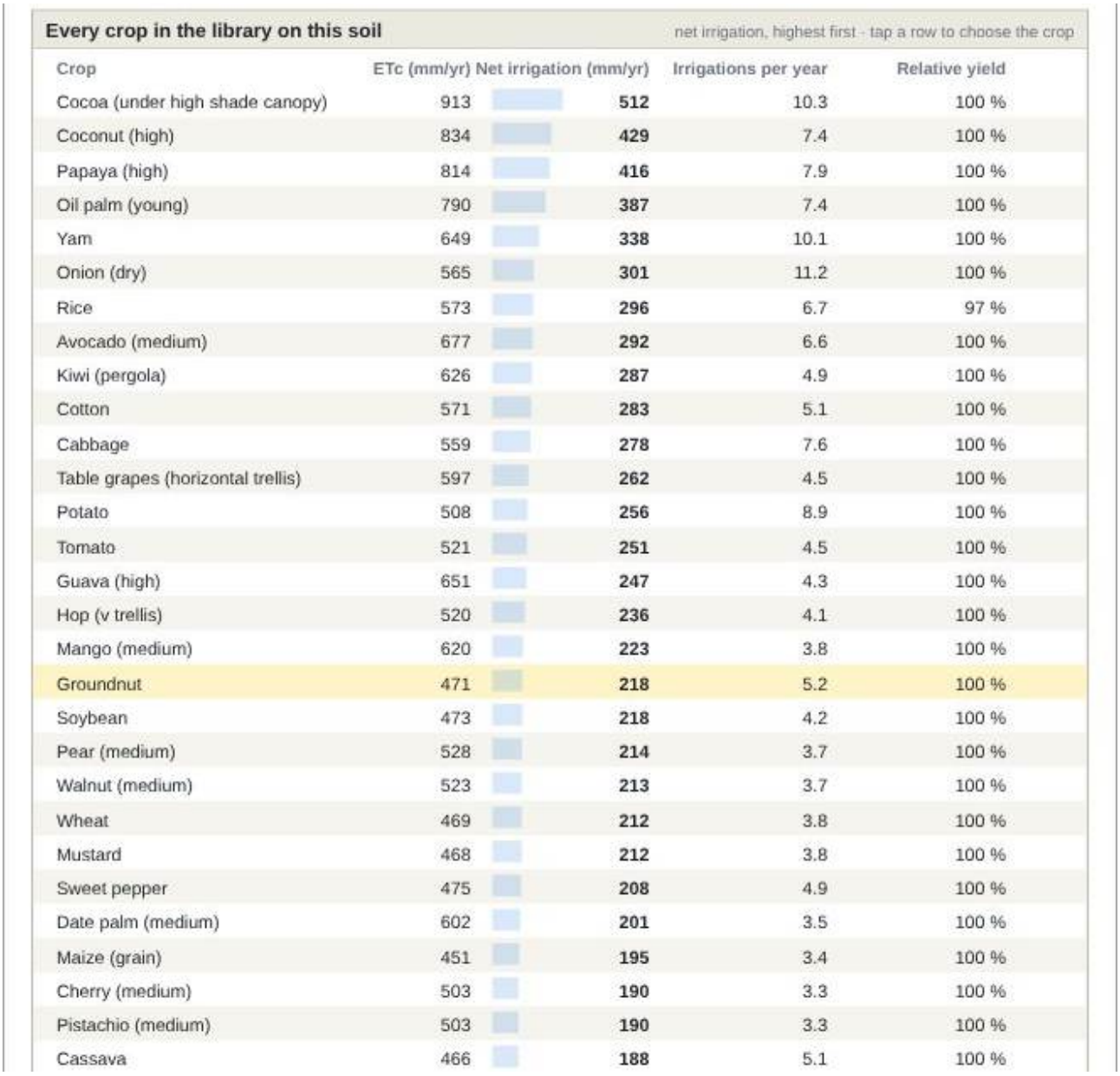


Figure 11.39: Crop water requirement, Groundnut, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
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Onion (dry)	565	**301**	11.2	100 %
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Avocado (medium)	677	**292**	6.6	100 %
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Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
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Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.40: Crop water requirement, Groundnut, part 2.

11.21 Guava

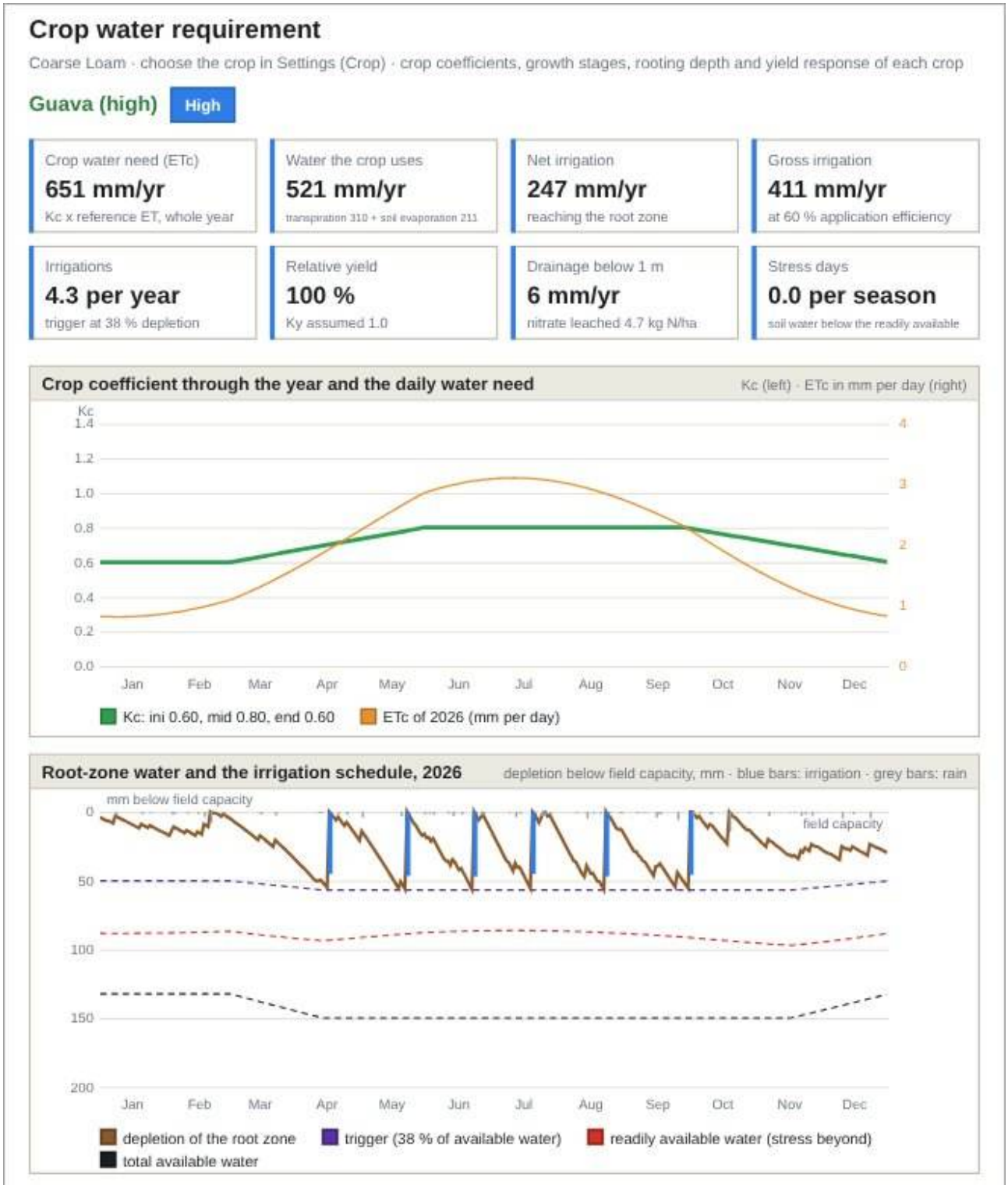

Figure 11.41: Crop water requirement, Guava, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
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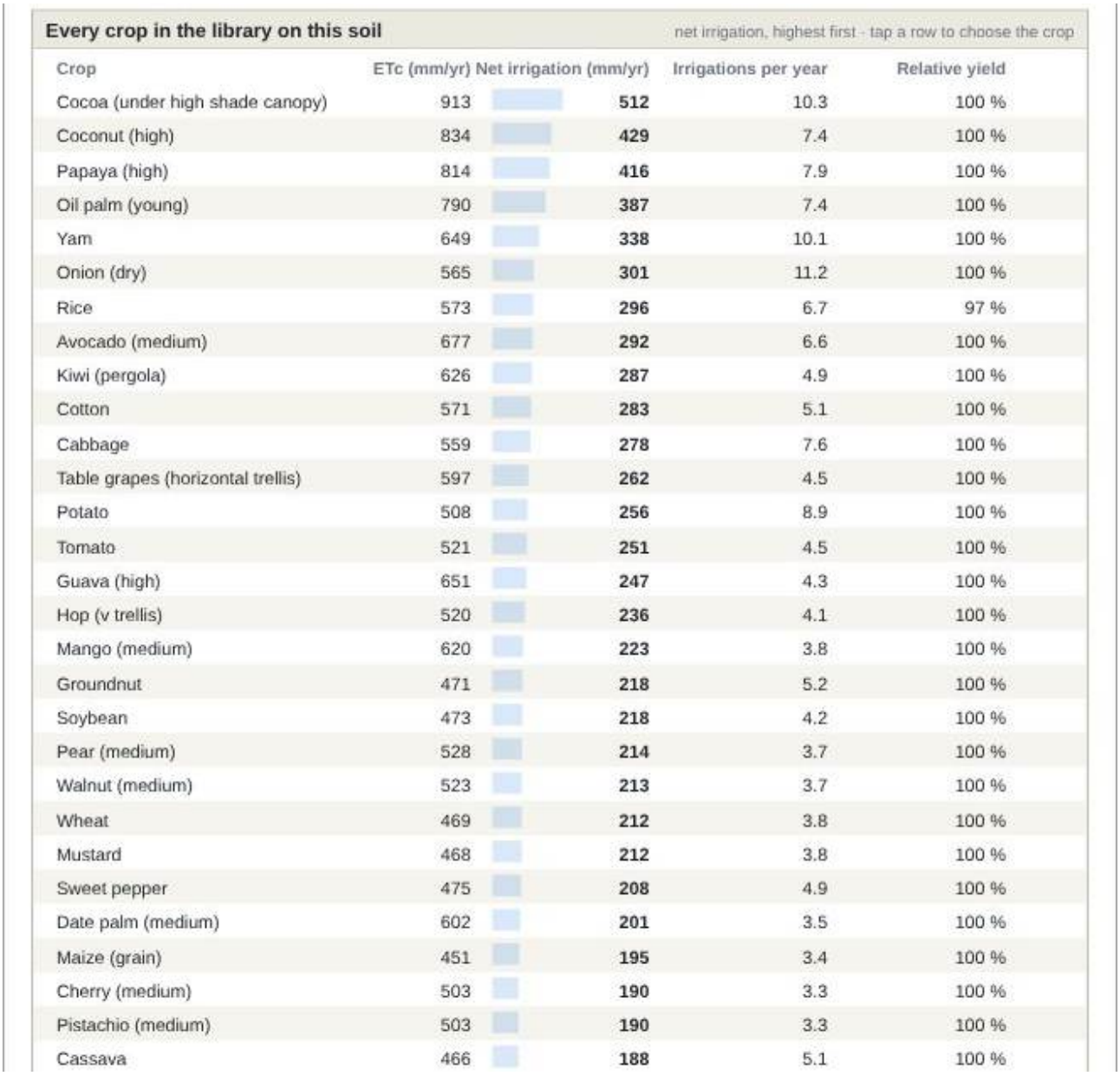
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.42: Crop water requirement, Guava, part 2.

11.22 Hazelnut

Figure 11.43: Crop water requirement, Hazelnut, part 1.



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Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
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Oil palm (young)	790	**387**	7.4	100 %
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Onion (dry)	565	**301**	11.2	100 %
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Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
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Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.44: Crop water requirement, Hazelnut, part 2.

11.23 Hop

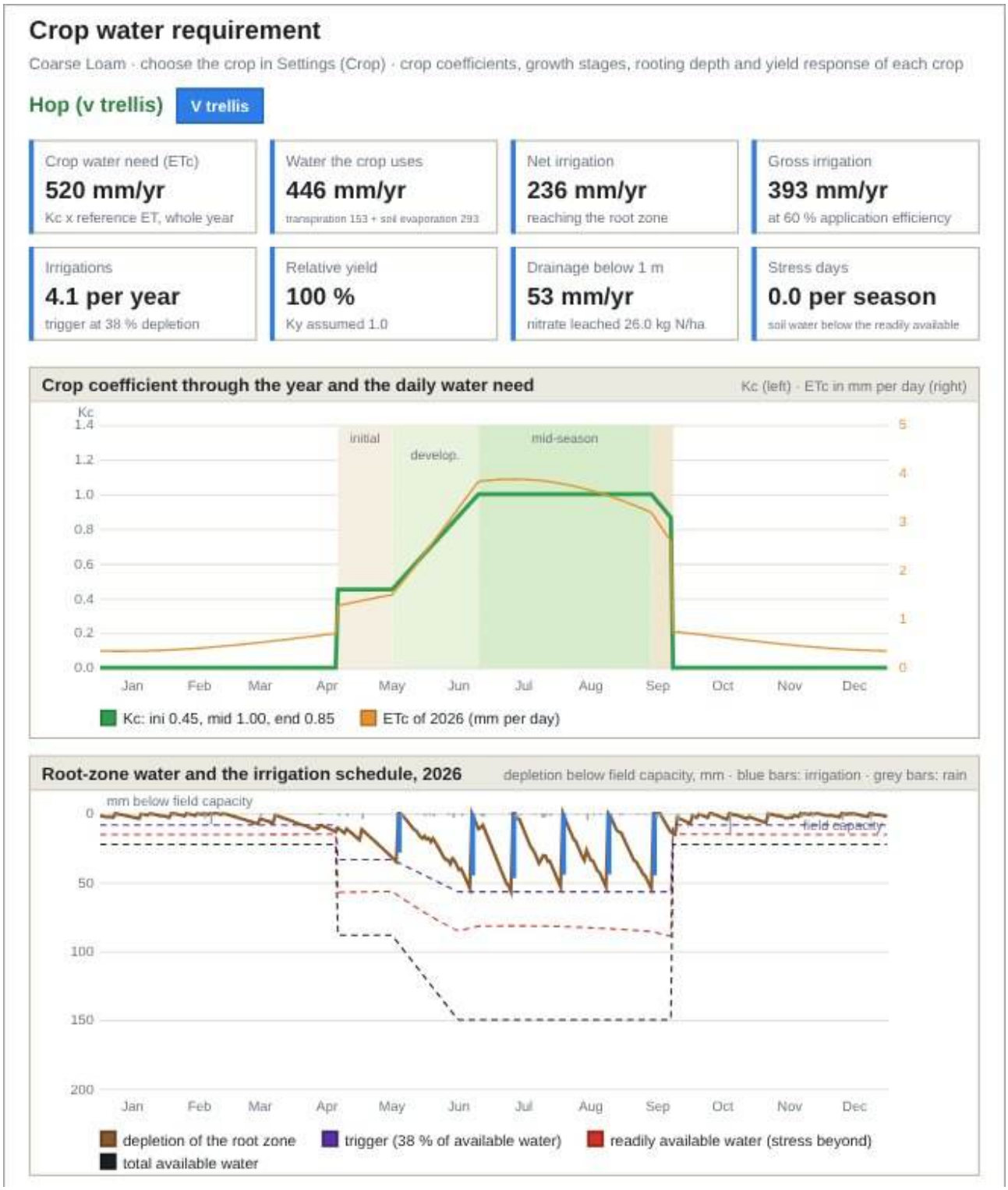

Figure 11.45: Crop water requirement, Hop, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
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Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.46: Crop water requirement, Hop, part 2.

11.24 Kiwi

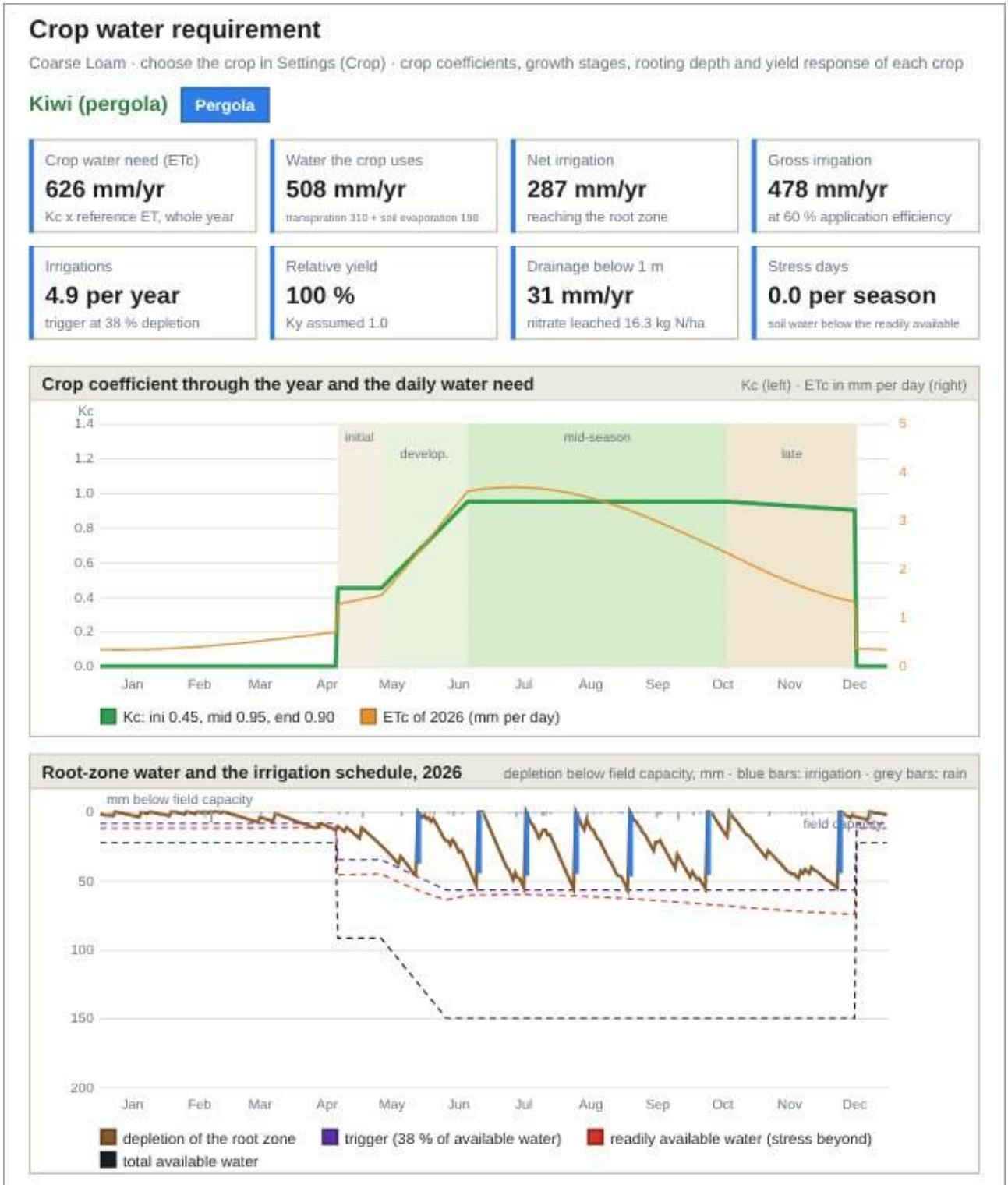
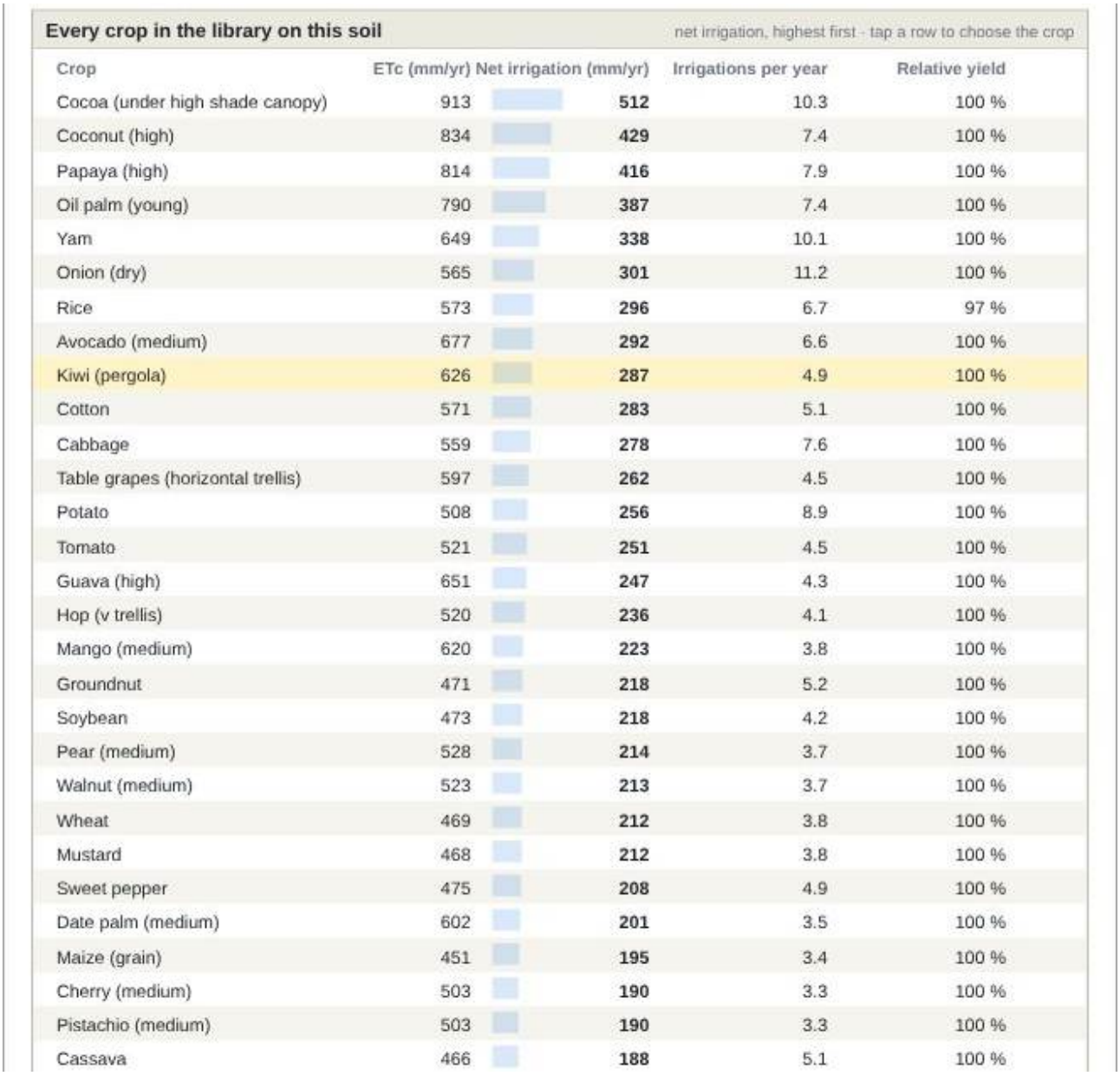


Figure 11.47: Crop water requirement, Kiwi, part 1.



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Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
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Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
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Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.48: Crop water requirement, Kiwi, part 2.

11.25 Maize (grain)

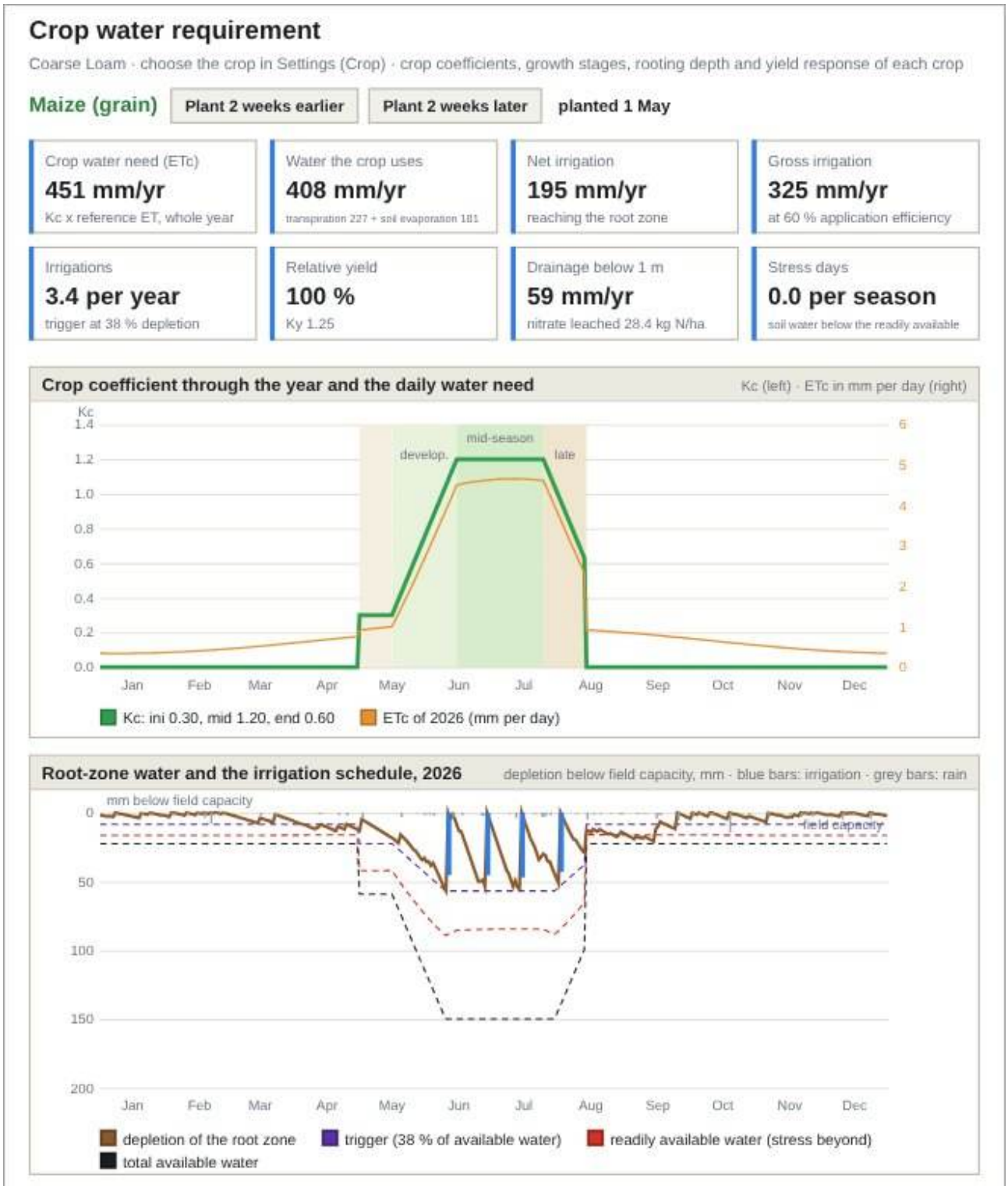

Figure 11.49: Crop water requirement, Maize (grain), part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
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Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.50: Crop water requirement, Maize (grain), part 2.

11.26 Mango

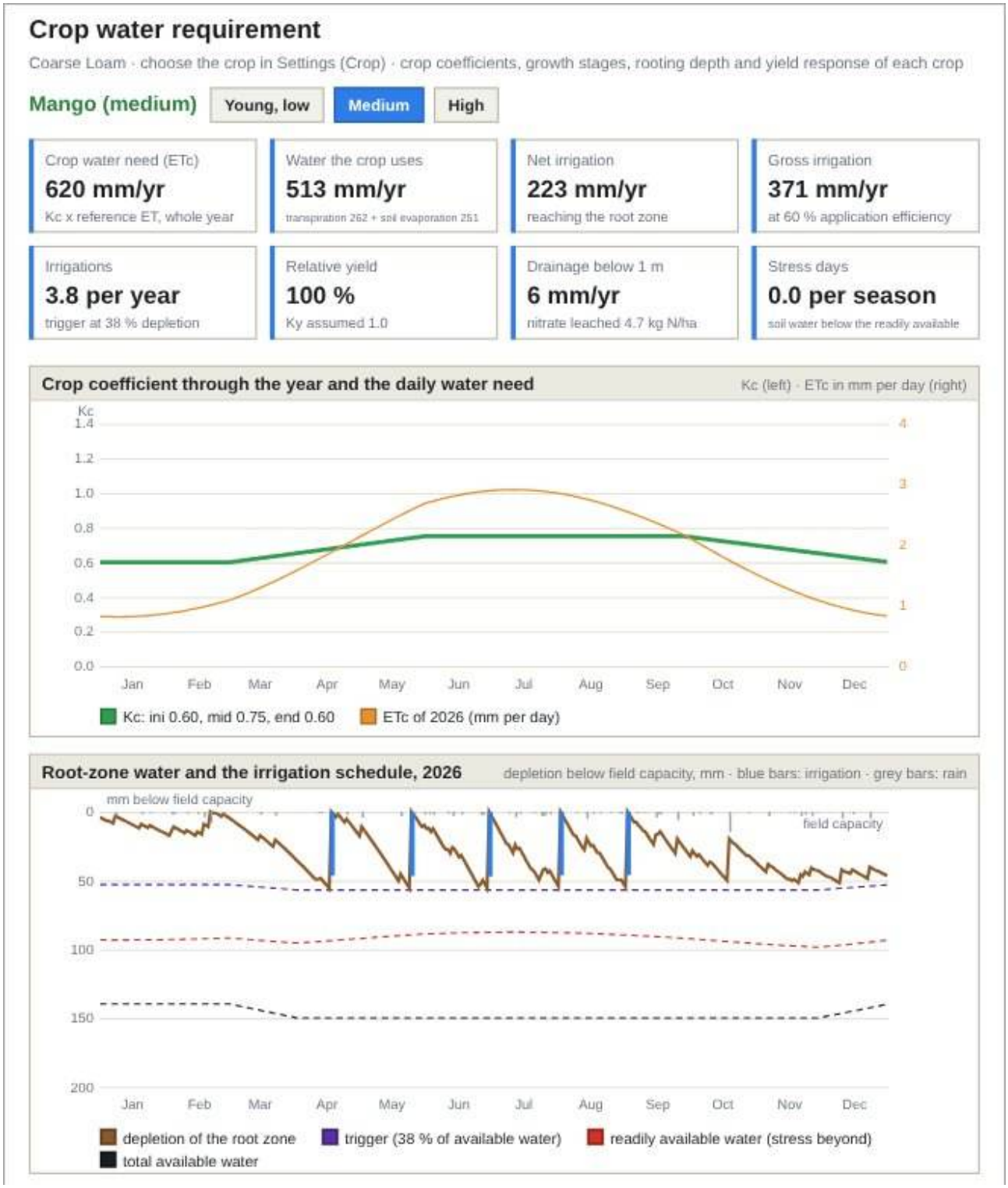
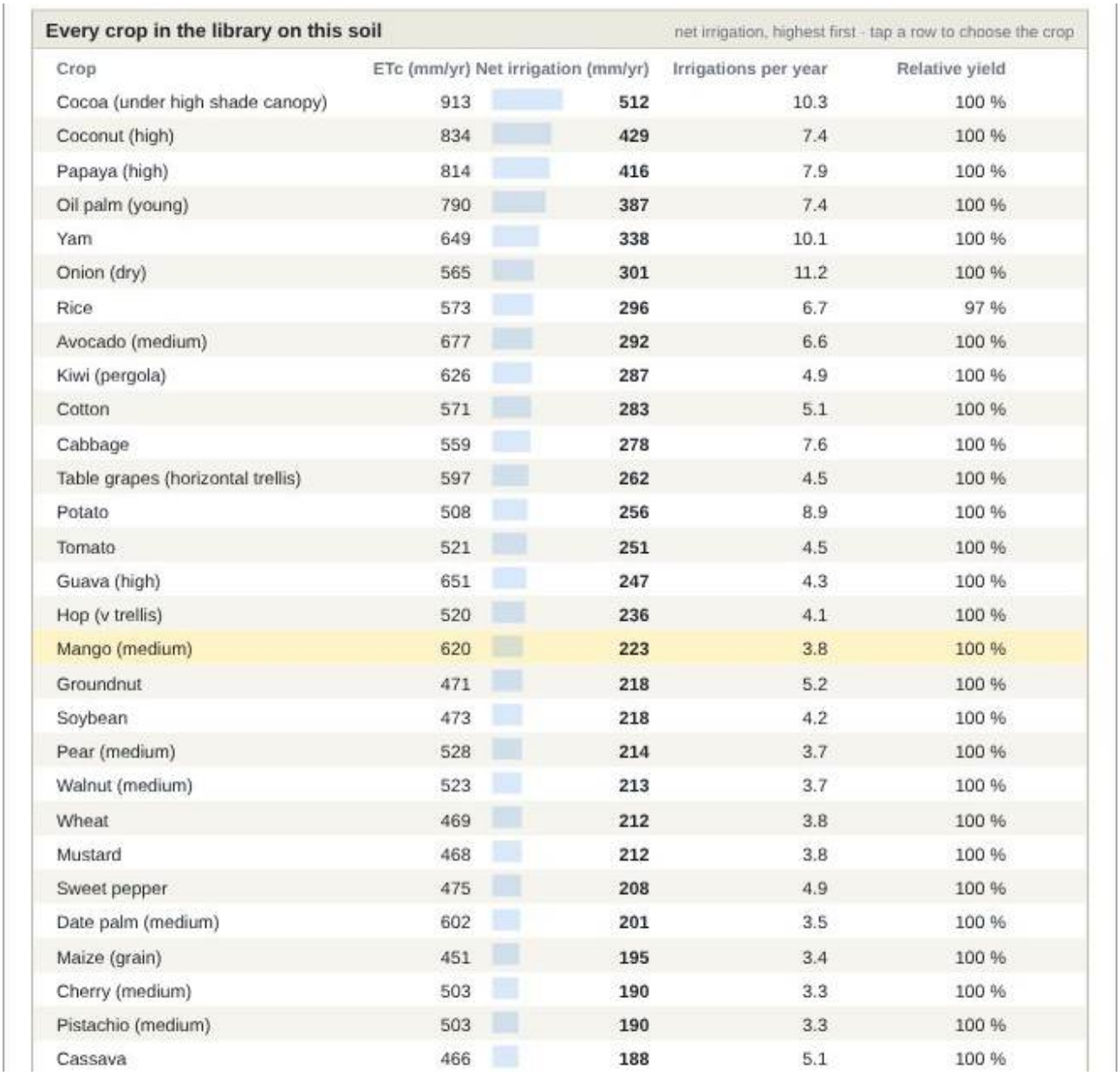


Figure 11.51: Crop water requirement, Mango, part 1.



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Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
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Rice	573	**296**	6.7	97 %
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Kiwi (pergola)	626	**287**	4.9	100 %
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Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
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Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
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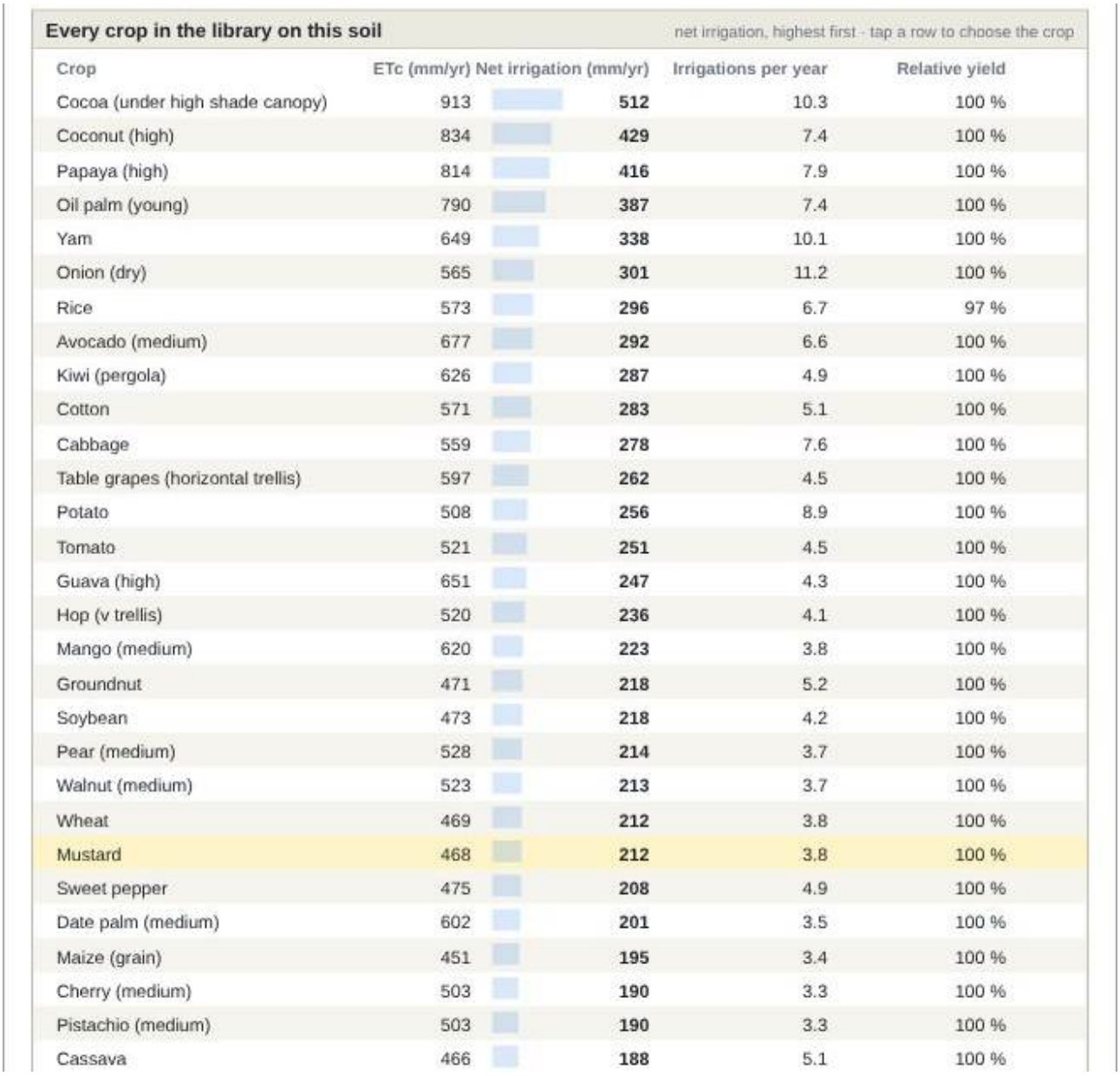
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.52: Crop water requirement, Mango, part 2.

11.27 Mustard

Figure 11.53: Crop water requirement, Mustard, part 1.



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Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
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Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.54: Crop water requirement, Mustard, part 2.

11.28 Oil palm

Figure 11.55: Crop water requirement, Oil palm, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.56: Crop water requirement, Oil palm, part 2.

11.29 Olive

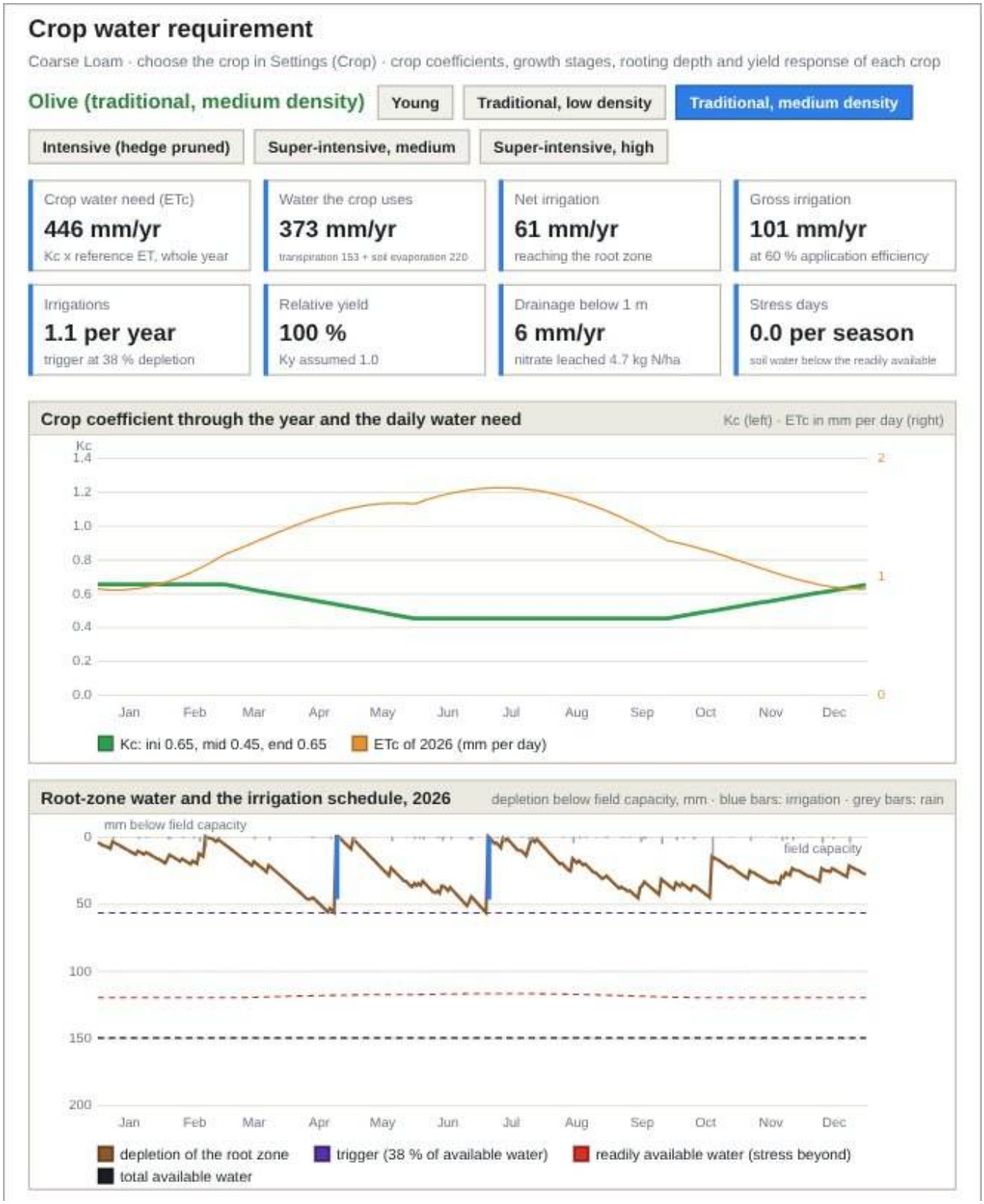


Figure 11.57: Crop water requirement, Olive, part 1.

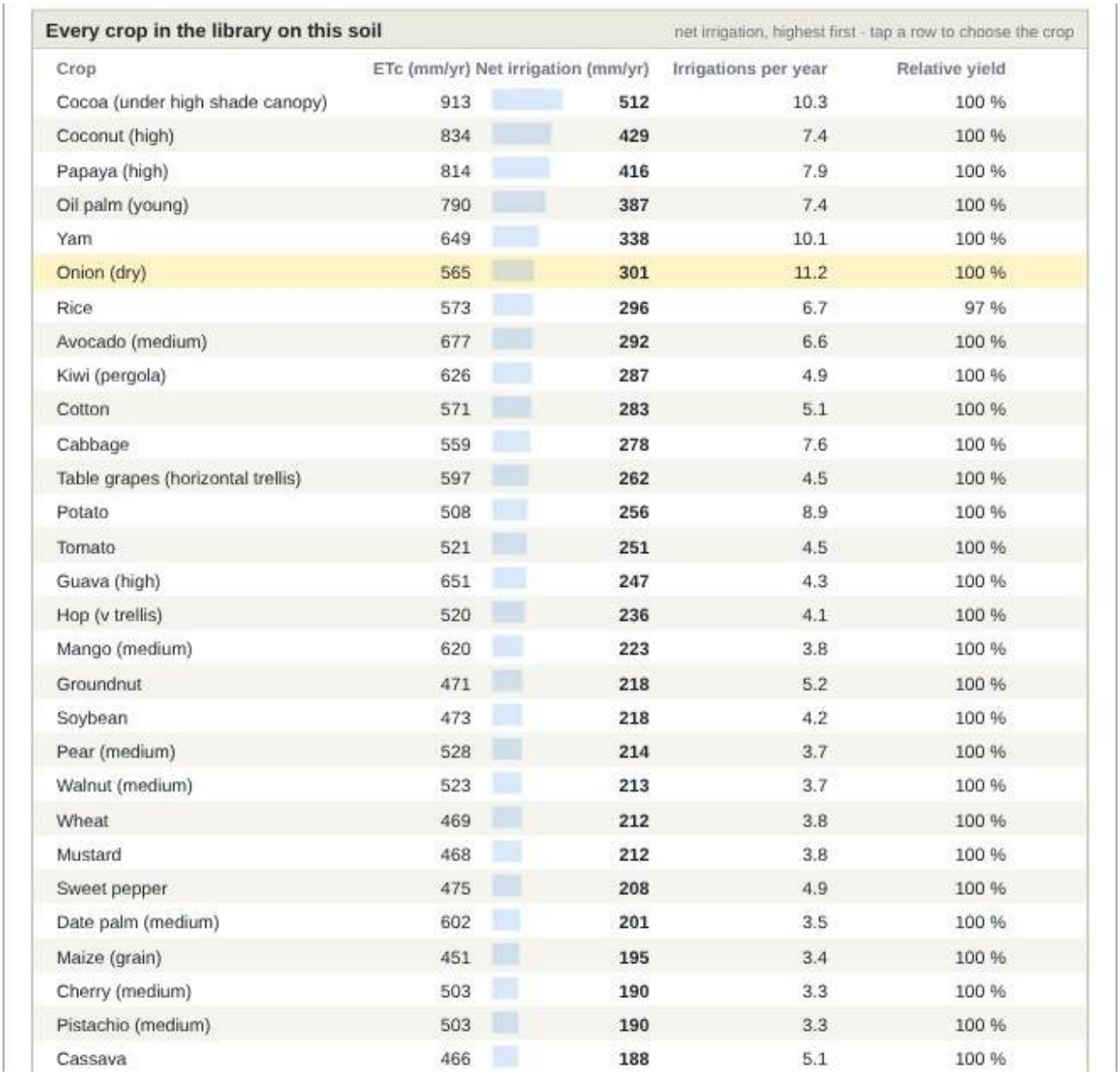
Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.58: Crop water requirement, Olive, part 2.

11.30 Onion (dry)

Figure 11.59: Crop water requirement, Onion (dry), part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
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Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
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Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.60: Crop water requirement, Onion (dry), part 2.

11.31 Papaya

Figure 11.61: Crop water requirement, Papaya, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
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Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.62: Crop water requirement, Papaya, part 2.

11.32 Peach

Figure 11.63: Crop water requirement, Peach, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
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Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.64: Crop water requirement, Peach, part 2.

11.33 Pear

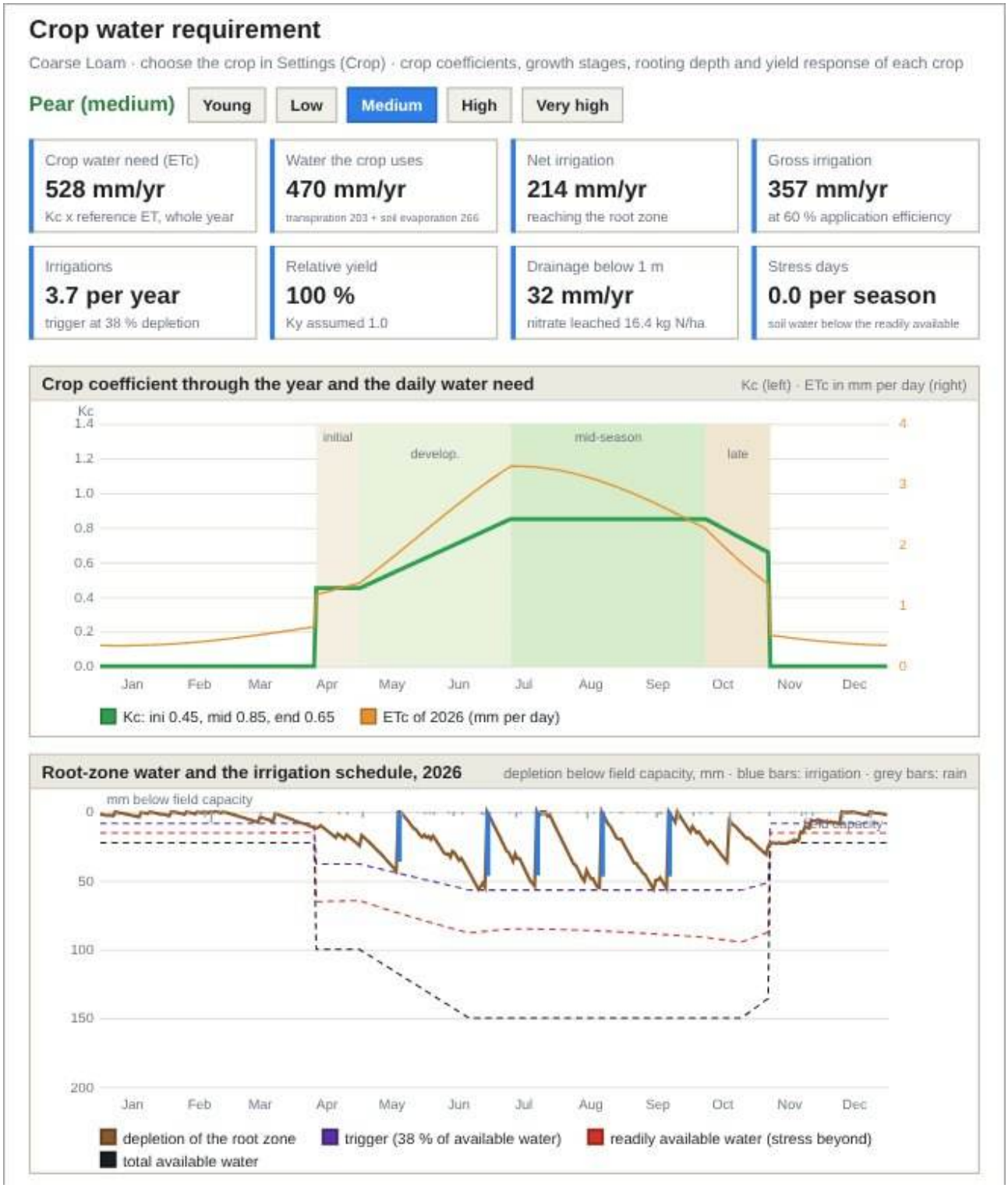

Figure 11.65: Crop water requirement, Pear, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
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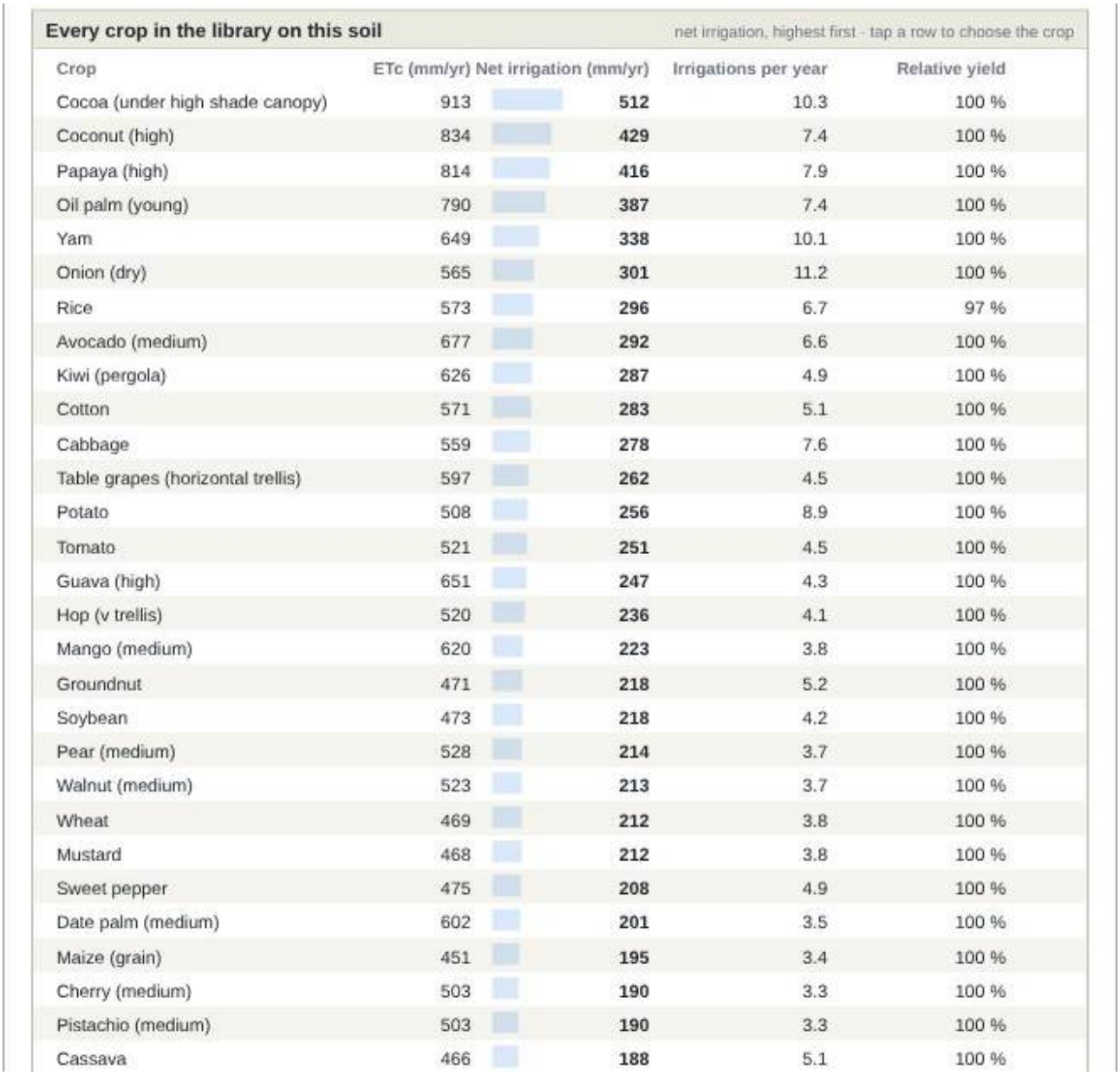
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.66: Crop water requirement, Pear, part 2.

11.34 Millet

Figure 11.67: Crop water requirement, Millet, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
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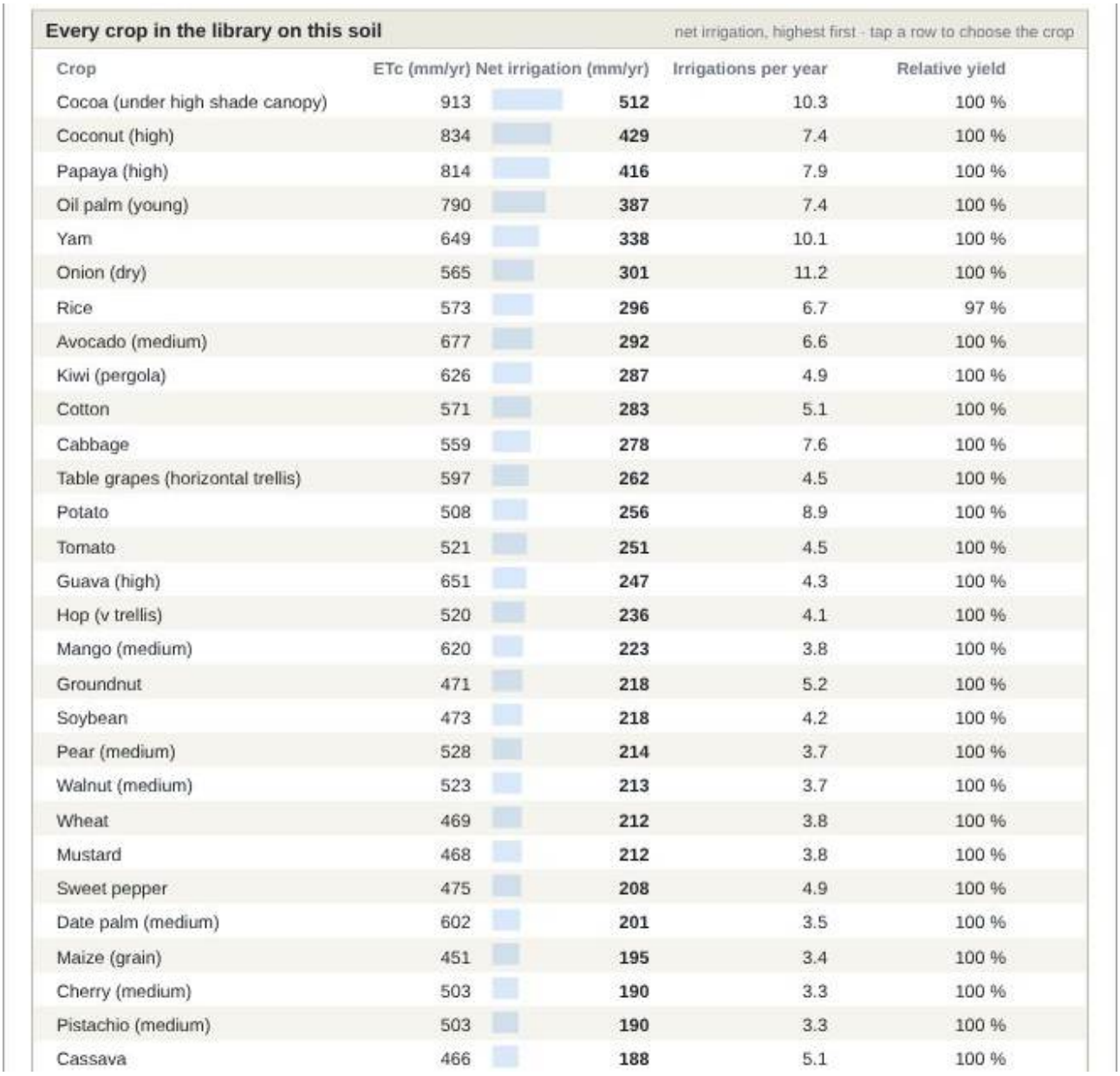
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.68: Crop water requirement, Millet, part 2.

11.35 Pecan

Figure 11.69: Crop water requirement, Pecan, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.70: Crop water requirement, Pecan, part 2.

11.36 Sweet pepper

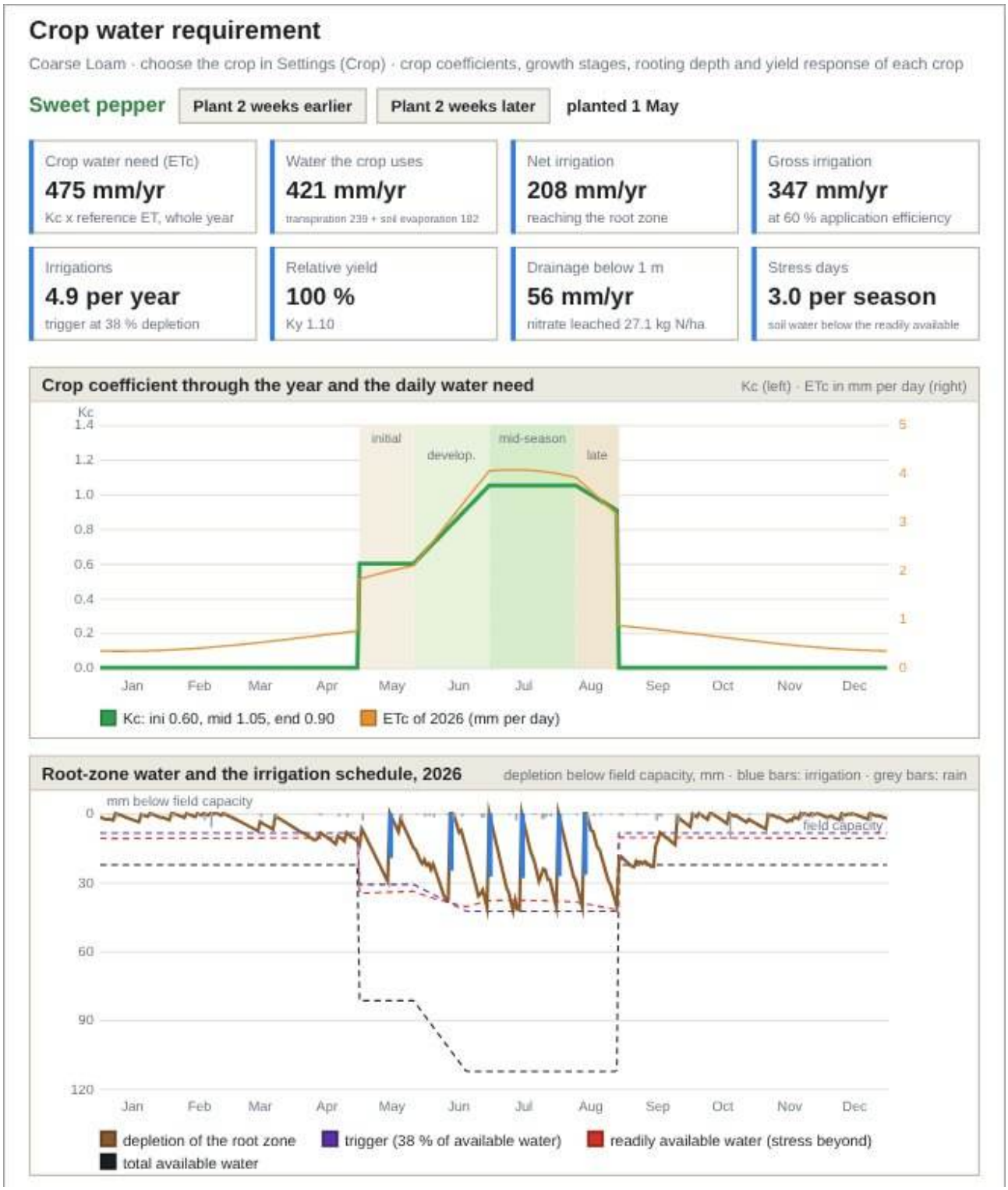


Figure 11.71: Crop water requirement, Sweet pepper, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
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Kiwi (pergola)	626	**287**	4.9	100 %
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Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
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Guava (high)	651	**247**	4.3	100 %
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Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
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Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.72: Crop water requirement, Sweet pepper, part 2.

11.37 Persimmon

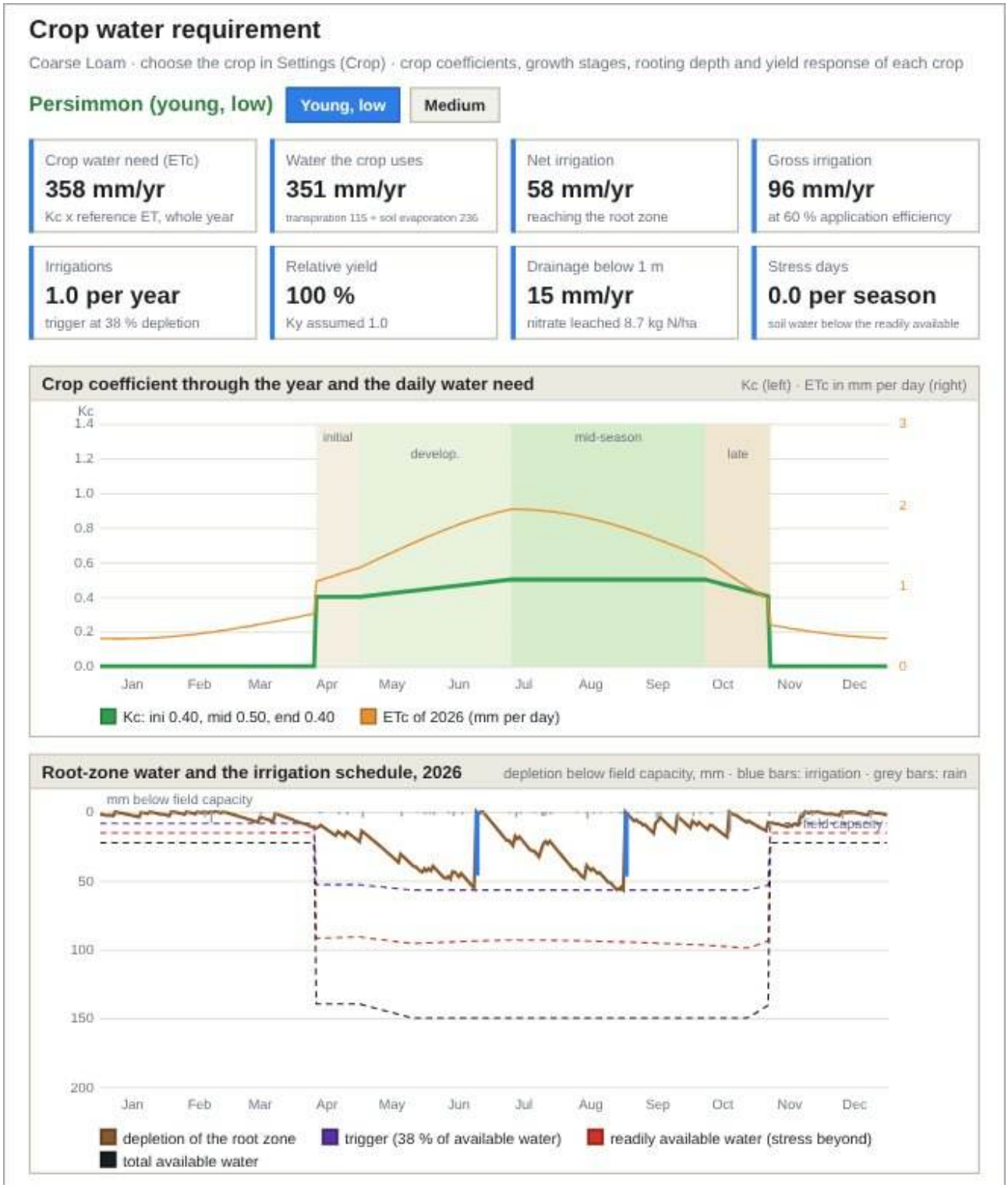

Figure 11.73: Crop water requirement, Persimmon, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
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Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
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Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.74: Crop water requirement, Persimmon, part 2.

11.38 Pistachio

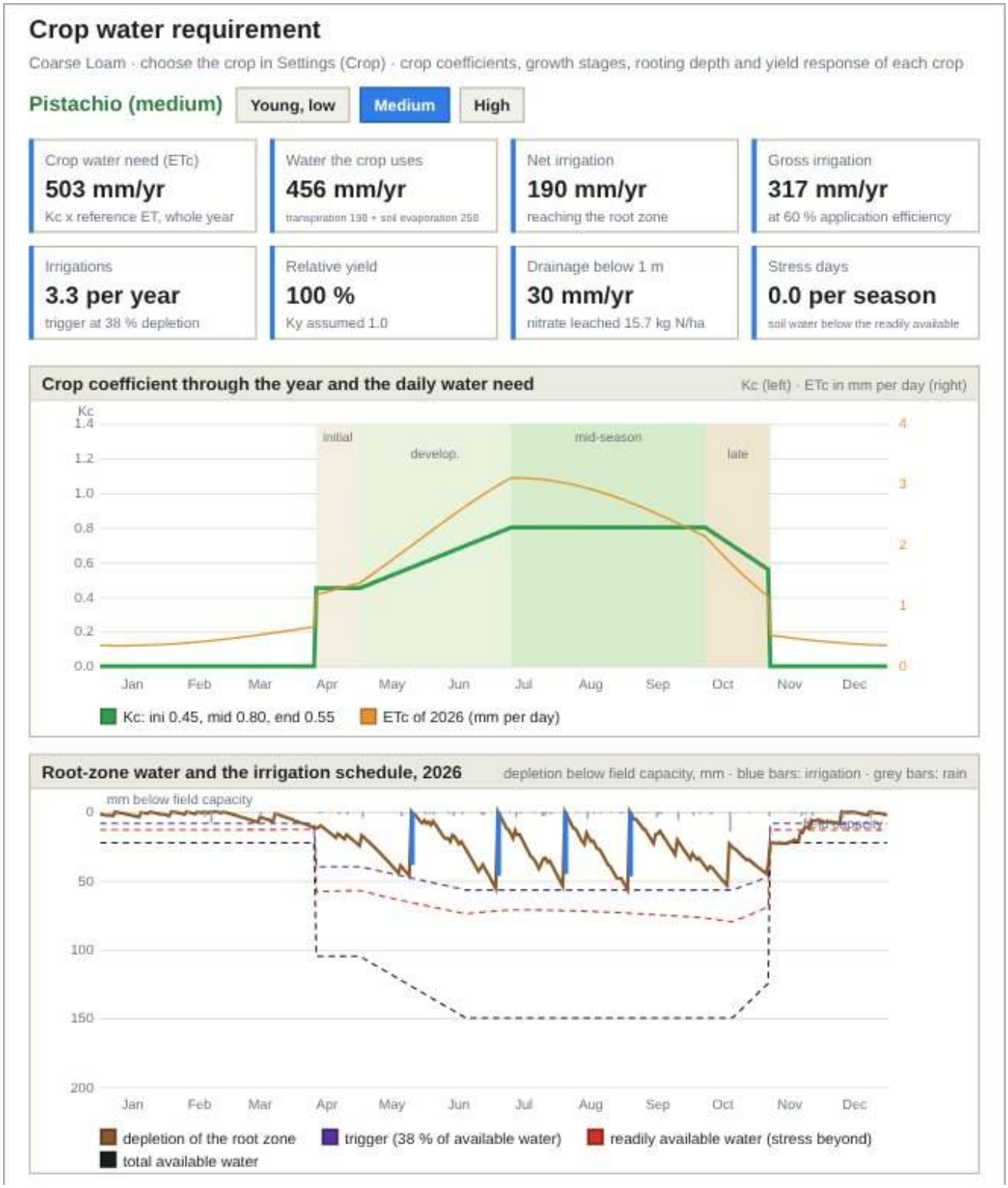
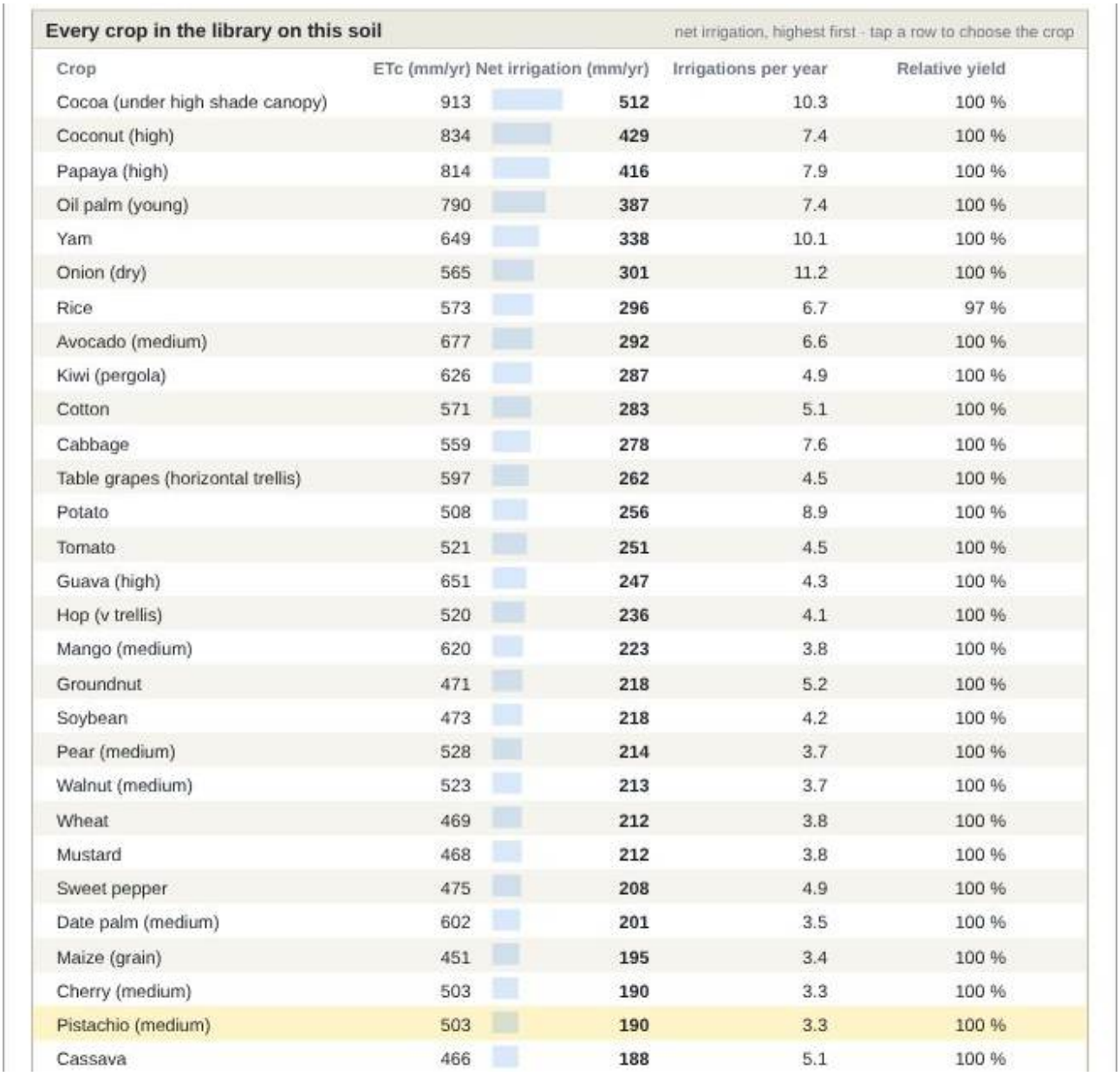


Figure 11.75: Crop water requirement, Pistachio, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.76: Crop water requirement, Pistachio, part 2.

11.39 Pomegranate

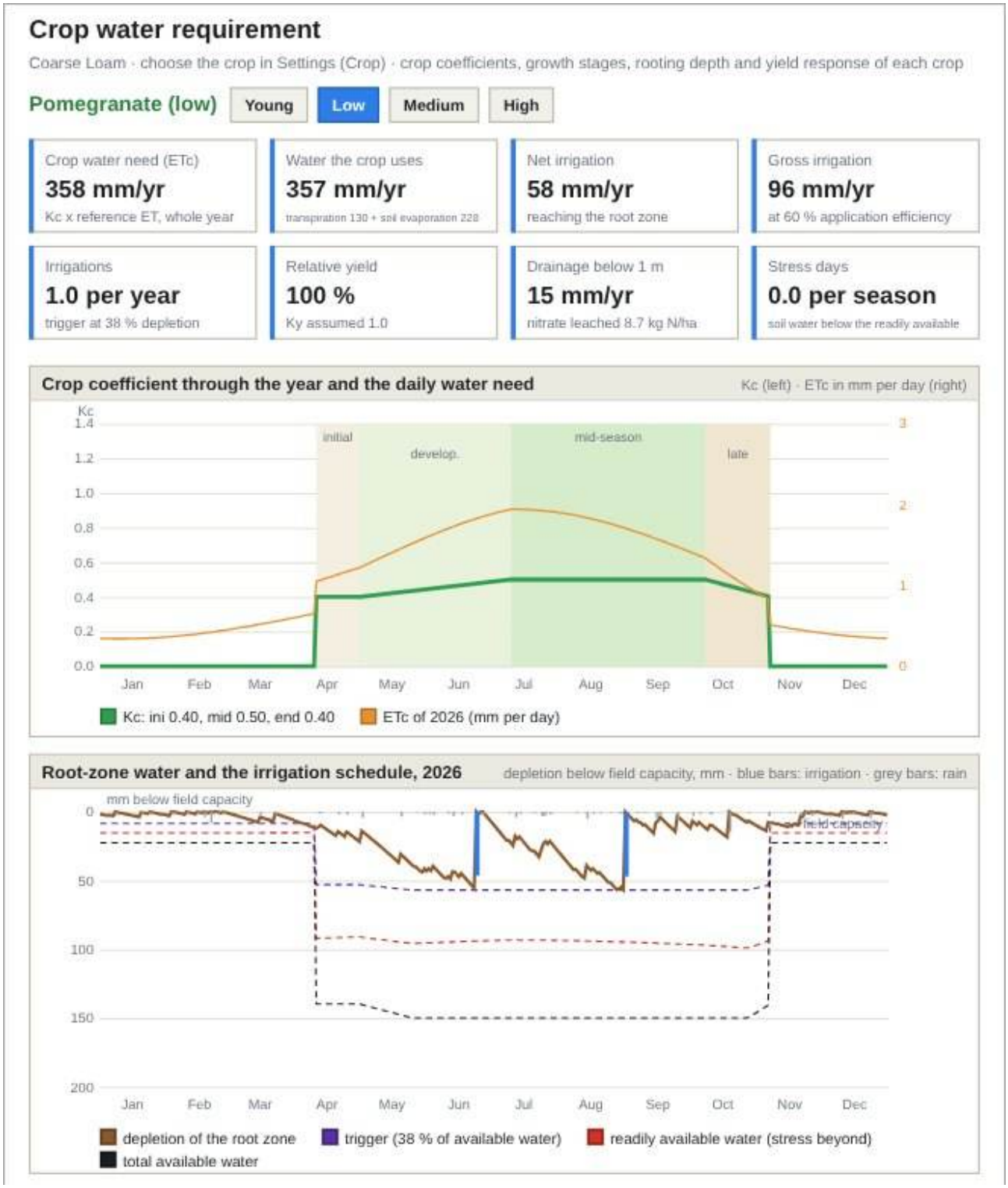


Figure 11.77: Crop water requirement, Pomegranate, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.78: Crop water requirement, Pomegranate, part 2.

11.40 Potato

Figure 11.79: Crop water requirement, Potato, part 1.

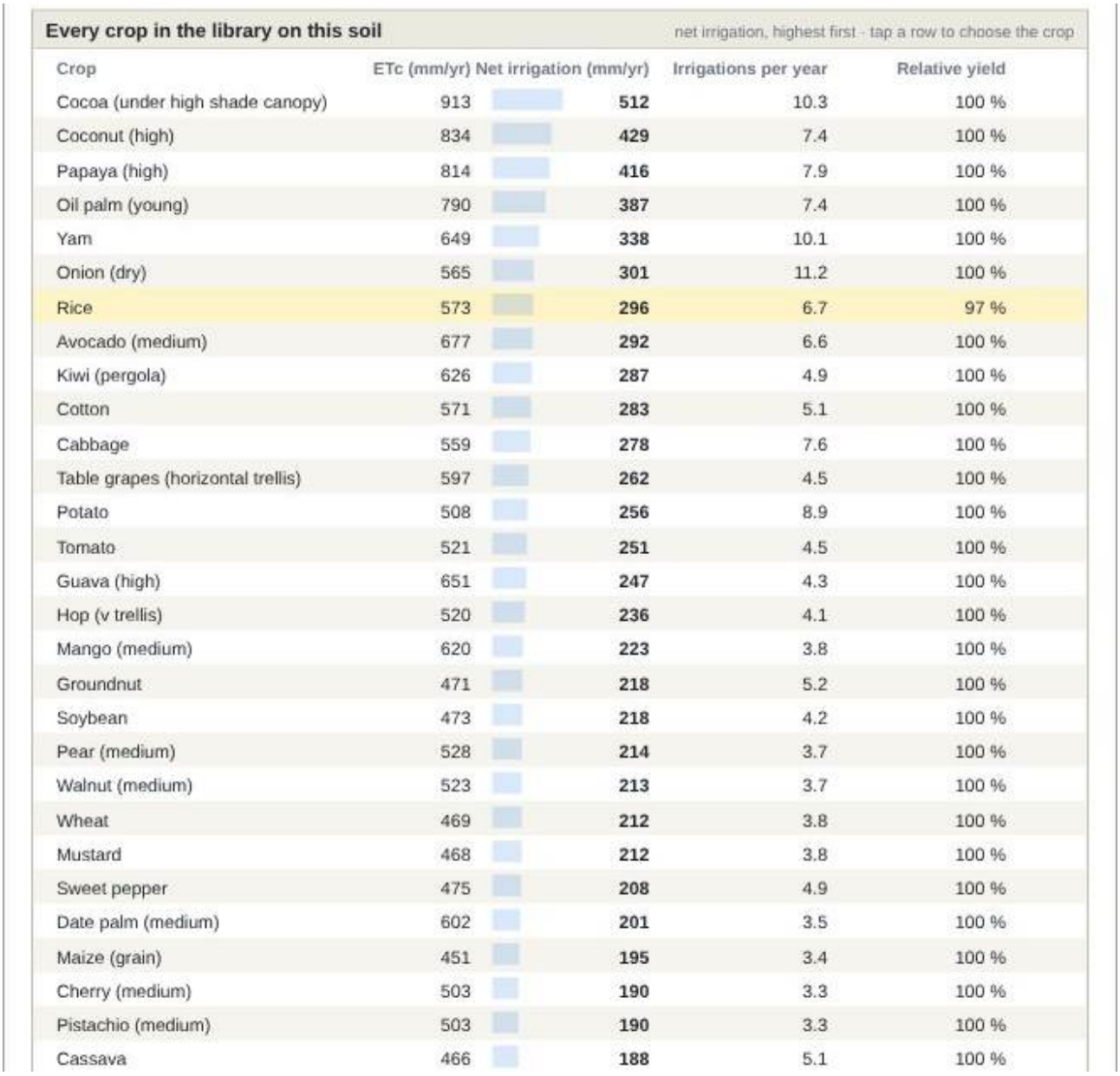
Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.80: Crop water requirement, Potato, part 2.

11.41 Rice

Figure 11.81: Crop water requirement, Rice, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.82: Crop water requirement, Rice, part 2.

11.42 Sorghum (grain)

Figure 11.83: Crop water requirement, Sorghum (grain), part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.84: Crop water requirement, Sorghum (grain), part 2.

11.43 Soybean

Figure 11.85: Crop water requirement, Soybean, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.86: Crop water requirement, Soybean, part 2.

11.44 Table grapes

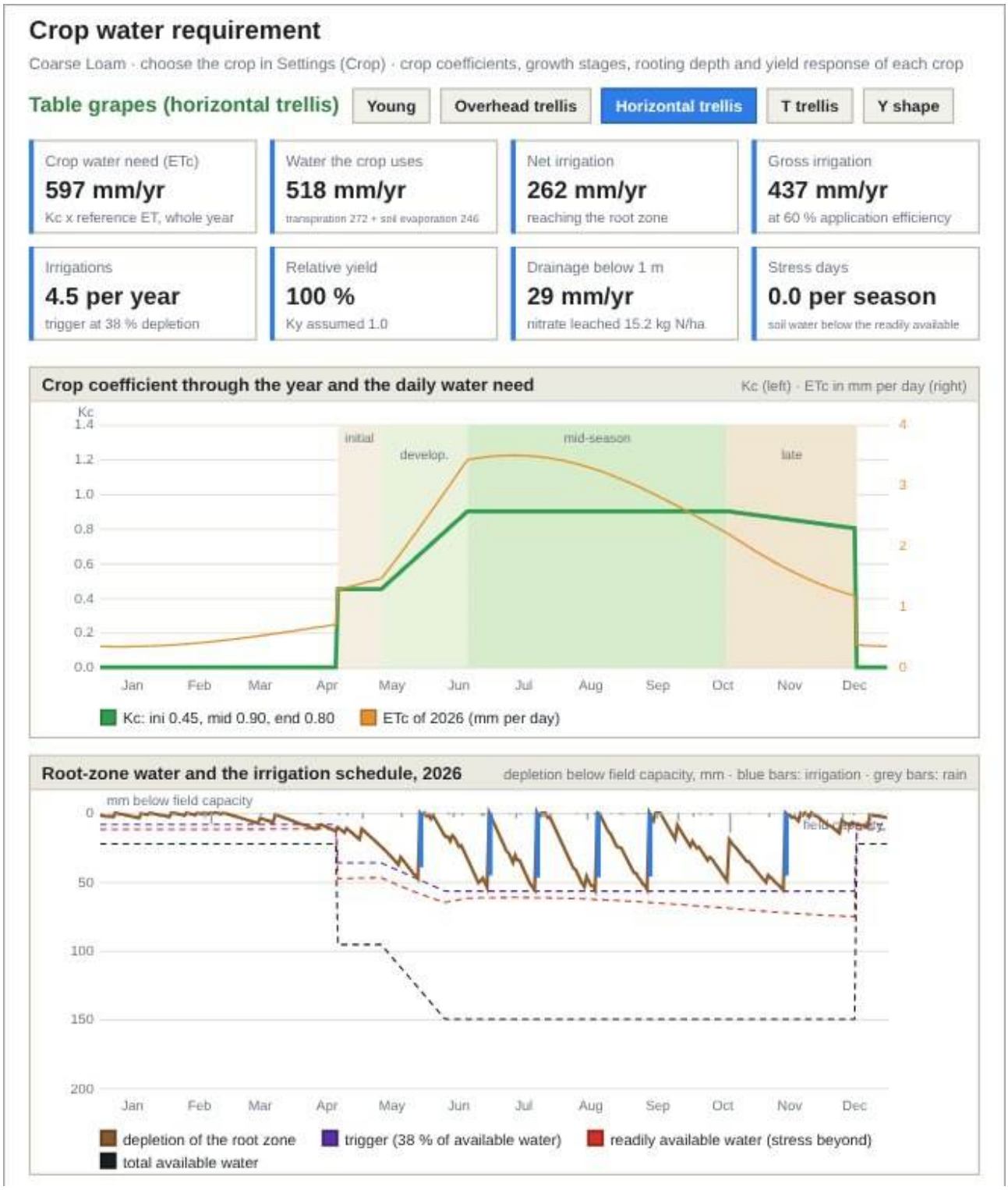


Figure 11.87: Crop water requirement, Table grapes, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.88: Crop water requirement, Table grapes, part 2.

11.45 Tomato

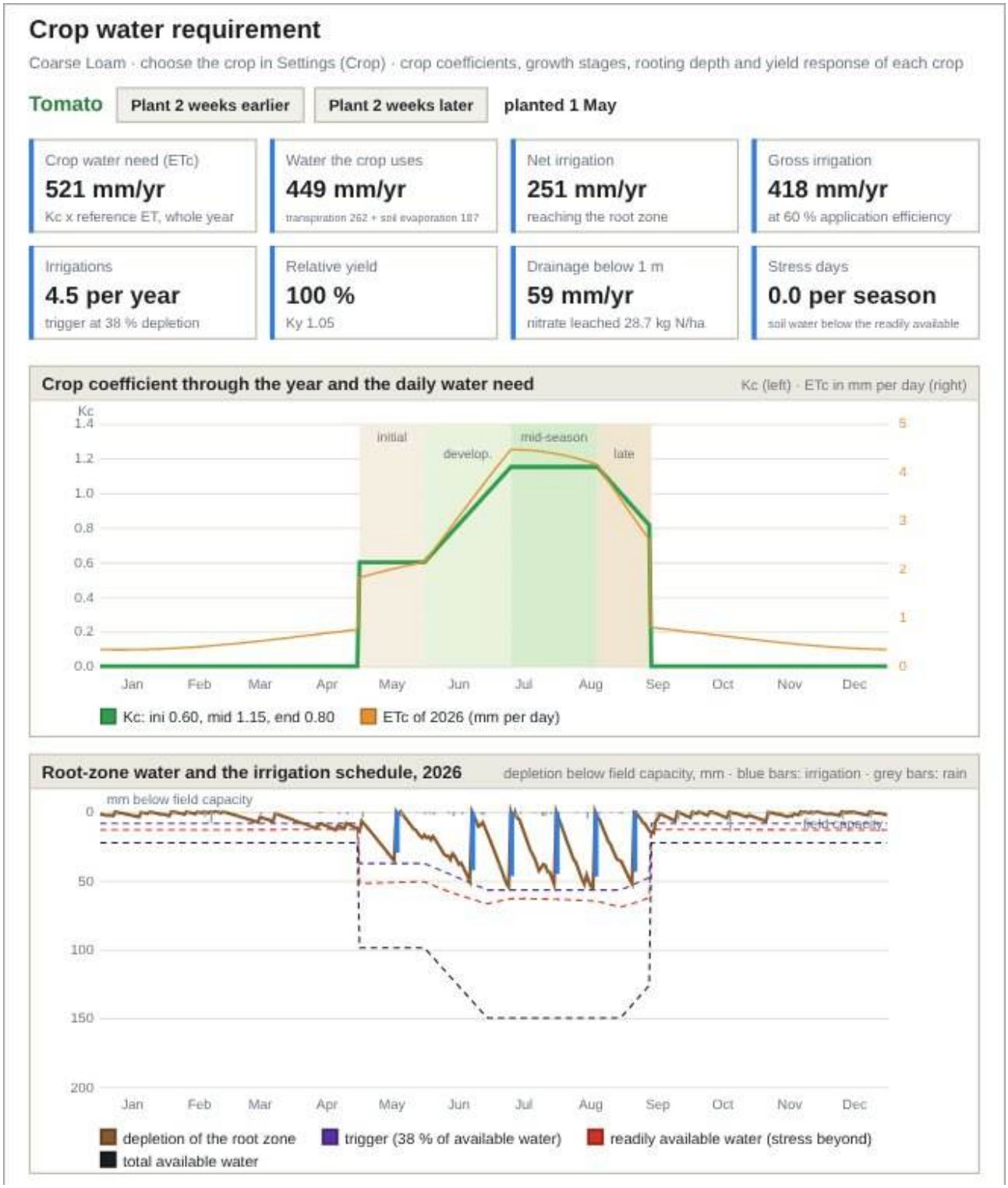


Figure 11.89: Crop water requirement, Tomato, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.90: Crop water requirement, Tomato, part 2.

11.46 Walnut

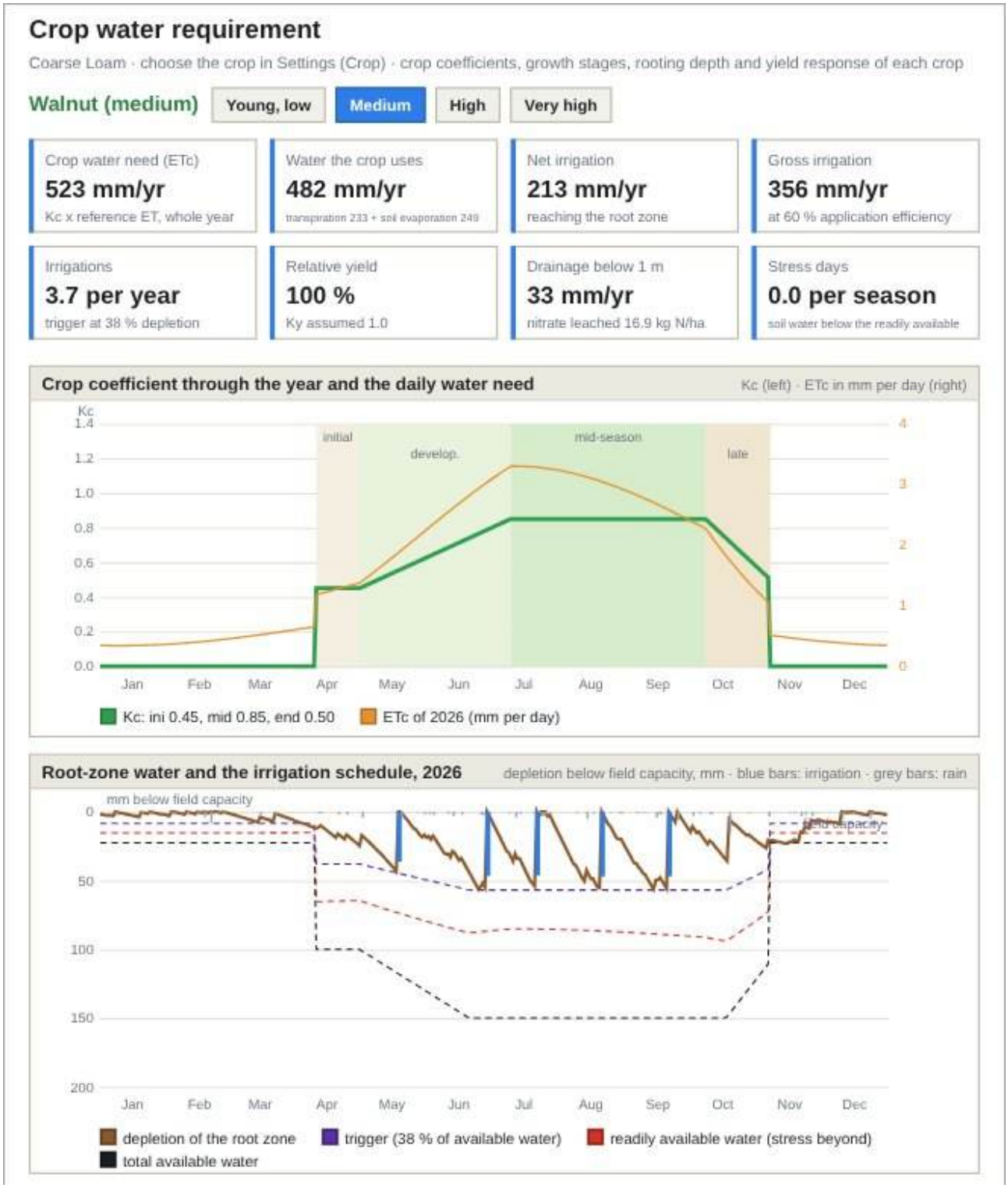


Figure 11.91: Crop water requirement, Walnut, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
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Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.92: Crop water requirement, Walnut, part 2.

11.47 Wheat

Figure 11.93: Crop water requirement, Wheat, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
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Cabbage	559	**278**	7.6	100 %
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Groundnut	471	**218**	5.2	100 %
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Pear (medium)	528	**214**	3.7	100 %
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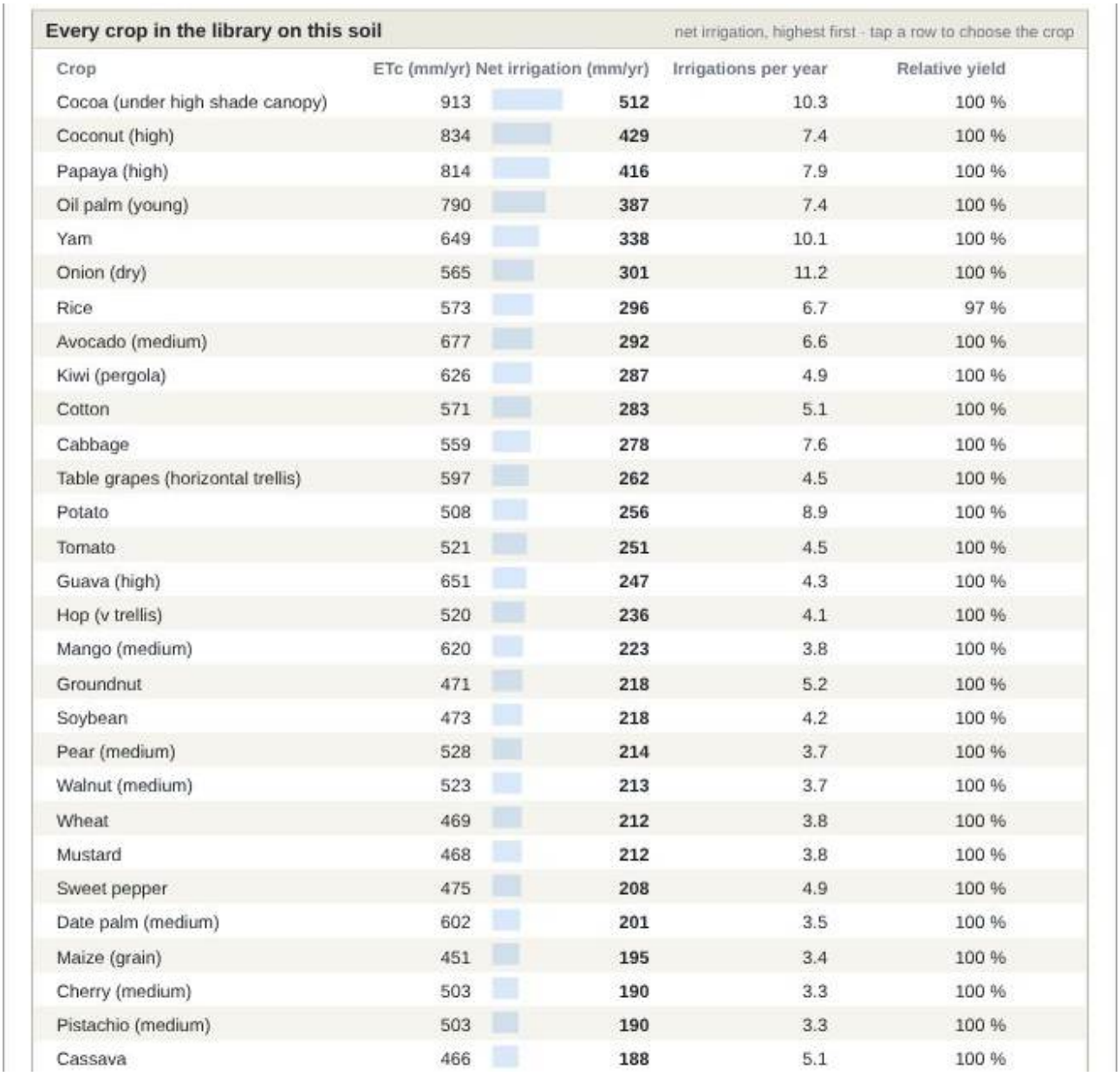
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.94: Crop water requirement, Wheat, part 2.

11.48 Wine grapes

Figure 11.95: Crop water requirement, Wine grapes, part 1.



Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
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Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.96: Crop water requirement, Wine grapes, part 2.

11.49 Yam

Figure 11.97: Crop water requirement, Yam, part 1.

Every crop in the library on this soil — net irrigation, highest first · tap a row to choose the crop

Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
Cocoa (under high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
Cotton	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %

Figure 11.98: Crop water requirement, Yam, part 2.